IHP

SG13G2

SG13G2 DC CV PMOS HV

Rev. 1.2 (190924)

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1 Documentation

1.1 Included Effects in Simulation Model

General	Date, Version	24.09.2019
	Processversion	SG13G2
	Simulator	ADS 2019, Spectre 18.1
	Simulation model	PSP 103.6
	Warning Boundaries	On

Statistic	Corner	x
	Process variations for Monte Carlo Simulation	e
	Mismatch	x
Transistor	DC behavior	x
	Drain diode	x
	Overlap cap.	x
	Junction cap.	x
	1/f Noise	x_1
	HF / Thermal Noise	-
	Substrate resistance (RF)	x_2
	Gate resistance (RF)	x_2
Temperature	-40°C, 25°C, 70°C, 125°C	x

- x measured and modelled
- x_1 parameters taken from SG13
- x_2 measurements are done from a different wafer run than DC model which leads to slight offsets
- e estimated and modelled
- o not included
- - not applicable

1.2 Time range measurement

The verification of the time range measurements is done by the standard IHP verification procedure.

2 Setup

2.1 Measurement Conditions

2.1.1 Measurement Conditions for Transistors

	Sweep	Start[V]	Stop[V]	No. Points	List or f(Vth)
idvg	vg	-500m	3.6	83	
	vb	0	-3.6	5	
	vd	100m	3.3	3	
	vs	0		1	
idvd	vd	0	3.6	37	
	vg	Lin f(Vth)	3.6	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	0		1	
	vs	0		1	
idvd_vbmin	vd	0	3.6	37	
	vg	Lin f(Vth)	3.6	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	-3.6		1	
	vs	0		1	
idvd_vth	vd	0	3.6	37	
	vg	List f(Vth)		3	Ref. Setup: idvg Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	0		1	
	vs	0		1	
idvd_vth_vbmin	vd	0	3.6	37	
	vg	List		3	Ref. Setup: idvg

		f(Vth)			Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	-3.6		1	
	vs	0		1	
RF idvg	vg	-500m	3.6	83	
	vd	100m	3.3	3	
RF idvd	vd	0	3.6	37	
	vg	0	3.6	13	
RF sdut	freq	100MEG	47.6G	31	
	vd	0	3.6	13	
	vg	0	3.6	13	

2.1.2 Measurement Conditions for Capacitance

	Sweep	Start[V]	Step[V]	Stop[V]	High Node	Low Node
Junction Bulk-Drain	1 VHI	5	-200m	-200m	B	D
Junction Bulk-Source	1 VHI	1	-20m	-300m	B	S
Oxide	1 VHI	-3.6	100m	3.6	B	G
Overlap	1 VHI	3.6	-100m	-3.6	G	D

2.2 Devices

2.2.1 Setup of DC Test Devices

Name	W[um]	L[um]	AD[um²]	AS[um²]	PD[um]	PS[um]	NRD	NRS	NF	Comment	Size Category
W10u0_L10u0_S562_3	10	10	3.4	3.4	20.7	20.7	0	0	1		Large

W0u3_L10u0_S555_4	300 m	10	102 m	102 m	1.28	1.28	0	0	1		Narrow
W10u0_L0u35_S561_3	10	350 m	3.4	3.4	20.7	20.7	0	0	1		Short
W0u3_L0u35_S565_3	300 m	350 m	102 m	102 m	1.28	1.28	0	0	1		Small
W10u0_L0u4_T356_S387_1	10	400 m	0	0	0	0	0	0	1		L Scale
W10u0_L3u0_S562_1	10	3	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L5u0_S562_2	10	5	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L0u5_S561_4	10	500 m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W05u0_L10u0_S563_1	5	10	1.7	1.7	10.7	10.7	0	0	1		W Scale
W0u6_L10u0_S564_5	600 m	10	204 m	204 m	1.88	1.88	0	0	1		W Scale
W02u0_L10u0_S563_5	2	10	680 m	680 m	4.68	4.68	0	0	1		W Scale
W0u6_L1u2_S564_4	600 m	1.2	204 m	204 m	1.88	1.88	0	0	1		LW Scale
W0u3_L0u5_S565_4	300 m	500 m	102 m	102 m	1.28	1.28	0	0	1		LW Scale
W0u3_L1u2_S565_5	300 m	1.2	102 m	102 m	1.28	1.28	0	0	1		LW Scale
W0u6_L0u35_S564_2	600 m	350 m	204 m	204 m	1.88	1.88	0	0	1		LW Scale
W0u3_L0u4_T356_S385M_3	300 m	400 m	0	0	0	0	0	0	1		LW Scale
W02u0_L1u2_S563_3	2	1.2	680 m	680 m	4.68	4.68	0	0	1		LW Scale
W0u6_L0u5_S564_3	600 m	500 m	204 m	204 m	1.88	1.88	0	0	1		LW Scale
W05u0_L5u0_S562_5	5	5	1.7	1.7	10.7	10.7	0	0	1		LW Scale
W02u0_L3u0_S563_4	2	3	680 m	680 m	4.68	4.68	0	0	1		LW Scale
W10u0_L0u33_S561_2	10	330 m	3.4	3.4	20.7	20.7	0	0	1		Additional
W10u0_L0u3_S561_1	10	300	3.4	3.4	20.	20.	0	0	1		Addition

		m			7	7					al
W0u3_L0u3_S565_1	300 m	300 m	102 m	102 m	1.28	1.28	0	0	1		Additional
W10u0_L0u4_S387_1	10	400 m	3.4	3.4	20.7	20.7	0	0	1		Additional
W0u6_L0u33_S564_1	600 m	330 m	204 m	204 m	1.88	1.88	0	0	1		Additional
W05u0_L0u33_S562_4	5	330 m	1.7	1.7	10.7	10.7	0	0	1		Additional
W0u15_L0u33_S555_5	150 m	330 m	101 m	101 m	1.34	1.34	0	0	1		Additional
W02u0_L0u33_S563_2	2	330 m	680 m	680 m	4.68	4.68	0	0	1		Additional
W0u3_L0u33_S565_2	300 m	330 m	102 m	102 m	1.28	1.28	0	0	1		Additional

2.2.2 Setup of CV Test Devices

Name	W[um]	L[um]	NF	AD[um²]	AS[um²]	PD[um]	PS[um]	Comment	Size Category
Junction Capacitance Bulk - Drain									
CJBDarea_S351B_5			1	37.5K		850		B-High, S/D-Low	BD Area
CJBDperim_S351B_6			1	28.6K		102K		B-High, S/D-Low	BD Perim
Junction Capacitance Bulk - Source									
CJBSarea_S351B_5			1	0		0		B-High, S/D-Low	BS Area
CJBSperim_S351B_6			1	0		0		B-High, S/D-	BS Perim

								Low	
Oxide Capacitance									
Cox_S342K_4	160	250	1					B-High, G-Low	Oxide
Overlap Capacitance									
Cgds1_S351B_9	6K	300m	600	1.14K	1.14K	6.25K	6.25K	G-High, S/D-Low, B+Sub-GND	Overlap GDS
Cgdsb1_S351B_9	6K	300m	600	1.14K	1.14K	6.25K	6.25K	G-High, S/D+B-Low	Overlap GDSB
Cgds2_S351B_10	400	10	40	79.8	79.8	436	436	G-High, S/D-Low, B+Sub-GND	Overlap 1 GDS
Cgdsb2_S351B_10	400	10	40	79.8	79.8	436	436	G-High, S/D+B-Low	Overlap 1 GDSB

3 Corner Devices

3.1 Target Values vs. Simulated Values

3.1.1 W = 10.00u, L = 400.0n, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	-710m	-706m	0.59	-650m	-656m	-0.88	-590m	-594m	-0.67
Ron*W	-	-	-	4.8	4.75	-0.96	-	-	-
Ion/W	-190	-190	0	-235	-236	-0.24	-280	-281	-0.24
log(Ioff/W)	-13.5	-12.4	3.12	-12.5	-12.3	2.39	-11.5	-12.1	2.09

3.1.2 W = 10.00u, L = 10.00u, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	-740m	-739m	0.19	-690m	-688m	0.28	-640m	-637m	0.43
Ion/W	-7.4	-7.37	0.37	-8	-7.98	0.31	-8.6	-8.61	0

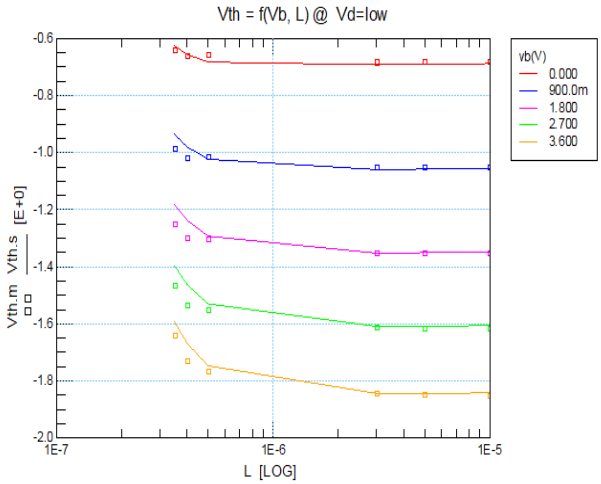
3.1.3 W = 300.0n, L = 400.0n, NF = 1

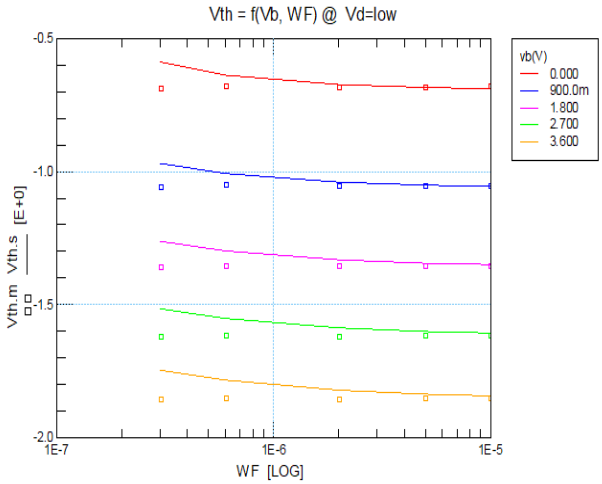
	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	-720m	-719m	0	-640m	-645m	-0.79	-560m	-563m	-0.55
Ion/W	-220	-221	-0.32	-265	-264	0.23	-310	-311	-0.43

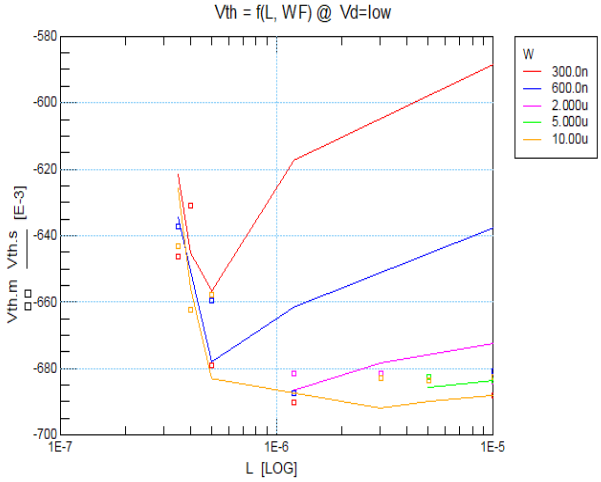
4 DC Transistors

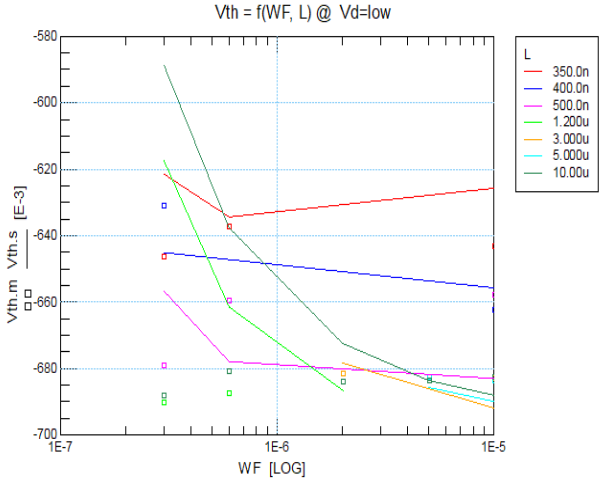
4.1 Summarized Results

4.1.1 Threshold Voltage Linear Region

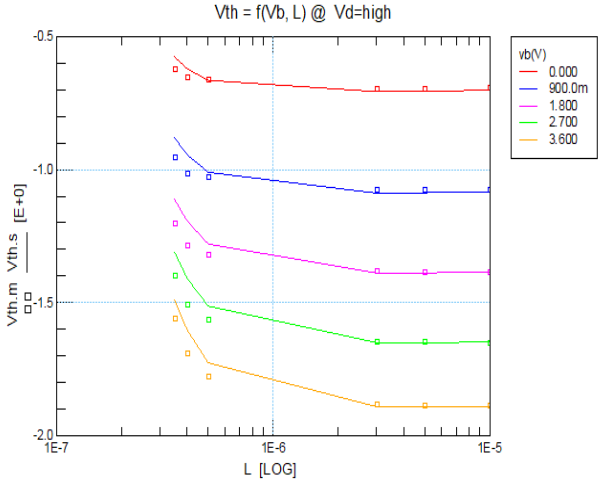


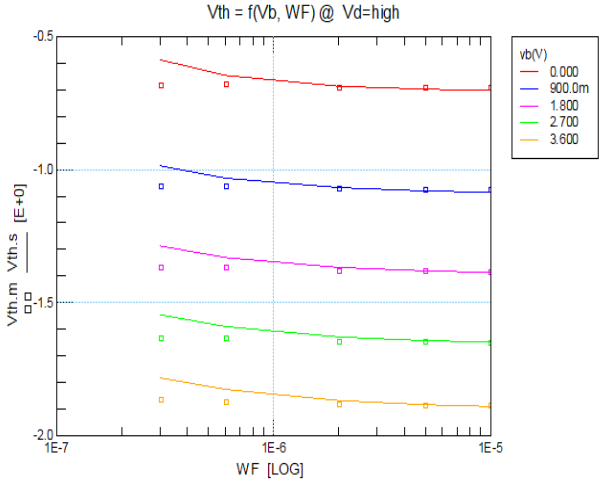


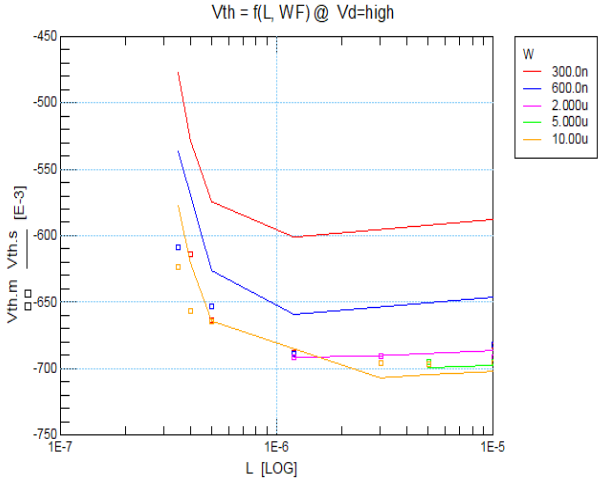


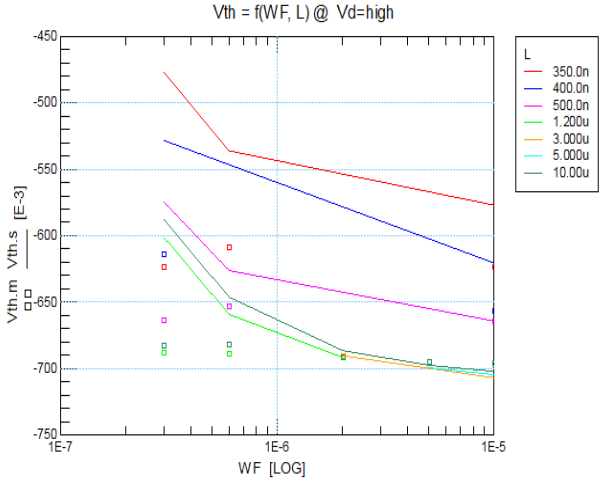


4.1.2 Threshold Voltage in Saturation

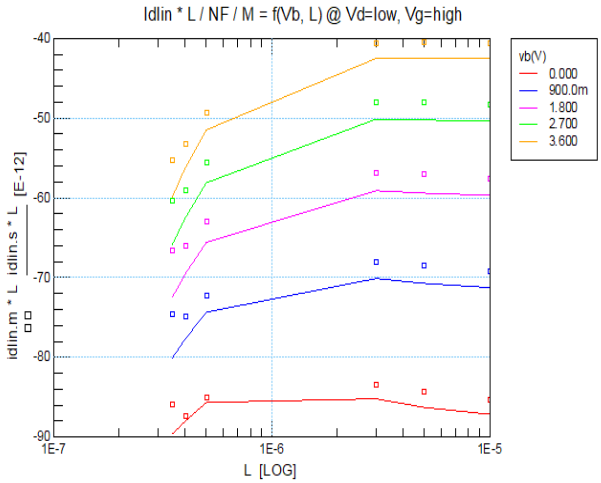


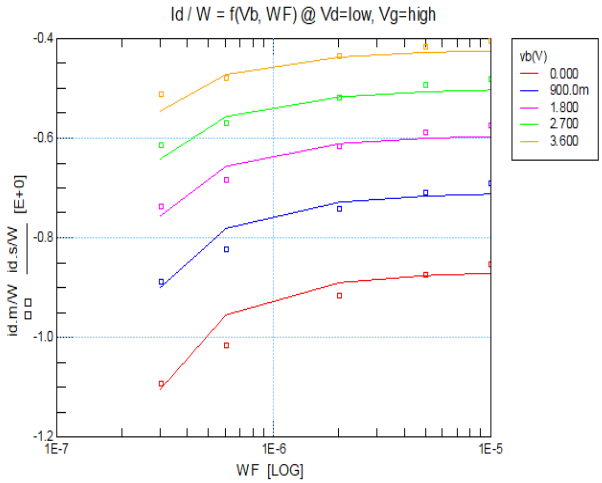


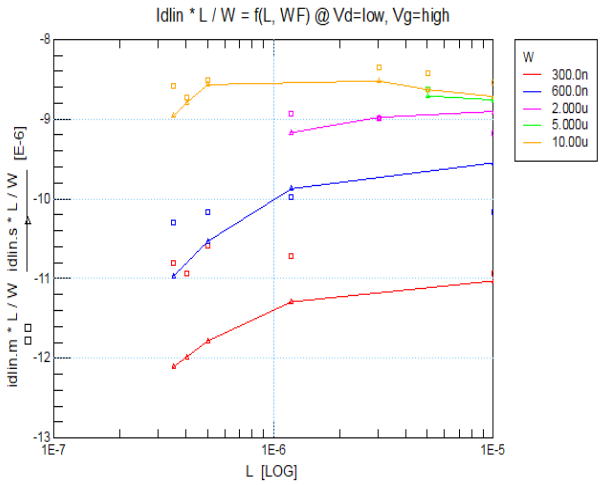


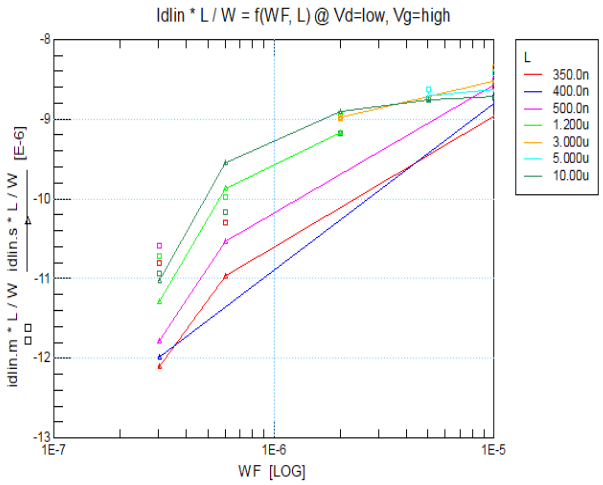


4.1.3 Maximum Drain Current in Linear Region

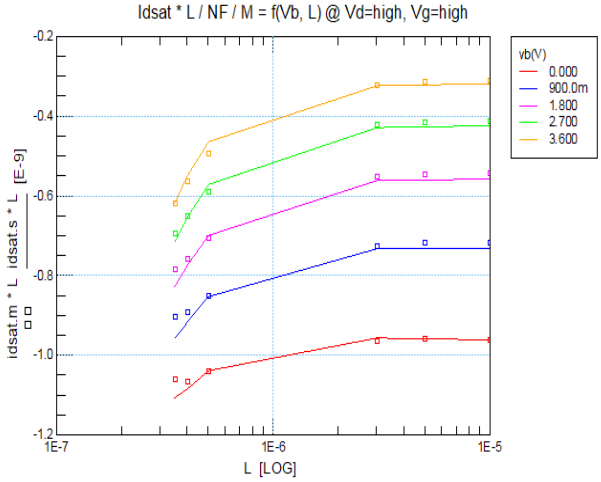


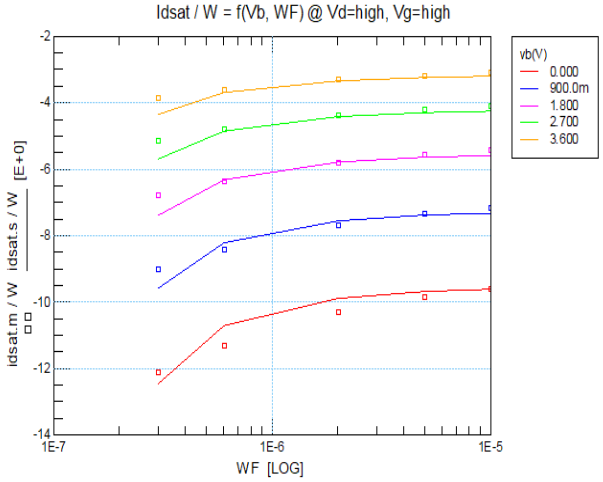


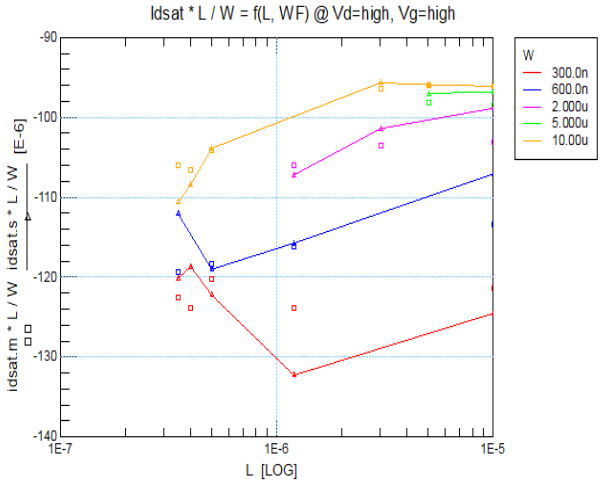


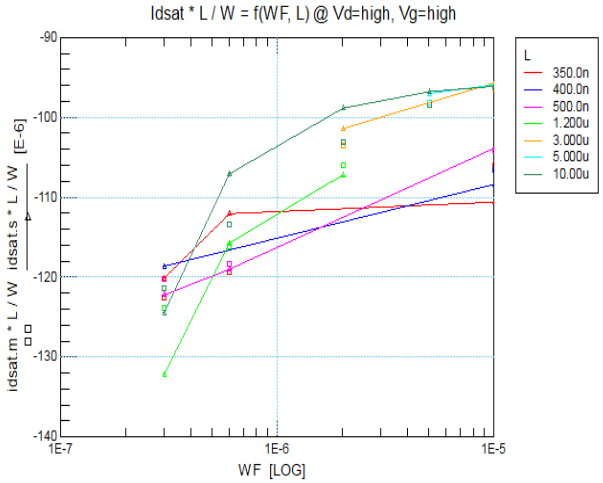


4.1.4 Maximum Drain Current in Saturation Region





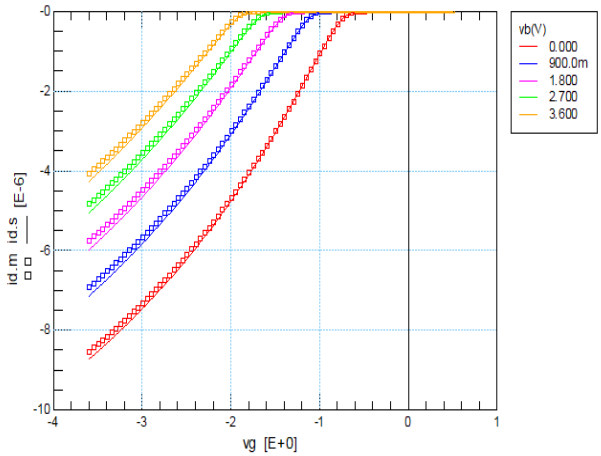




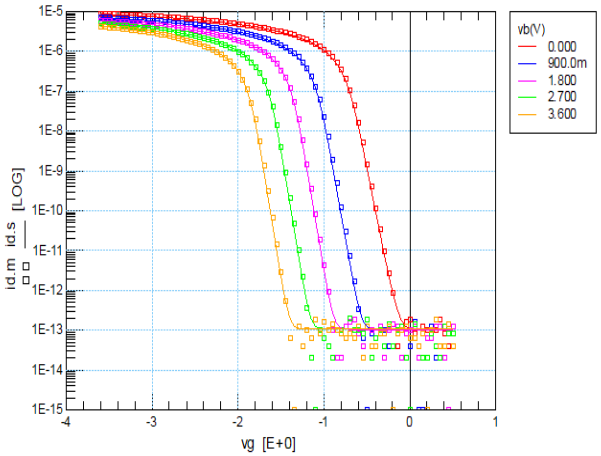
4.2 W10u0_L10u0_S562_3

idvg, Vd = -0.1V, T = 27°C

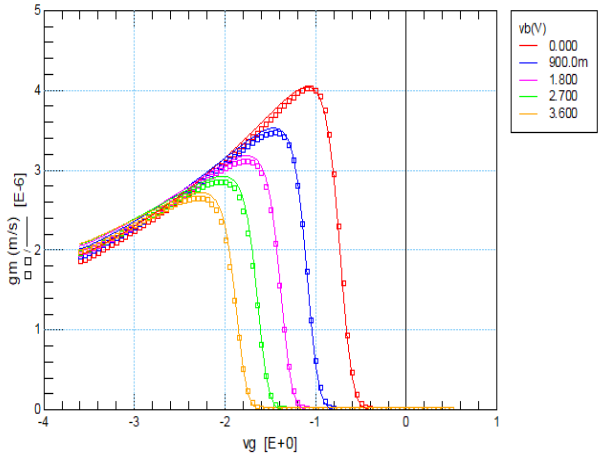
Id = f(Vg, Vb) @ Vd=low



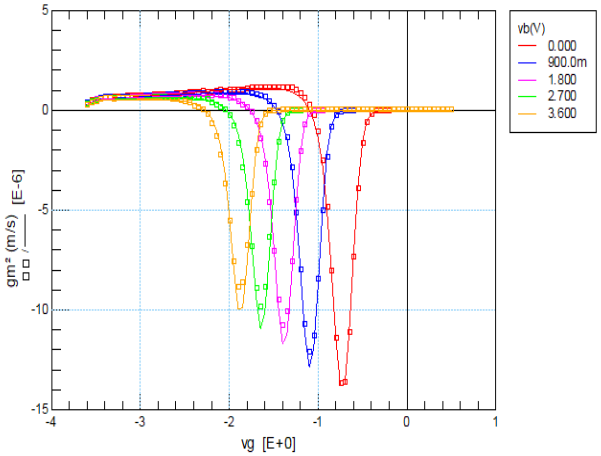
Id = f(Vg, Vb) @ Vd=low



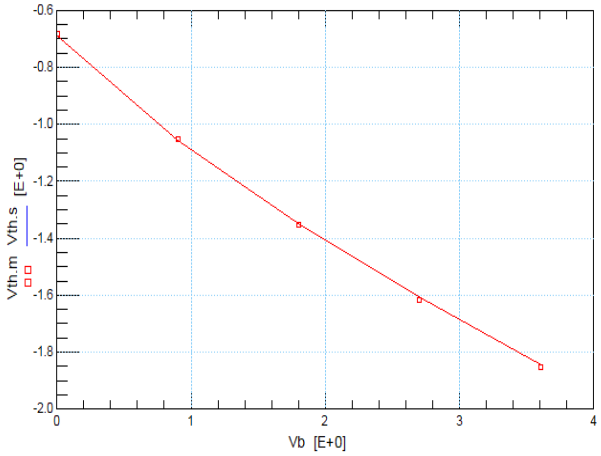
gm = f(Vg, Vb) @ Vd=low



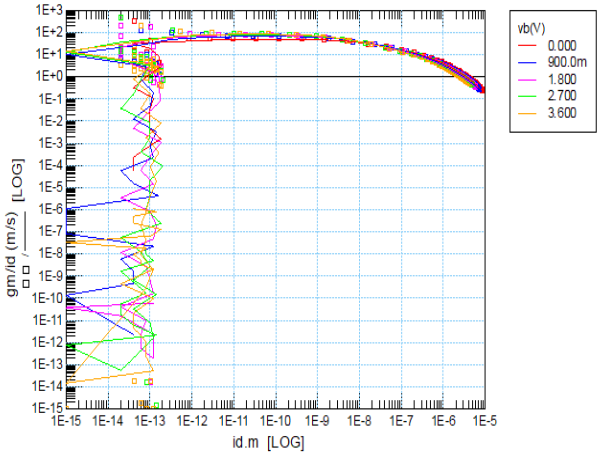
gm² = f(Vg, Vb) @ Vd=low

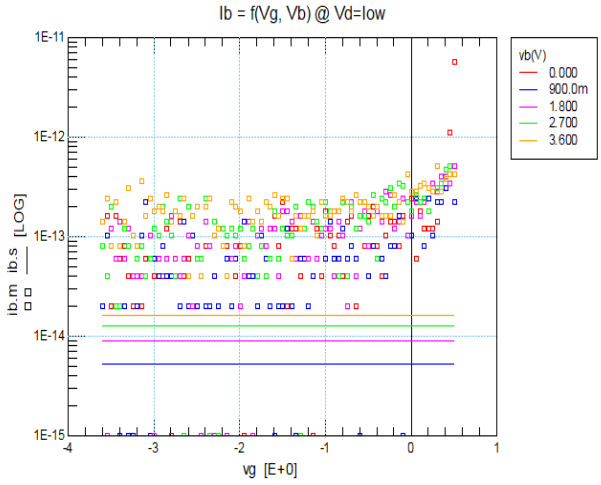


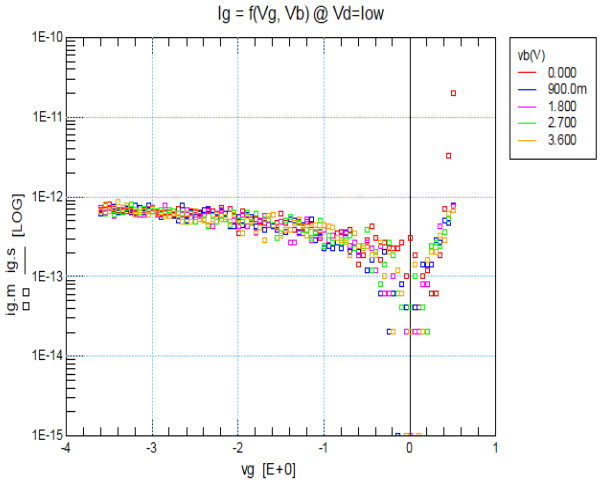
Vth @ Vd=low



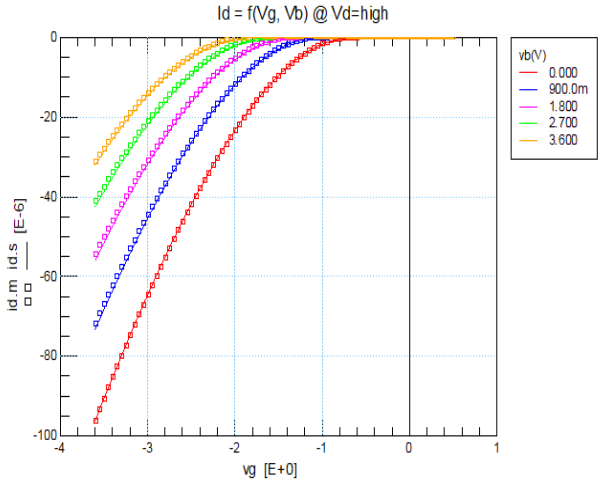
gm/id=f(id.m) @ Vd=low

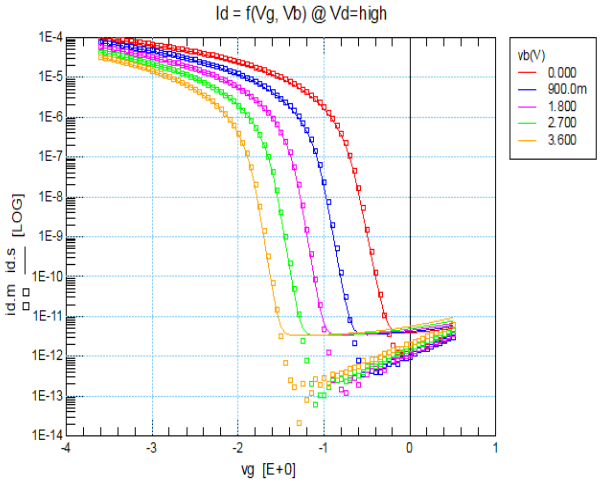


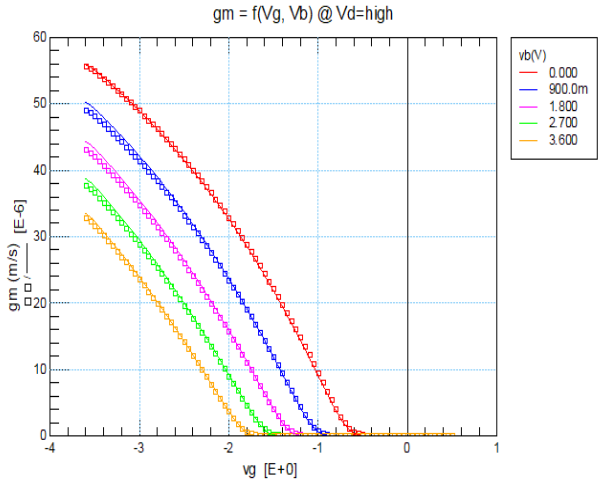


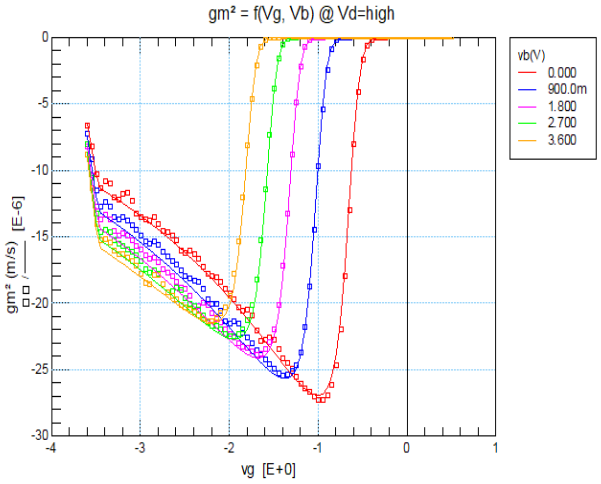


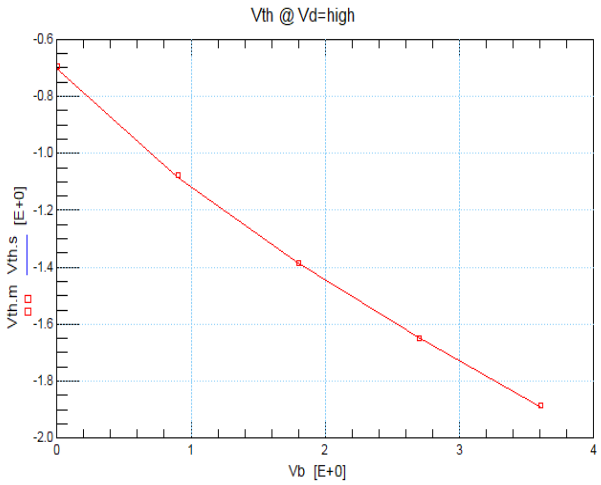
idvg, Vd = -3.3V, T = 27°C

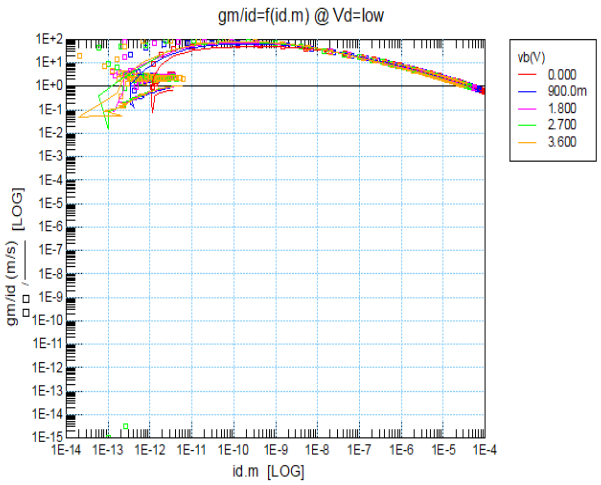


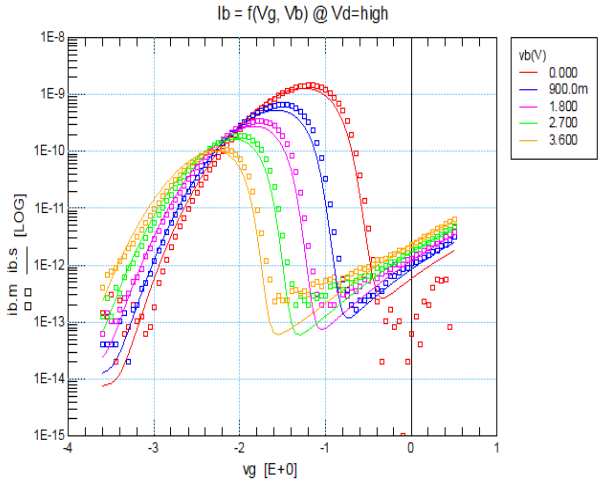


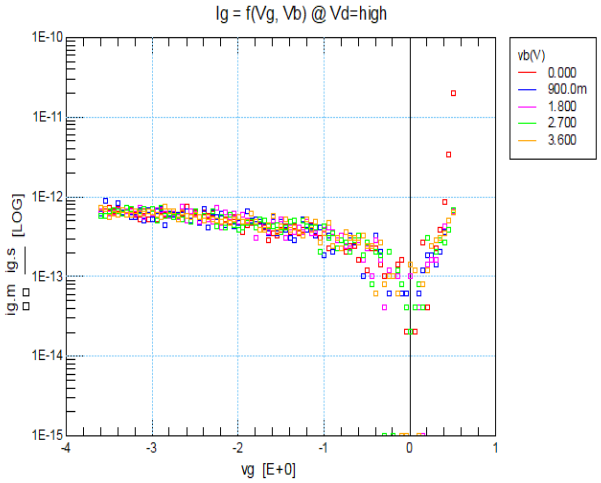




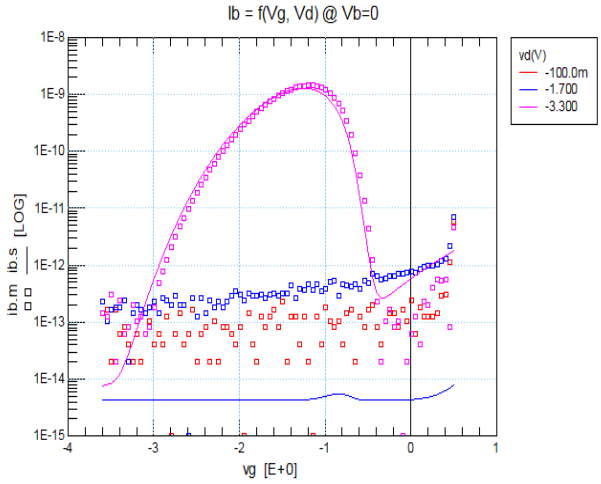


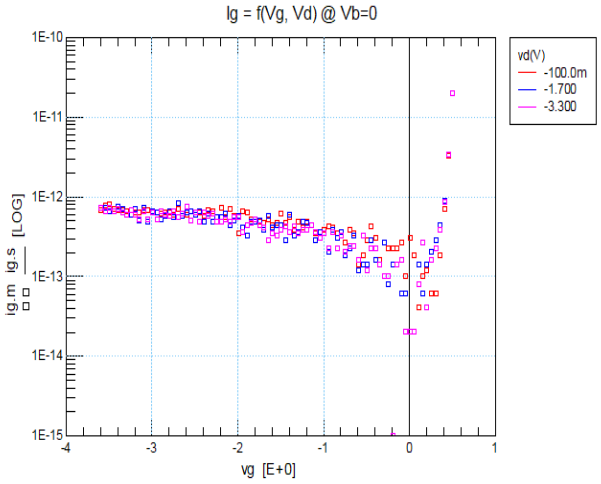




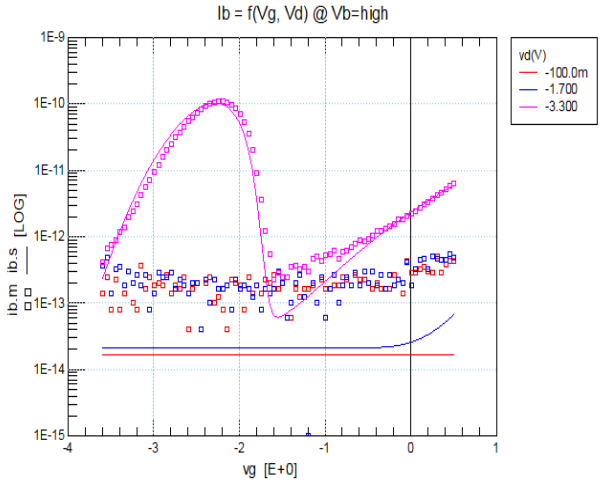


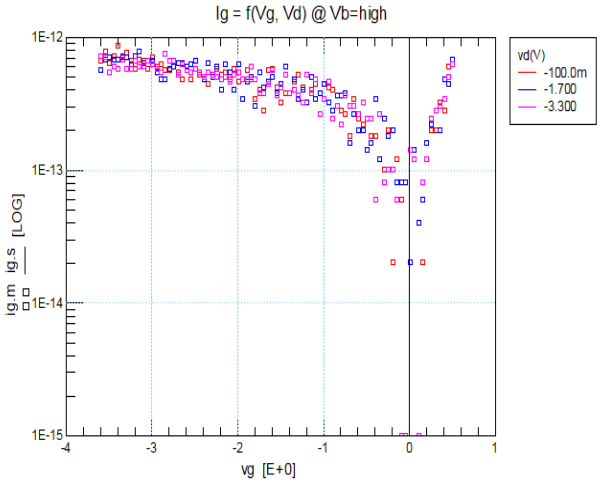
idvg, Ib,Ig, Vb = 0V, T = 27°C



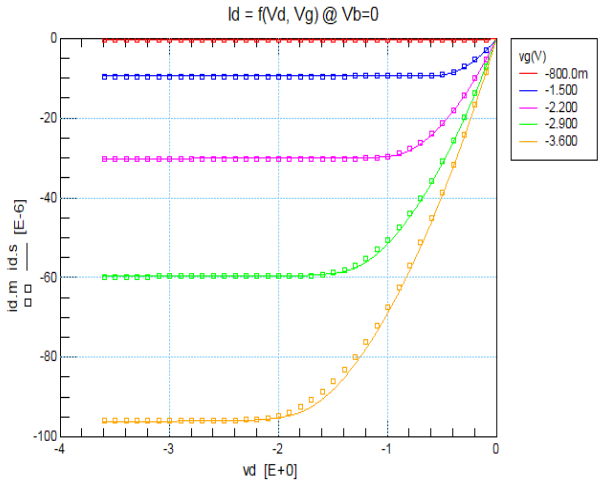


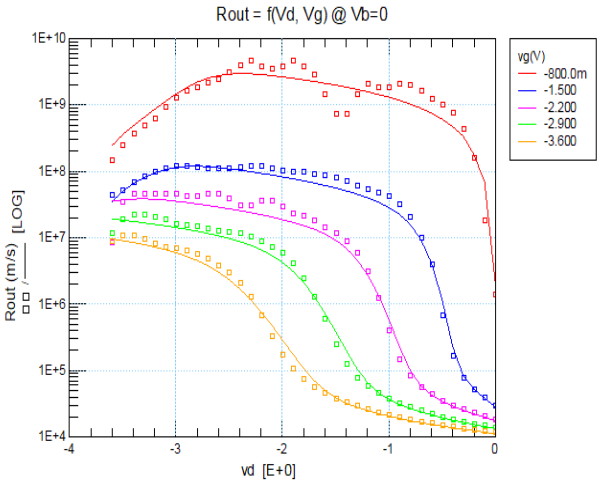
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

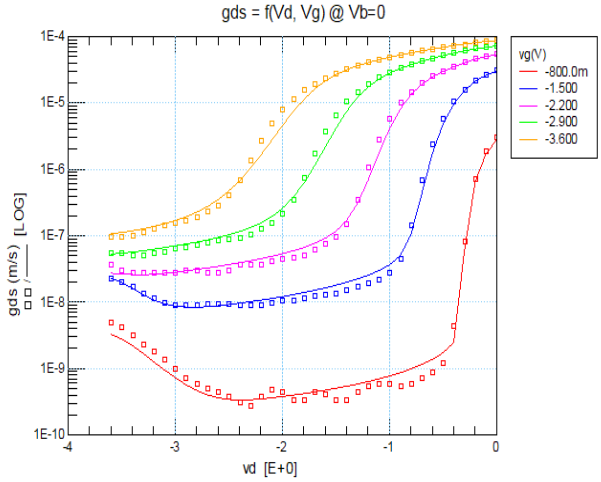




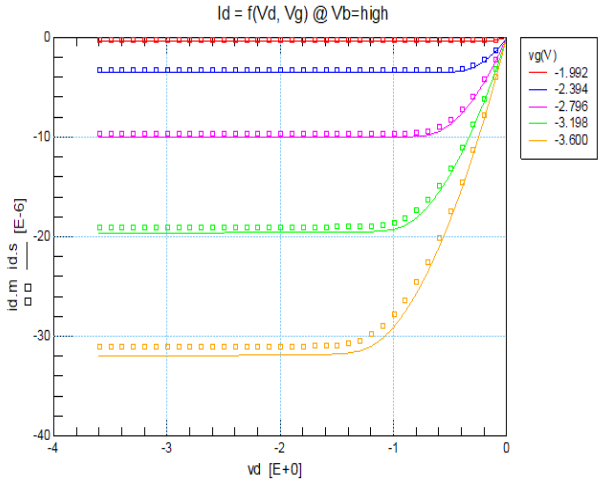
idvd, Vb = 0V, T = 27°C

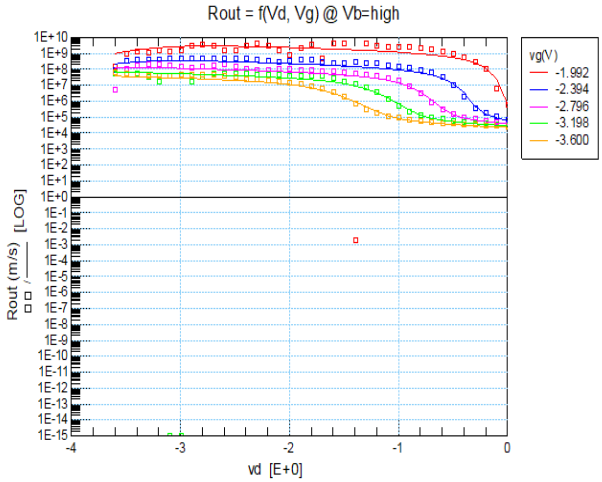


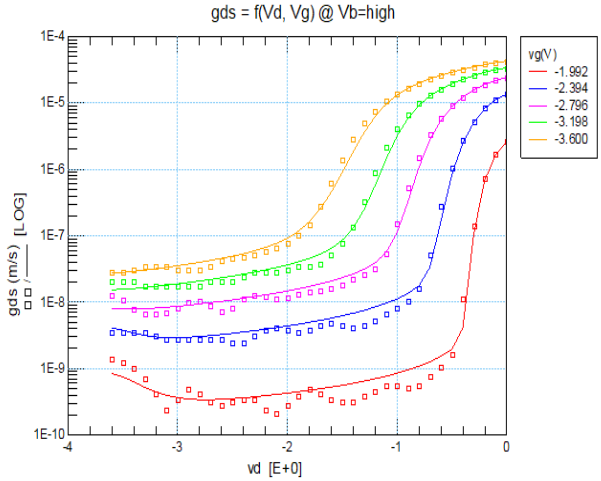




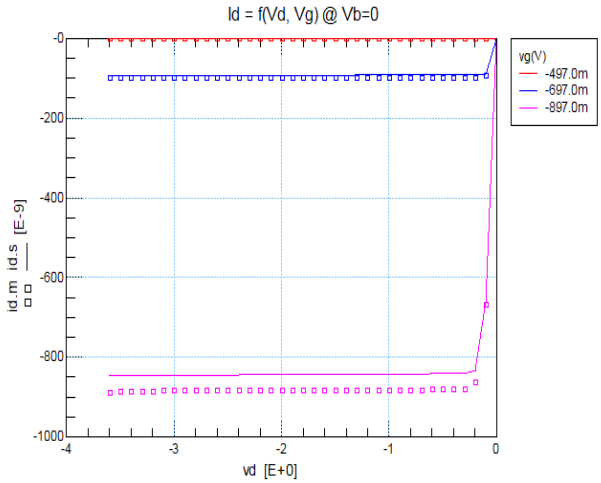
idvd_vbmin, Vb = 3.6V, T = 27°C

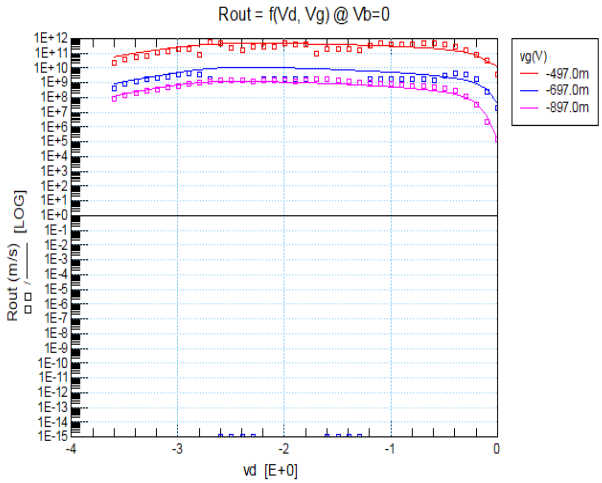


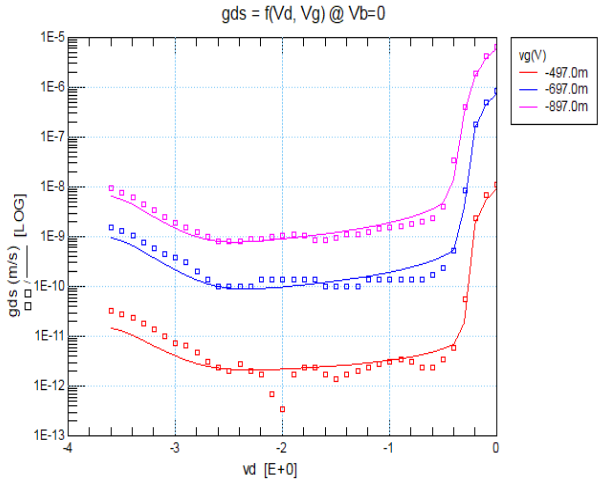




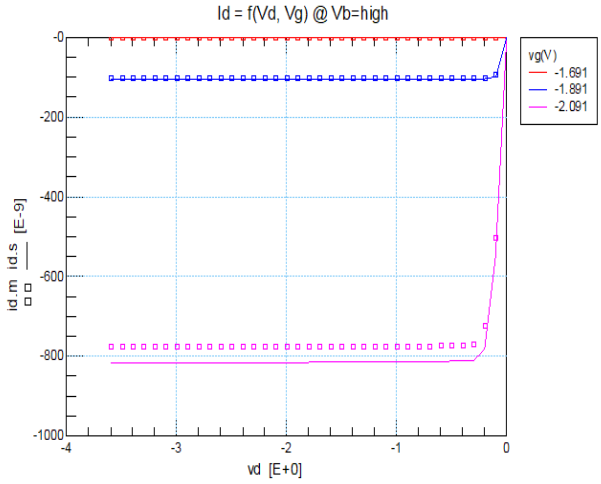
idvd_vth, Vb = 0V, T = 27°C

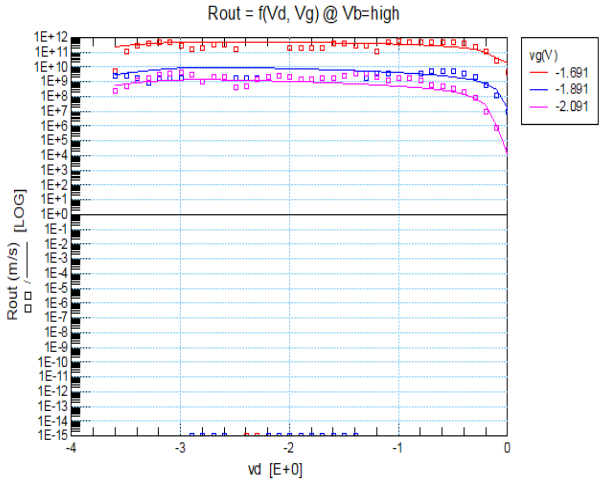


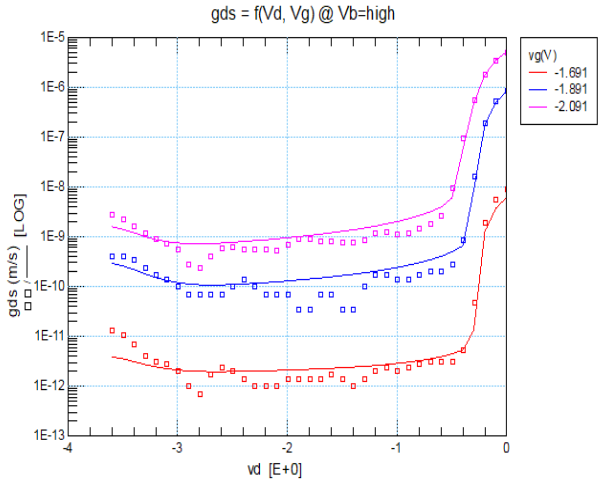




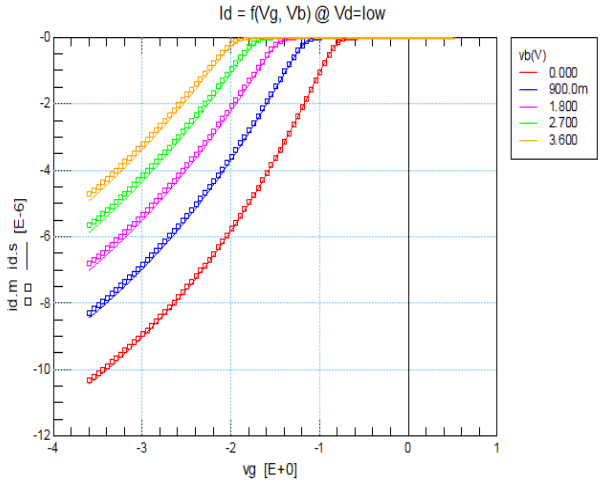
idvd_vth_vbmin, Vb = 3.6V, T = 27°C

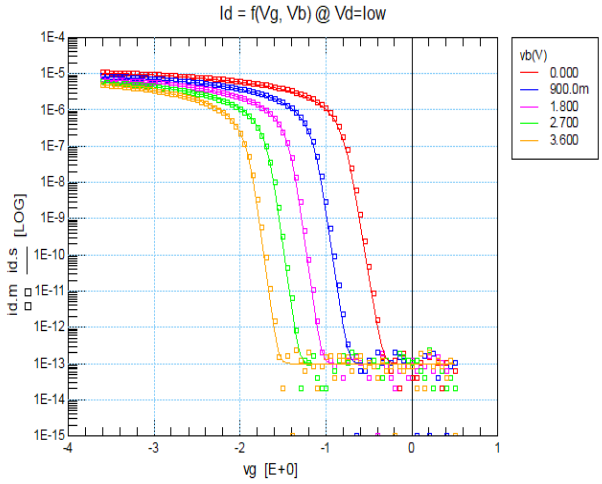


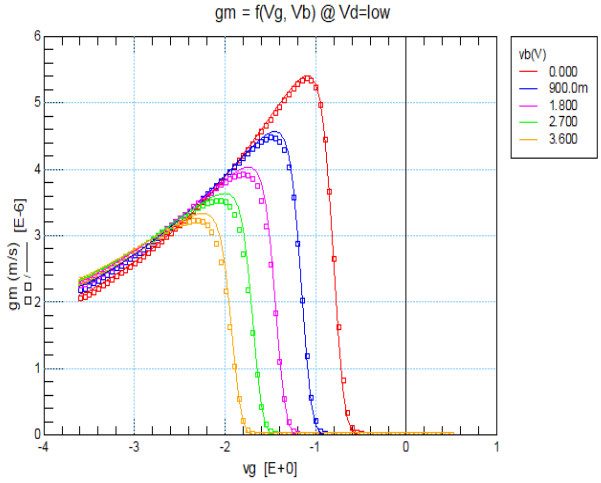


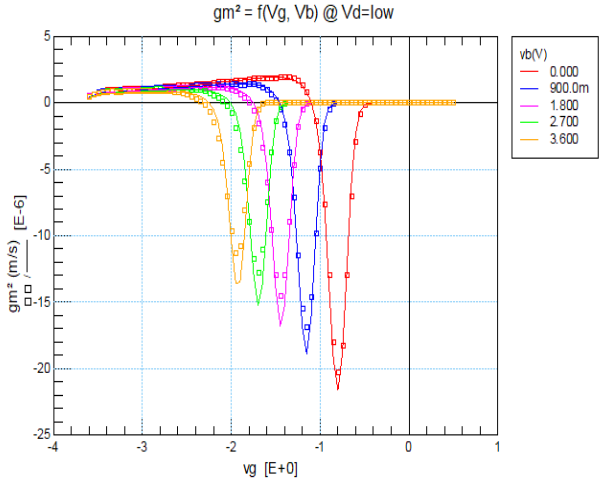


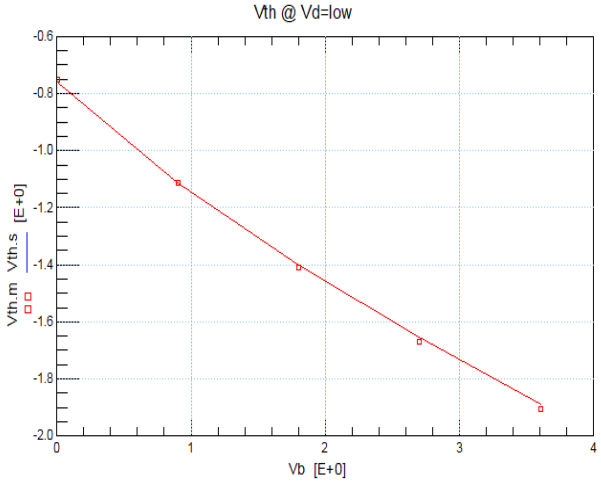
idvg, Vd = -0.1V, T = -40°C

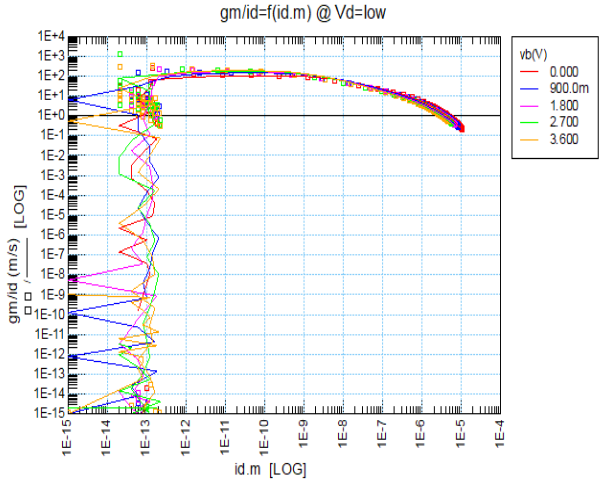


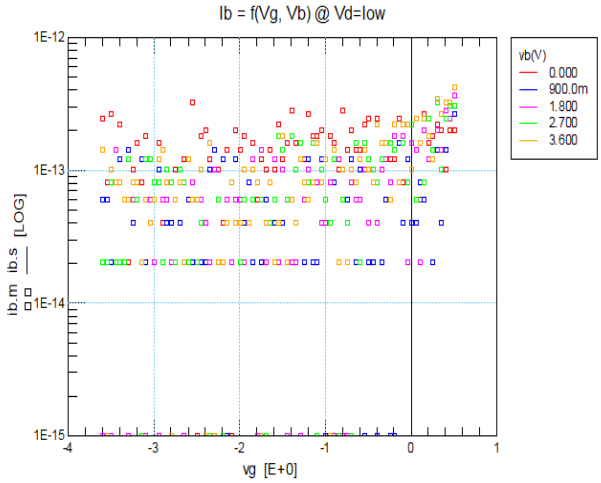


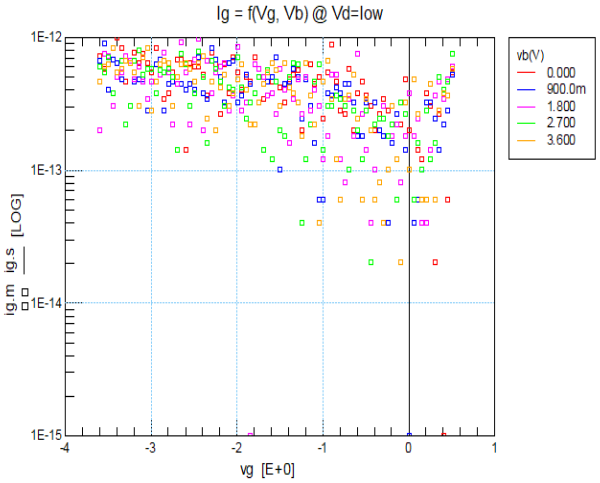




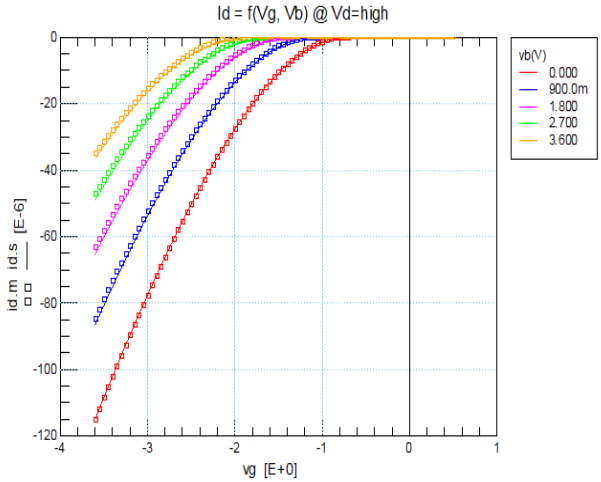


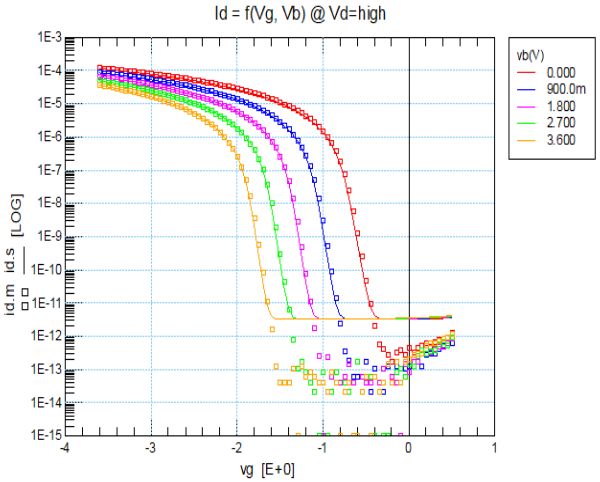


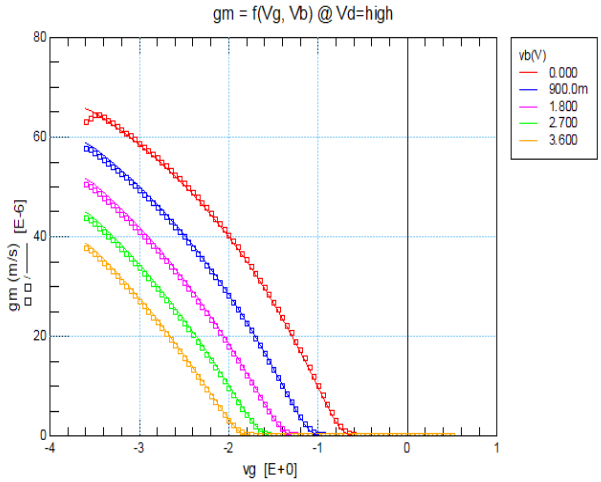


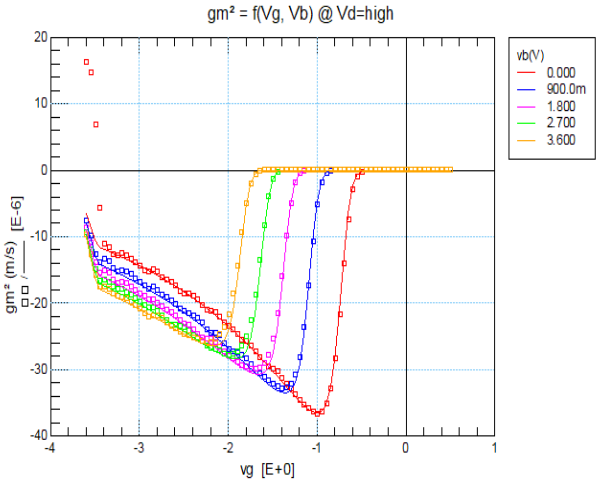


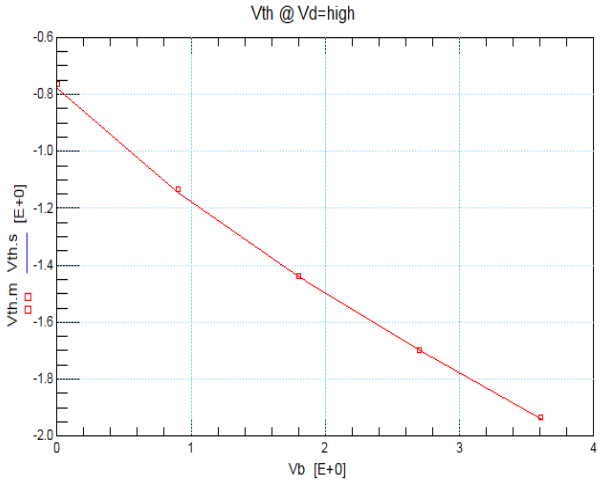
idvg, Vd = -3.3V, T = -40°C

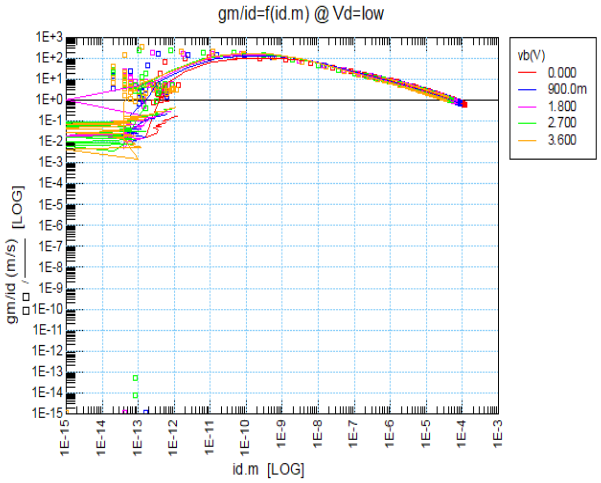


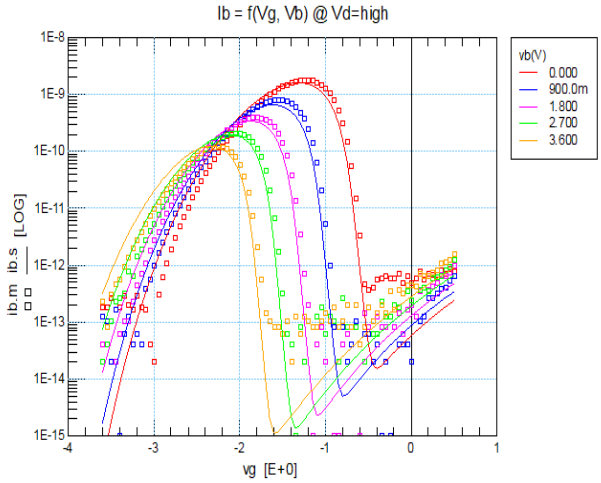


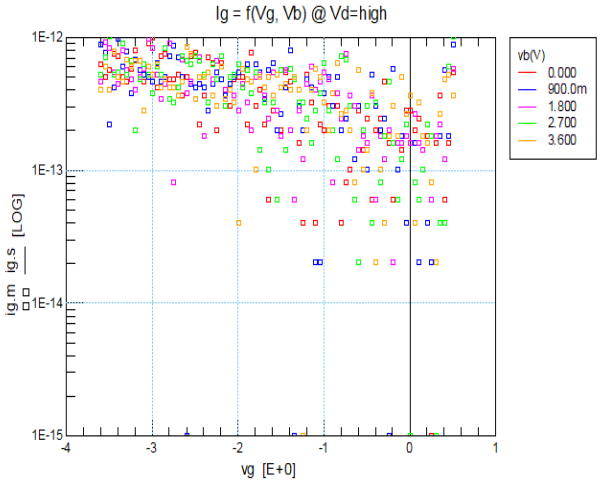




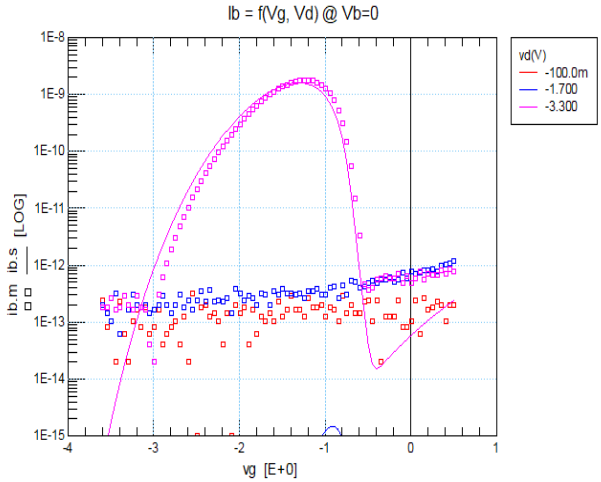


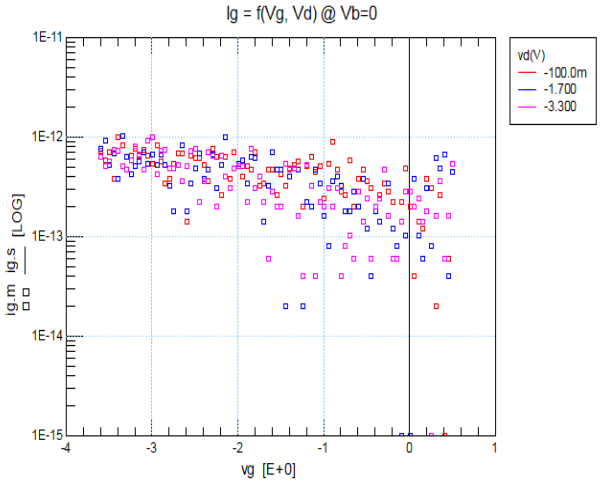




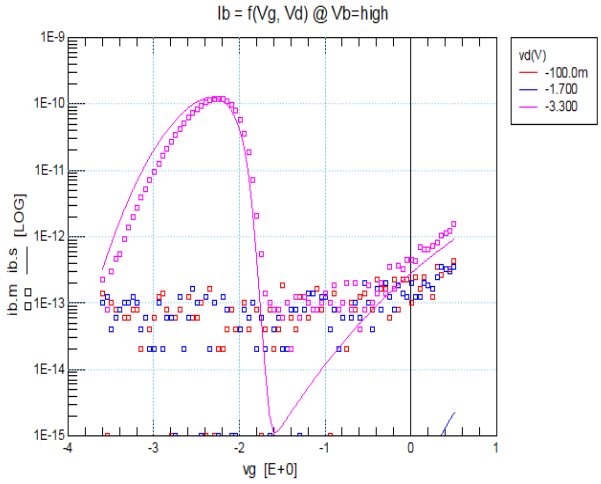


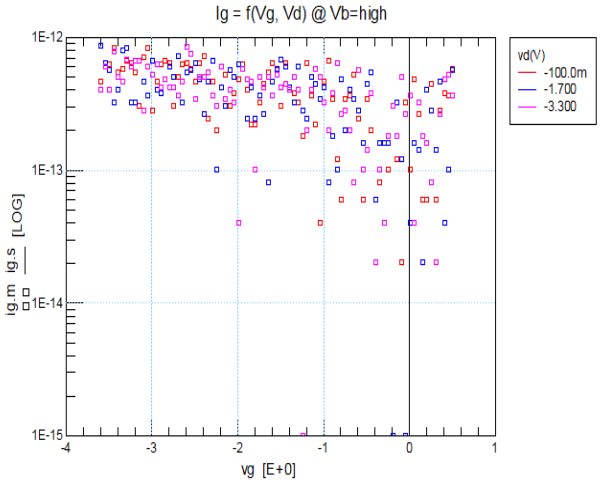
idvg, Ib,Ig, Vb = 0V, T = -40°C



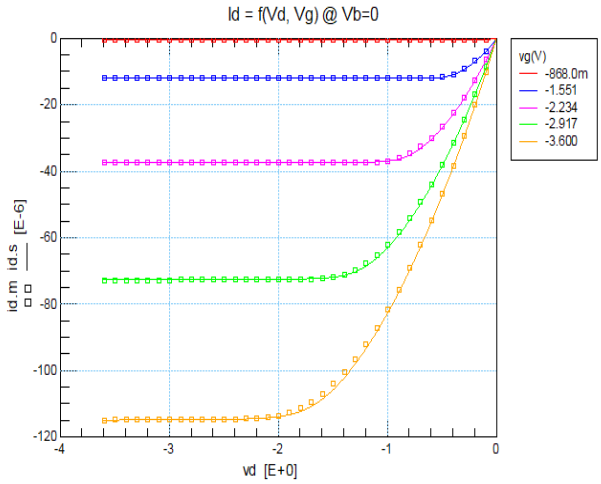


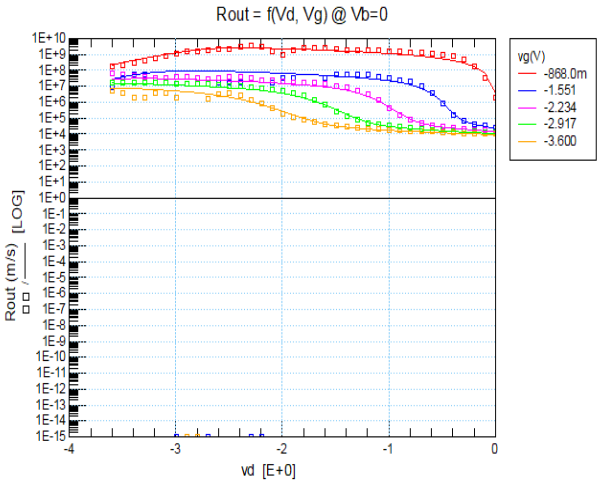
idvg, Ib,Ig, Vb = 3.6V, T = -40°C

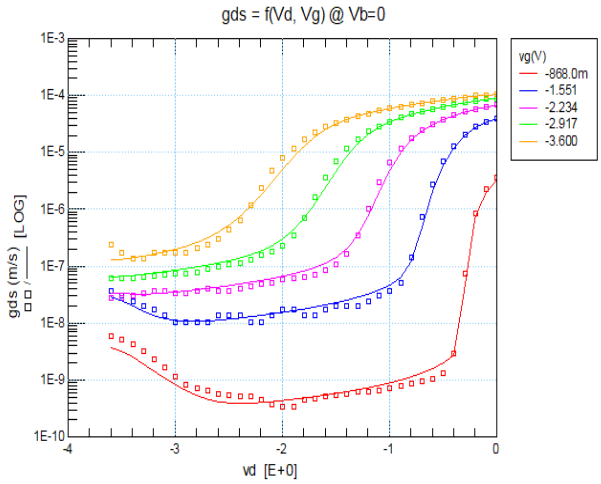




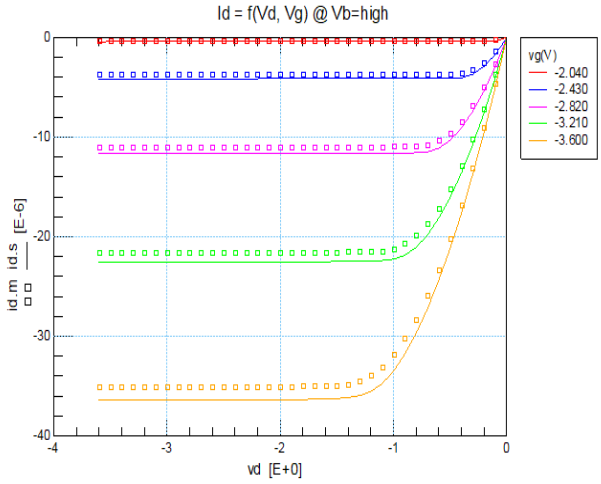
idvd, Vb = 0V, T = -40°C

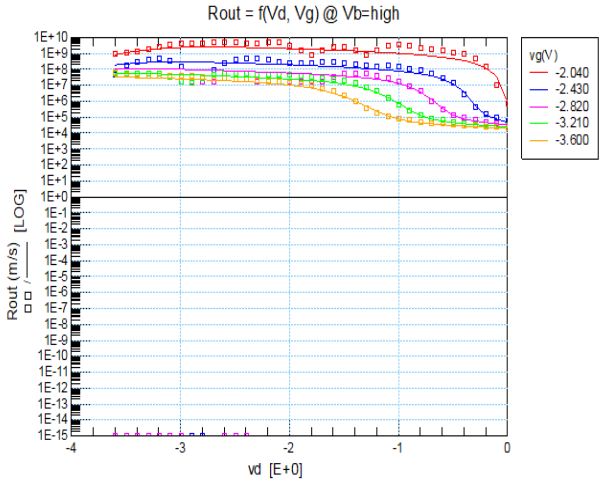


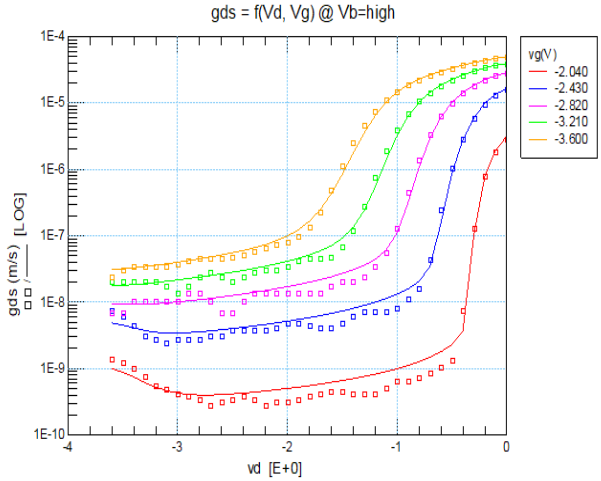




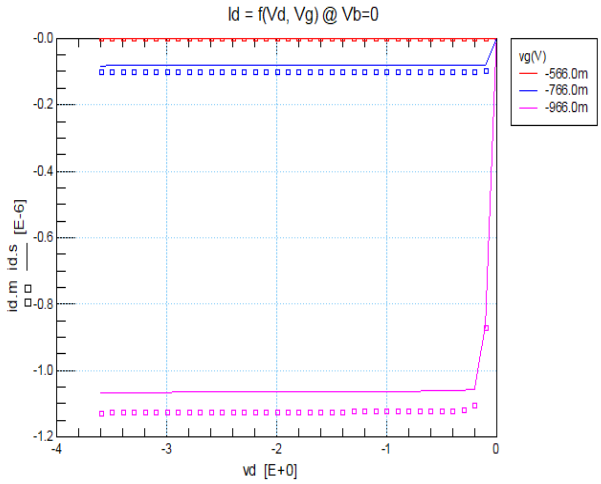
idvd_vbmin, Vb = 3.6V, T = -40°C

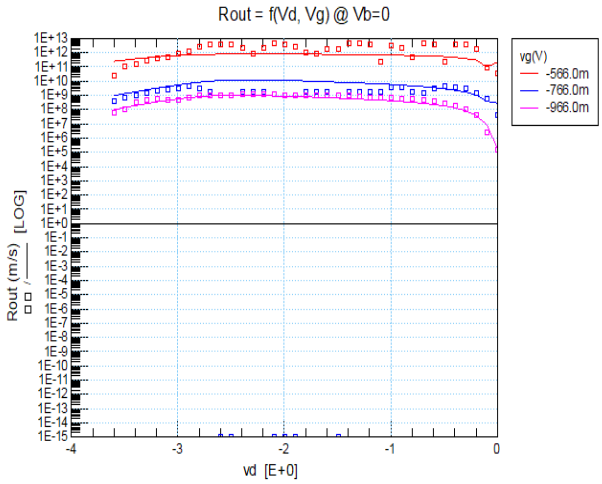


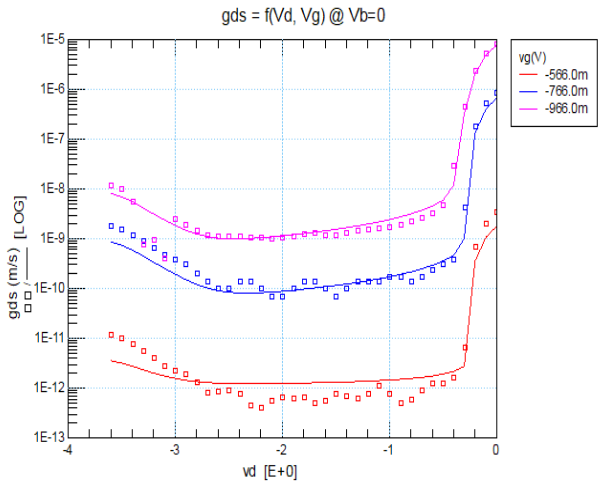




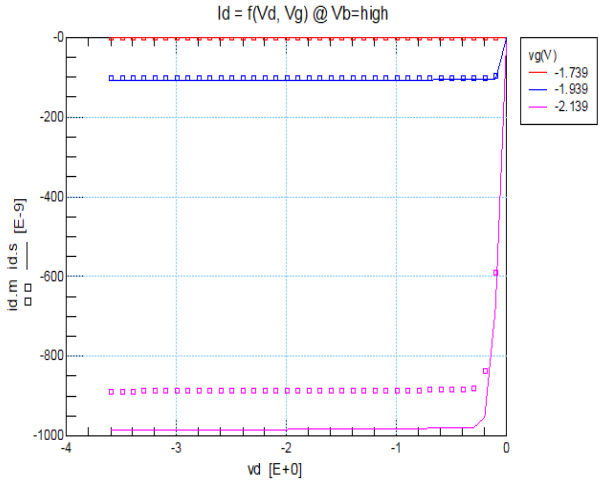
idvd_vth, Vb = 0V, T = -40°C

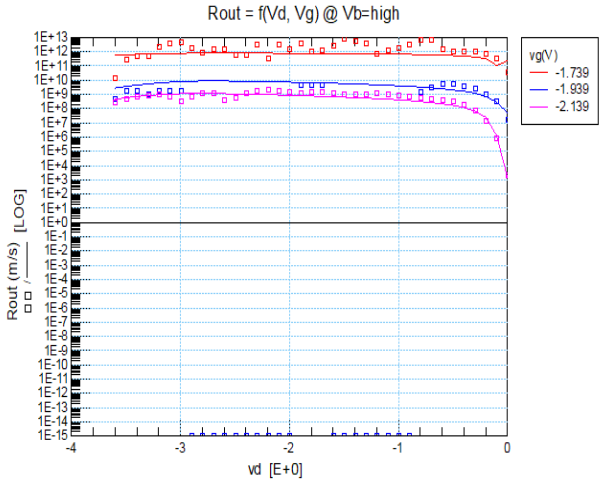


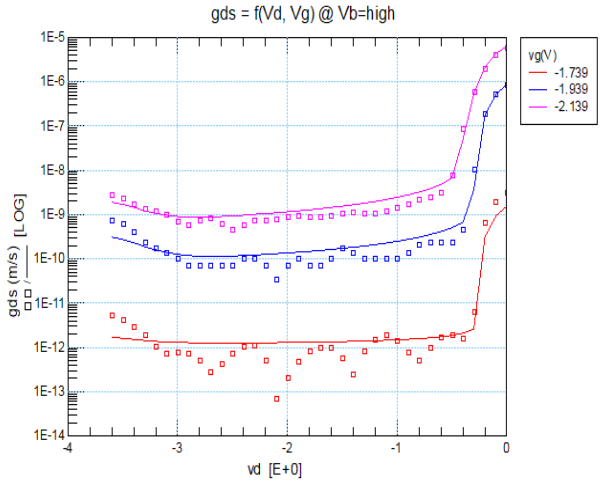




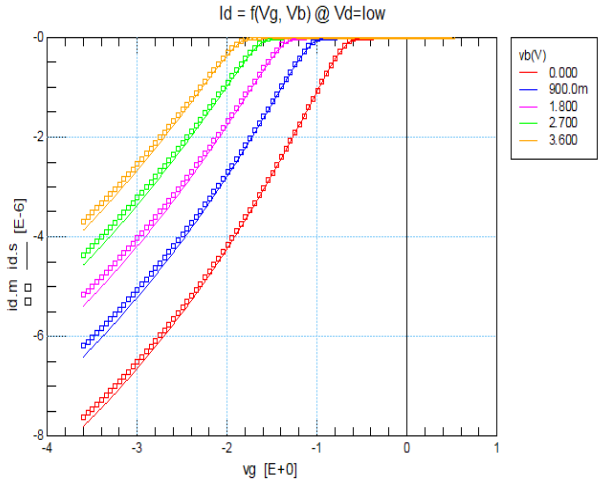
idvd_vth_vbmin, Vb = 3.6V, T = -40°C

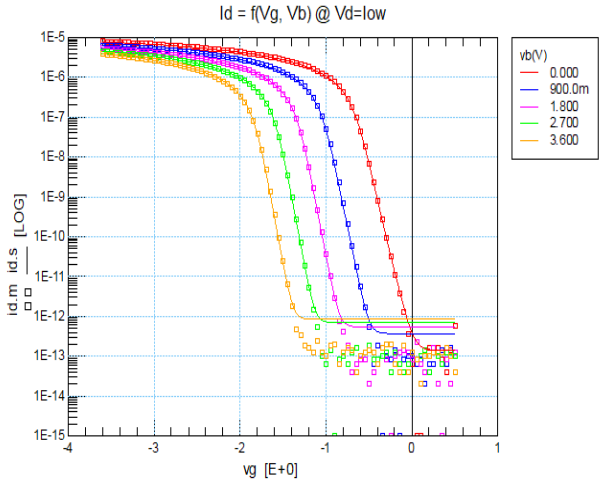


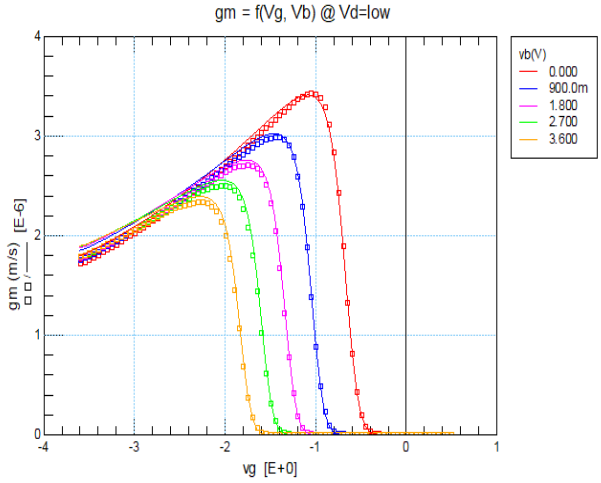


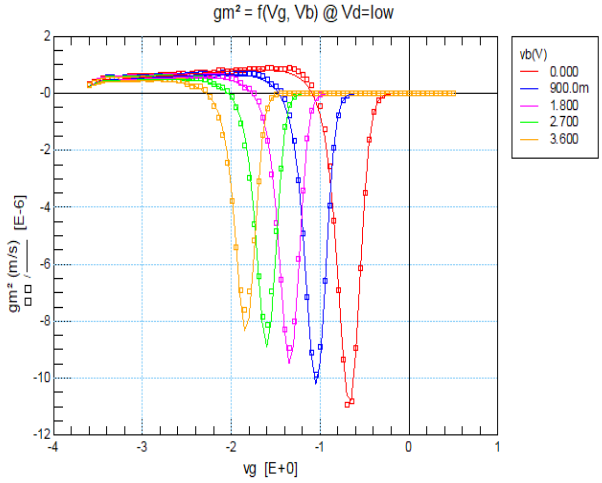


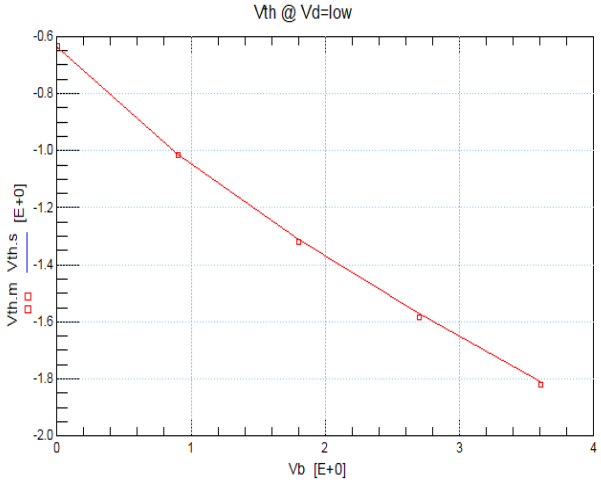
idvg, Vd = -0.1V, T = 70°C

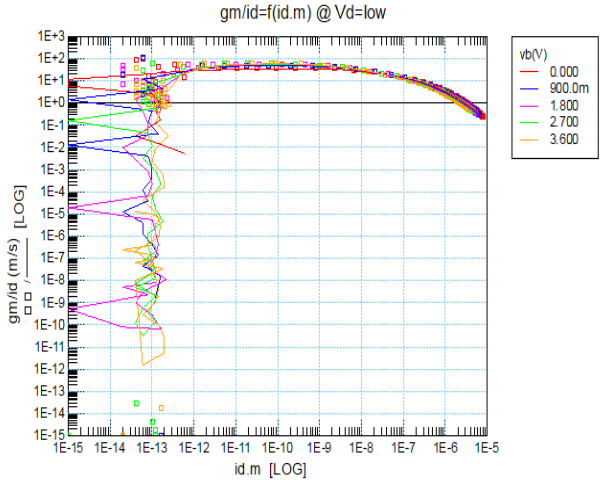


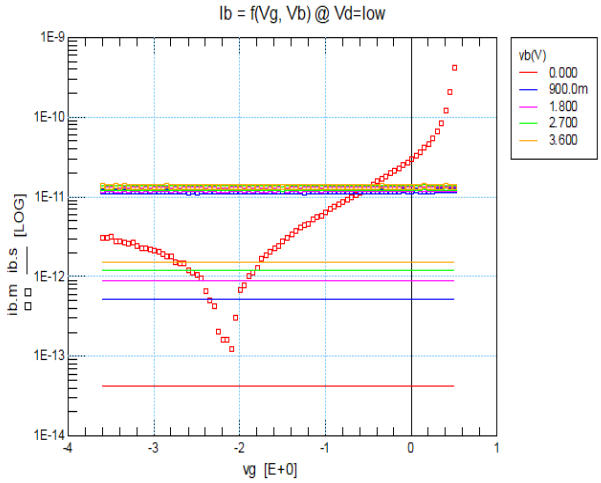


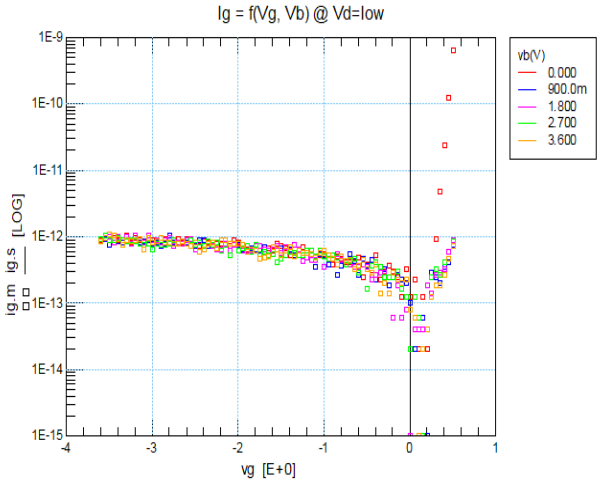




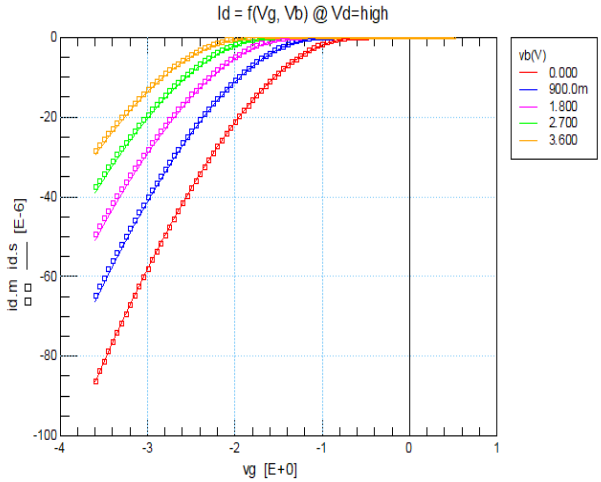


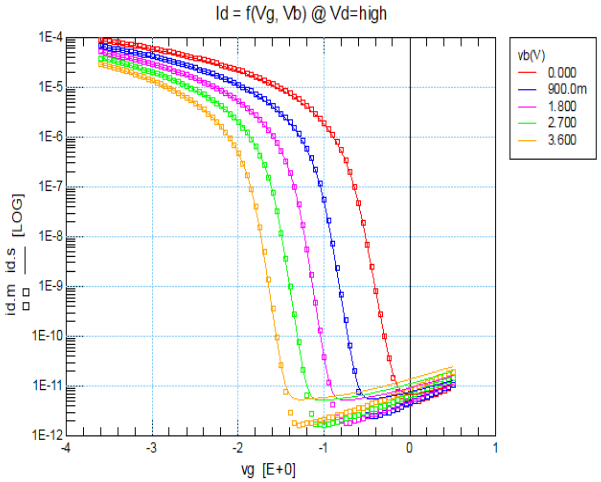


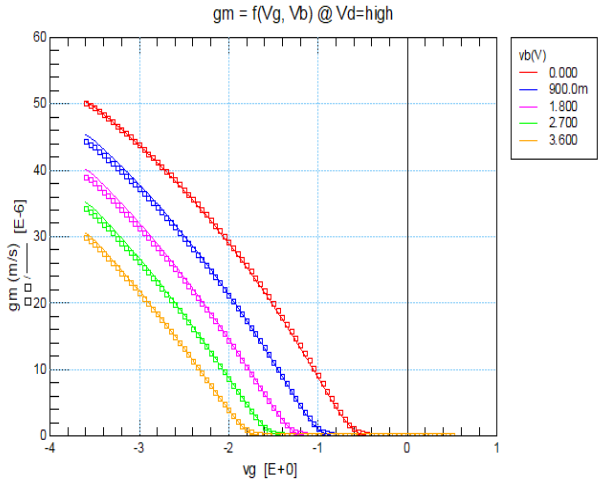


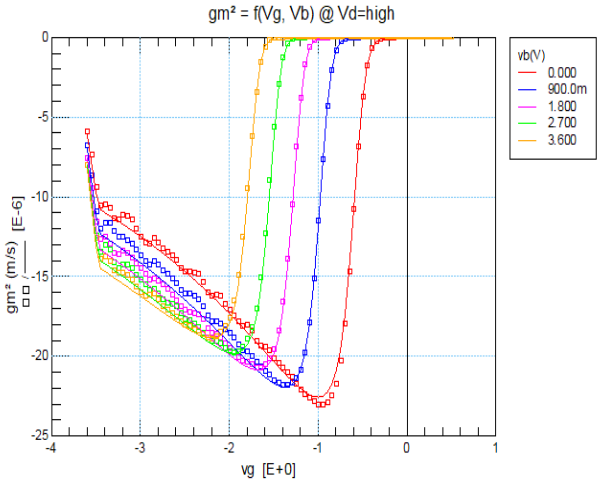


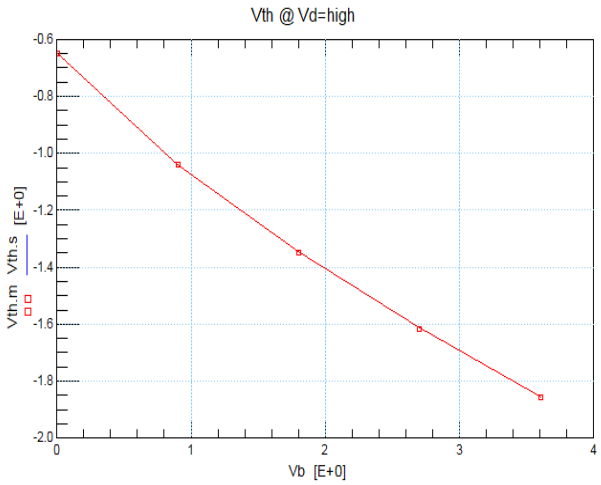
idvg, Vd = -3.3V, T = 70°C

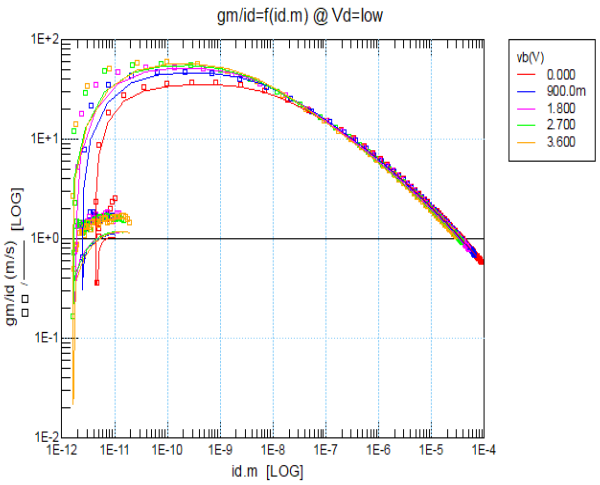


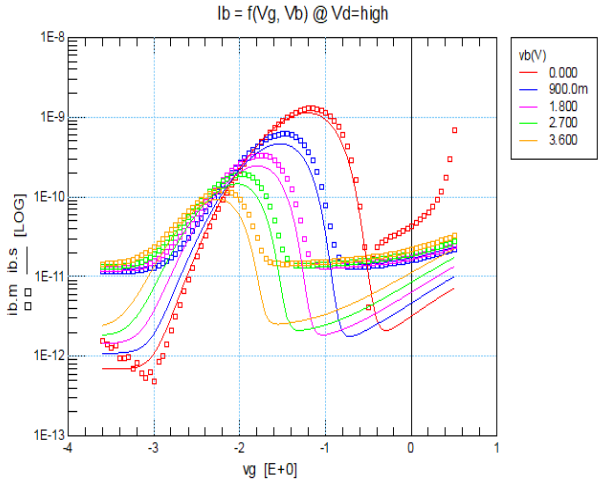


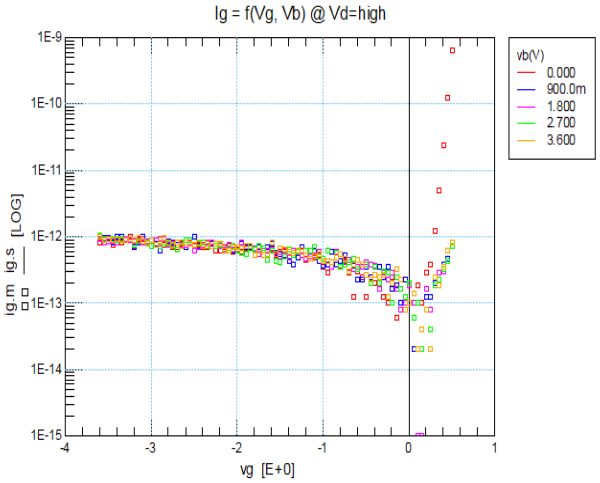




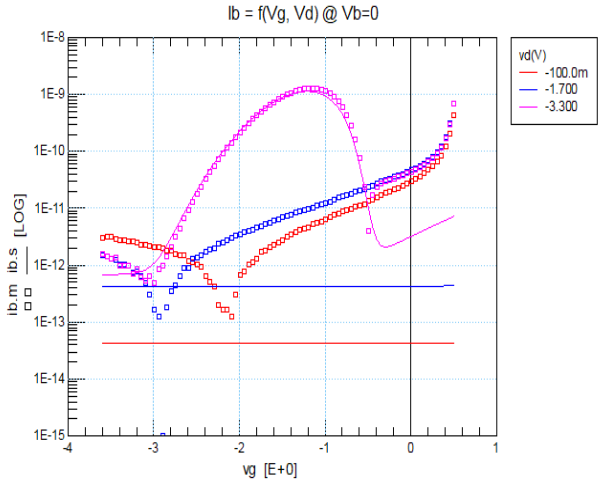


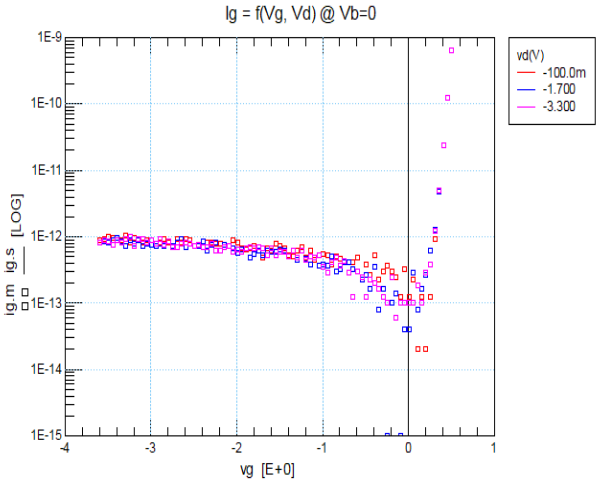




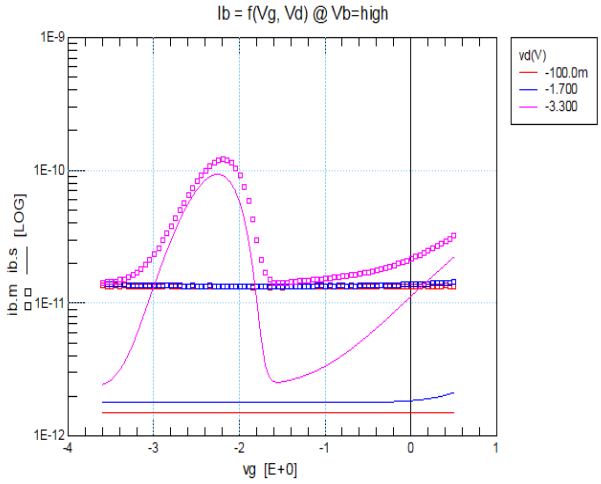


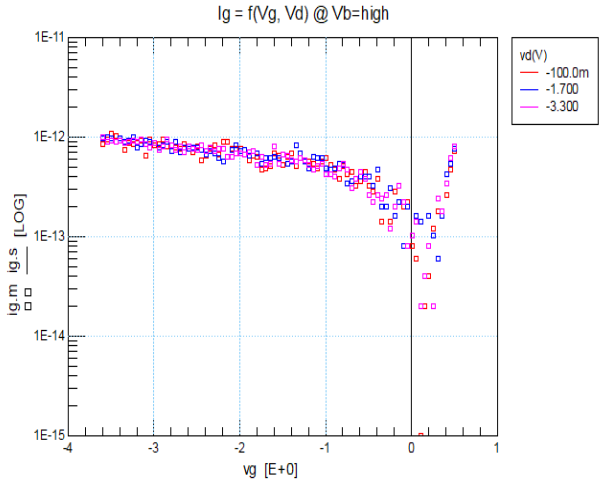
idvg, Ib,Ig, Vb = 0V, T = 70°C



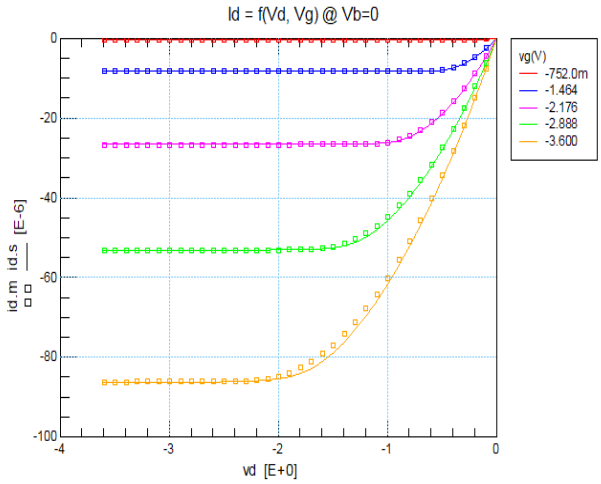


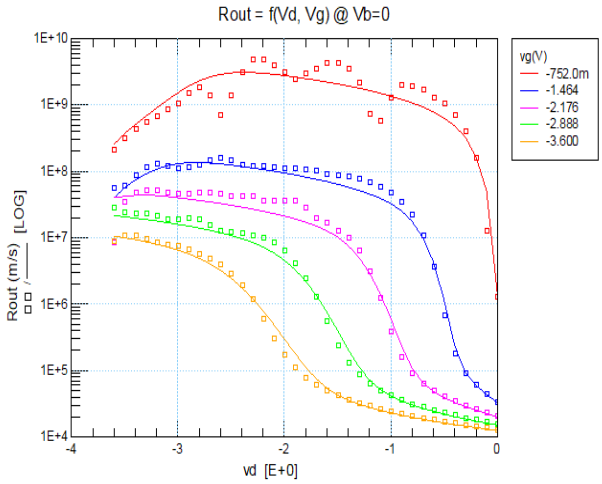
idvg, Ib,Ig, Vb = 3.6V, T = 70°C

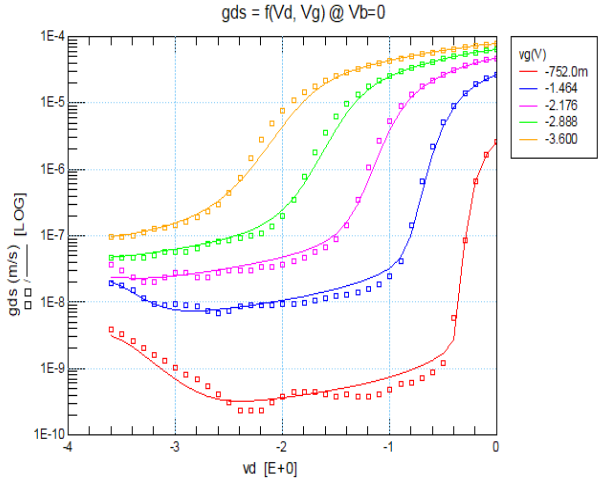




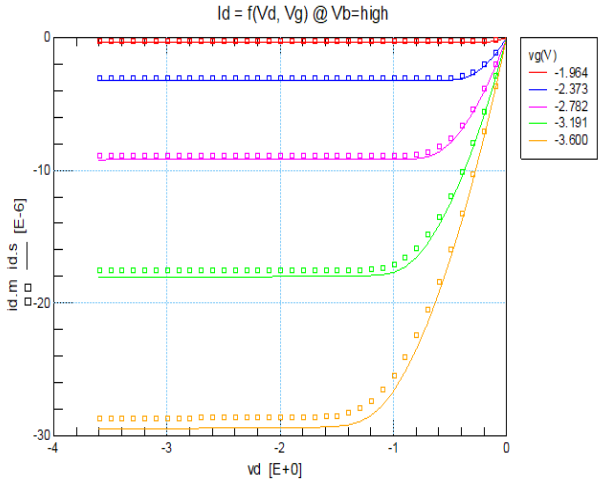
idvd, Vb = 0V, T = 70°C

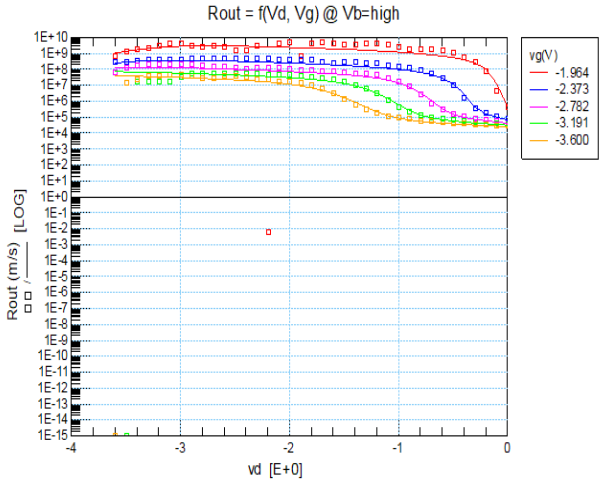


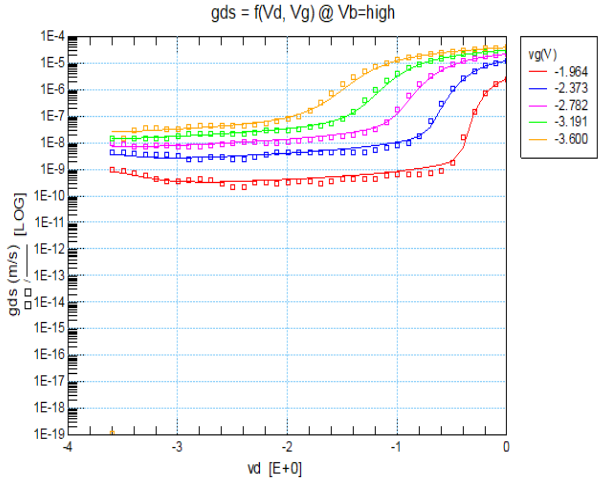




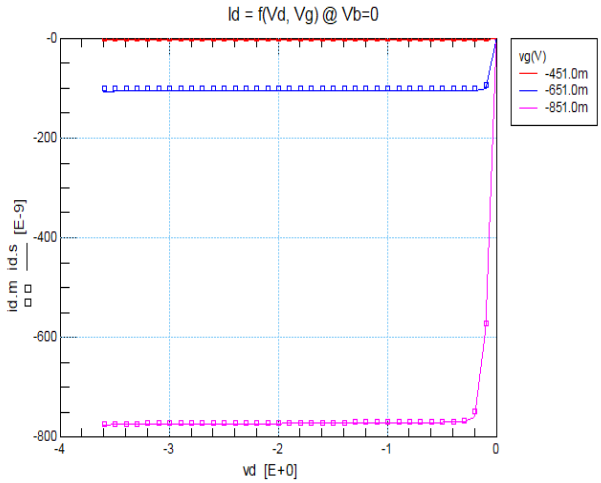
idvd_vbmin, Vb = 3.6V, T = 70°C

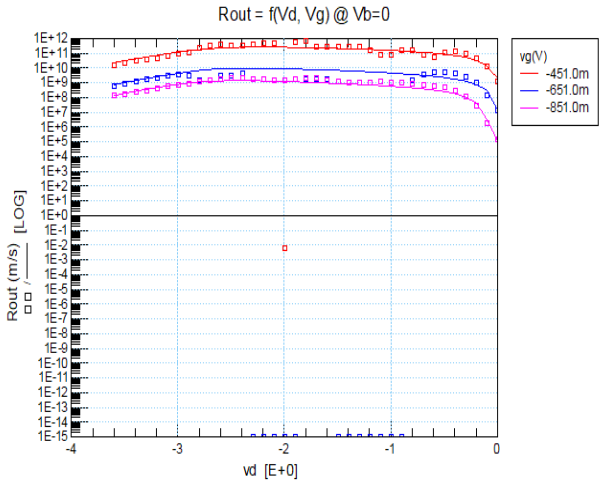


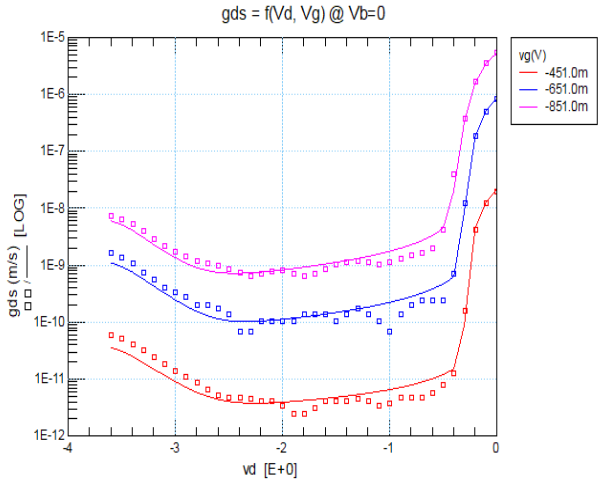




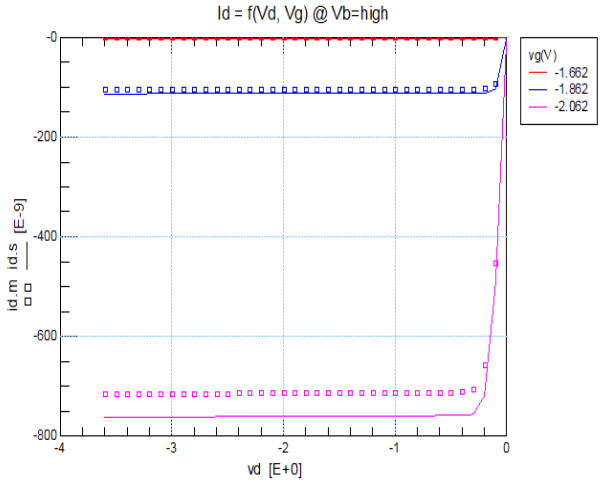
idvd_vth, Vb = 0V, T = 70°C

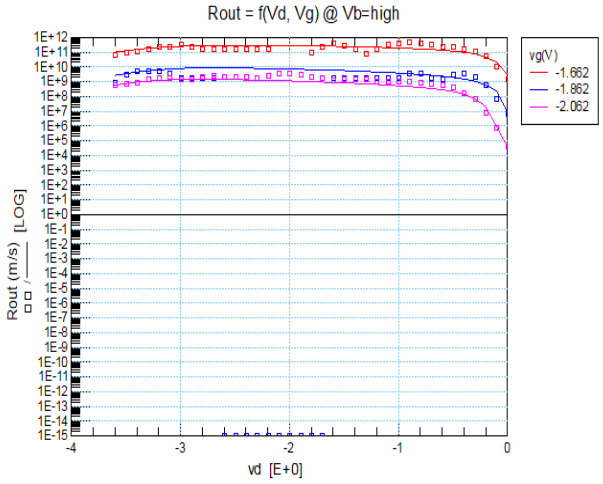


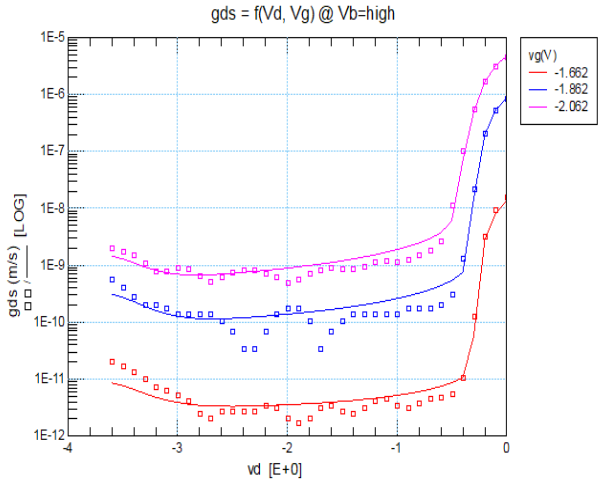




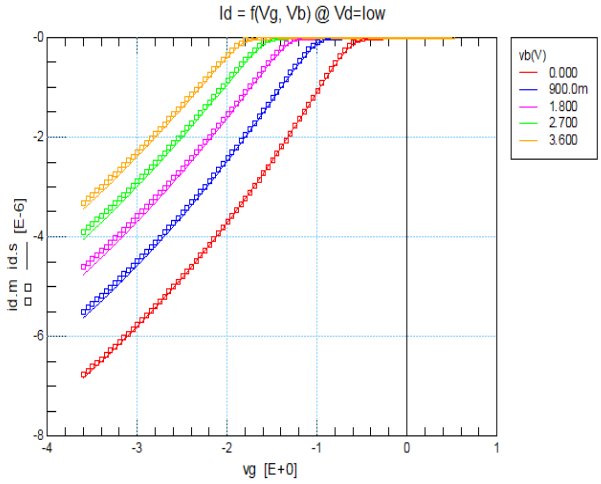
idvd_vth_vbmin, Vb = 3.6V, T = 70°C

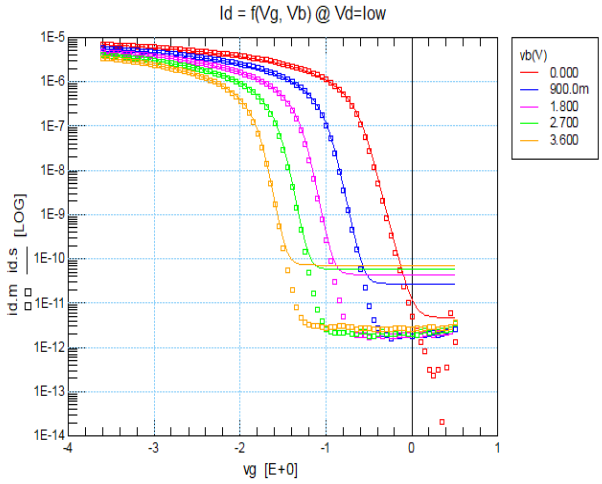


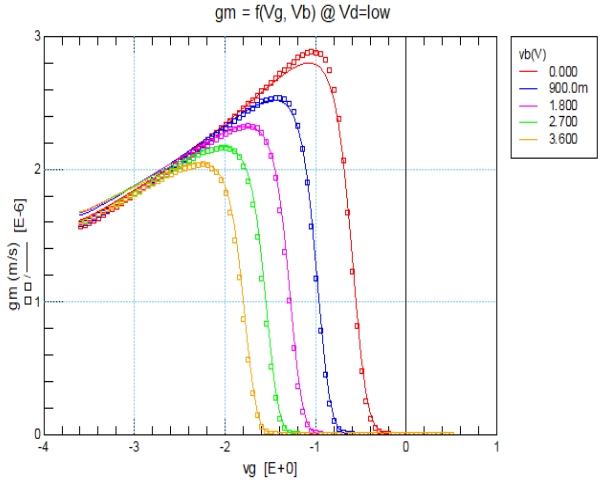


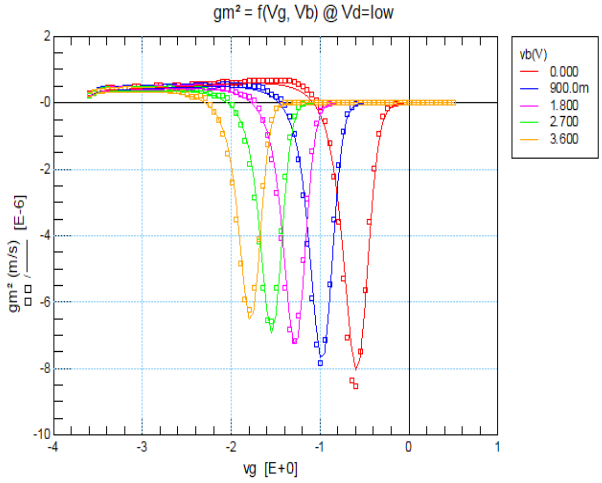


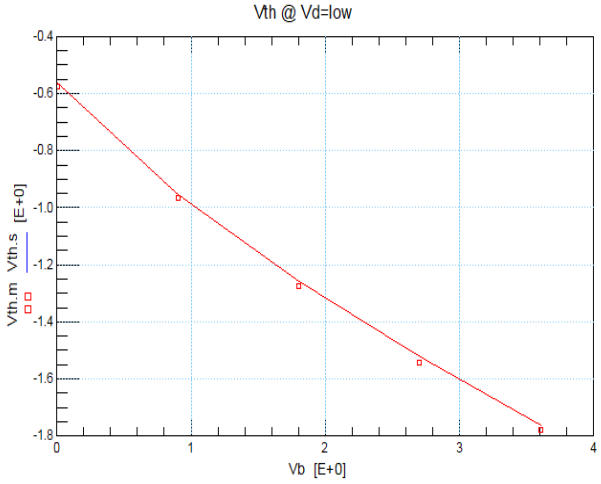
idvg, Vd = -0.1V, T = 125°C

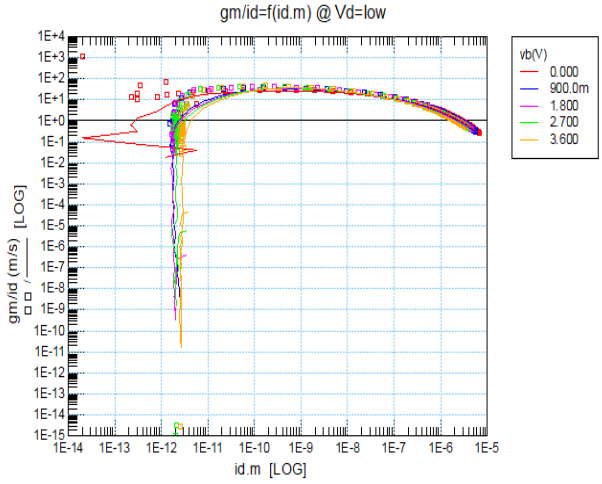


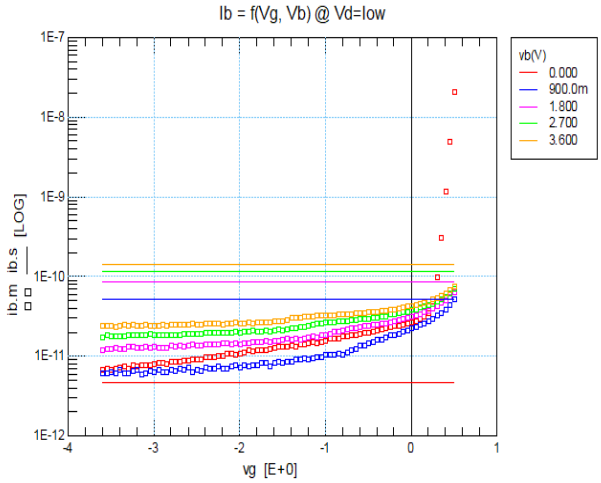


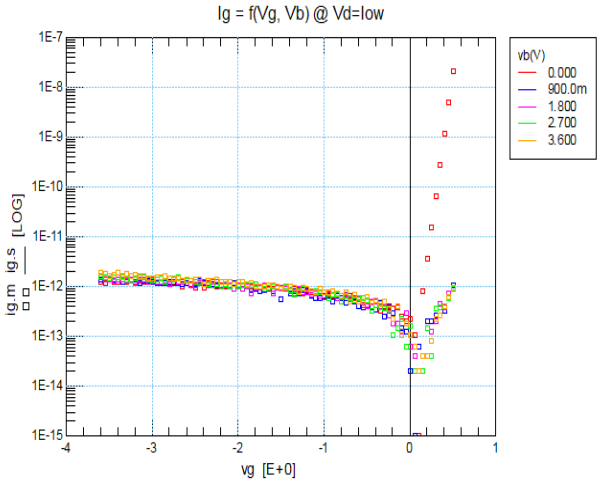




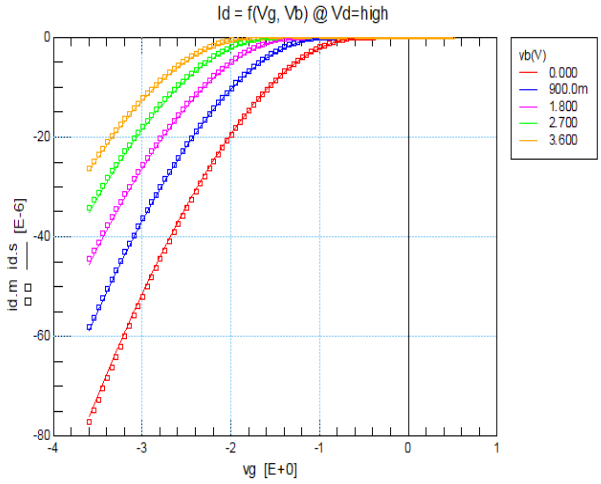


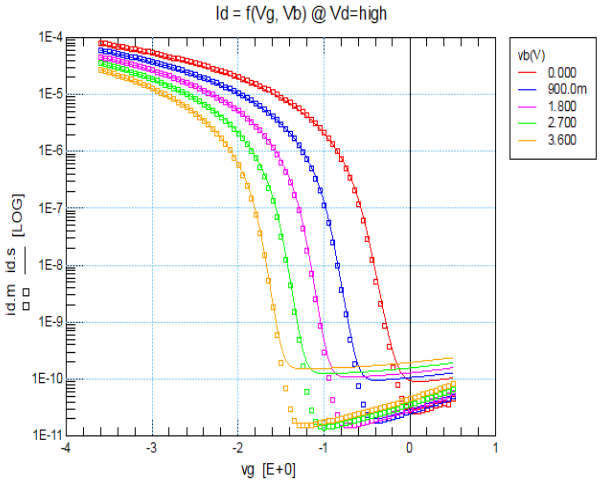


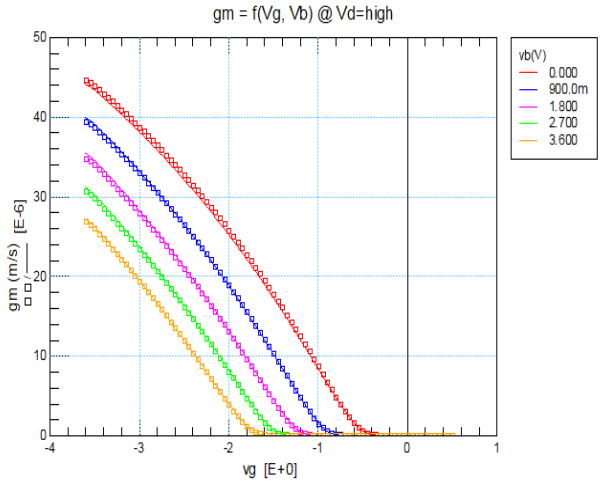


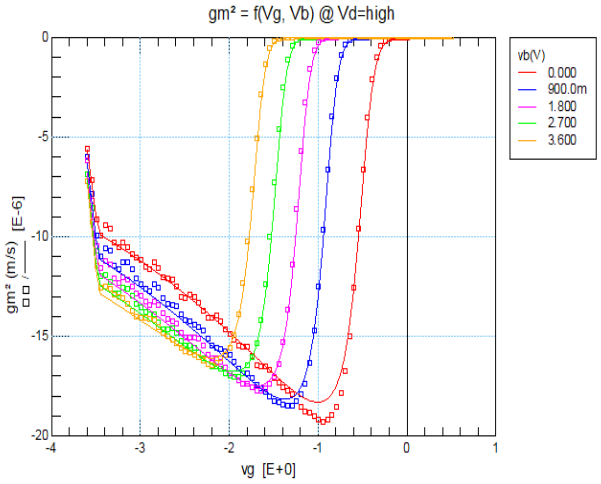


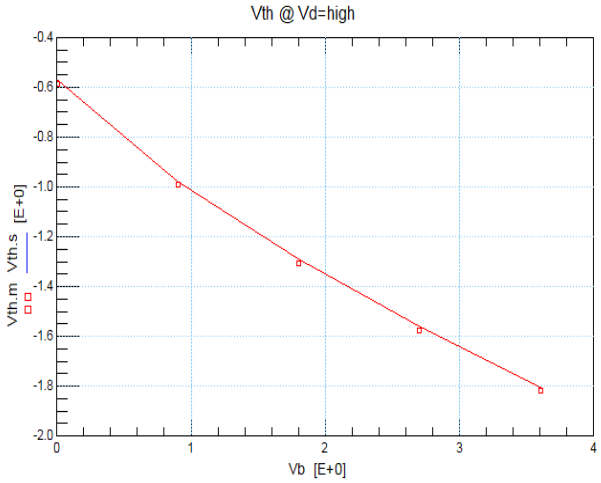
idvg, Vd = -3.3V, T = 125°C

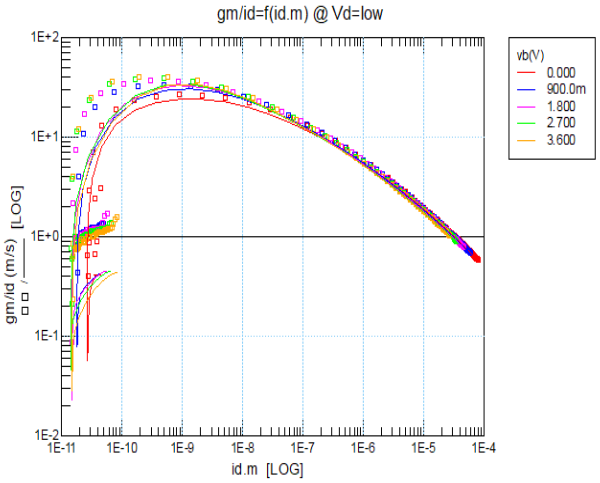


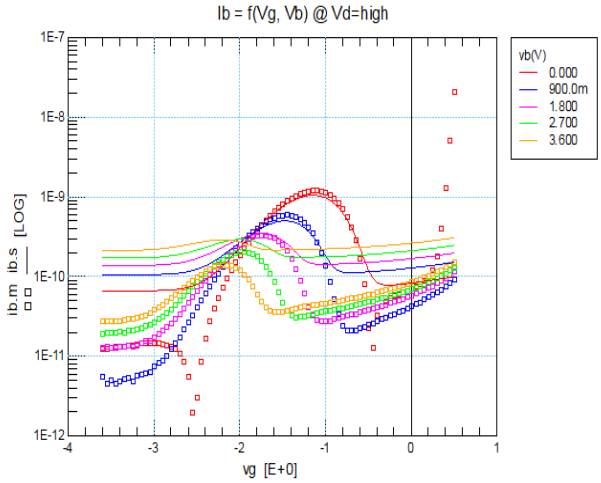


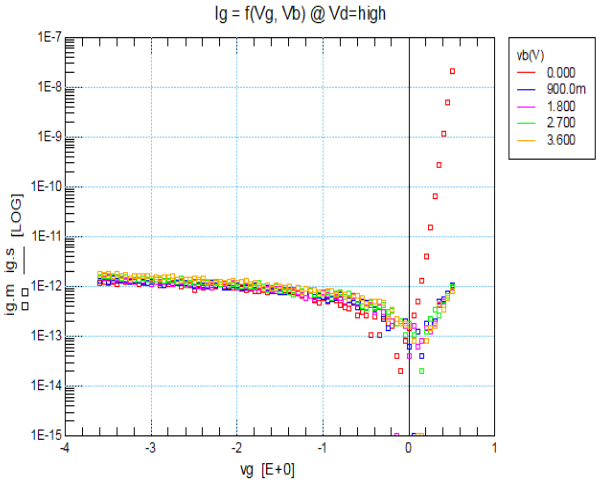




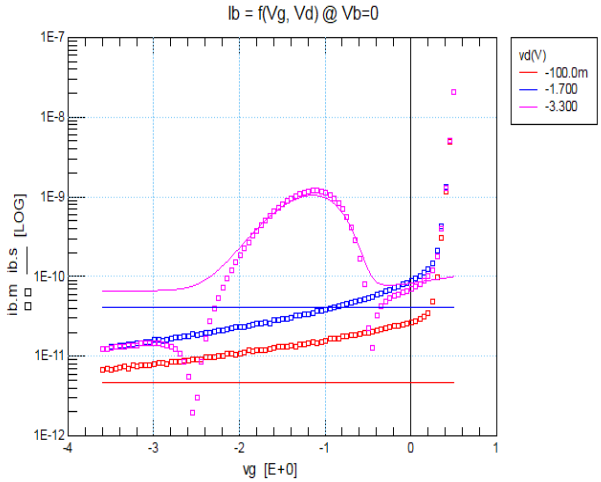


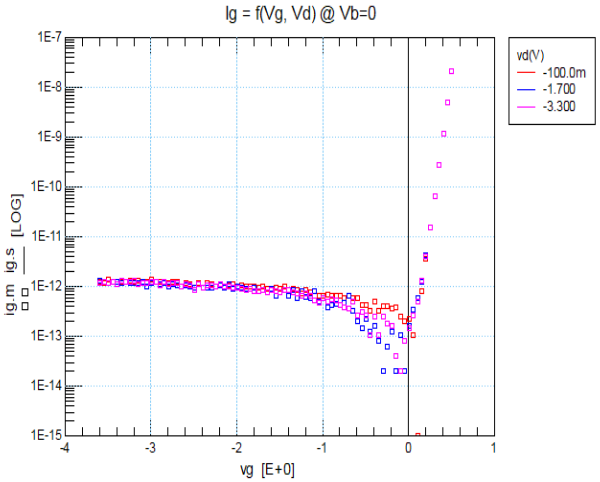




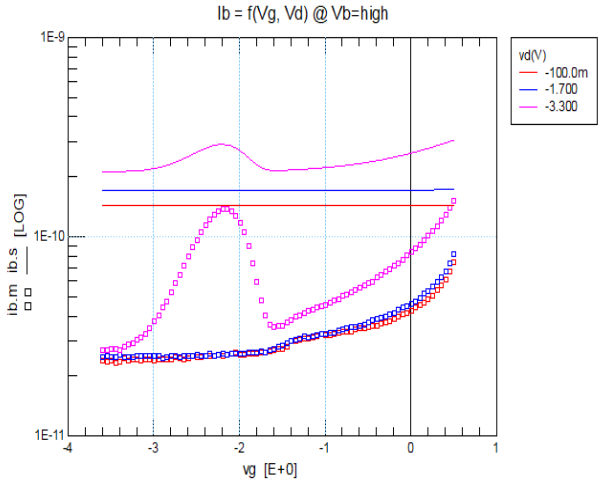


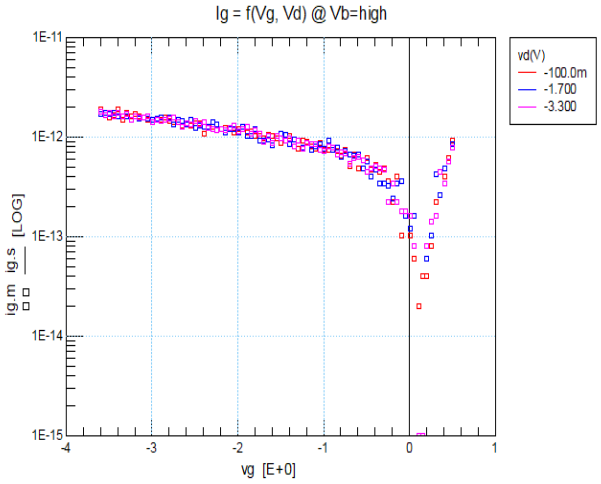
idvg, Ib,Ig, Vb = 0V, T = 125°C



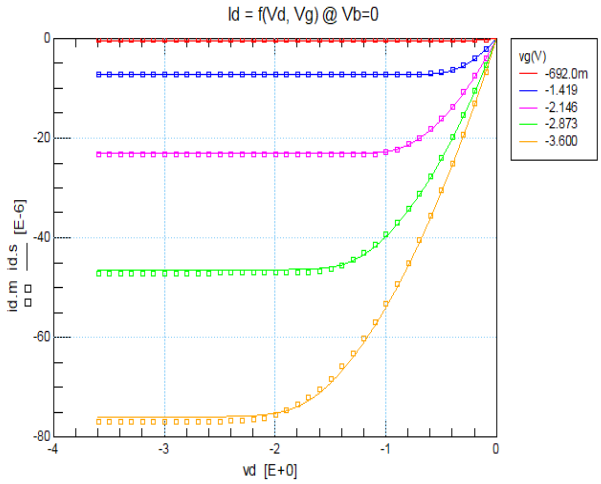


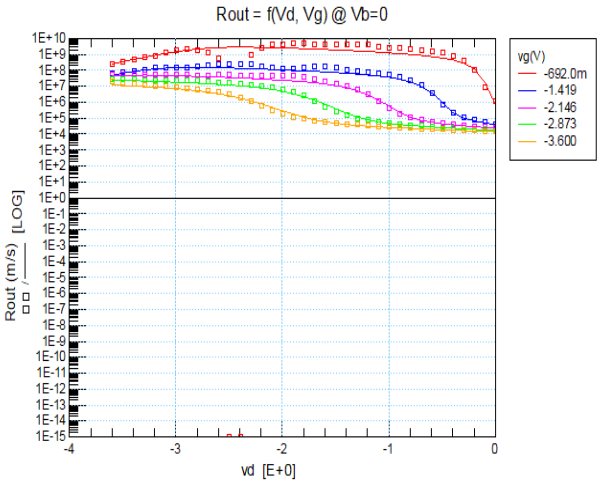
idvg, Ib,Ig, Vb = 3.6V, T = 125°C

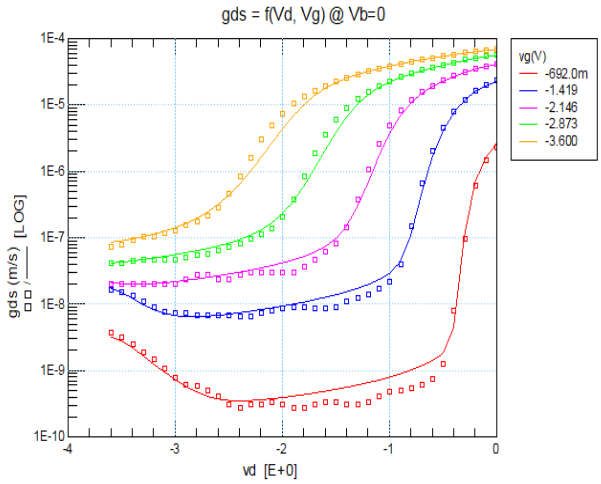




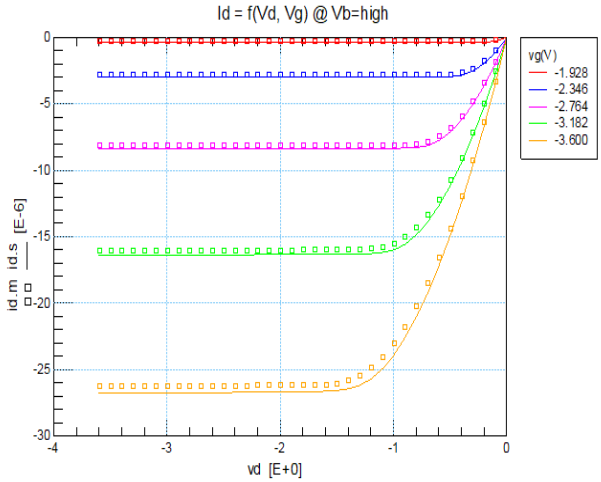
idvd, Vb = 0V, T = 125°C

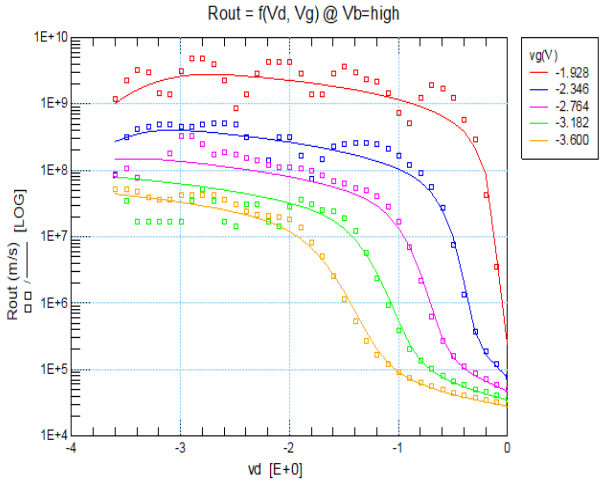


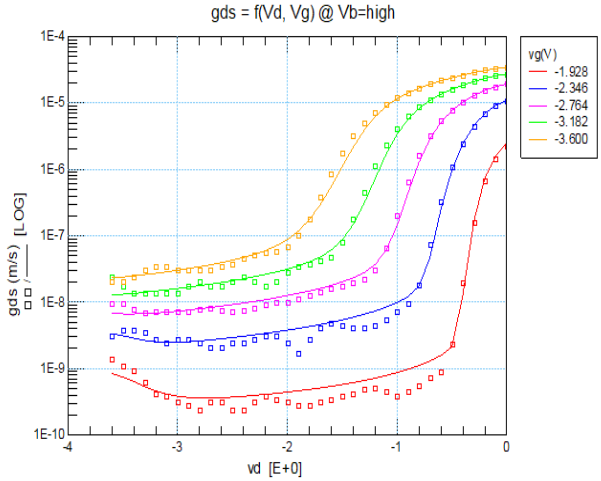




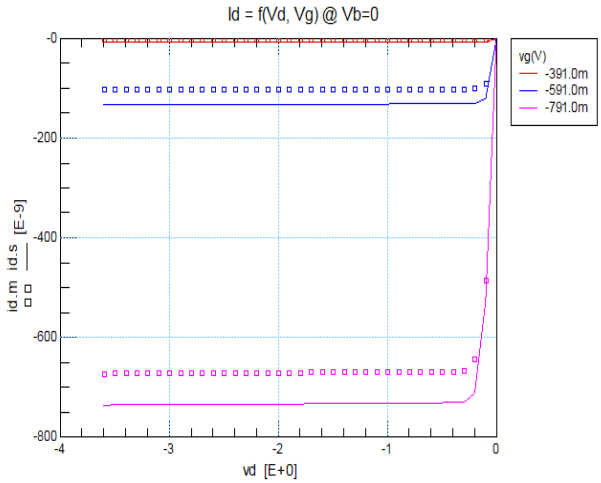
idvd_vbmin, Vb = 3.6V, T = 125°C

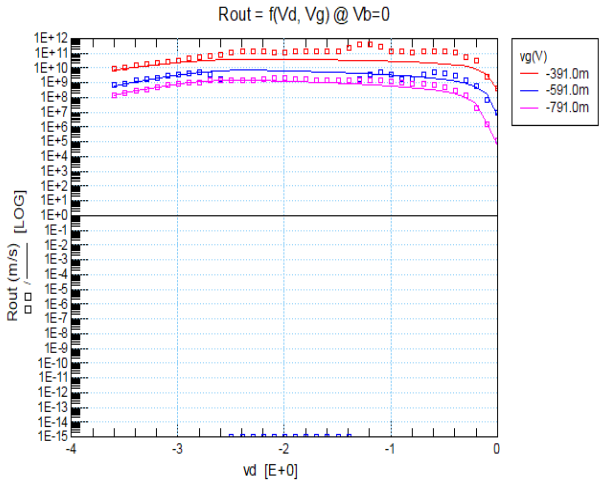


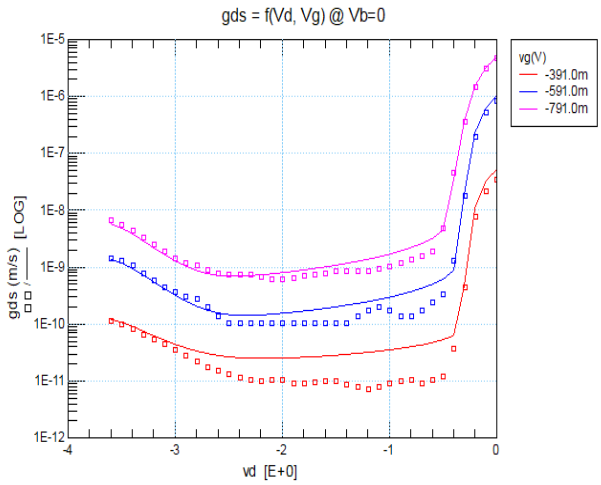




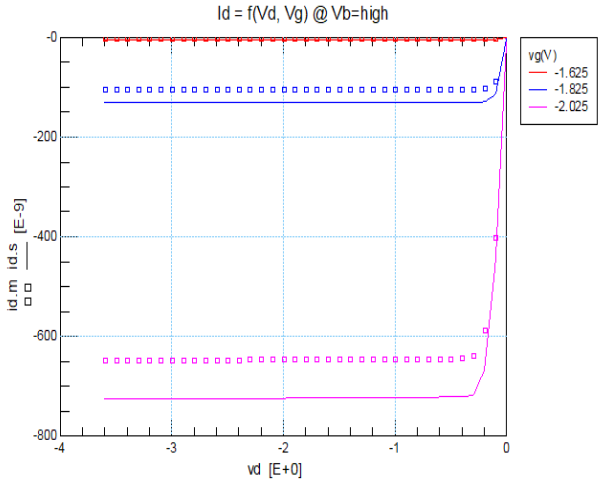
idvd_vth, Vb = 0V, T = 125°C

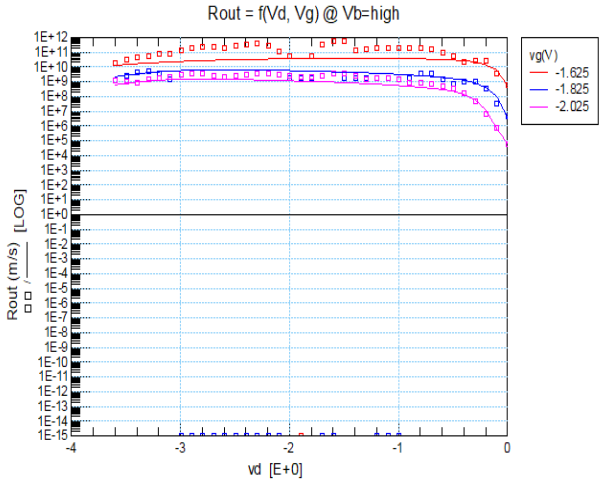


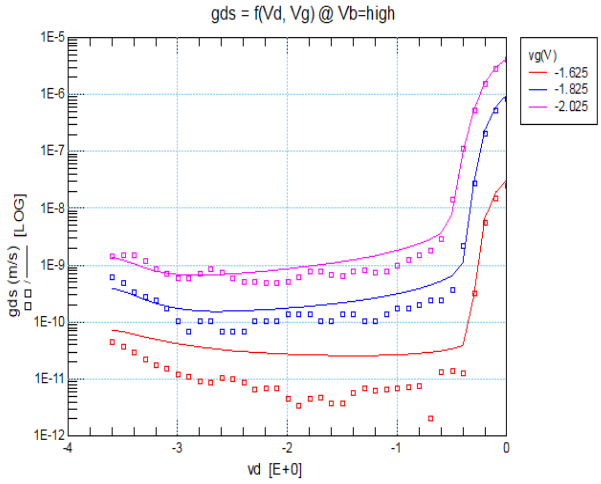




idvd_vth_vbmin, Vb = 3.6V, T = 125°C



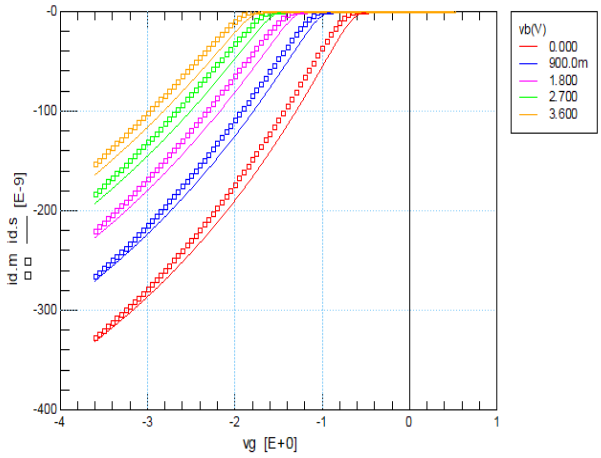




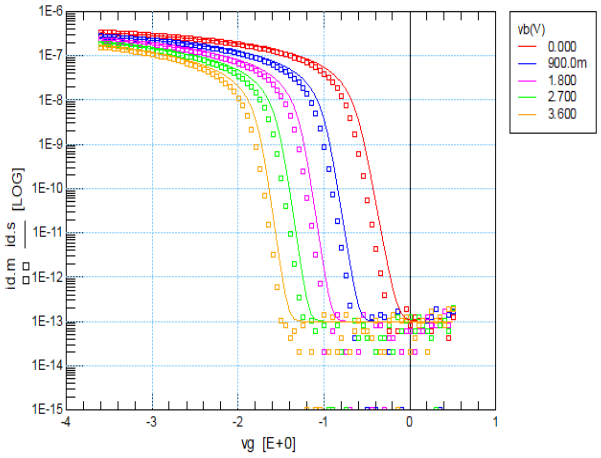
4.3 W0u3_L10u0_S555_4

idvg, Vd = -0.1V, T = 27°C

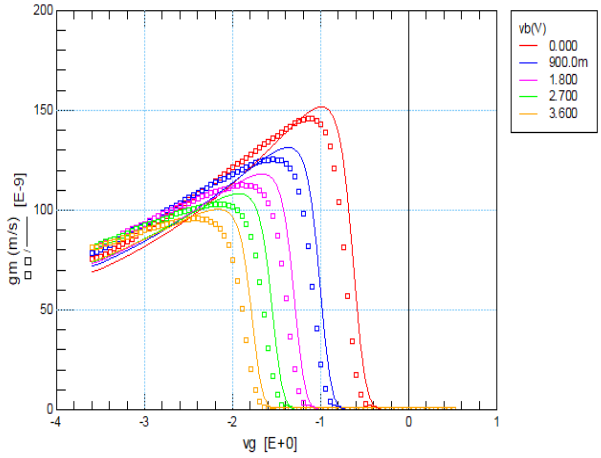
Id = f(Vg, Vb) @ Vd=low



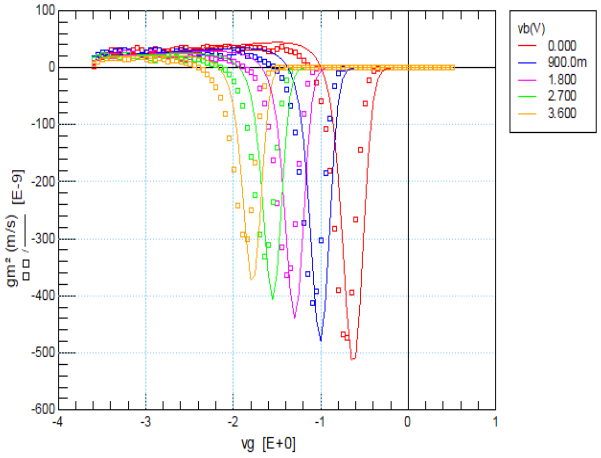
Id = f(Vg, Vb) @ Vd=low



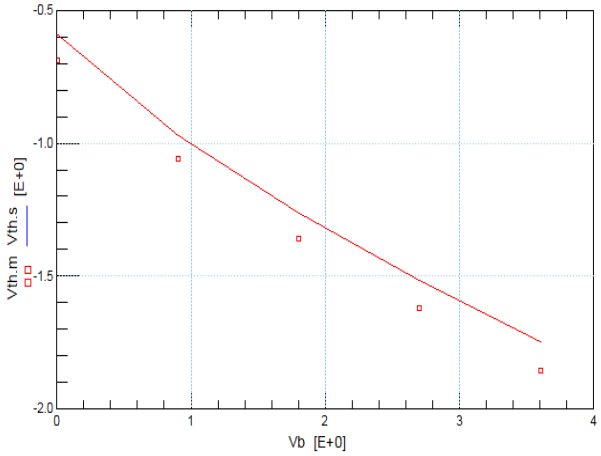
gm = f(Vg, Vb) @ Vd=low



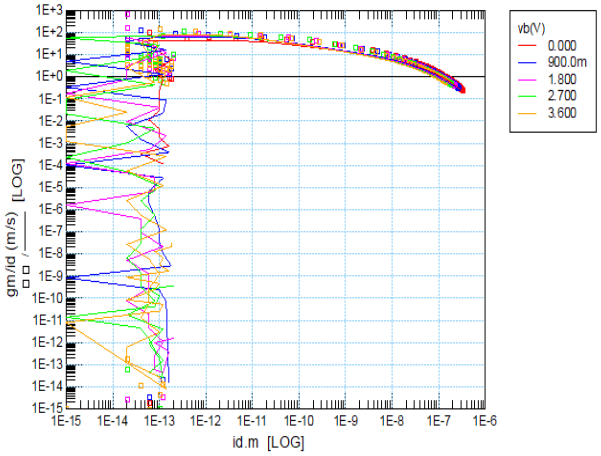
gm² = f(Vg, Vb) @ Vd=low

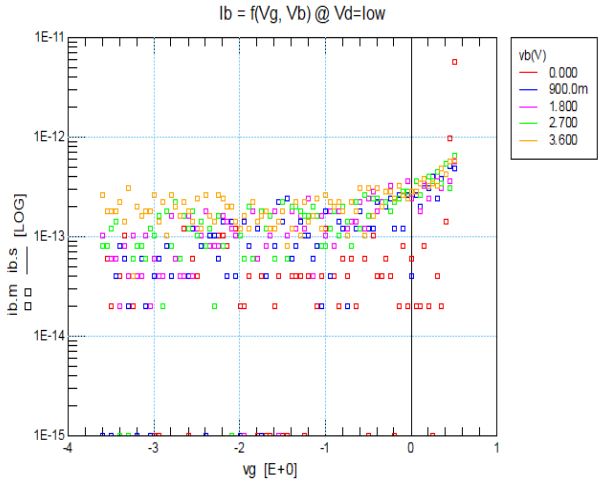


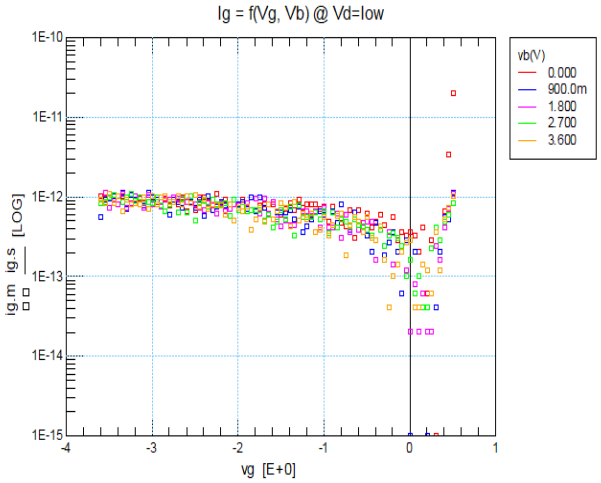
Vth @ Vd=low



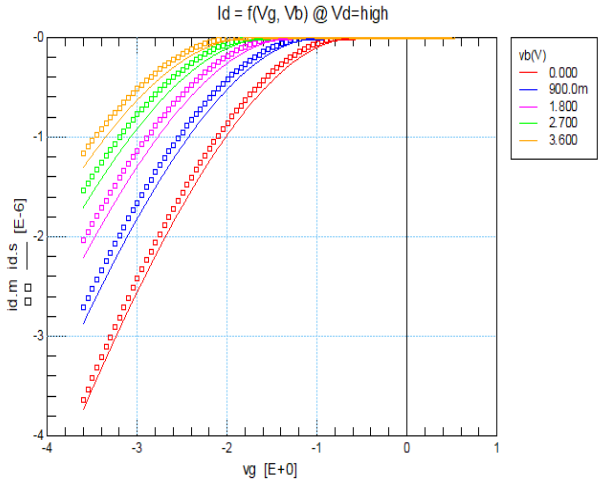
gm/id=f(id.m) @ Vd=low

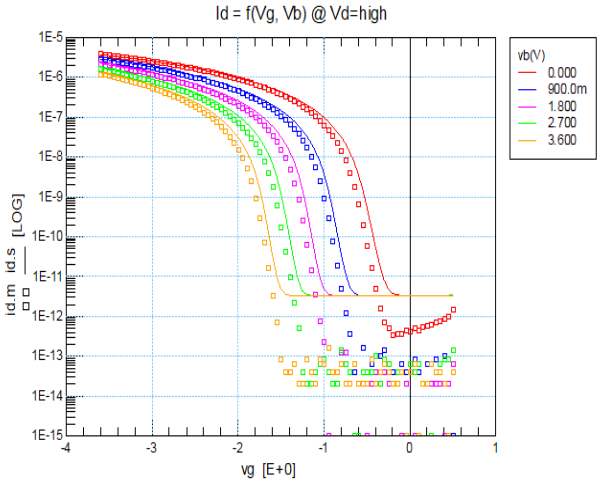


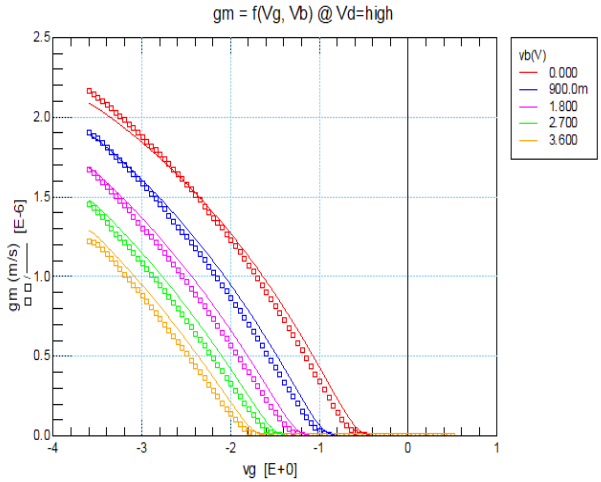


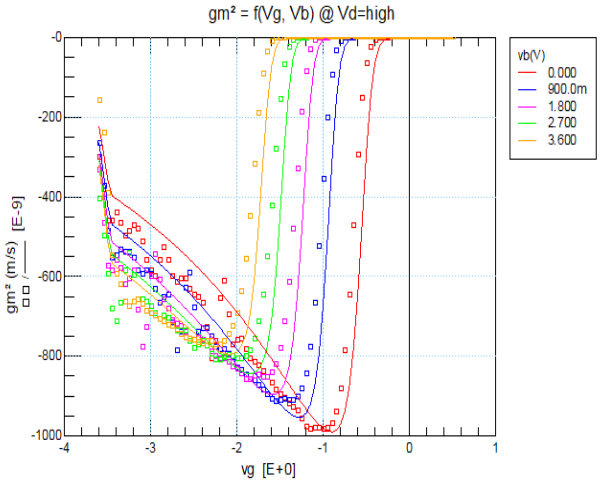


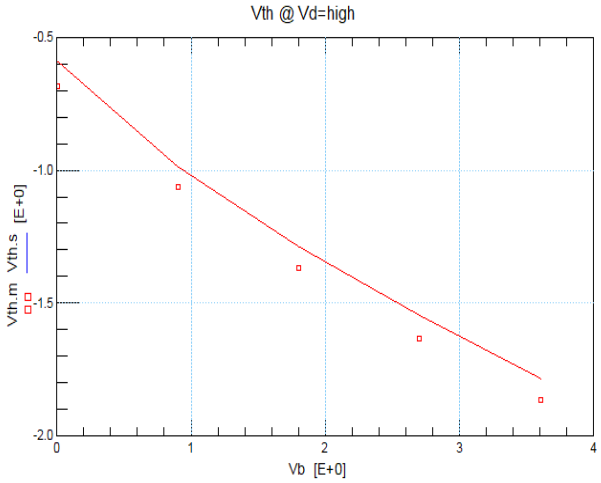
idvg, Vd = -3.3V, T = 27°C

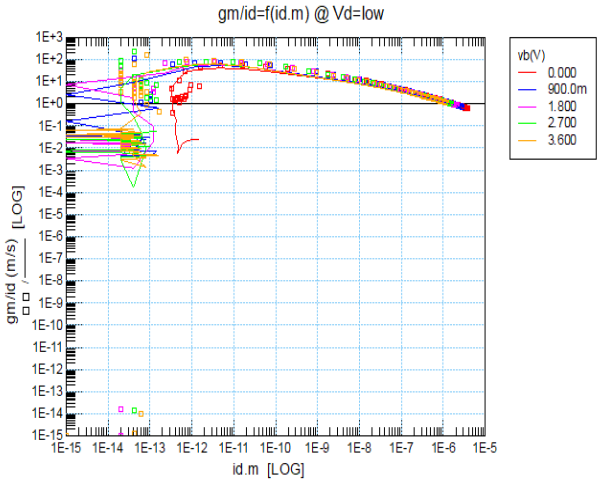


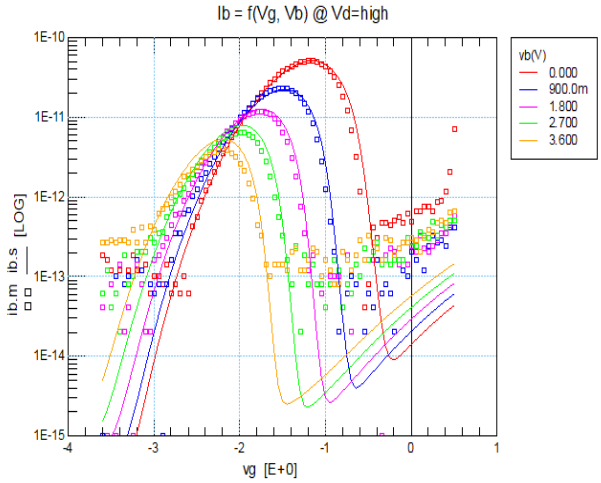


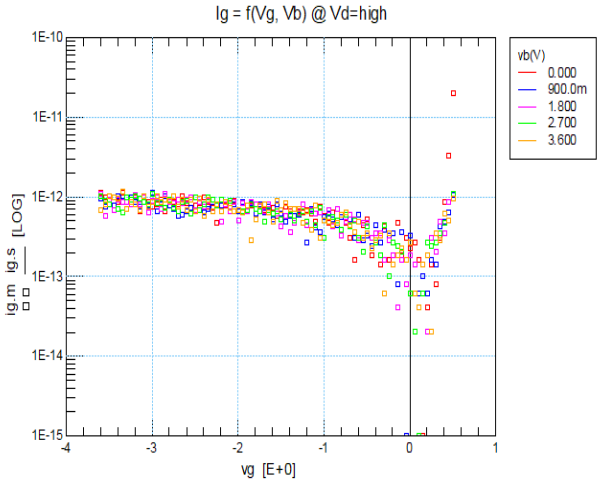




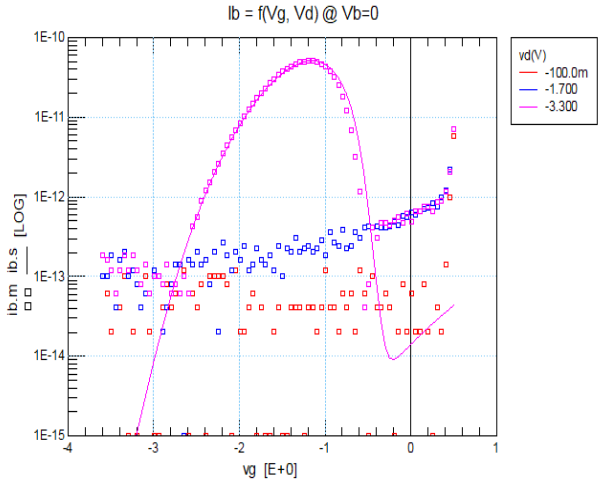


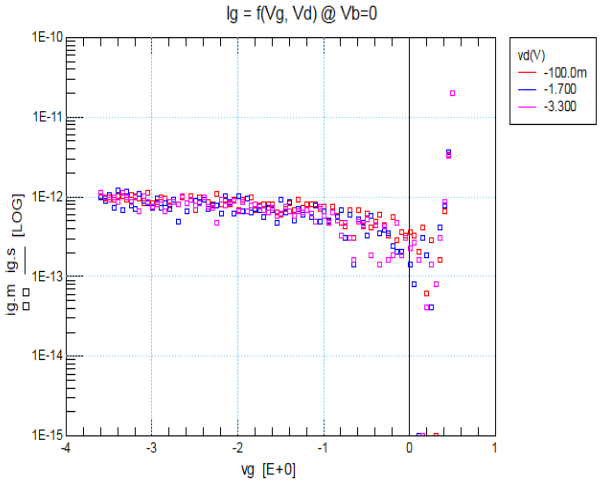




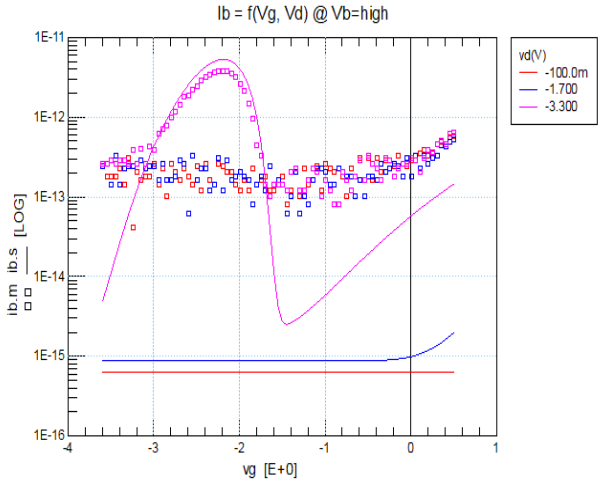


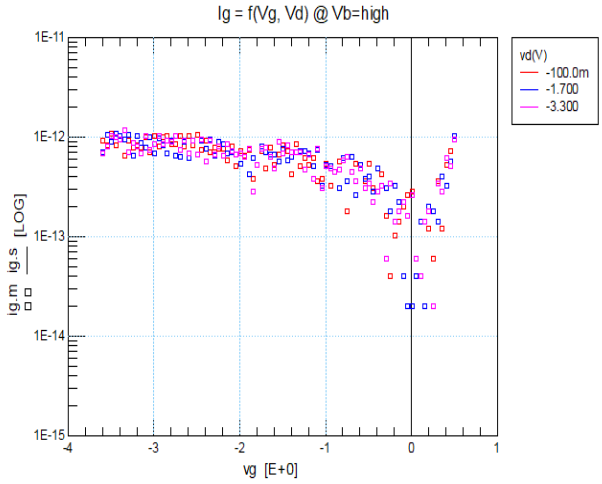
idvg, Ib,Ig, Vb = 0V, T = 27°C



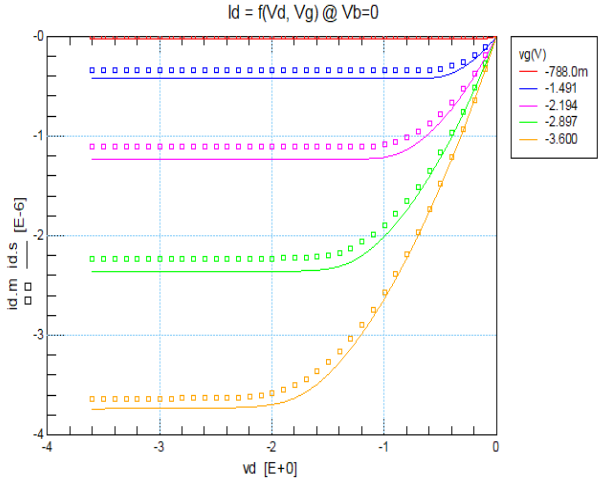


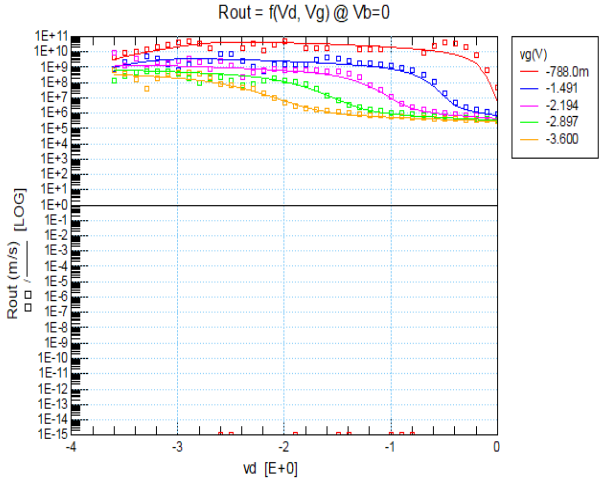
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

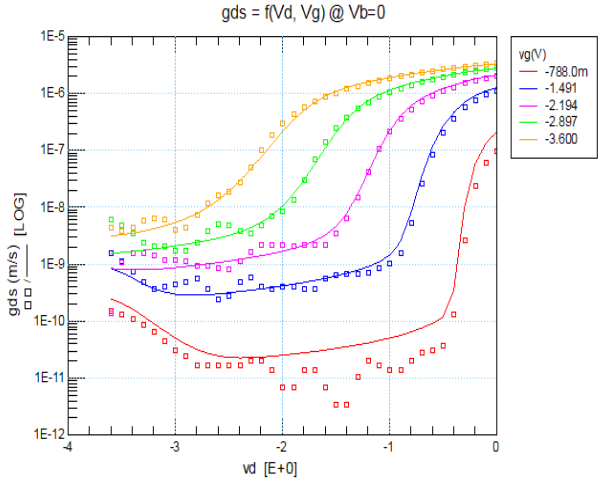




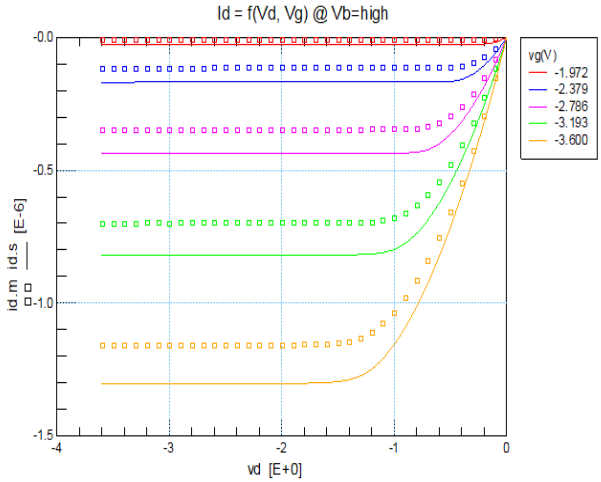
idvd, Vb = 0V, T = 27°C

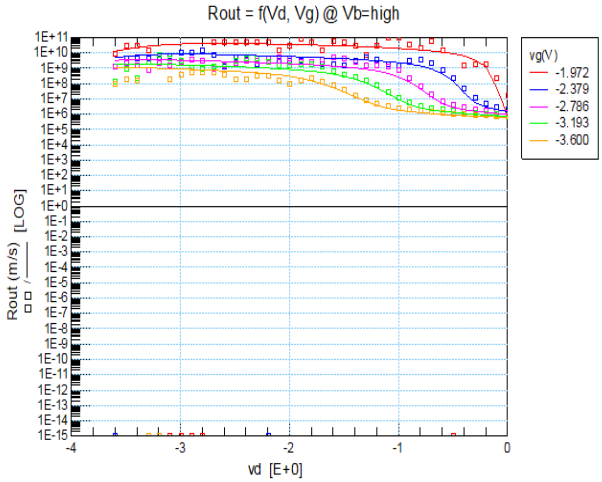


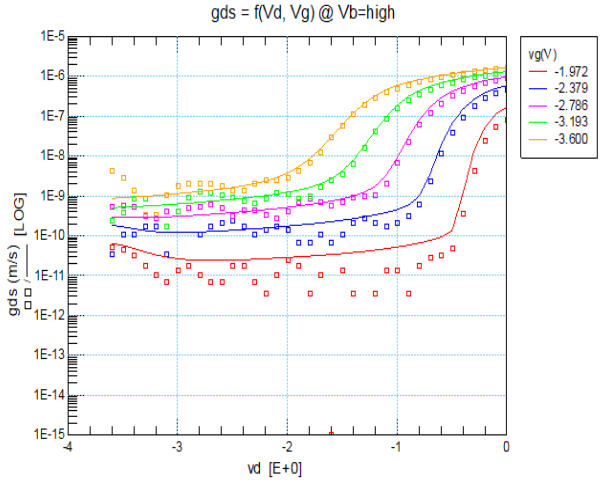




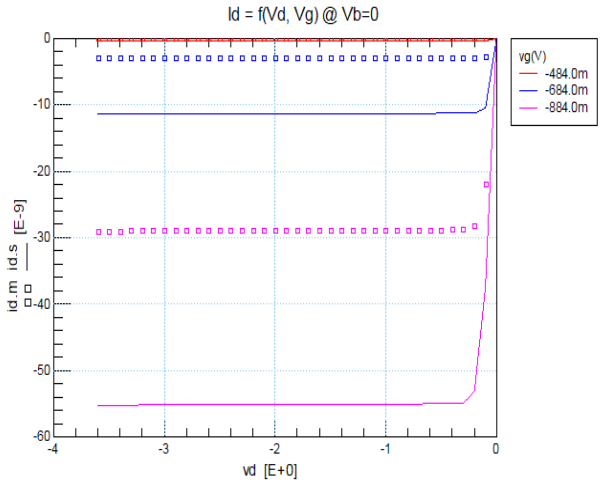
idvd_vbmin, Vb = 3.6V, T = 27°C

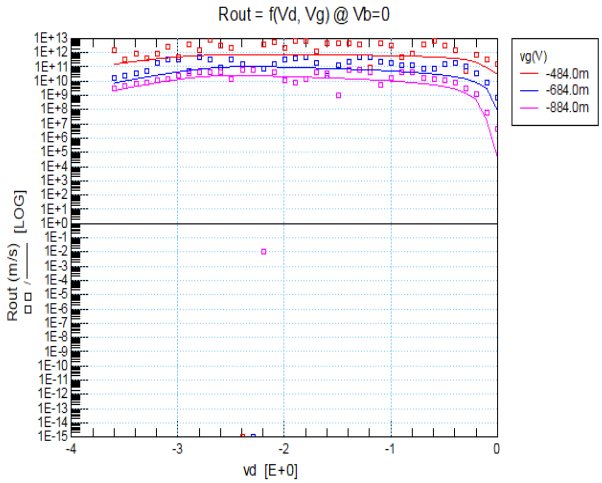


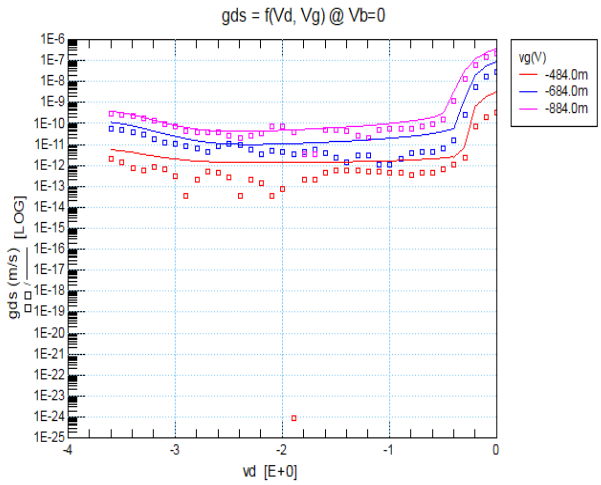




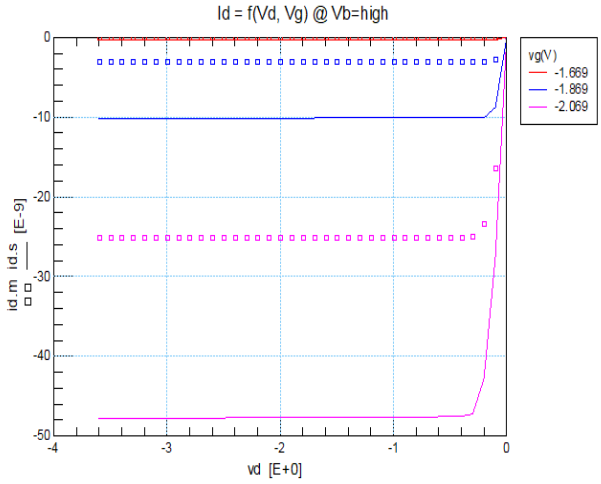
idvd_vth, Vb = 0V, T = 27°C

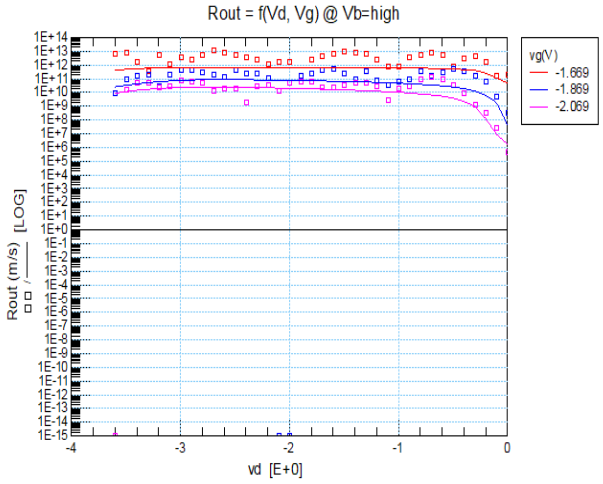


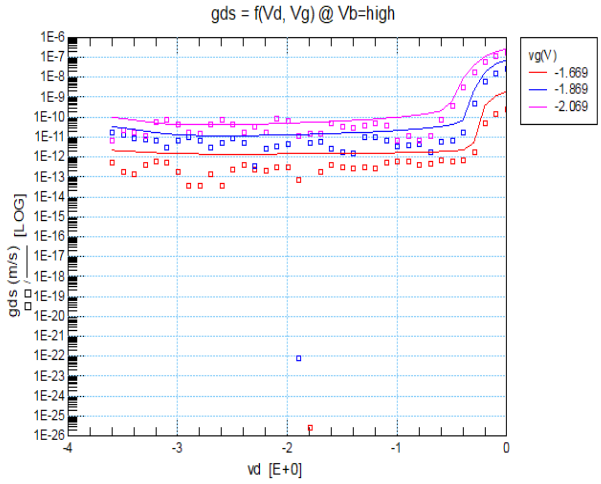




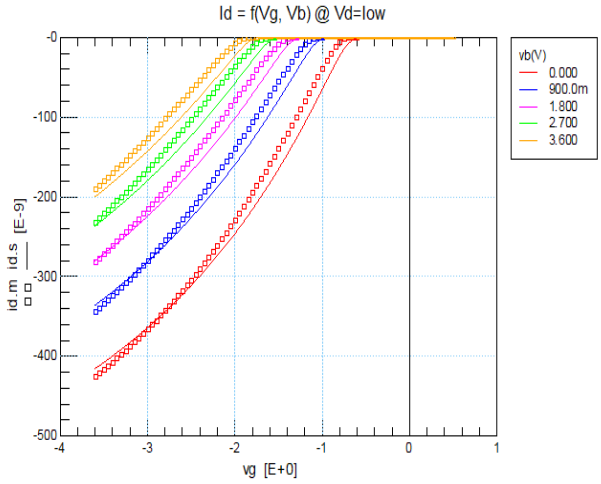
idvd_vth_vbmin, Vb = 3.6V, T = 27°C

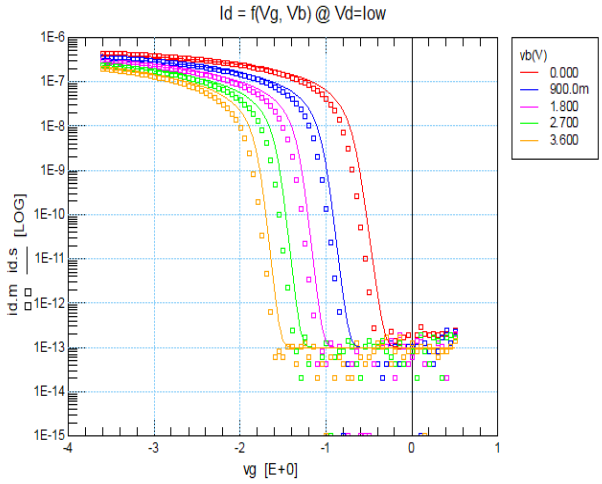


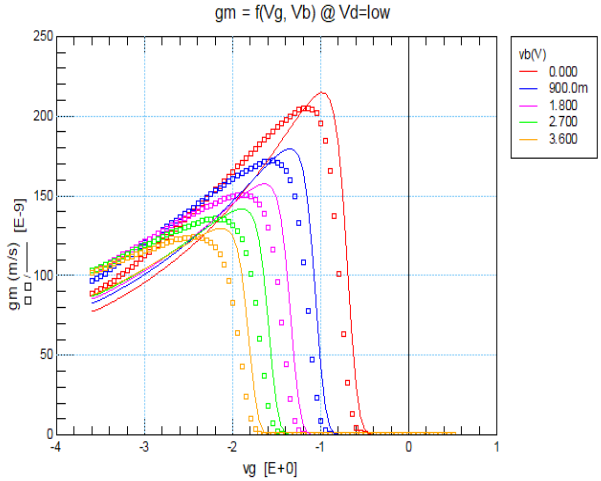


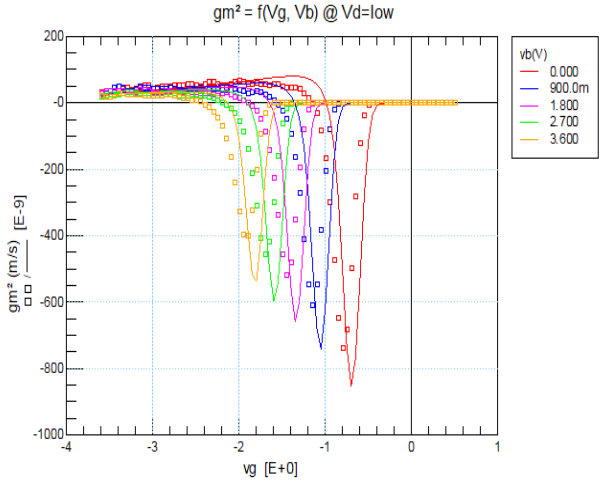


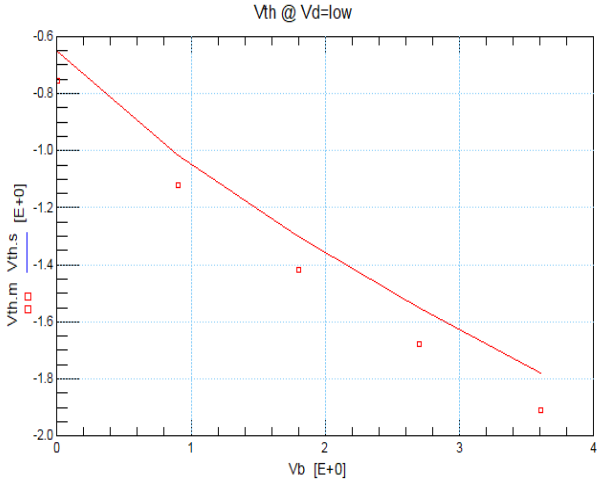
idvg, Vd = -0.1V, T = -40°C

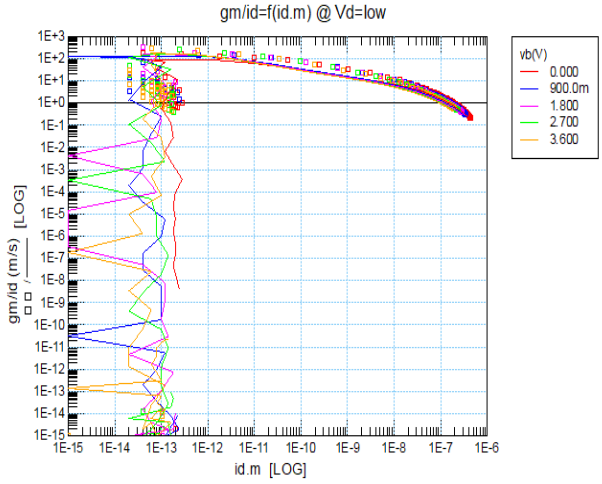


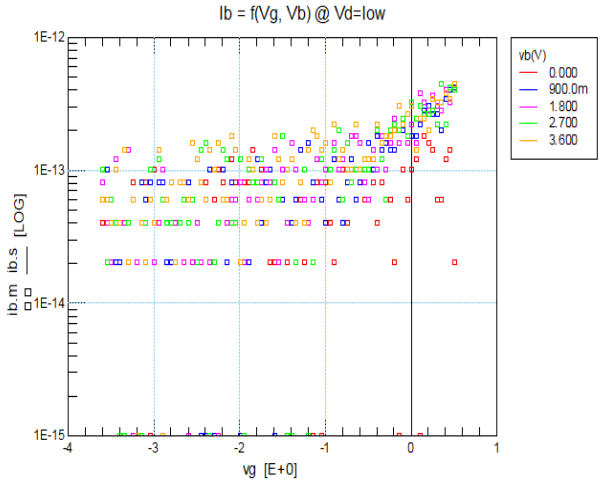


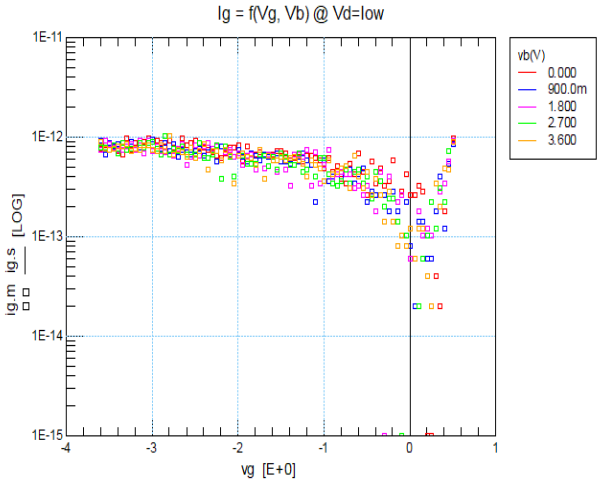




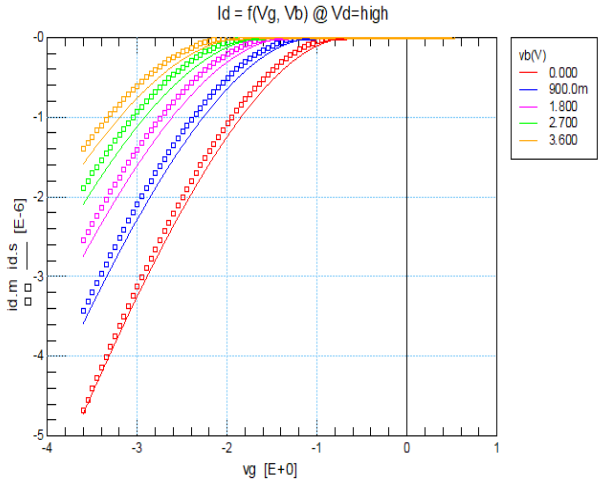


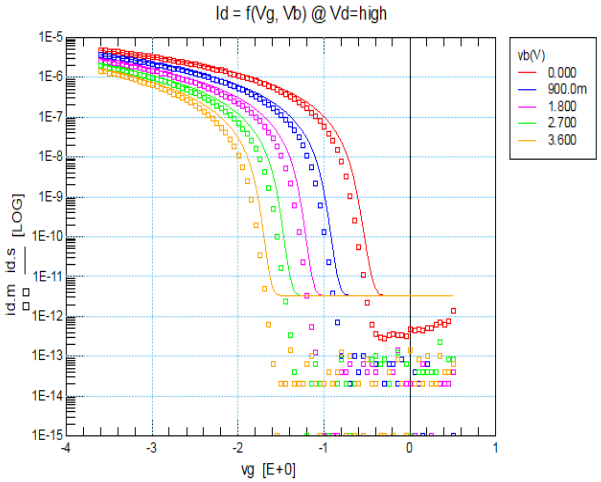


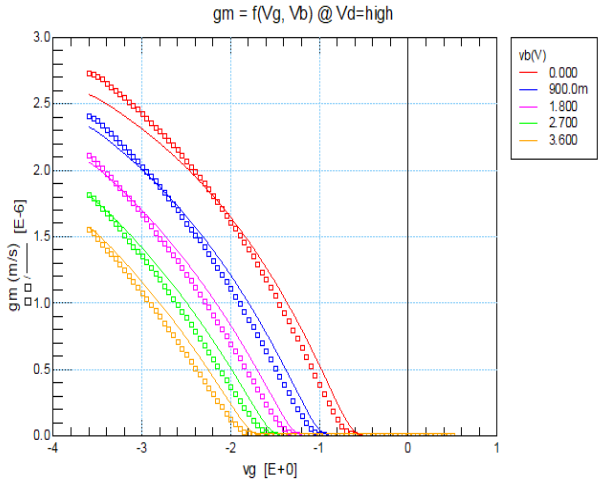


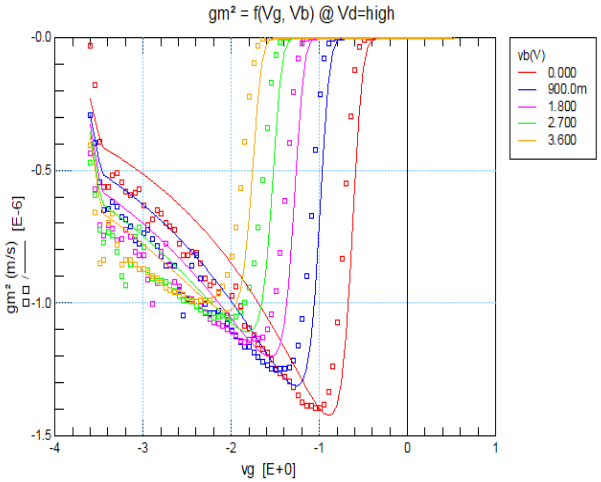


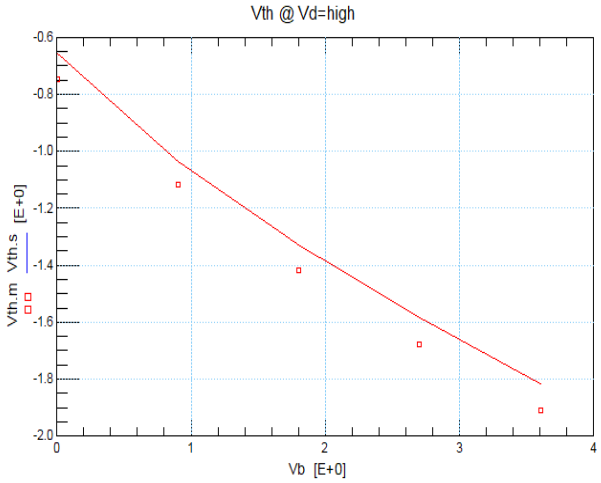
idvg, Vd = -3.3V, T = -40°C

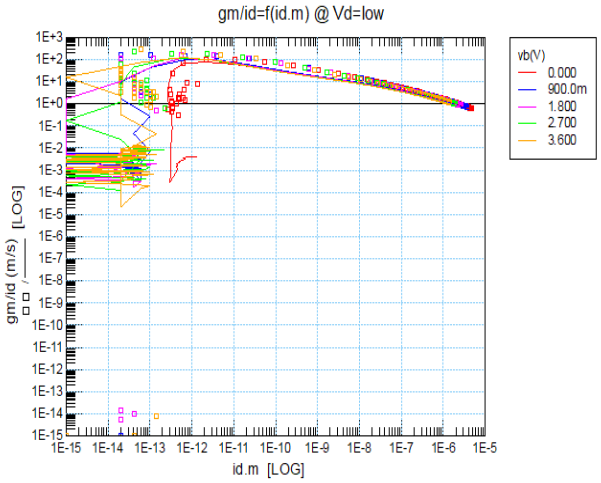


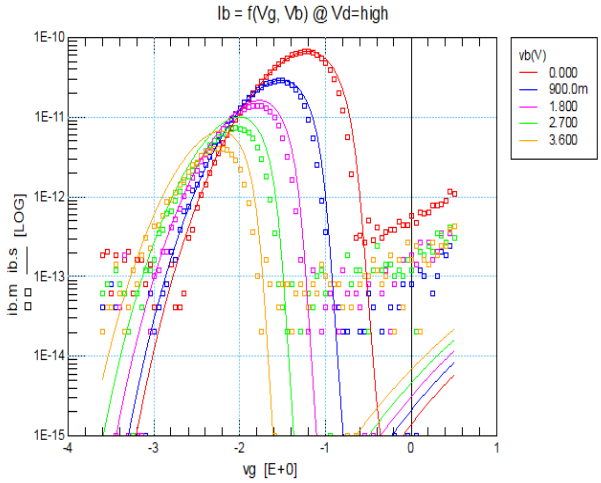


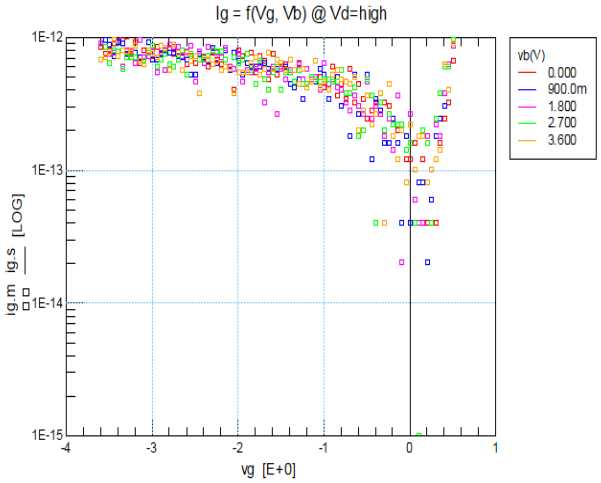




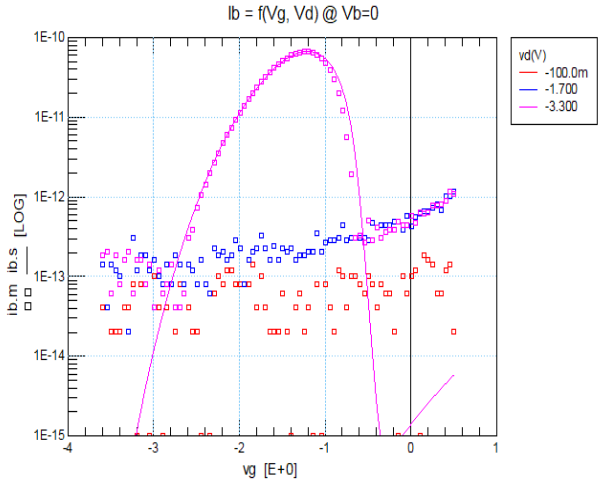


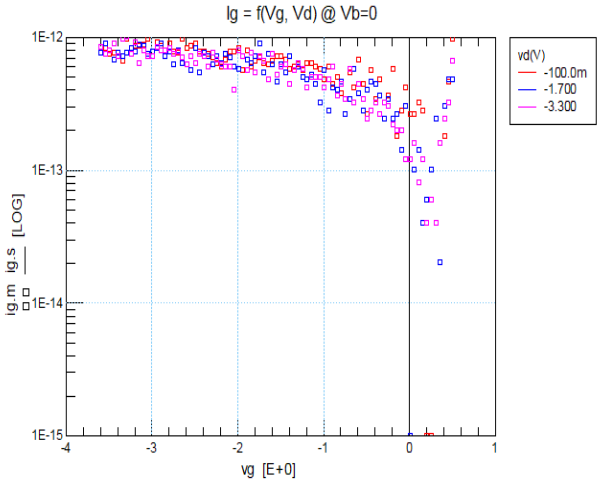




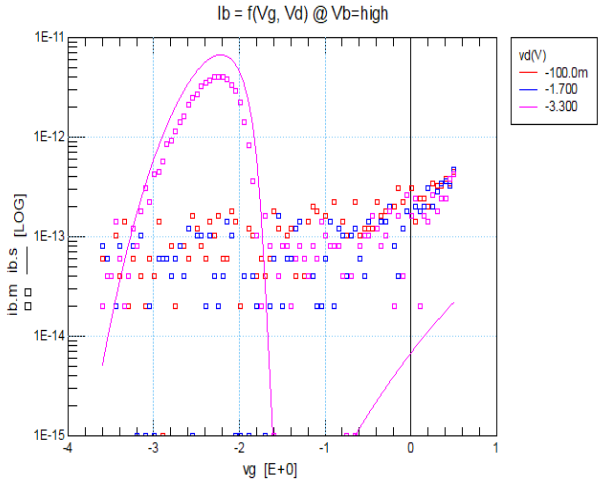


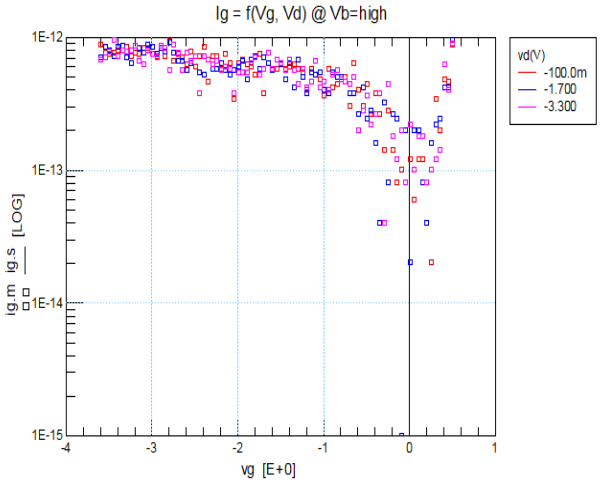
idvg, Ib,Ig, Vb = 0V, T = -40°C



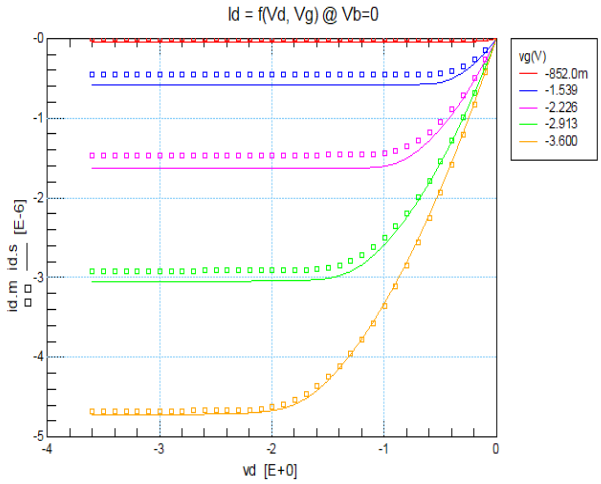


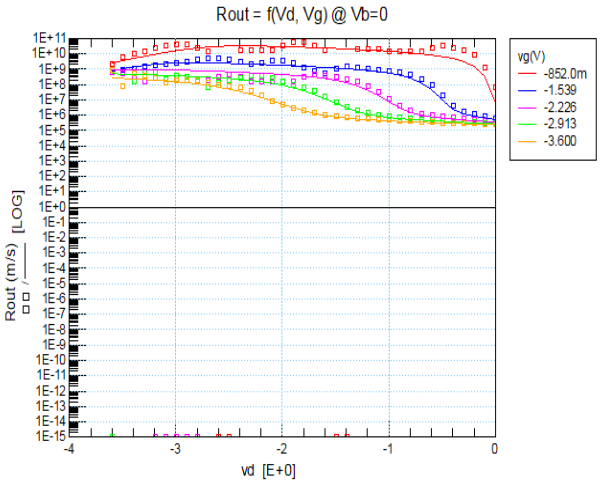
idvg, Ib,Ig, Vb = 3.6V, T = -40°C

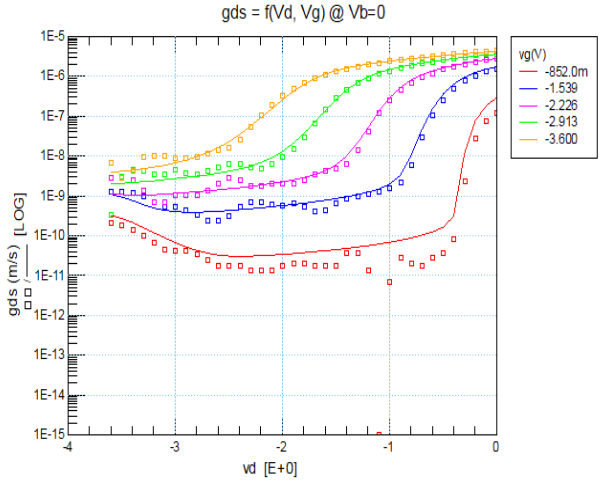




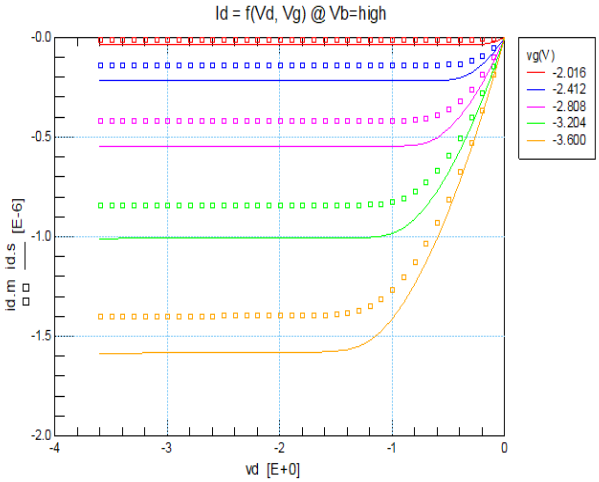
idvd, Vb = 0V, T = -40°C

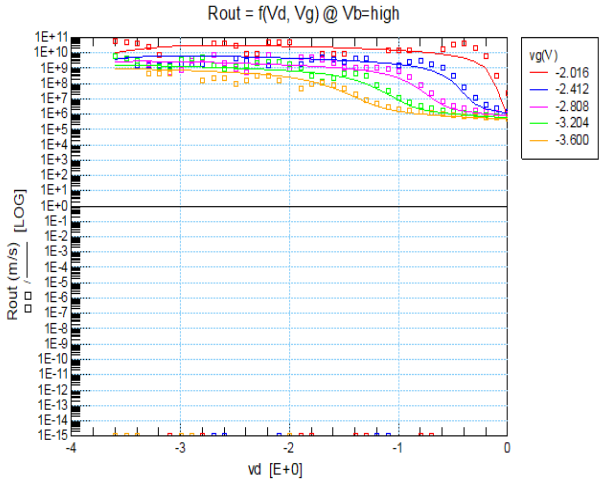


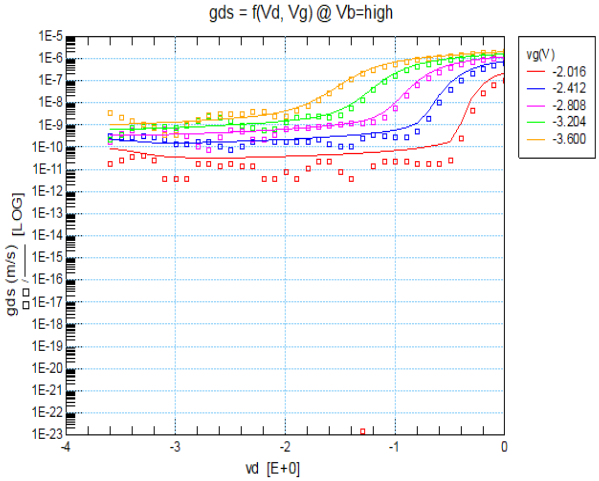




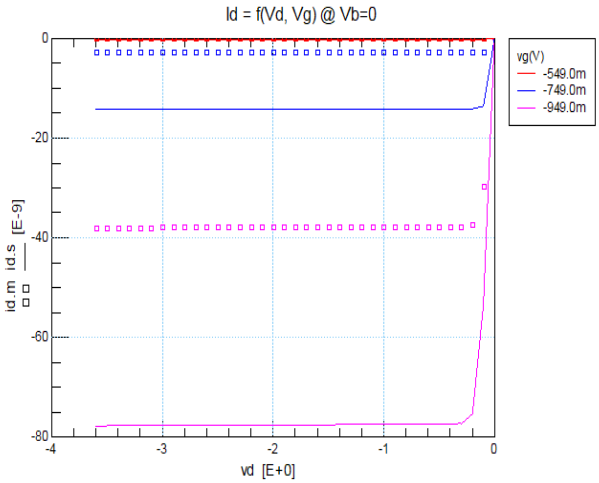
idvd_vbmin, Vb = 3.6V, T = -40°C

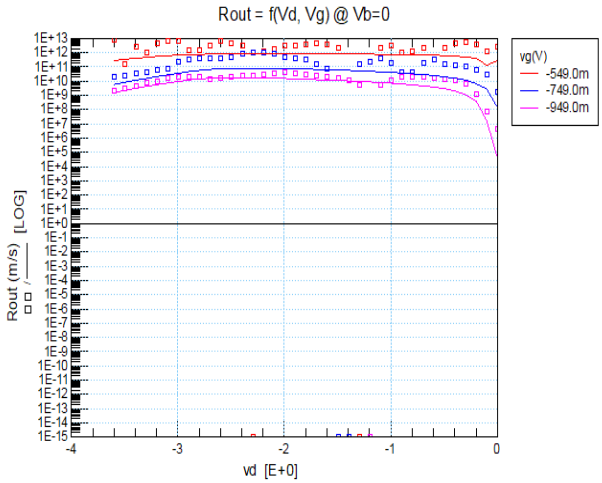


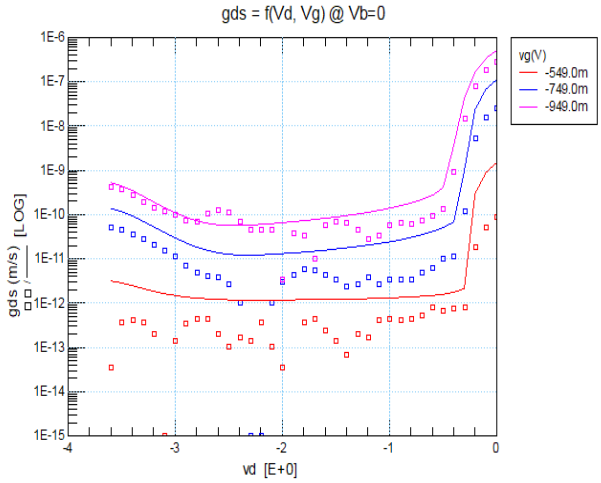




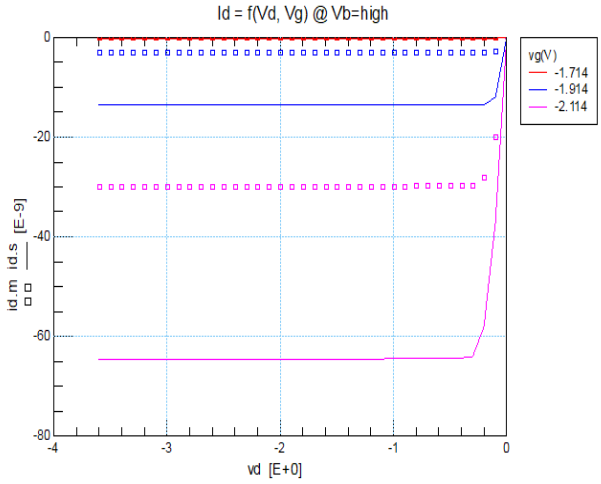
idvd_vth, Vb = 0V, T = -40°C

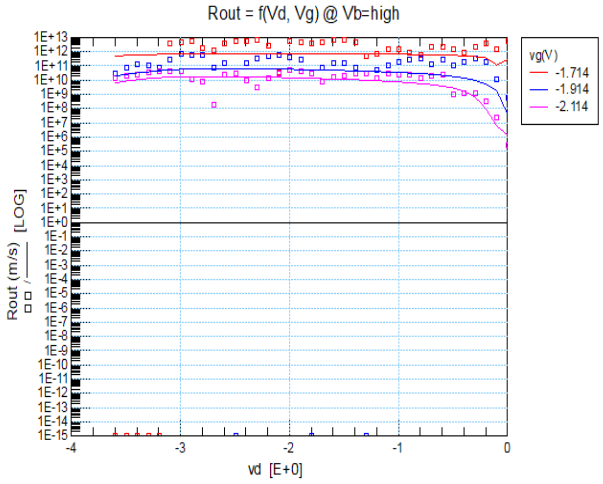


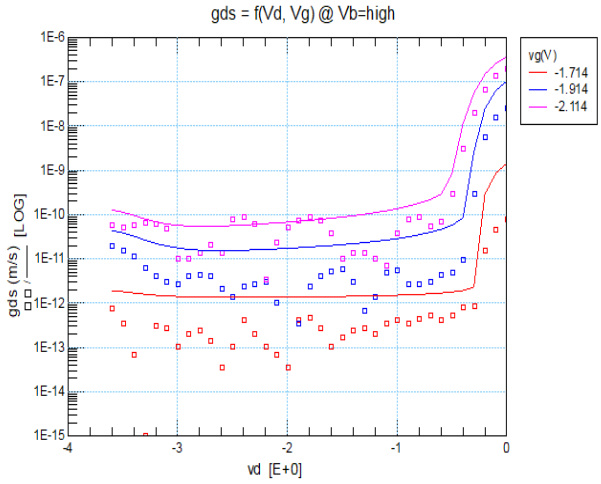




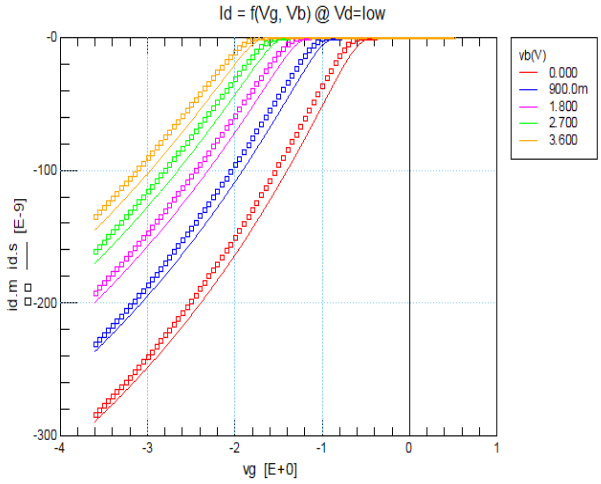
idvd_vth_vbmin, Vb = 3.6V, T = -40°C

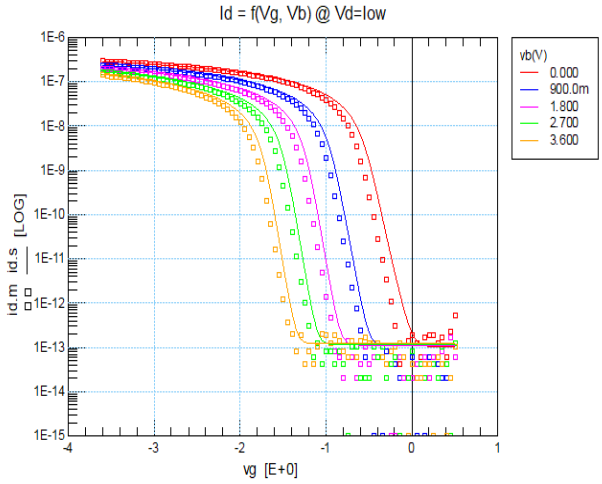


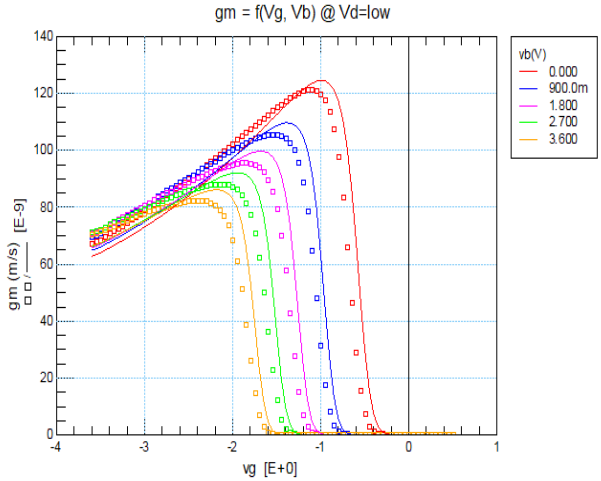


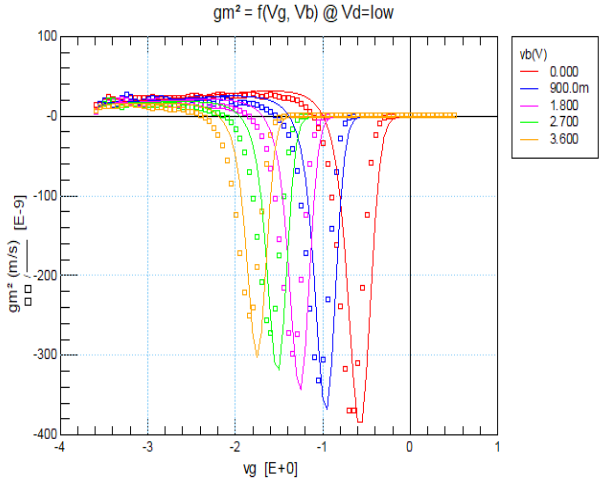


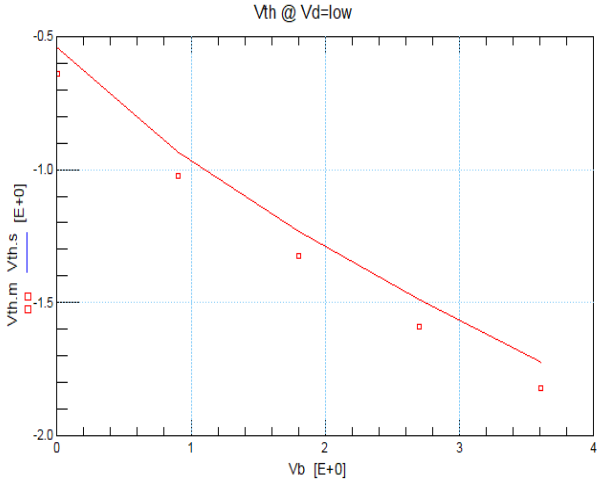
idvg, Vd = -0.1V, T = 70°C

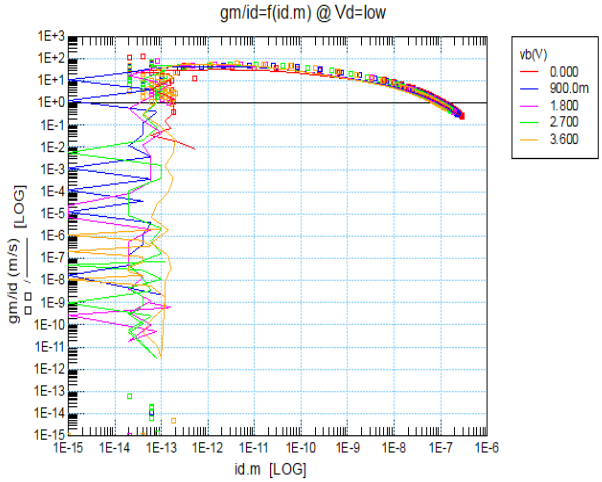


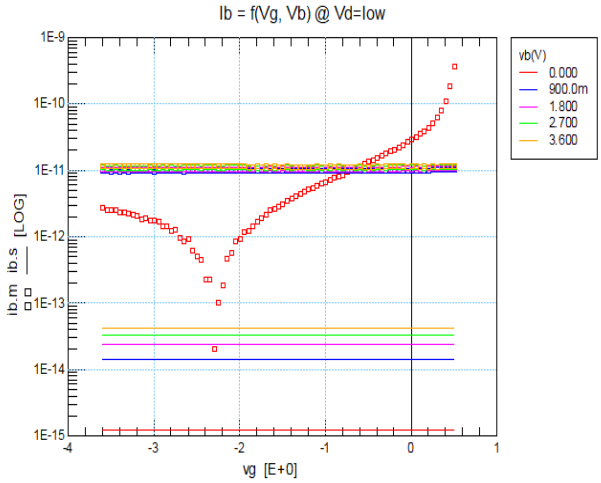


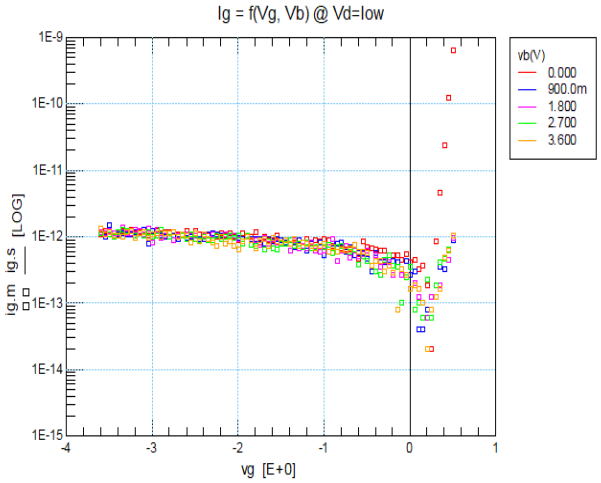




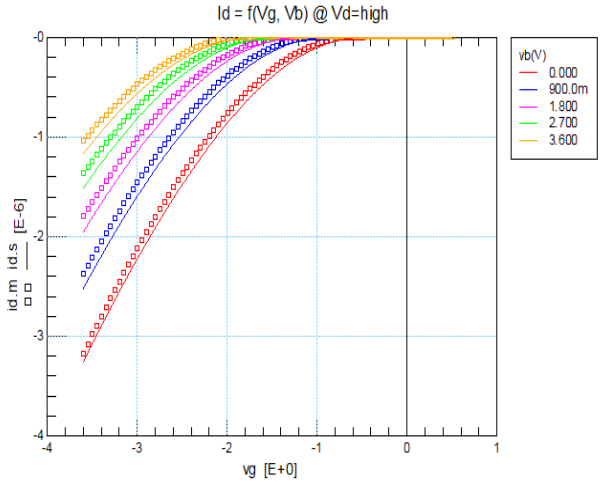


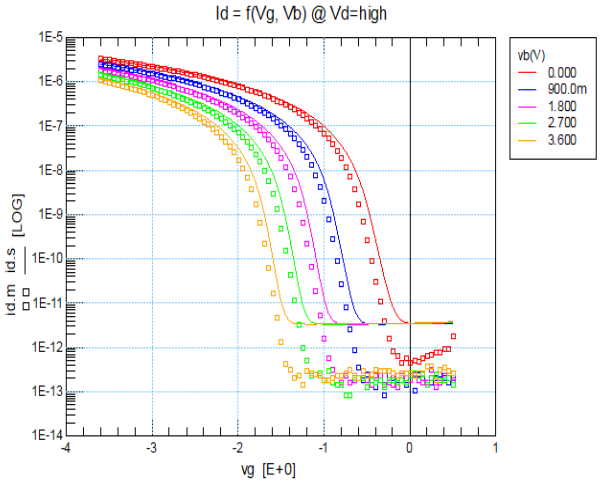


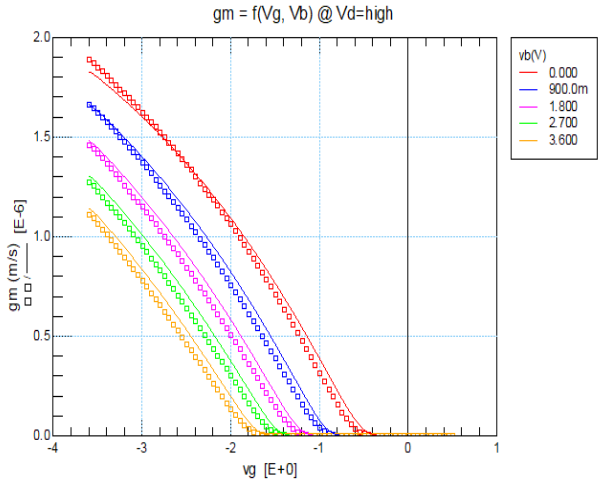


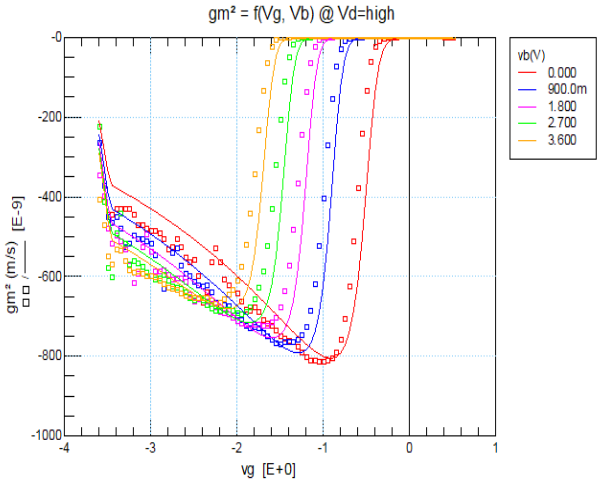


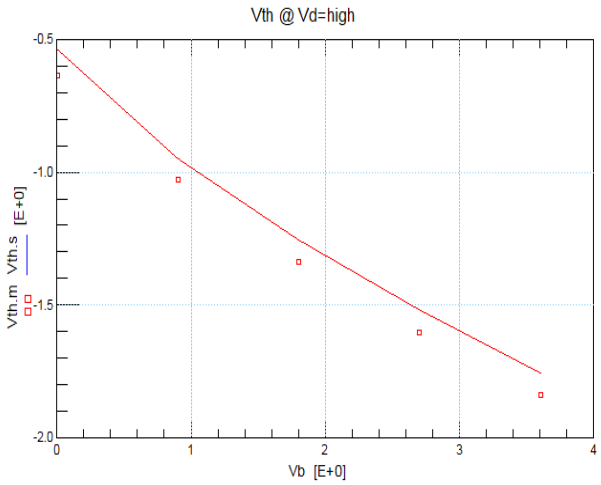
idvg, Vd = -3.3V, T = 70°C

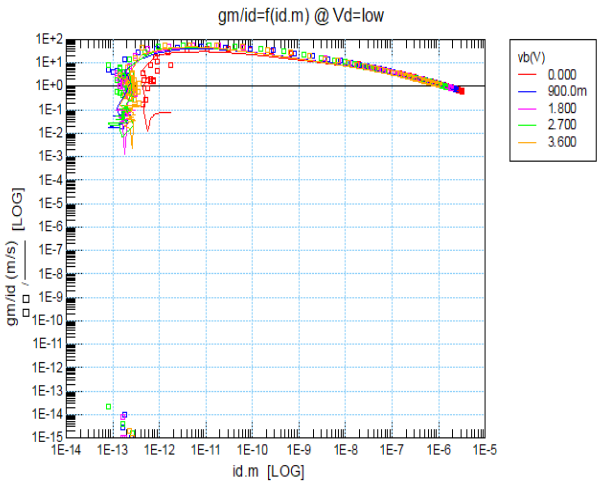


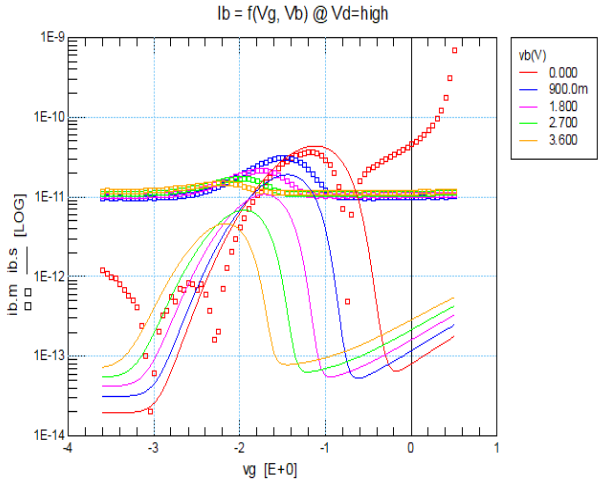


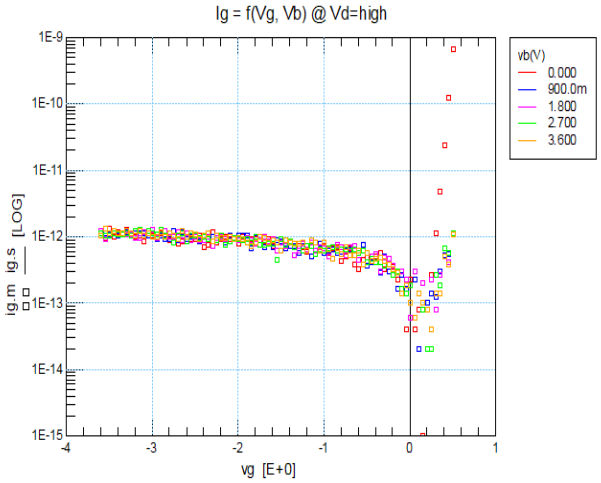




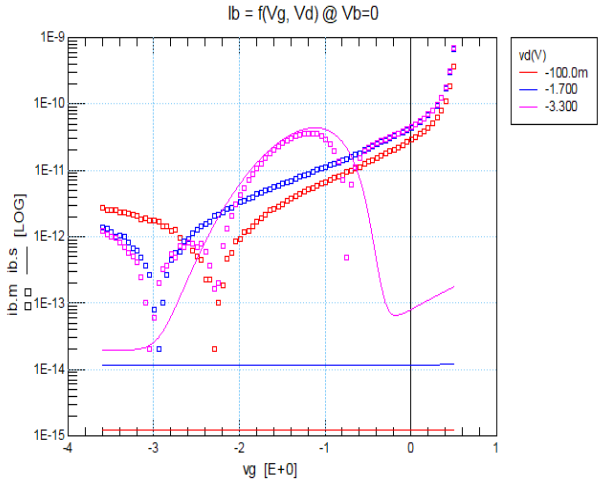


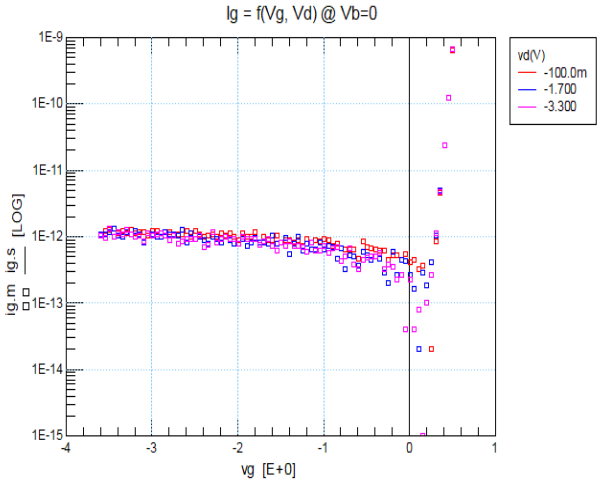




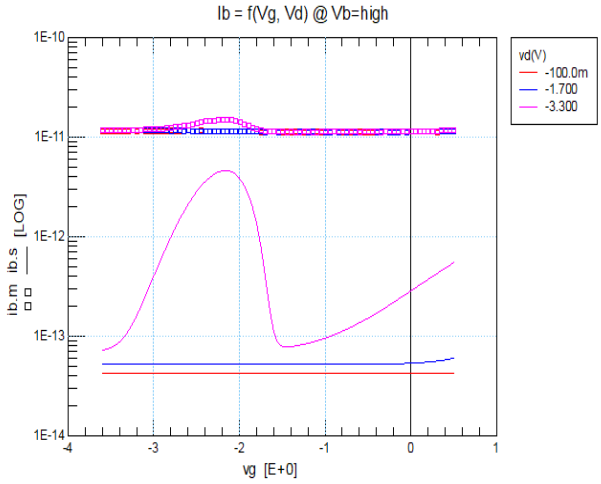


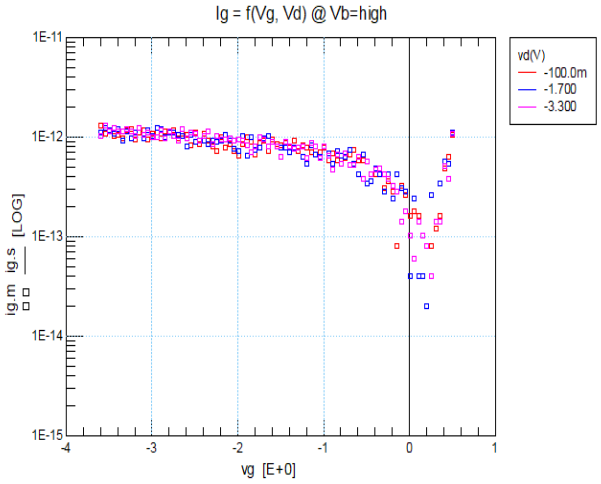
idvg, Ib,Ig, Vb = 0V, T = 70°C



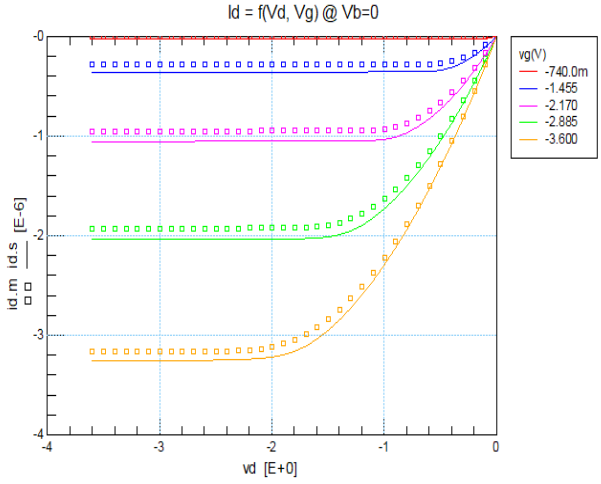


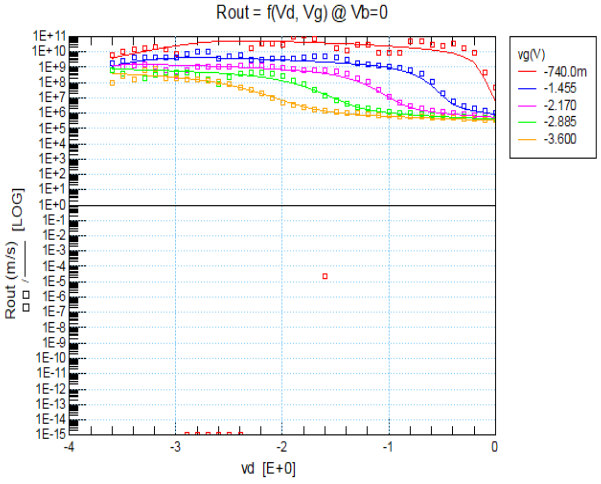
idvg, Ib,Ig, Vb = 3.6V, T = 70°C

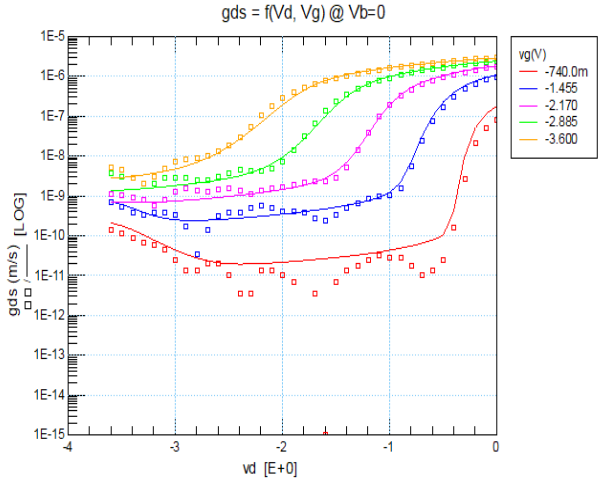




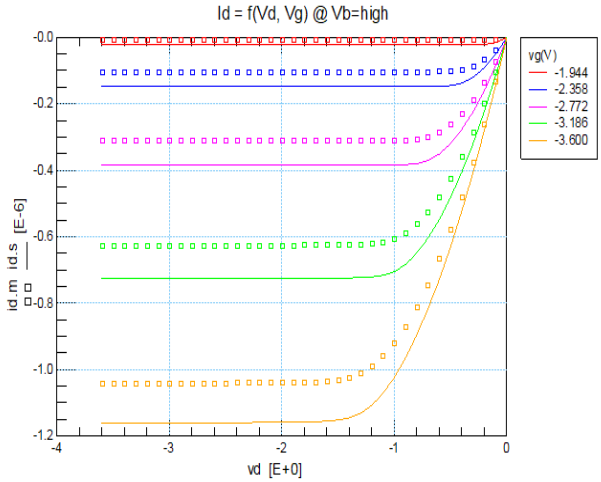
idvd, Vb = 0V, T = 70°C

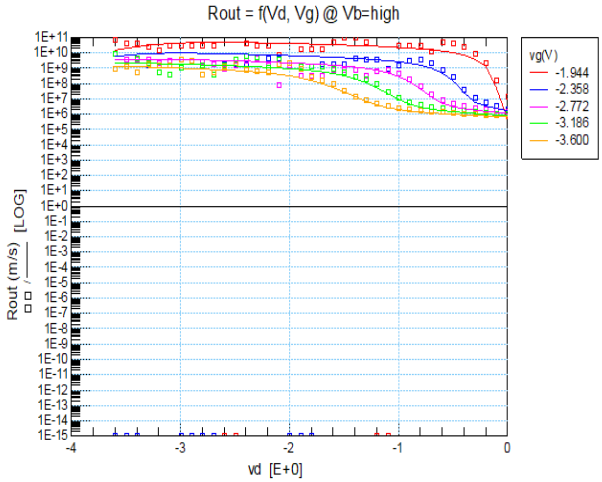


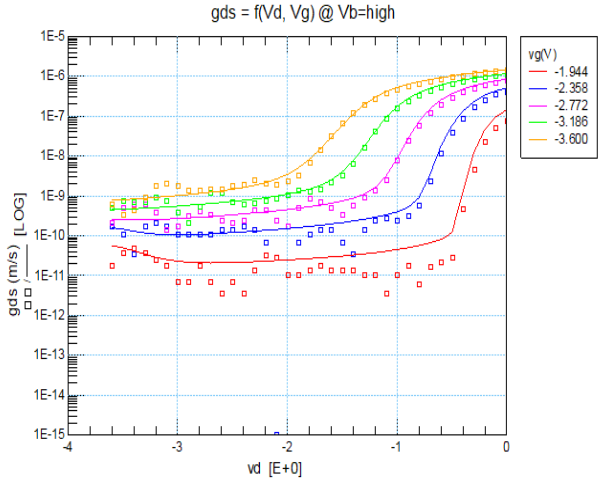




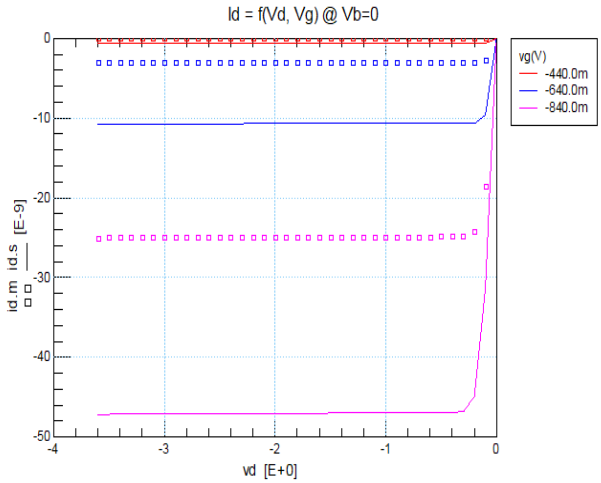
idvd_vbmin, Vb = 3.6V, T = 70°C

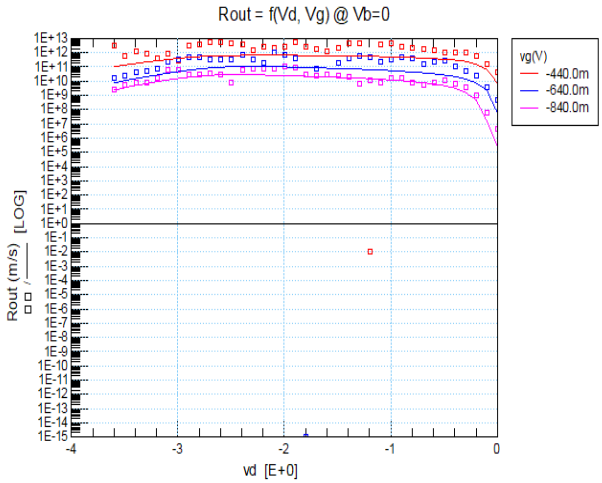


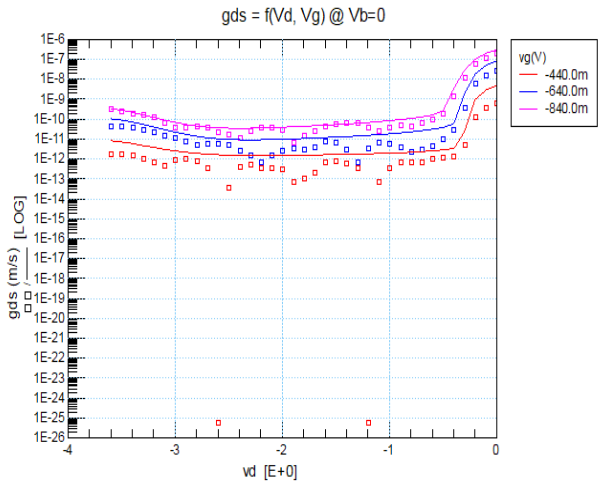




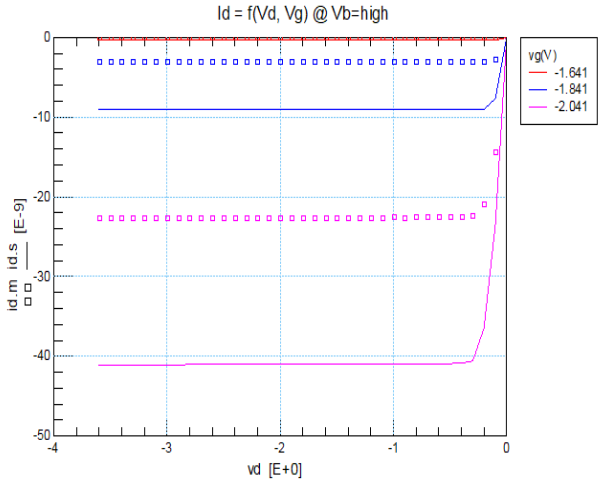
idvd_vth, Vb = 0V, T = 70°C

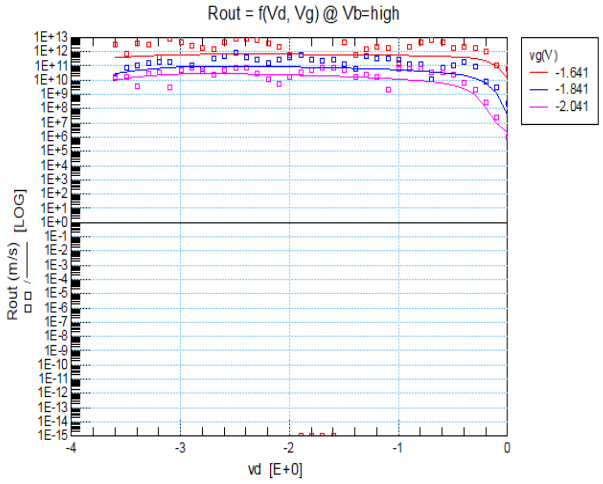


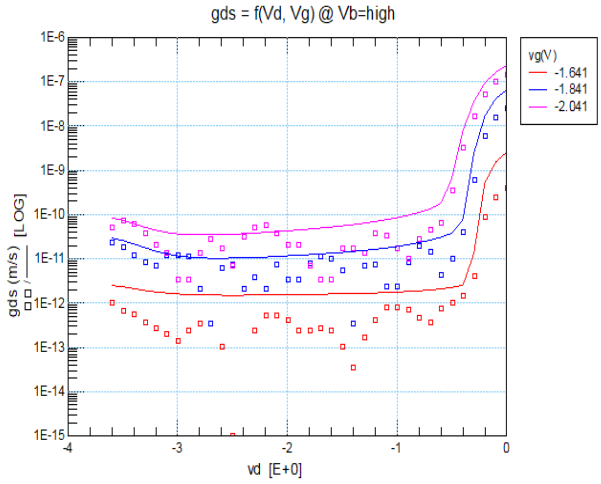




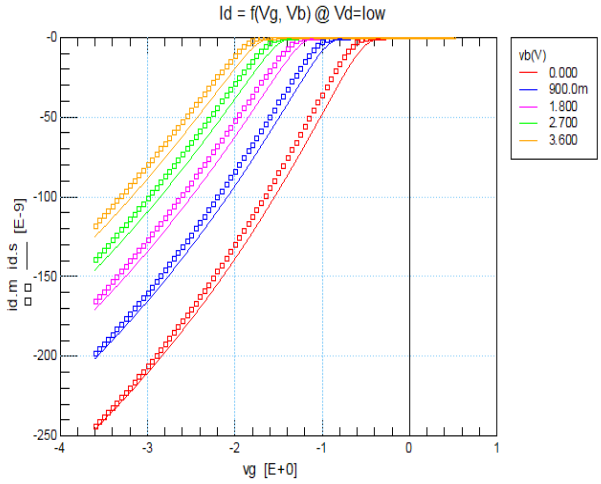
idvd_vth_vbmin, Vb = 3.6V, T = 70°C

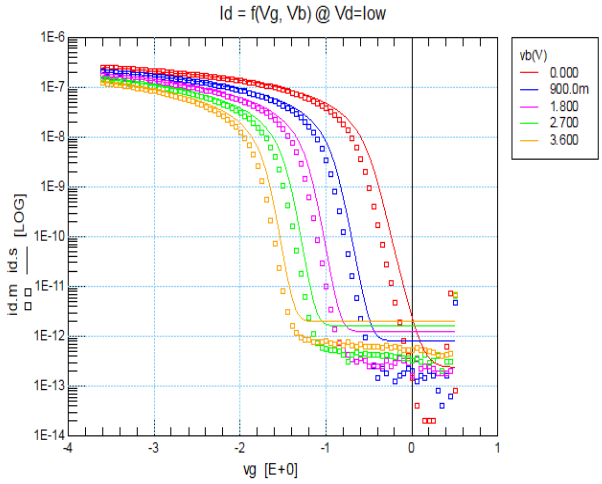


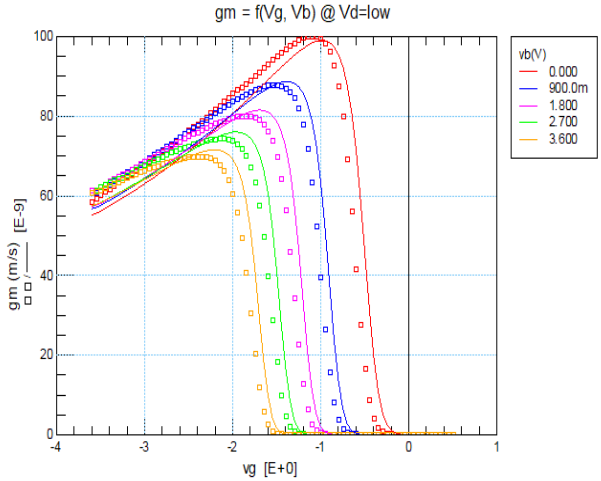


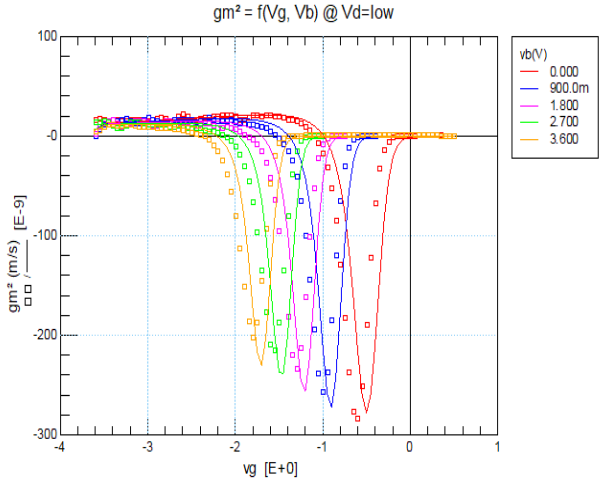


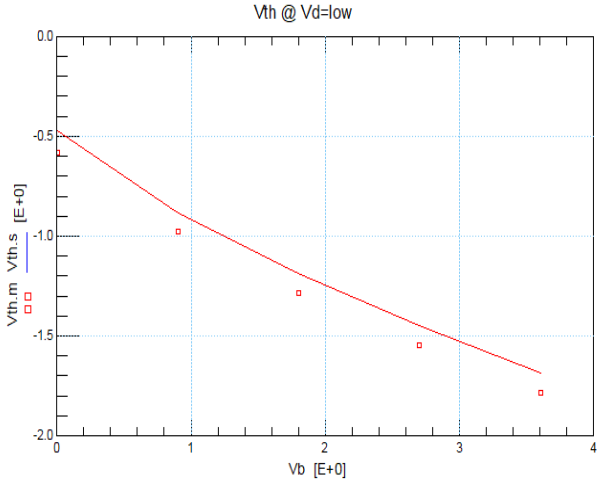
idvg, Vd = -0.1V, T = 125°C

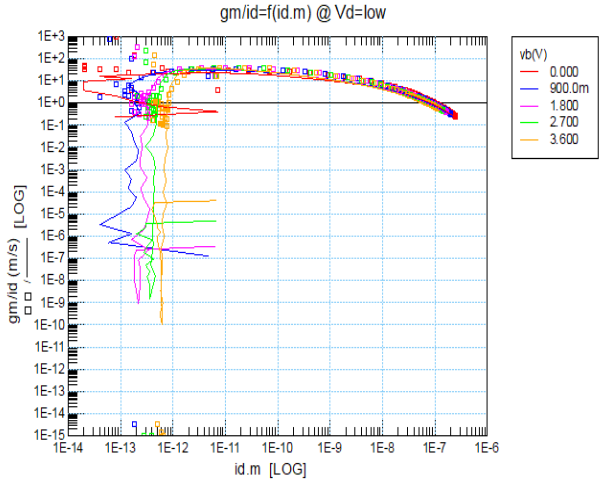


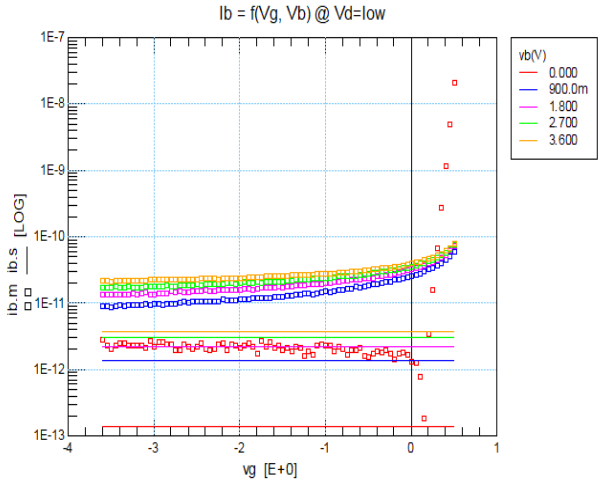


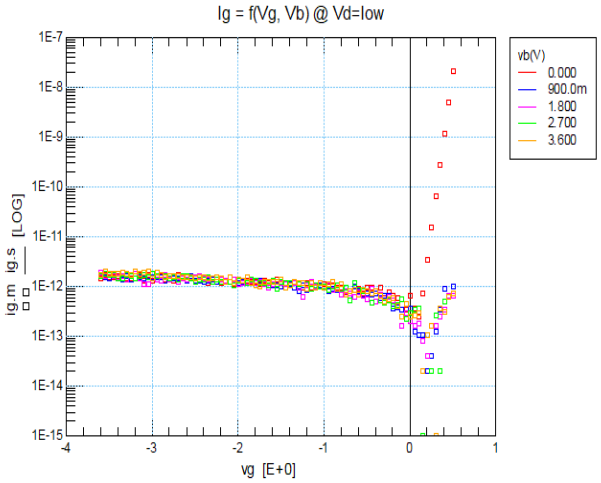




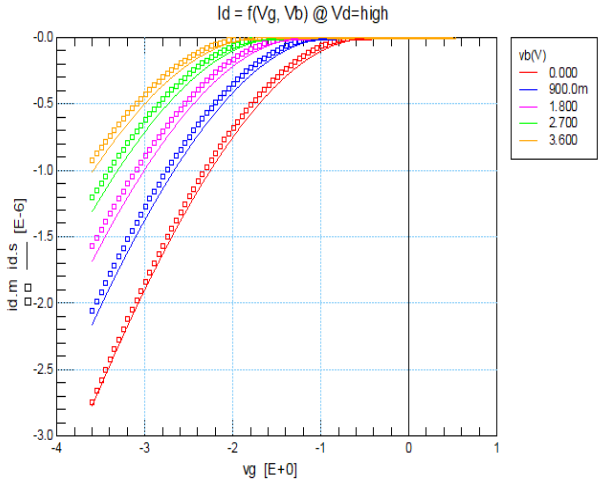


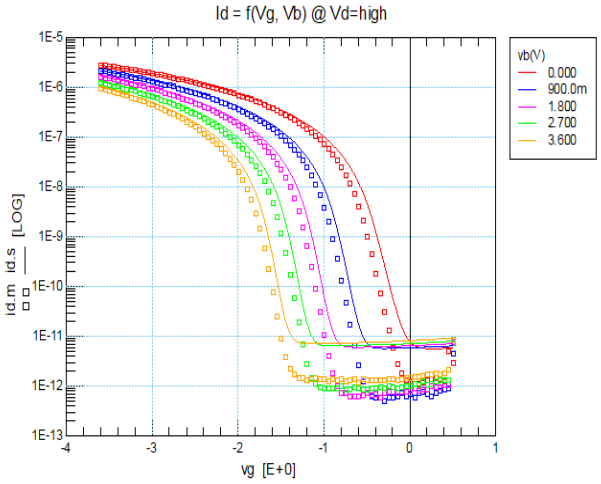


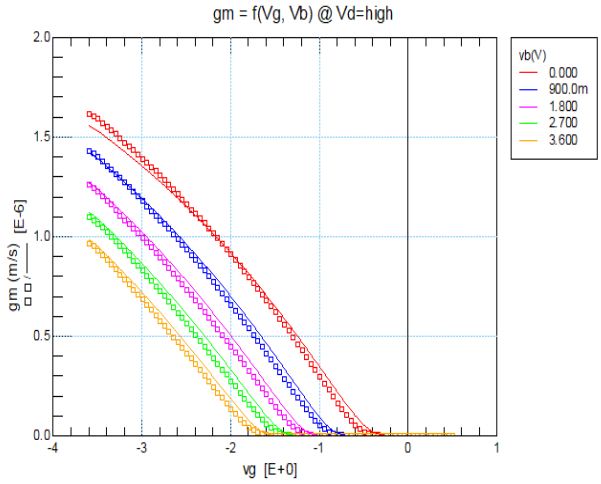


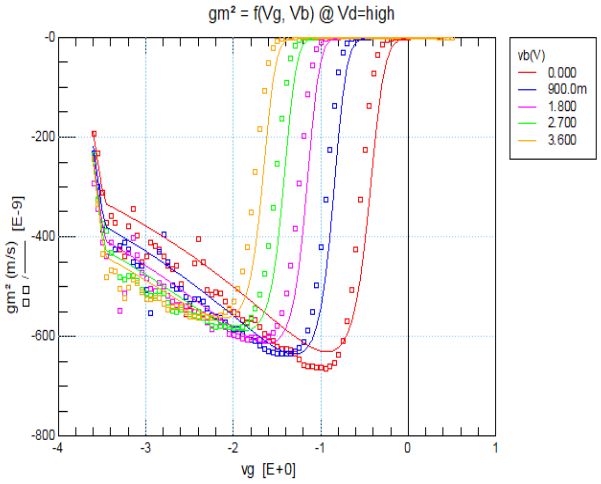


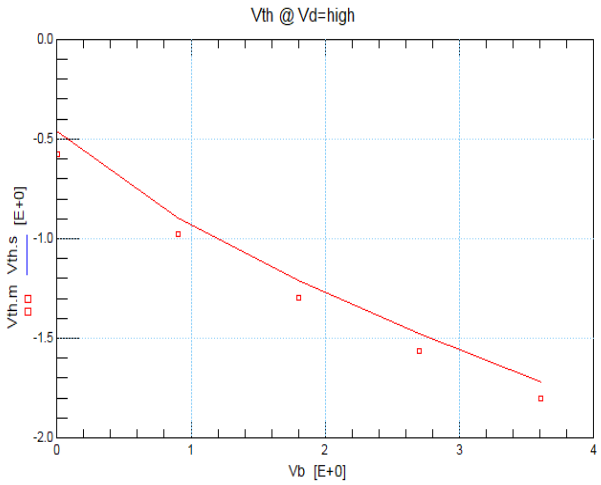
idvg, Vd = -3.3V, T = 125°C

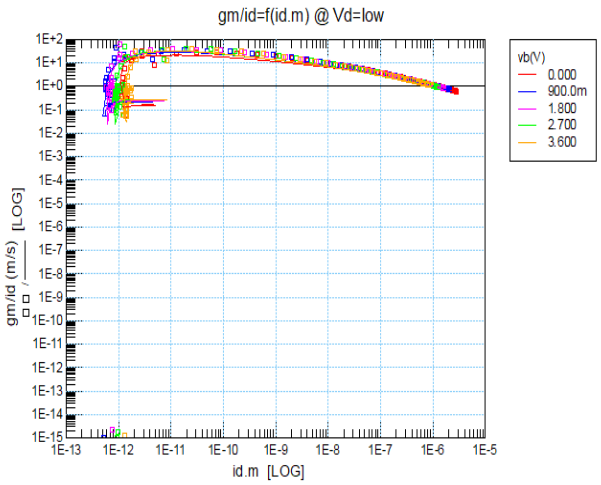


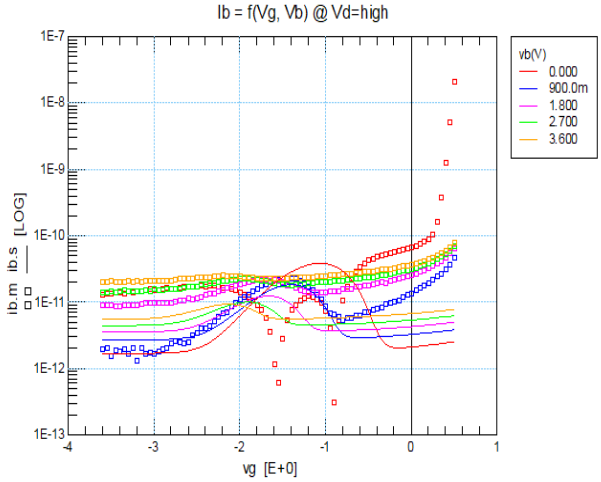


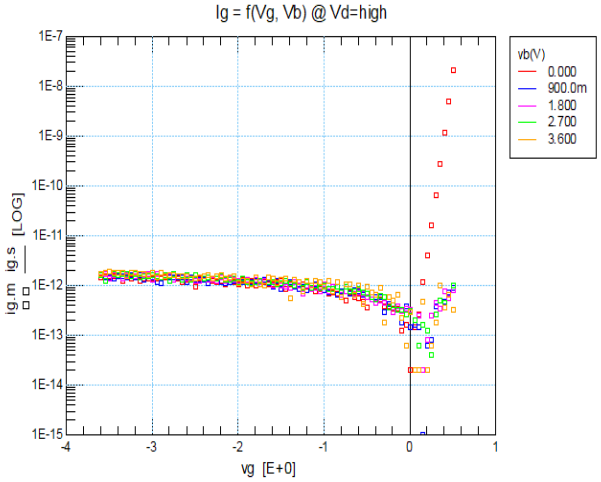




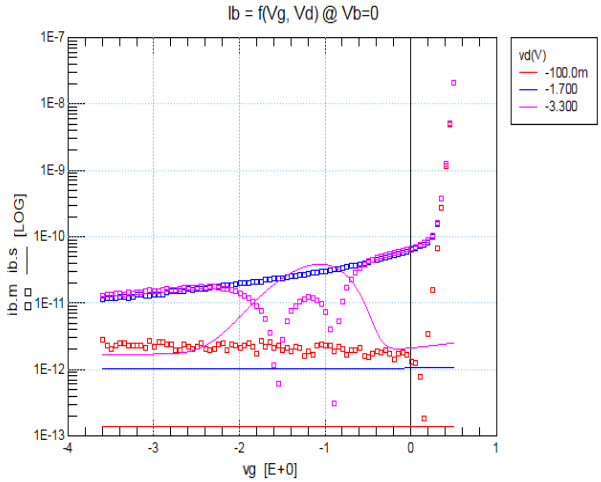


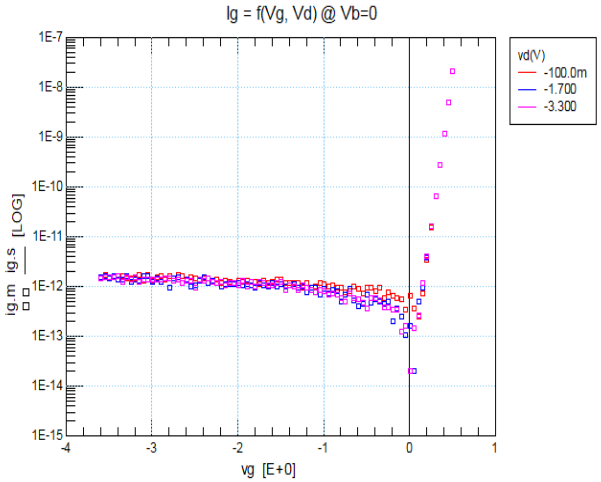




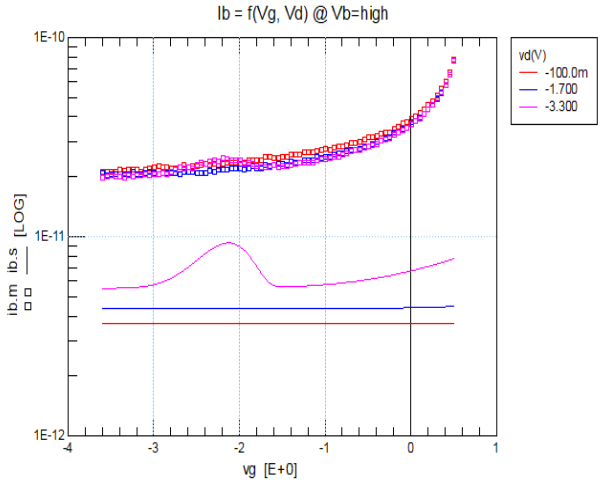


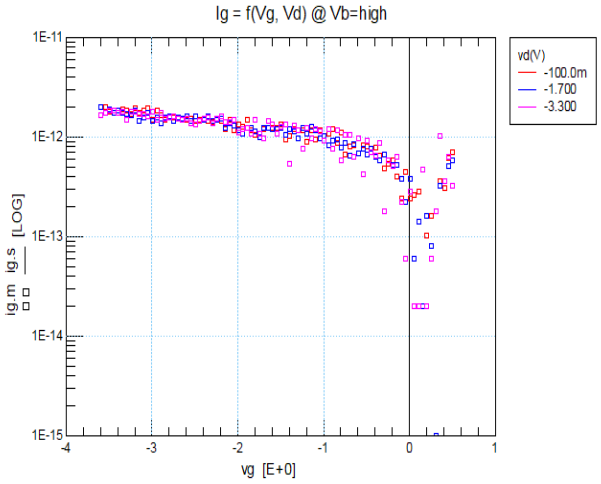
idvg, Ib,Ig, Vb = 0V, T = 125°C



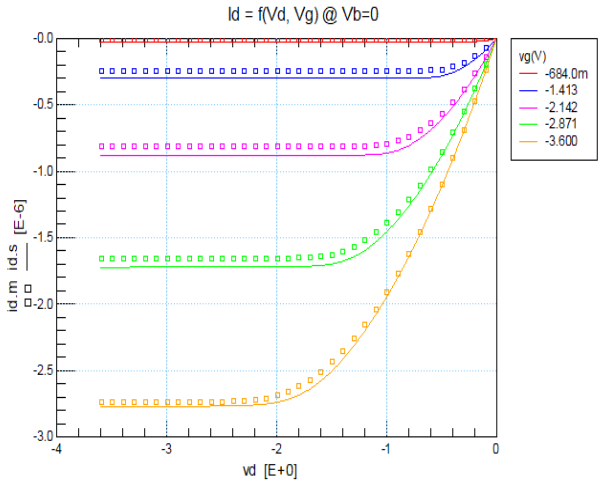


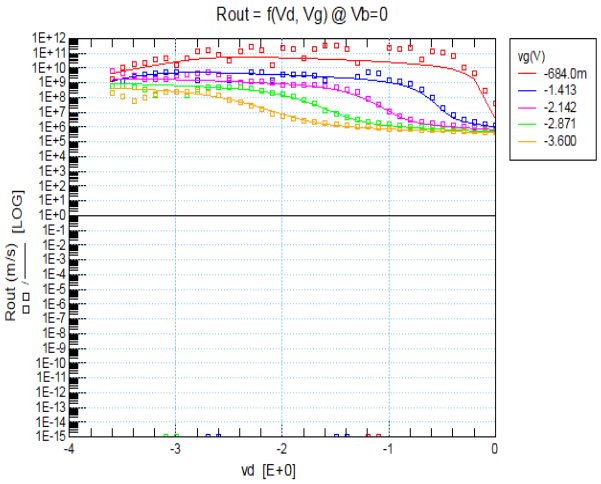
idvg, Ib,Ig, Vb = 3.6V, T = 125°C

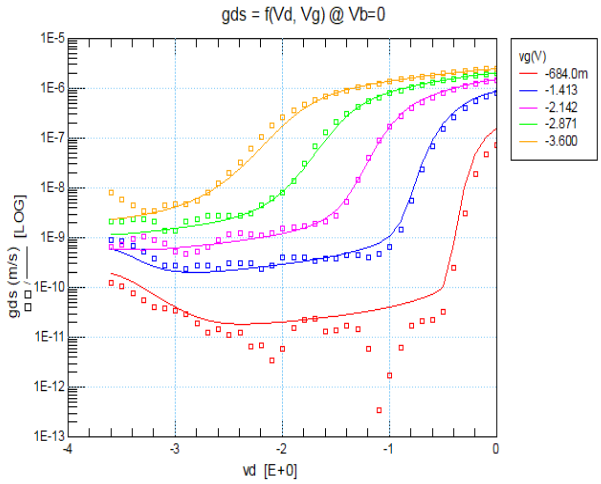




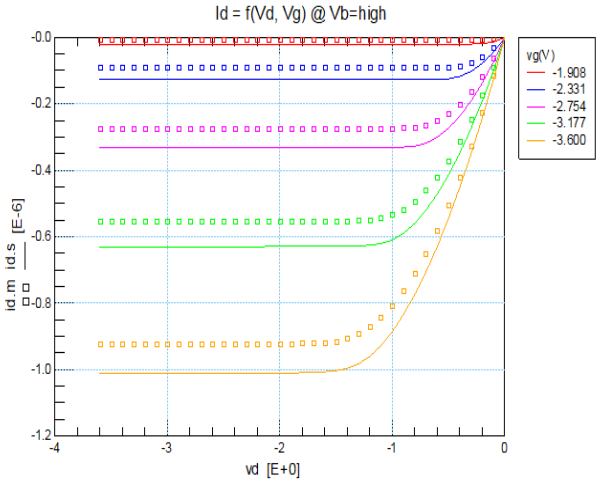
idvd, Vb = 0V, T = 125°C

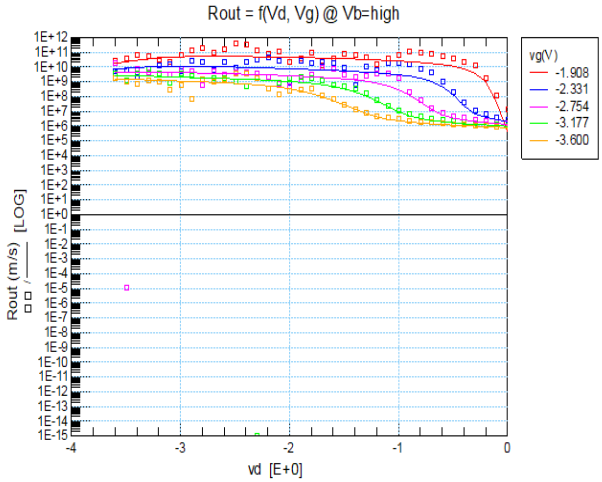


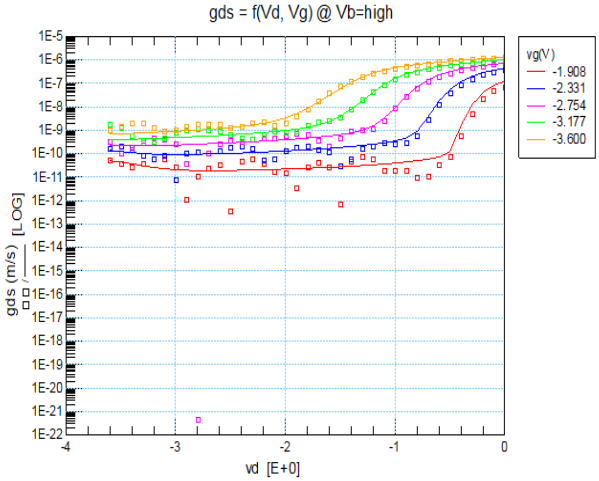




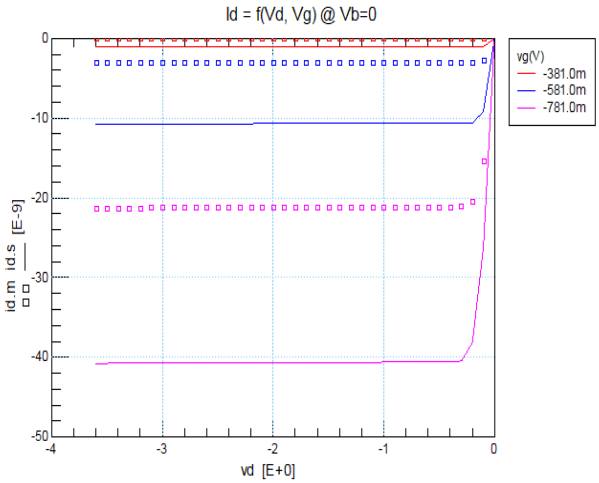
idvd_vbmin, Vb = 3.6V, T = 125°C

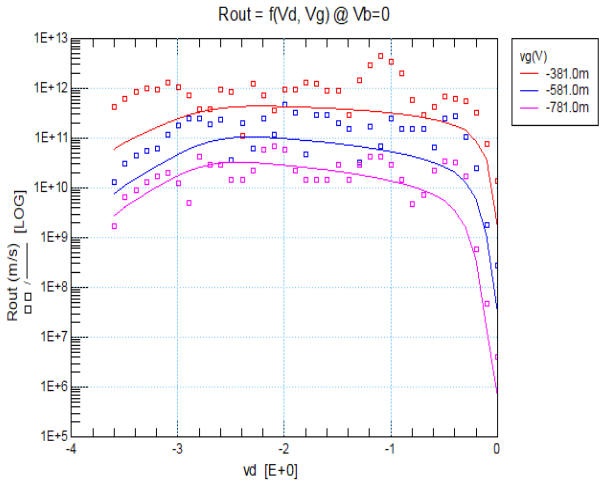


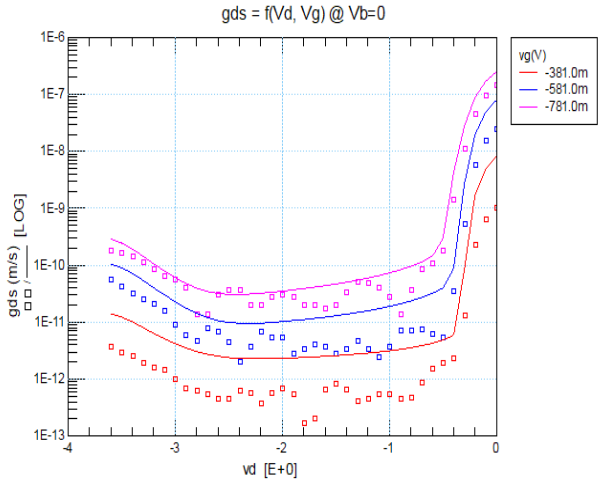




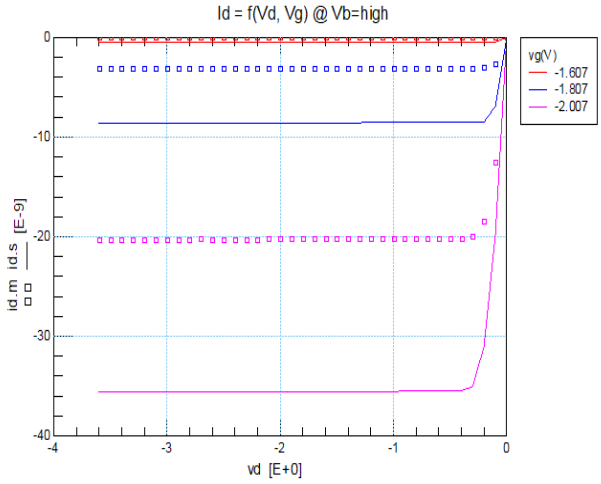
idvd_vth, Vb = 0V, T = 125°C

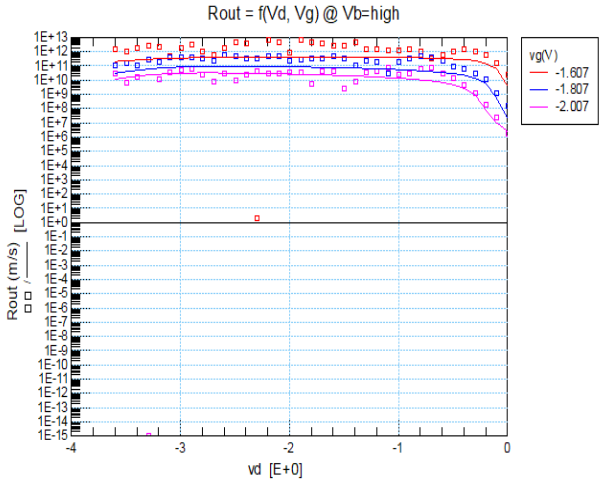


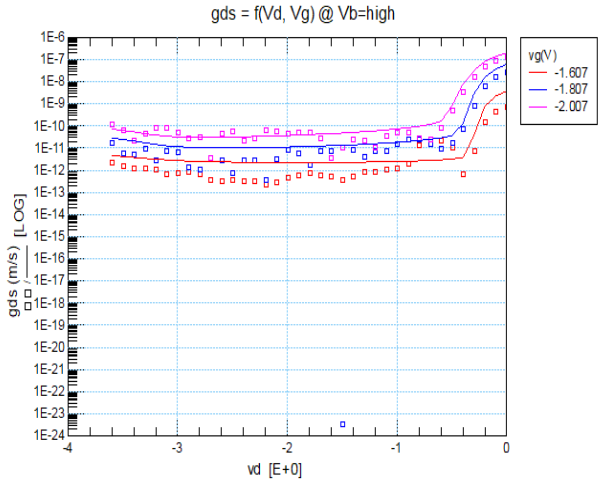




idvd_vth_vbmin, Vb = 3.6V, T = 125°C



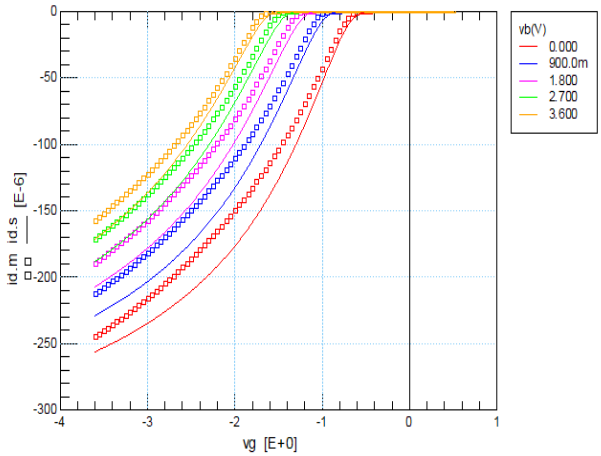




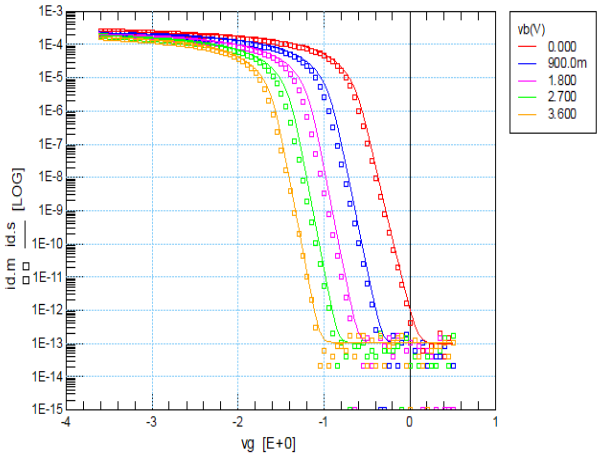
4.4 W10u0_L0u35_S561_3

idvg, Vd = -0.1V, T = 27°C

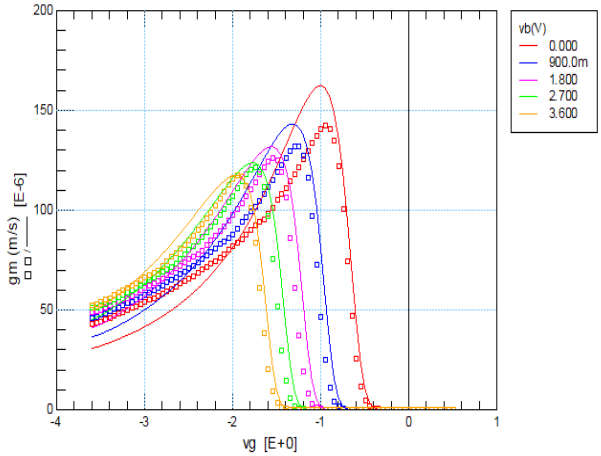
Id = f(Vg, Vb) @ Vd=low



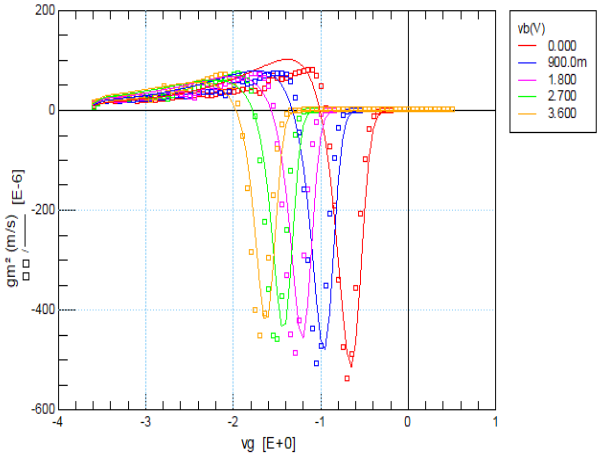
Id = f(Vg, Vb) @ Vd=low



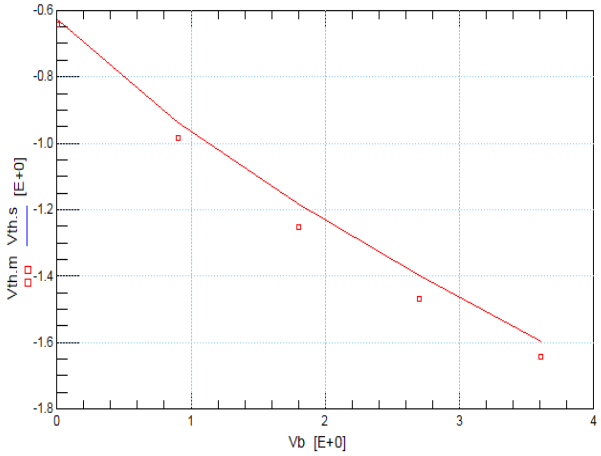
gm = f(Vg, Vb) @ Vd=low



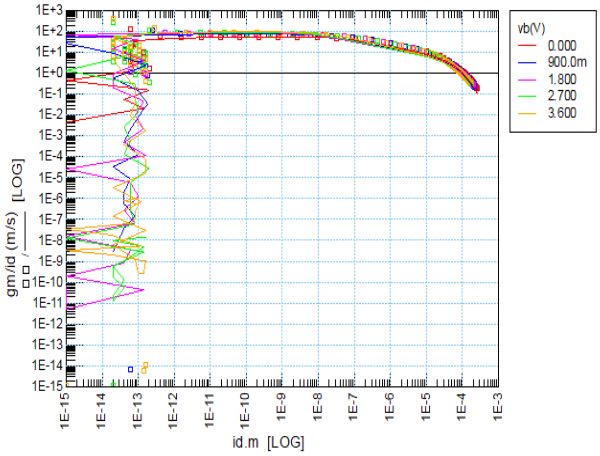
gm² = f(Vg, Vb) @ Vd=low

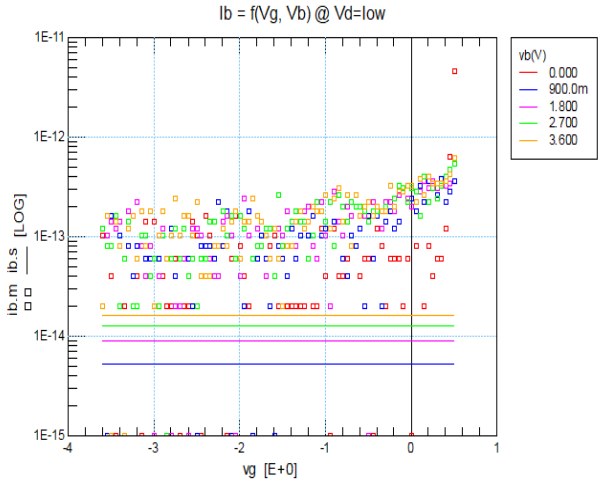


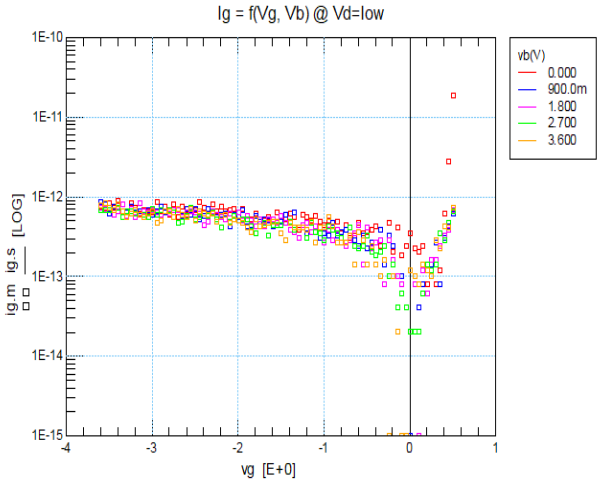
Vth @ Vd=low



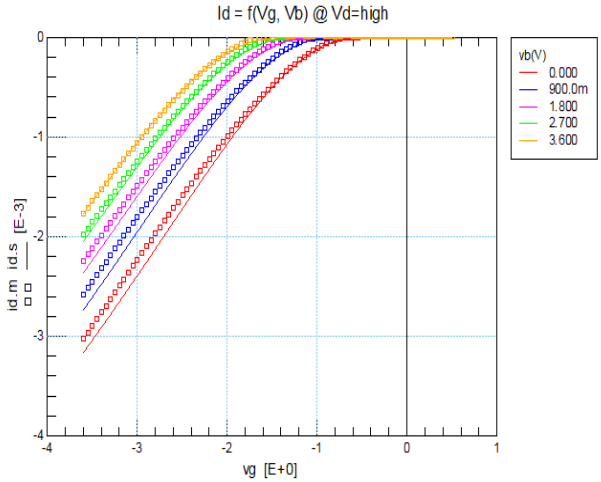
gm/id=f(id.m) @ Vd=low

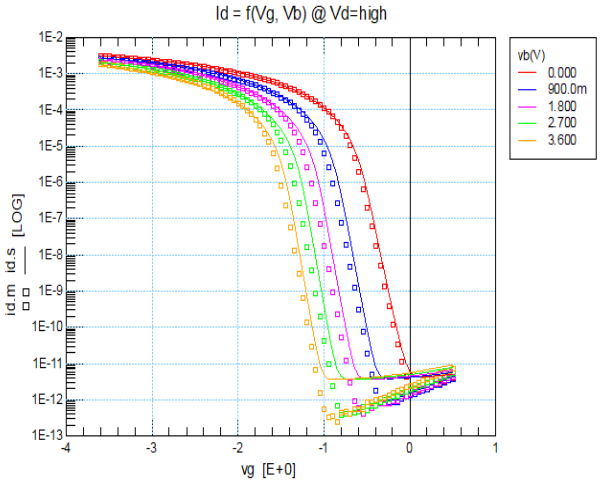


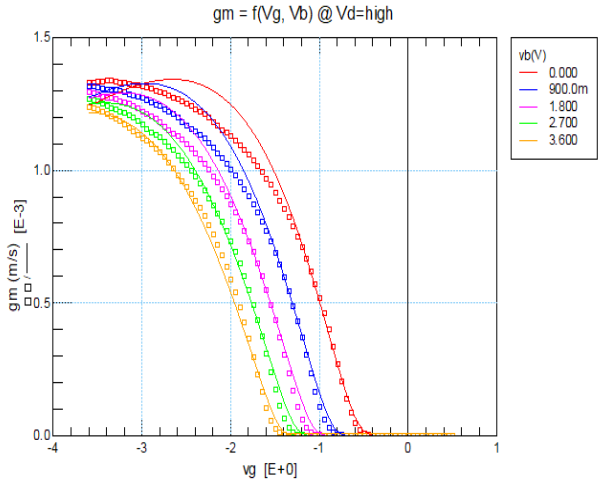


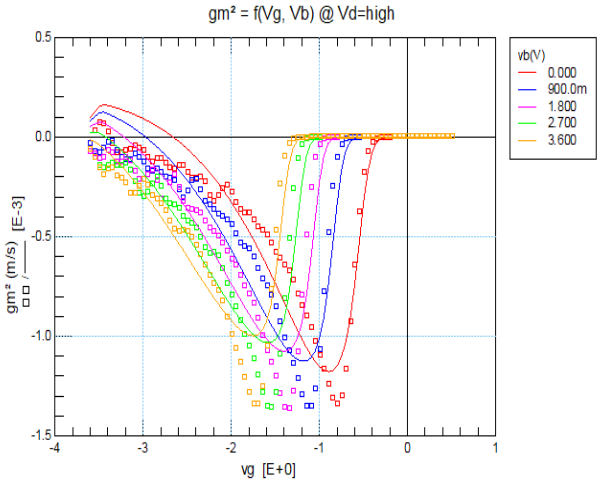


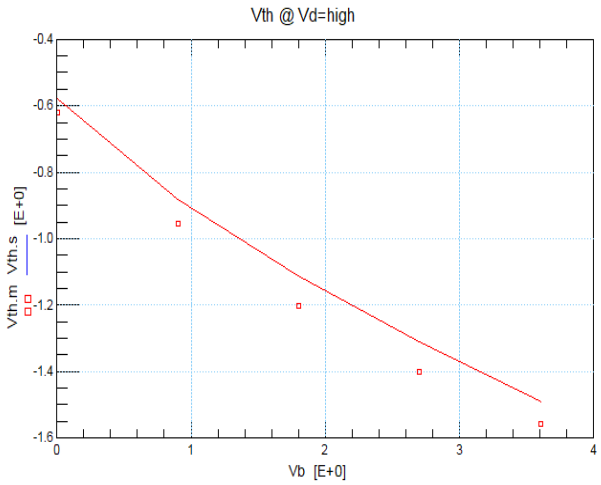
idvg, Vd = -3.3V, T = 27°C

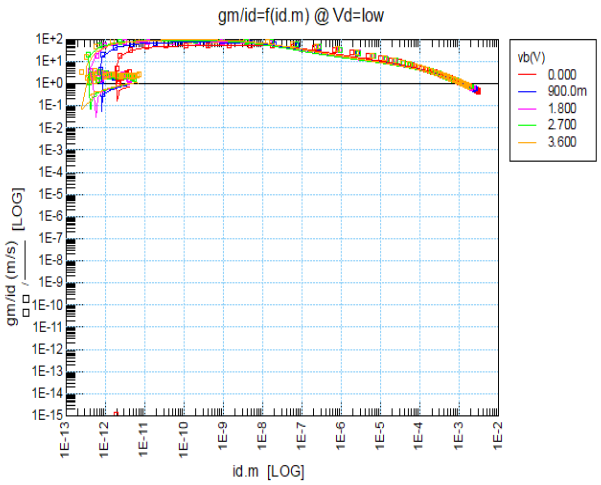


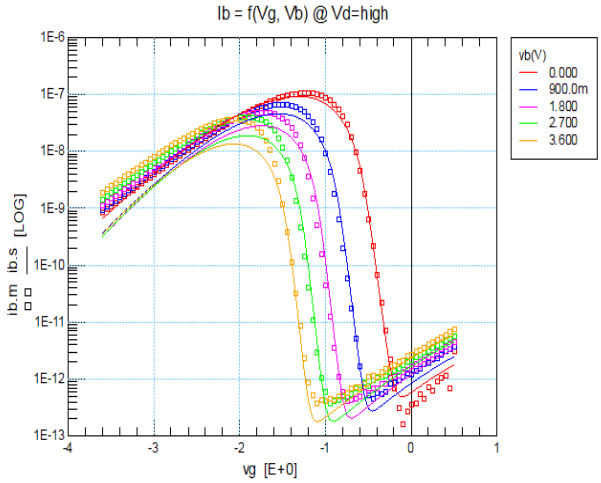


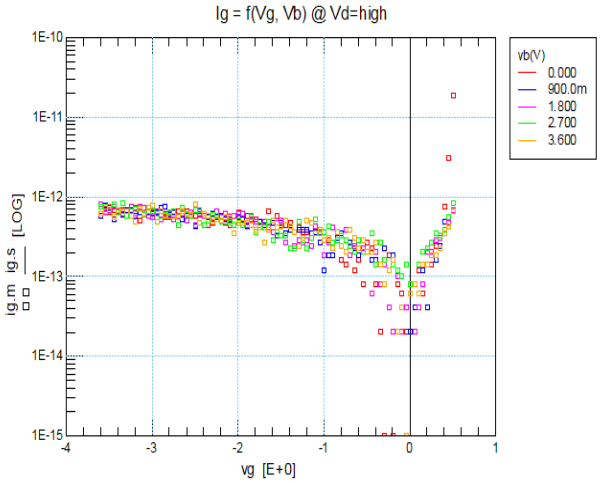




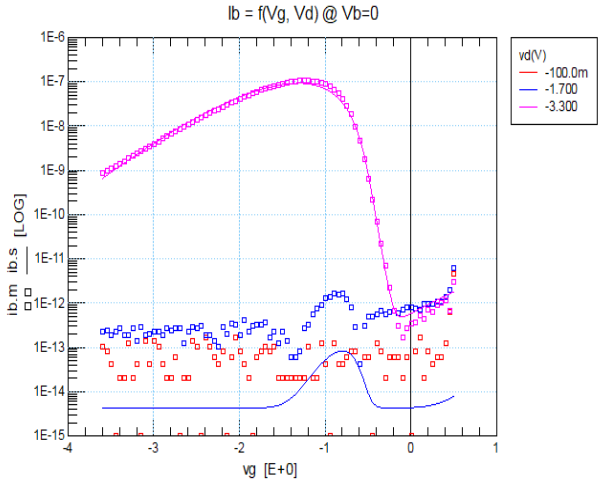


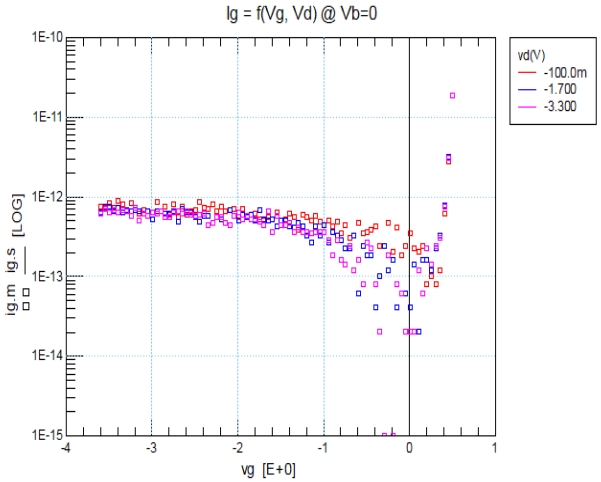




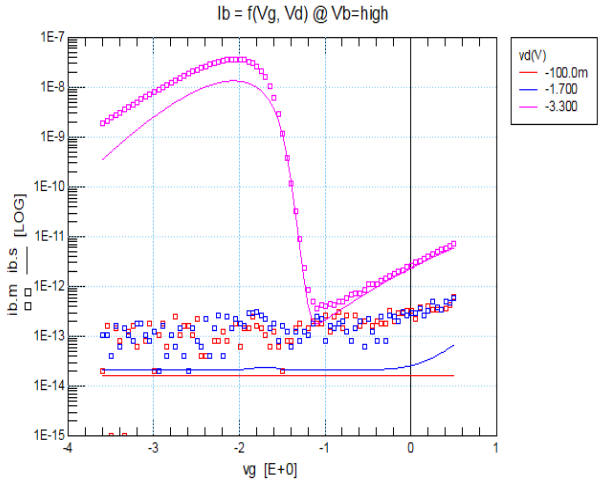


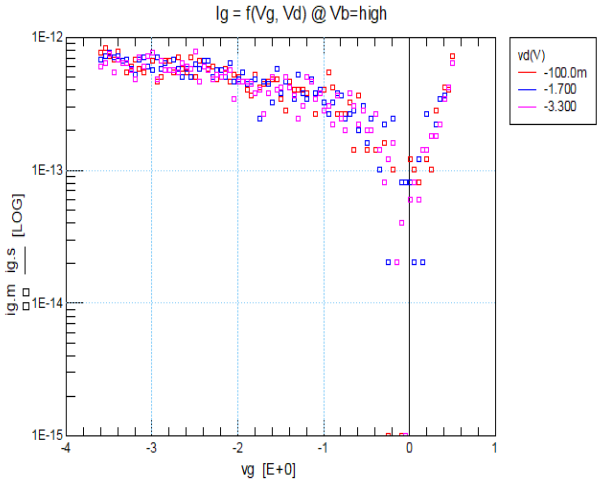
idvg, Ib,Ig, Vb = 0V, T = 27°C



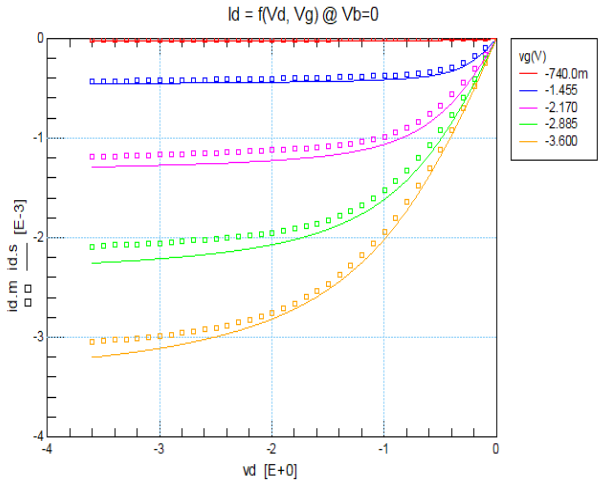


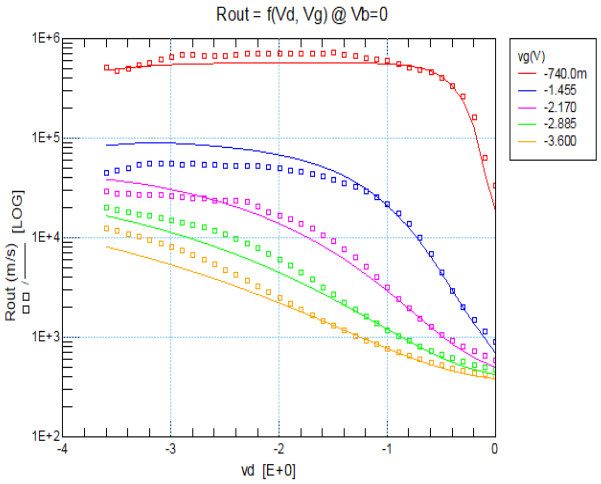
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

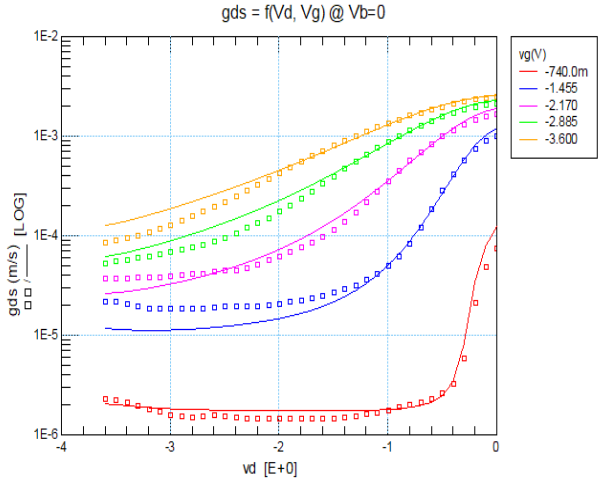




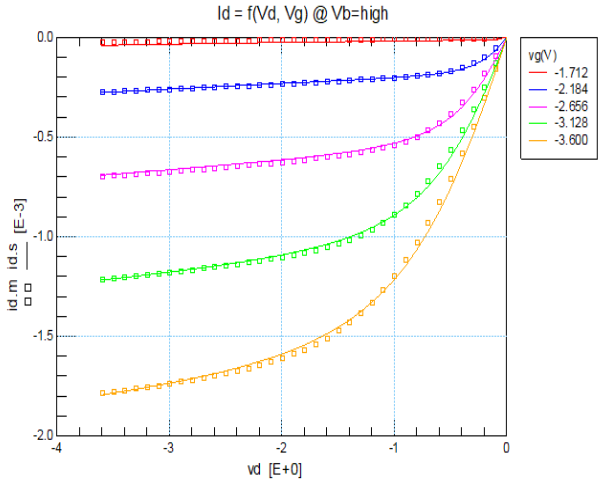
idvd, Vb = 0V, T = 27°C

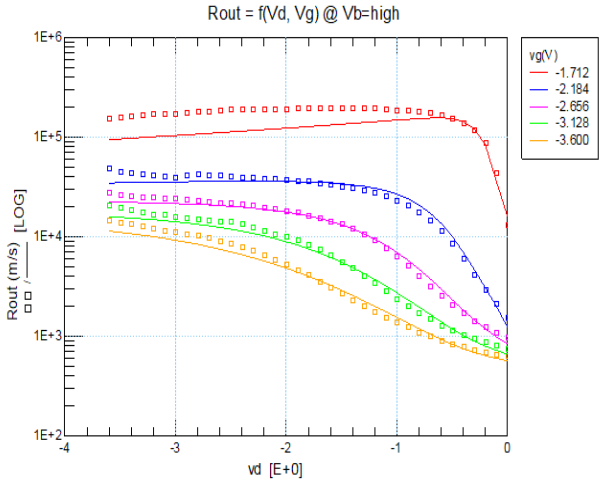


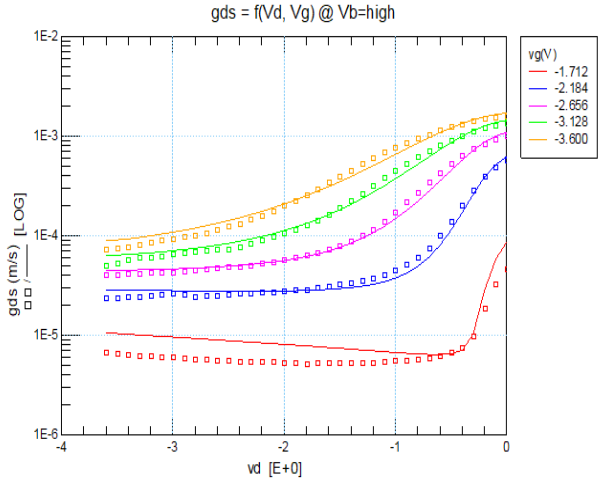




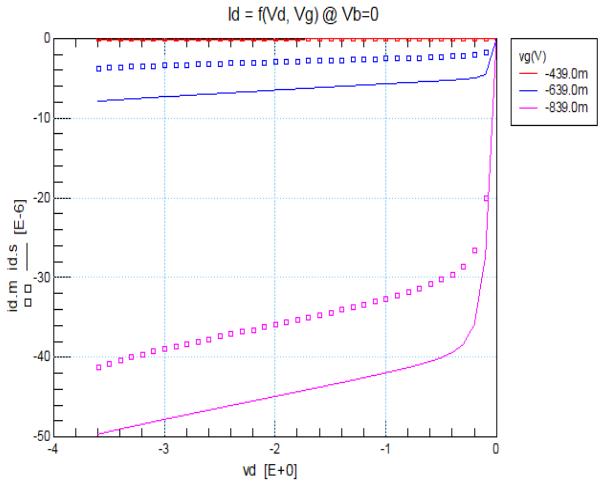
idvd_vbmin, Vb = 3.6V, T = 27°C

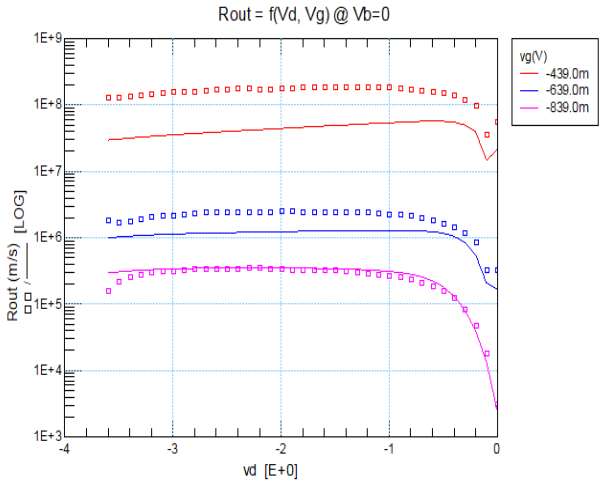


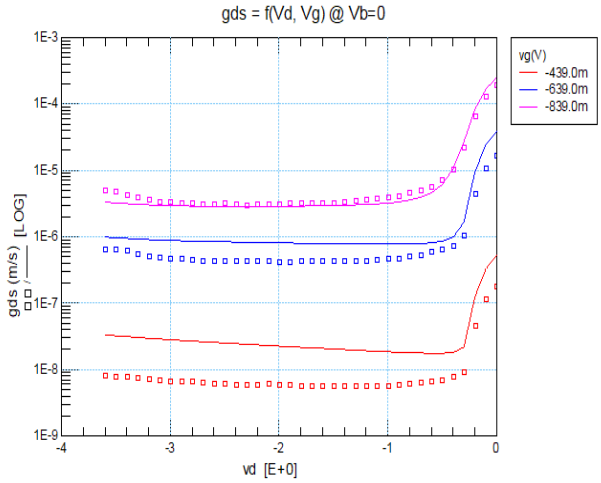




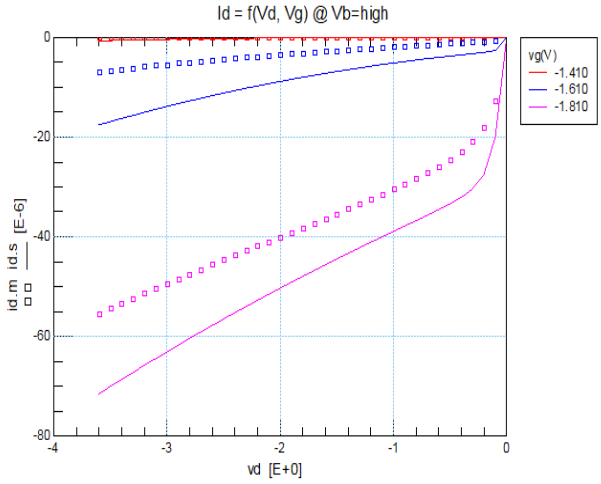
idvd_vth, Vb = 0V, T = 27°C

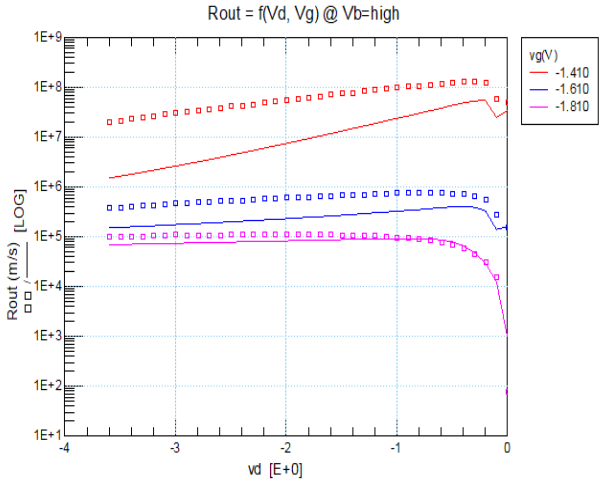


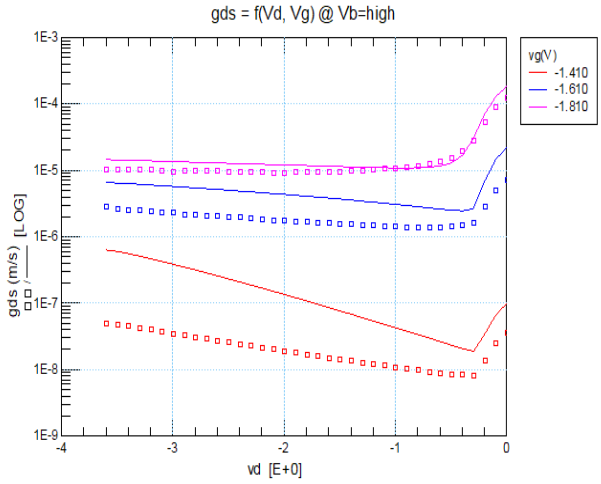




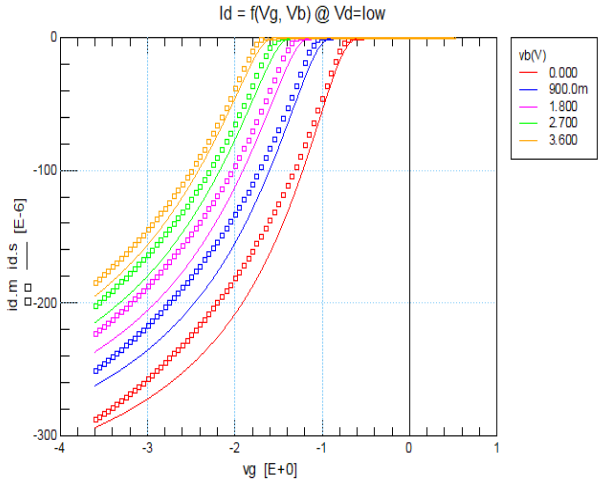
idvd_vth_vbmin, Vb = 3.6V, T = 27°C

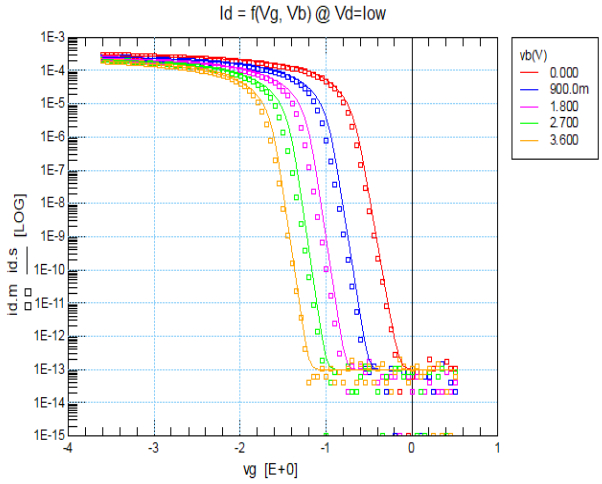


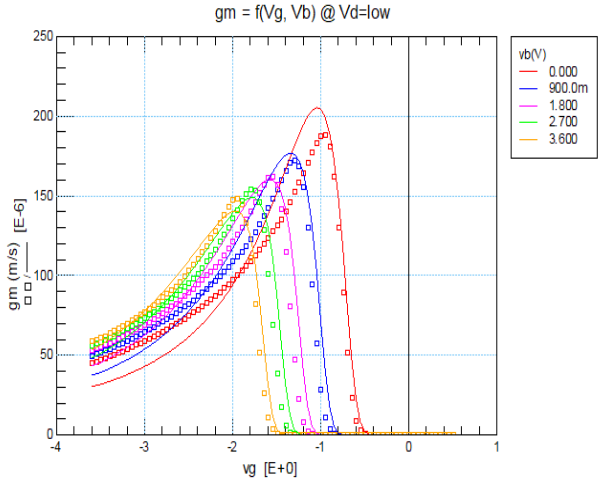


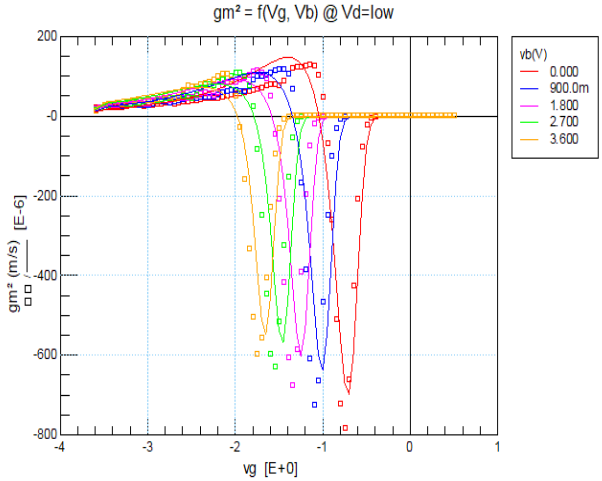


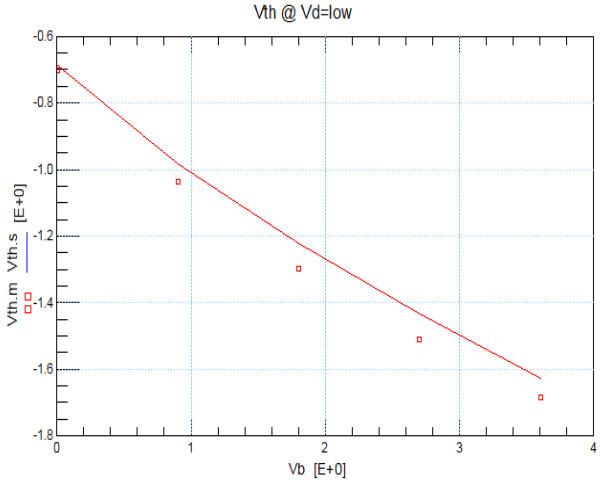
idvg, Vd = -0.1V, T = -40°C

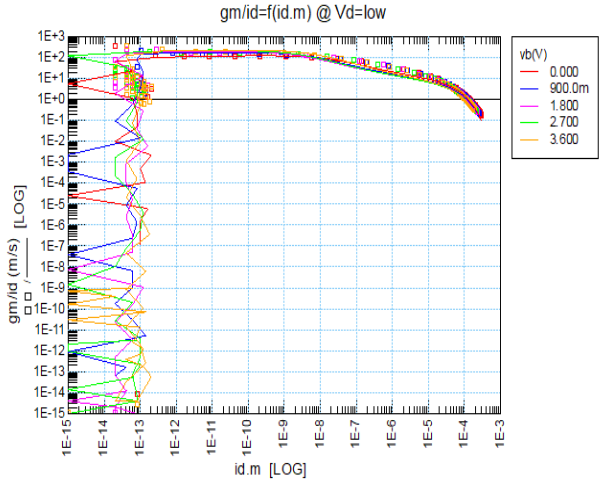


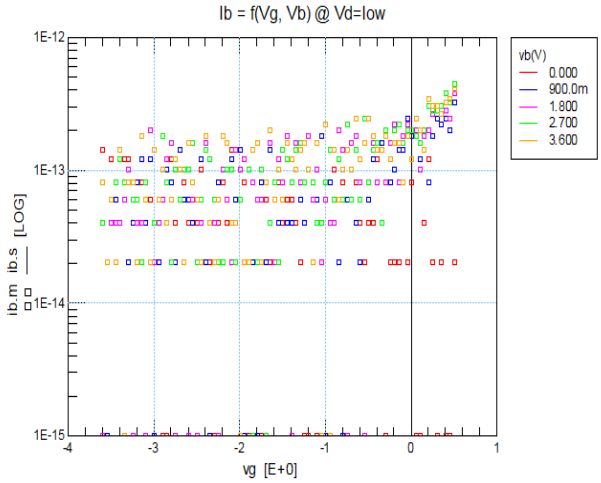


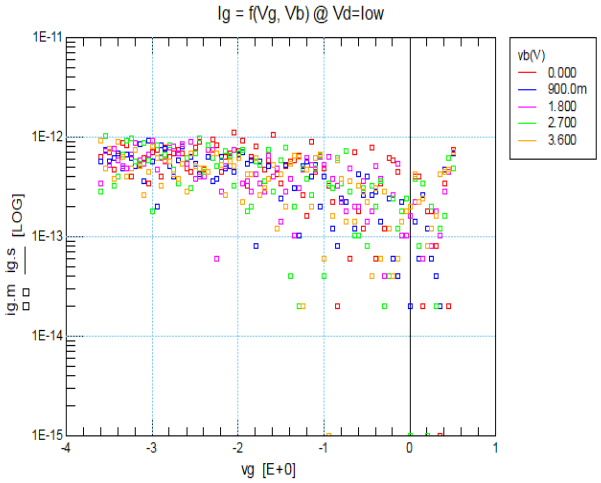




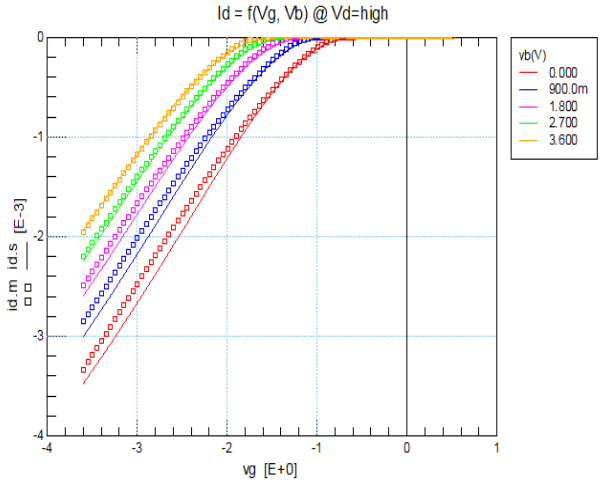


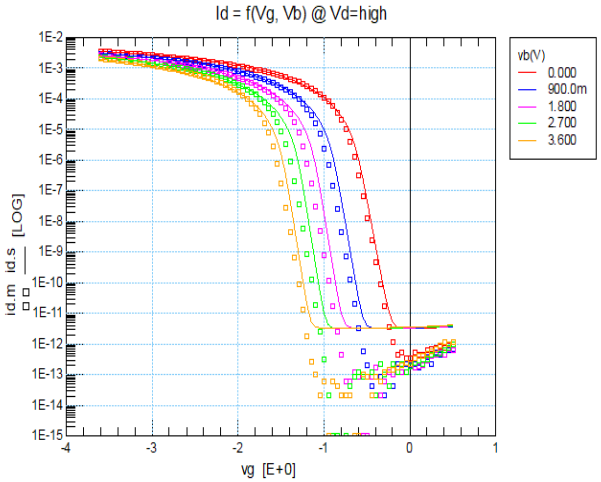


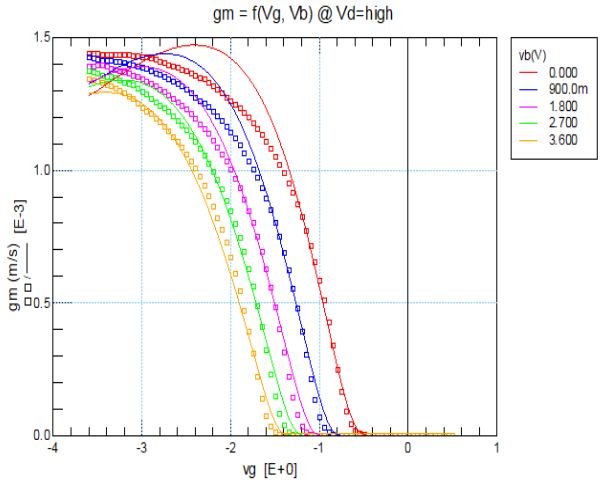


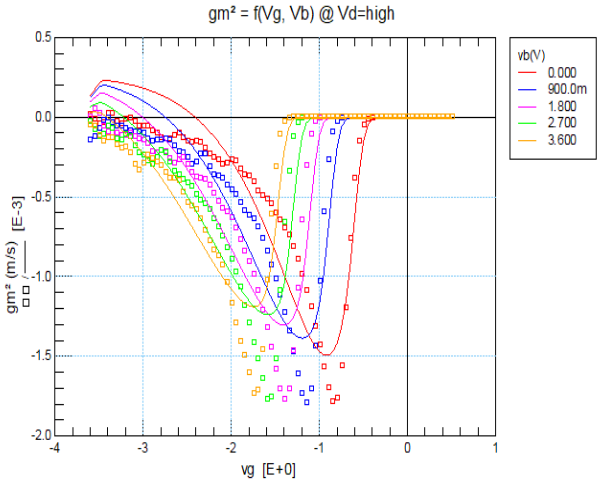


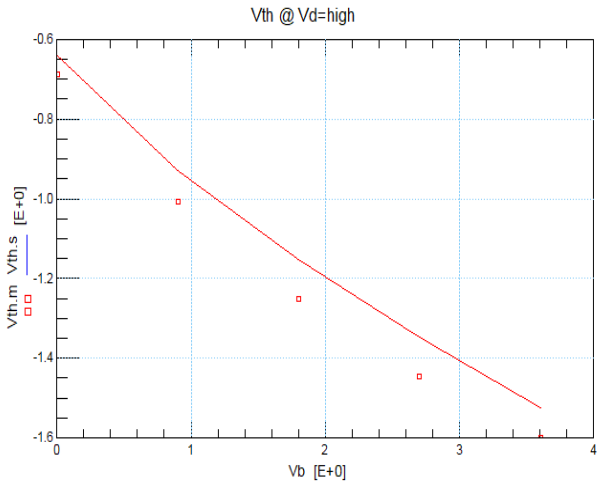
idvg, Vd = -3.3V, T = -40°C

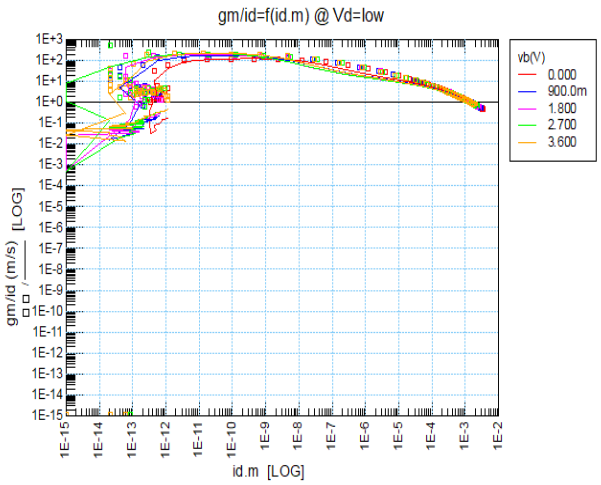


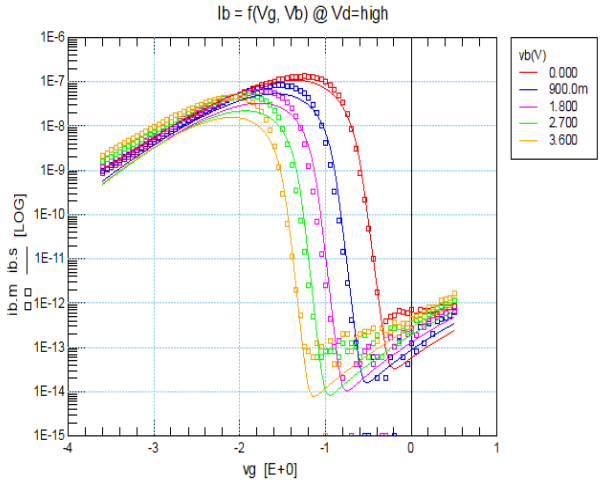


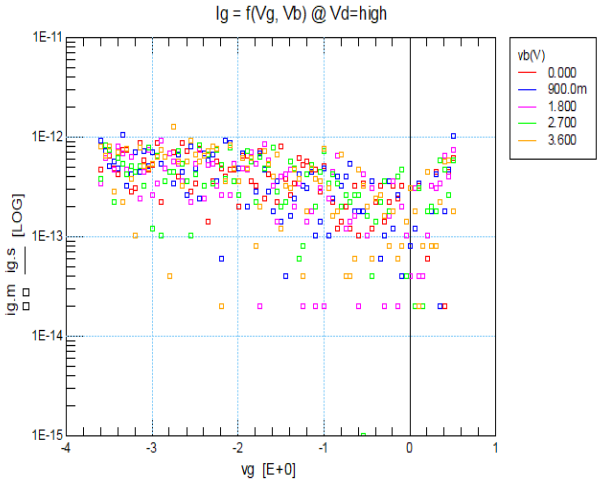




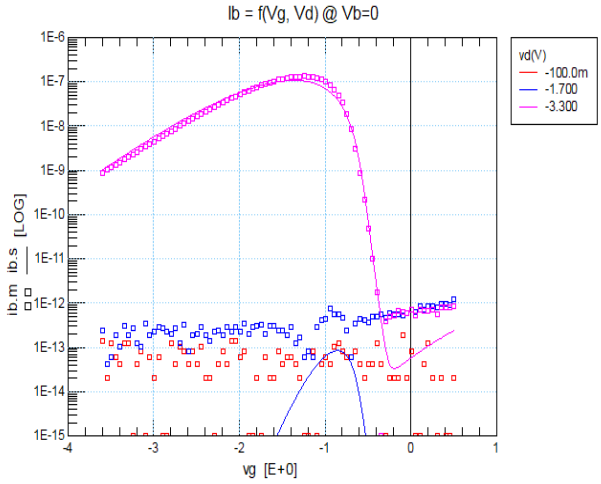


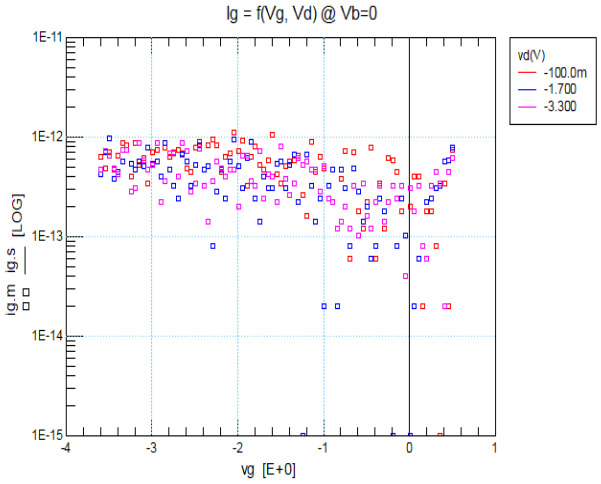




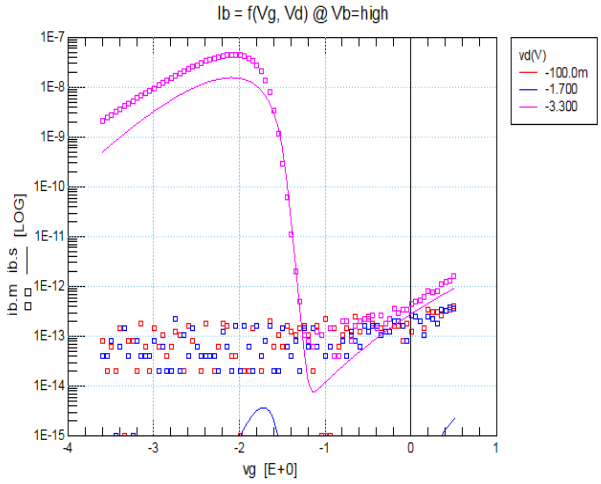


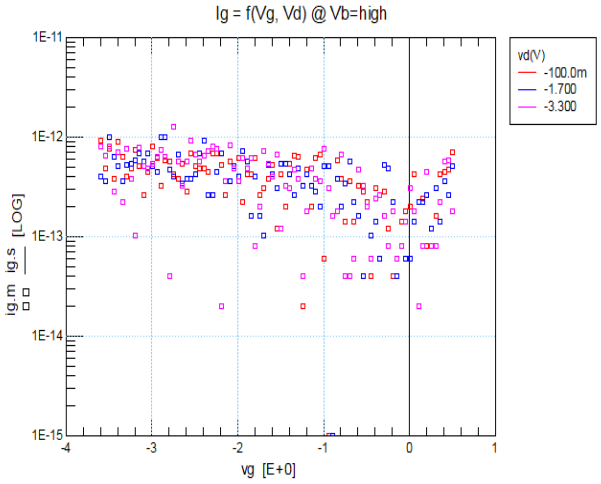
idvg, Ib,Ig, Vb = 0V, T = -40°C



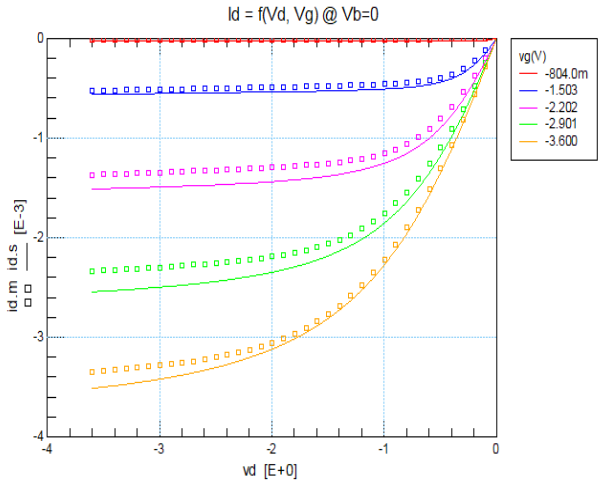


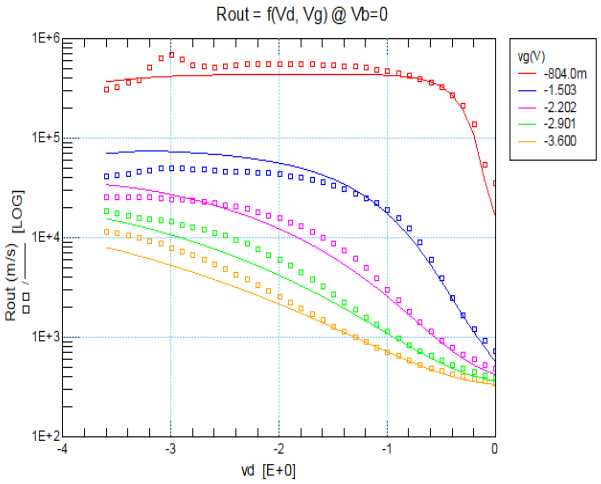
idvg, Ib,Ig, Vb = 3.6V, T = -40°C

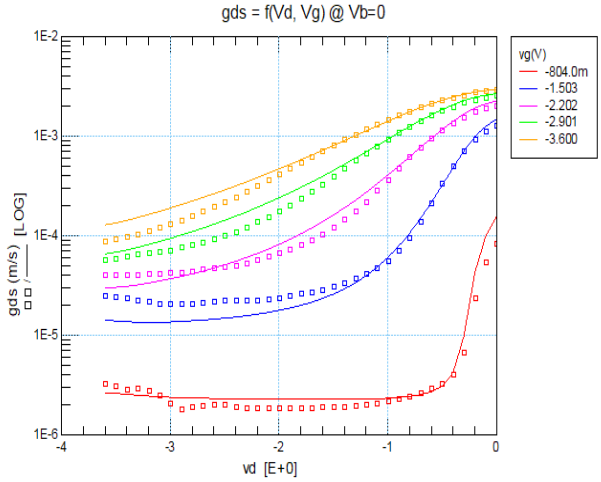




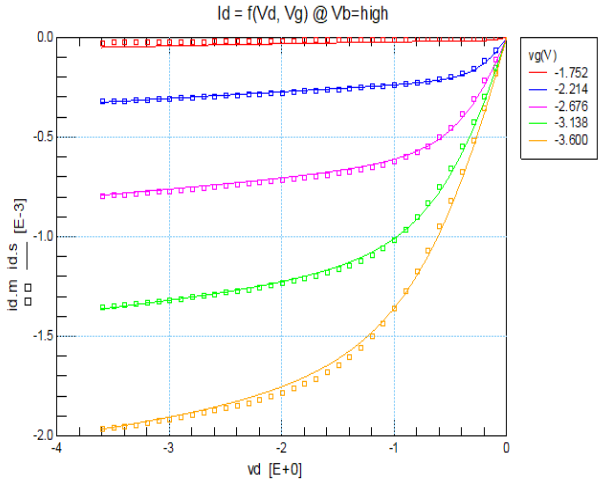
idvd, Vb = 0V, T = -40°C

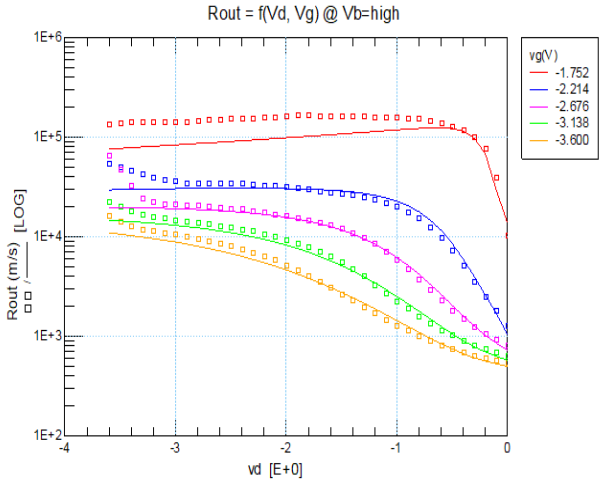


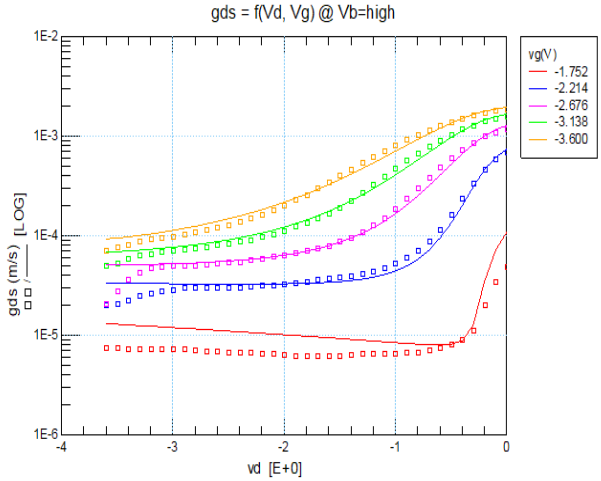




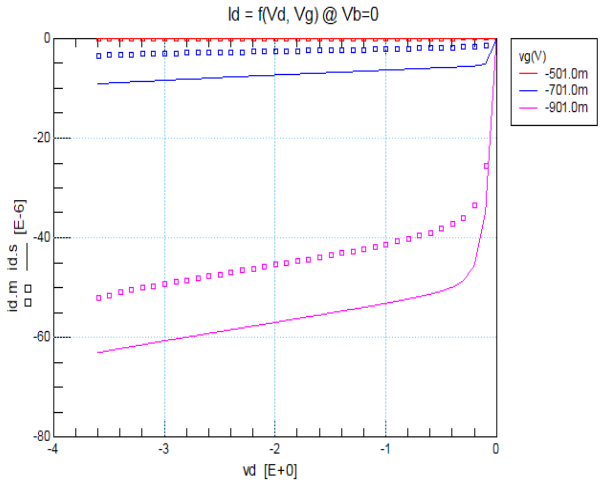
idvd_vbmin, Vb = 3.6V, T = -40°C

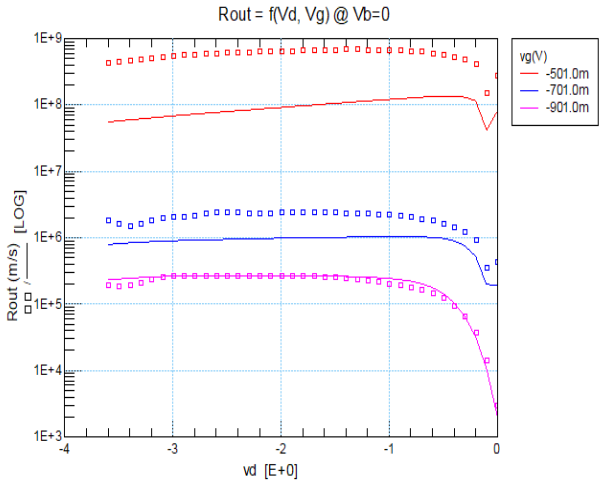


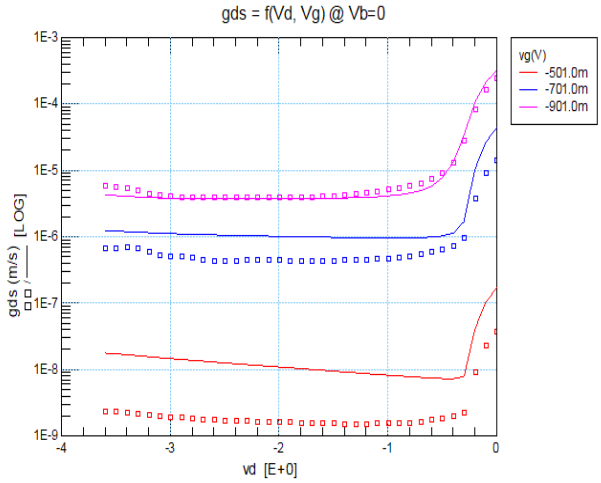




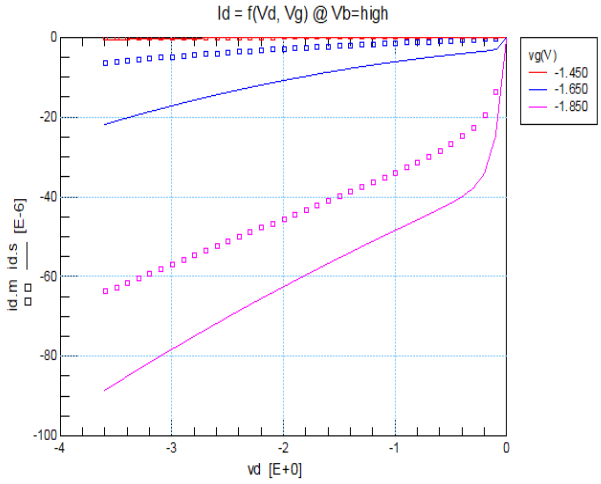
idvd_vth, Vb = 0V, T = -40°C

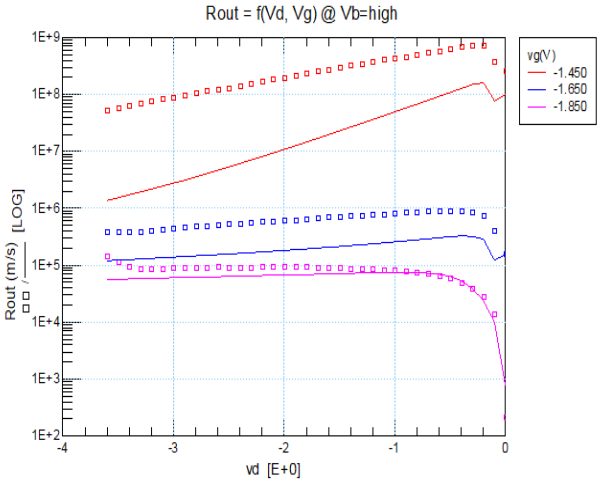


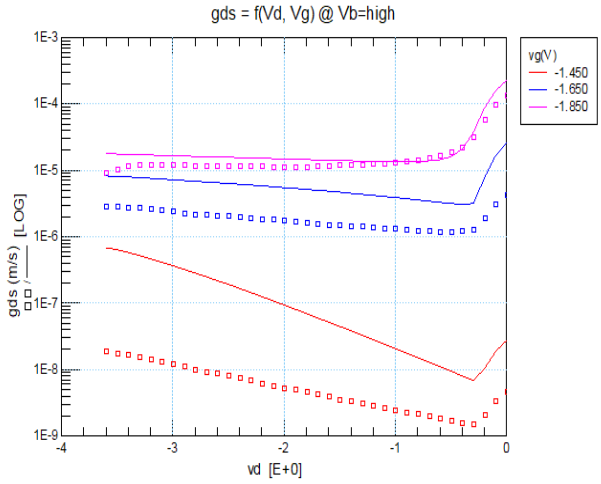




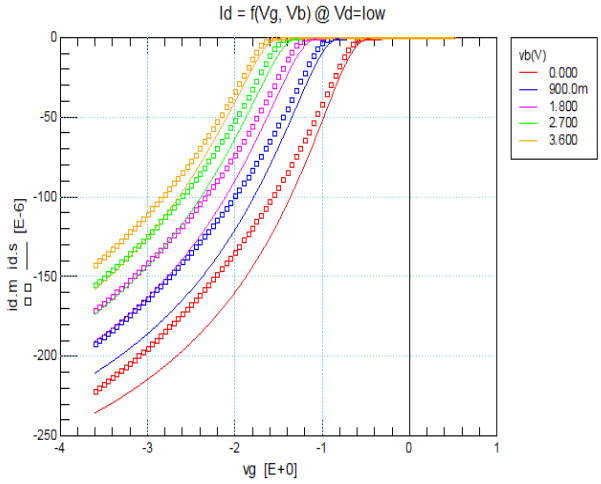
idvd_vth_vbmin, Vb = 3.6V, T = -40°C

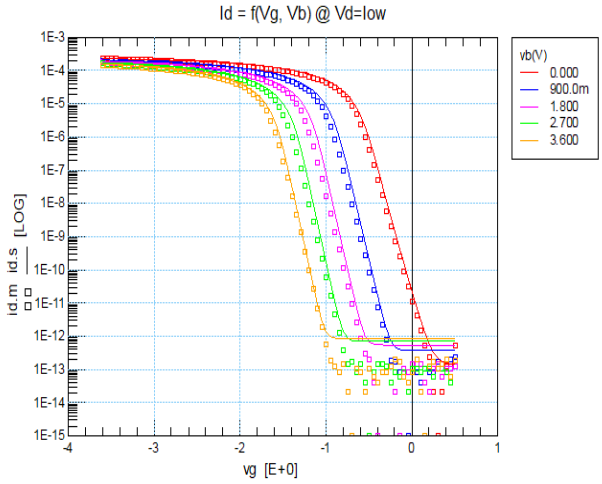


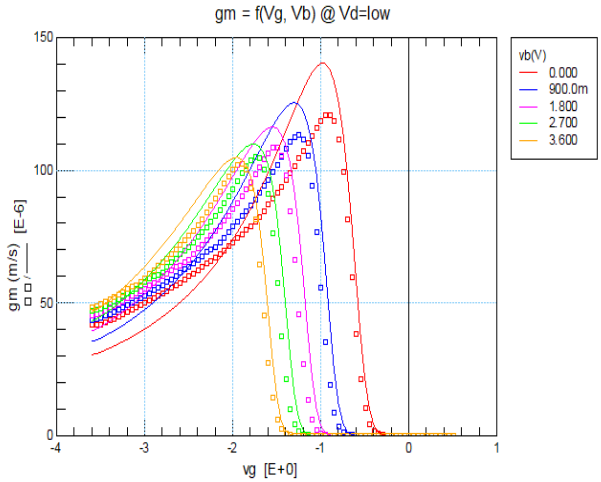


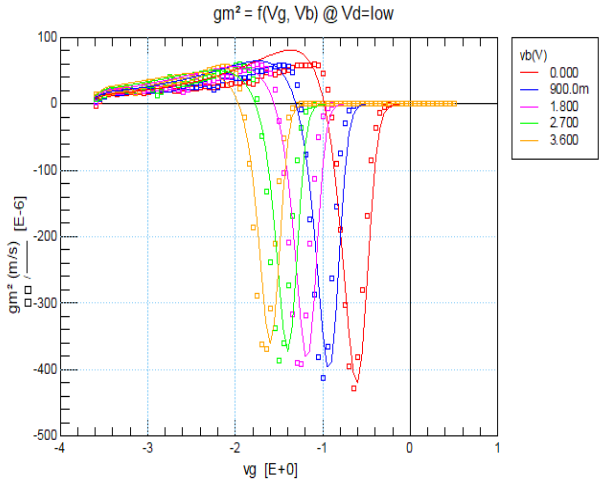


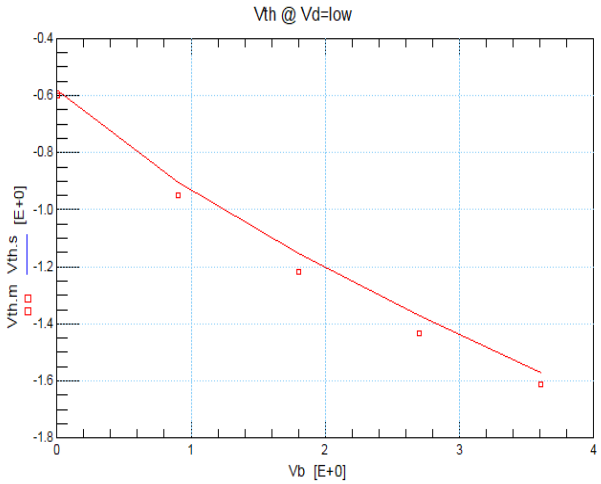
idvg, Vd = -0.1V, T = 70°C

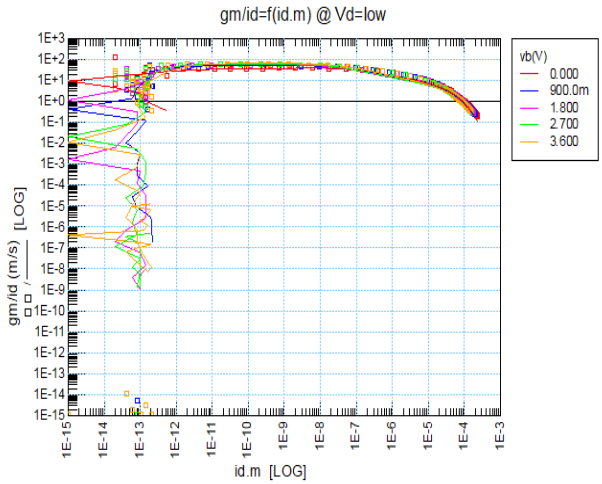


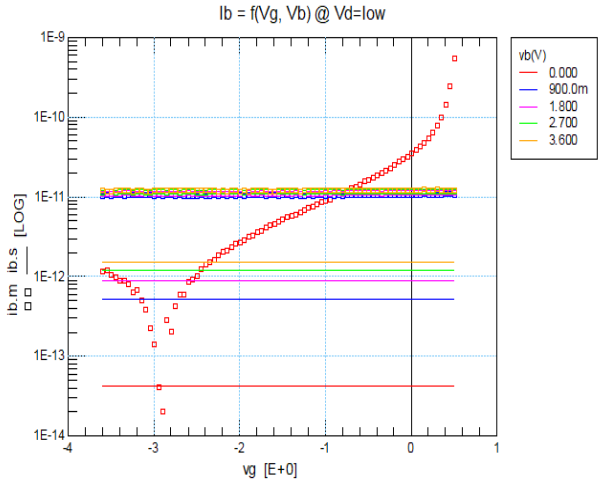


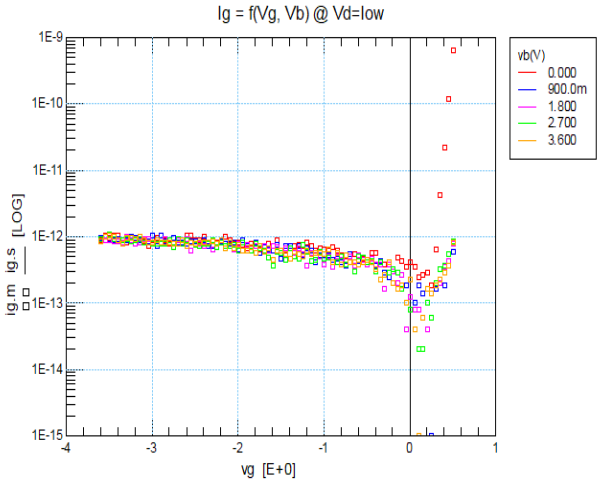




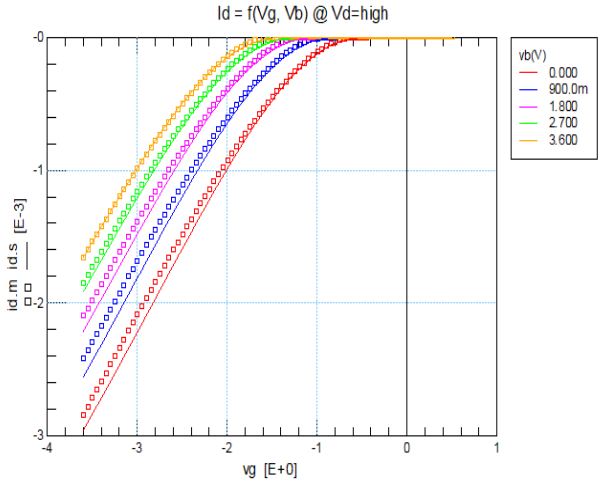


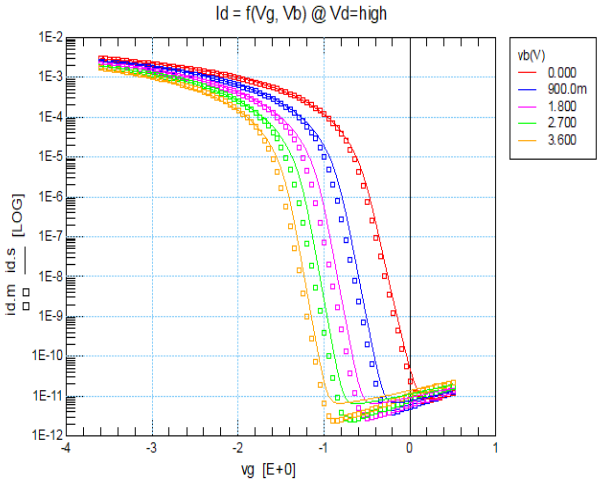


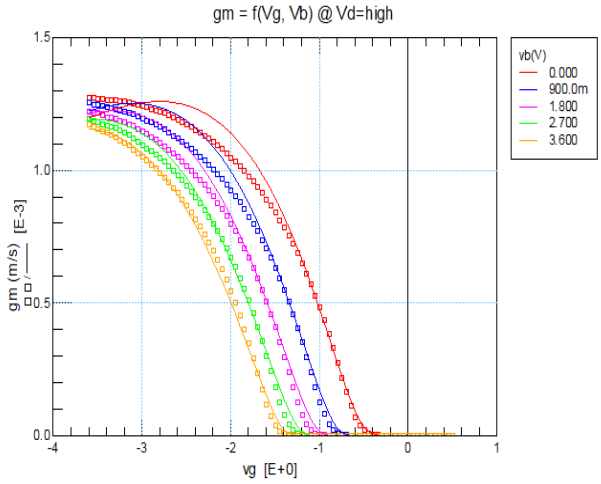


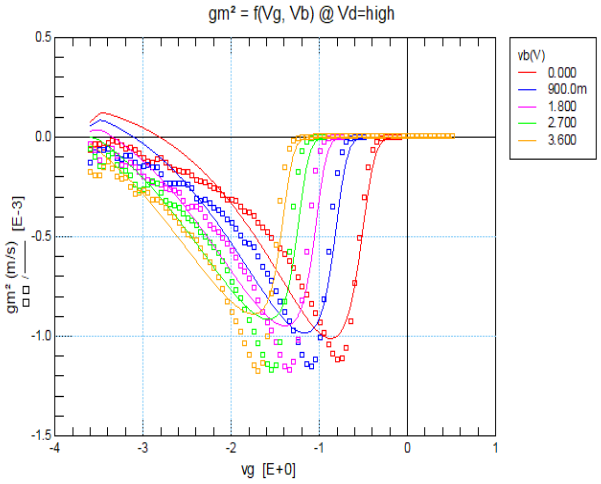


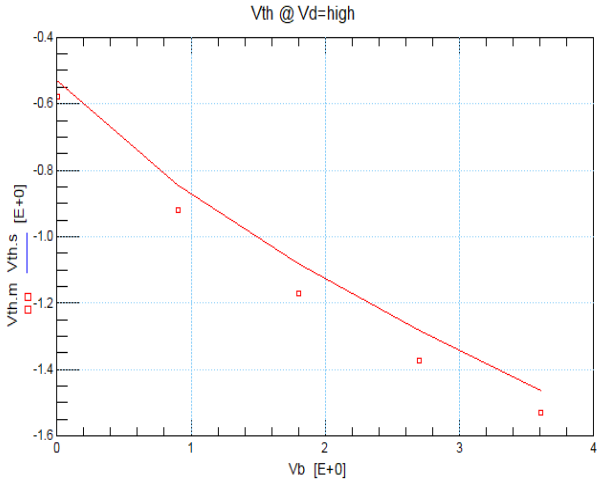
idvg, Vd = -3.3V, T = 70°C

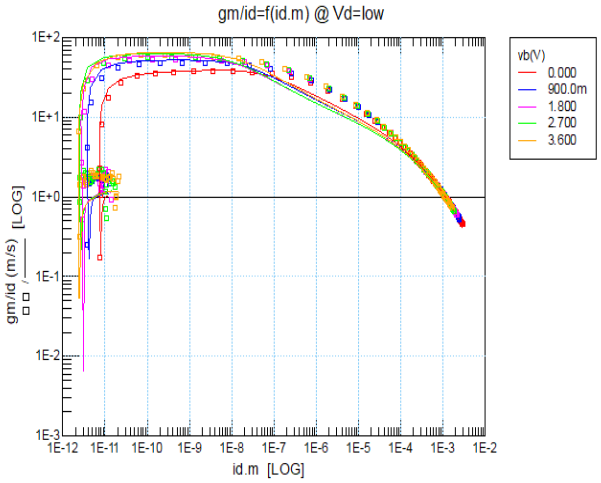


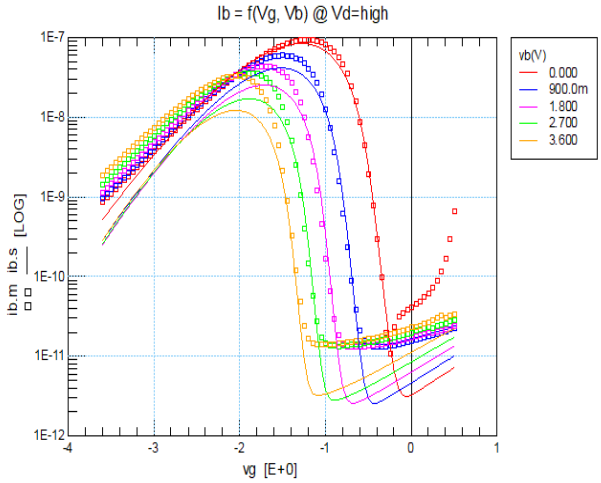


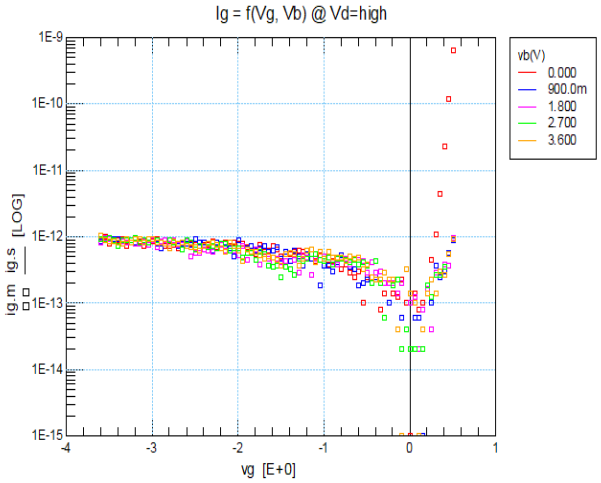




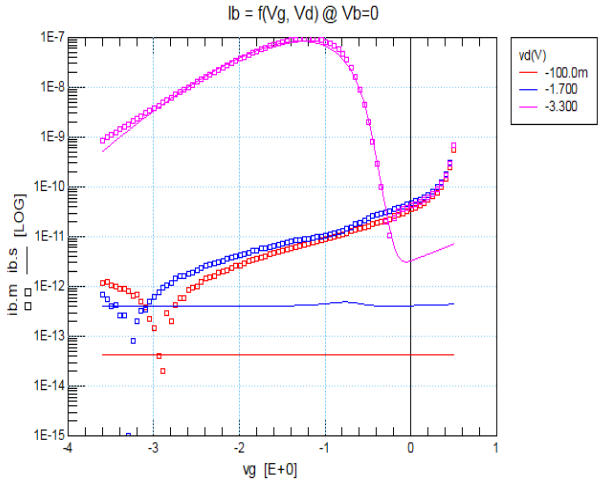


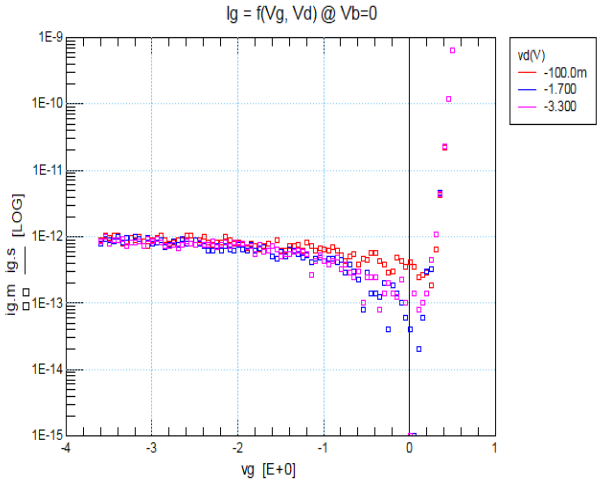




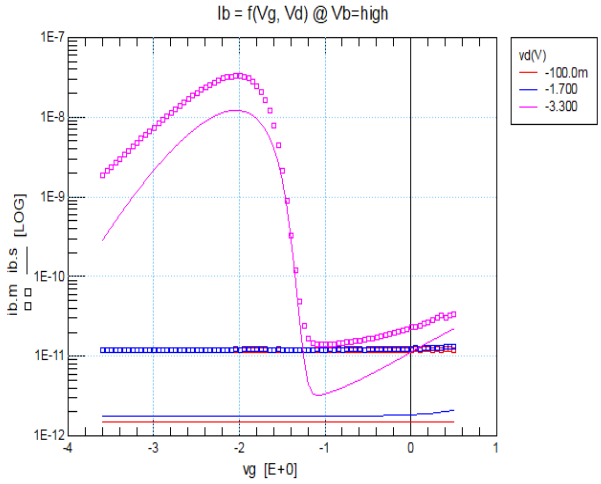


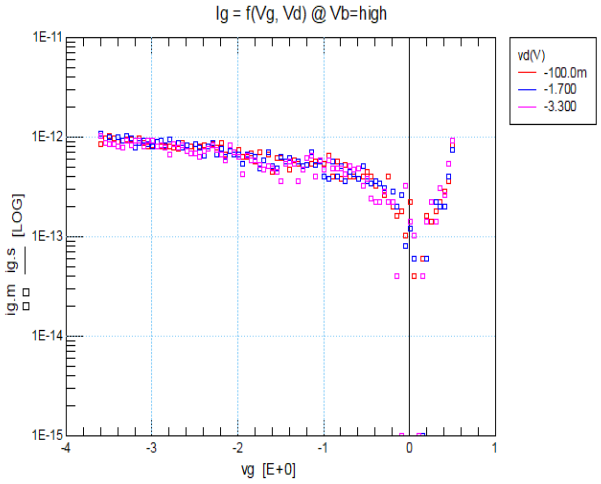
idvg, Ib,Ig, Vb = 0V, T = 70°C



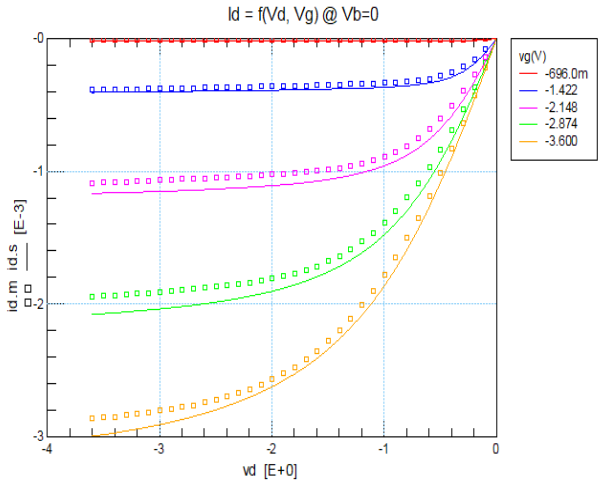


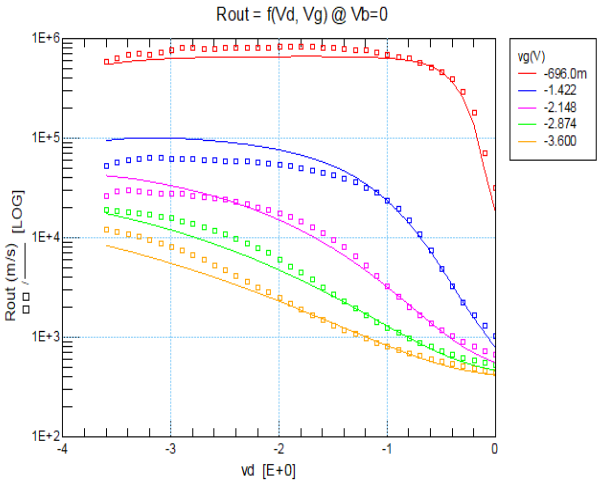
idvg, Ib,Ig, Vb = 3.6V, T = 70°C

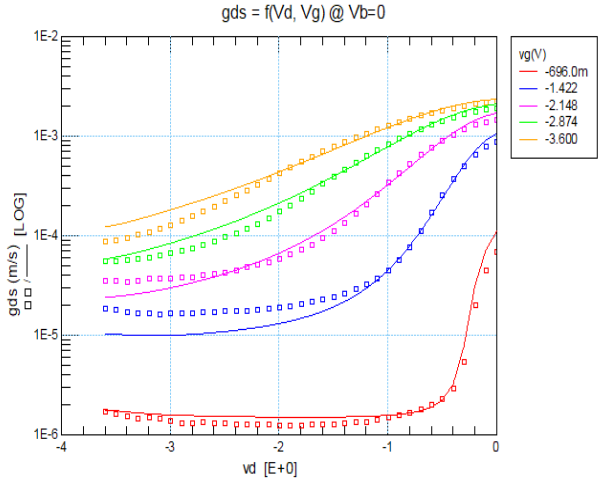




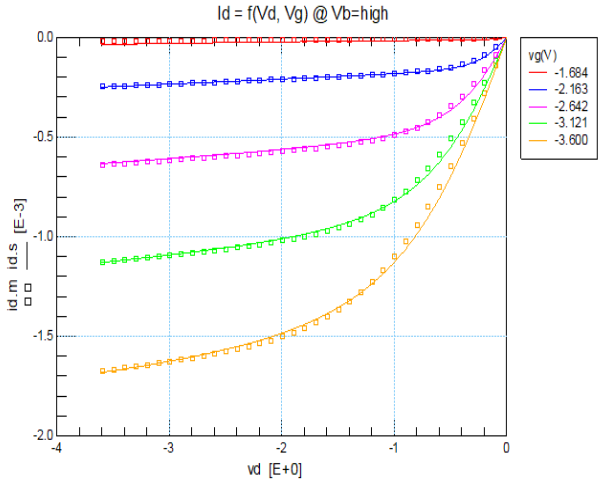
idvd, Vb = 0V, T = 70°C

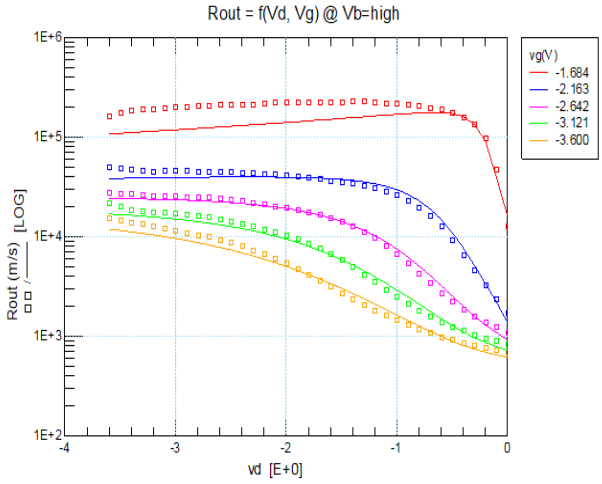


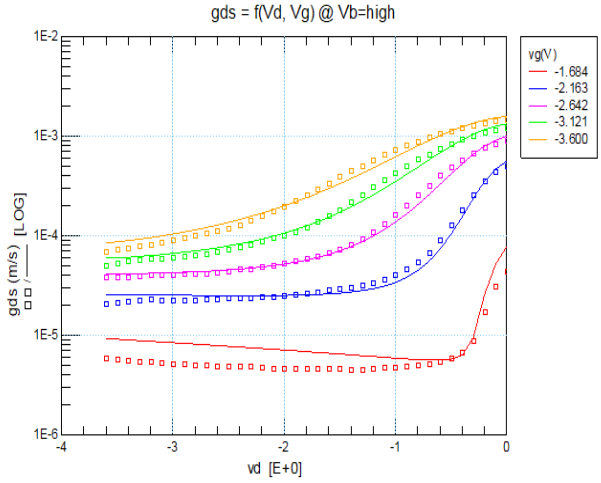




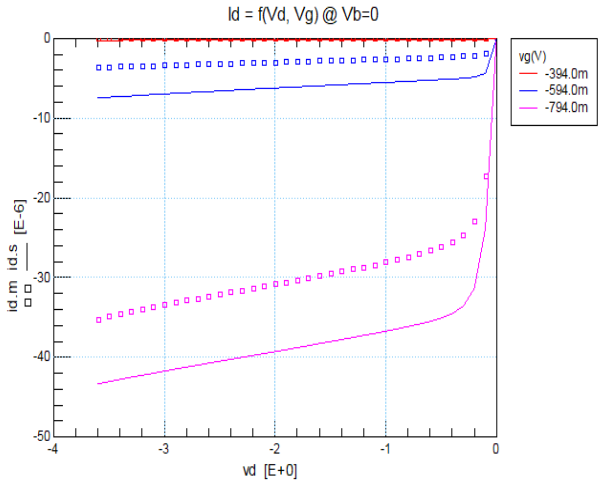
idvd_vbmin, Vb = 3.6V, T = 70°C

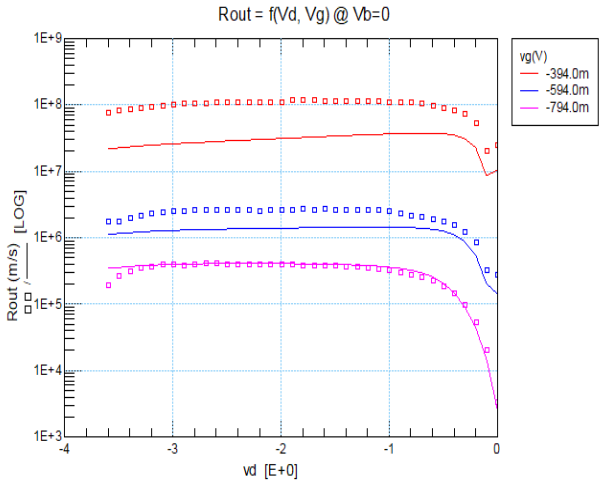


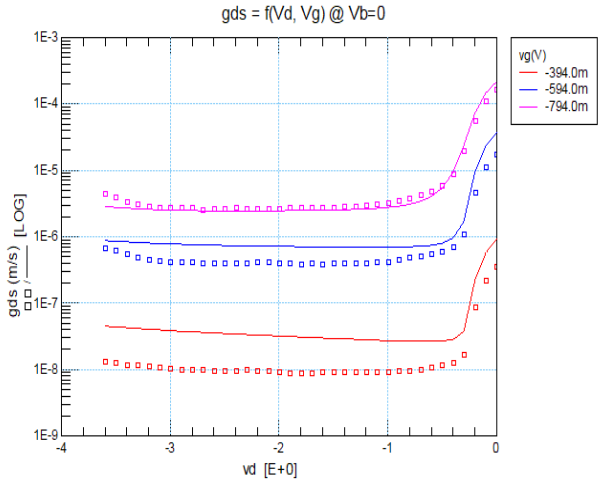




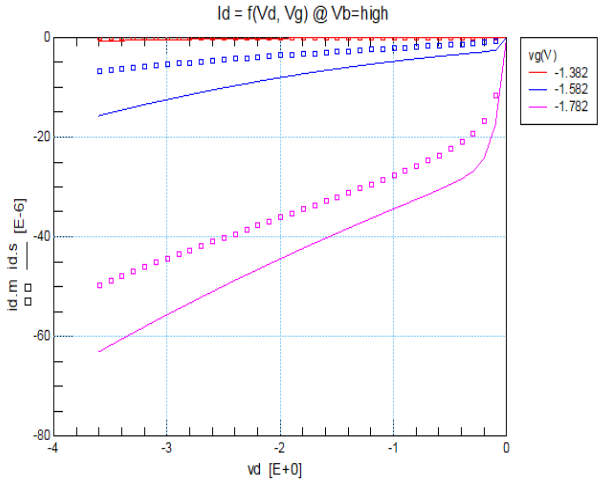
idvd_vth, Vb = 0V, T = 70°C

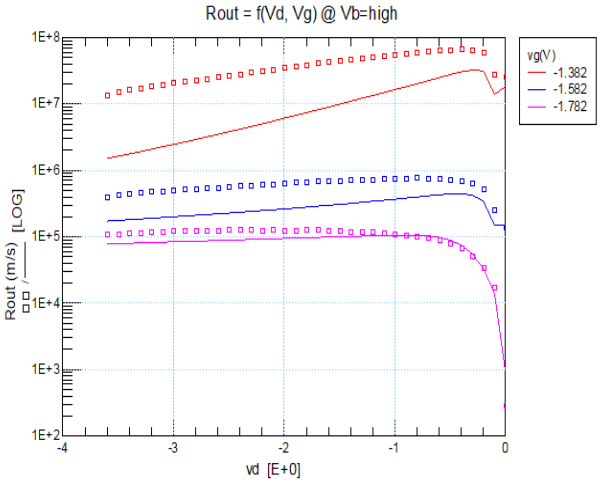


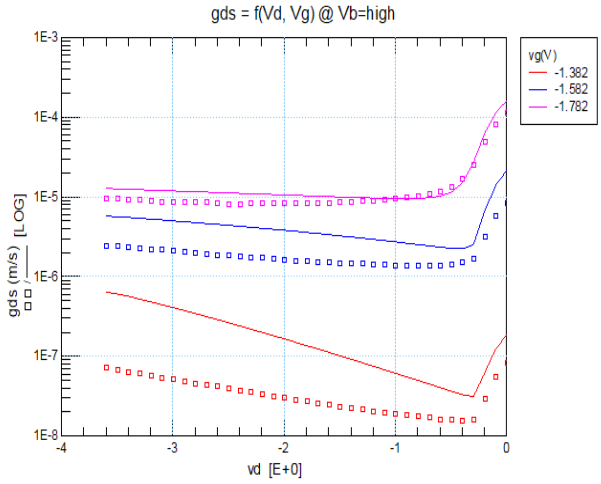




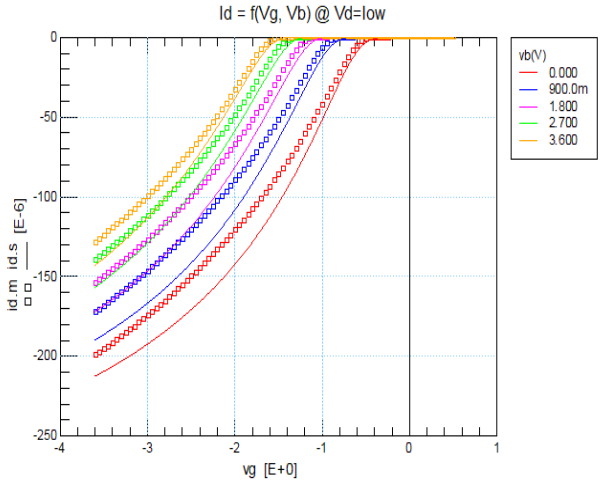
idvd_vth_vbmin, Vb = 3.6V, T = 70°C

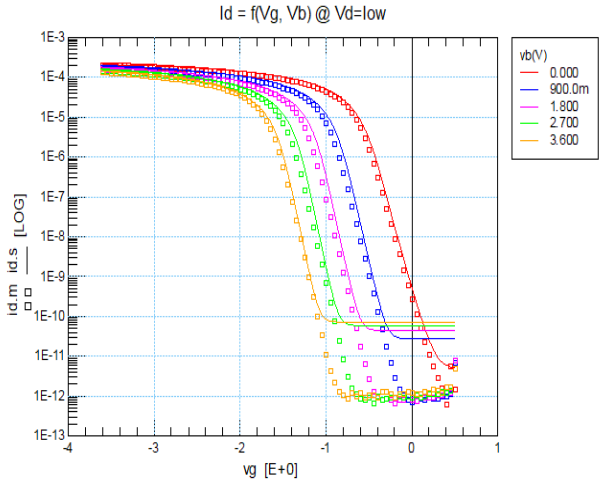


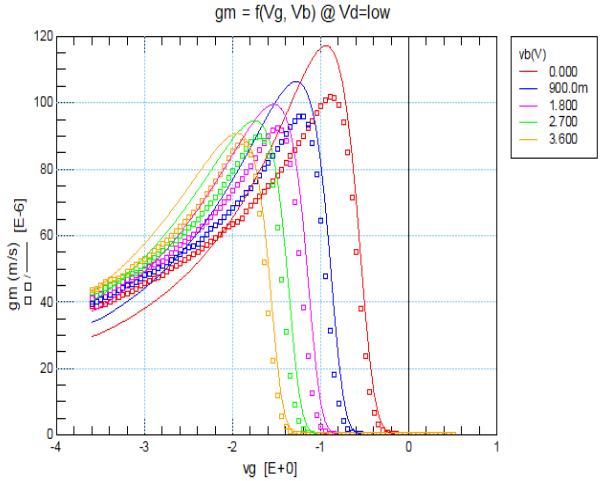


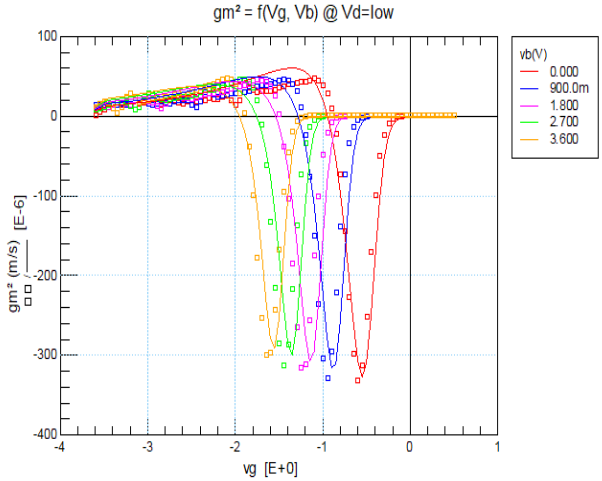


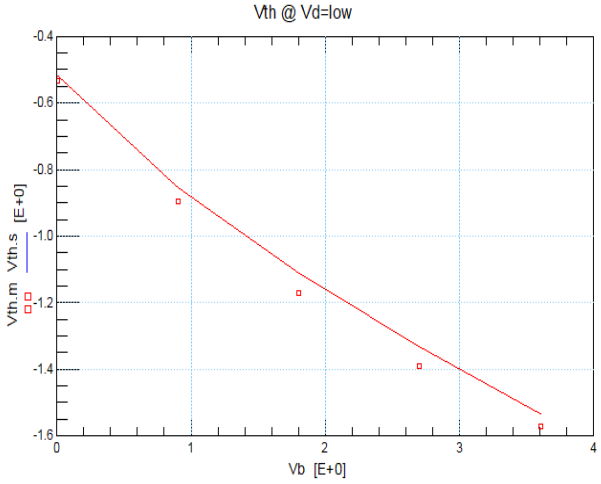
idvg, Vd = -0.1V, T = 125°C

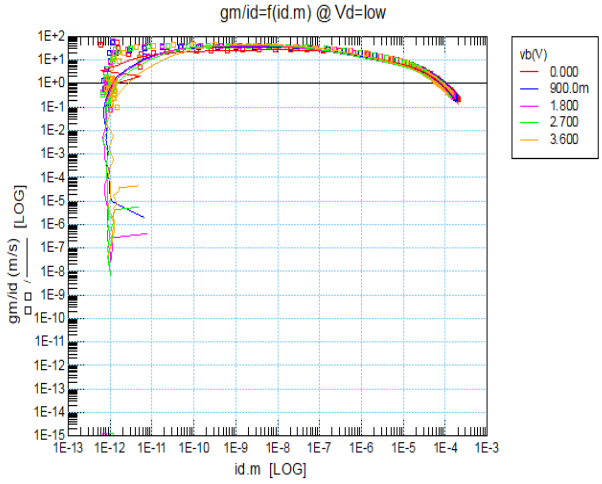


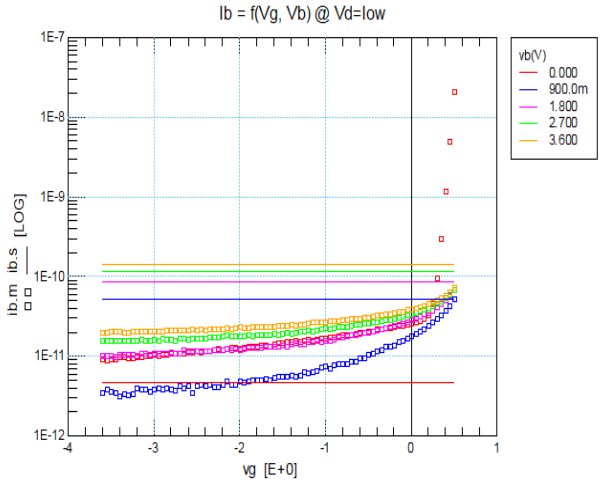


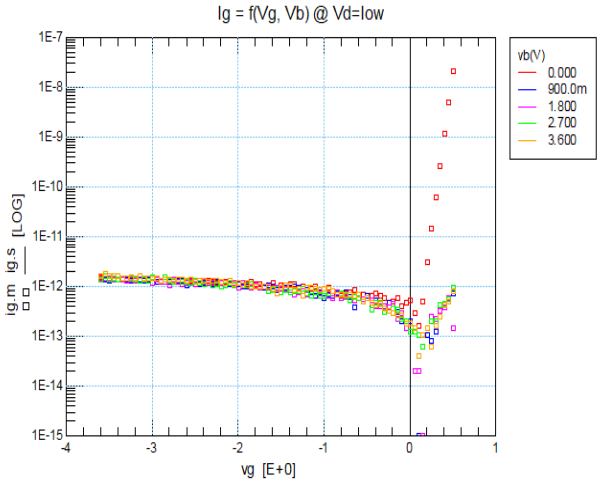




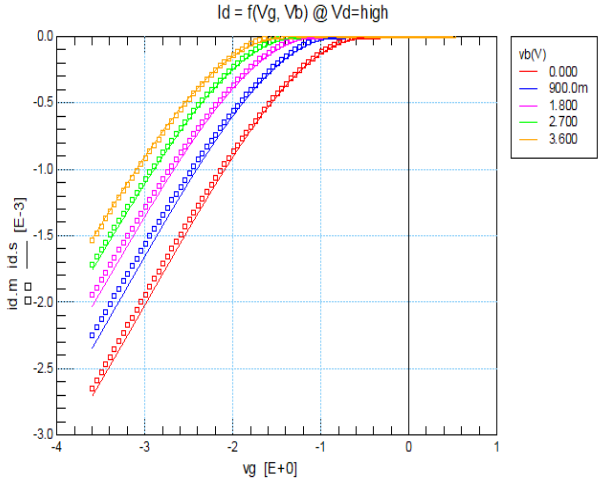


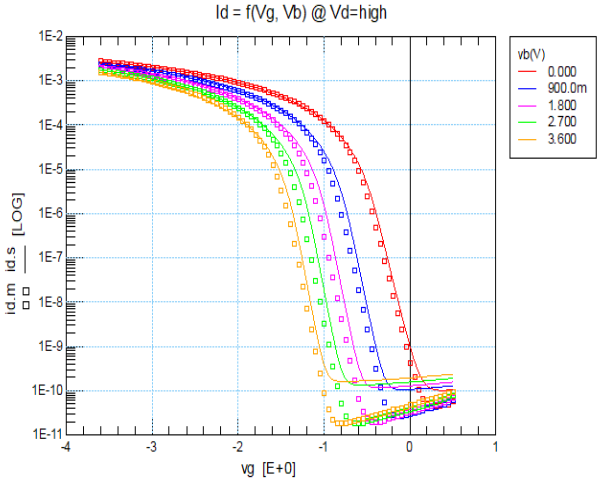


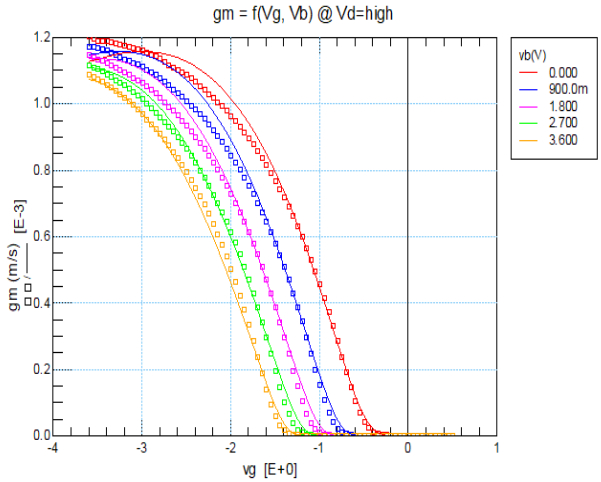


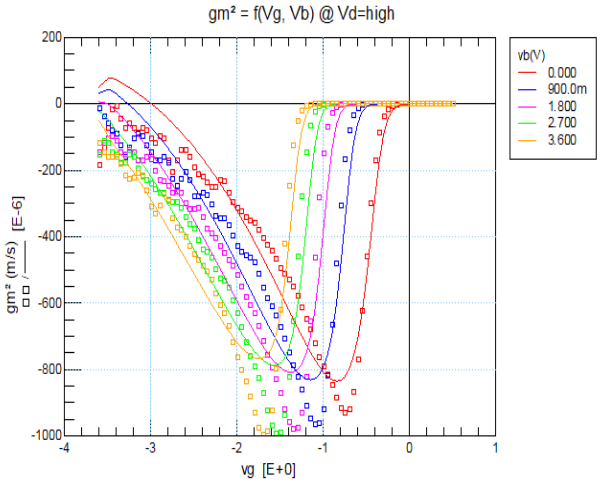


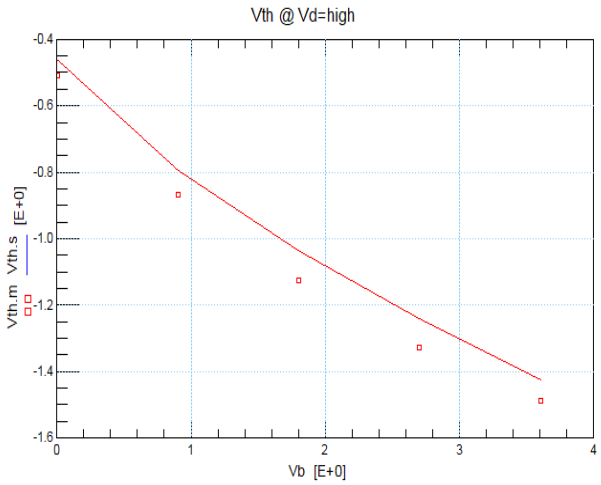
idvg, Vd = -3.3V, T = 125°C

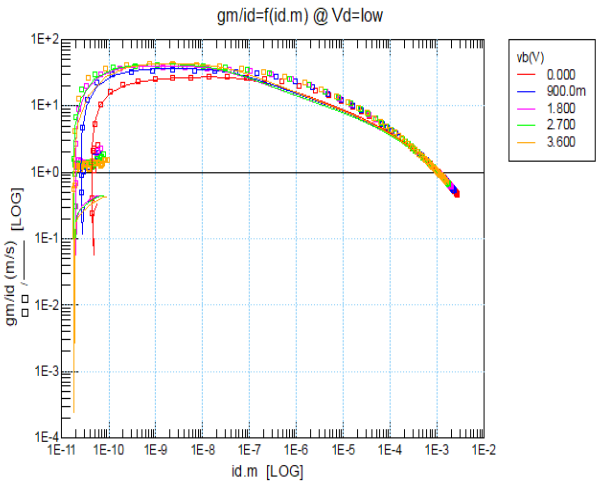


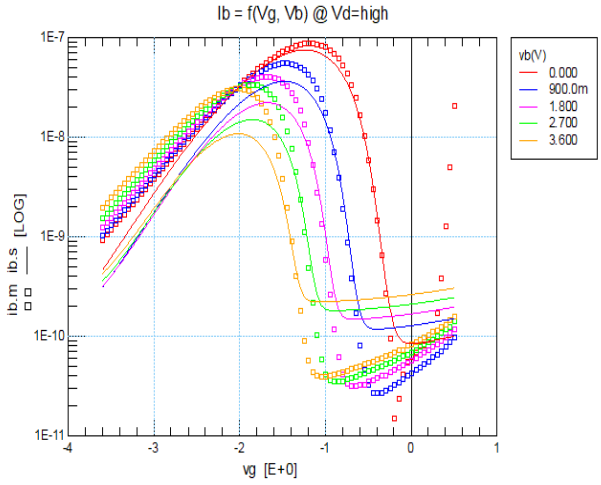


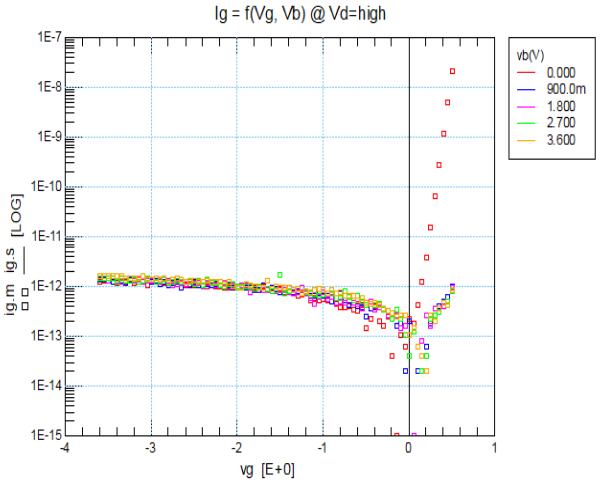




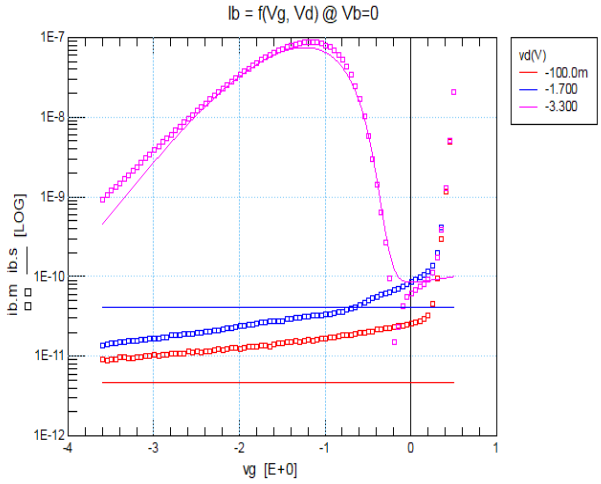


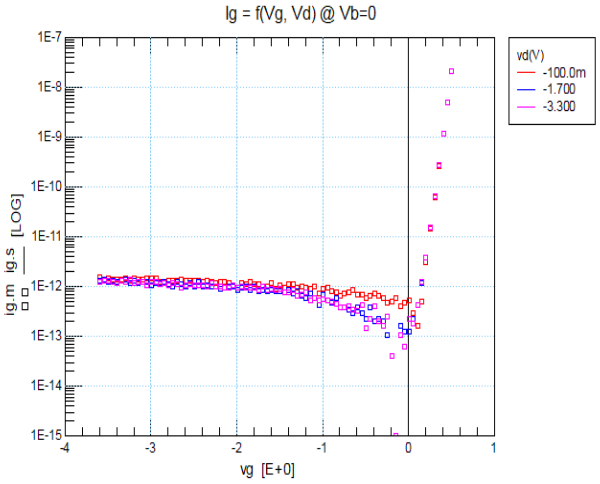




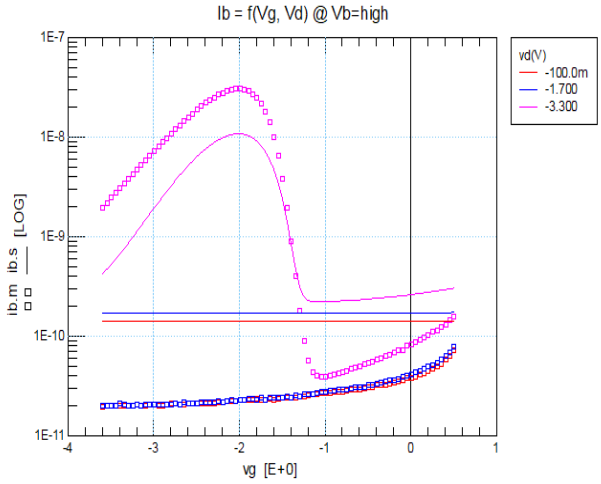


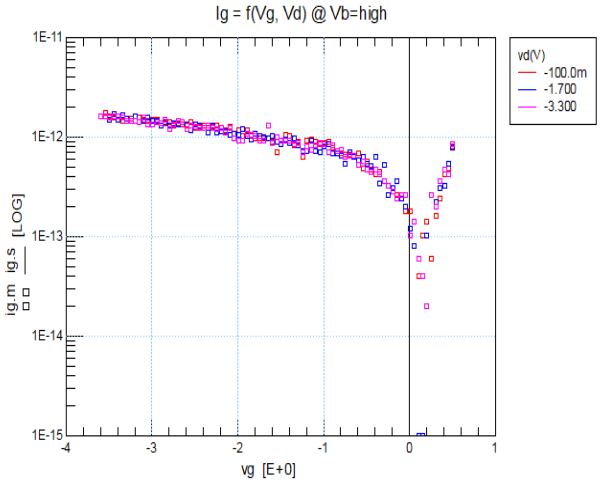
idvg, Ib,Ig, Vb = 0V, T = 125°C



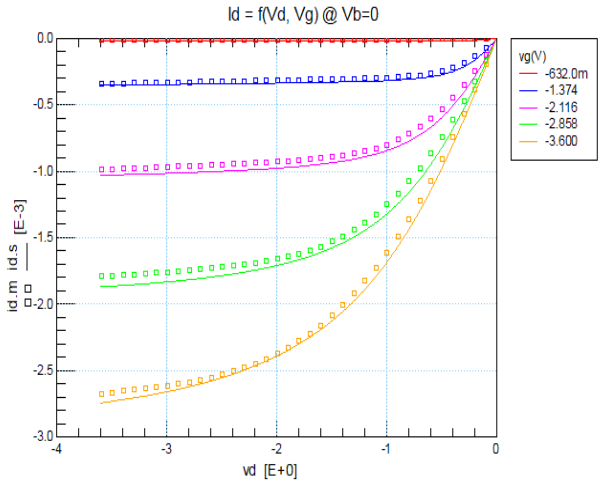


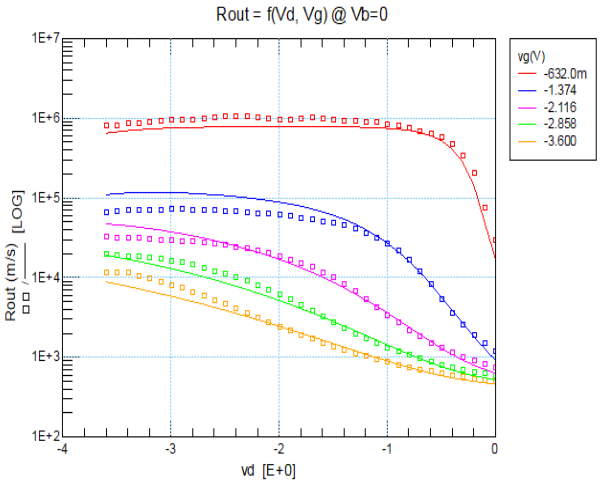
idvg, Ib,Ig, Vb = 3.6V, T = 125°C

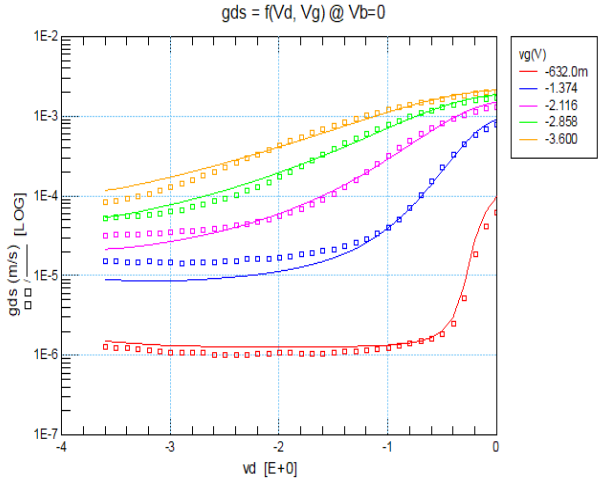




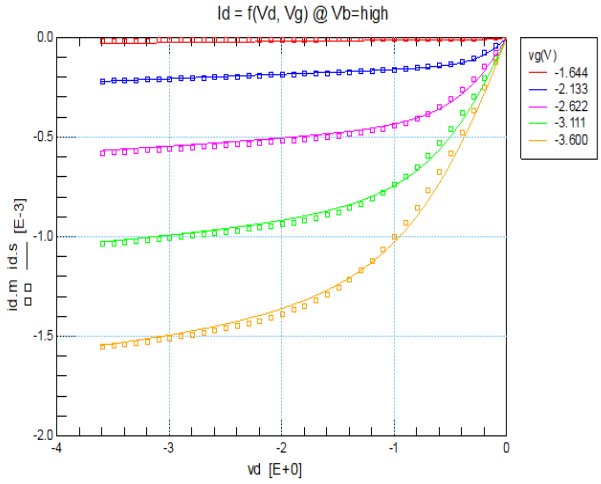
idvd, Vb = 0V, T = 125°C

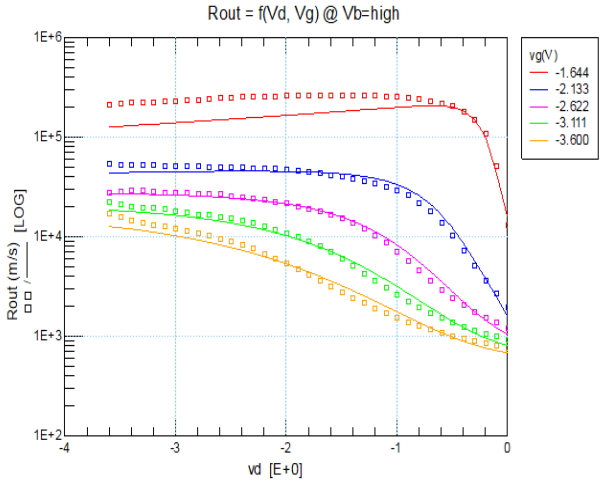


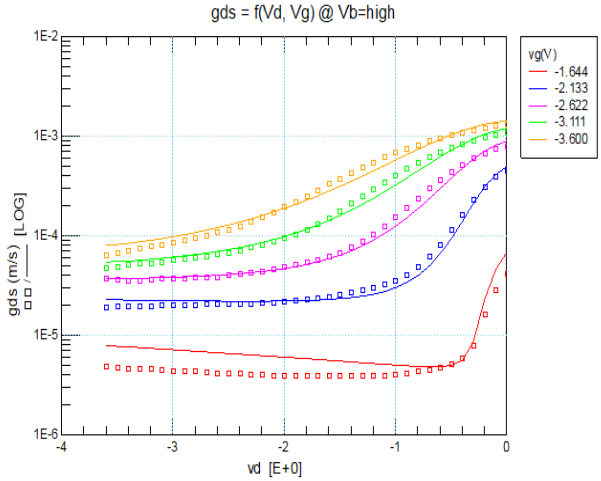




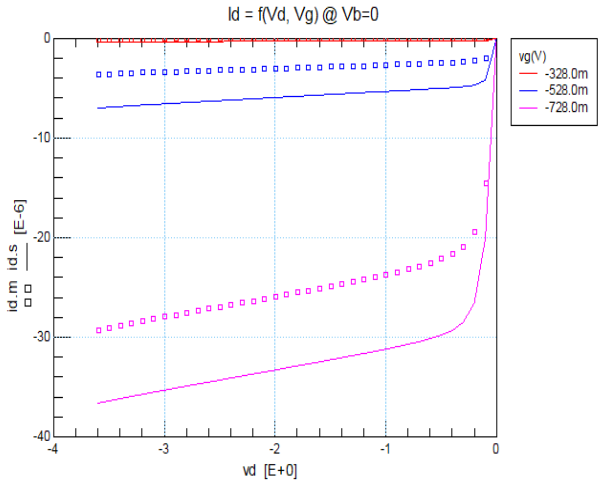
idvd_vbmin, Vb = 3.6V, T = 125°C

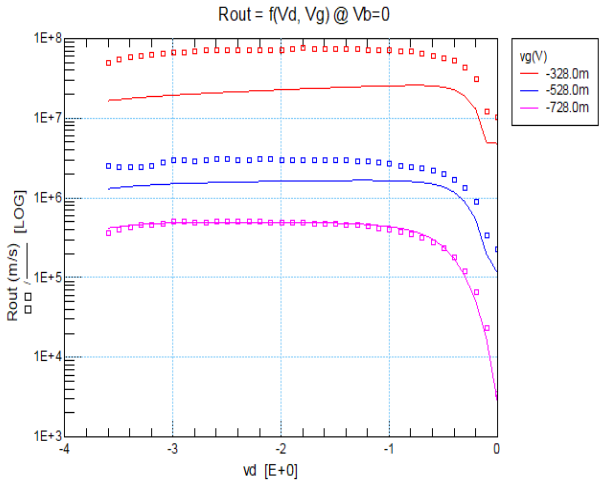


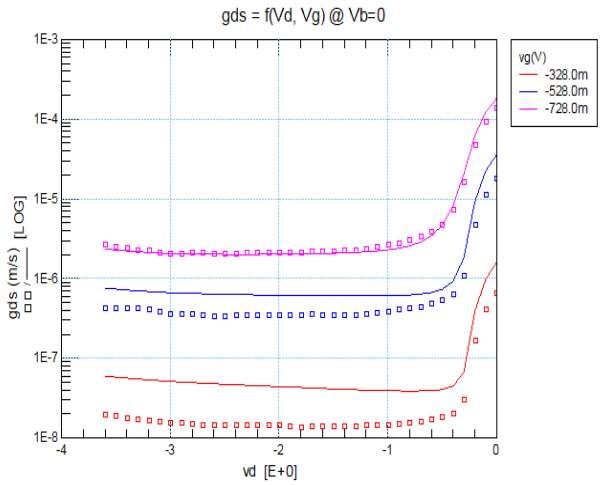




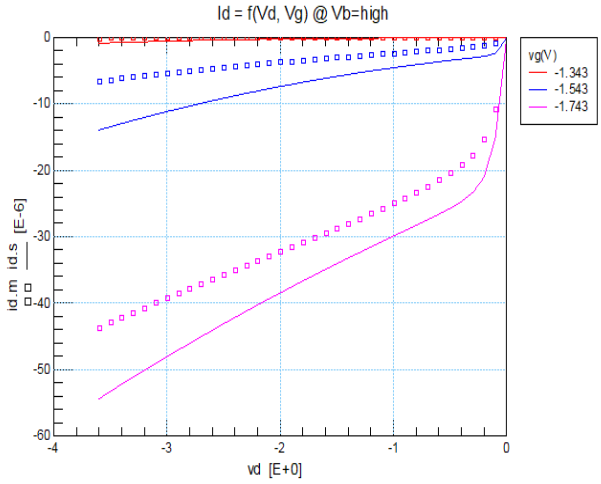
idvd_vth, Vb = 0V, T = 125°C

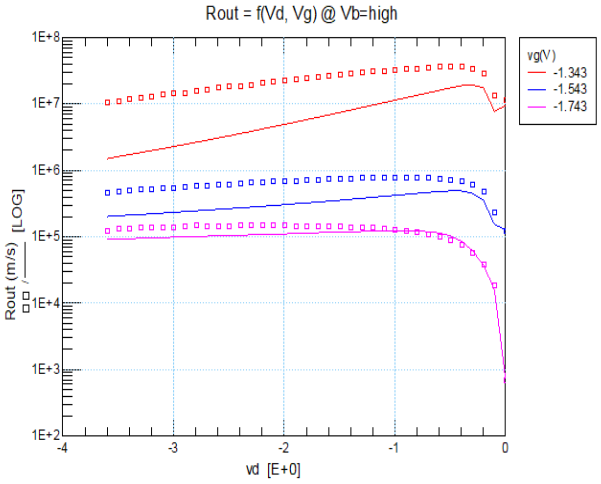


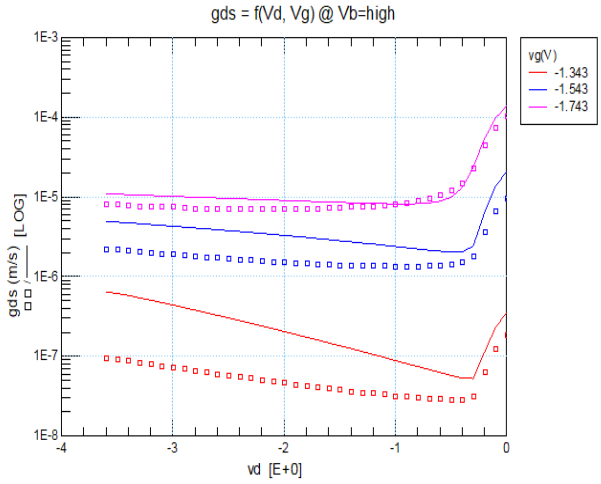




idvd_vth_vbmin, Vb = 3.6V, T = 125°C

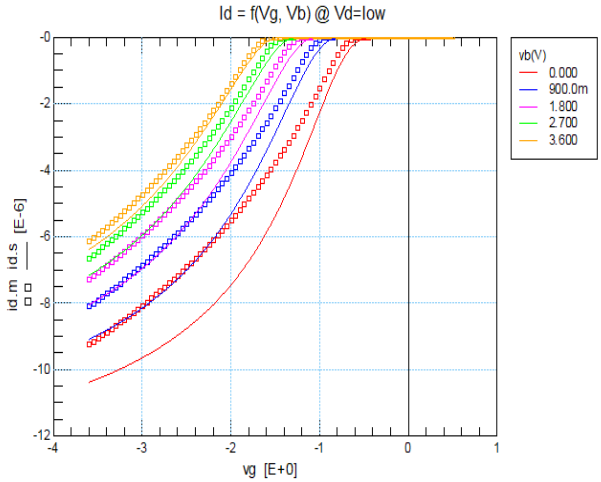


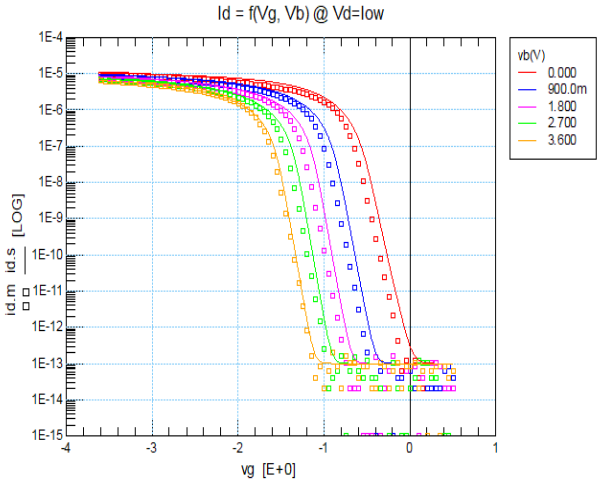


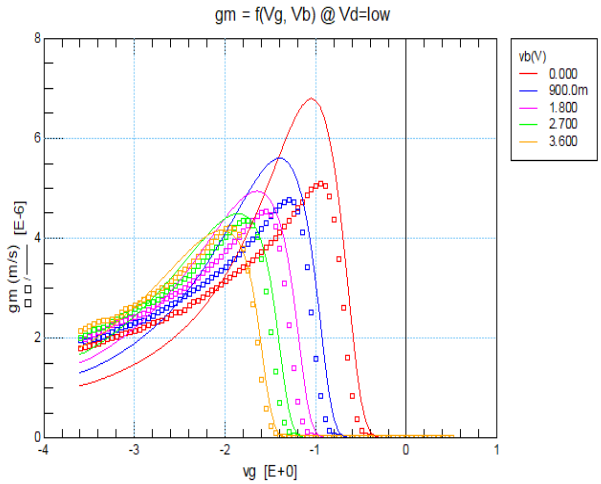


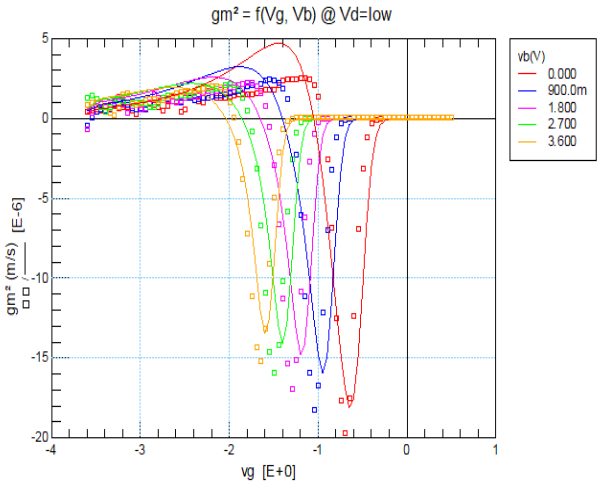
4.5 W0u3_L0u35_S565_3

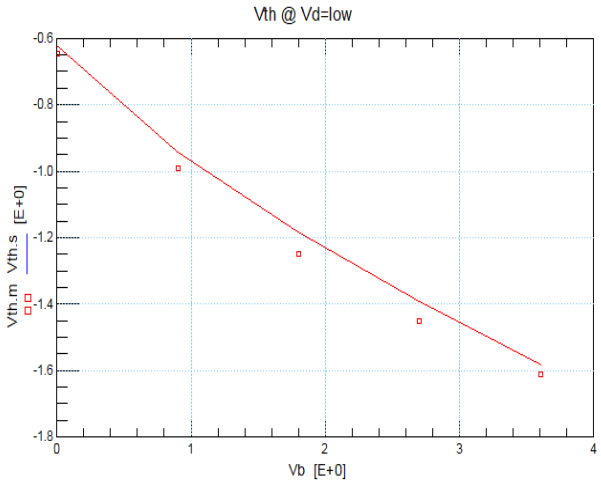
idvg, Vd = -0.1V, T = 27°C

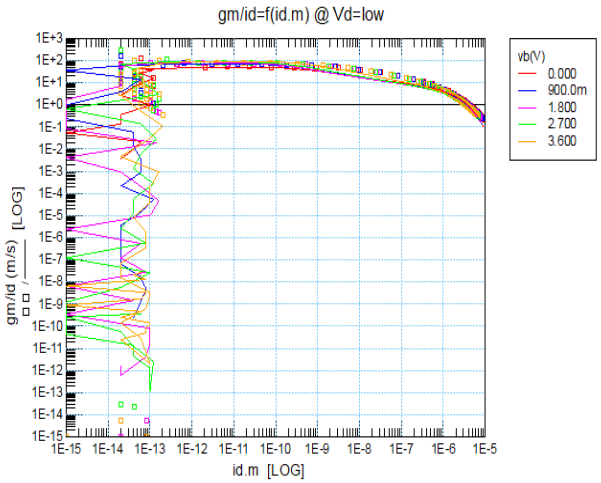


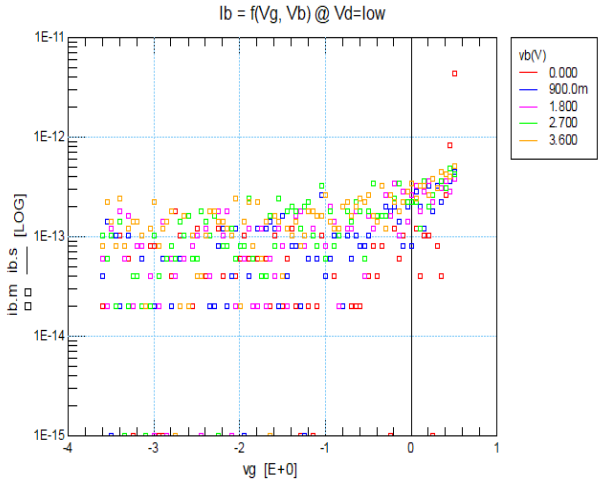


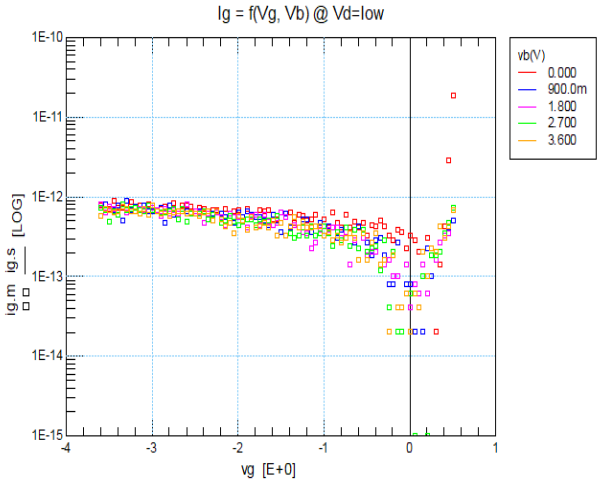




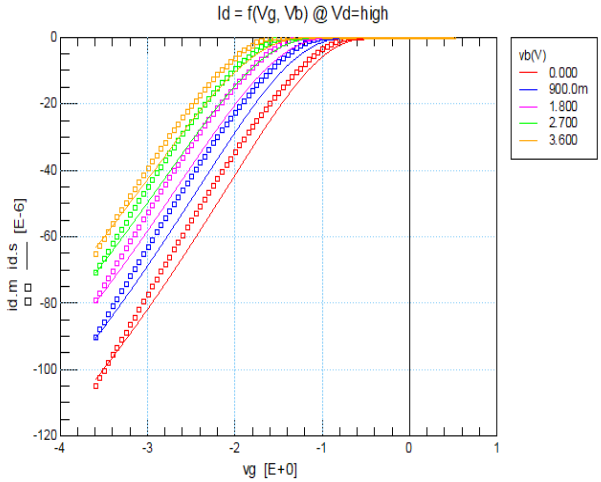


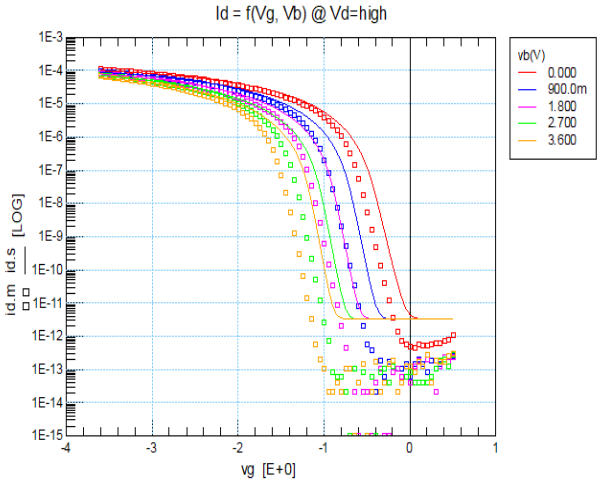


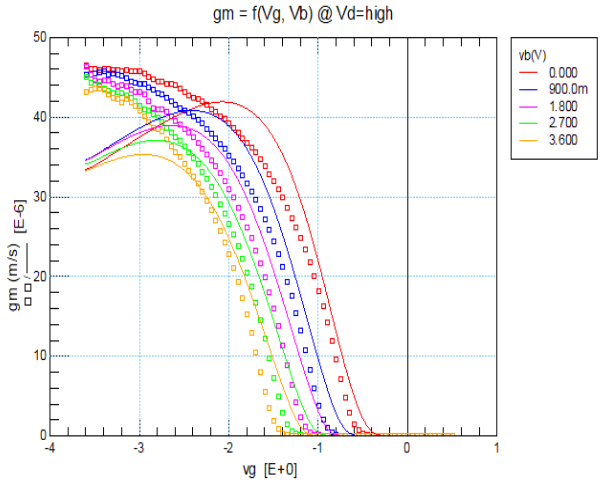


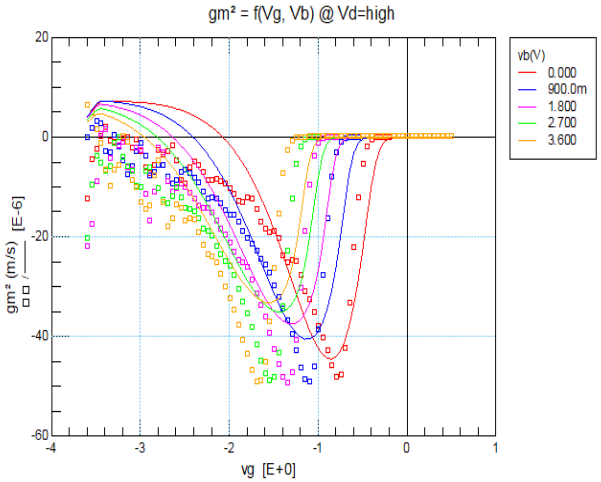


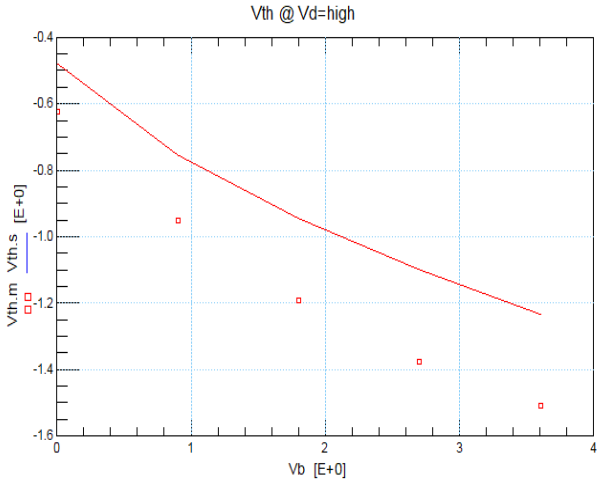
idvg, Vd = -3.3V, T = 27°C

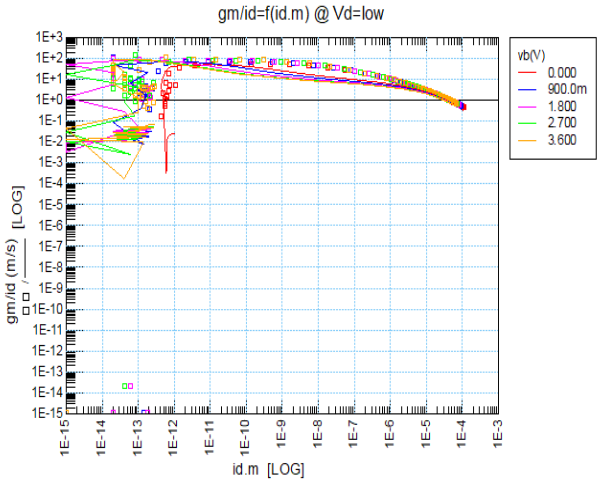


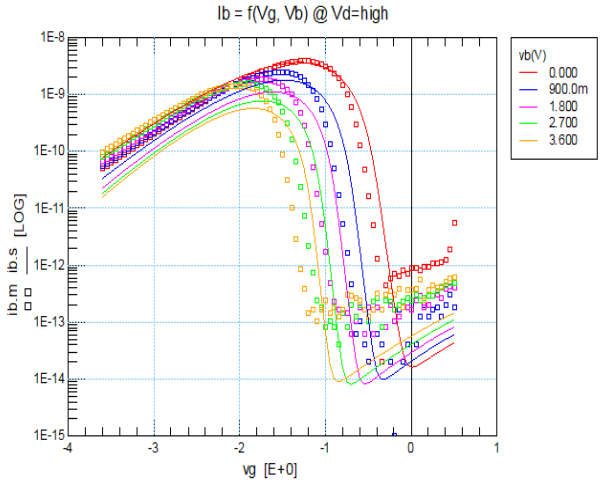


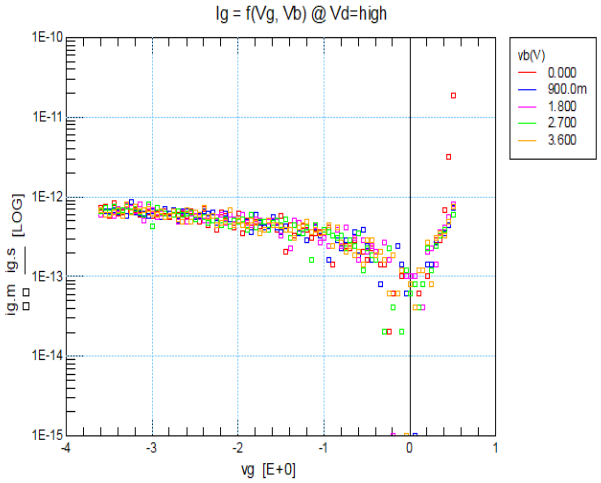




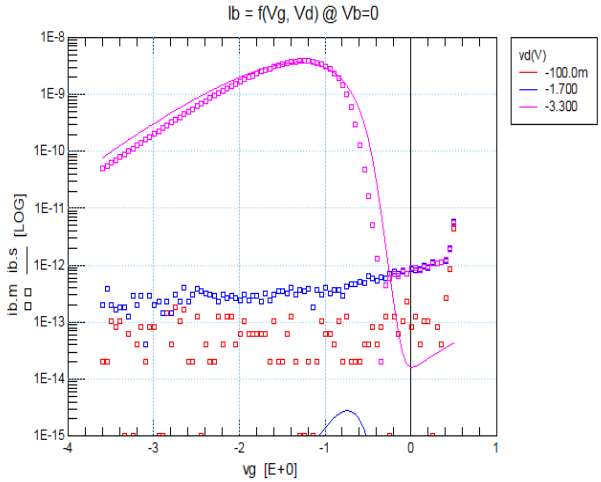


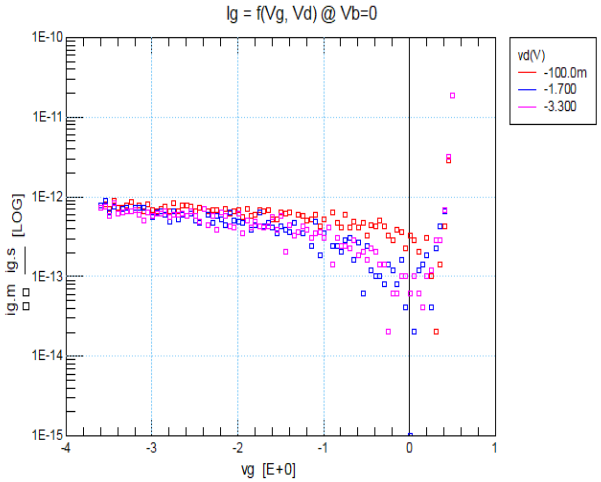




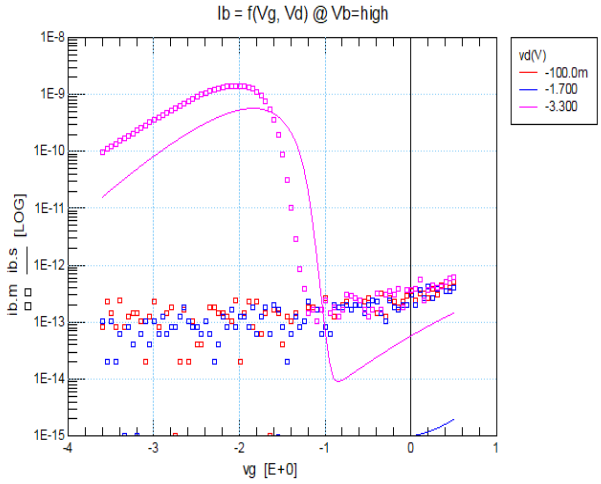


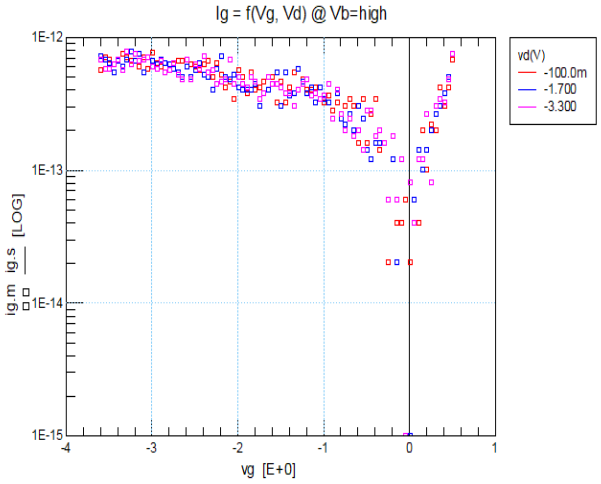
idvg, Ib,Ig, Vb = 0V, T = 27°C



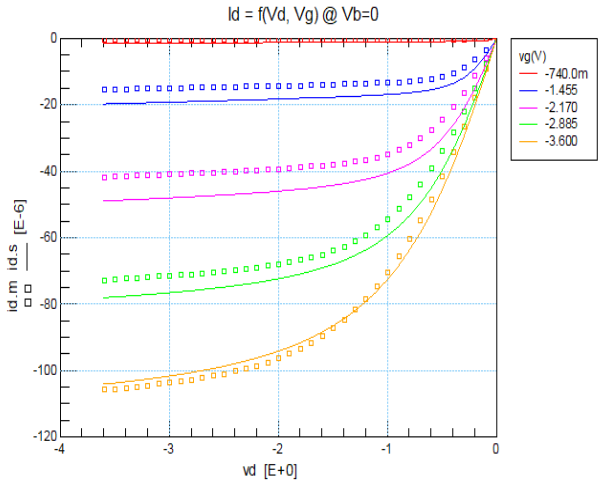


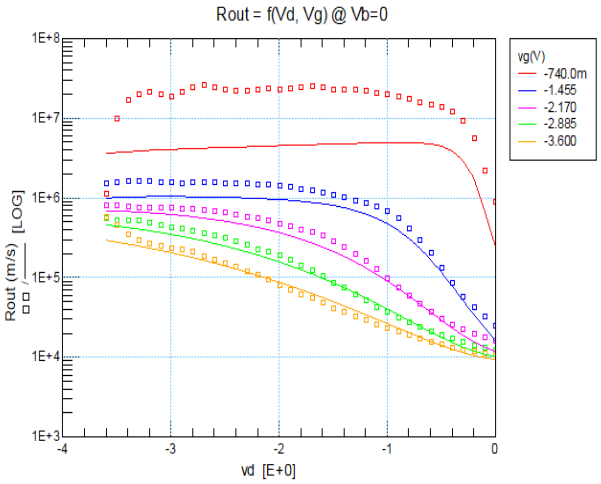
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

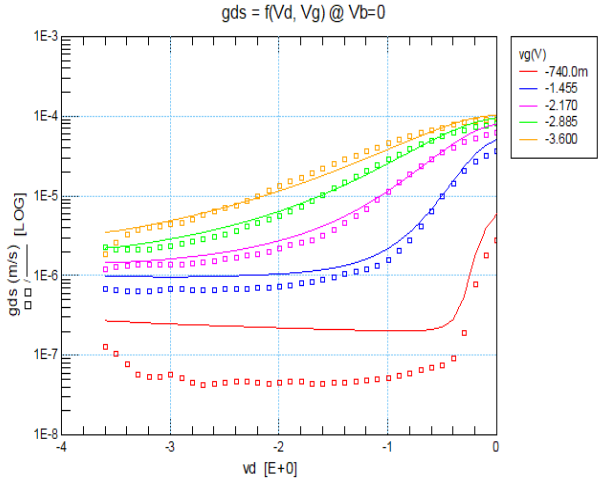




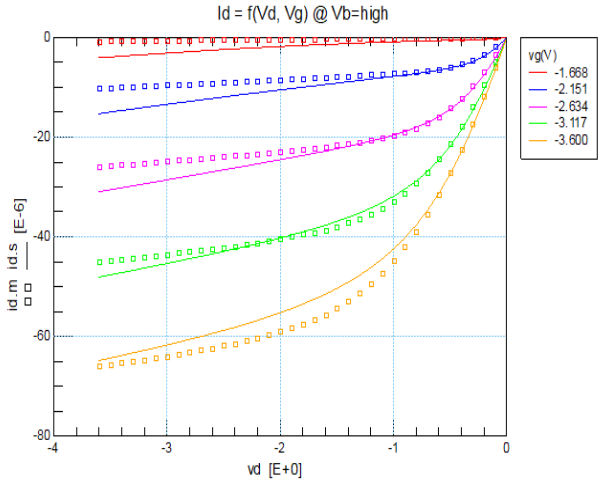
idvd, Vb = 0V, T = 27°C

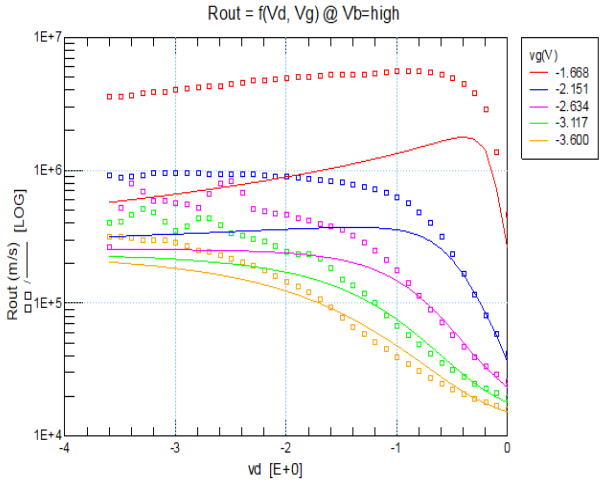


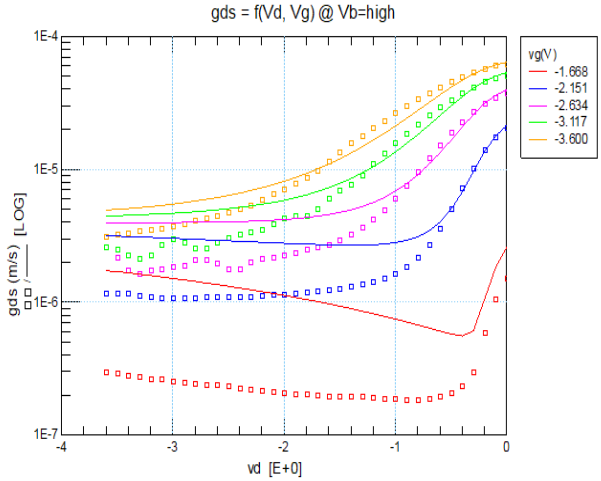




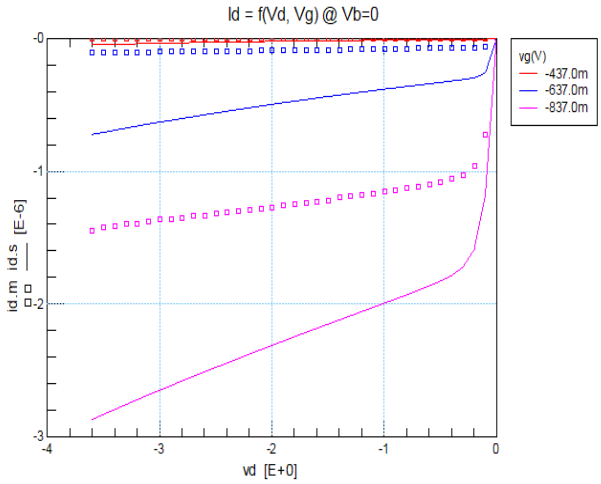
idvd_vbmin, Vb = 3.6V, T = 27°C

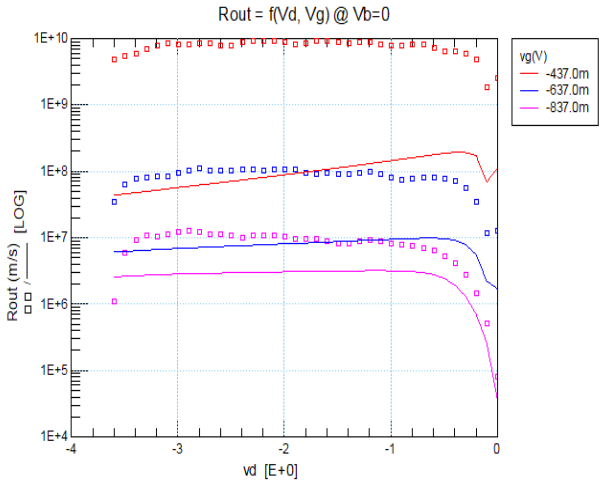


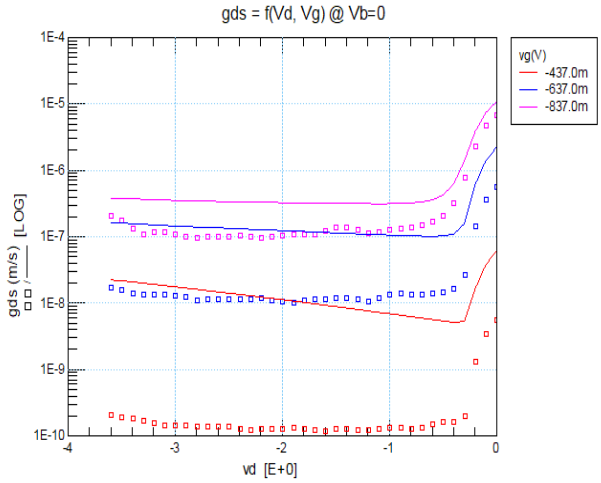




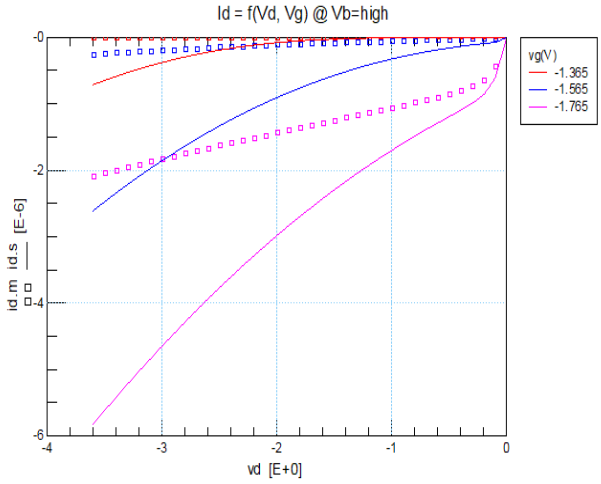
idvd_vth, Vb = 0V, T = 27°C

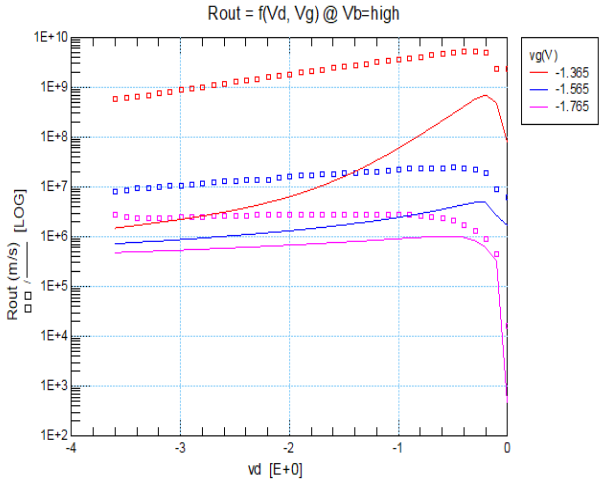


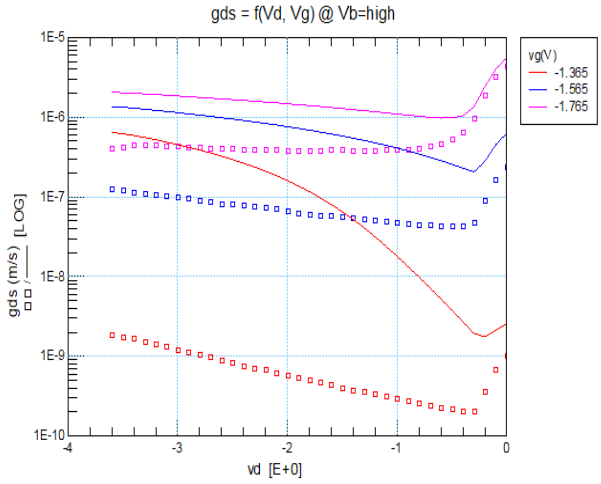




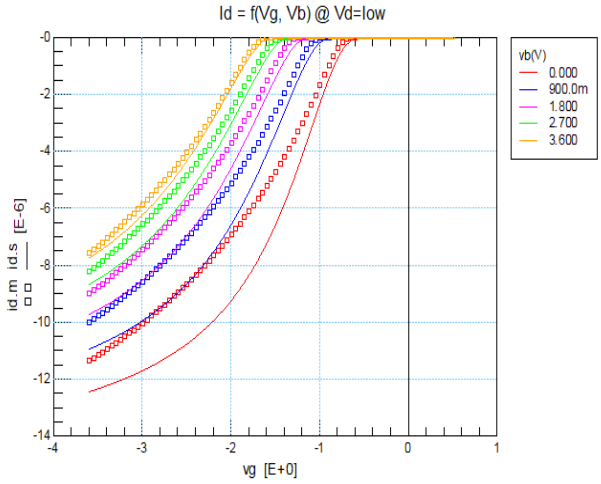
idvd_vth_vbmin, Vb = 3.6V, T = 27°C

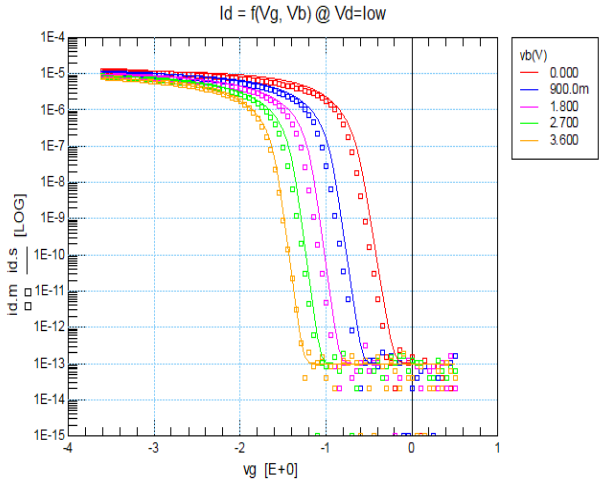


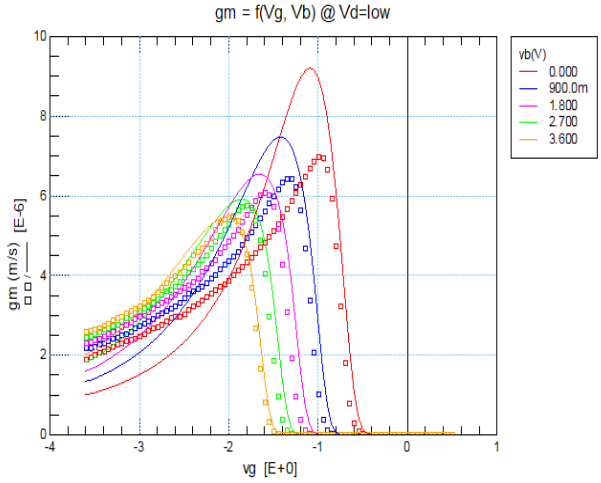


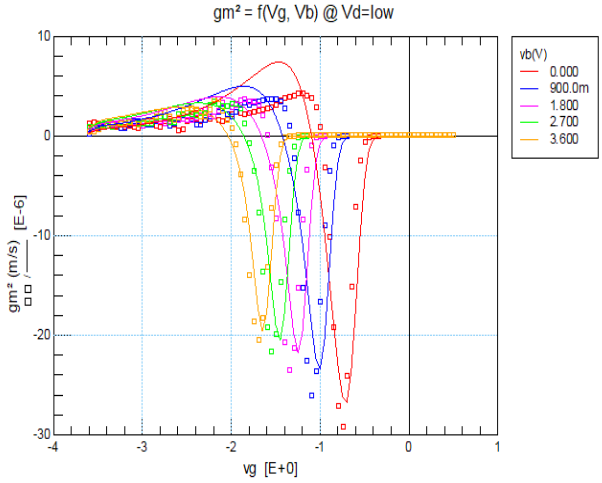


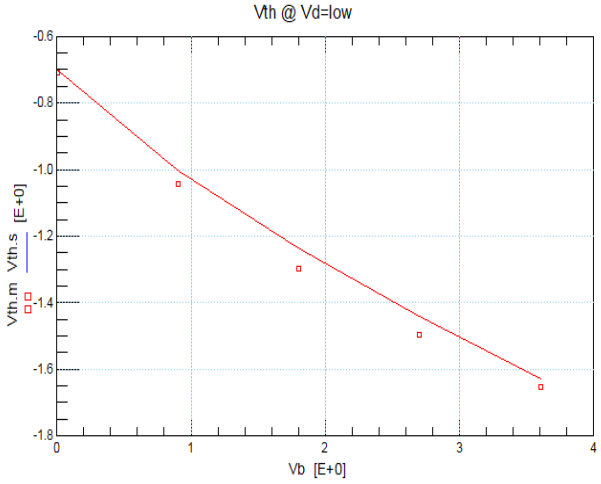
idvg, Vd = -0.1V, T = -40°C

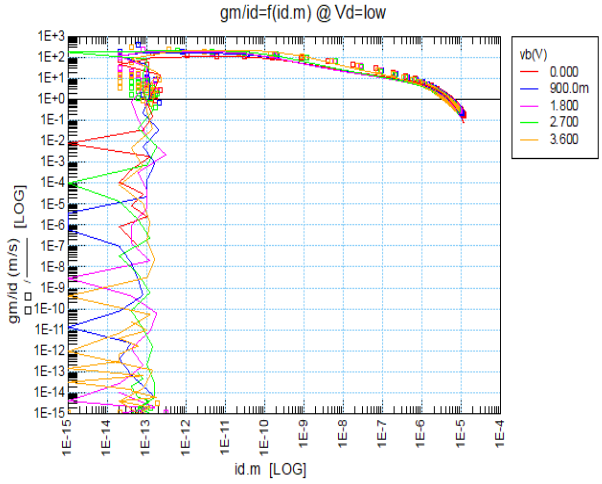


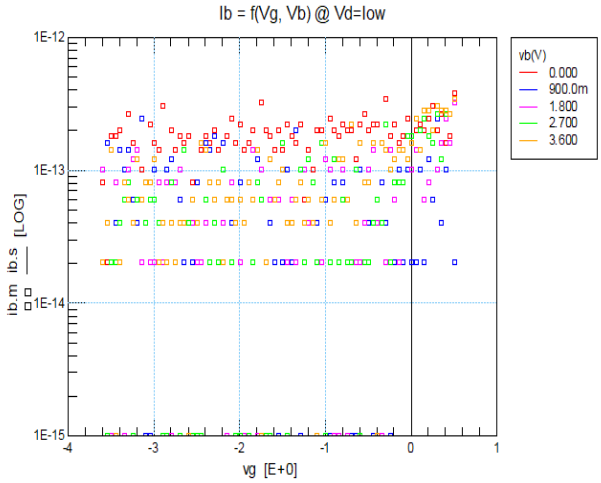


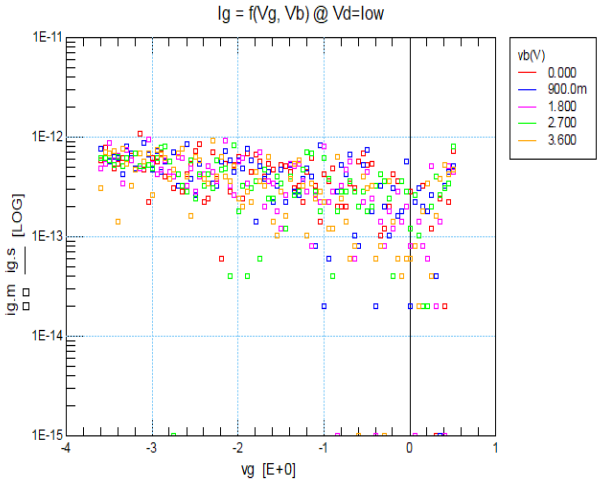




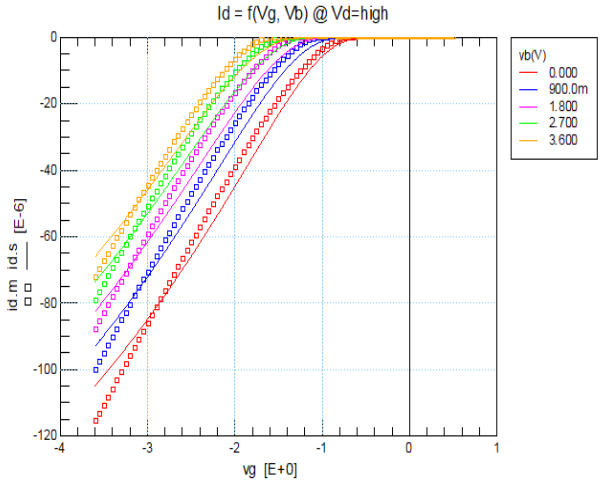


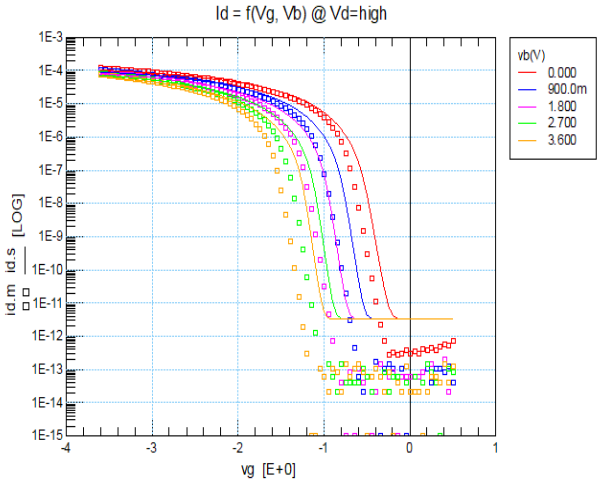


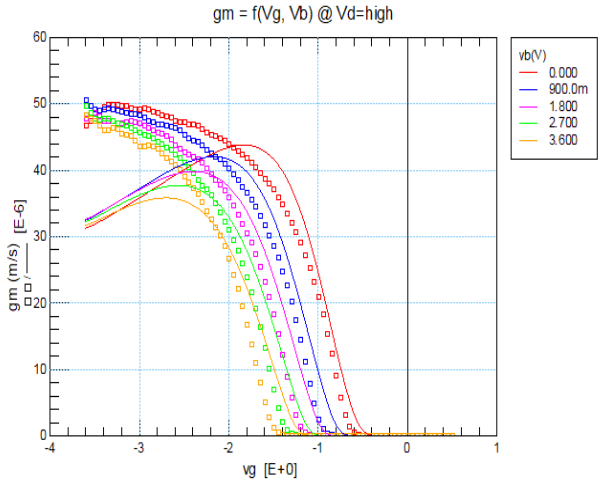


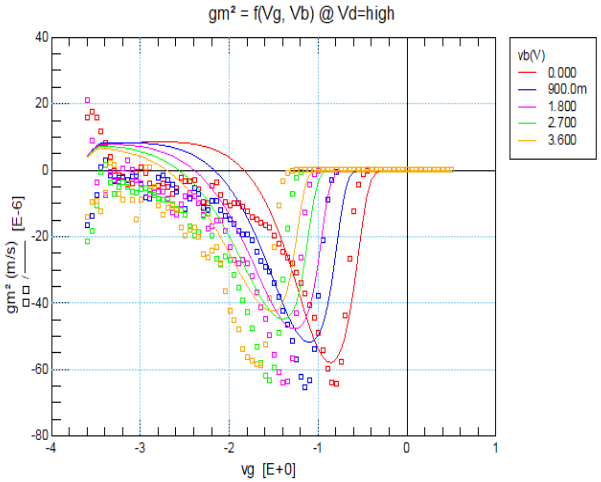


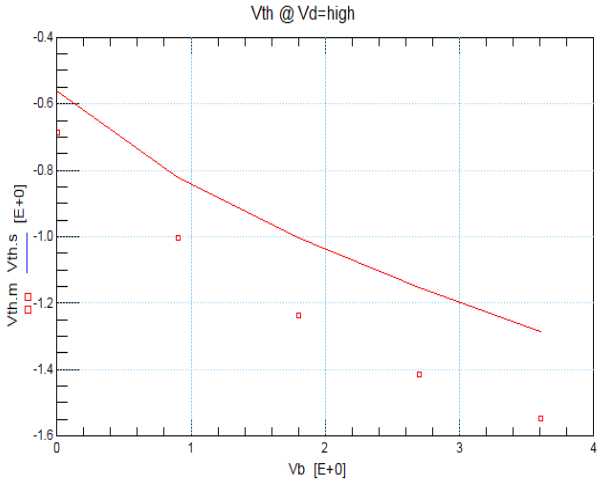
idvg, Vd = -3.3V, T = -40°C

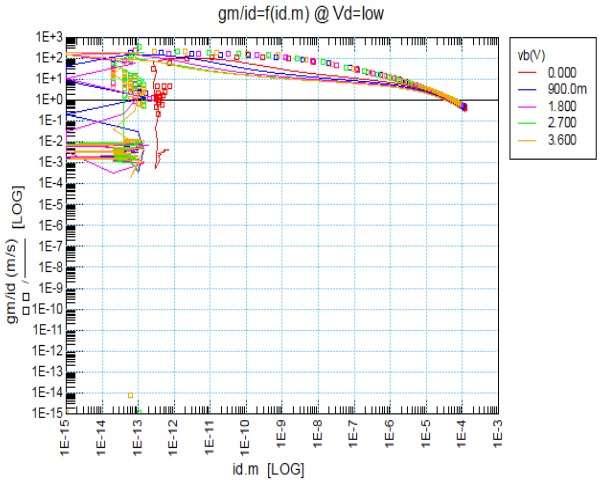


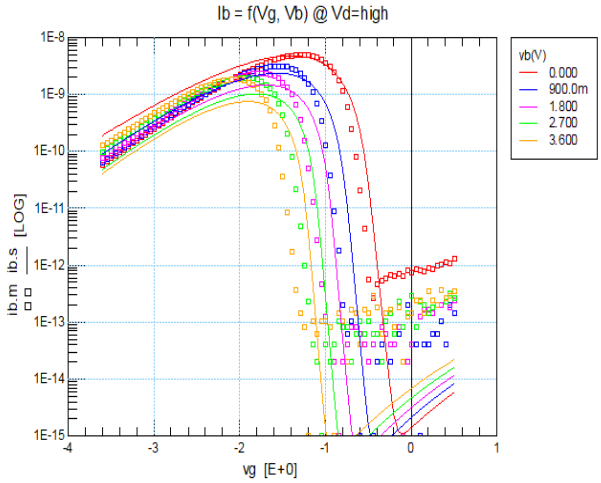


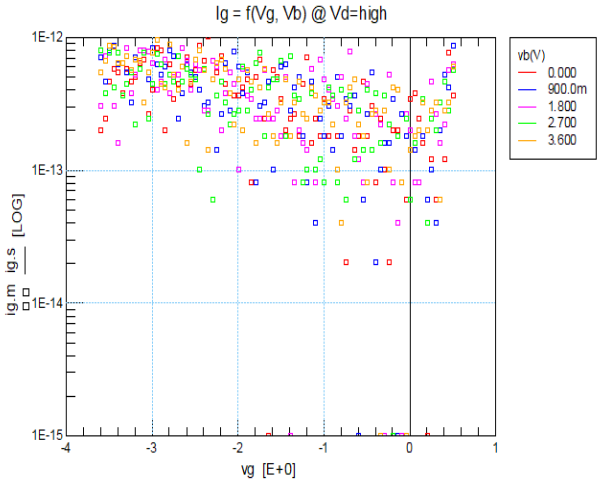




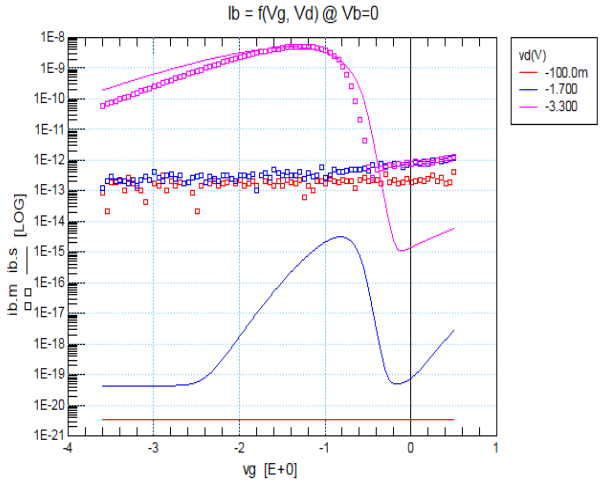


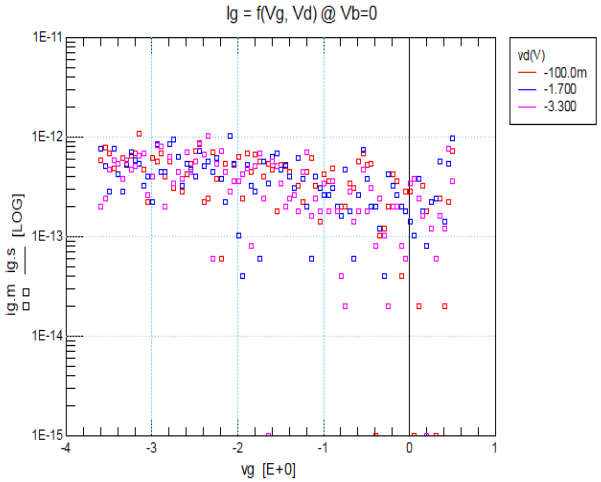




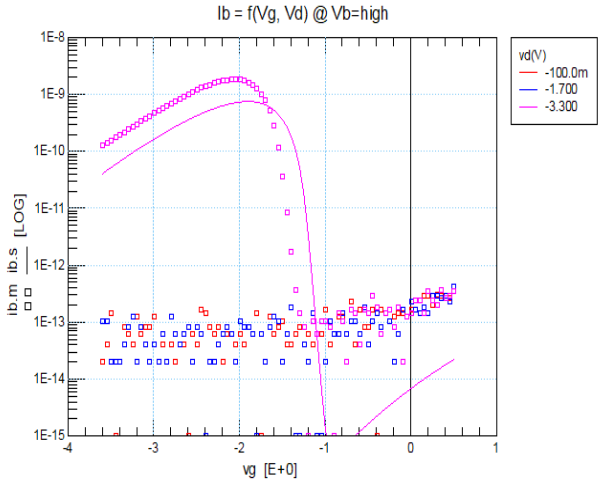


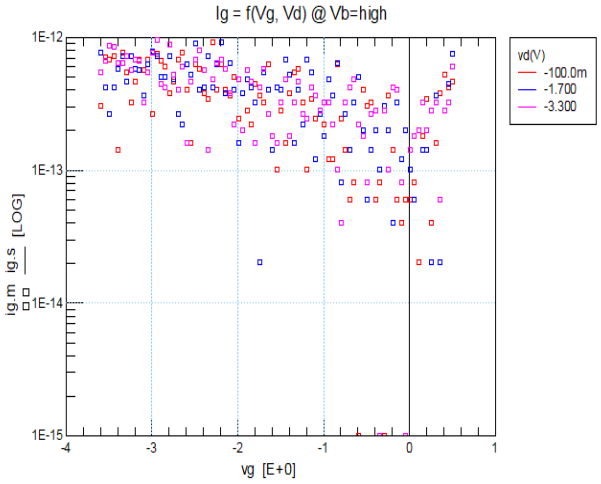
idvg, Ib,Ig, Vb = 0V, T = -40°C



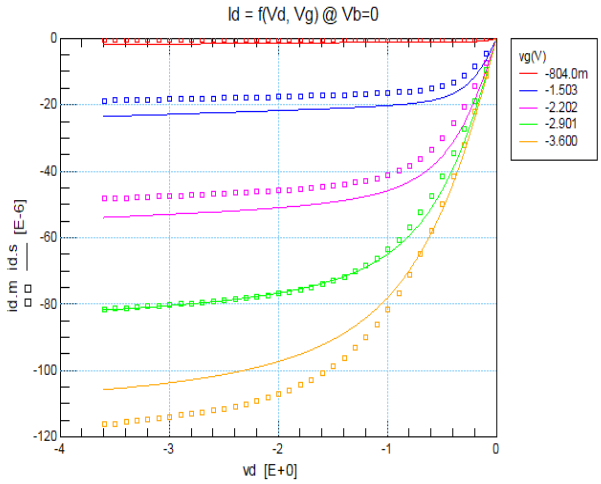


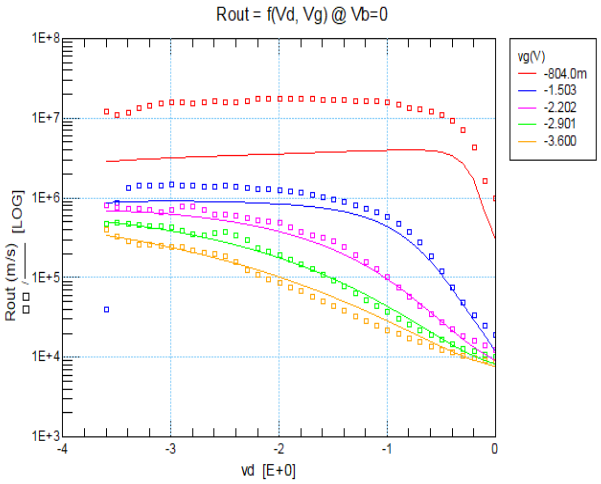
idvg, Ib,Ig, Vb = 3.6V, T = -40°C

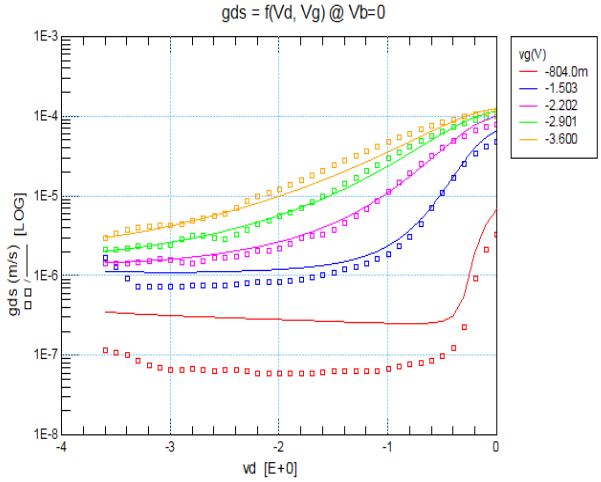




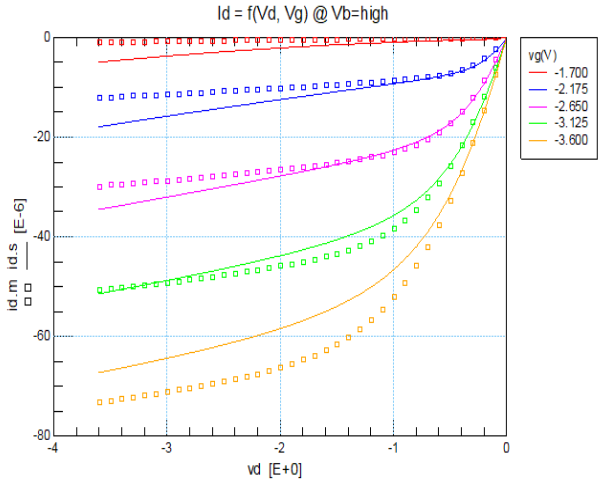
idvd, Vb = 0V, T = -40°C

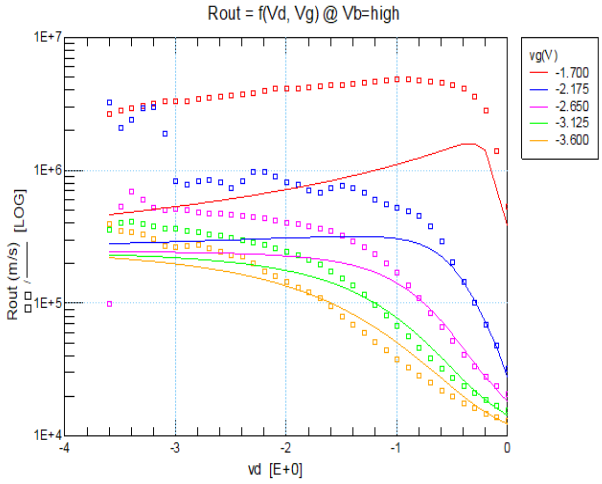


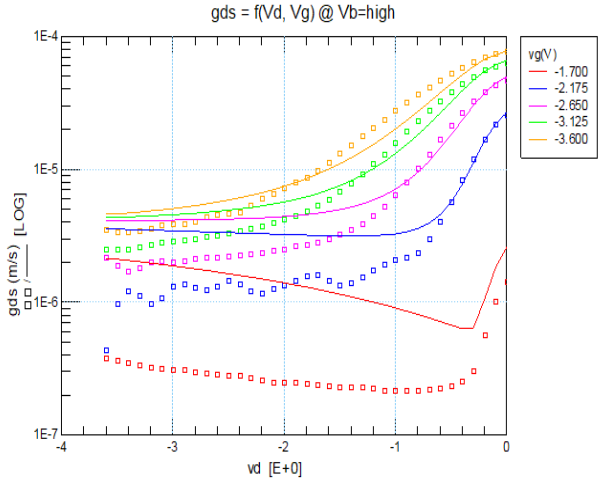




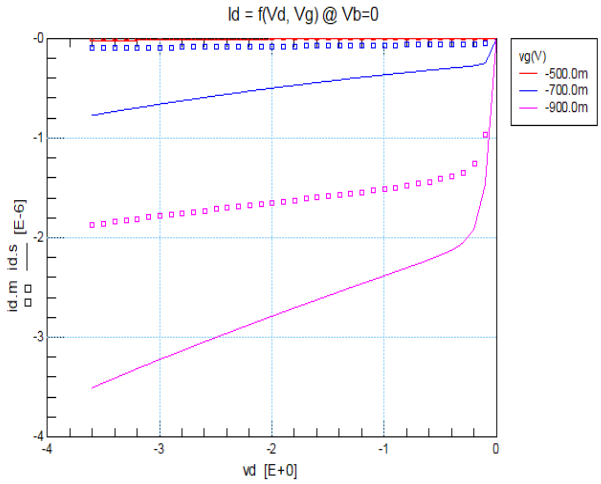
idvd_vbmin, Vb = 3.6V, T = -40°C

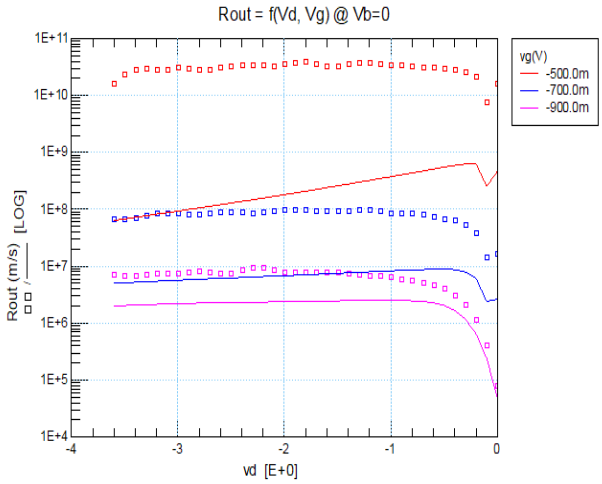


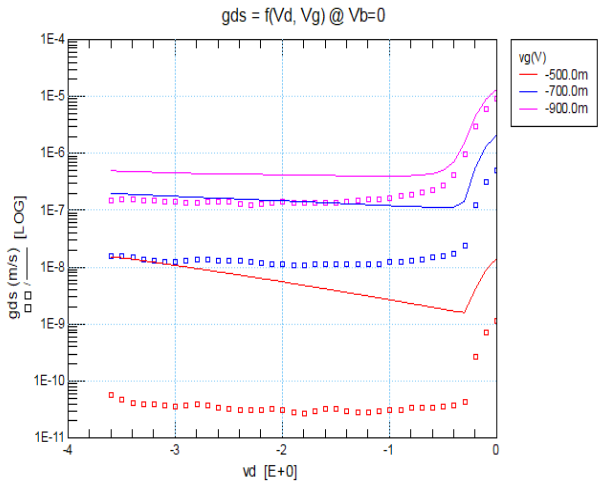




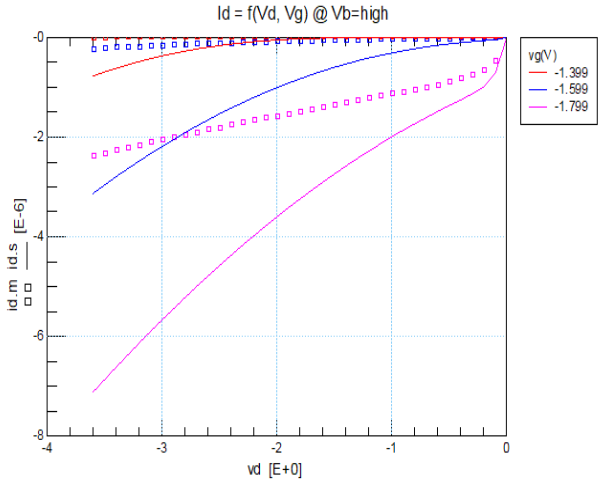
idvd_vth, Vb = 0V, T = -40°C

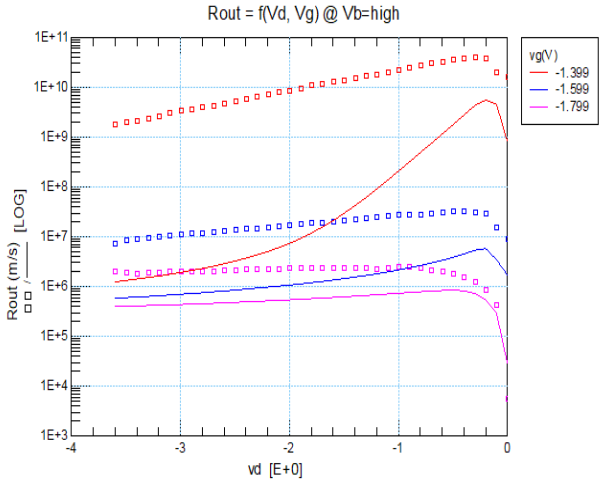


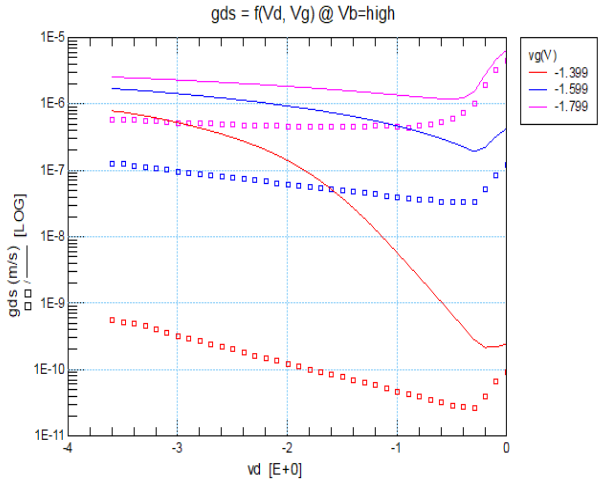




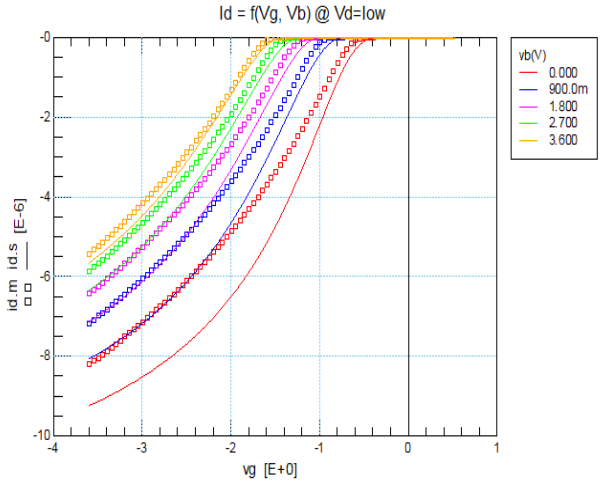
idvd_vth_vbmin, Vb = 3.6V, T = -40°C

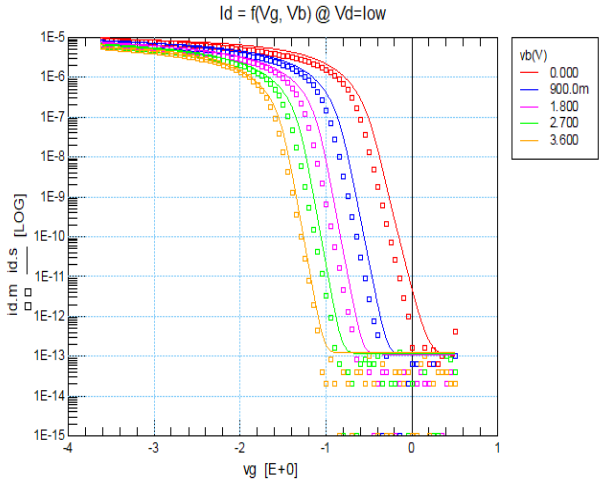


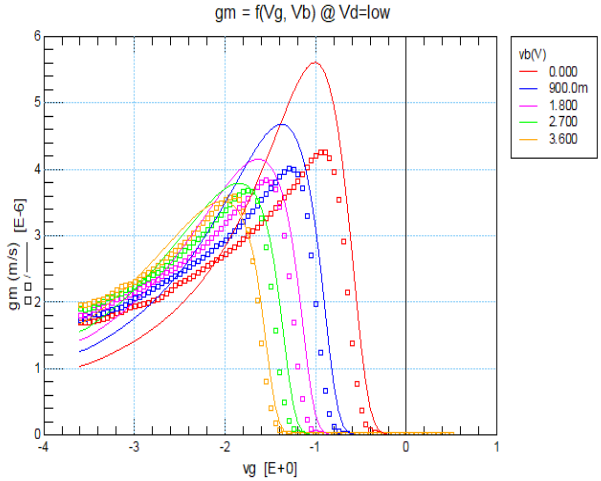


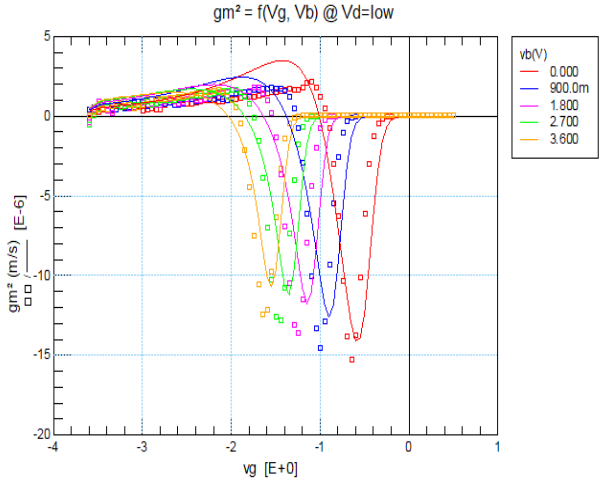


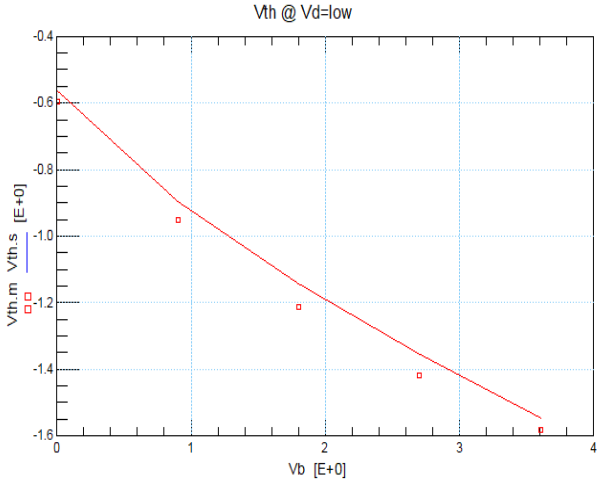
idvg, Vd = -0.1V, T = 70°C

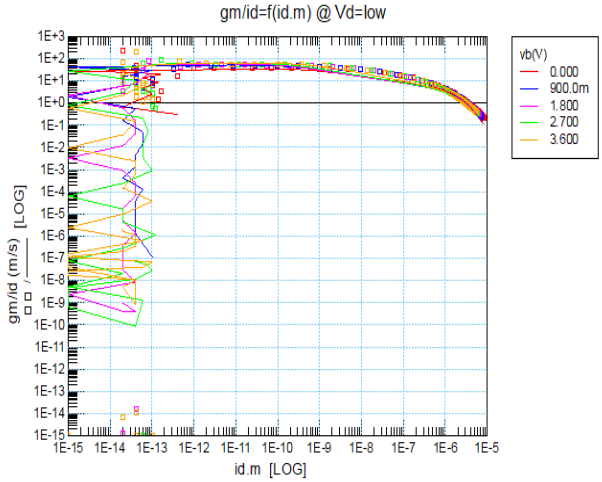


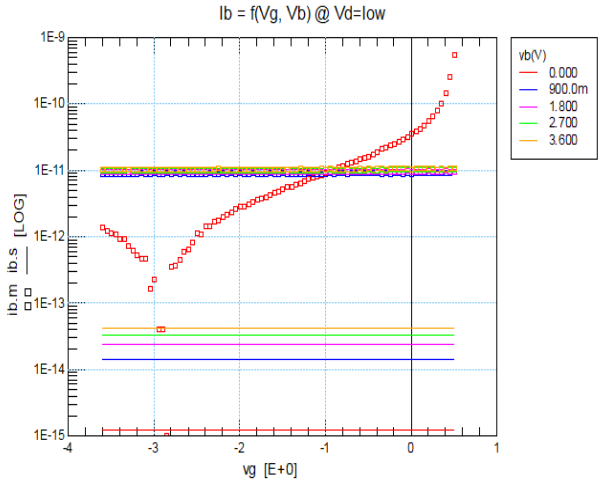


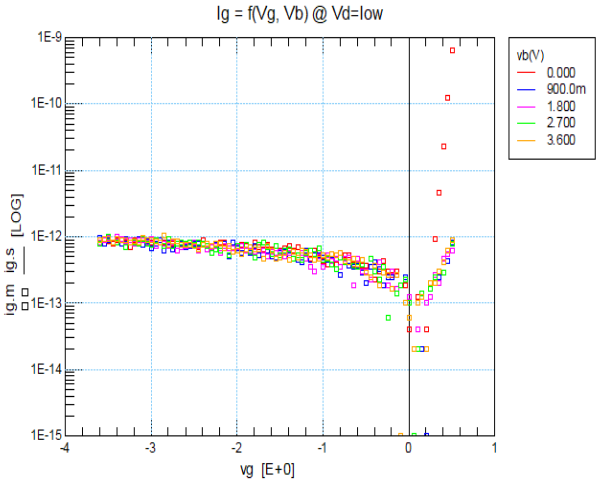




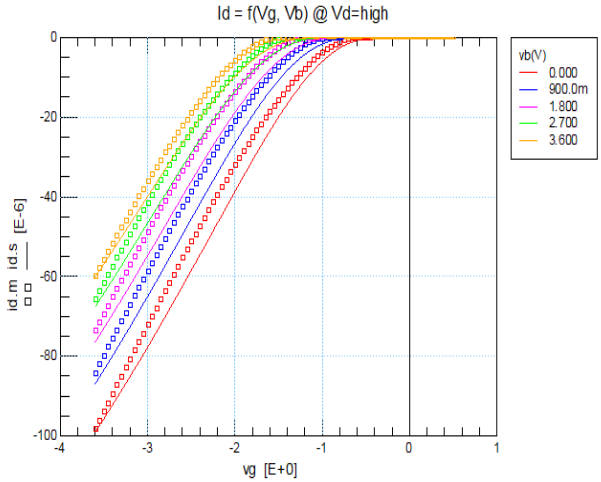


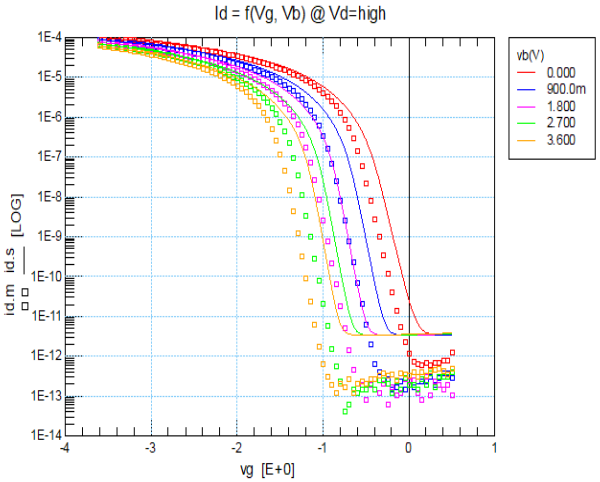


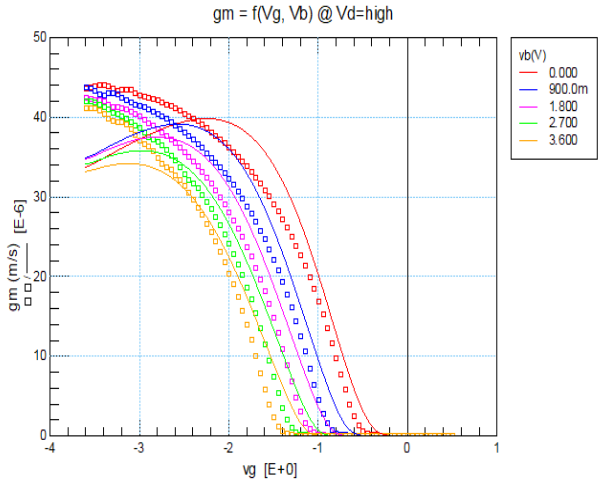


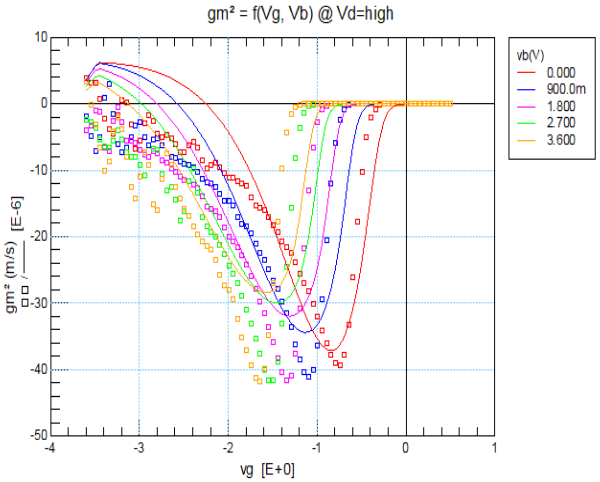


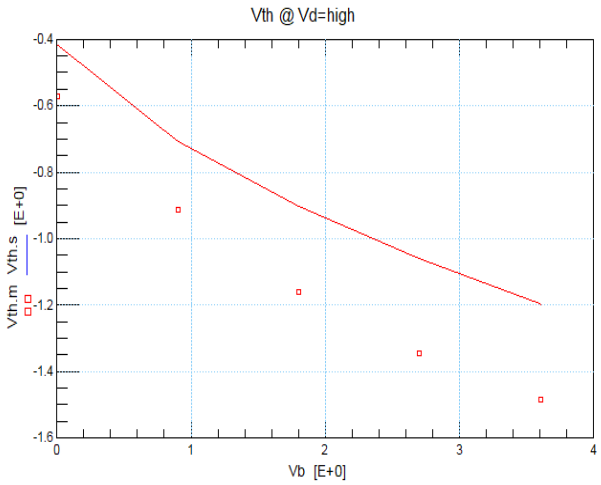
idvg, Vd = -3.3V, T = 70°C

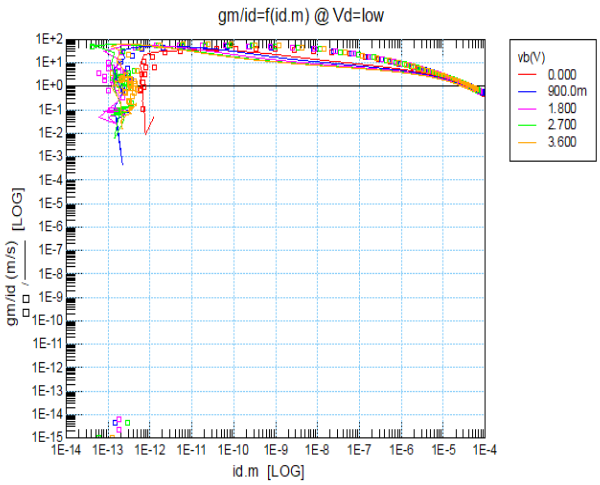


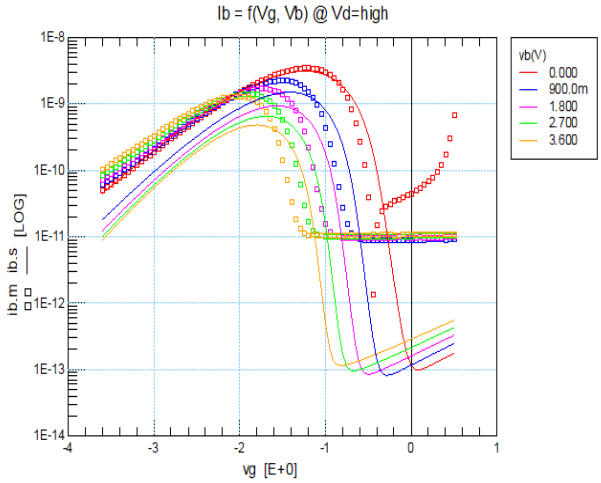


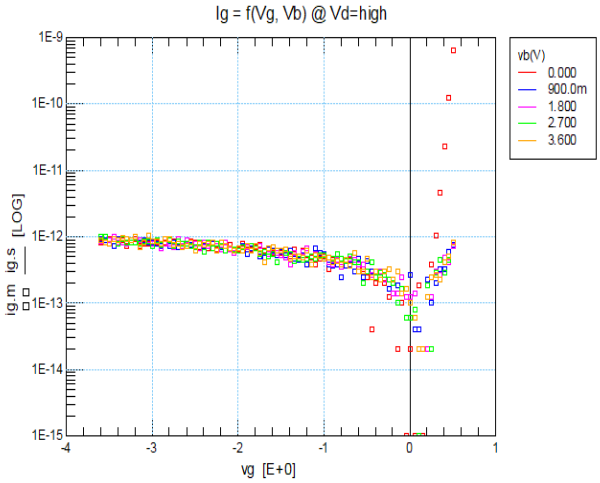




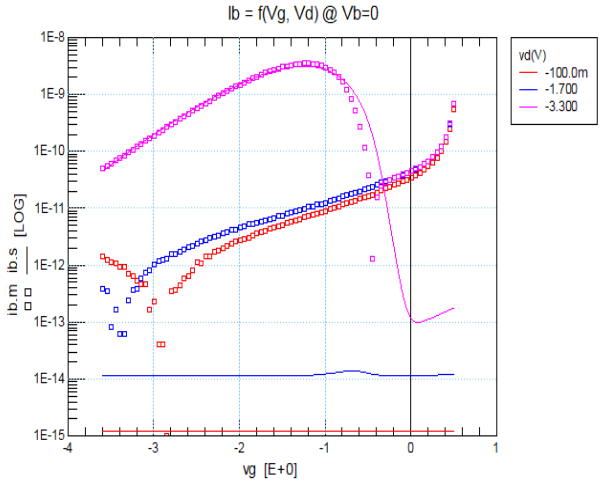


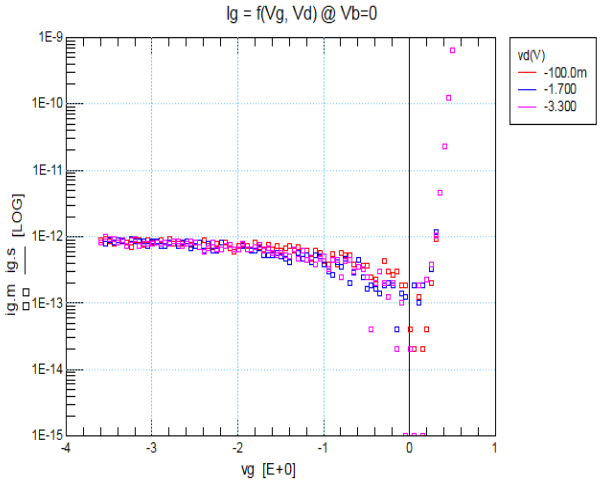




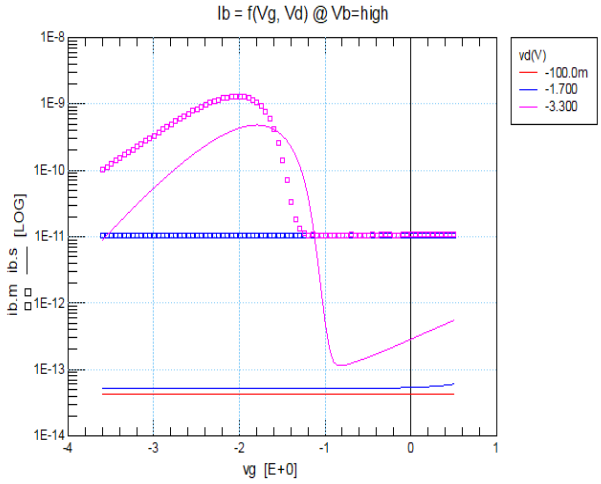


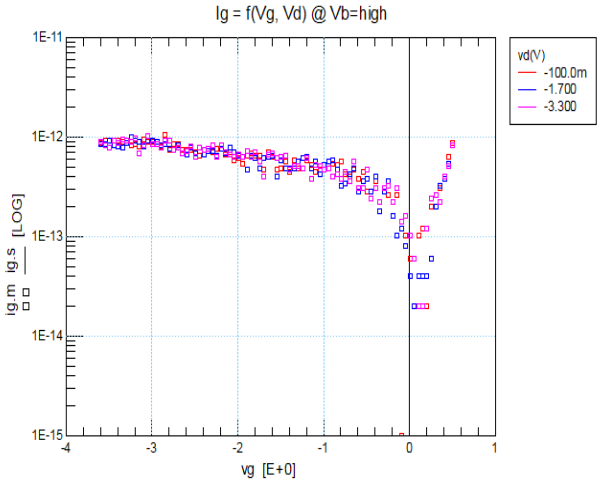
idvg, Ib,Ig, Vb = 0V, T = 70°C



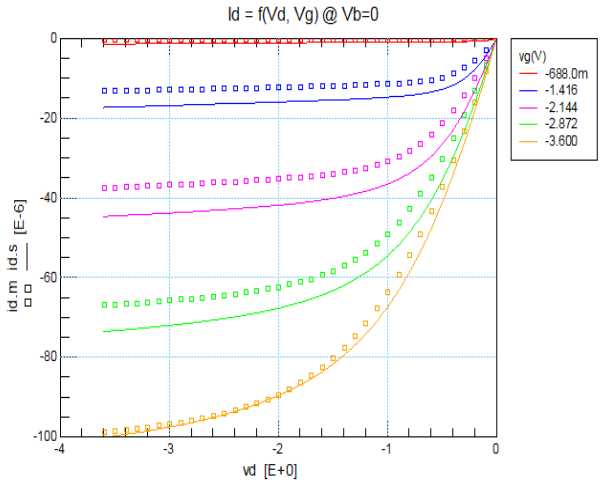


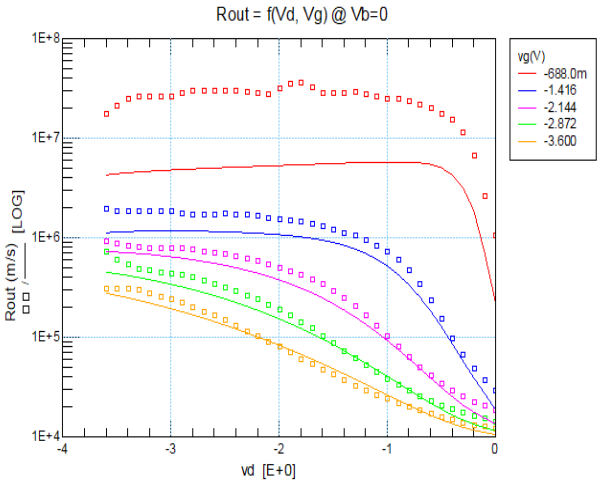
idvg, Ib,Ig, Vb = 3.6V, T = 70°C

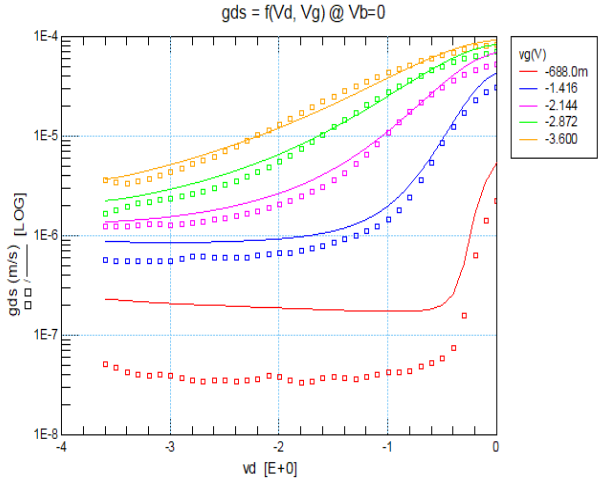




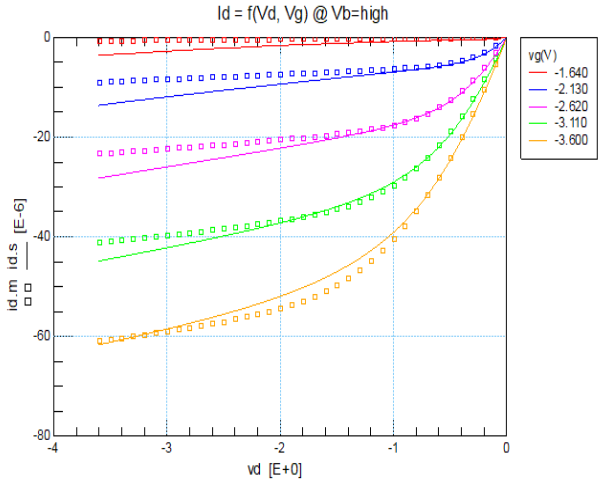
idvd, Vb = 0V, T = 70°C

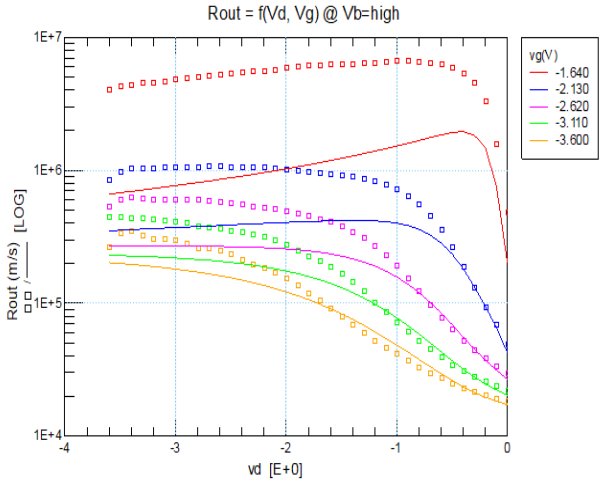


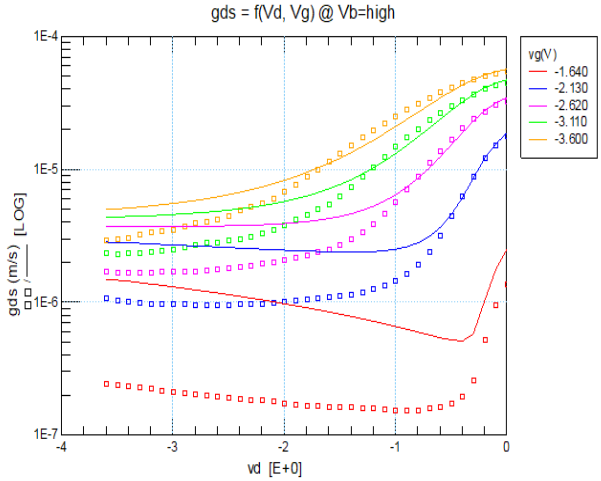




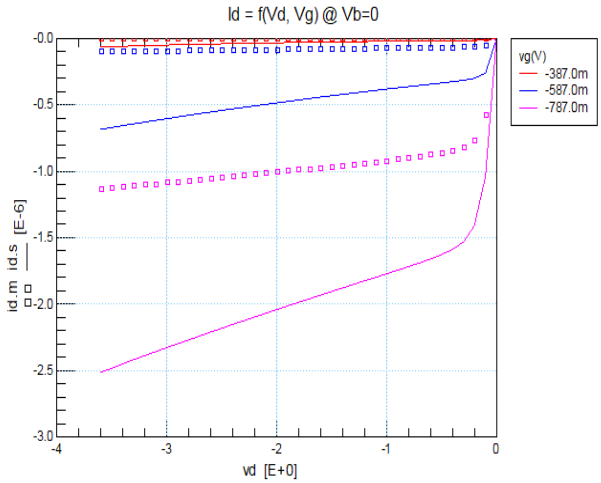
idvd_vbmin, Vb = 3.6V, T = 70°C

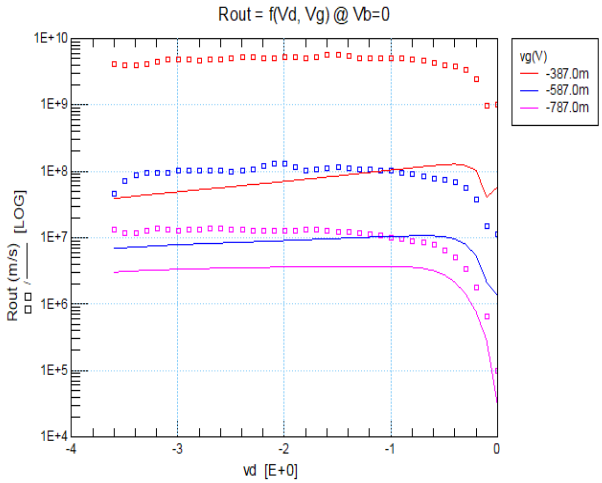


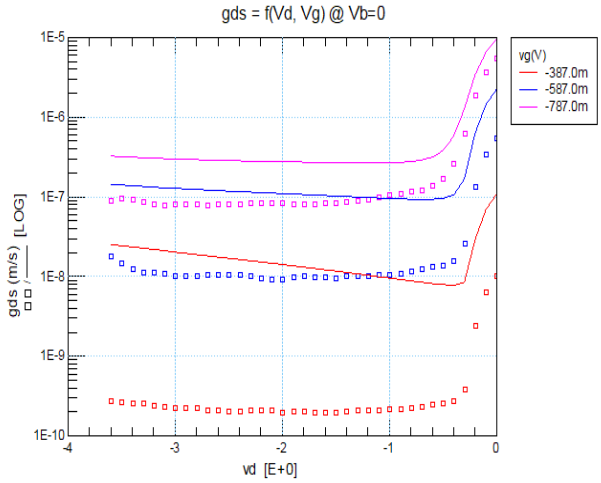




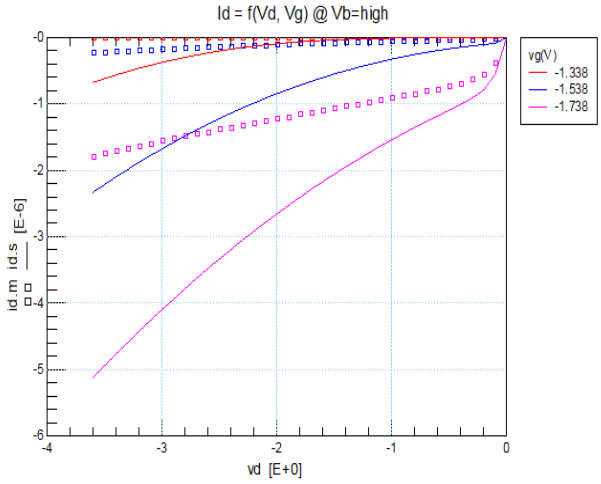
idvd_vth, Vb = 0V, T = 70°C

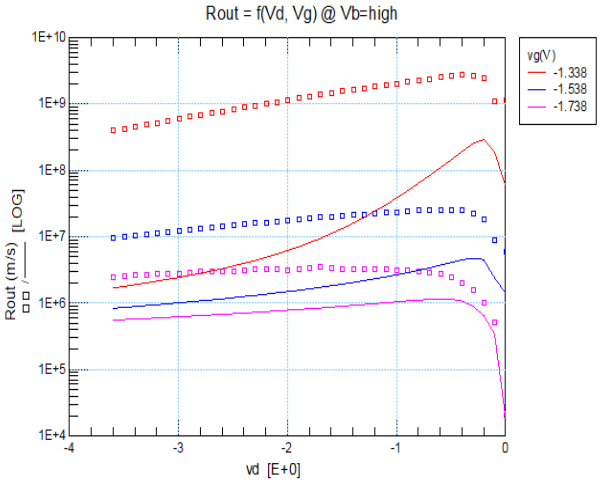


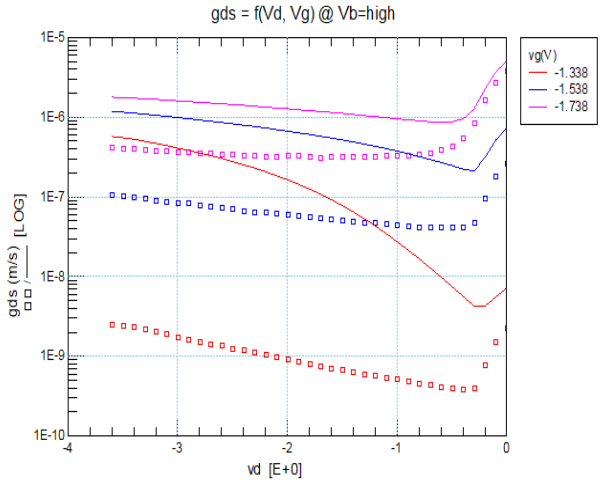




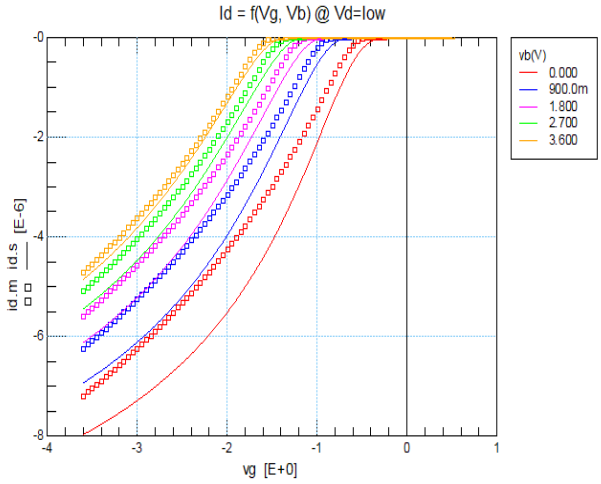
idvd_vth_vbmin, Vb = 3.6V, T = 70°C

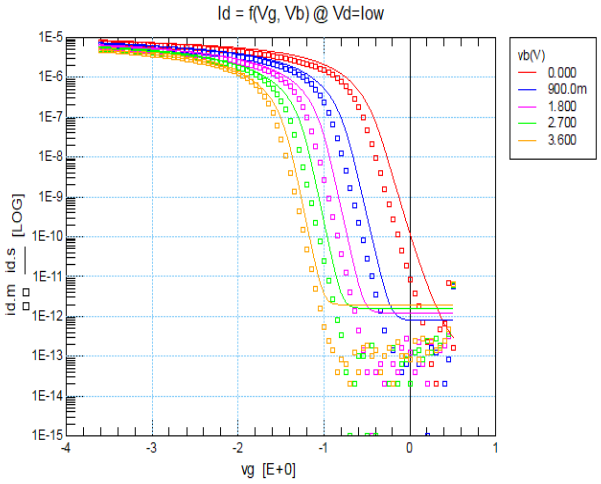


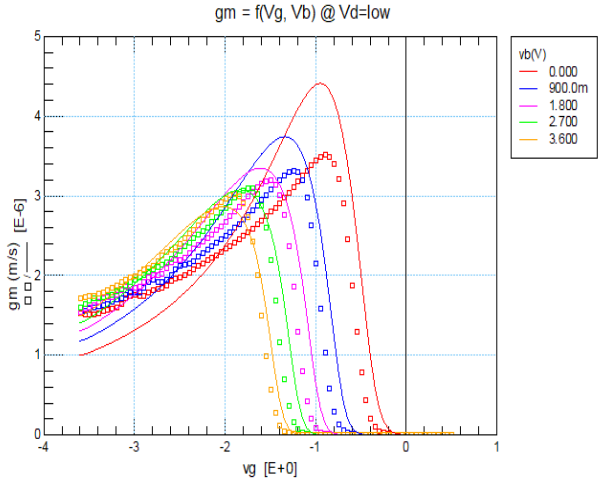


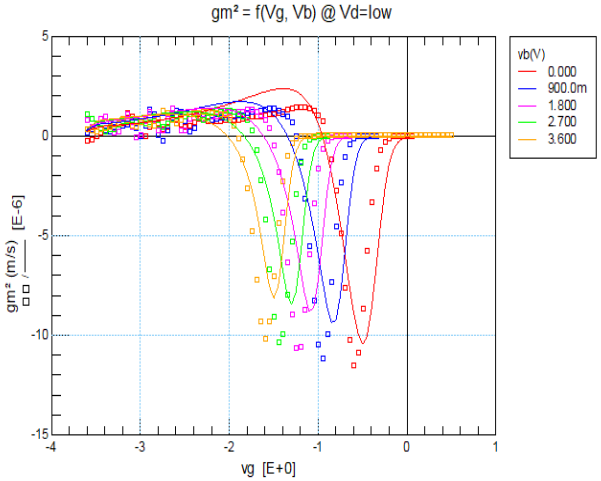


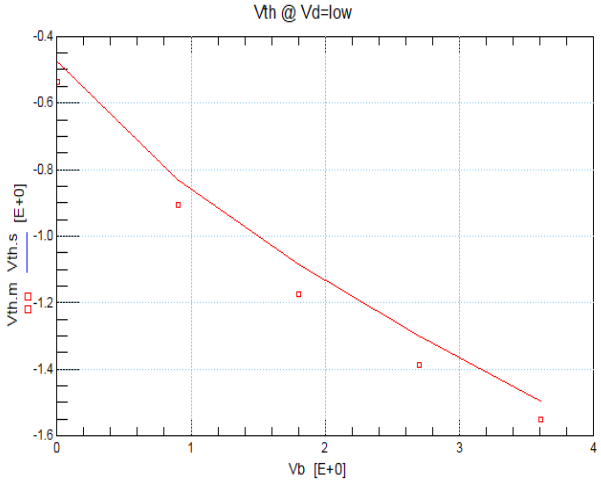
idvg, Vd = -0.1V, T = 125°C

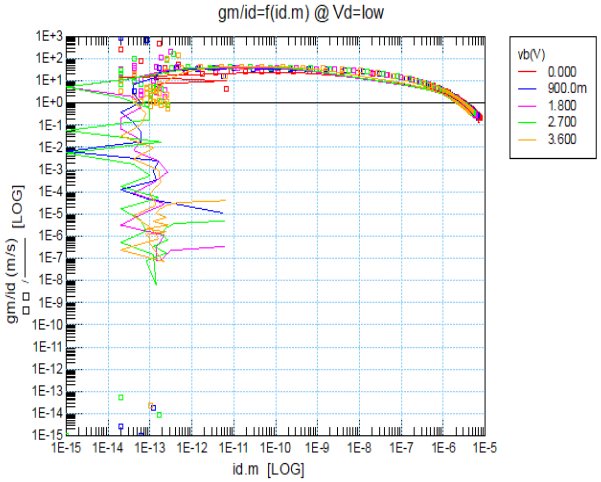


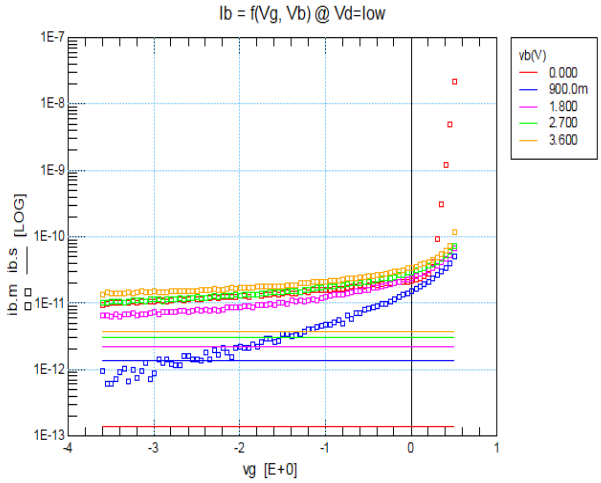


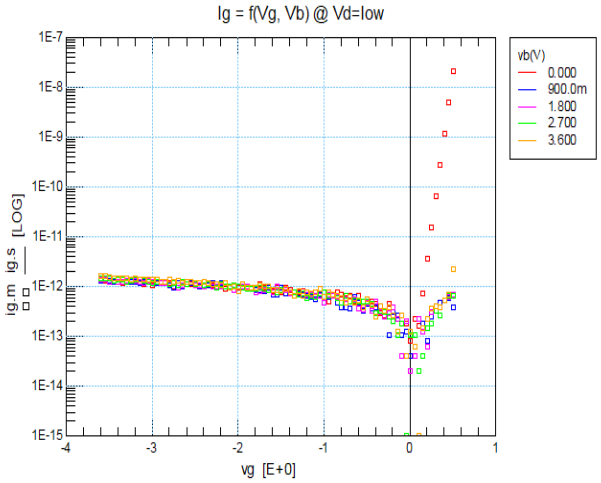




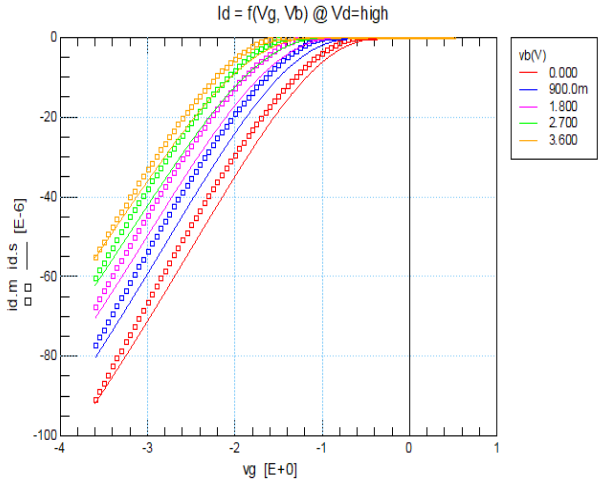


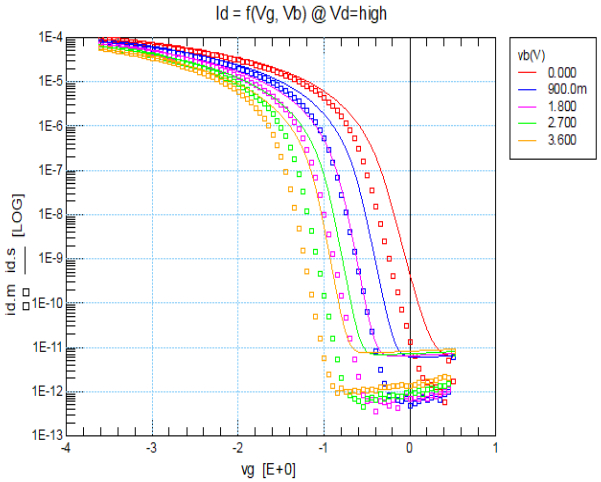


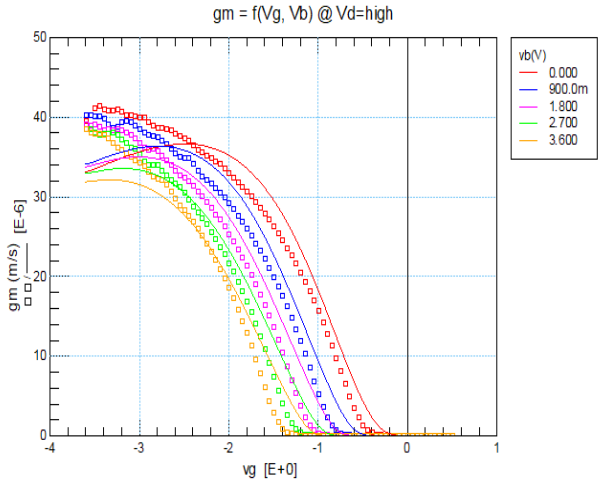


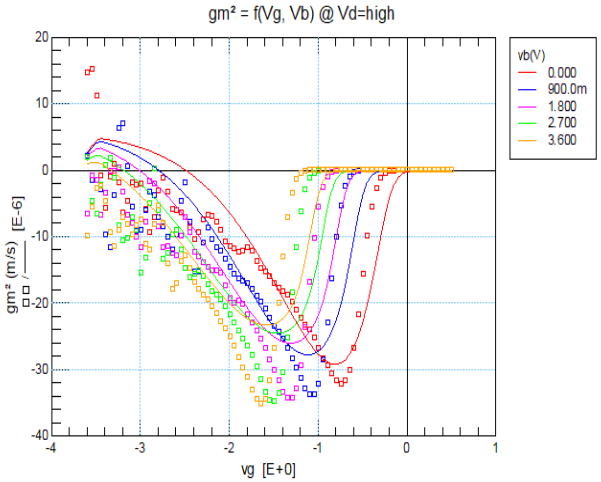


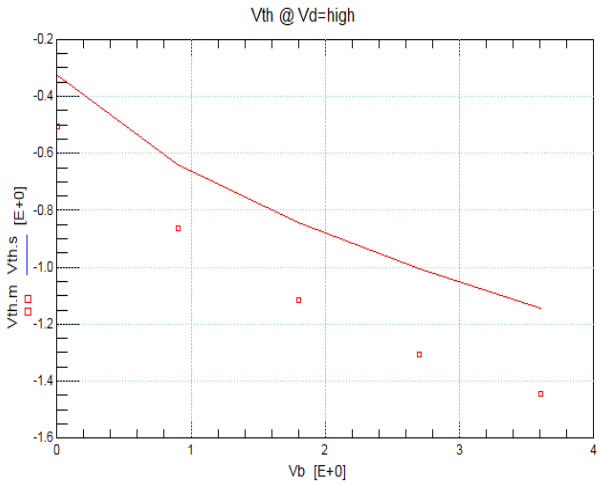
idvg, Vd = -3.3V, T = 125°C

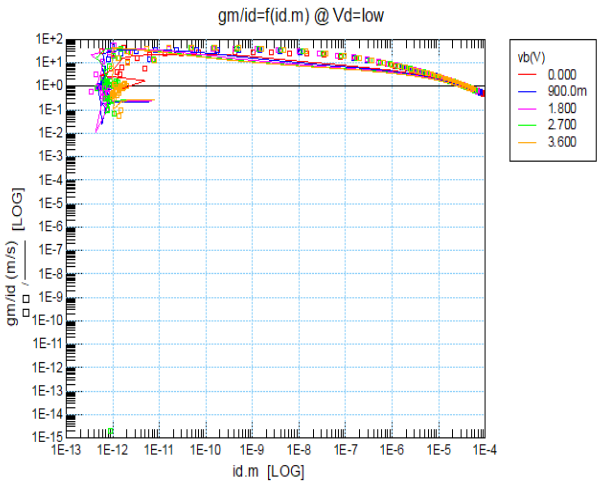


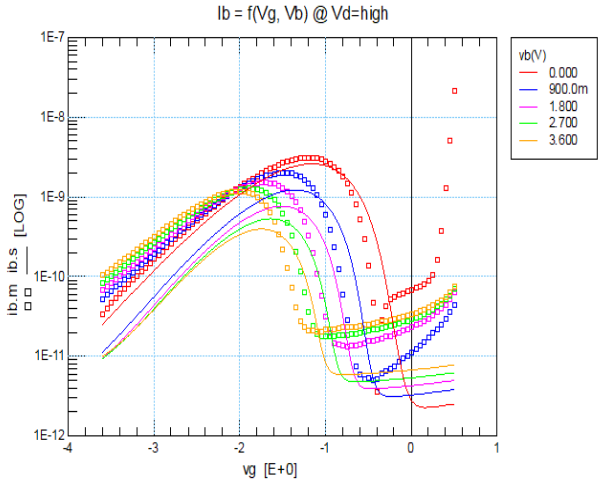


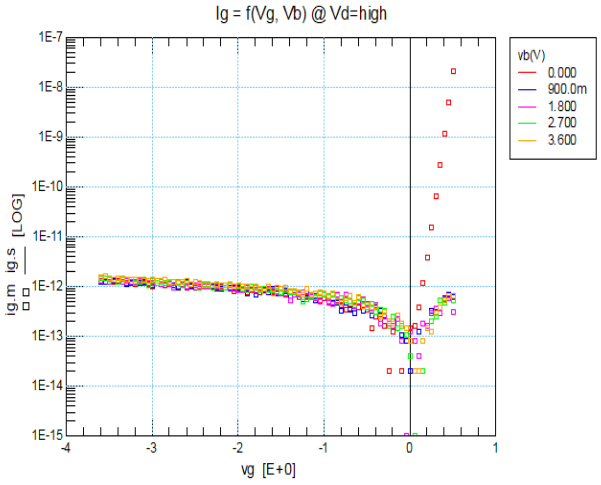




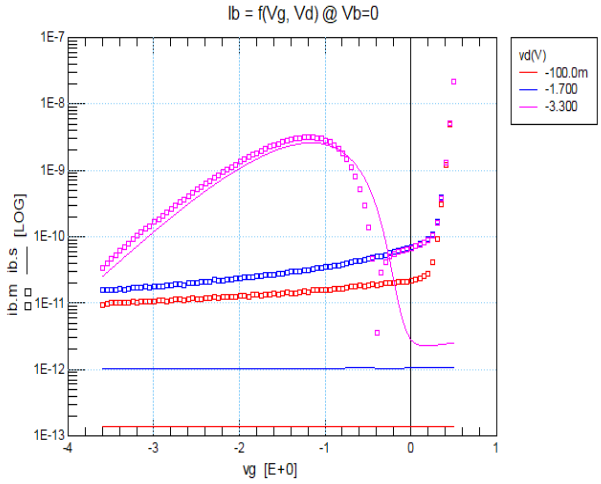


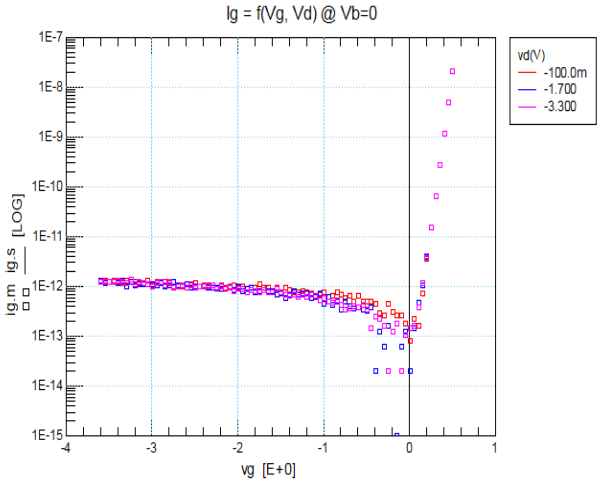




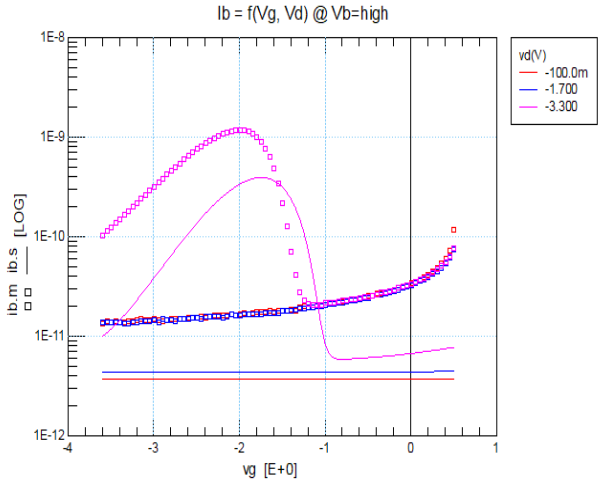


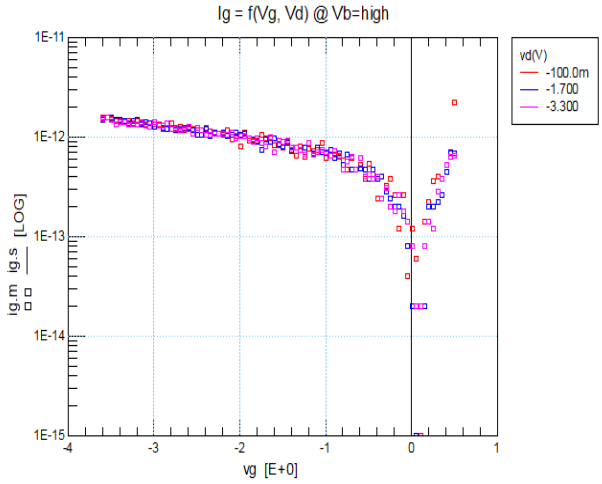
idvg, Ib,Ig, Vb = 0V, T = 125°C



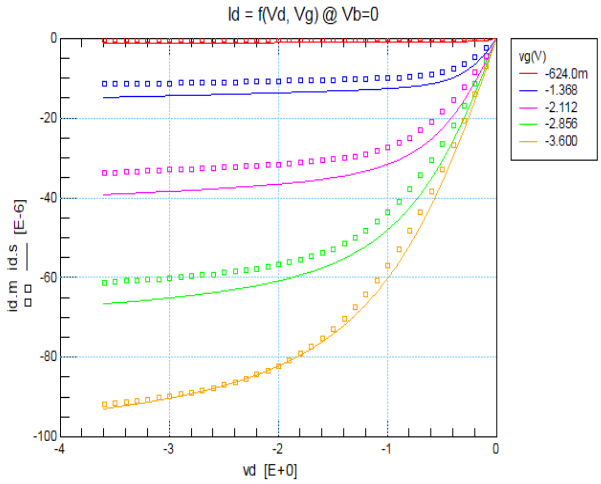


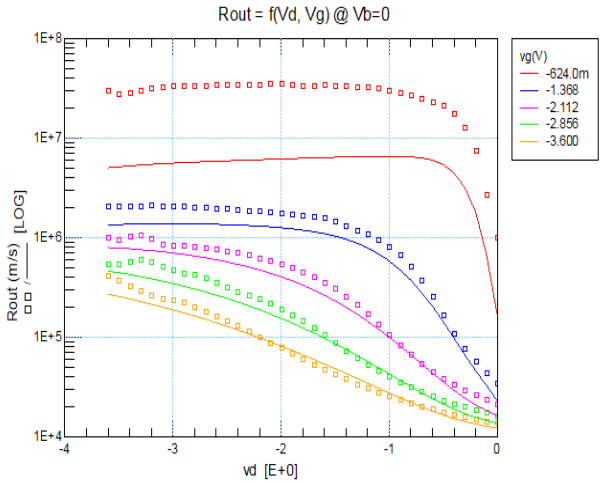
idvg, Ib,Ig, Vb = 3.6V, T = 125°C

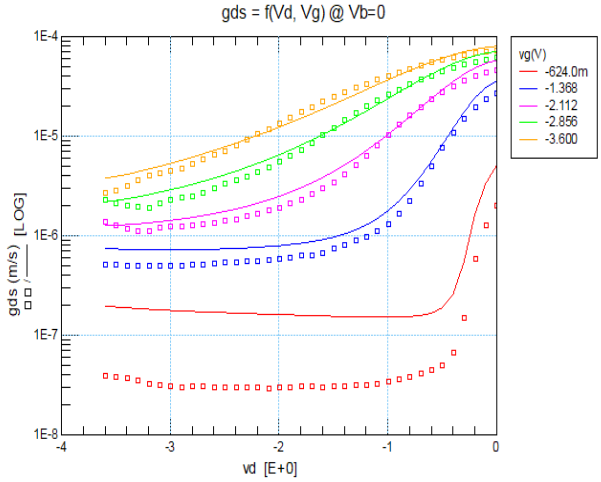




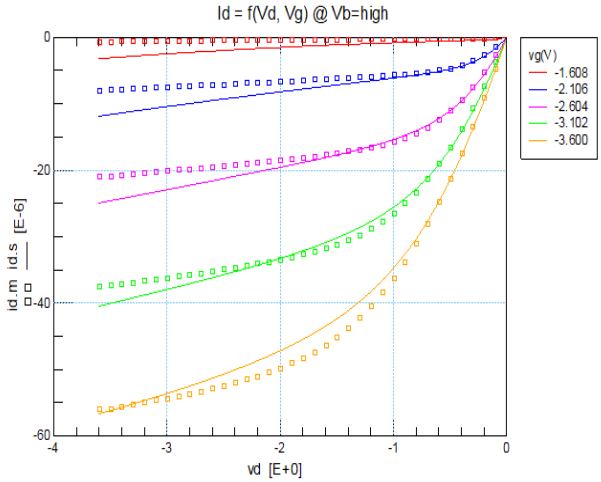
idvd, Vb = 0V, T = 125°C

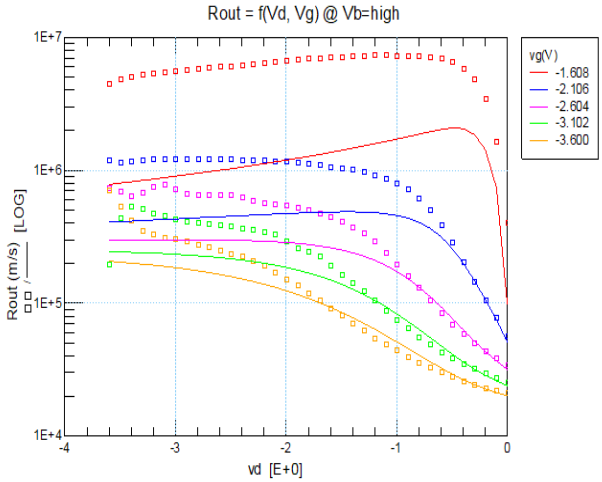


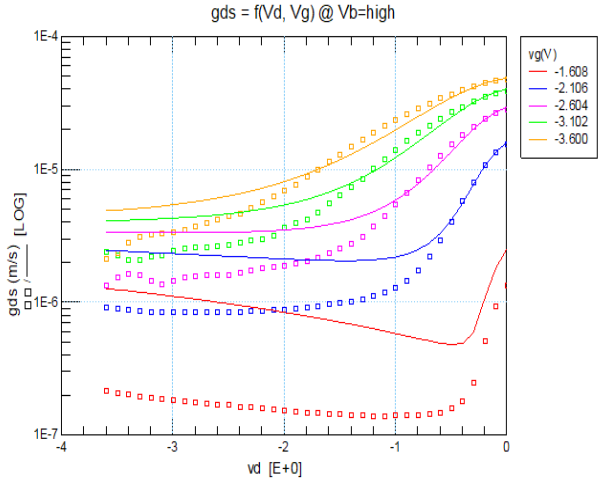




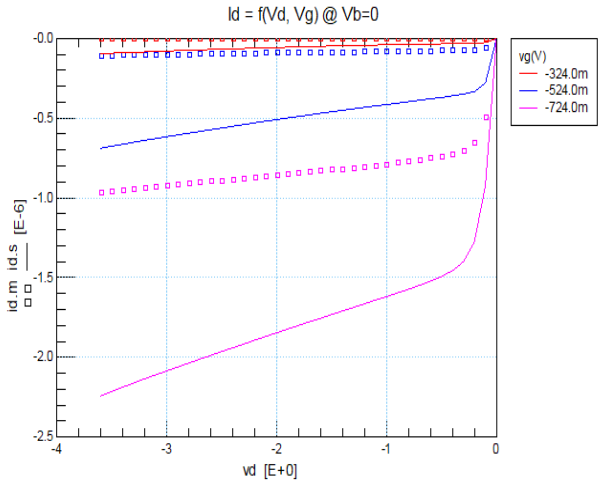
idvd_vbmin, Vb = 3.6V, T = 125°C

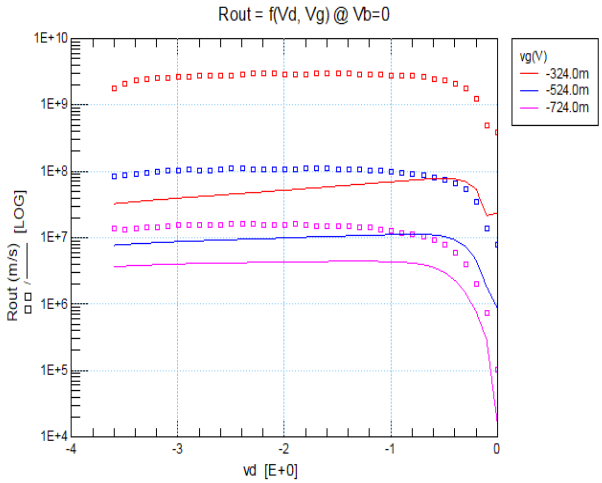


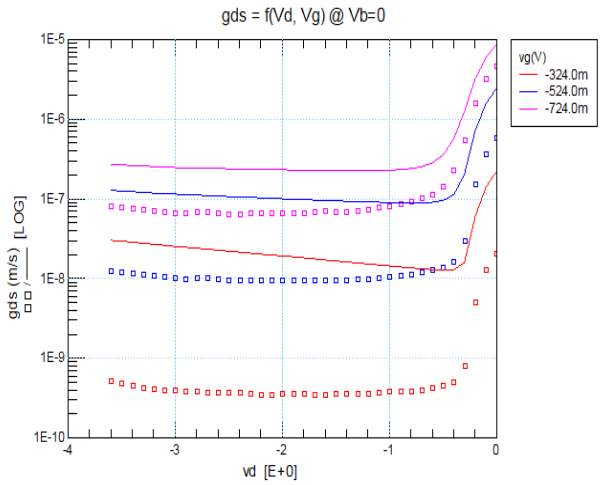




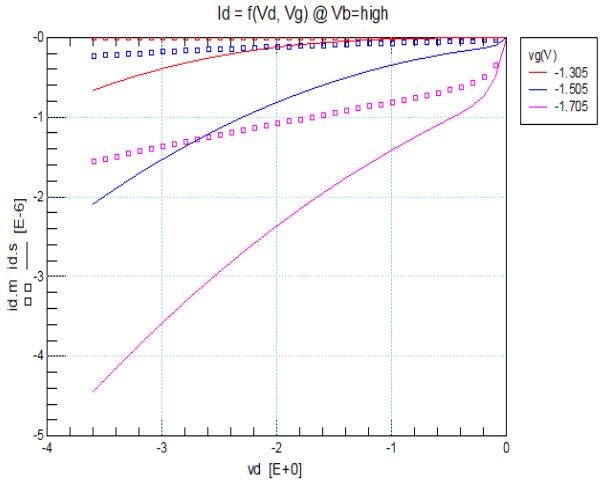
idvd_vth, Vb = 0V, T = 125°C

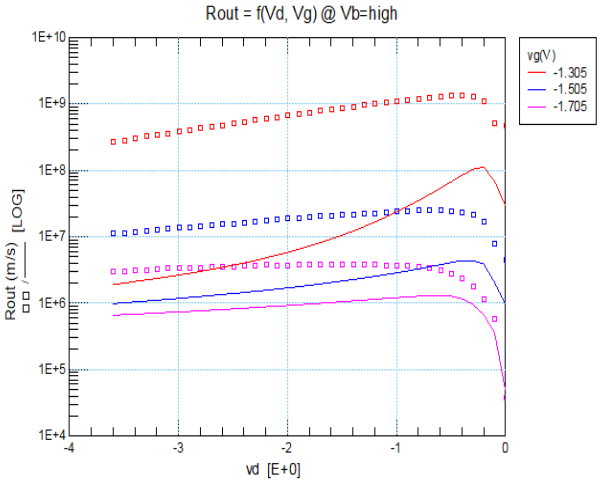


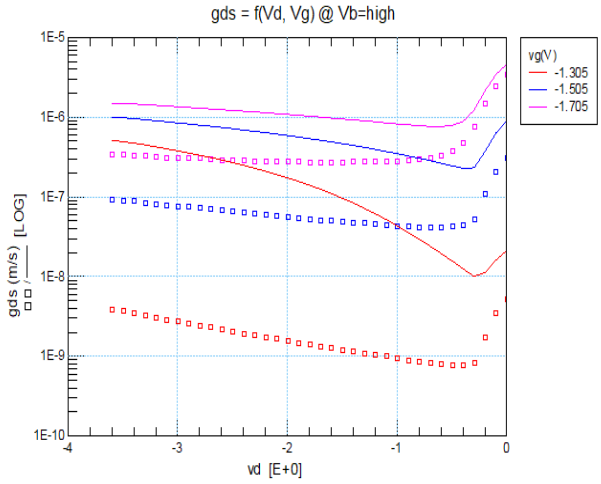




idvd_vth_vbmin, Vb = 3.6V, T = 125°C

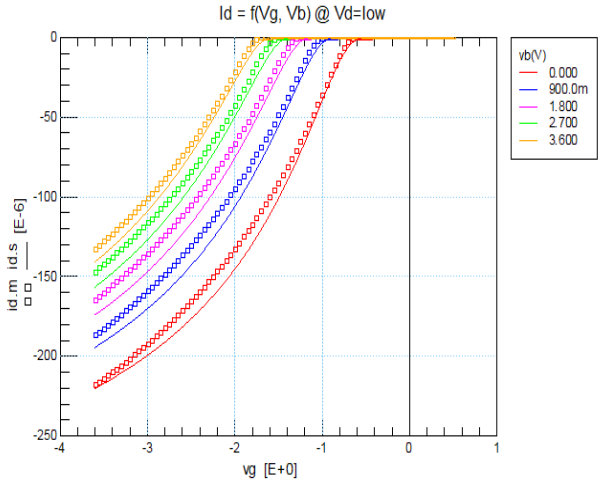


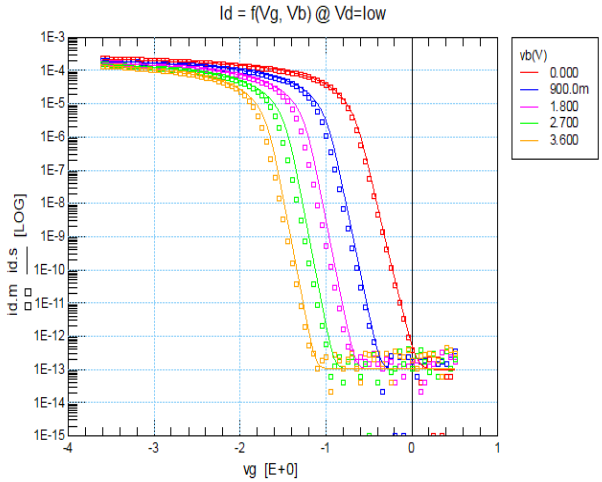


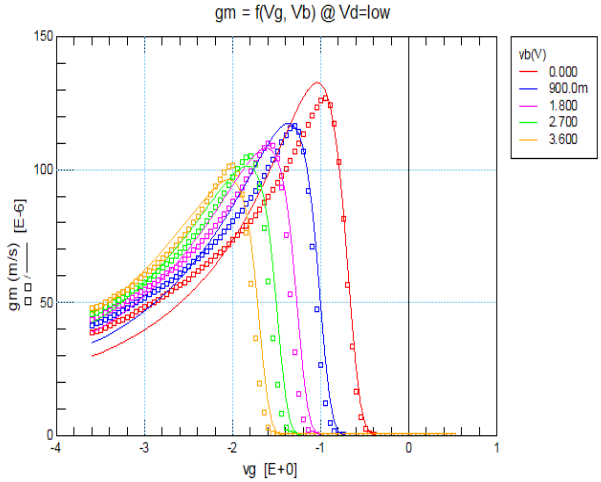


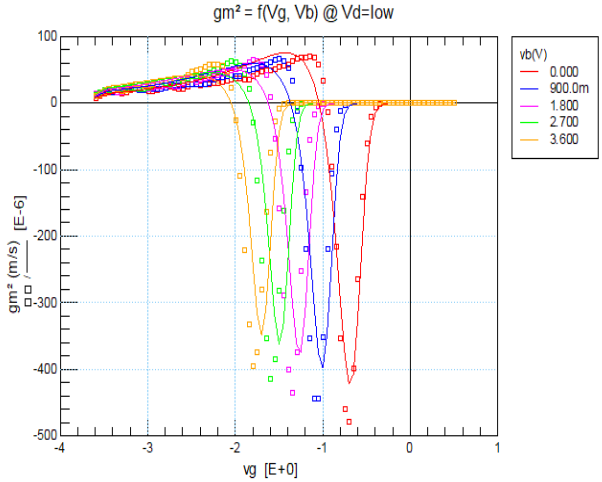
4.6 W10u0_L0u4_T356_S387_1

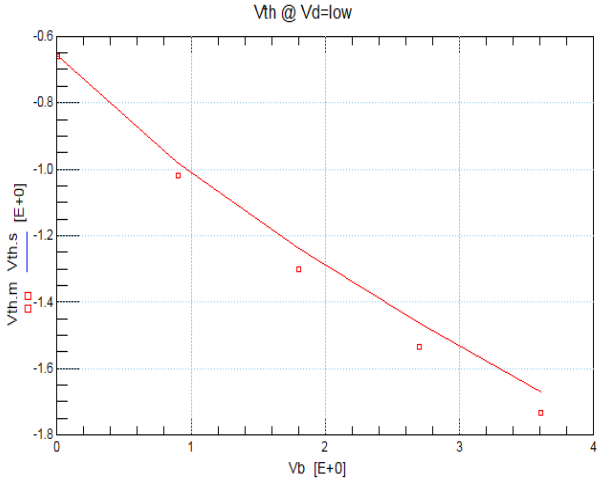
idvg, Vd = -0.1V, T = 27°C

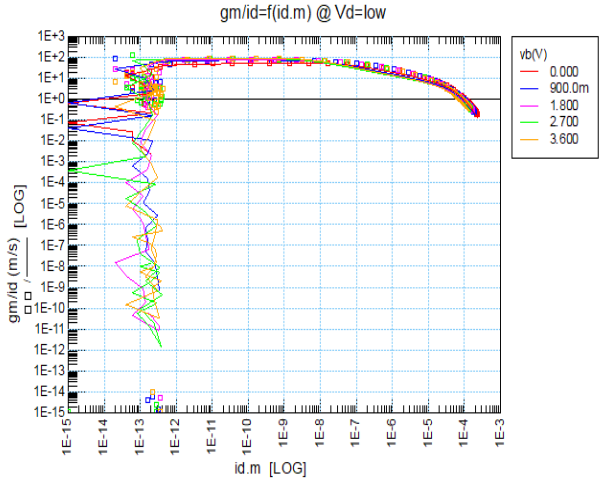


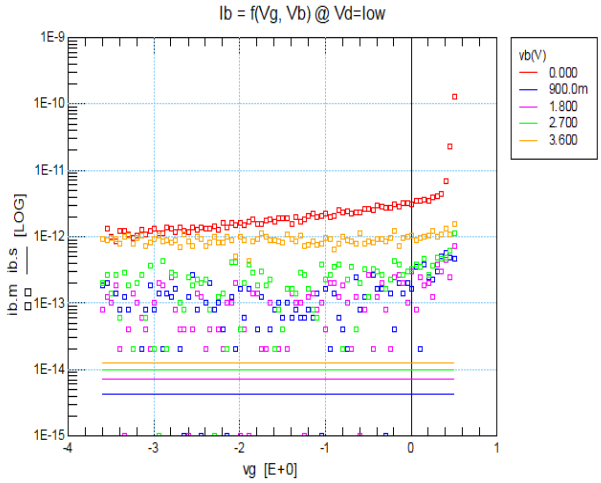


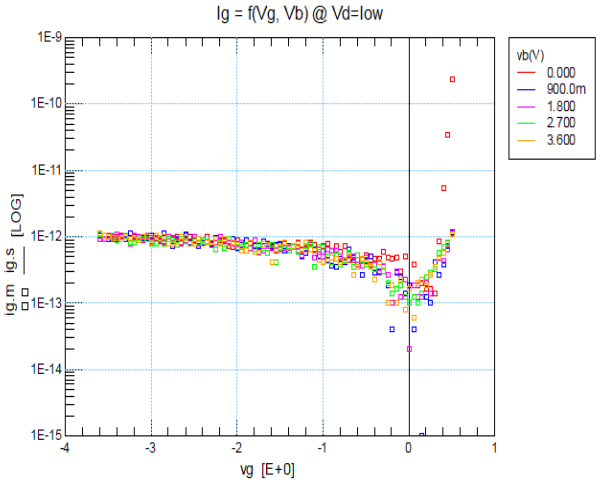




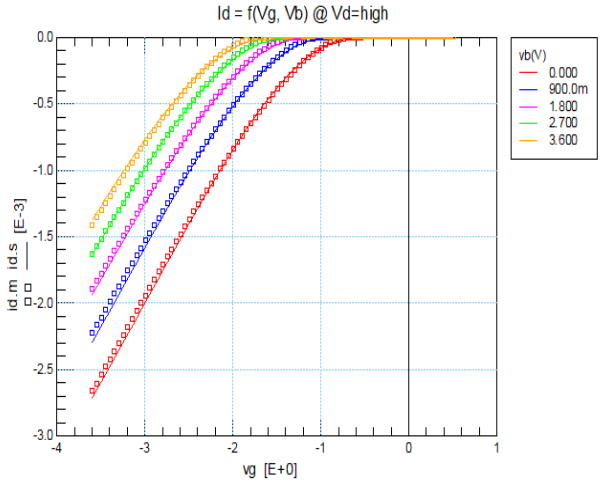


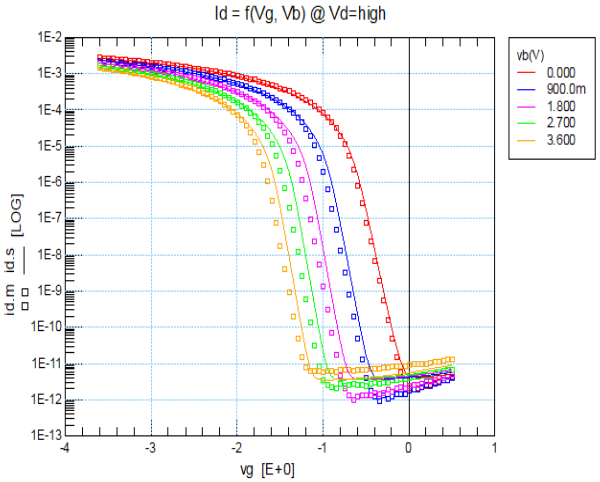


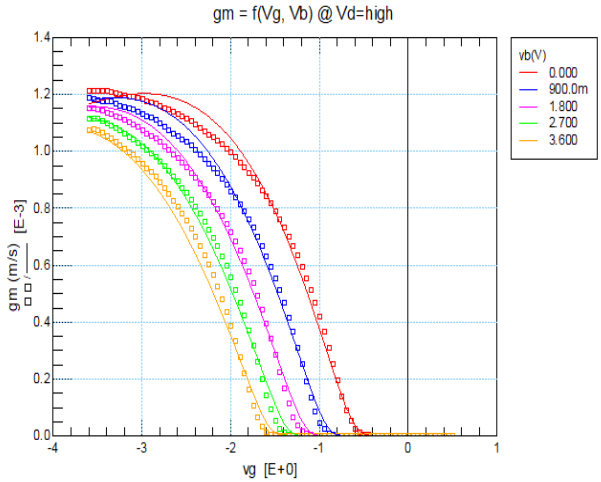


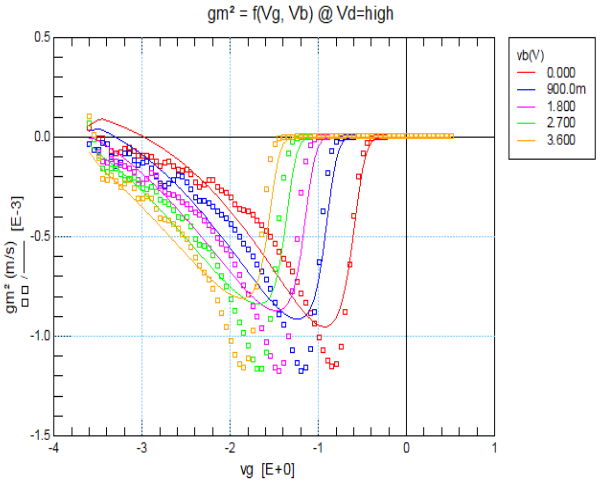


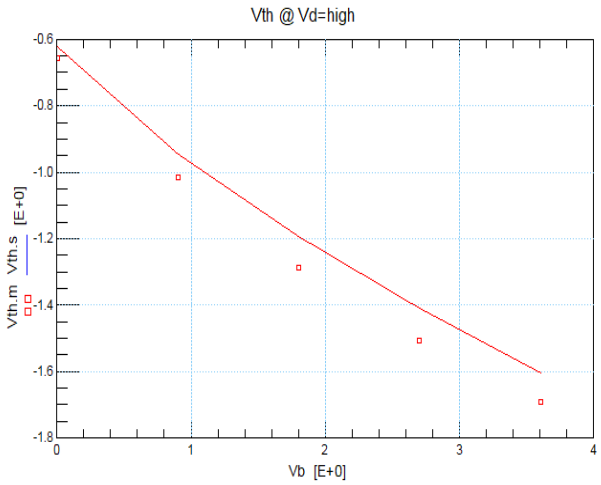
idvg, Vd = -3.3V, T = 27°C

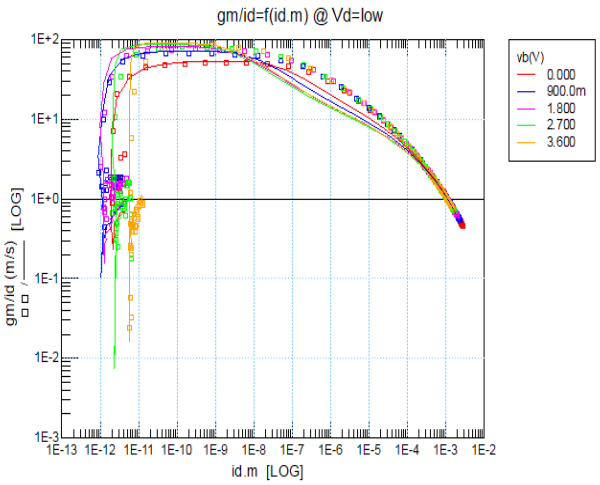


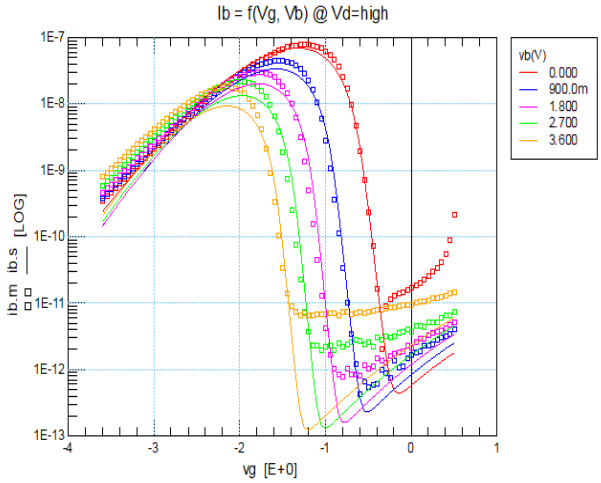


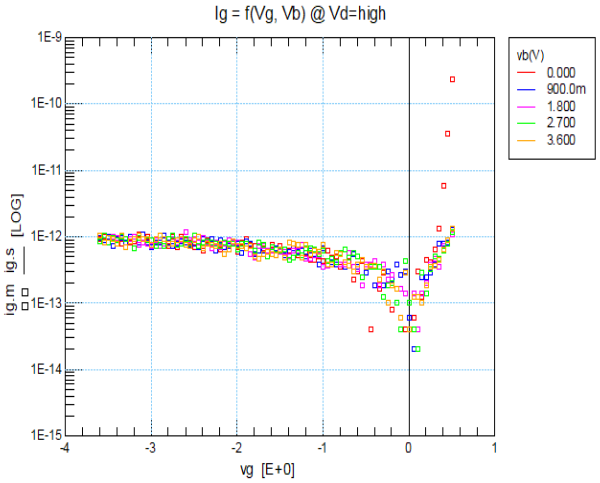




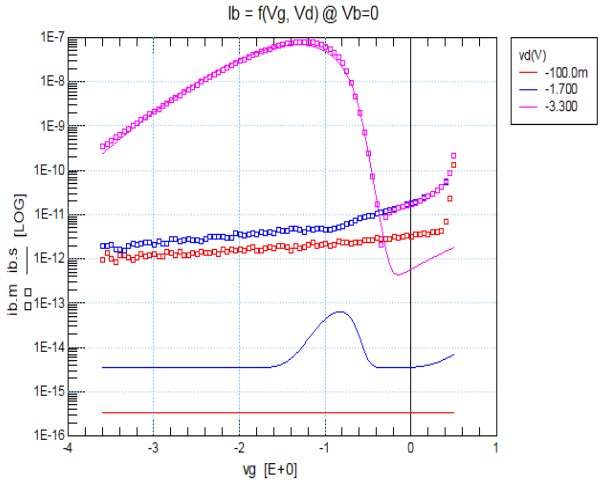


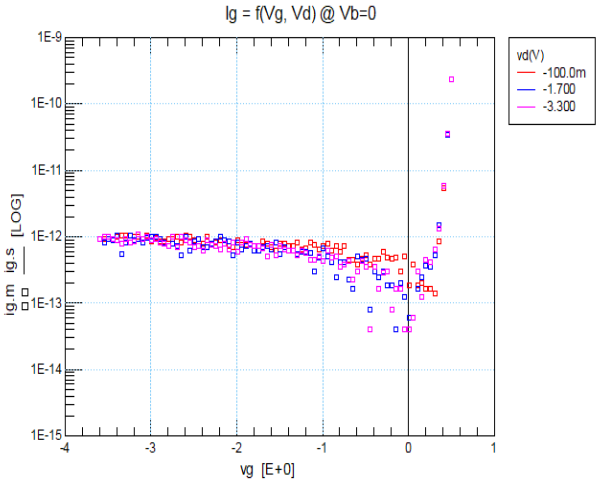




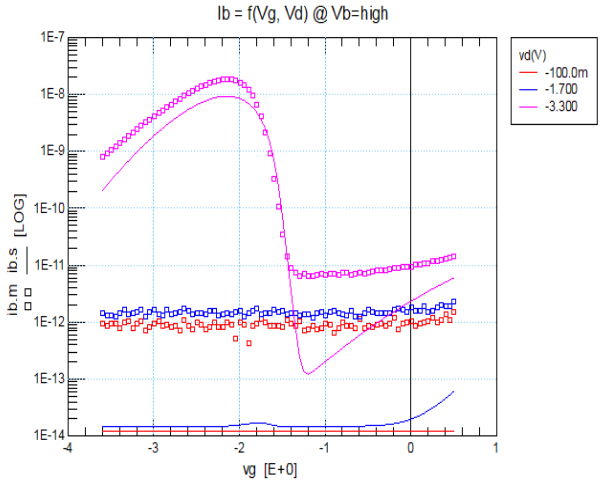


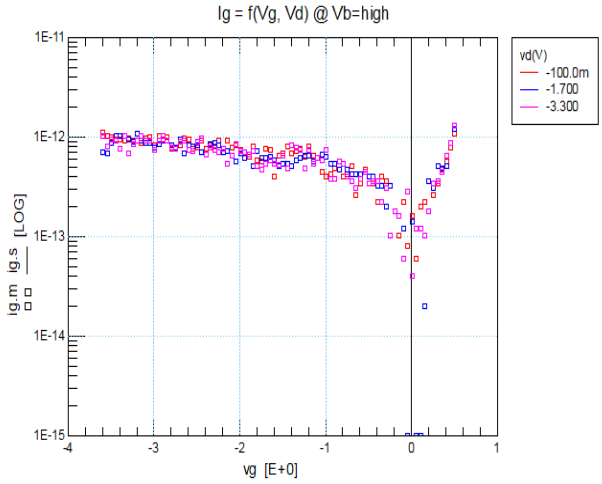
idvg, Ib,Ig, Vb = 0V, T = 27°C



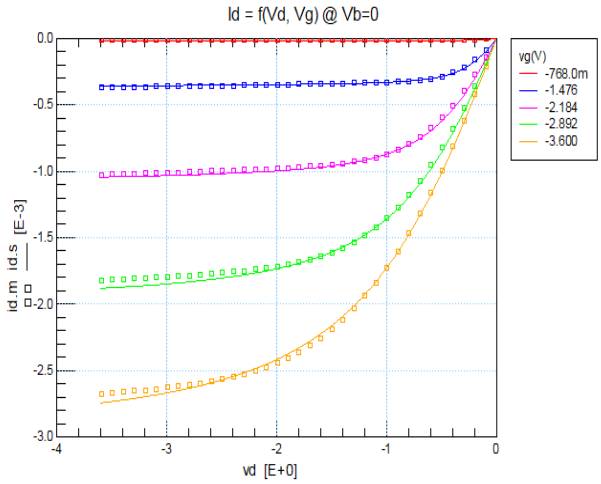


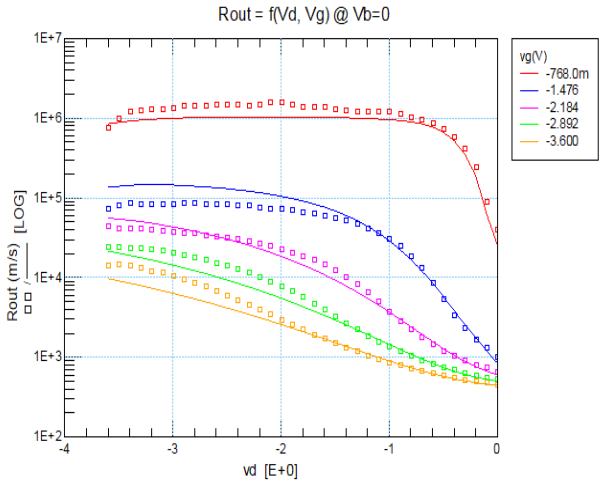
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

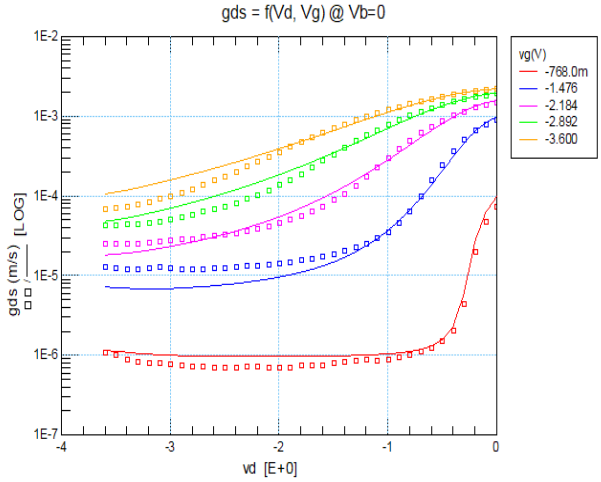




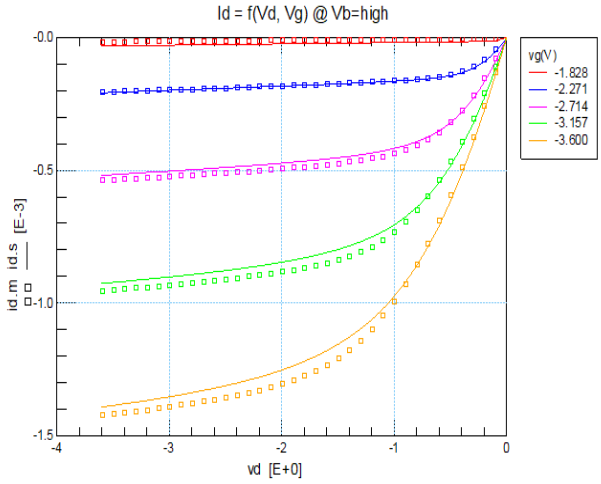
idvd, Vb = 0V, T = 27°C

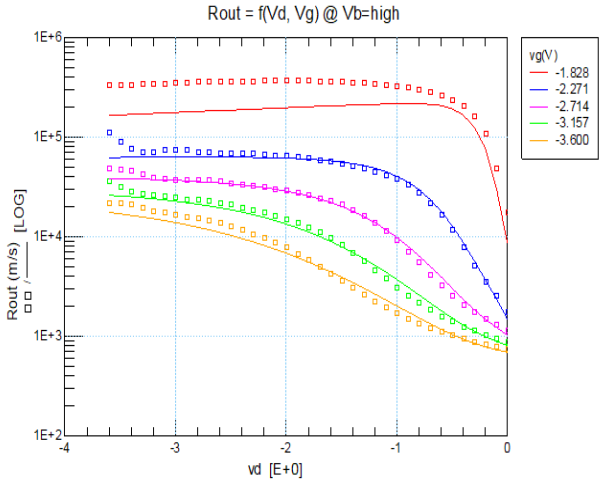


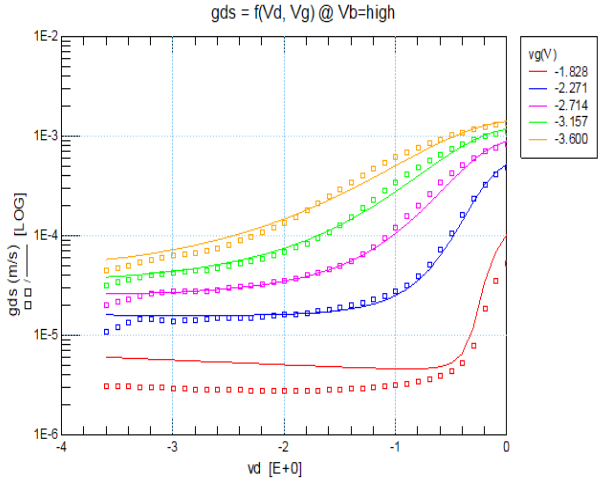




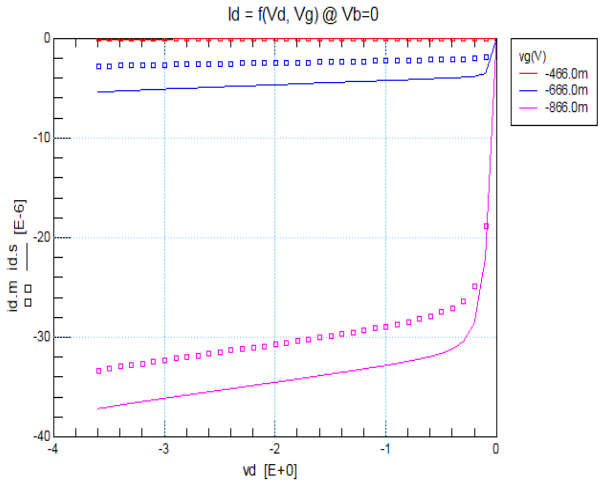
idvd_vbmin, Vb = 3.6V, T = 27°C

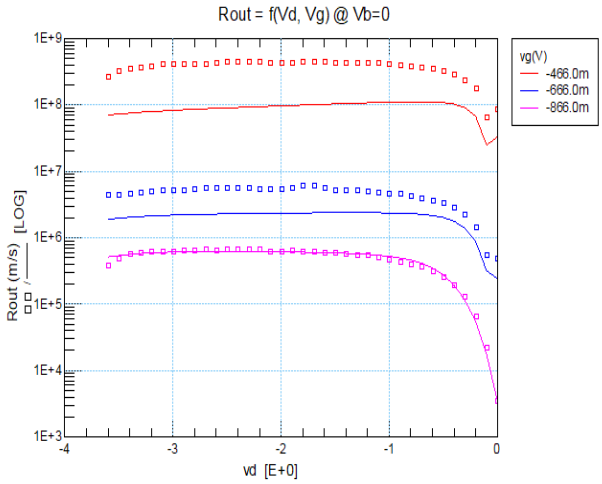


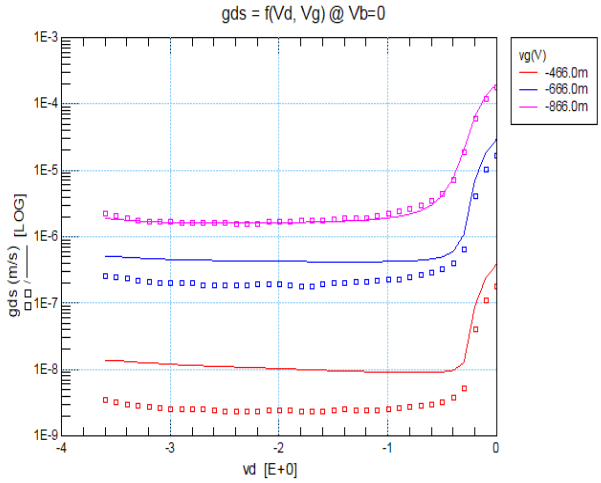




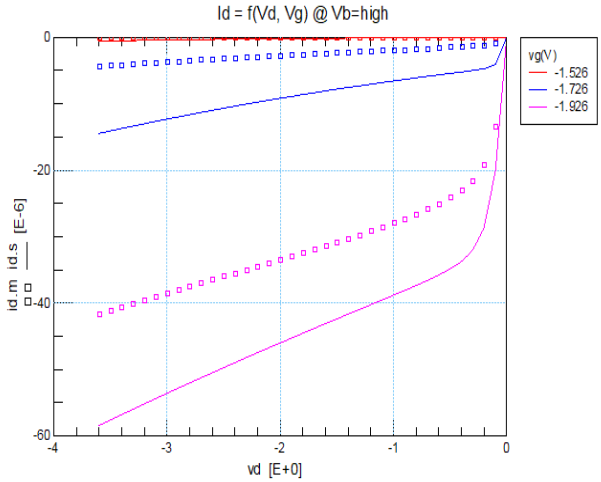
idvd_vth, Vb = 0V, T = 27°C

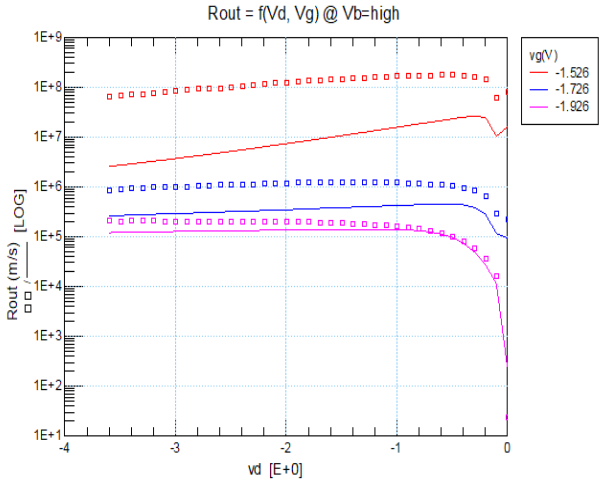


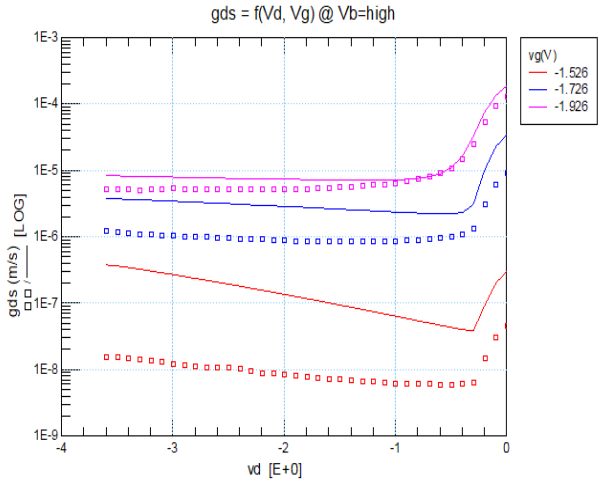




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

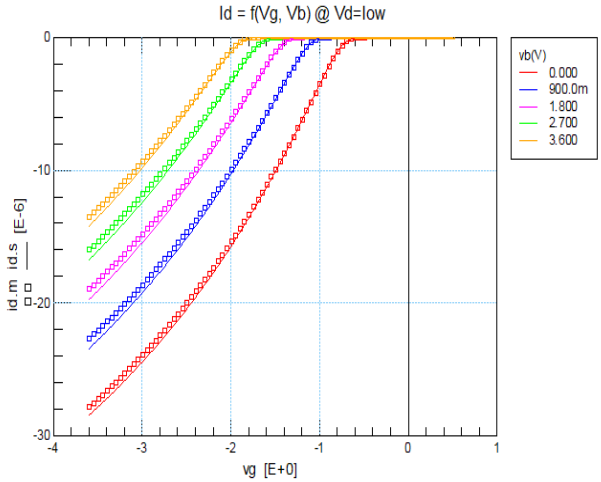


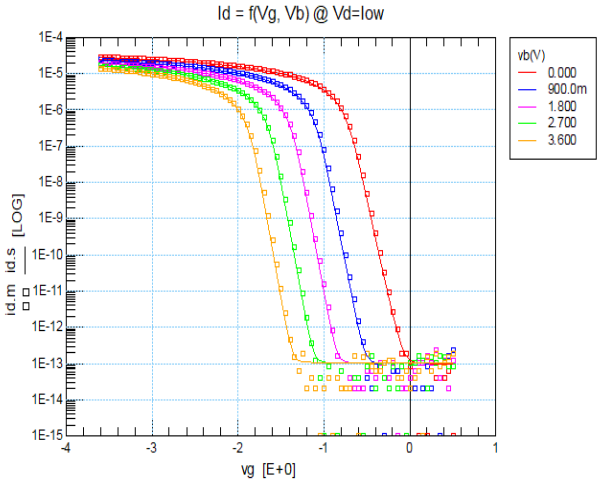


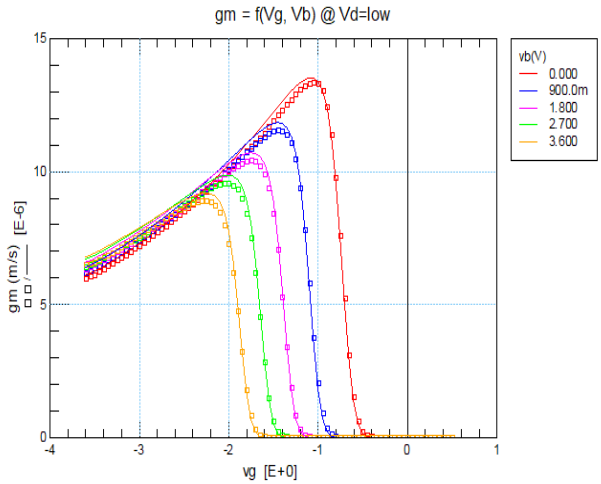


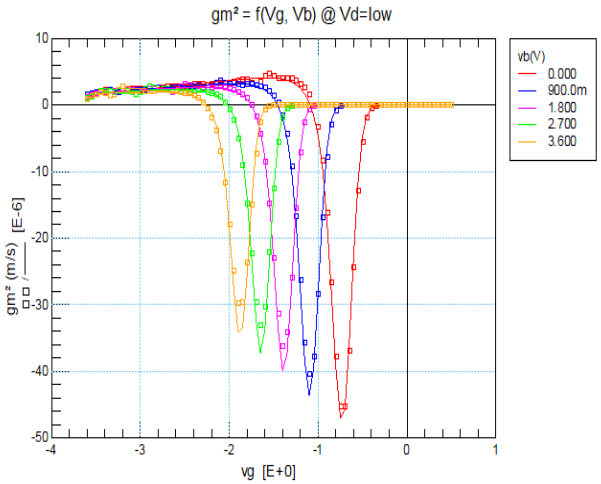
4.7 W10u0_L3u0_S562_1

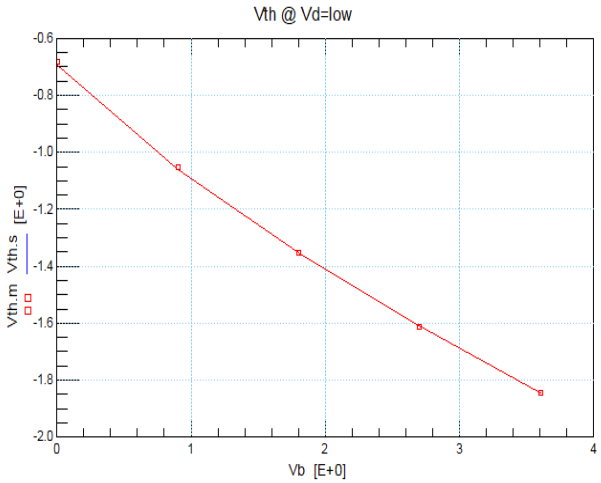
idvg, Vd = -0.1V, T = 27°C

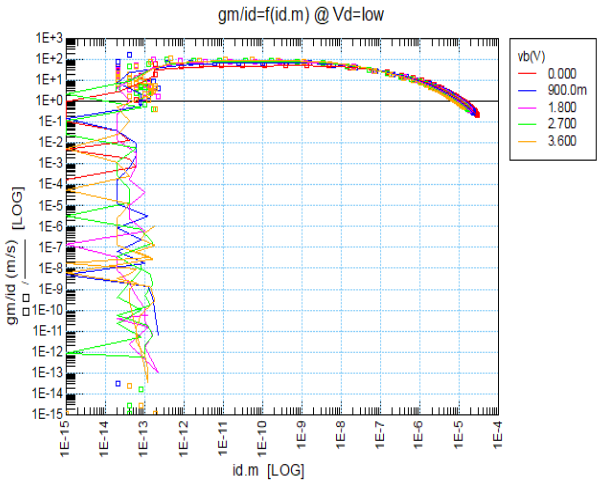


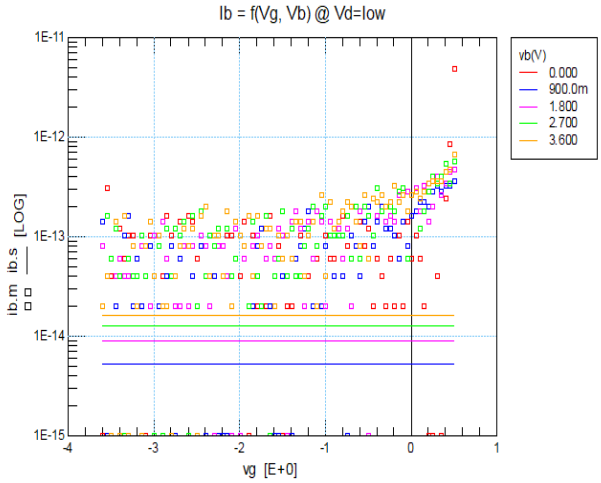


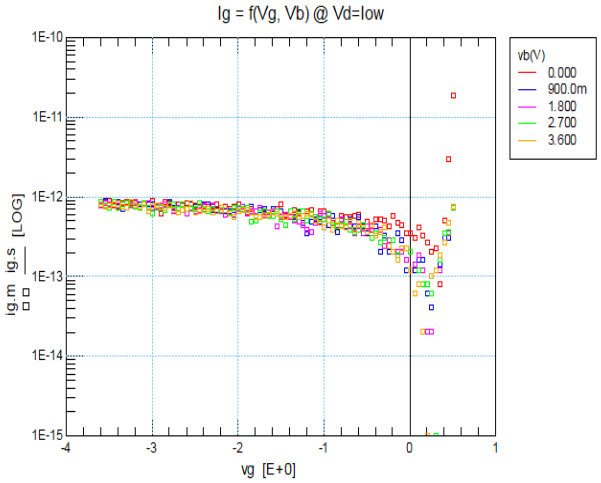




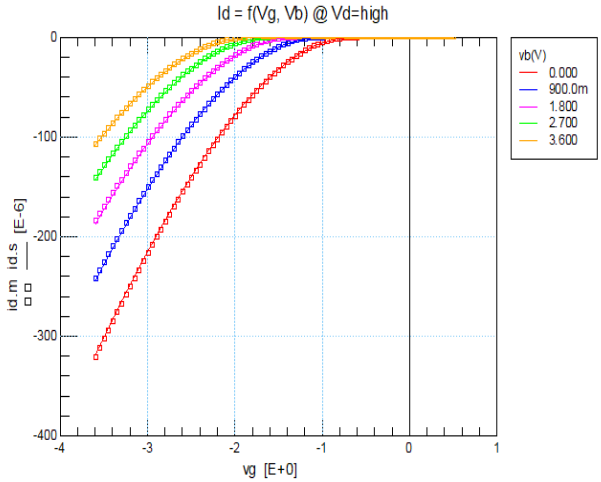


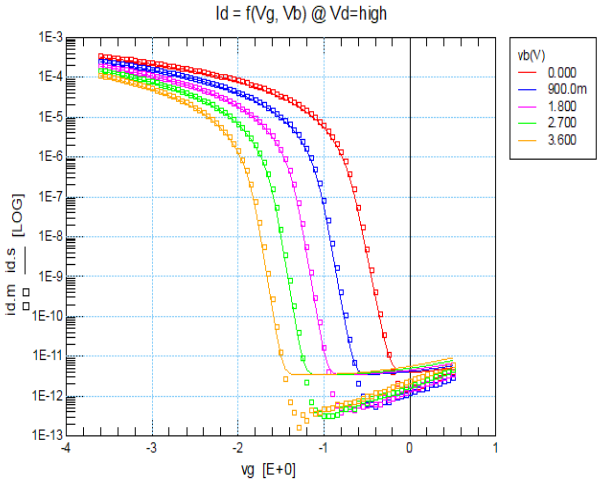


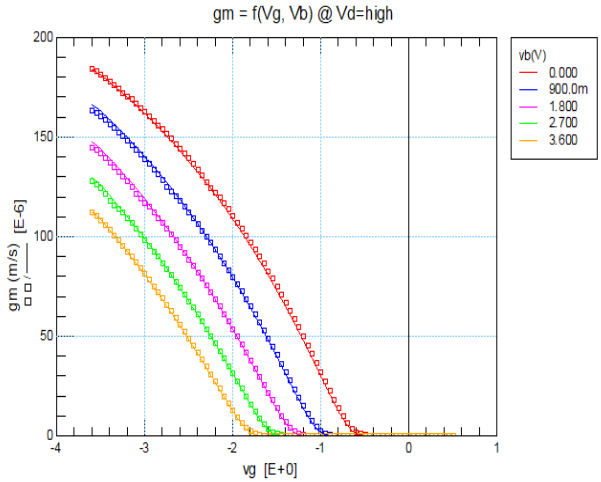


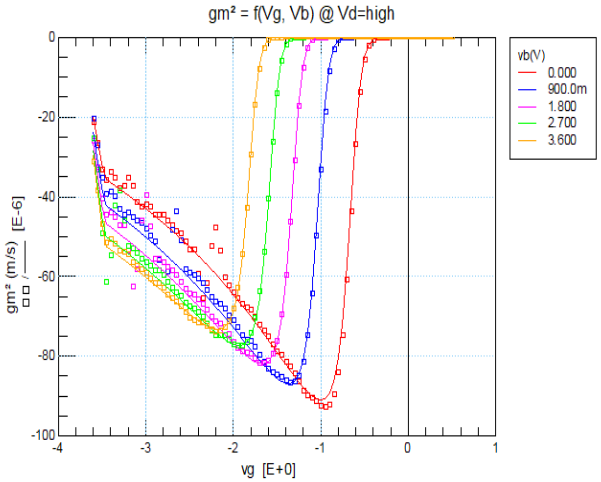


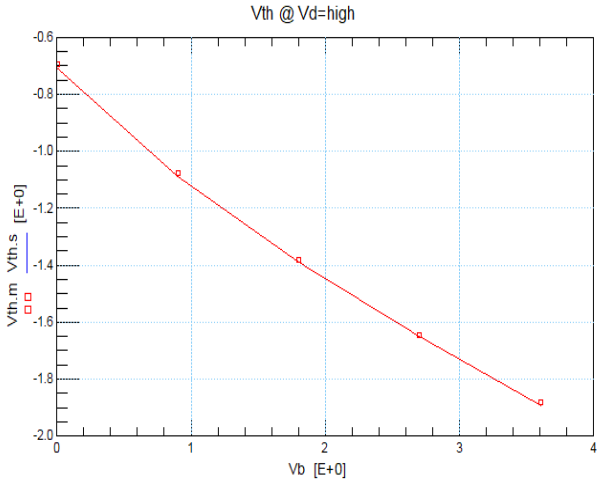
idvg, Vd = -3.3V, T = 27°C

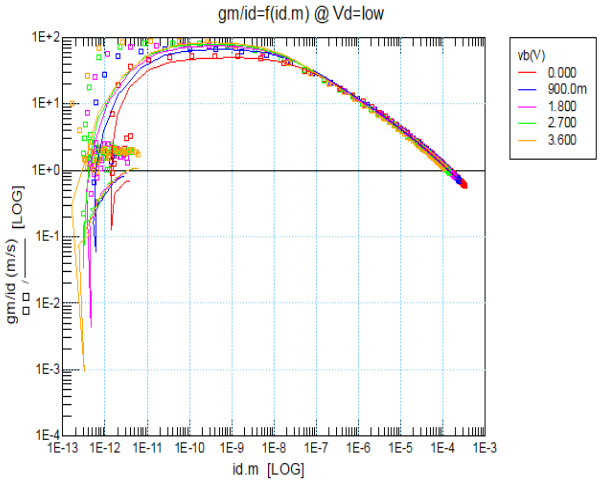


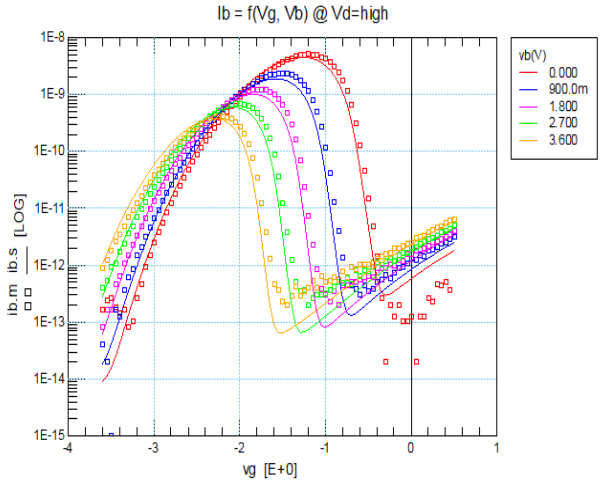


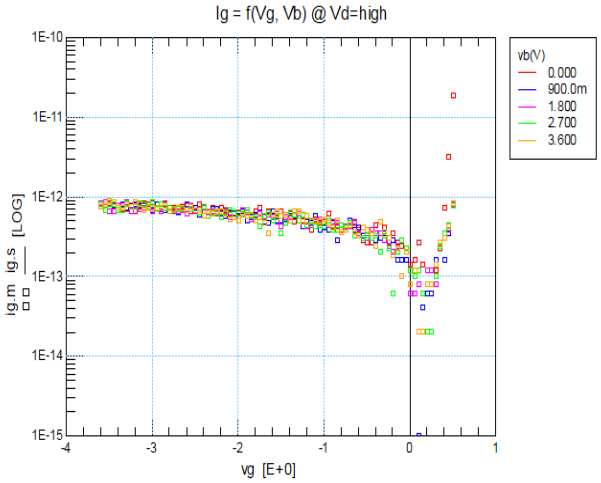




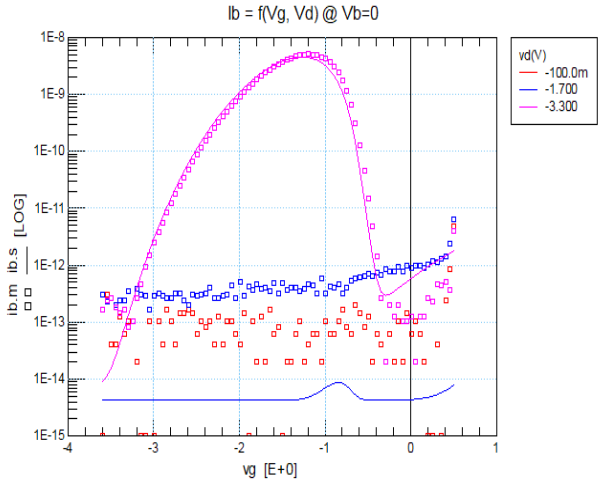


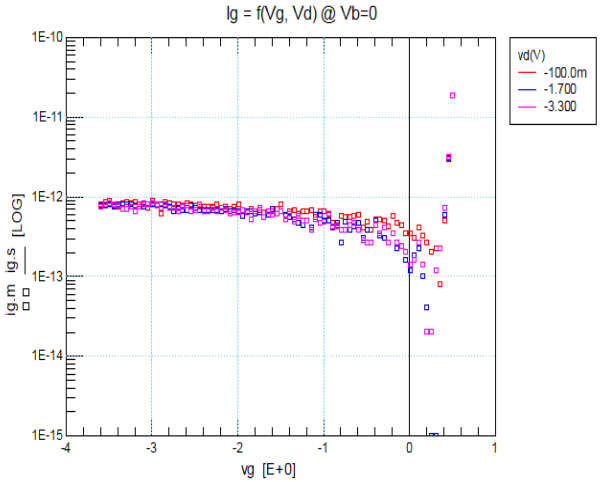




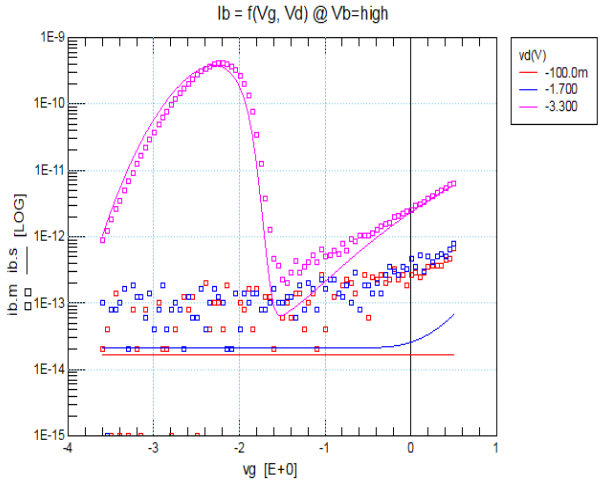


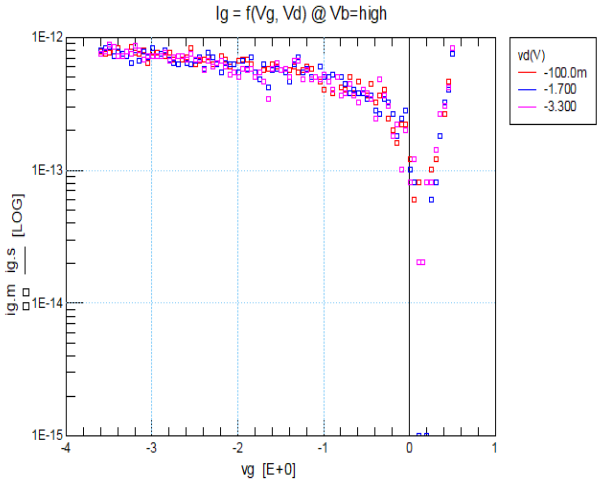
idvg, Ib,Ig, Vb = 0V, T = 27°C



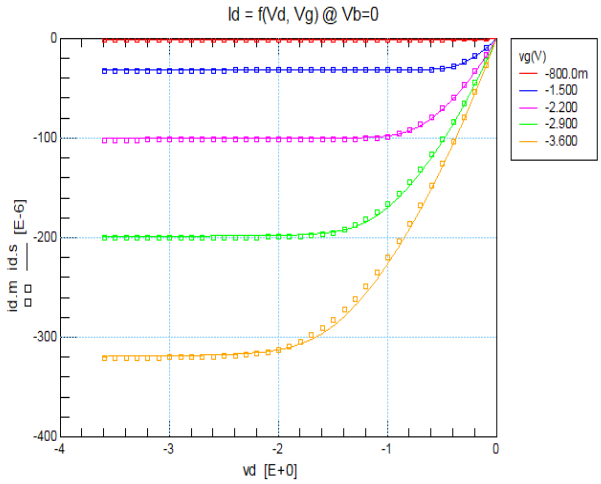


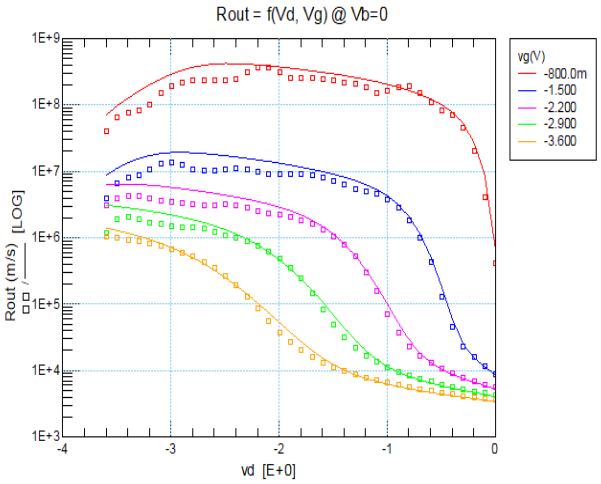
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

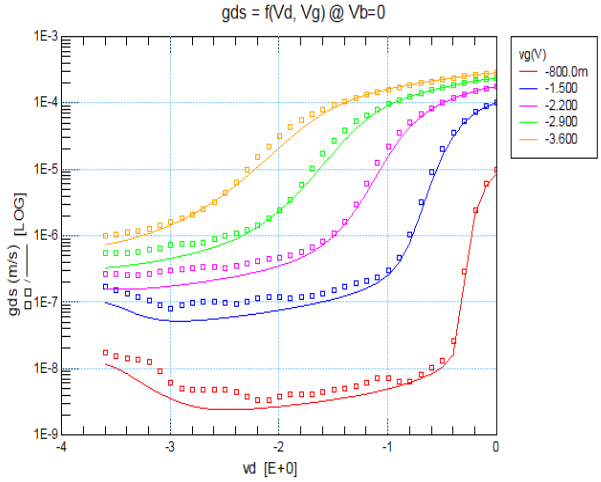




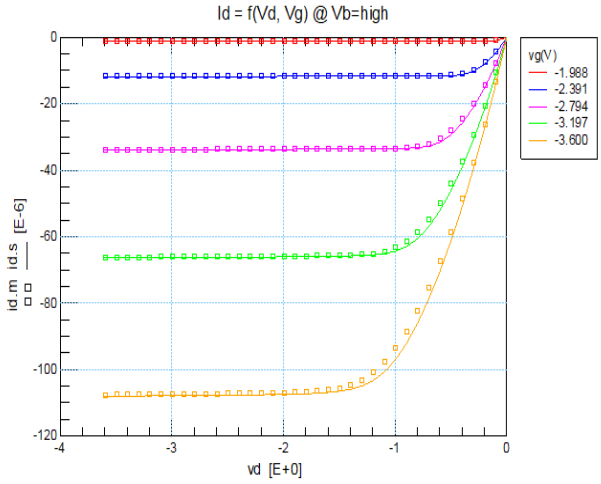
idvd, Vb = 0V, T = 27°C

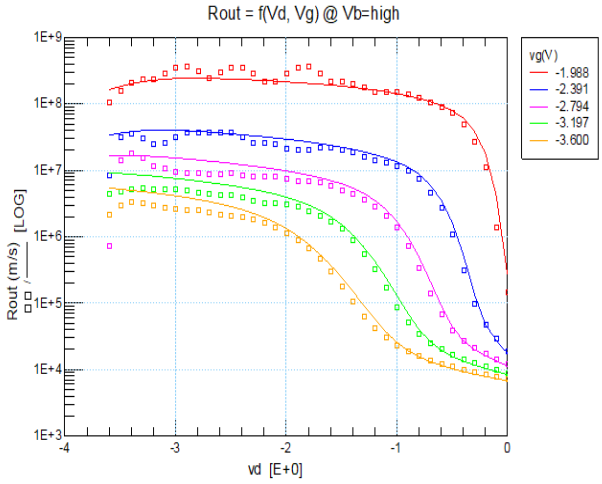


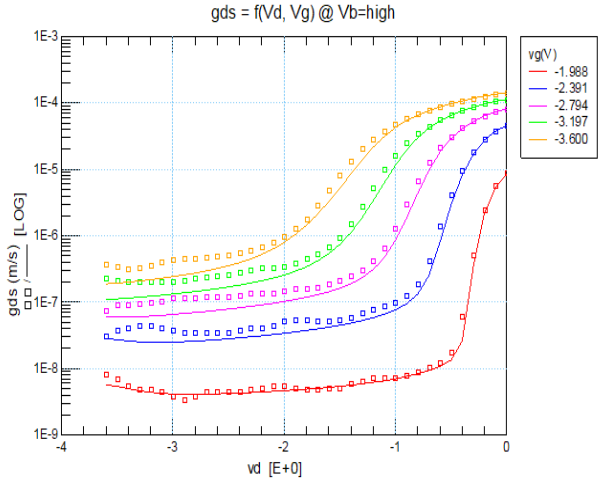




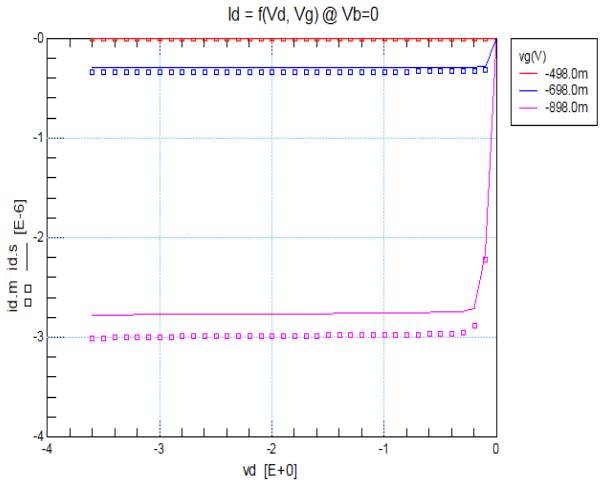
idvd_vbmin, Vb = 3.6V, T = 27°C

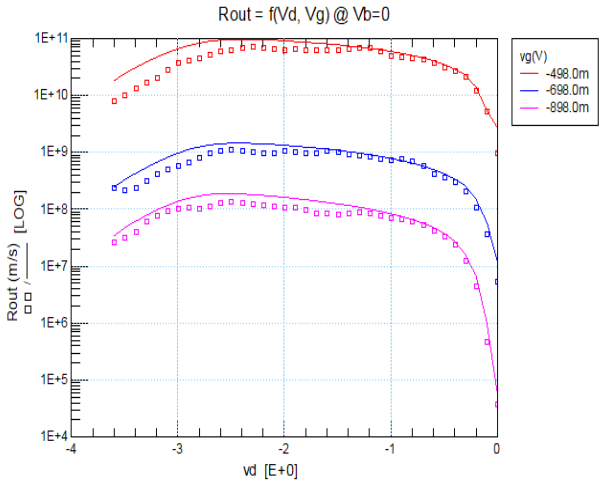


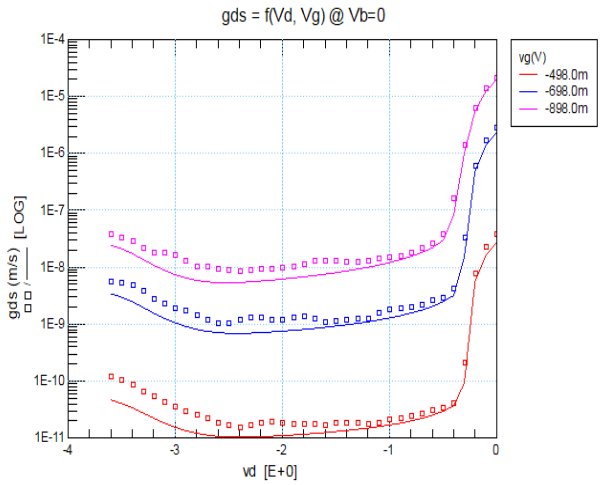




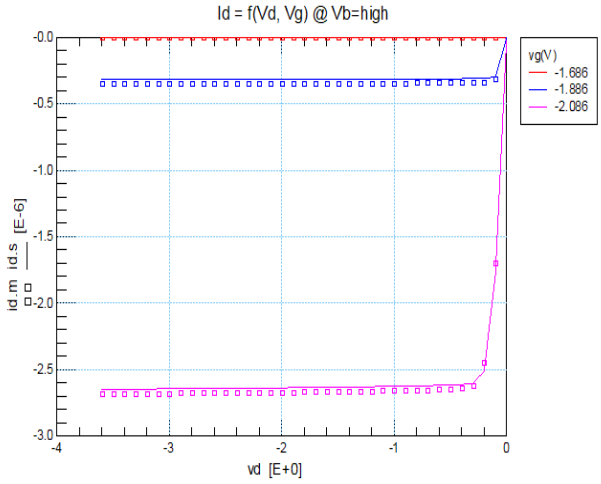
idvd_vth, Vb = 0V, T = 27°C

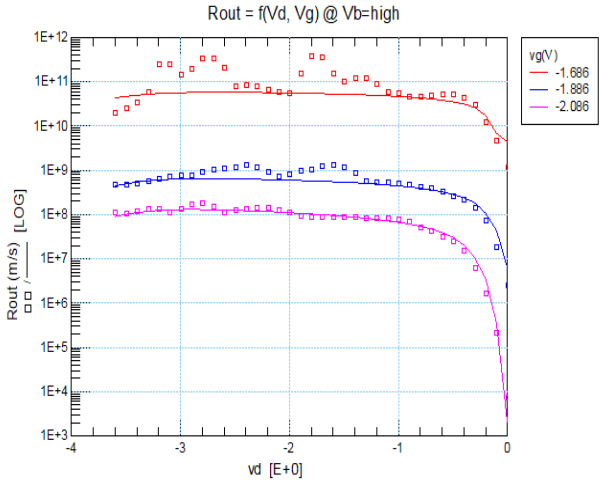


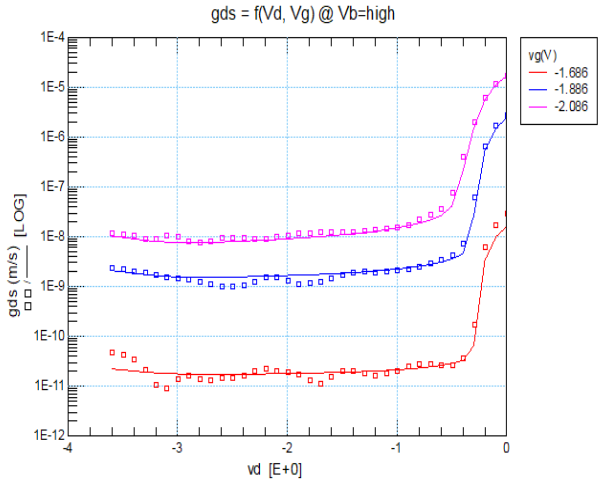




idvd_vth_vbmin, Vb = 3.6V, T = 27°C



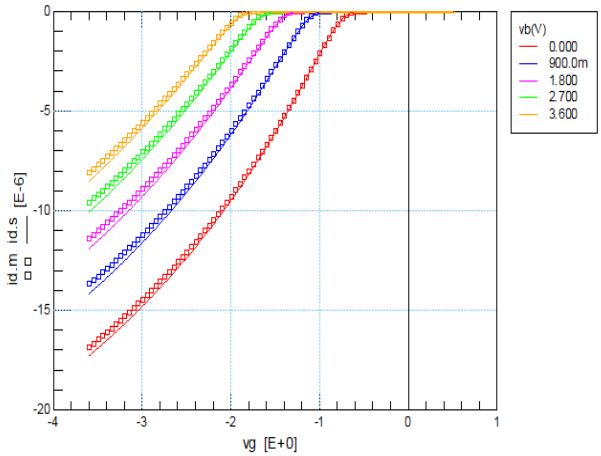




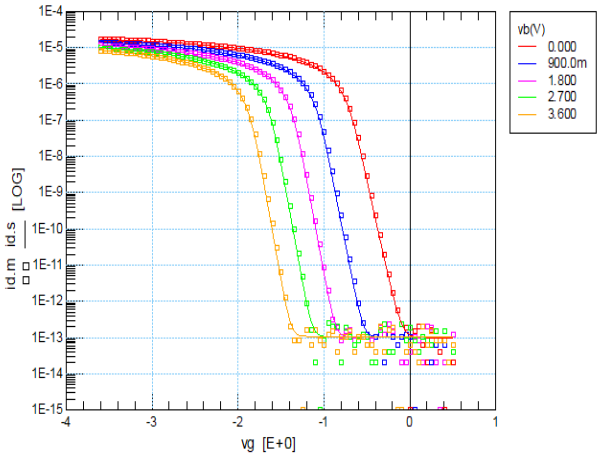
4.8 W10u0_L5u0_S562_2

idvg, Vd = -0.1V, T = 27°C

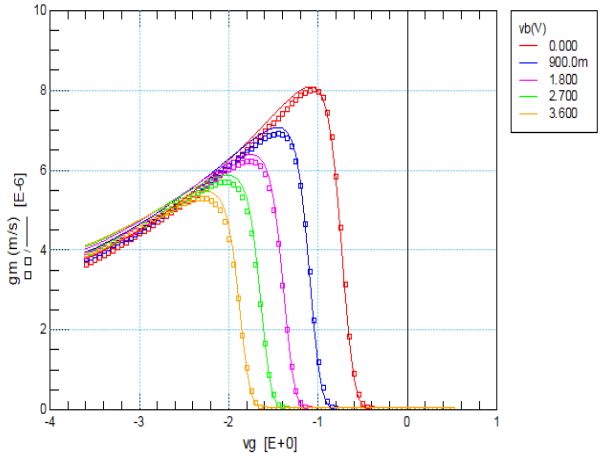
Id = f(Vg, Vb) @ Vd=low



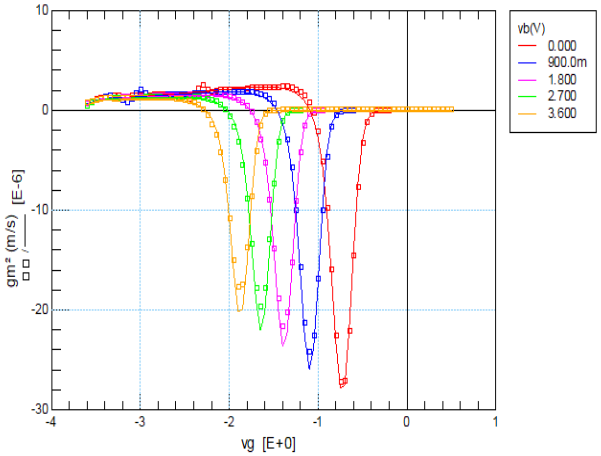
Id = f(Vg, Vb) @ Vd=low



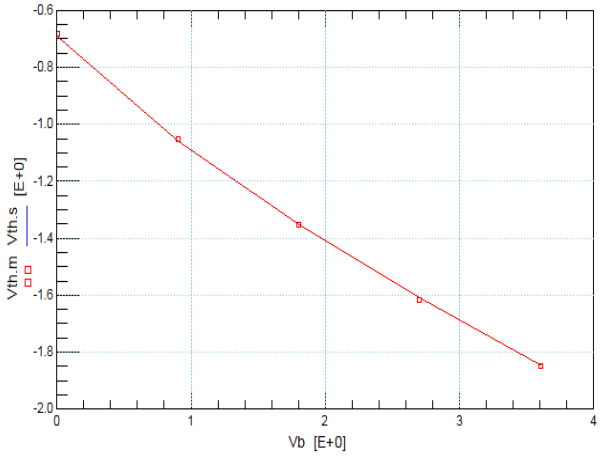
gm = f(Vg, Vb) @ Vd=low



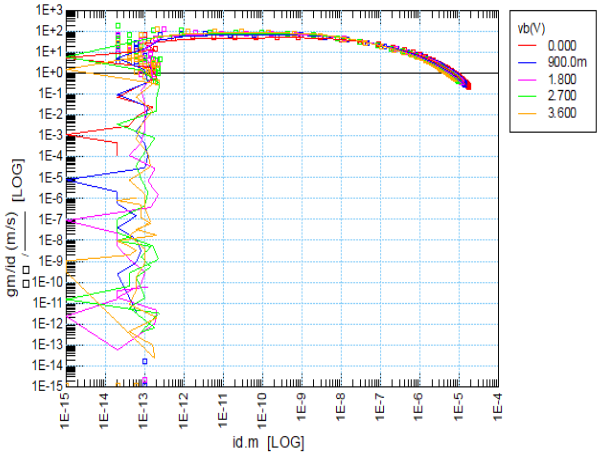
gm² = f(Vg, Vb) @ Vd=low

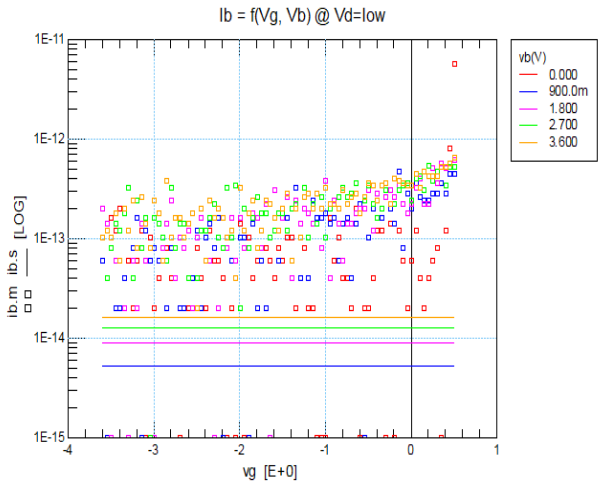


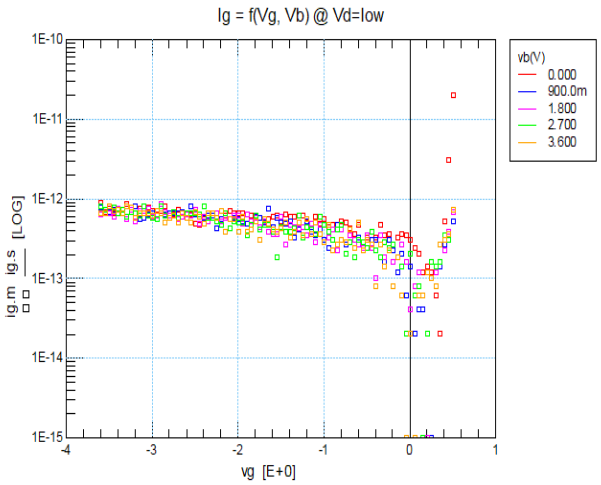
Vth @ Vd=low



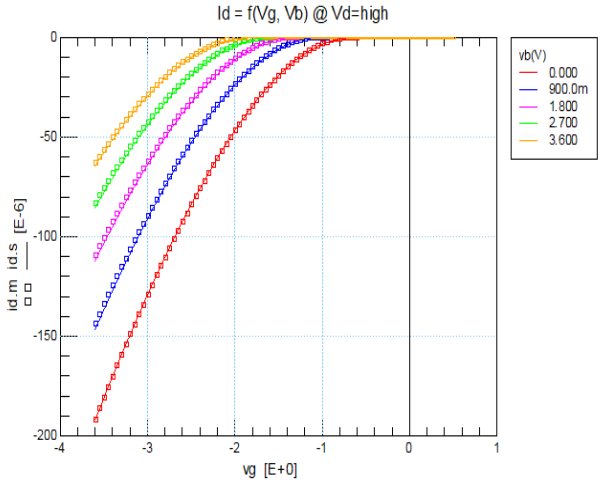
gm/id=f(id.m) @ Vd=low

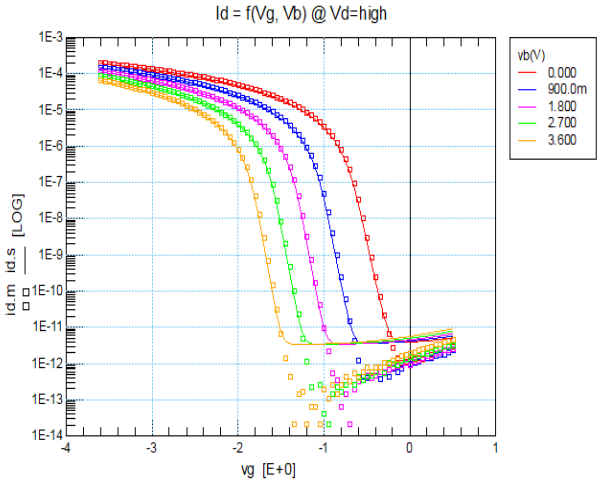


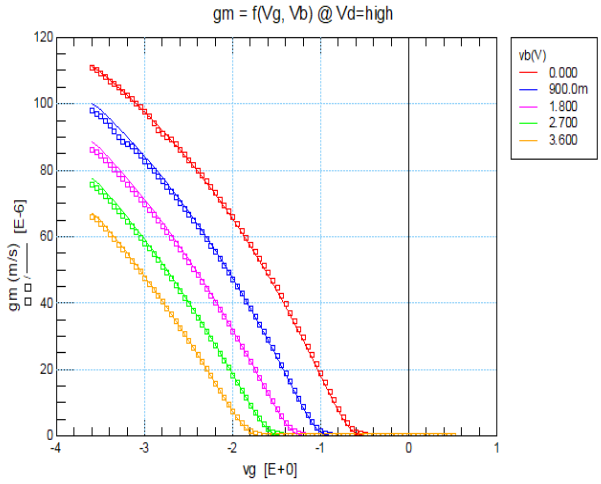


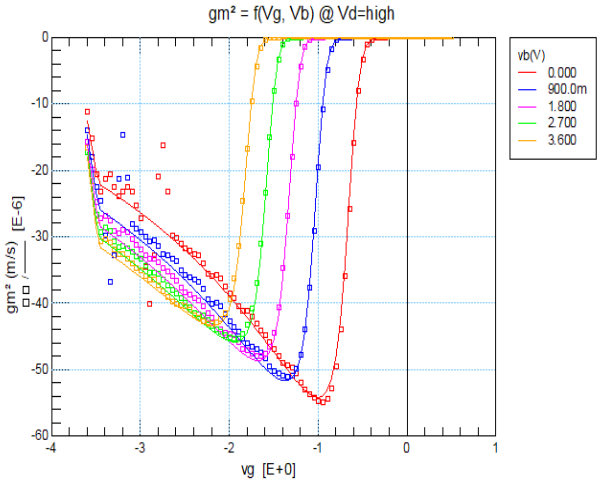


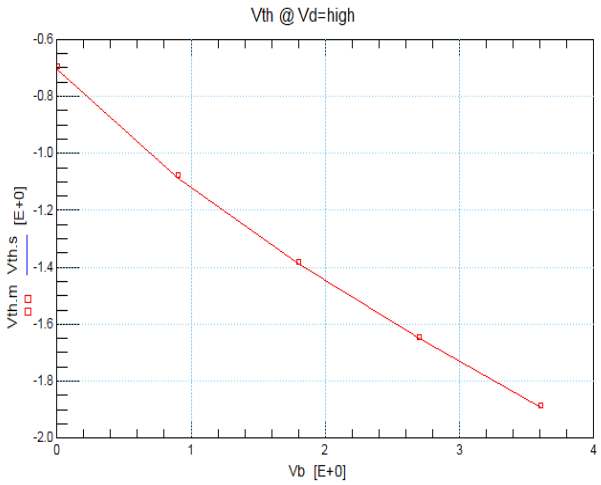
idvg, Vd = -3.3V, T = 27°C

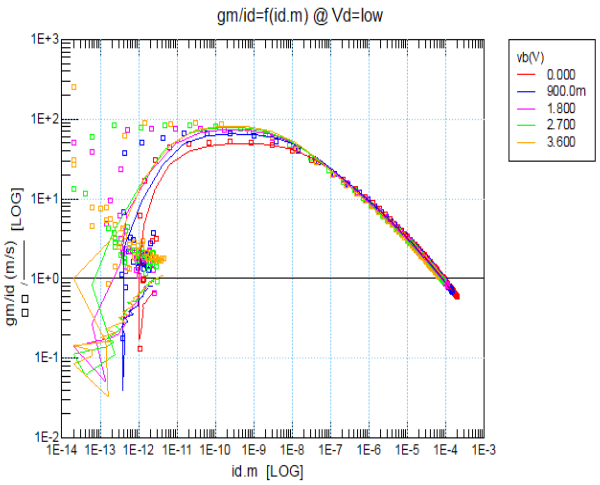


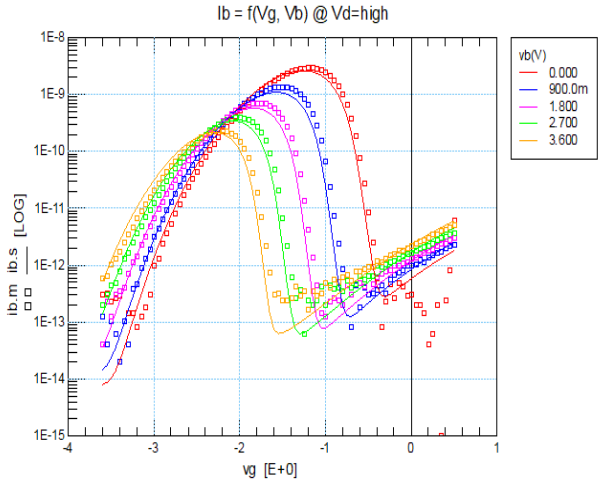


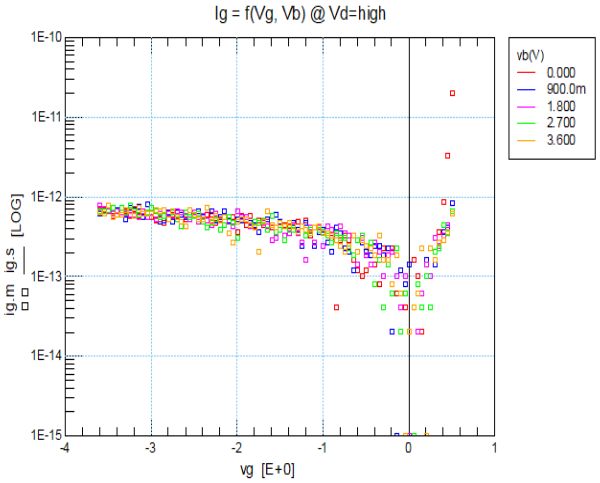




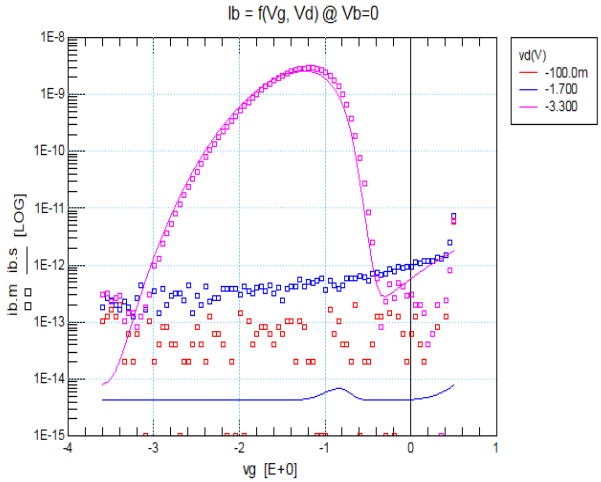


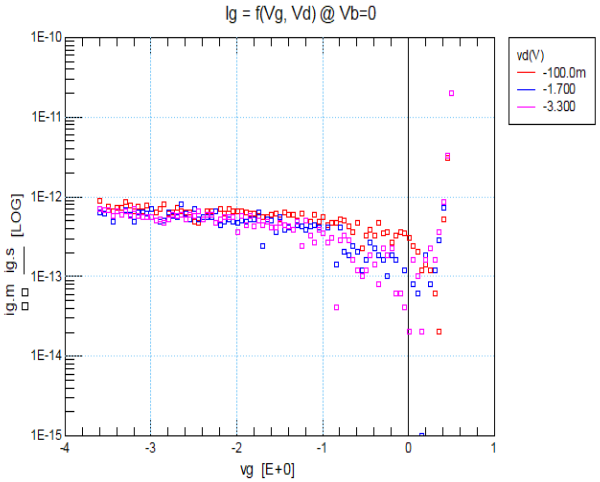




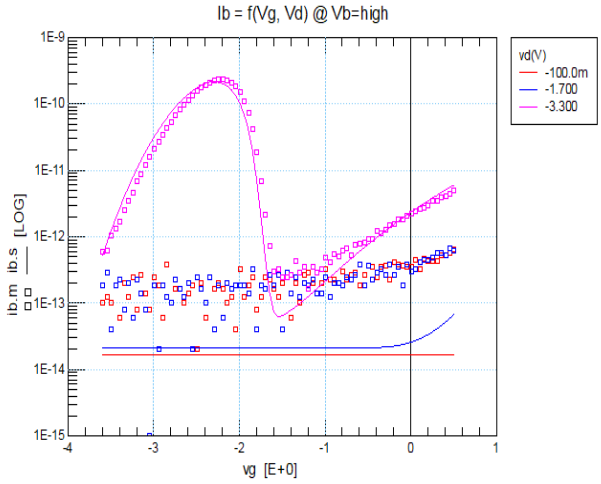


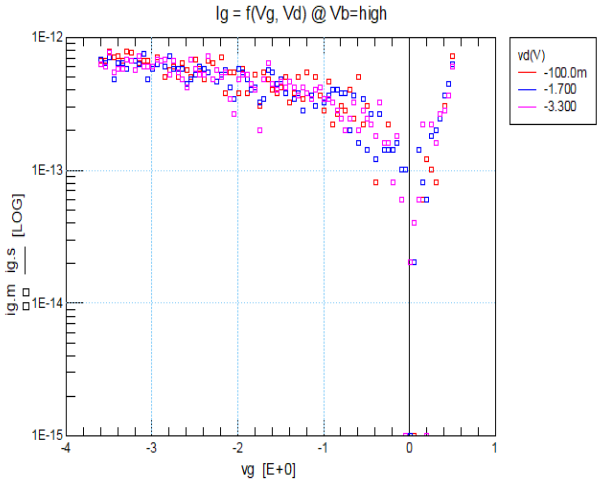
idvg, Ib,Ig, Vb = 0V, T = 27°C



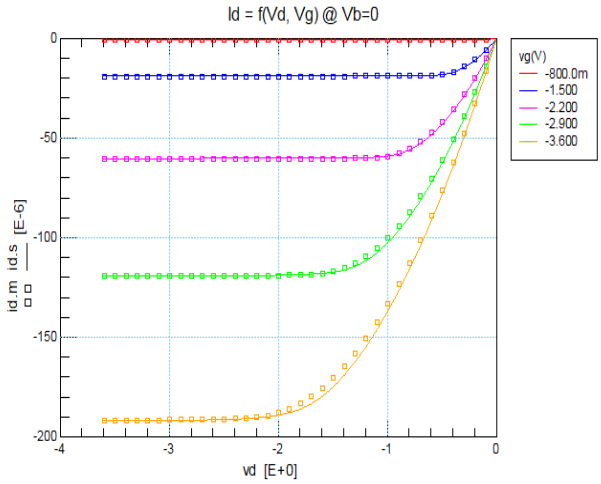


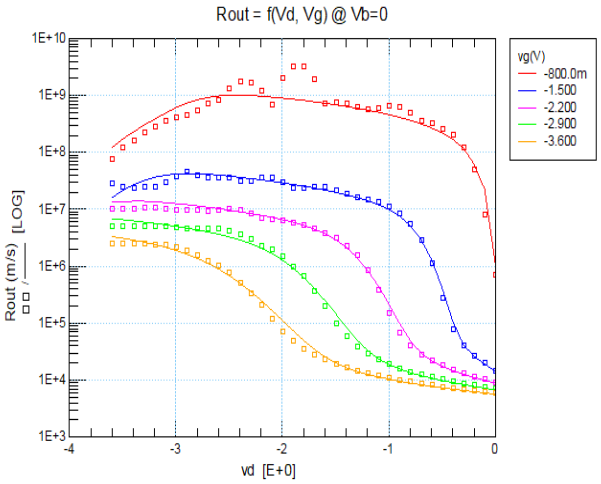
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

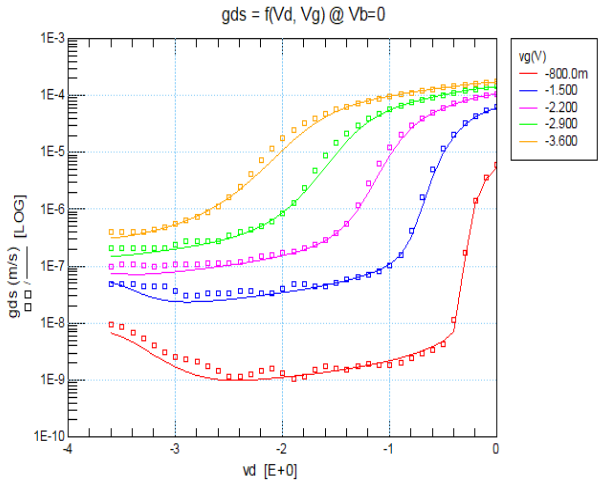




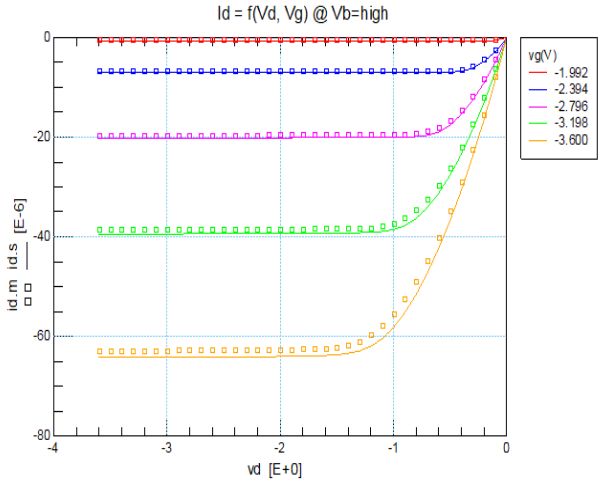
idvd, Vb = 0V, T = 27°C

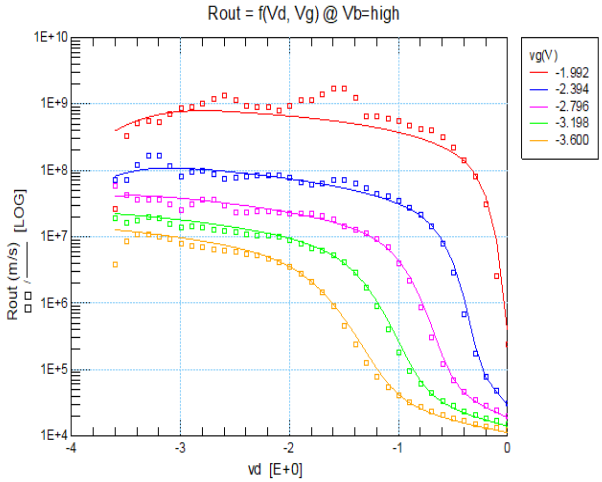


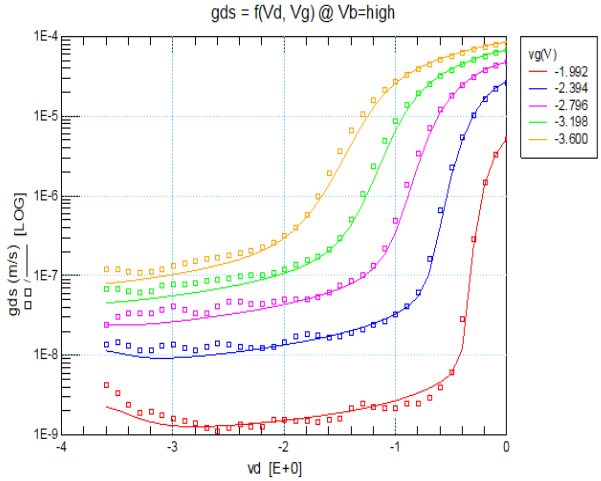




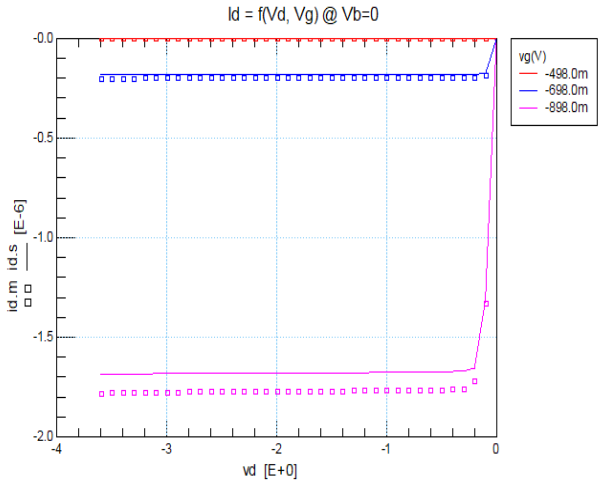
idvd_vbmin, Vb = 3.6V, T = 27°C

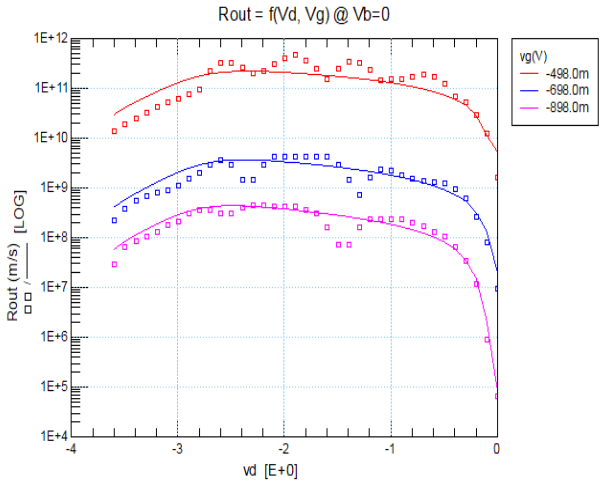


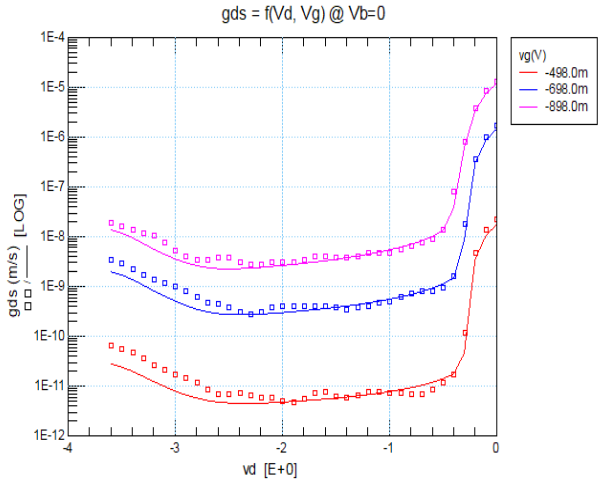




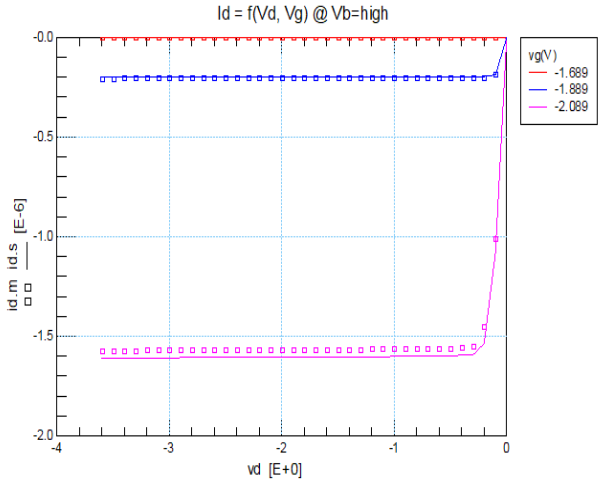
idvd_vth, Vb = 0V, T = 27°C

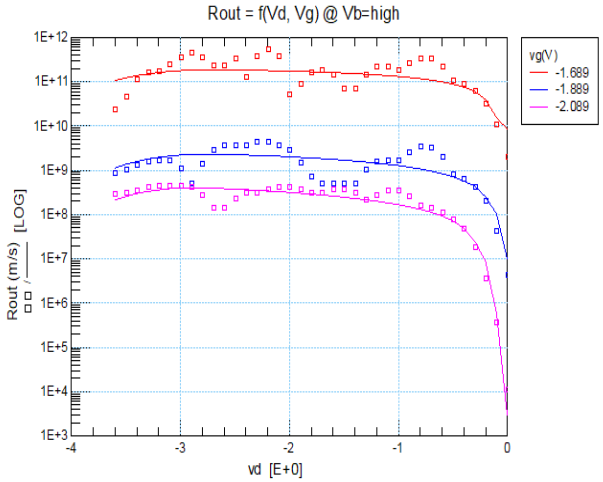


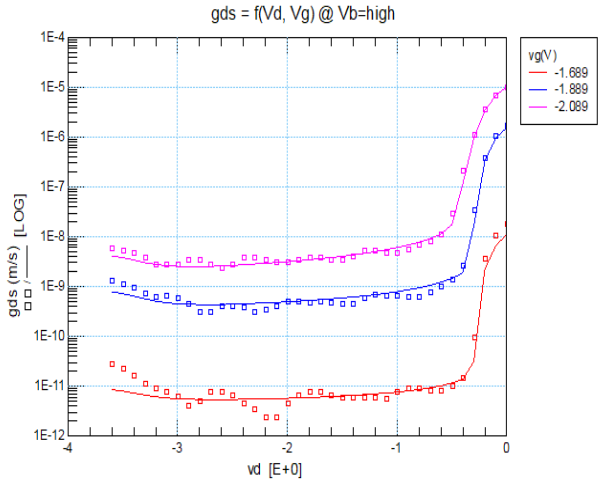




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

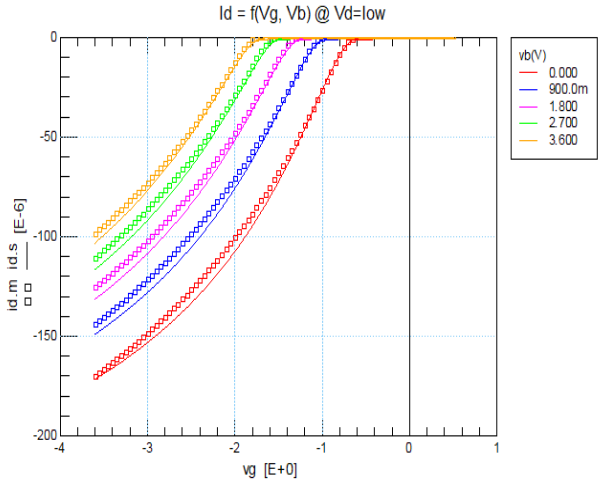


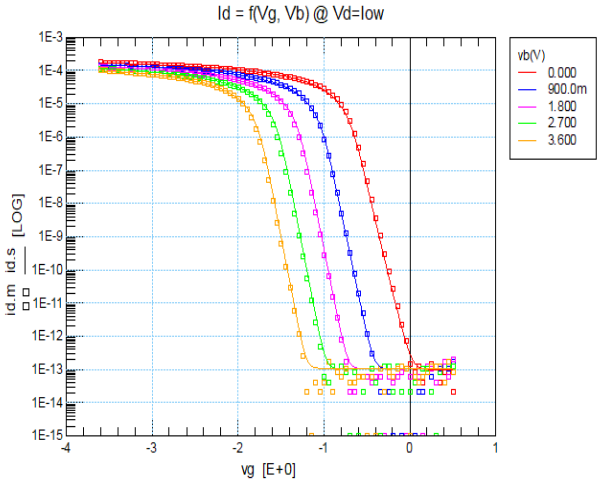


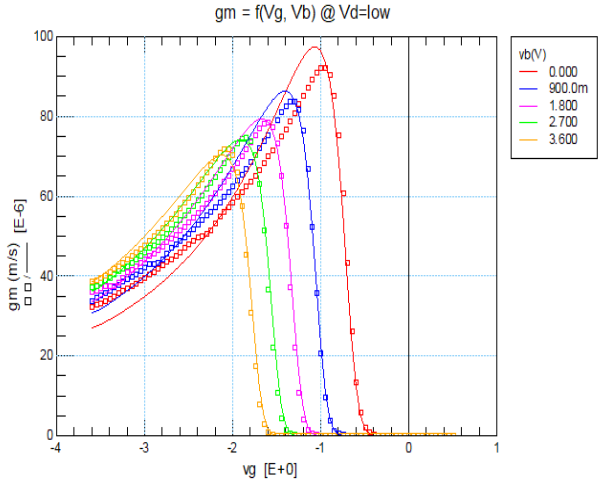


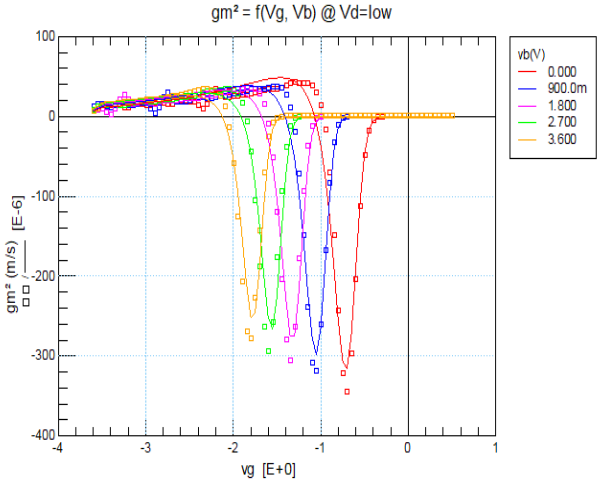
4.9 W10u0_L0u5_S561_4

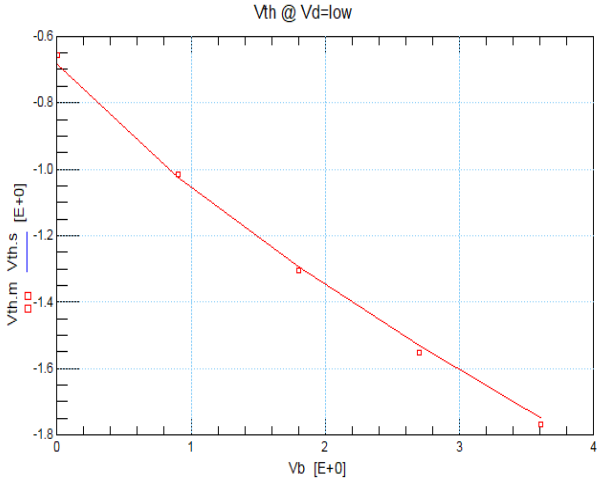
idvg, Vd = -0.1V, T = 27°C

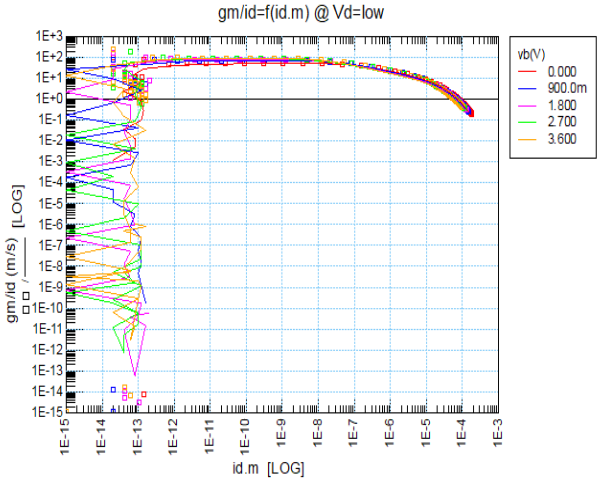


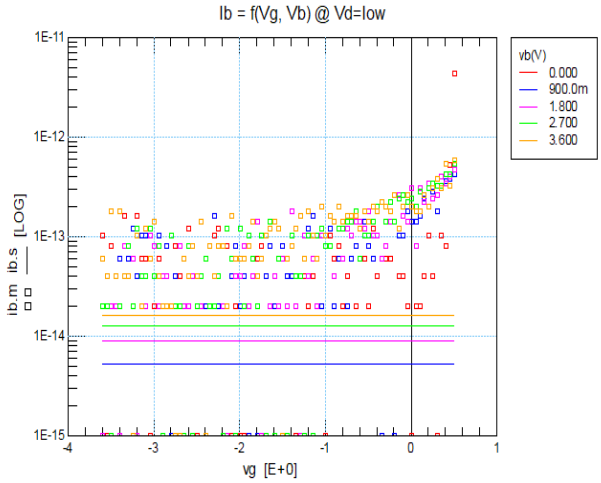


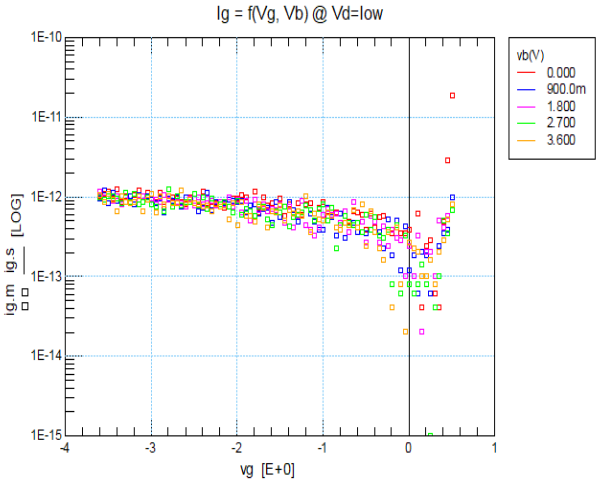




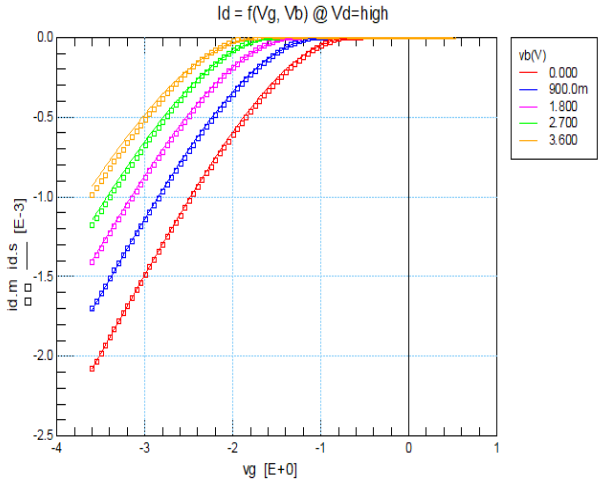


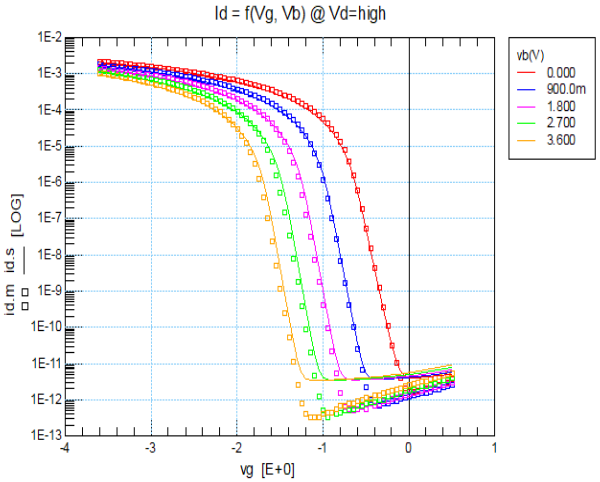


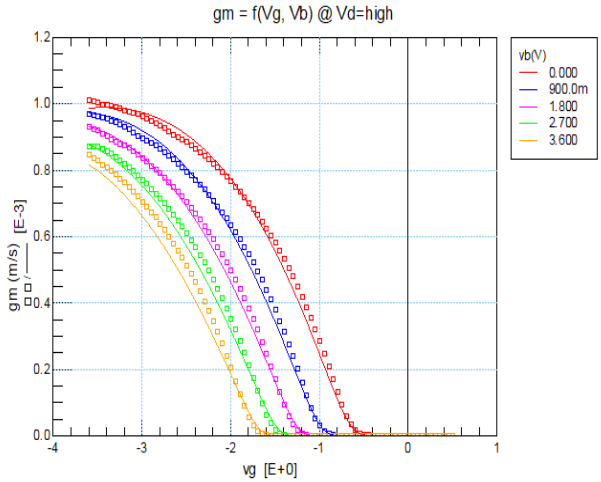


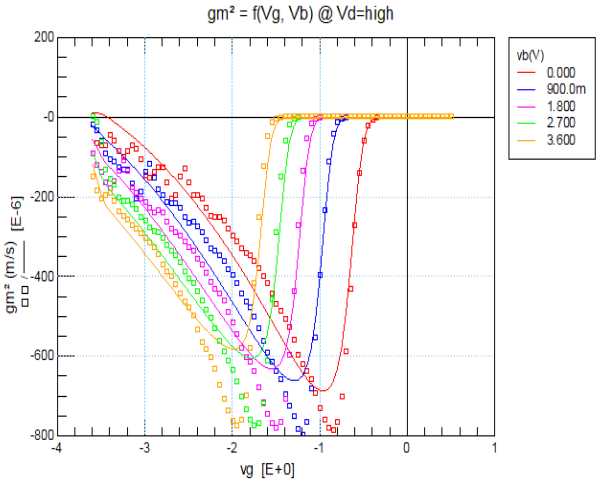


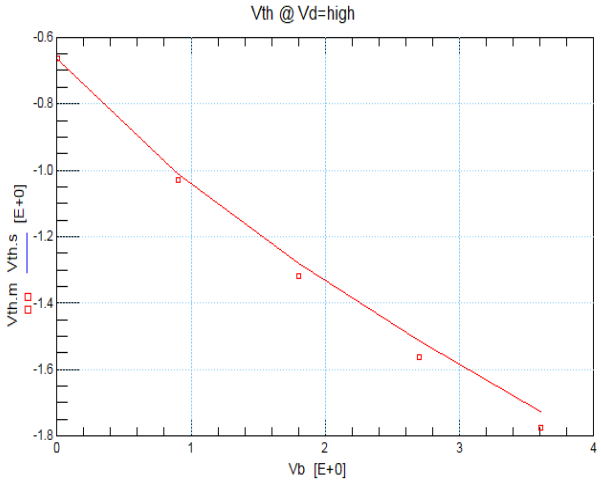
idvg, Vd = -3.3V, T = 27°C

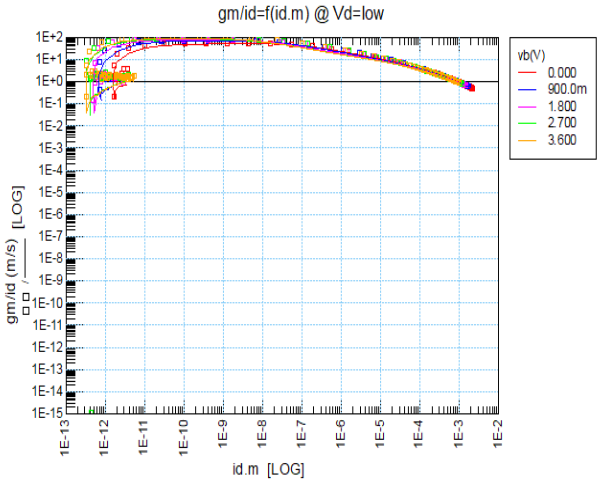


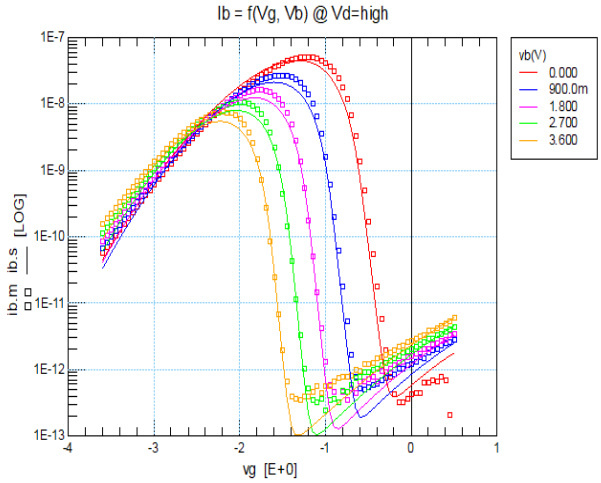


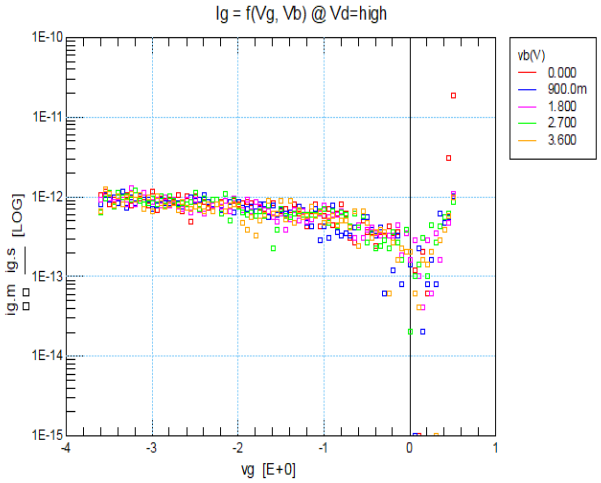




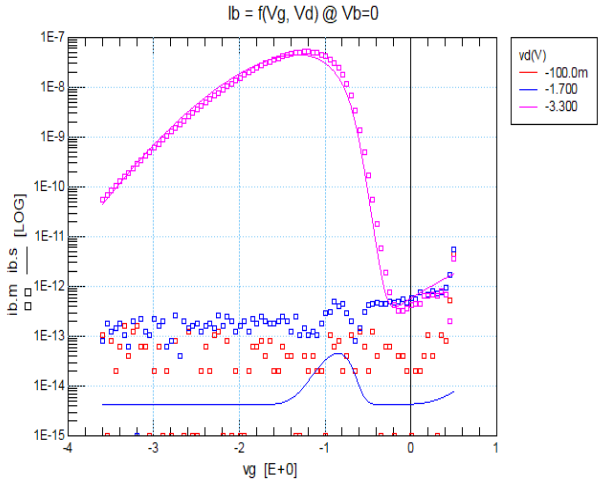


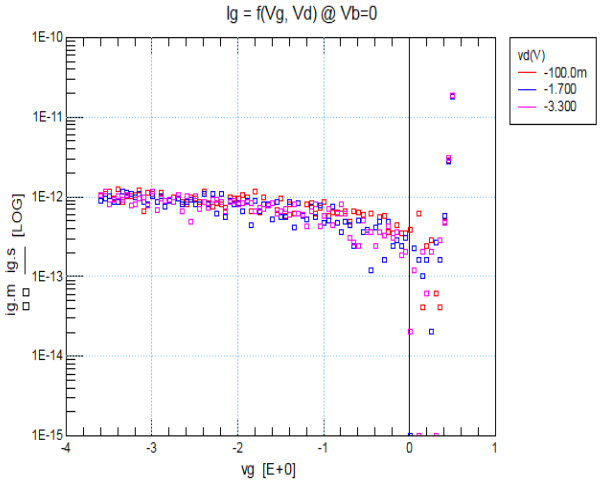




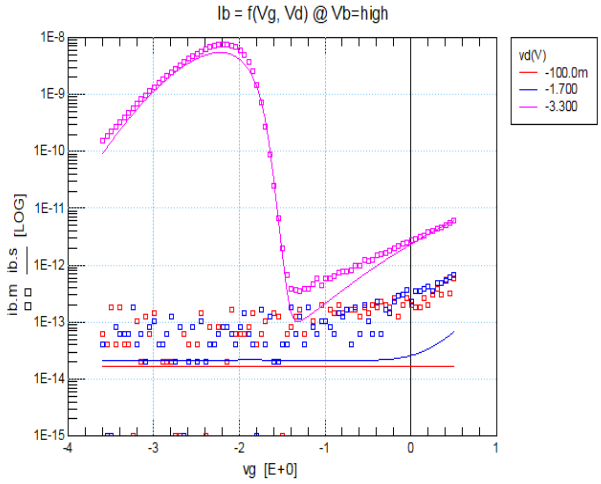


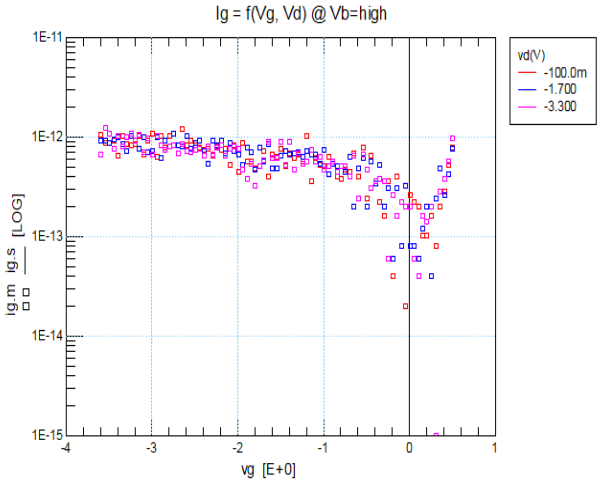
idvg, Ib,Ig, Vb = 0V, T = 27°C



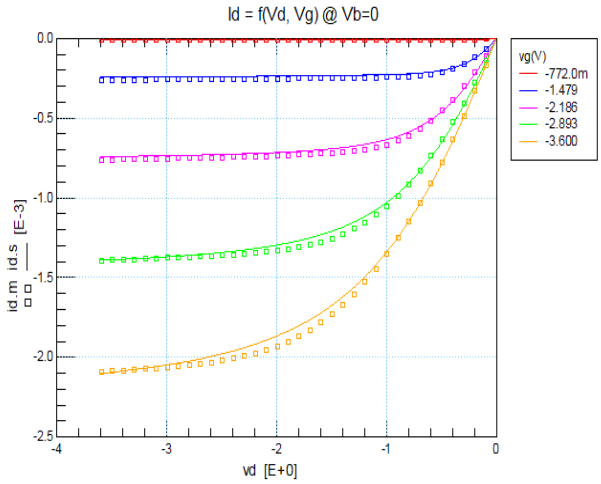


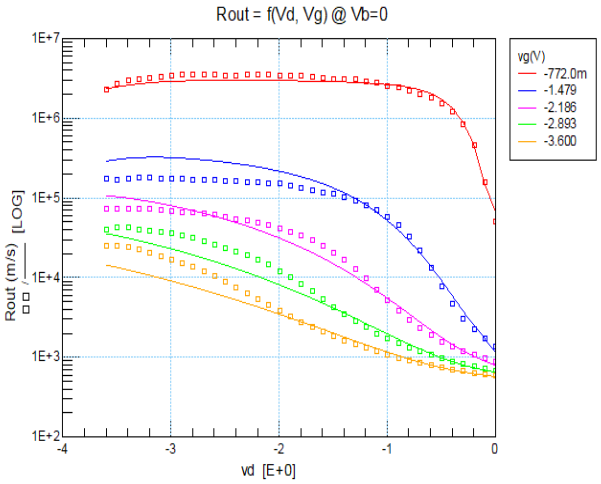
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

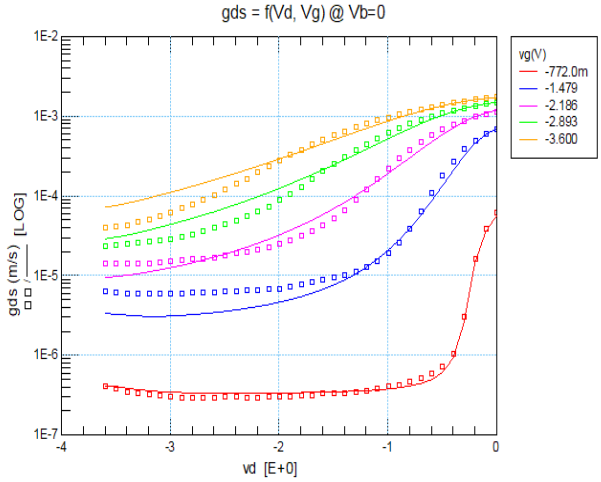




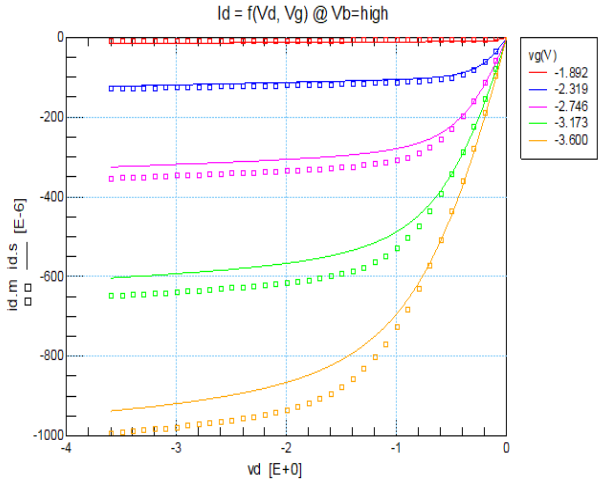
idvd, Vb = 0V, T = 27°C

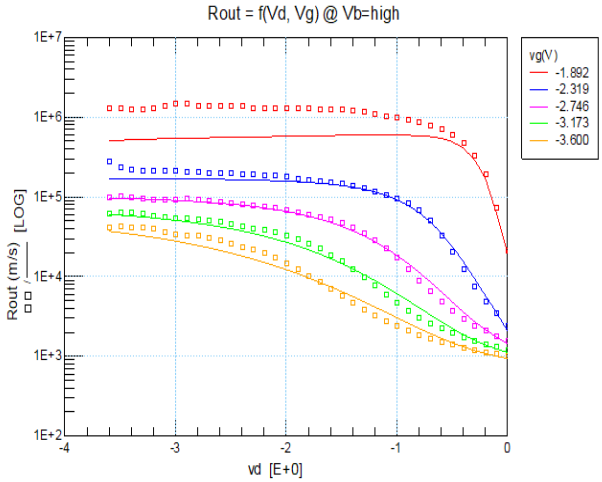


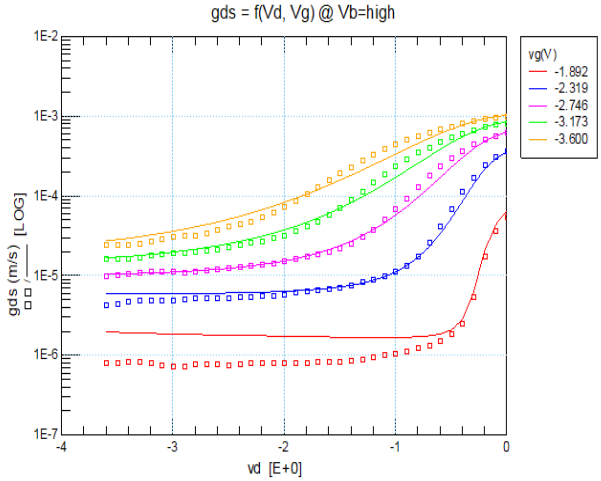




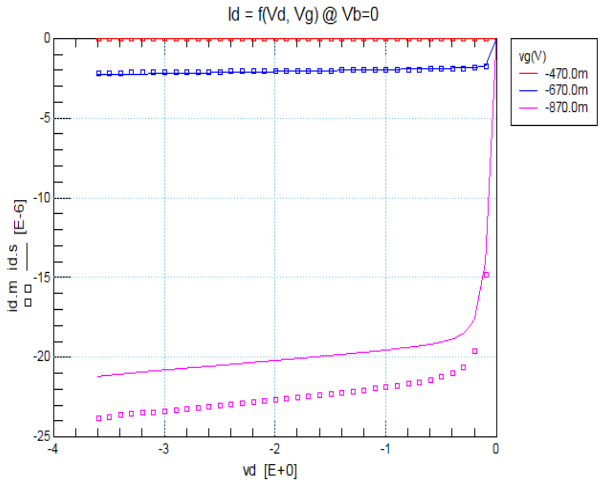
idvd_vbmin, Vb = 3.6V, T = 27°C

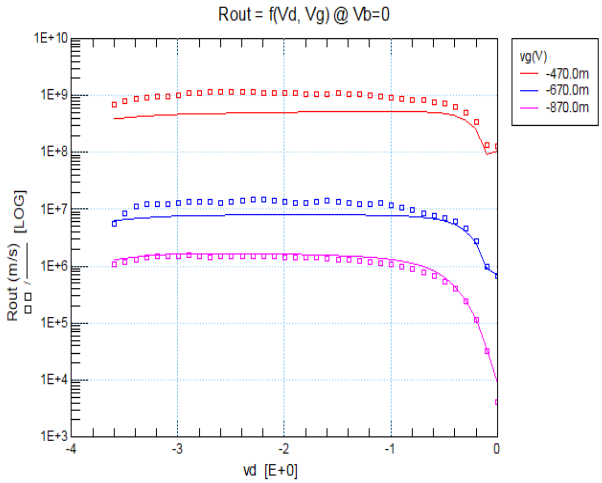


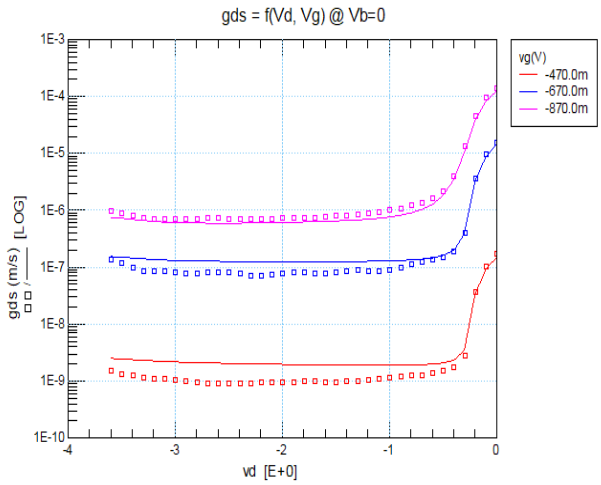




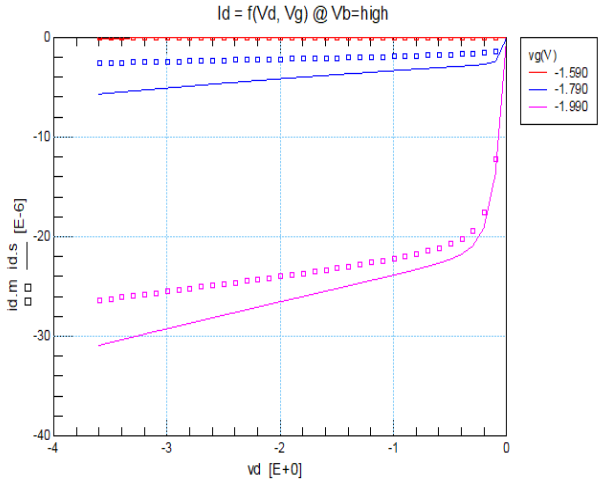
idvd_vth, Vb = 0V, T = 27°C

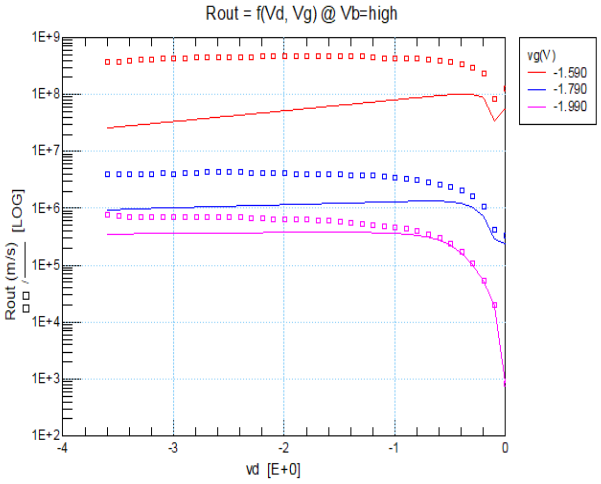


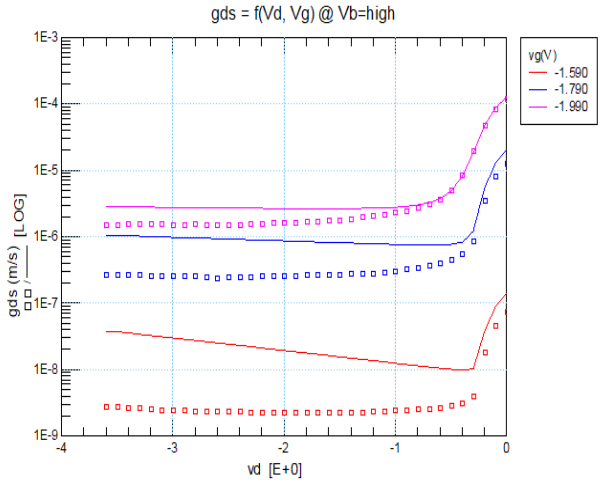




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

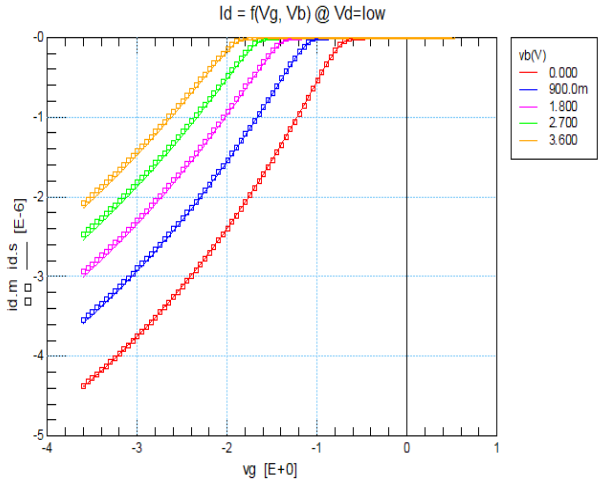


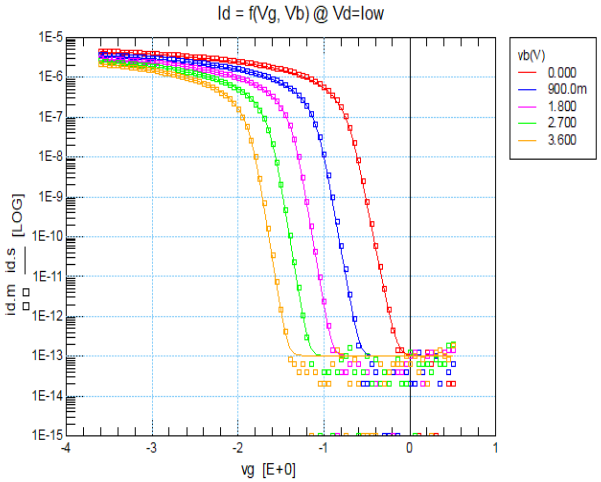


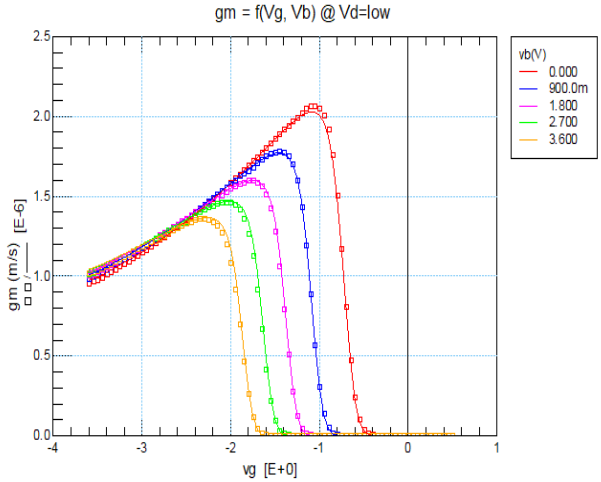


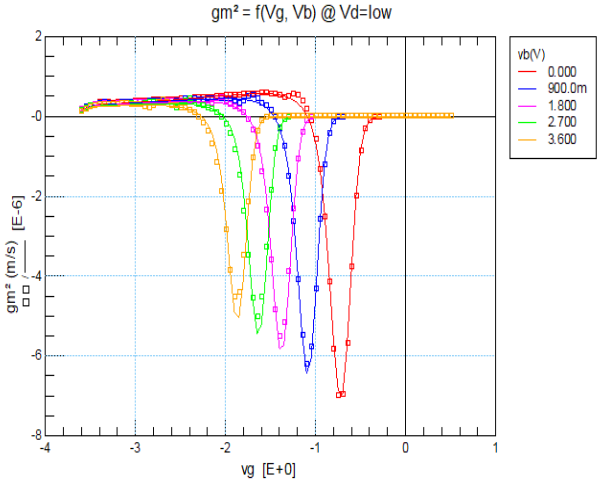
4.10 W05u0_L10u0_S563_1

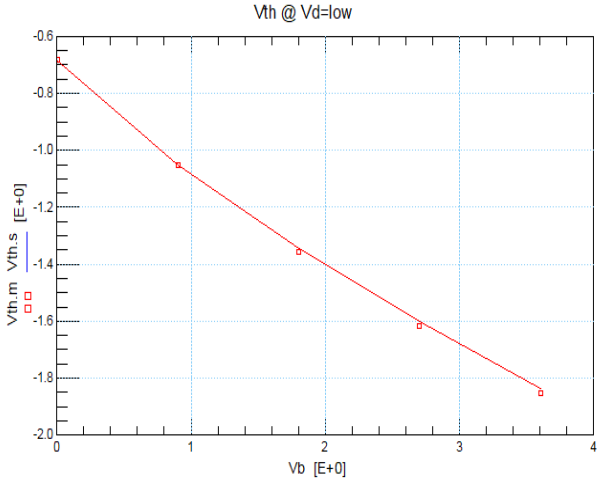
idvg, Vd = -0.1V, T = 27°C

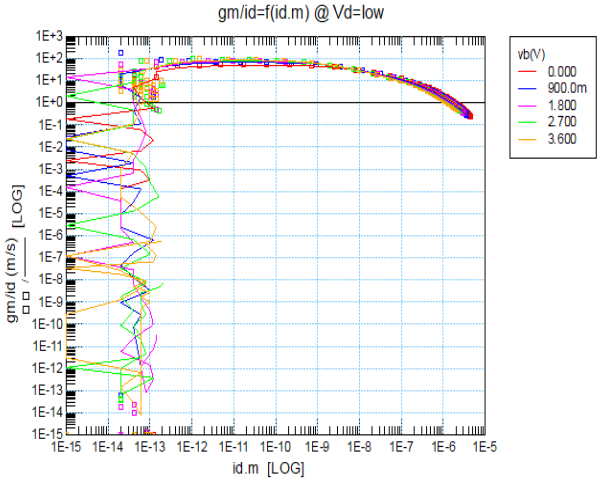


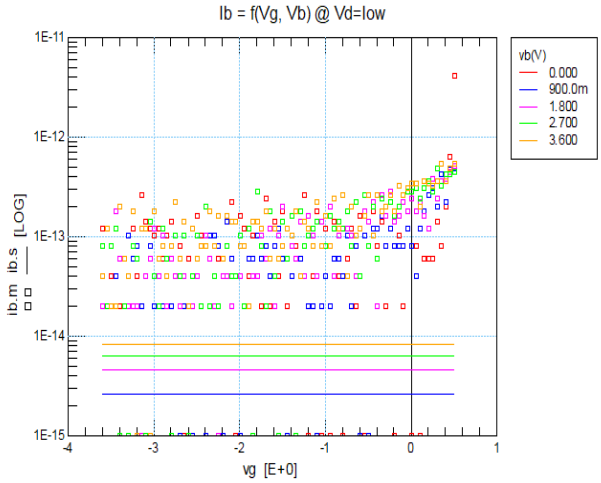


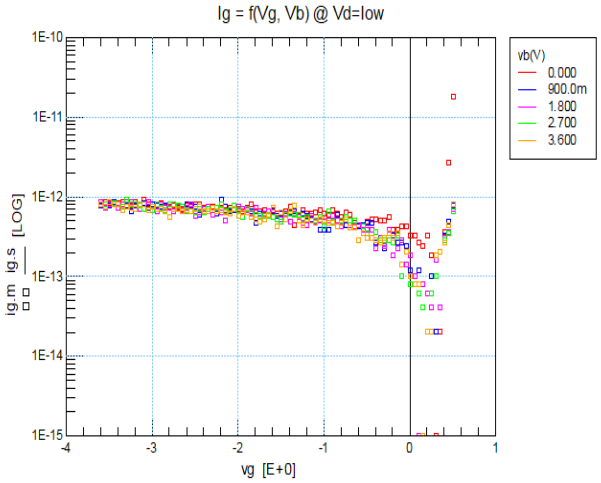




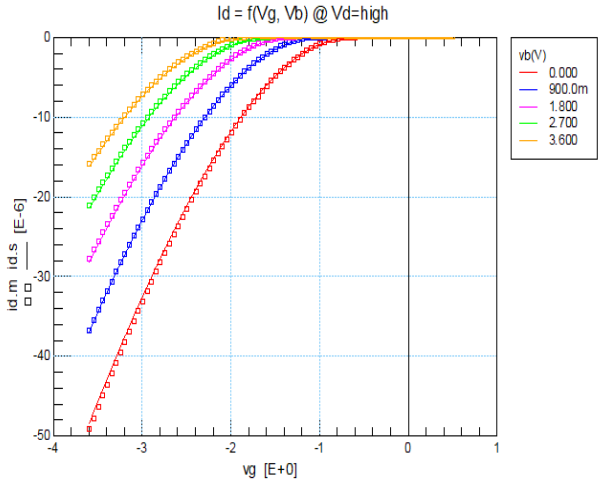


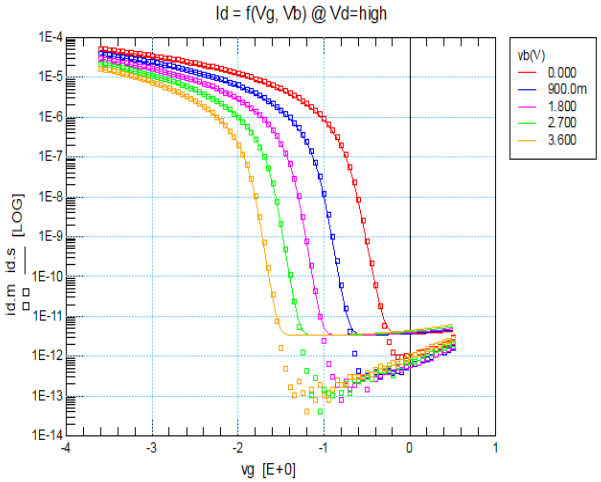


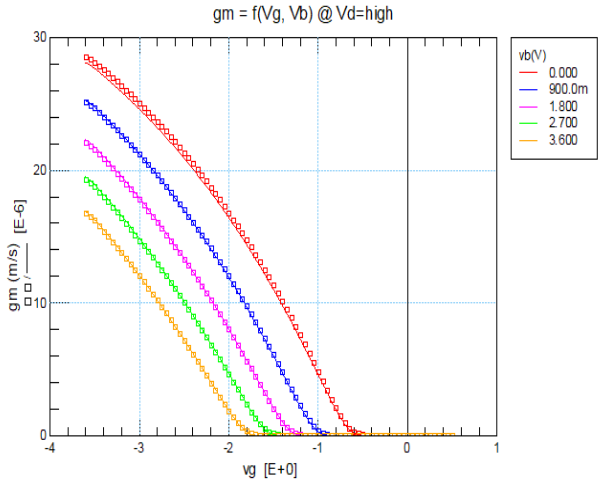


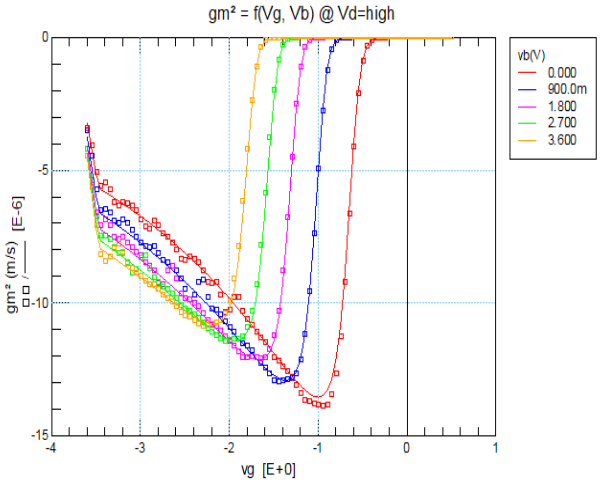


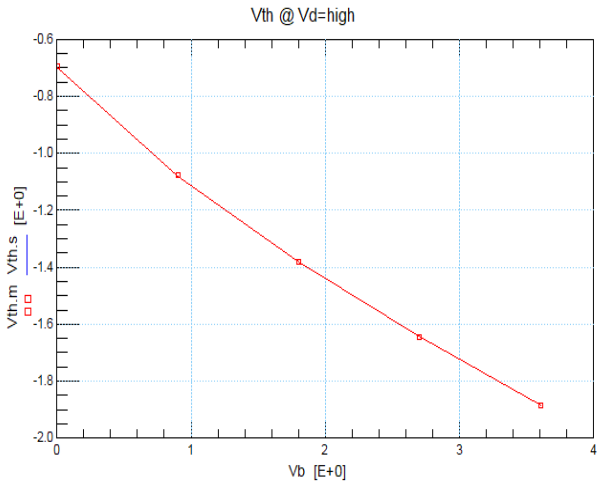
idvg, Vd = -3.3V, T = 27°C

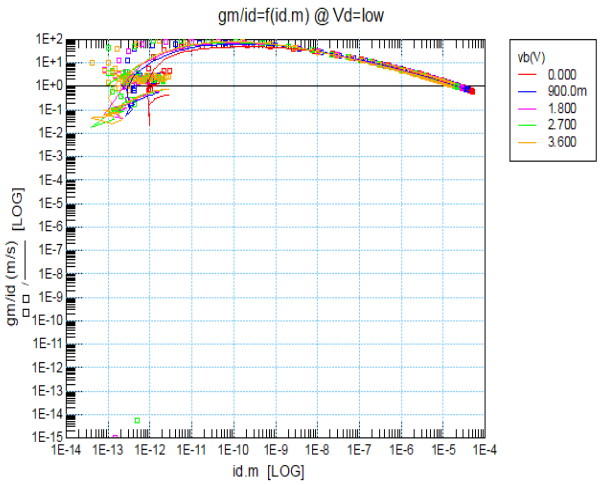


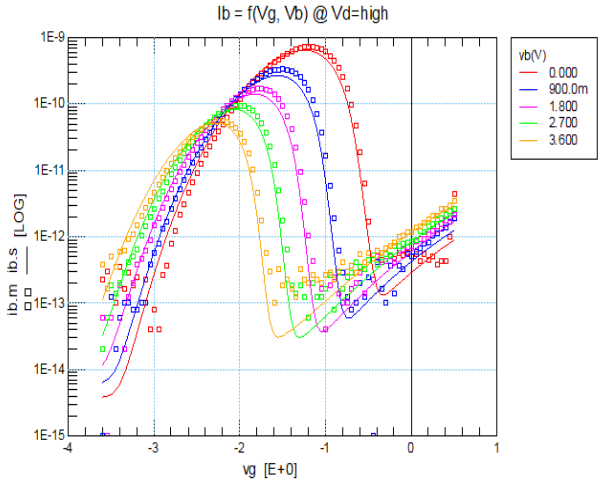


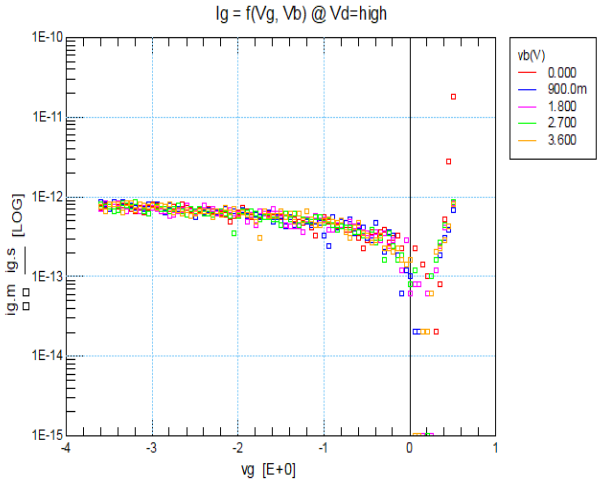




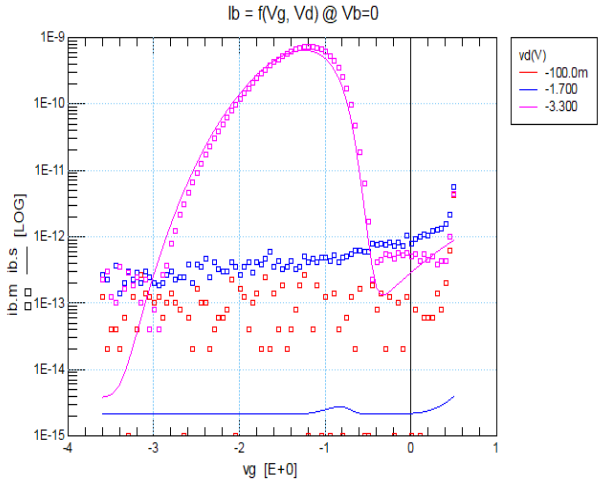


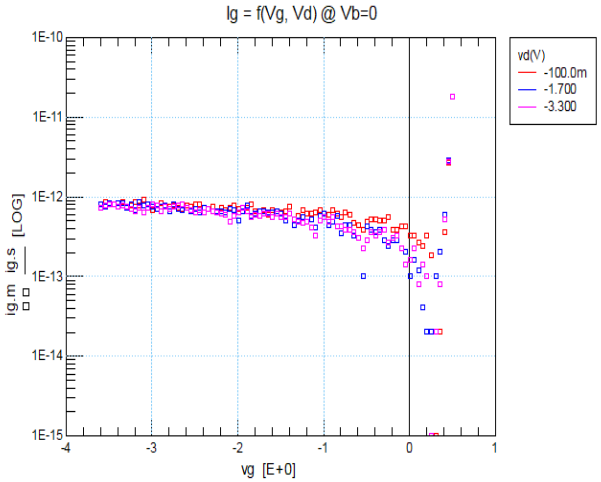




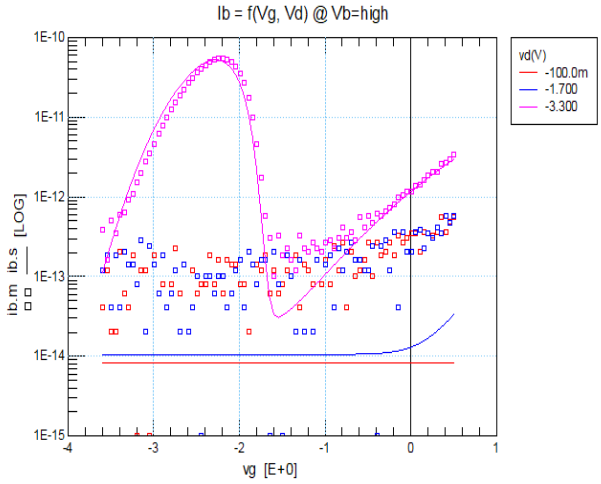


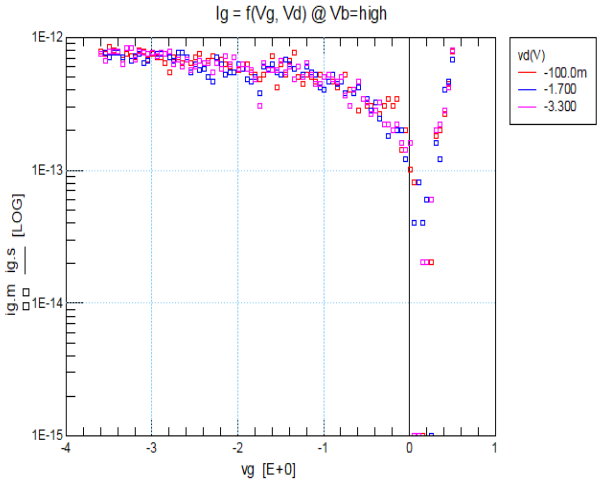
idvg, Ib,Ig, Vb = 0V, T = 27°C



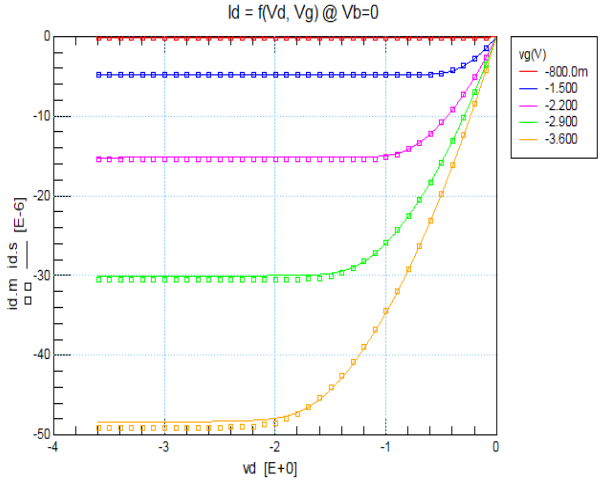


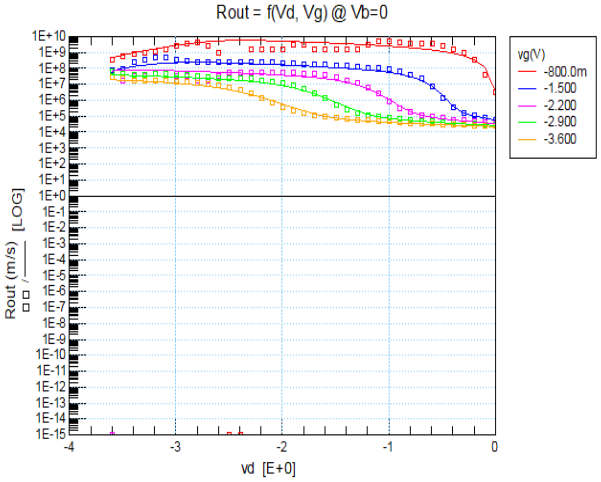
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

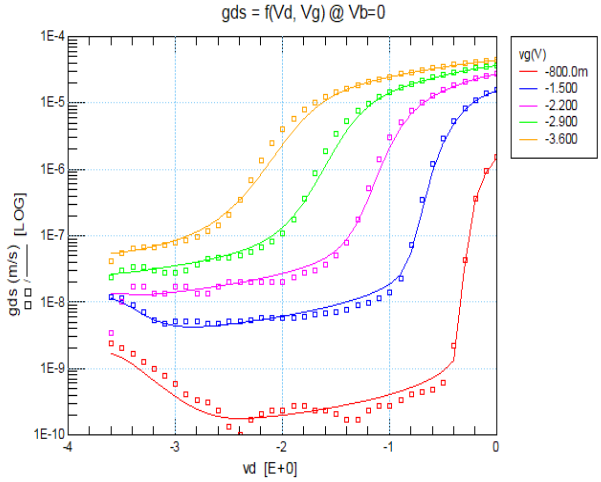




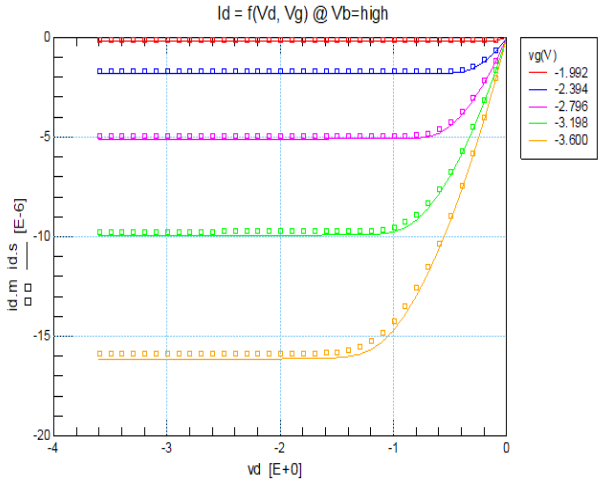
idvd, Vb = 0V, T = 27°C

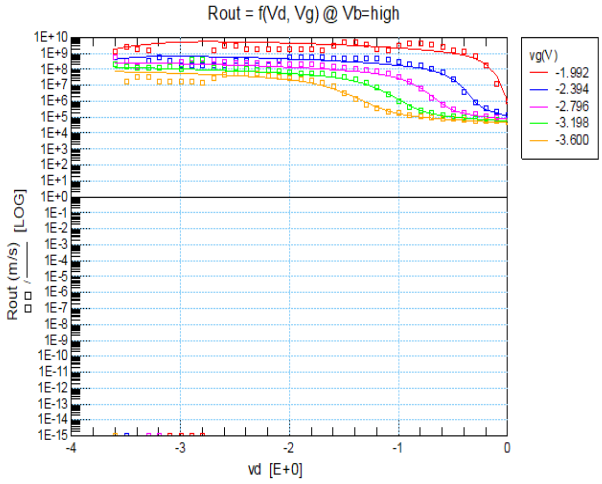


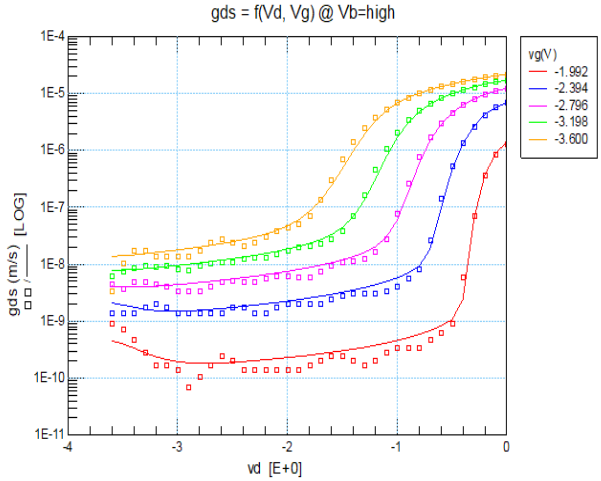




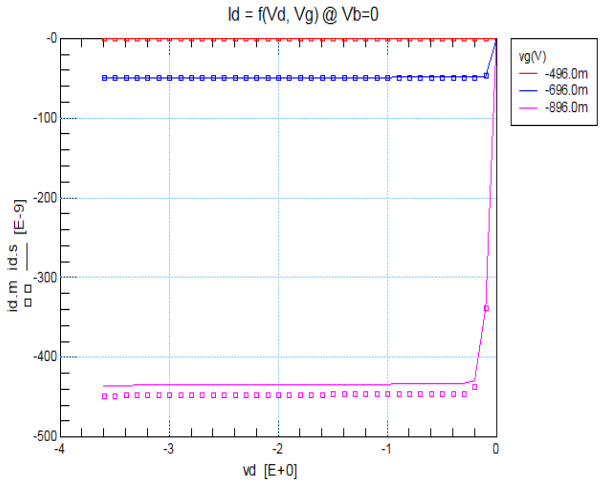
idvd_vbmin, Vb = 3.6V, T = 27°C

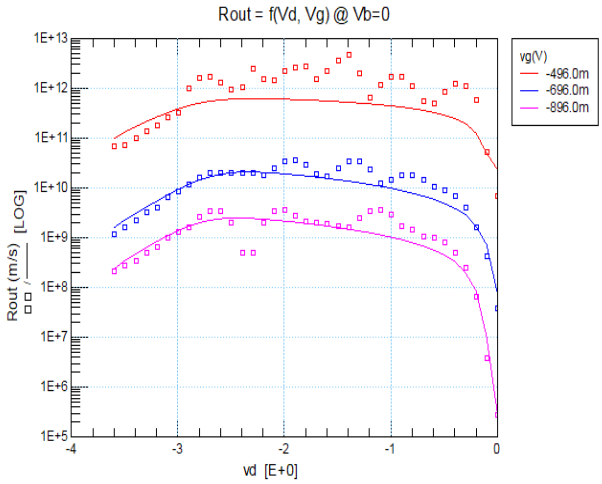


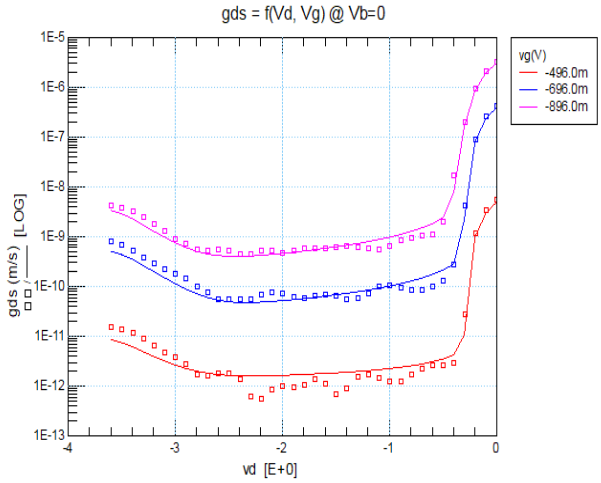




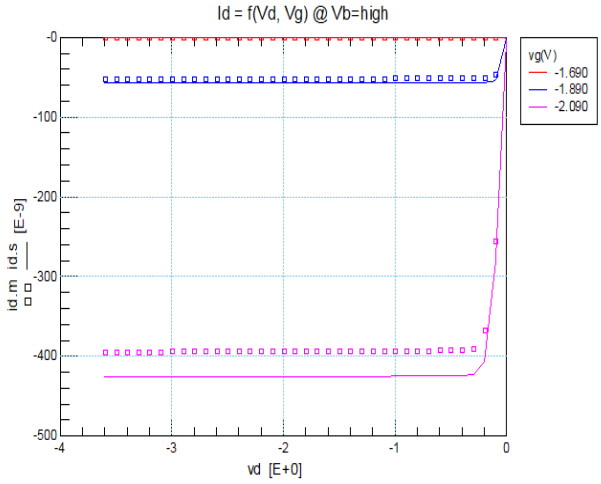
idvd_vth, Vb = 0V, T = 27°C

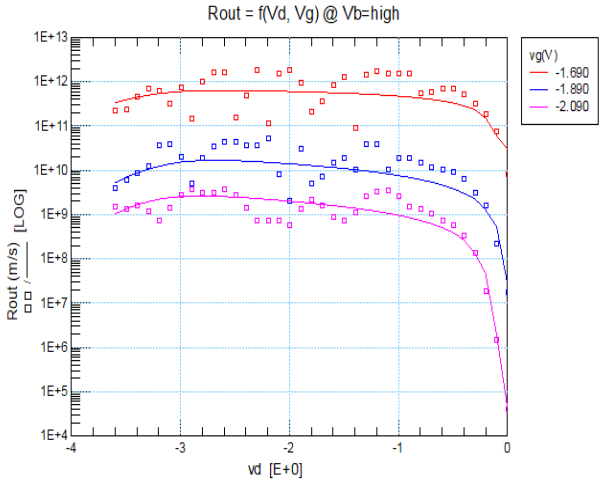


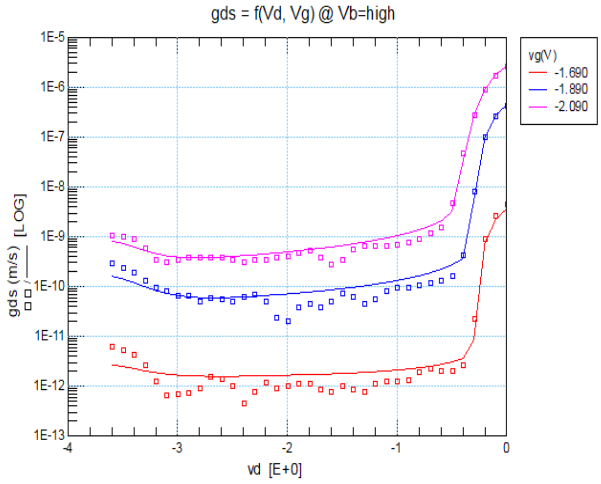




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

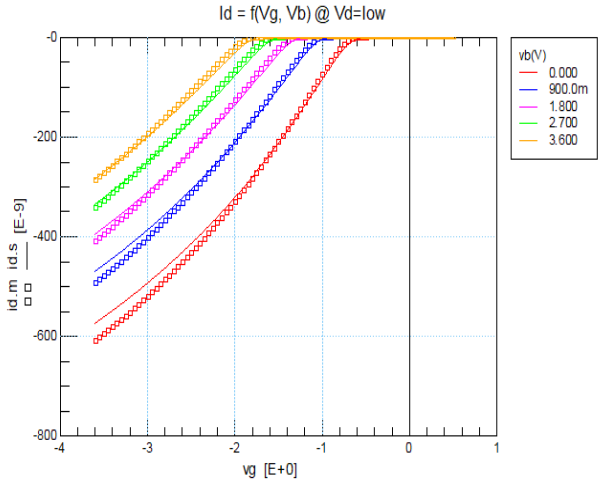


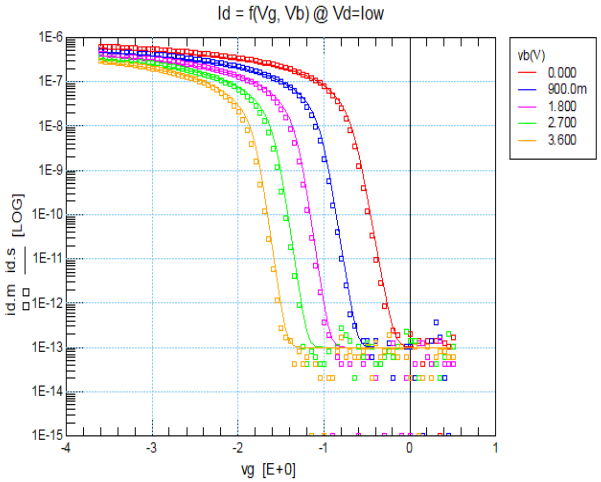


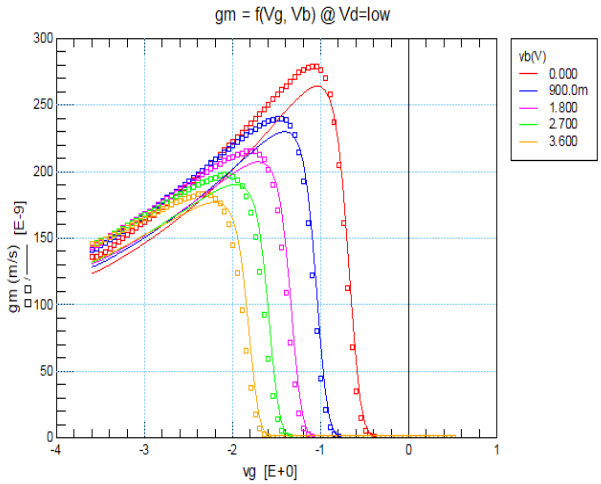


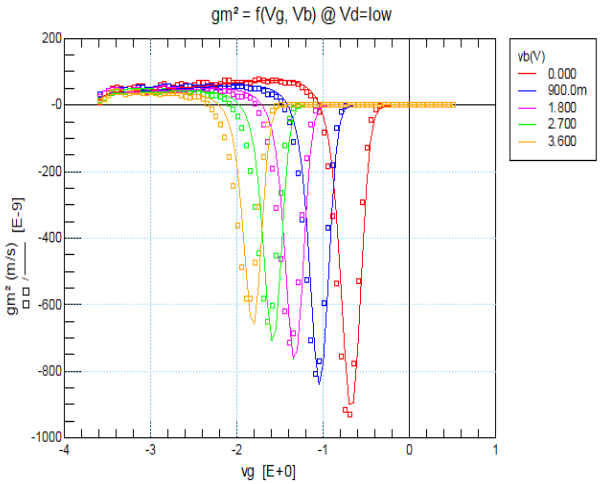
4.11 W0u6_L10u0_S564_5

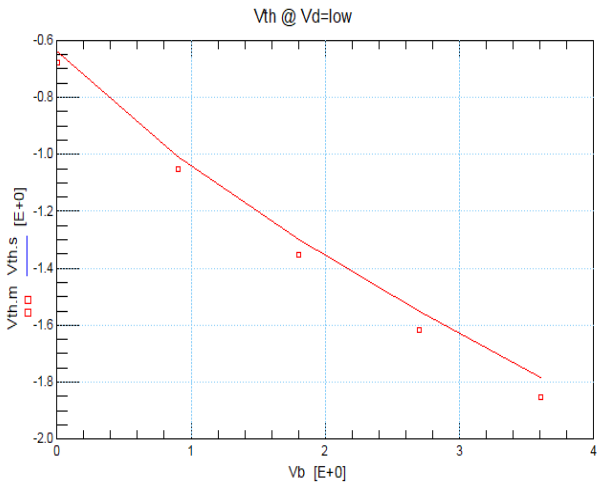
idvg, Vd = -0.1V, T = 27°C

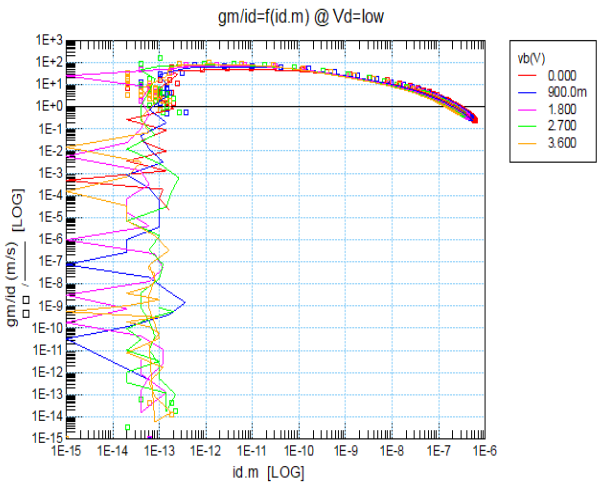


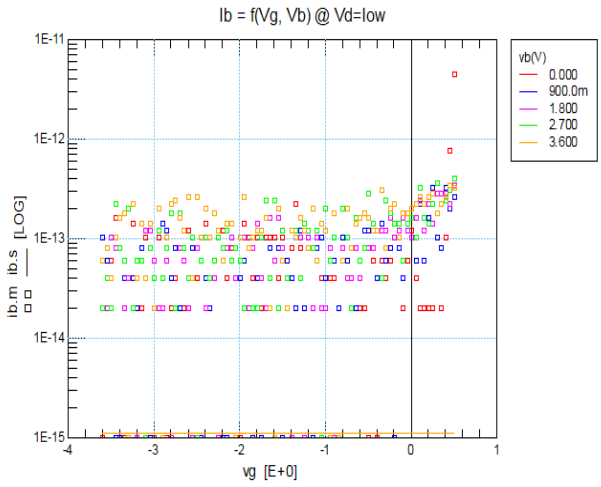


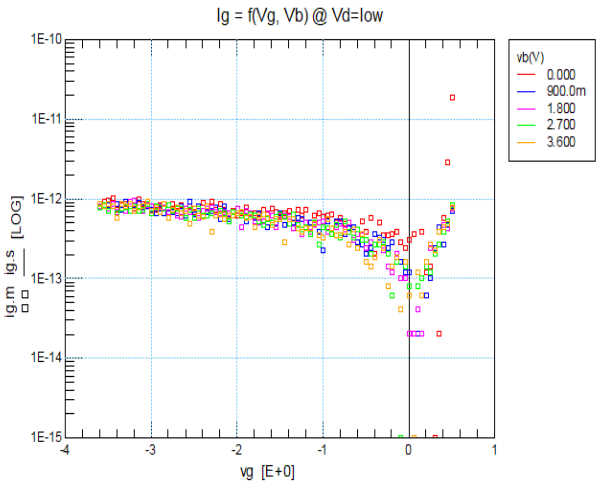




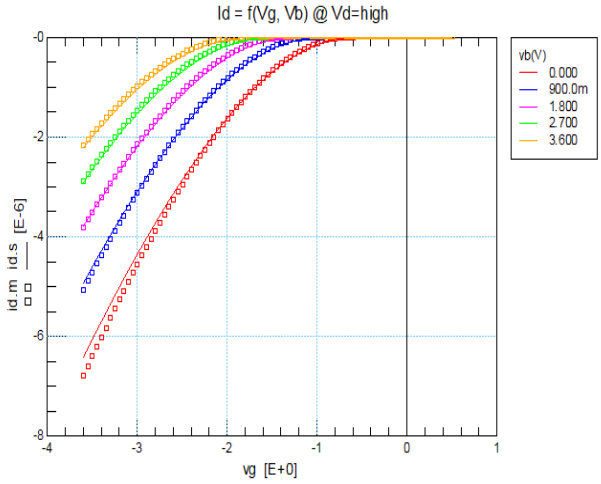


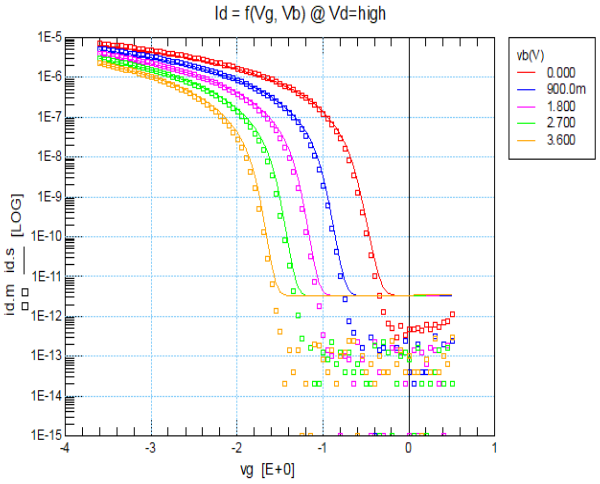


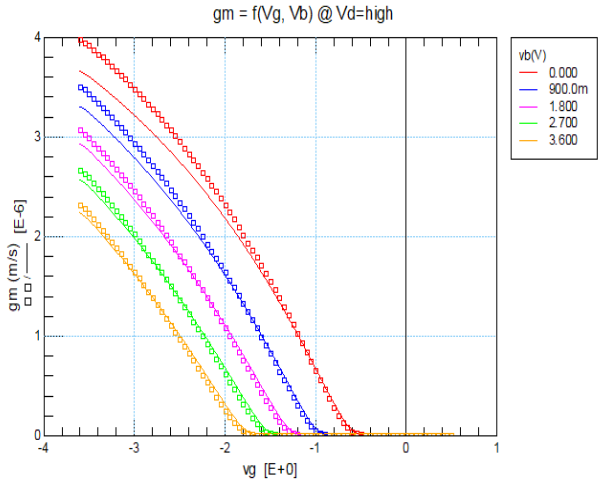


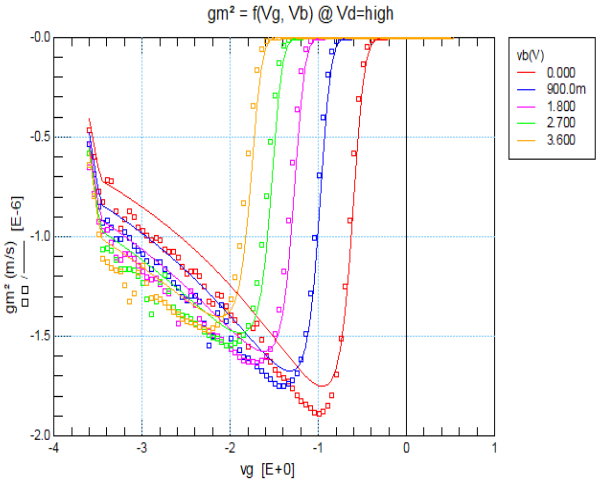


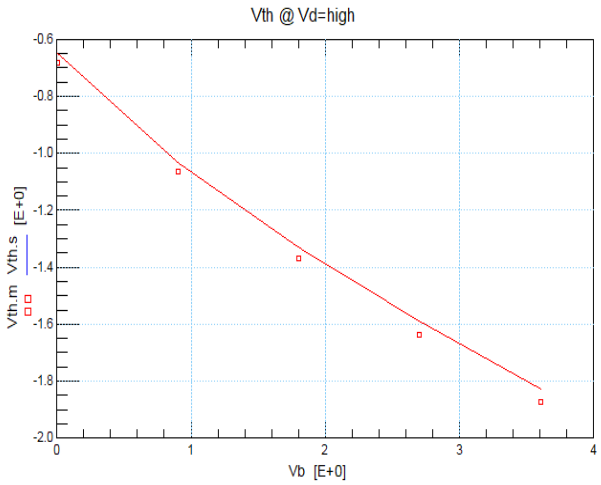
idvg, Vd = -3.3V, T = 27°C

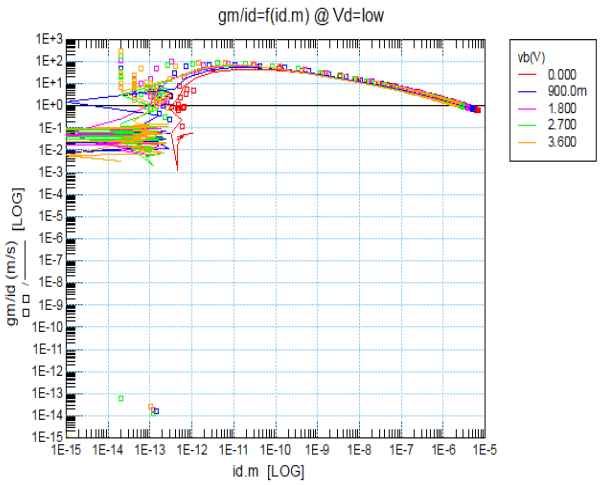


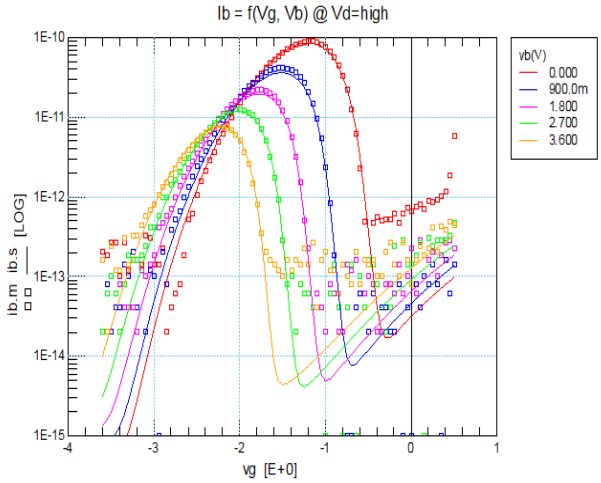


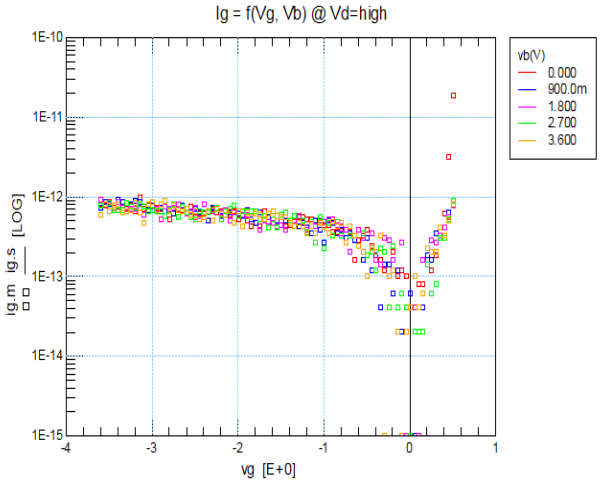




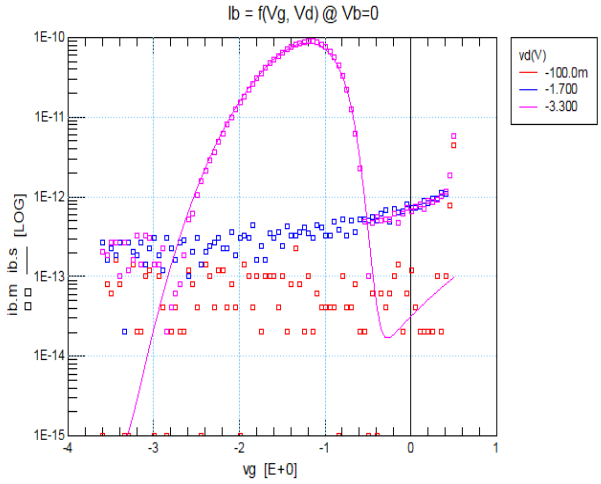


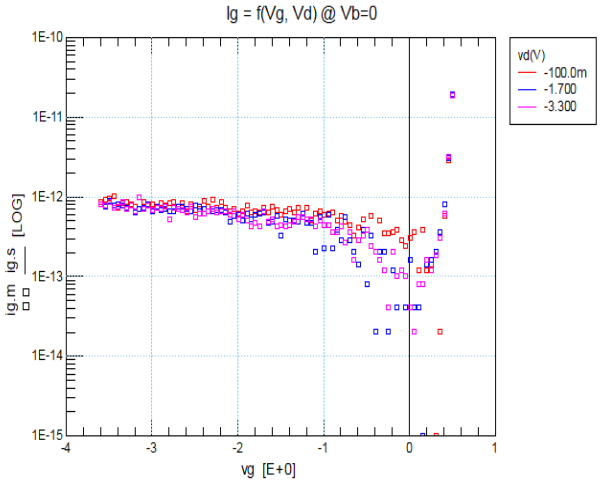




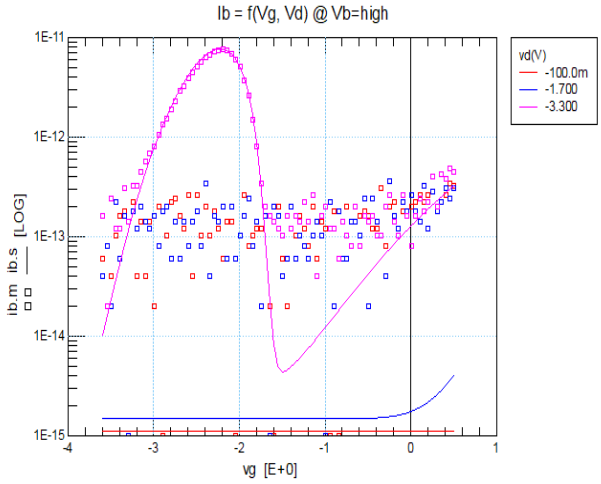


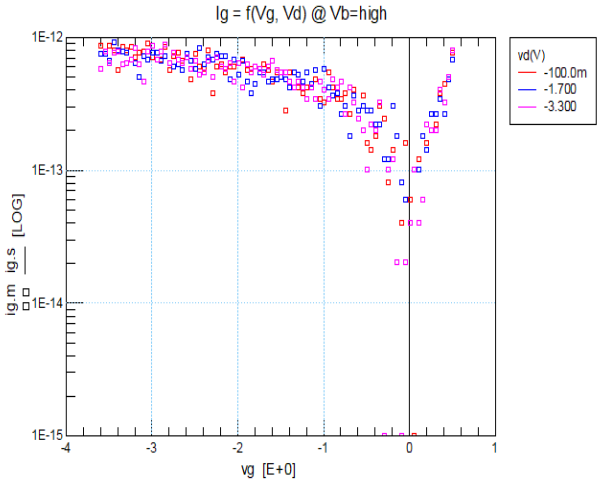
idvg, Ib,Ig, Vb = 0V, T = 27°C



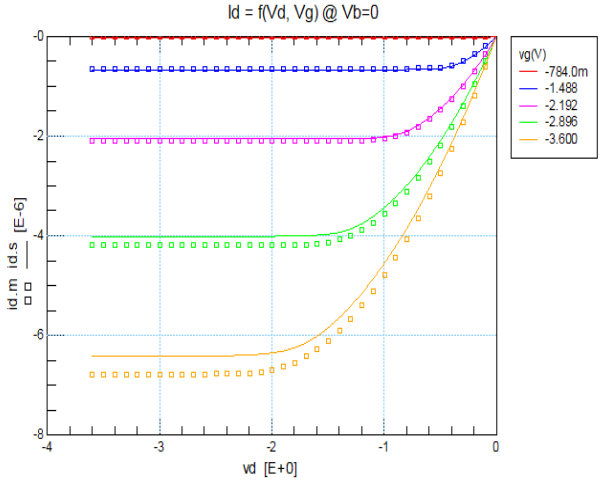


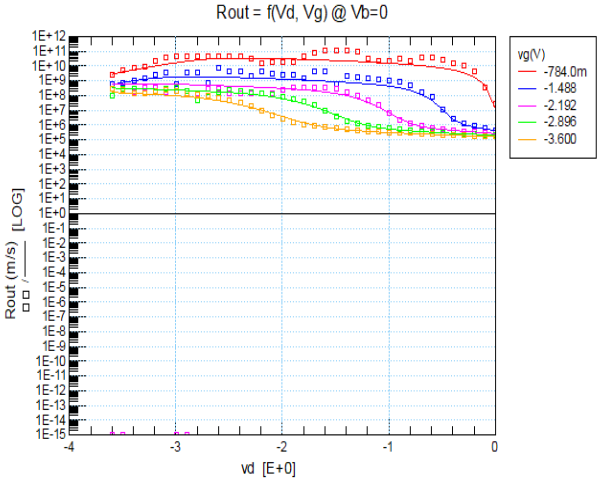
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

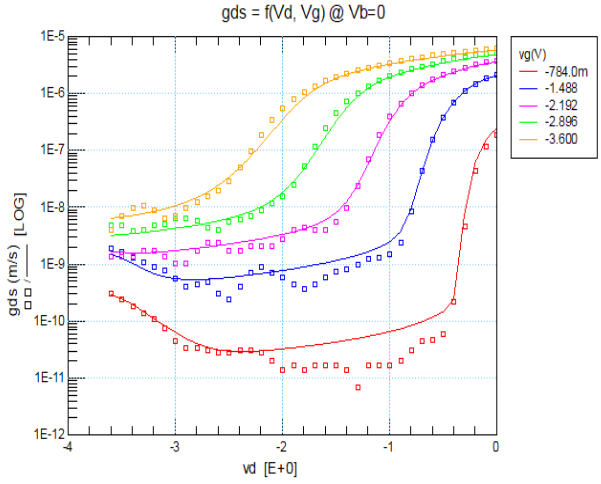




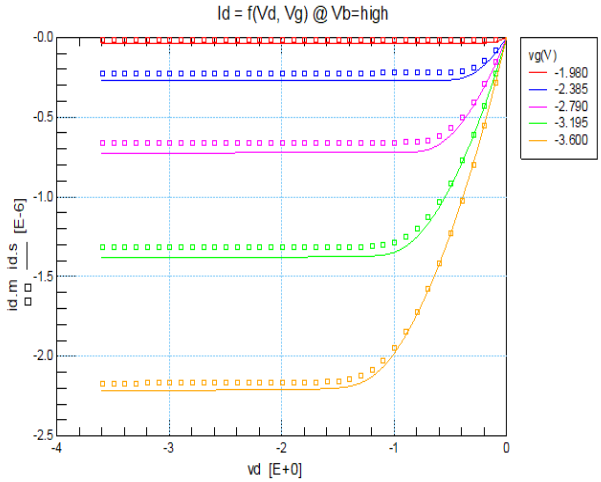
idvd, Vb = 0V, T = 27°C

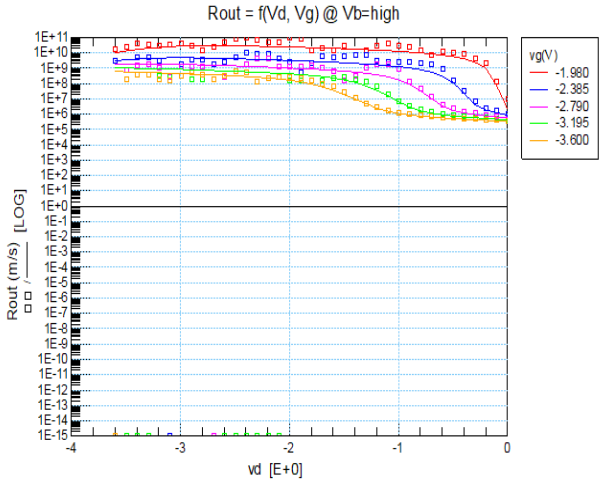


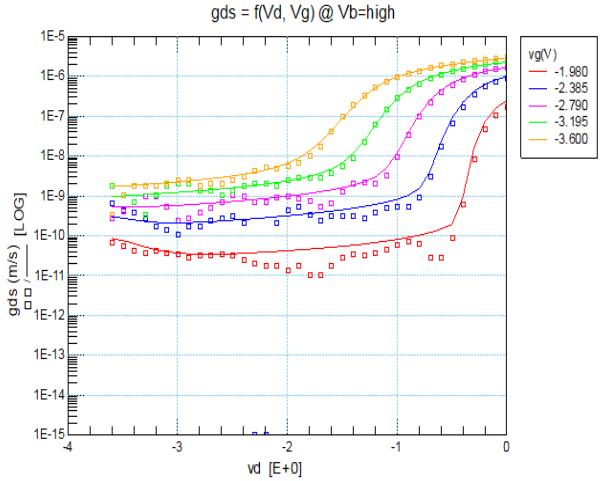




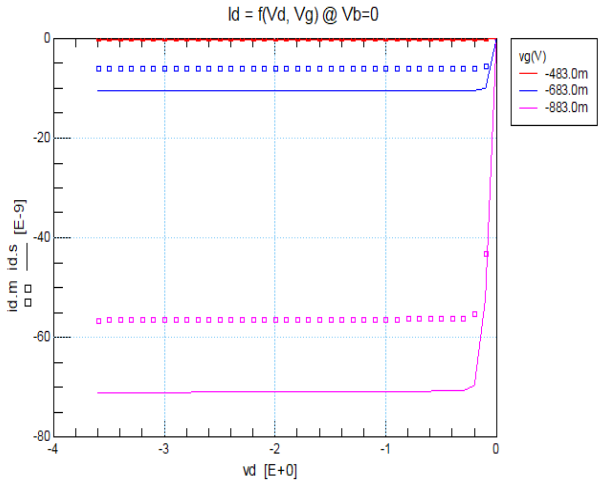
idvd_vbmin, Vb = 3.6V, T = 27°C

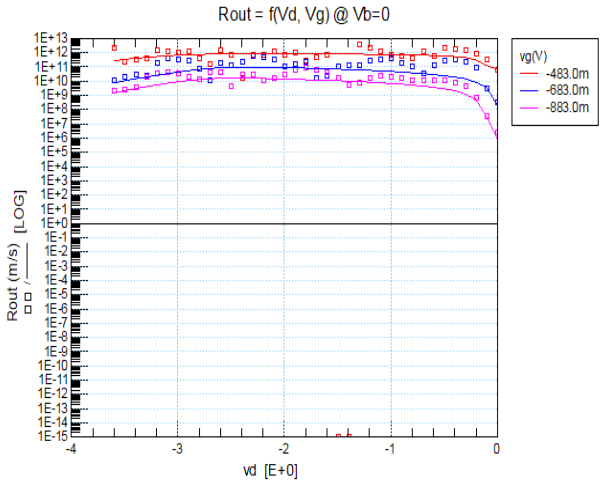


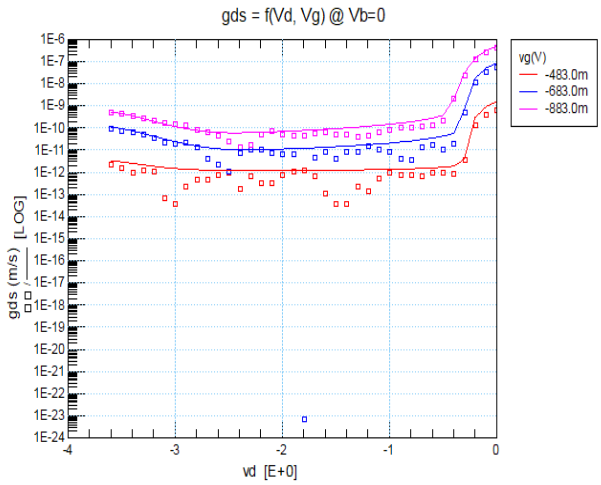




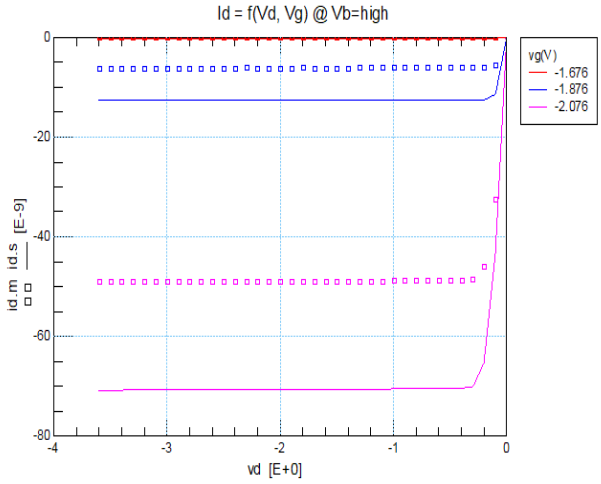
idvd_vth, Vb = 0V, T = 27°C

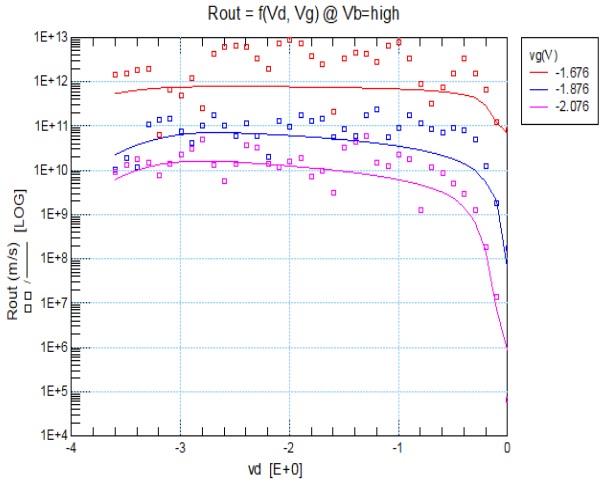


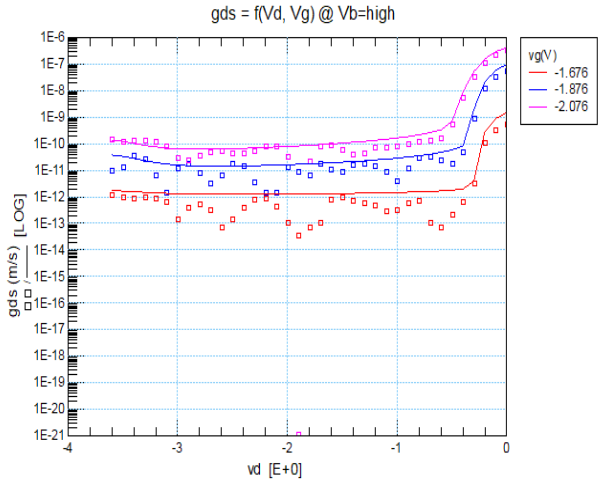




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

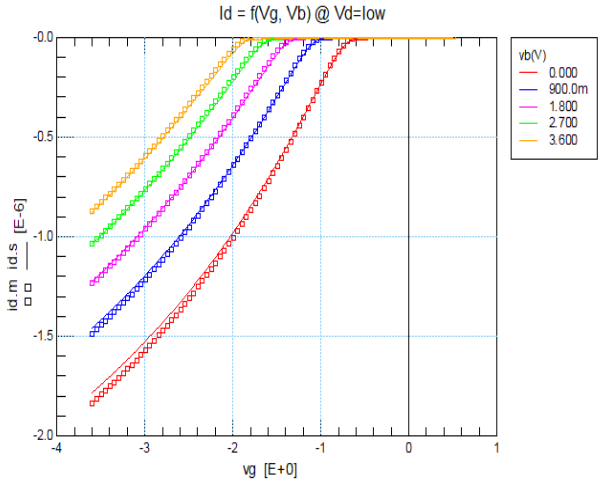


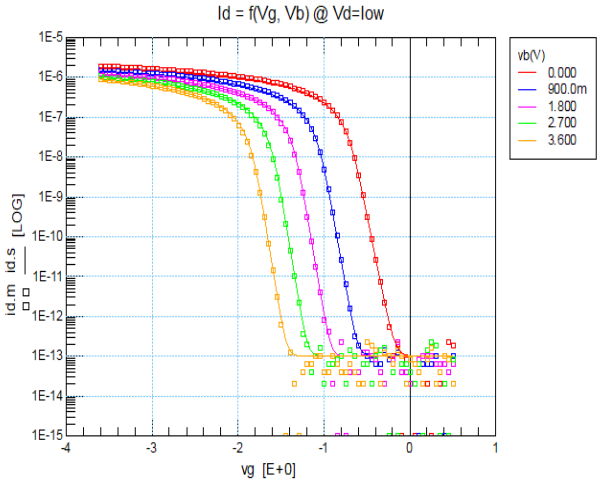


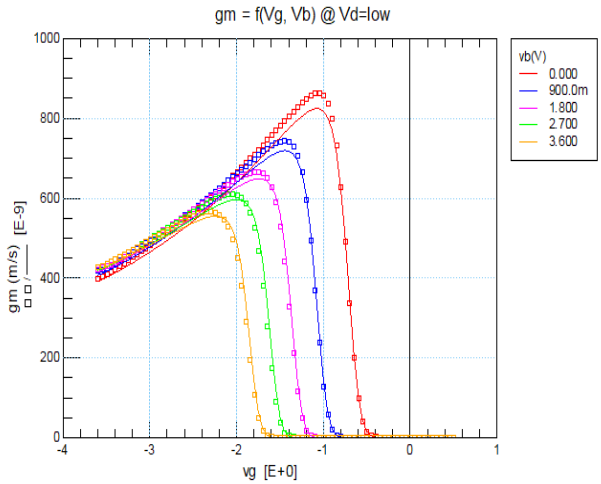


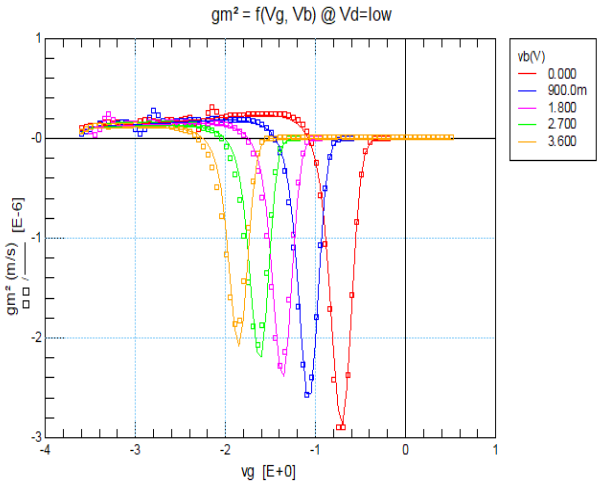
4.12 W02u0_L10u0_S563_5

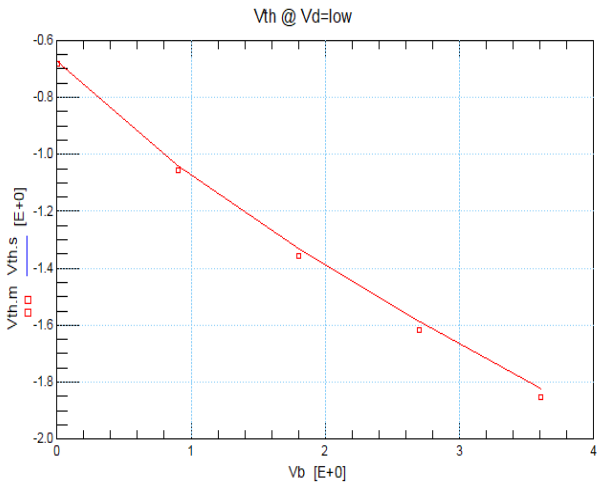
idvg, Vd = -0.1V, T = 27°C

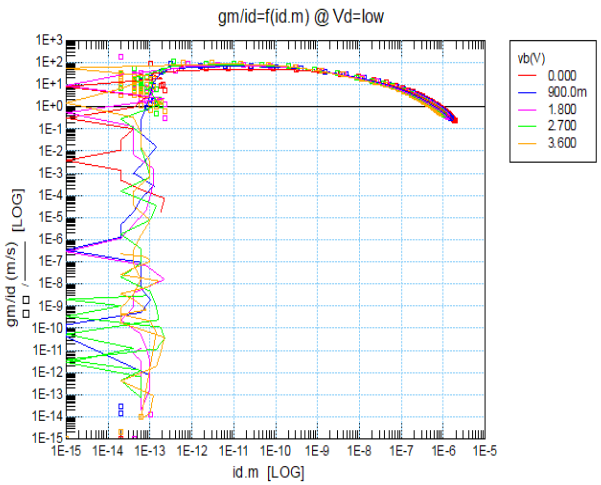


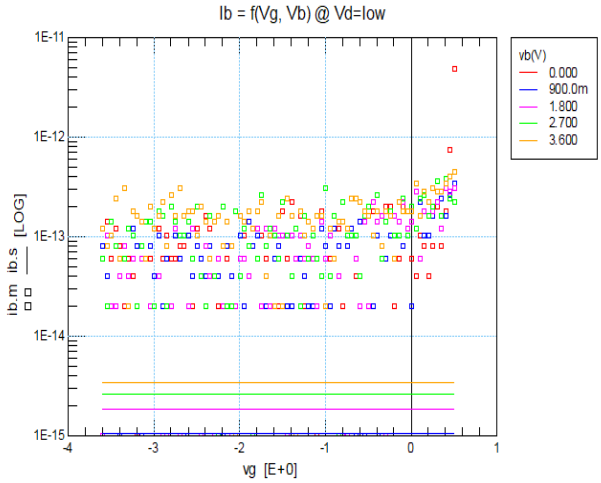


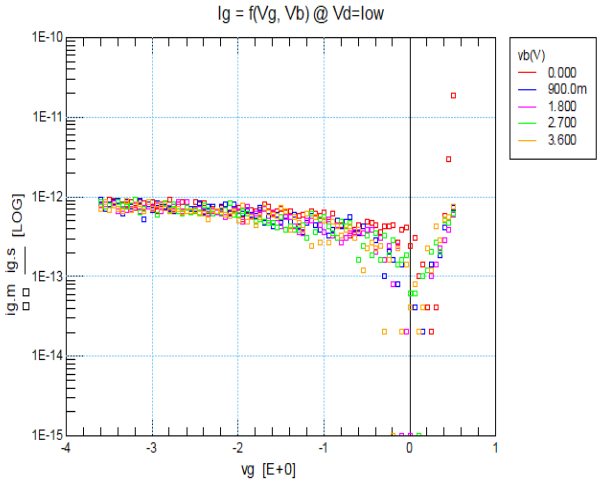




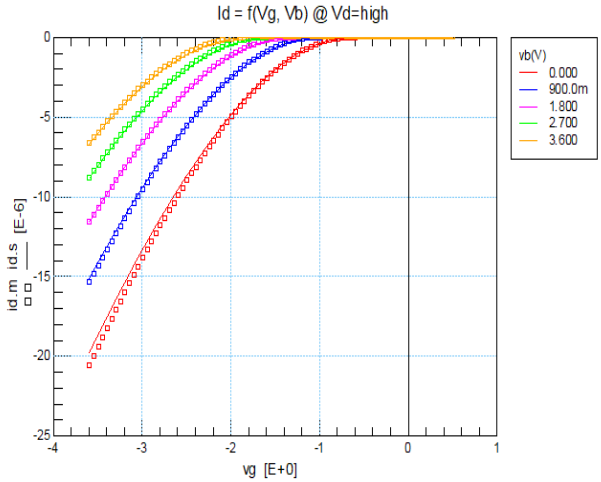


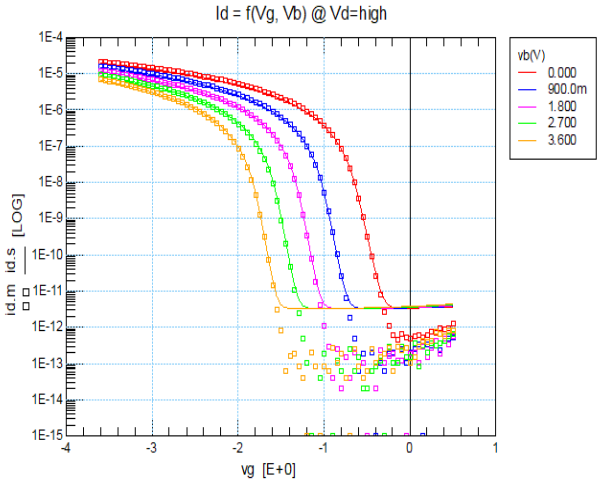


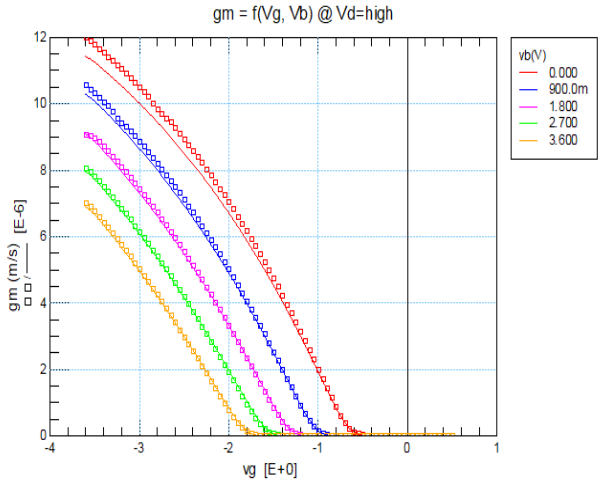


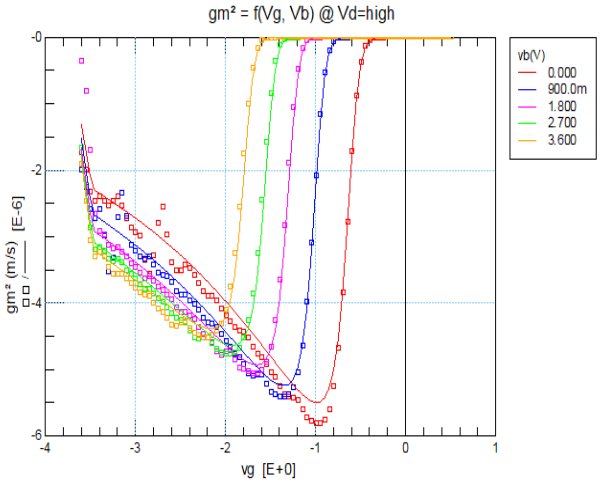


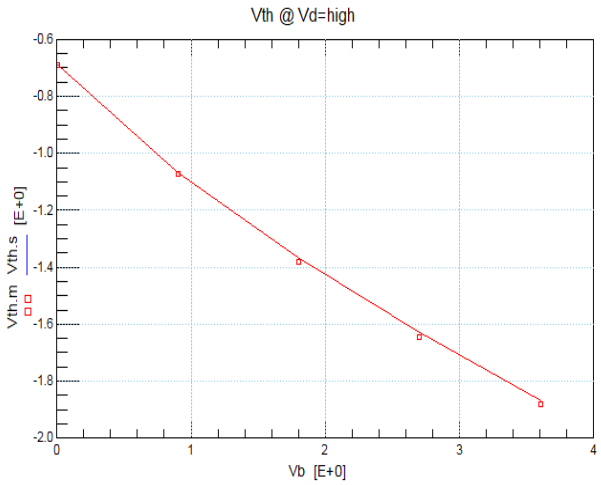
idvg, Vd = -3.3V, T = 27°C

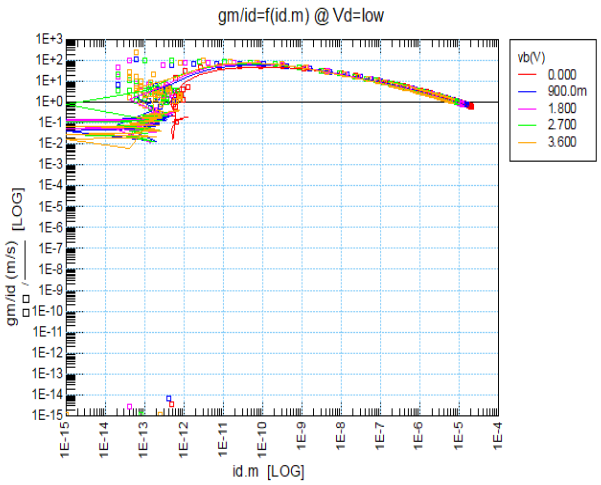


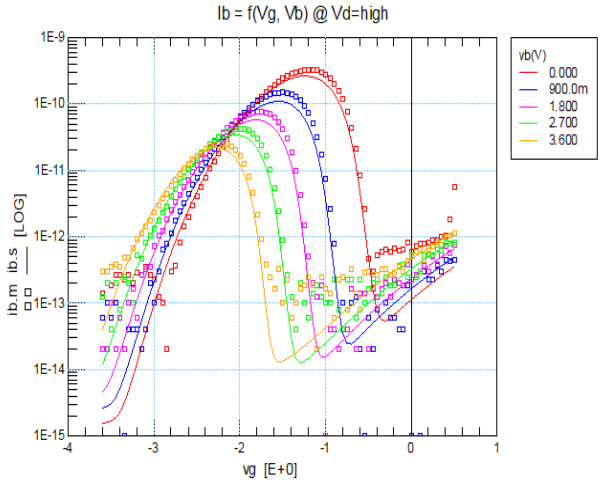


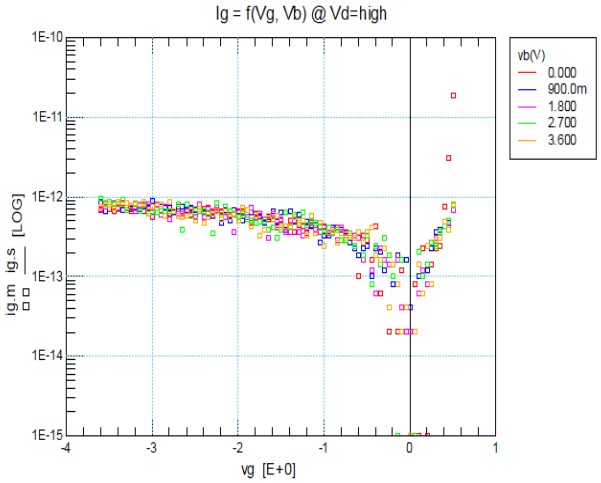




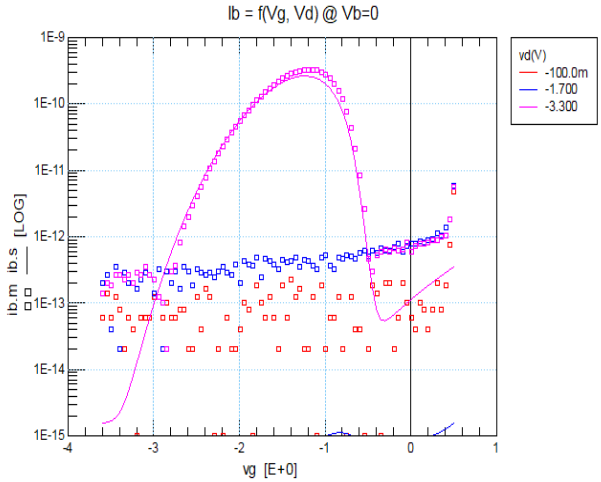


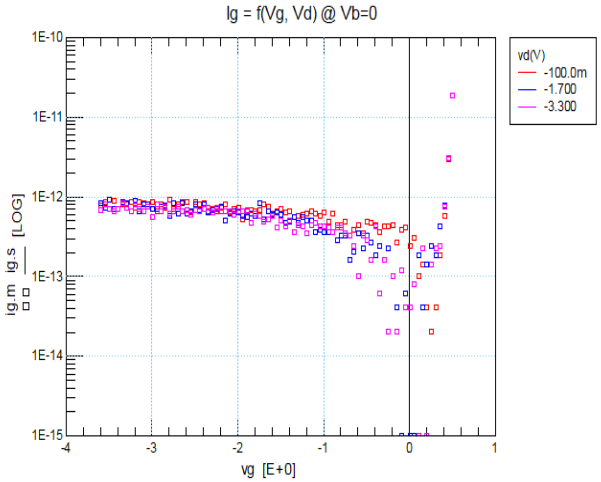




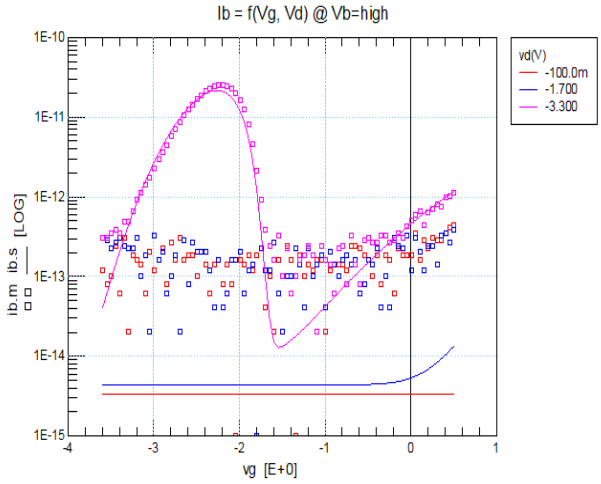


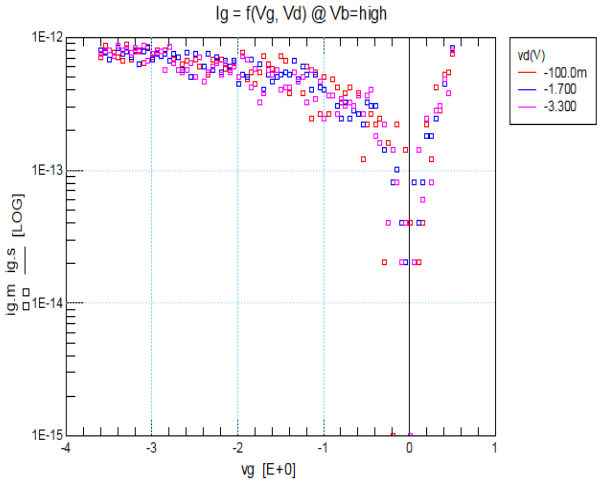
idvg, Ib,Ig, Vb = 0V, T = 27°C



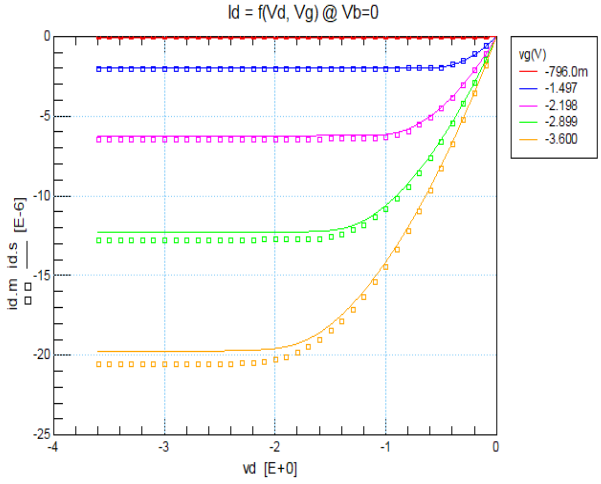


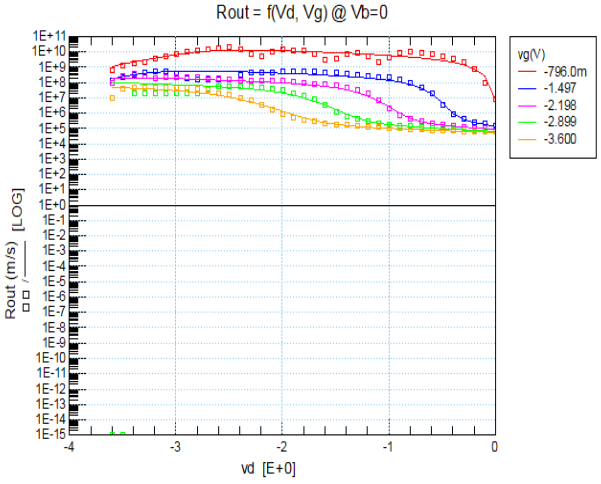
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

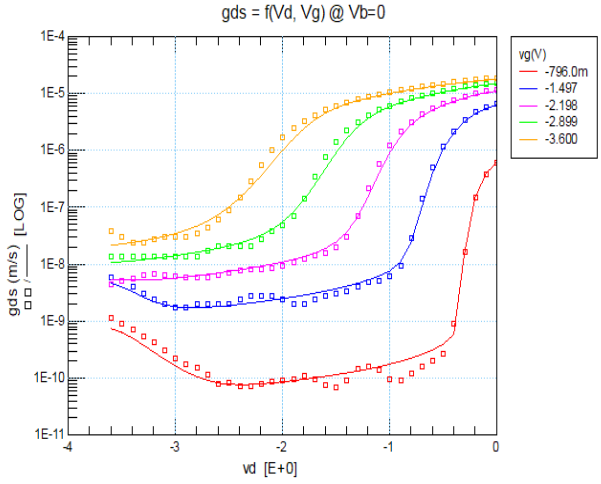




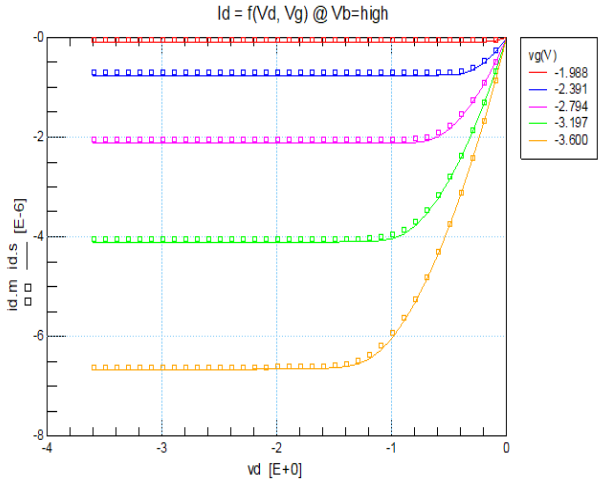
idvd, Vb = 0V, T = 27°C

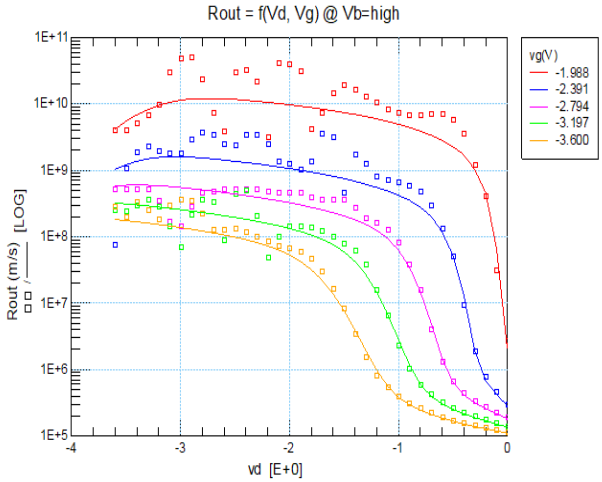


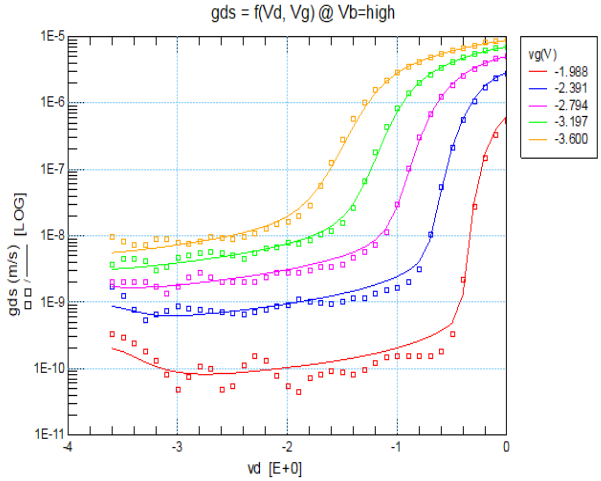




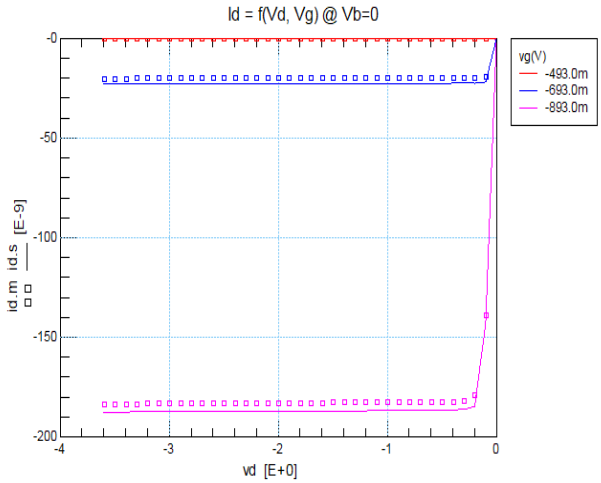
idvd_vbmin, Vb = 3.6V, T = 27°C

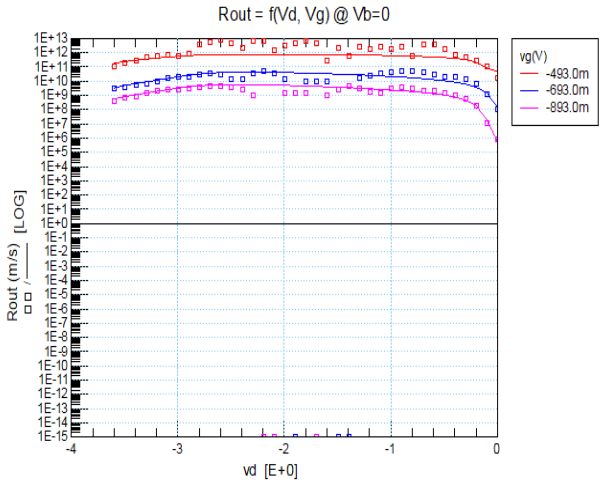


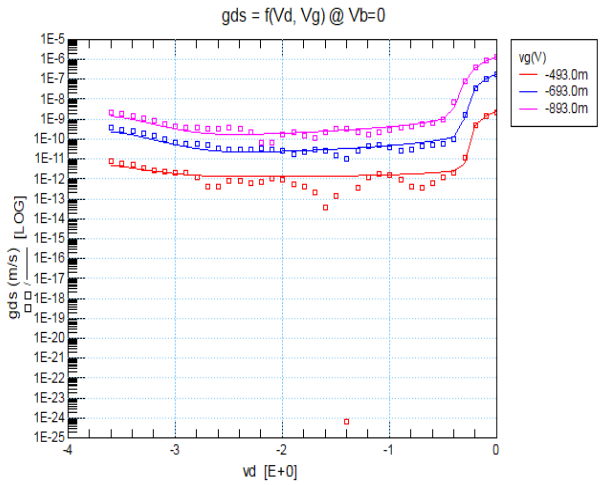




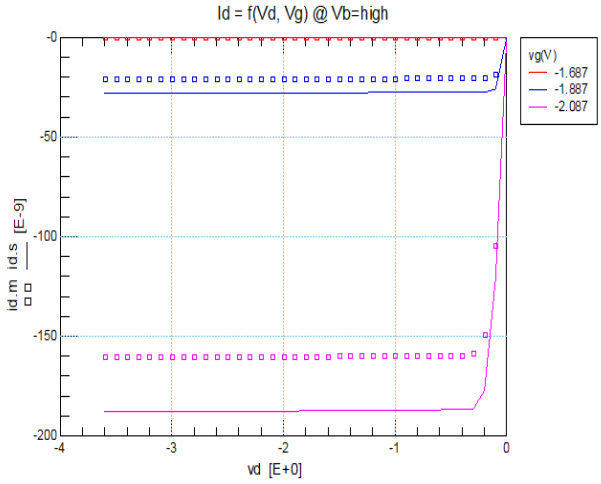
idvd_vth, Vb = 0V, T = 27°C

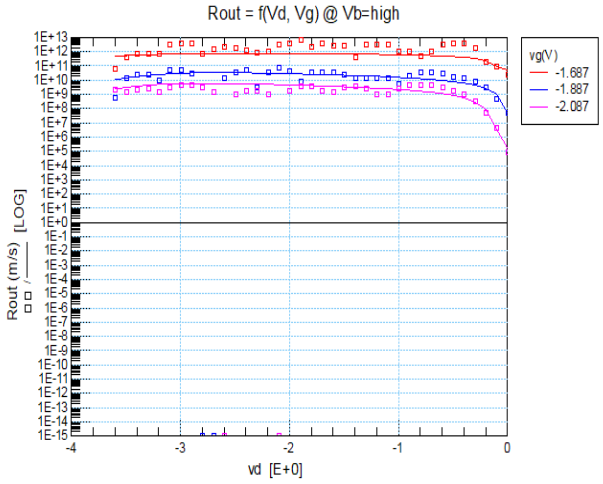


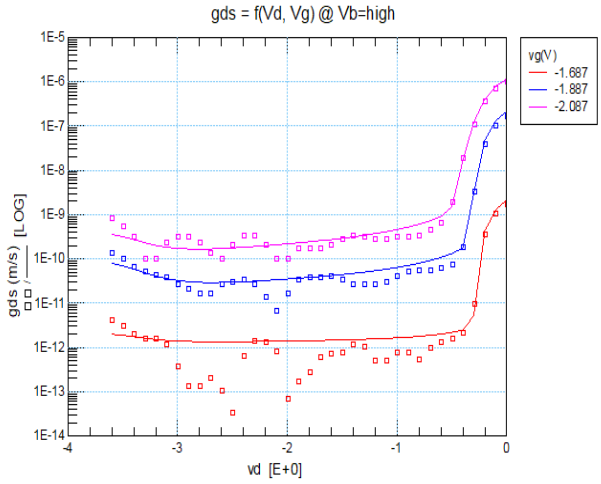




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

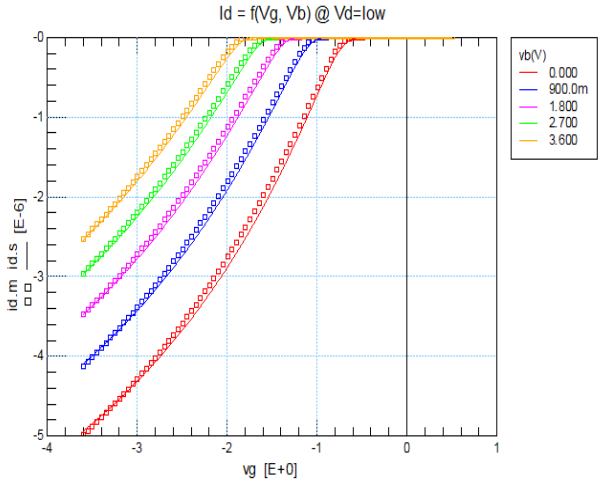


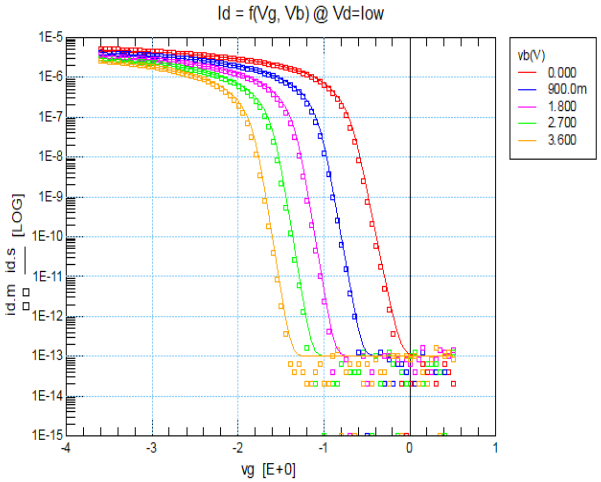


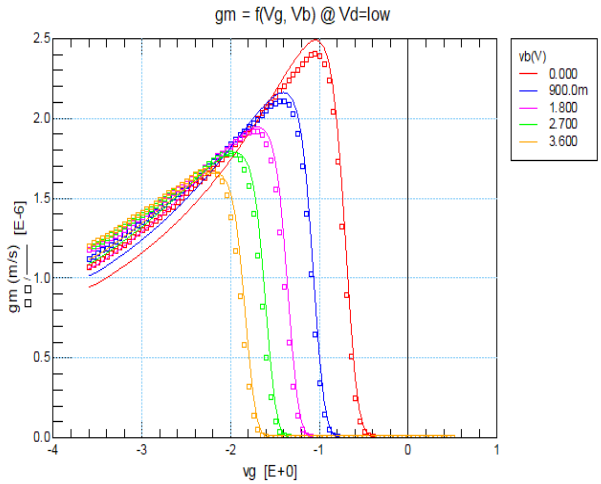


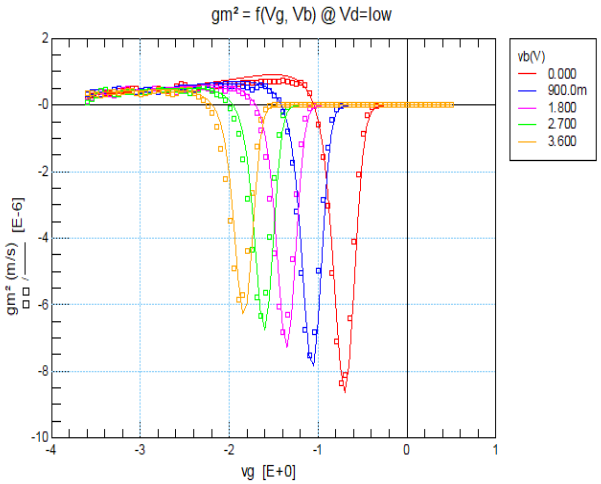
4.13 W0u6_L1u2_S564_4

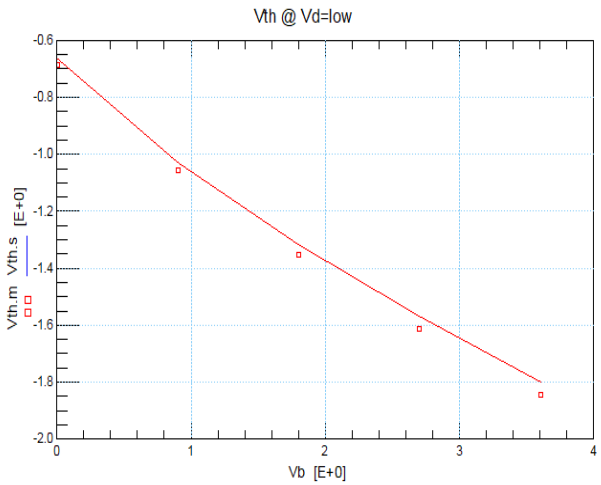
idvg, Vd = -0.1V, T = 27°C

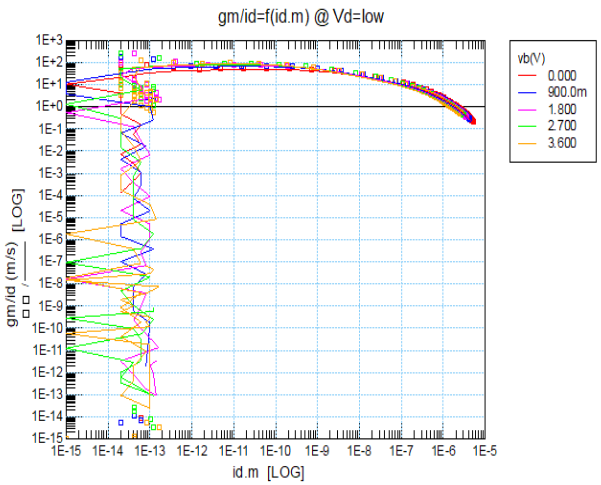


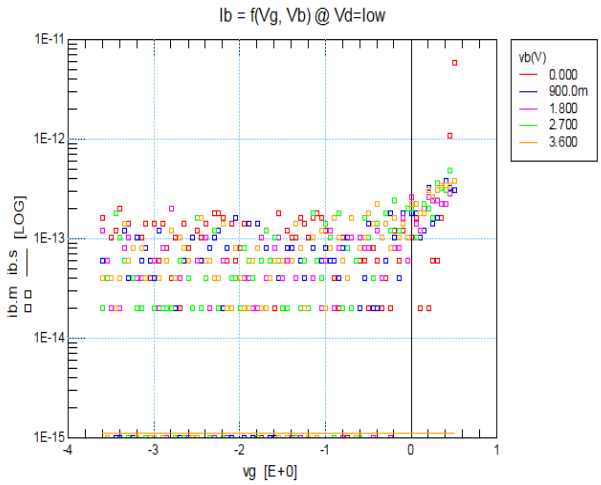


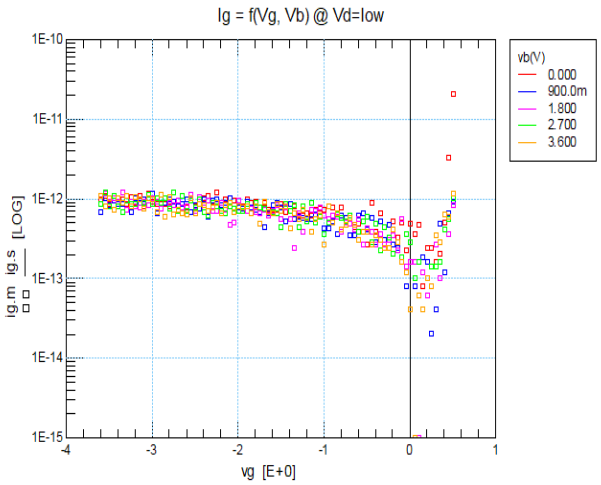




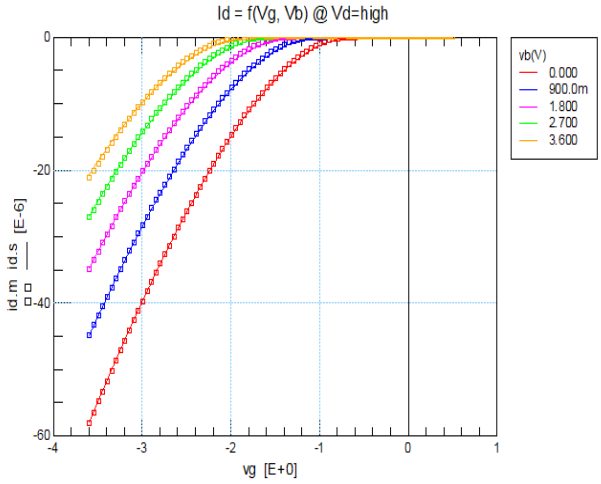


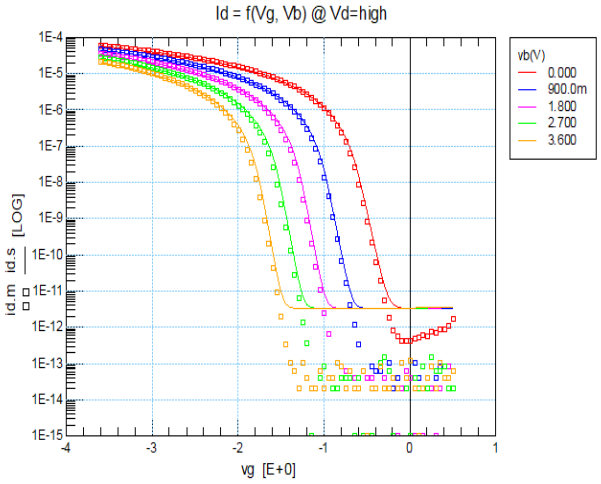


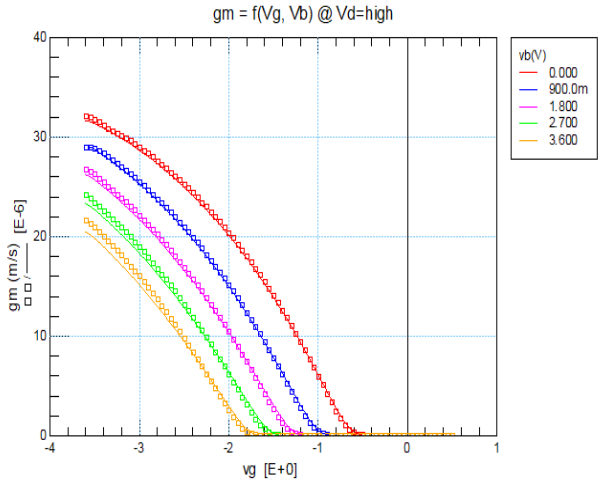


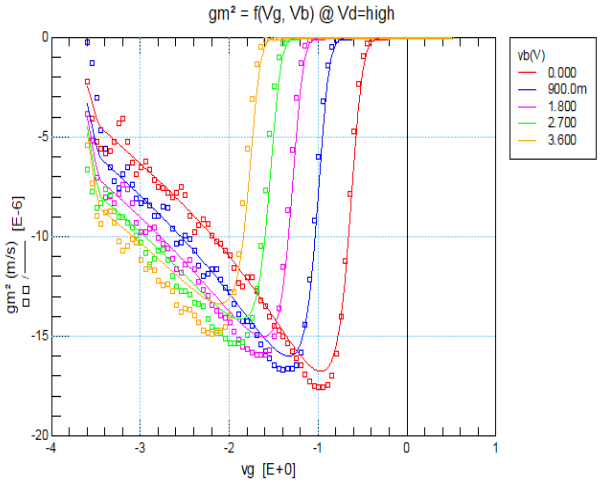


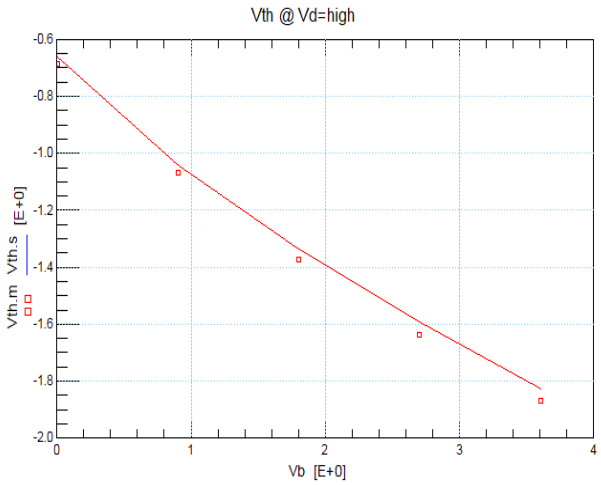
idvg, Vd = -3.3V, T = 27°C

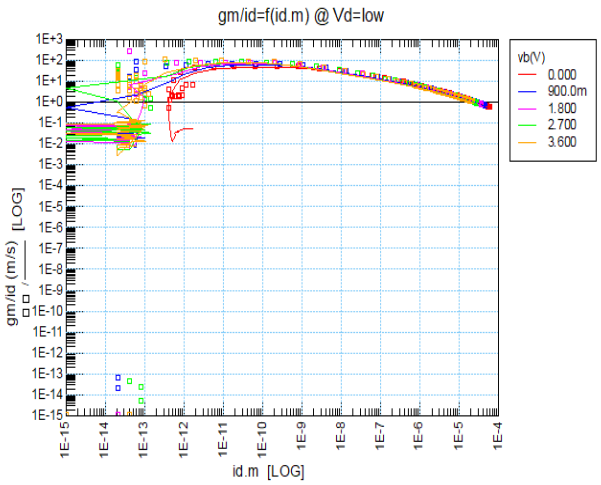


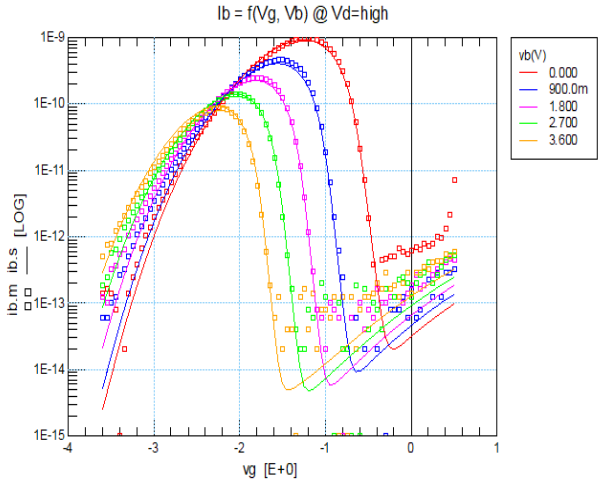


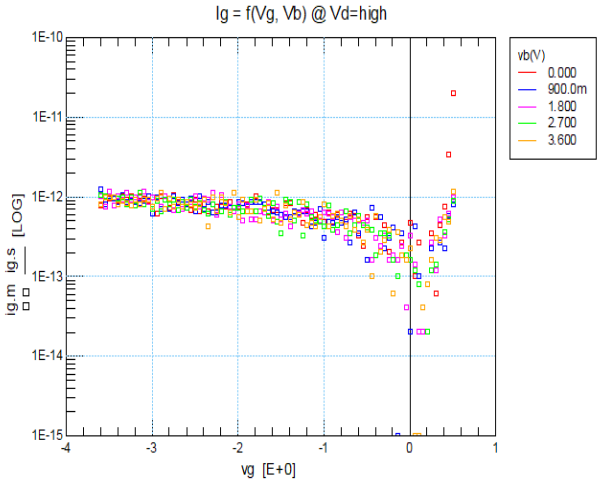




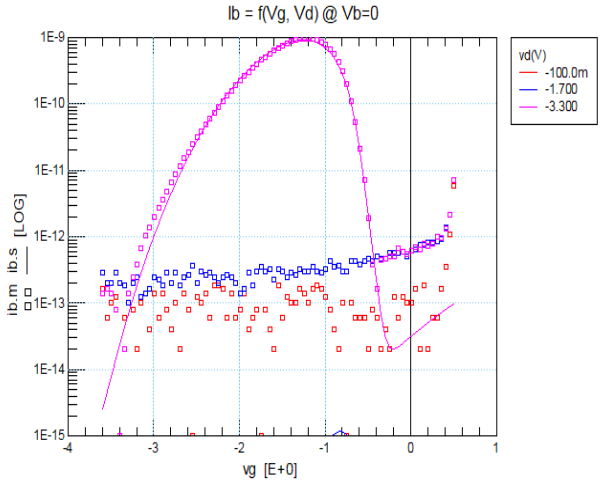


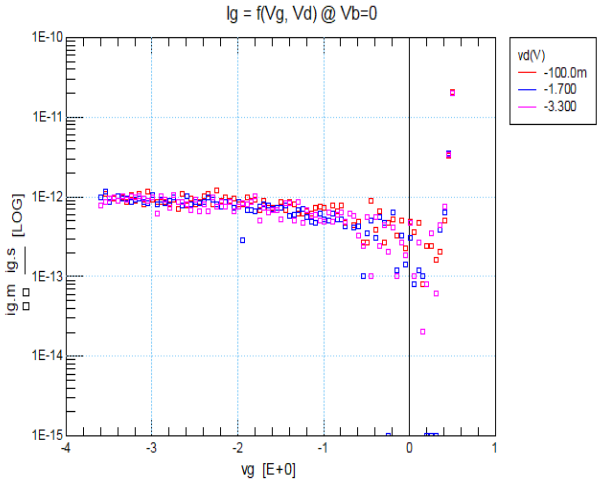




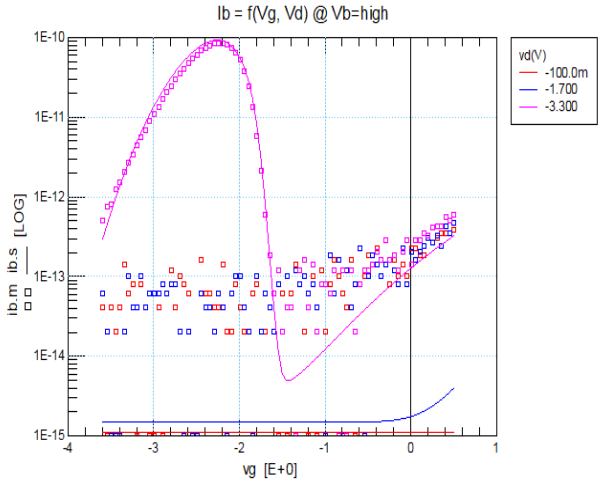


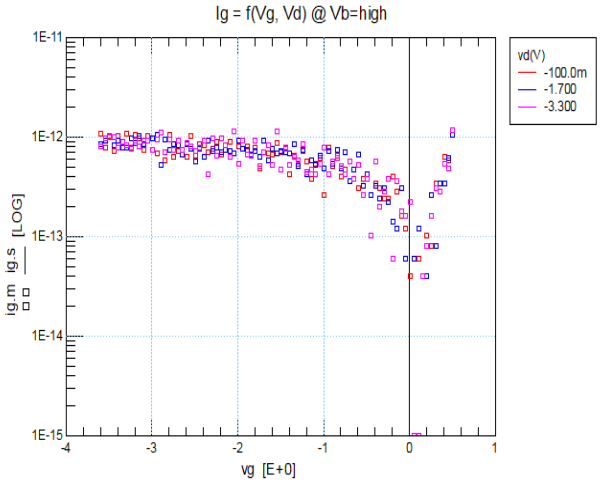
idvg, Ib,Ig, Vb = 0V, T = 27°C



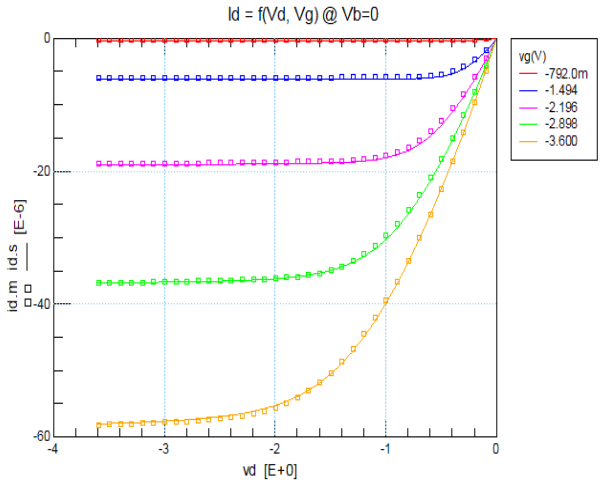


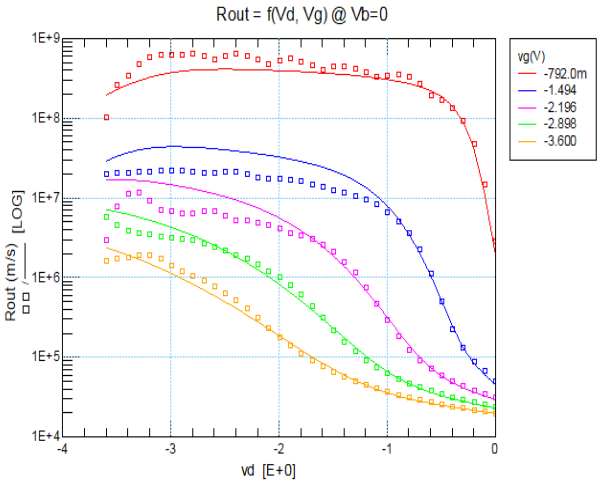
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

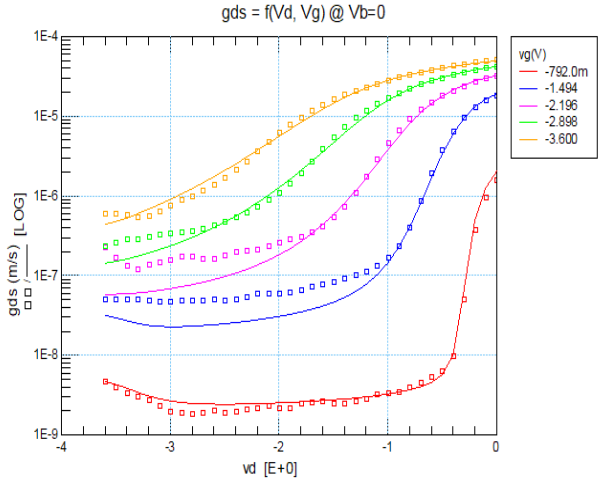




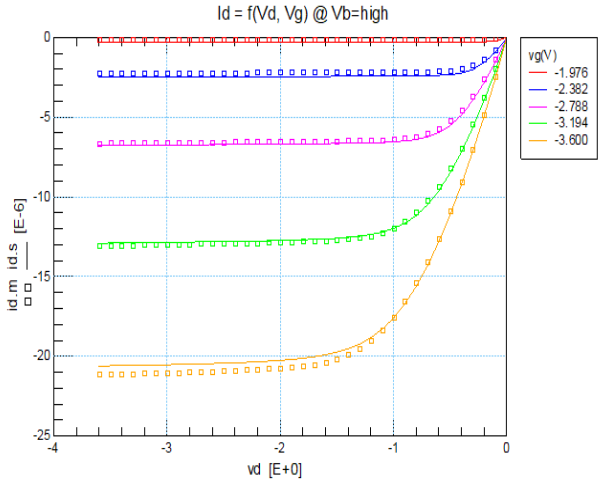
idvd, Vb = 0V, T = 27°C

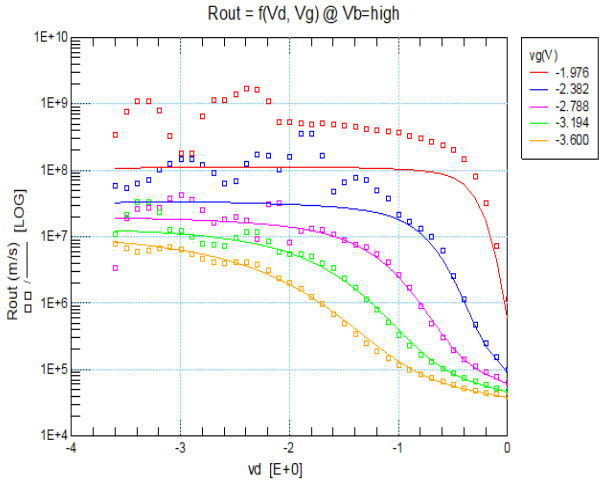


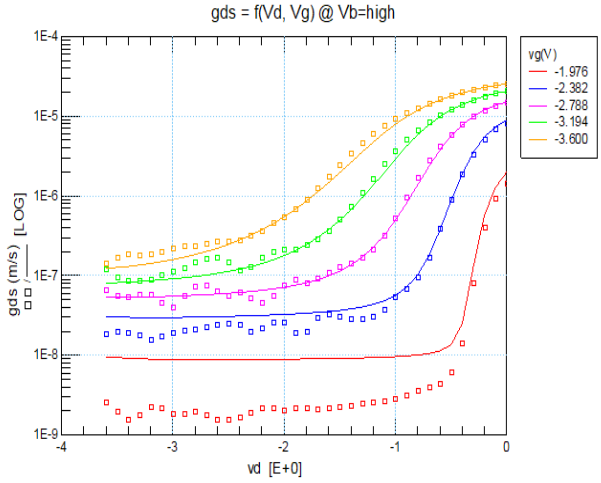




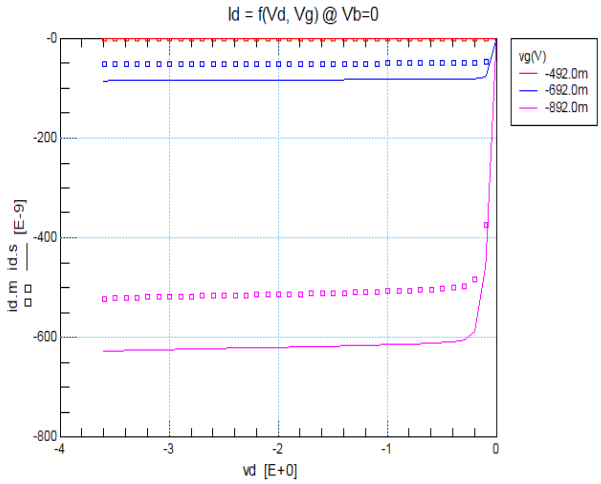
idvd_vbmin, Vb = 3.6V, T = 27°C

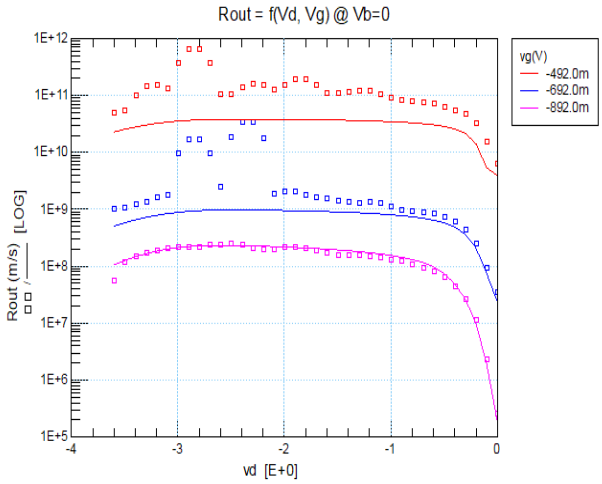


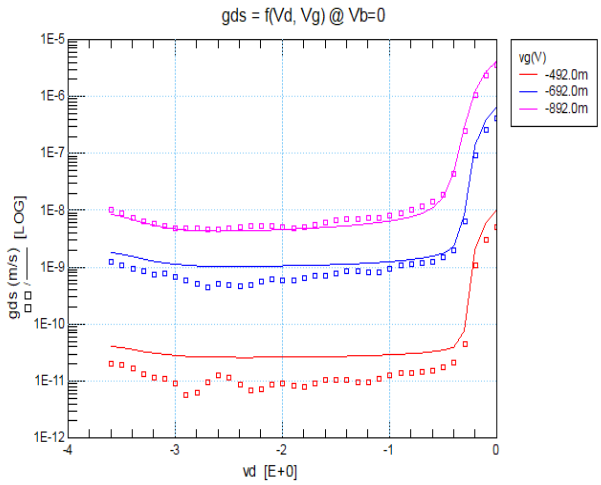




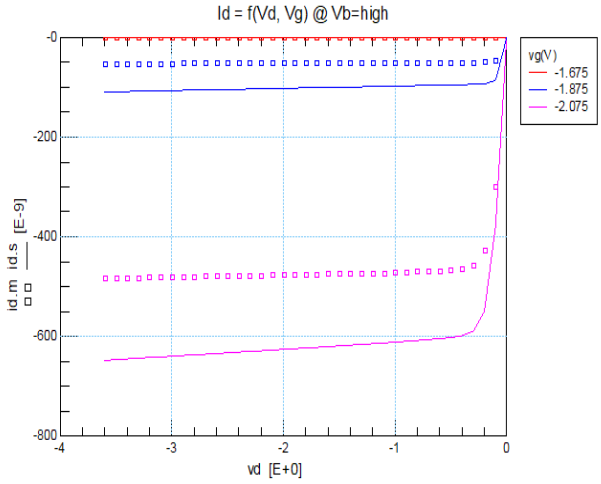
idvd_vth, Vb = 0V, T = 27°C

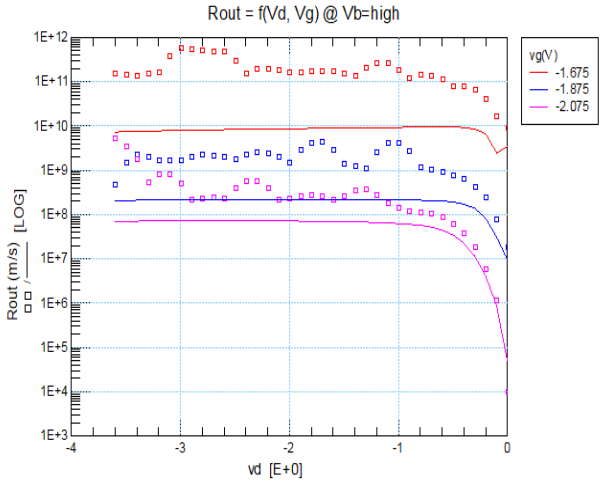


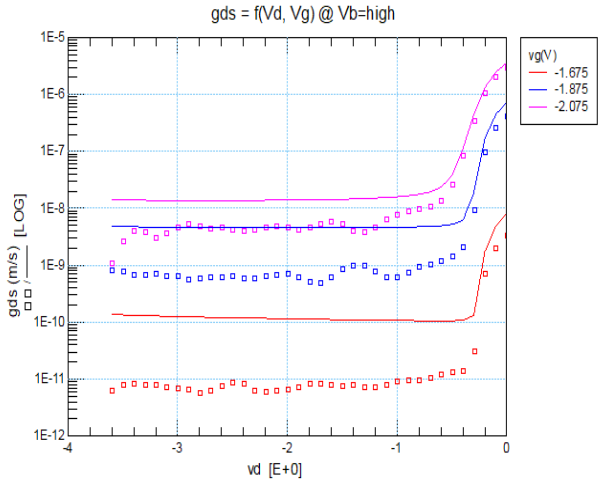




idvd_vth_vbmin, Vb = 3.6V, T = 27°C



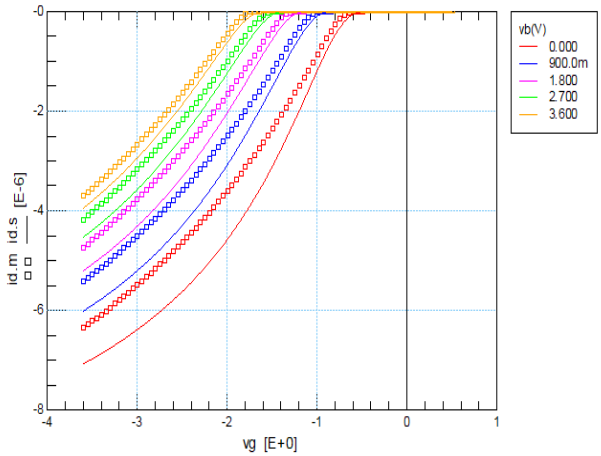




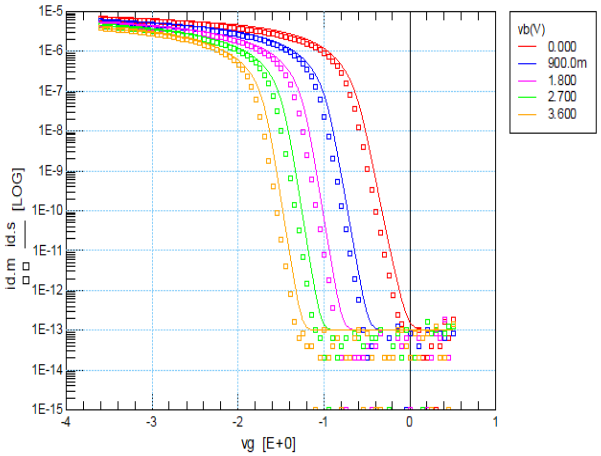
4.14 W0u3_L0u5_S565_4

idvg, Vd = -0.1V, T = 27°C

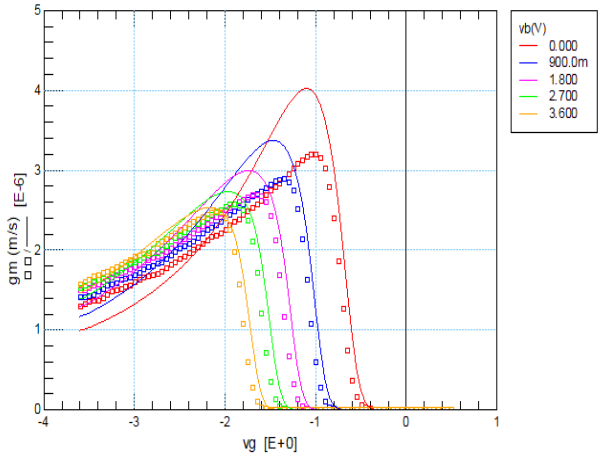
Id = f(Vg, Vb) @ Vd=low



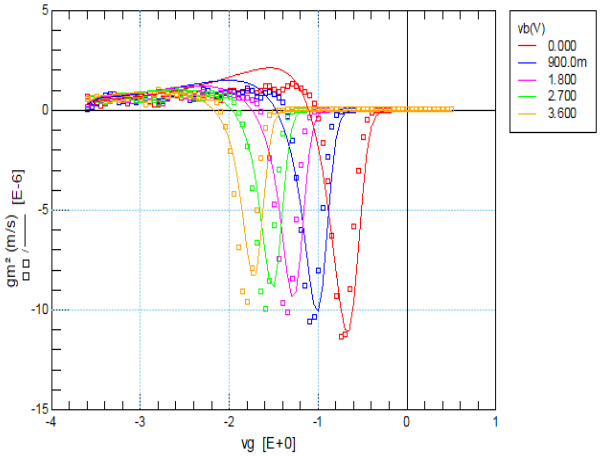
Id = f(Vg, Vb) @ Vd=low



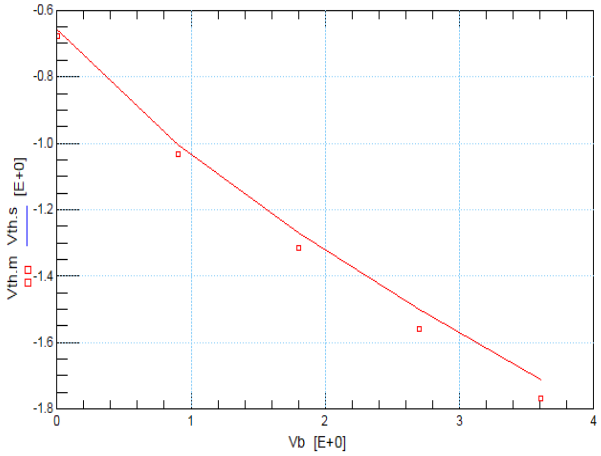
gm = f(Vg, Vb) @ Vd=low



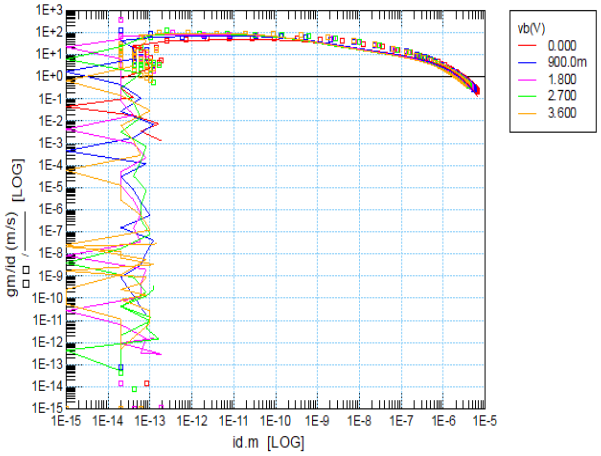
gm² = f(Vg, Vb) @ Vd=low

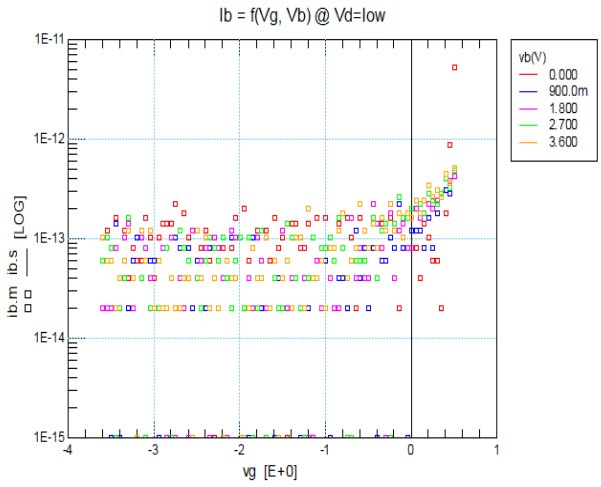


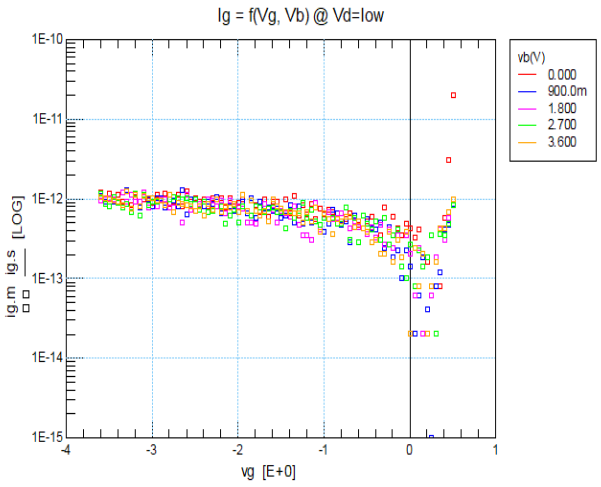
Vth @ Vd=low



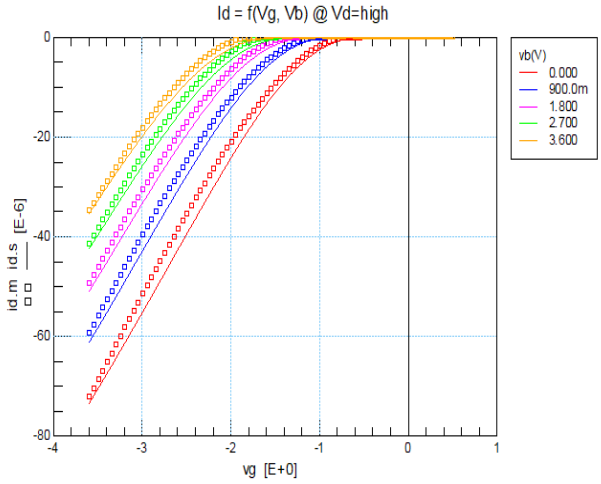
gm/id=f(id.m) @ Vd=low

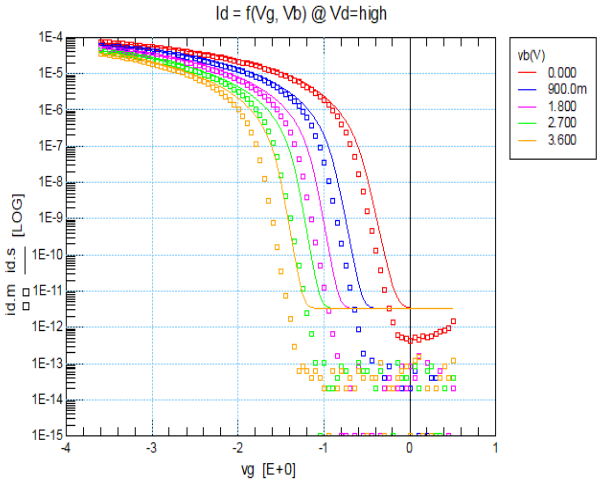


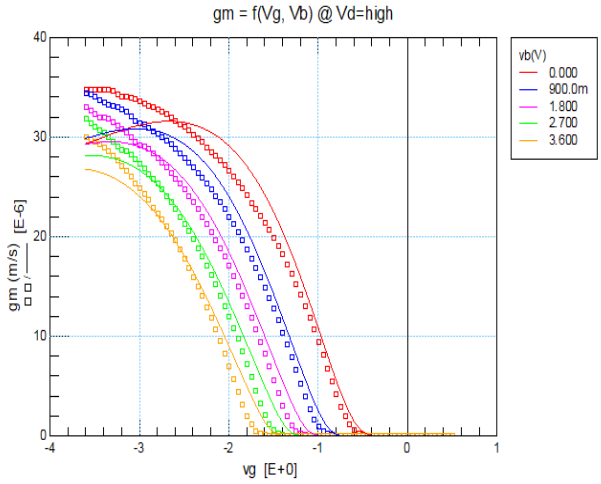


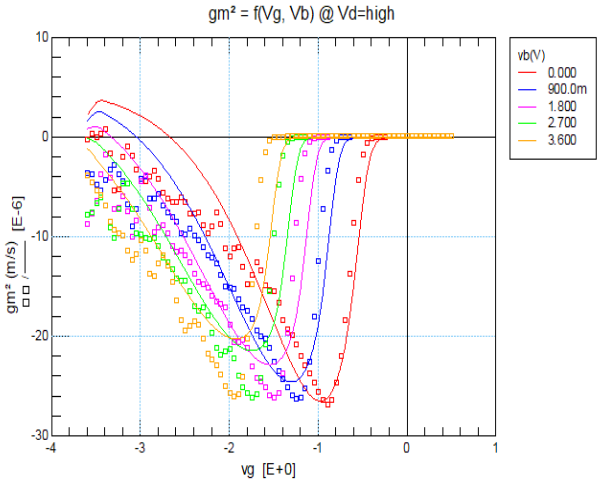


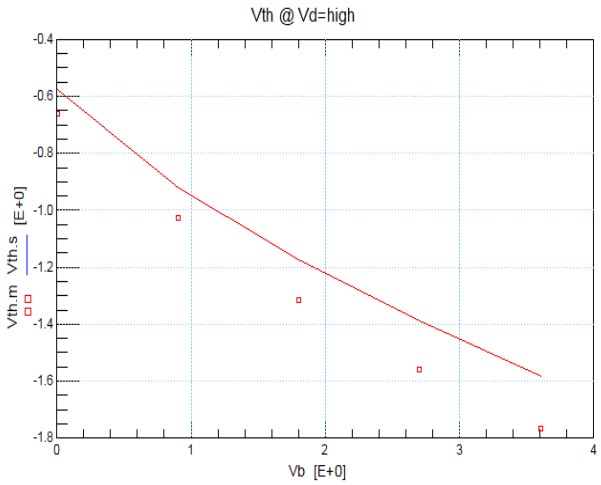
idvg, Vd = -3.3V, T = 27°C

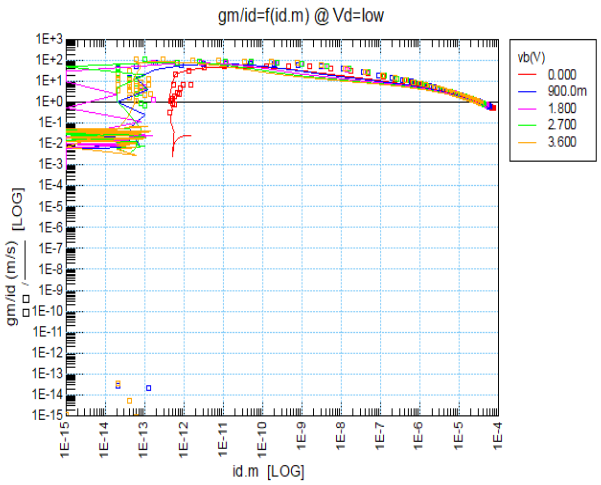


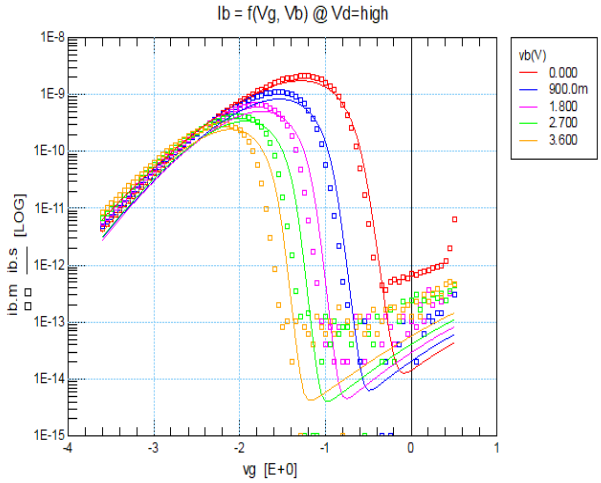


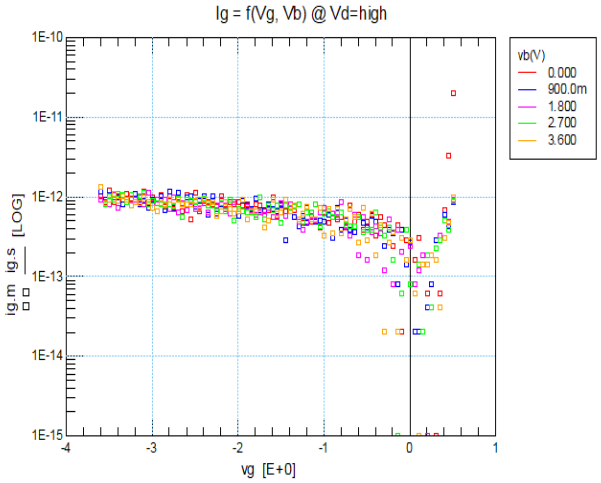




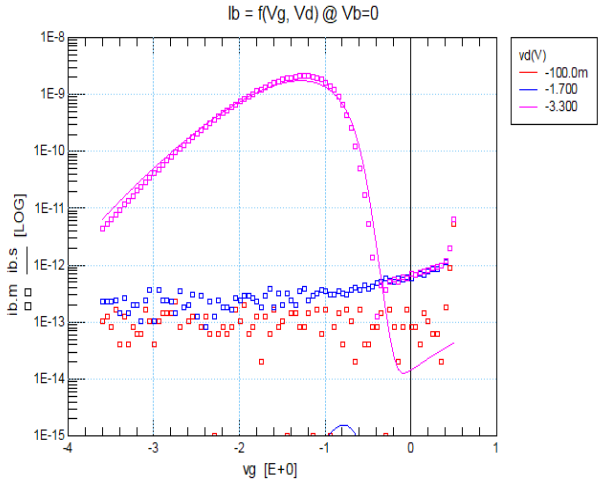


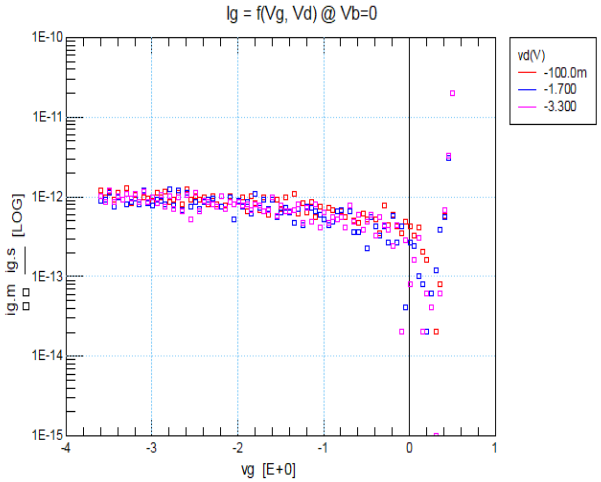




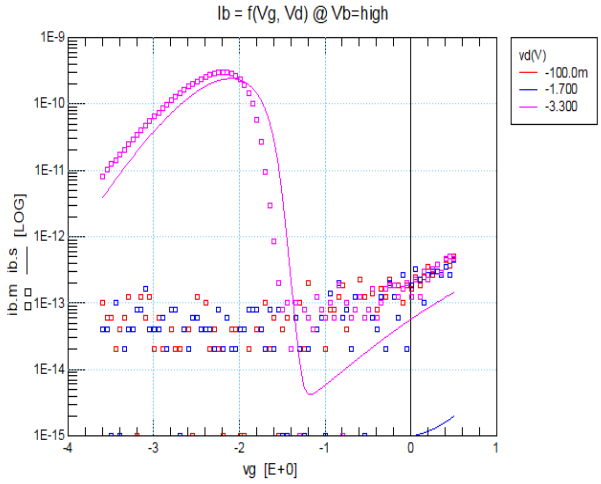


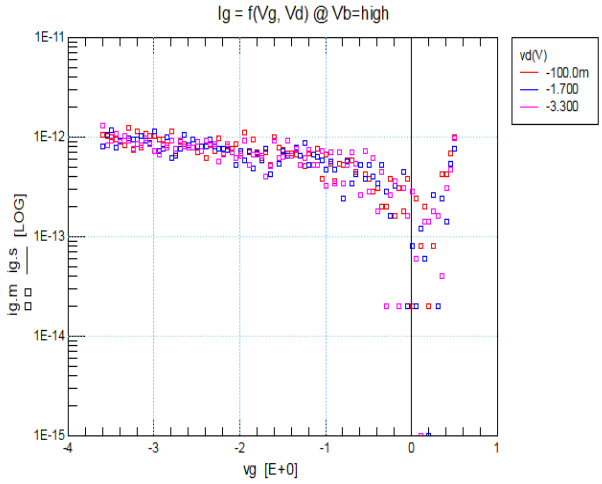
idvg, Ib,Ig, Vb = 0V, T = 27°C



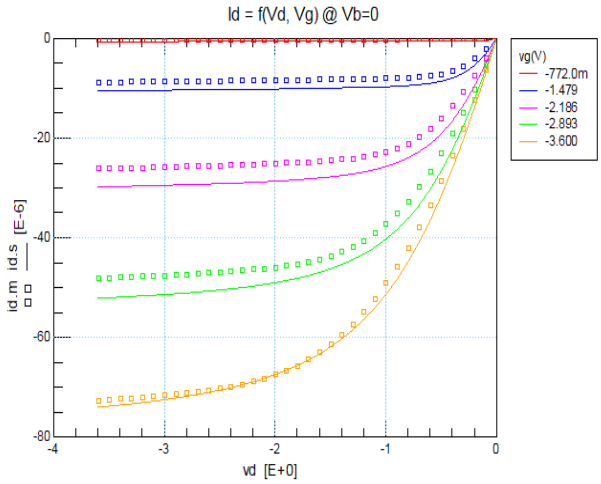


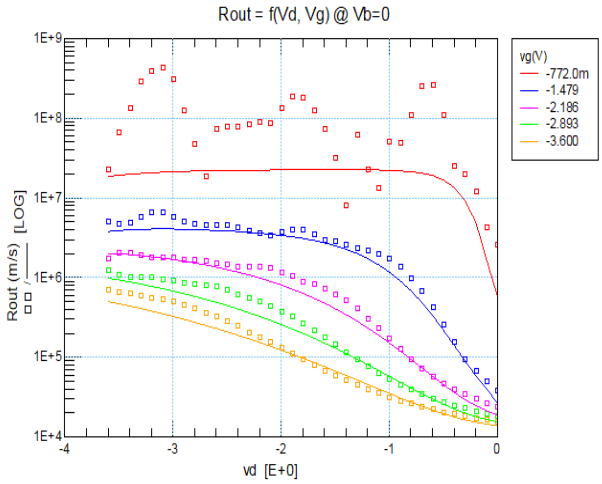
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

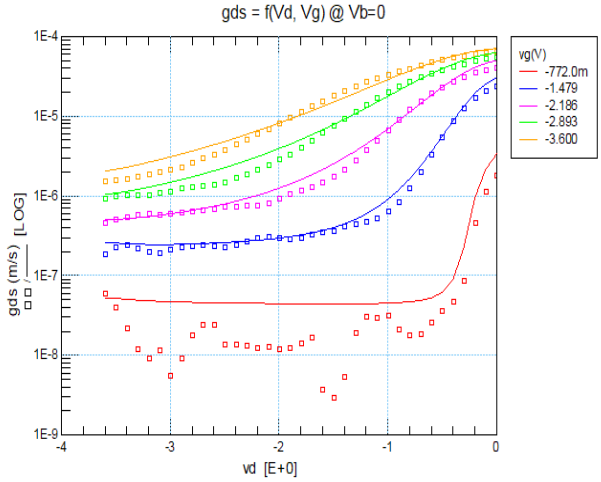




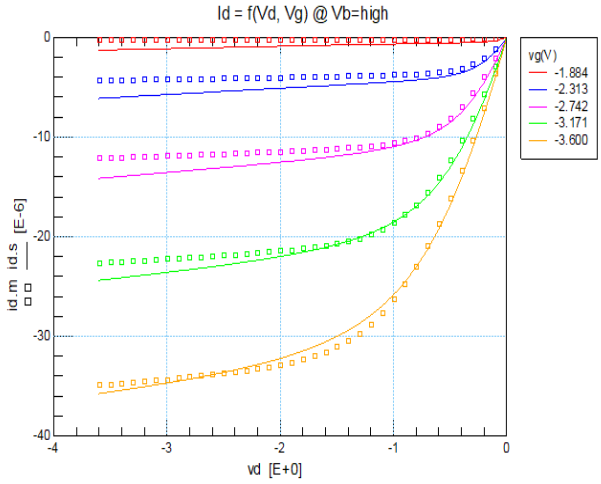
idvd, Vb = 0V, T = 27°C

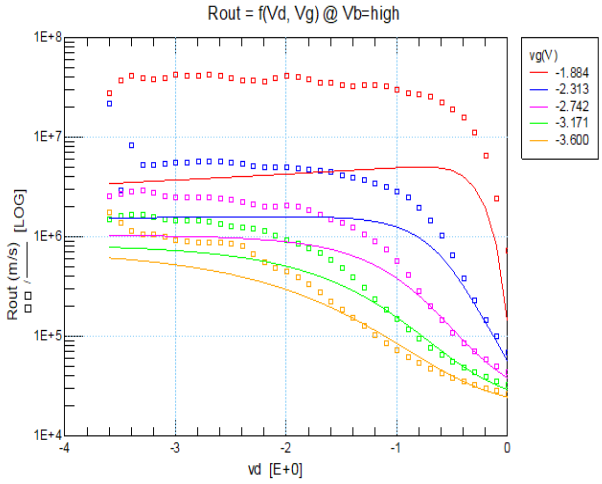


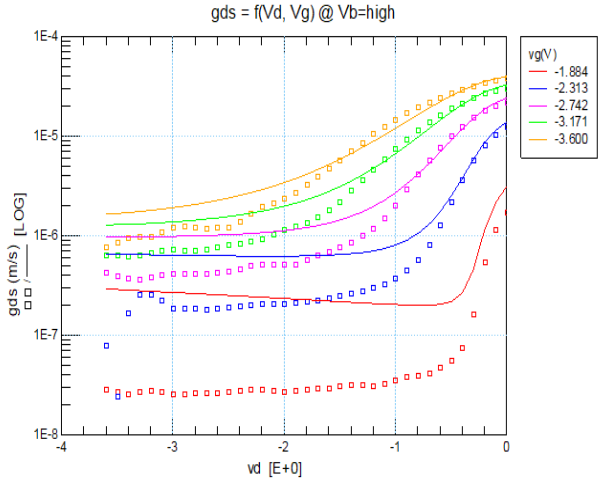




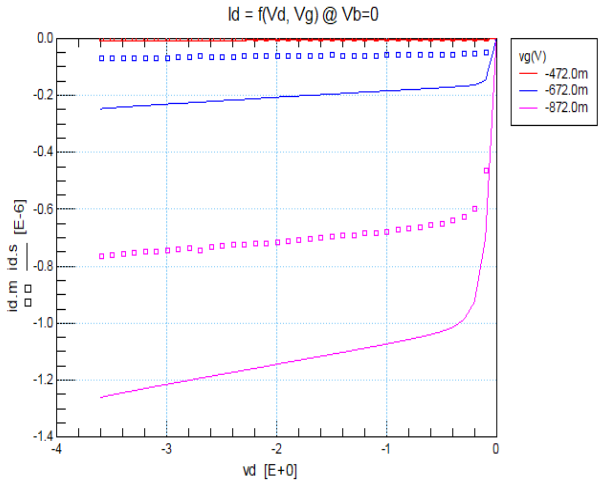
idvd_vbmin, Vb = 3.6V, T = 27°C

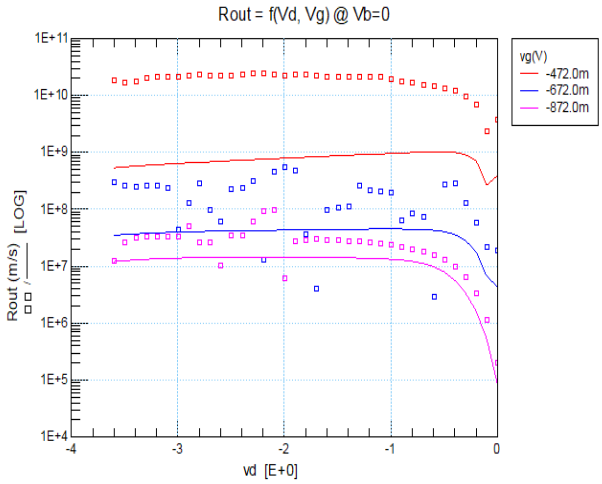


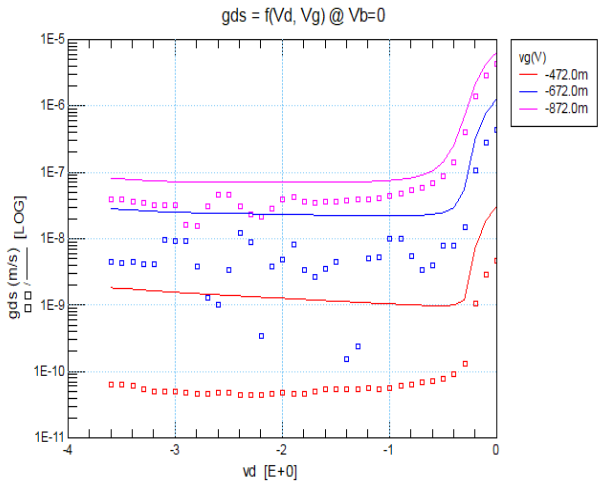




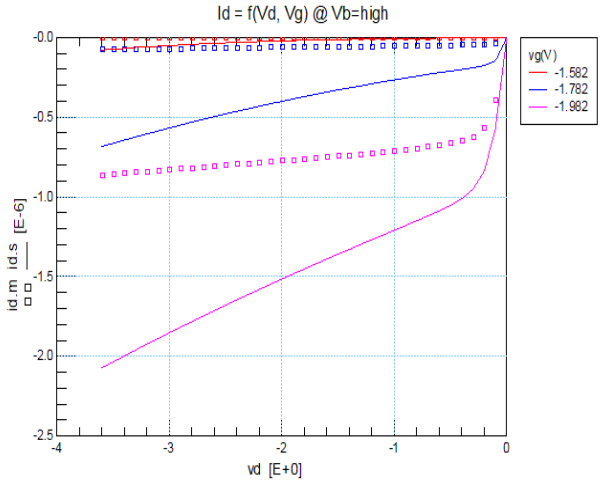
idvd_vth, Vb = 0V, T = 27°C

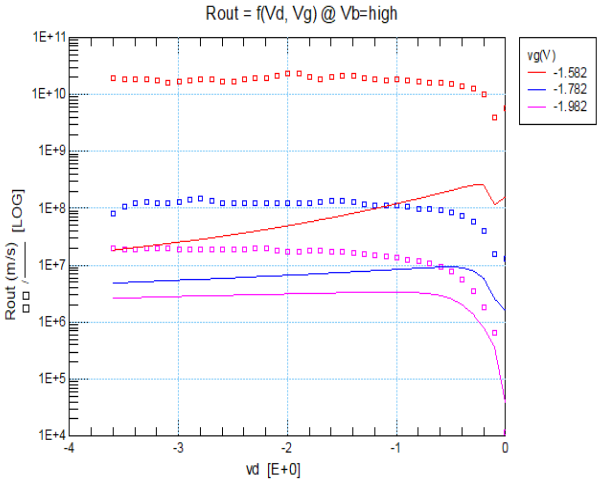


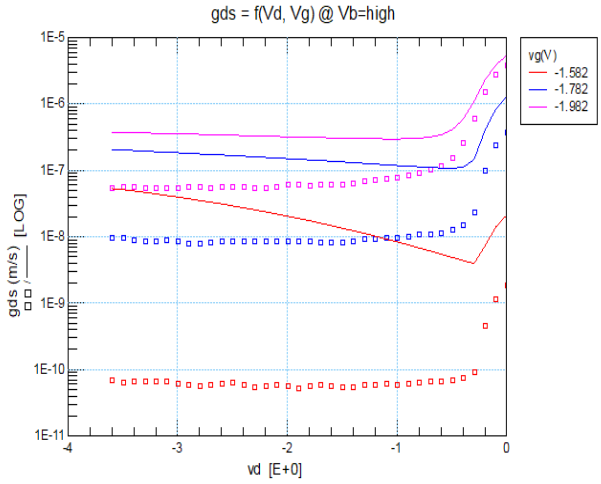




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

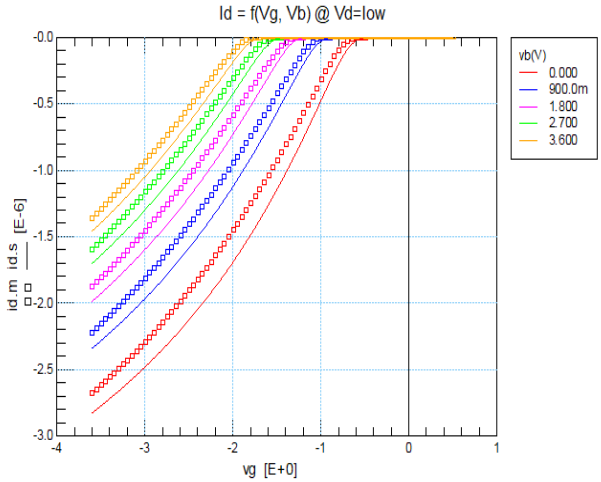


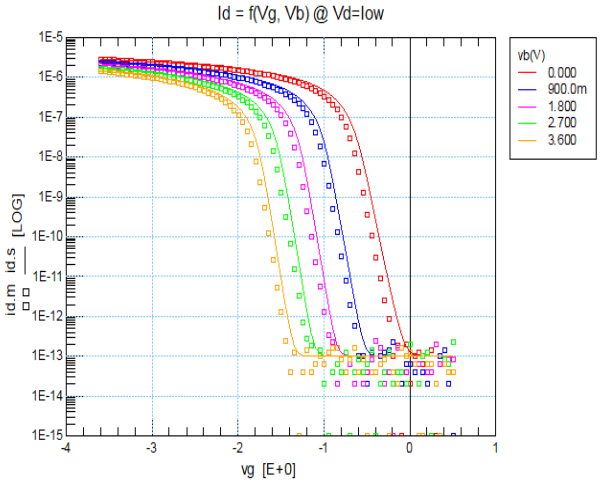


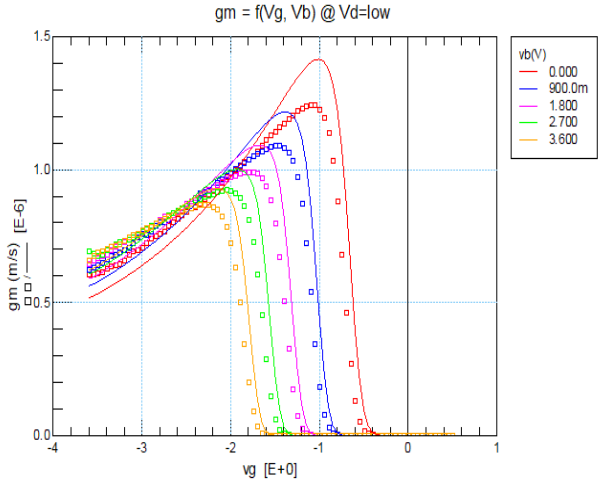


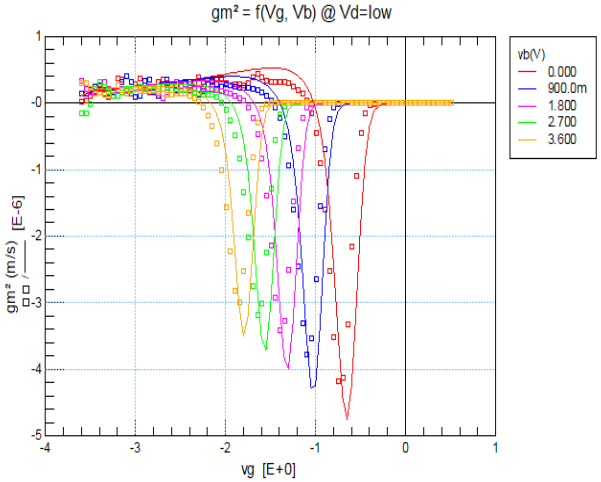
4.15W0u3_L1u2_S565_5

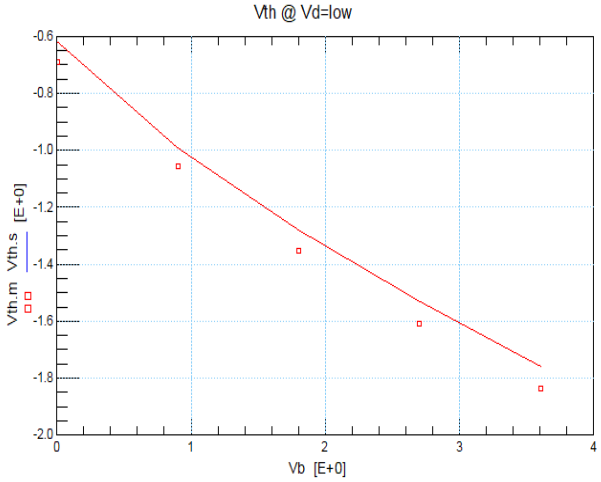
idvg, Vd = -0.1V, T = 27°C

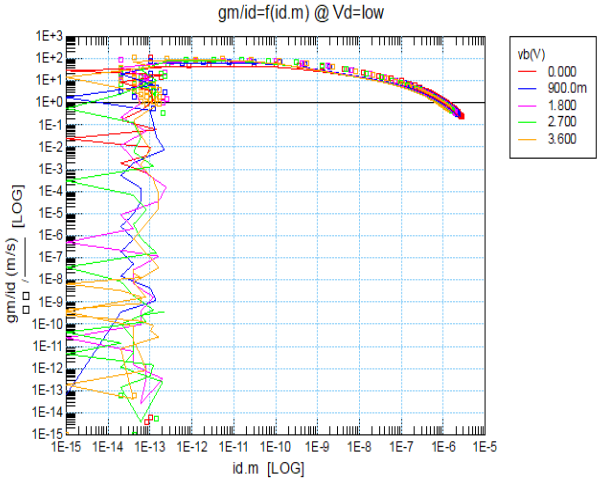


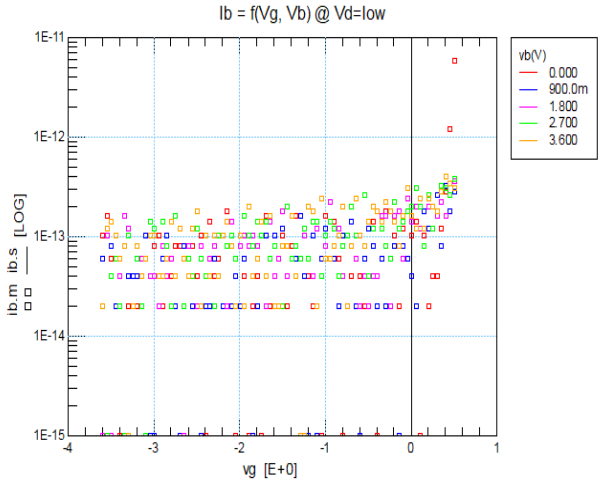


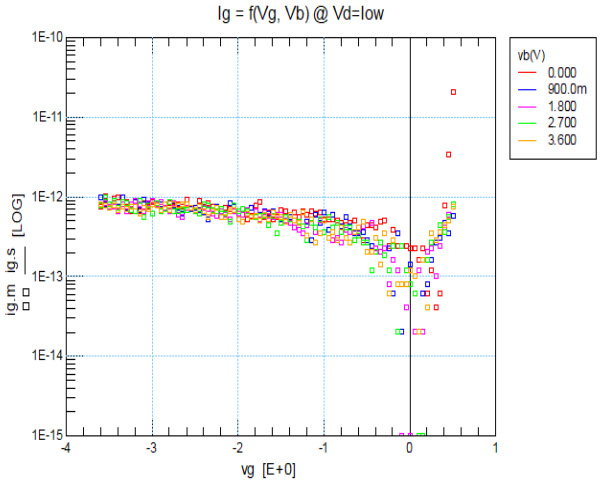




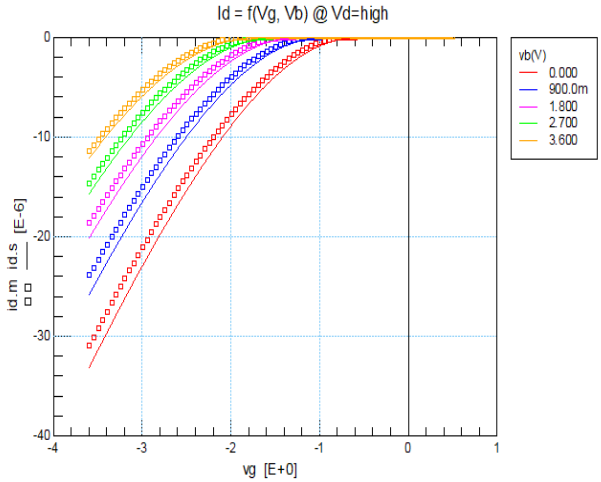


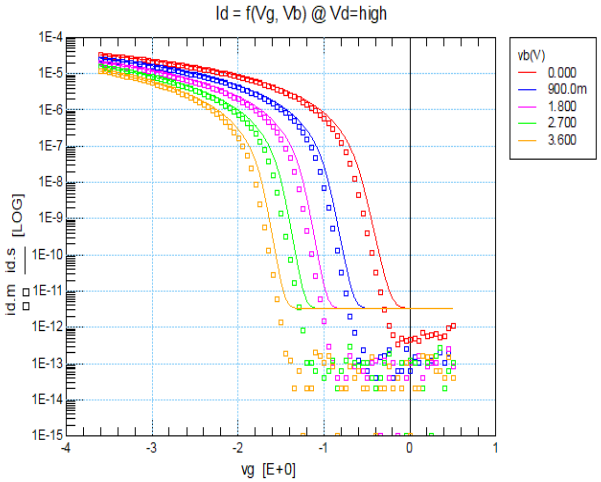


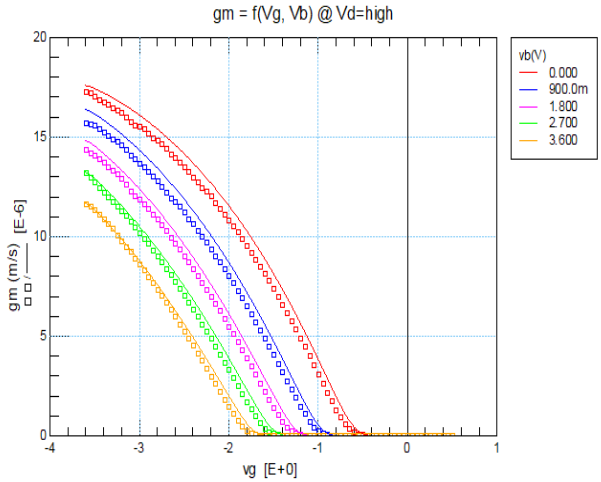


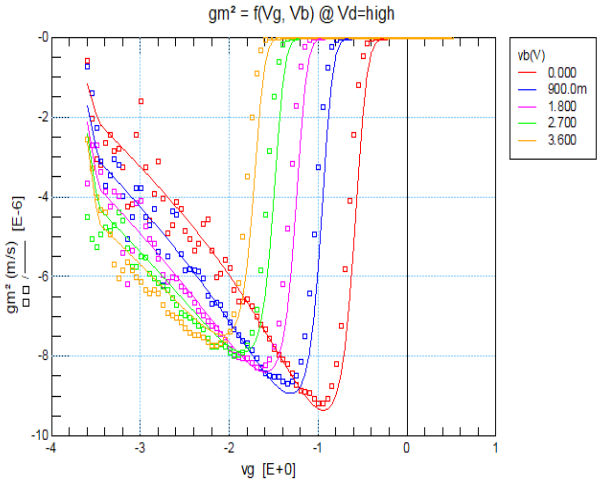


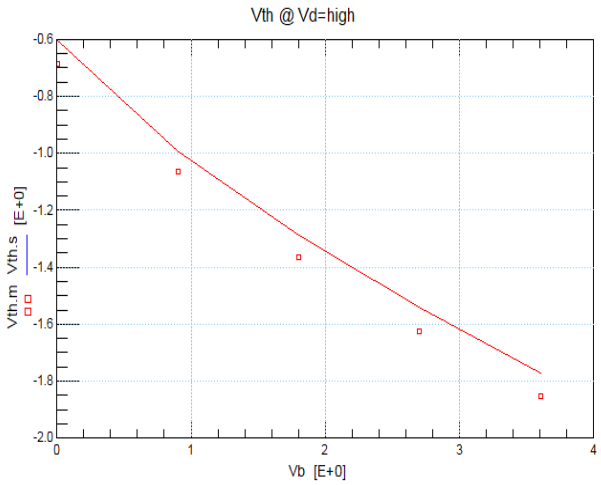
idvg, Vd = -3.3V, T = 27°C

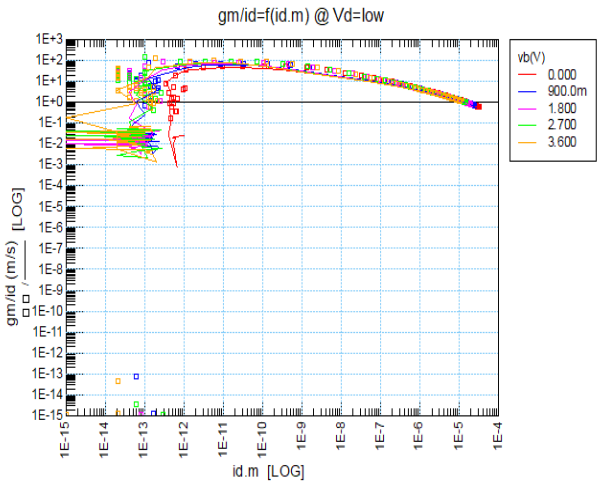


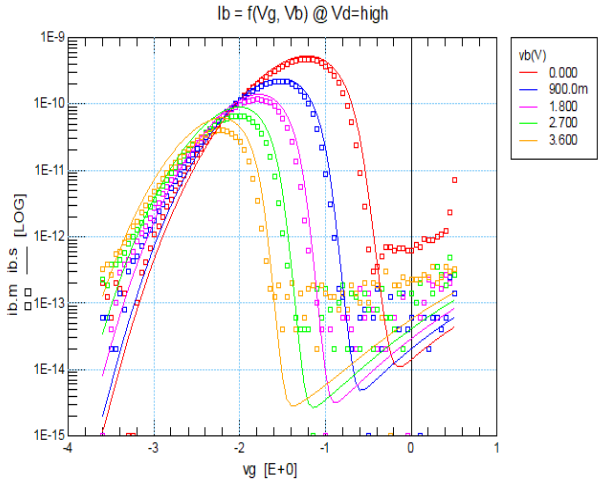


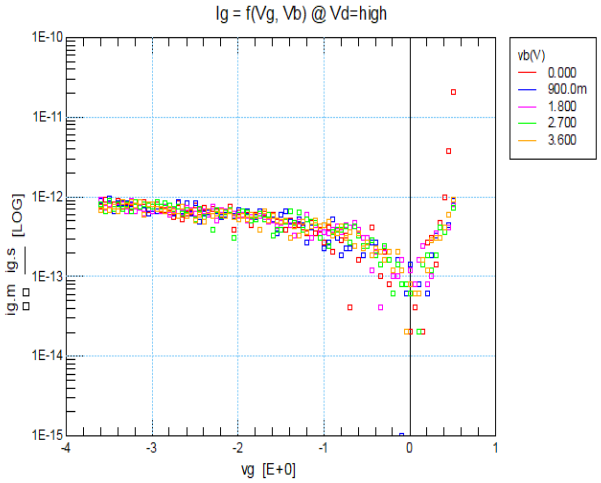




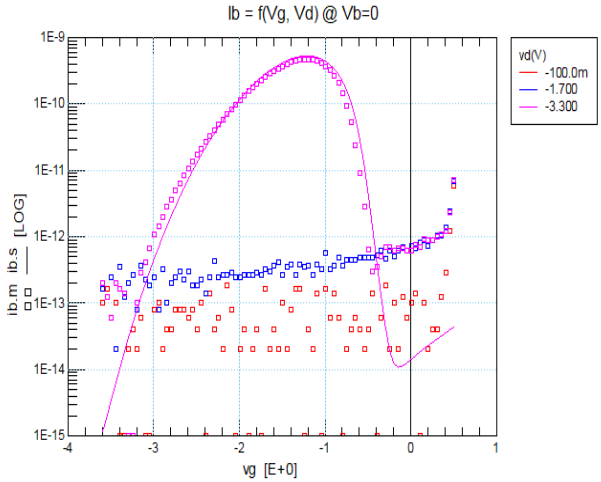


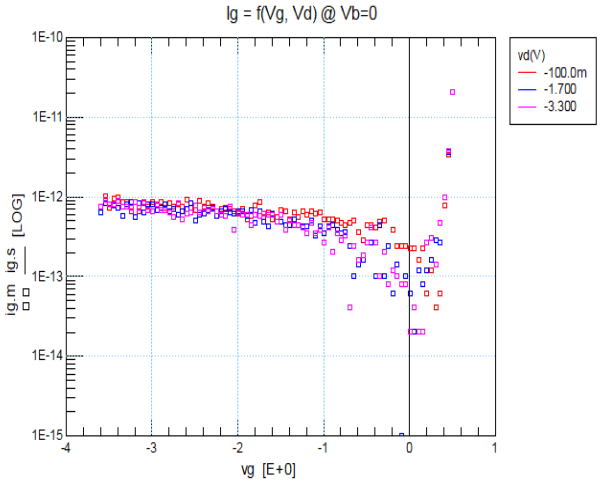




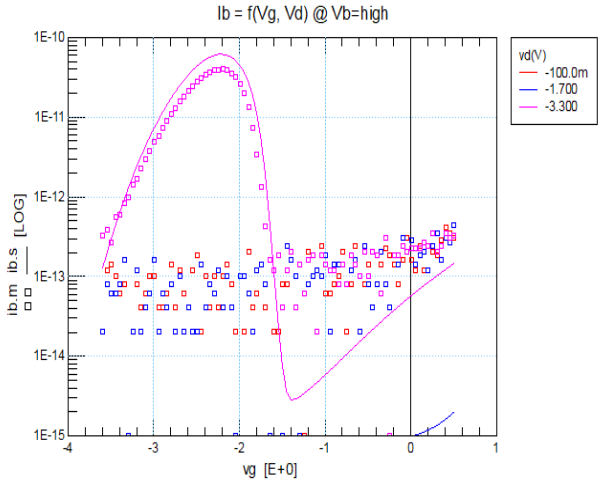


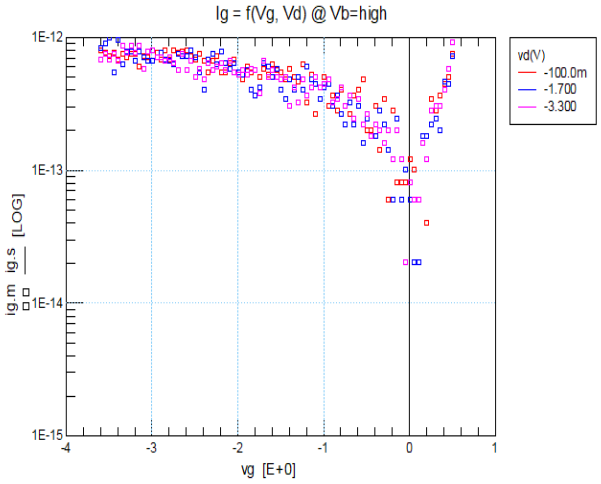
idvg, Ib,Ig, Vb = 0V, T = 27°C



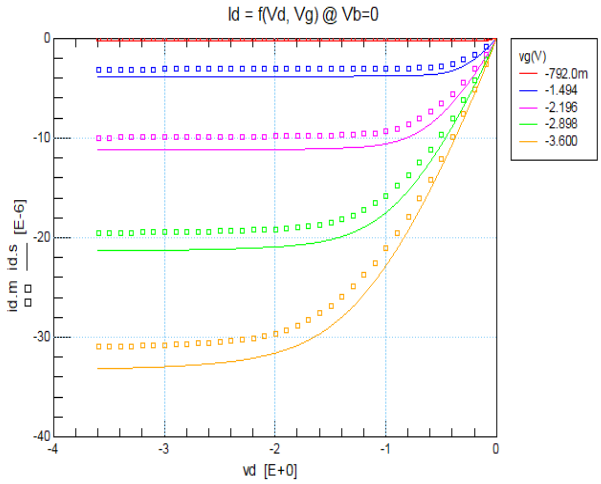


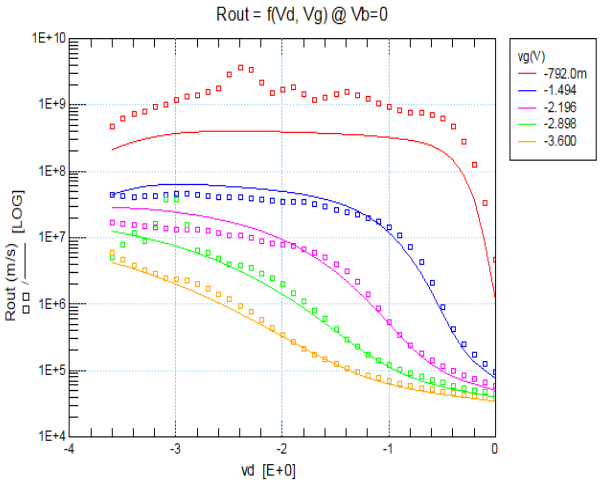
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

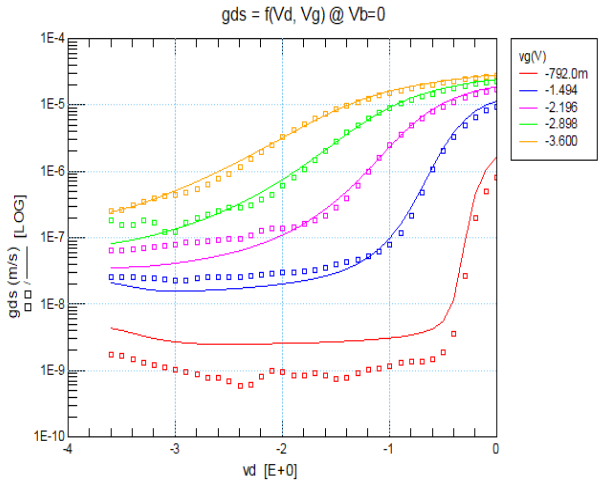




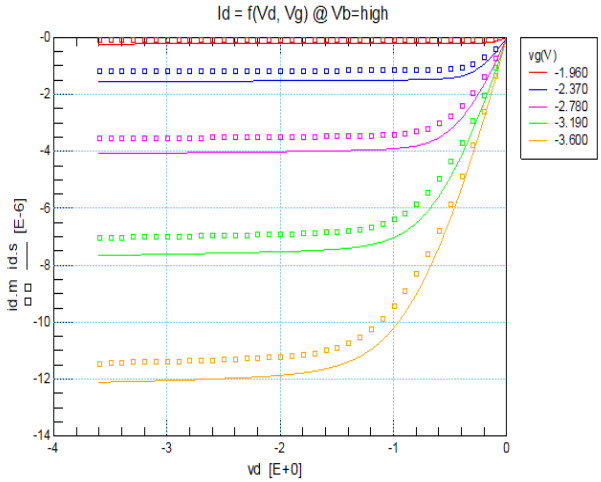
idvd, Vb = 0V, T = 27°C

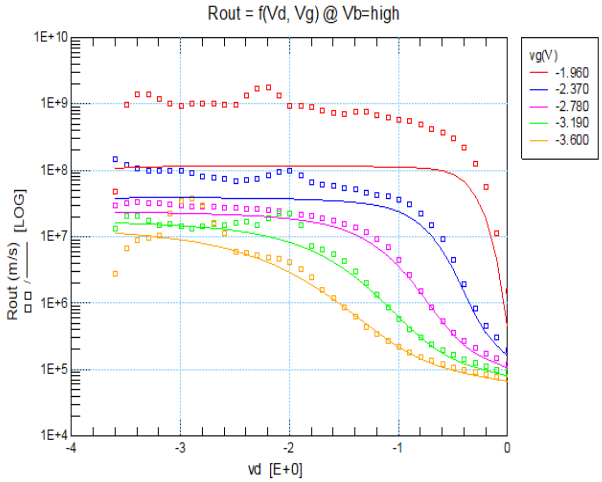


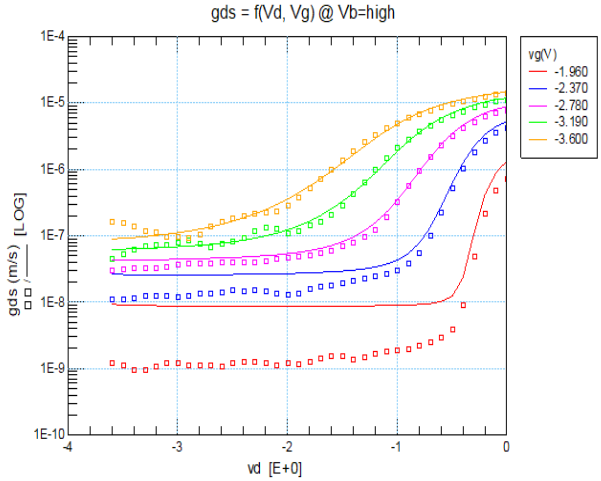




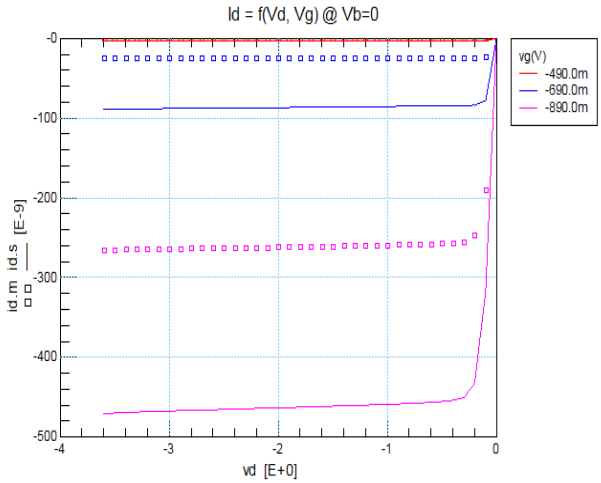
idvd_vbmin, Vb = 3.6V, T = 27°C

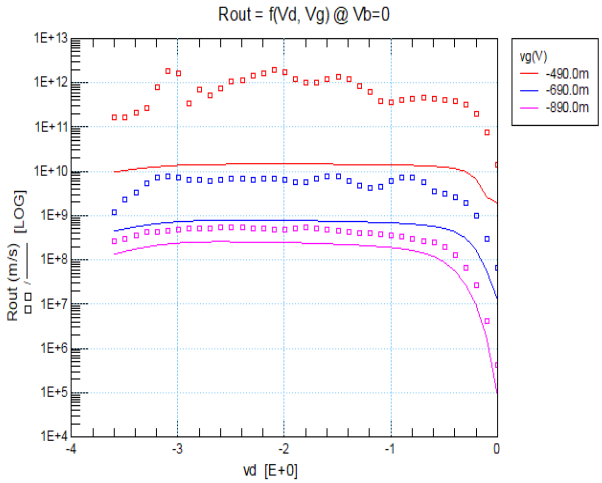


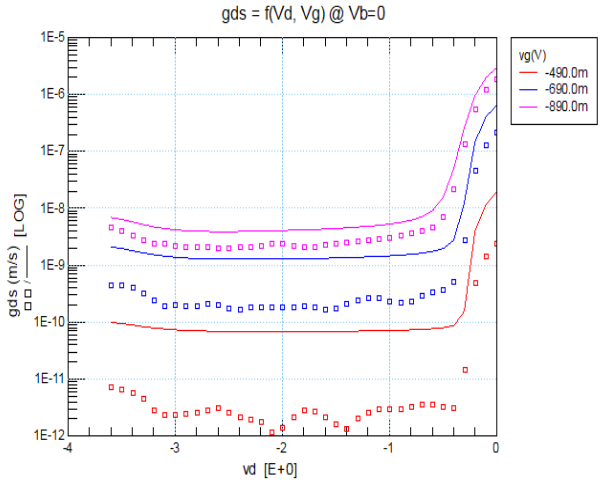




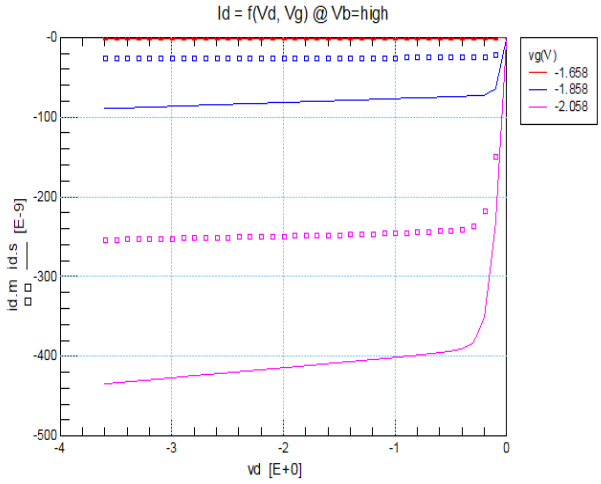
idvd_vth, Vb = 0V, T = 27°C

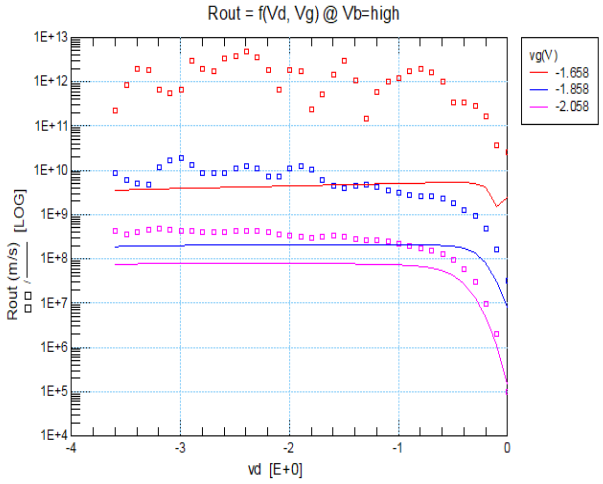


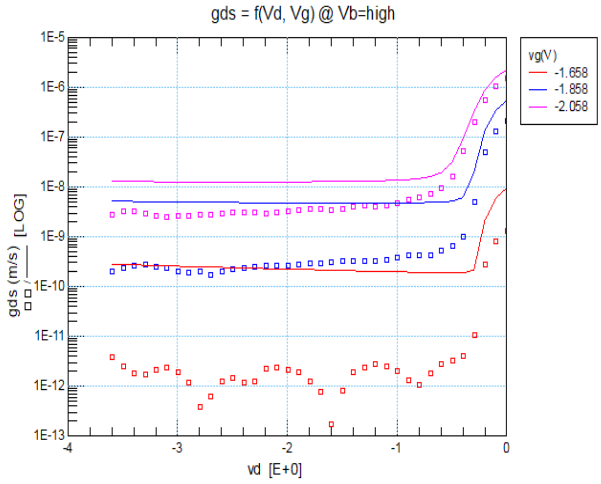




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

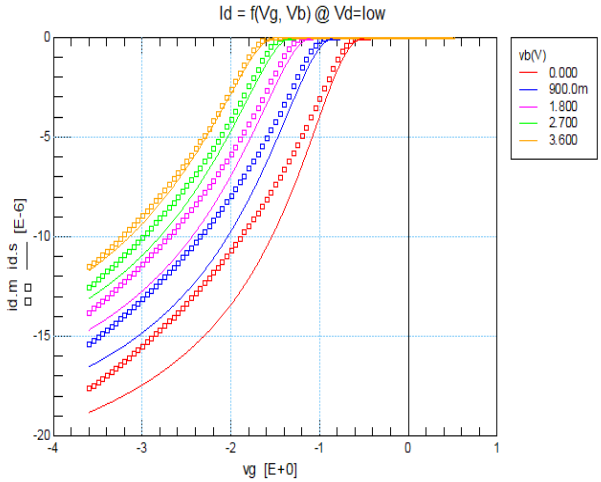


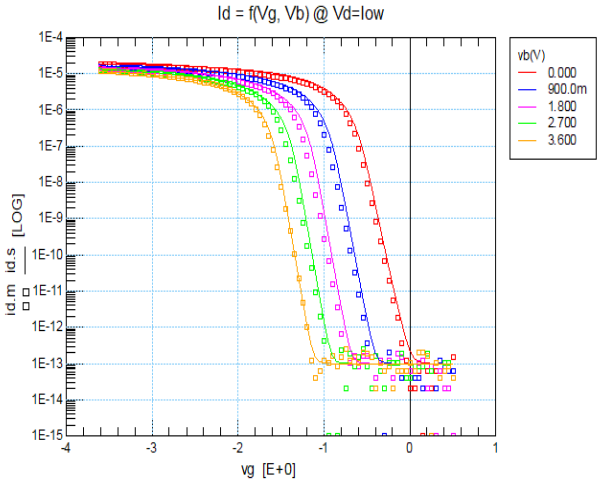


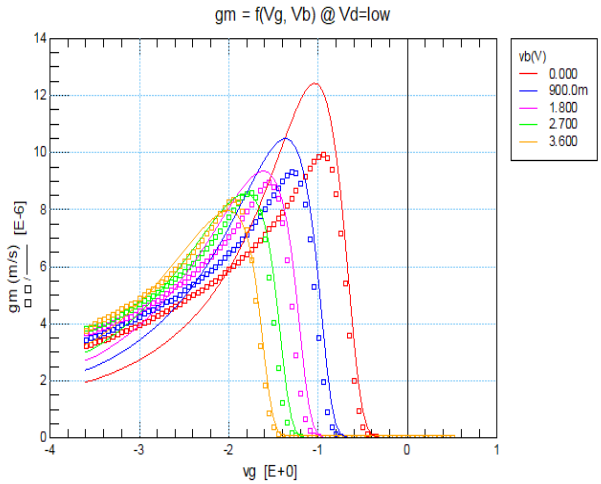


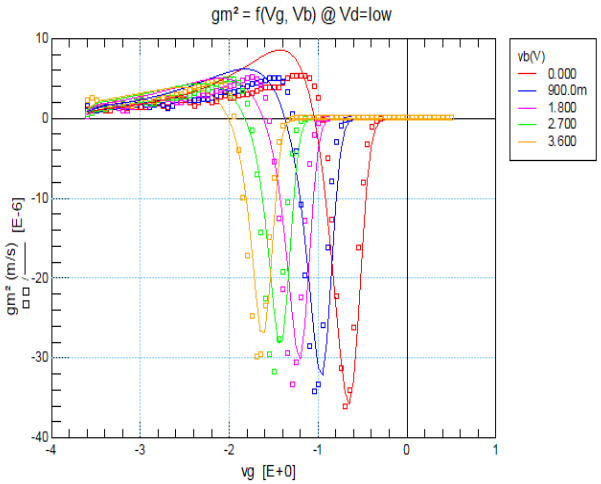
4.16W0u6_L0u35_S564_2

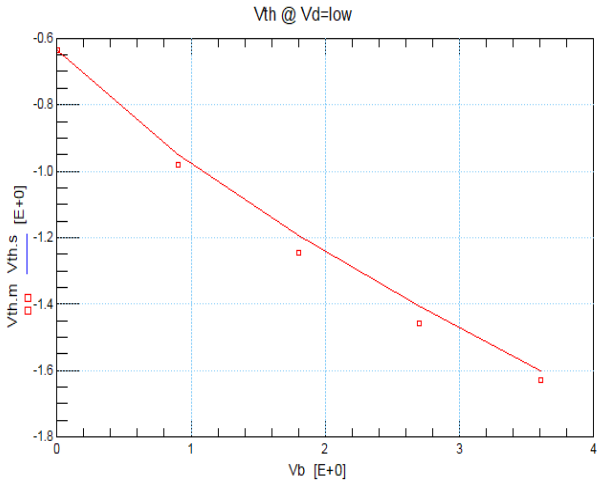
idvg, Vd = -0.1V, T = 27°C

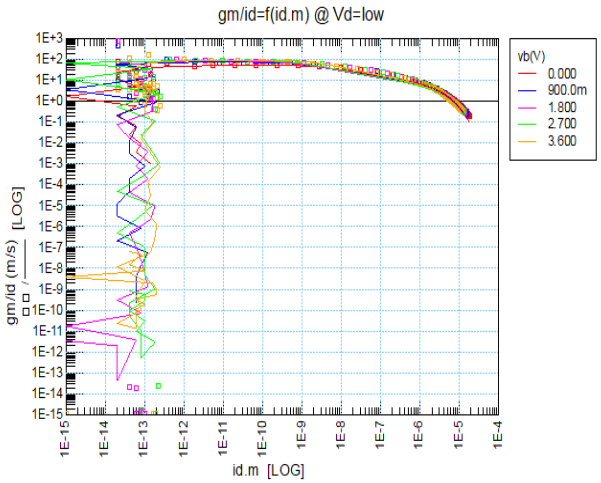


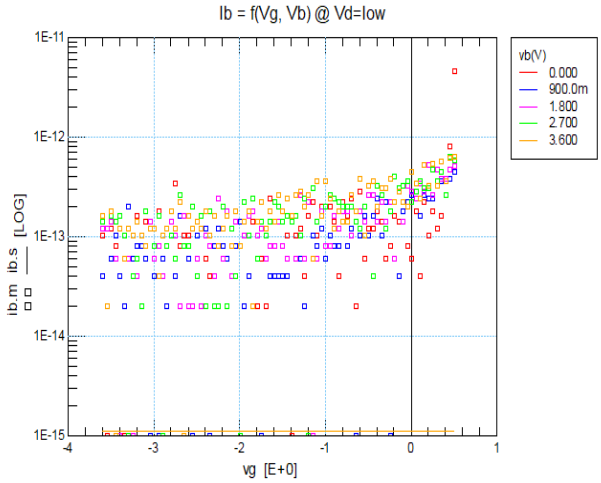


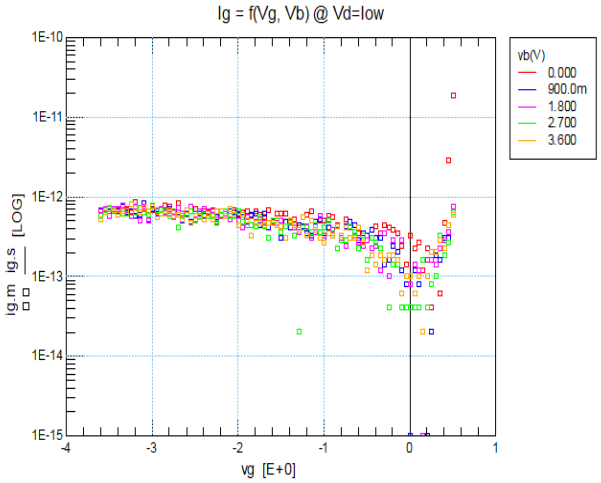




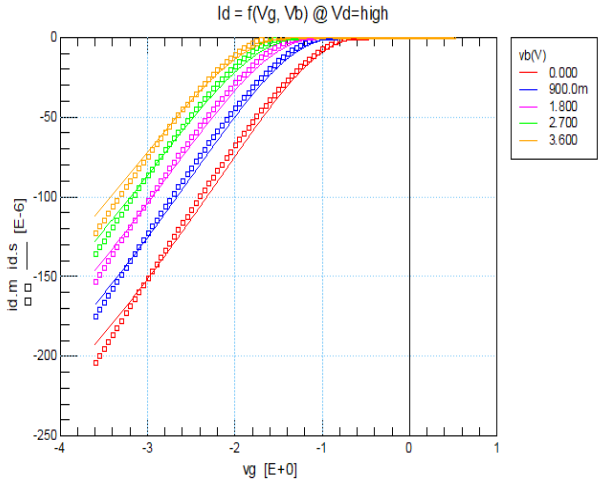


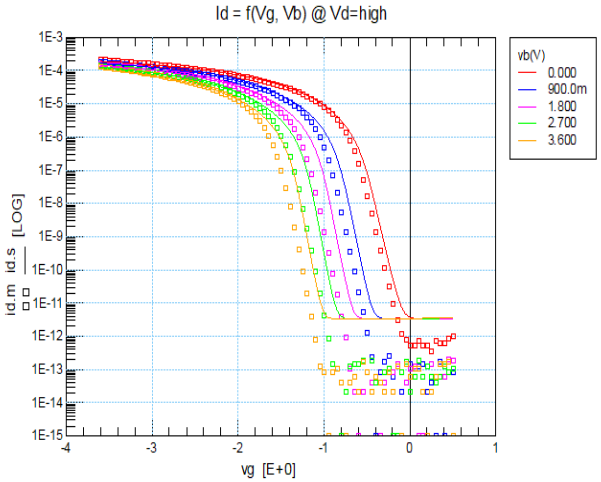


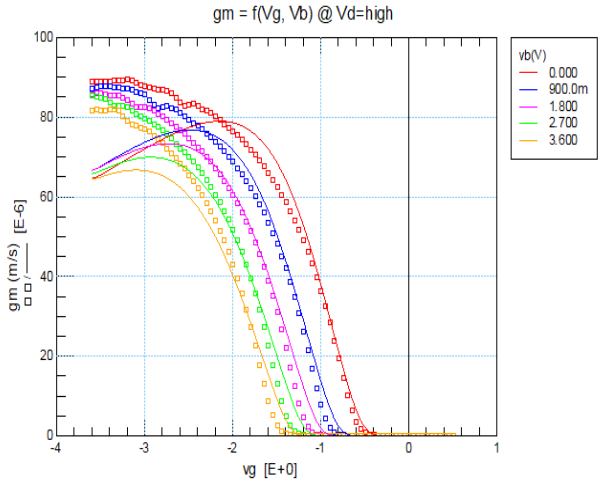


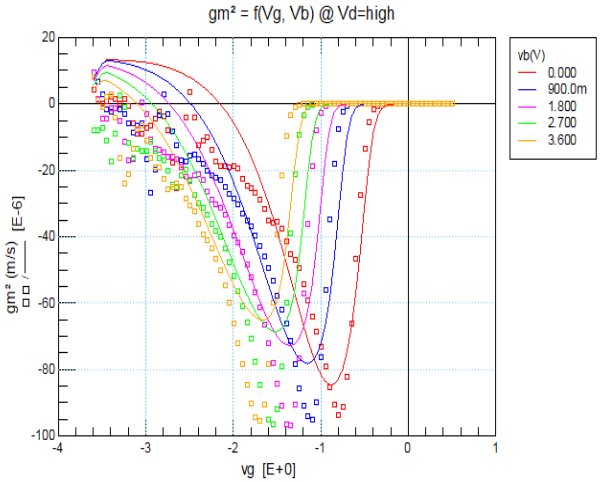


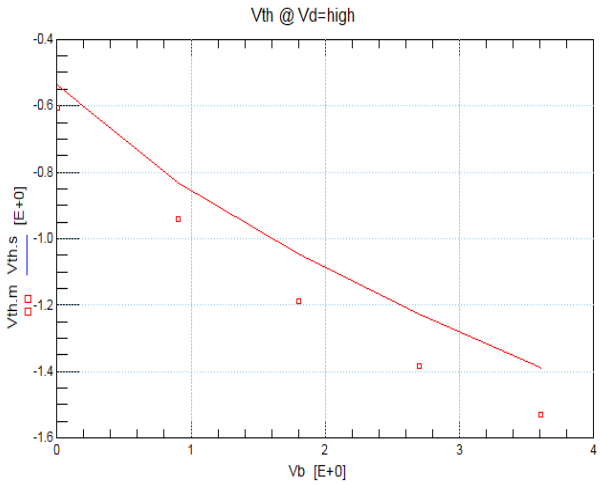
idvg, Vd = -3.3V, T = 27°C

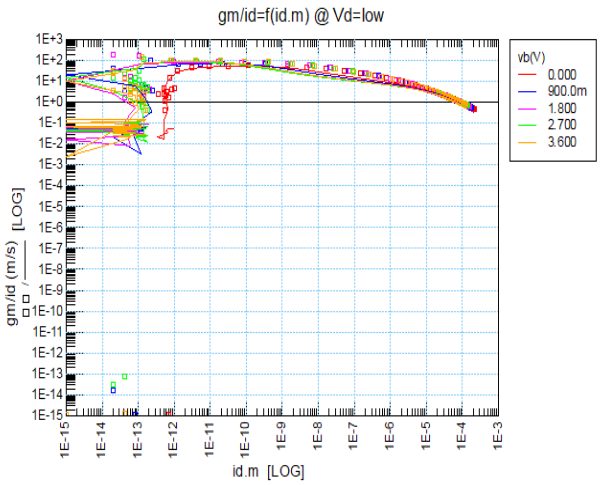


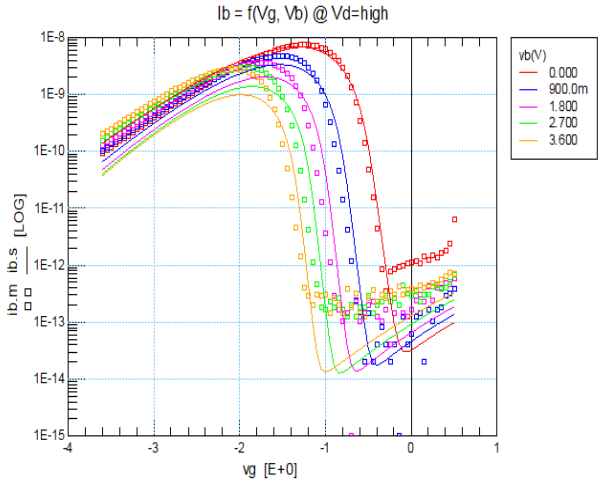


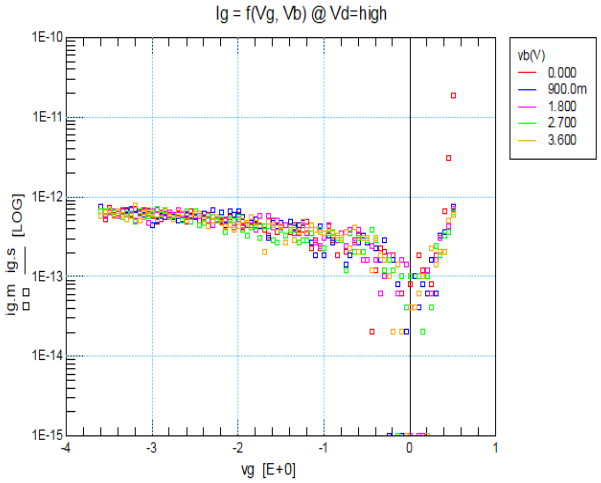




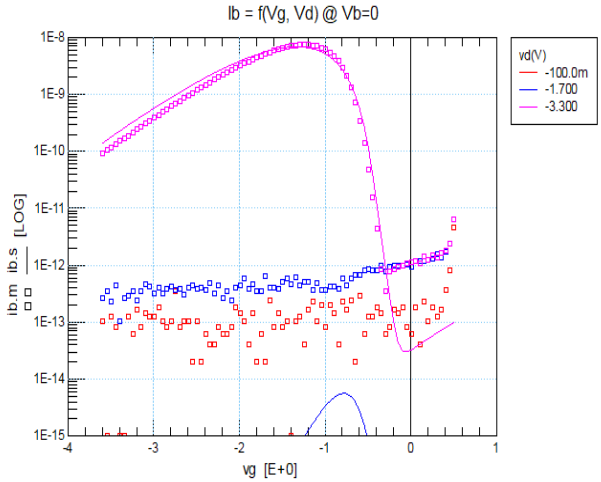


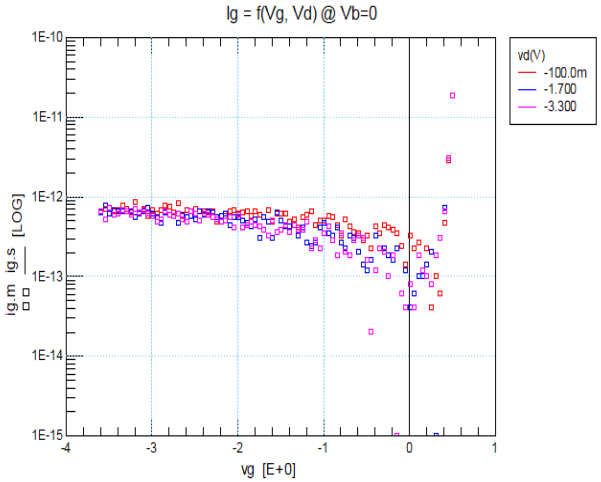




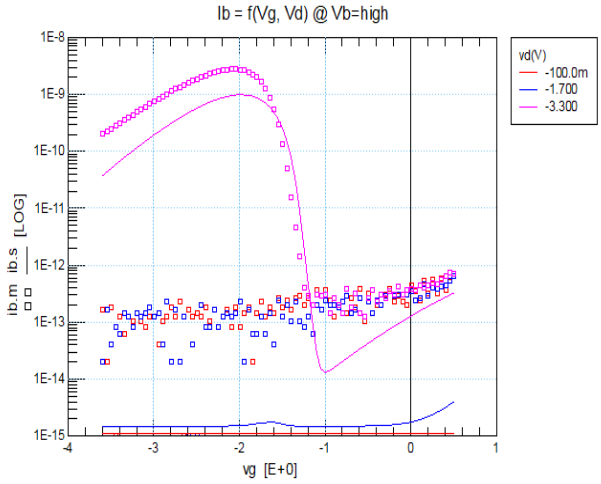


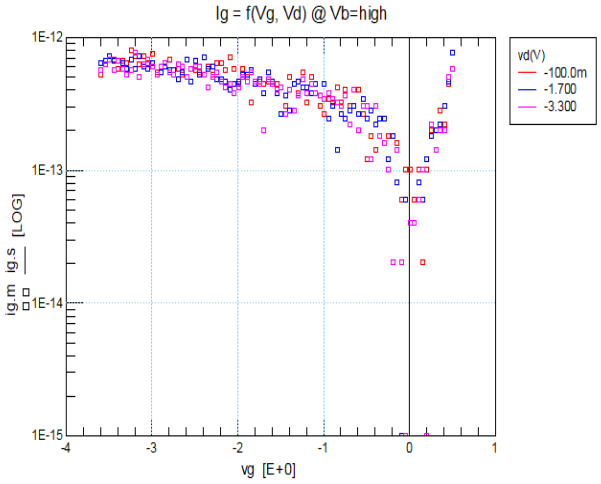
idvg, Ib,Ig, Vb = 0V, T = 27°C



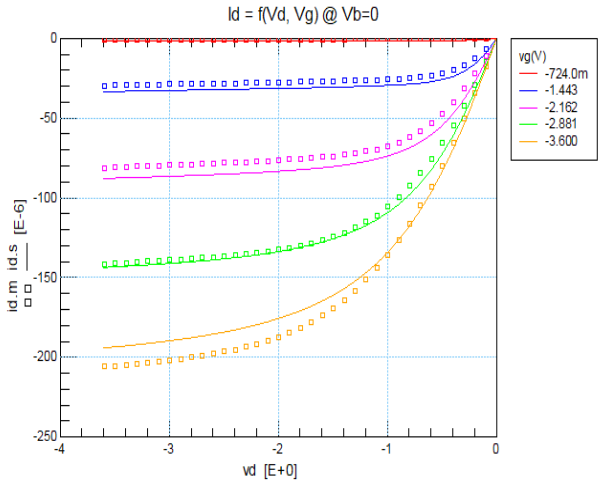


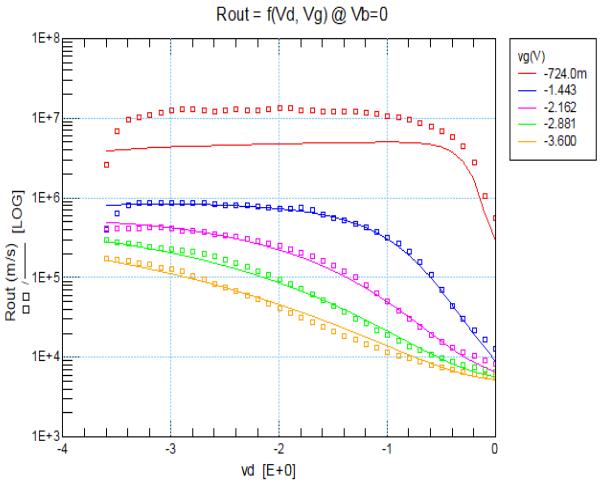
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

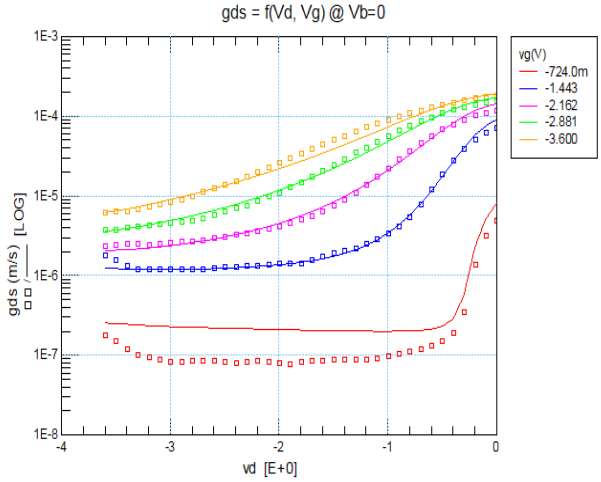




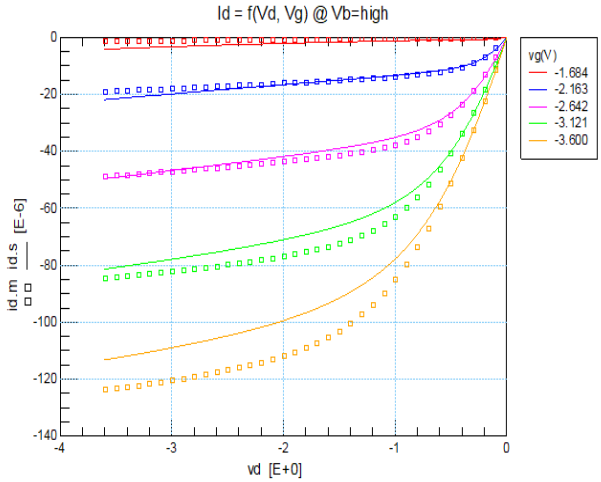
idvd, Vb = 0V, T = 27°C

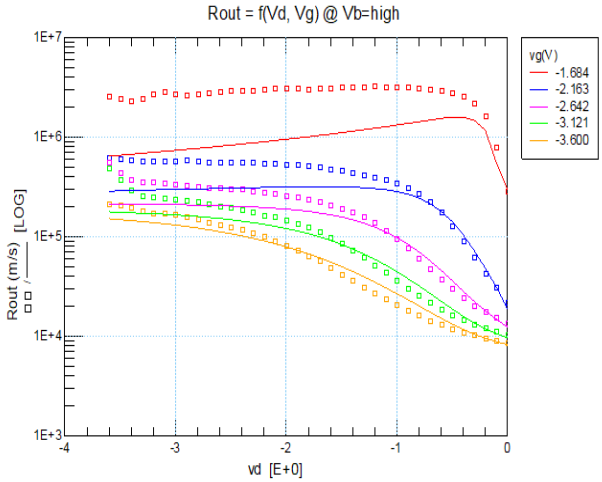


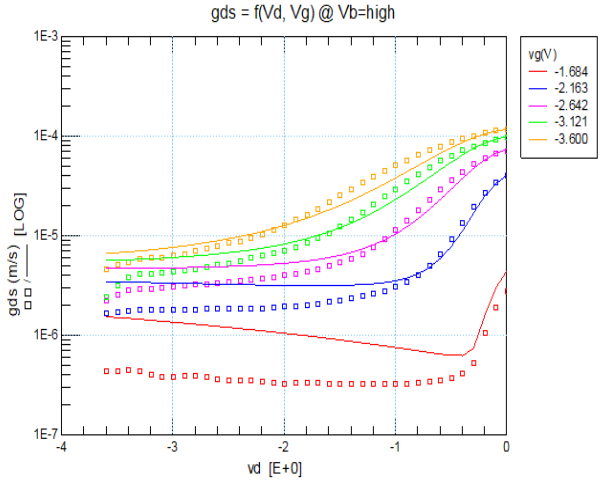




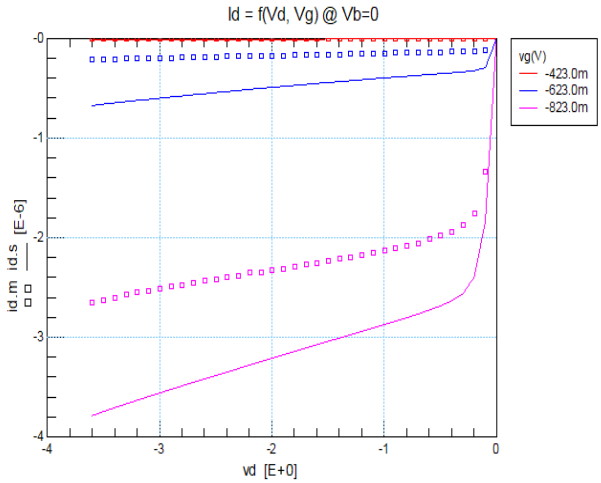
idvd_vbmin, Vb = 3.6V, T = 27°C

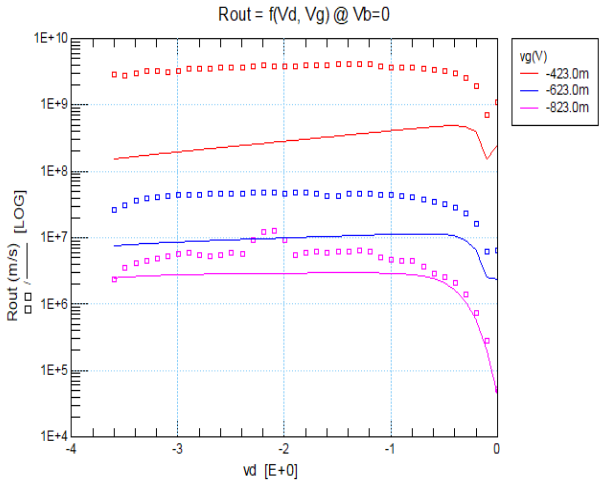


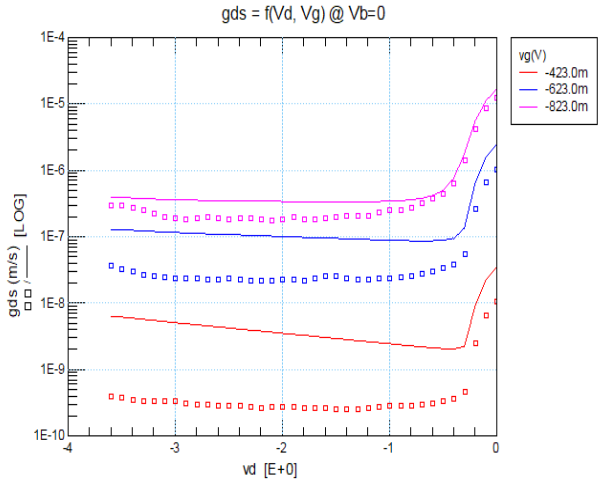




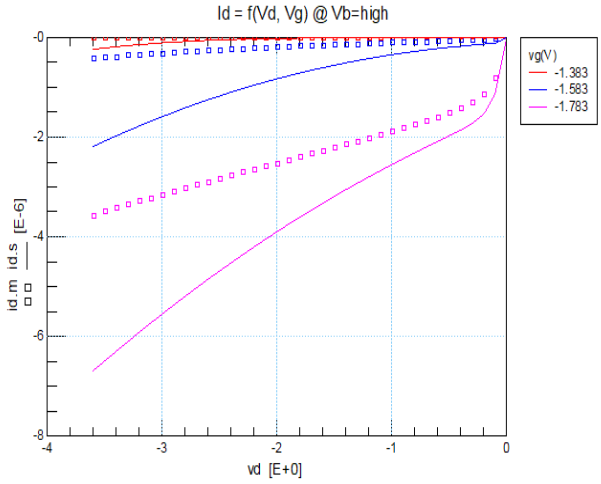
idvd_vth, Vb = 0V, T = 27°C

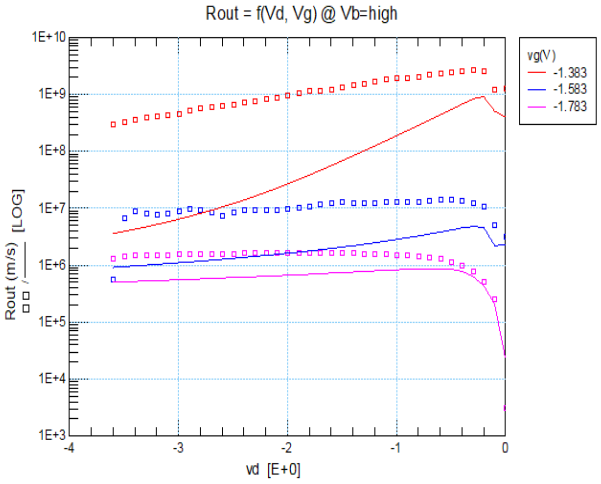


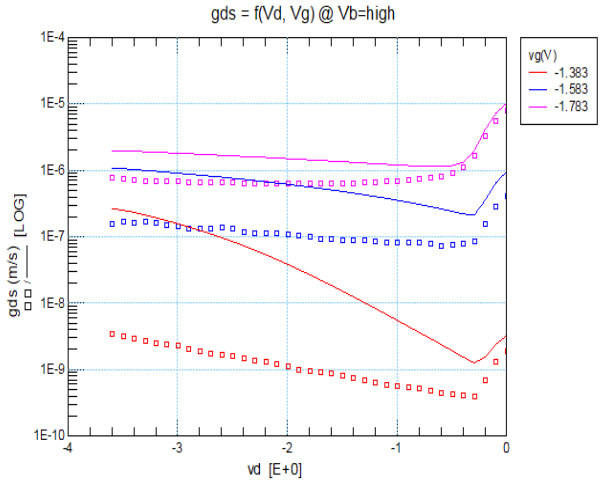




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

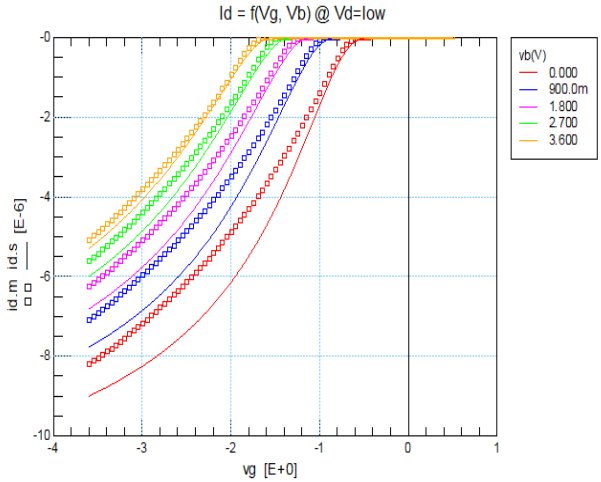


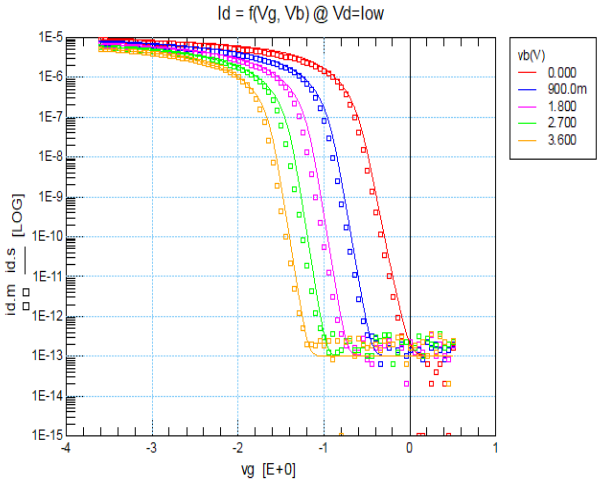


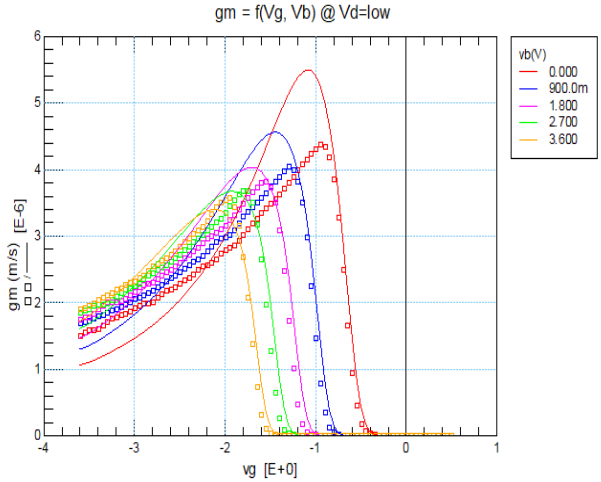


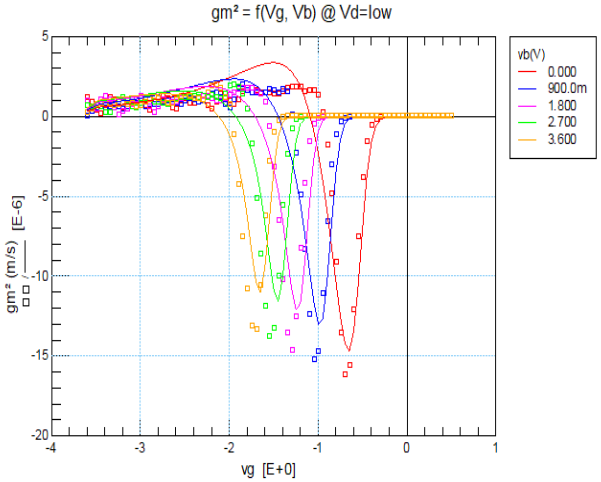
4.17W0u3_L0u4_T356_S385M_3

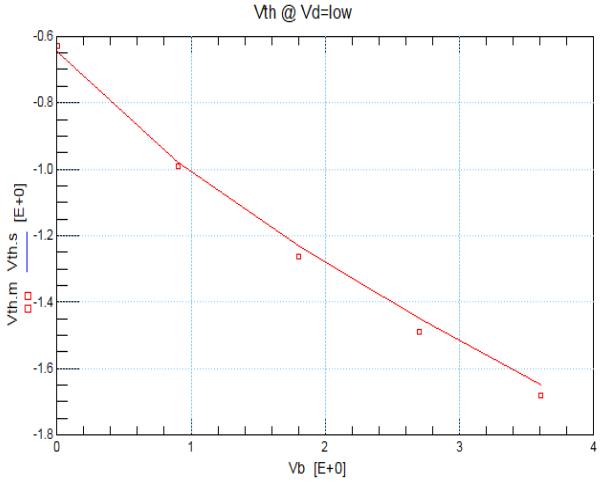
idvg, Vd = -0.1V, T = 27°C

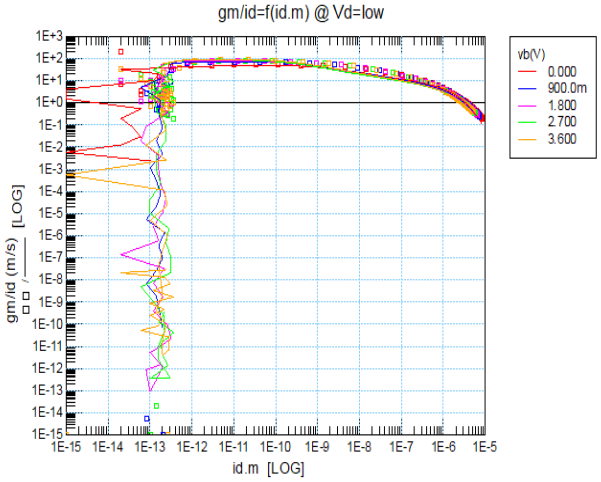


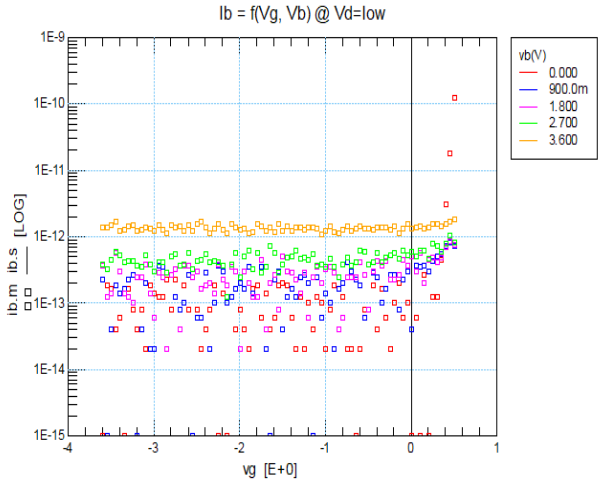


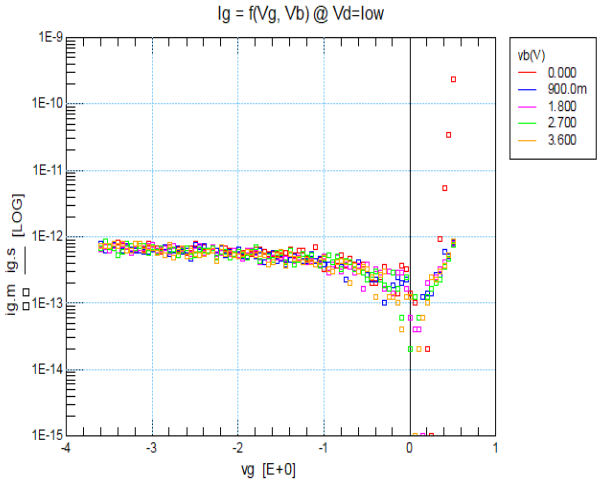




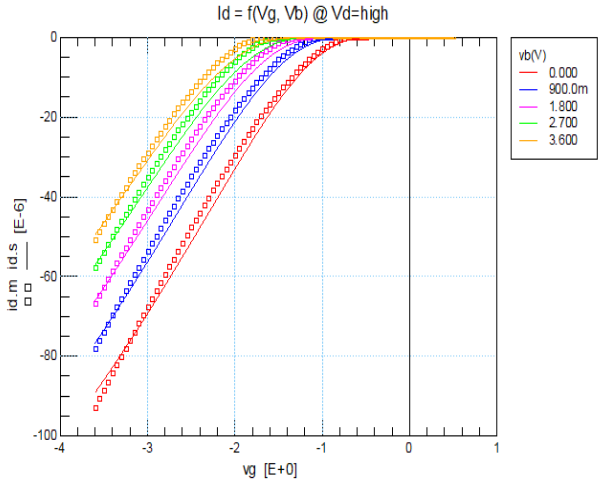


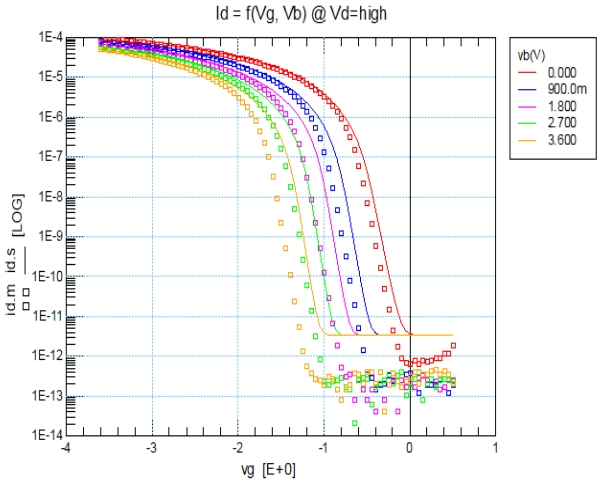


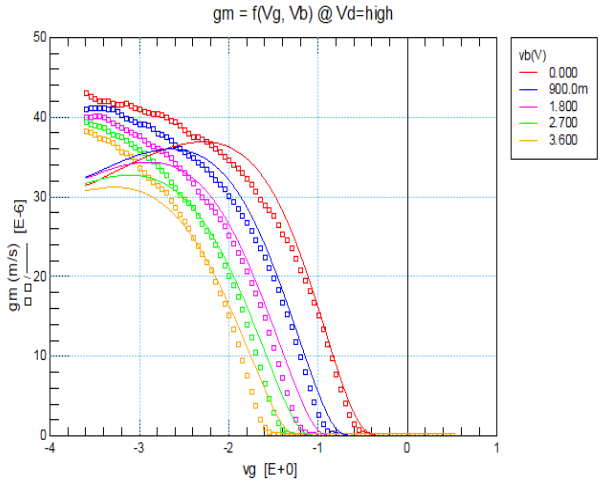


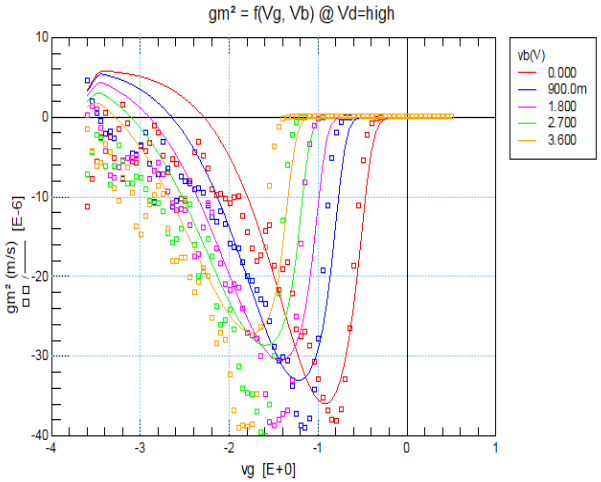


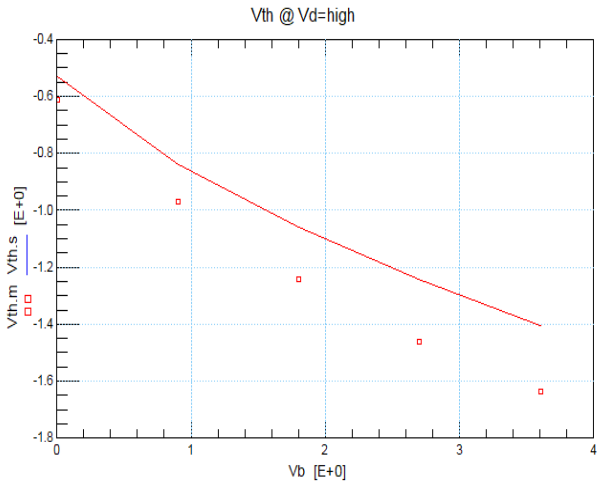
idvg, Vd = -3.3V, T = 27°C

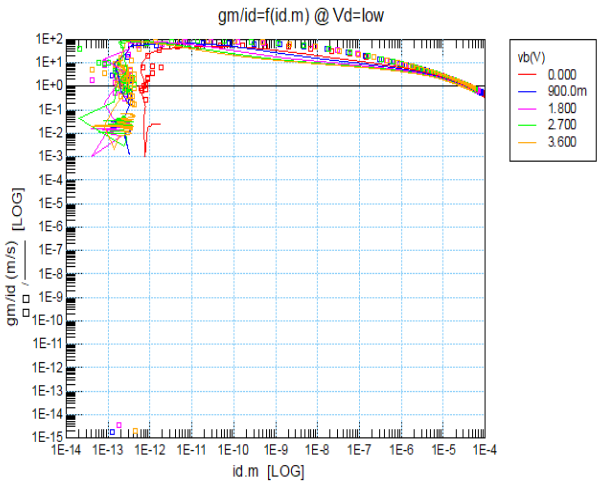


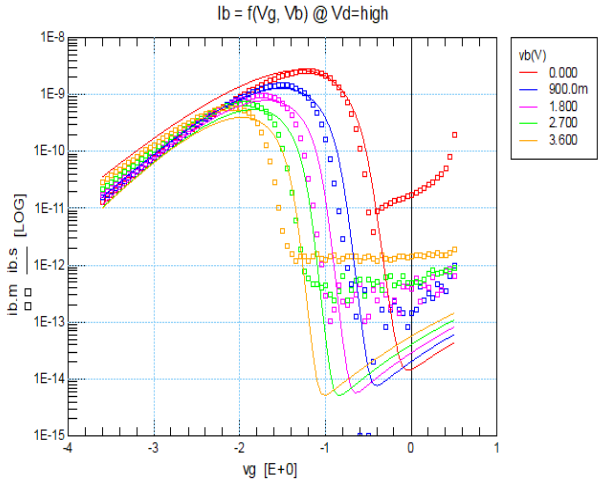


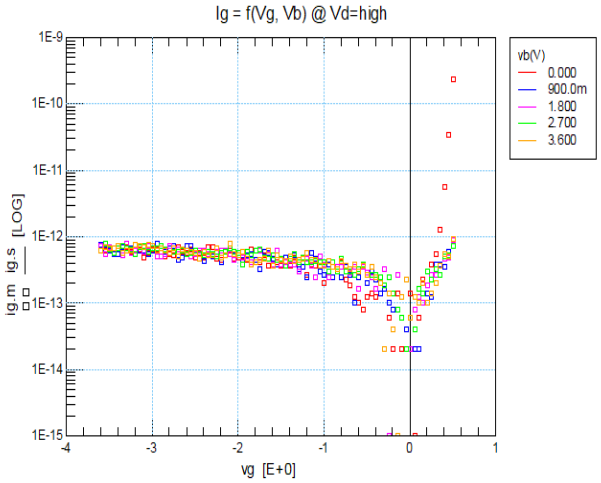




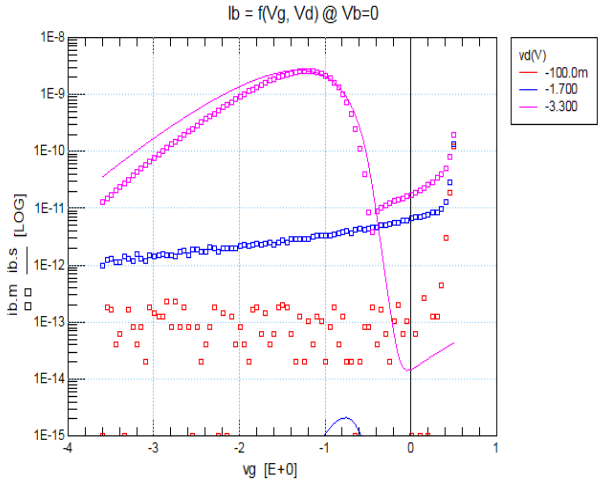


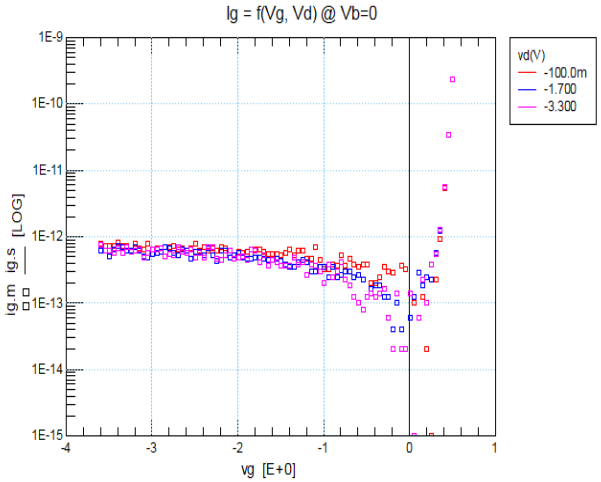




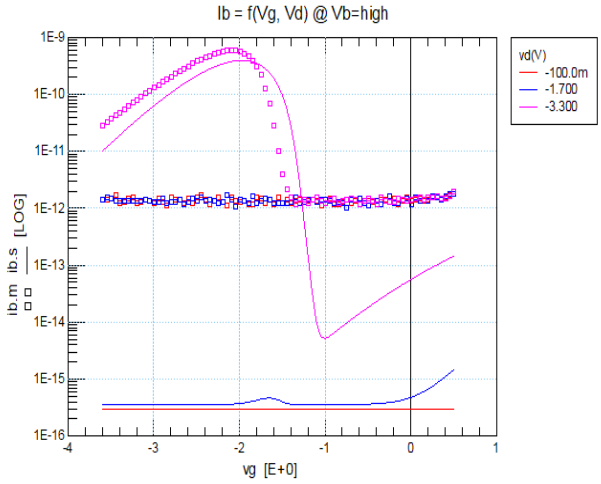


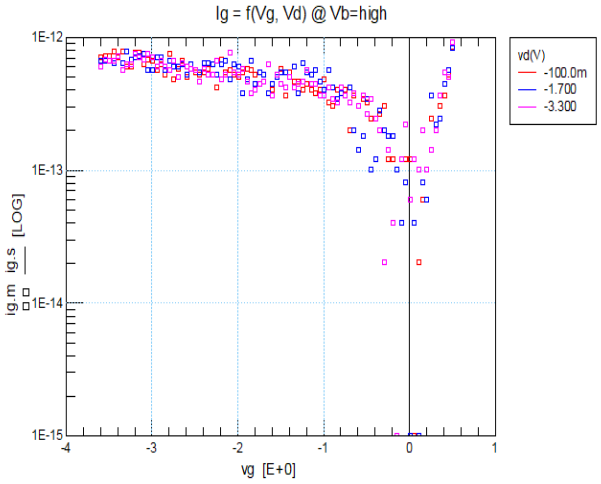
idvg, Ib,Ig, Vb = 0V, T = 27°C



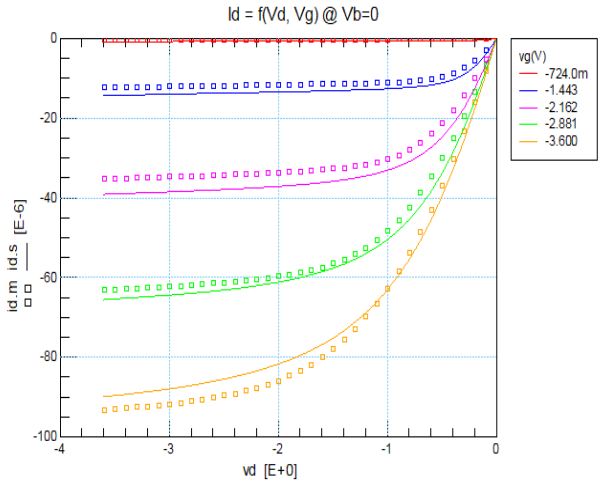


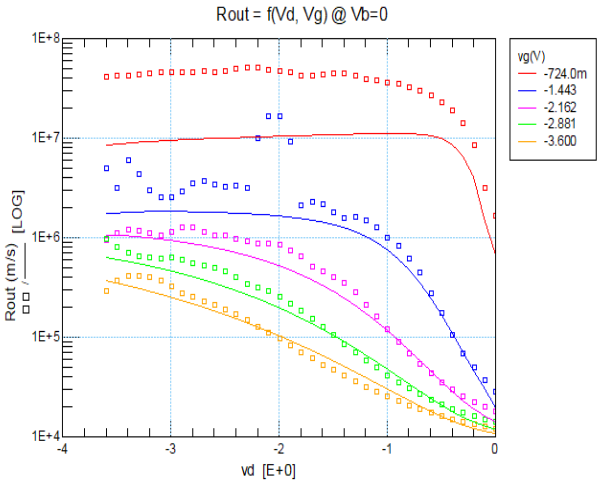
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

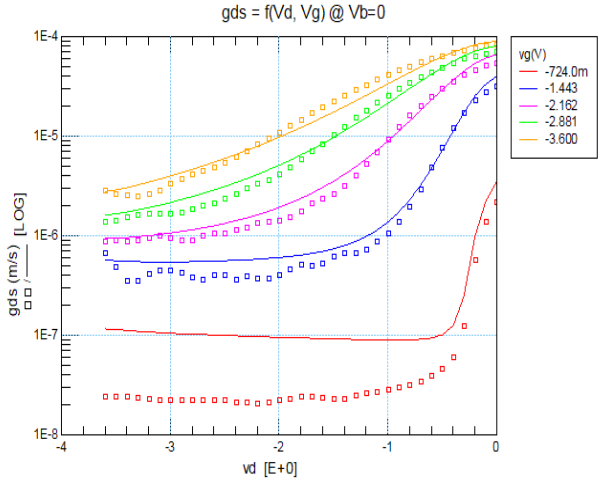




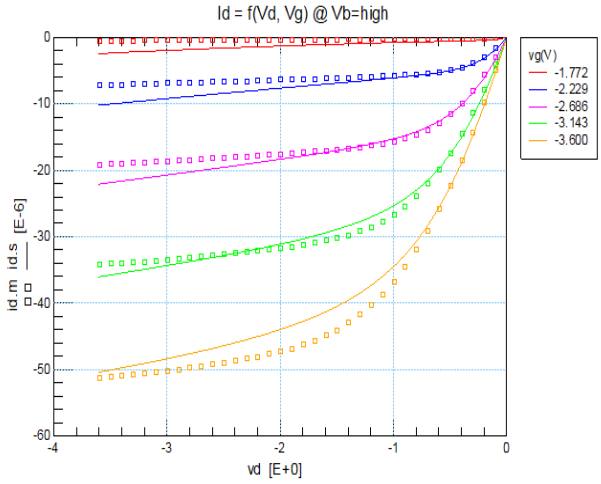
idvd, Vb = 0V, T = 27°C

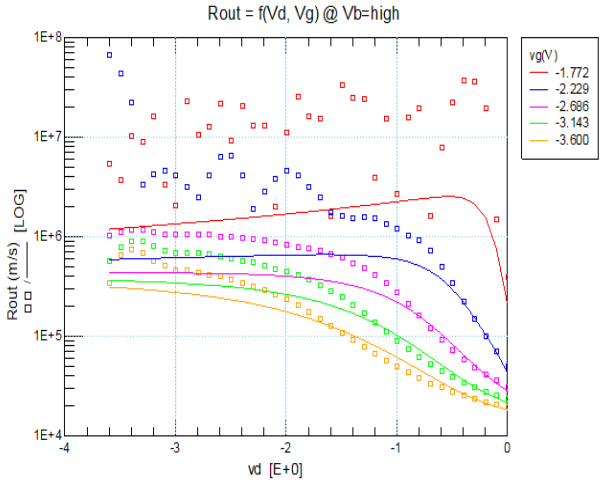


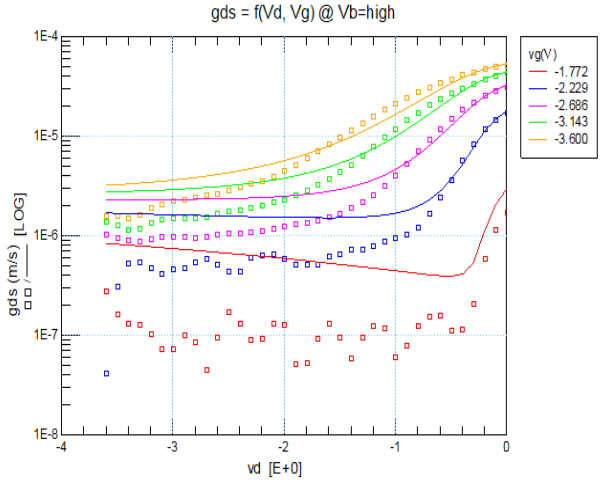




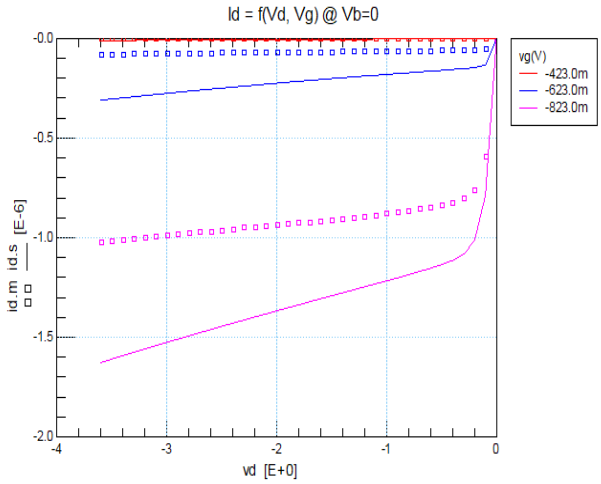
idvd_vbmin, Vb = 3.6V, T = 27°C

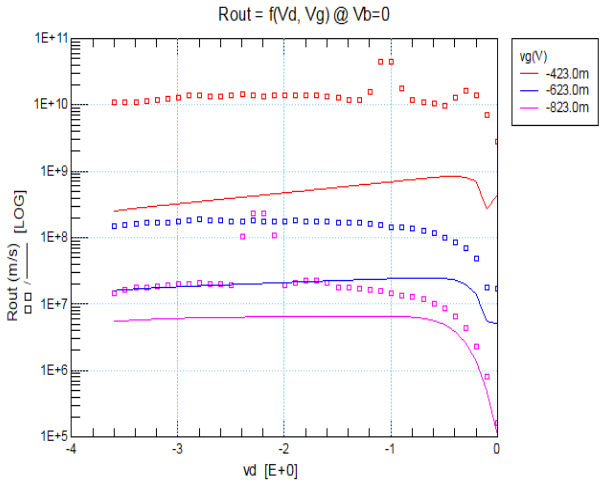


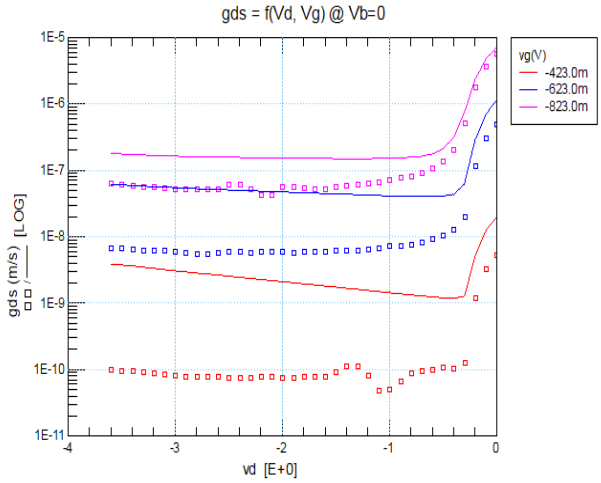




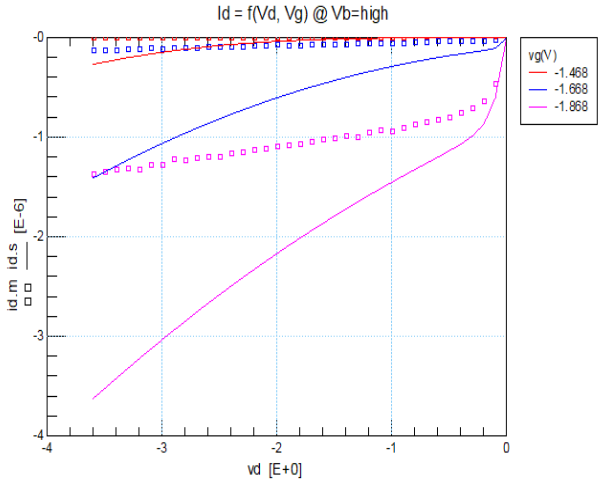
idvd_vth, Vb = 0V, T = 27°C

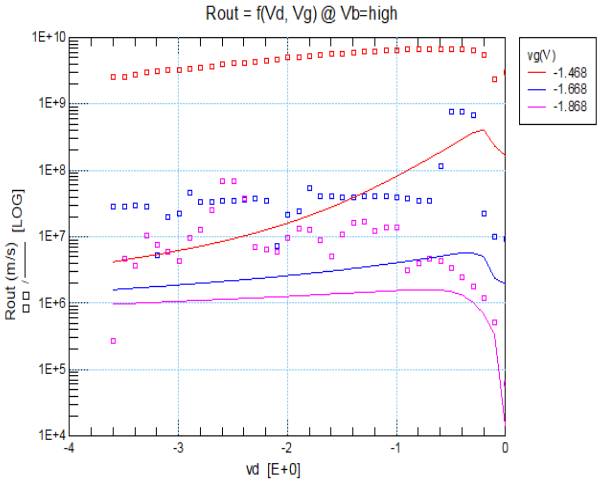


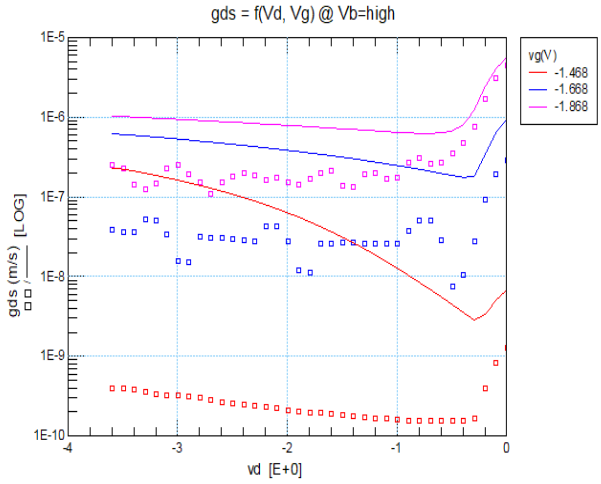




idvd_vth_vbmin, Vb = 3.6V, T = 27°C



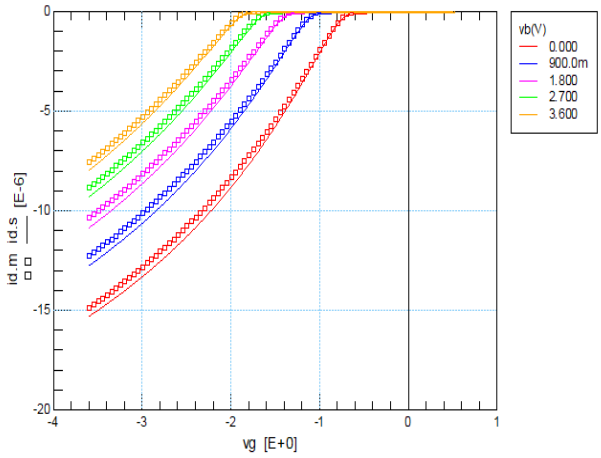




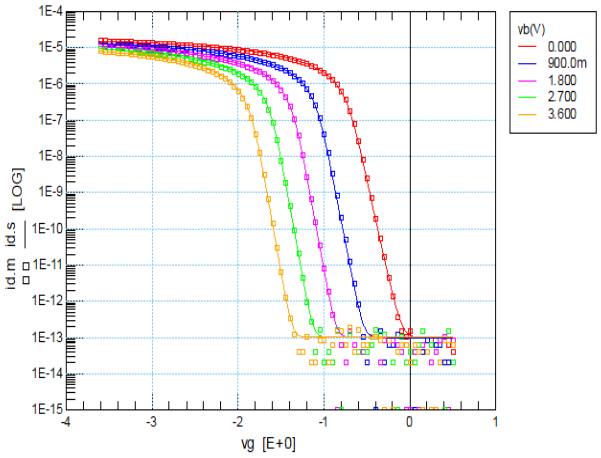
4.18 W02u0_L1u2_S563_3

idvg, Vd = -0.1V, T = 27°C

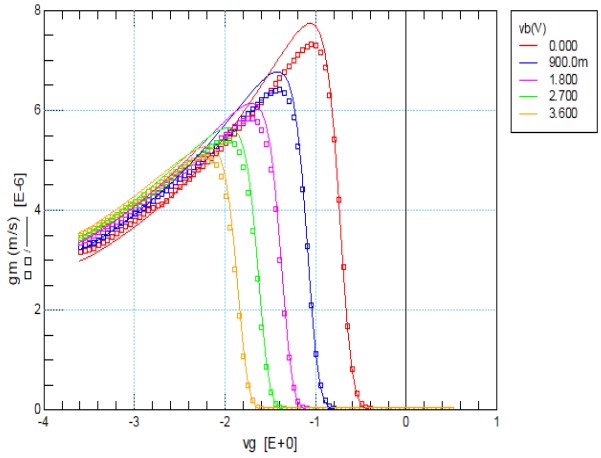
Id = f(Vg, Vb) @ Vd=low



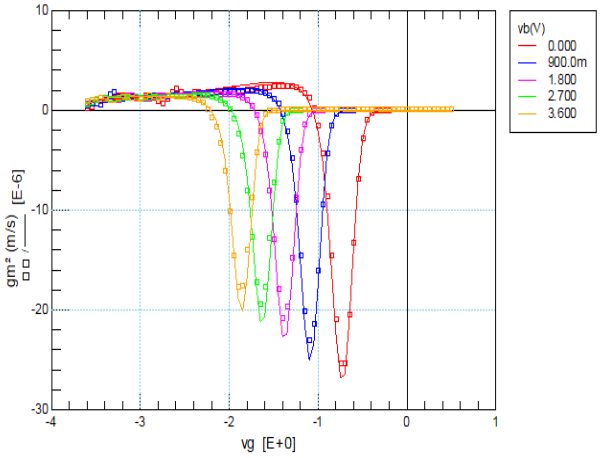
Id = f(Vg, Vb) @ Vd=low



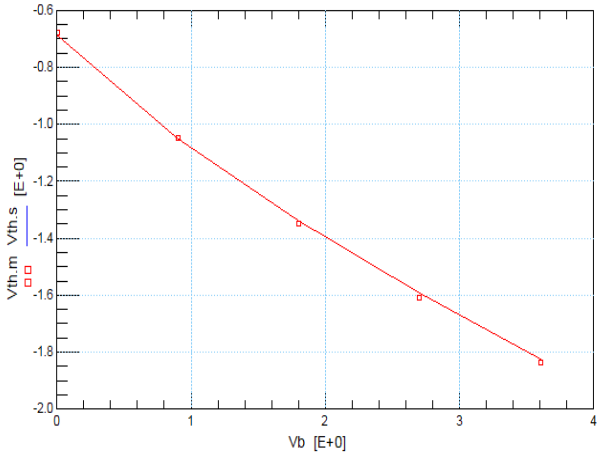
gm = f(Vg, Vb) @ Vd=low



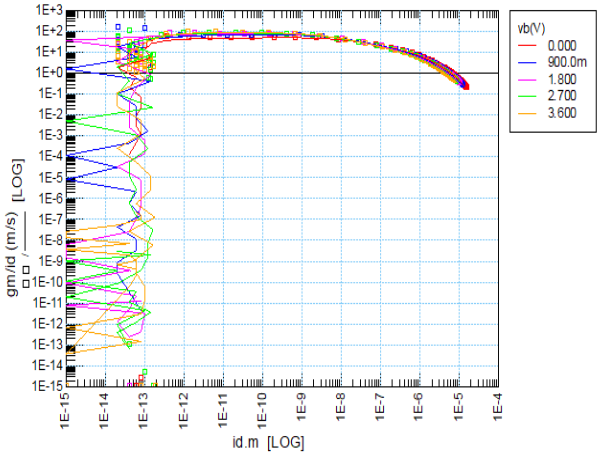
gm² = f(Vg, Vb) @ Vd=low

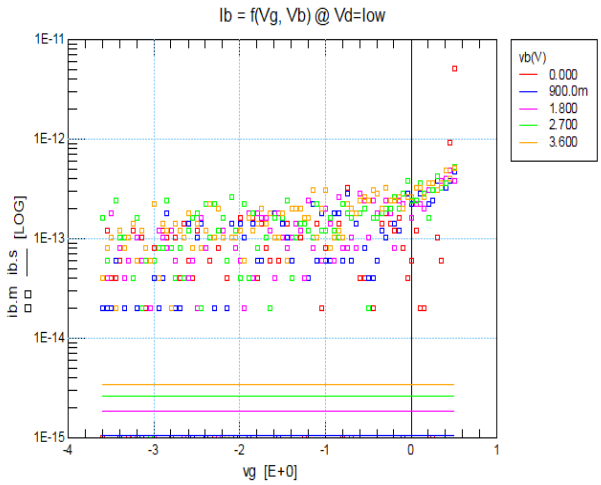


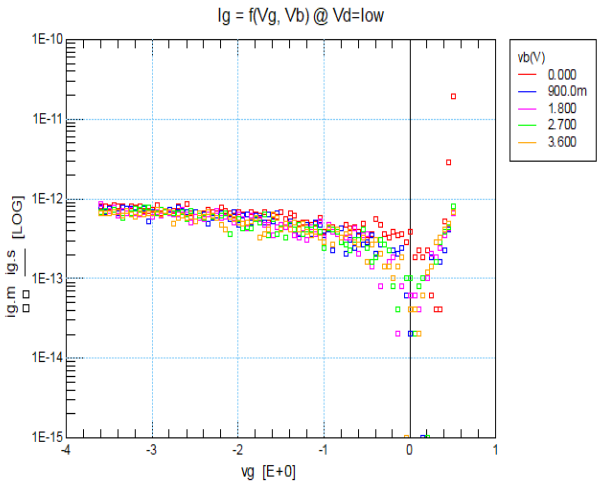
Vth @ Vd=low



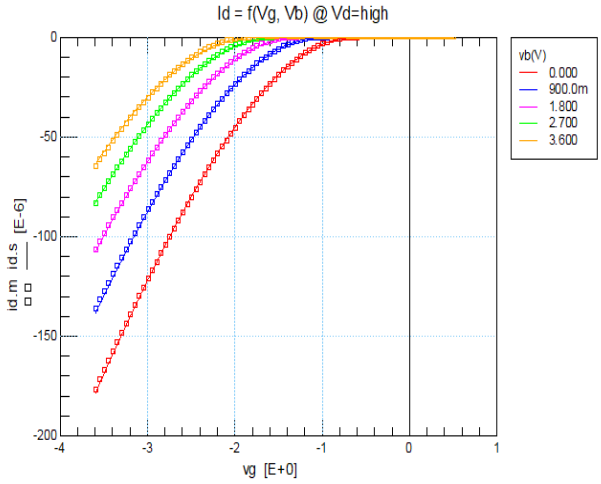
gm/id=f(id.m) @ Vd=low

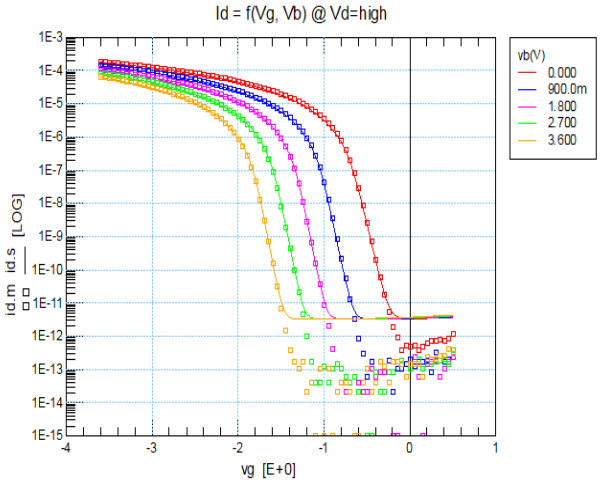


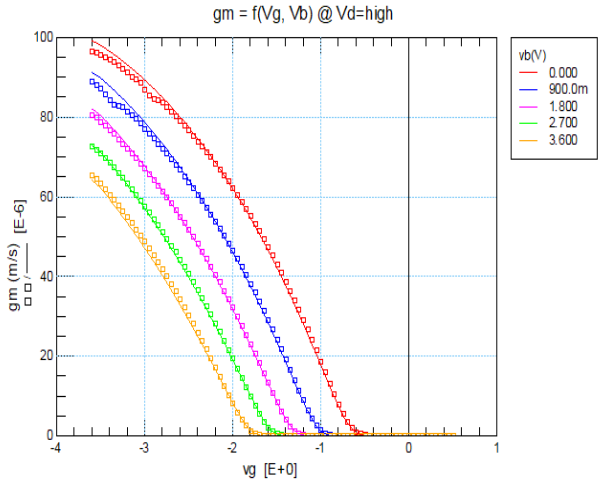


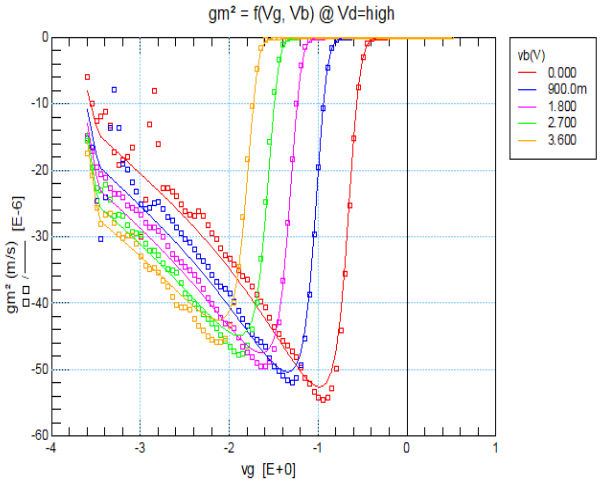


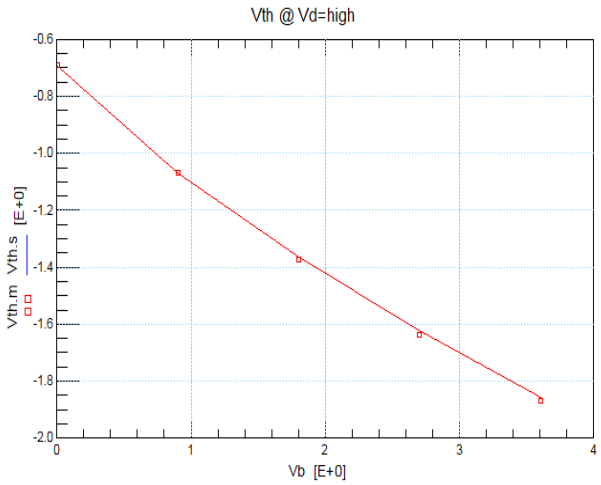
idvg, Vd = -3.3V, T = 27°C

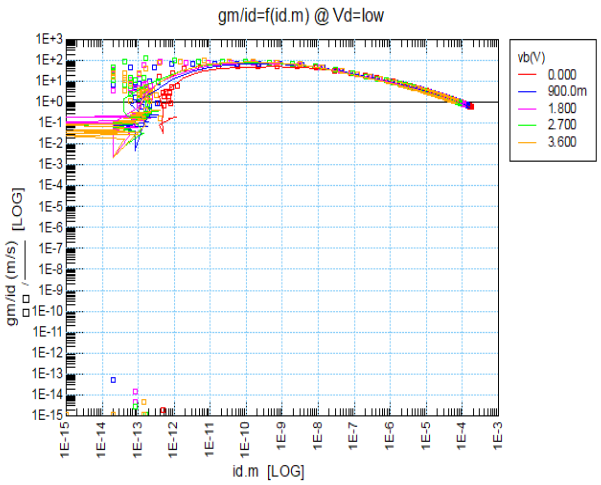


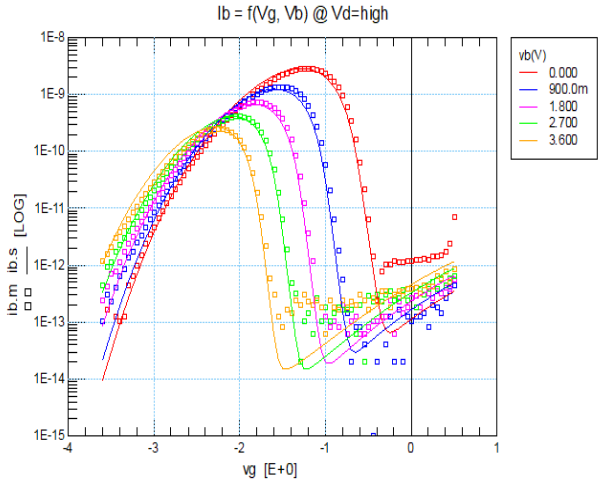


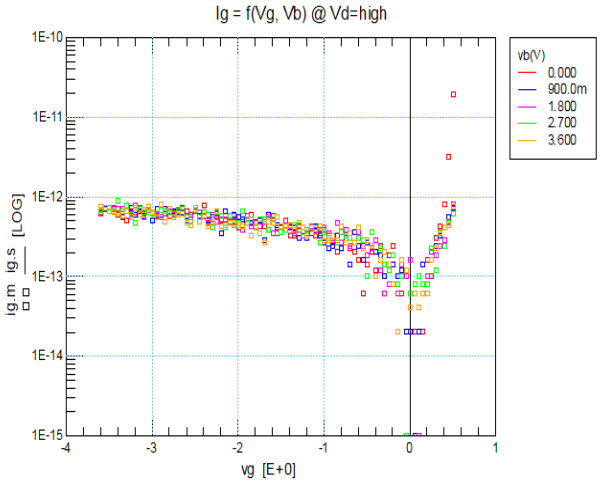




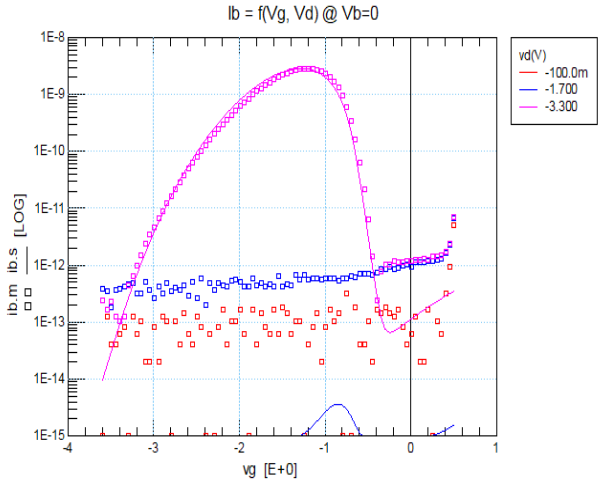


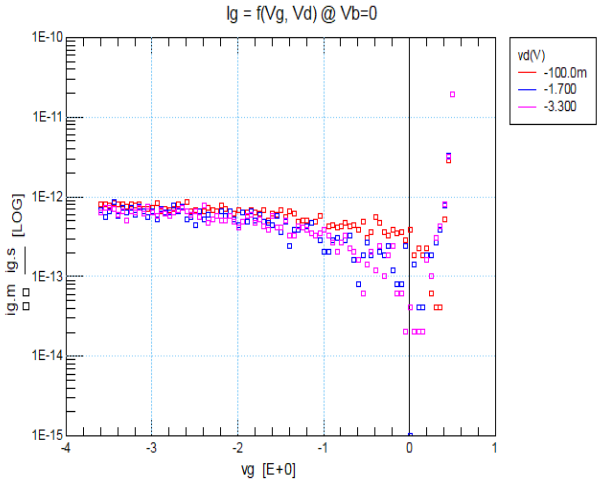




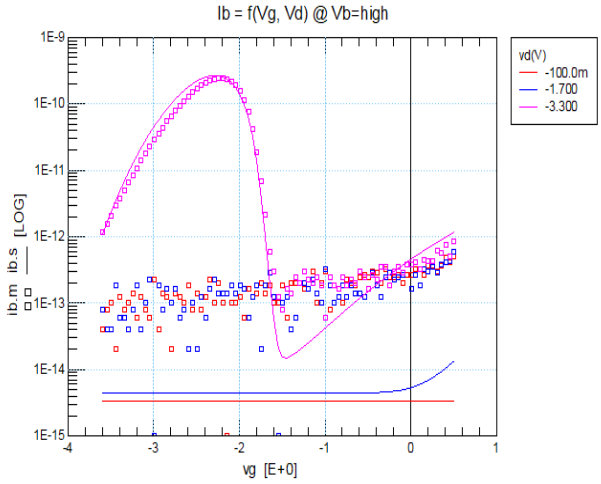


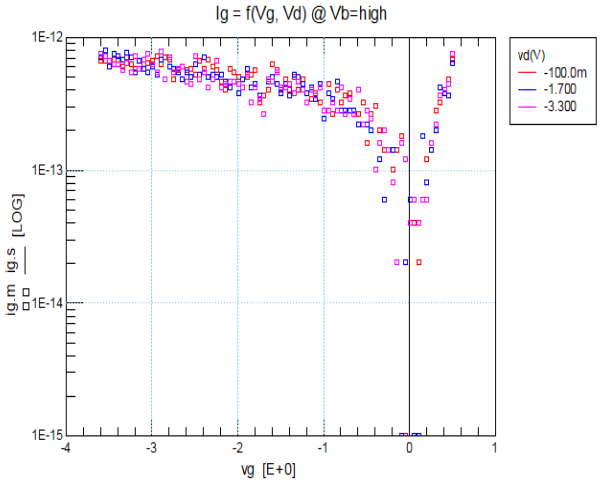
idvg, Ib,Ig, Vb = 0V, T = 27°C



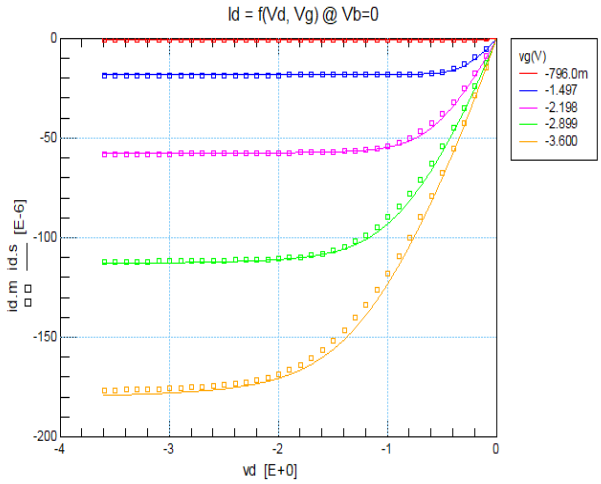


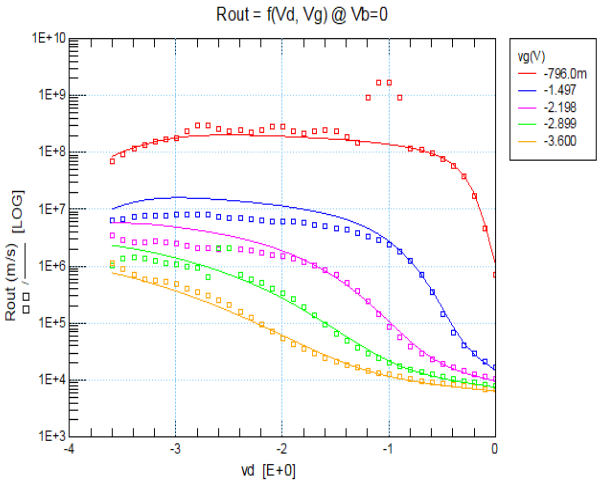
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

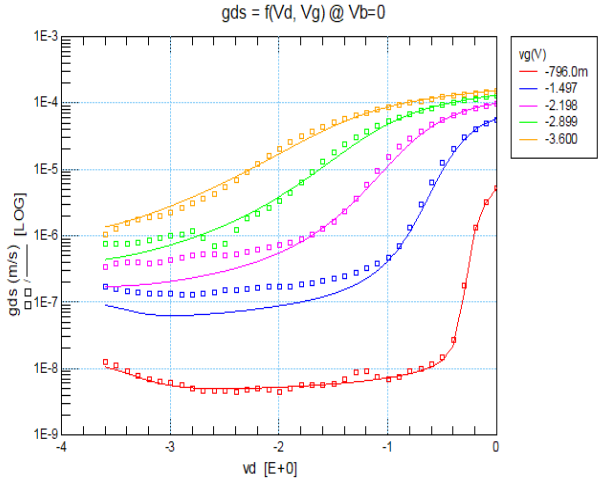




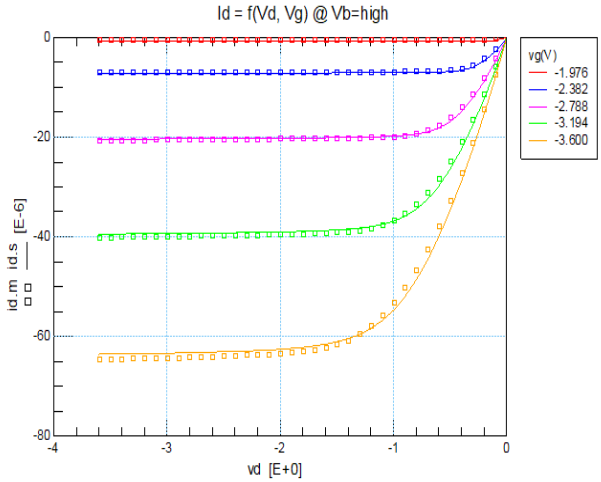
idvd, Vb = 0V, T = 27°C

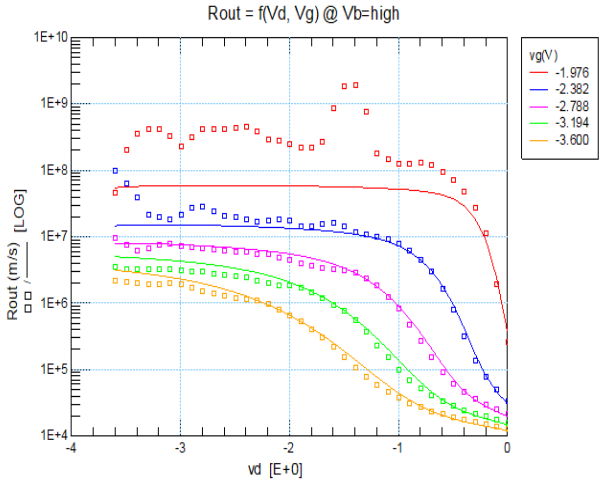


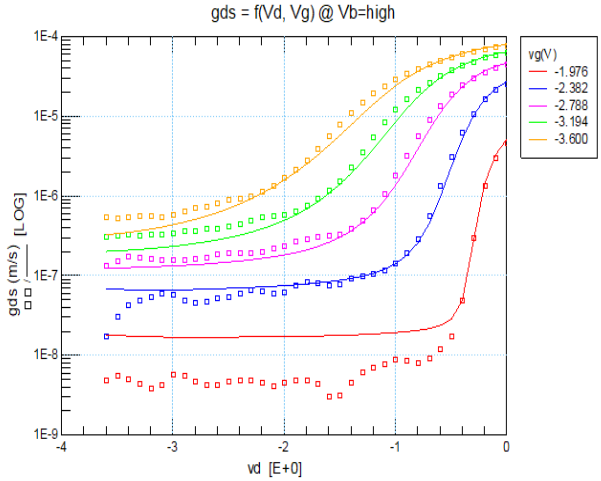




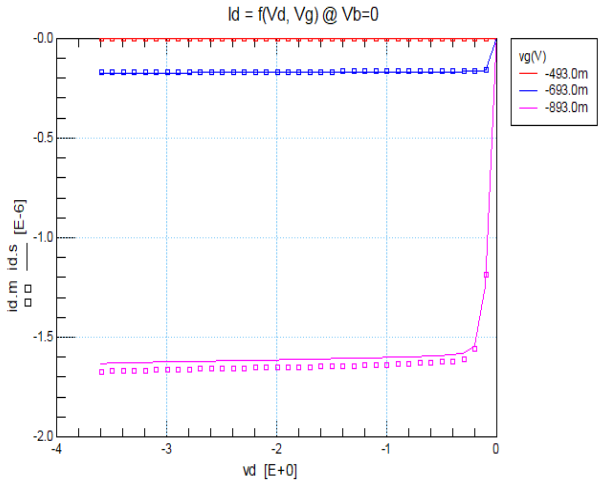
idvd_vbmin, Vb = 3.6V, T = 27°C

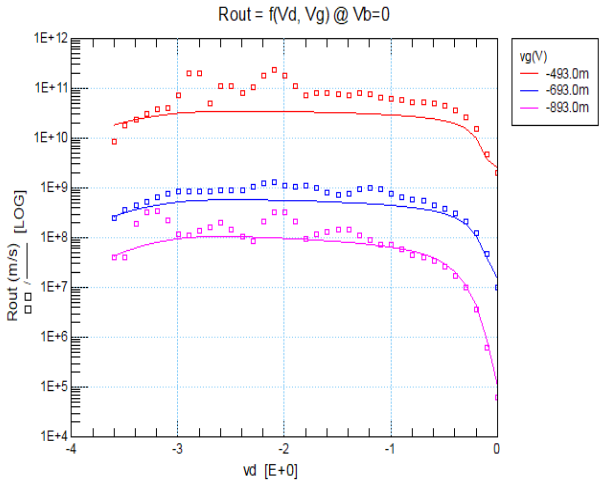


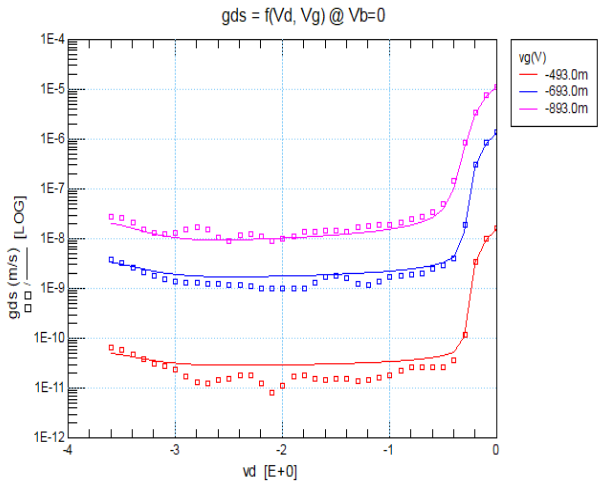




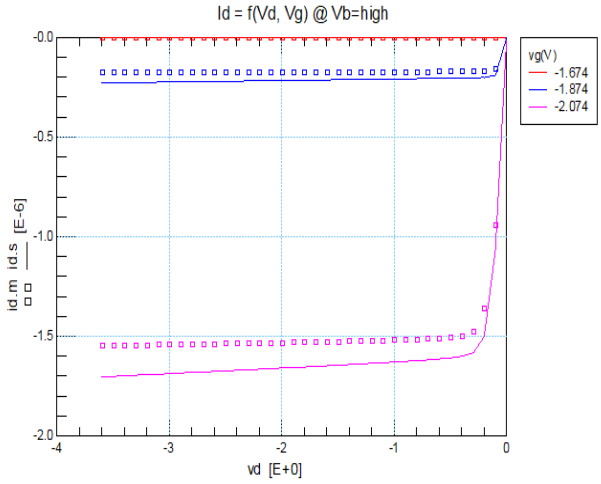
idvd_vth, Vb = 0V, T = 27°C

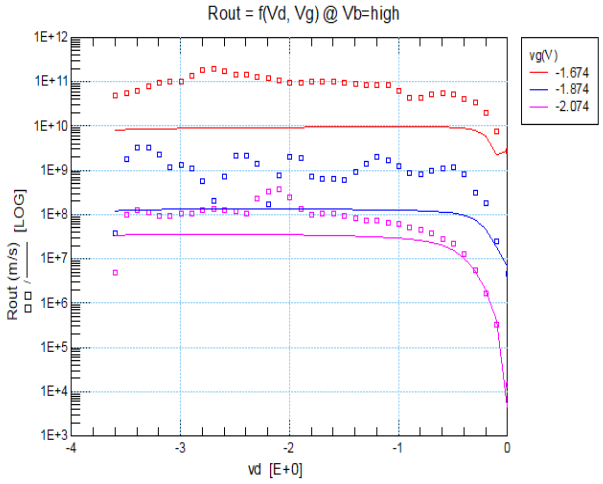


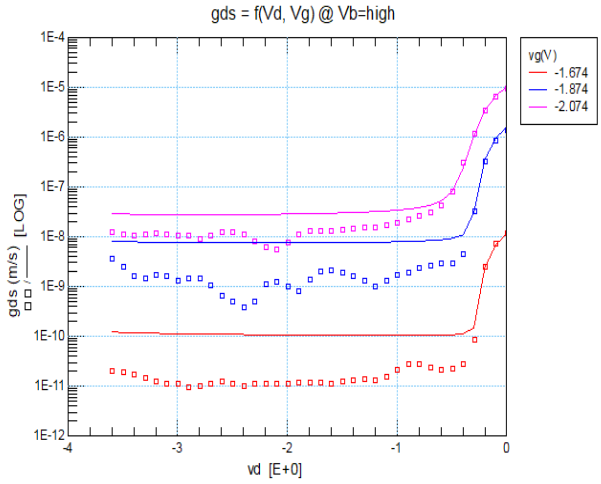




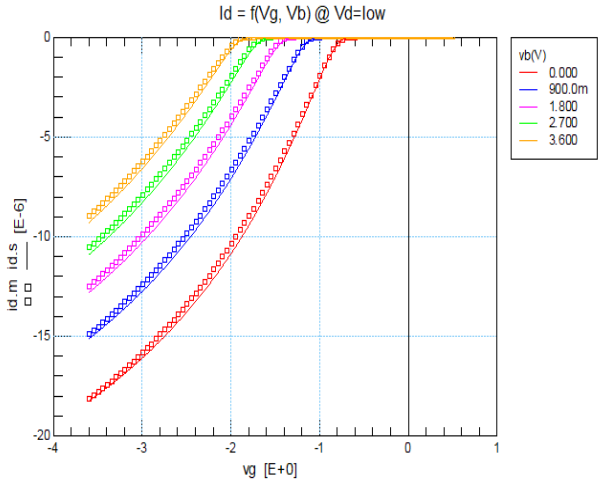
idvd_vth_vbmin, Vb = 3.6V, T = 27°C

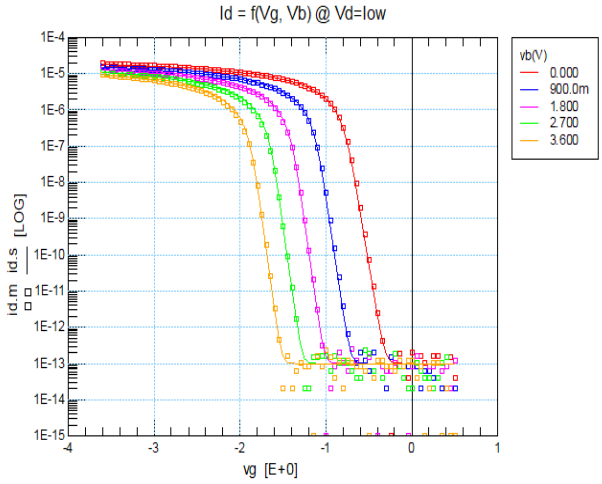


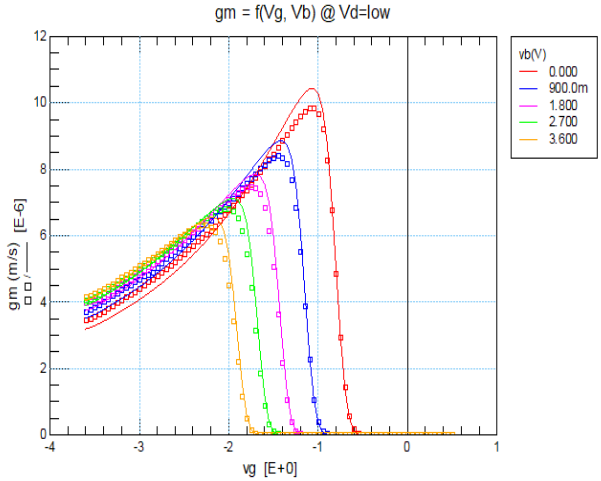


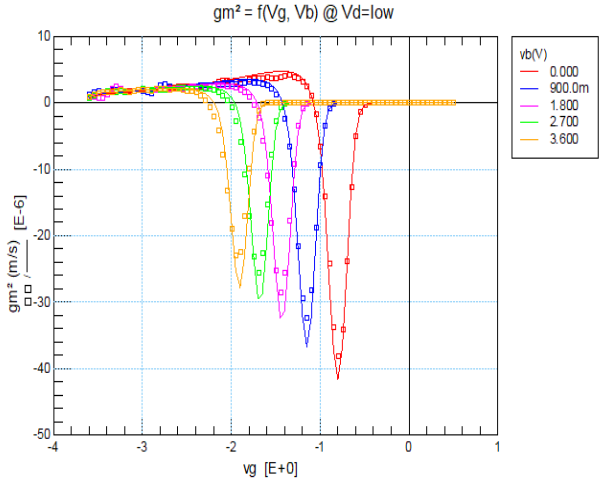


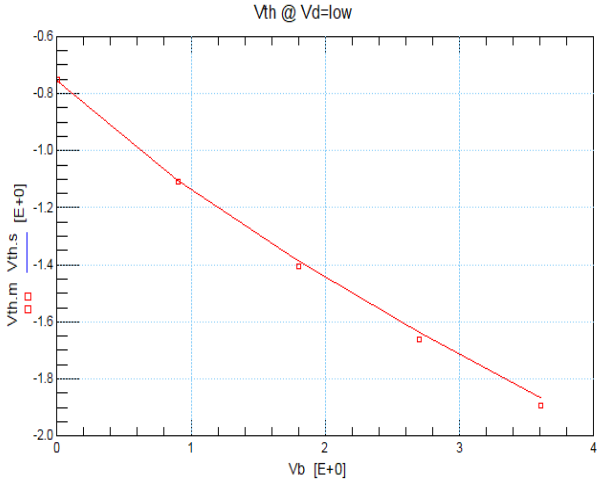
idvg, Vd = -0.1V, T = -40°C

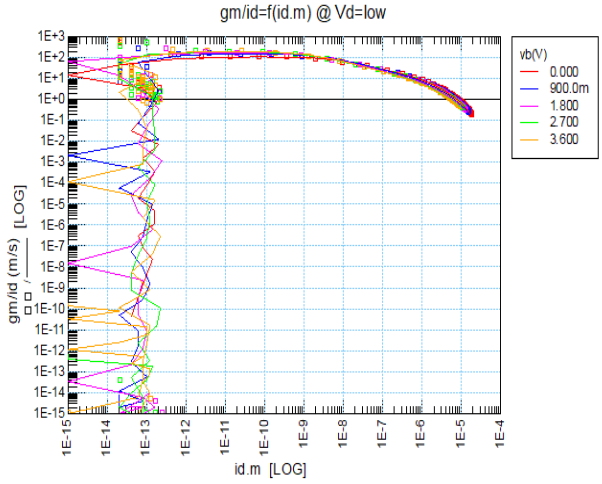


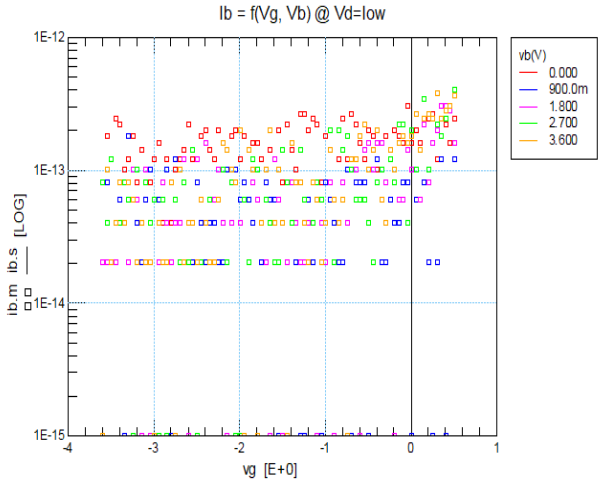


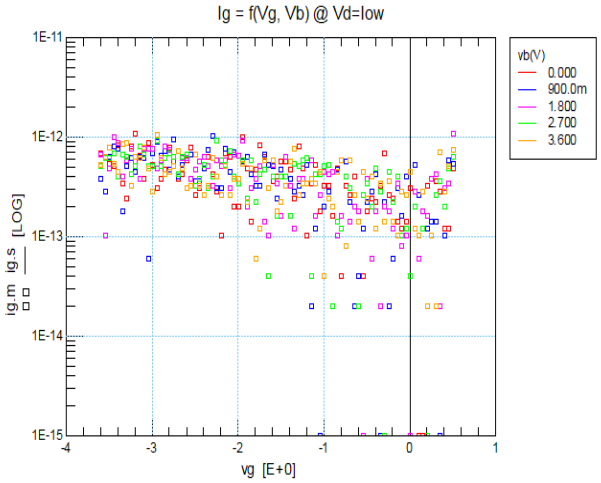




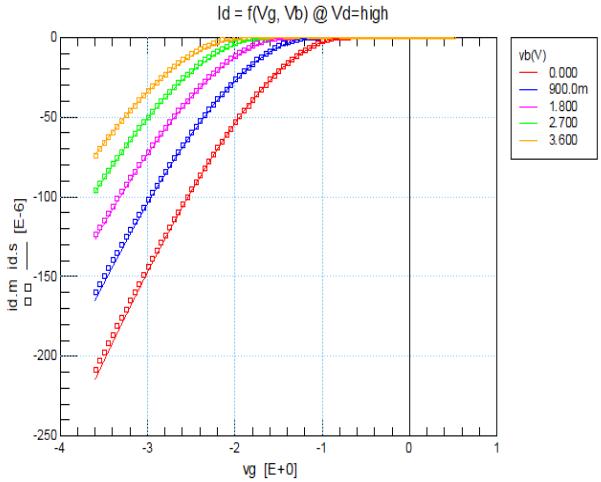


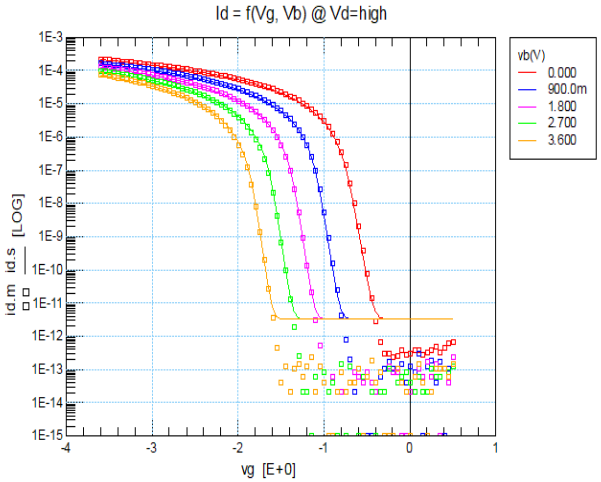


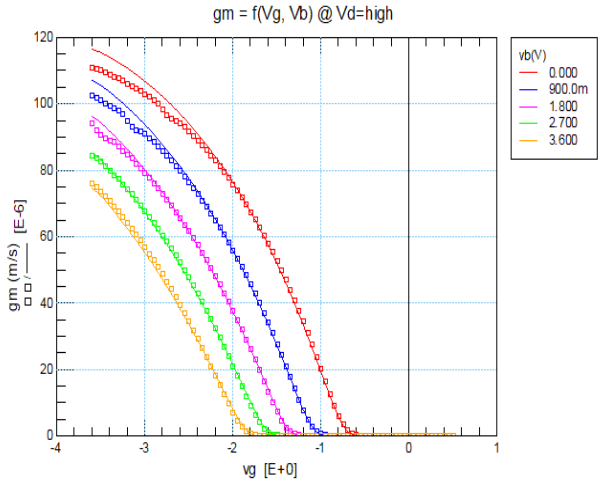


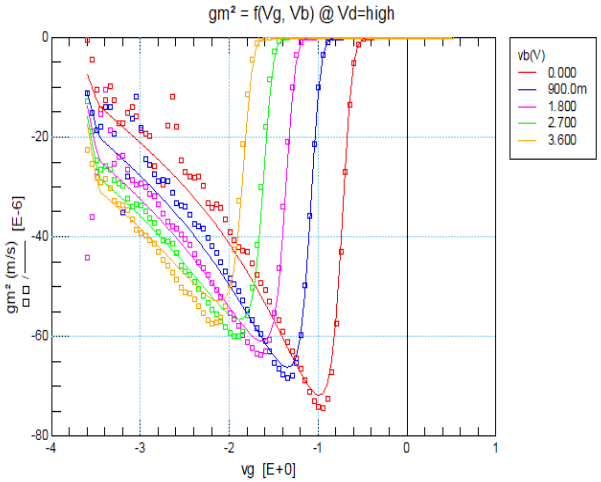


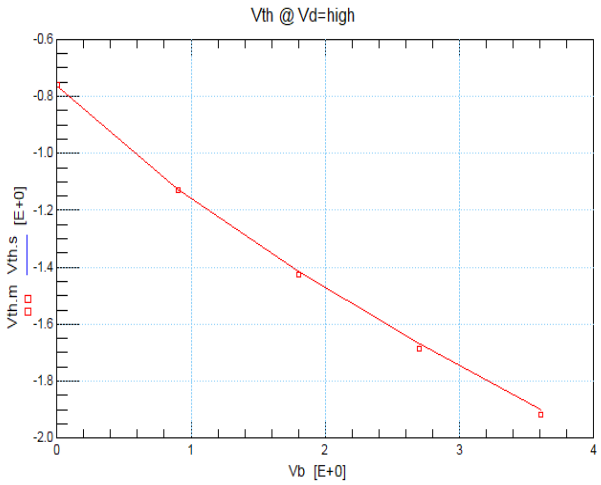
idvg, Vd = -3.3V, T = -40°C

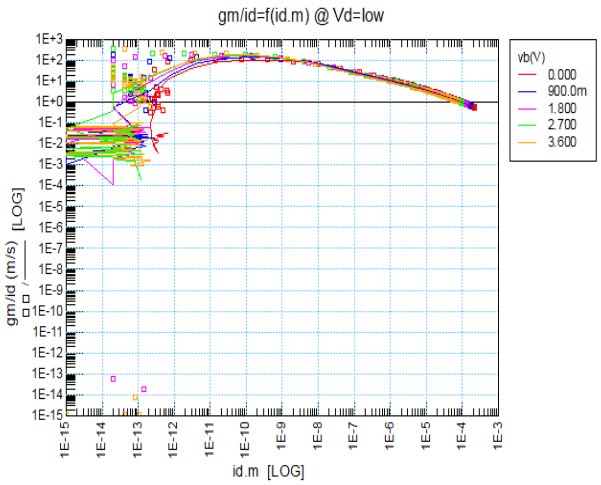


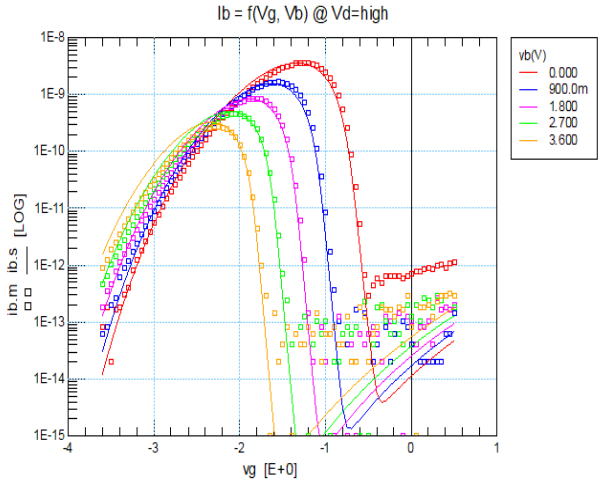


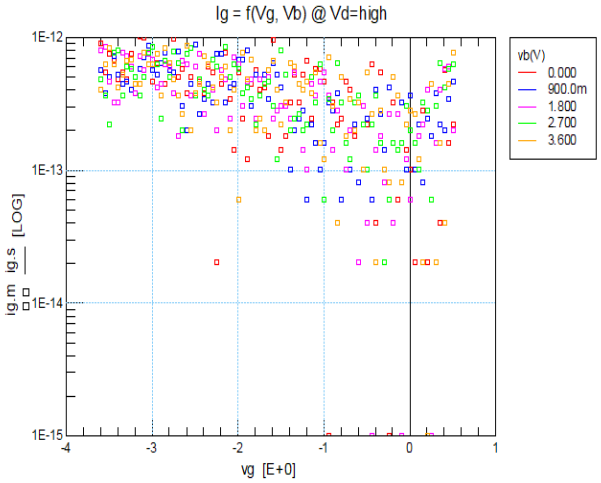




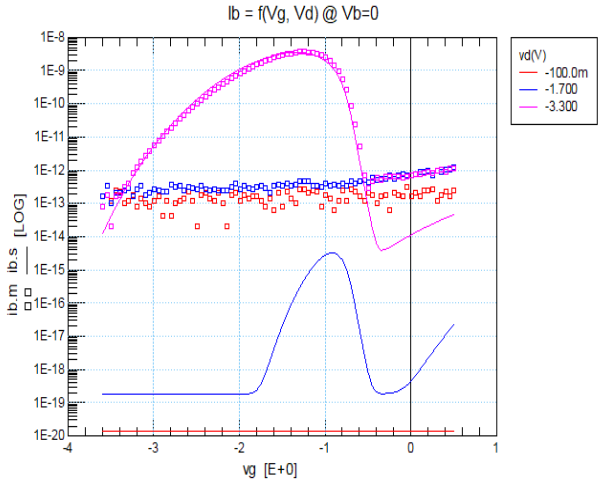


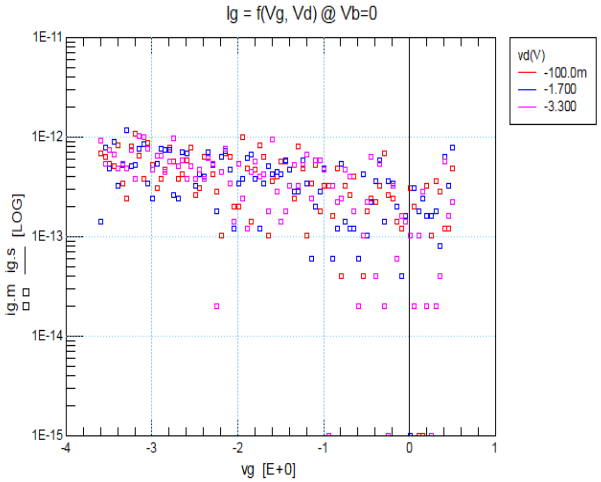




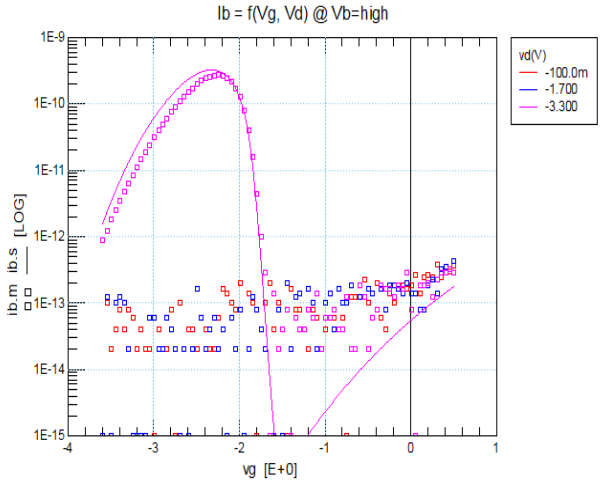


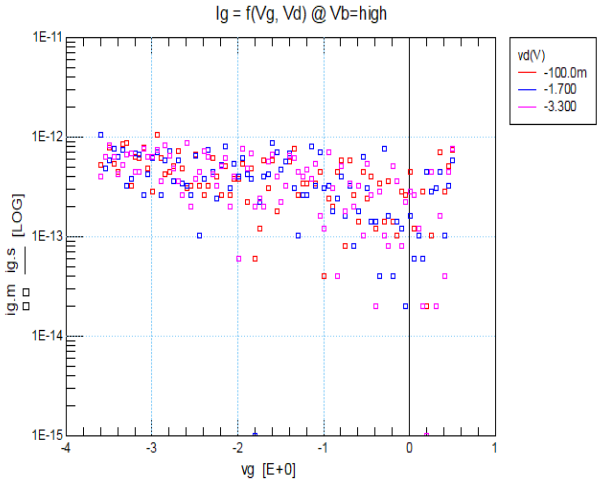
idvg, Ib,Ig, Vb = 0V, T = -40°C



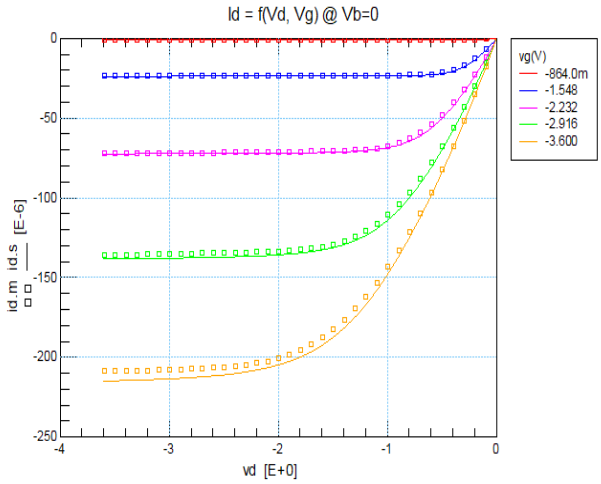


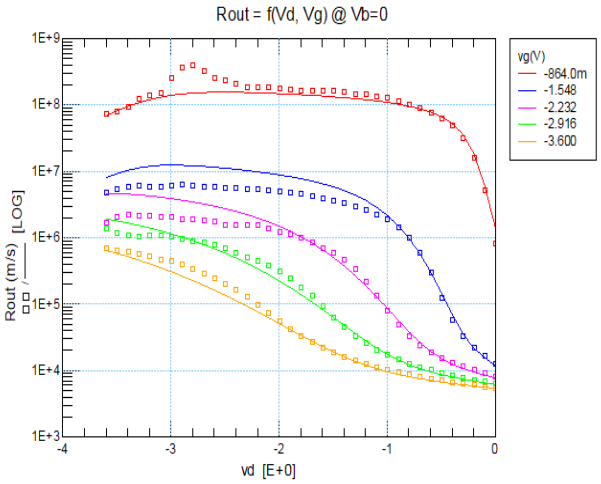
idvg, Ib,Ig, Vb = 3.6V, T = -40°C

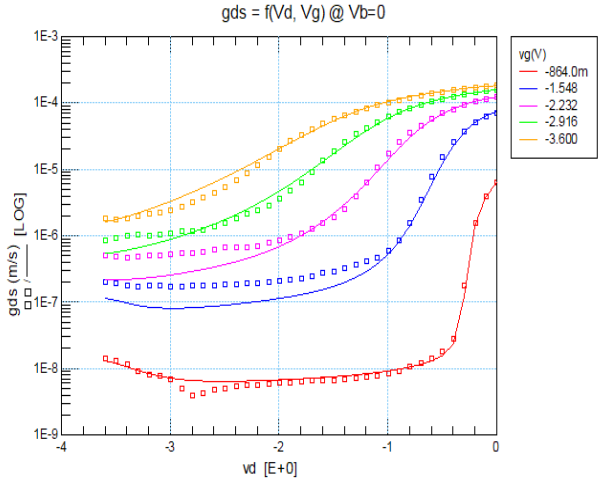




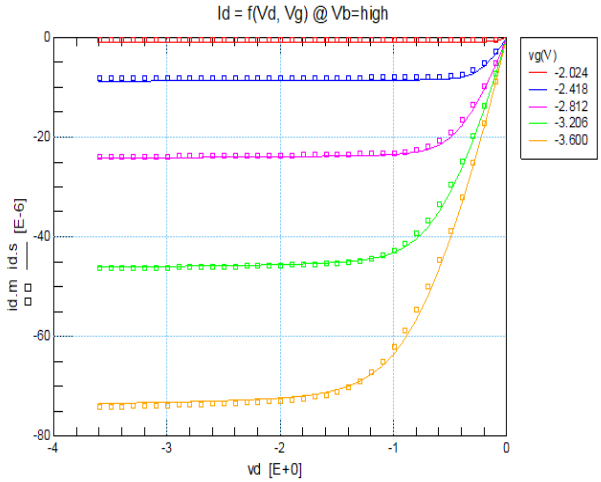
idvd, Vb = 0V, T = -40°C

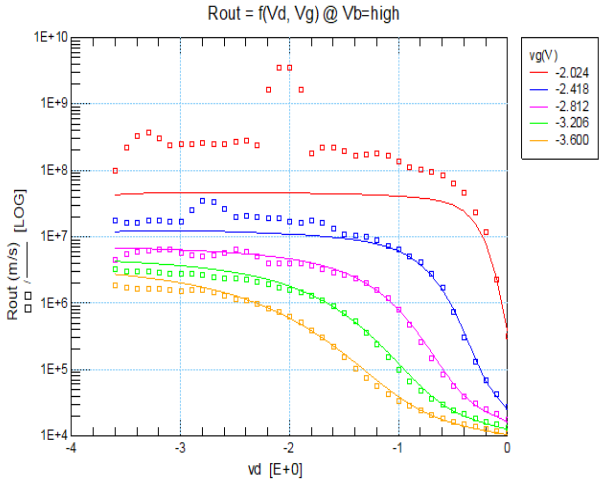


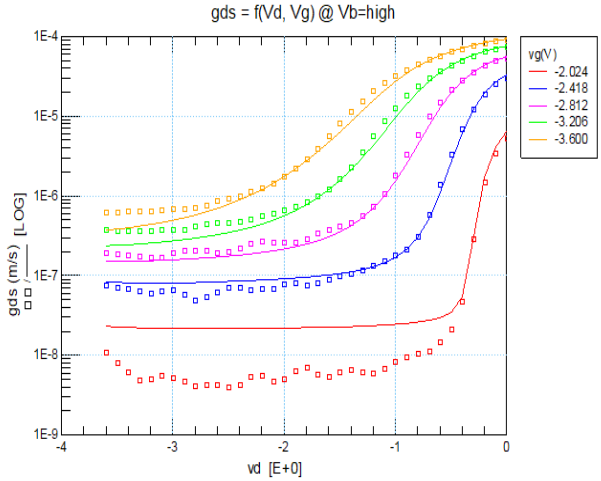




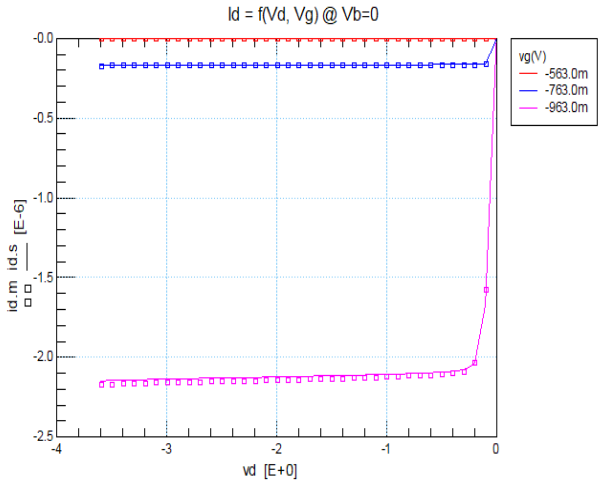
idvd_vbmin, Vb = 3.6V, T = -40°C

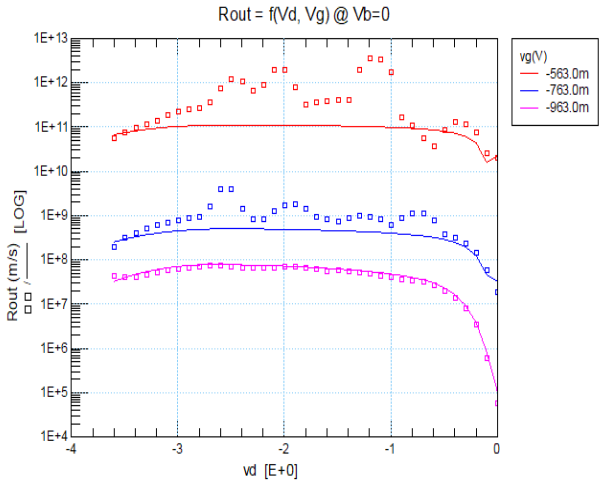


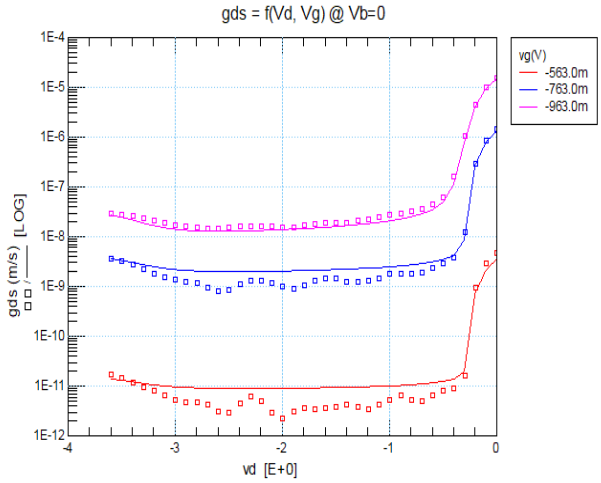




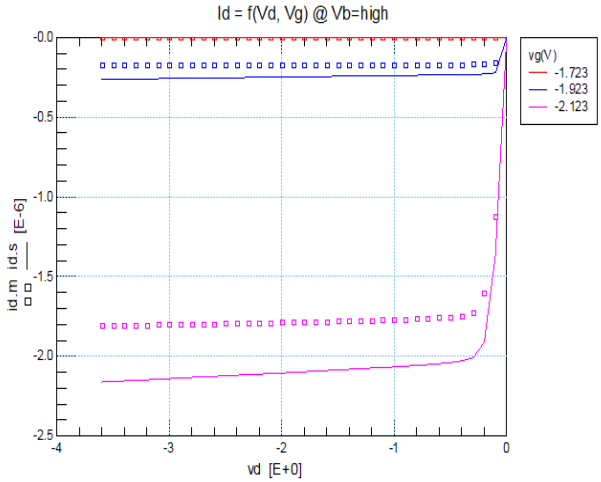
idvd_vth, Vb = 0V, T = -40°C

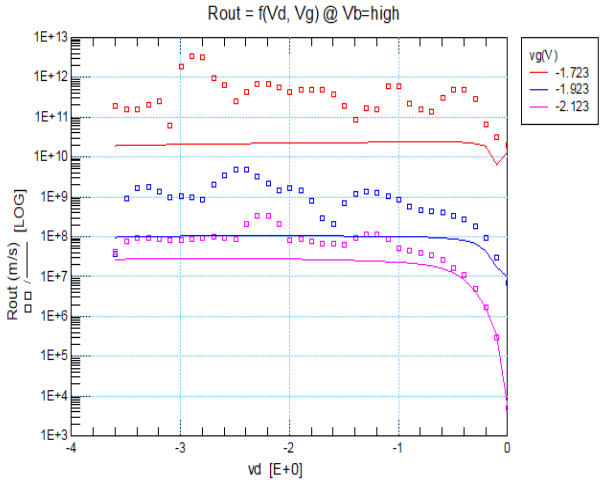


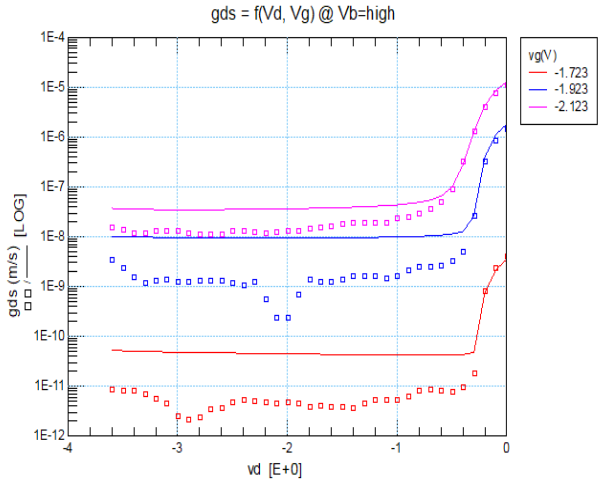




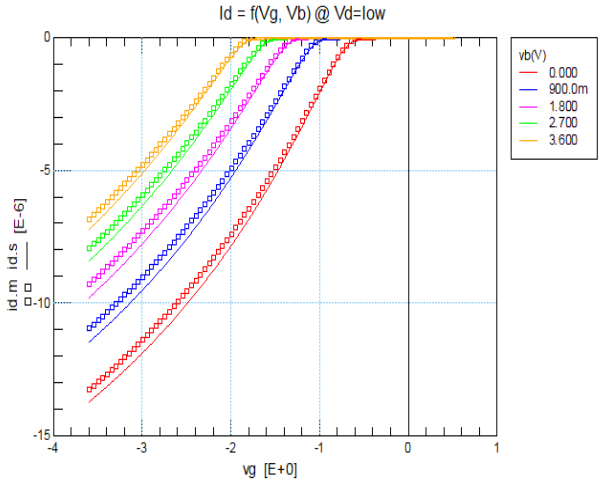
idvd_vth_vbmin, Vb = 3.6V, T = -40°C

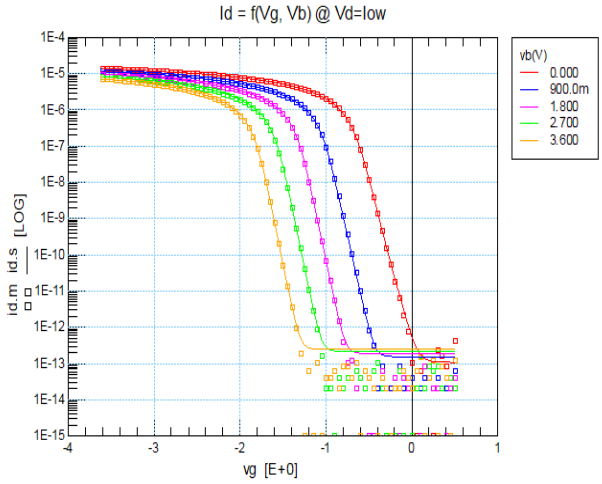


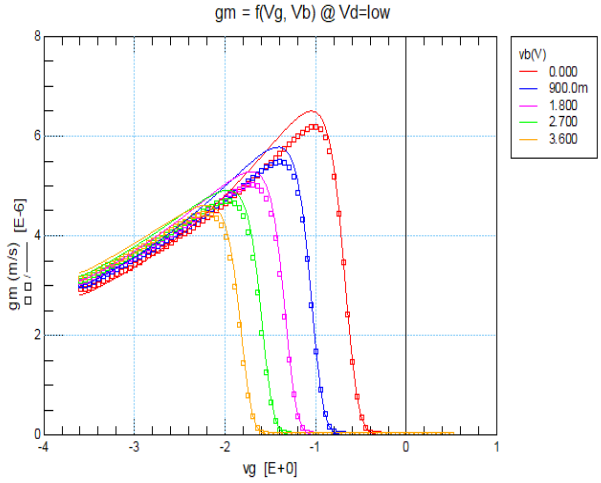


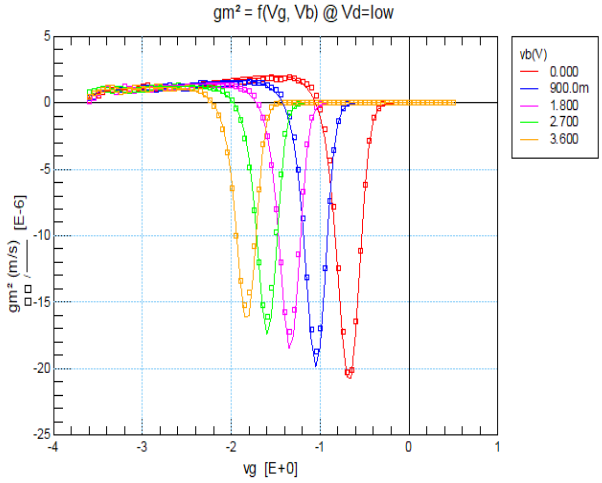


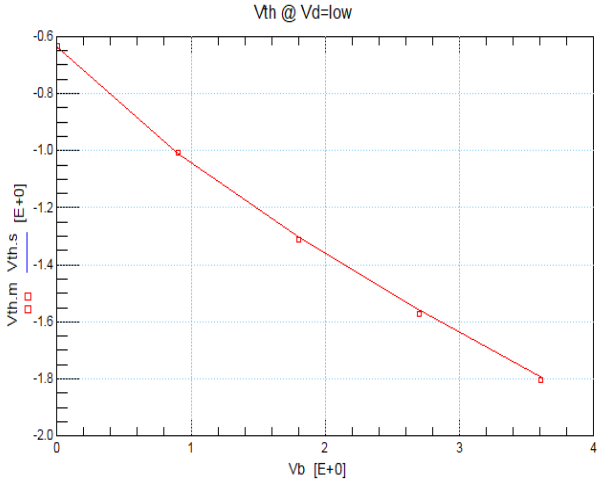
idvg, Vd = -0.1V, T = 70°C

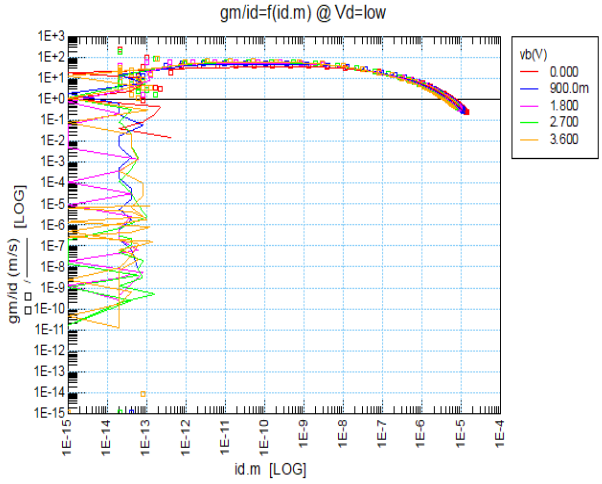


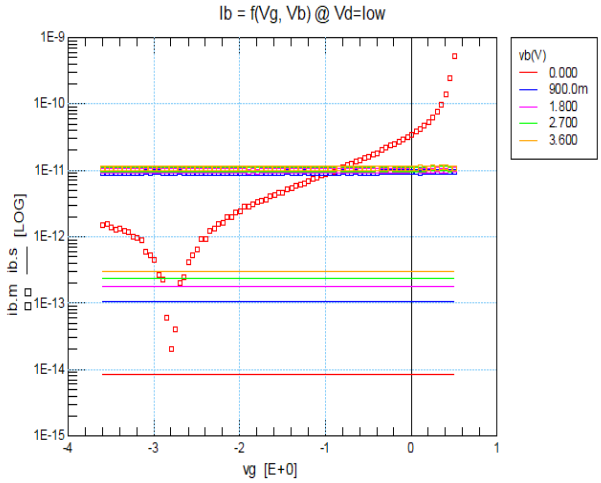


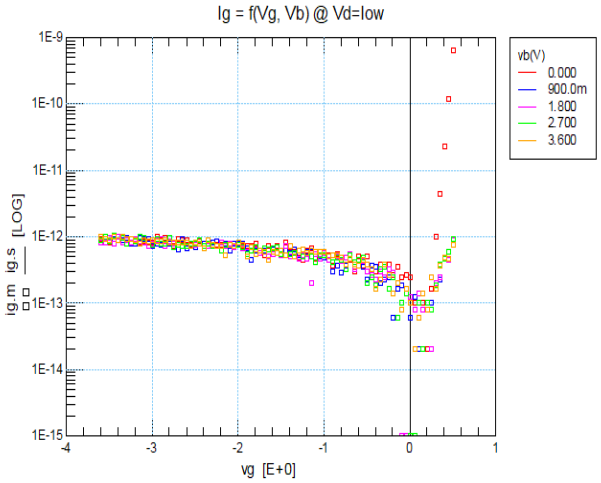




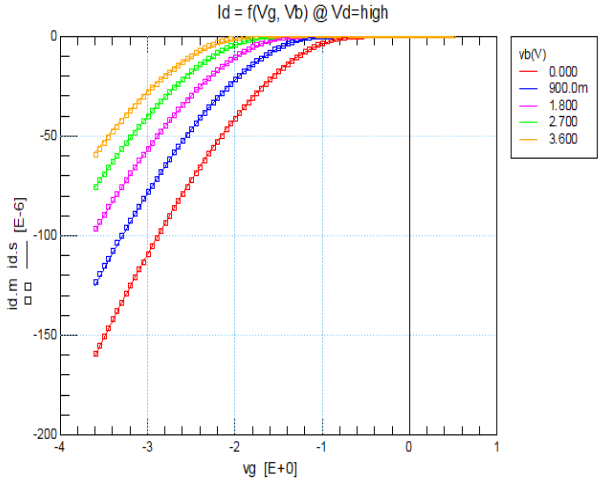


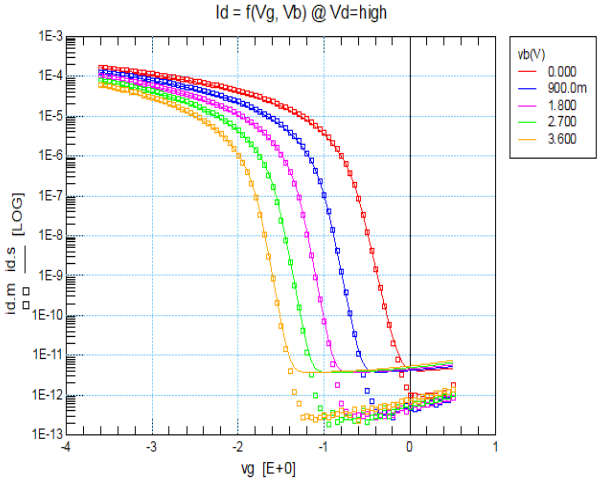


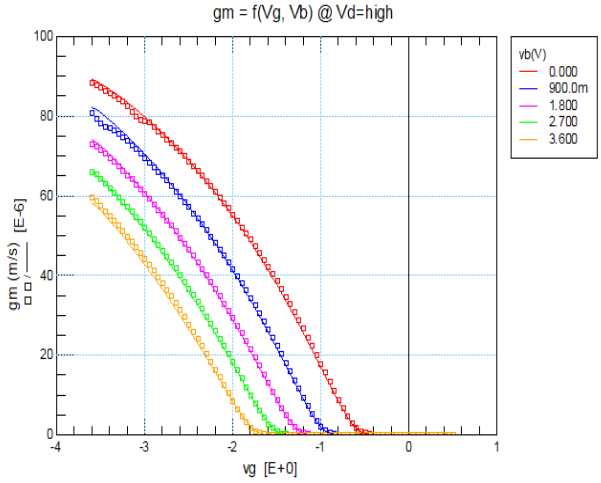


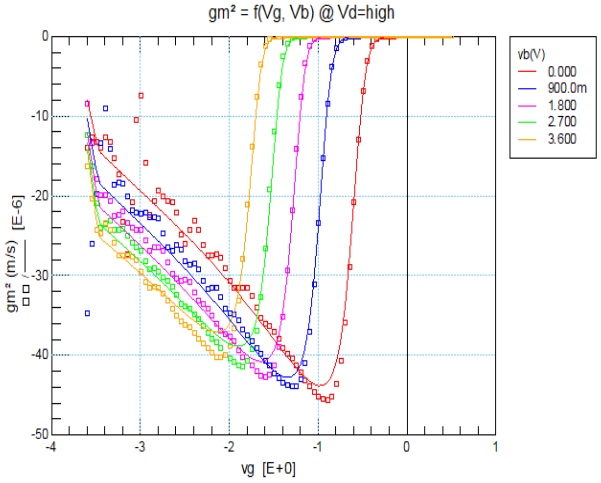


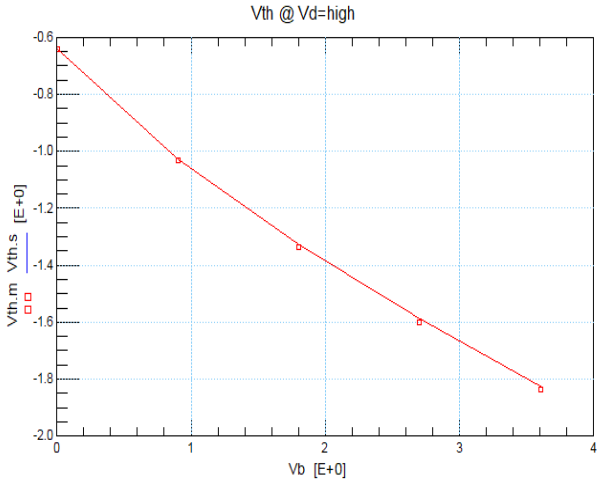
idvg, Vd = -3.3V, T = 70°C

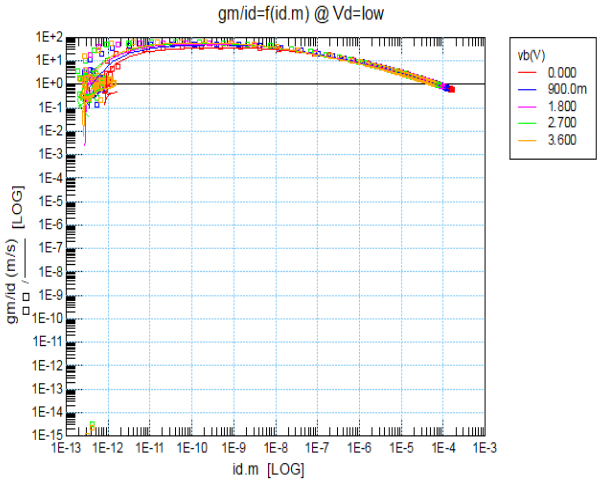


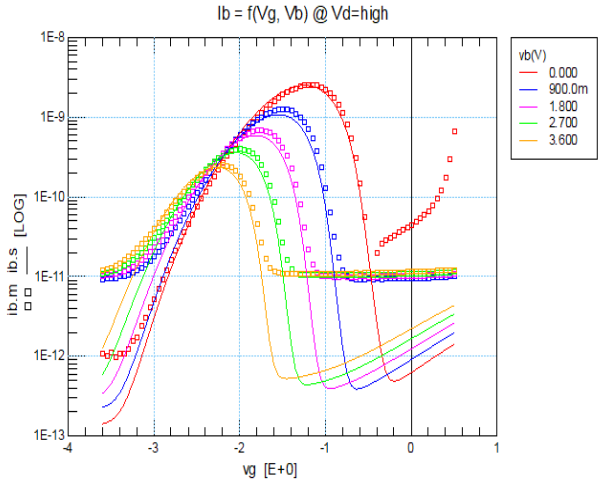


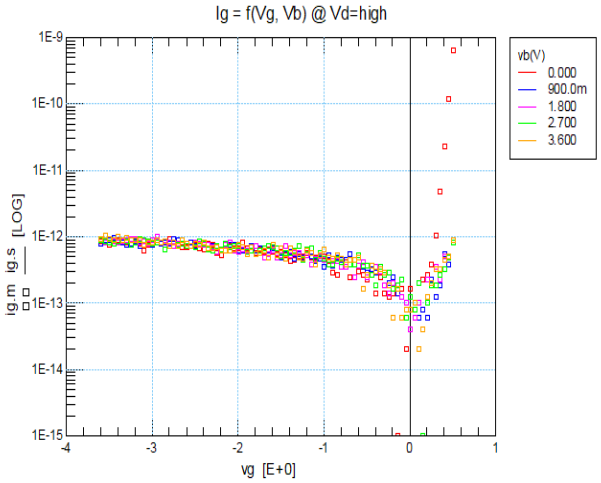




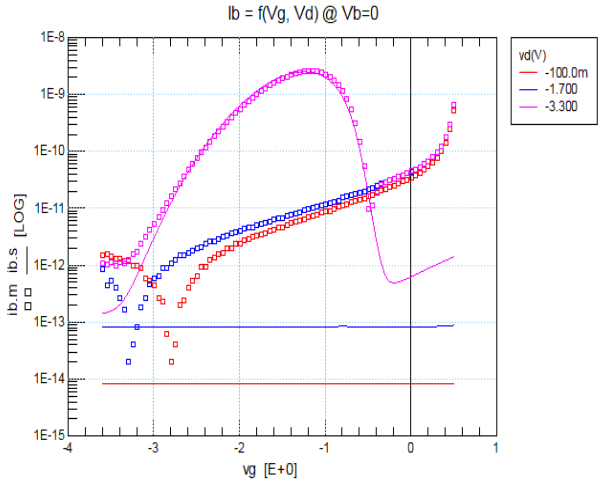


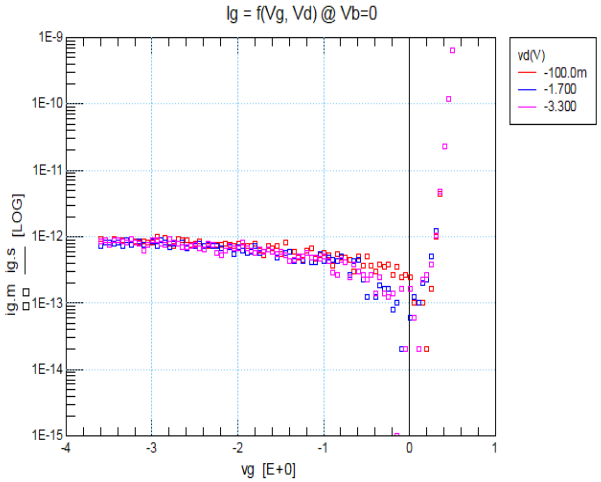




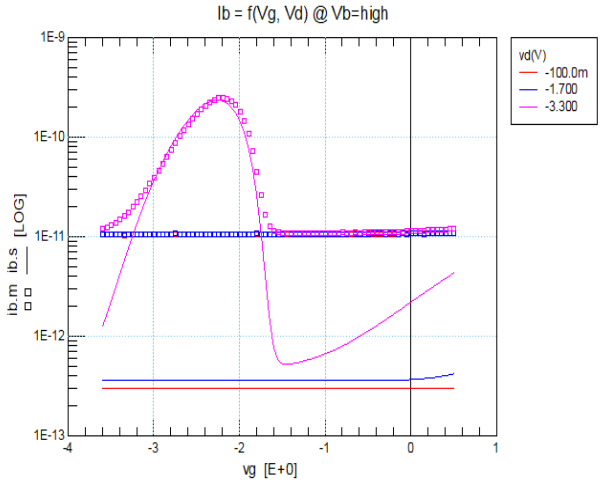


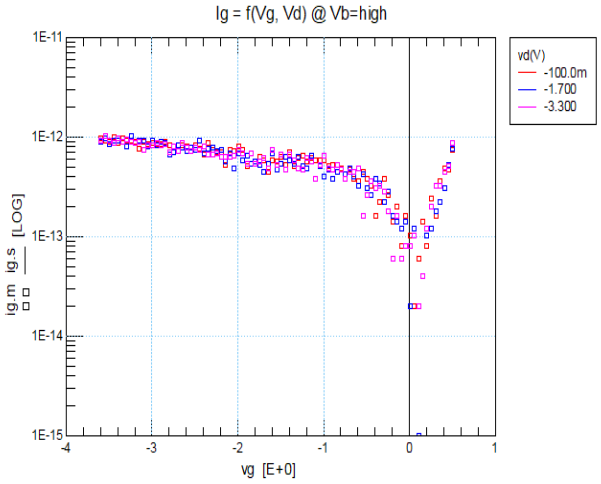
idvg, Ib,Ig, Vb = 0V, T = 70°C



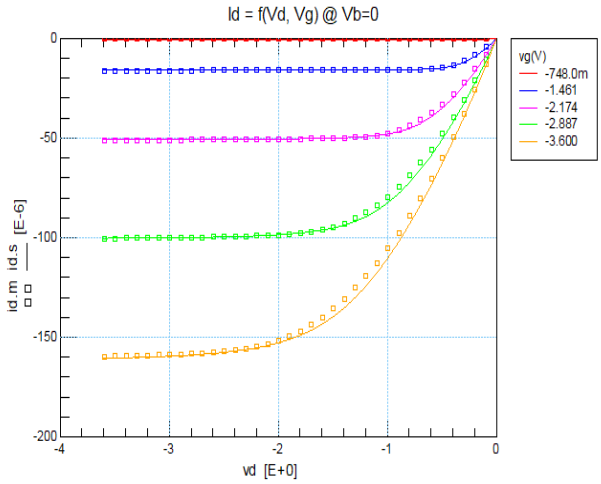


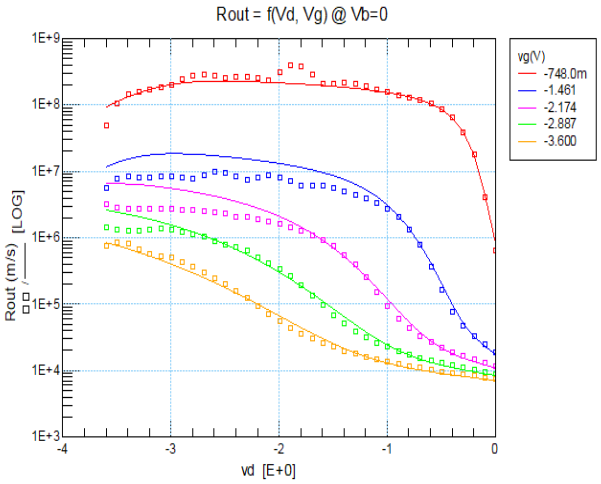
idvg, Ib,Ig, Vb = 3.6V, T = 70°C

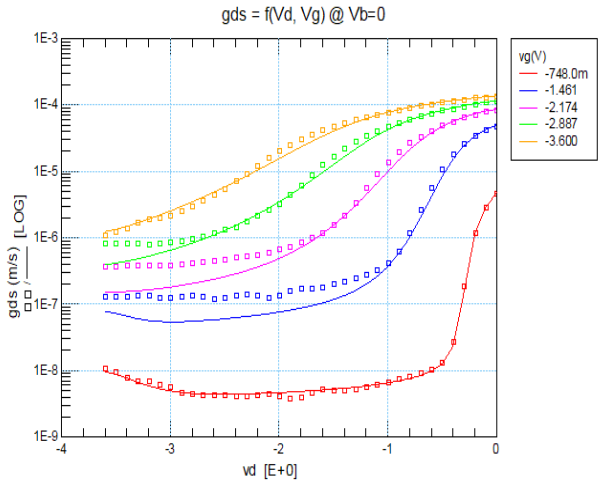




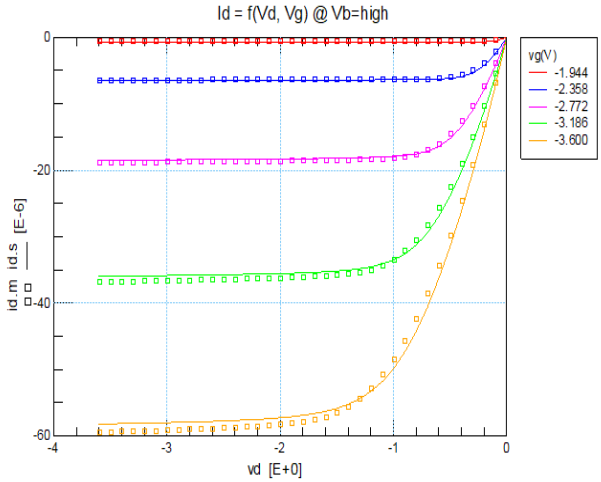
idvd, Vb = 0V, T = 70°C

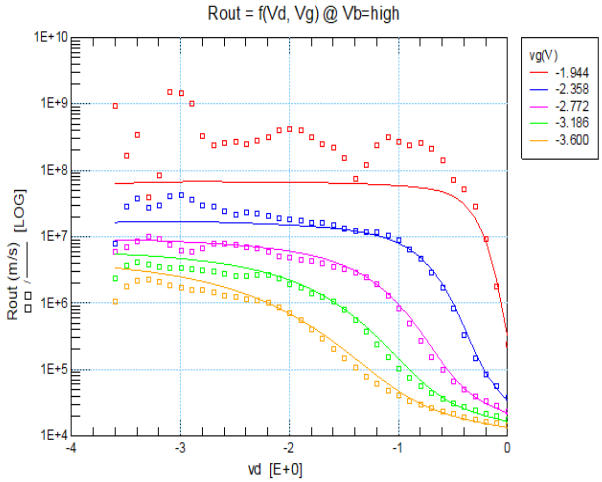


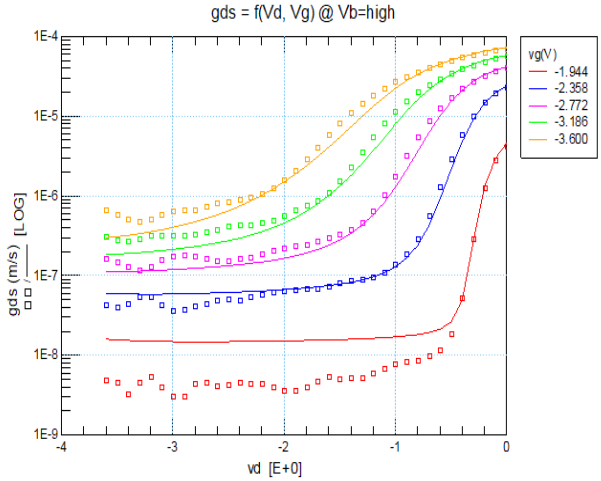




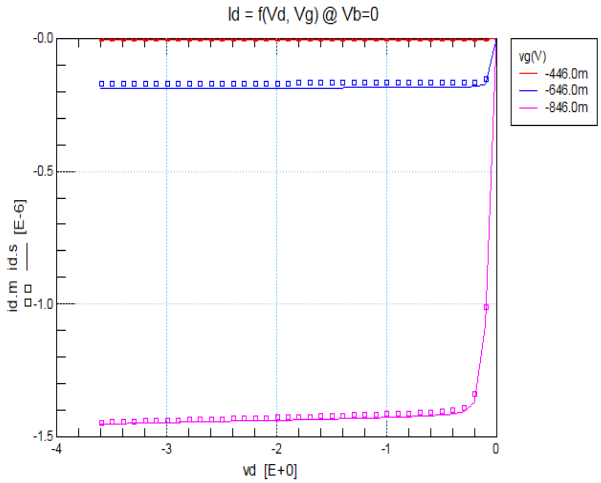
idvd_vbmin, Vb = 3.6V, T = 70°C

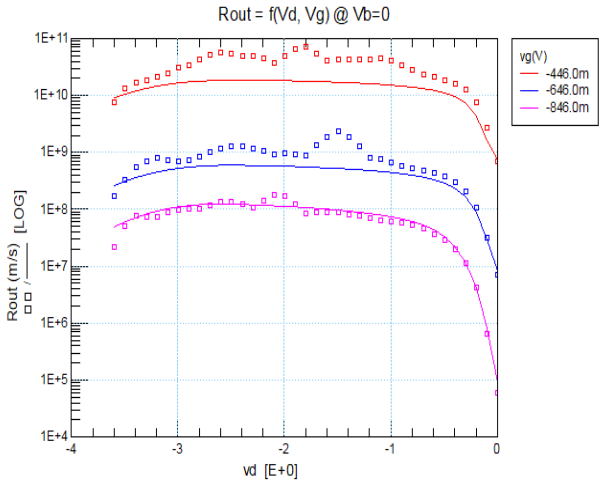


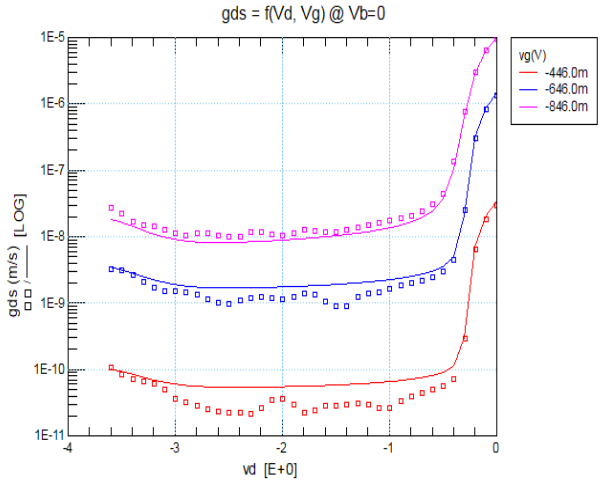




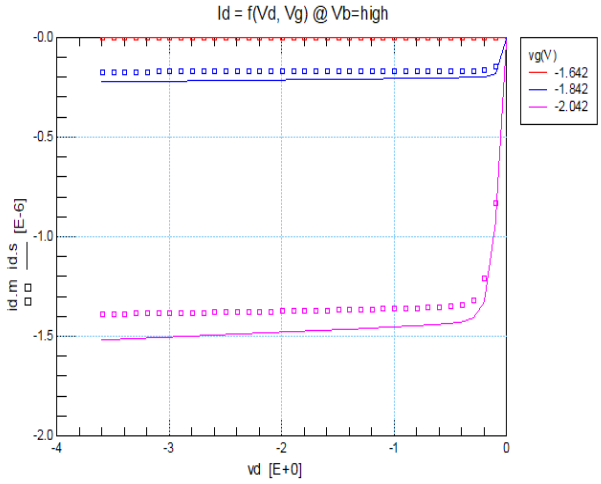
idvd_vth, Vb = 0V, T = 70°C

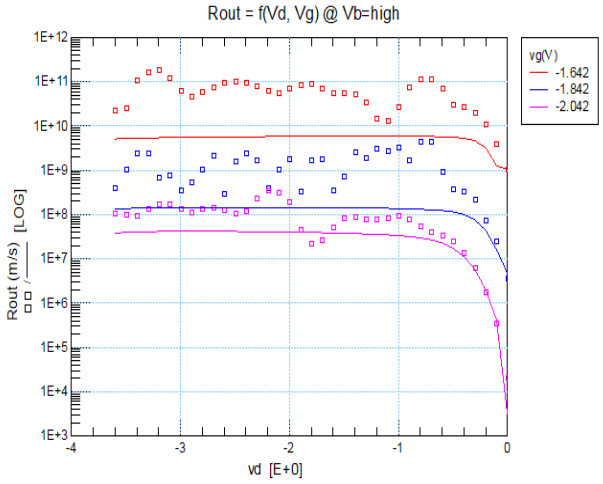


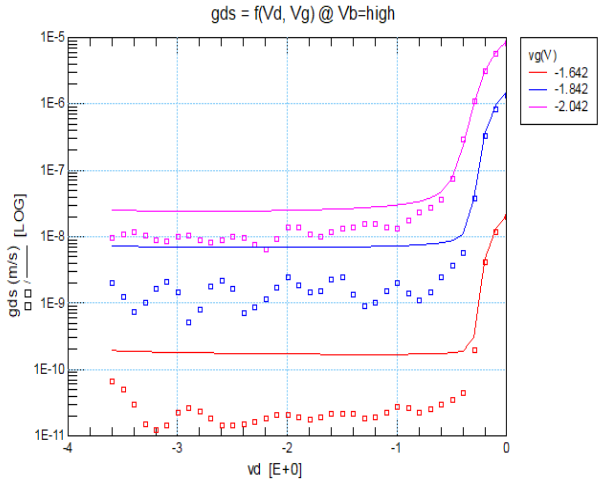




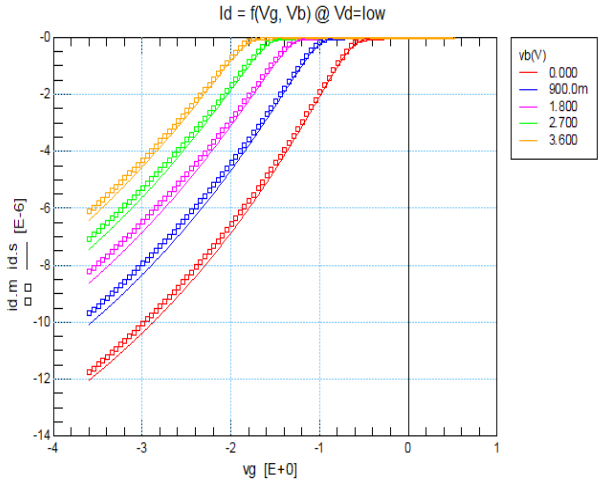
idvd_vth_vbmin, Vb = 3.6V, T = 70°C

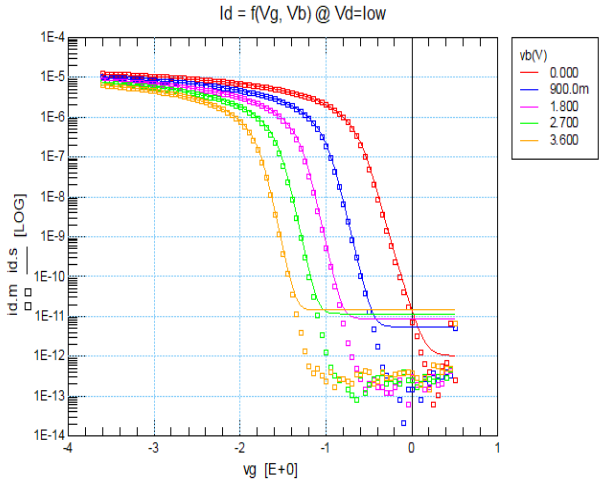


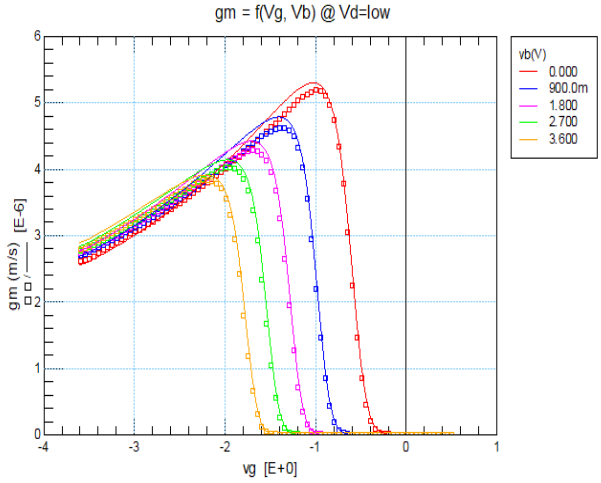


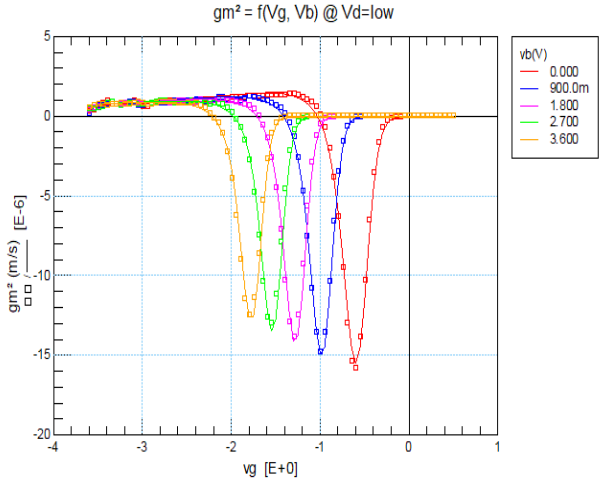


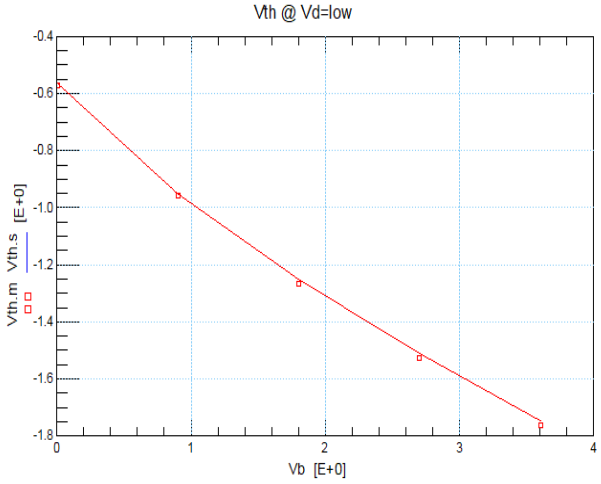
idvg, Vd = -0.1V, T = 125°C

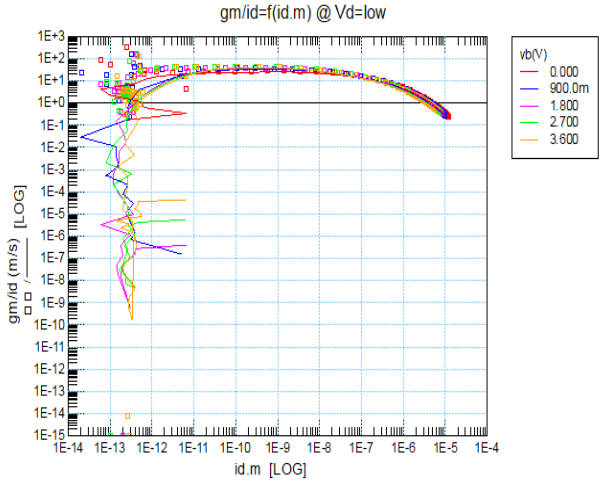


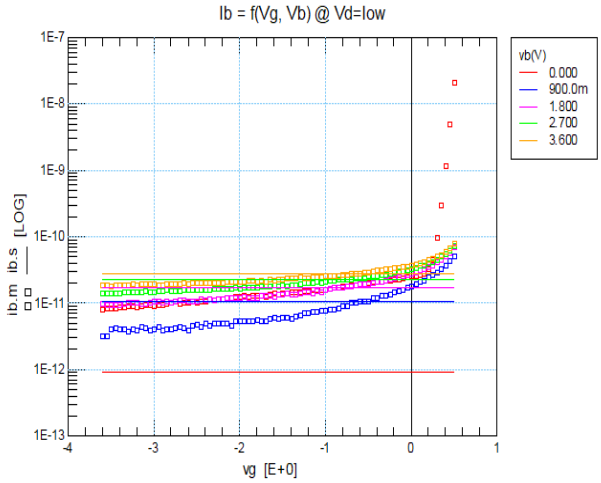


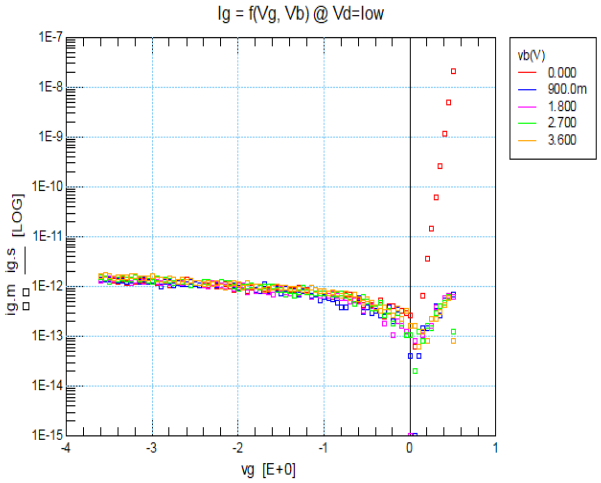




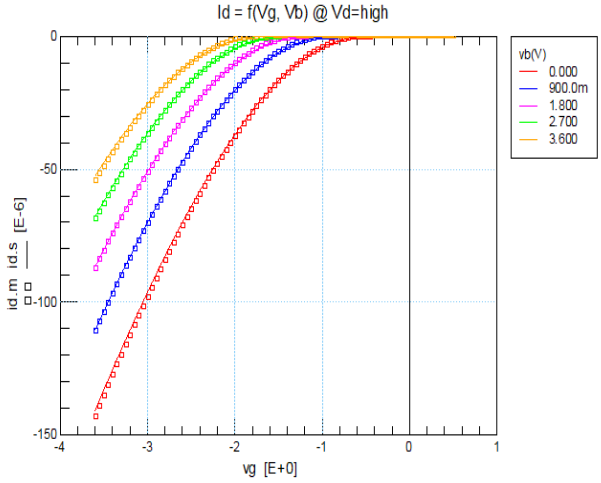


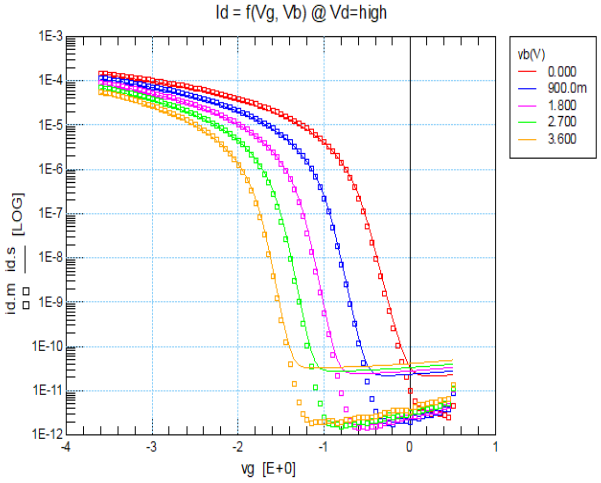


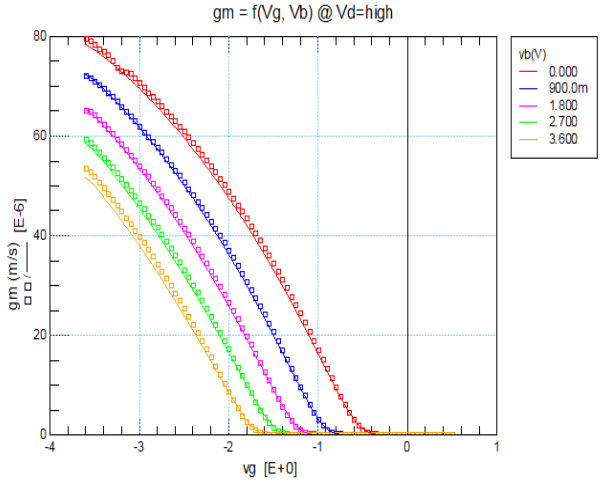


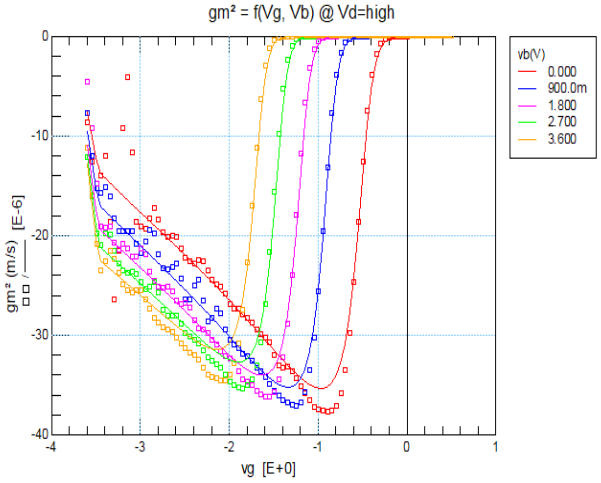


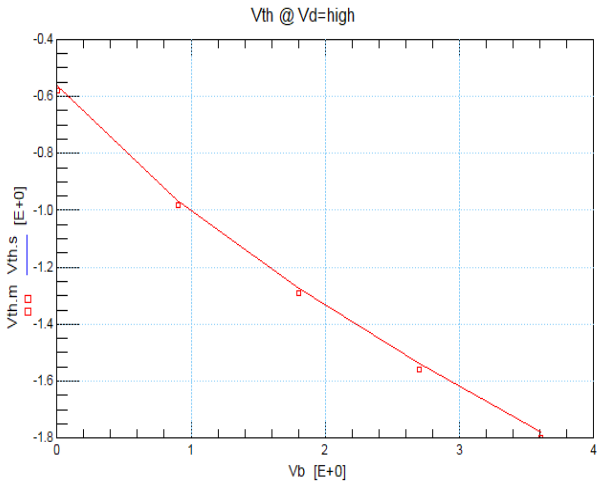
idvg, Vd = -3.3V, T = 125°C

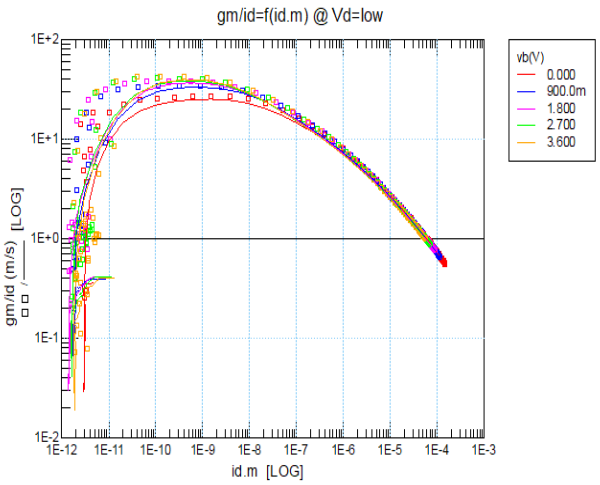


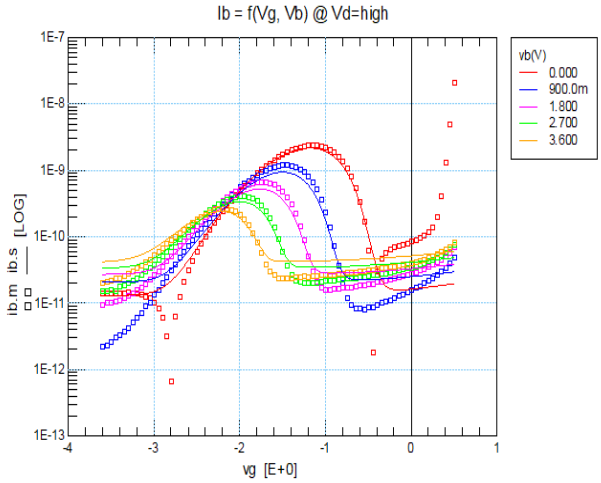


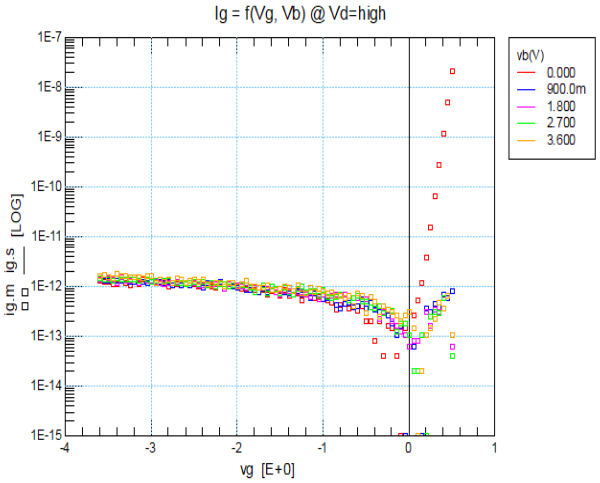




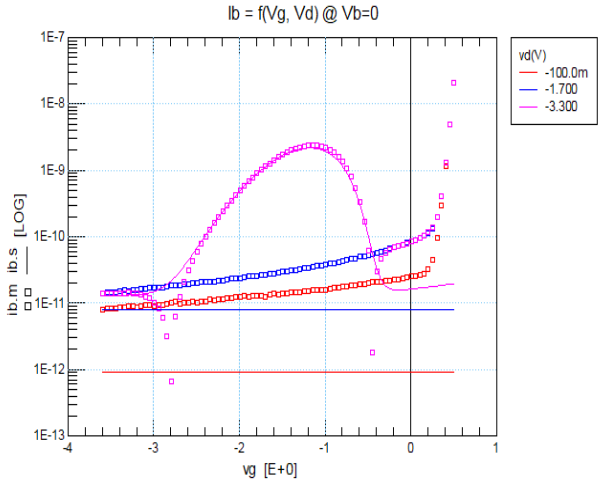


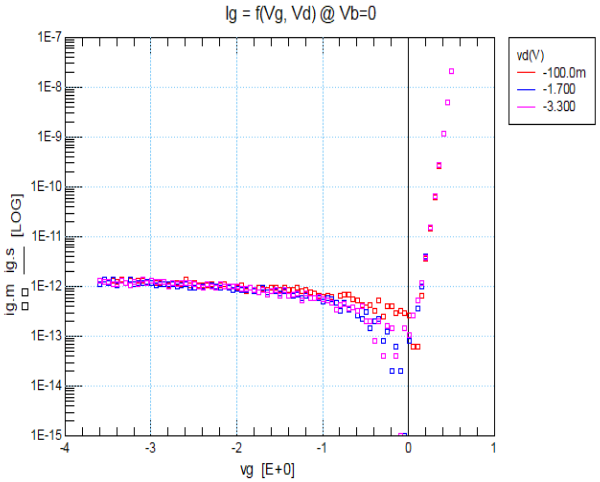




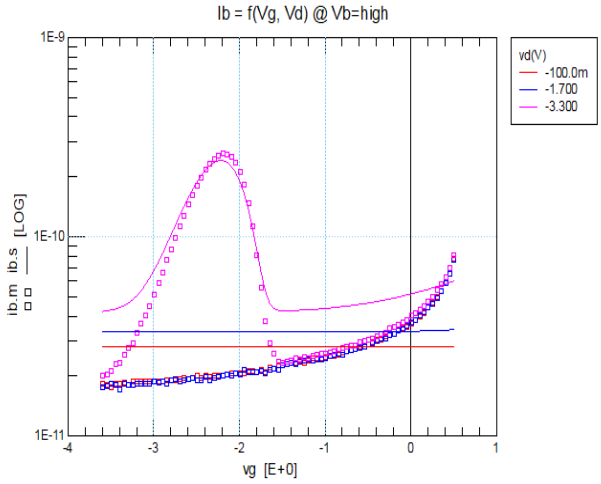


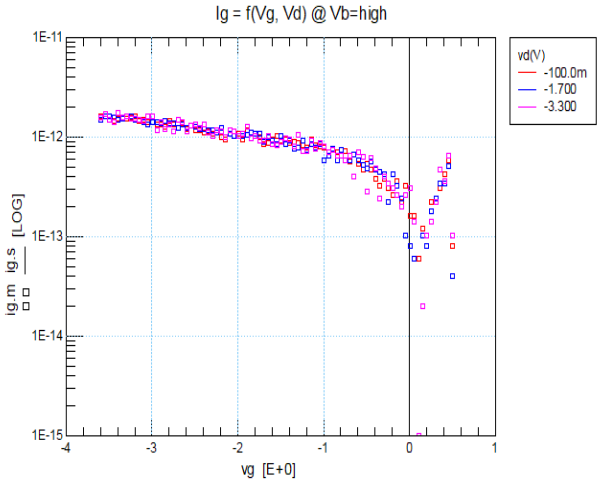
idvg, Ib,Ig, Vb = 0V, T = 125°C



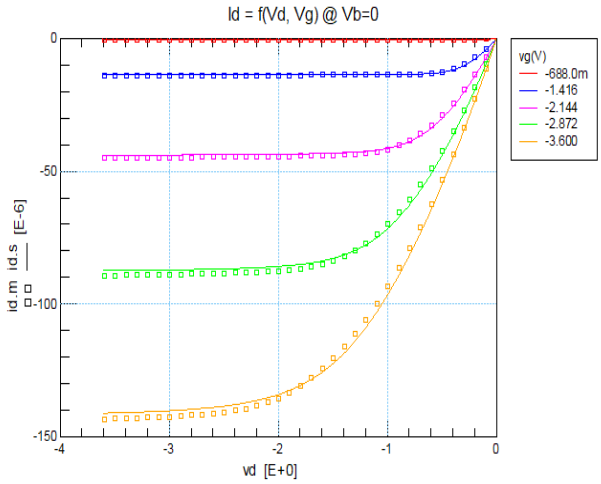


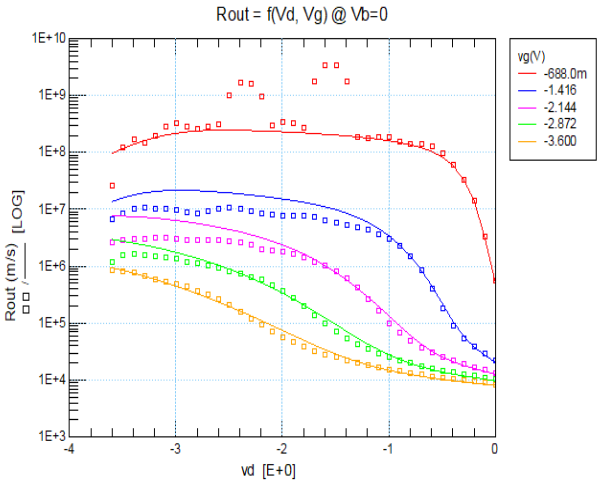
idvg, Ib,Ig, Vb = 3.6V, T = 125°C

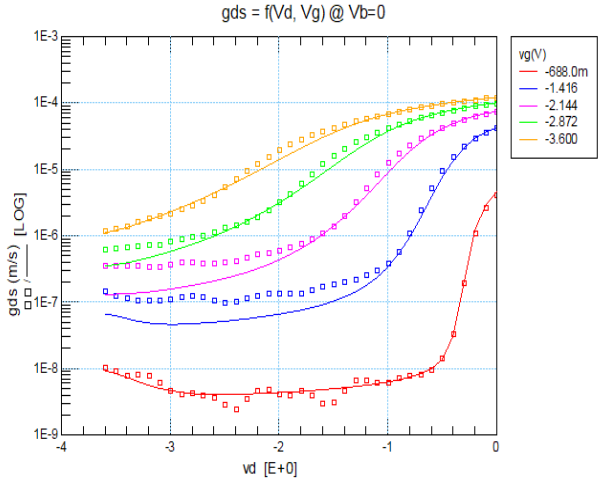




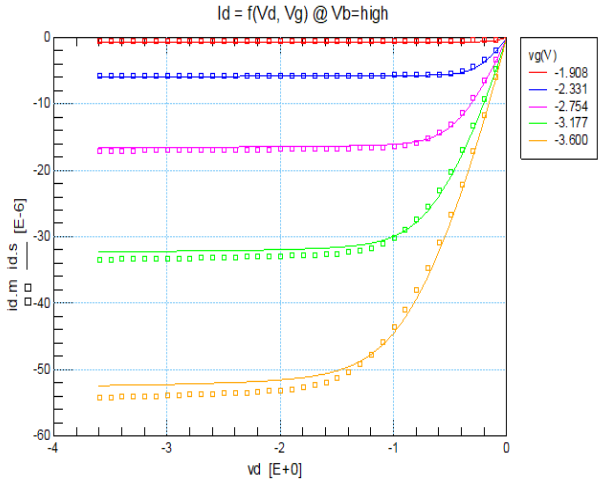
idvd, Vb = 0V, T = 125°C

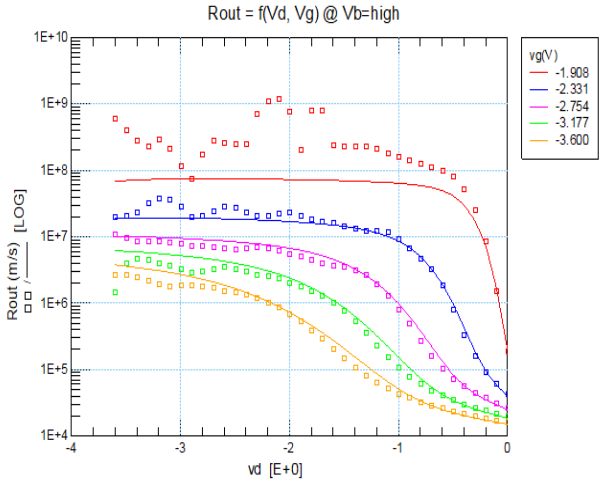


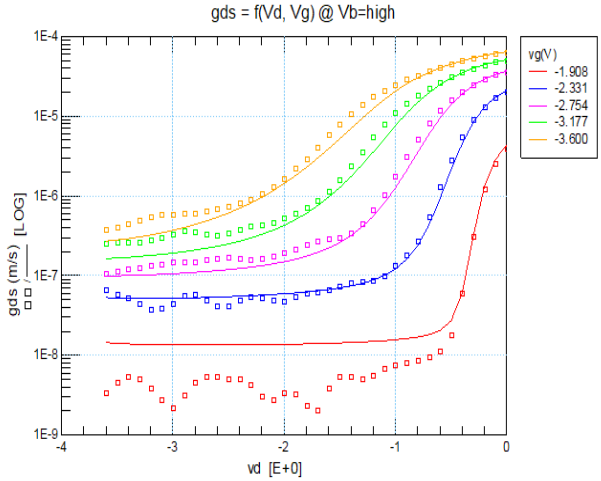




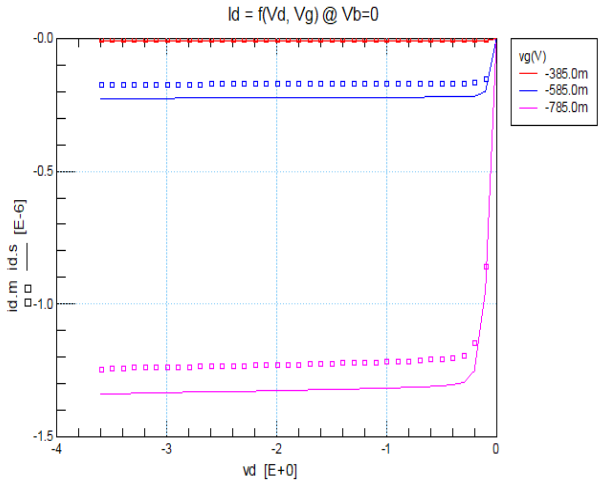
idvd_vbmin, Vb = 3.6V, T = 125°C

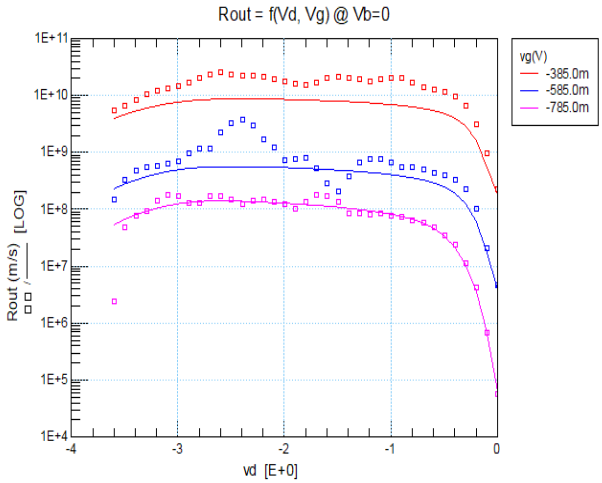


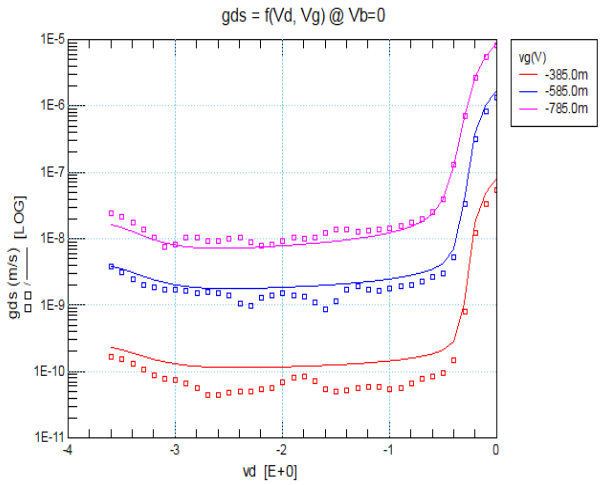




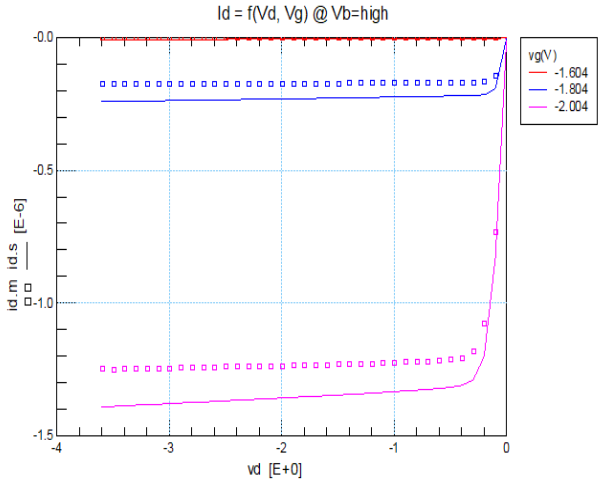
idvd_vth, Vb = 0V, T = 125°C

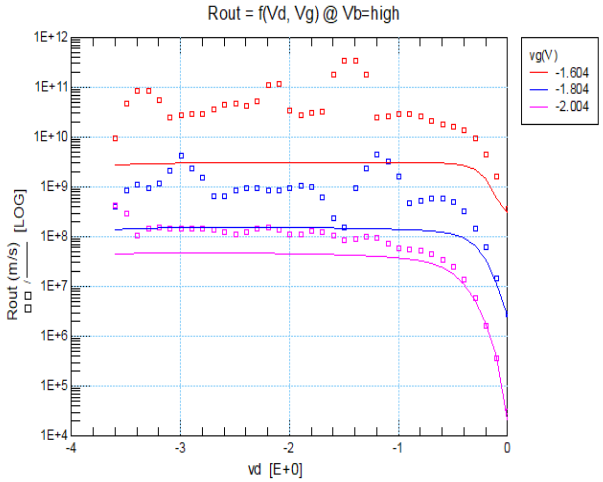


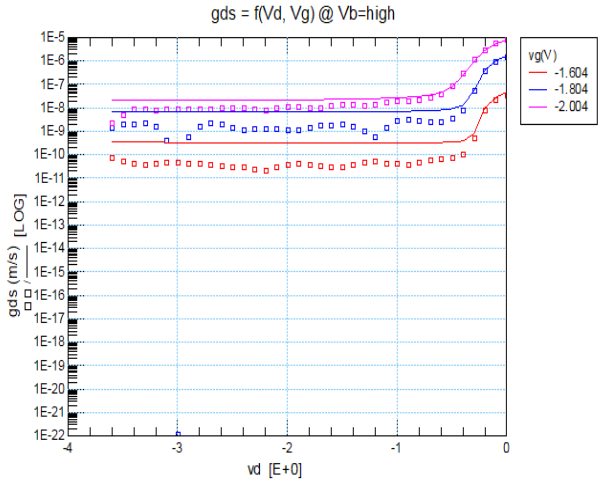




idvd_vth_vbmin, Vb = 3.6V, T = 125°C

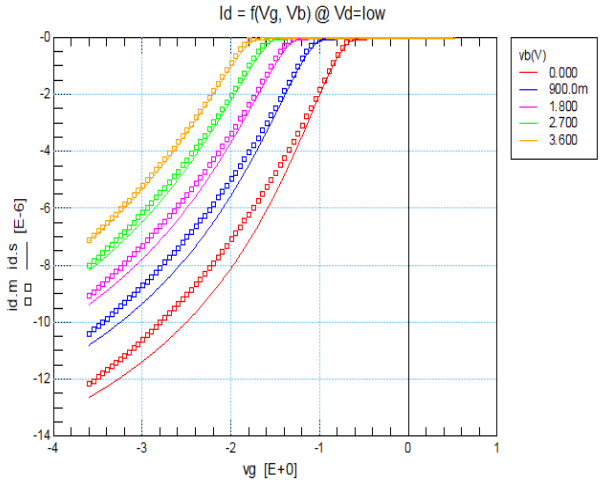


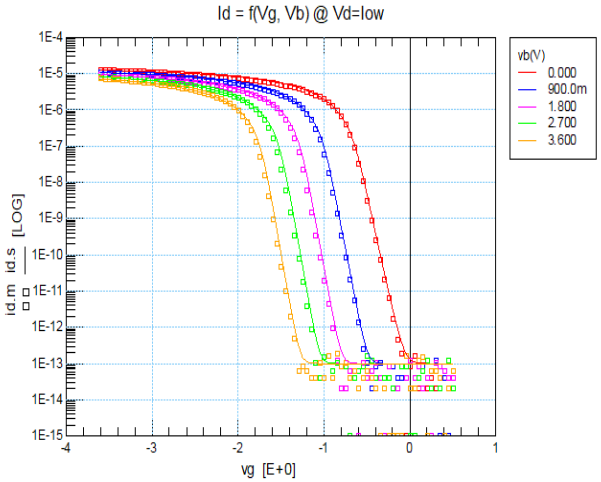


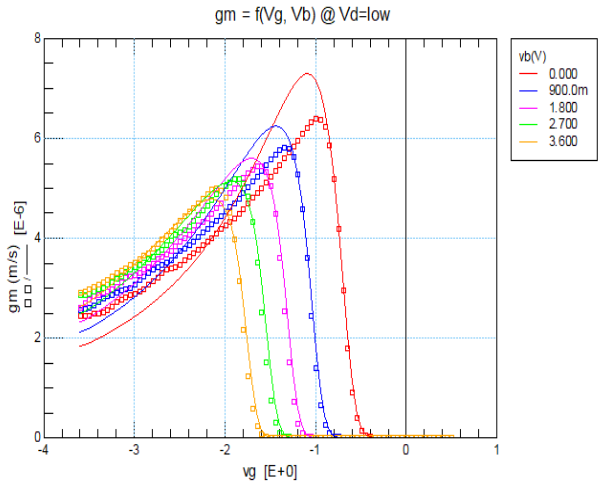


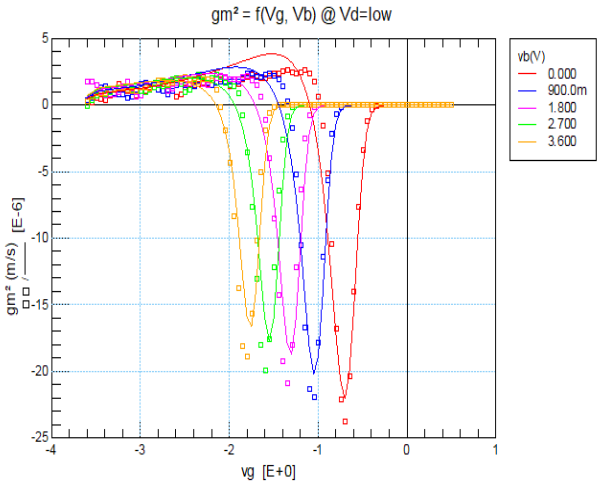
4.19W0u6_L0u5_S564_3

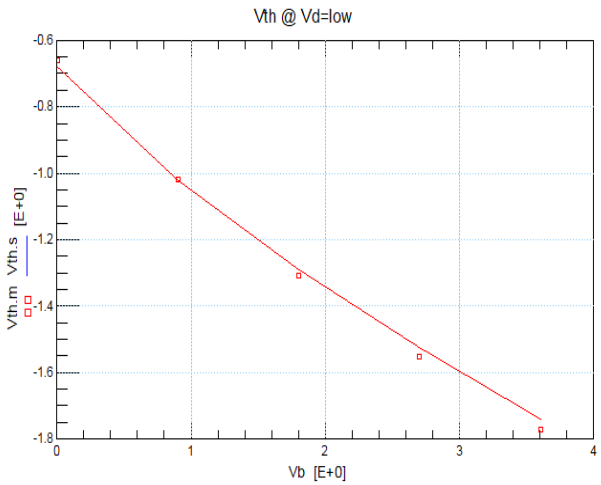
idvg, Vd = -0.1V, T = 27°C

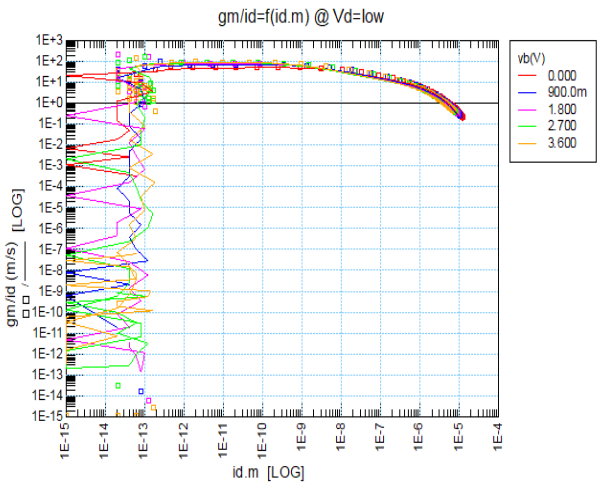


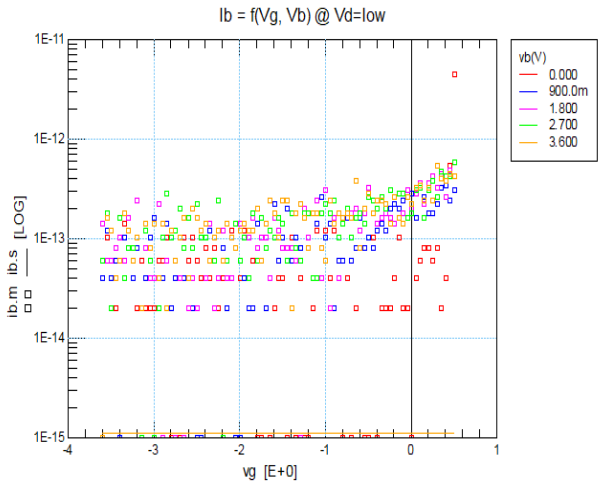


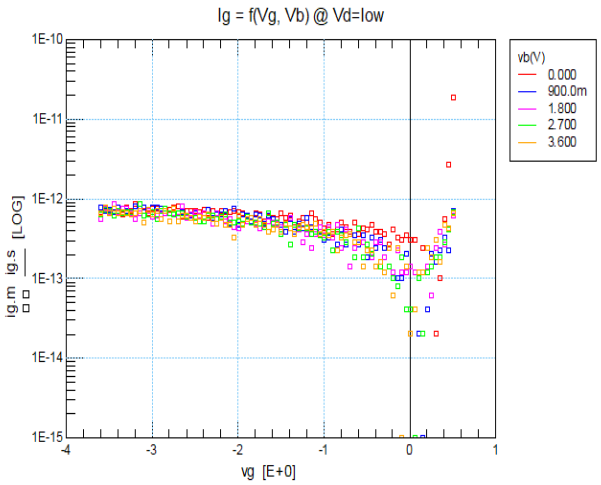




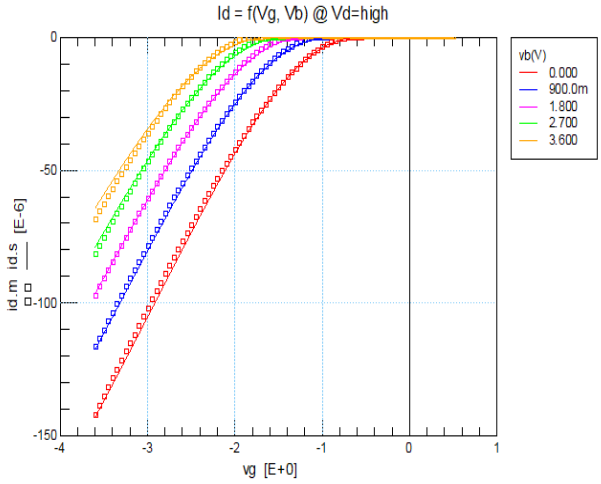


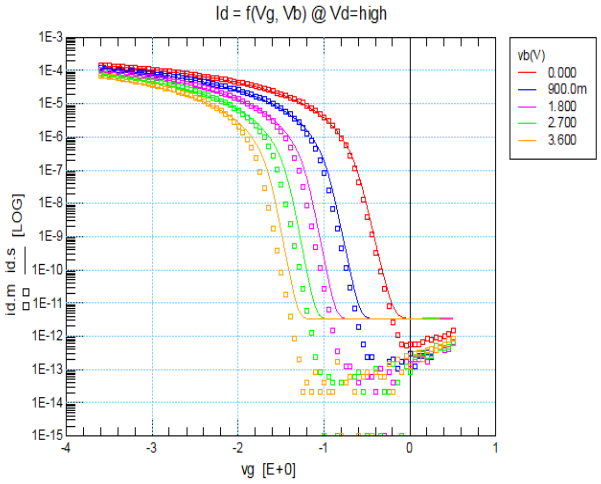


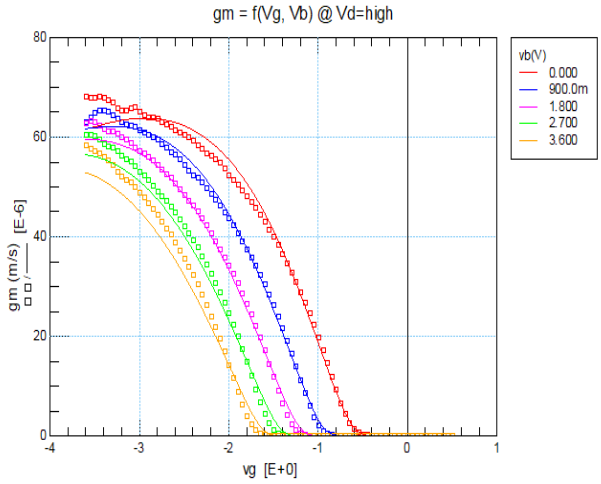


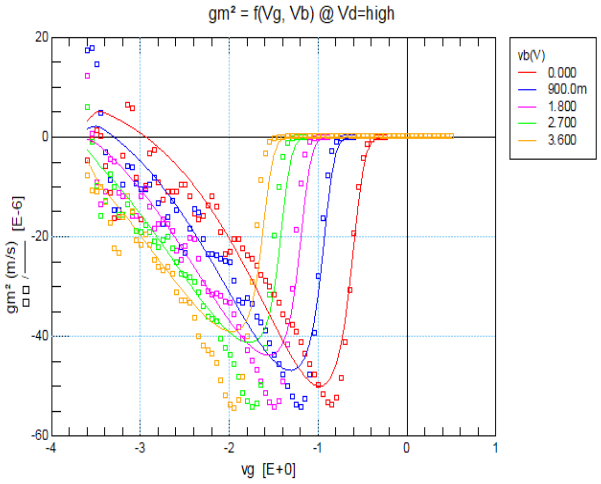


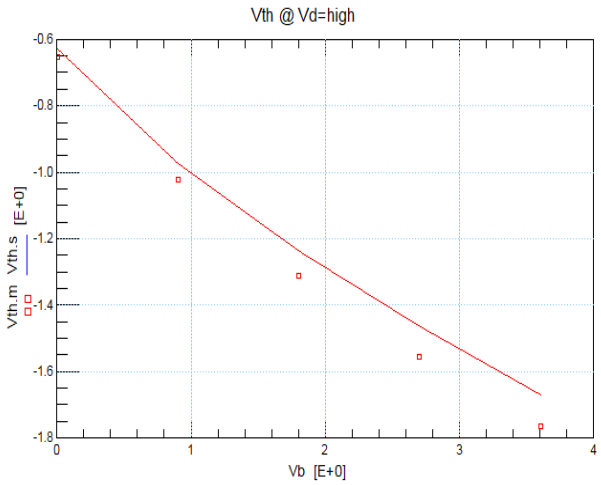
idvg, Vd = -3.3V, T = 27°C

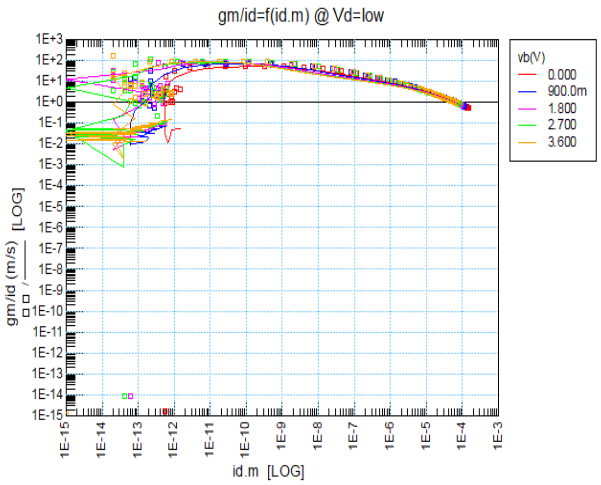


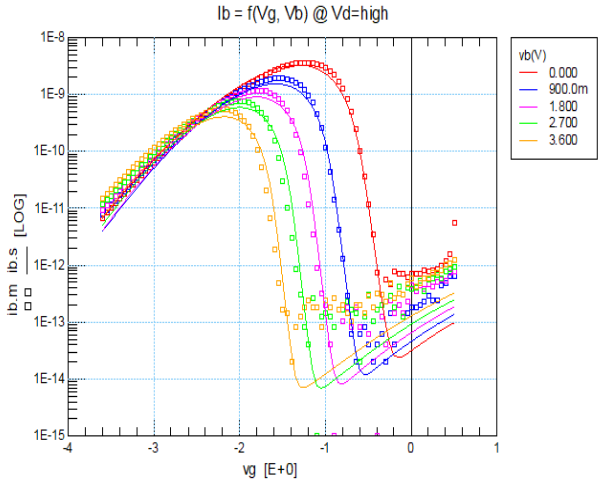


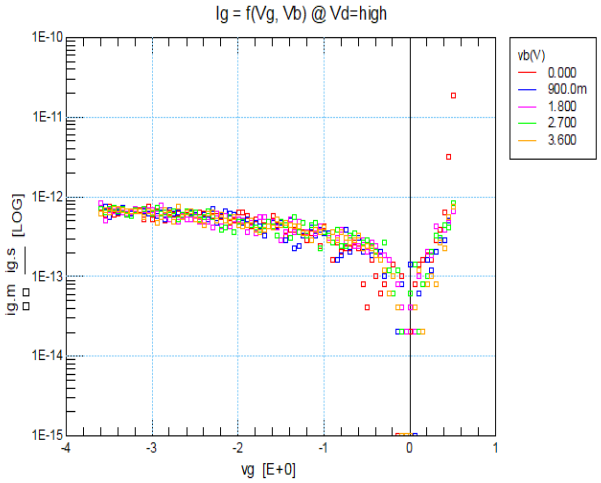




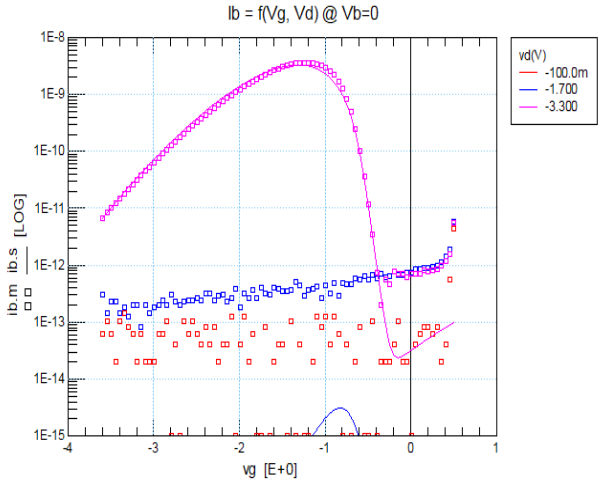


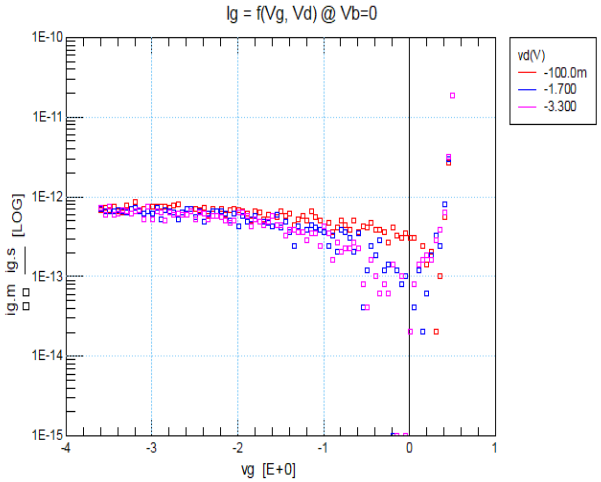




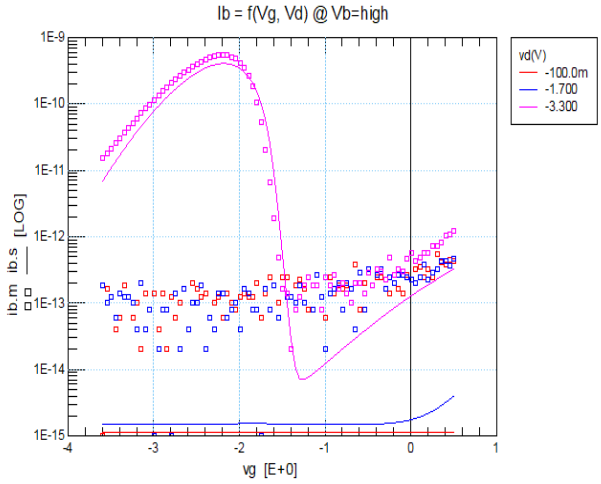


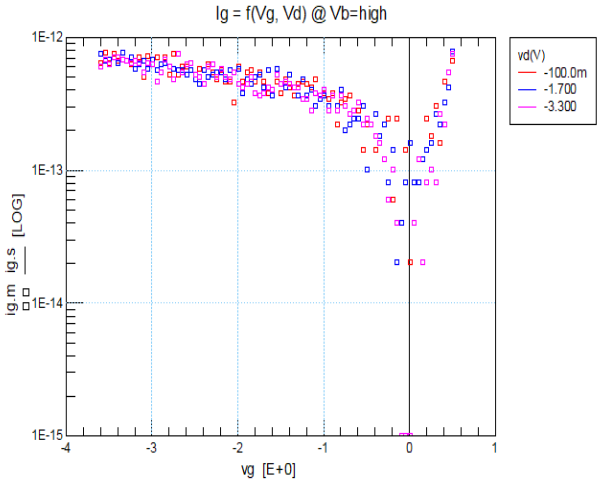
idvg, Ib,Ig, Vb = 0V, T = 27°C



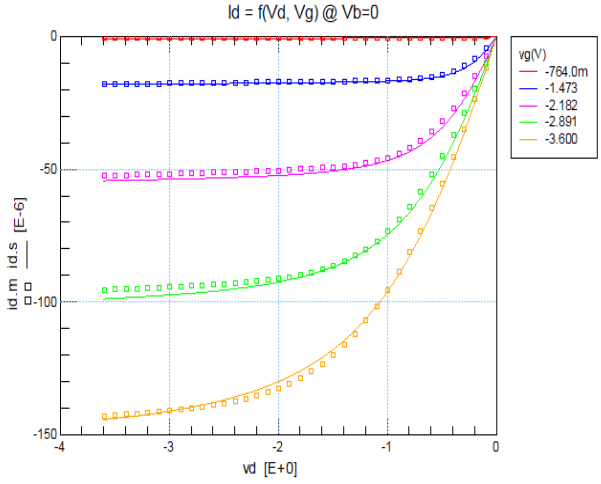


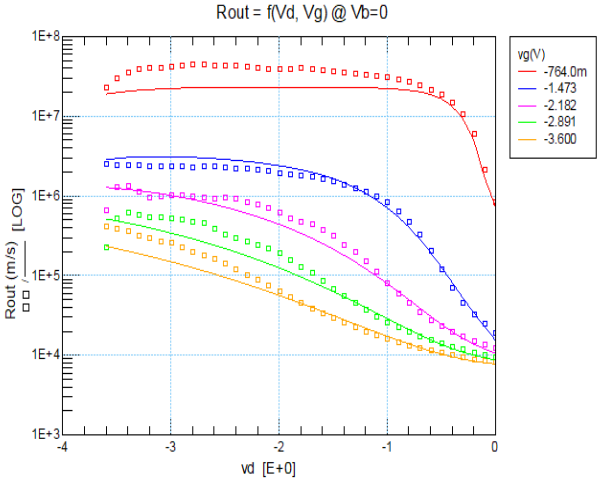
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

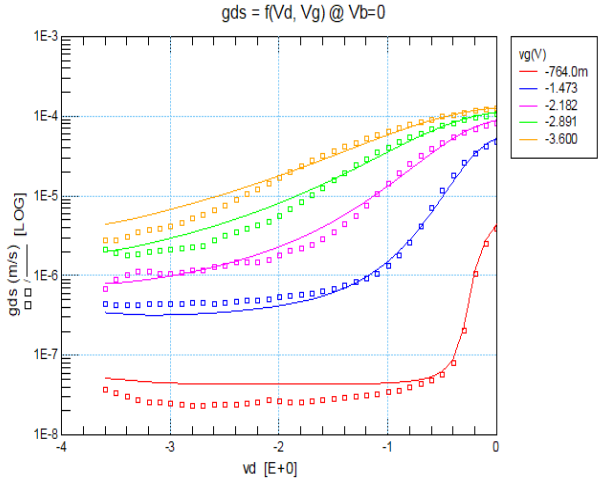




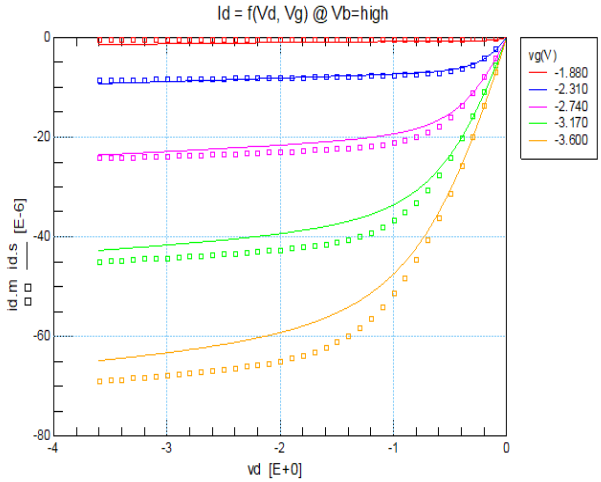
idvd, Vb = 0V, T = 27°C

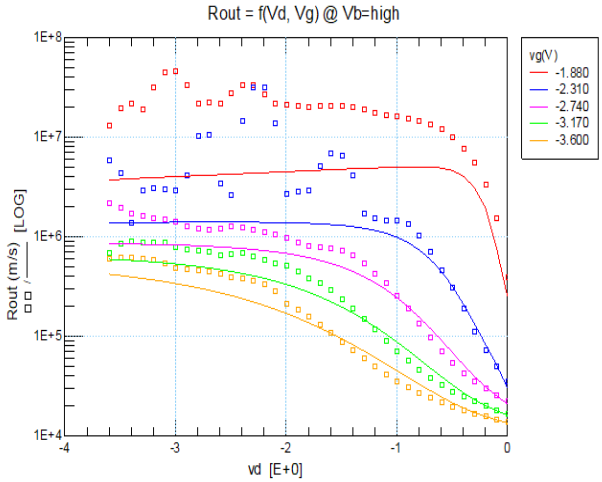


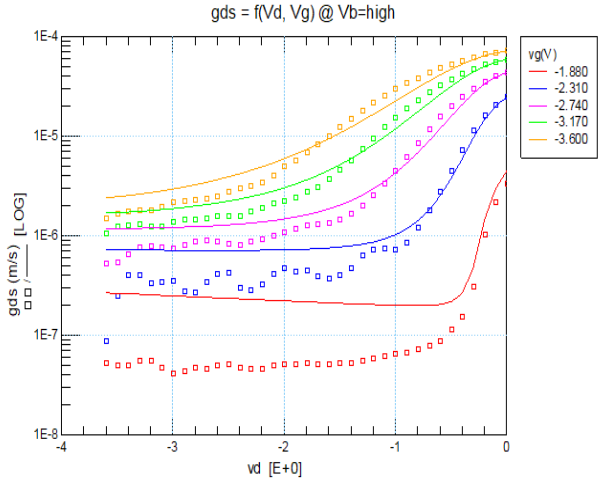




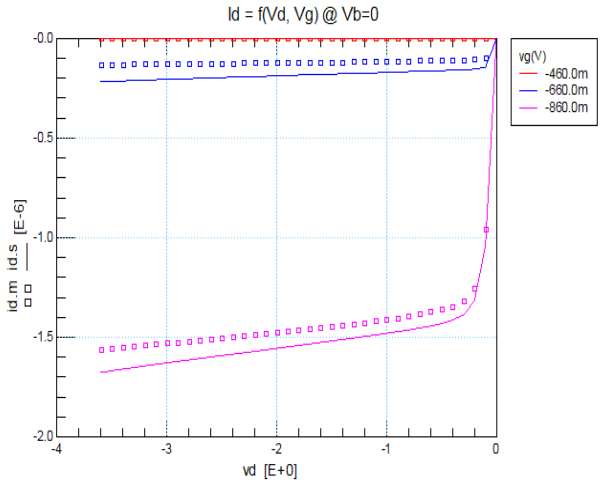
idvd_vbmin, Vb = 3.6V, T = 27°C

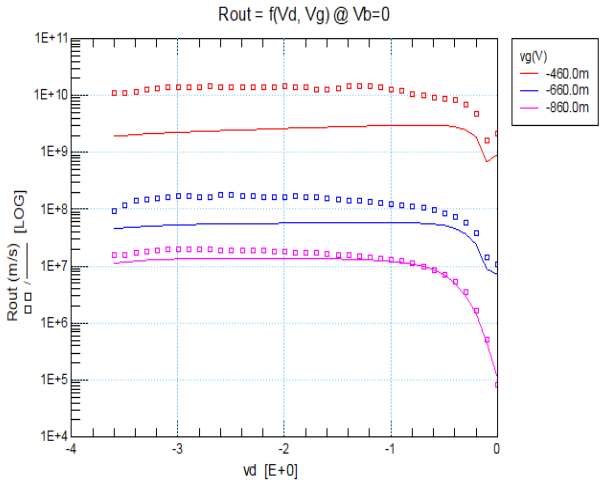


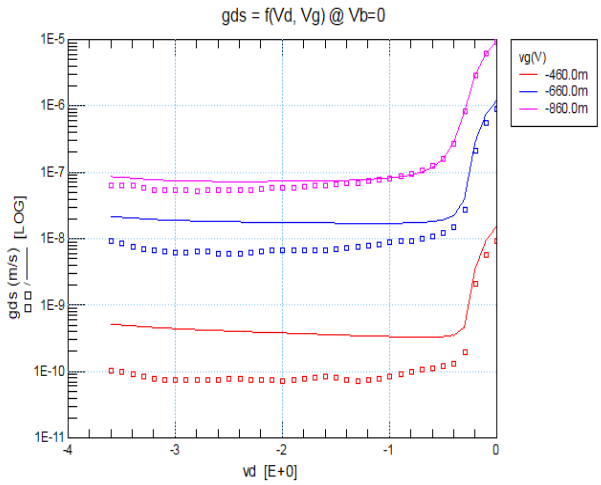




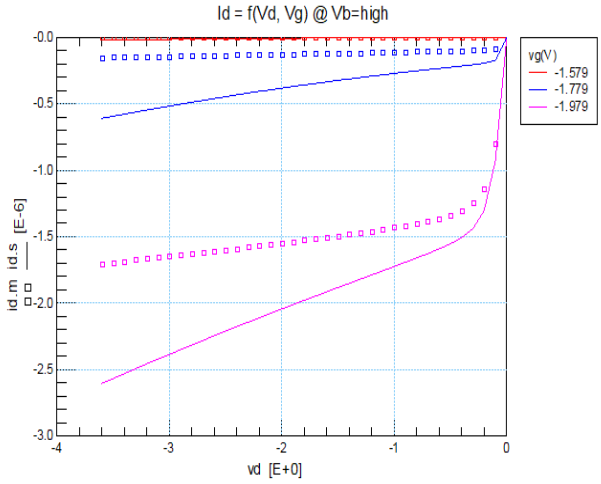
idvd_vth, Vb = 0V, T = 27°C

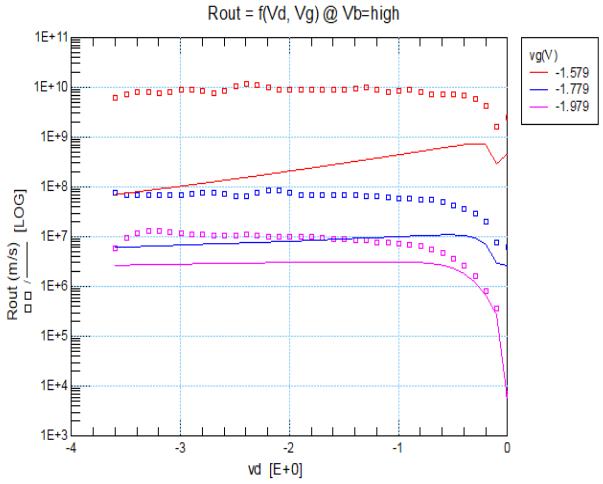


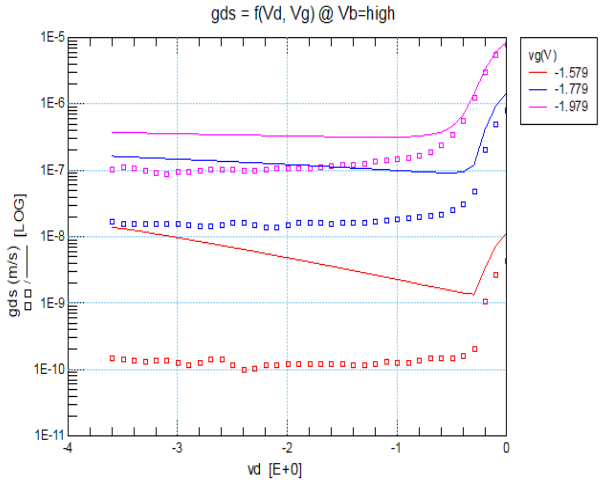




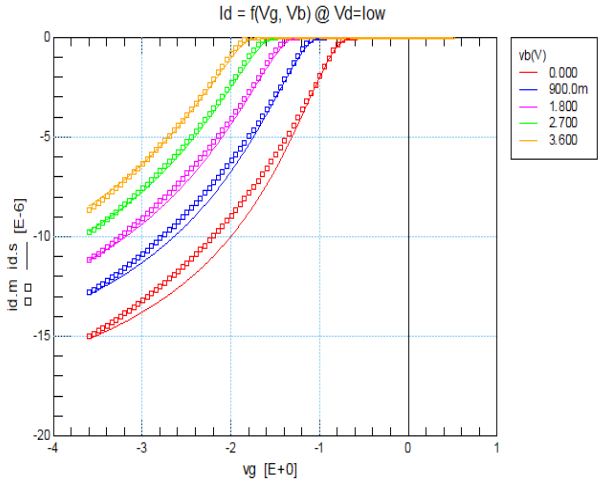
idvd_vth_vbmin, Vb = 3.6V, T = 27°C

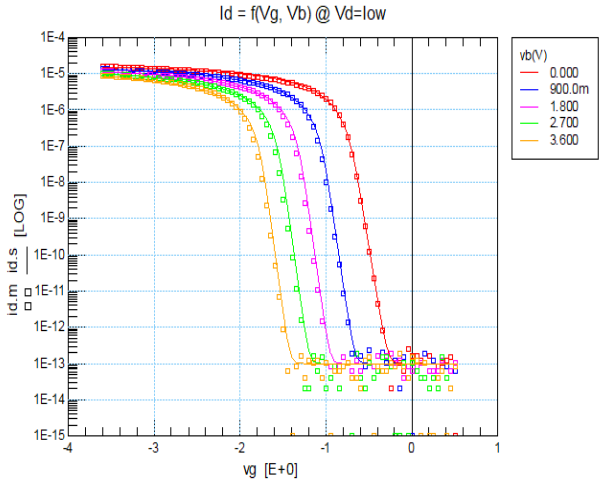


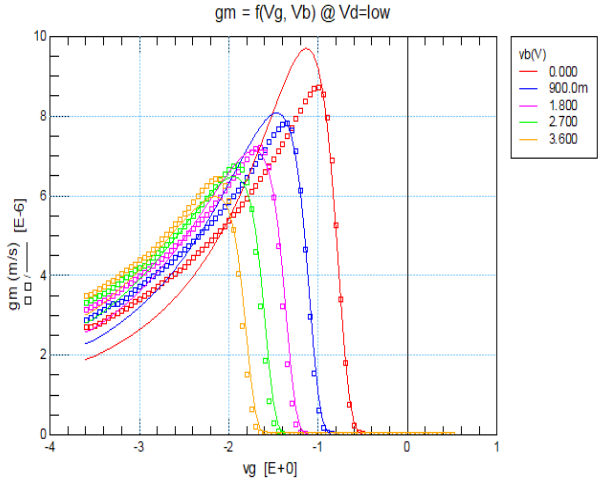


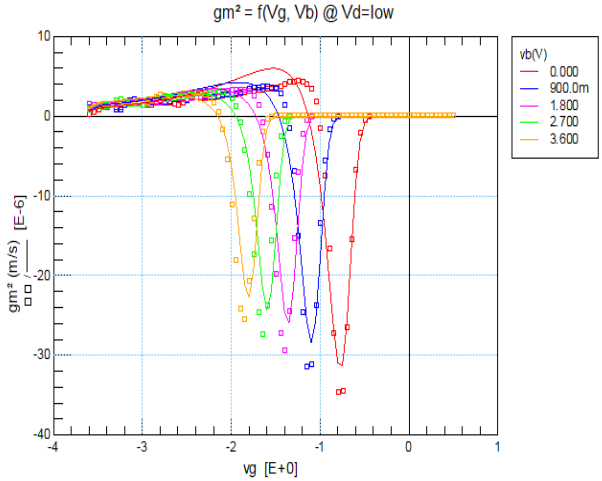


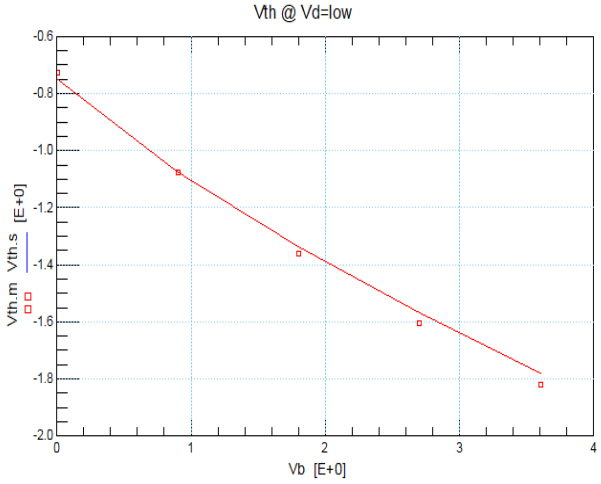
idvg, Vd = -0.1V, T = -40°C

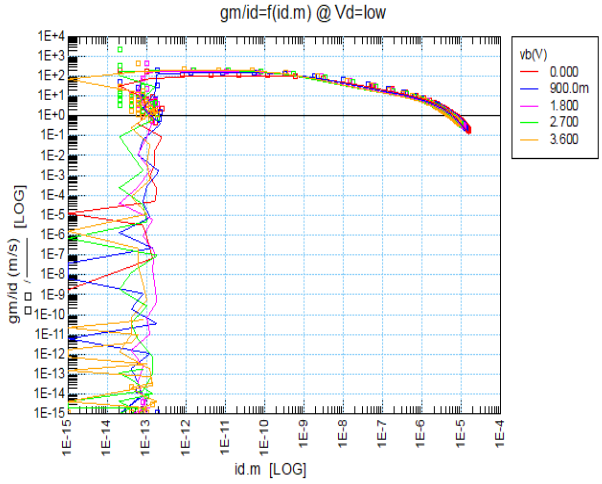


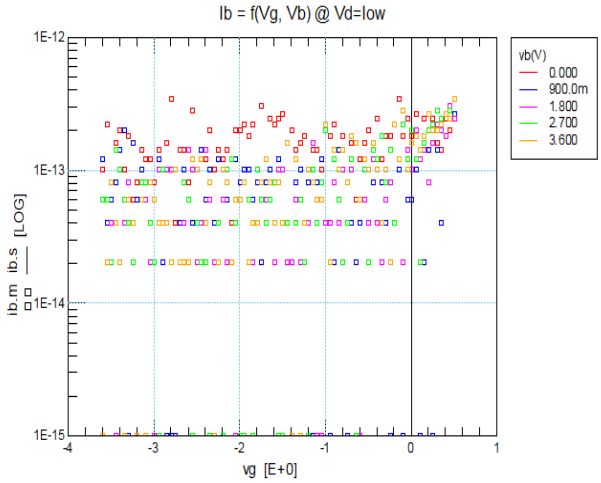


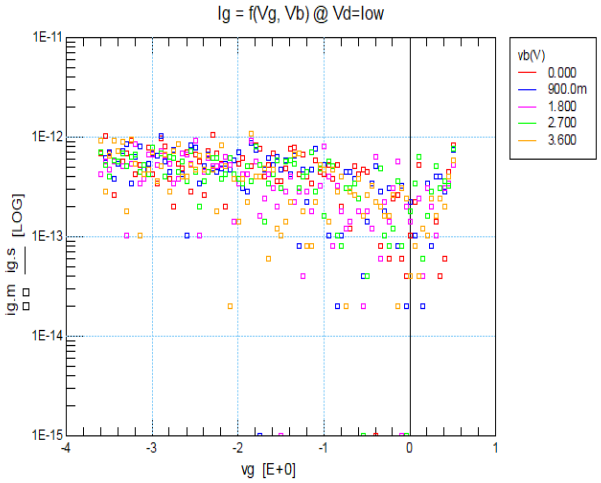




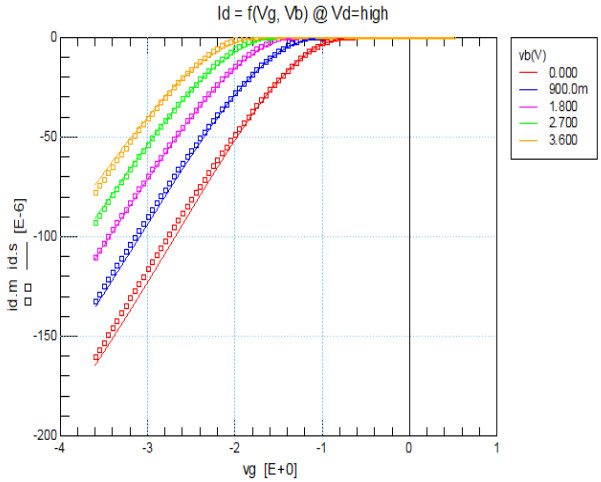


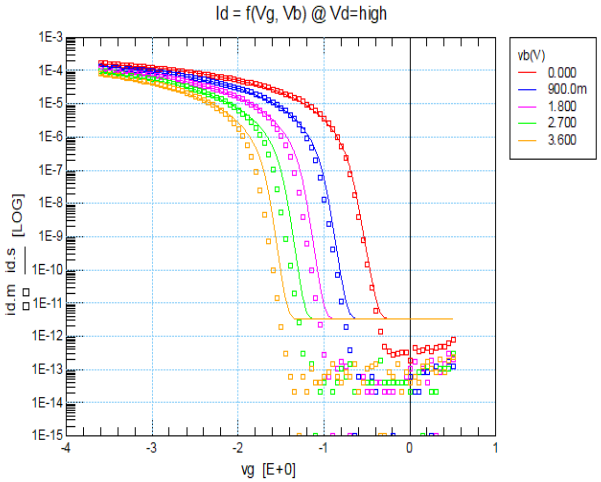


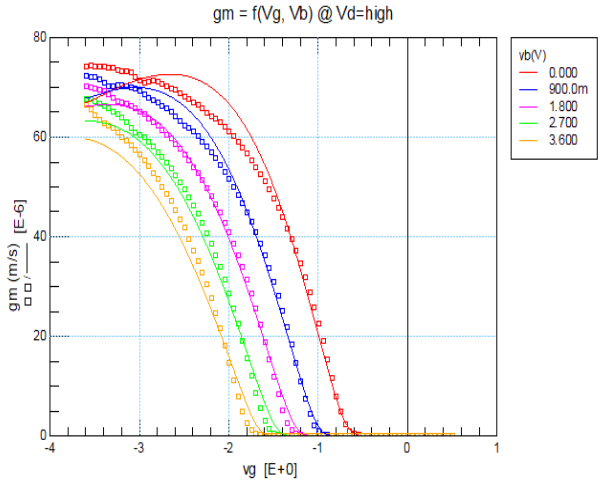


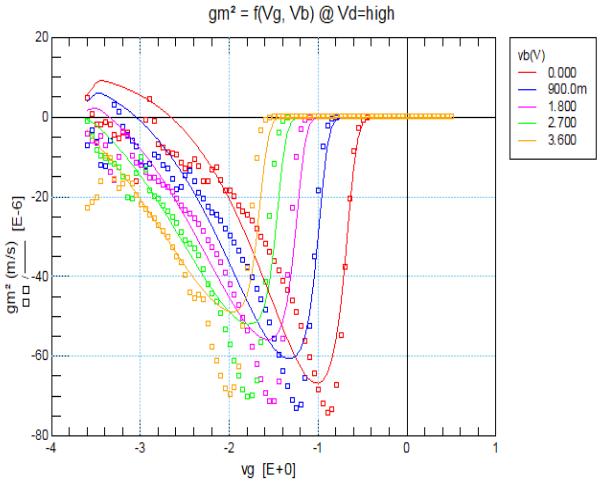


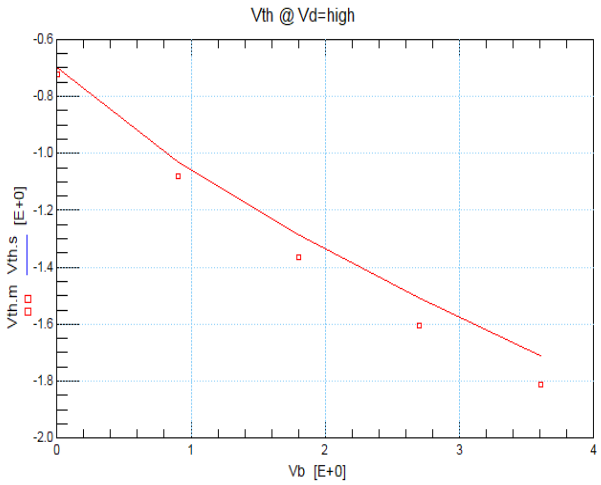
idvg, Vd = -3.3V, T = -40°C

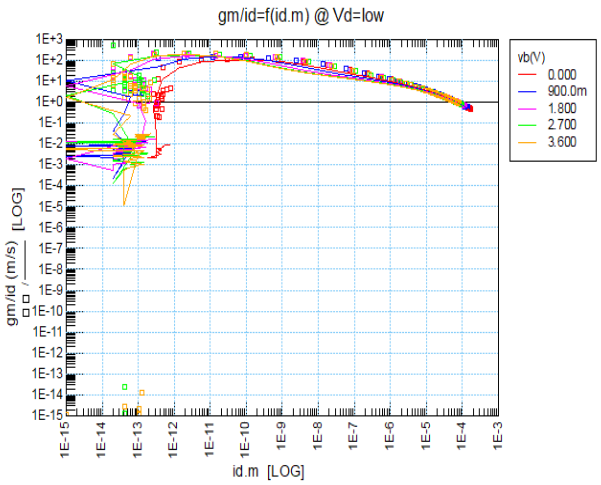


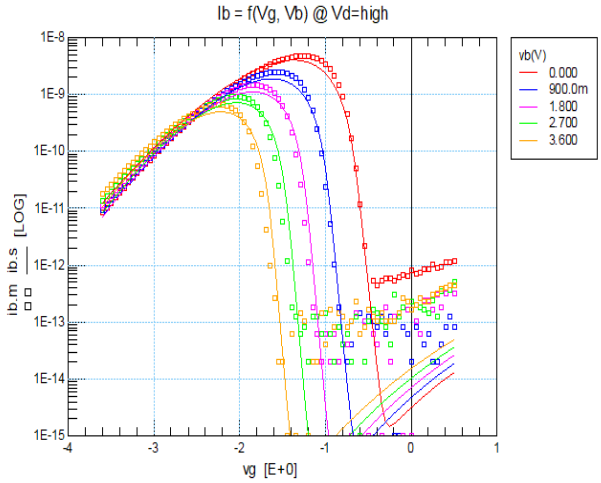


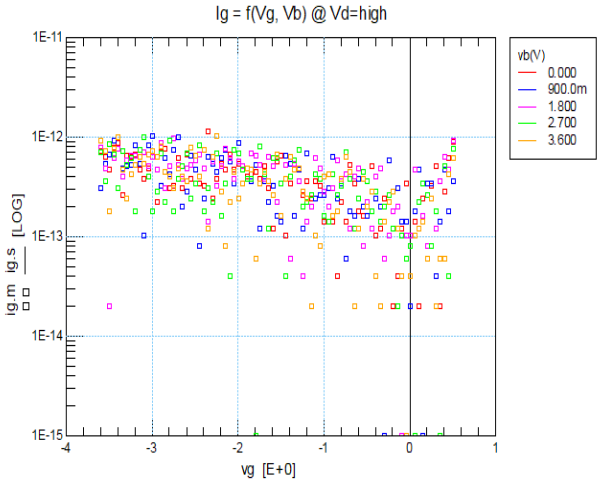




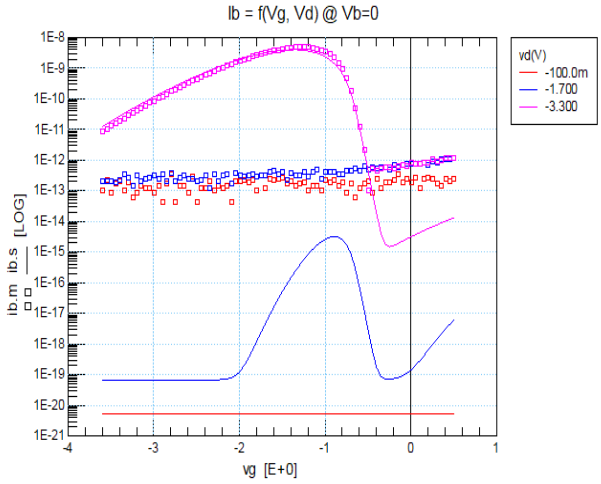


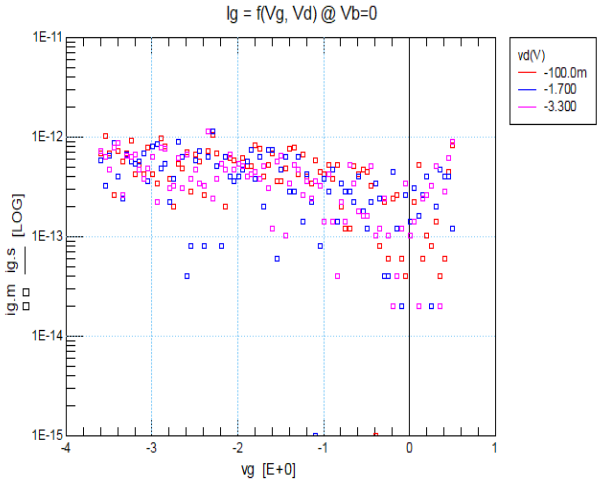




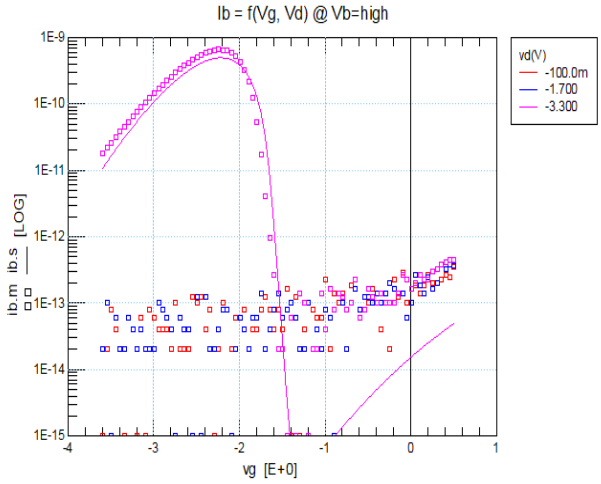


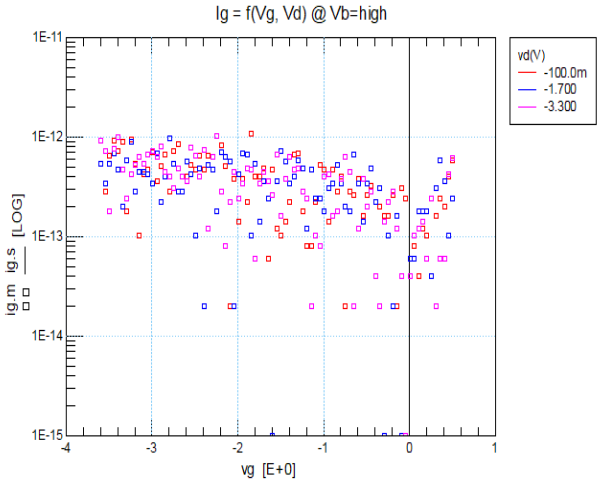
idvg, Ib,Ig, Vb = 0V, T = -40°C



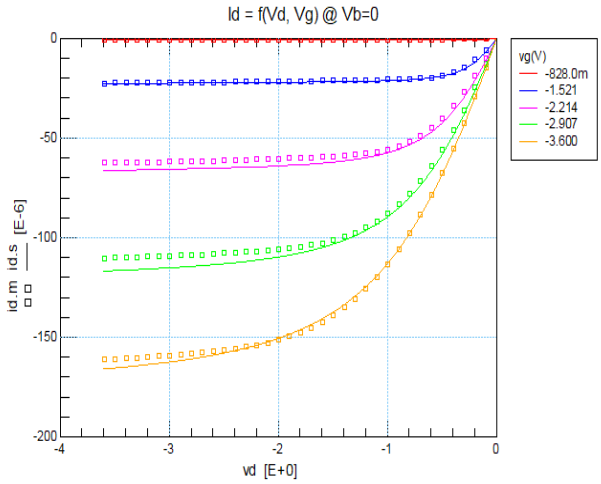


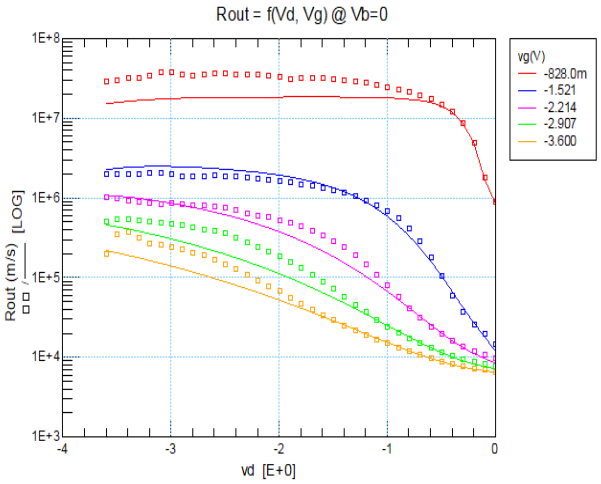
idvg, Ib,Ig, Vb = 3.6V, T = -40°C

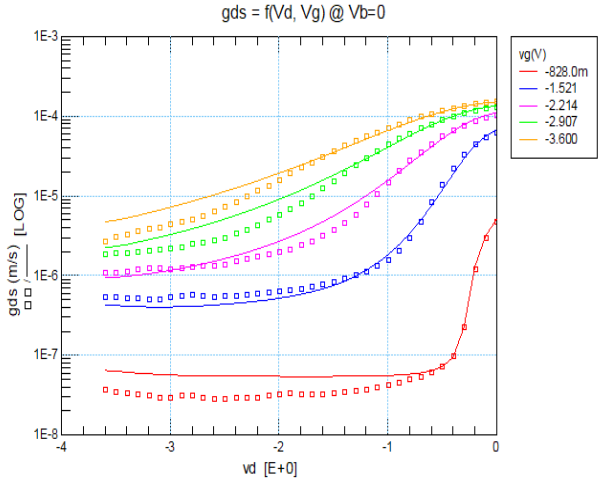




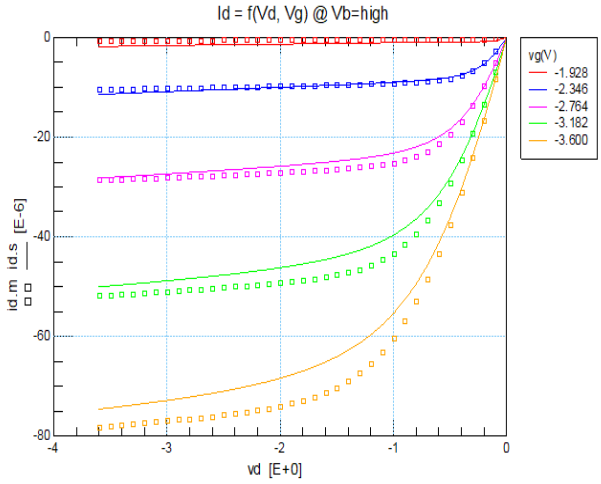
idvd, Vb = 0V, T = -40°C

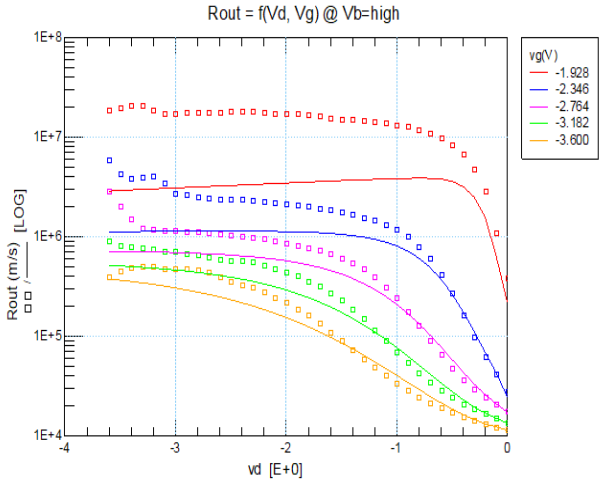


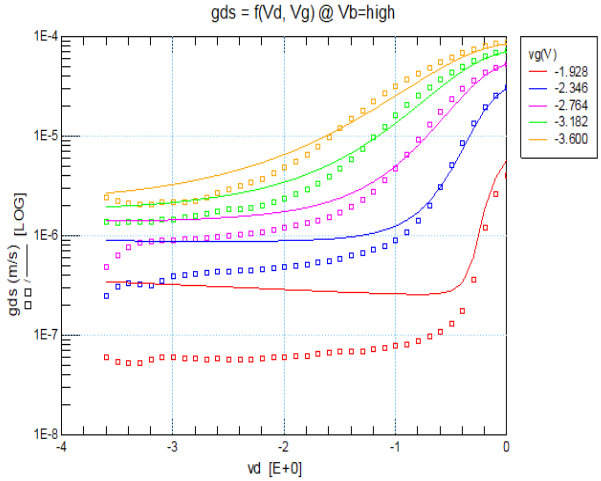




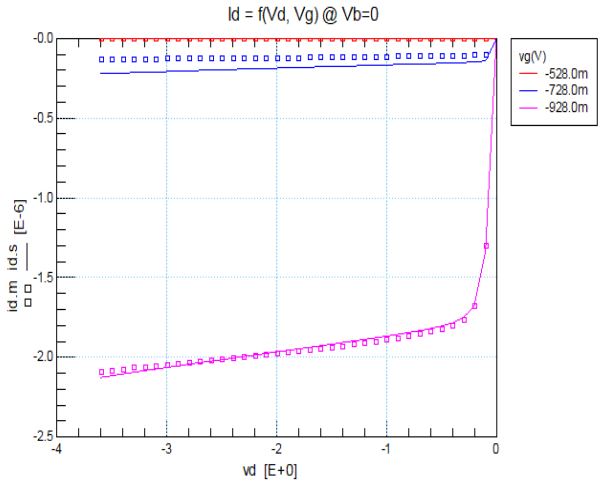
idvd_vbmin, Vb = 3.6V, T = -40°C

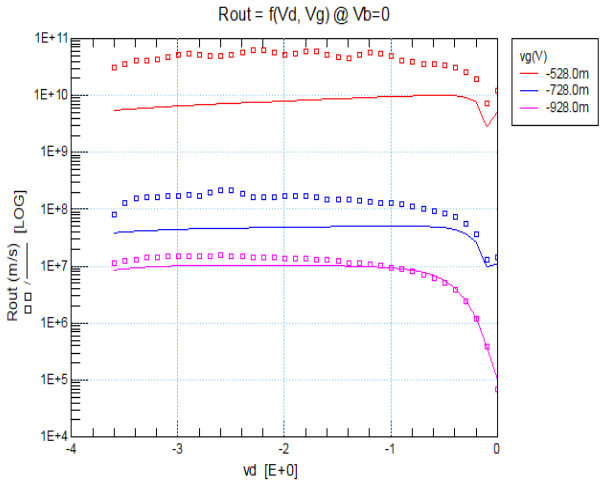


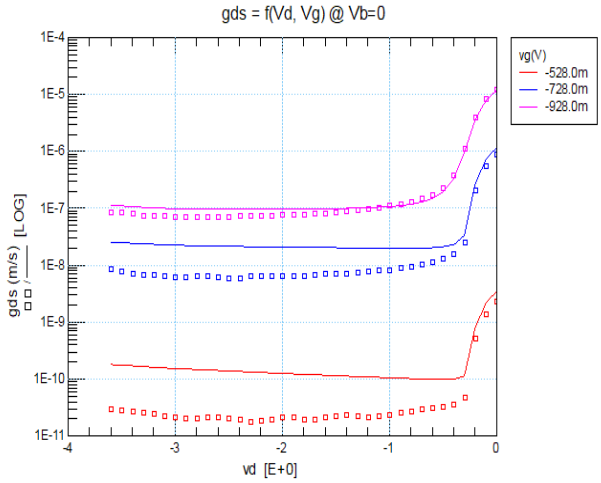




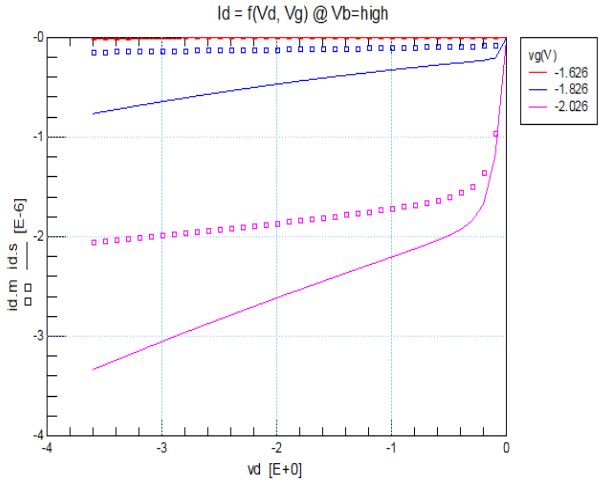
idvd_vth, Vb = 0V, T = -40°C

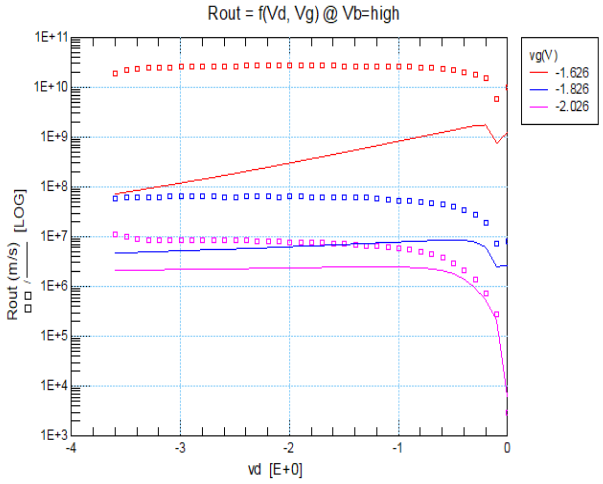


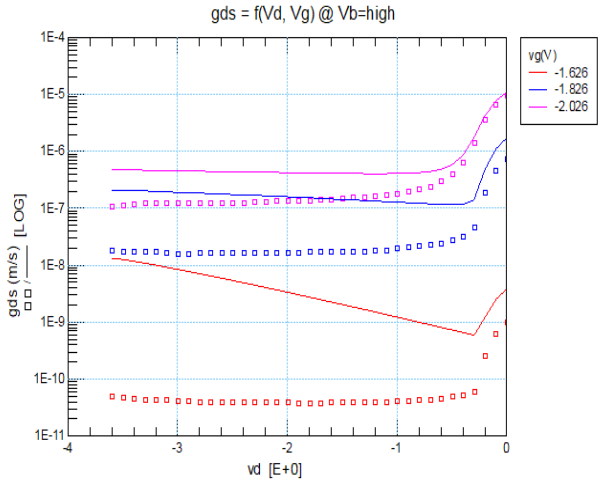




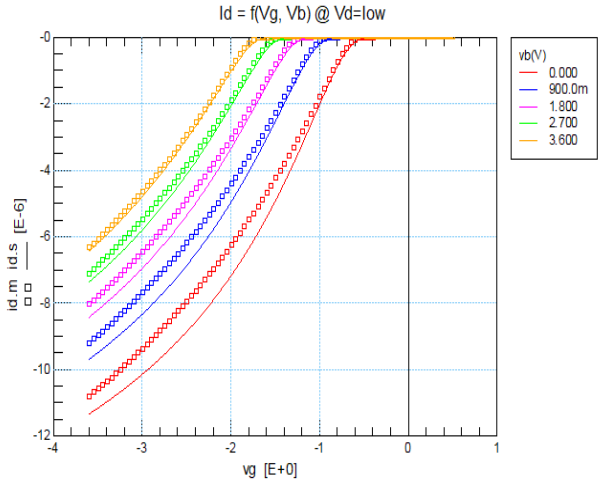
idvd_vth_vbmin, Vb = 3.6V, T = -40°C

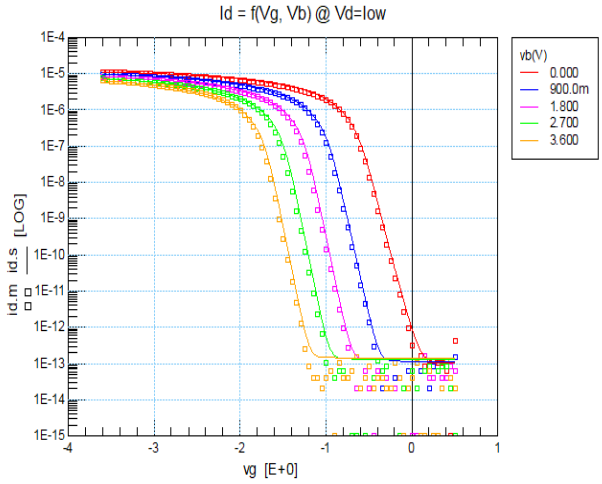


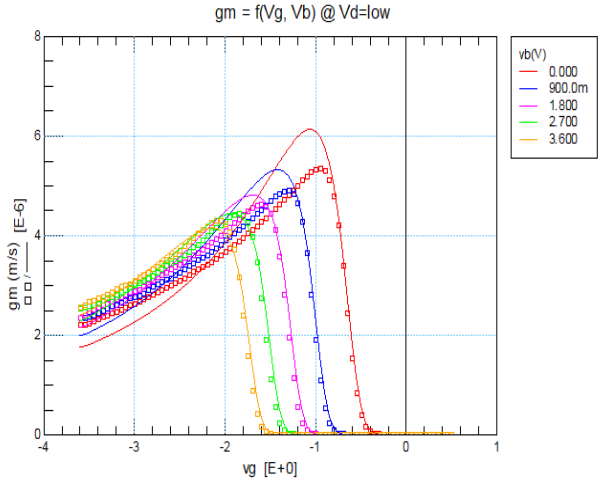


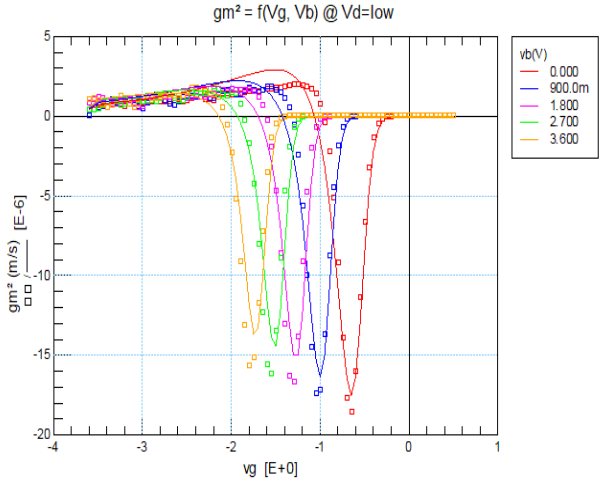


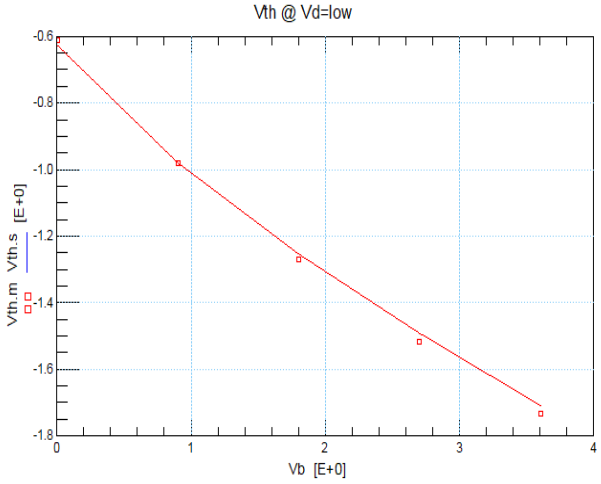
idvg, Vd = -0.1V, T = 70°C

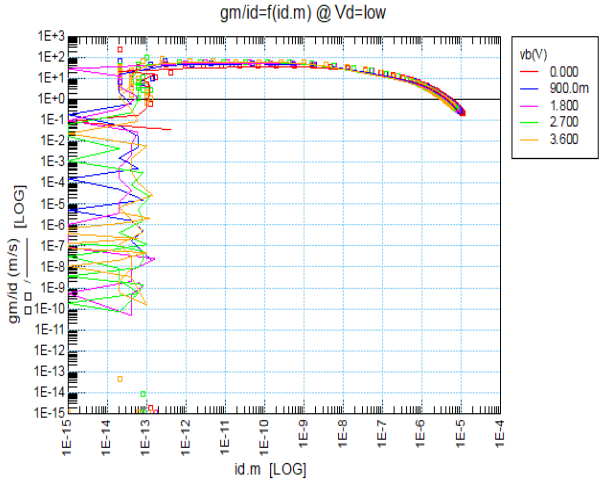


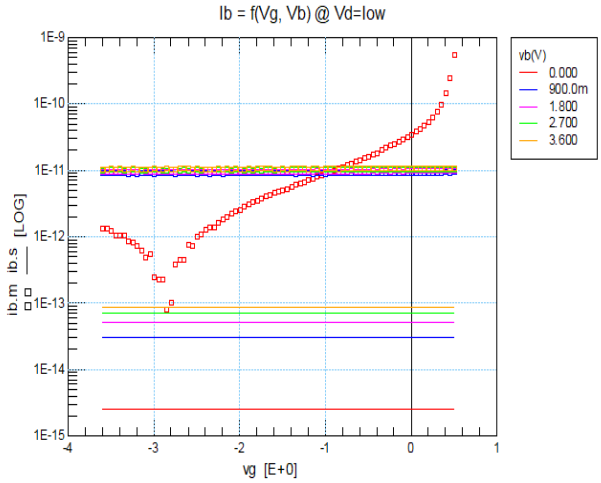


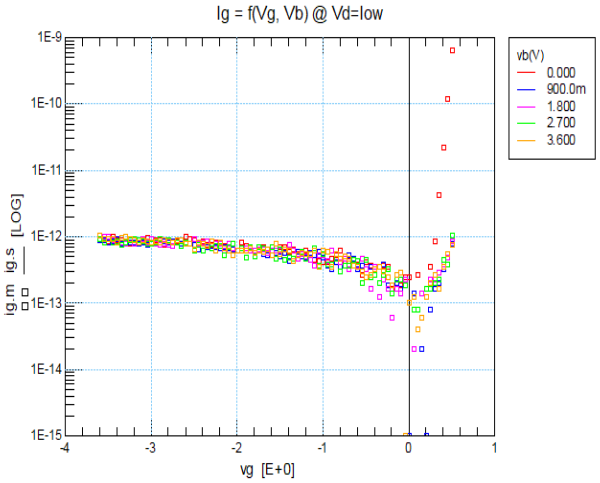




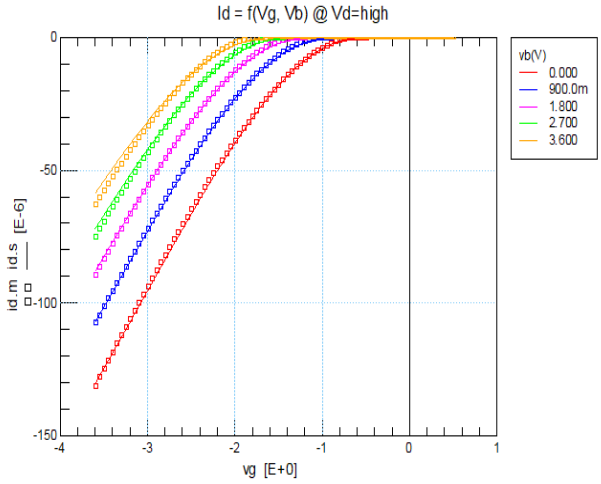


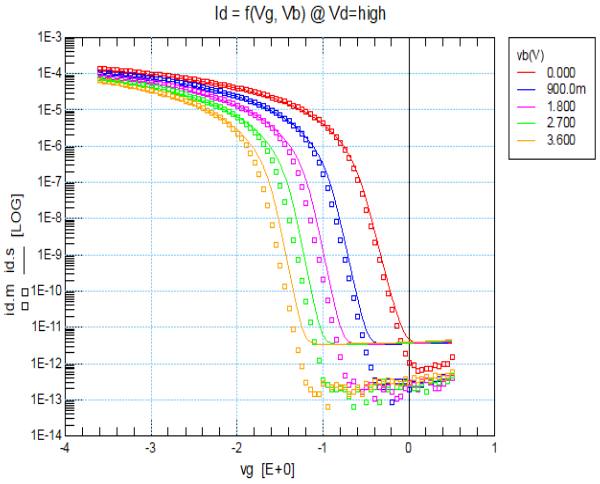


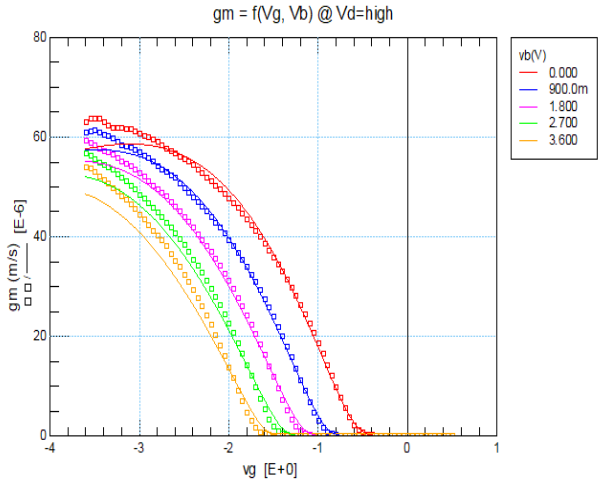


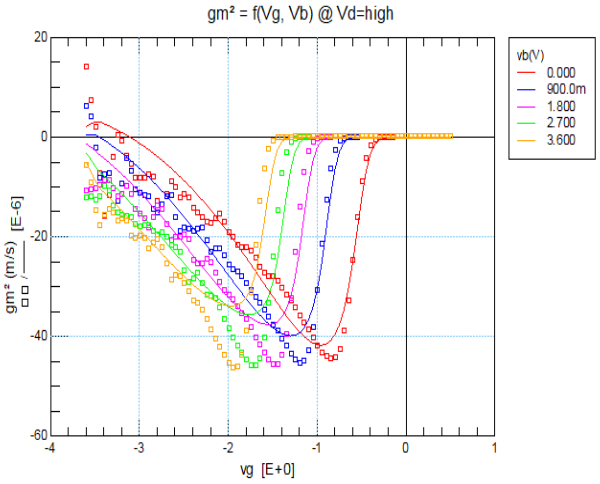


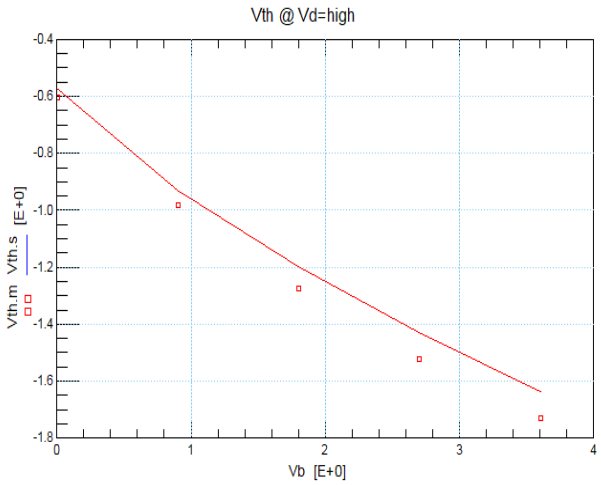
idvg, Vd = -3.3V, T = 70°C

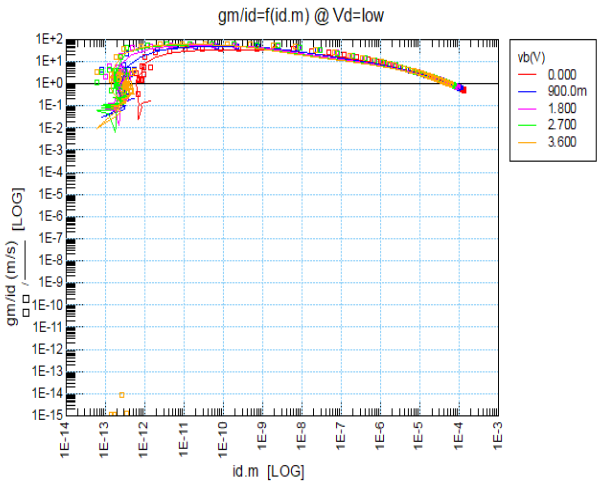


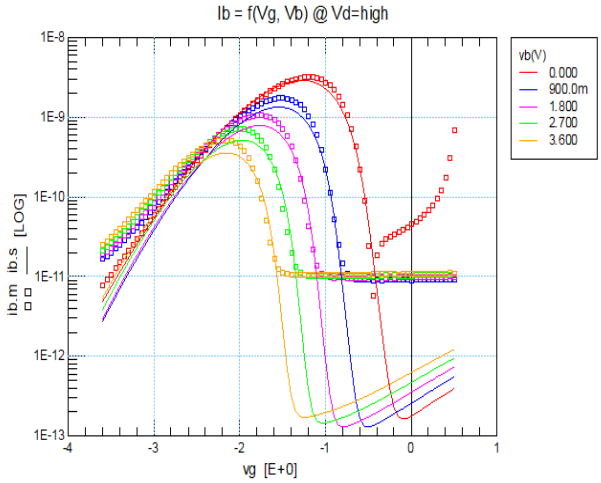


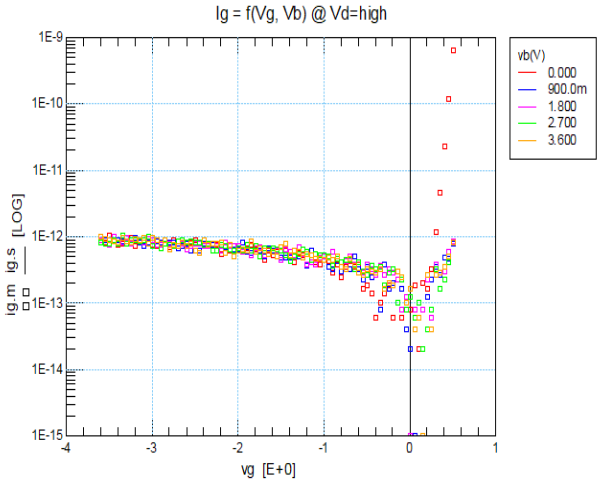




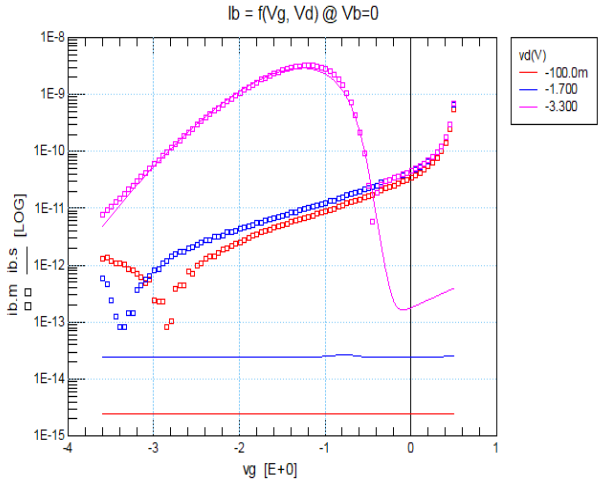


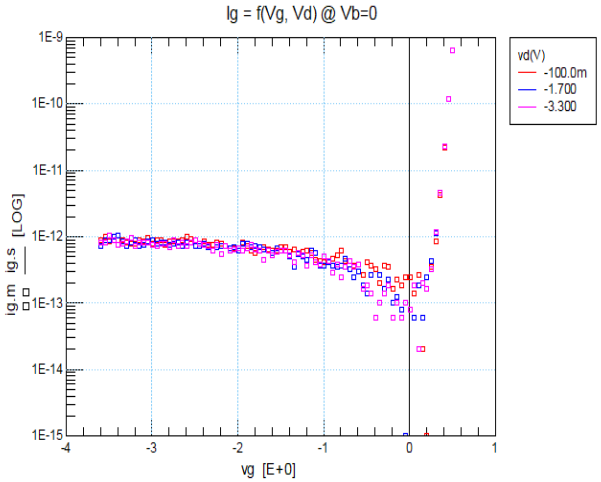




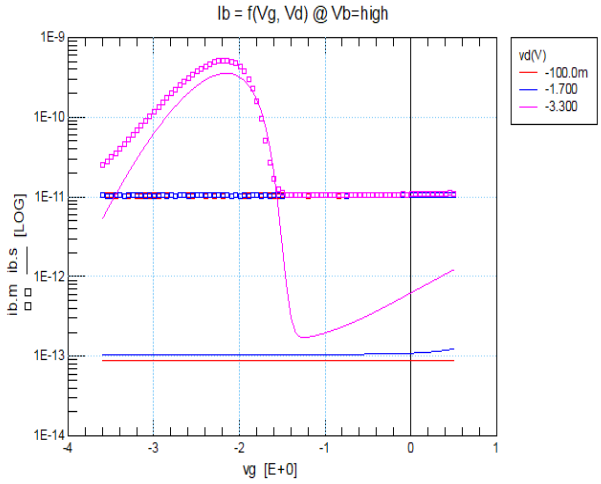


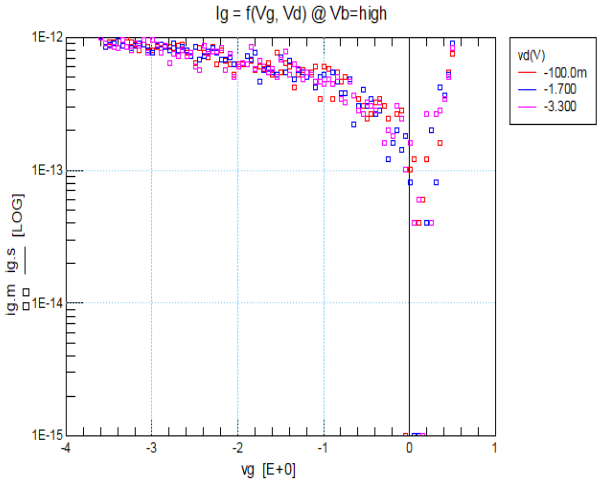
idvg, Ib,Ig, Vb = 0V, T = 70°C



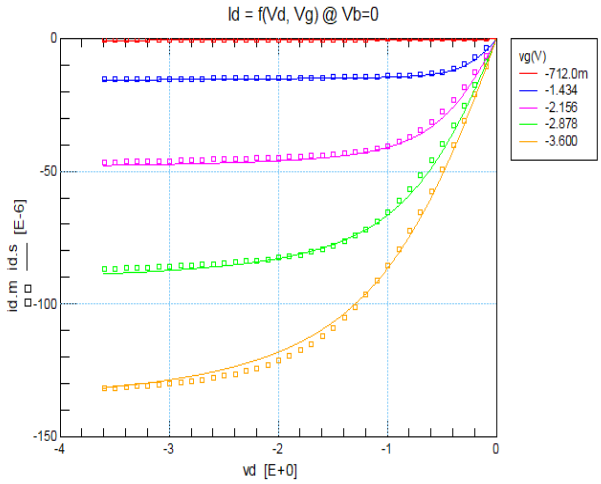


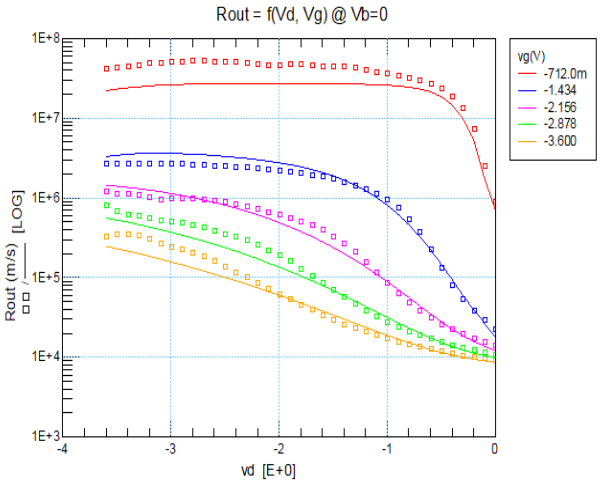
idvg, Ib,Ig, Vb = 3.6V, T = 70°C

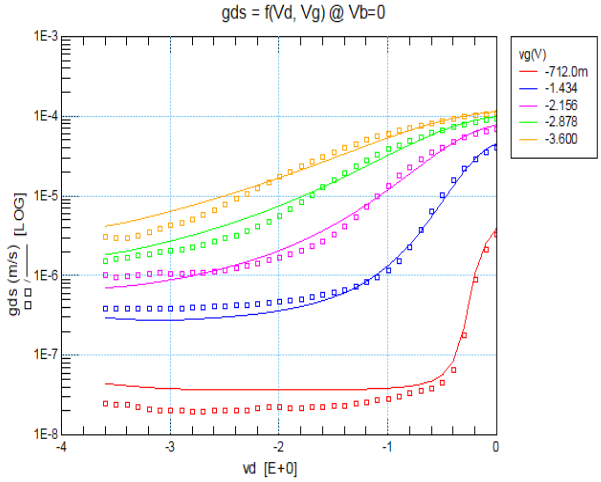




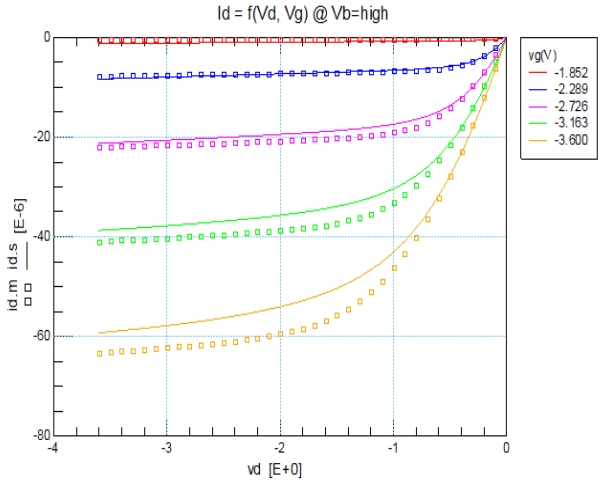
idvd, Vb = 0V, T = 70°C

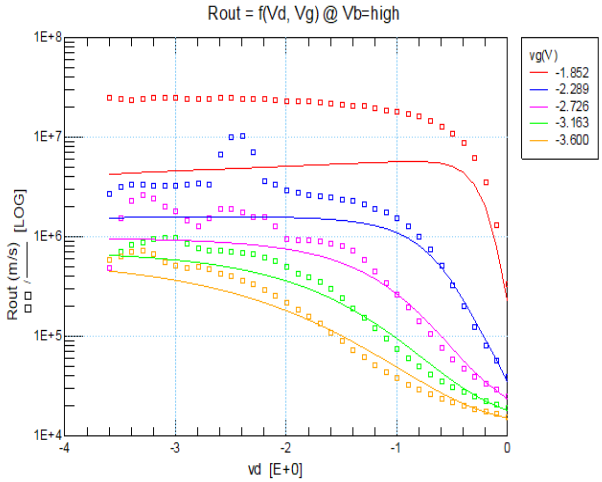


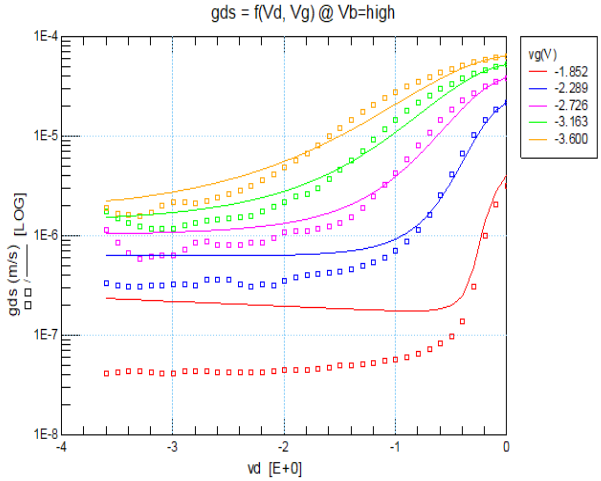




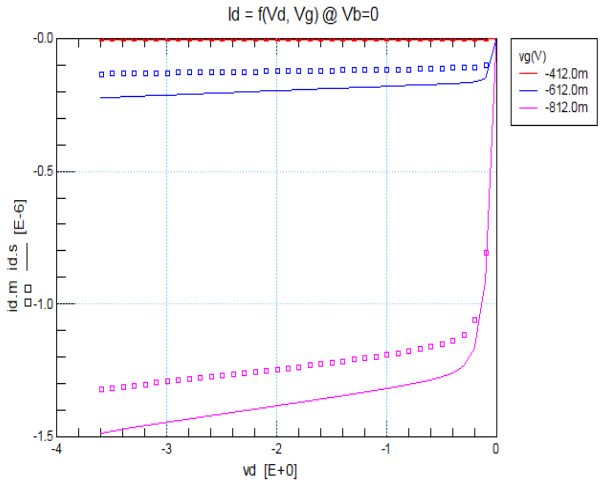
idvd_vbmin, Vb = 3.6V, T = 70°C

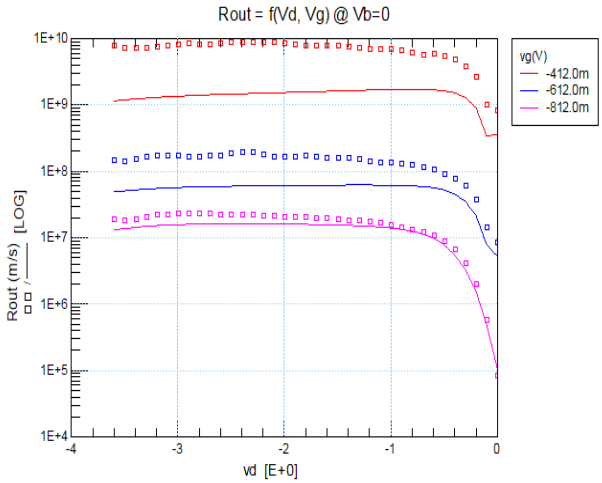


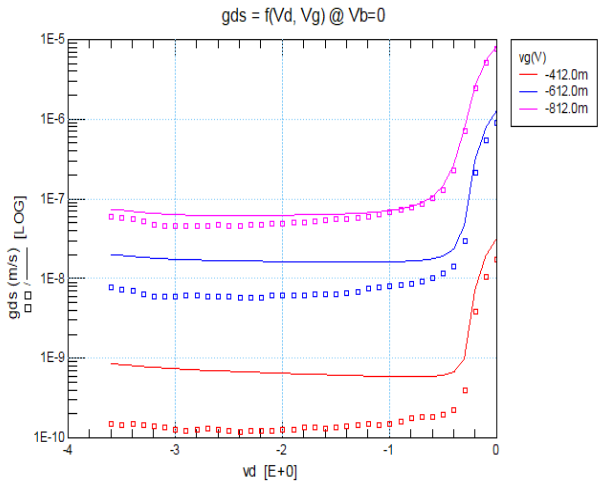




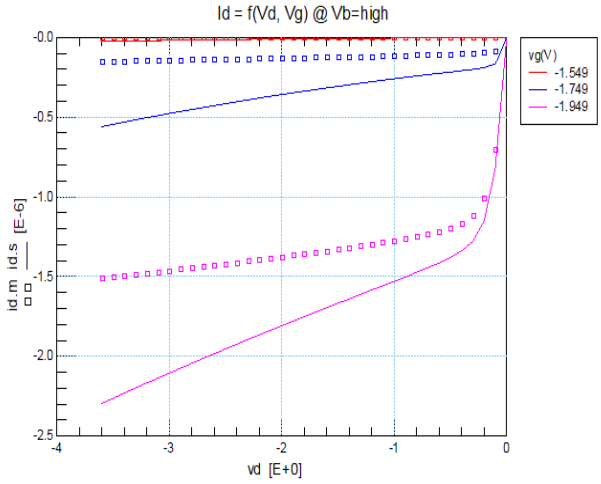
idvd_vth, Vb = 0V, T = 70°C

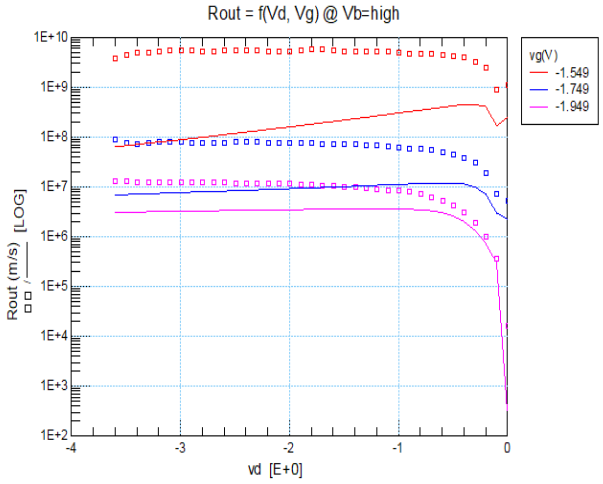


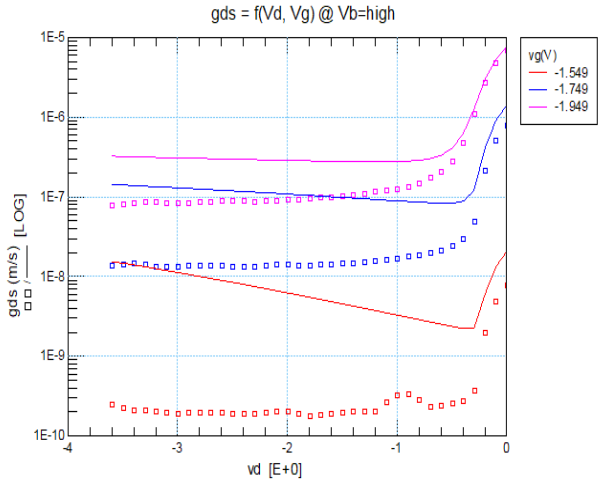




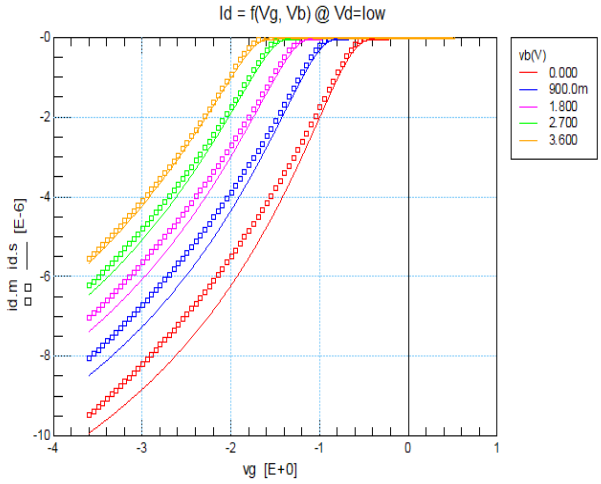
idvd_vth_vbmin, Vb = 3.6V, T = 70°C

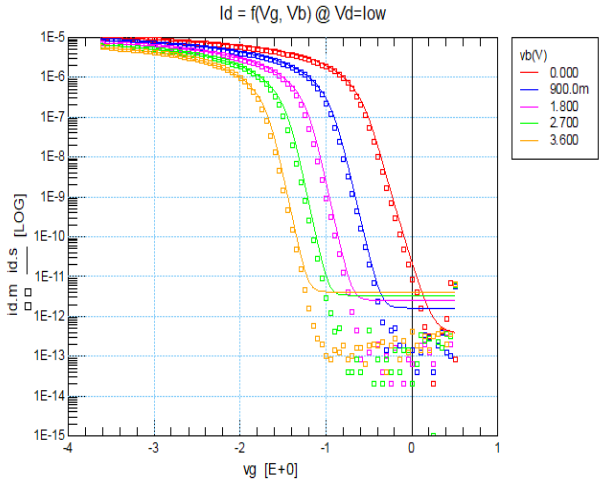


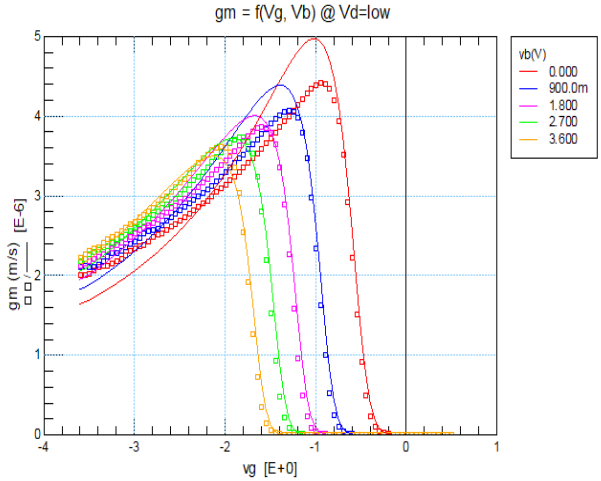


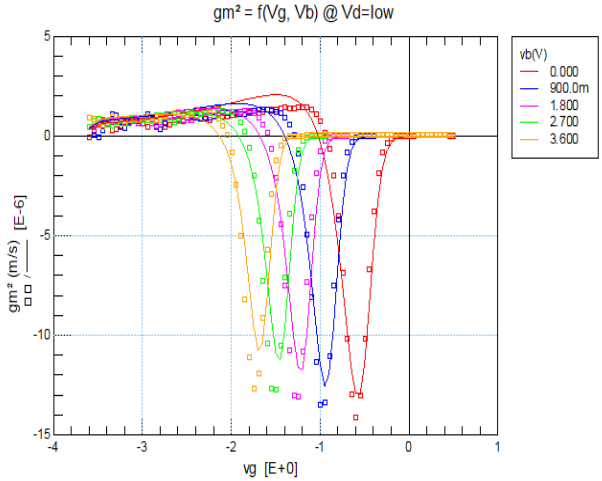


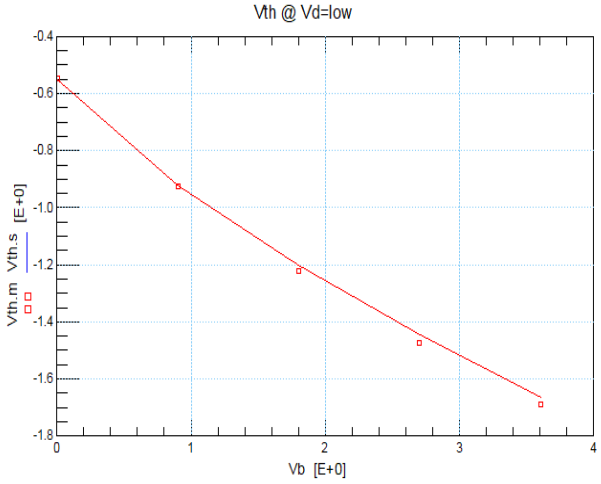
idvg, Vd = -0.1V, T = 125°C

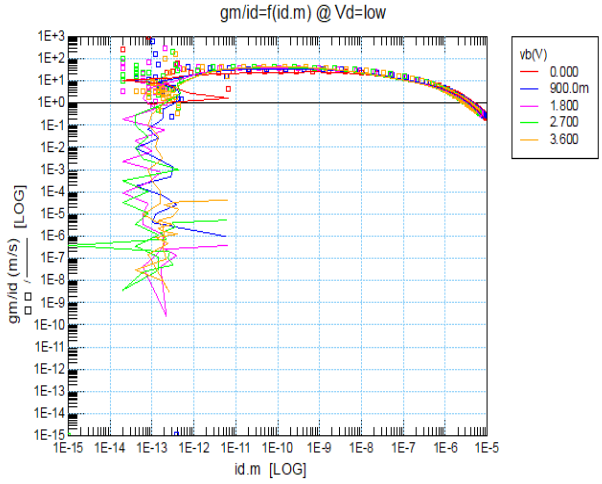


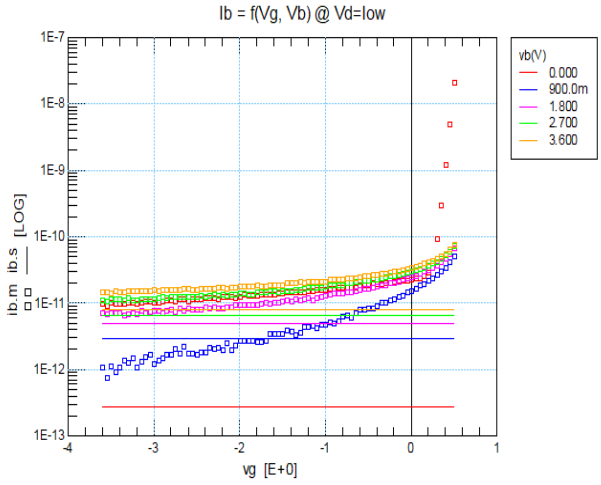


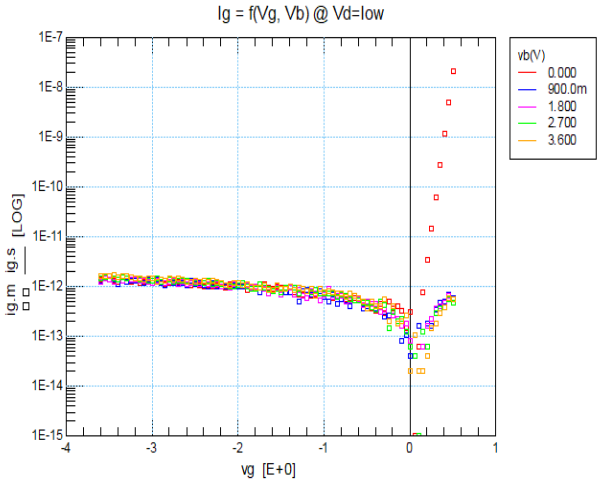




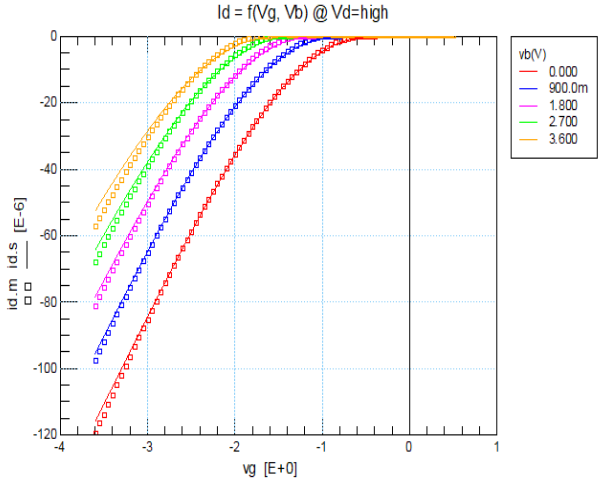


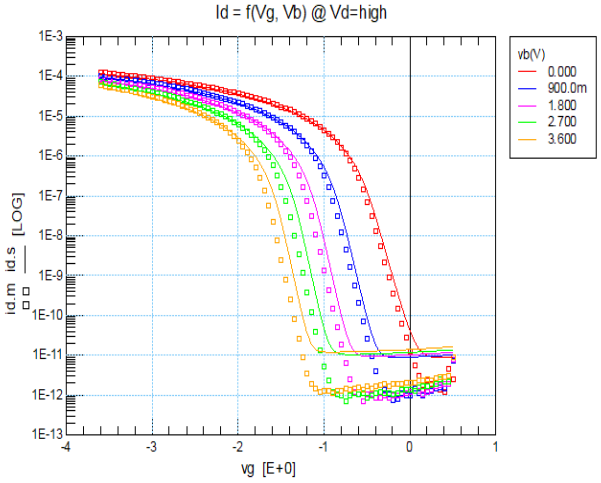


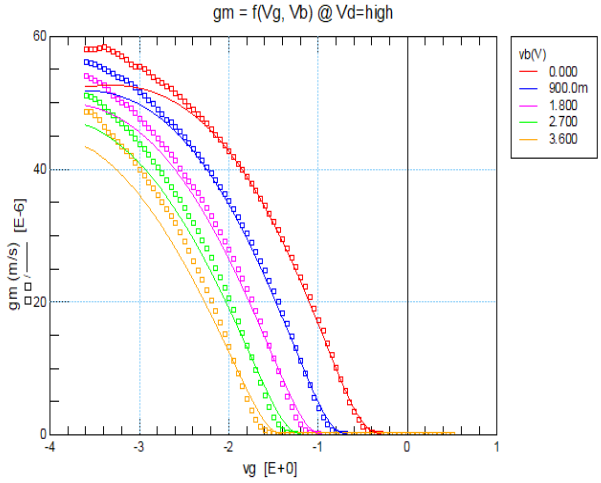


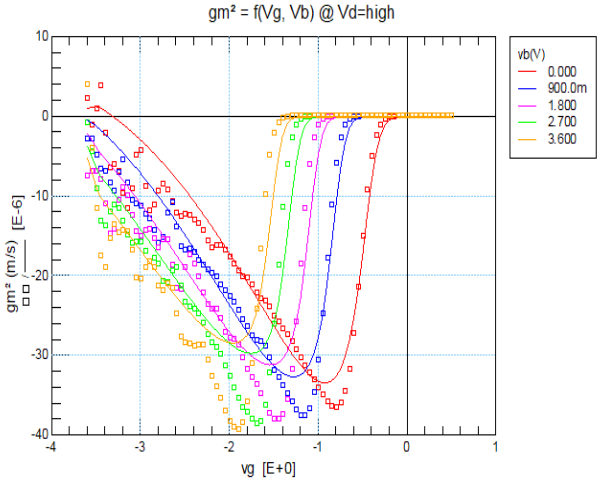


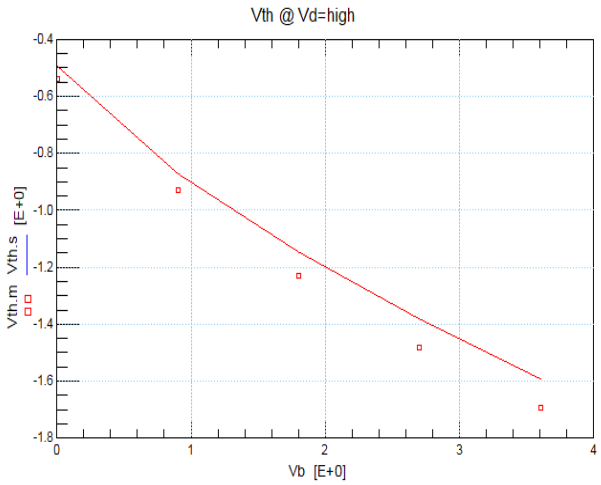
idvg, Vd = -3.3V, T = 125°C

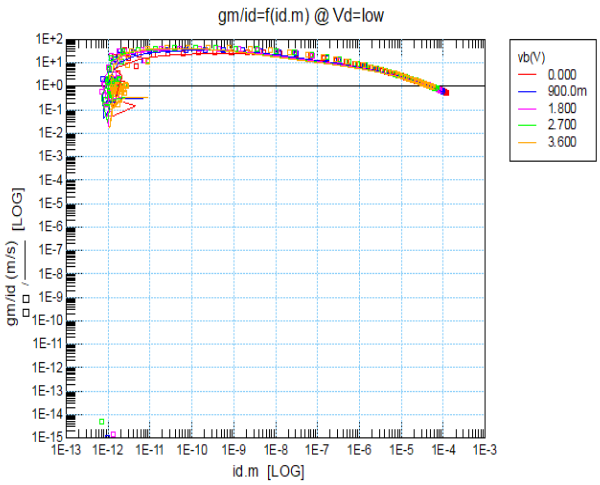


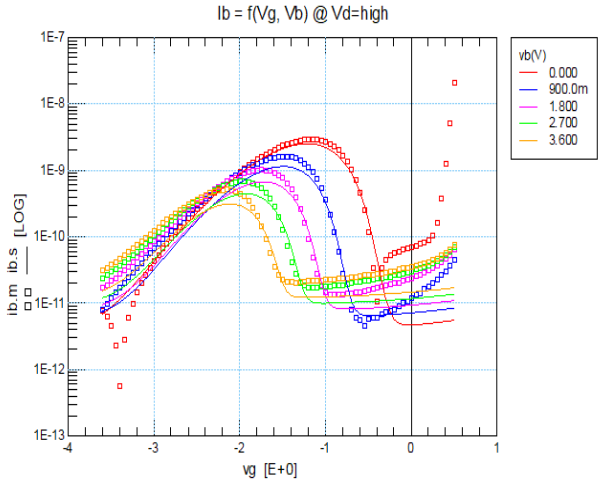


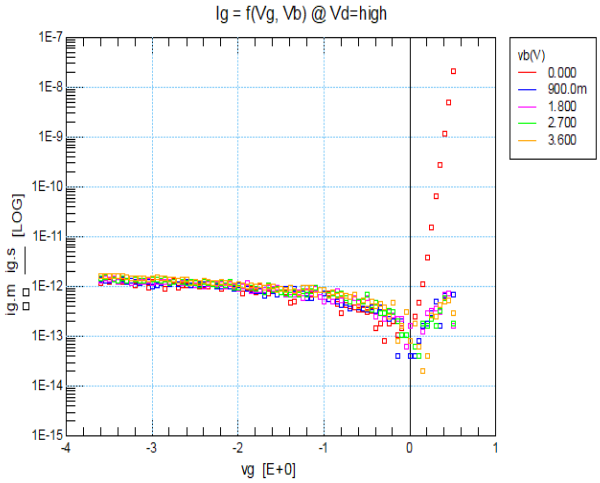




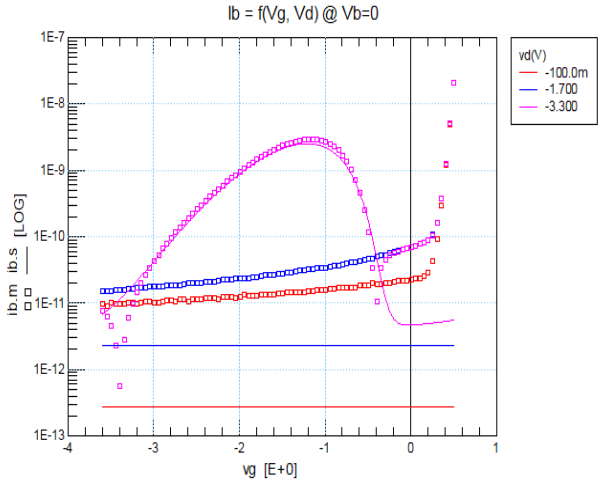


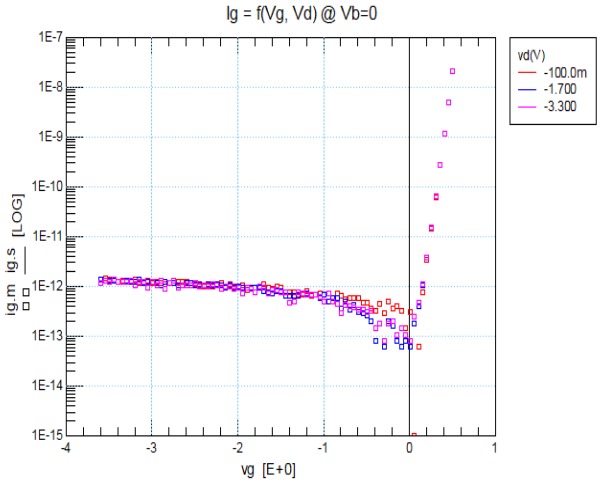




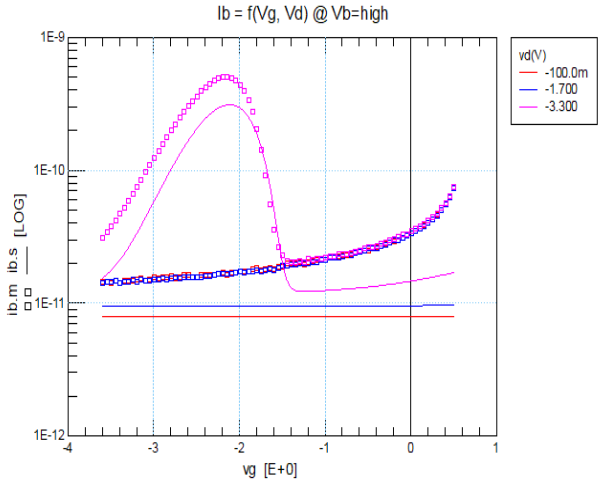


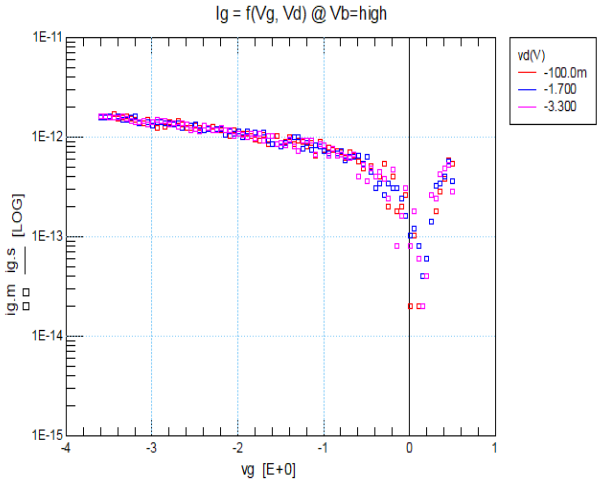
idvg, Ib,Ig, Vb = 0V, T = 125°C



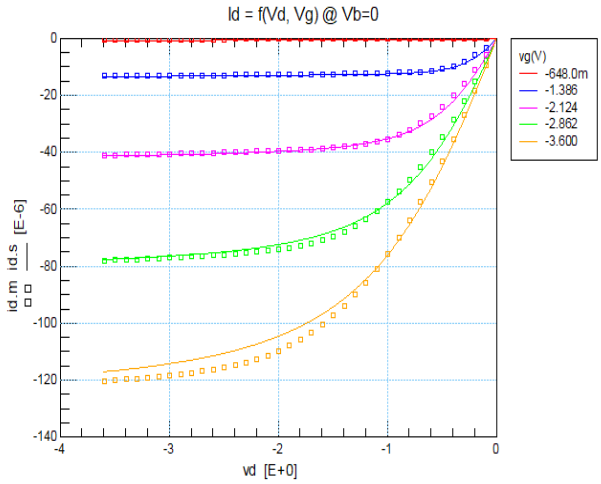


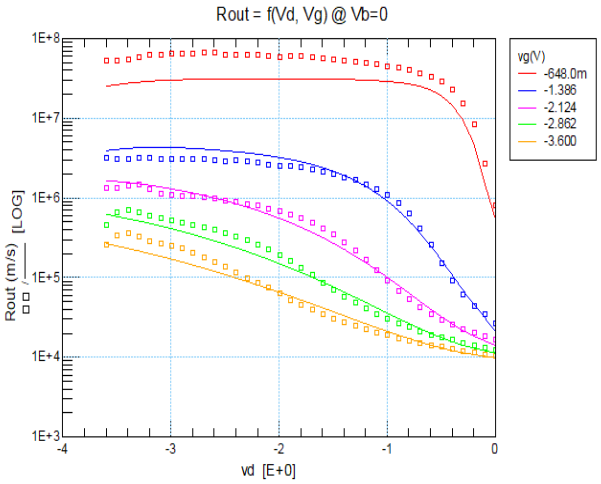
idvg, Ib,Ig, Vb = 3.6V, T = 125°C

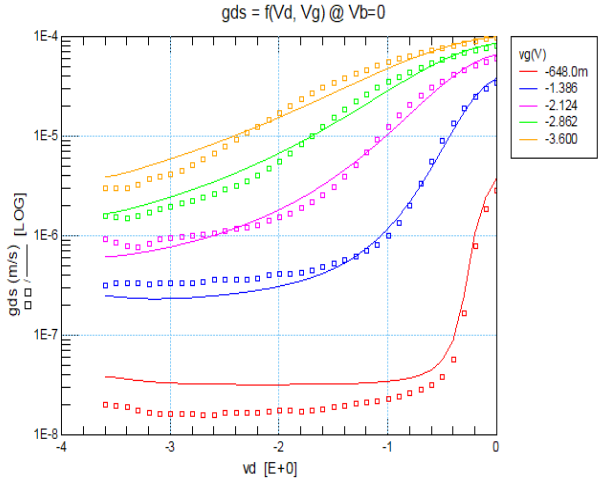




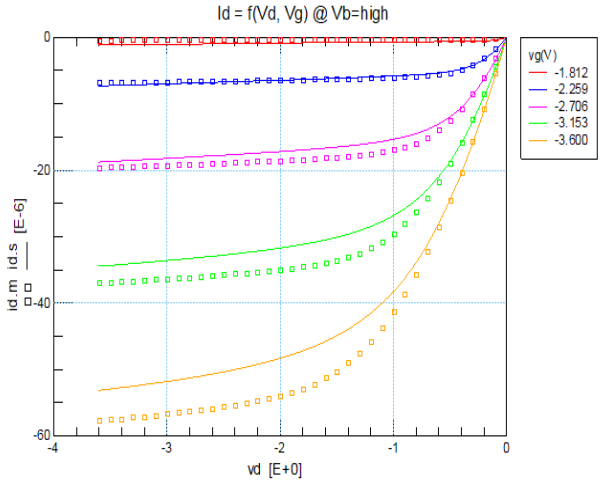
idvd, Vb = 0V, T = 125°C

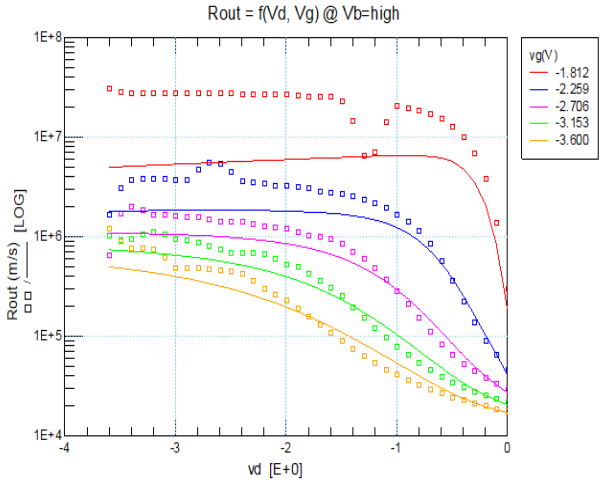


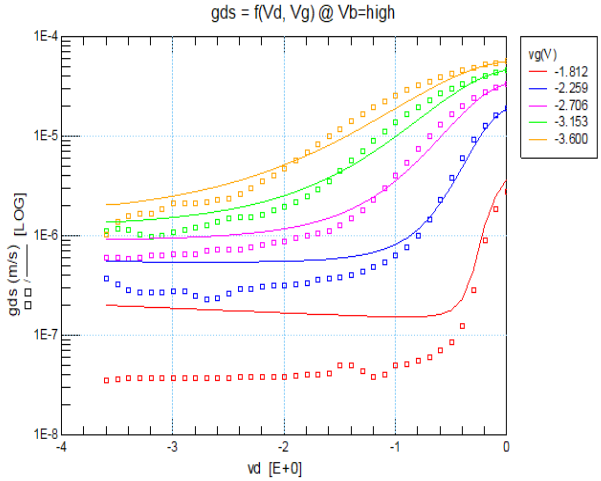




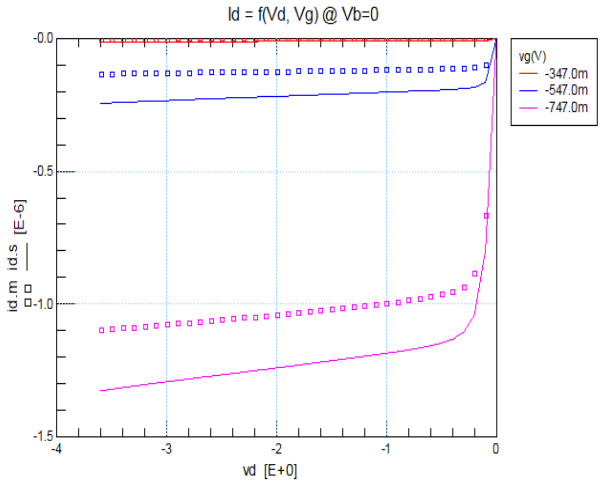
idvd_vbmin, Vb = 3.6V, T = 125°C

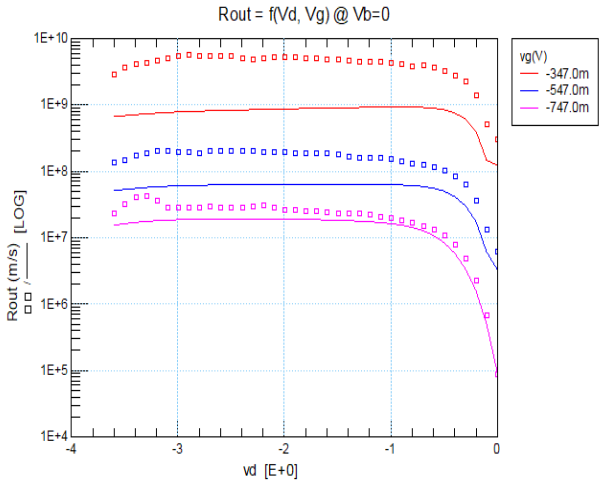


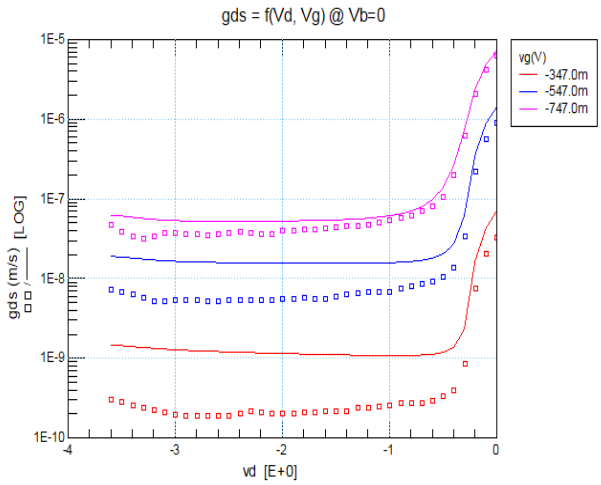




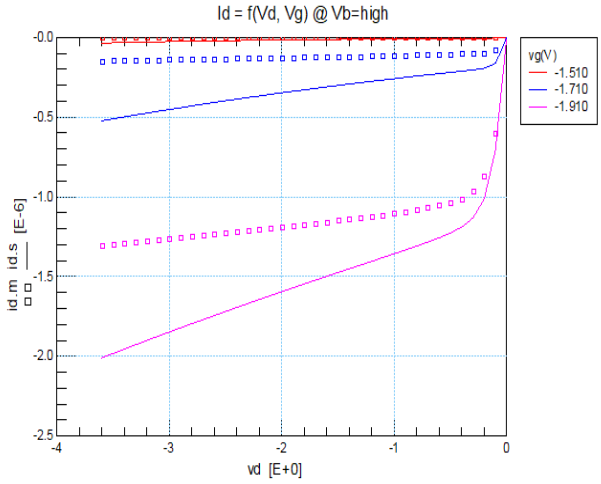
idvd_vth, Vb = 0V, T = 125°C

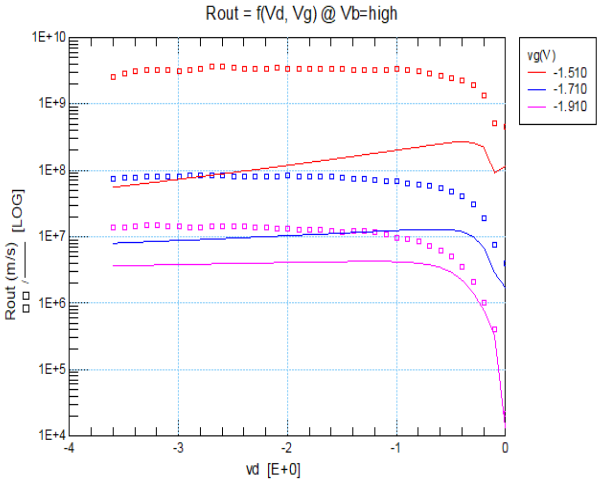


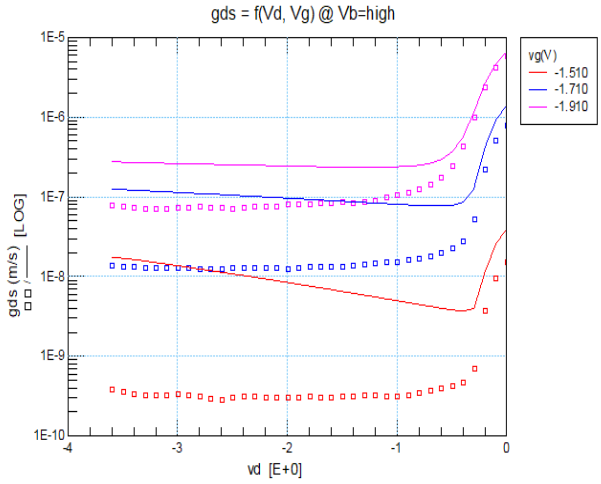




idvd_vth_vbmin, Vb = 3.6V, T = 125°C



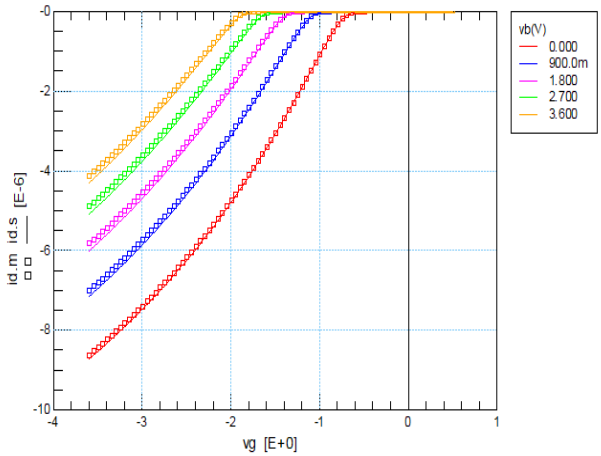




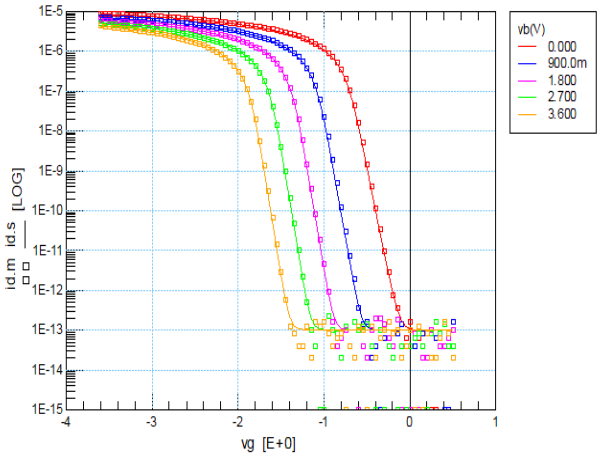
4.20 W05u0_L5u0_S562_5

idvg, Vd = -0.1V, T = 27°C

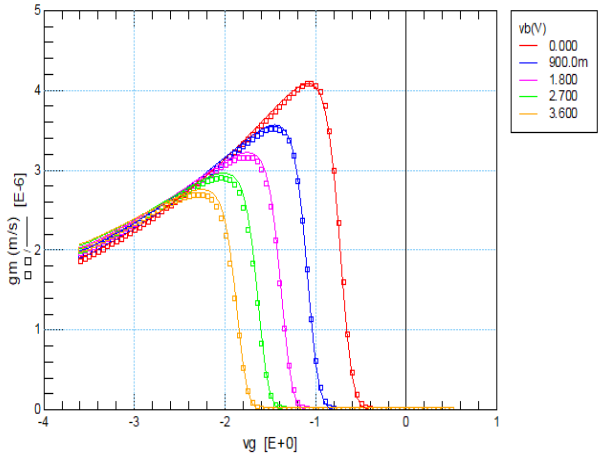
Id = f(Vg, Vb) @ Vd=low



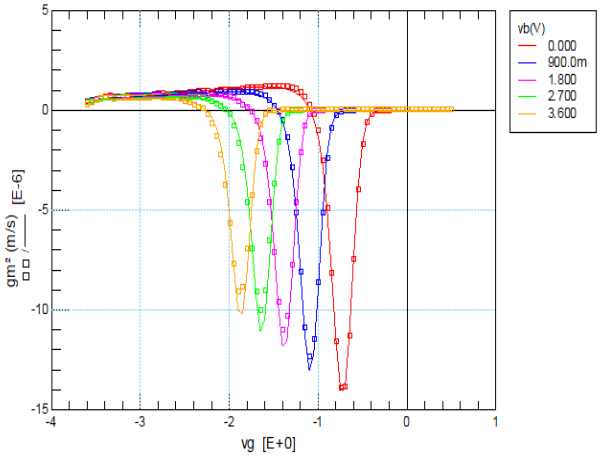
Id = f(Vg, Vb) @ Vd=low



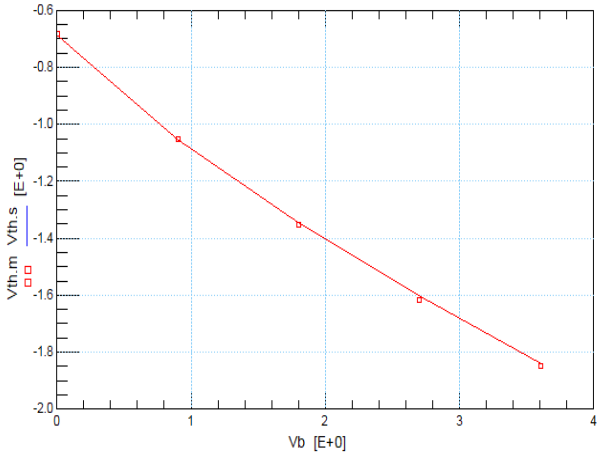
gm = f(Vg, Vb) @ Vd=low



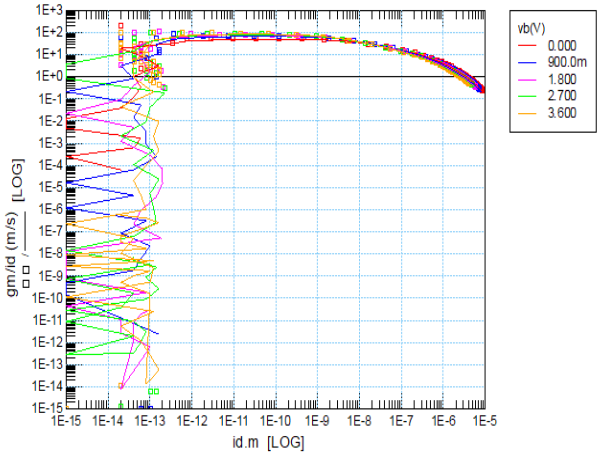
gm² = f(Vg, Vb) @ Vd=low

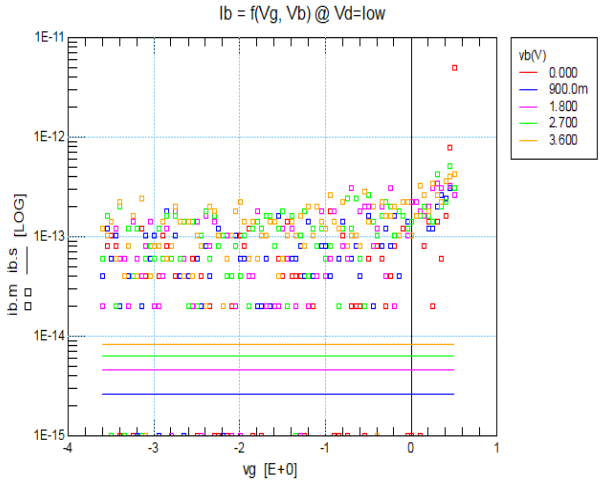


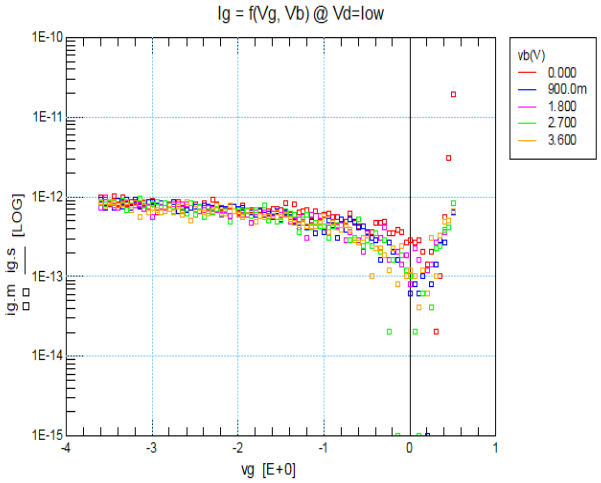
Vth @ Vd=low



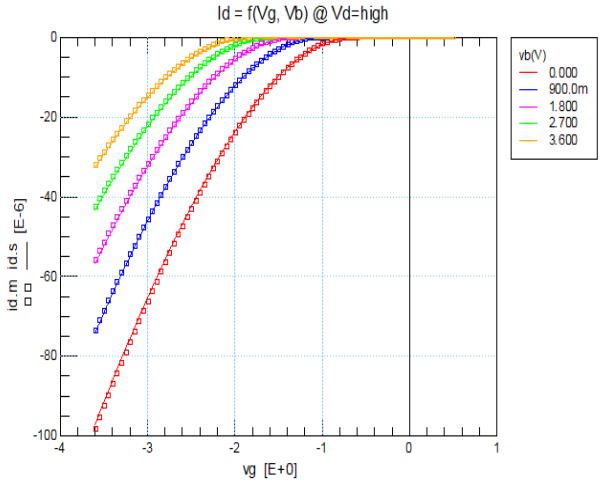
gm/id=f(id.m) @ Vd=low

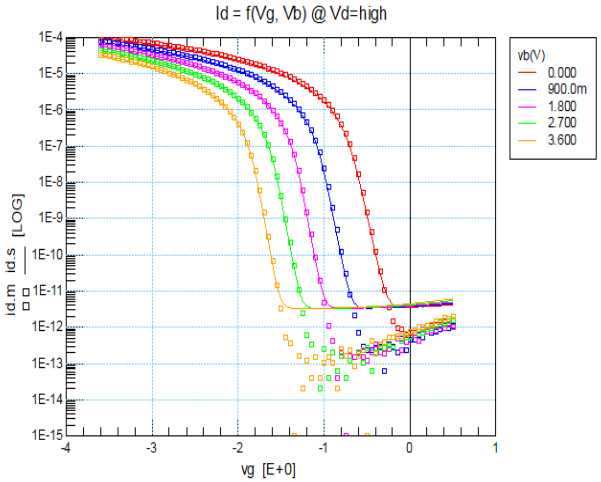


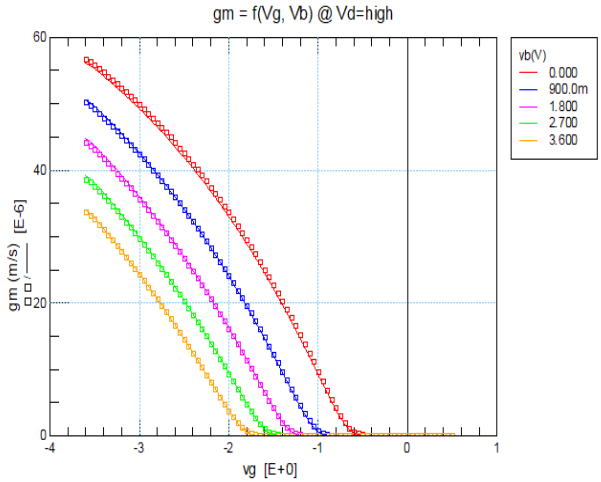


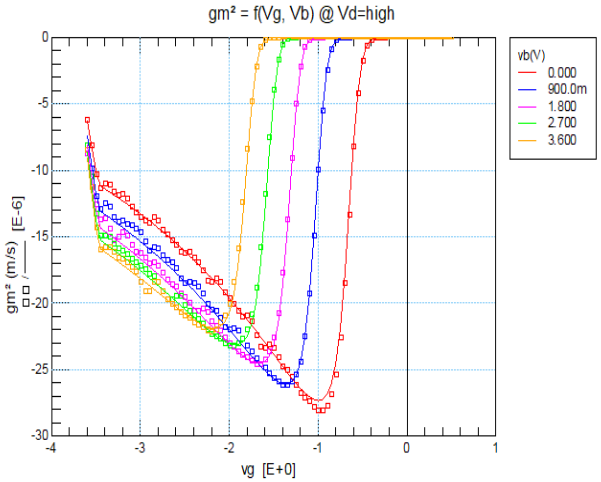


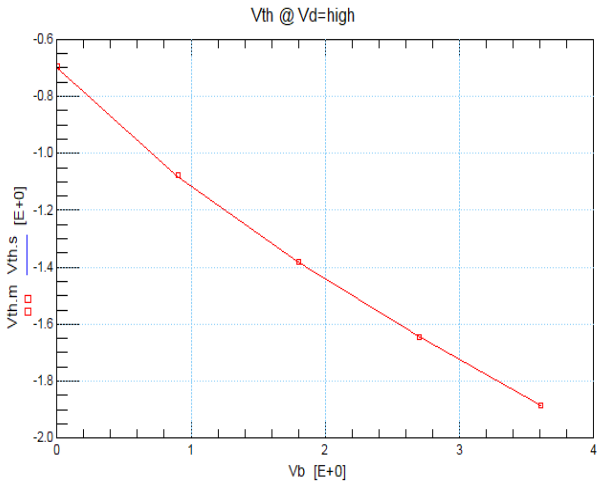
idvg, Vd = -3.3V, T = 27°C

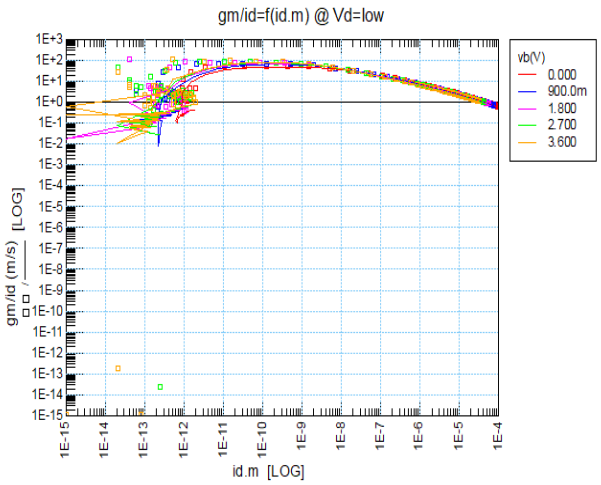


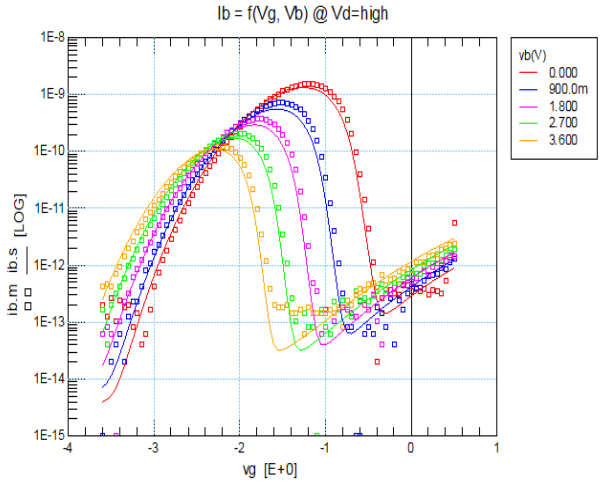


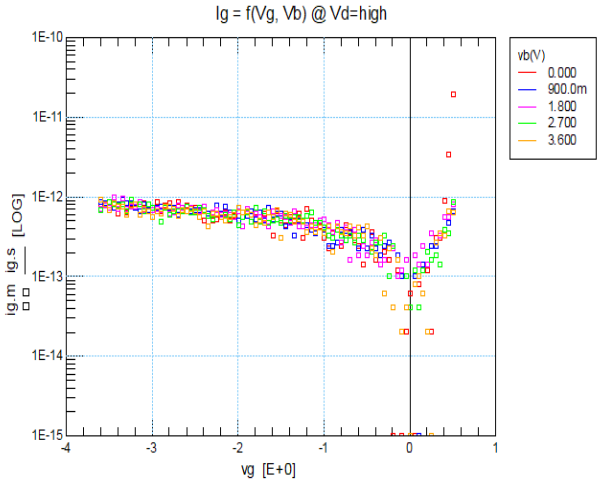




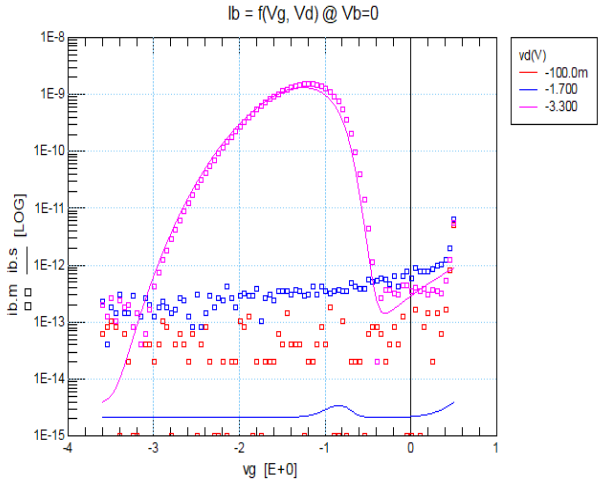


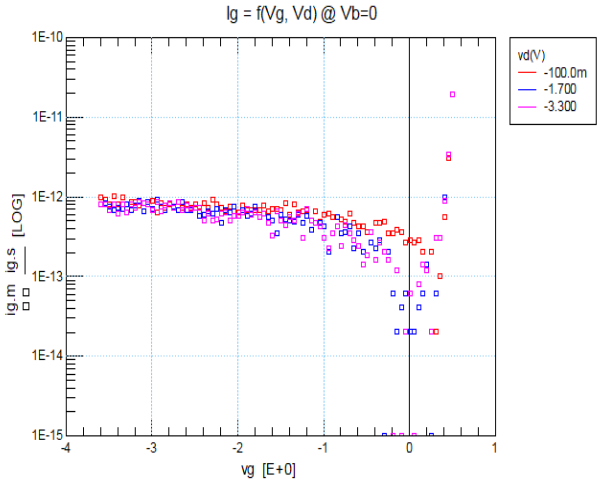




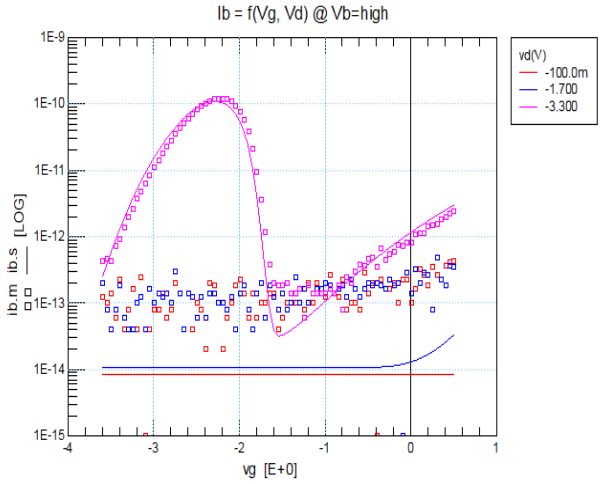


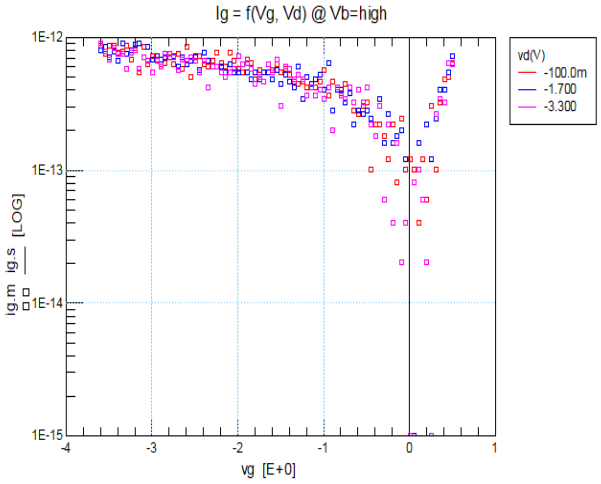
idvg, Ib,Ig, Vb = 0V, T = 27°C



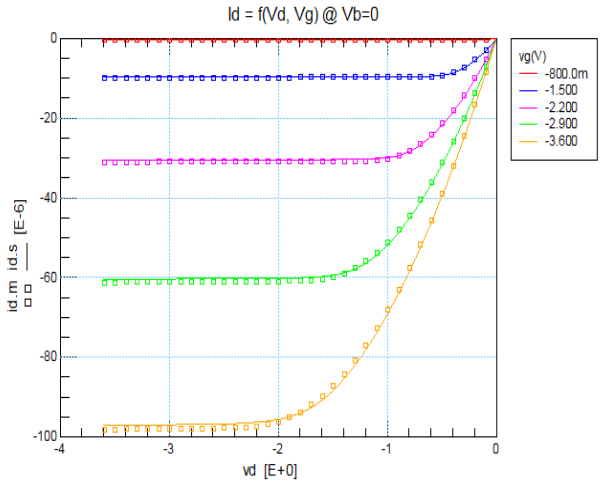


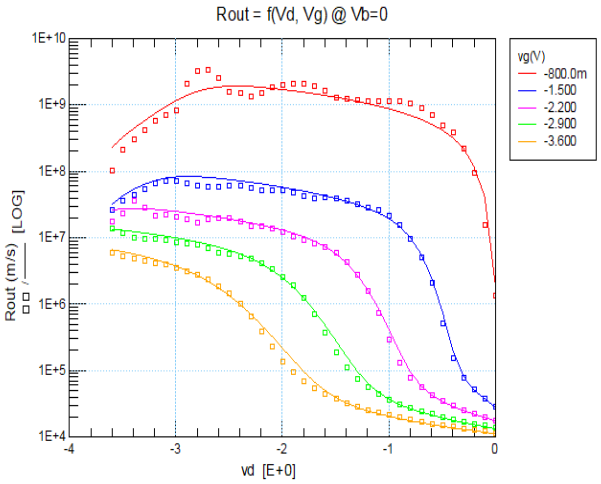
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

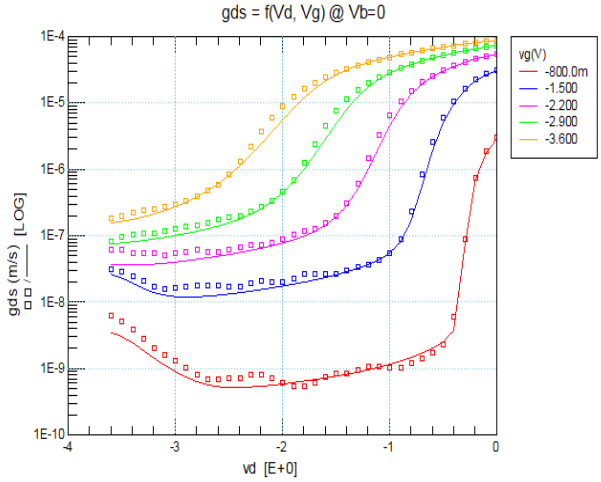




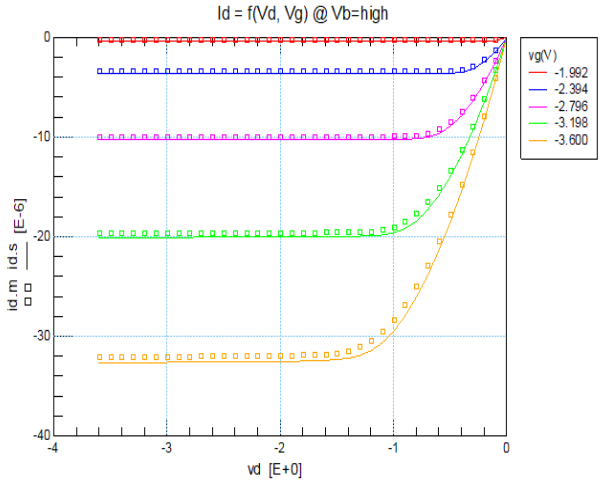
idvd, Vb = 0V, T = 27°C

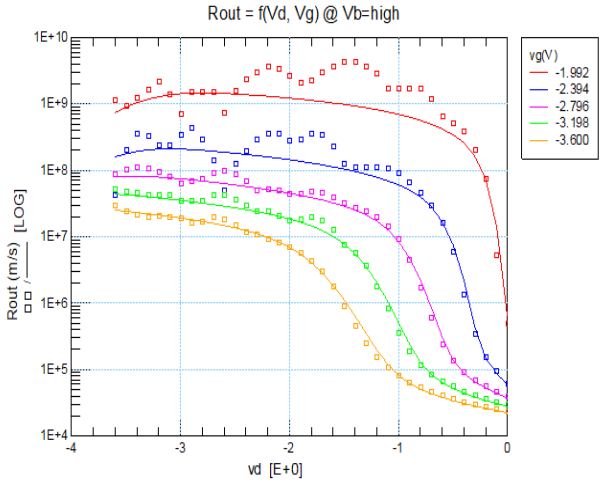


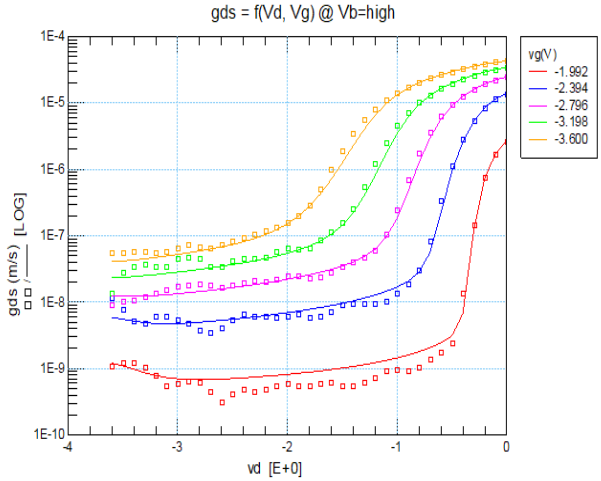




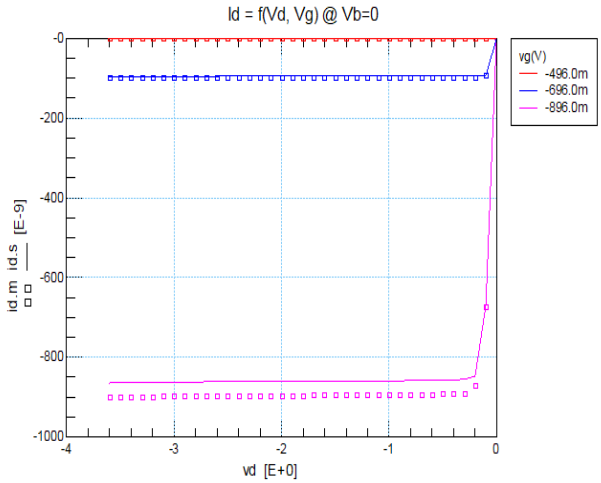
idvd_vbmin, Vb = 3.6V, T = 27°C

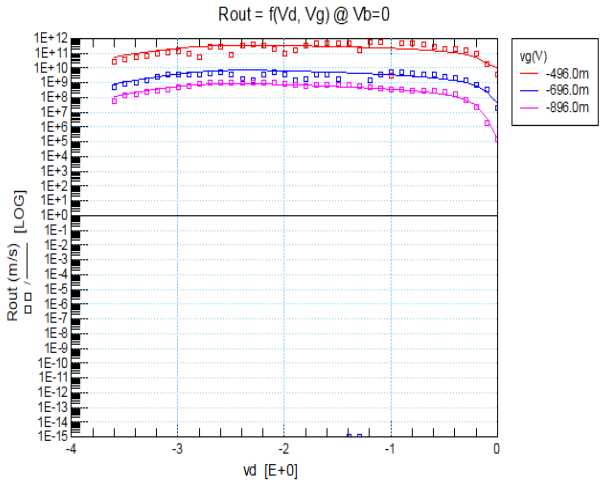


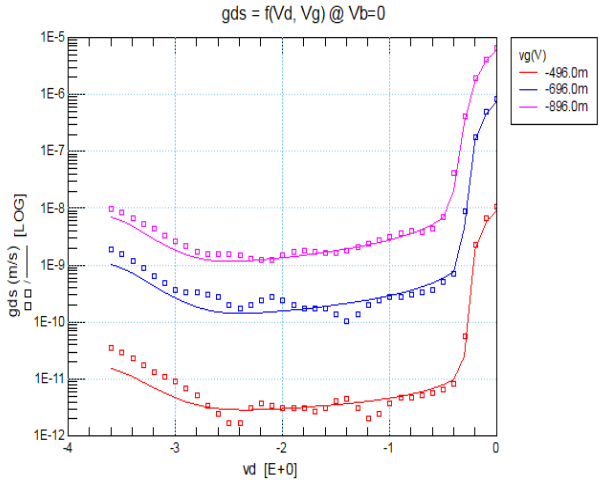




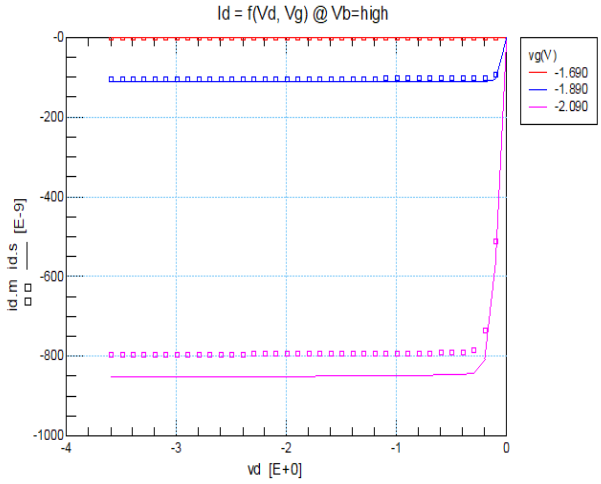
idvd_vth, Vb = 0V, T = 27°C

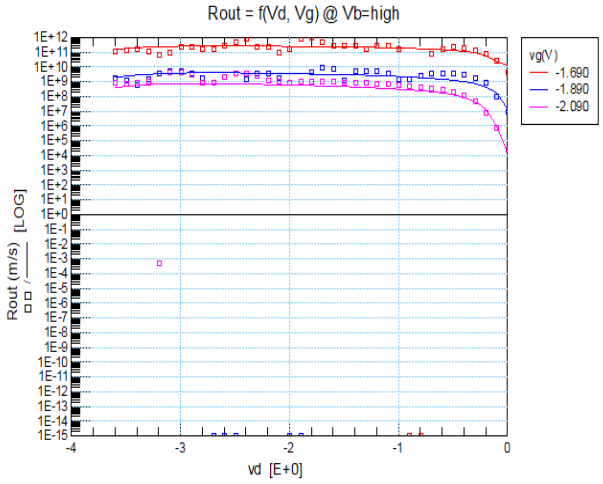


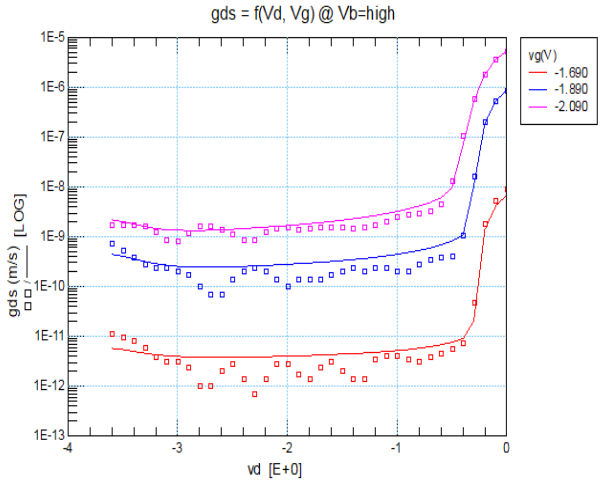




idvd_vth_vbmin, Vb = 3.6V, T = 27°C



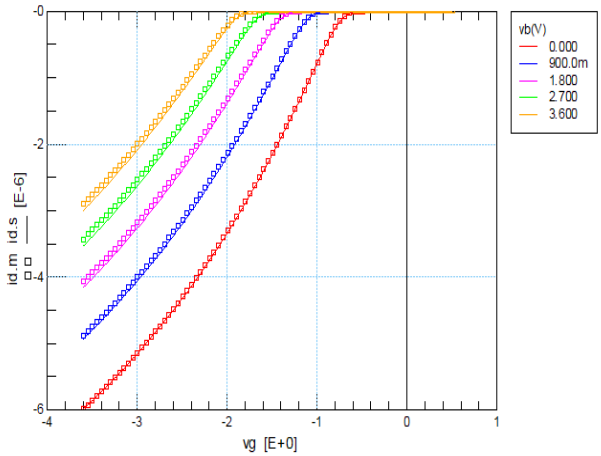




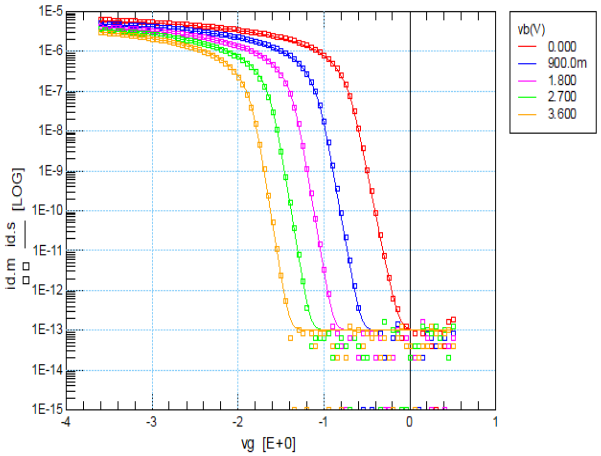
4.21 W02u0_L3u0_S563_4

idvg, Vd = -0.1V, T = 27°C

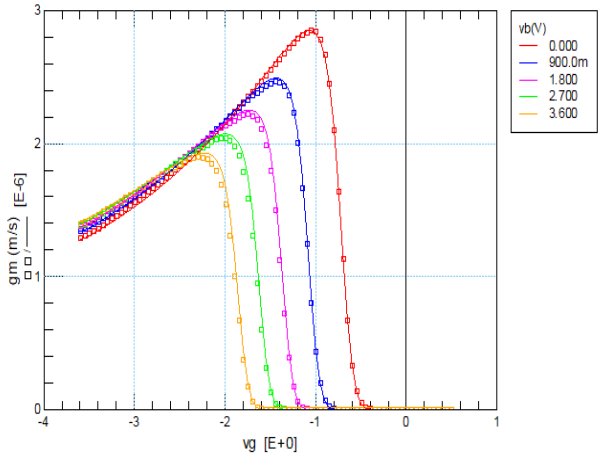
Id = f(Vg, Vb) @ Vd=low



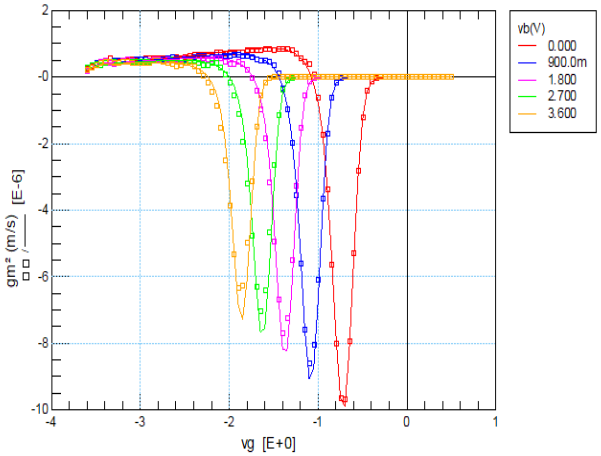
Id = f(Vg, Vb) @ Vd=low



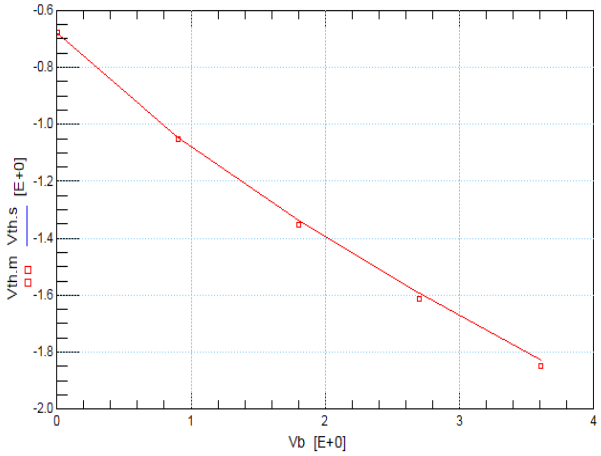
gm = f(Vg, Vb) @ Vd=low



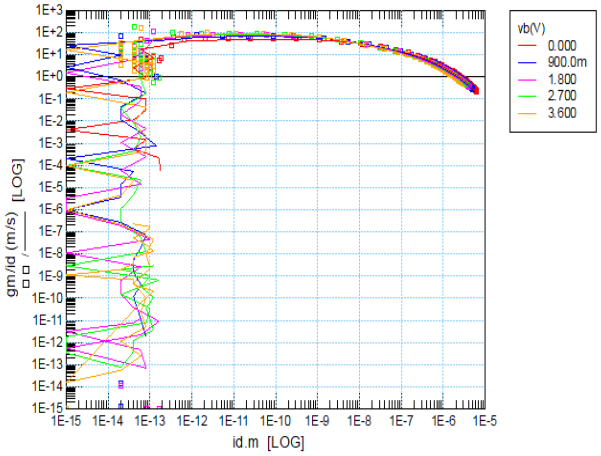
gm² = f(Vg, Vb) @ Vd=low

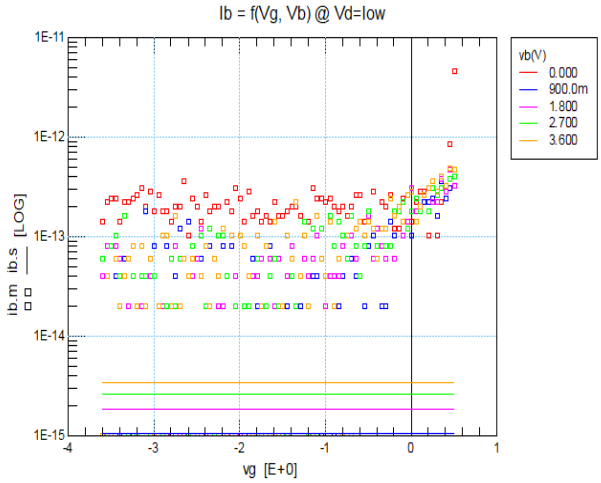


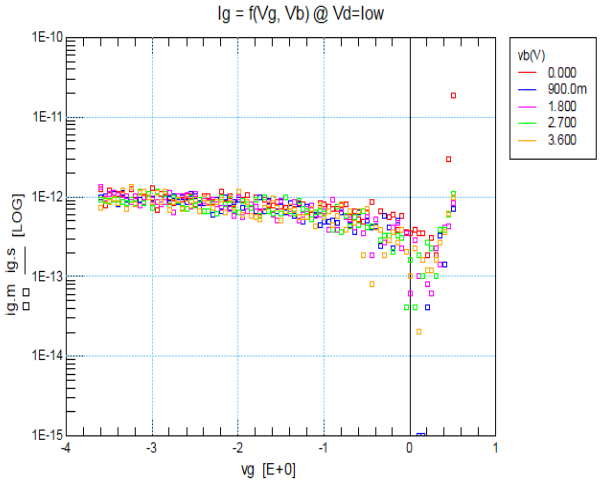
Vth @ Vd=low



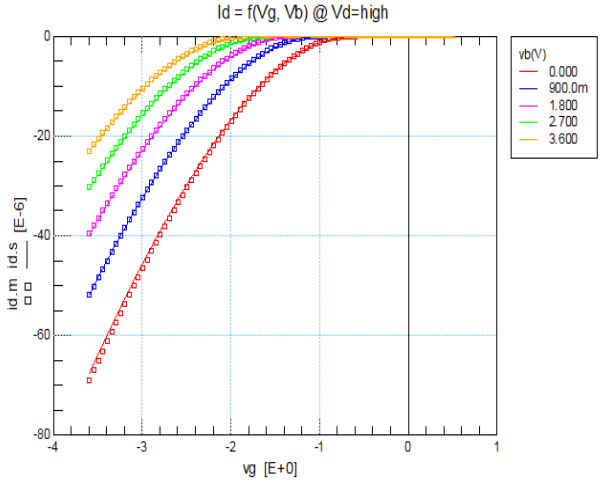
gm/id=f(id.m) @ Vd=low

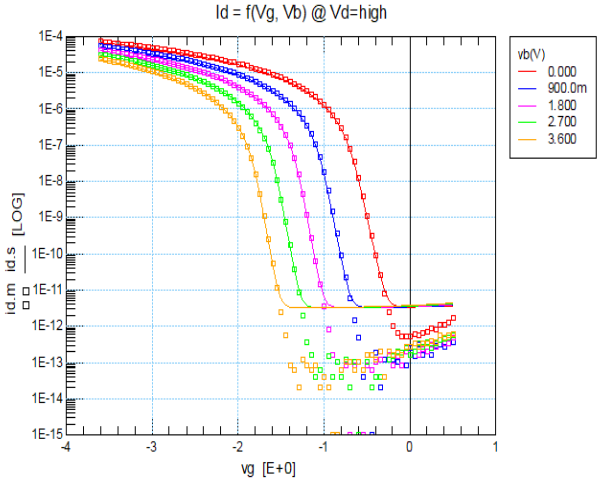


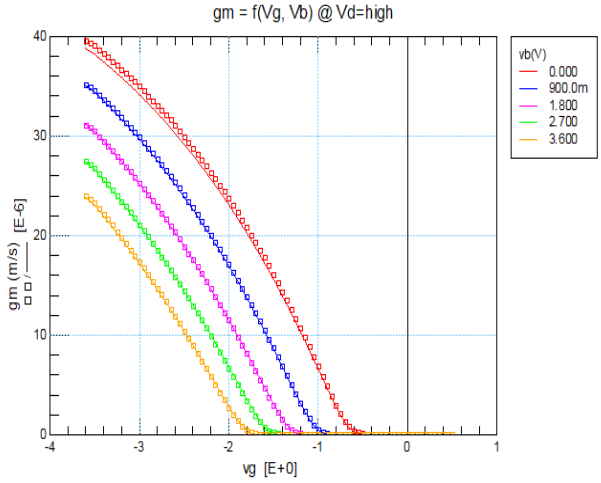


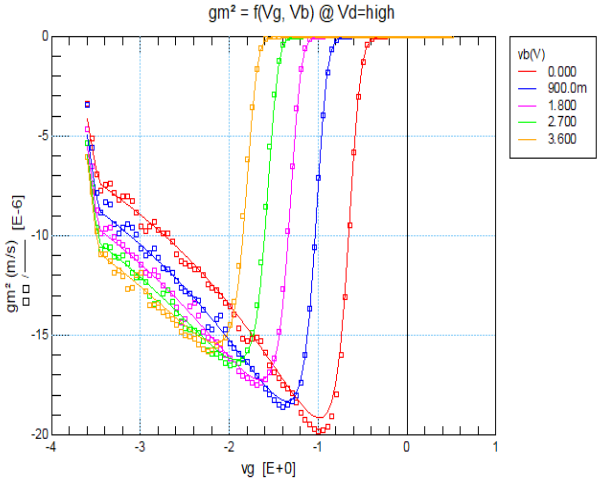


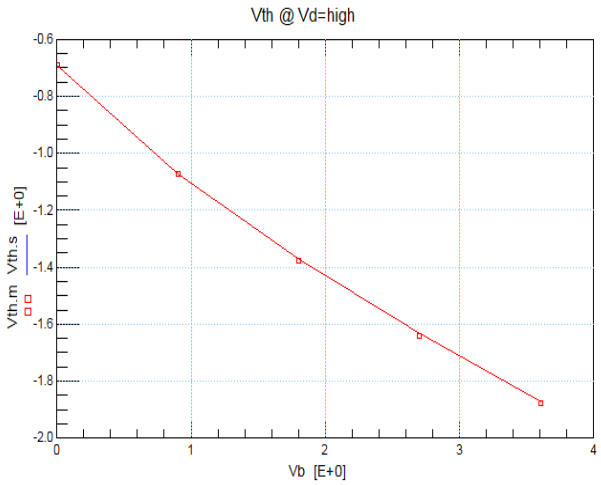
idvg, Vd = -3.3V, T = 27°C

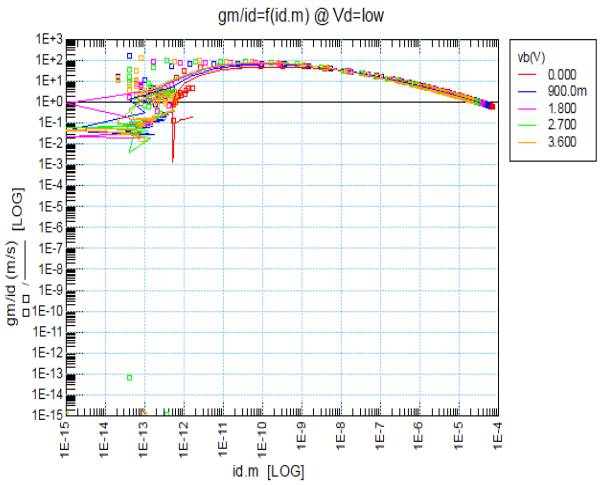


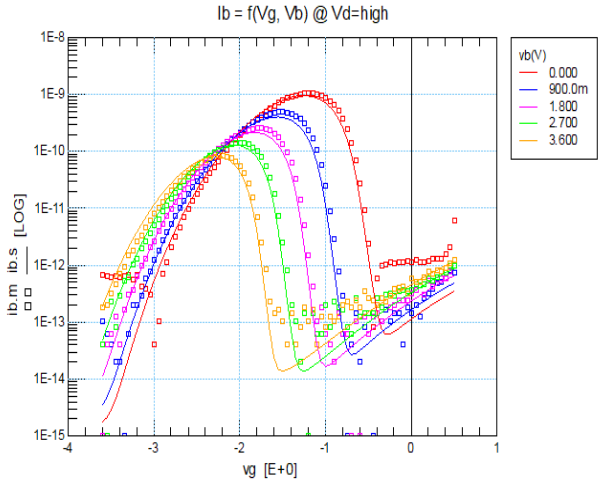


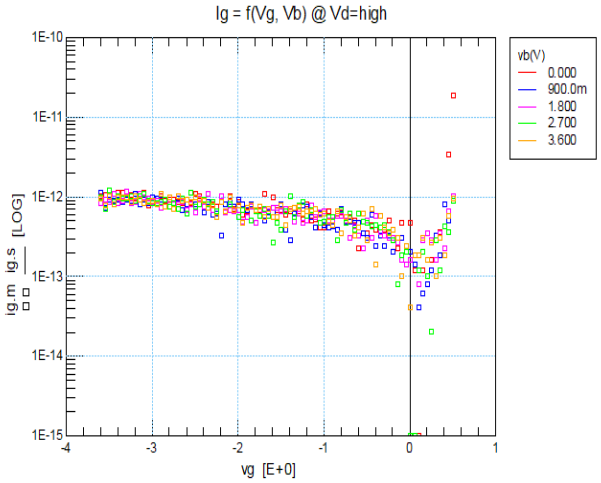




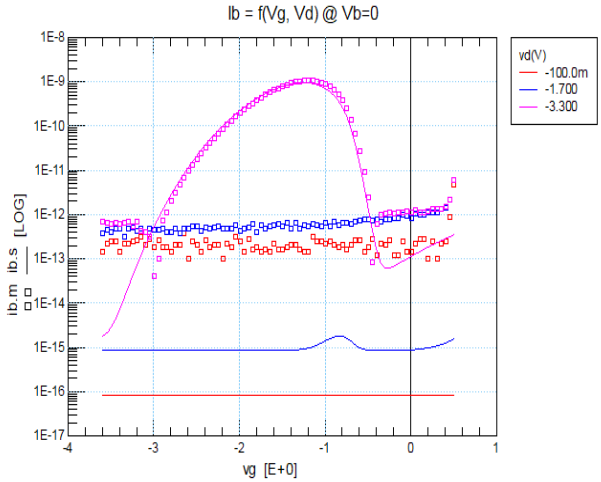


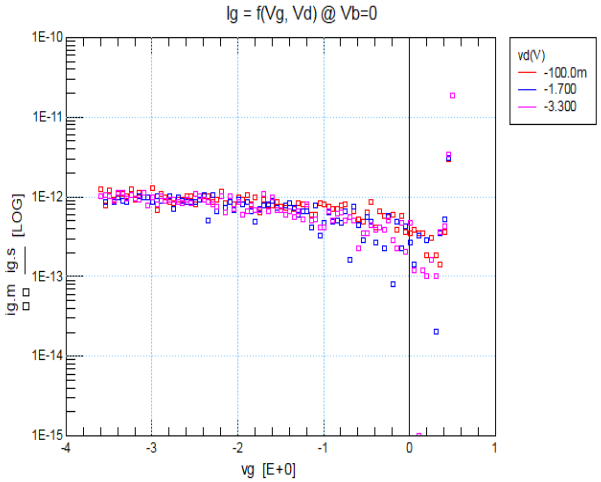




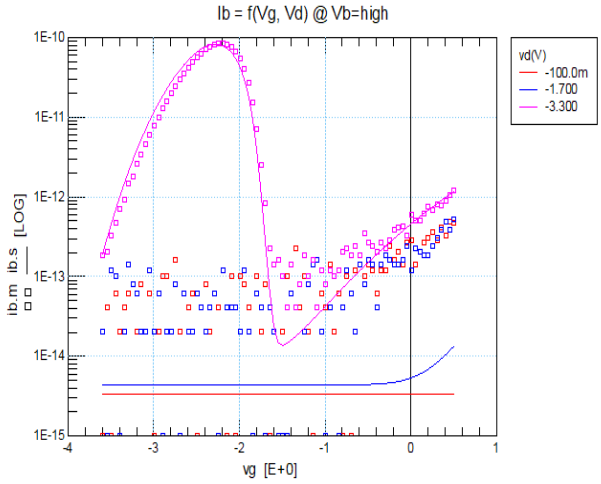


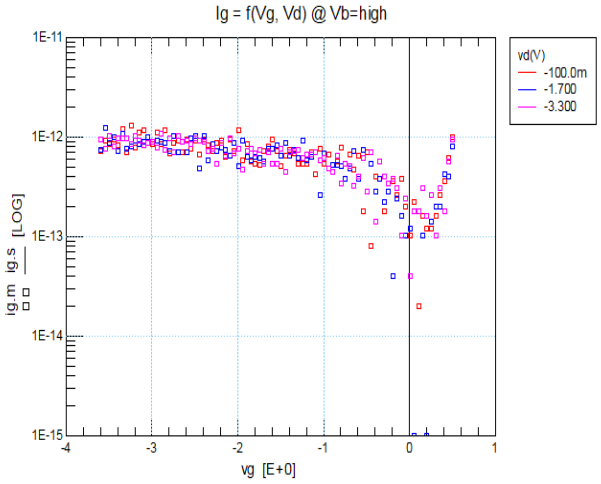
idvg, Ib,Ig, Vb = 0V, T = 27°C



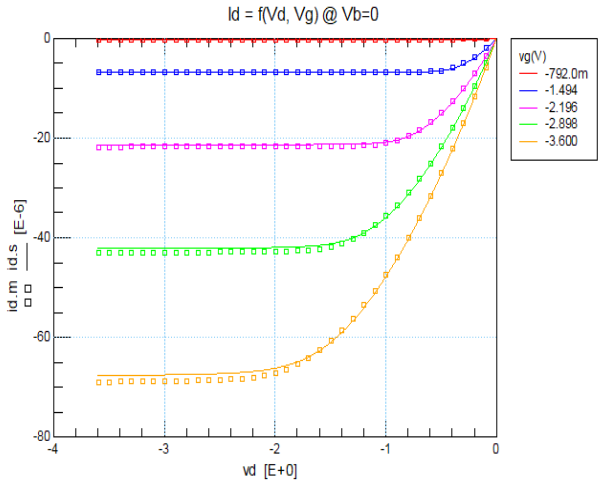


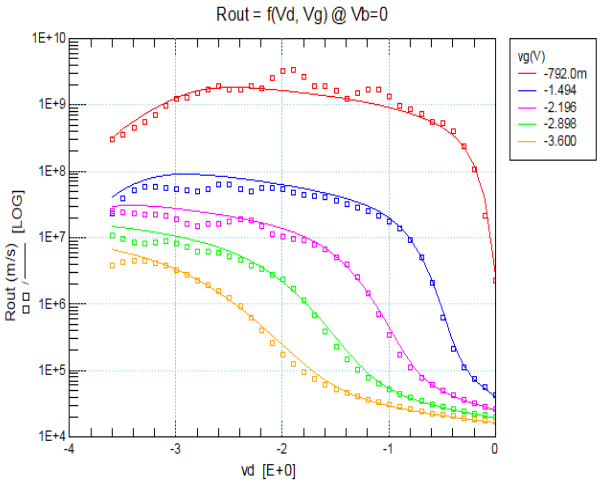
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

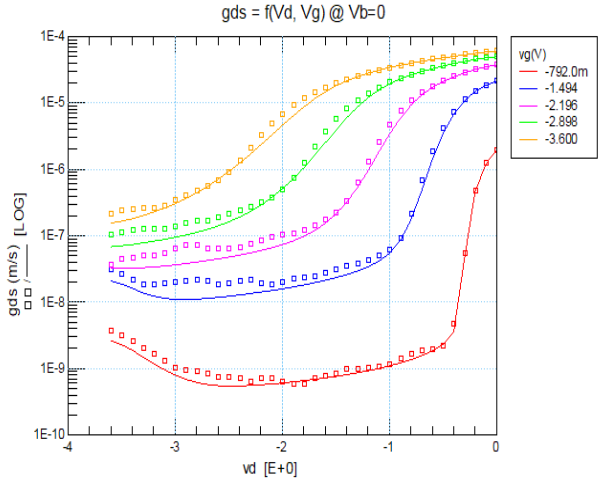




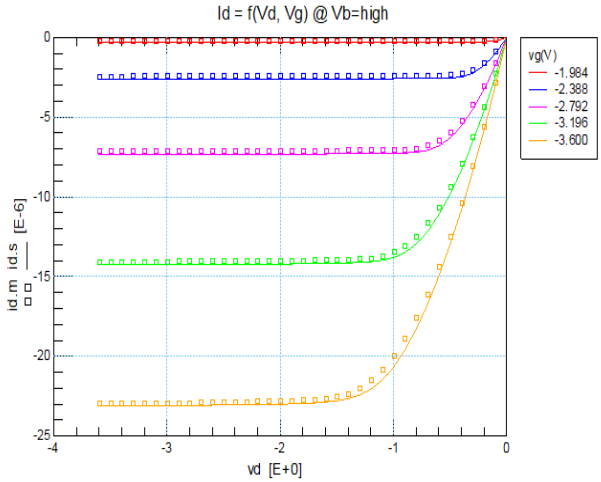
idvd, Vb = 0V, T = 27°C

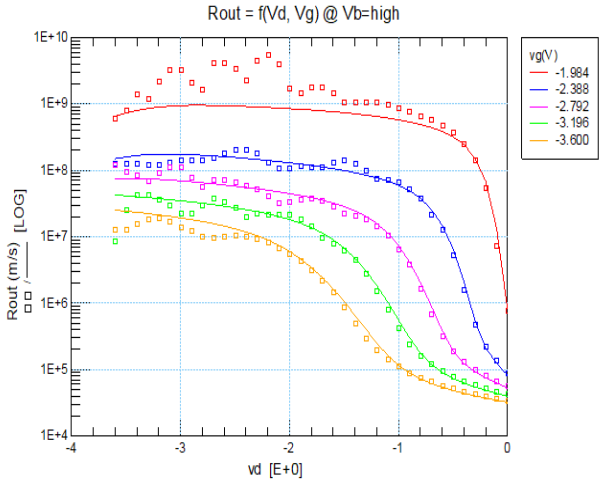


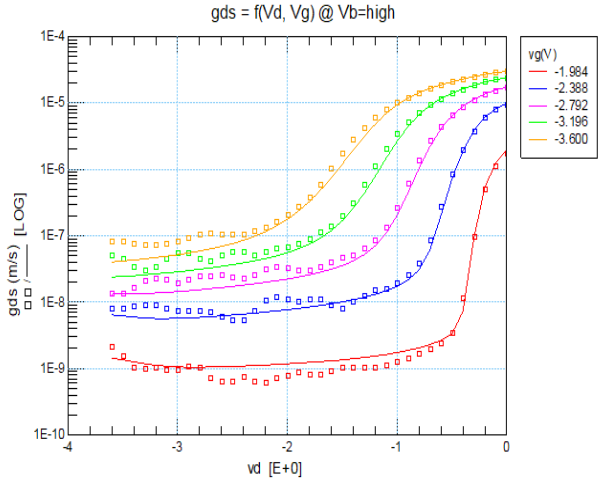




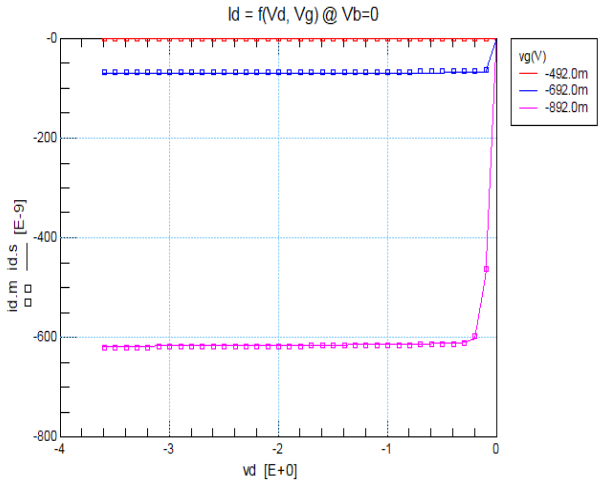
idvd_vbmin, Vb = 3.6V, T = 27°C

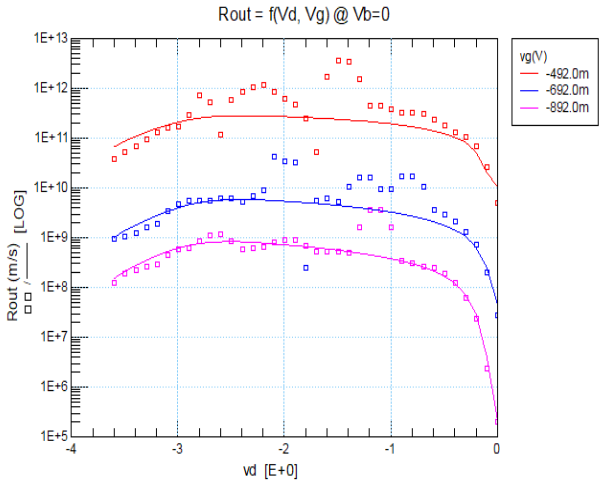


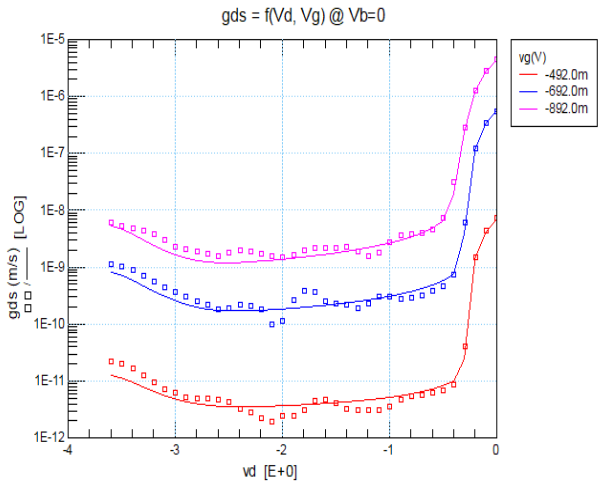




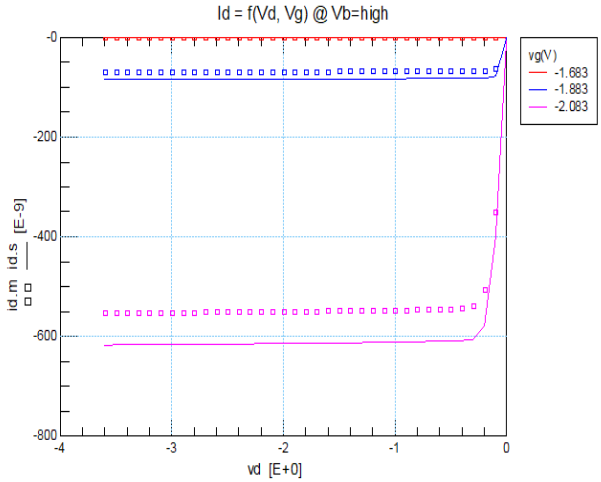
idvd_vth, Vb = 0V, T = 27°C

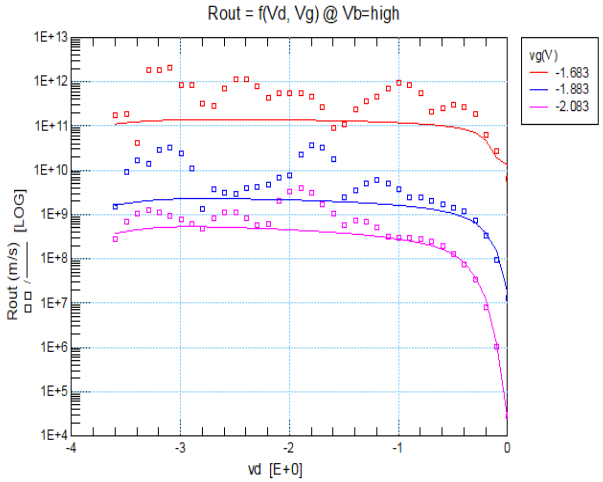


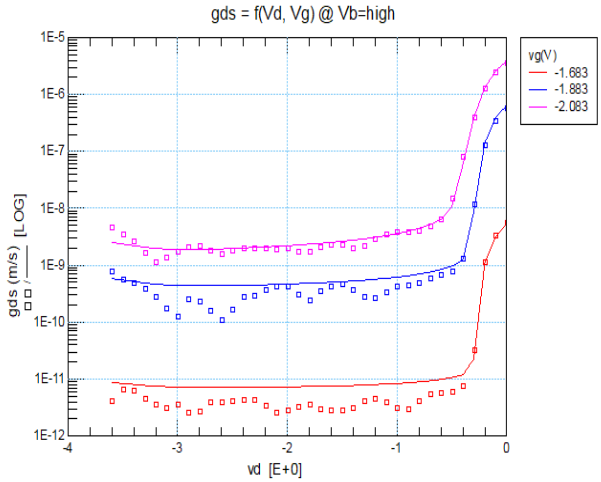




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

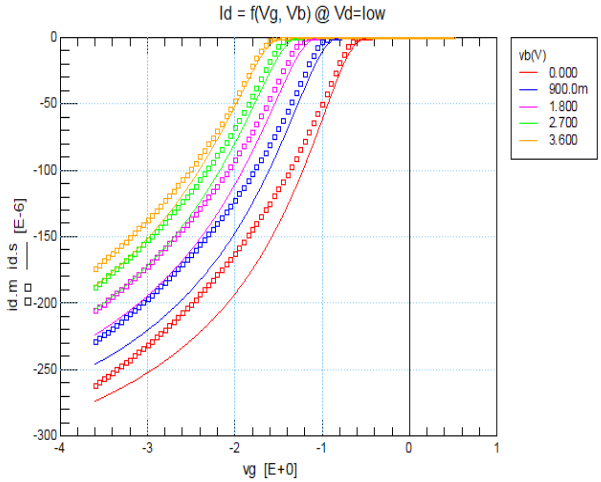


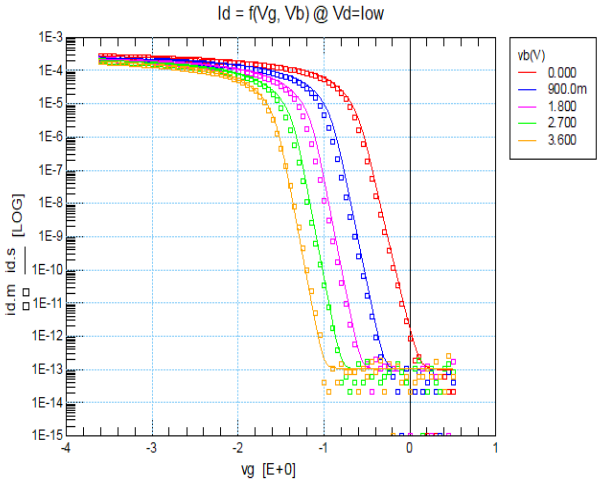


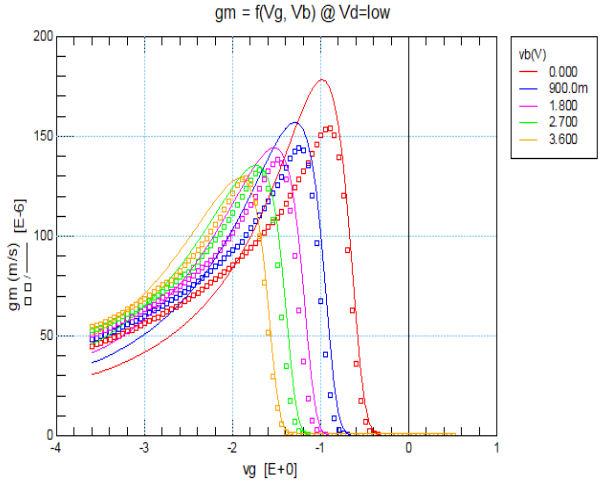


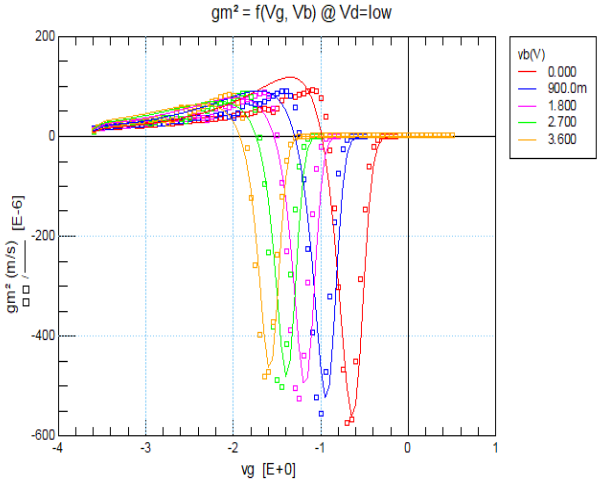
4.22 W10u0_L0u33_S561_2

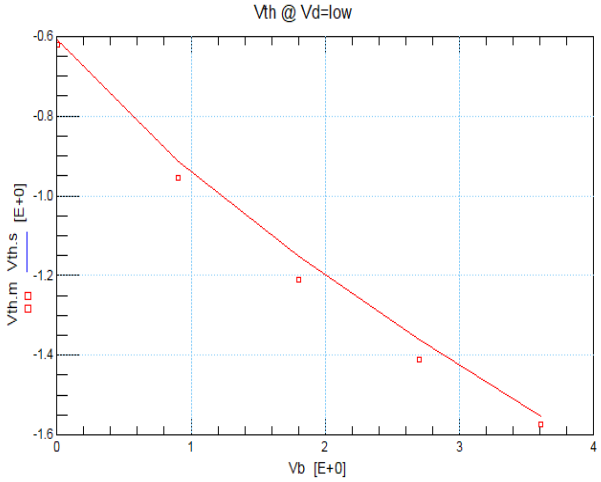
idvg, Vd = -0.1V, T = 27°C

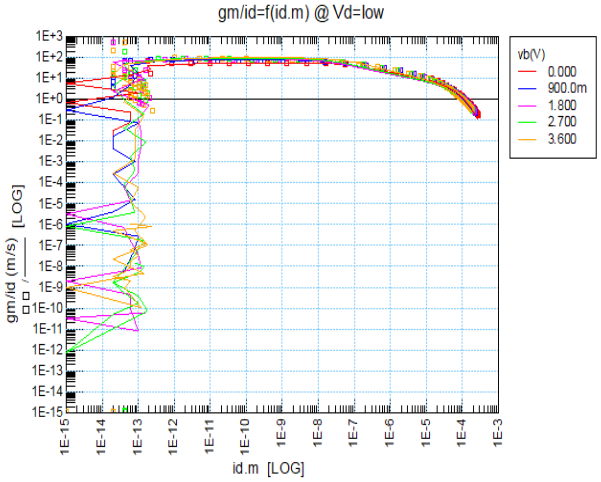


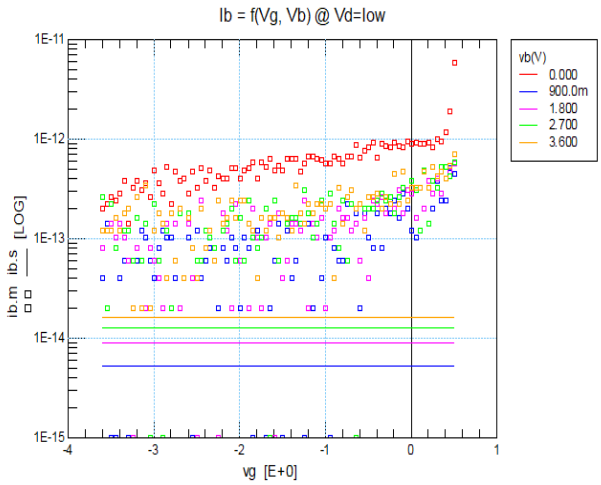


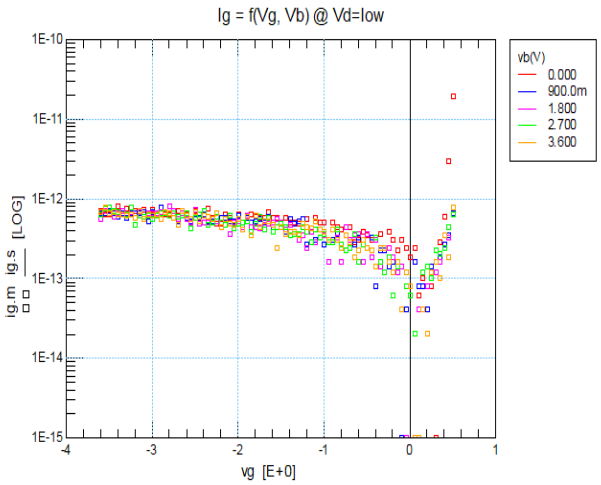




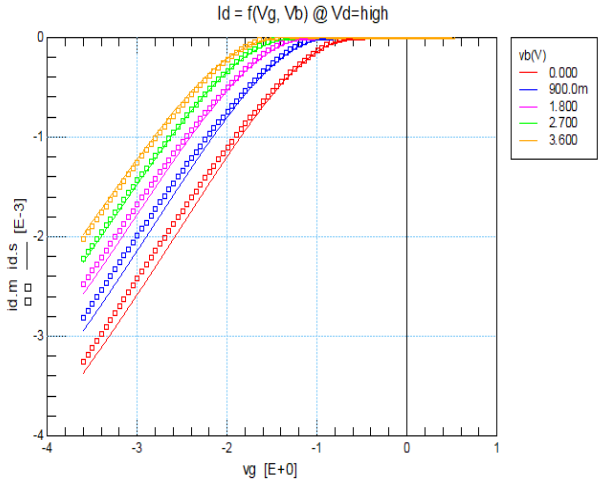


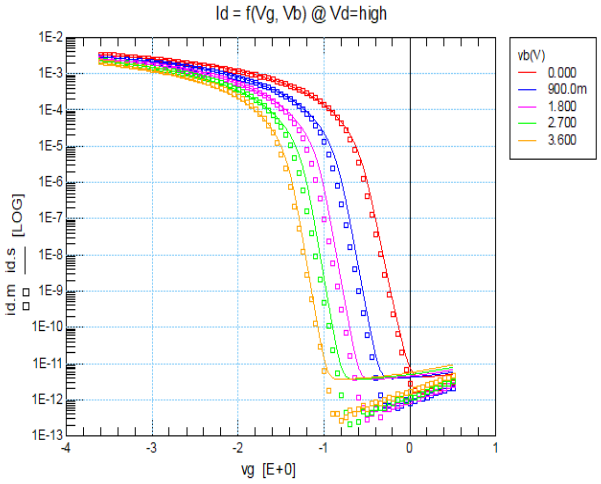


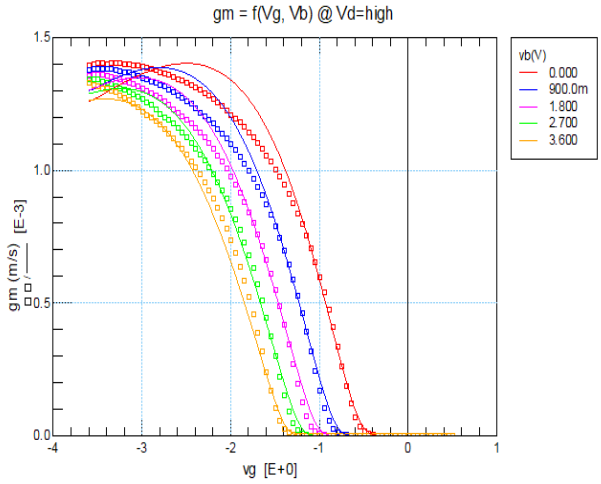


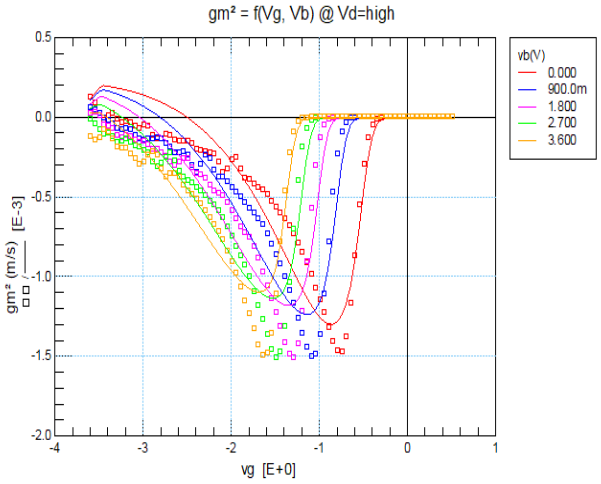


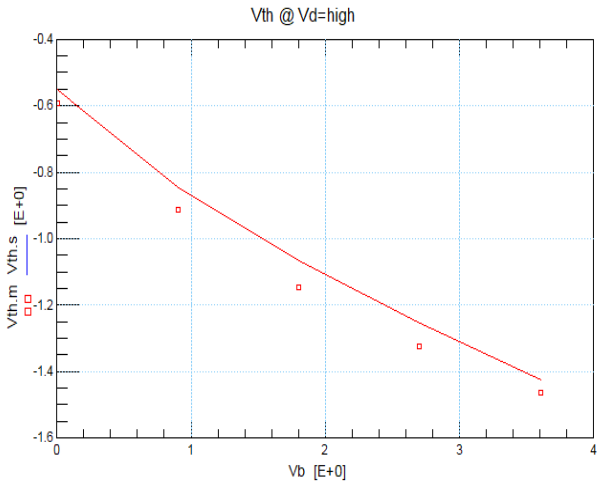
idvg, Vd = -3.3V, T = 27°C

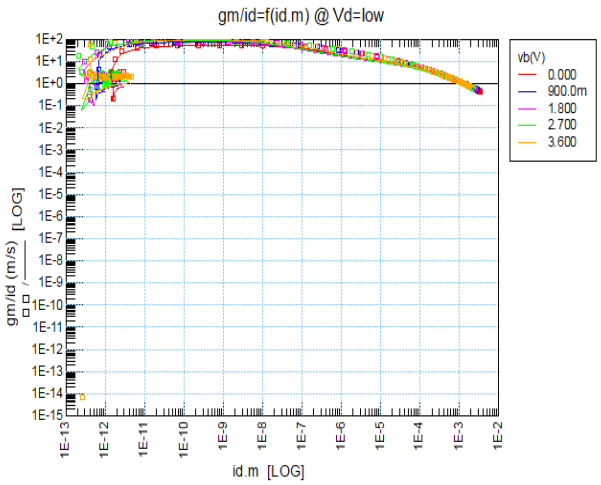


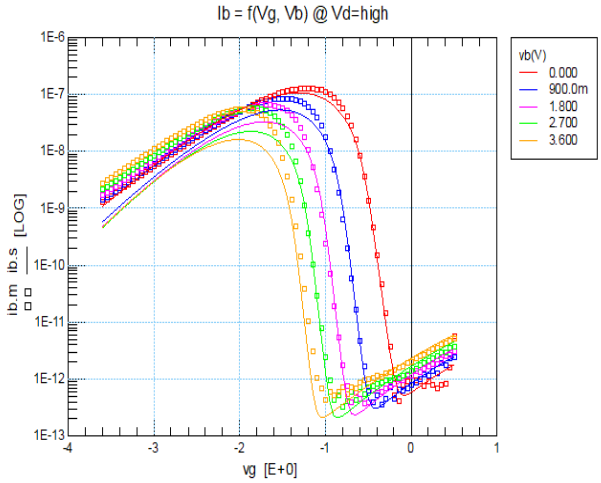


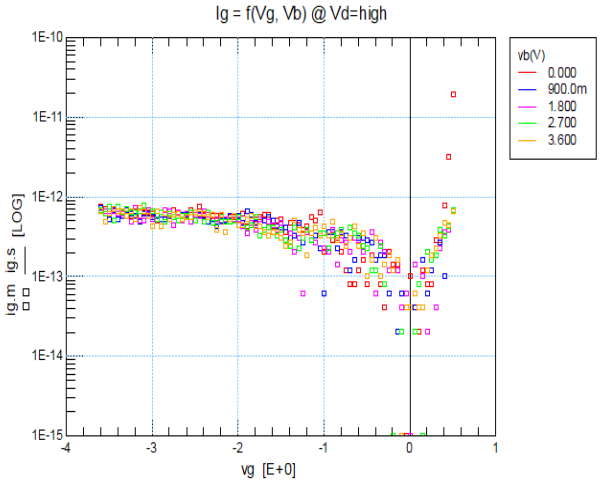




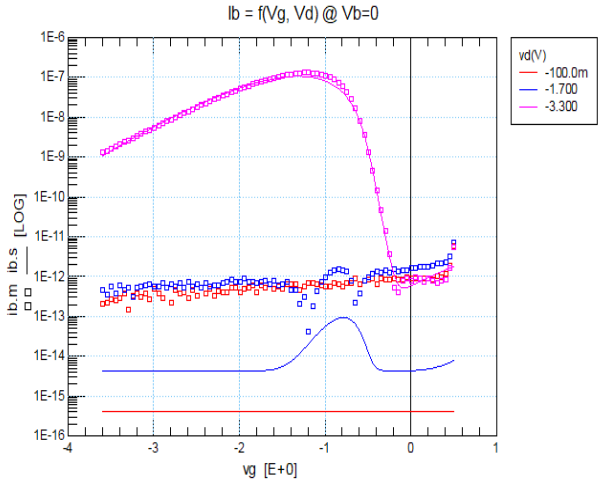


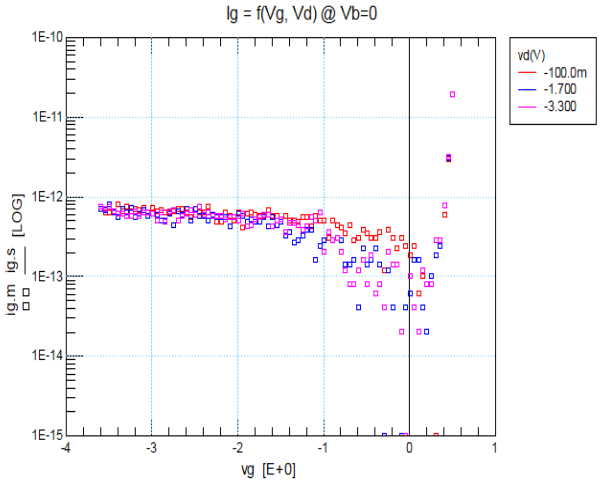




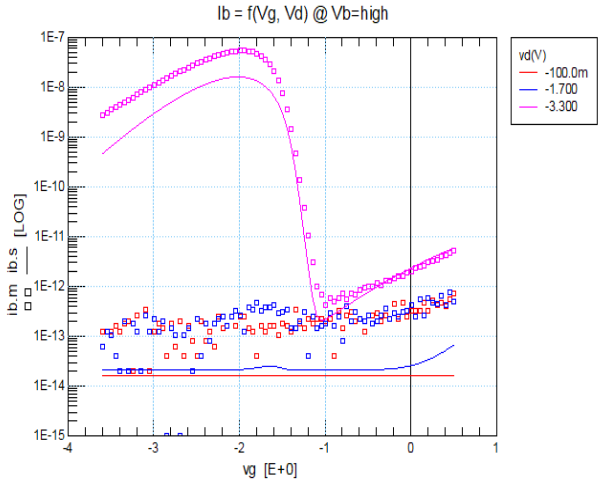


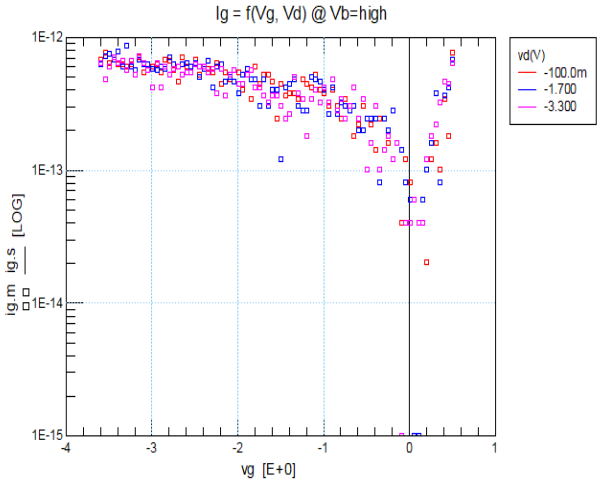
idvg, Ib,Ig, Vb = 0V, T = 27°C



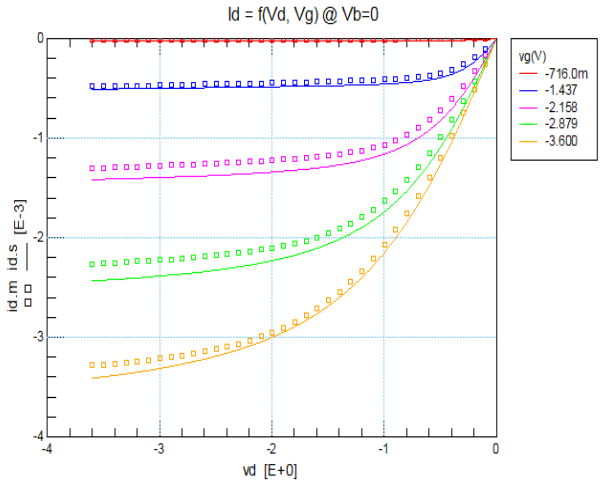


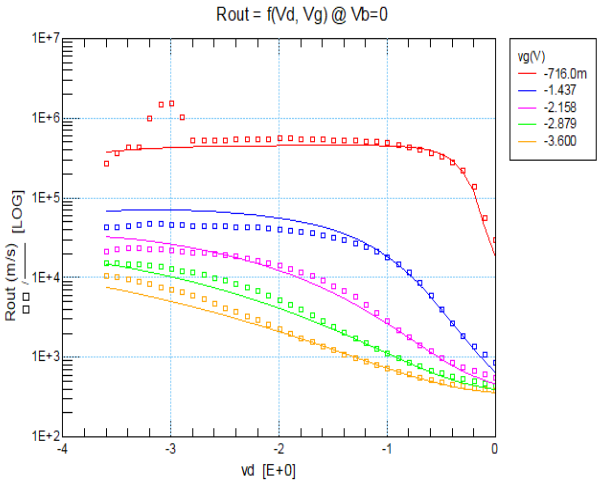
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

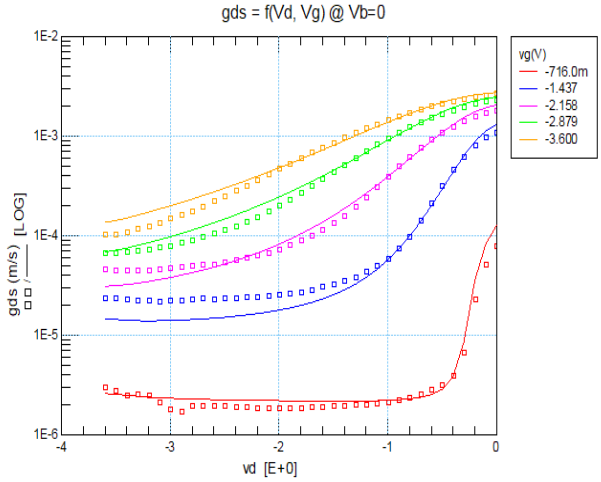




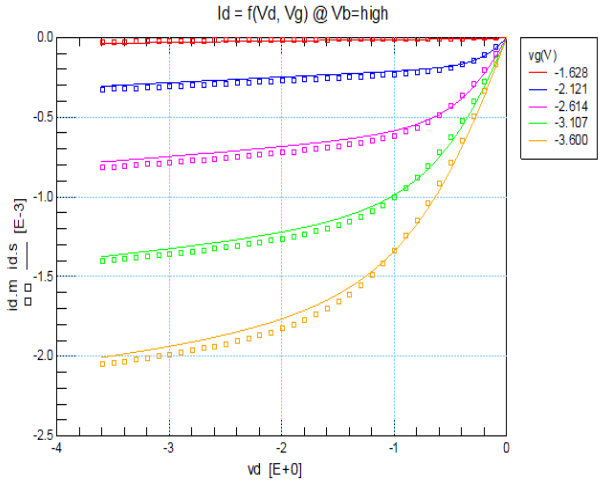
idvd, Vb = 0V, T = 27°C

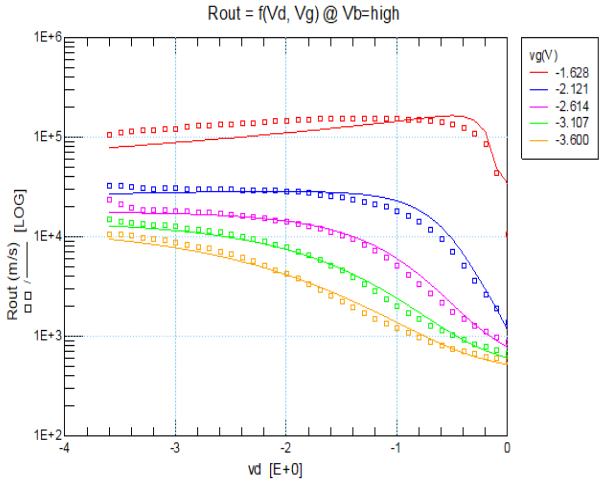


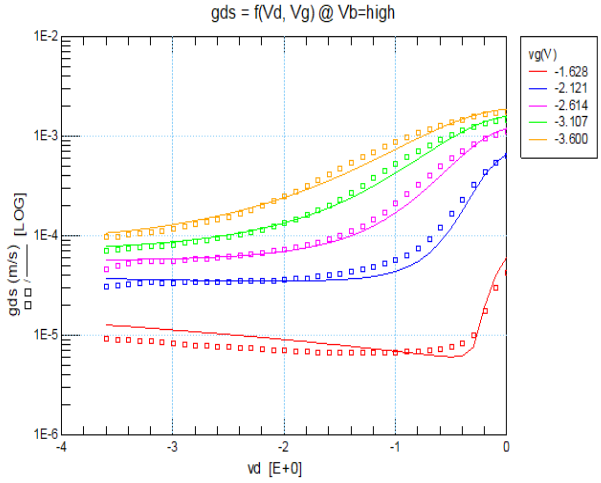




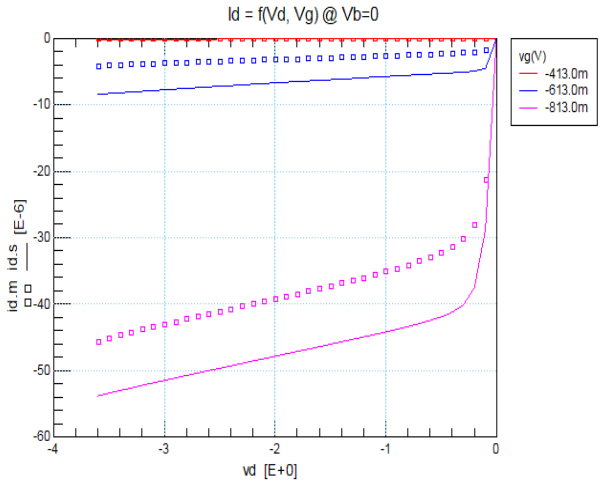
idvd_vbmin, Vb = 3.6V, T = 27°C

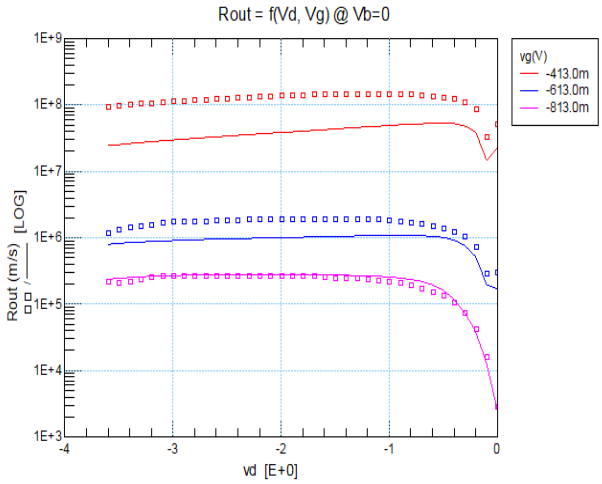


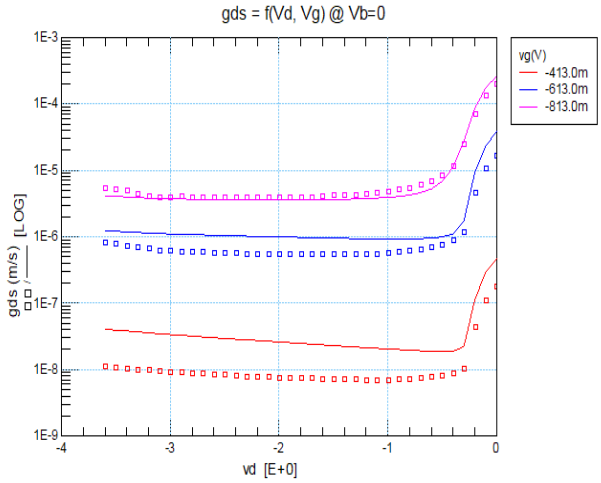




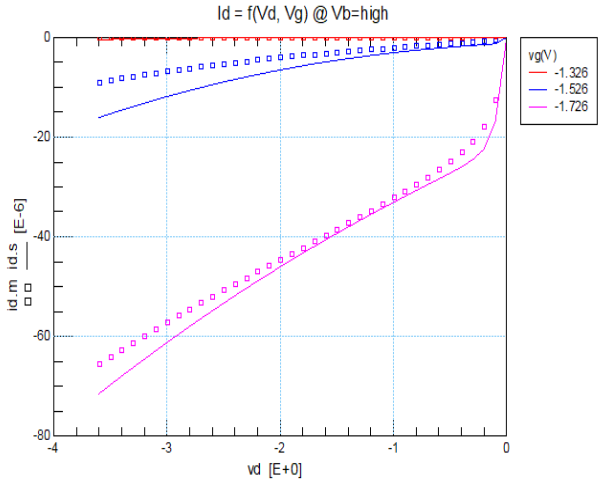
idvd_vth, Vb = 0V, T = 27°C

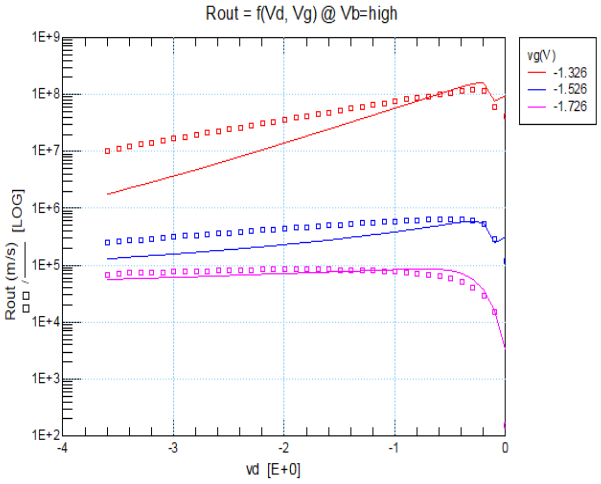


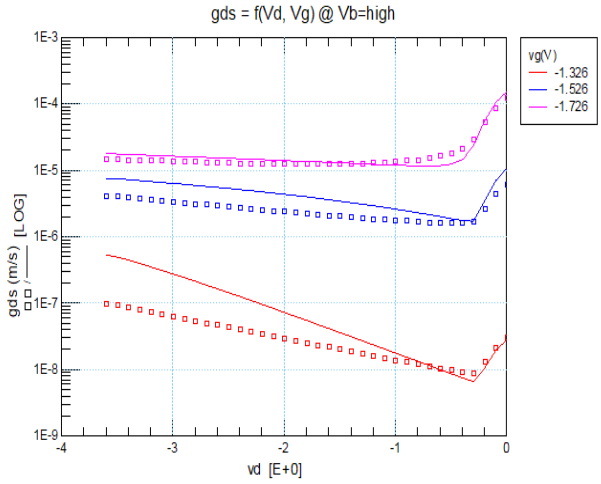




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

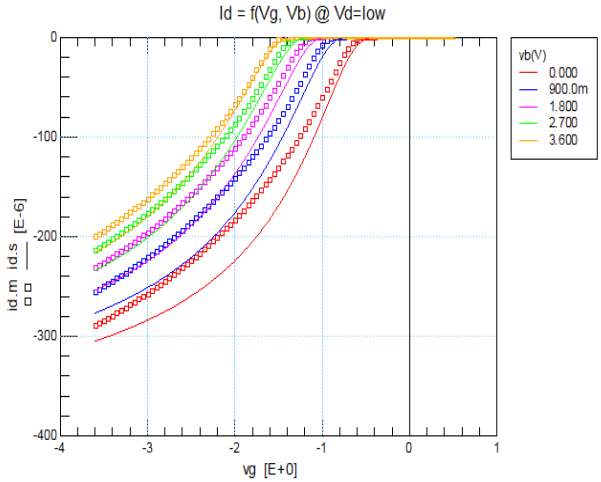


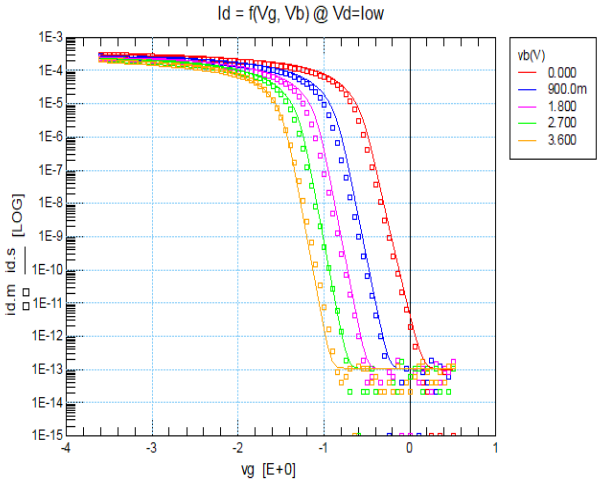


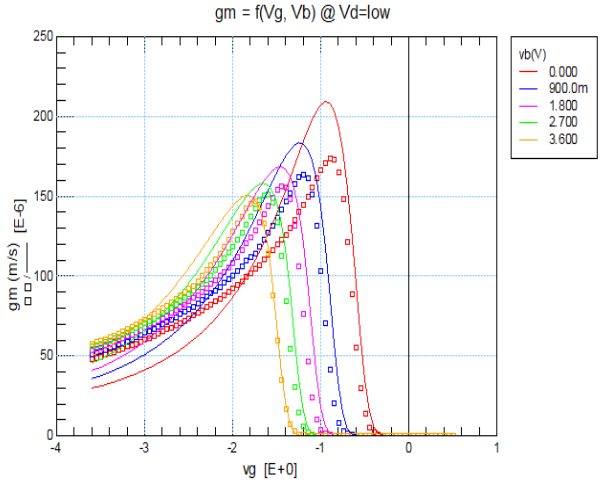


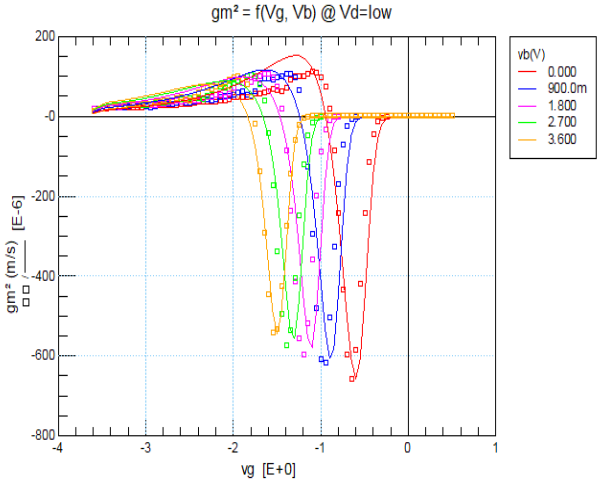
4.23W10u0_L0u3_S561_1

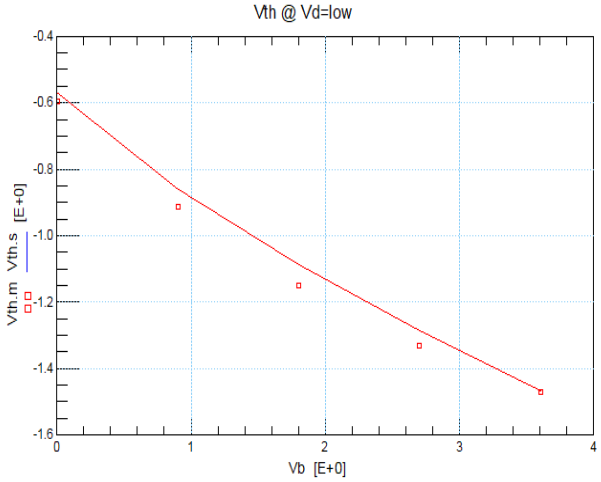
idvg, Vd = -0.1V, T = 27°C

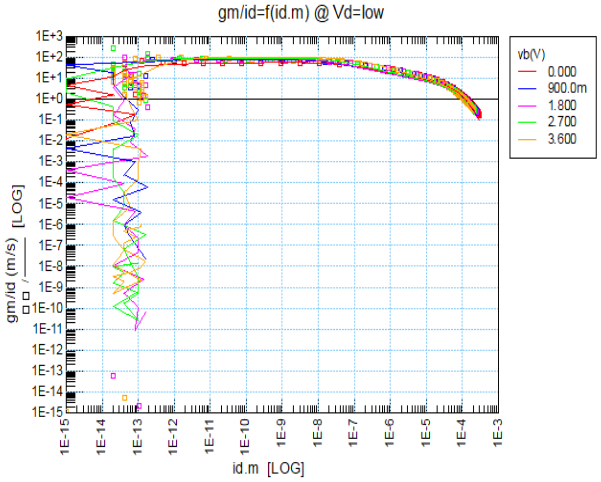


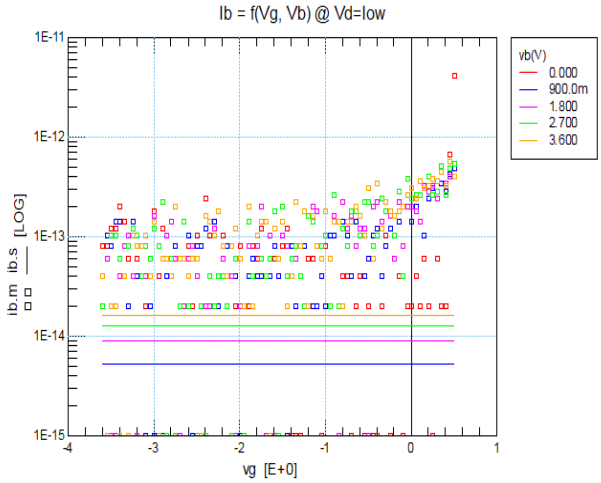


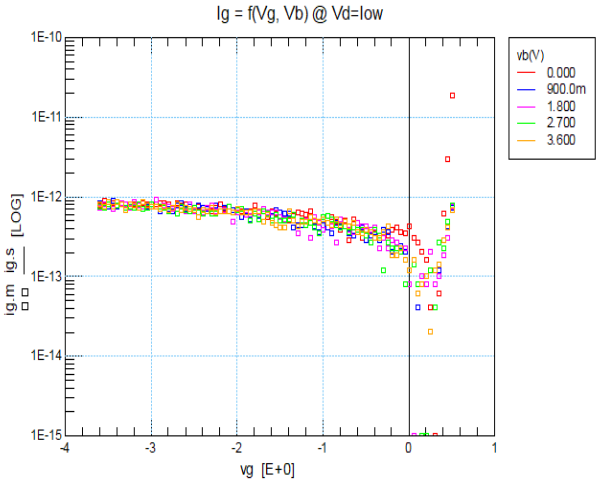




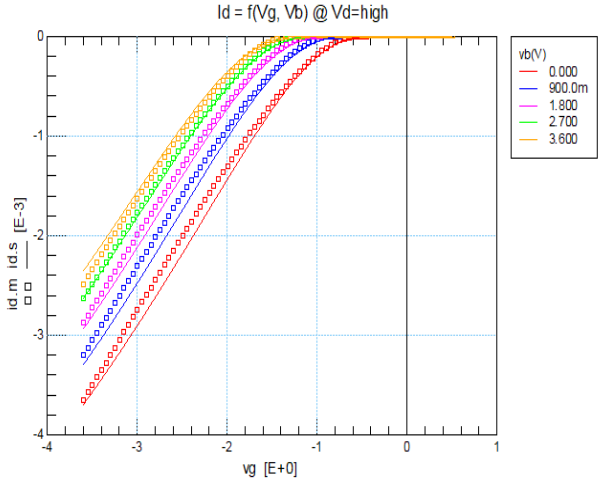


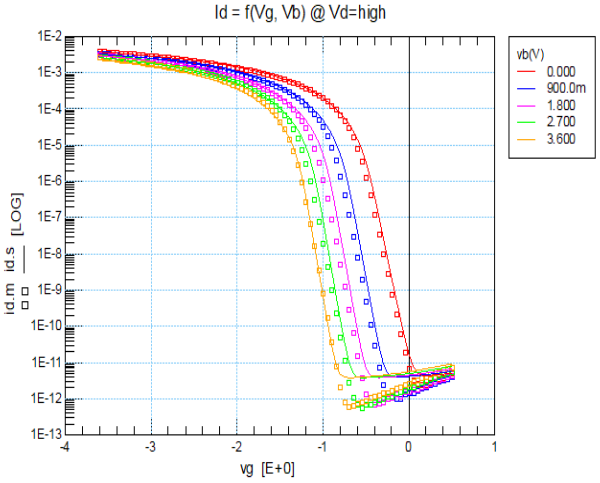


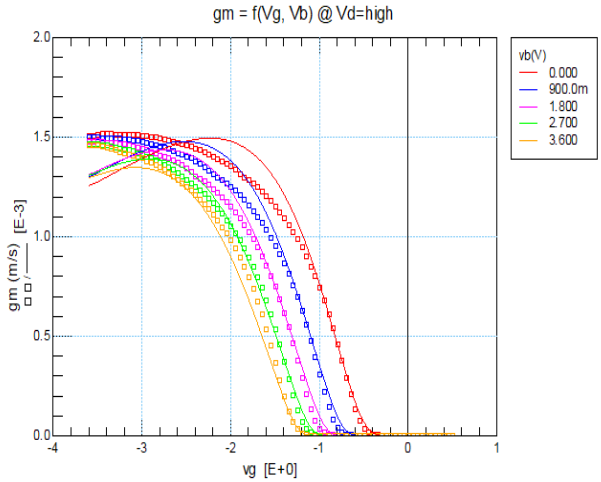


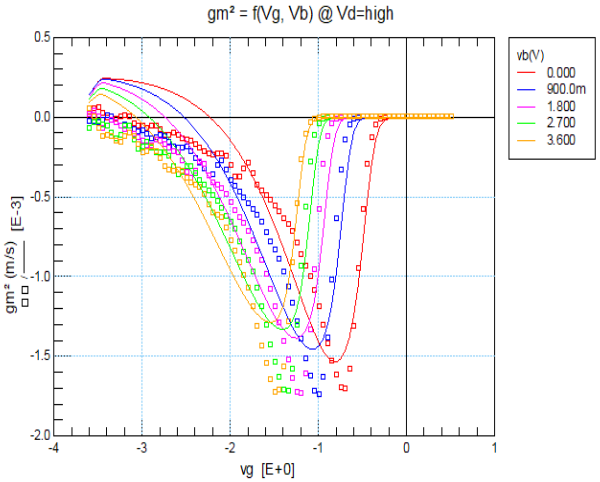


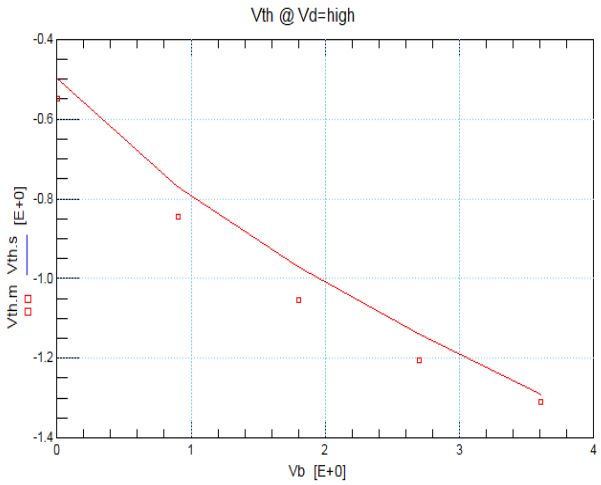
idvg, Vd = -3.3V, T = 27°C

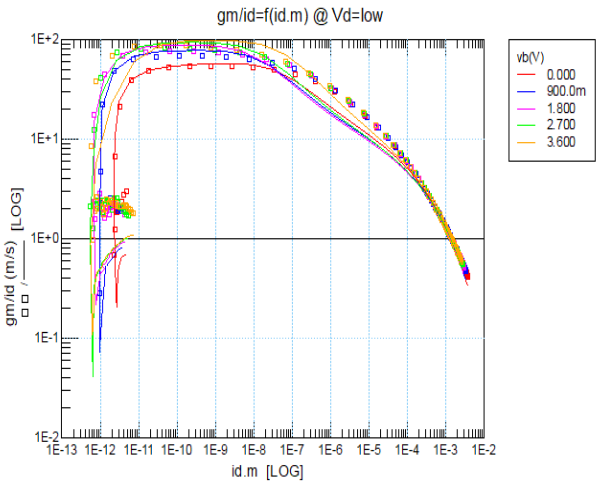


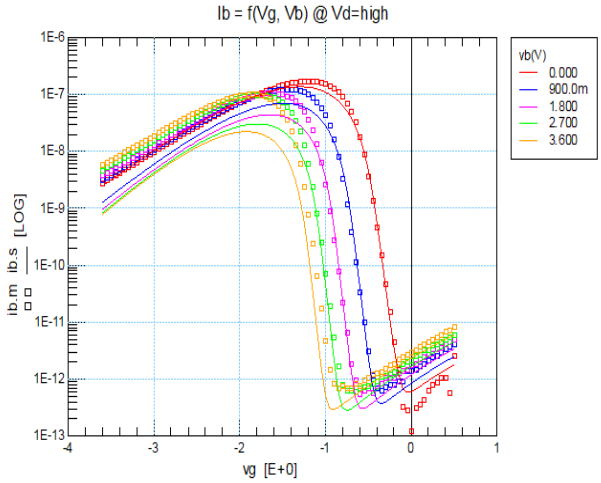


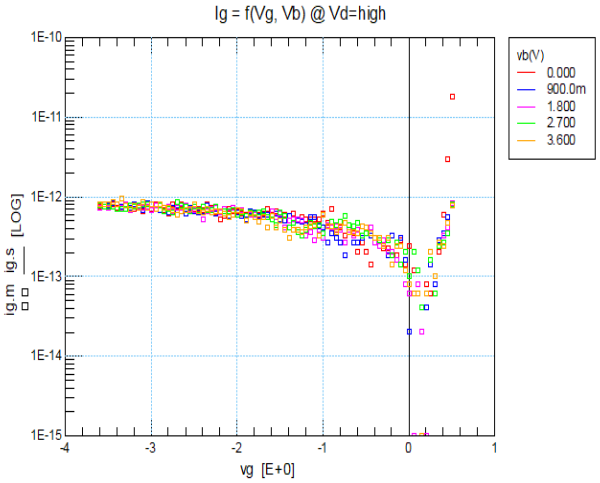




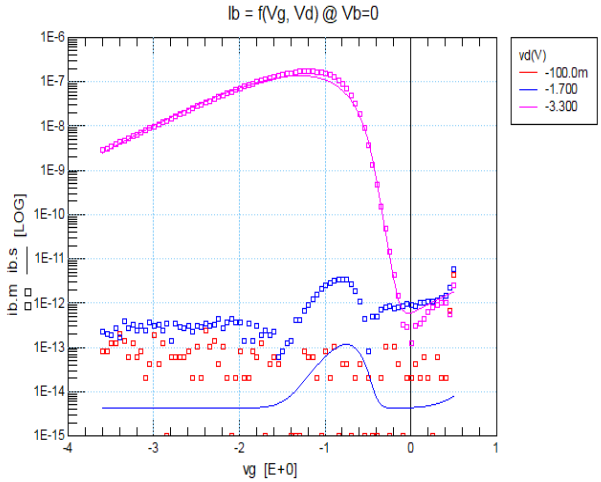


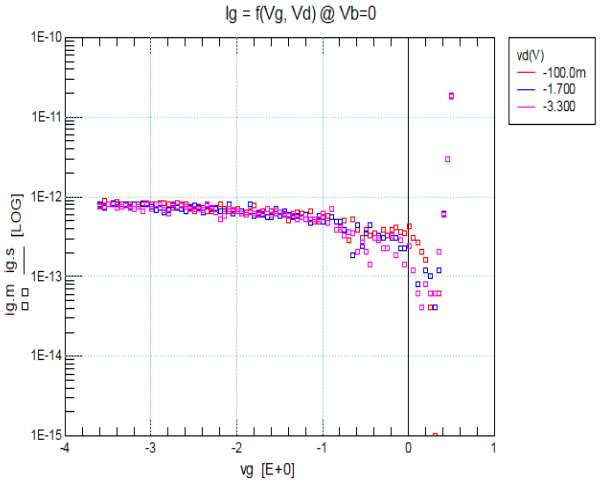




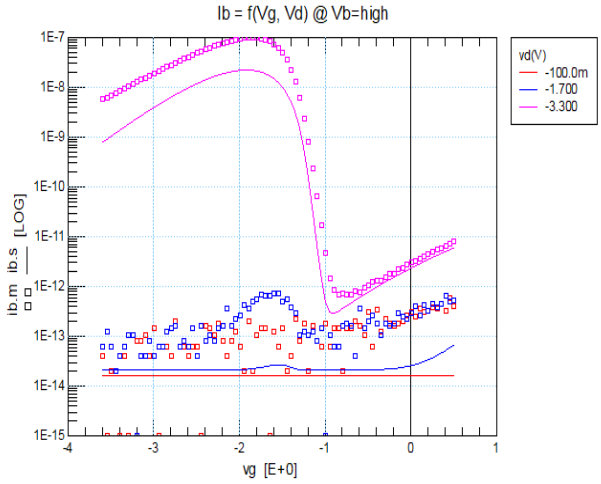


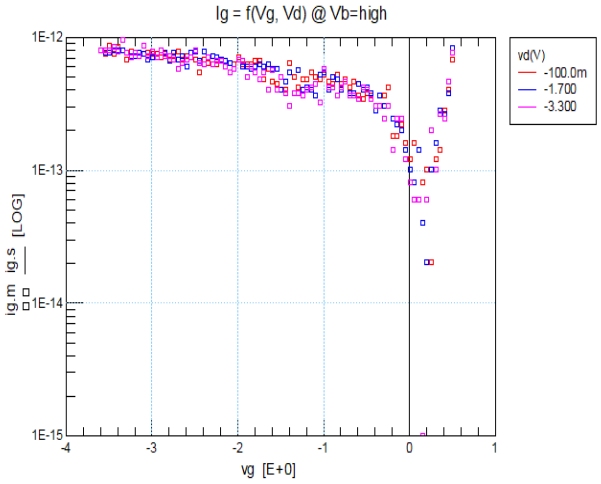
idvg, Ib,Ig, Vb = 0V, T = 27°C



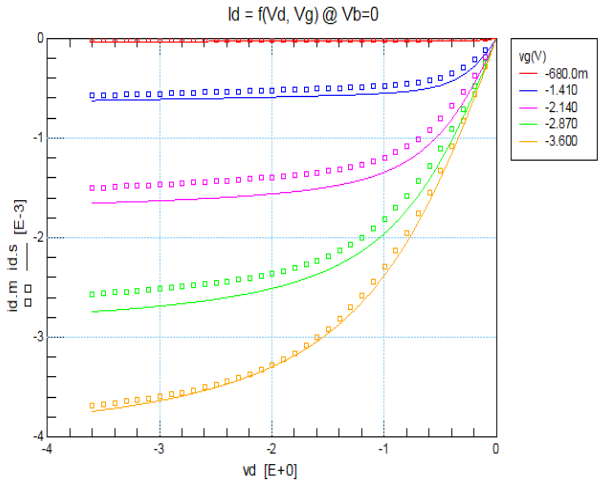


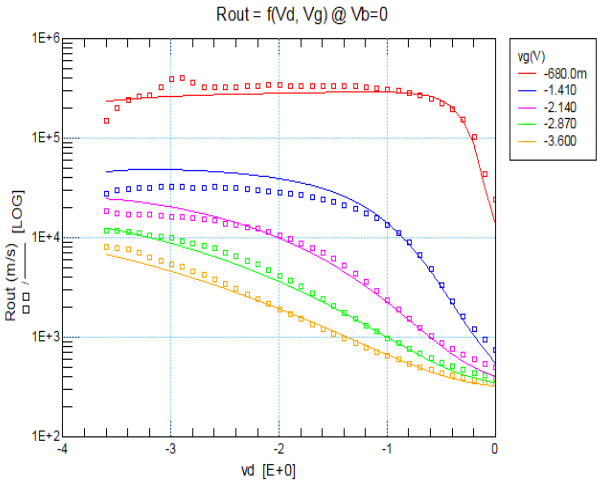
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

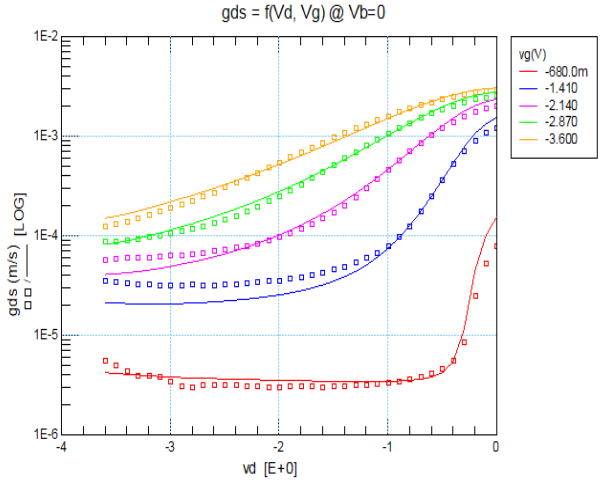




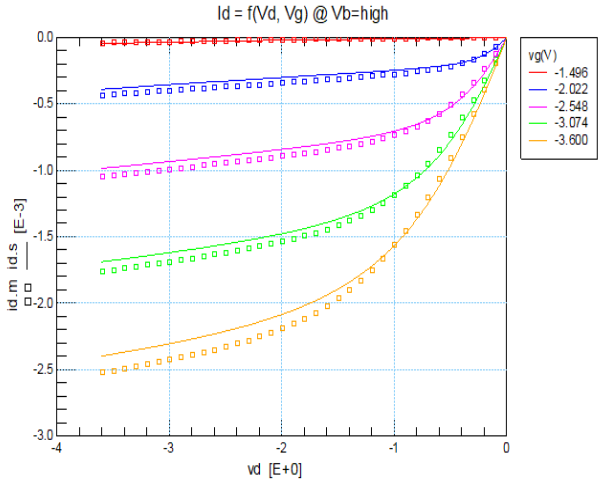
idvd, Vb = 0V, T = 27°C

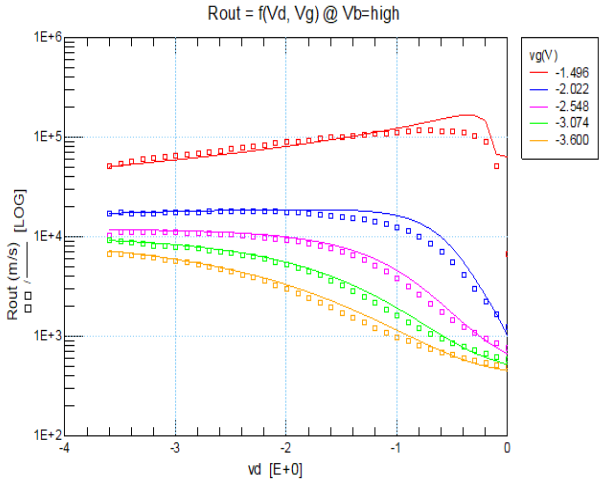


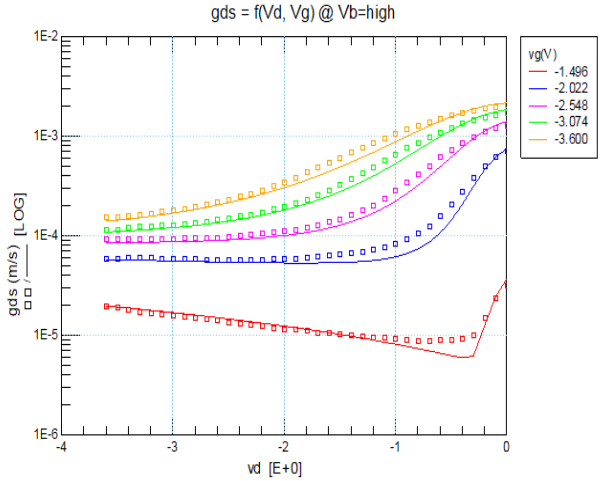




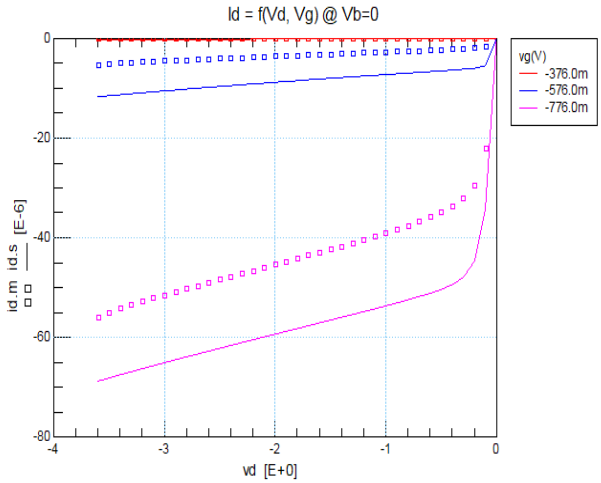
idvd_vbmin, Vb = 3.6V, T = 27°C

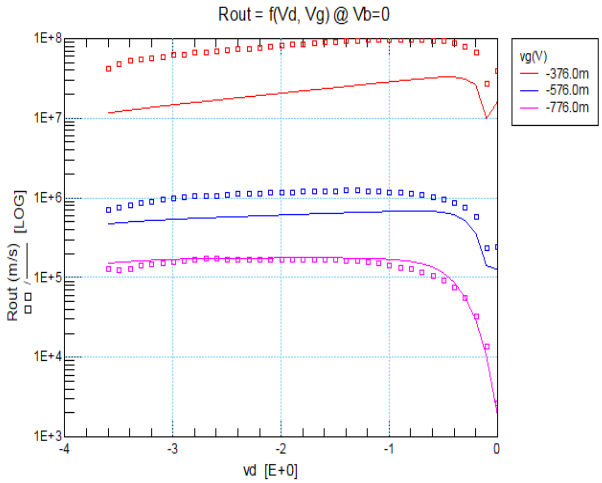


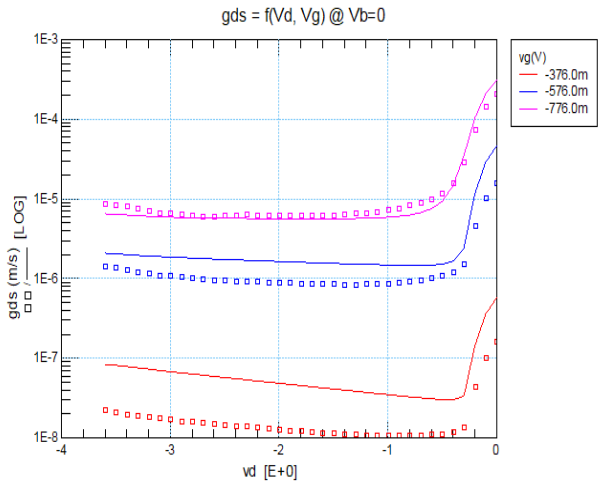




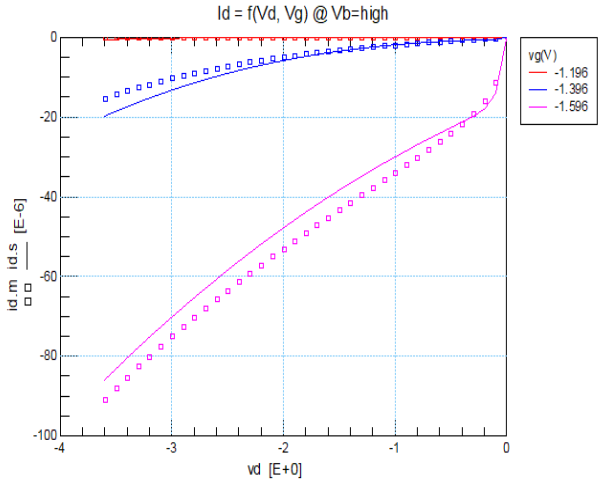
idvd_vth, Vb = 0V, T = 27°C

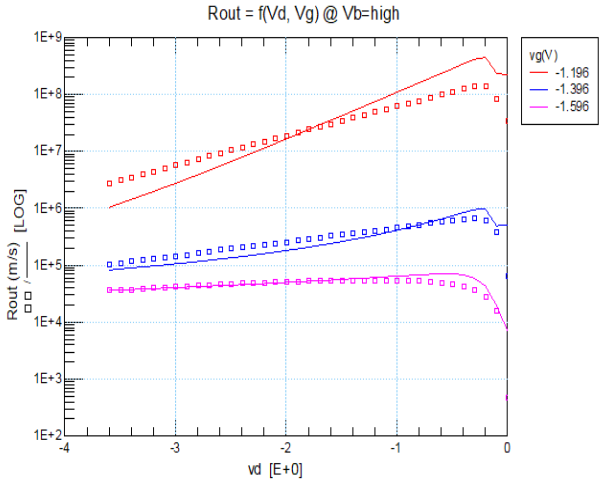


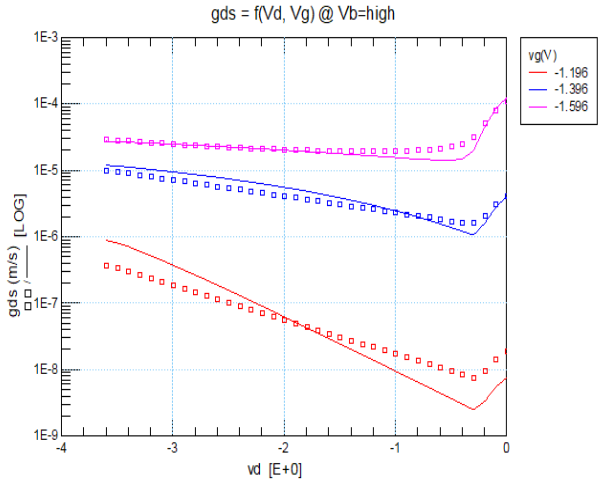




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

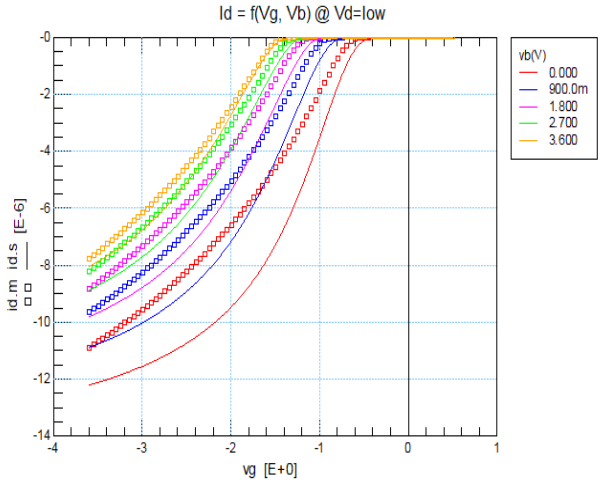


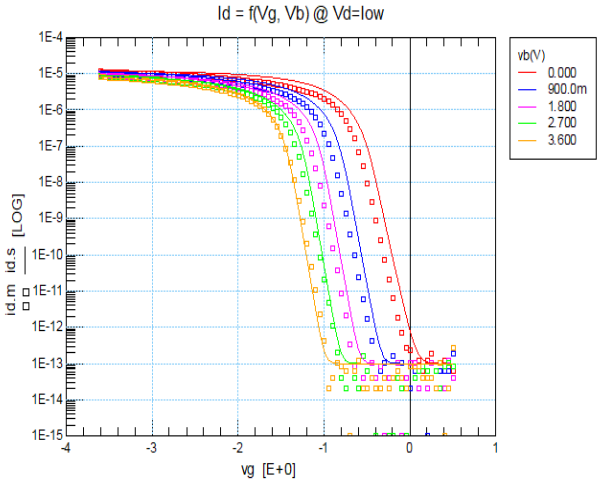


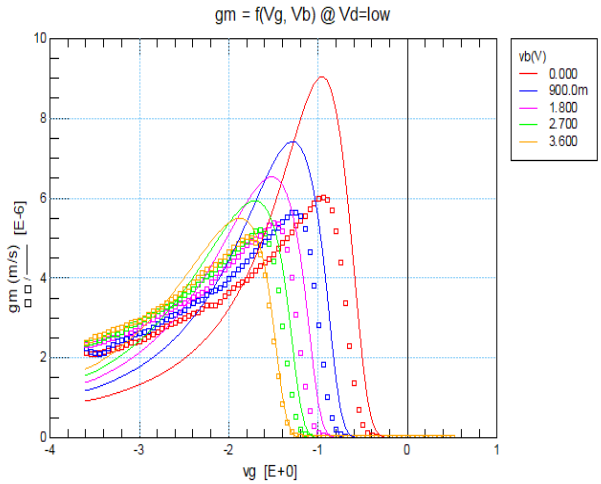


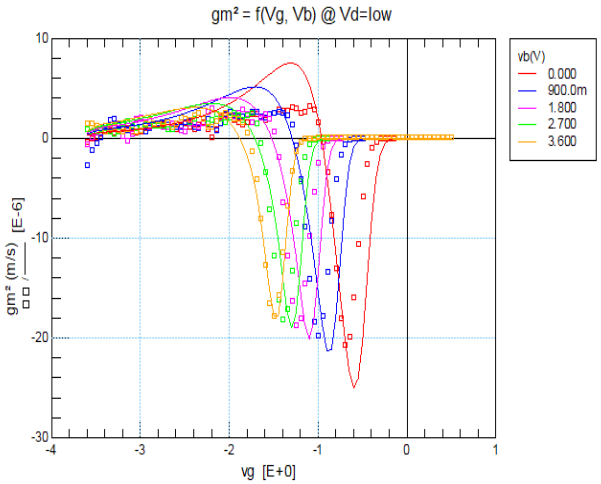
4.24 W0u3_L0u3_S565_1

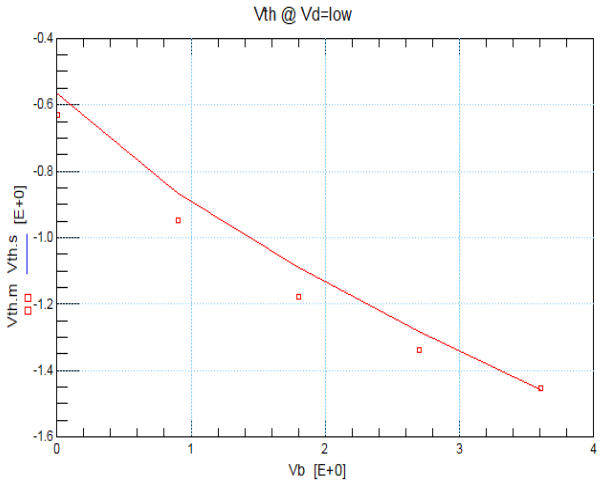
idvg, Vd = -0.1V, T = 27°C

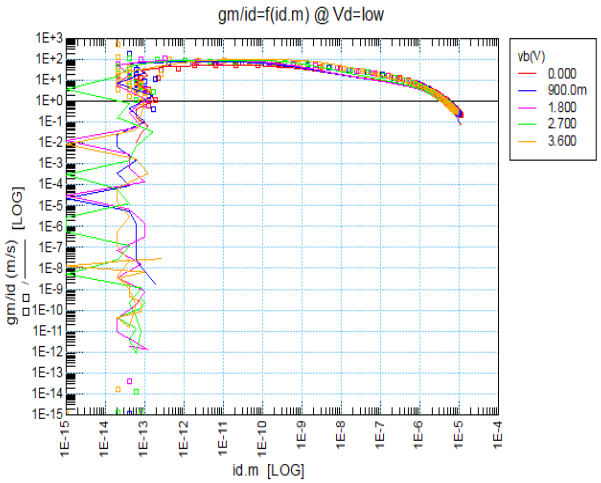


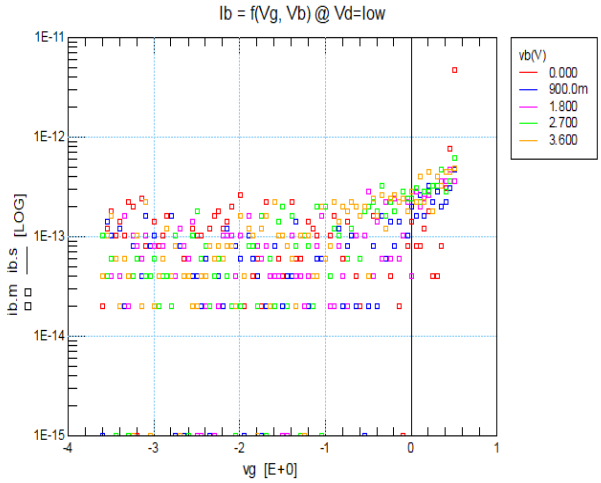


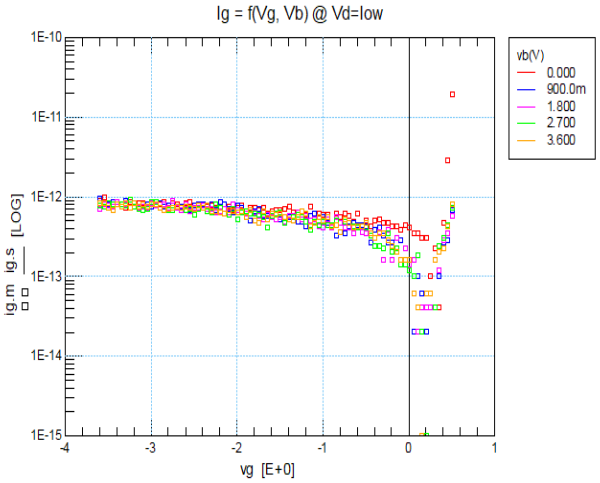




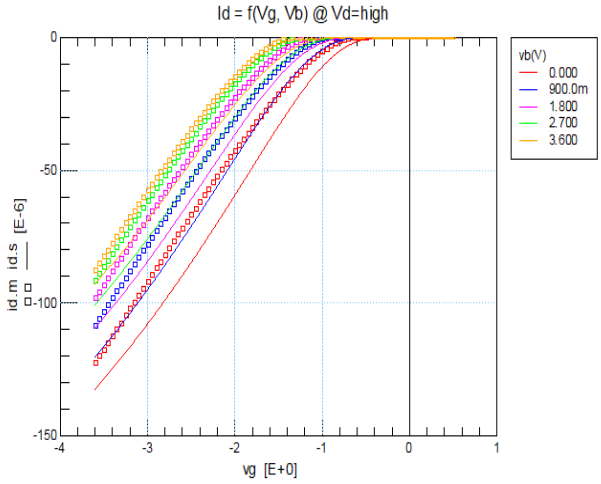


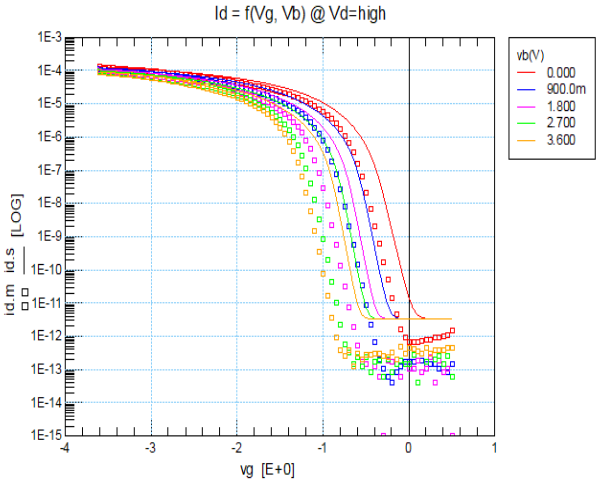


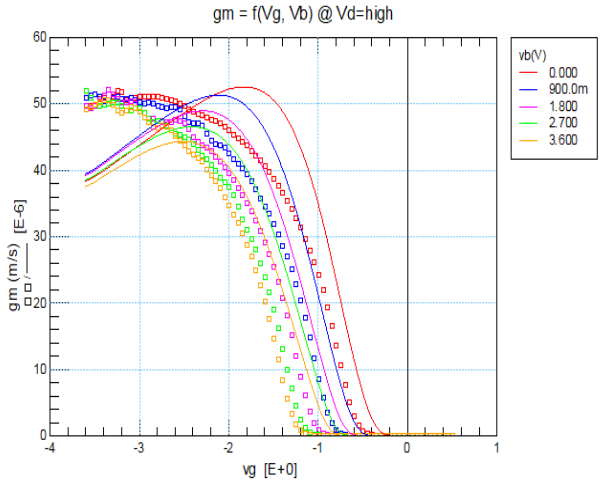


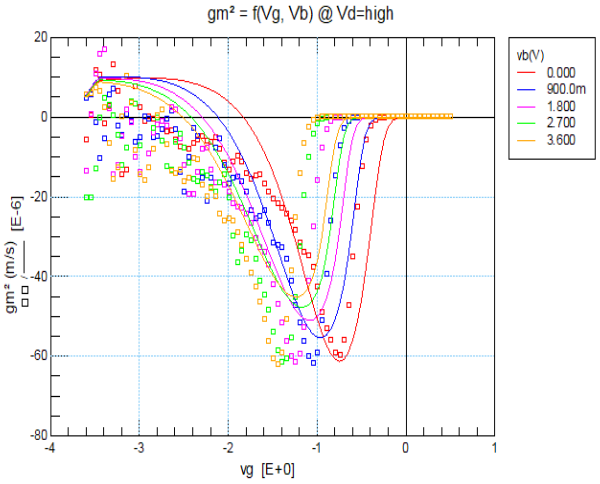


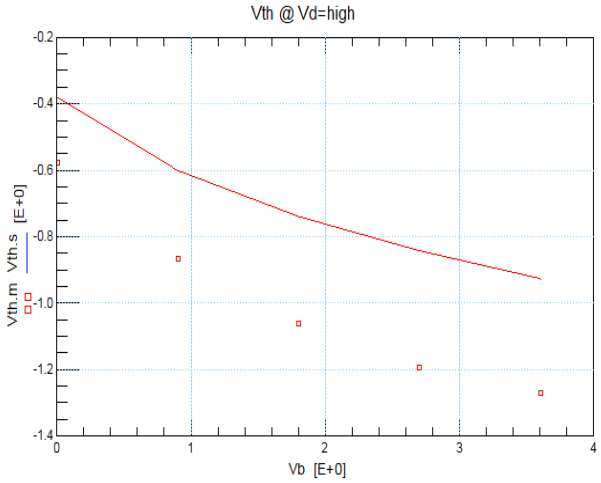
idvg, Vd = -3.3V, T = 27°C

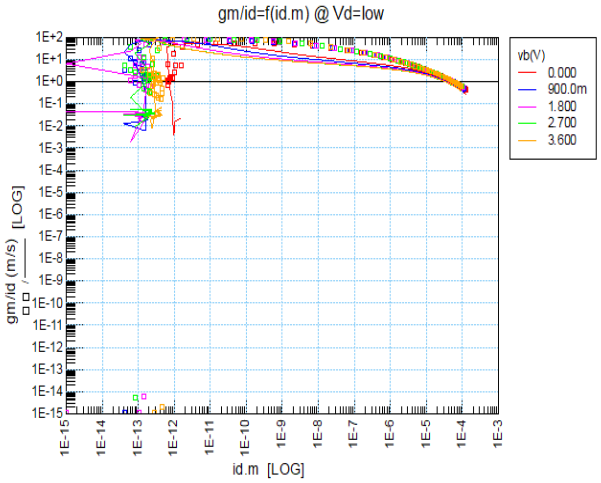


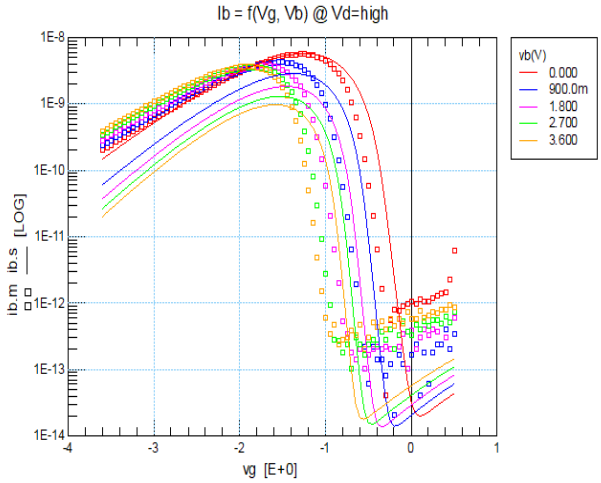


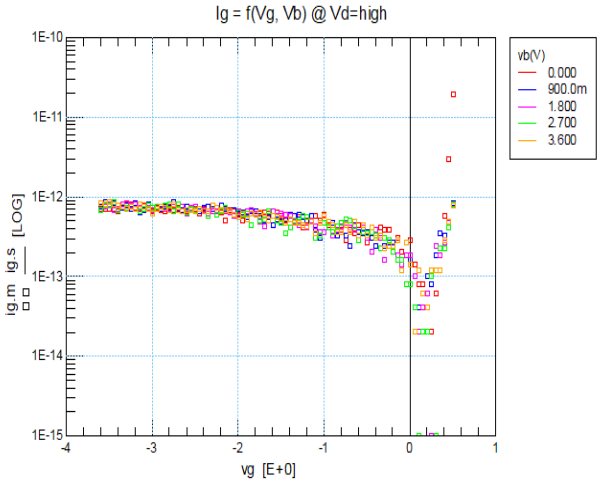




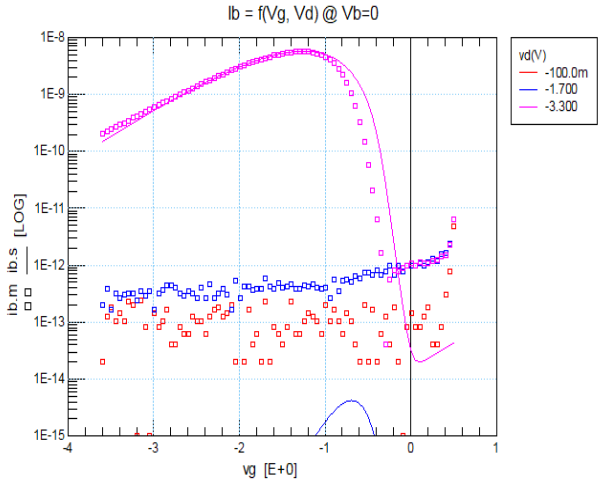


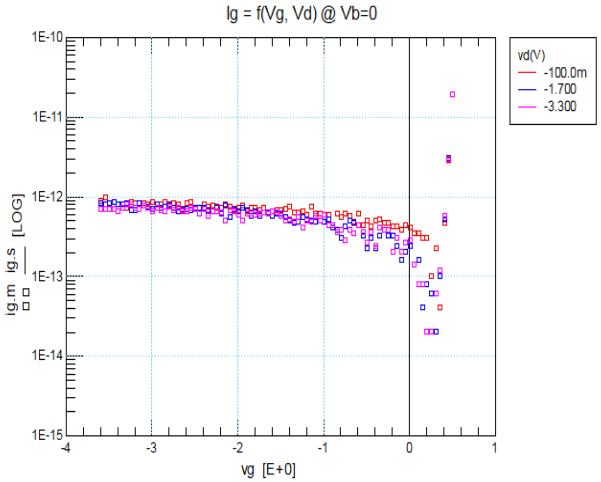




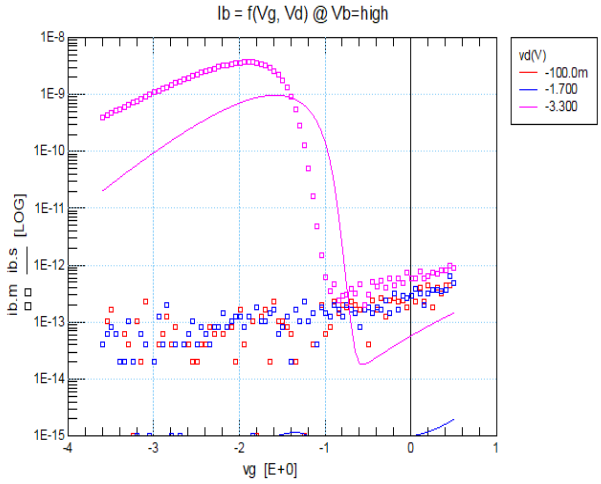


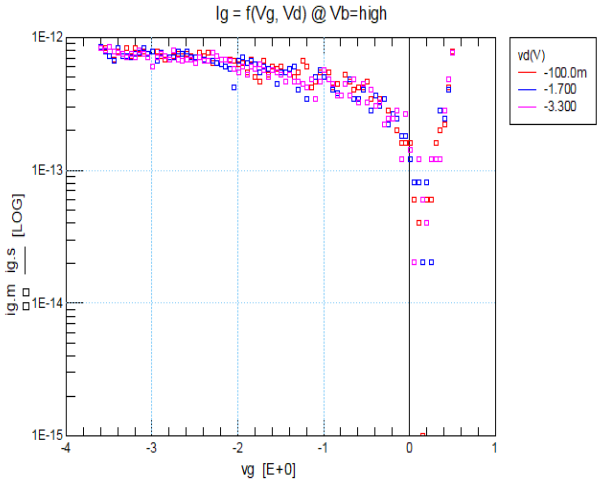
idvg, Ib,Ig, Vb = 0V, T = 27°C



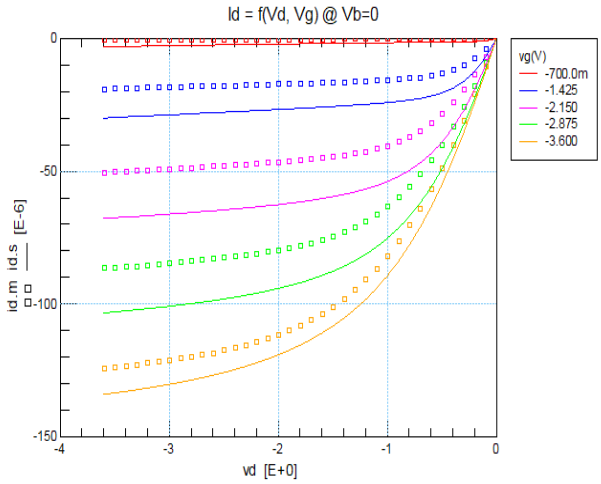


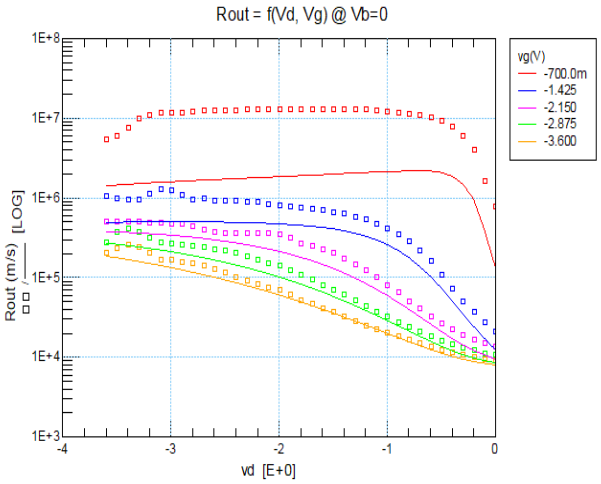
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

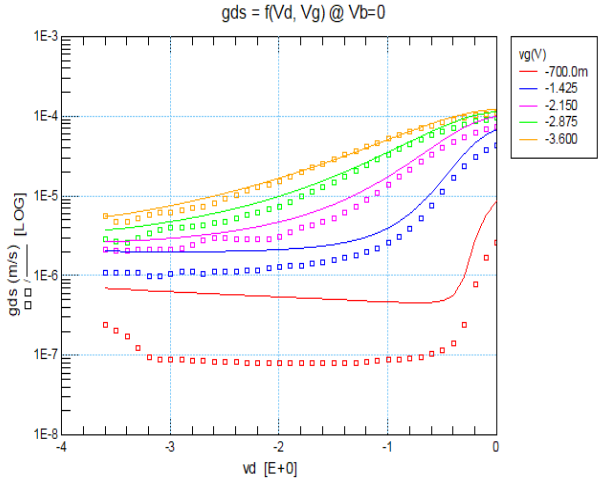




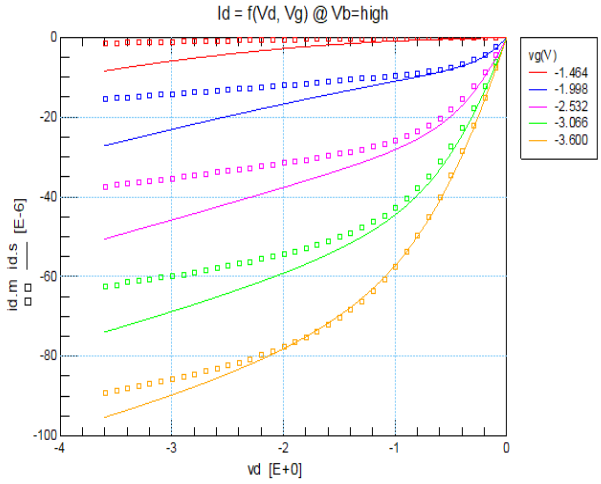
idvd, Vb = 0V, T = 27°C

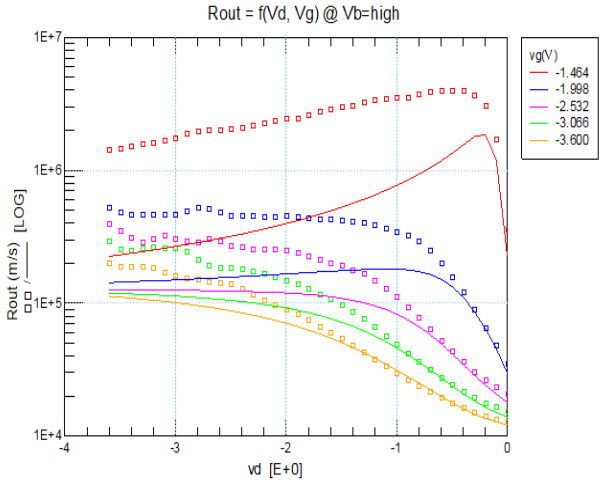


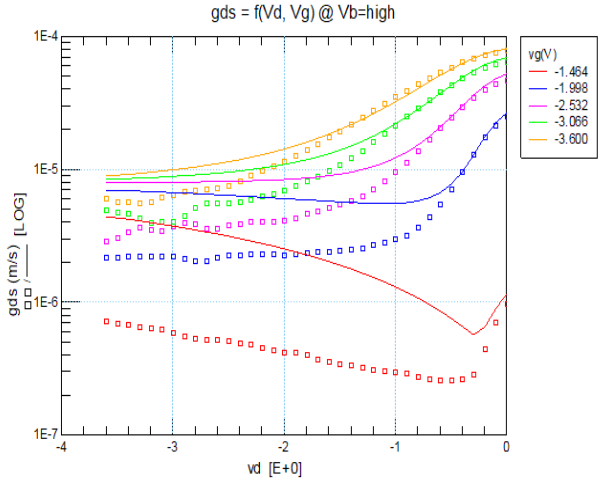




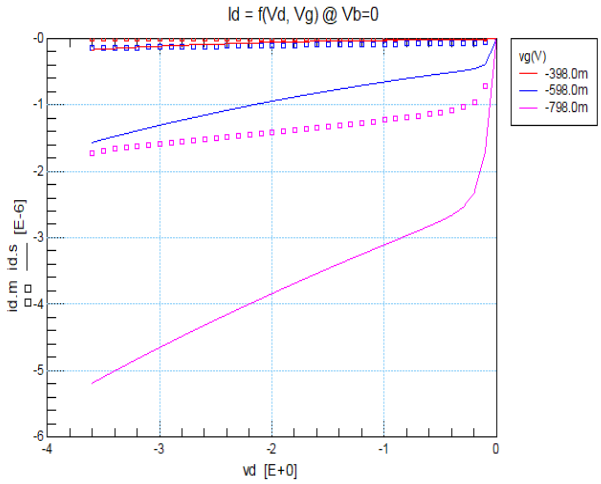
idvd_vbmin, Vb = 3.6V, T = 27°C

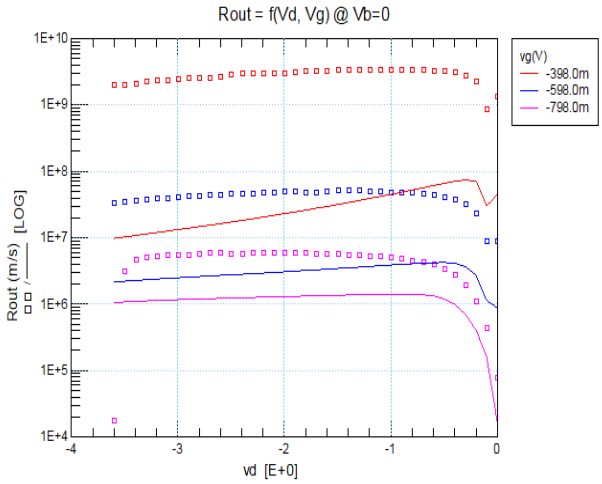


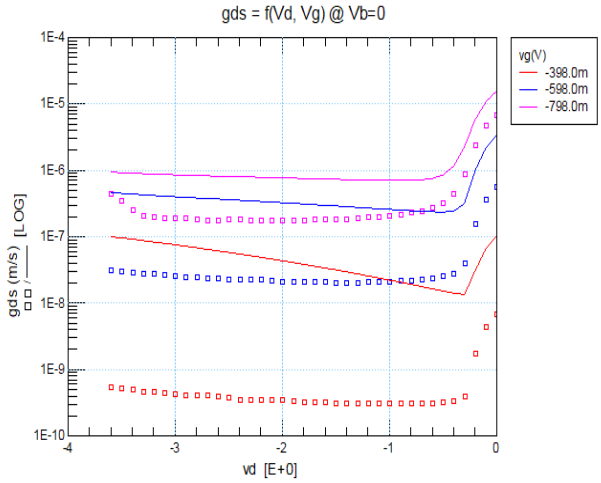




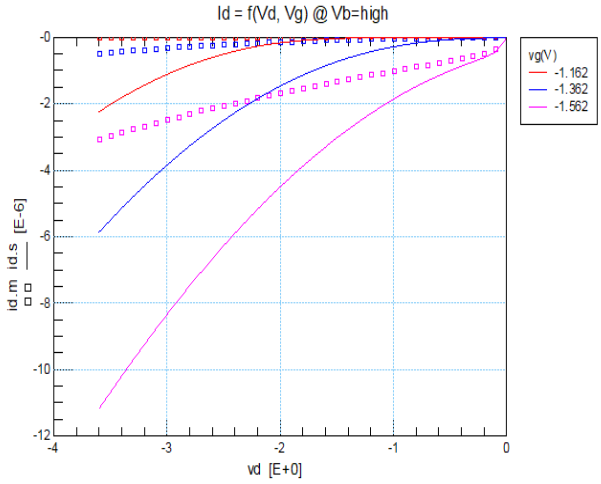
idvd_vth, Vb = 0V, T = 27°C

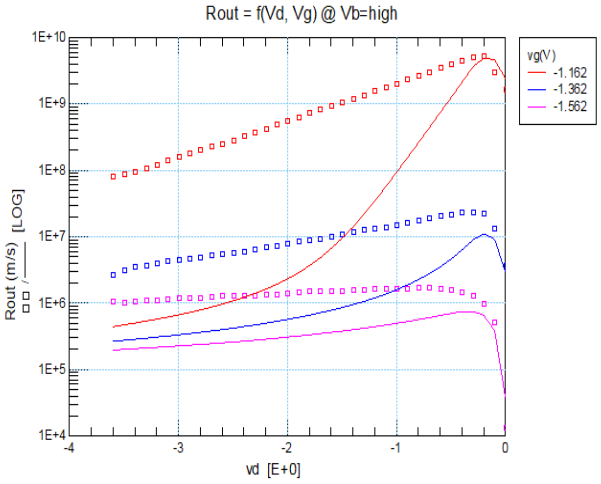


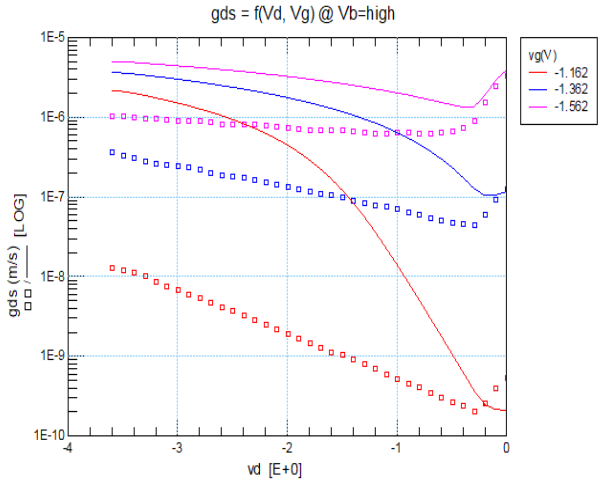




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

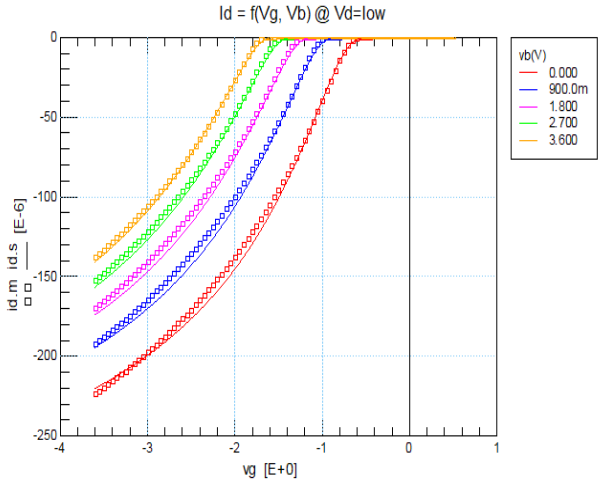


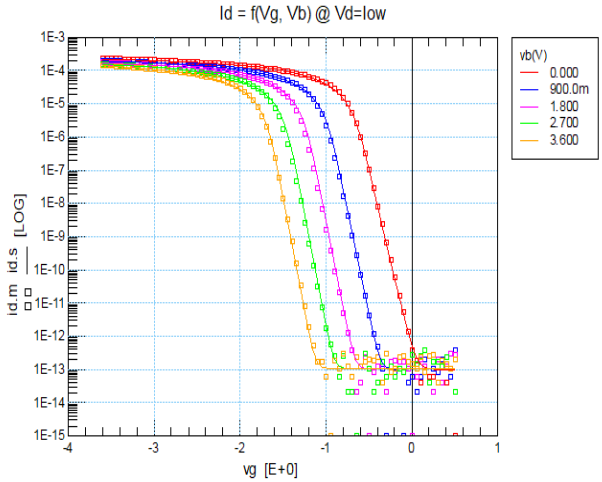


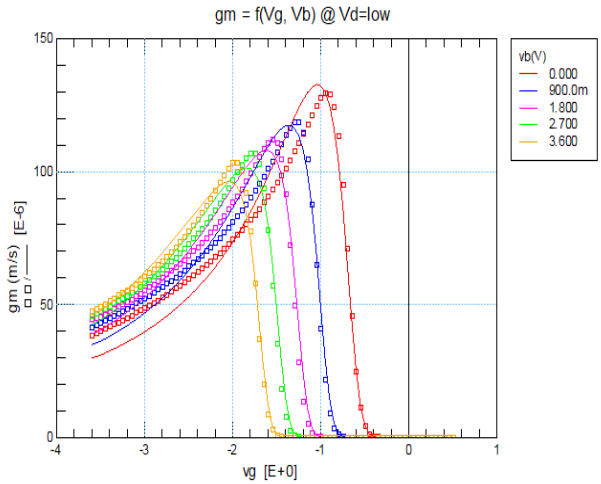


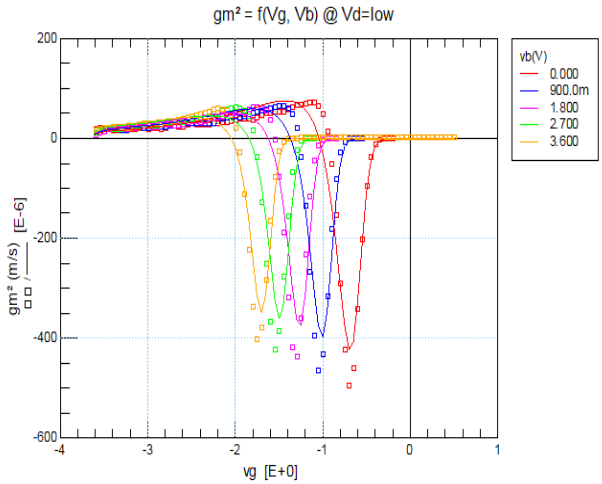
4.25W10u0_L0u4_S387_1

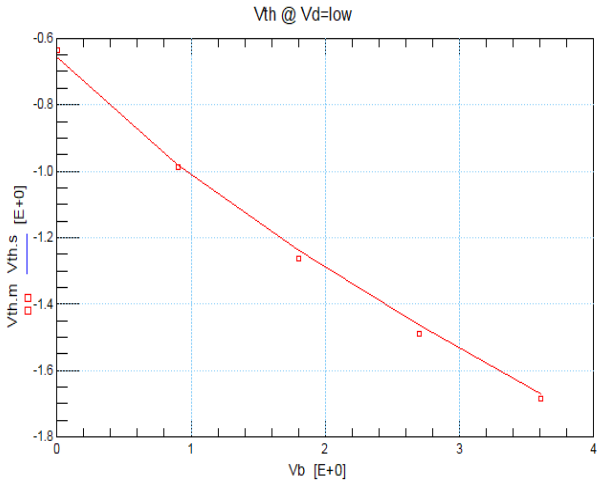
idvg, Vd = -0.1V, T = 27°C

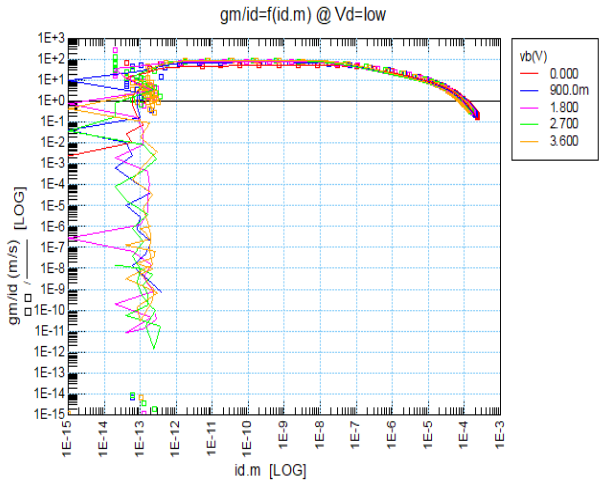


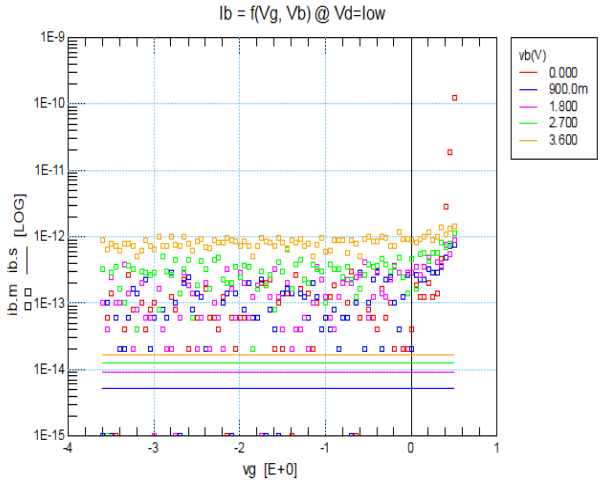


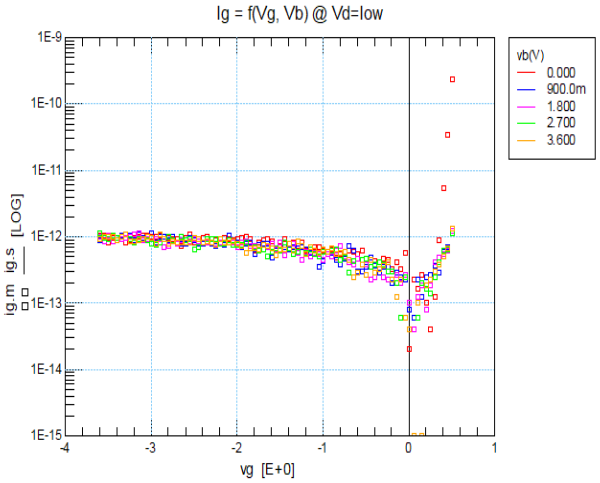




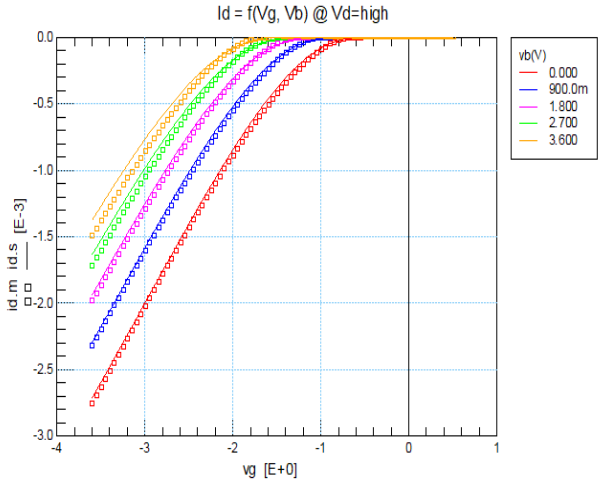


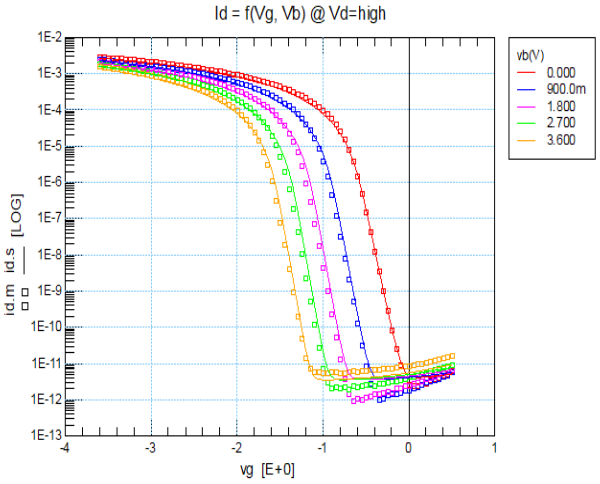


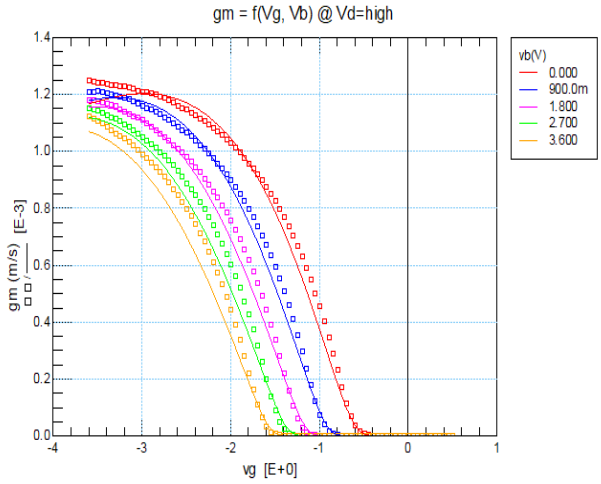


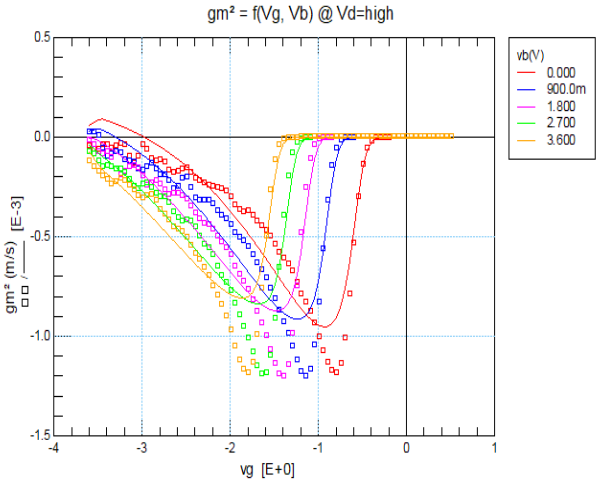


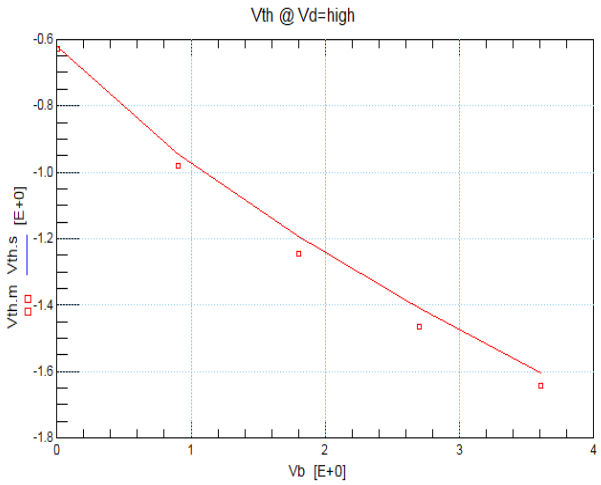
idvg, Vd = -3.3V, T = 27°C

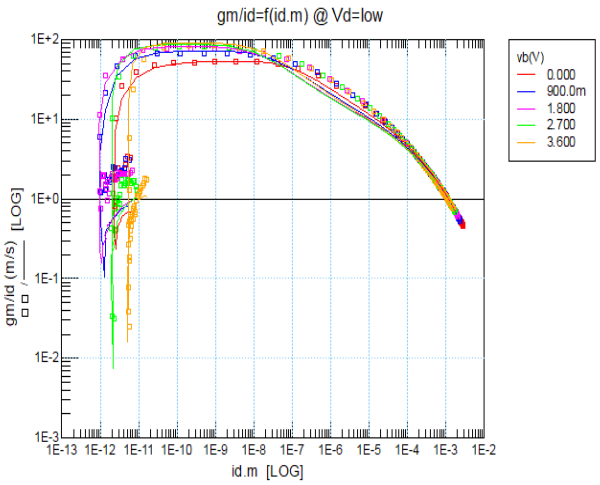


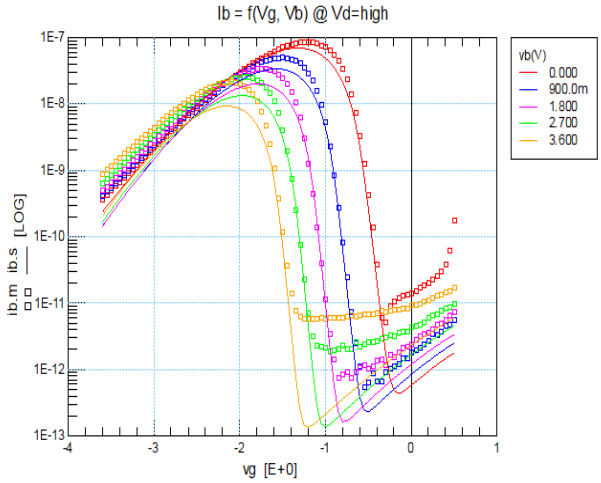


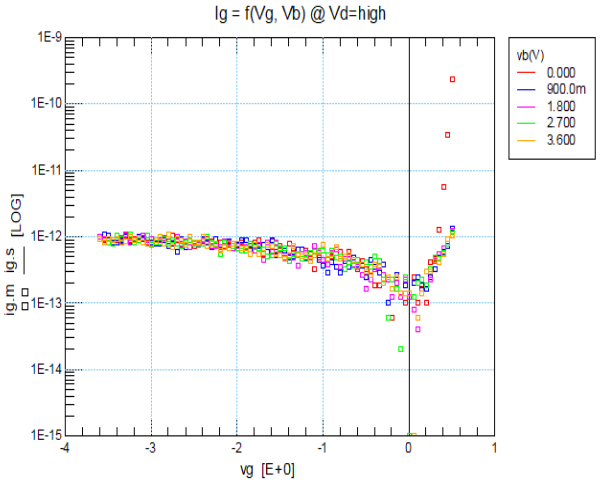




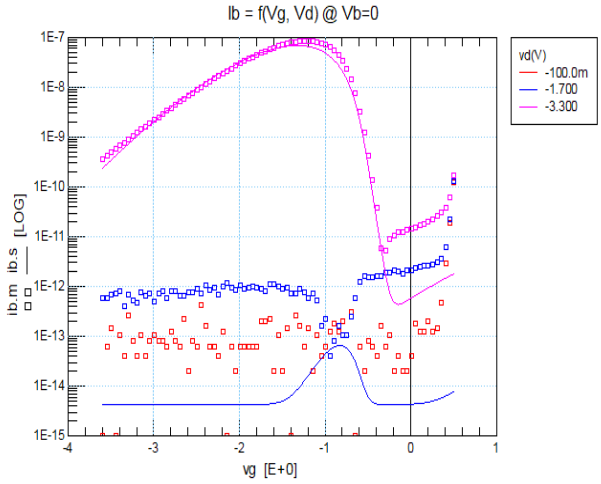


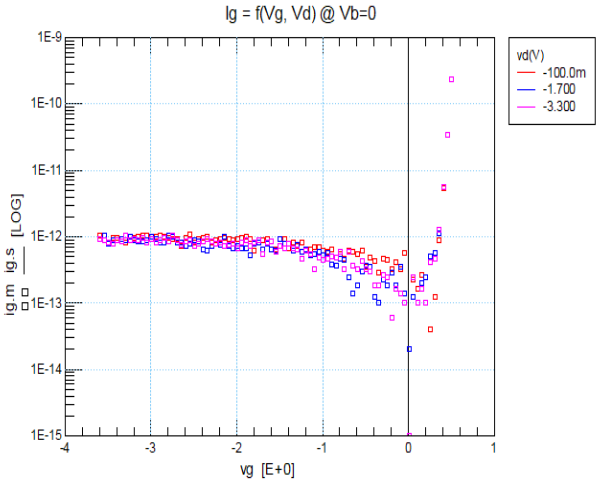




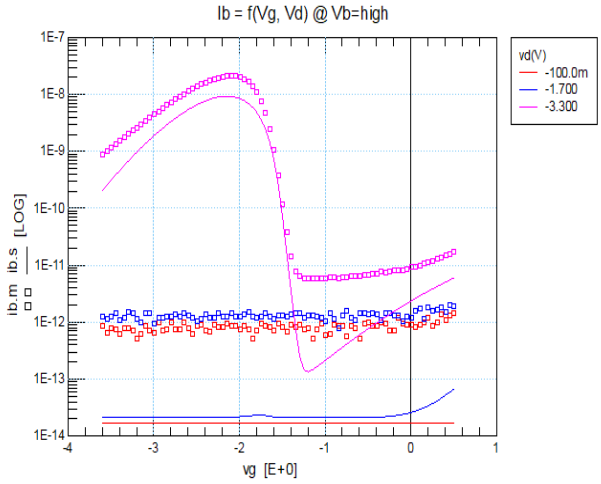


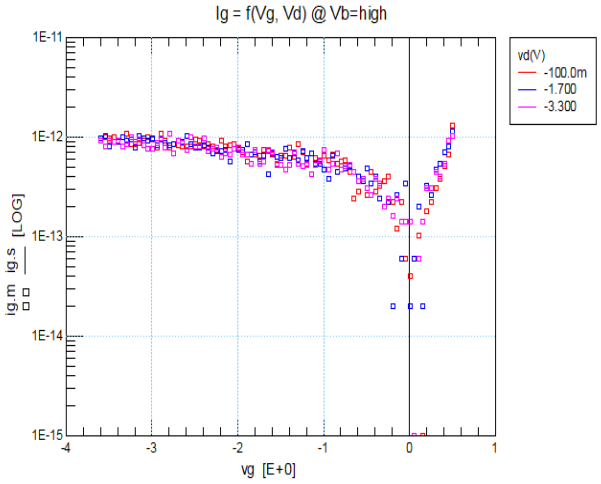
idvg, Ib,Ig, Vb = 0V, T = 27°C



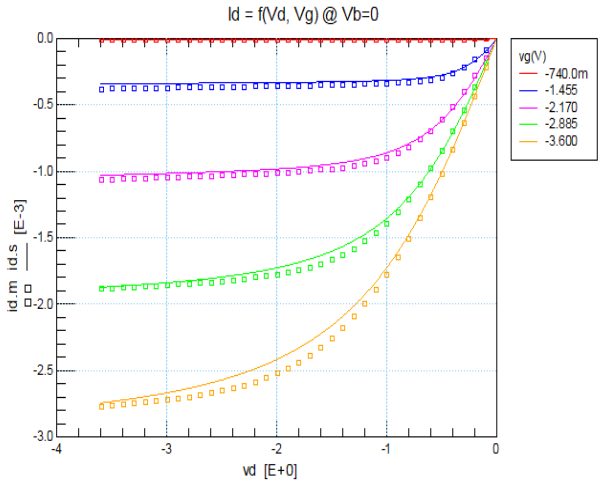


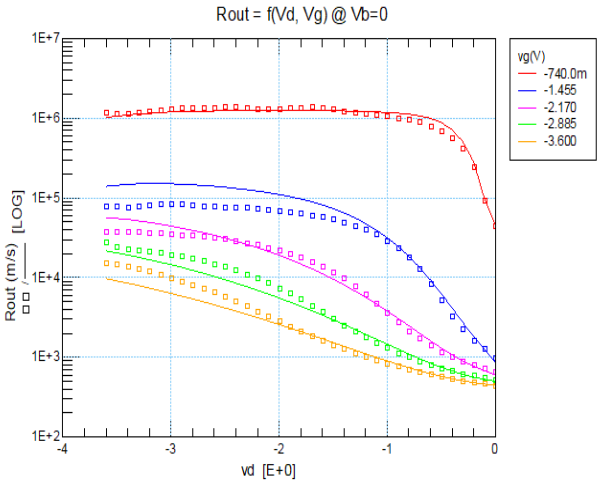
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

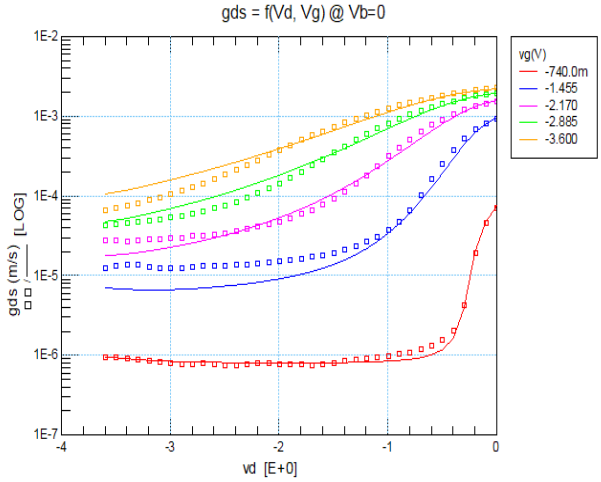




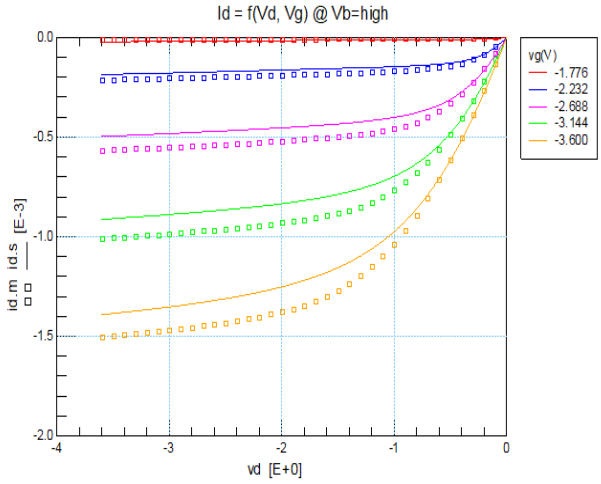
idvd, Vb = 0V, T = 27°C

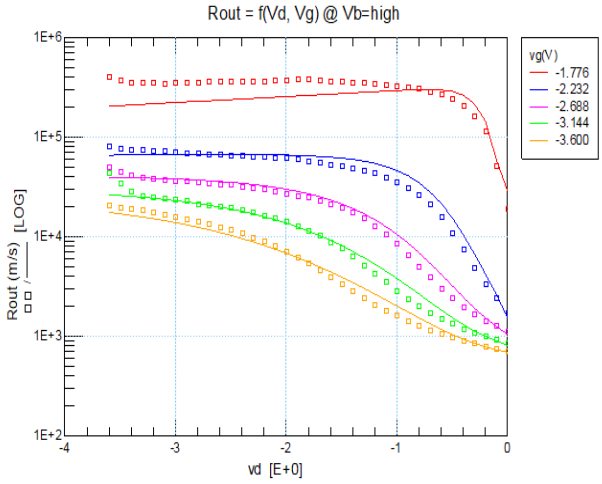


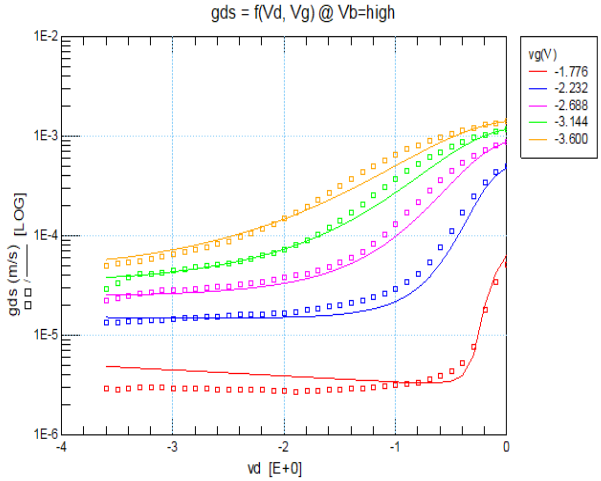




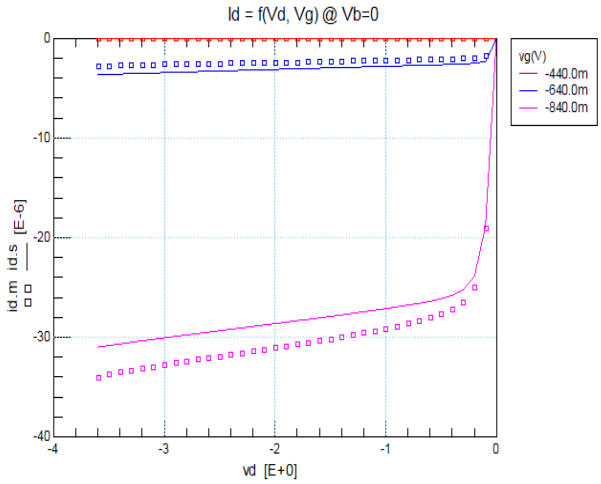
idvd_vbmin, Vb = 3.6V, T = 27°C

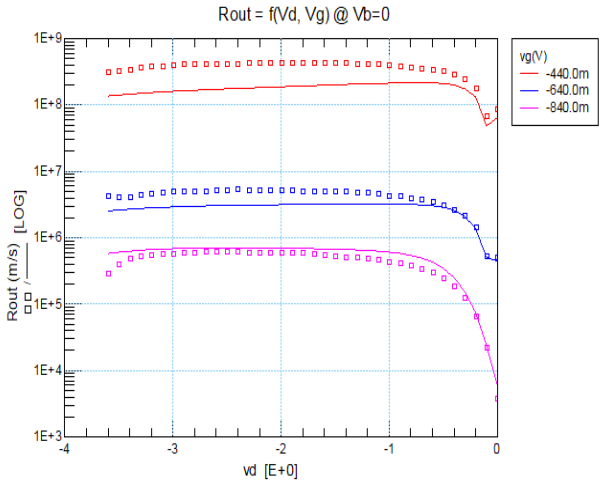


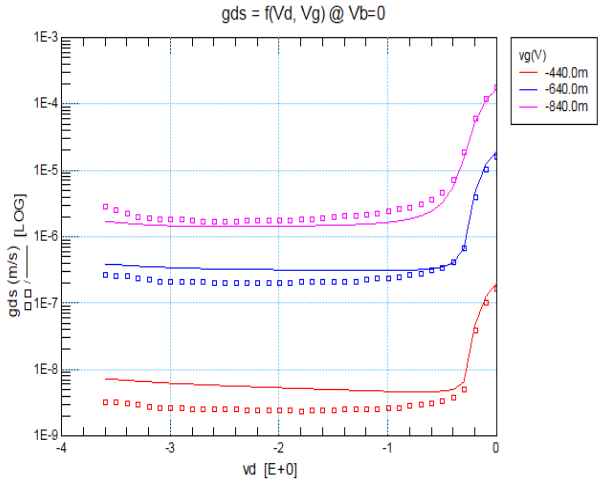




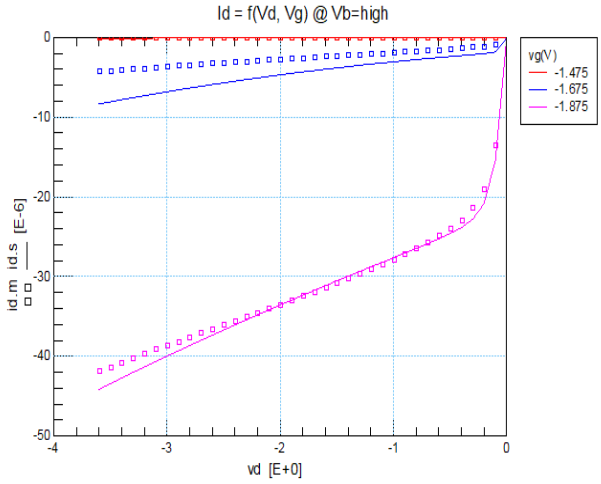
idvd_vth, Vb = 0V, T = 27°C

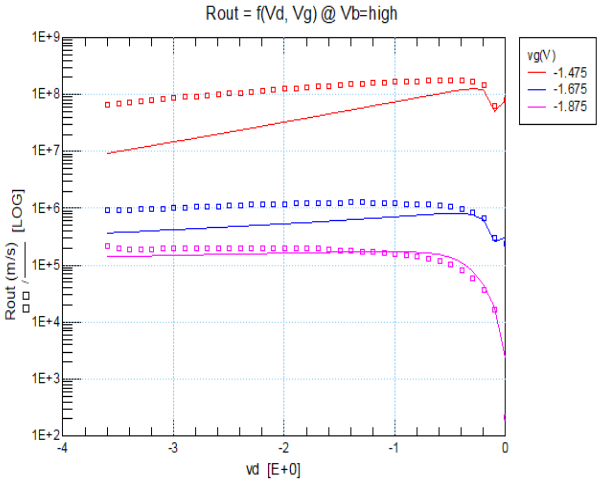


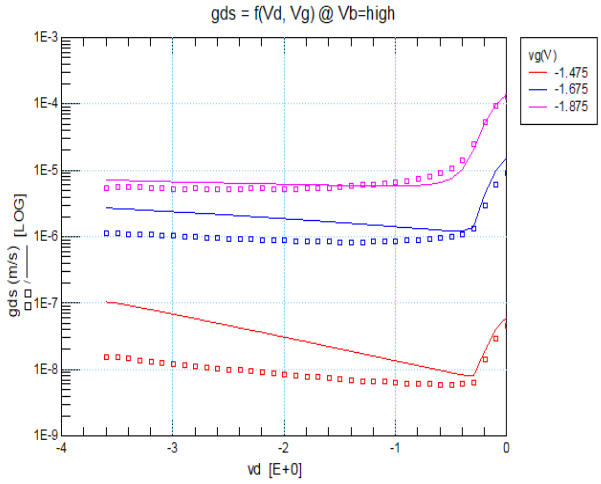




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

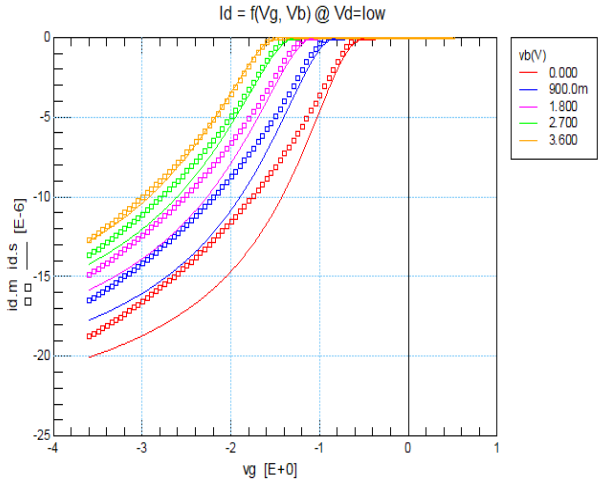


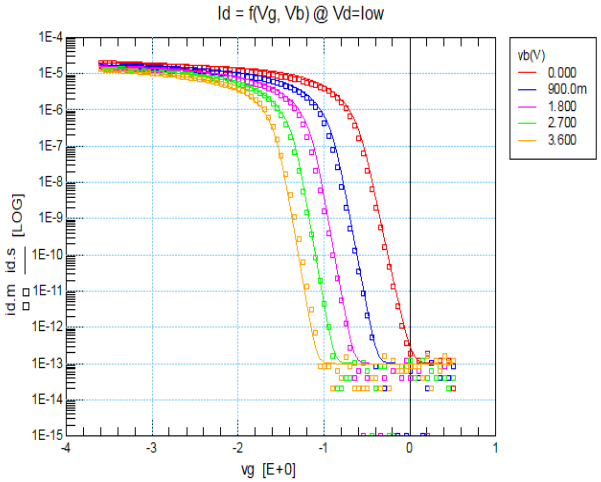


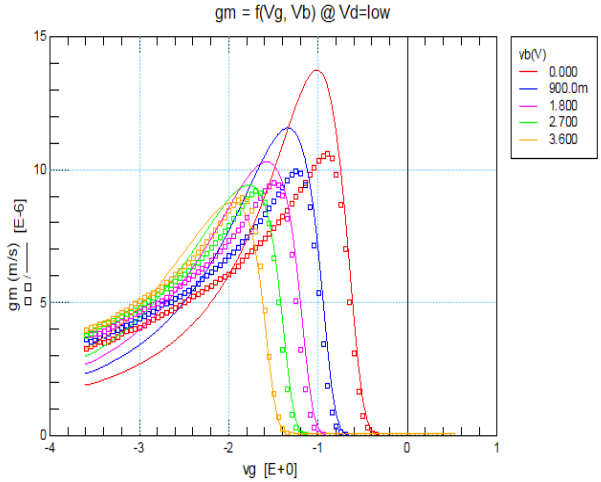


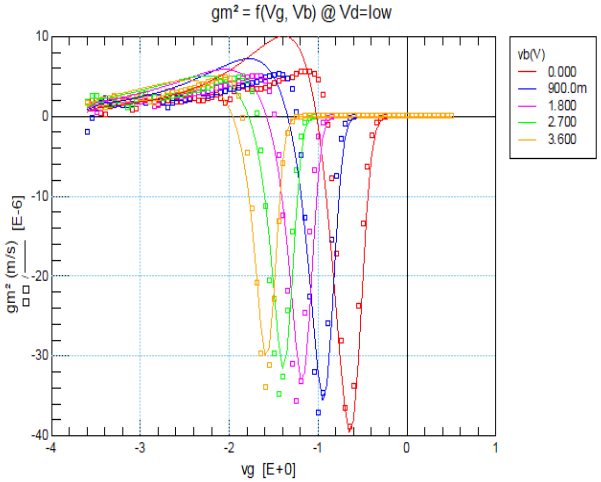
4.26W0u6_L0u33_S564_1

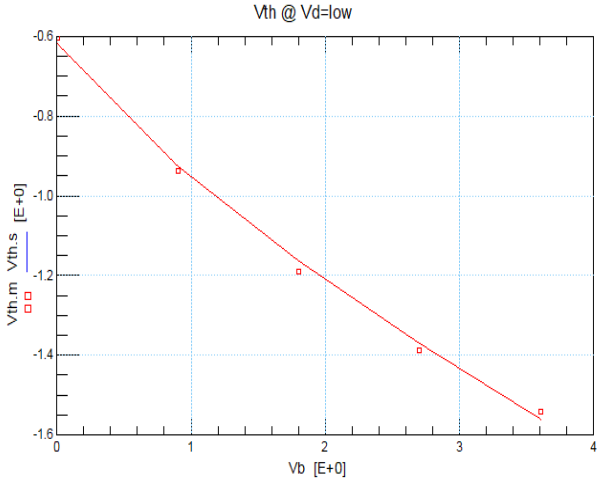
idvg, Vd = -0.1V, T = 27°C

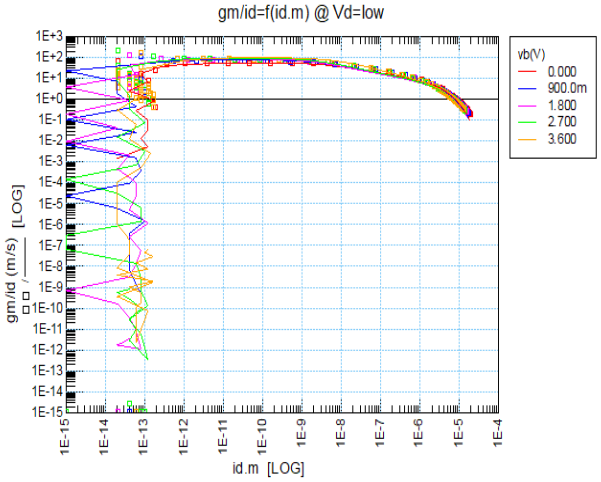


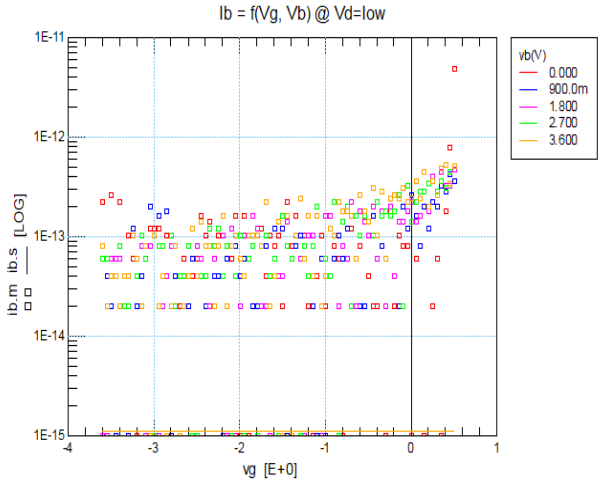


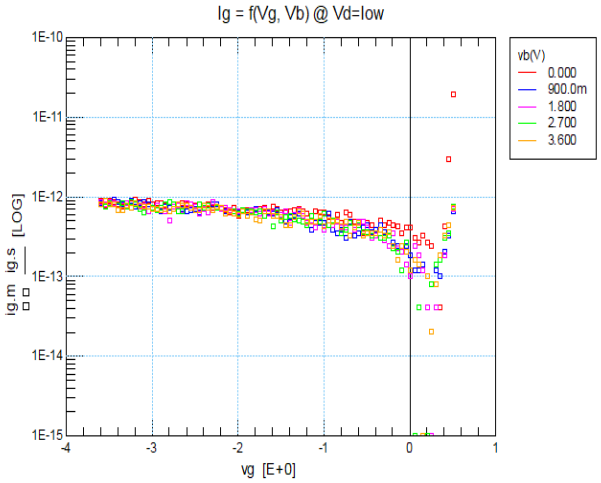




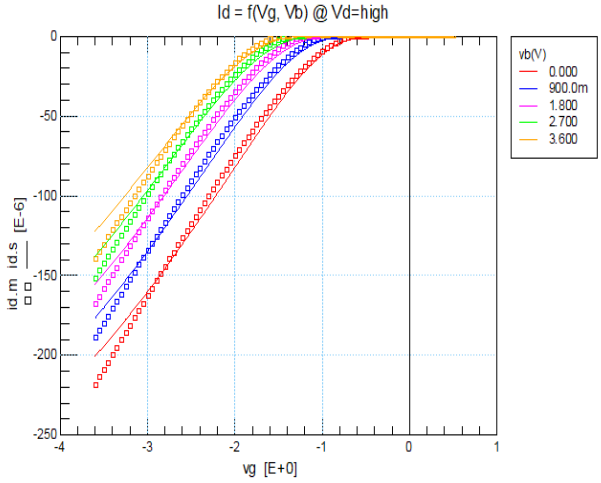


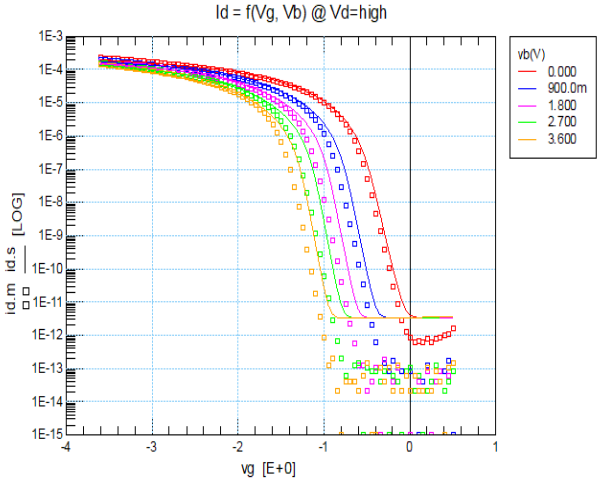


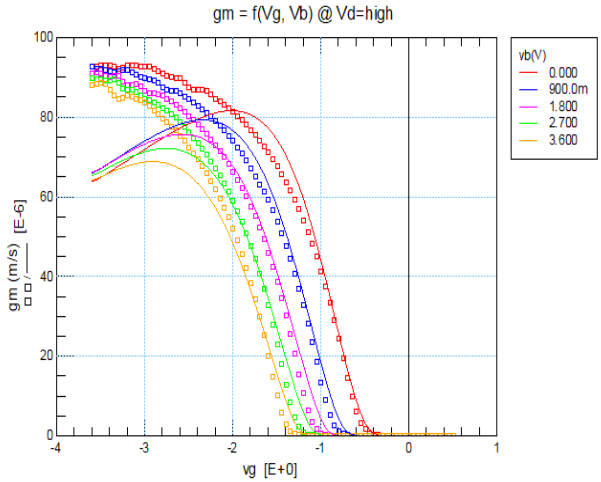


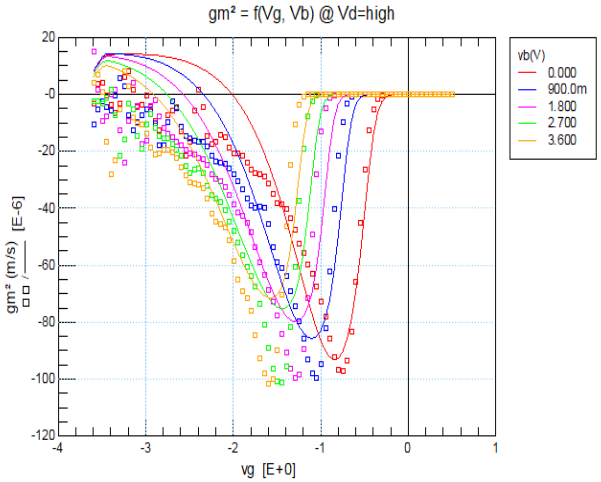


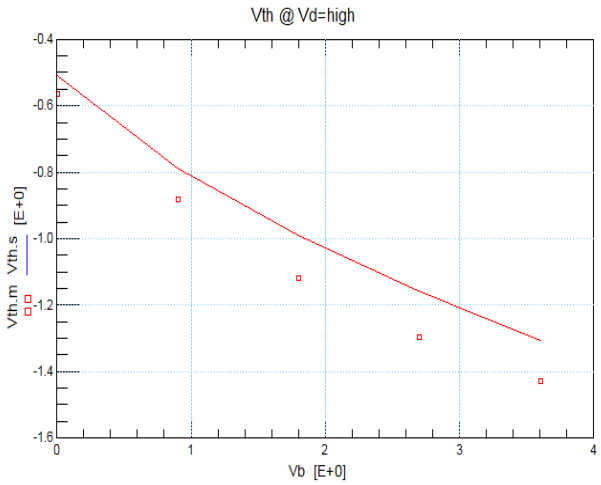
idvg, Vd = -3.3V, T = 27°C

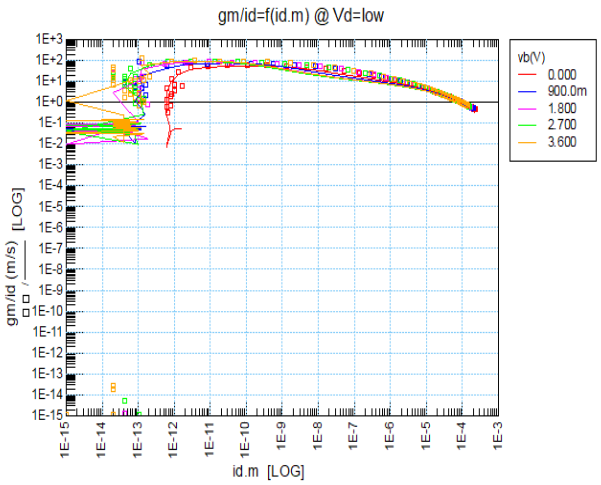


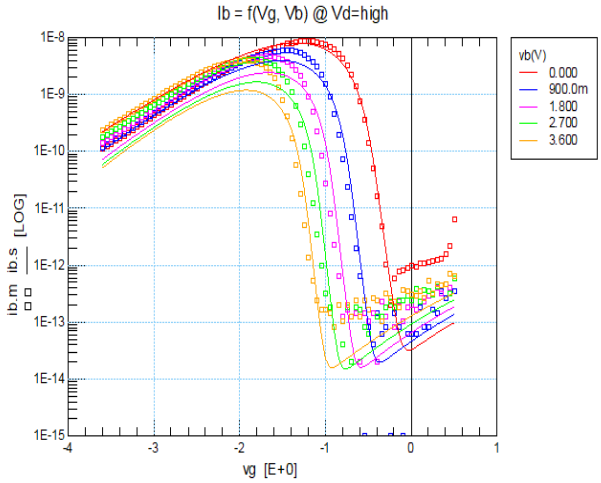


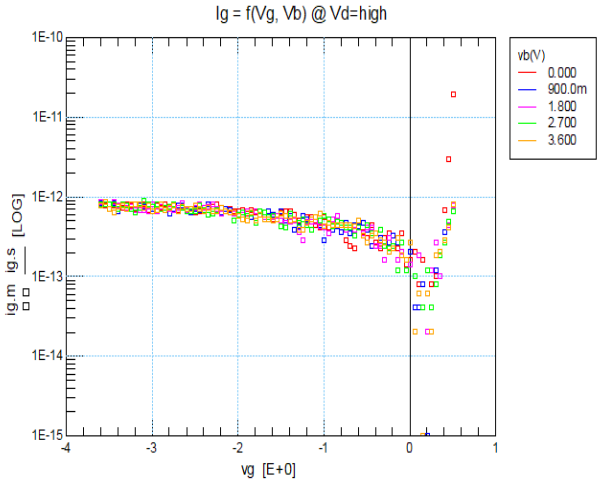




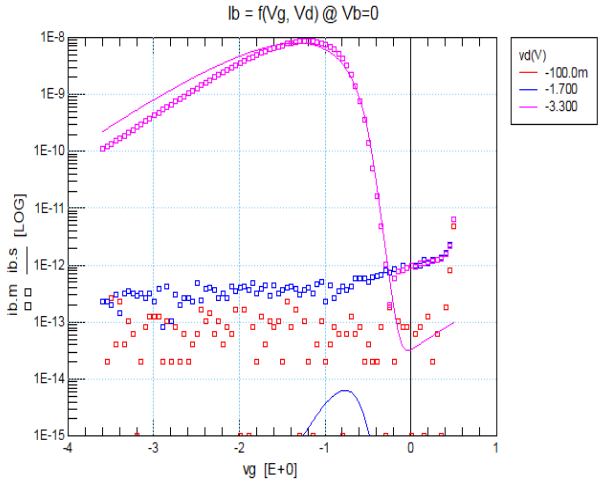


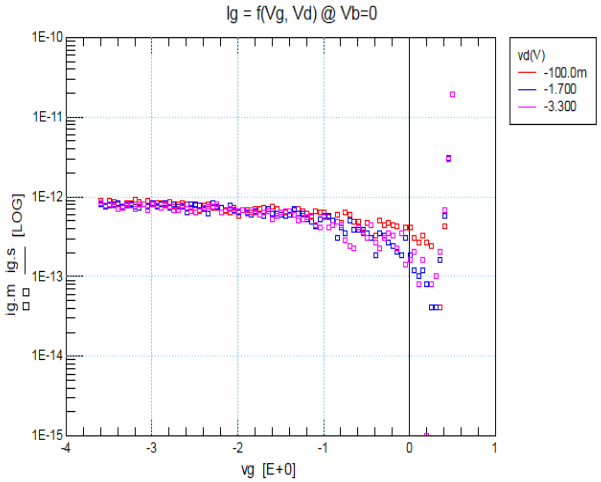




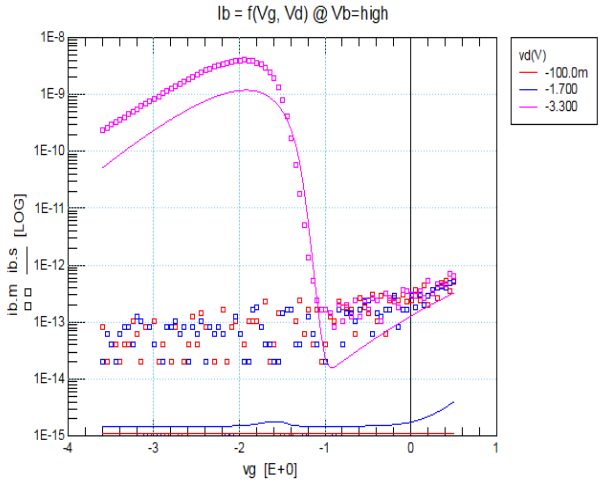


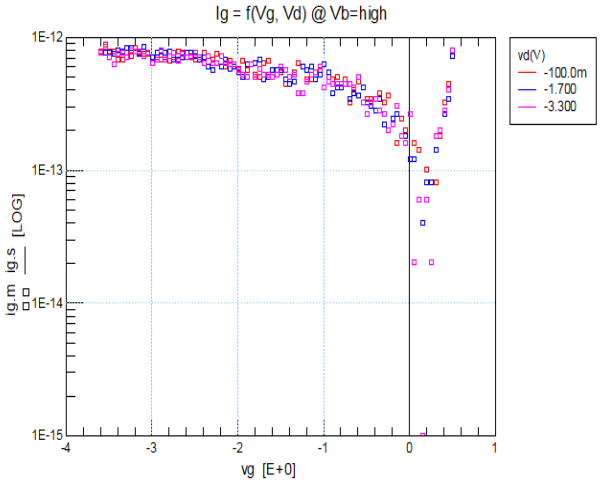
idvg, Ib,Ig, Vb = 0V, T = 27°C



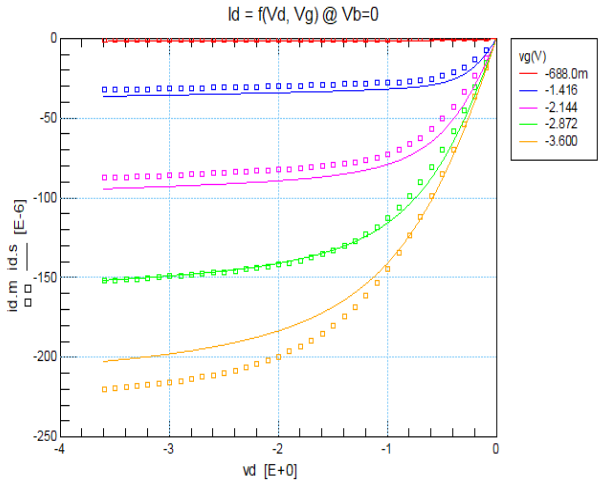


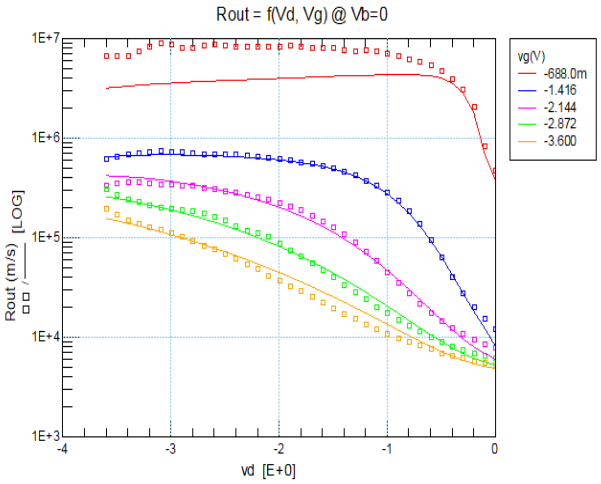
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

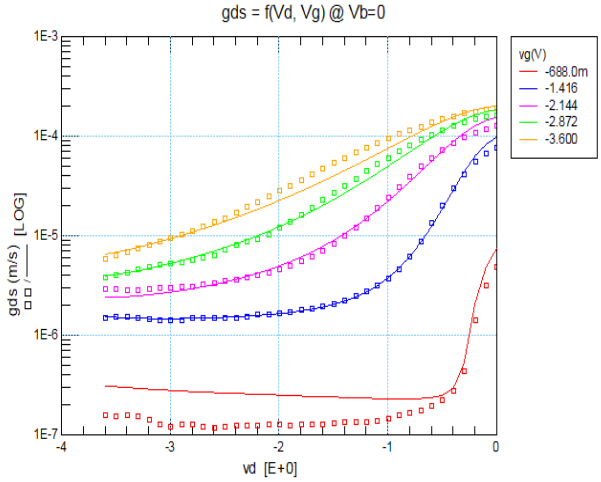




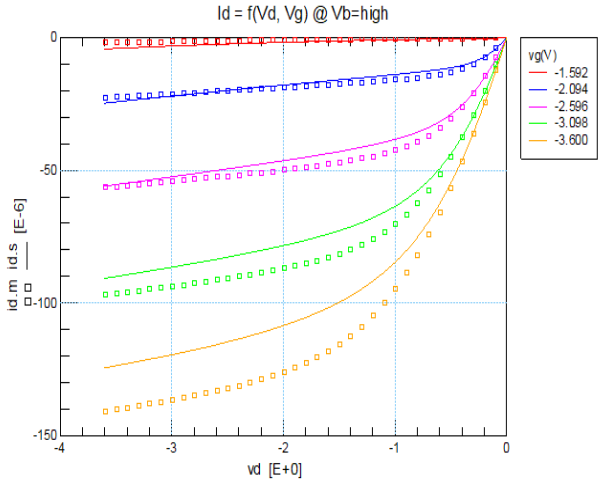
idvd, Vb = 0V, T = 27°C

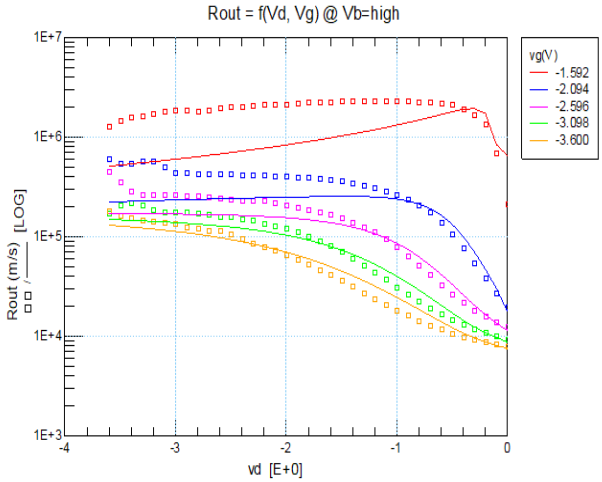


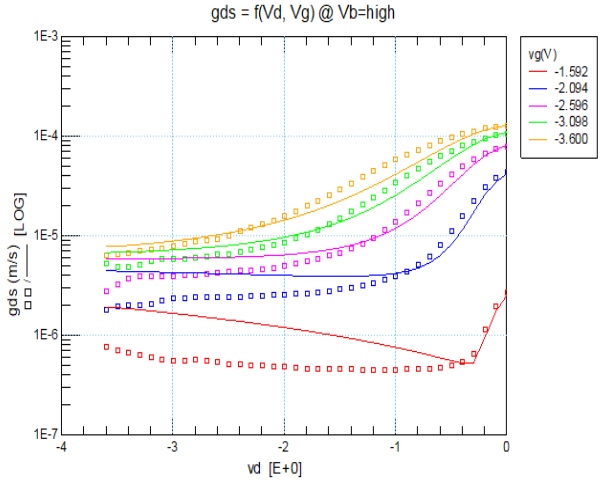




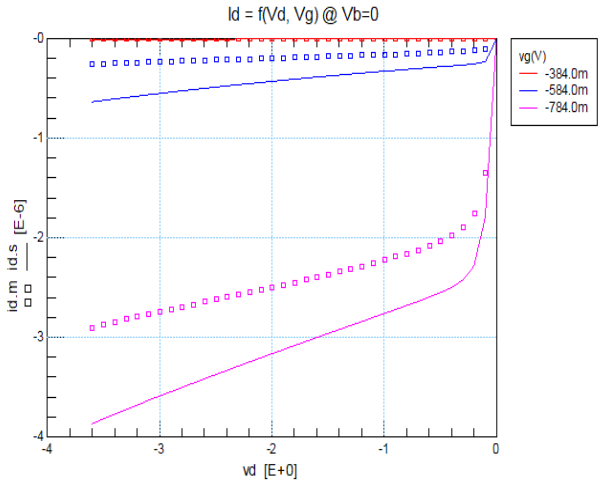
idvd_vbmin, Vb = 3.6V, T = 27°C

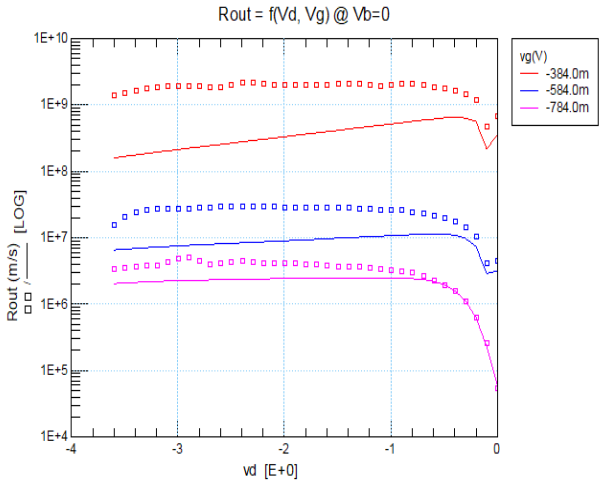


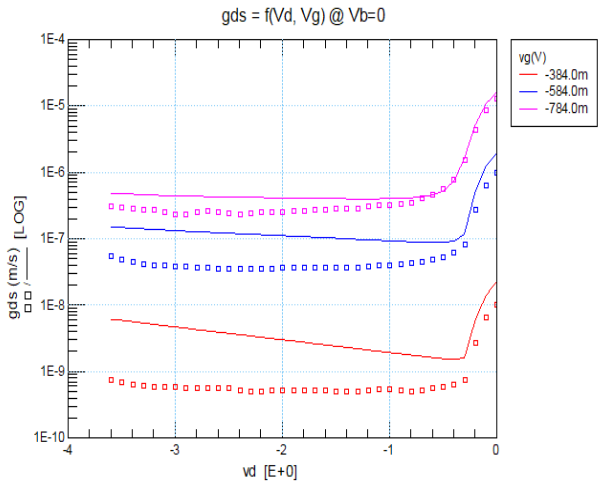




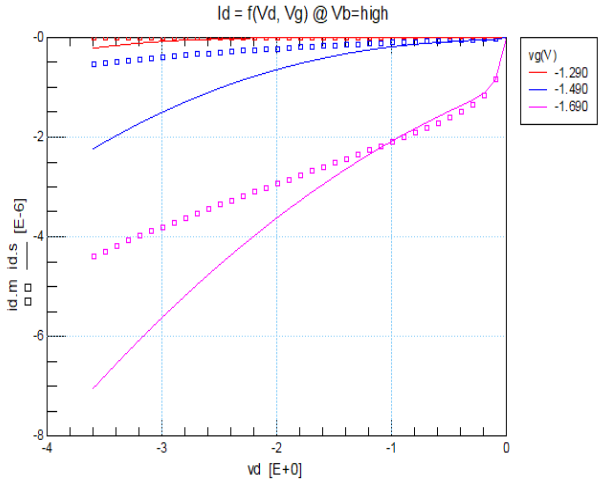
idvd_vth, Vb = 0V, T = 27°C

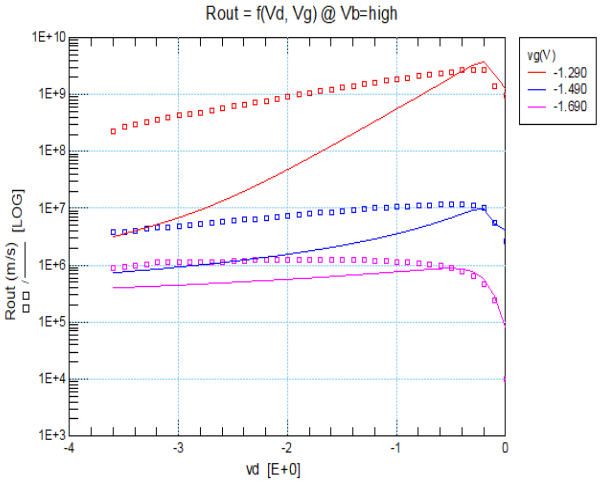


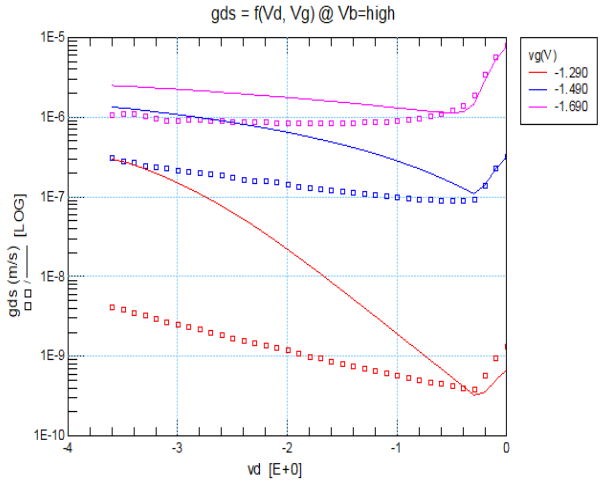




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

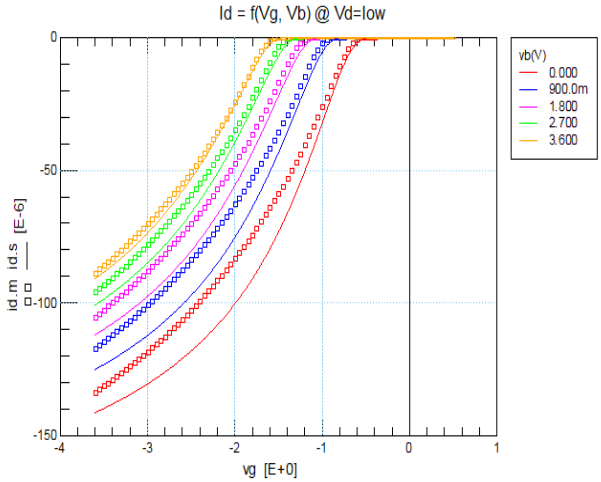


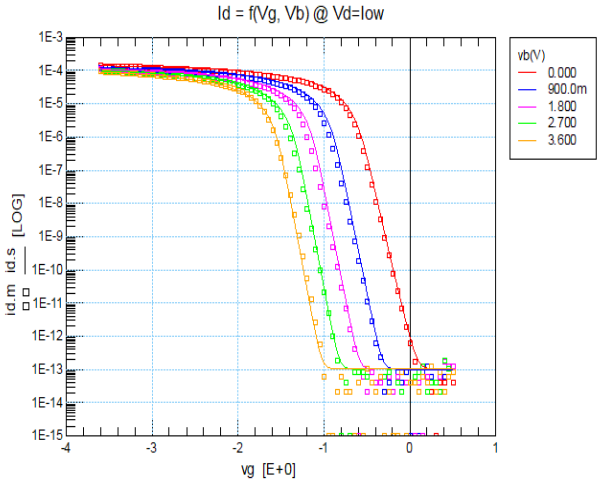


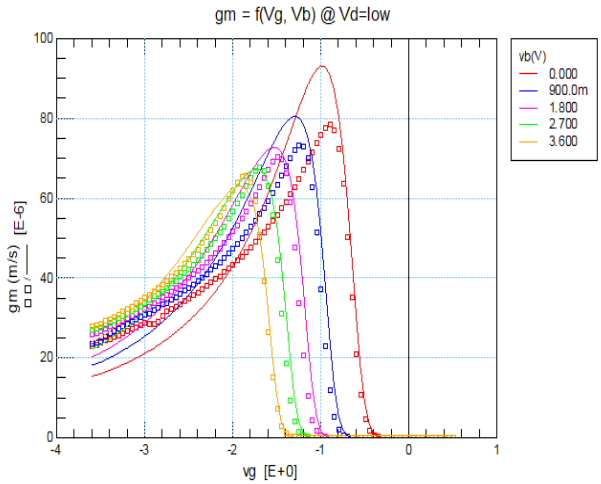


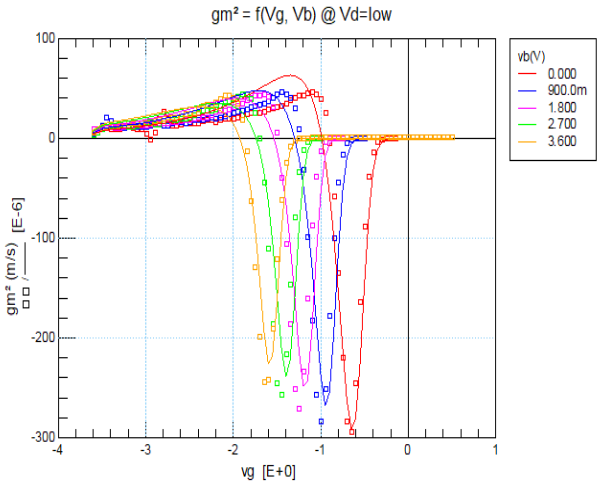
4.27W05u0_L0u33_S562_4

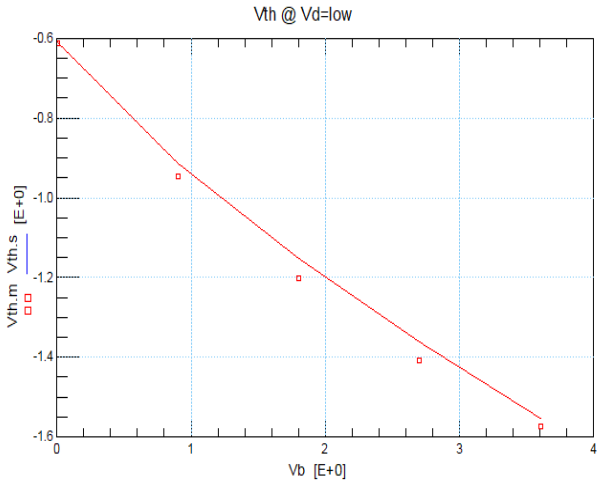
idvg, Vd = -0.1V, T = 27°C

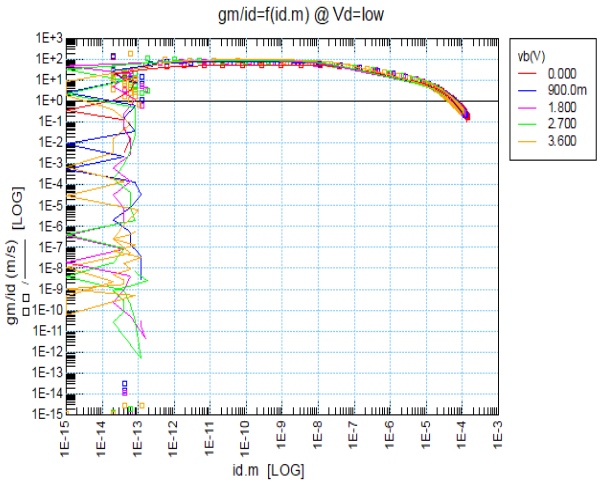


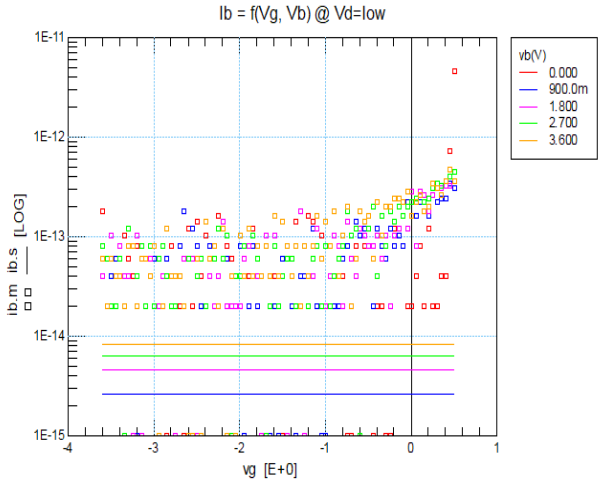


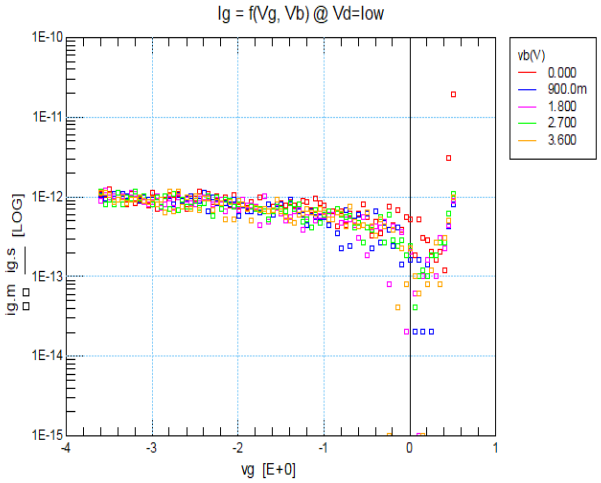




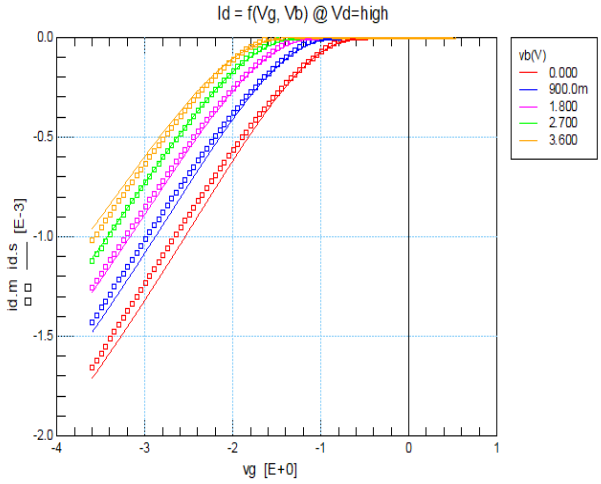


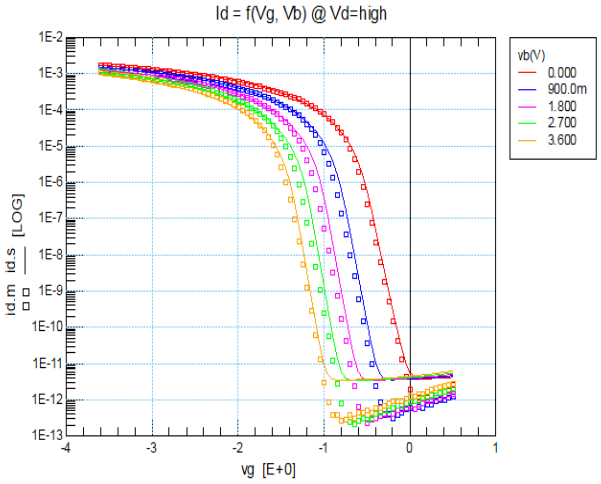


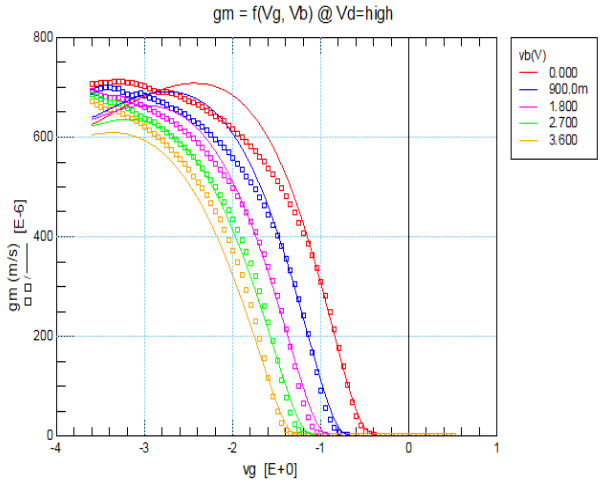


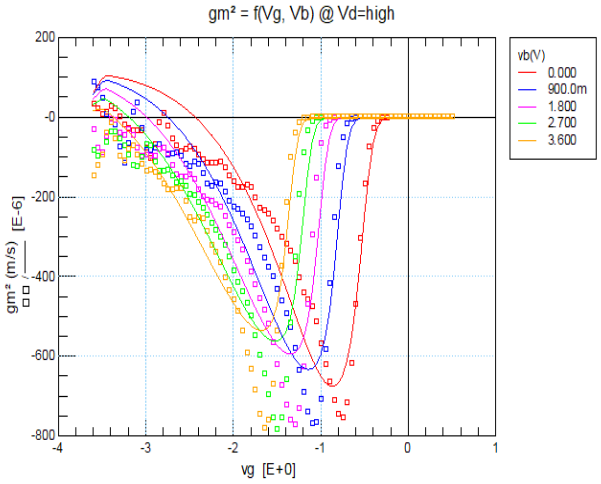


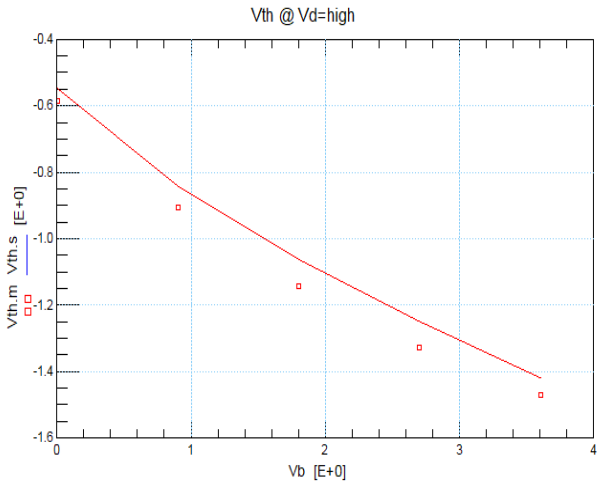
idvg, Vd = -3.3V, T = 27°C

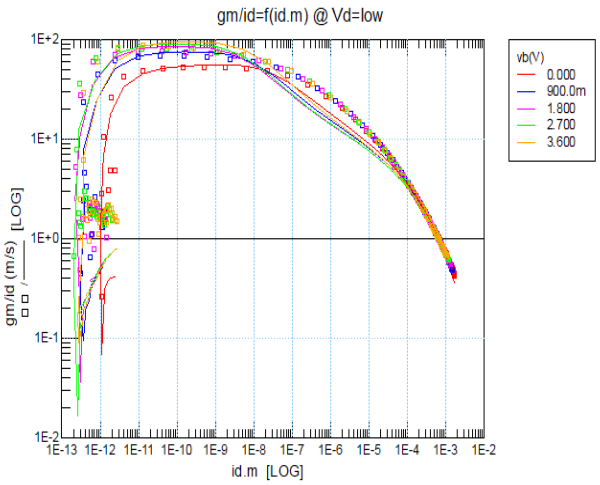


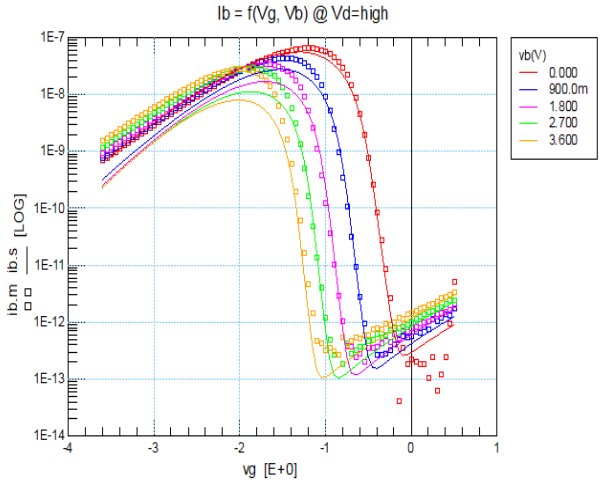


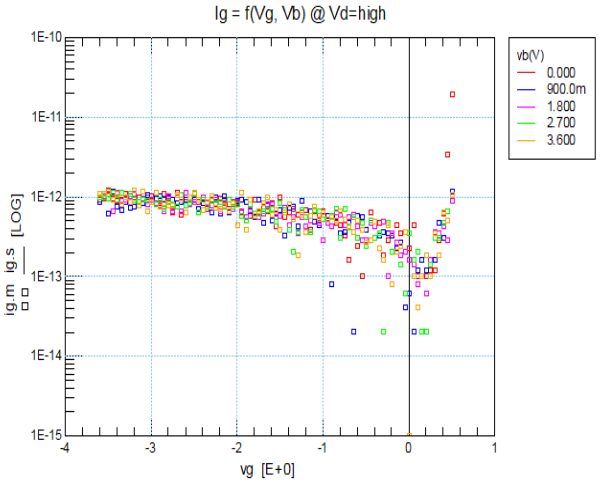




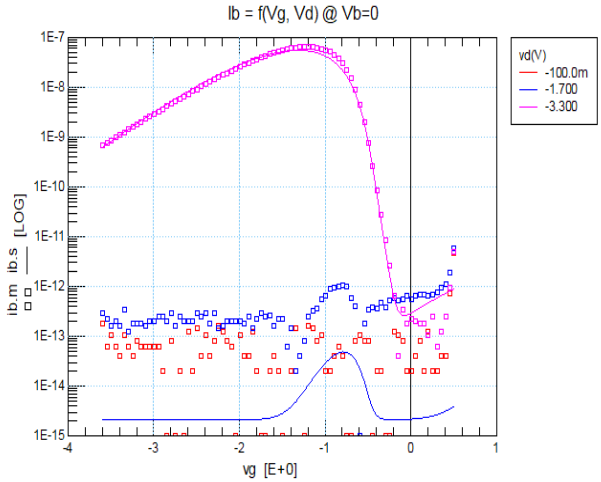


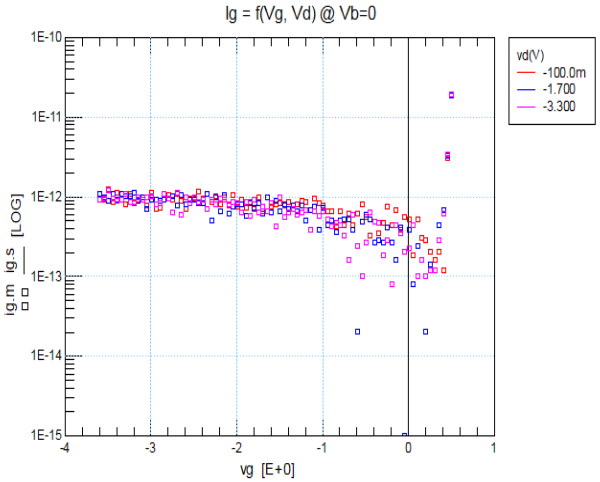




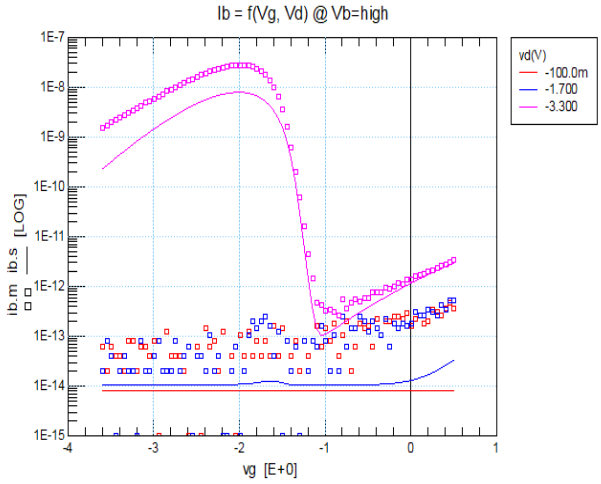


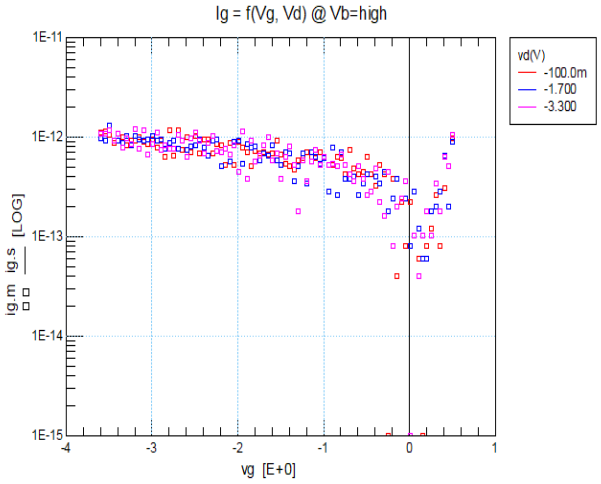
idvg, Ib,Ig, Vb = 0V, T = 27°C



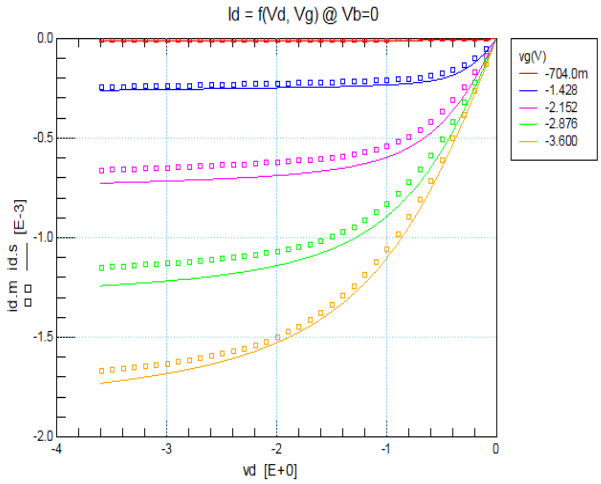


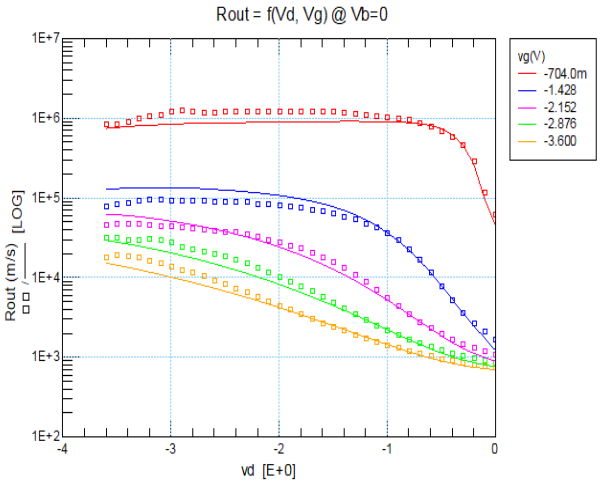
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

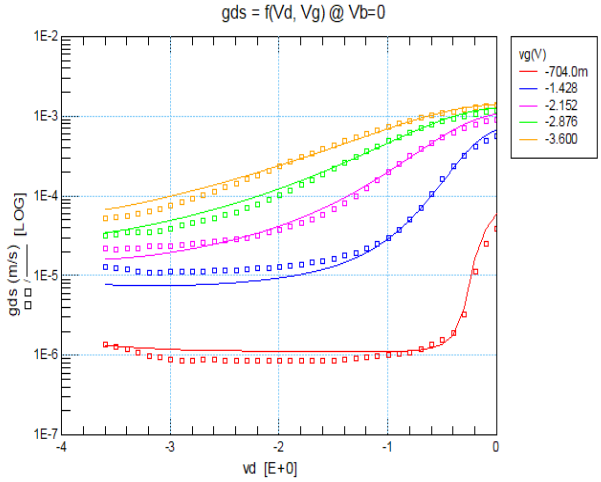




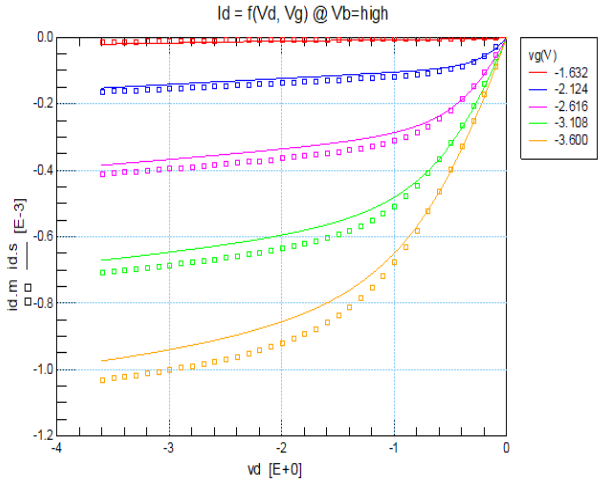
idvd, Vb = 0V, T = 27°C

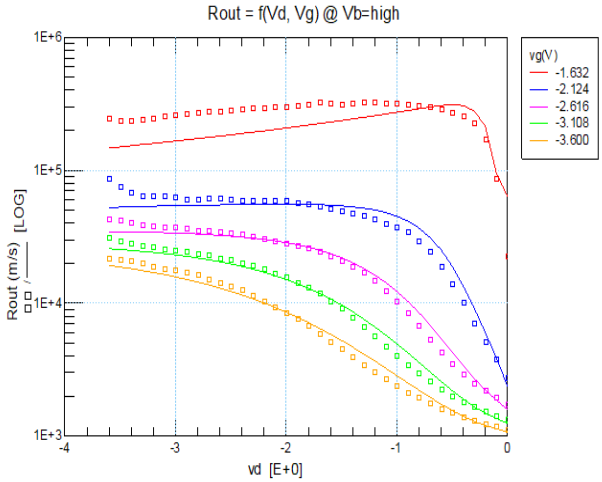


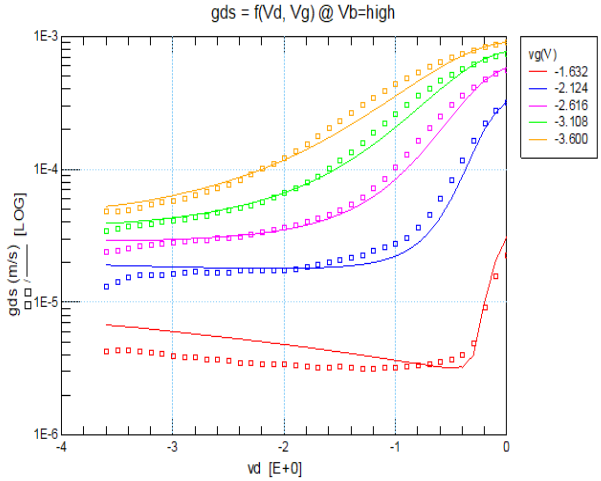




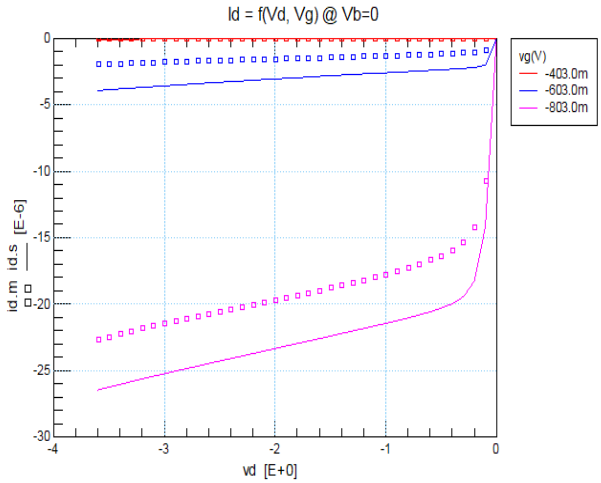
idvd_vbmin, Vb = 3.6V, T = 27°C

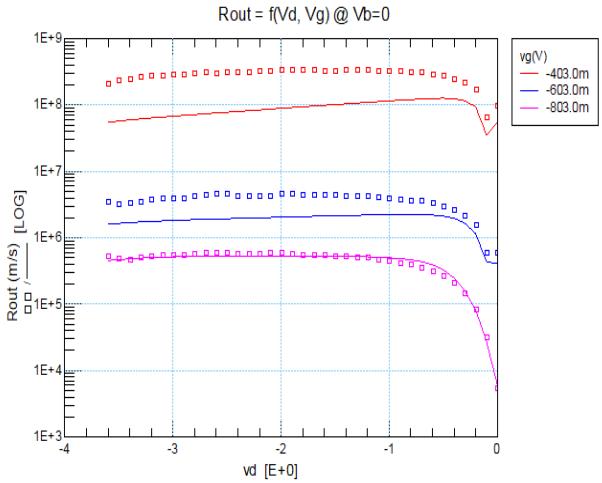


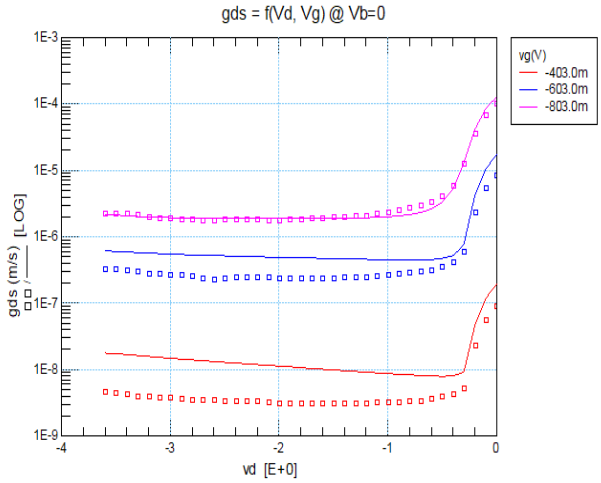




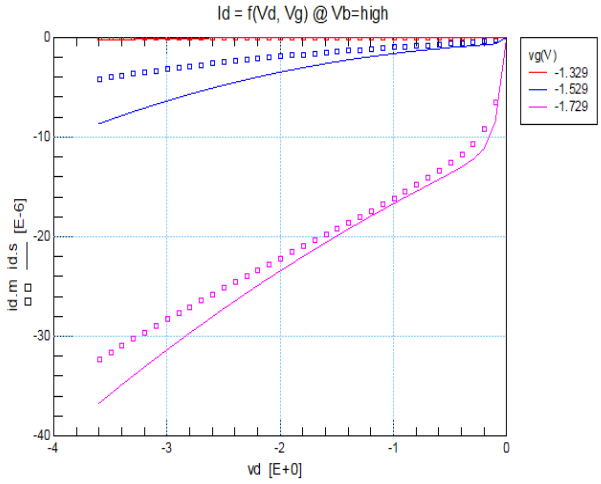
idvd_vth, Vb = 0V, T = 27°C

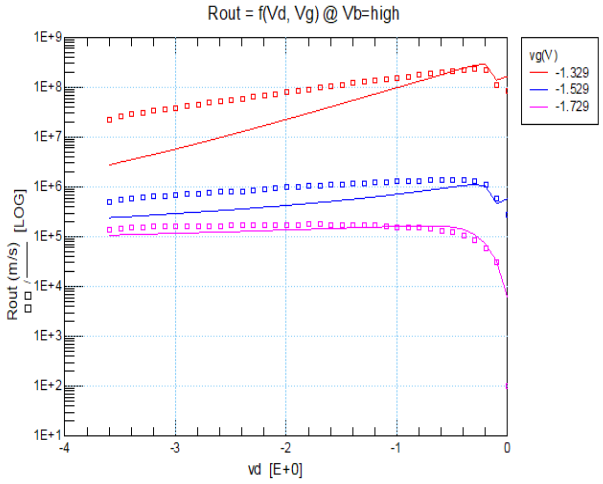


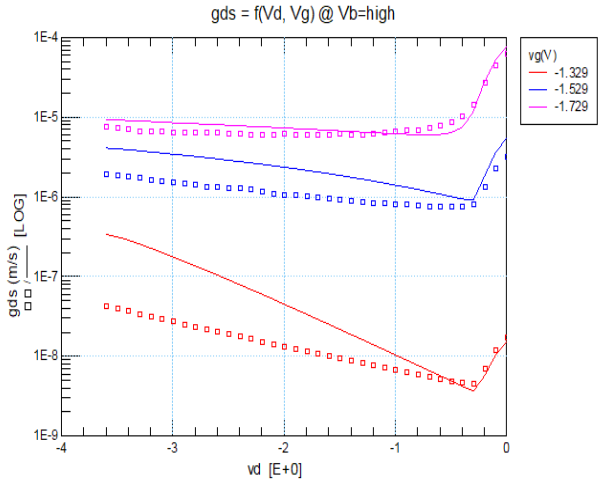




idvd_vth_vbmin, Vb = 3.6V, T = 27°C



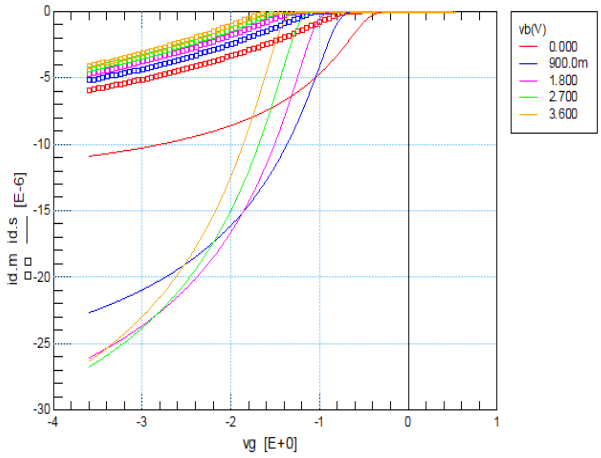




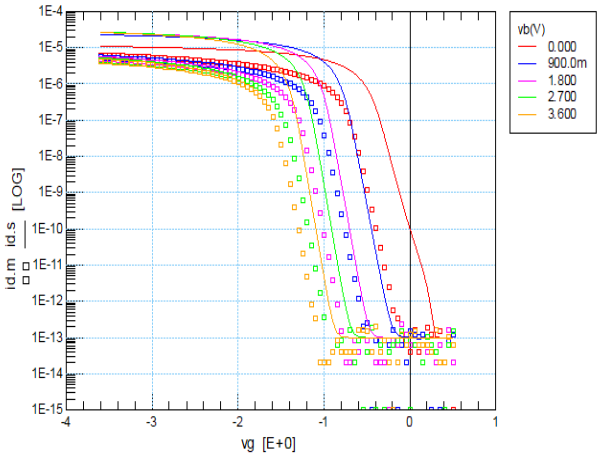
4.28 W0u15_L0u33_S555_5

idvg, Vd = -0.1V, T = 27°C

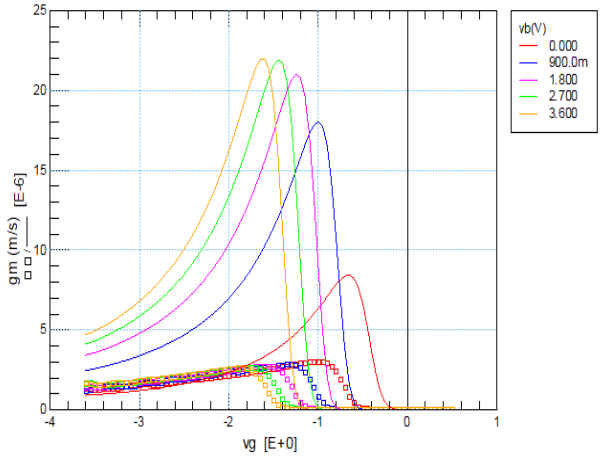
Id = f(Vg, Vb) @ Vd=low



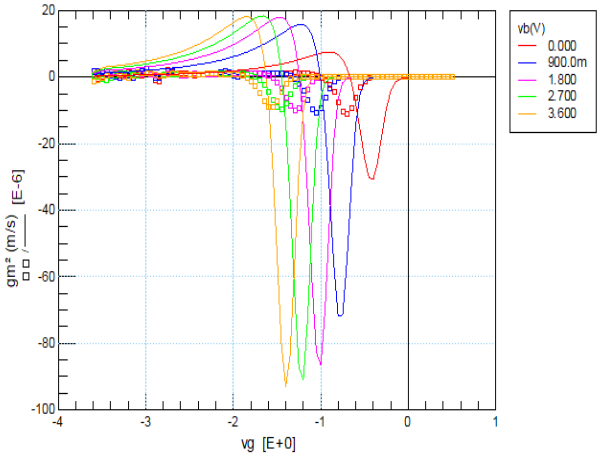
Id = f(Vg, Vb) @ Vd=low



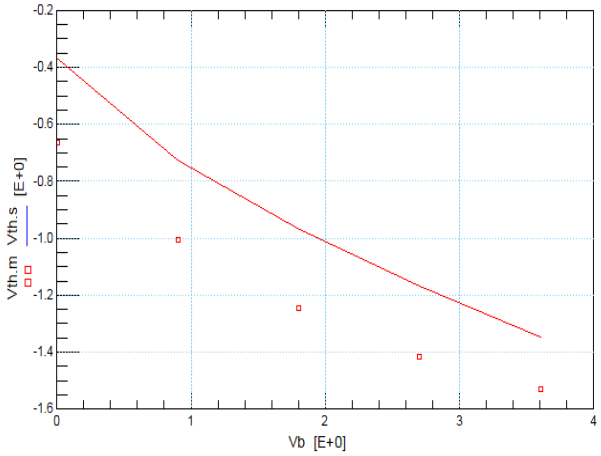
gm = f(Vg, Vb) @ Vd=low



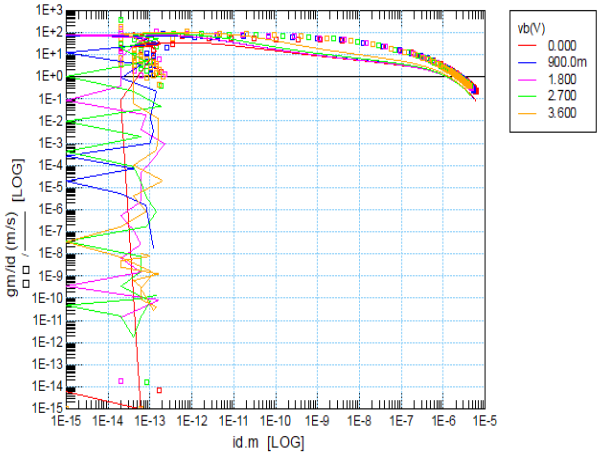
gm² = f(Vg, Vb) @ Vd=low

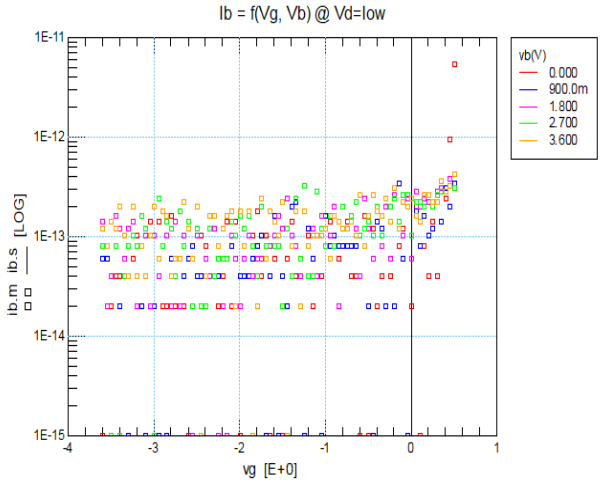


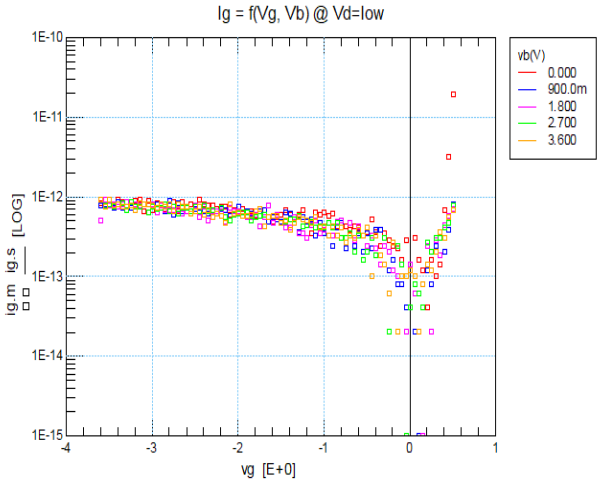
Vth @ Vd=low



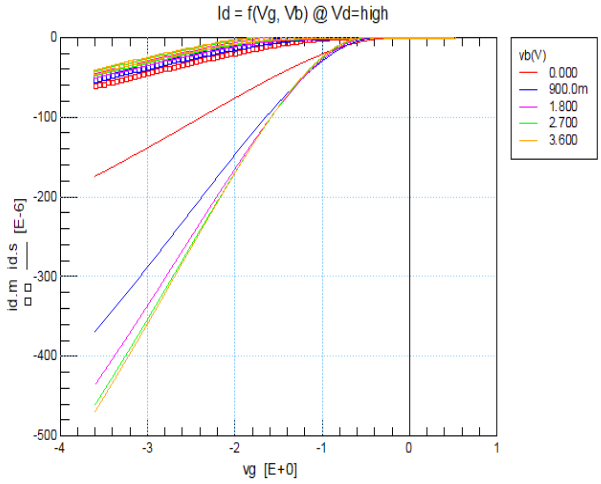
gm/id=f(id.m) @ Vd=low

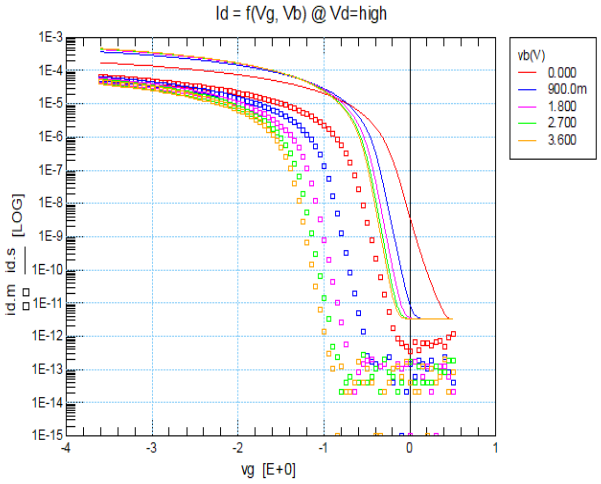


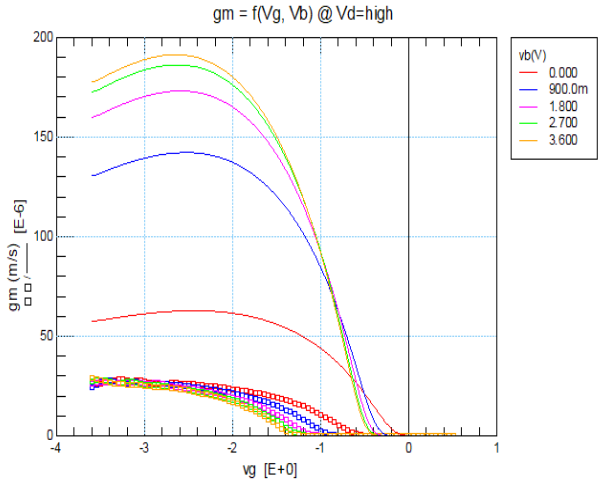


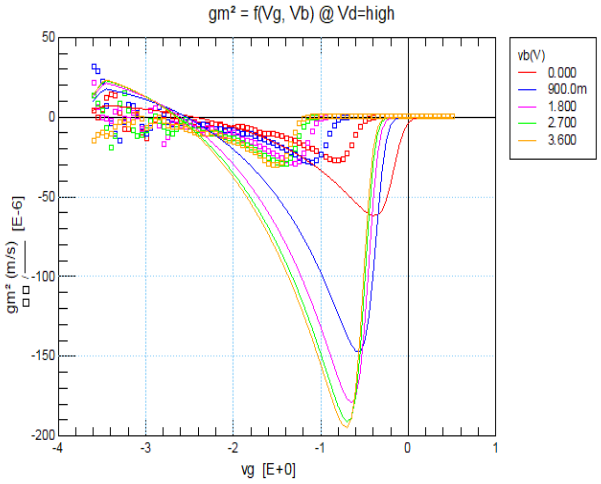


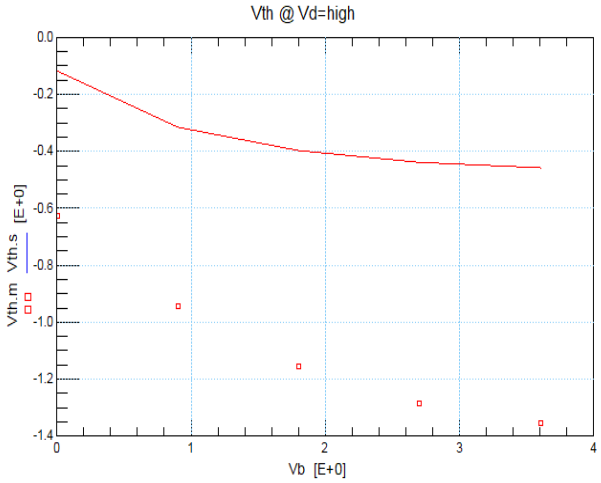
idvg, Vd = -3.3V, T = 27°C

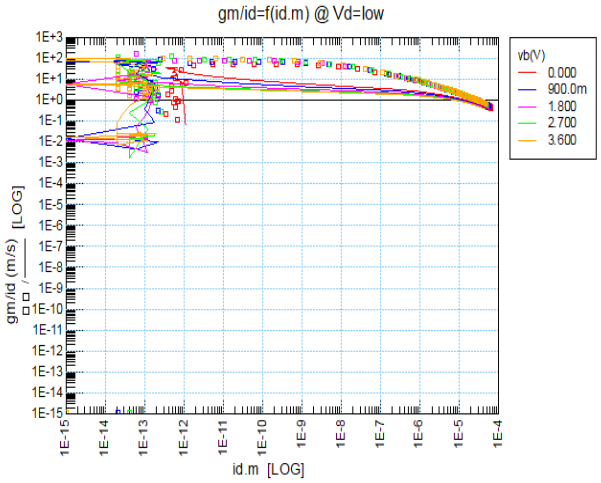


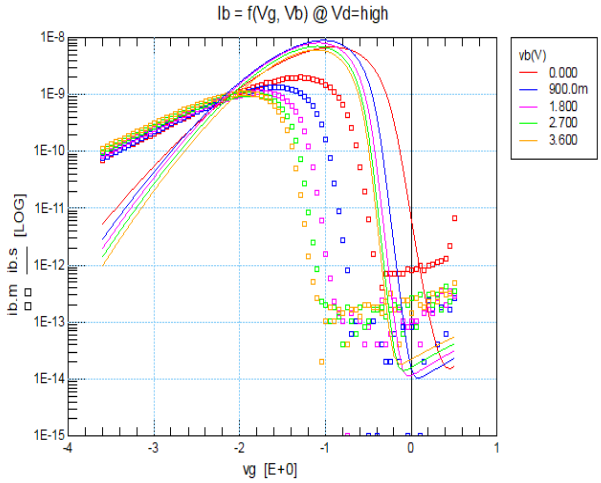


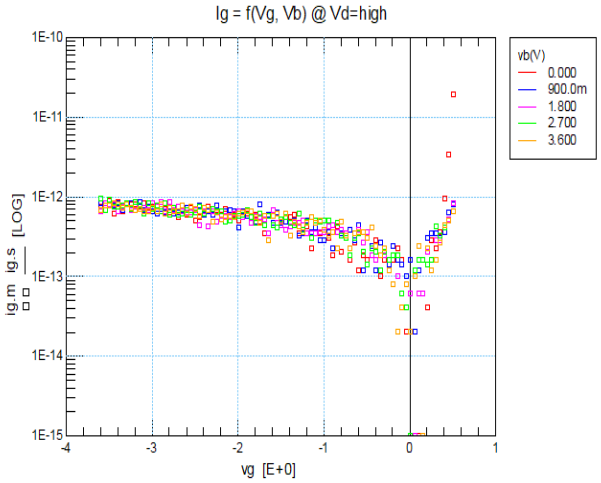




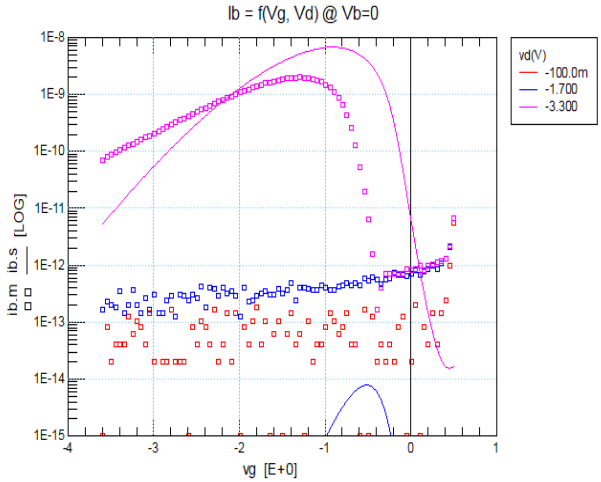


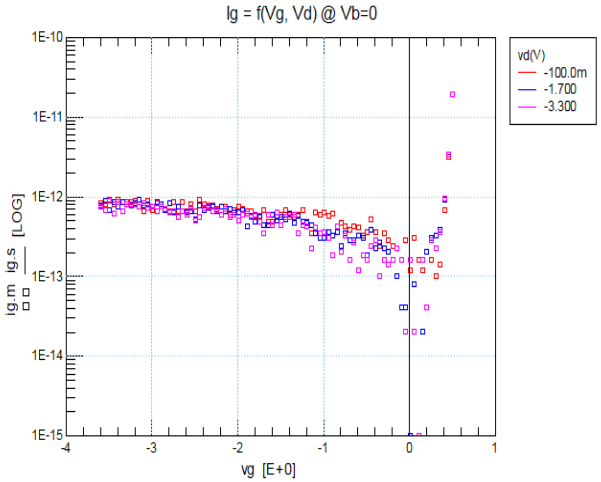




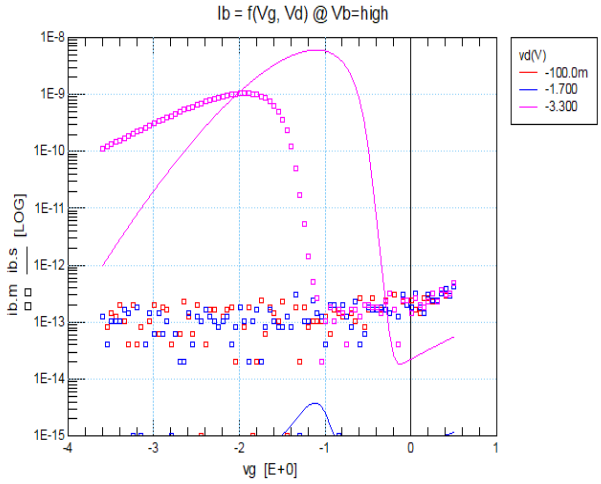


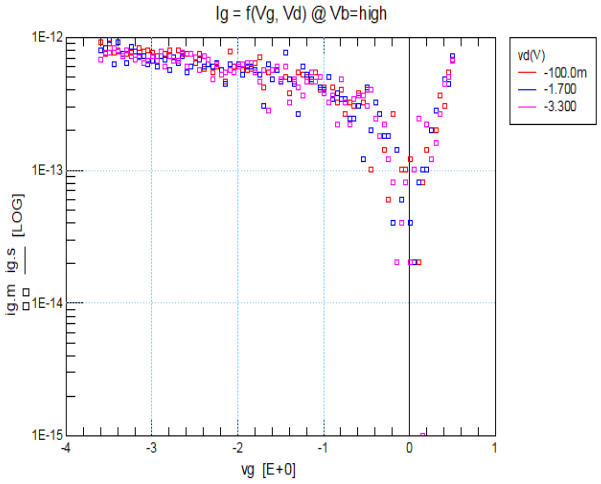
idvg, Ib,Ig, Vb = 0V, T = 27°C



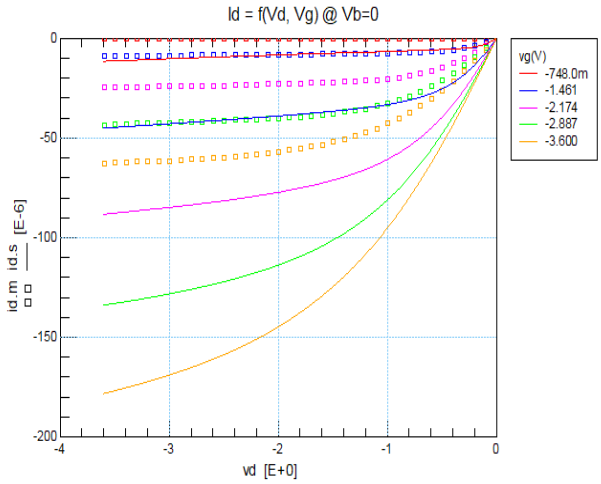


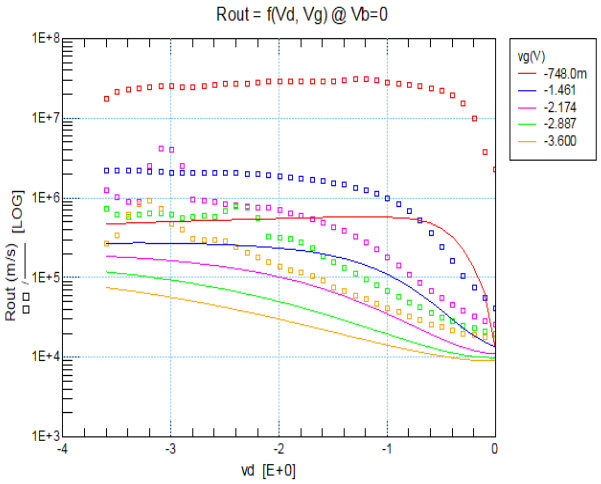
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

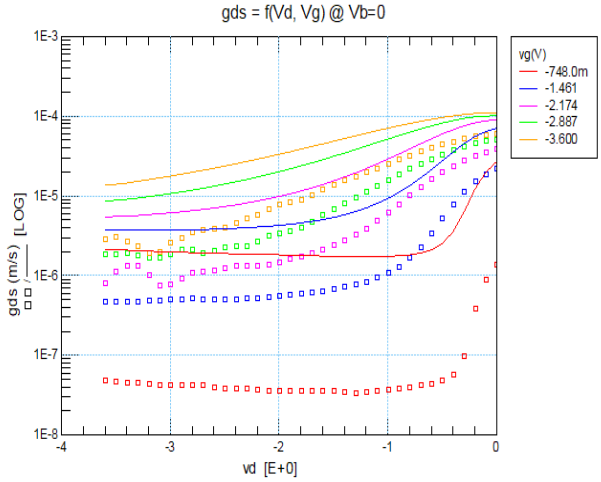




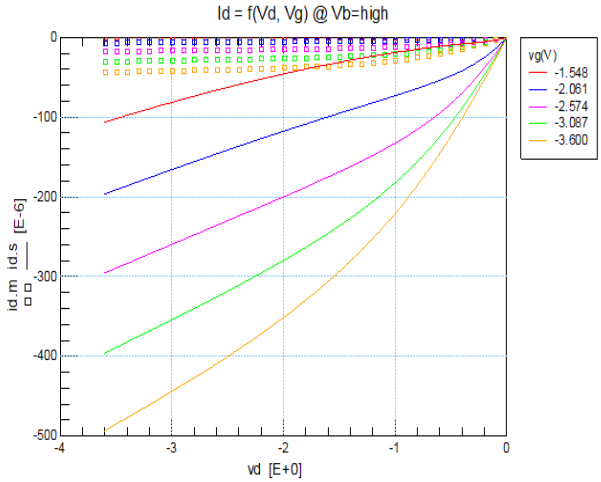
idvd, Vb = 0V, T = 27°C

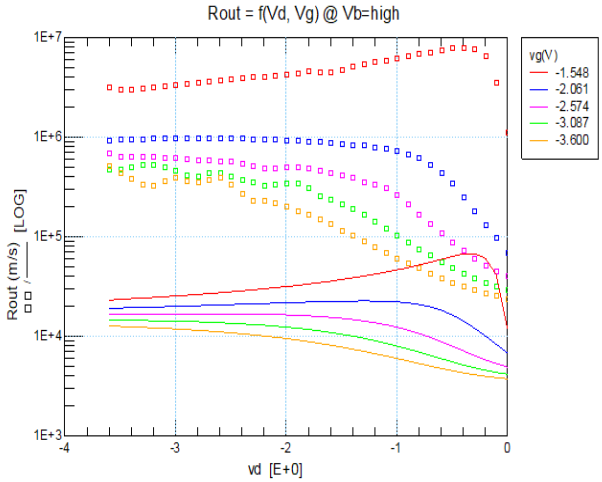


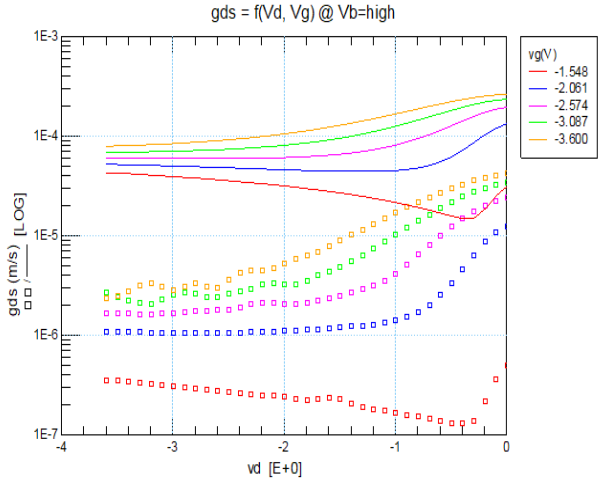




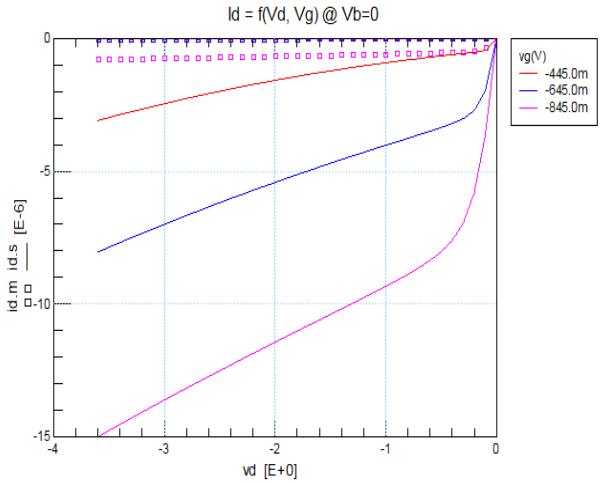
idvd_vbmin, Vb = 3.6V, T = 27°C

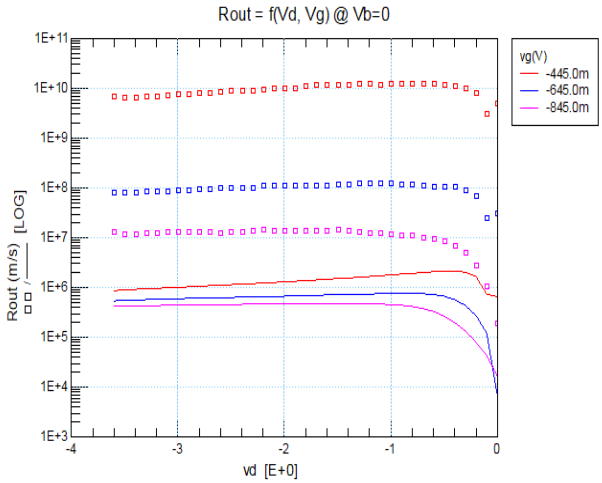


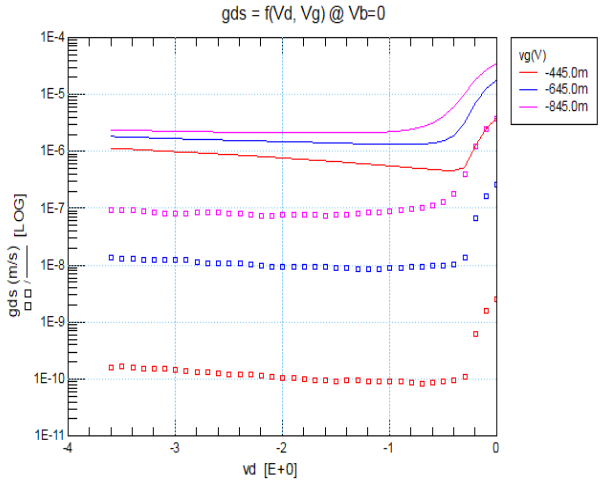




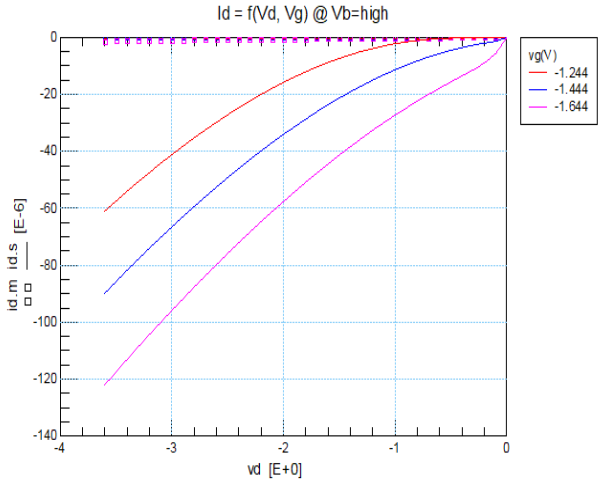
idvd_vth, Vb = 0V, T = 27°C

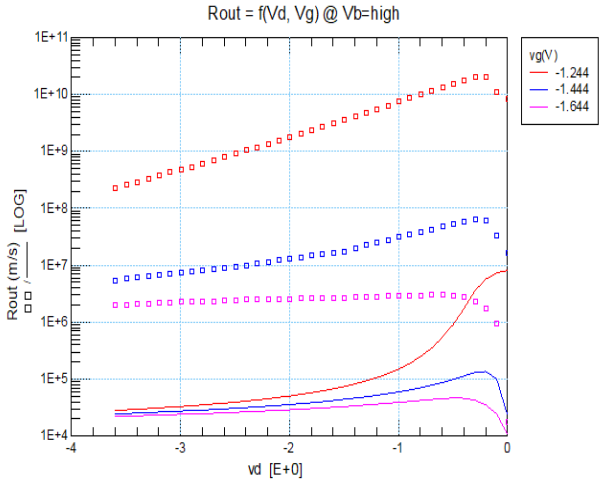


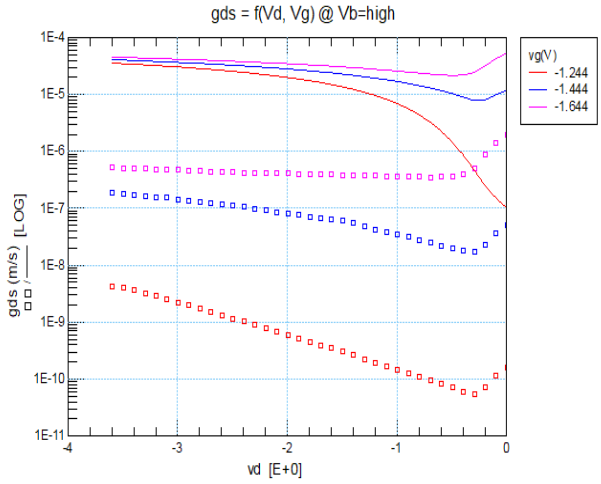




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

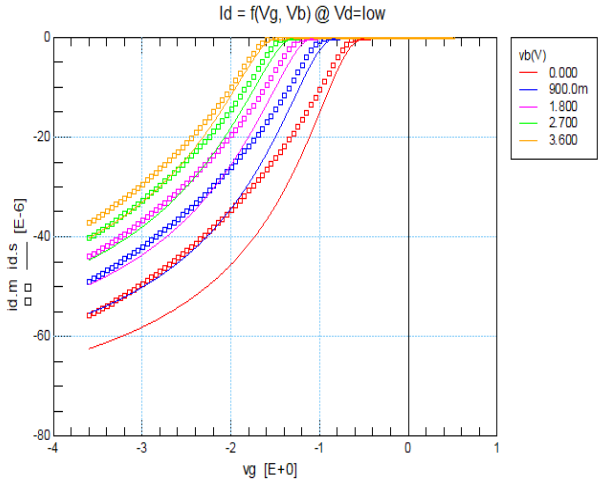


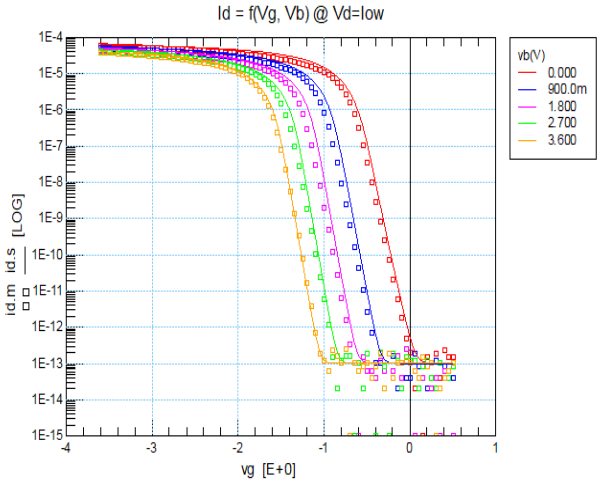


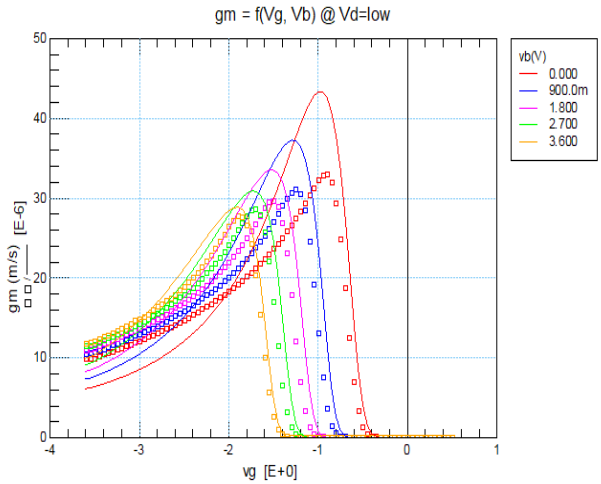


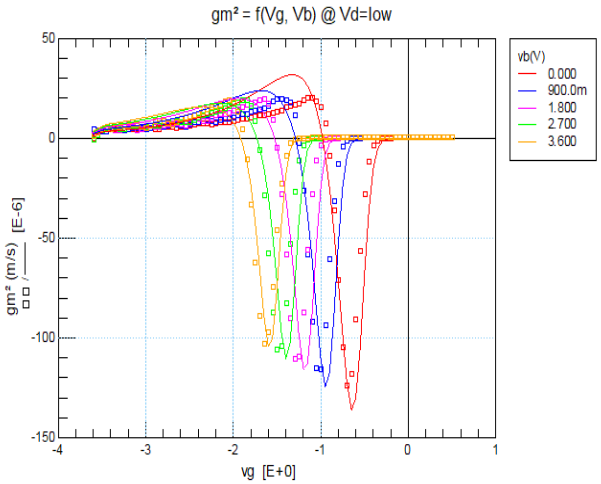
4.29W02u0_L0u33_S563_2

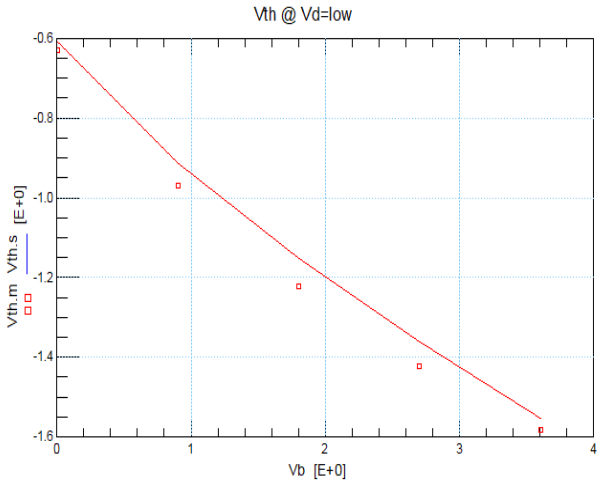
idvg, Vd = -0.1V, T = 27°C

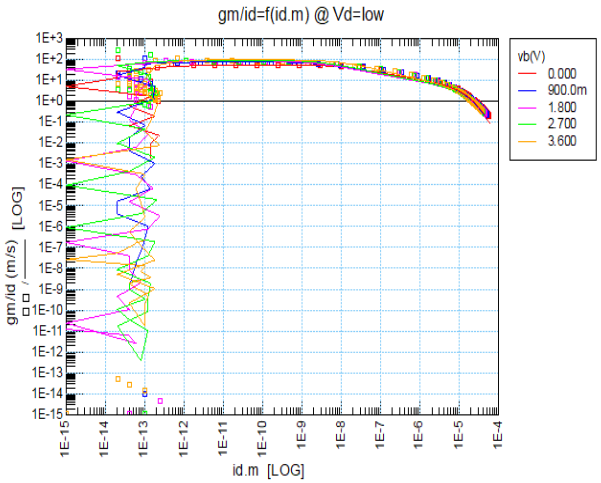


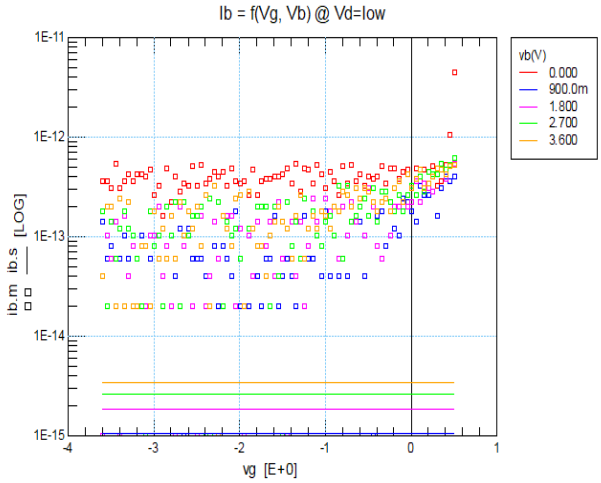


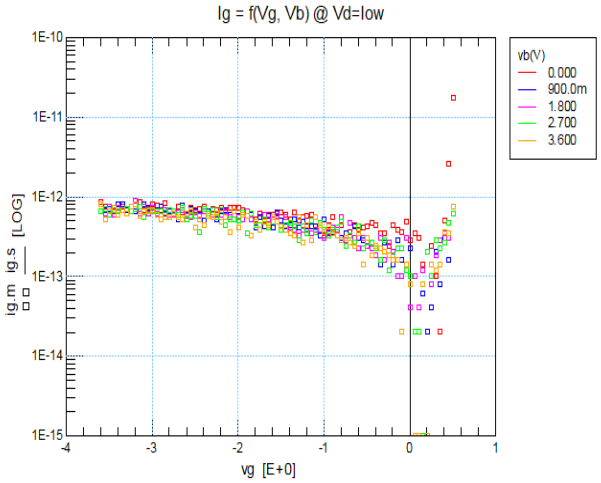




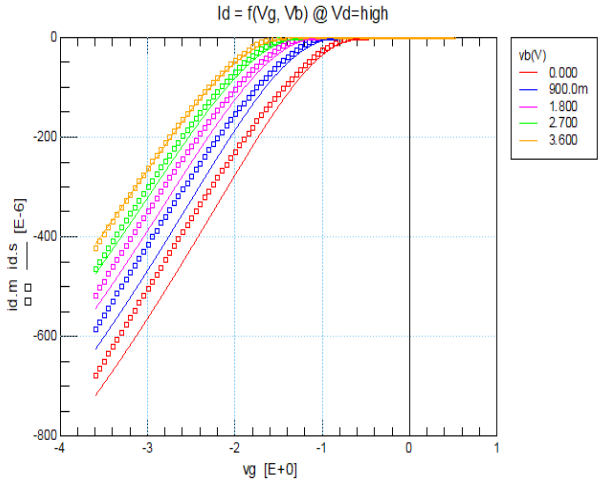


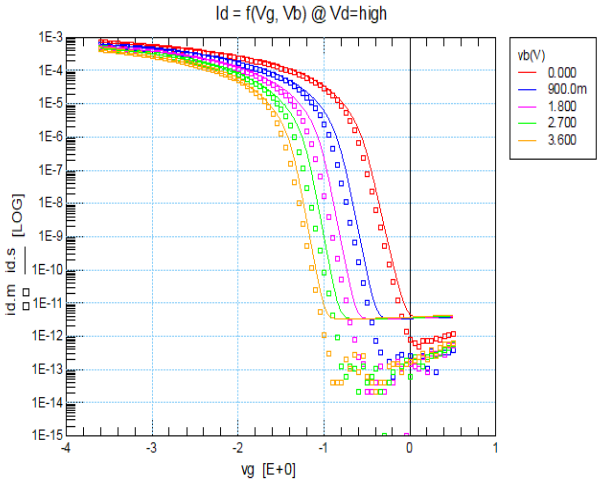


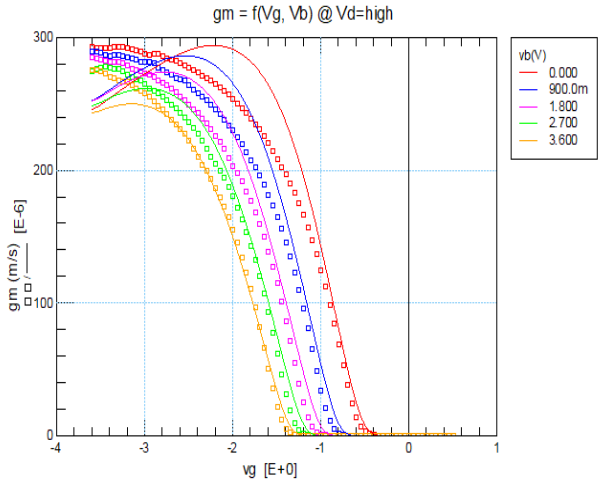


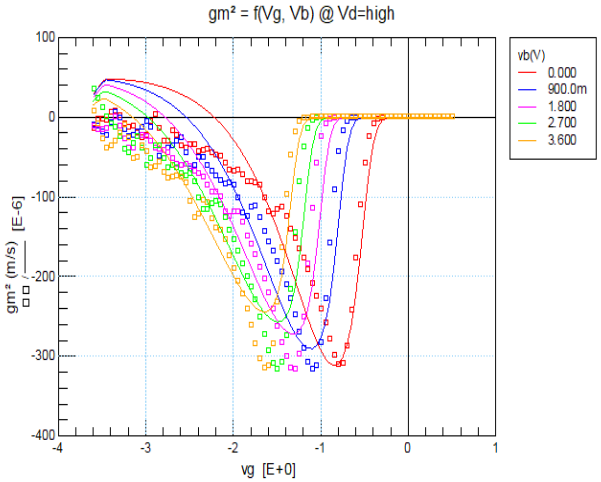


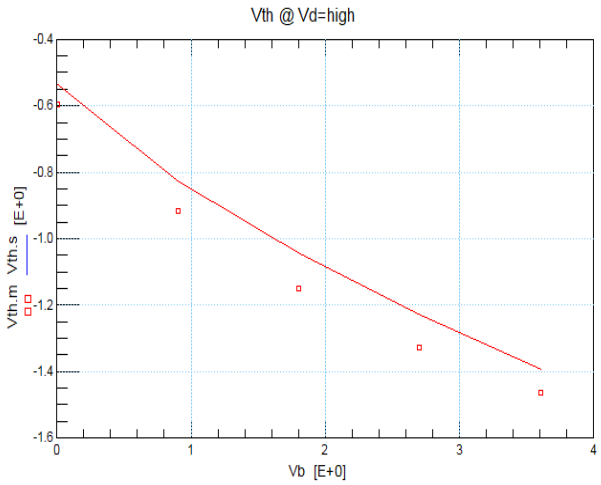
idvg, Vd = -3.3V, T = 27°C

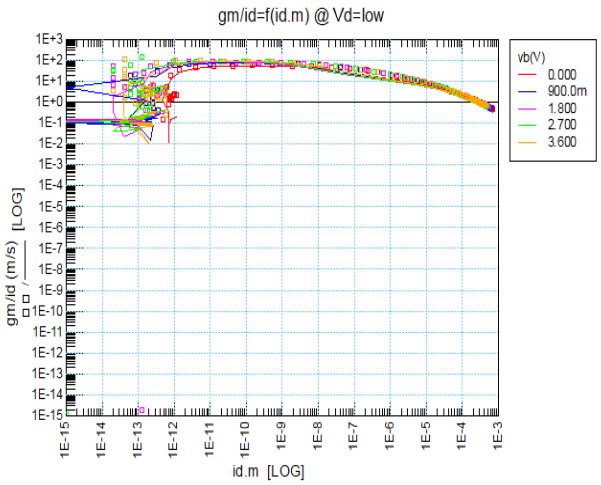


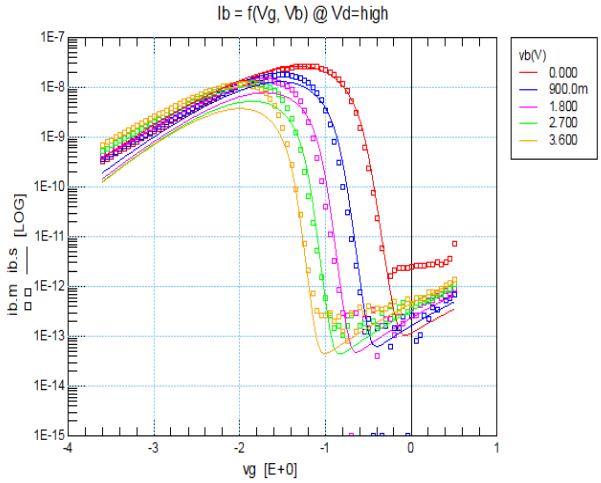


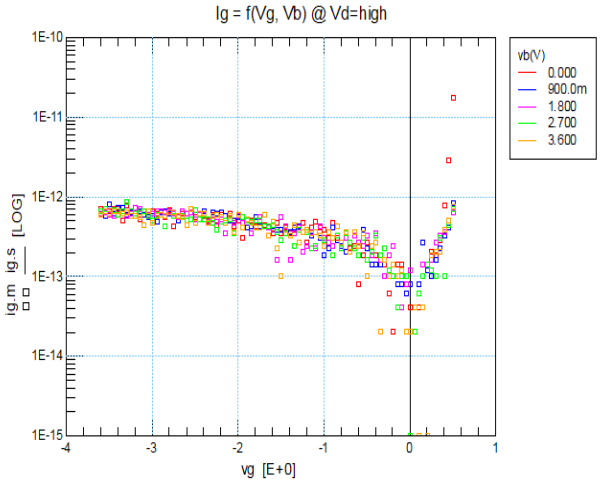




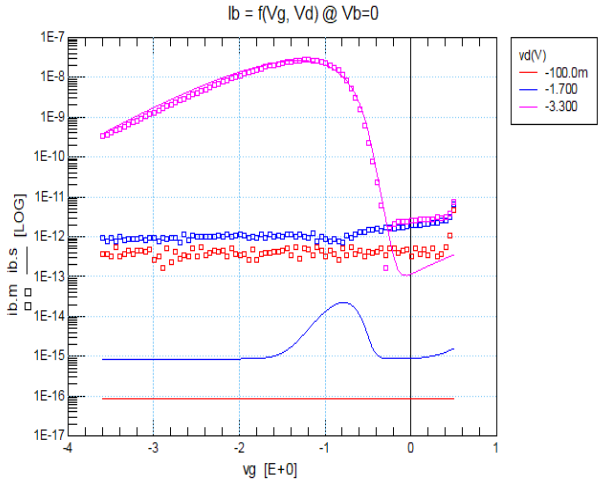


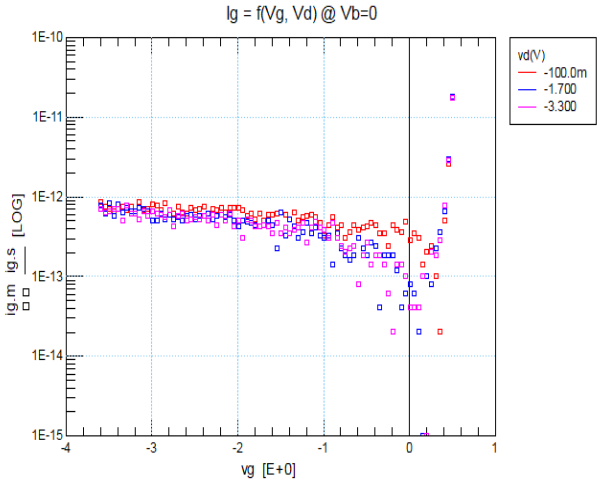




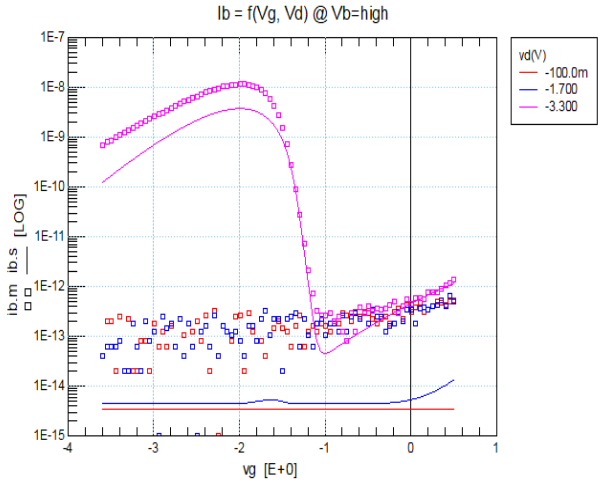


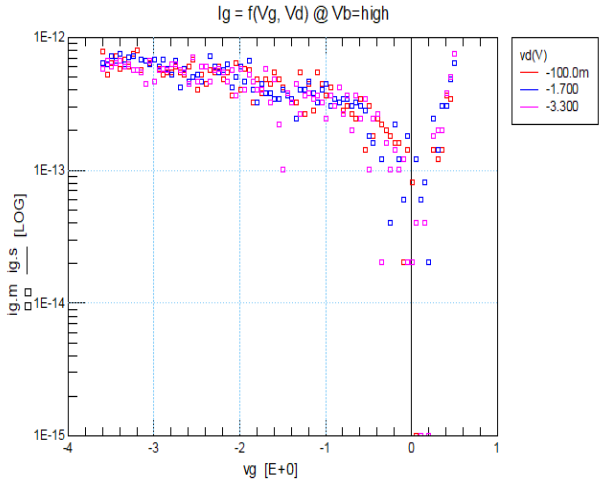
idvg, Ib,Ig, Vb = 0V, T = 27°C



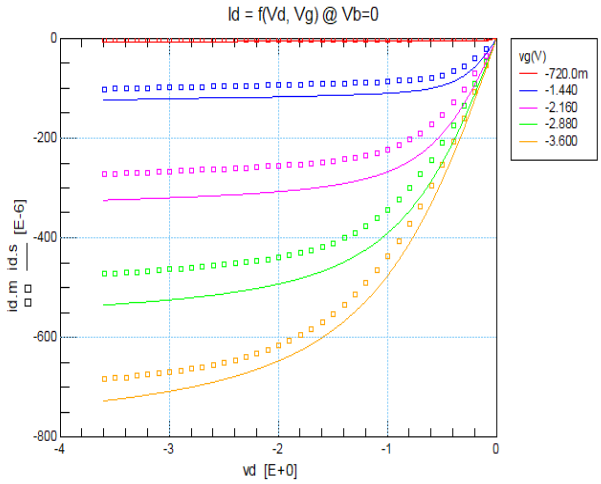


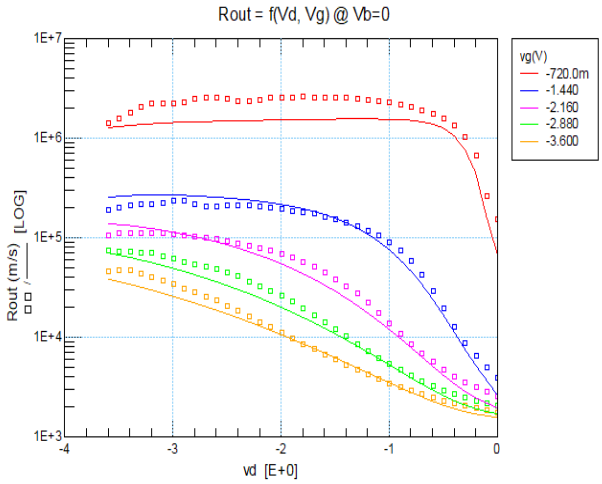
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

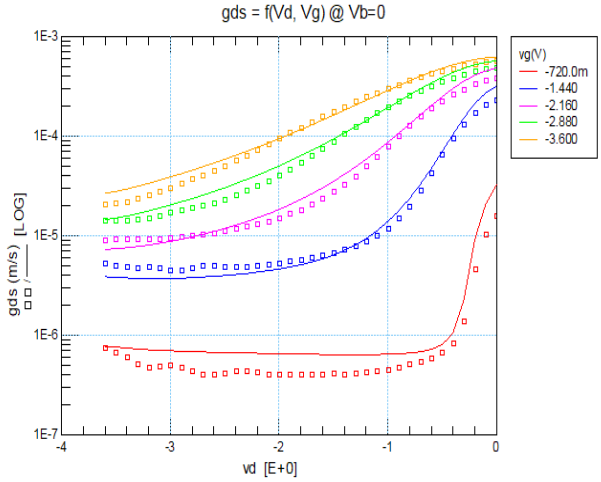




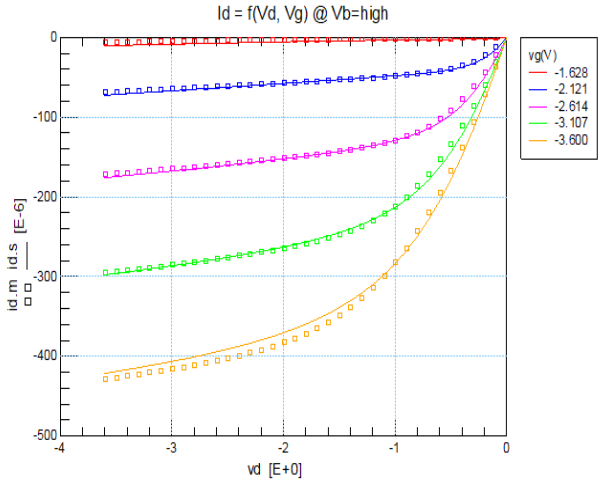
idvd, Vb = 0V, T = 27°C

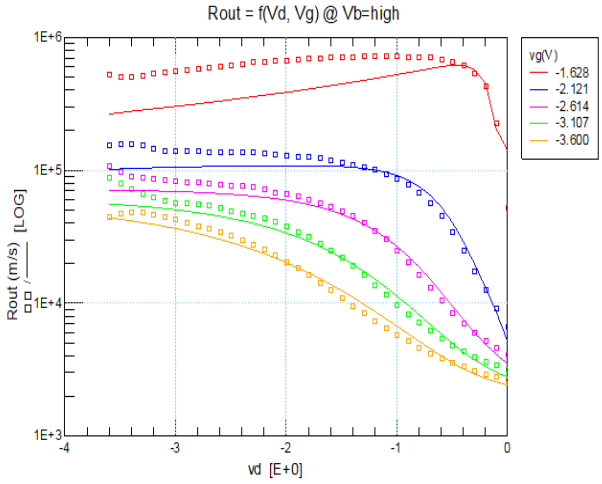


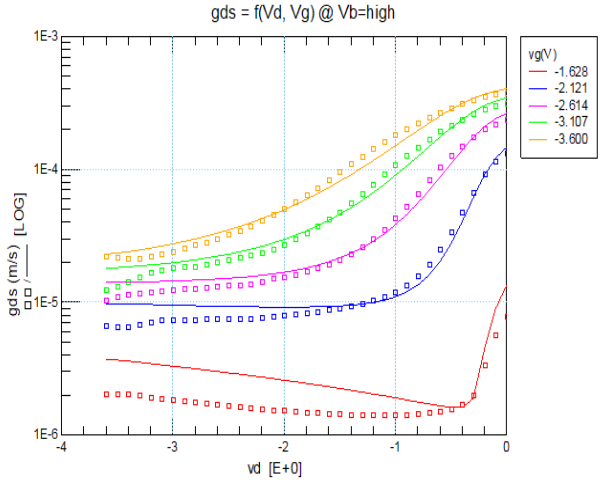




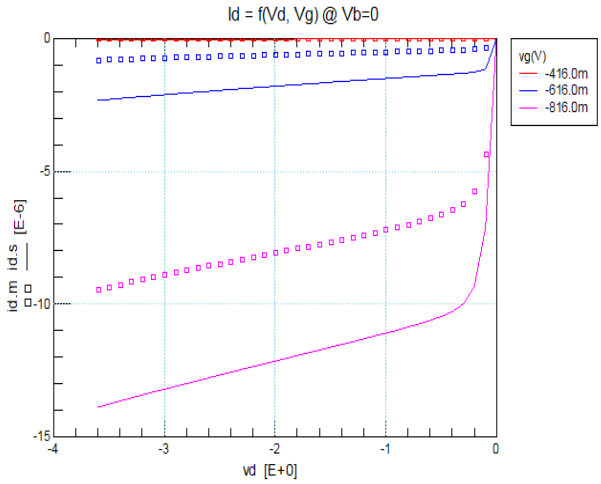
idvd_vbmin, Vb = 3.6V, T = 27°C

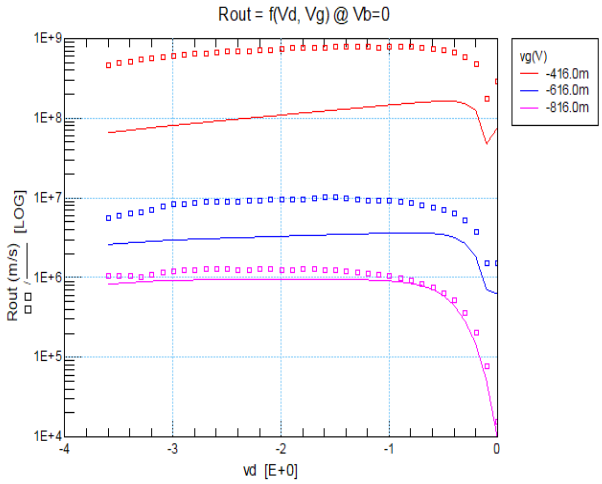


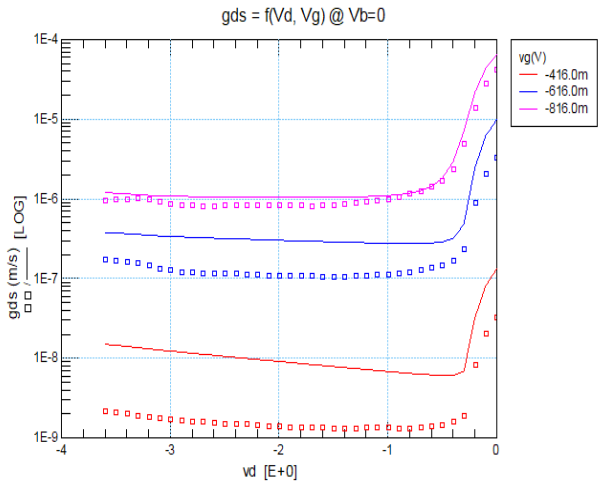




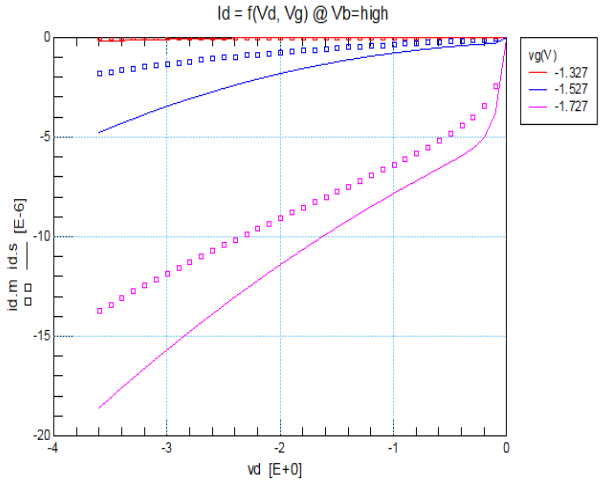
idvd_vth, Vb = 0V, T = 27°C

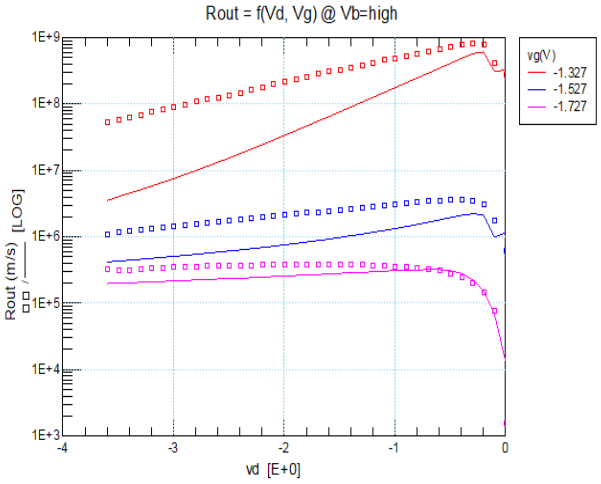


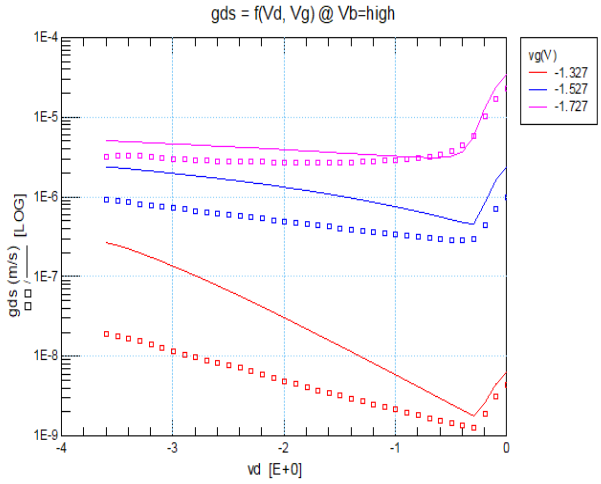




idvd_vth_vbmin, Vb = 3.6V, T = 27°C

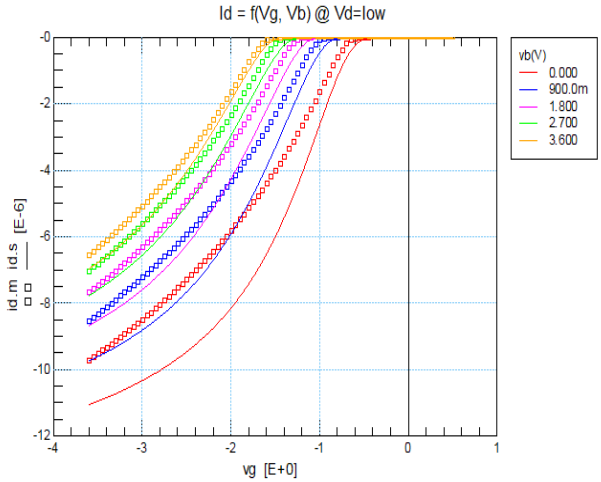


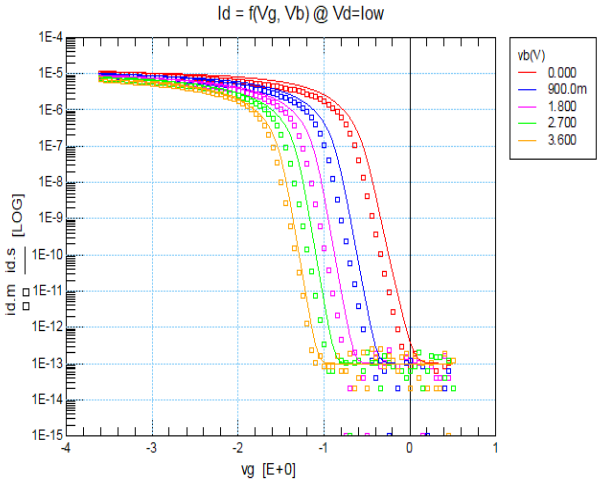


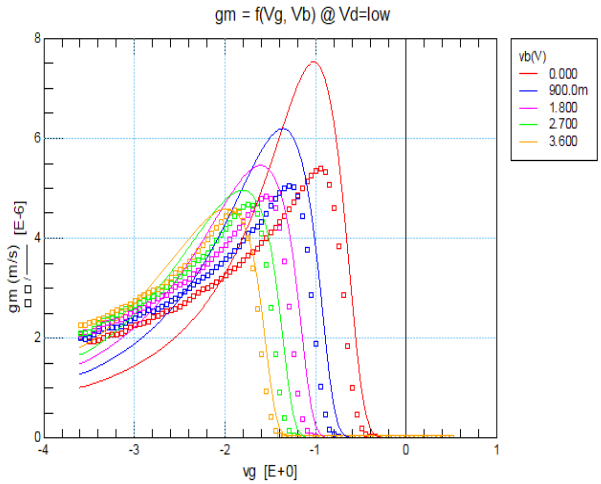


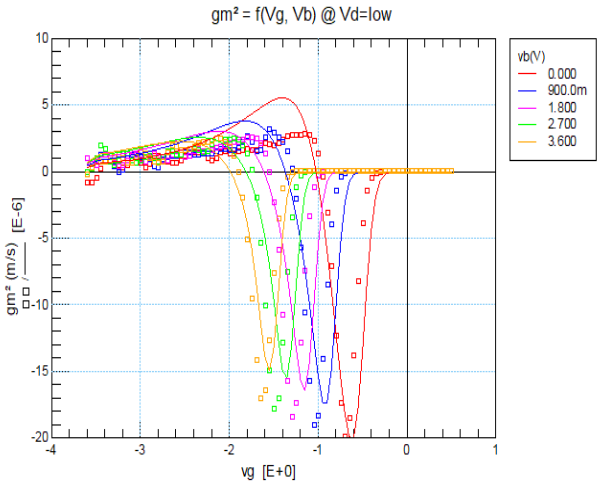
4.30 W0u3_L0u33_S565_2

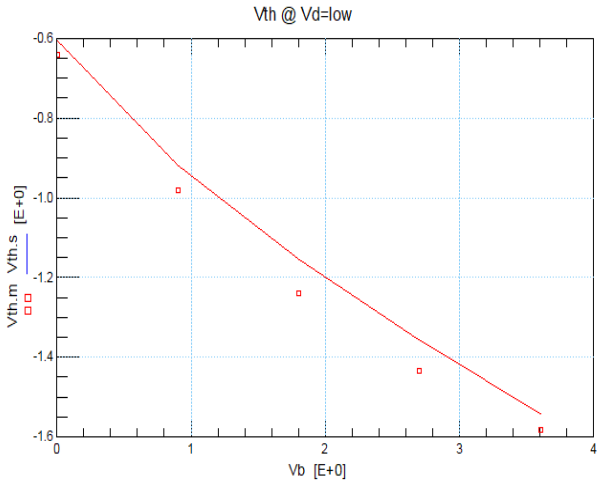
idvg, Vd = -0.1V, T = 27°C

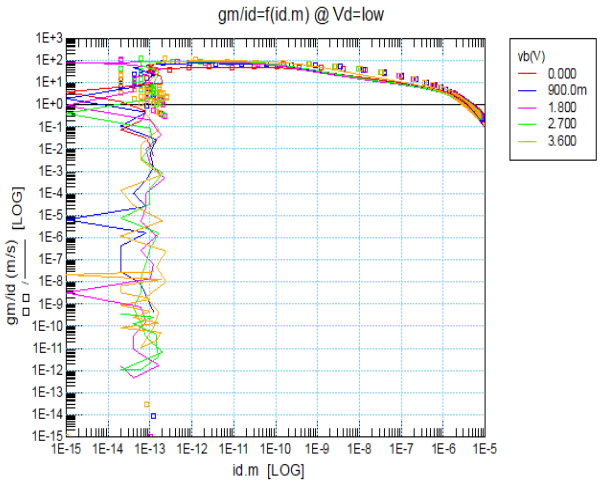


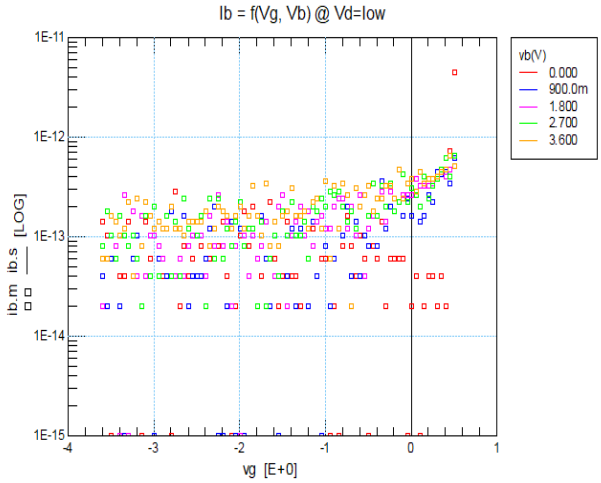


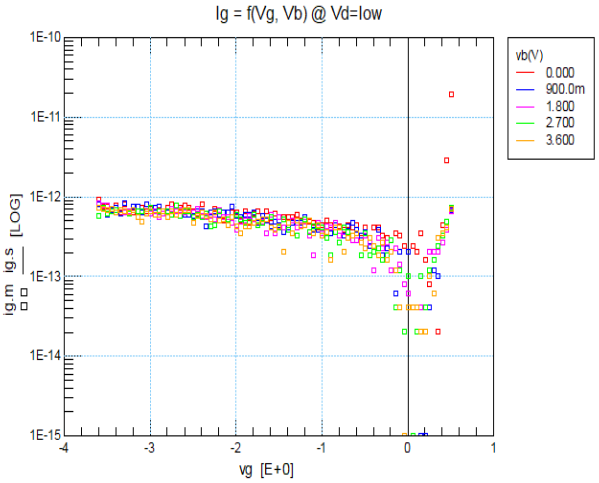




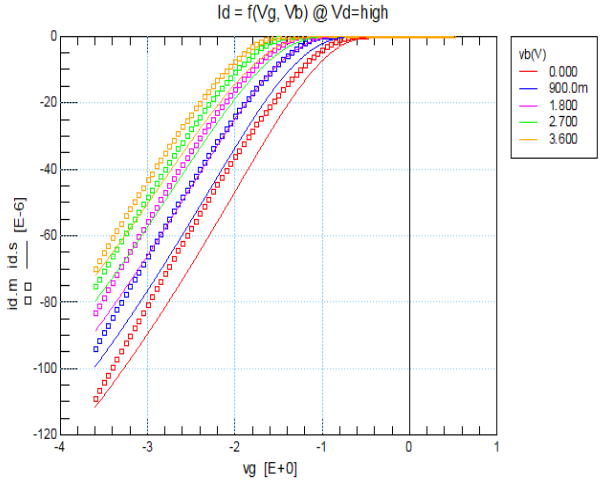


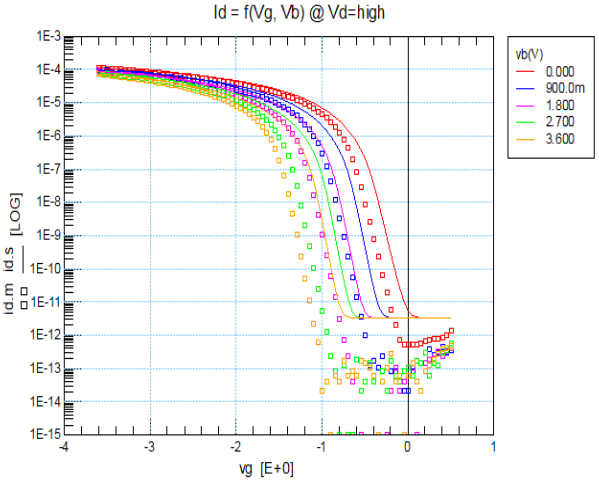


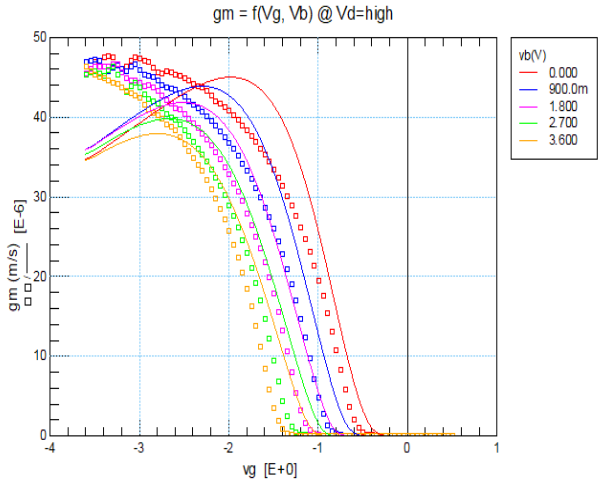


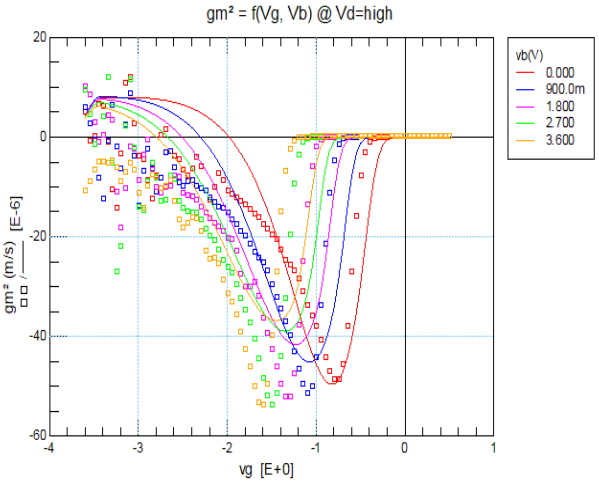


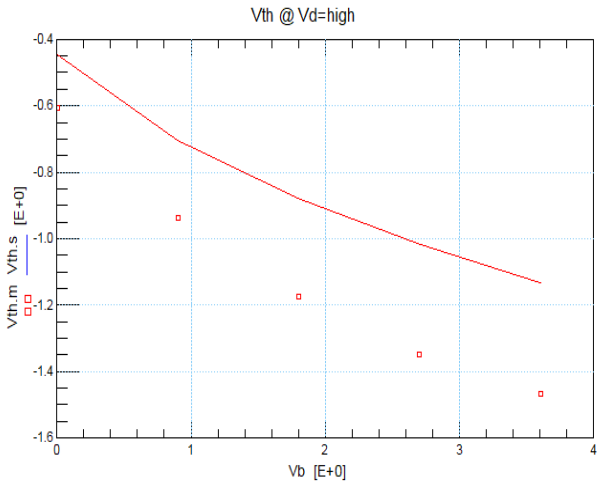
idvg, Vd = -3.3V, T = 27°C

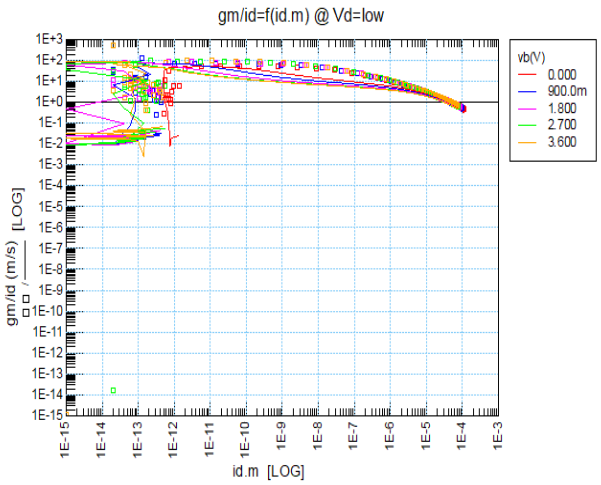


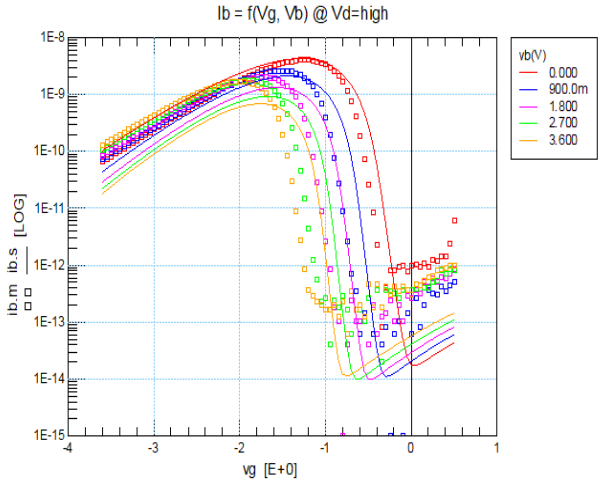


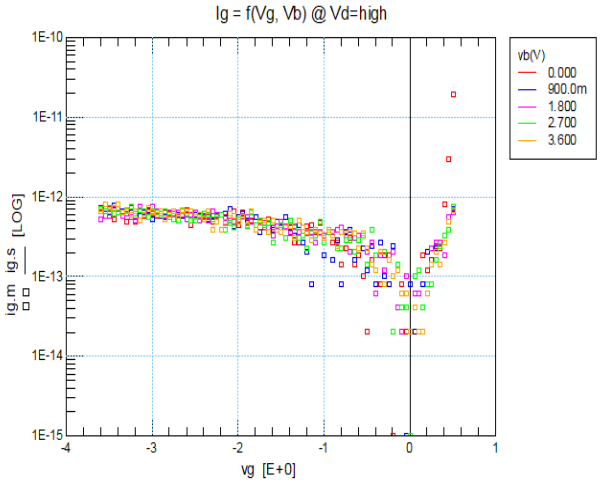




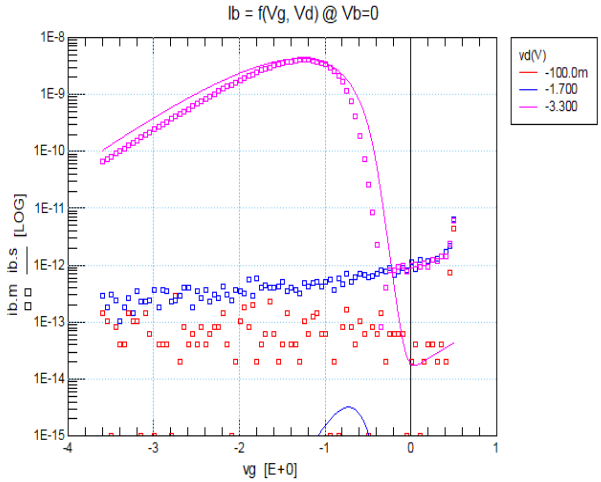


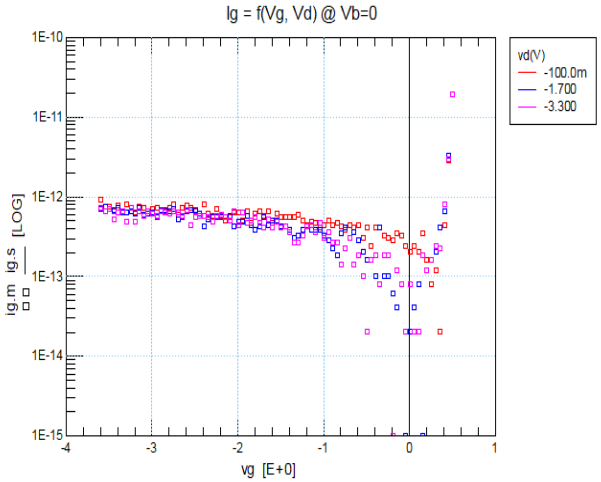




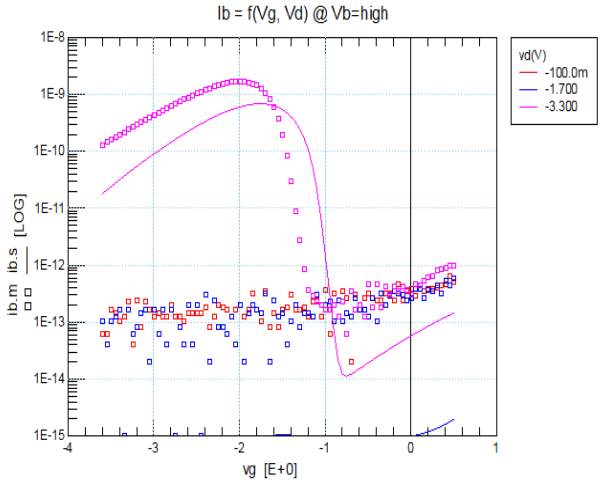


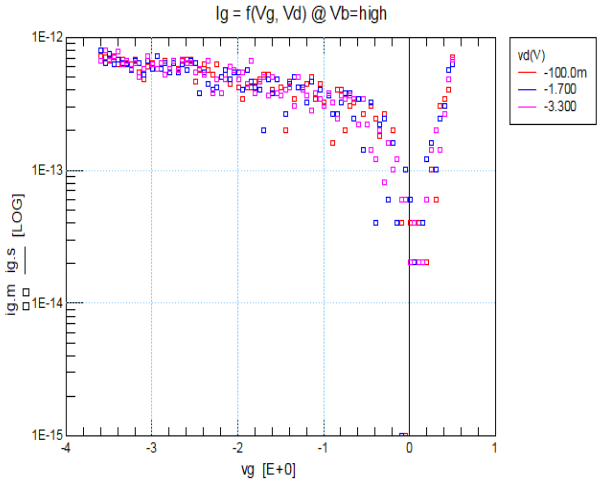
idvg, Ib,Ig, Vb = 0V, T = 27°C



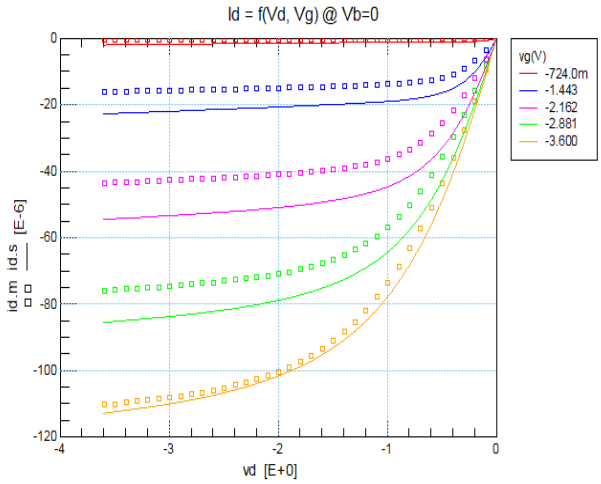


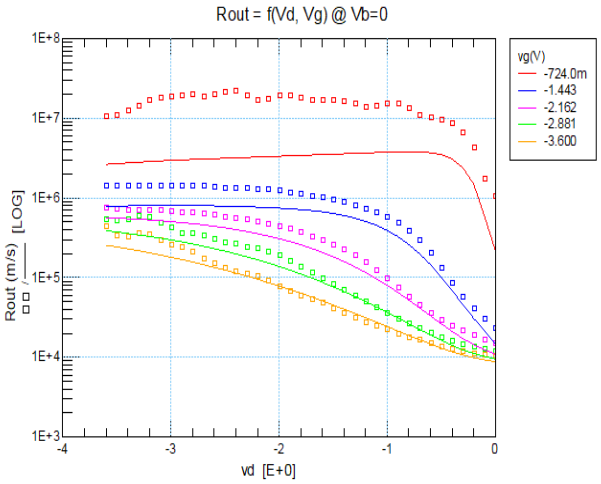
idvg, Ib,Ig, Vb = 3.6V, T = 27°C

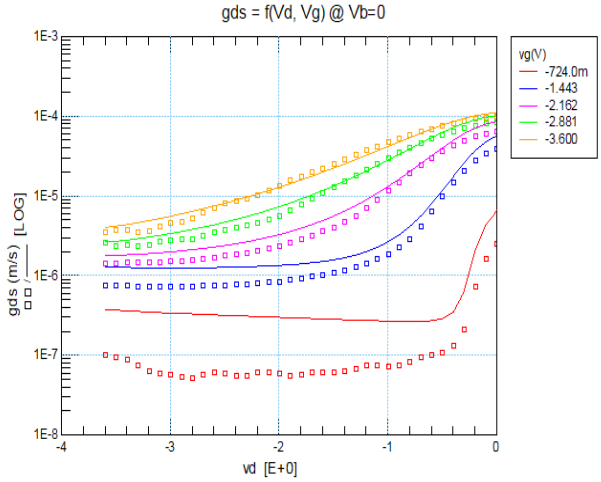




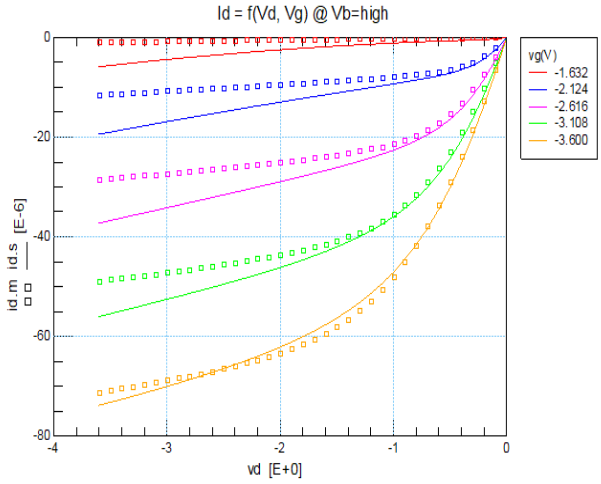
idvd, Vb = 0V, T = 27°C

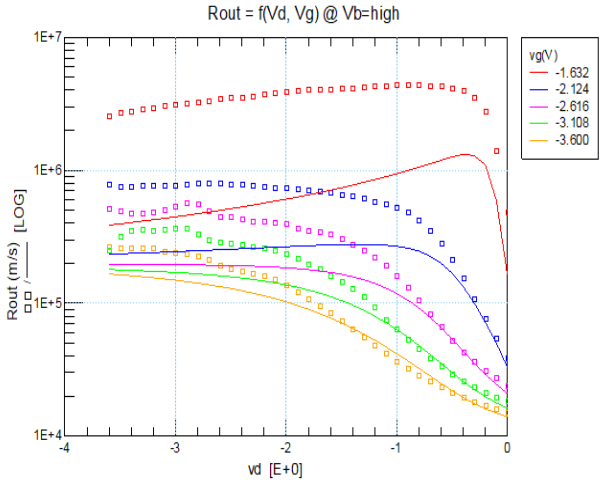


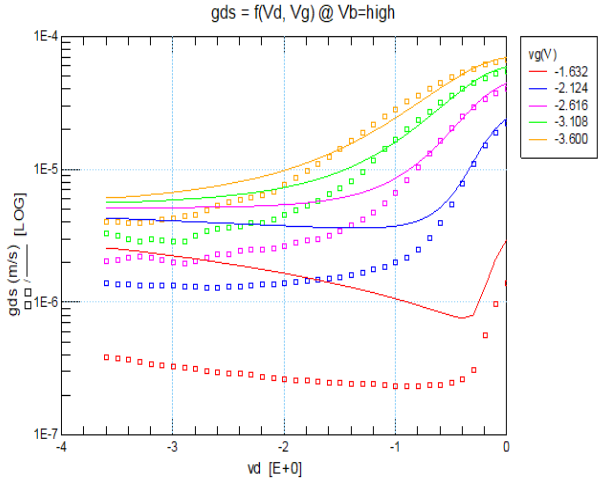




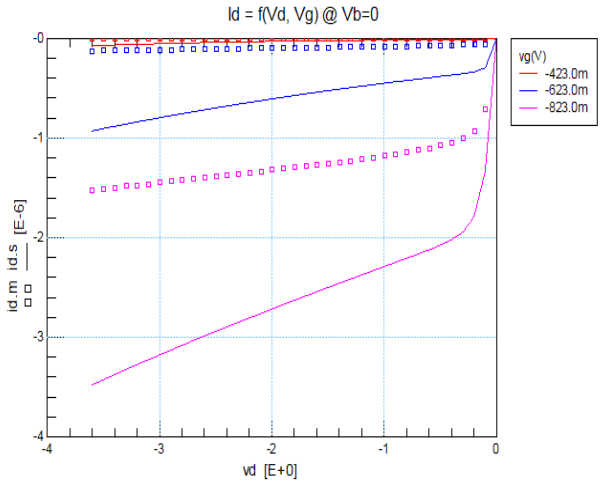
idvd_vbmin, Vb = 3.6V, T = 27°C

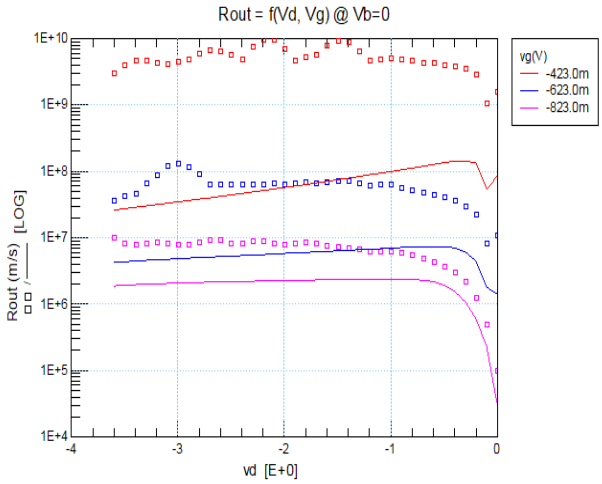


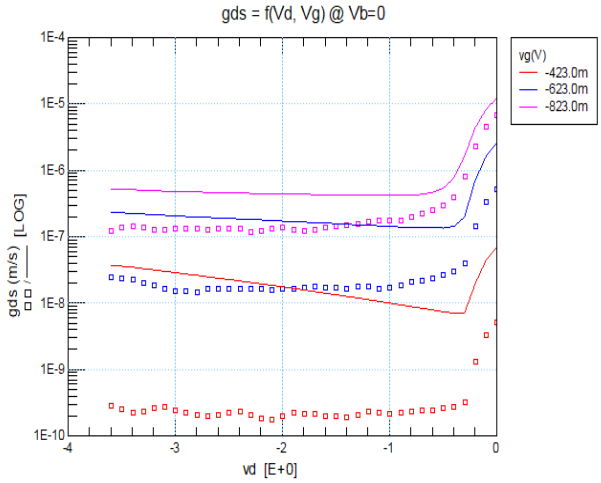




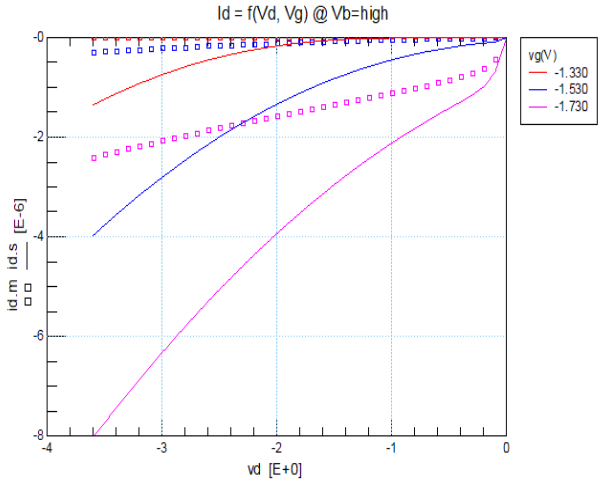
idvd_vth, Vb = 0V, T = 27°C

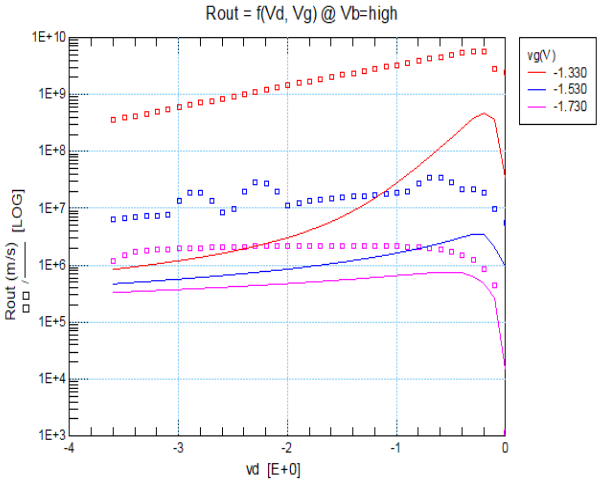


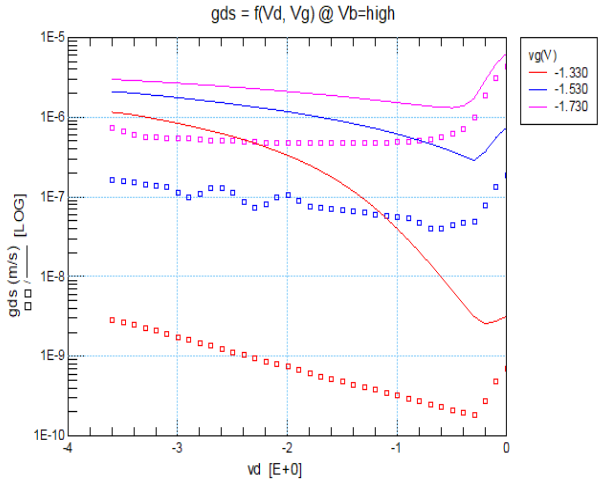




idvd_vth_vbmin, Vb = 3.6V, T = 27°C



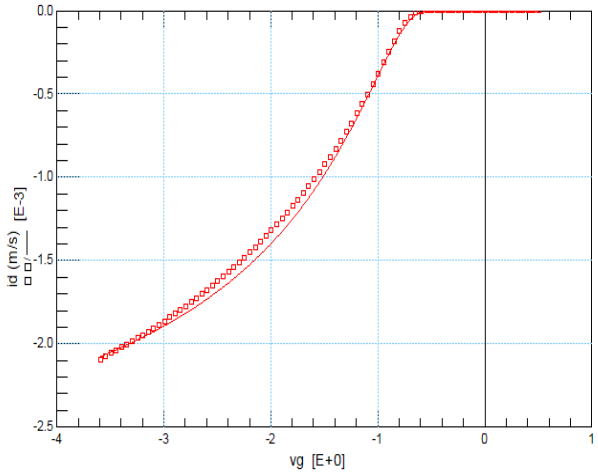




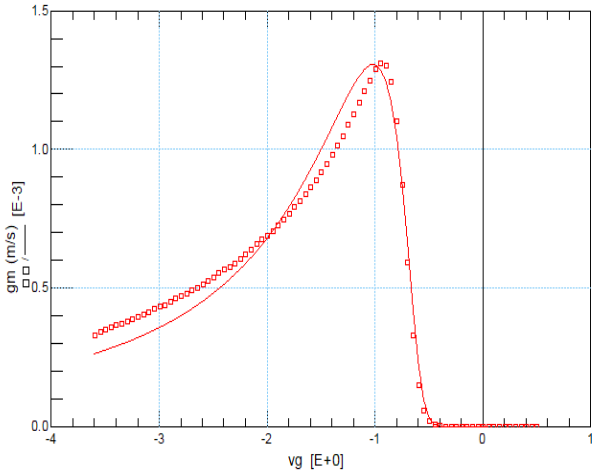
5 RF Transistors

5.1 NF10_WF10u0_L0u4_S453A_C4

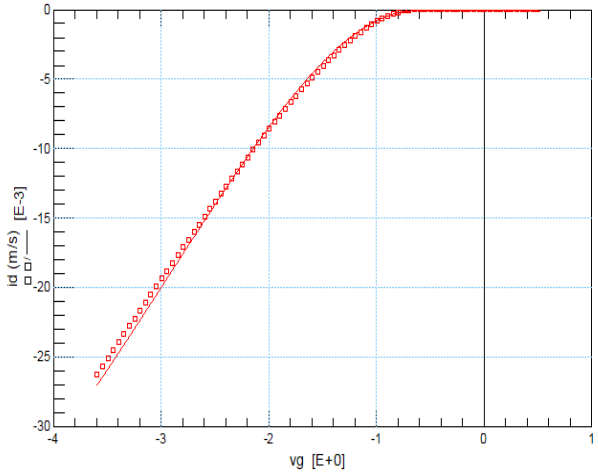
IdVg, T = 27°C



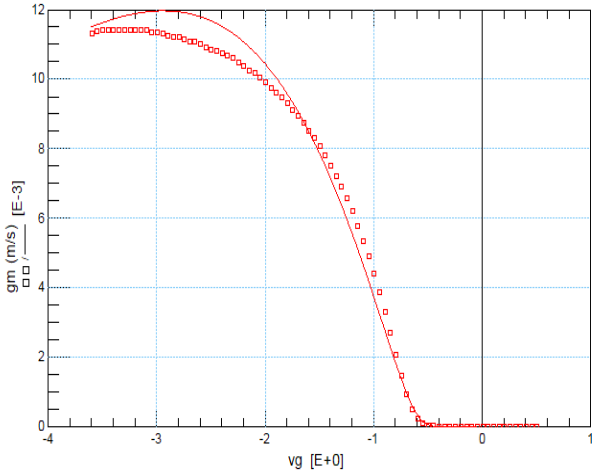
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

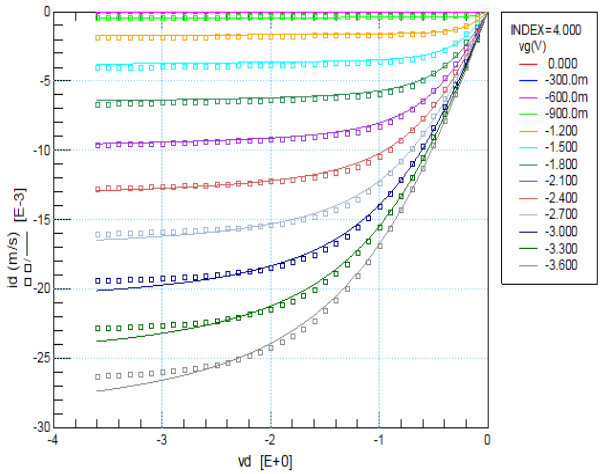


Id = f(Vg) @ Vd = -3.3V

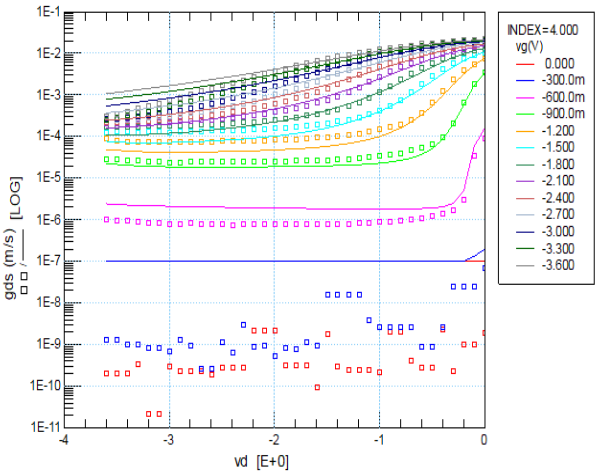


gm = f(Vg) @ Vd = -3.3V

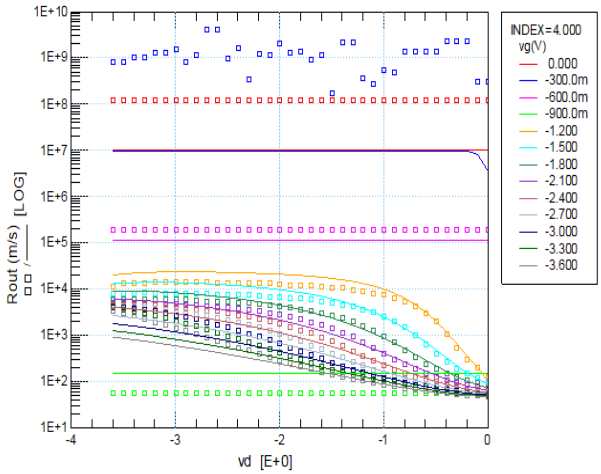
IdVd, T = 27°C



Id = f(Vd)

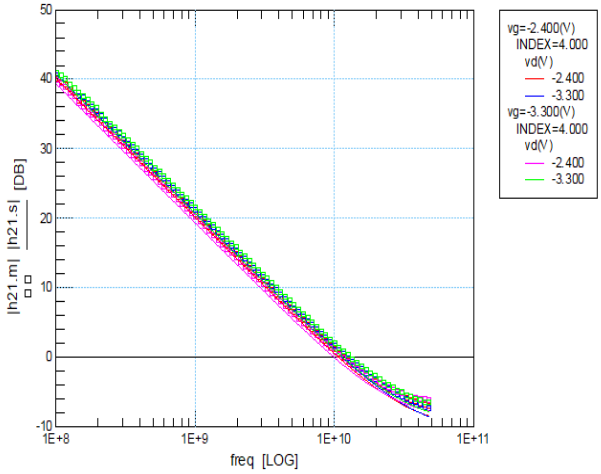


gds = f(Vd)

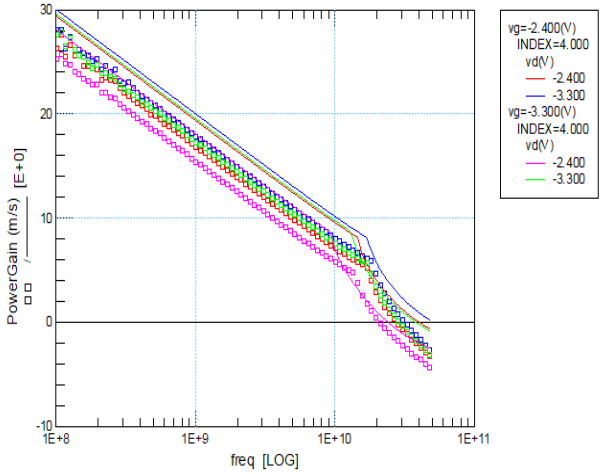


Rout = f(Vd)

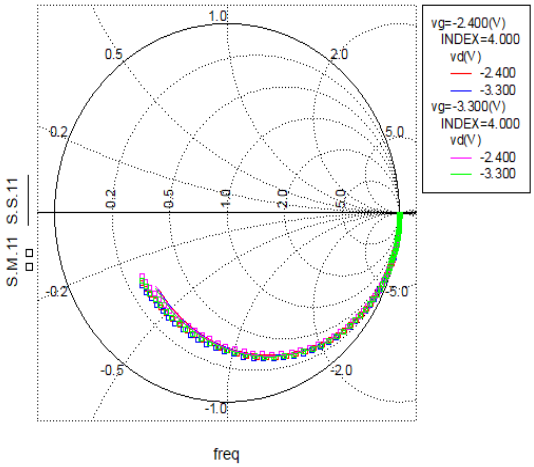
S-Parameter, T = 27°C



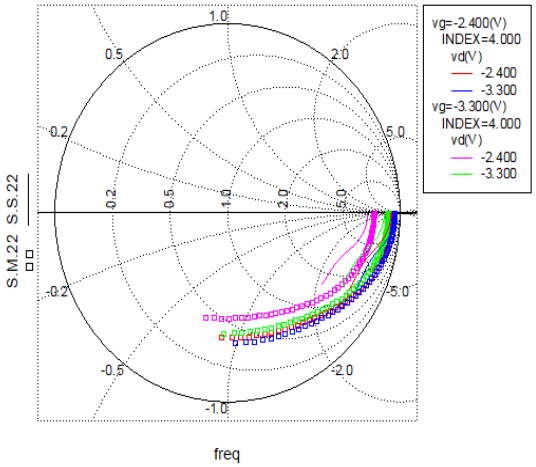
|h21| = f(Vg,Vd)



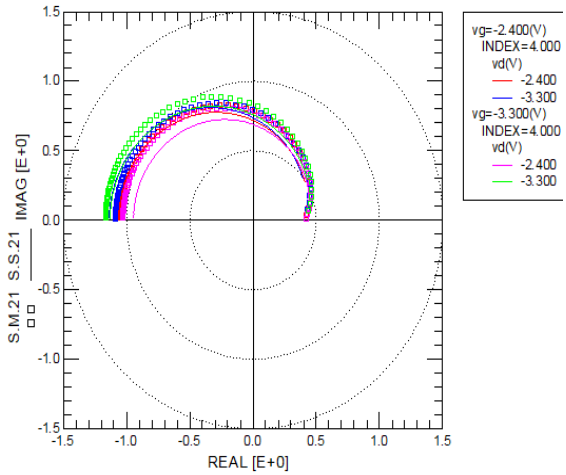
MSG/MAG = f(Vg,Vd)



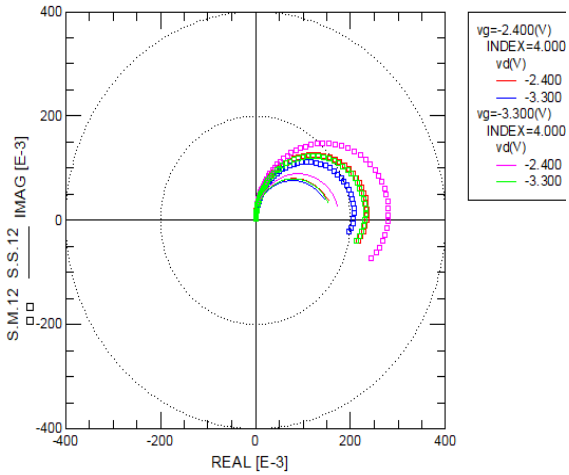
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

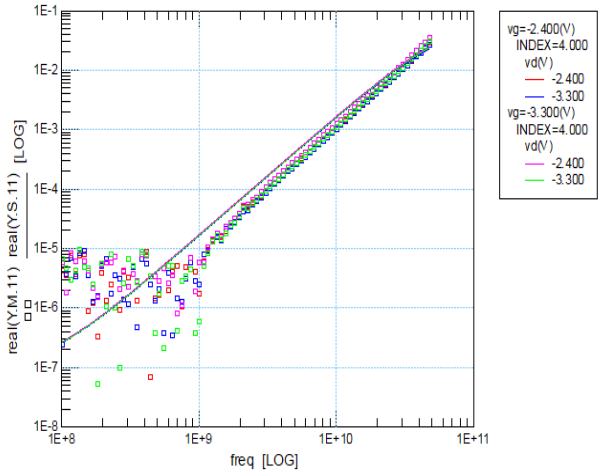


S21 = f(Vg,Vd)

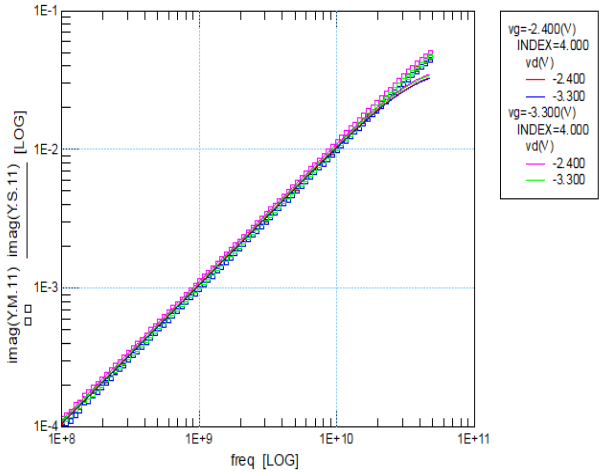


S12 = f(Vg,Vd)

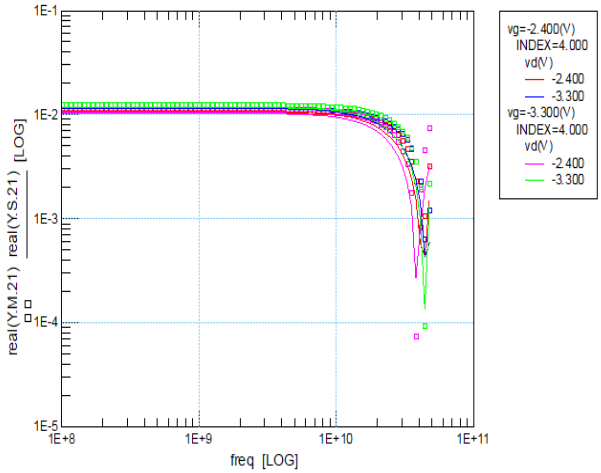
Y-Parameter, T = 27°C



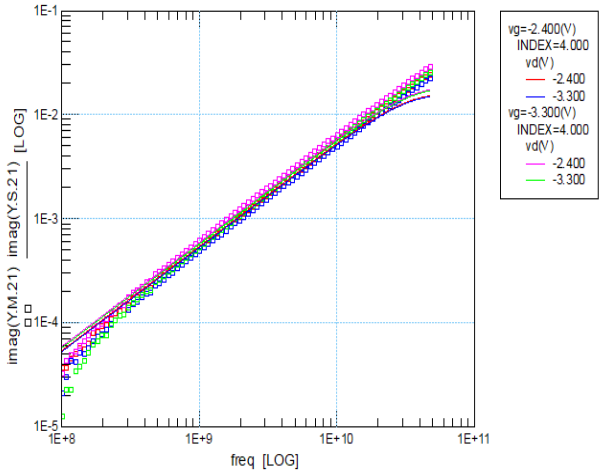
re(Y11) = f(Vg,Vd)



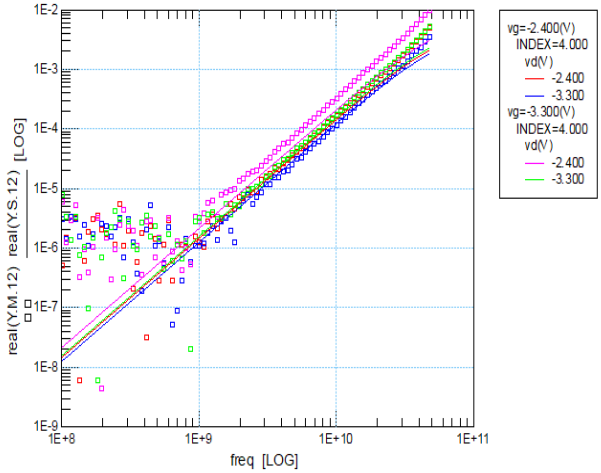
im(Y11) = f(Vg,Vd)



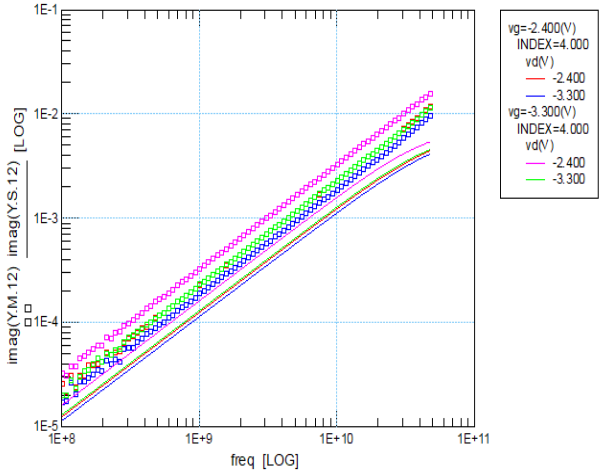
re(Y21) = f(Vg,Vd)



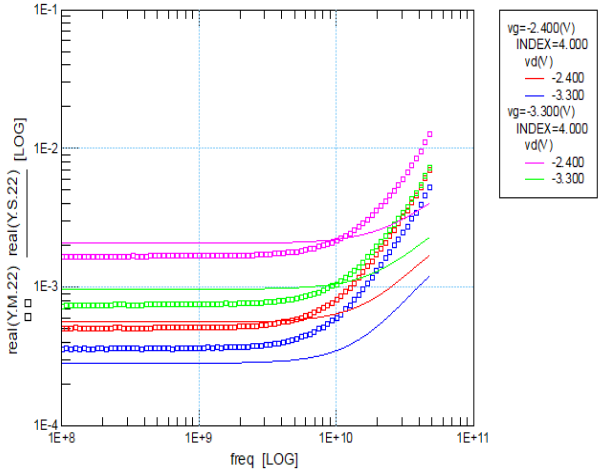
im(Y21) = f(Vg,Vd)



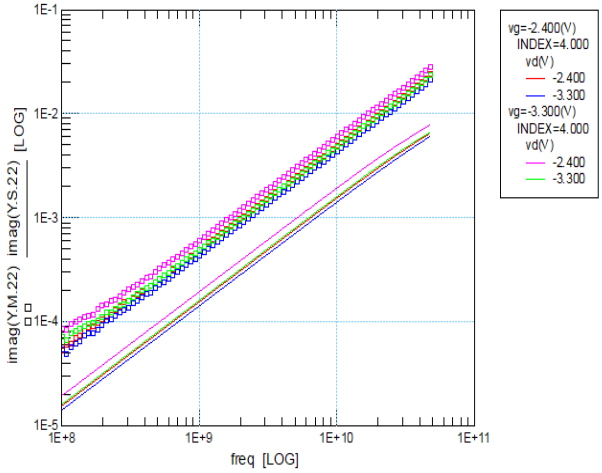
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

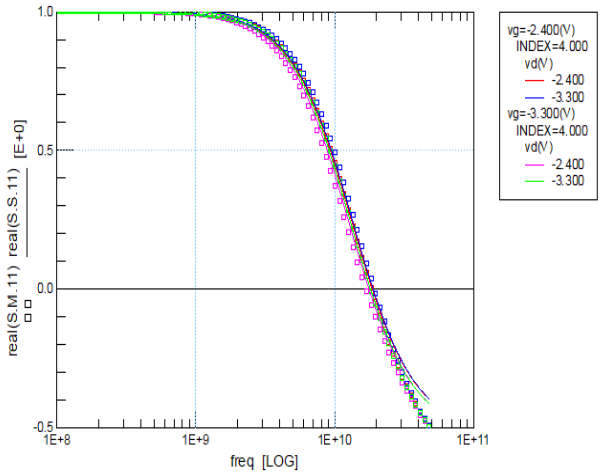


re(Y22) = f(Vg,Vd)

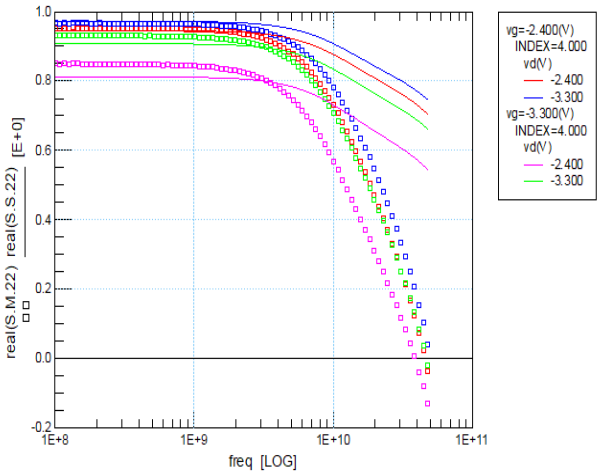


im(Y22) = f(Vg,Vd)

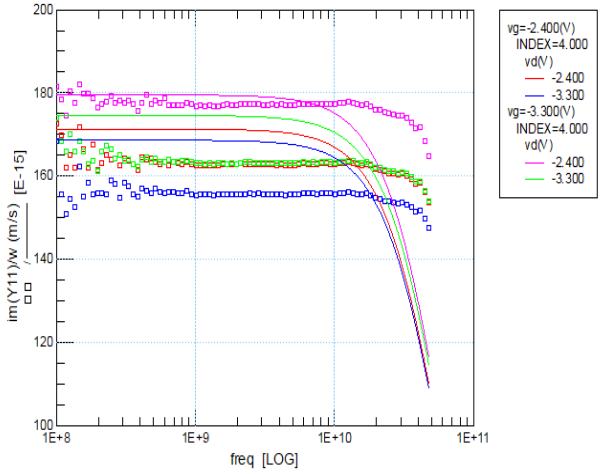
Verify Plots, T = 27°C



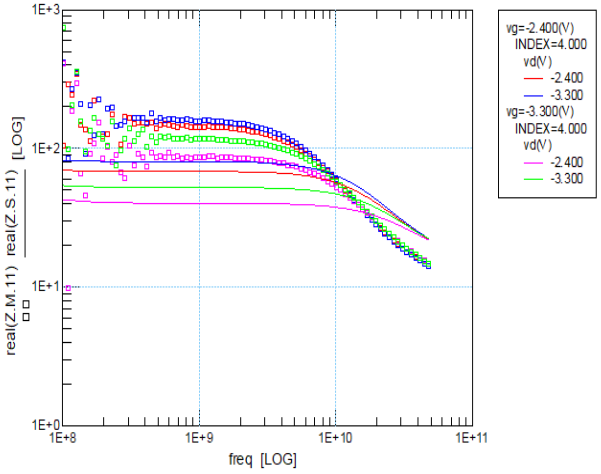
re(S11) = f(Vg,Vd)



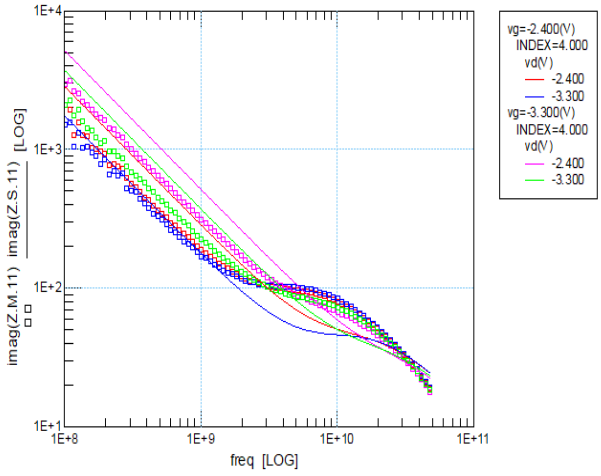
re(S22) = f(Vg,Vd)



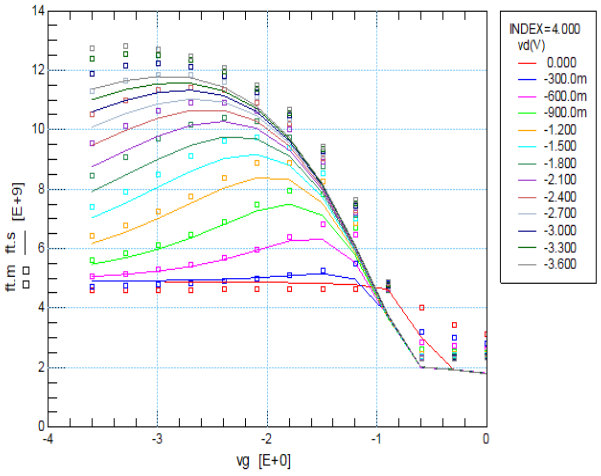
im(Y11) divided by ω



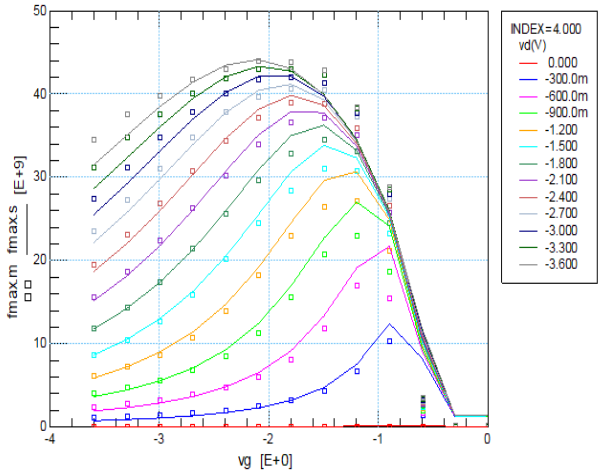
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



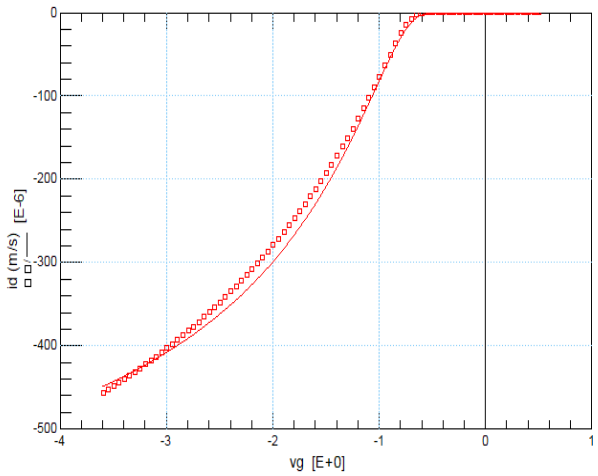
fT_vg = f(Vg,Vd)



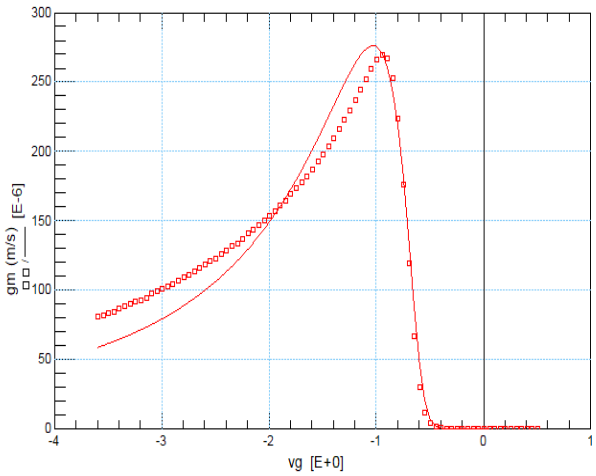
fmax_vg = f(Vg,Vd)

5.2 NF04_WF05u0_L0u4_S453A_C5

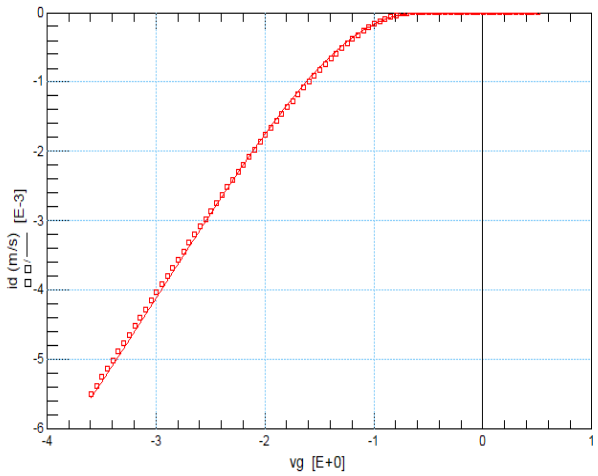
IdVg, T = 27°C



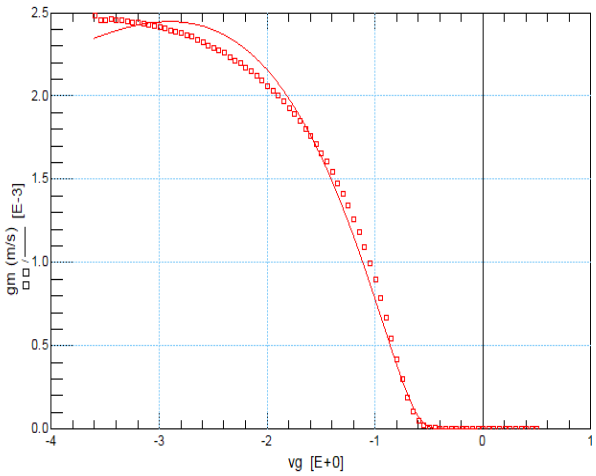
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

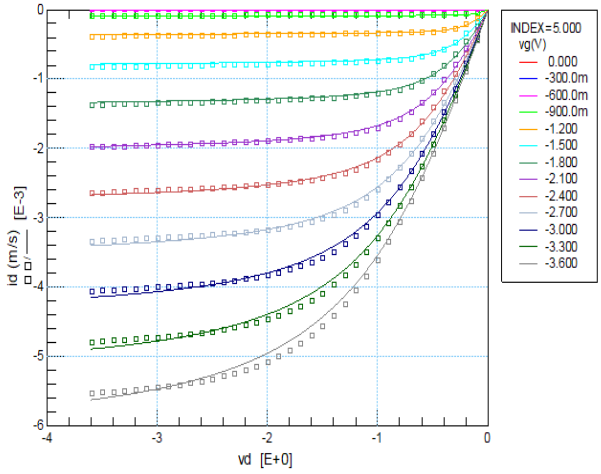


Id = f(Vg) @ Vd = -3.3V

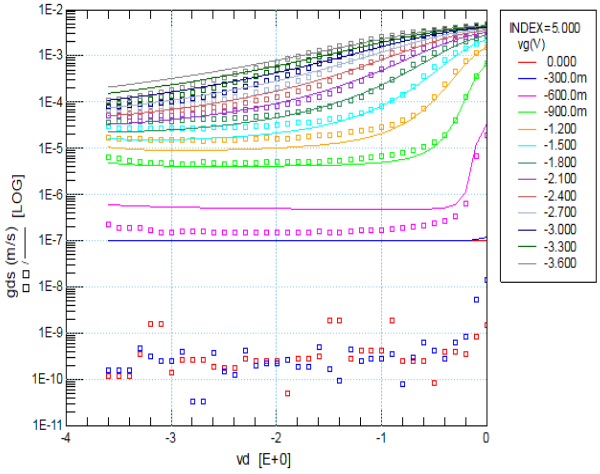


gm = f(Vg) @ Vd = -3.3V

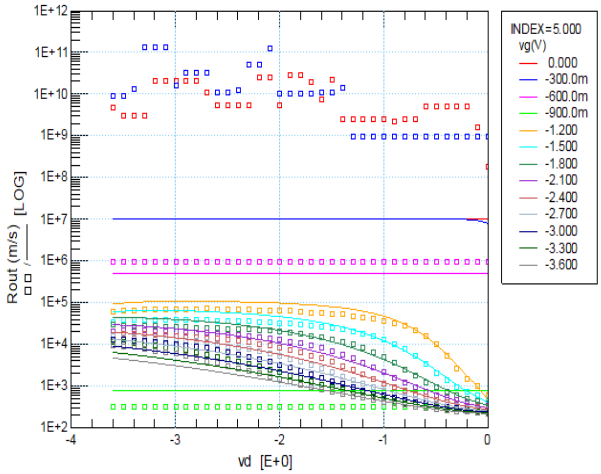
IdVd, T = 27°C



Id = f(Vd)

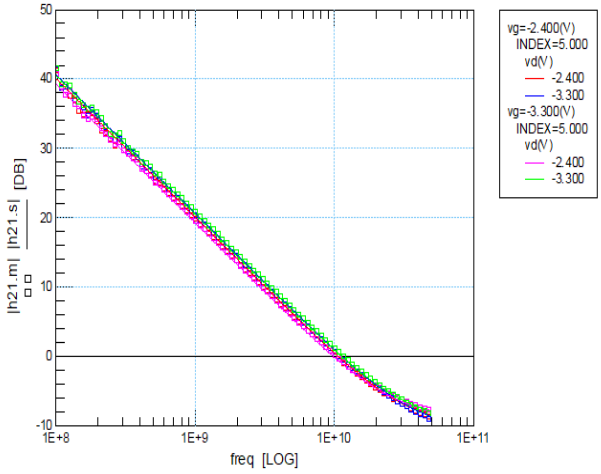


gds = f(Vd)

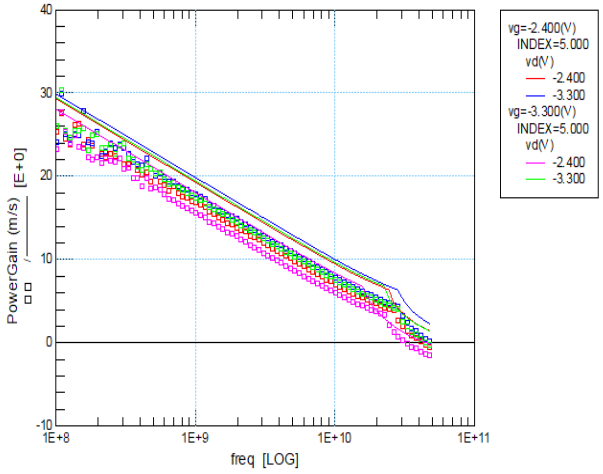


Rout = f(Vd)

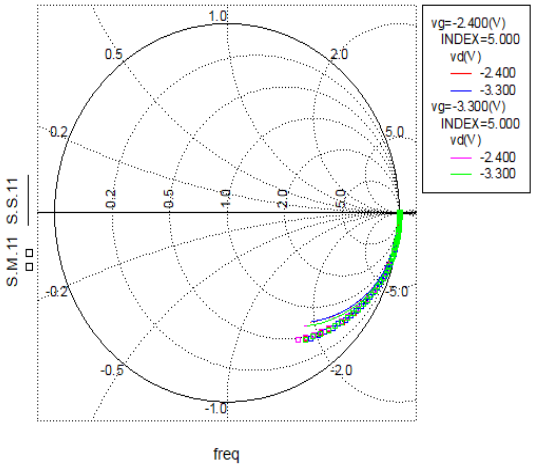
S-Parameter, T = 27°C



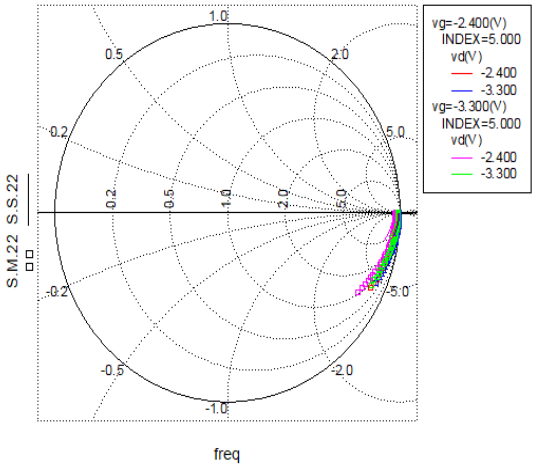
|h21| = f(Vg,Vd)



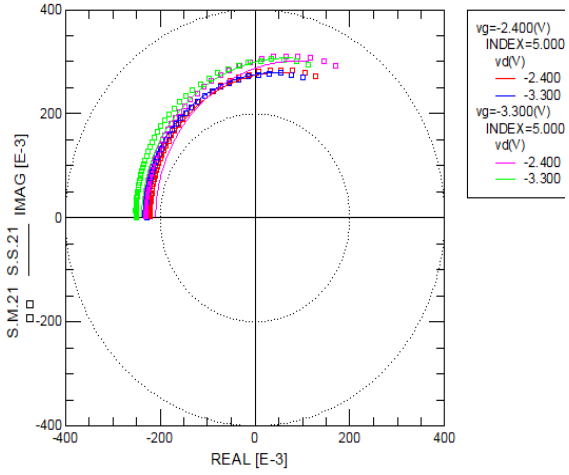
MSG/MAG = f(Vg,Vd)



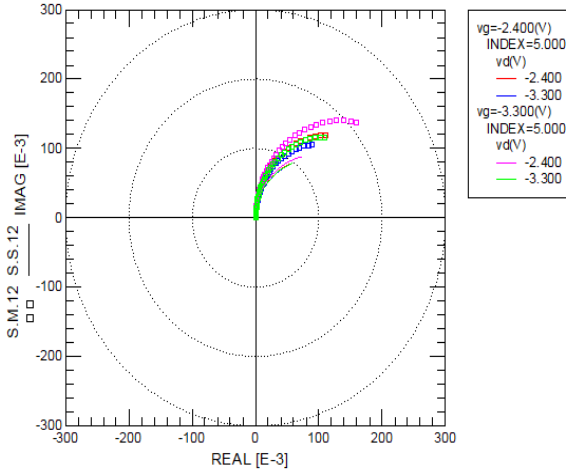
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

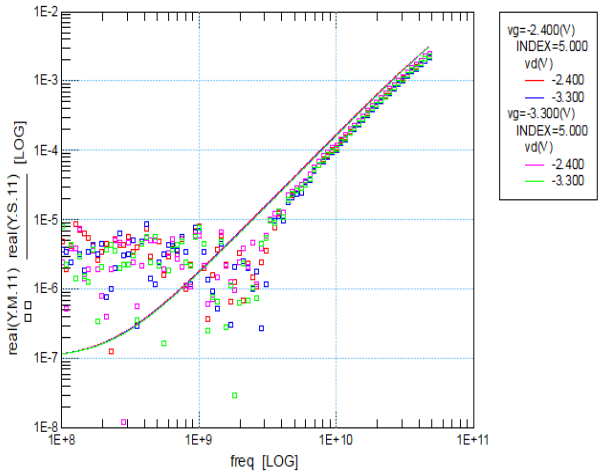


S21 = f(Vg,Vd)

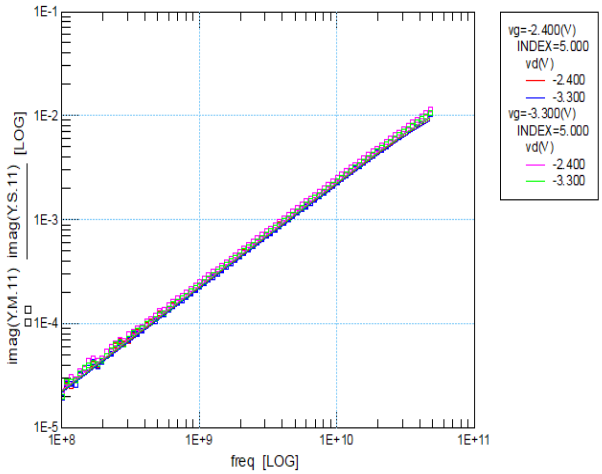


S12 = f(Vg,Vd)

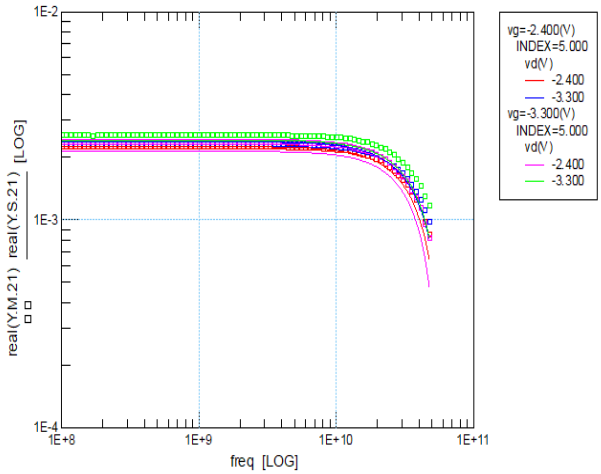
Y-Parameter, T = 27°C



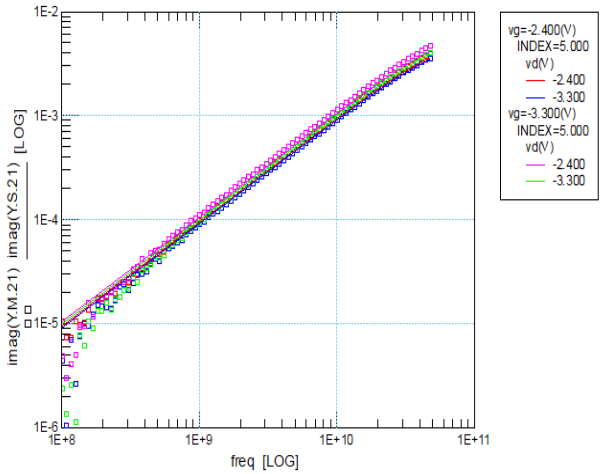
re(Y11) = f(Vg,Vd)



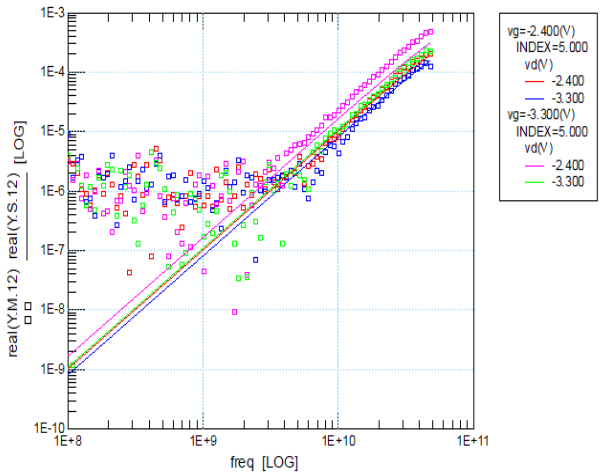
im(Y11) = f(Vg,Vd)



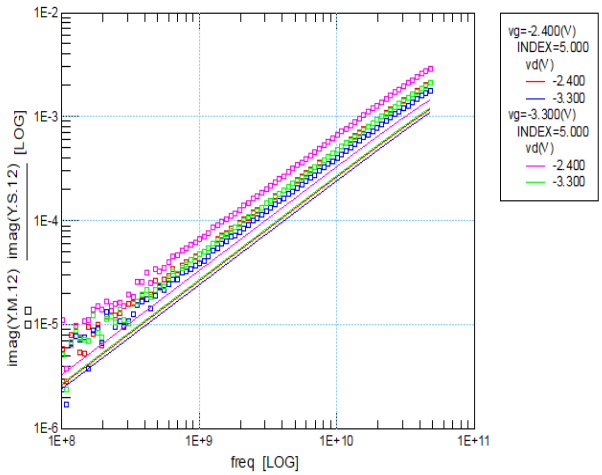
re(Y21) = f(Vg,Vd)



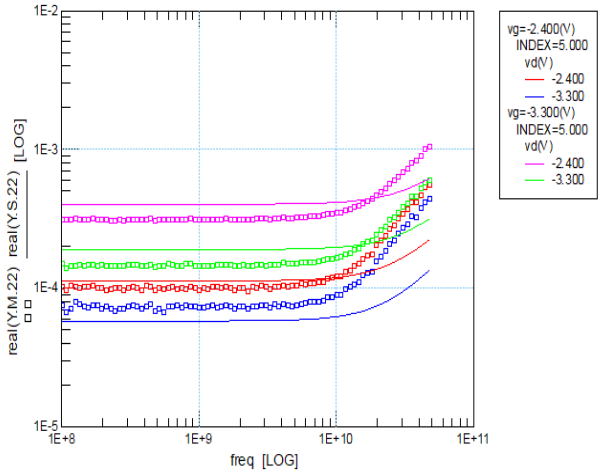
im(Y21) = f(Vg,Vd)



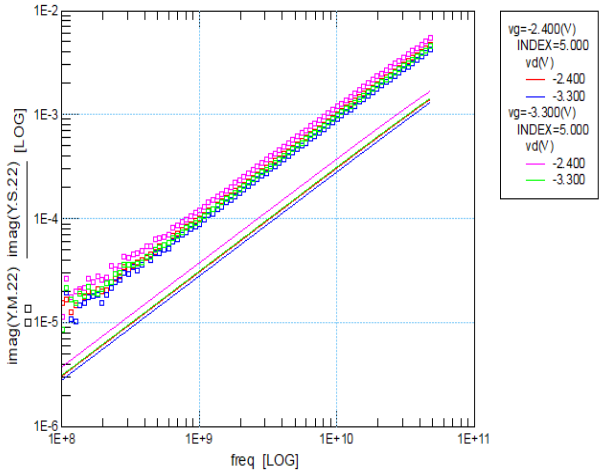
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

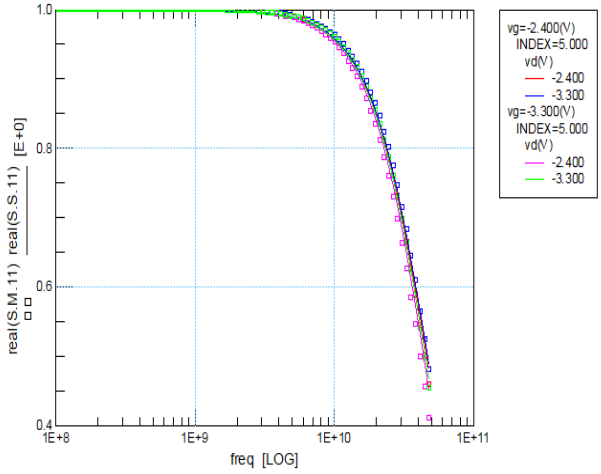


re(Y22) = f(Vg,Vd)

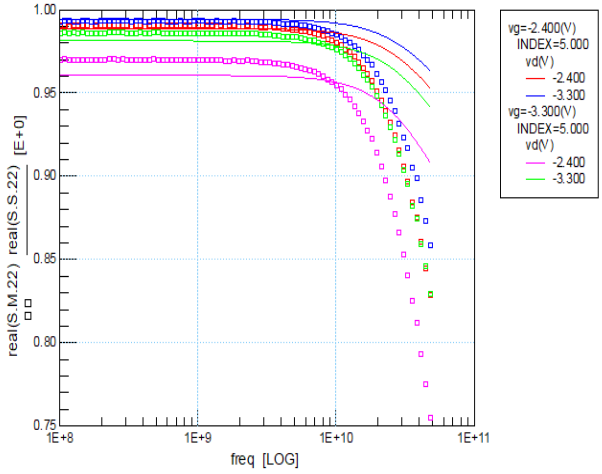


im(Y22) = f(Vg,Vd)

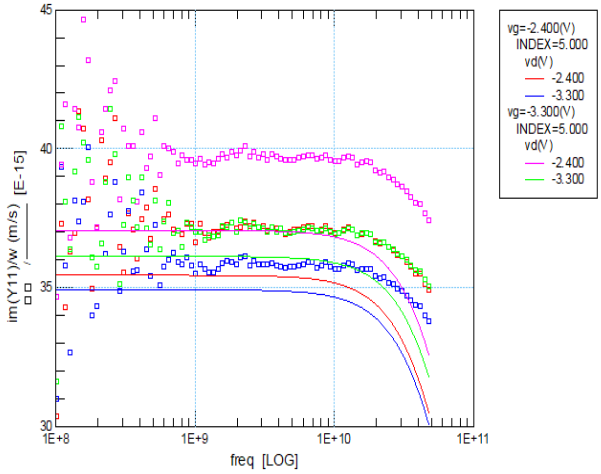
Verify Plots, T = 27°C



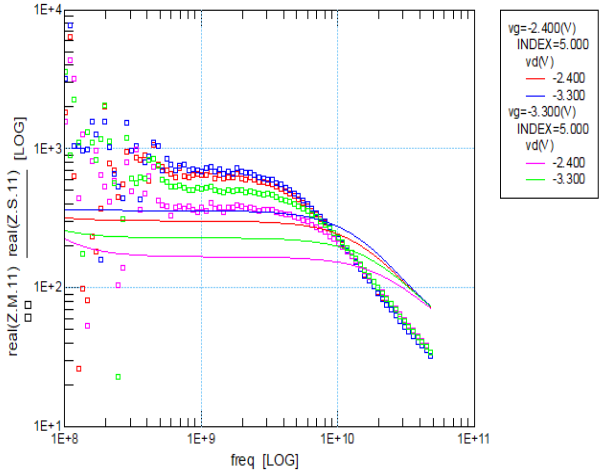
re(S11) = f(Vg,Vd)



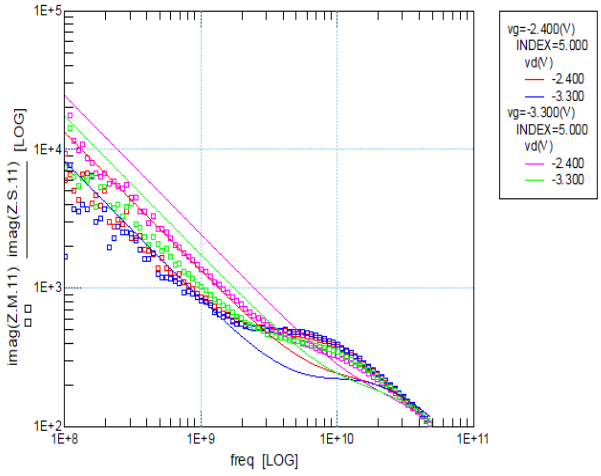
re(S22) = f(Vg,Vd)



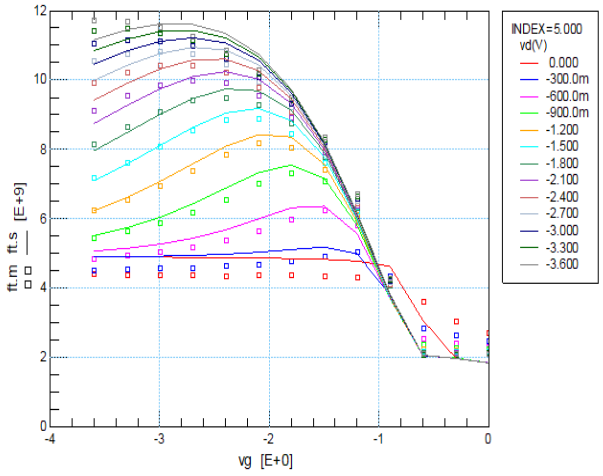
im(Y11) divided by ω



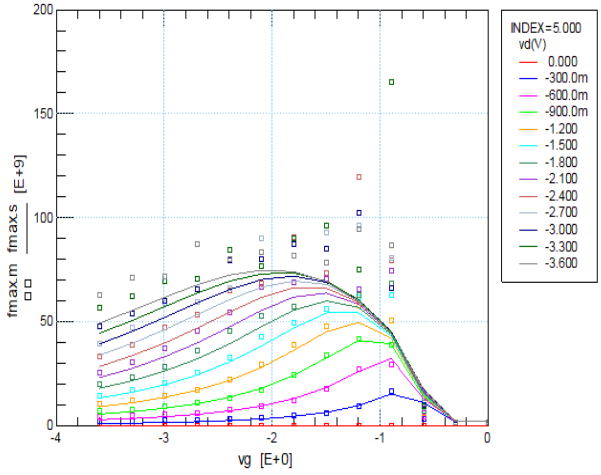
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



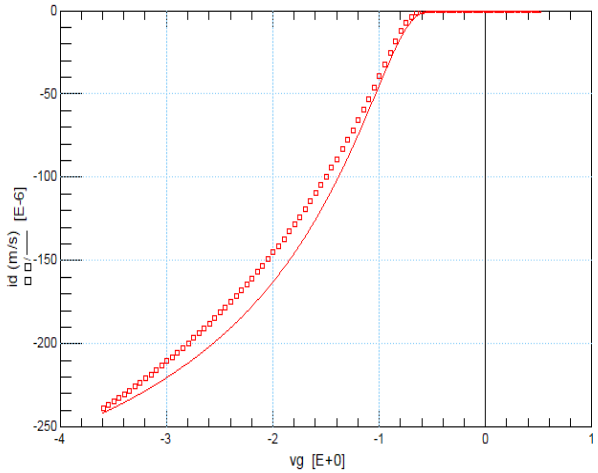
fT_vg = f(Vg,Vd)



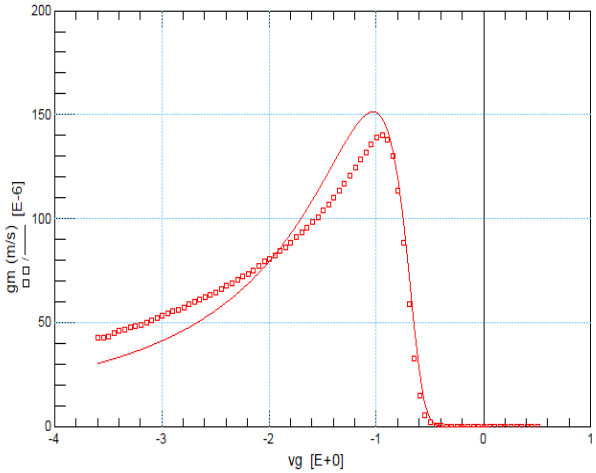
fmax_vg = f(Vg,Vd)

5.3 NF04_WF02u5_L0u4_S453A_D1

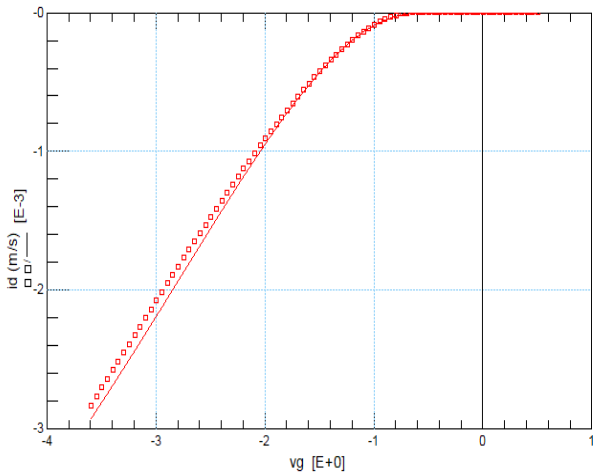
IdVg, T = 27°C



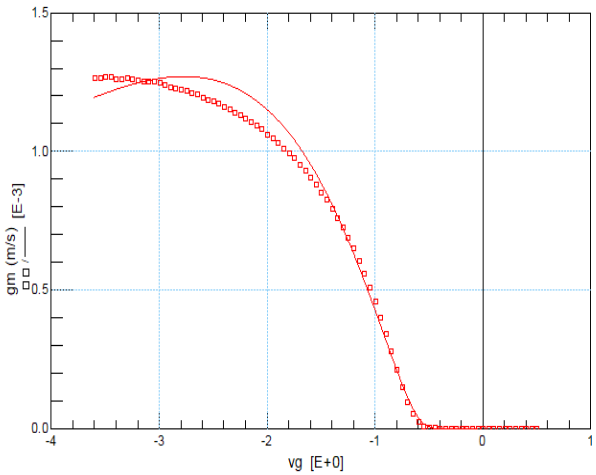
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

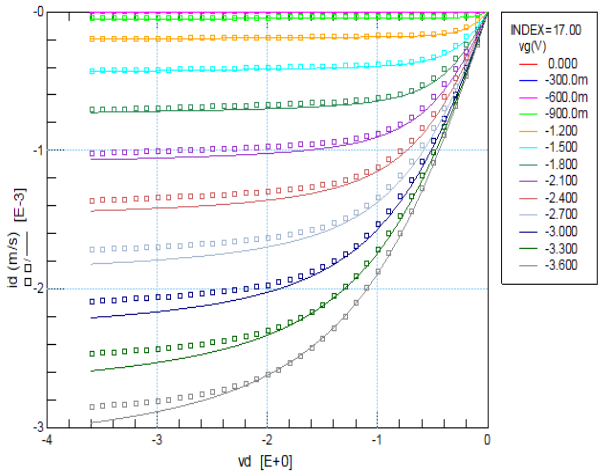


Id = f(Vg) @ Vd = -3.3V

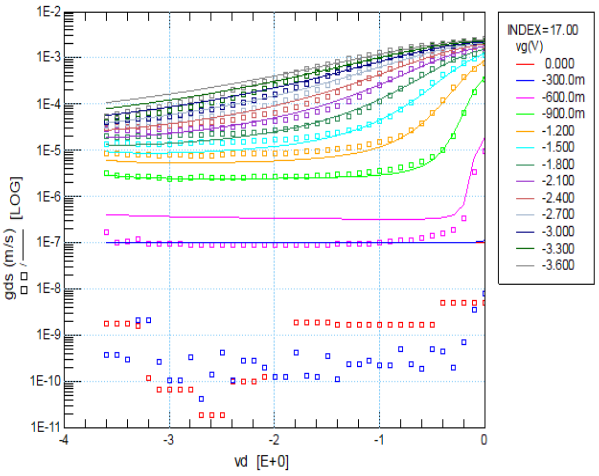


gm = f(Vg) @ Vd = -3.3V

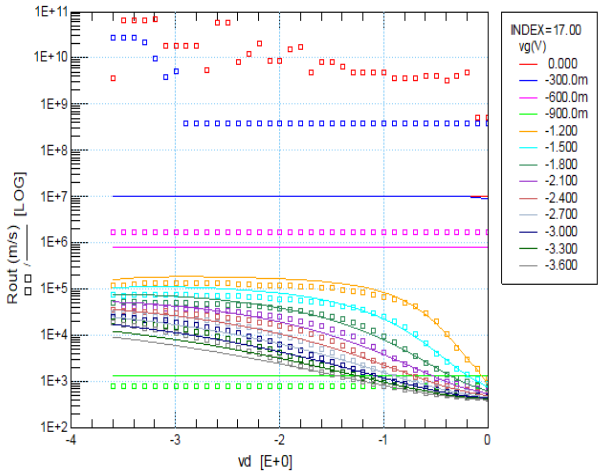
IdVd, T = 27°C



Id = f(Vd)

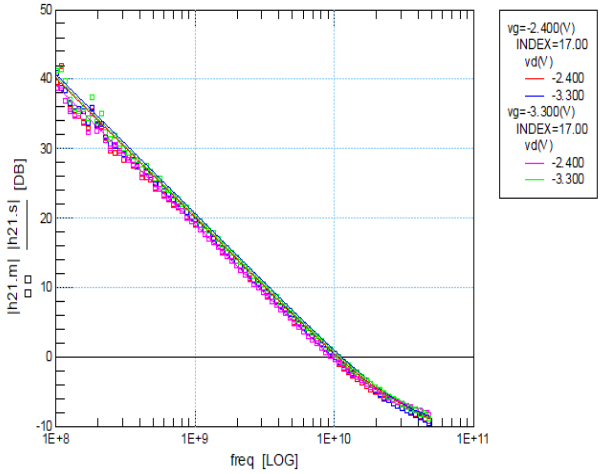


gds = f(Vd)

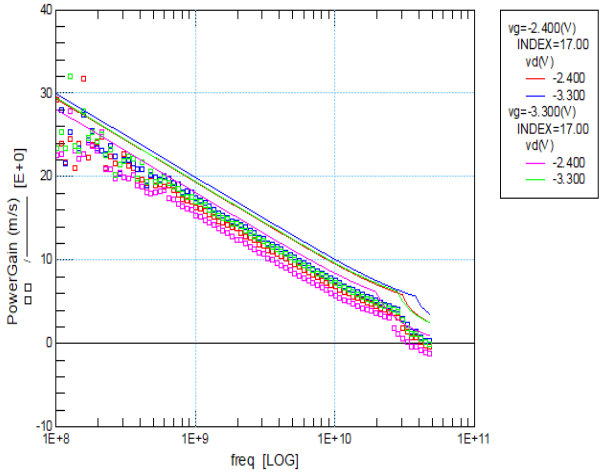


Rout = f(Vd)

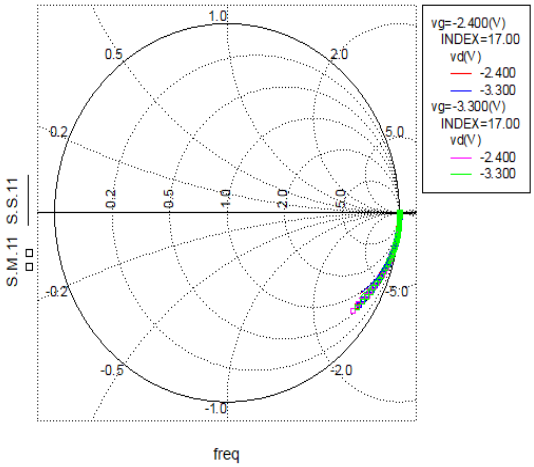
S-Parameter, T = 27°C



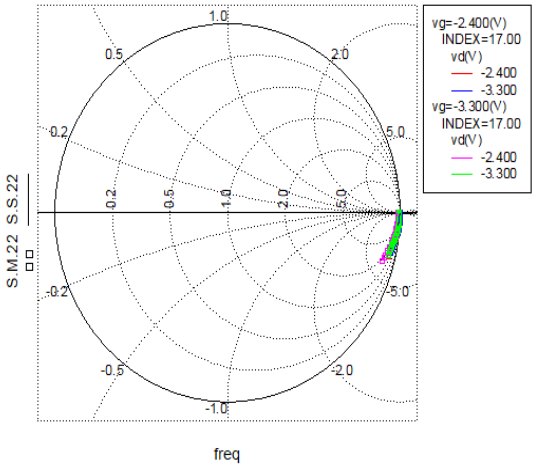
|h21| = f(Vg,Vd)



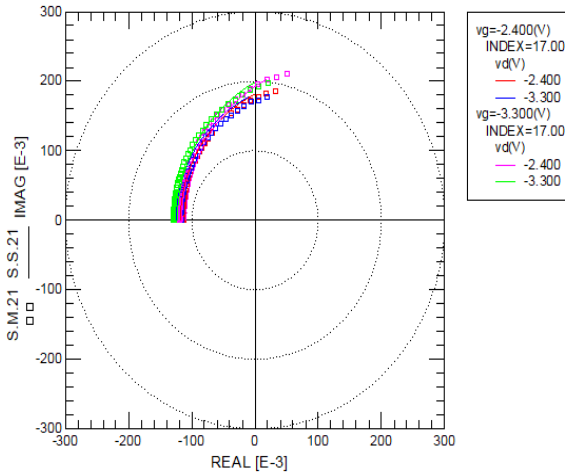
MSG/MAG = f(Vg,Vd)



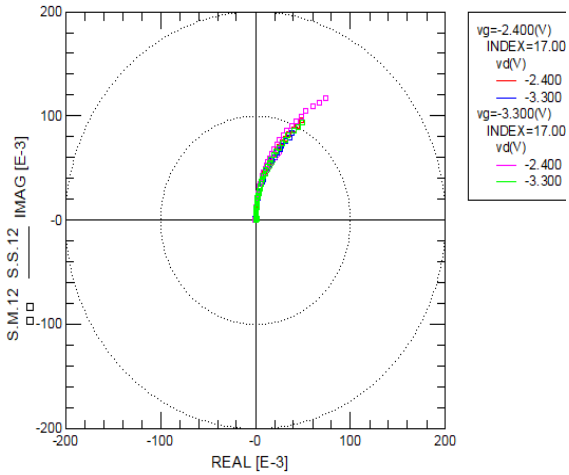
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

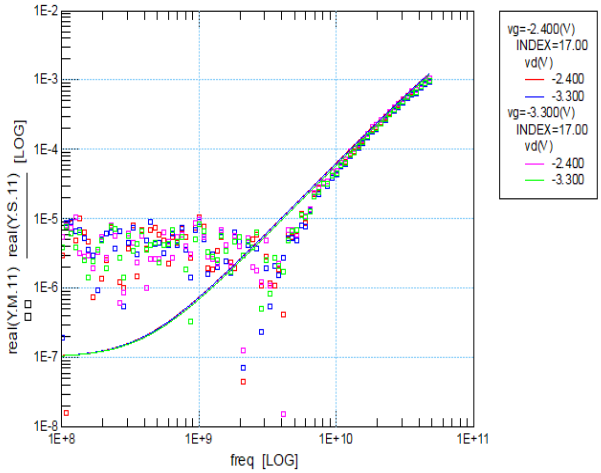


S21 = f(Vg,Vd)

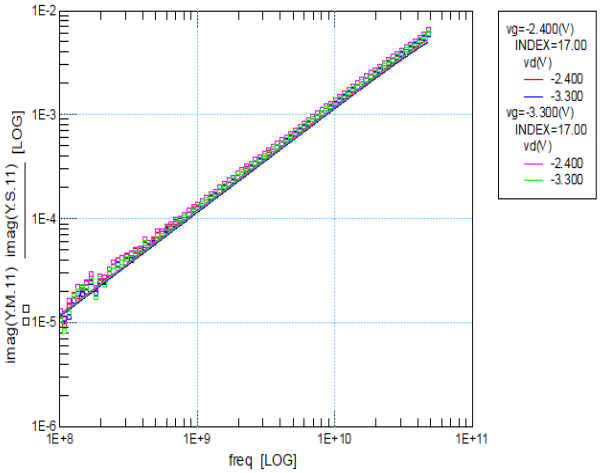


S12 = f(Vg,Vd)

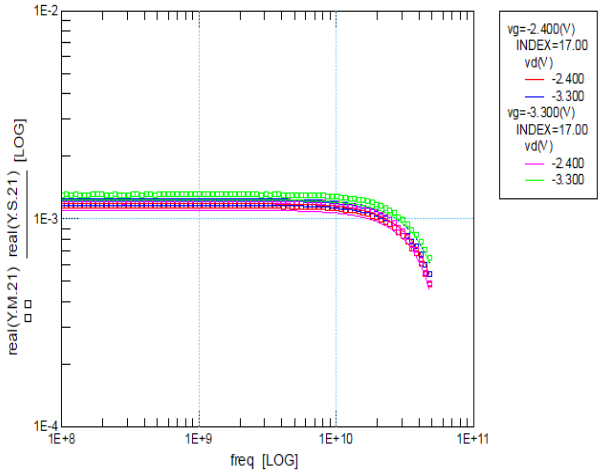
Y-Parameter, T = 27°C



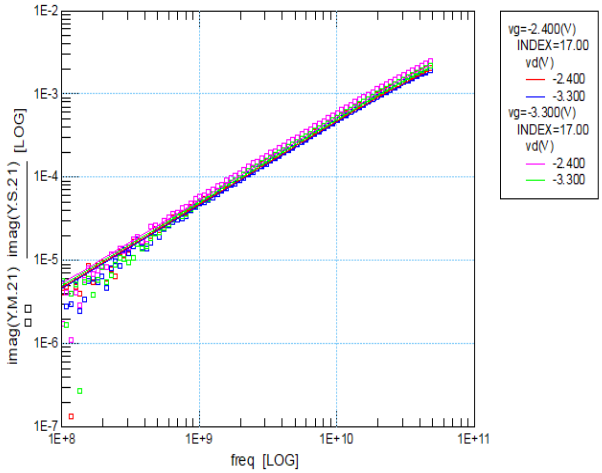
re(Y11) = f(Vg,Vd)



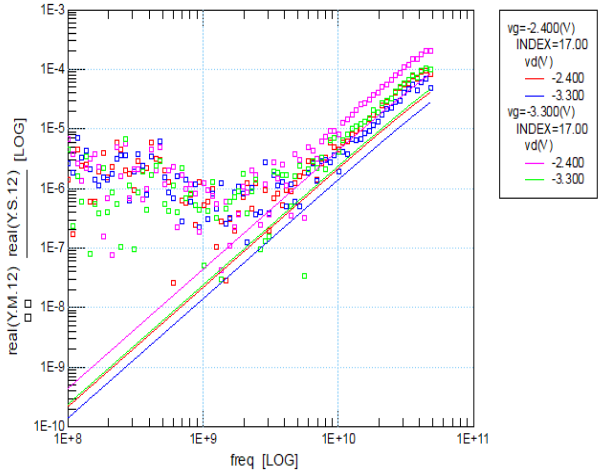
im(Y11) = f(Vg,Vd)



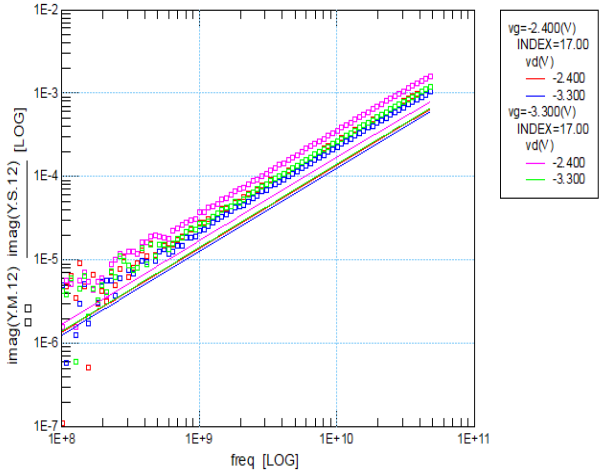
re(Y21) = f(Vg,Vd)



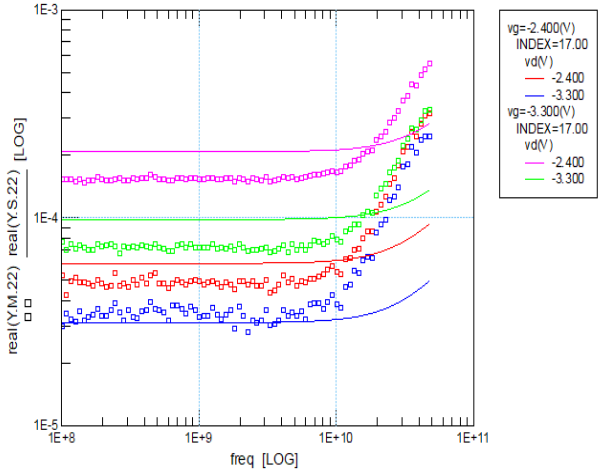
im(Y21) = f(Vg,Vd)



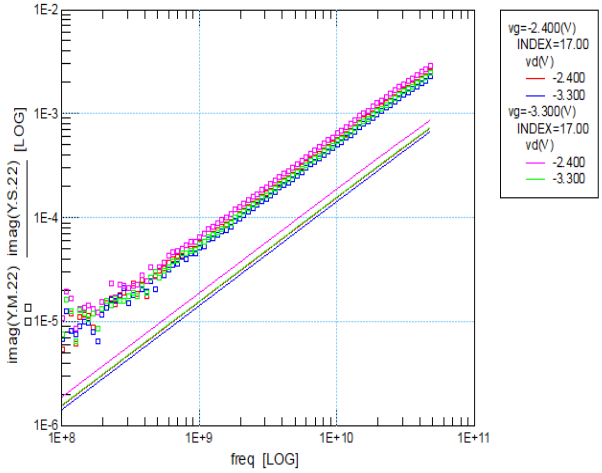
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

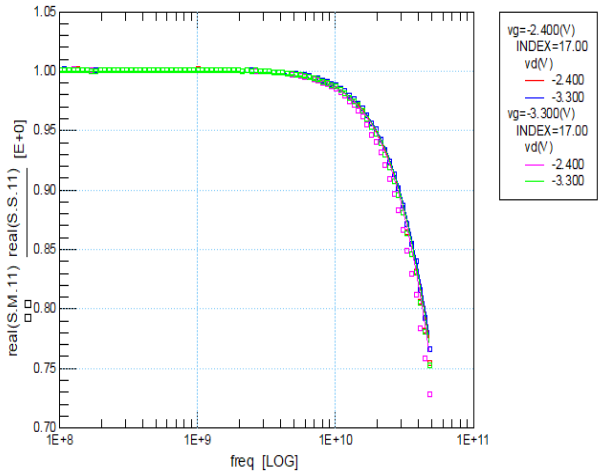


re(Y22) = f(Vg,Vd)

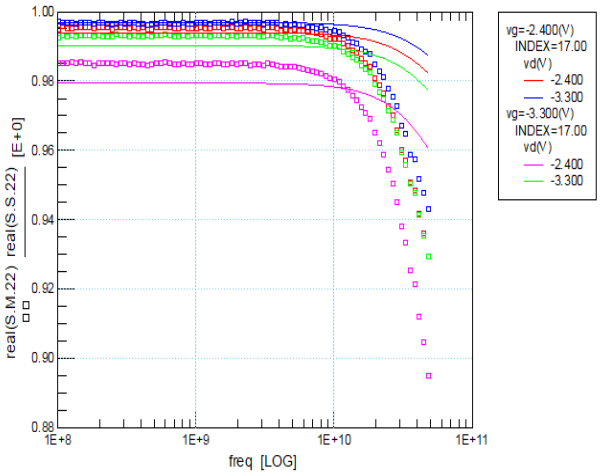


im(Y22) = f(Vg,Vd)

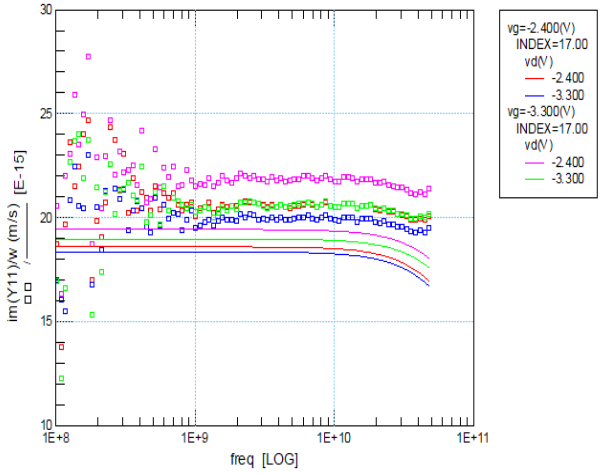
Verify Plots, T = 27°C



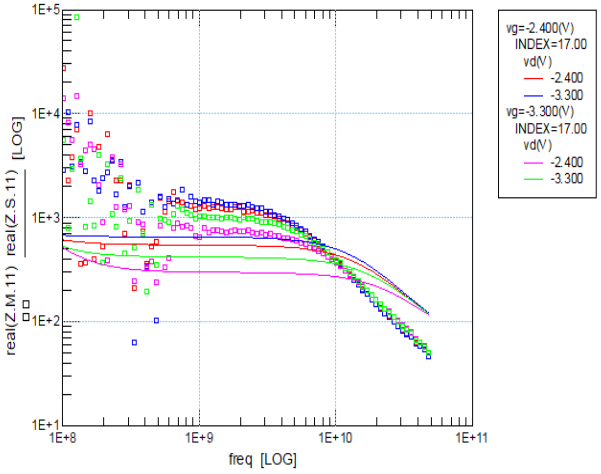
re(S11) = f(Vg,Vd)



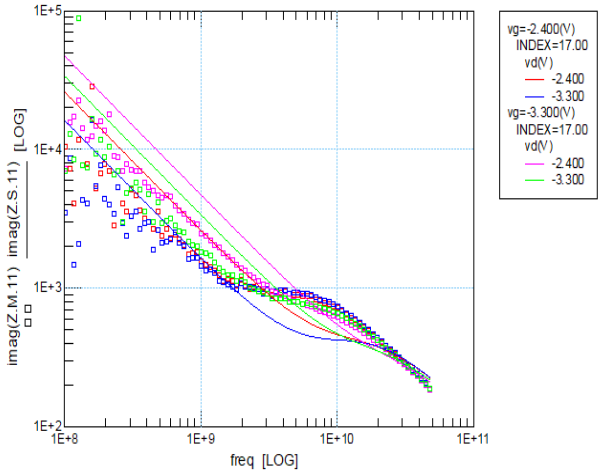
re(S22) = f(Vg,Vd)



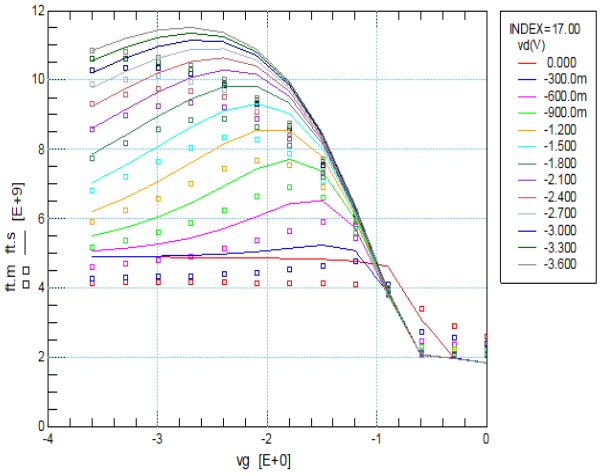
im(Y11) divided by ω



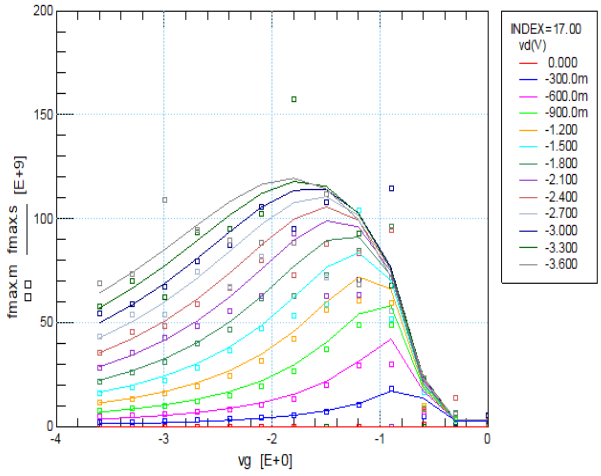
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



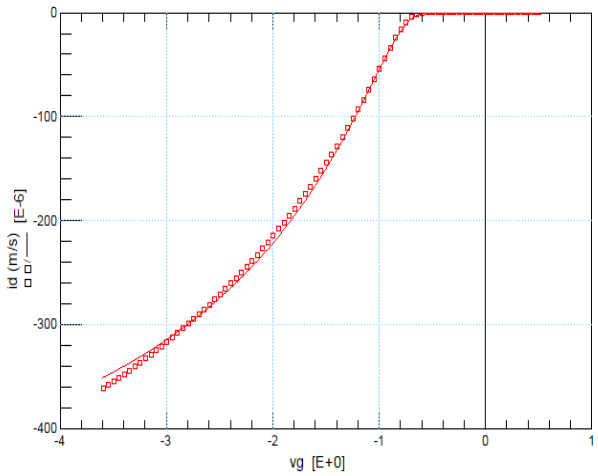
fT_vg = f(Vg,Vd)



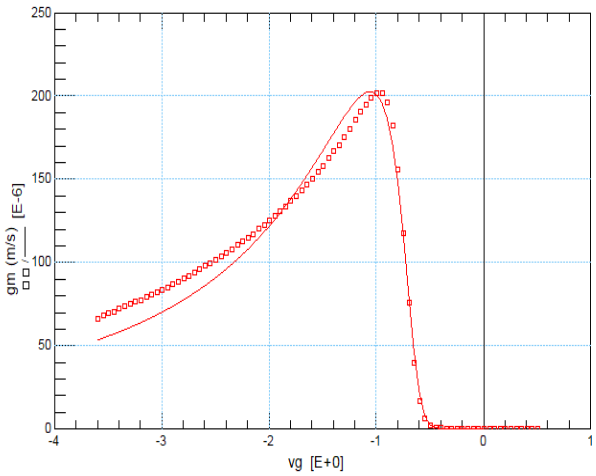
fmax_vg = f(Vg,Vd)

5.4 NF04_WF05u0_L0u5_post_D0

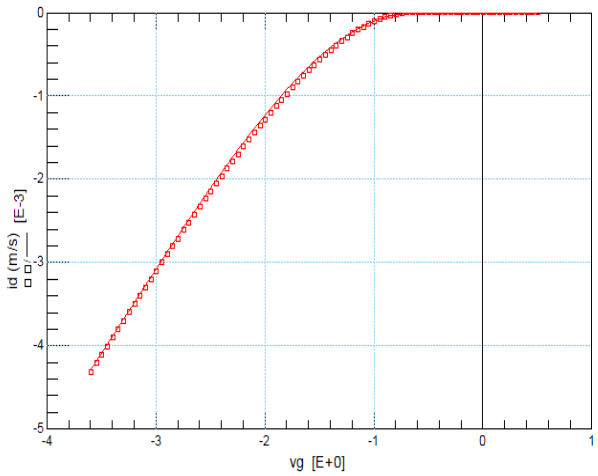
IdVg, T = 27°C



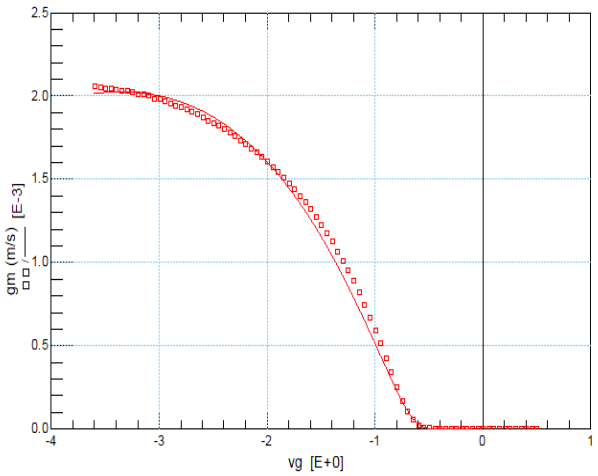
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

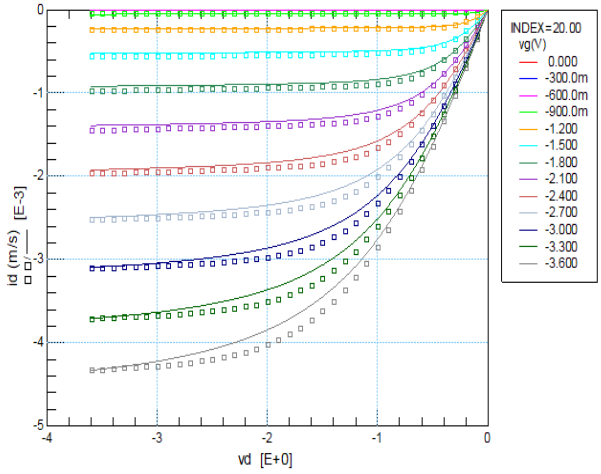


Id = f(Vg) @ Vd = -3.3V

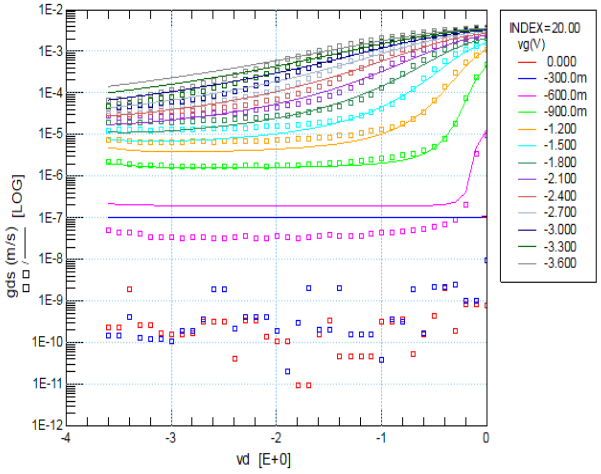


gm = f(Vg) @ Vd = -3.3V

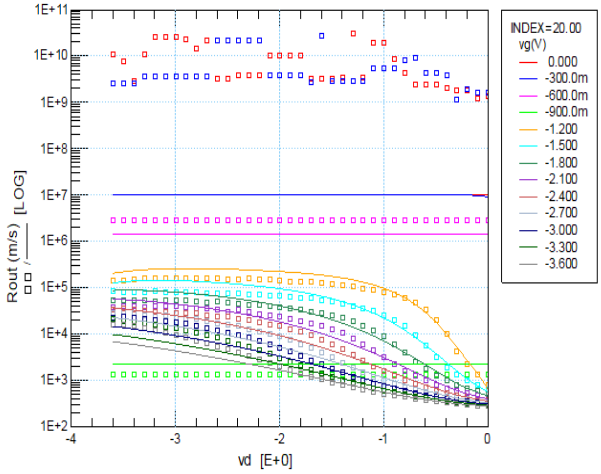
IdVd, T = 27°C



Id = f(Vd)

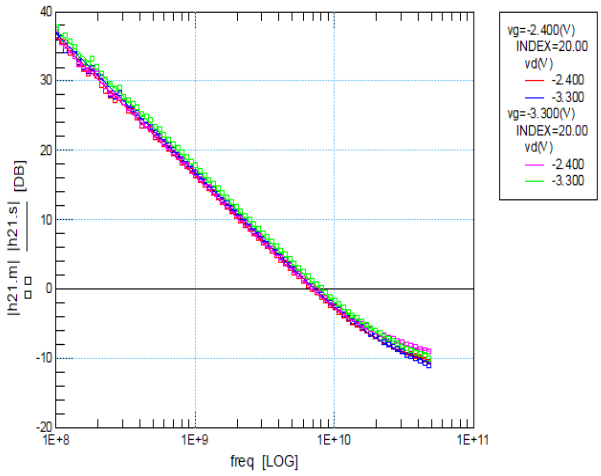


gds = f(Vd)

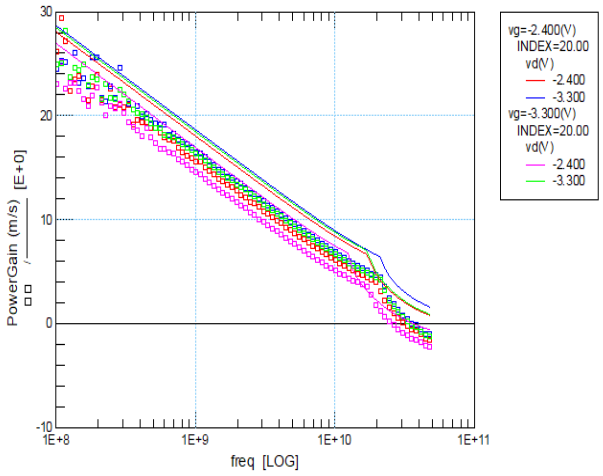


Rout = f(Vd)

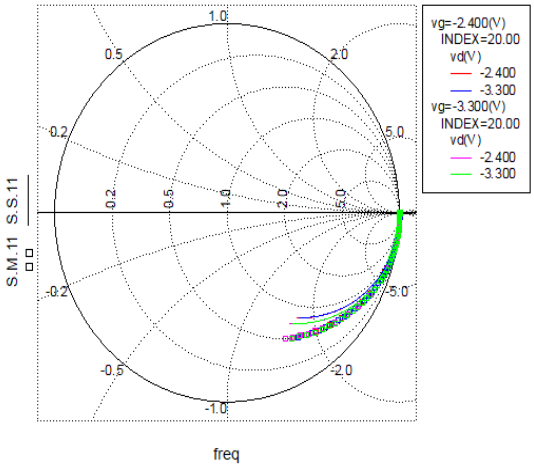
S-Parameter, T = 27°C



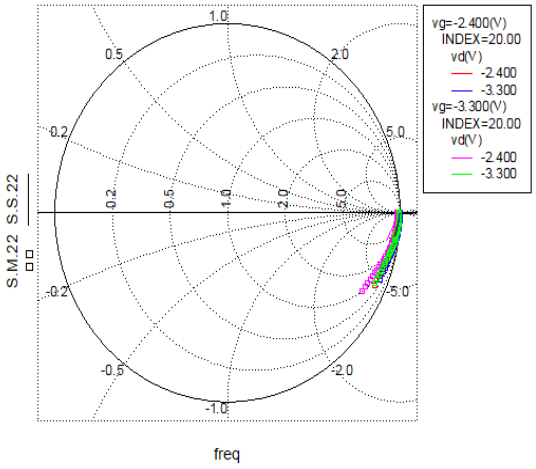
|h21| = f(Vg,Vd)



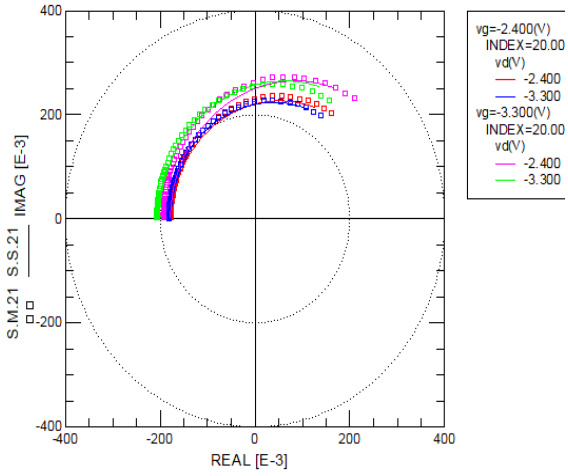
MSG/MAG = f(Vg,Vd)



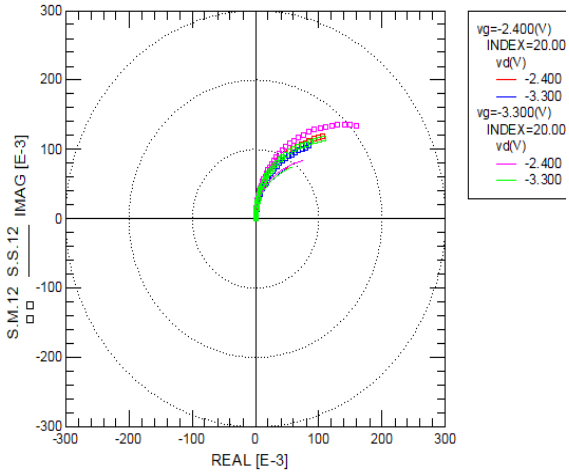
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

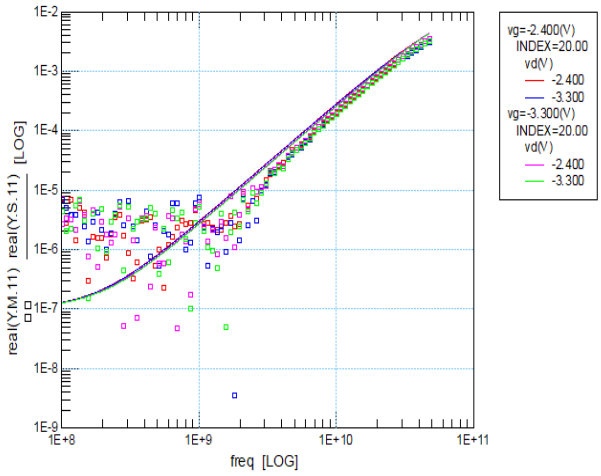


S21 = f(Vg,Vd)

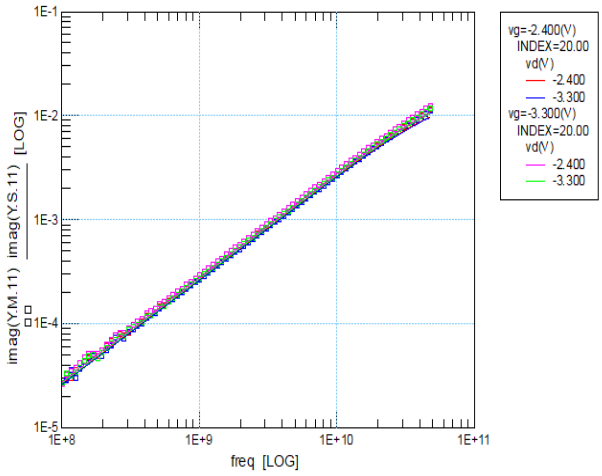


S12 = f(Vg,Vd)

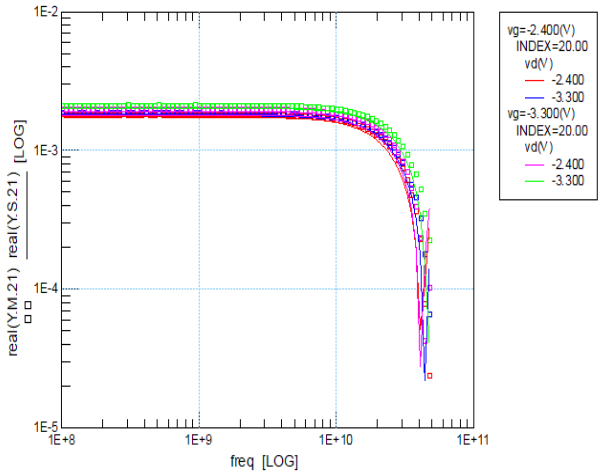
Y-Parameter, T = 27°C



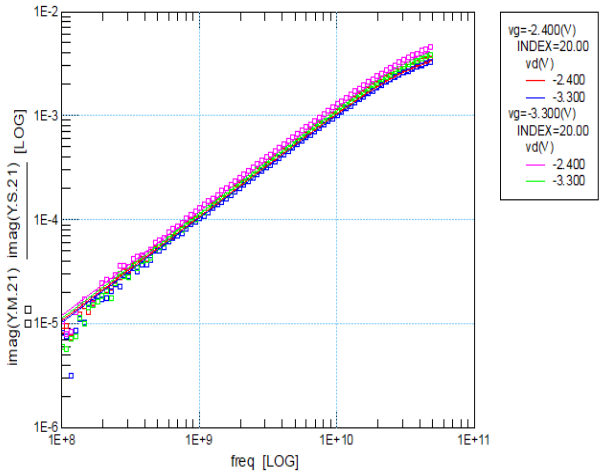
re(Y11) = f(Vg,Vd)



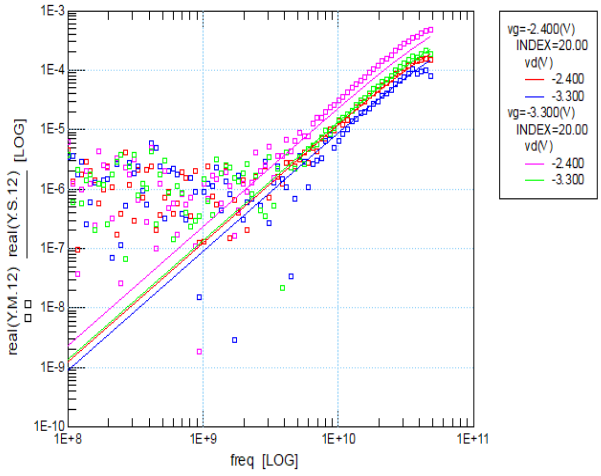
im(Y11) = f(Vg,Vd)



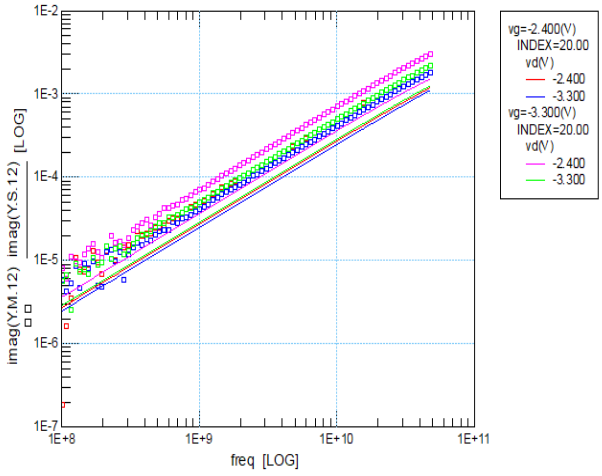
re(Y21) = f(Vg,Vd)



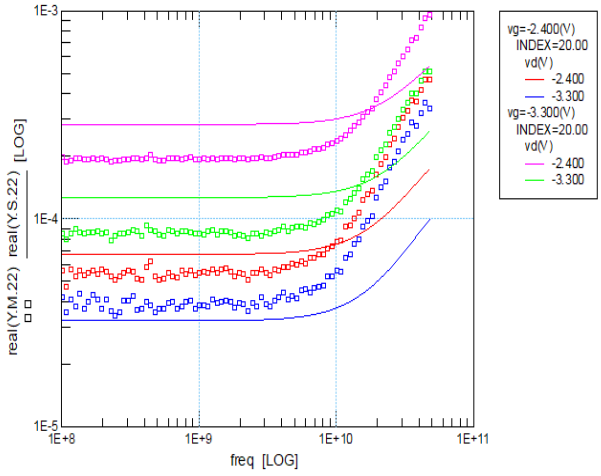
im(Y21) = f(Vg,Vd)



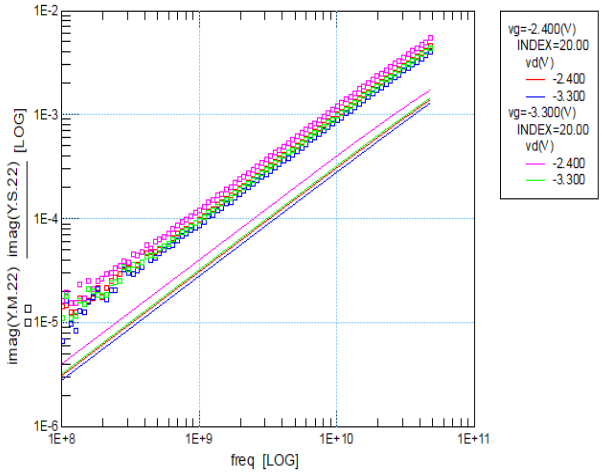
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

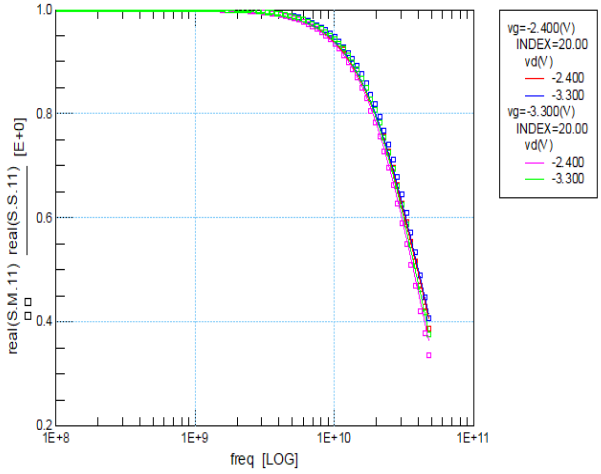


re(Y22) = f(Vg,Vd)

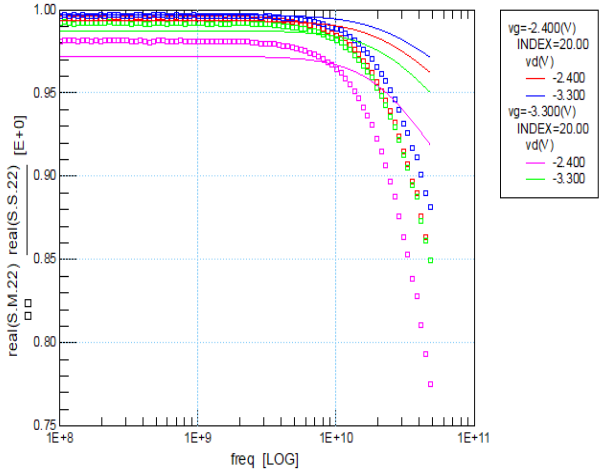


im(Y22) = f(Vg,Vd)

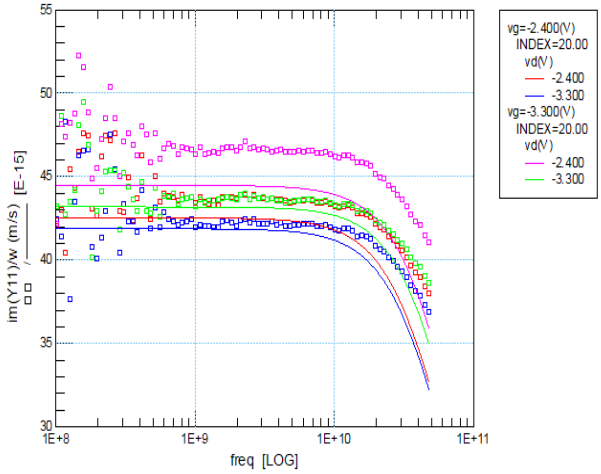
Verify Plots, T = 27°C



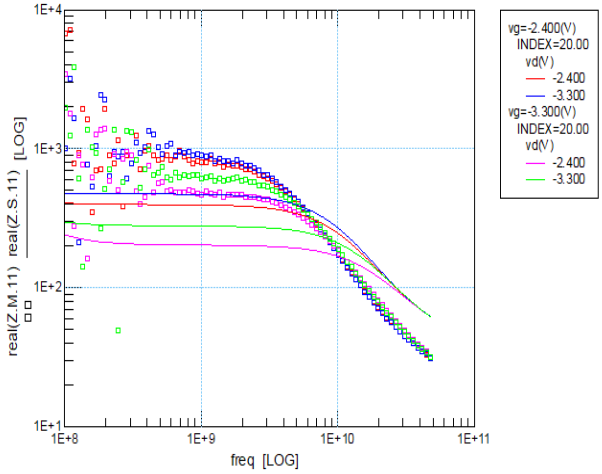
re(S11) = f(Vg,Vd)



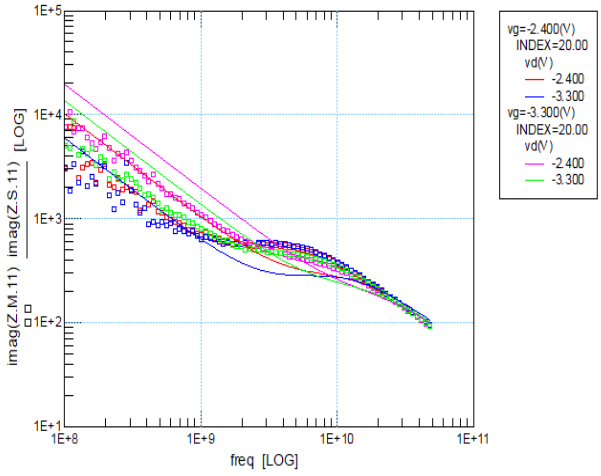
re(S22) = f(Vg,Vd)



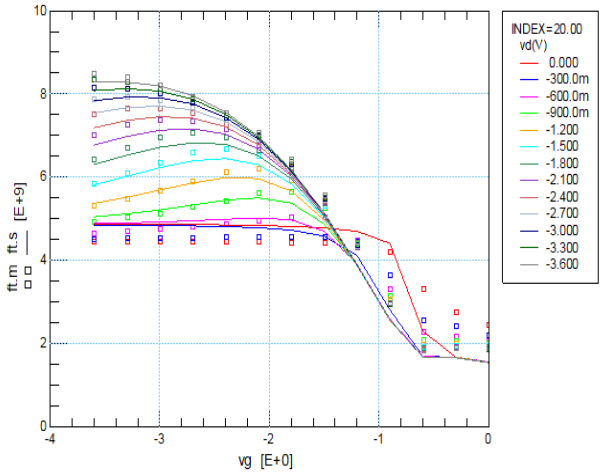
im(Y11) divided by ω



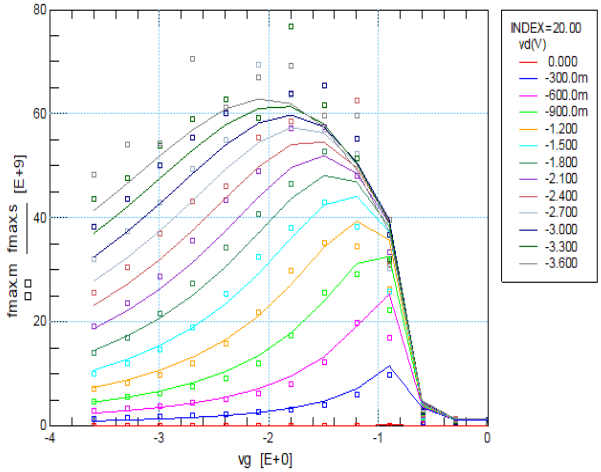
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



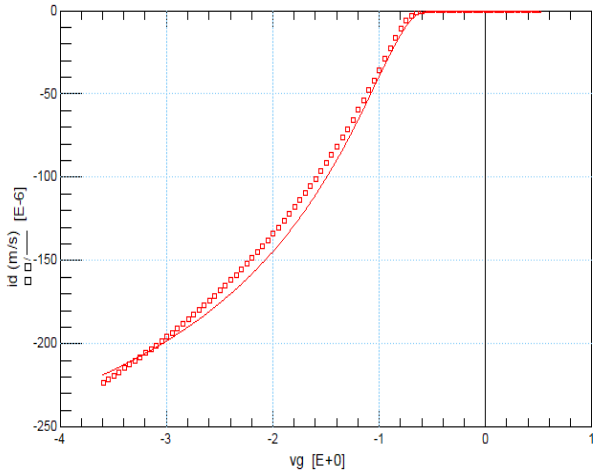
fT_vg = f(Vg,Vd)



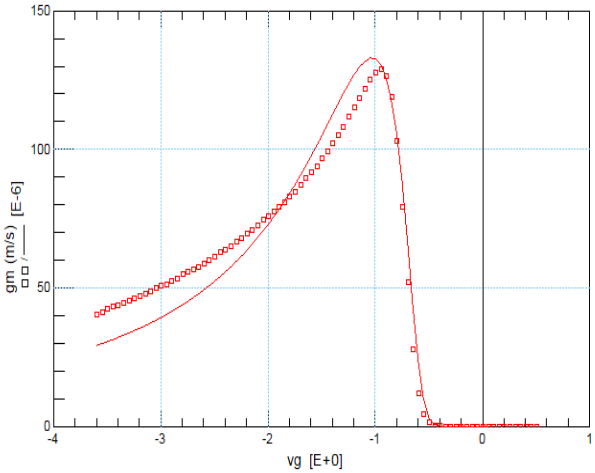
fmax_vg = f(Vg,Vd)

5.5 NF01_WF10u0_L0u4_S453A_D5

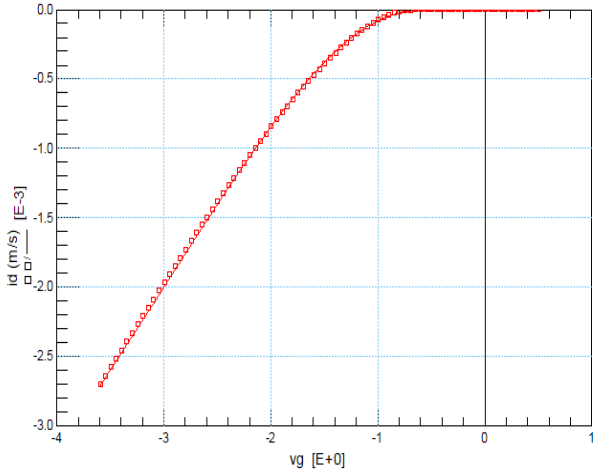
IdVg, T = 27°C



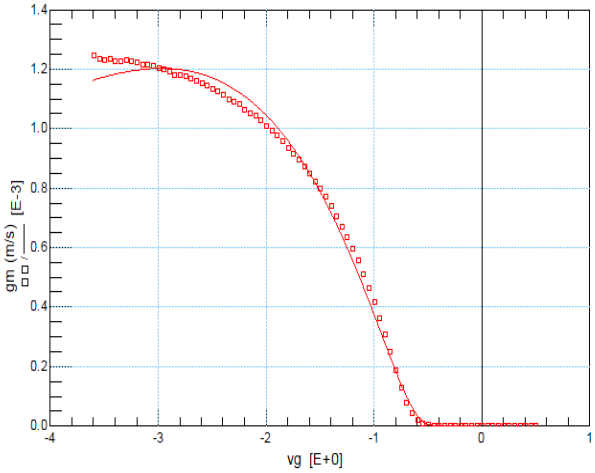
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

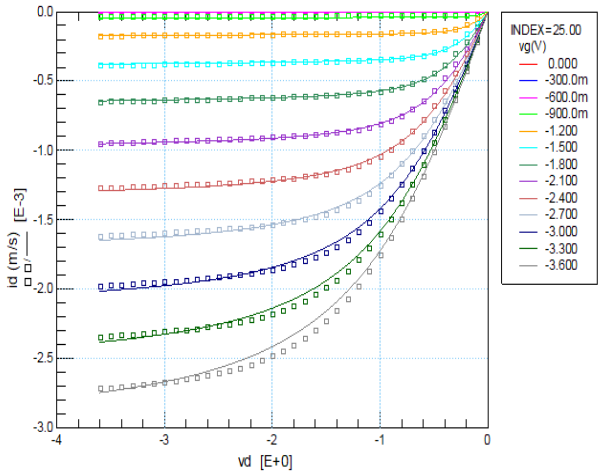


Id = f(Vg) @ Vd = -3.3V

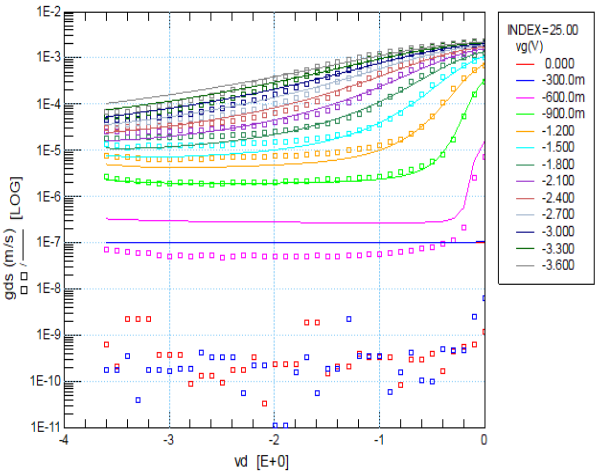


gm = f(Vg) @ Vd = -3.3V

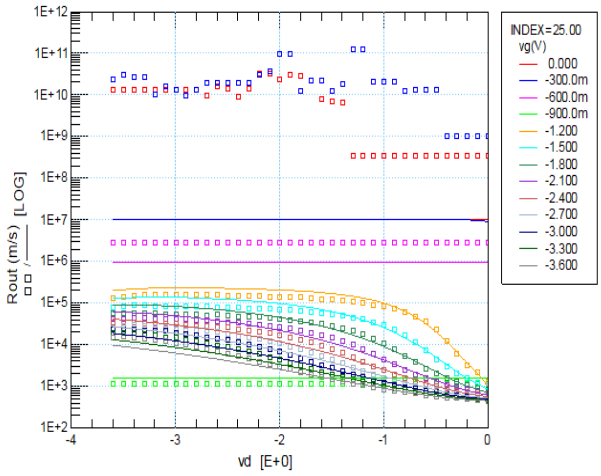
IdVd, T = 27°C



Id = f(Vd)

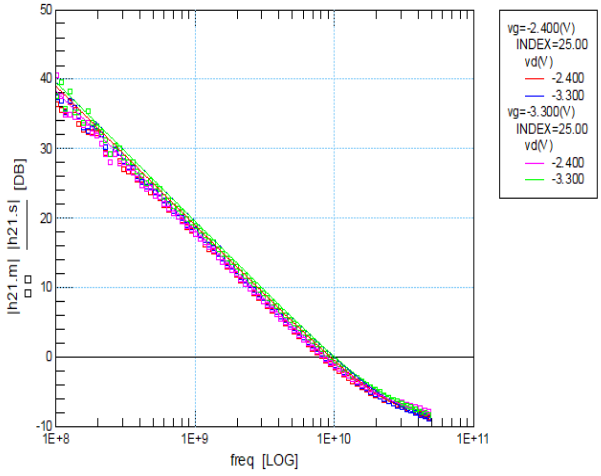


gds = f(Vd)

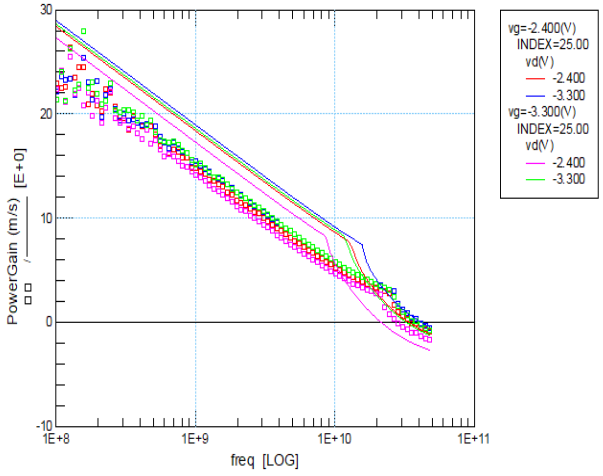


Rout = f(Vd)

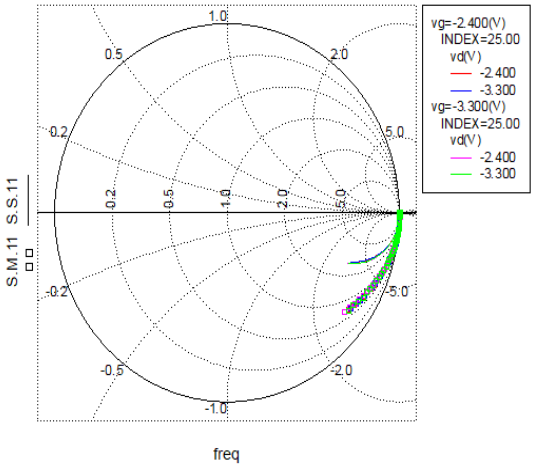
S-Parameter, T = 27°C



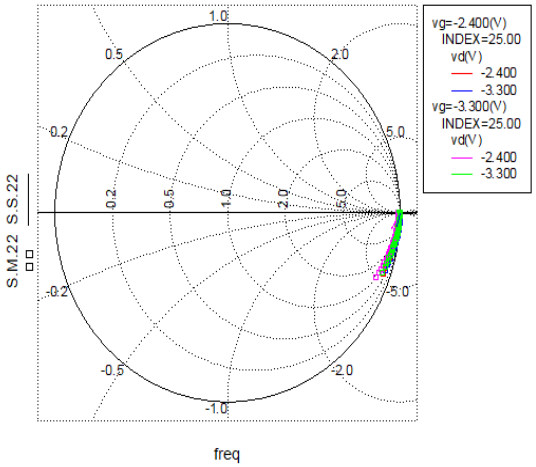
|h21| = f(Vg,Vd)



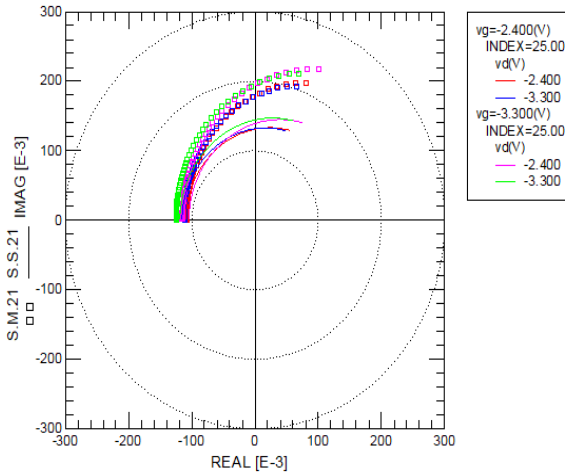
MSG/MAG = f(Vg,Vd)



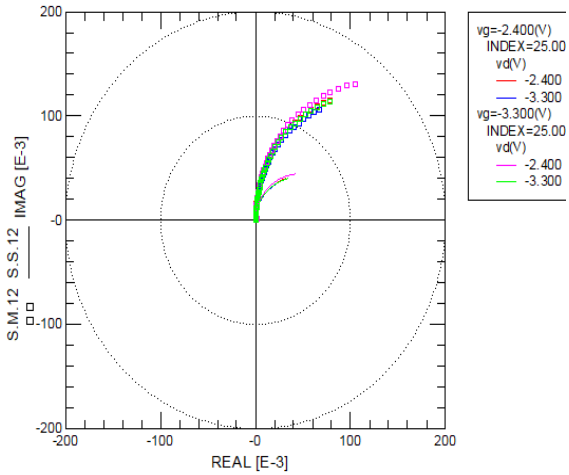
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

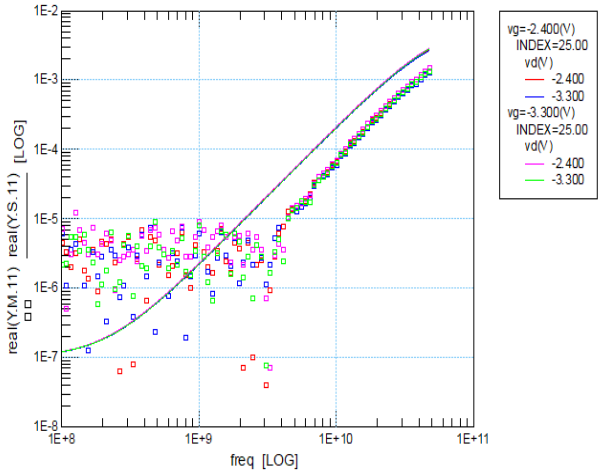


S21 = f(Vg,Vd)

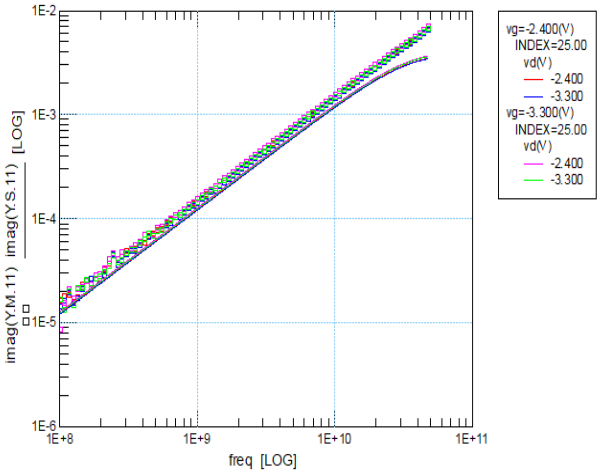


S12 = f(Vg,Vd)

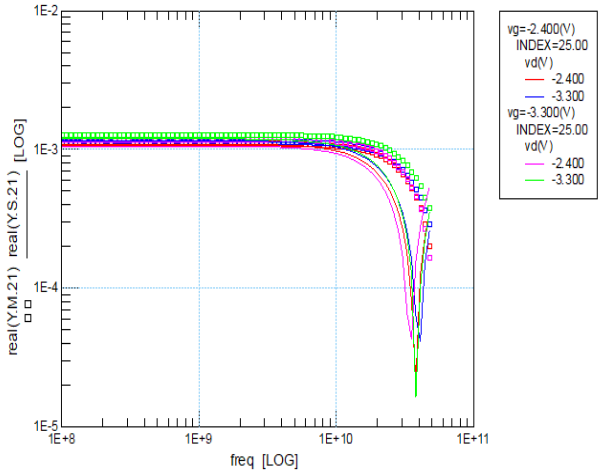
Y-Parameter, T = 27°C



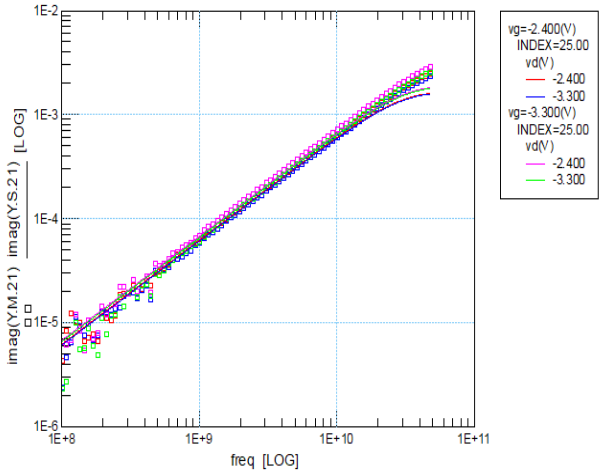
re(Y11) = f(Vg,Vd)



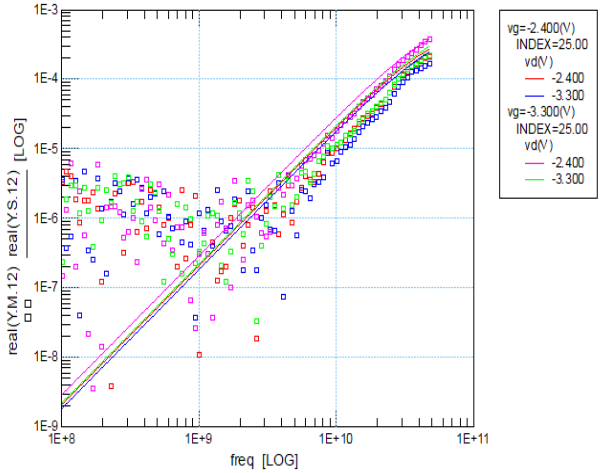
im(Y11) = f(Vg,Vd)



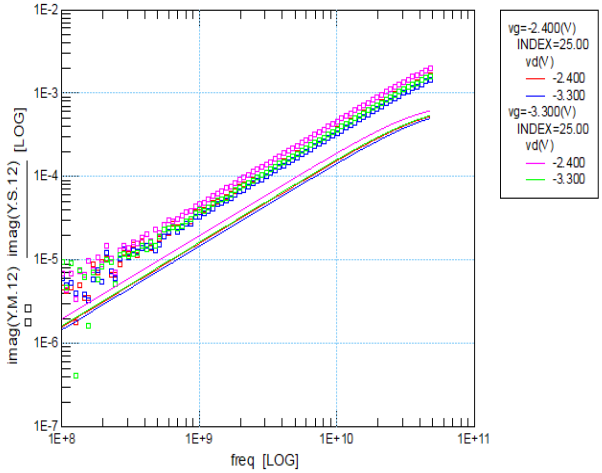
re(Y21) = f(Vg,Vd)



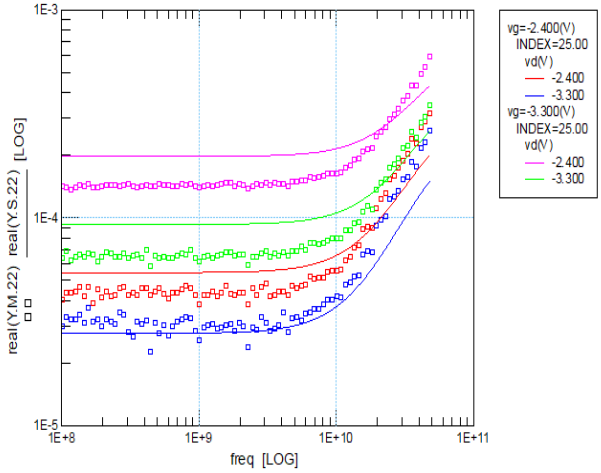
im(Y21) = f(Vg,Vd)



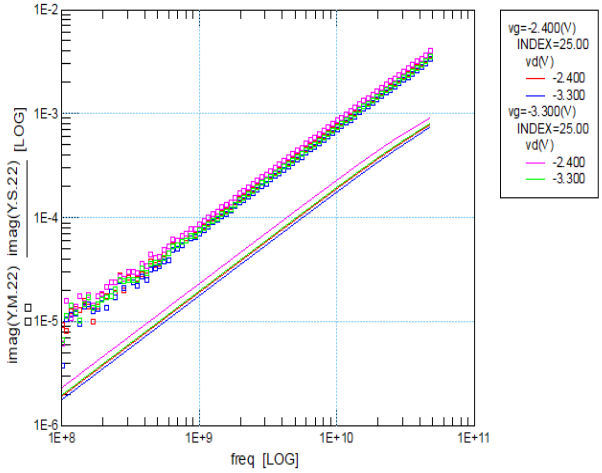
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

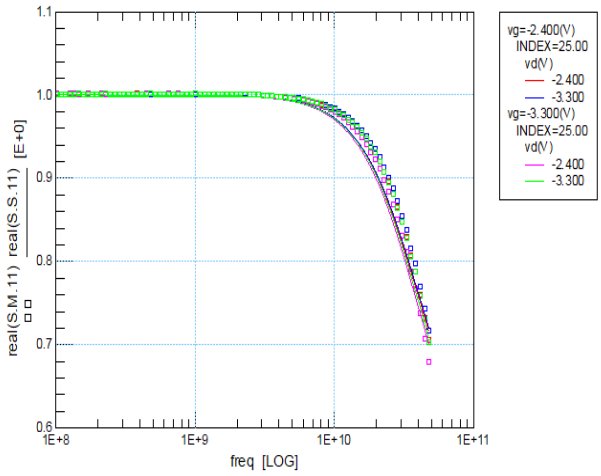


re(Y22) = f(Vg,Vd)

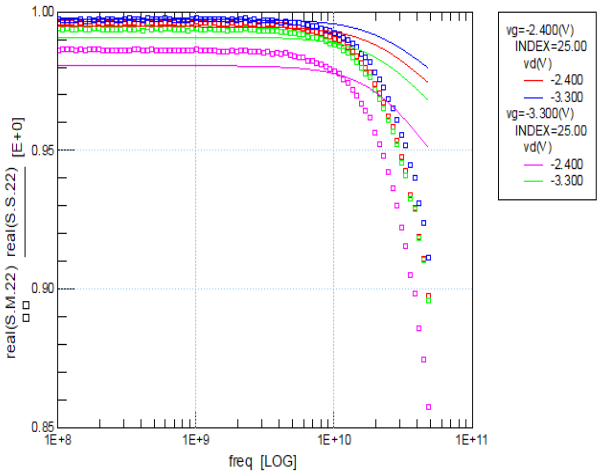


im(Y22) = f(Vg,Vd)

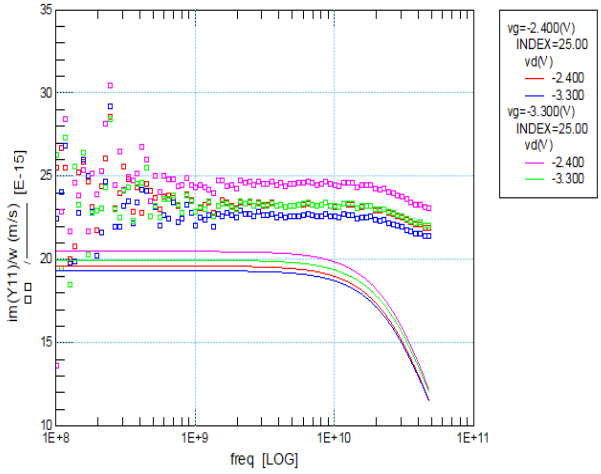
Verify Plots, T = 27°C



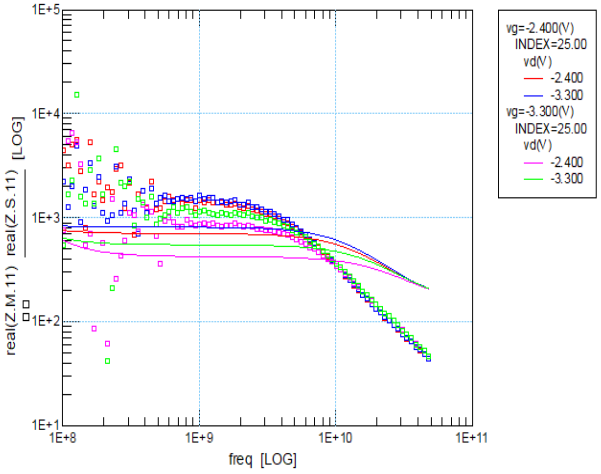
re(S11) = f(Vg,Vd)



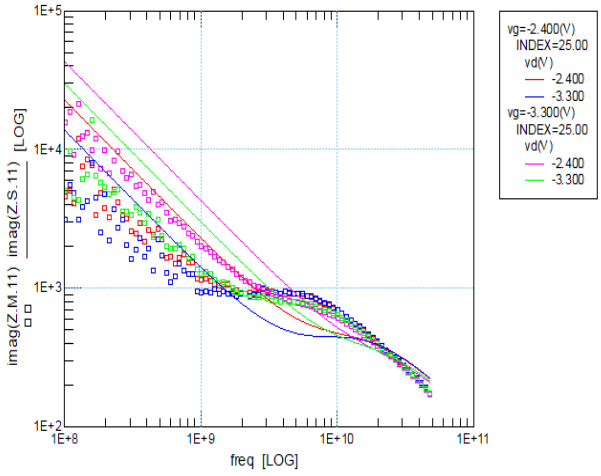
re(S22) = f(Vg,Vd)



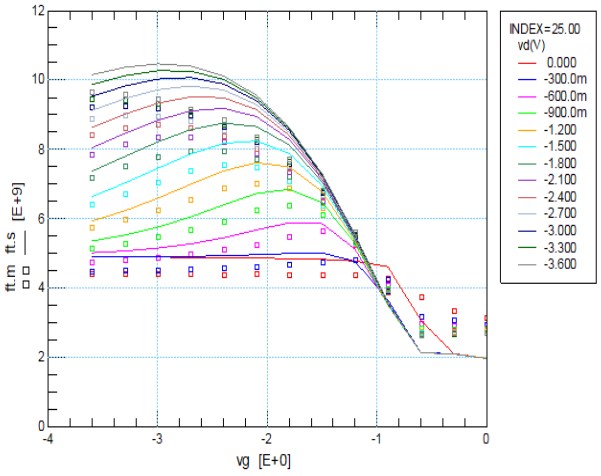
im(Y11) divided by ω



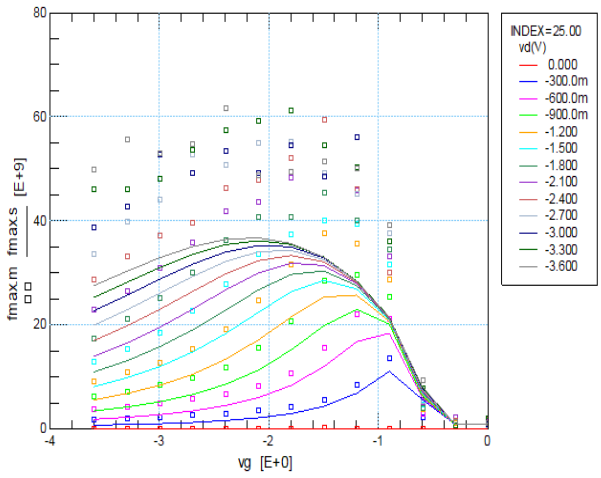
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



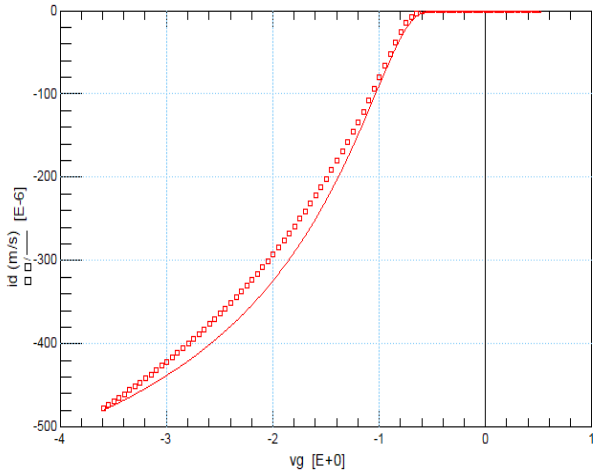
fT_vg = f(Vg,Vd)



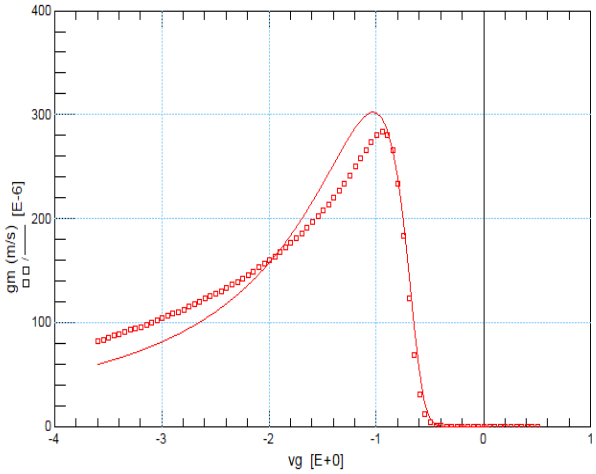
fmax_vg = f(Vg,Vd)

5.6 NF08_WF02u5_L0u4_S453A_D3

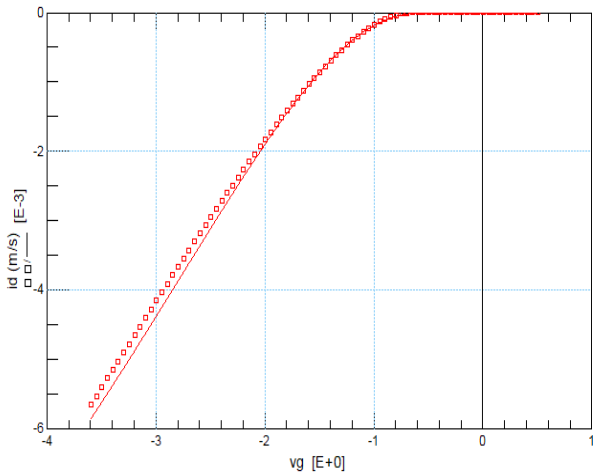
IdVg, T = 27°C



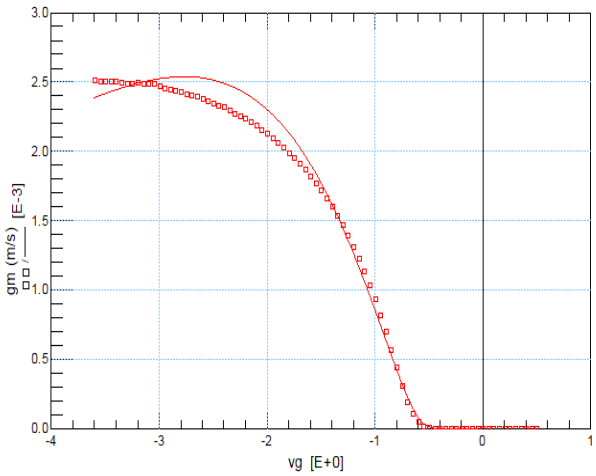
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

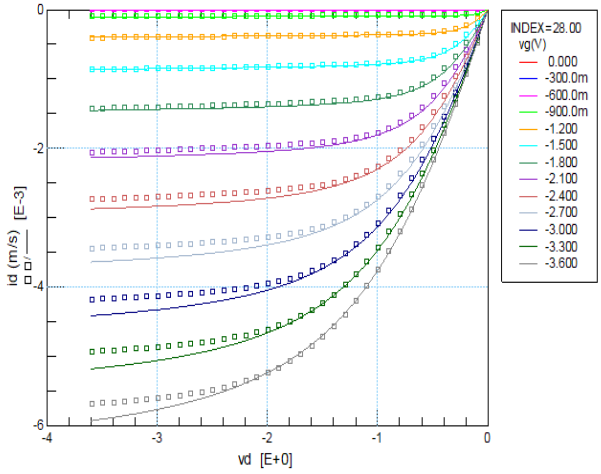


Id = f(Vg) @ Vd = -3.3V

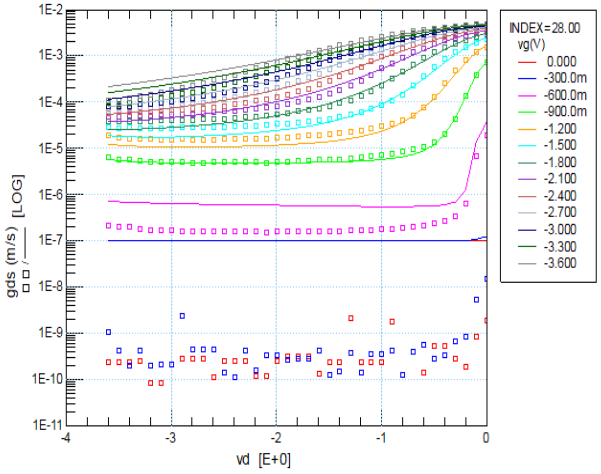


gm = f(Vg) @ Vd = -3.3V

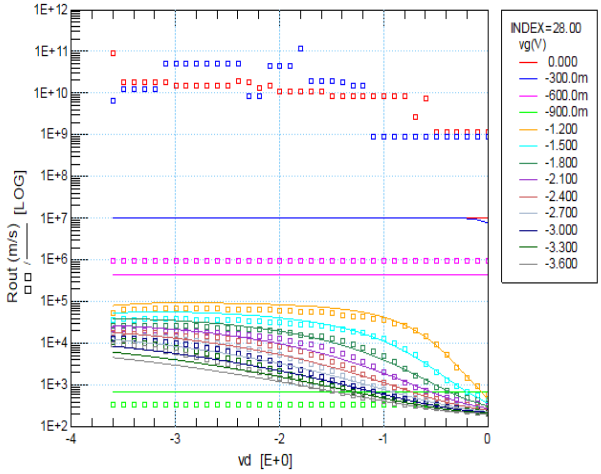
IdVd, T = 27°C



Id = f(Vd)

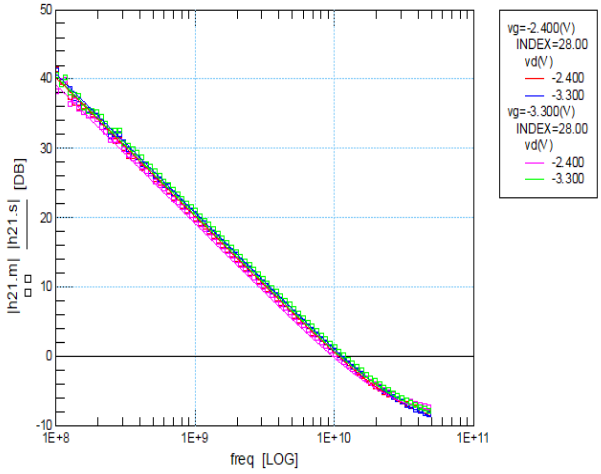


gds = f(Vd)

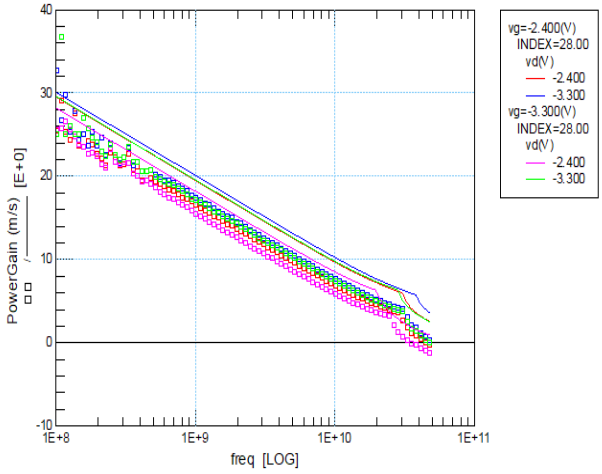


Rout = f(Vd)

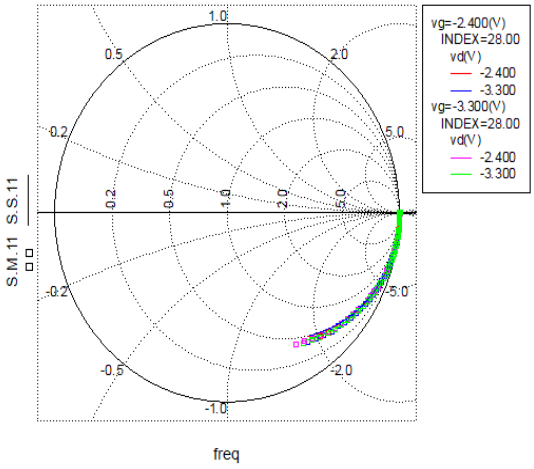
S-Parameter, T = 27°C



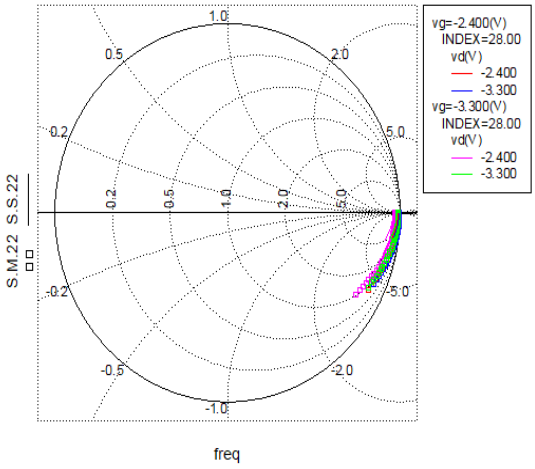
|h21| = f(Vg,Vd)



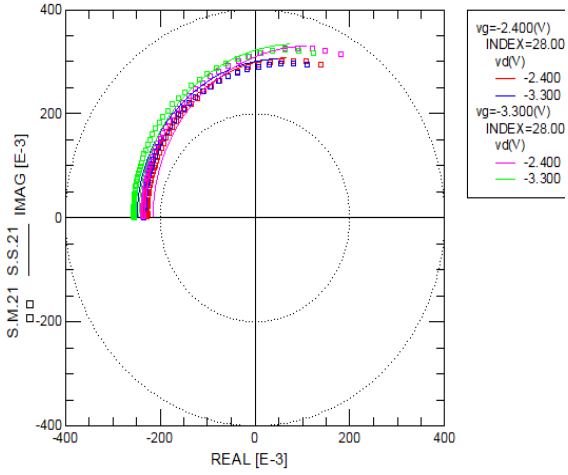
MSG/MAG = f(Vg,Vd)



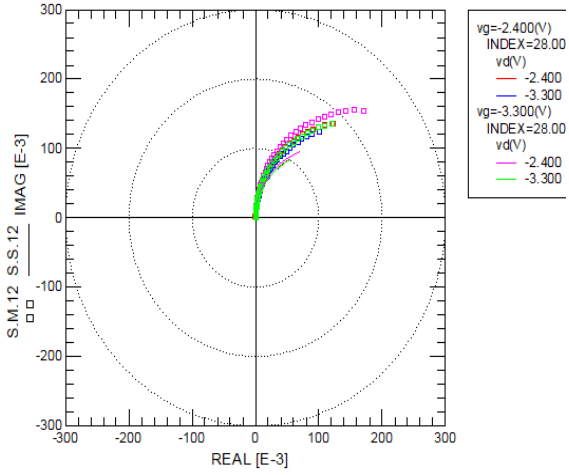
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

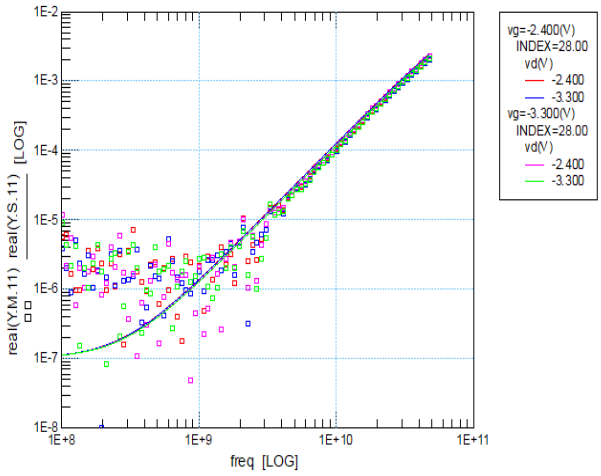


S21 = f(Vg,Vd)

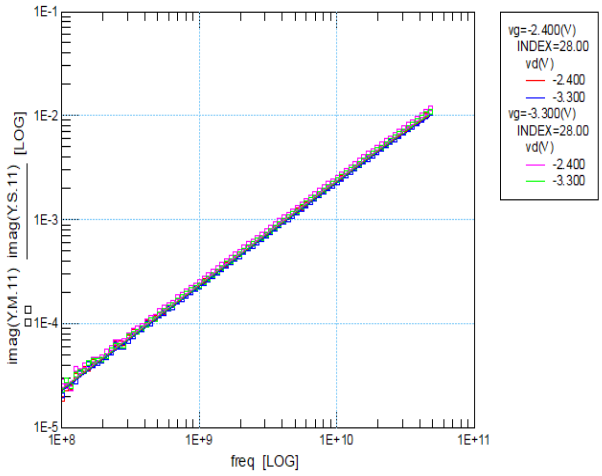


S12 = f(Vg,Vd)

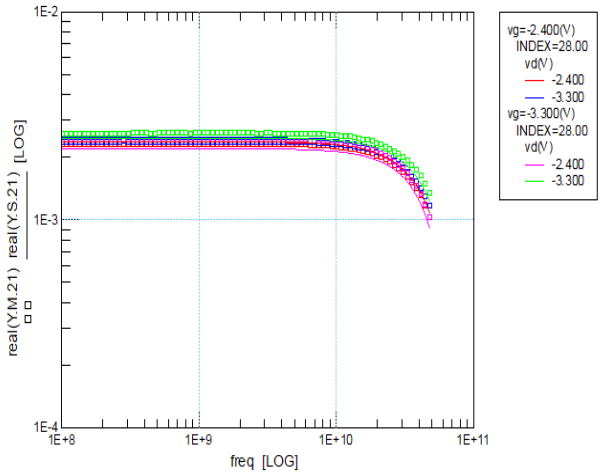
Y-Parameter, T = 27°C



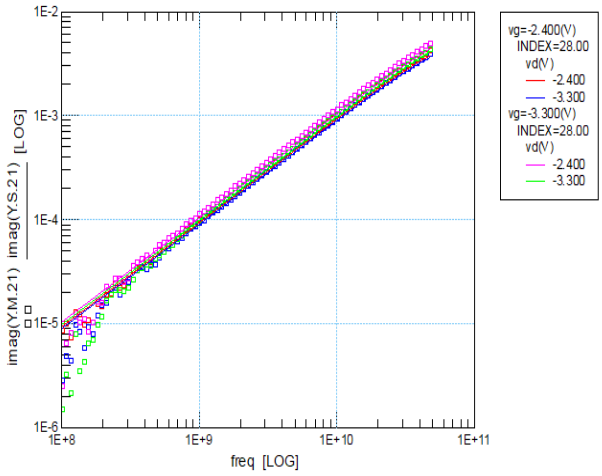
re(Y11) = f(Vg,Vd)



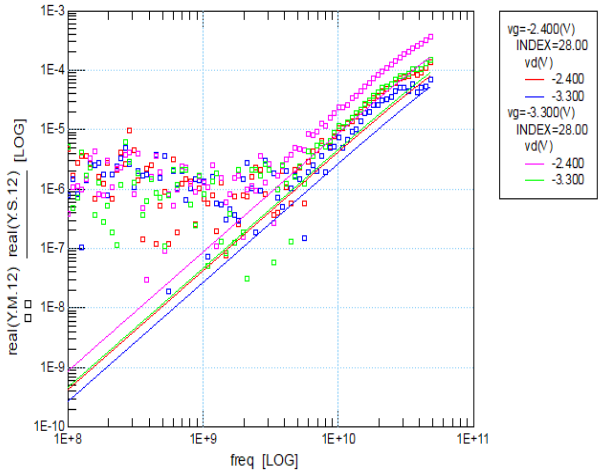
im(Y11) = f(Vg,Vd)



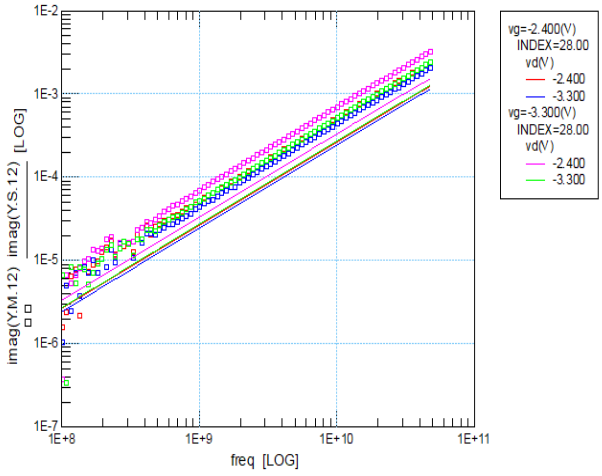
re(Y21) = f(Vg,Vd)



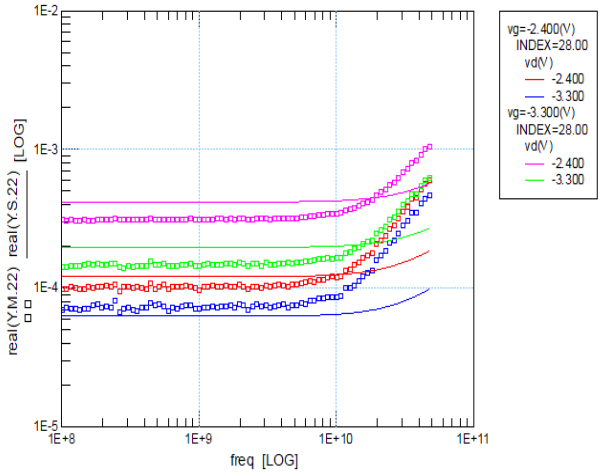
im(Y21) = f(Vg,Vd)



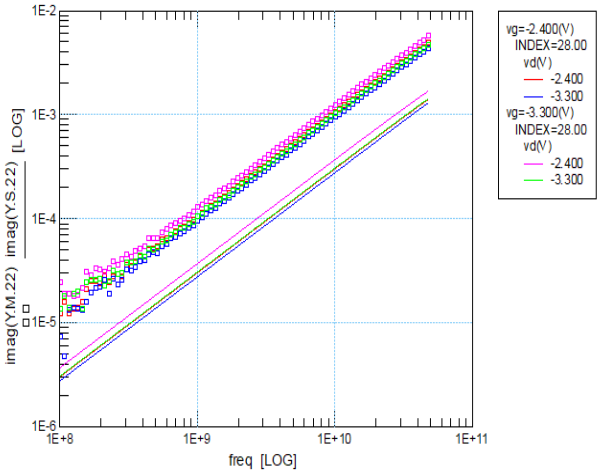
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

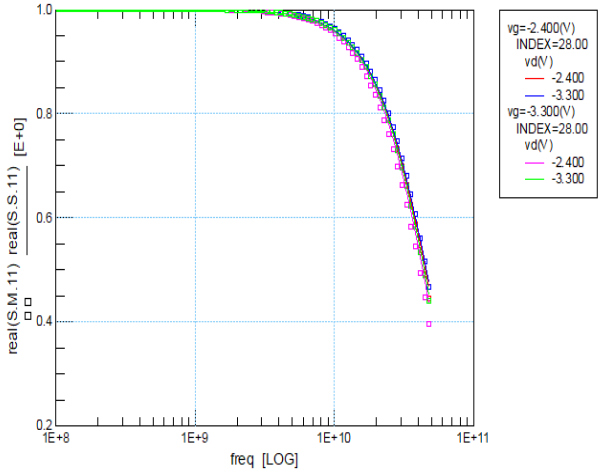


re(Y22) = f(Vg,Vd)

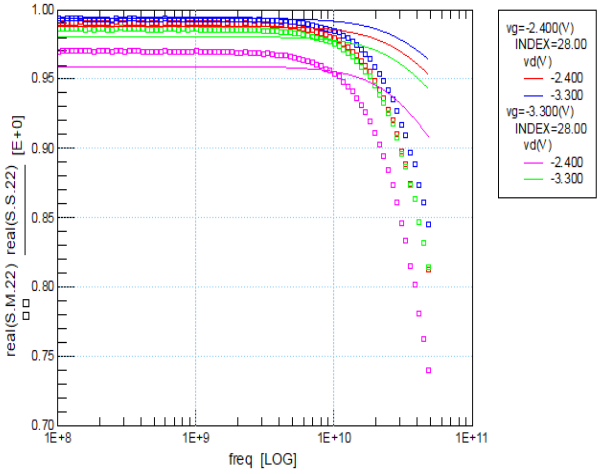


im(Y22) = f(Vg,Vd)

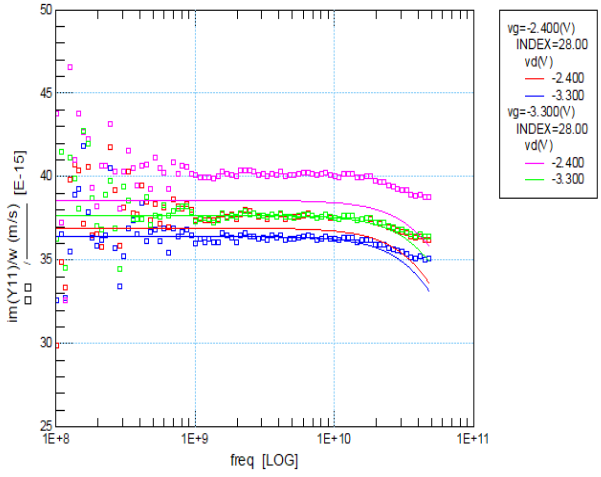
Verify Plots, T = 27°C



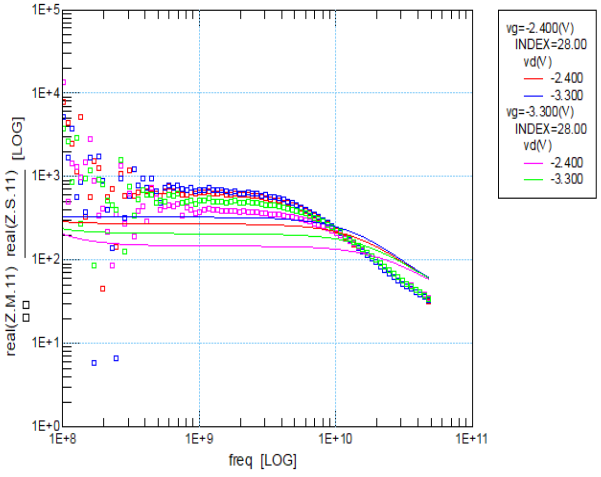
re(S11) = f(Vg,Vd)



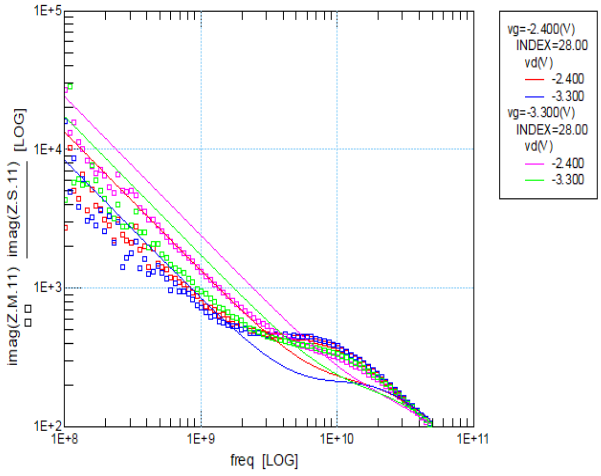
re(S22) = f(Vg,Vd)



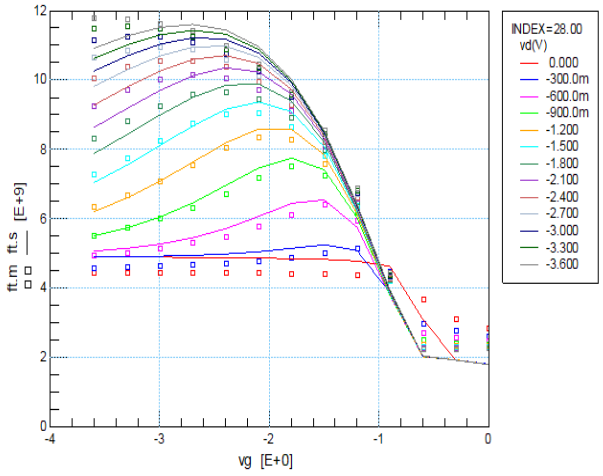
im(Y11) divided by ω



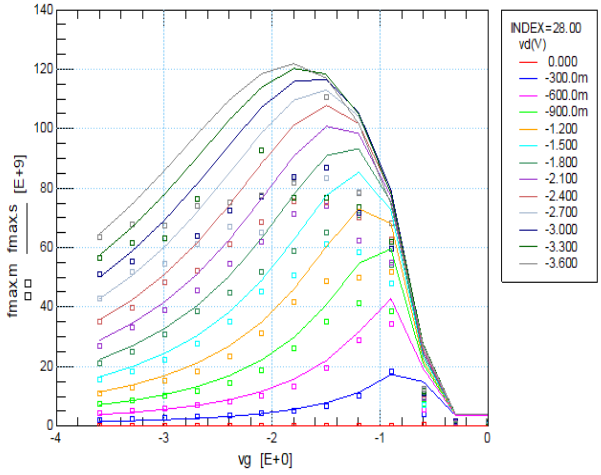
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



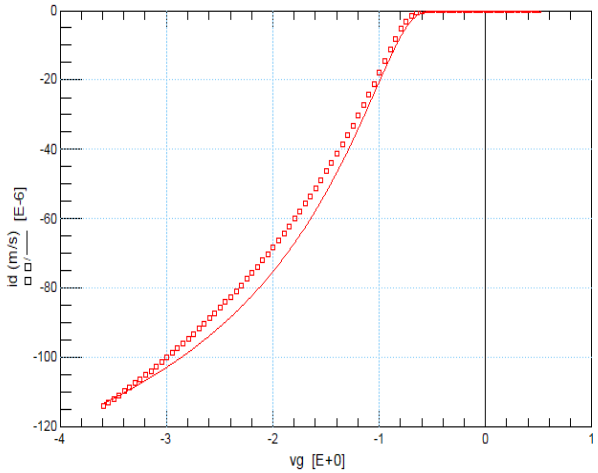
fT_vg = f(Vg,Vd)



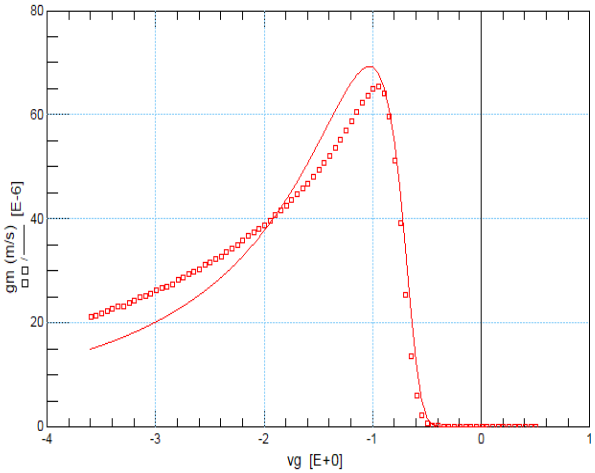
fmax_vg = f(Vg,Vd)

5.7 NF01_WF05u0_L0u4_S453A_E1

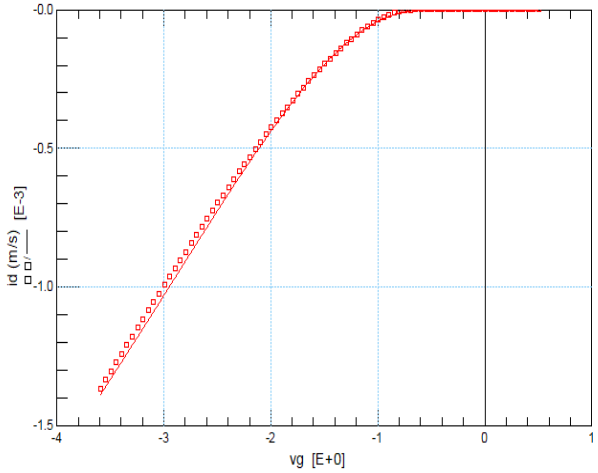
IdVg, T = 27°C



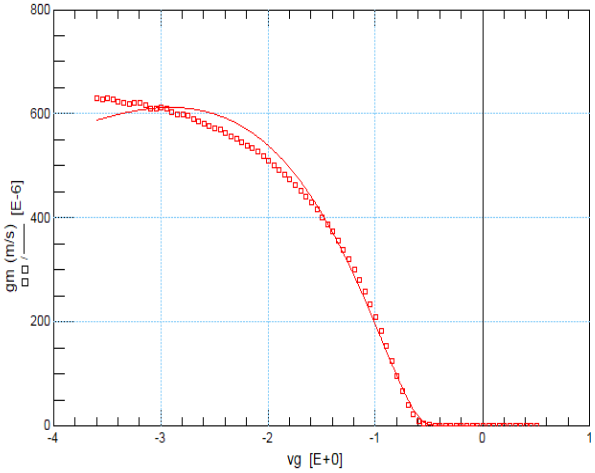
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

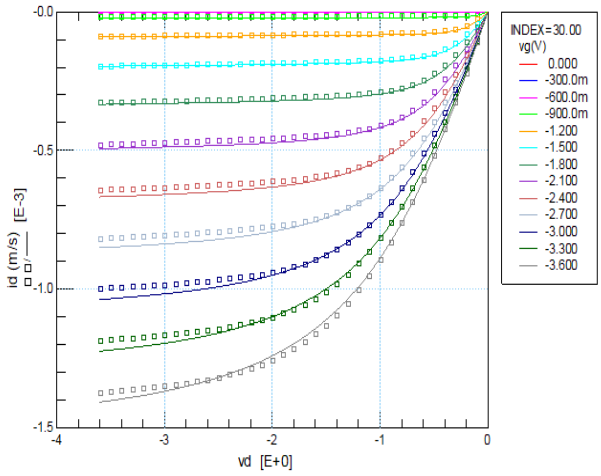


Id = f(Vg) @ Vd = -3.3V

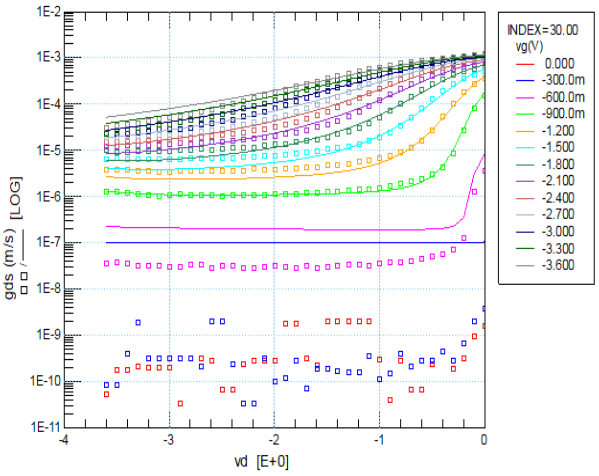


gm = f(Vg) @ Vd = -3.3V

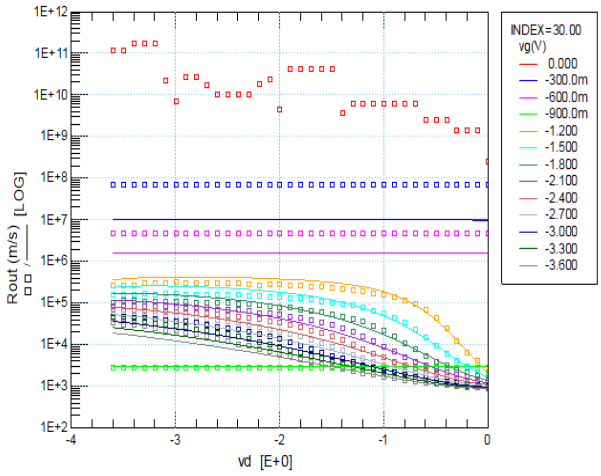
IdVd, T = 27°C



Id = f(Vd)

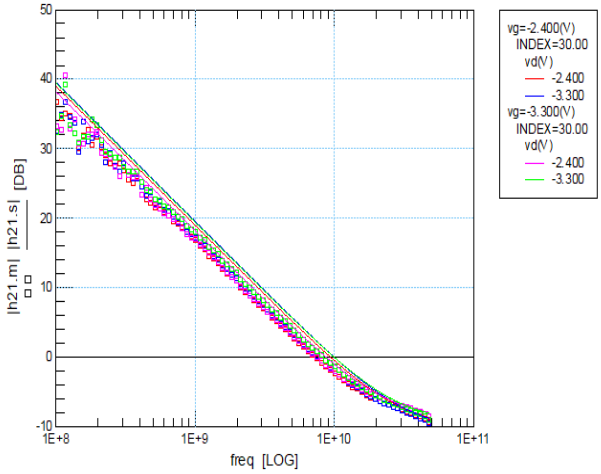


gds = f(Vd)

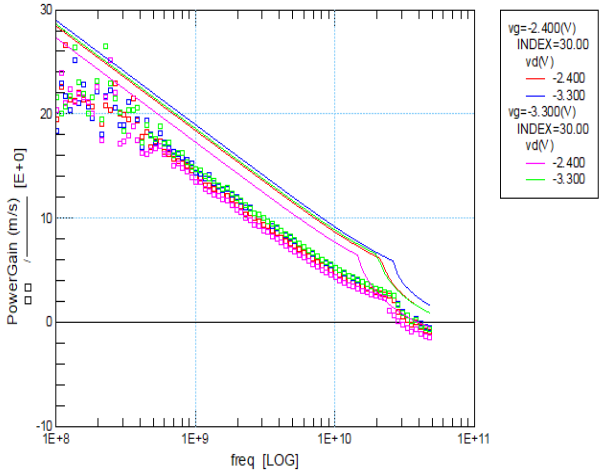


Rout = f(Vd)

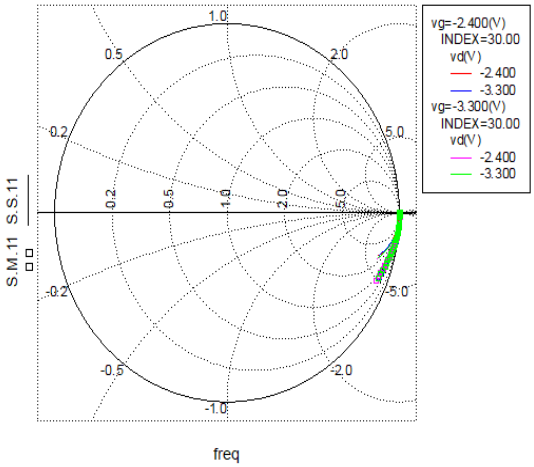
S-Parameter, T = 27°C



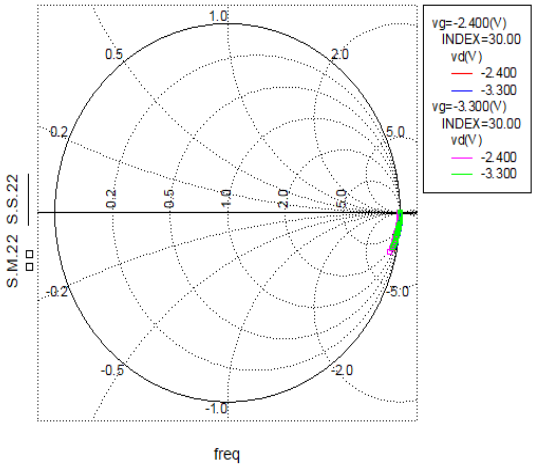
|h21| = f(Vg,Vd)



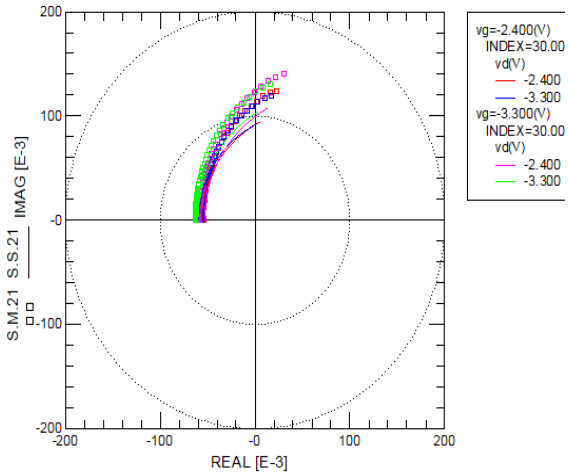
MSG/MAG = f(Vg,Vd)



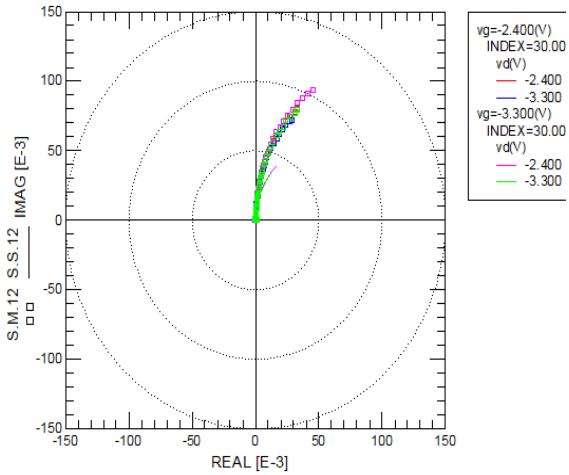
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

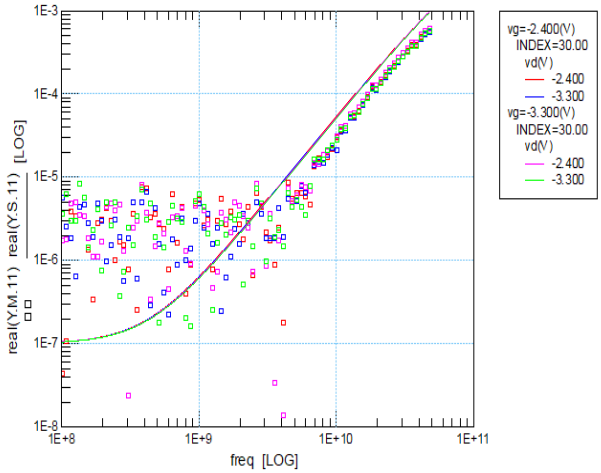


S21 = f(Vg,Vd)

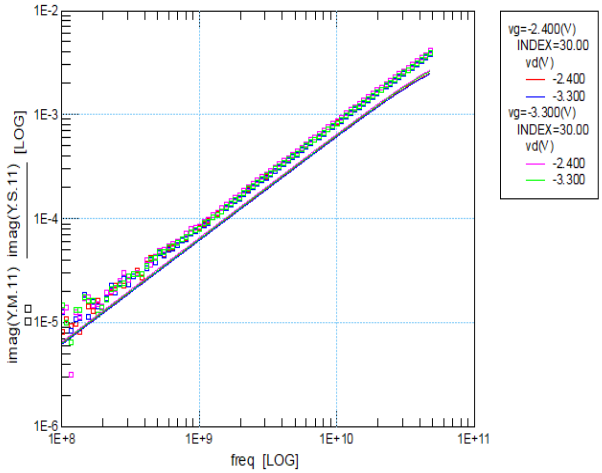


S12 = f(Vg,Vd)

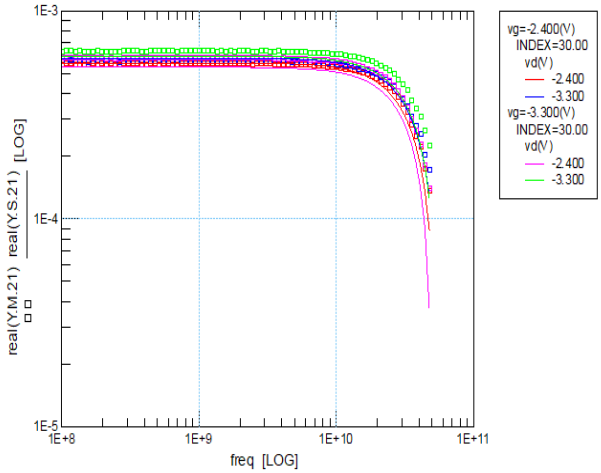
Y-Parameter, T = 27°C



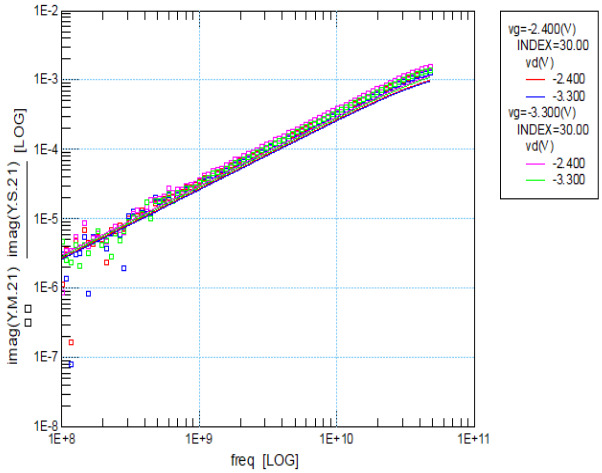
re(Y11) = f(Vg,Vd)



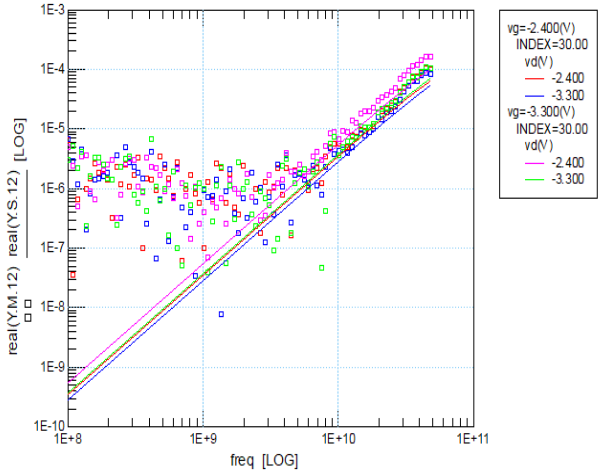
im(Y11) = f(Vg,Vd)



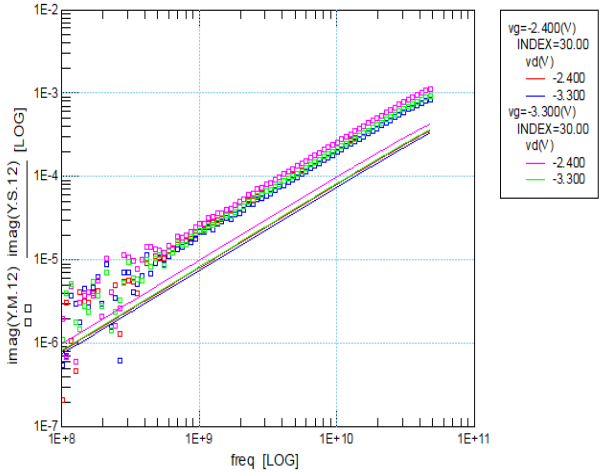
re(Y21) = f(Vg,Vd)



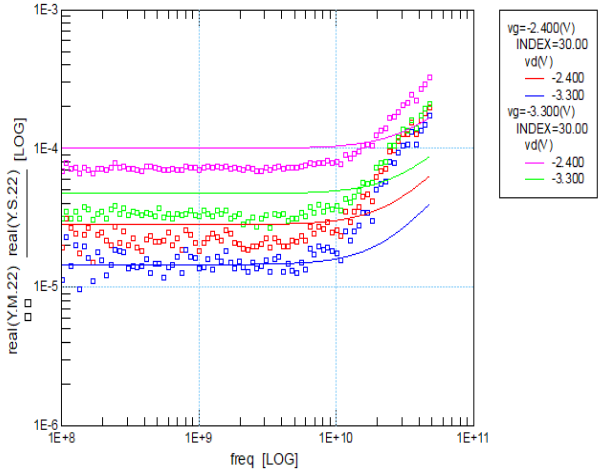
im(Y21) = f(Vg,Vd)



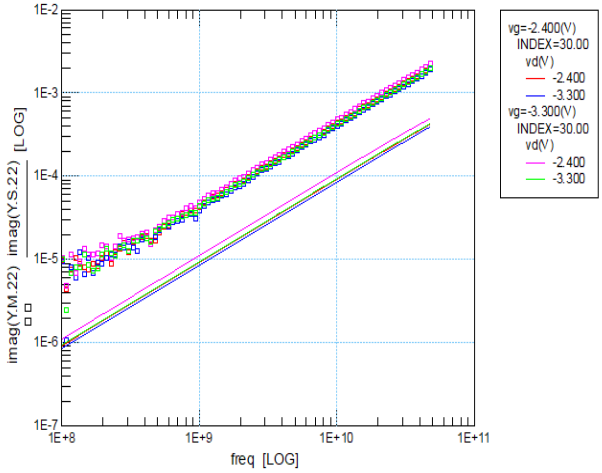
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

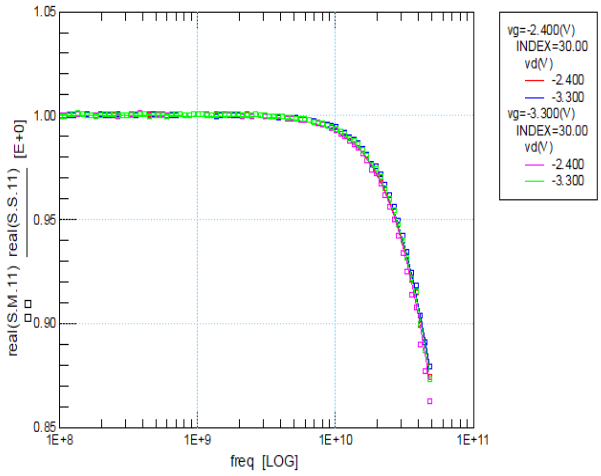


re(Y22) = f(Vg,Vd)

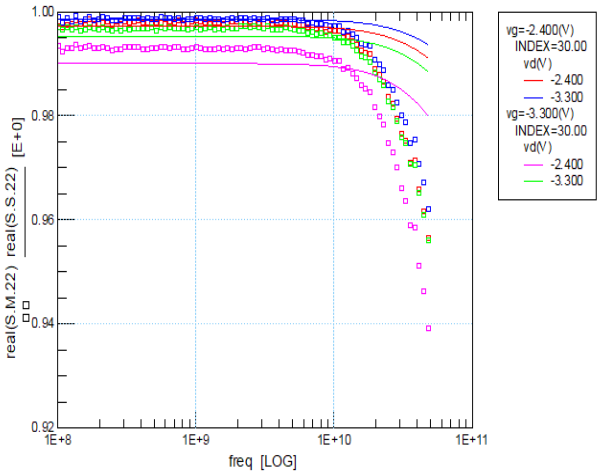


im(Y22) = f(Vg,Vd)

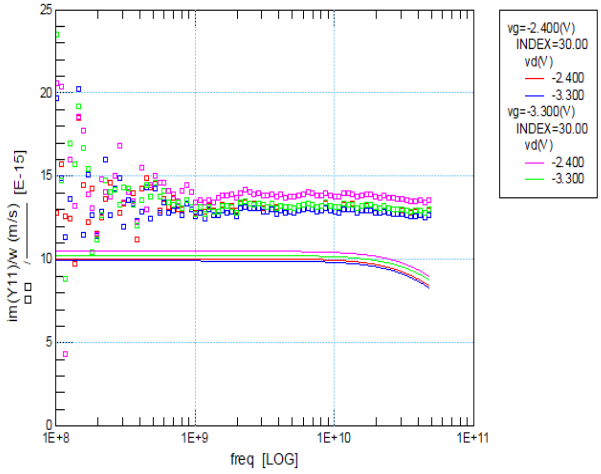
Verify Plots, T = 27°C



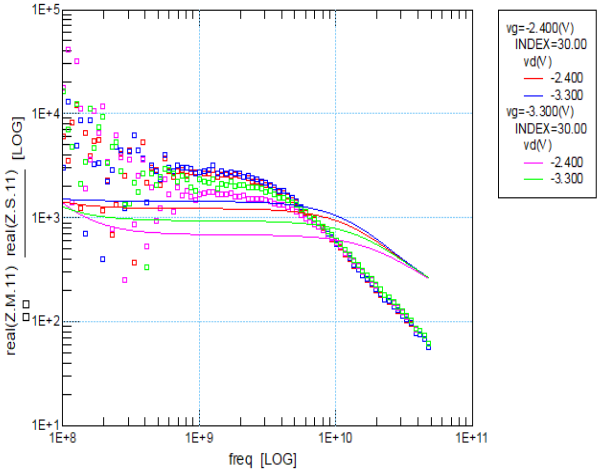
re(S11) = f(Vg,Vd)



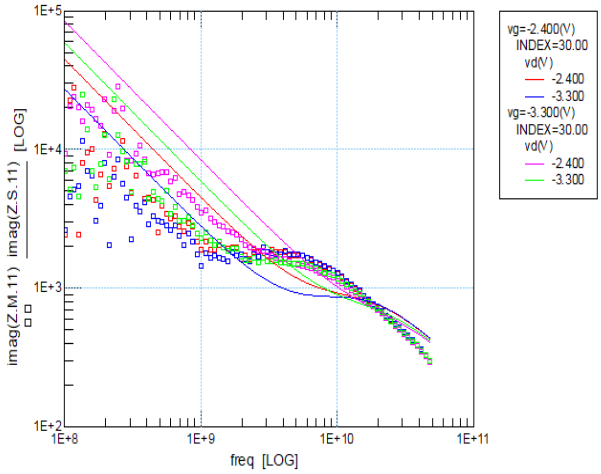
re(S22) = f(Vg,Vd)



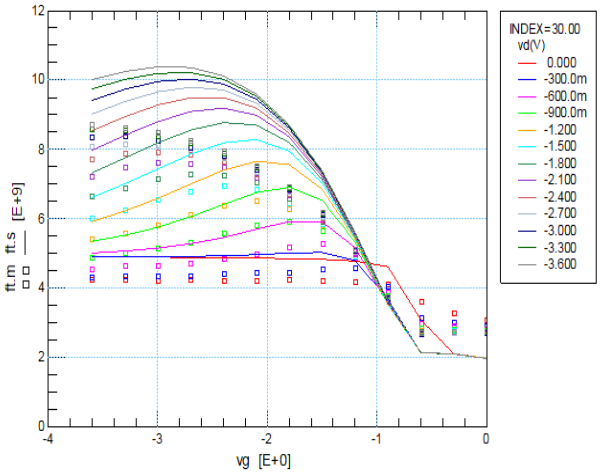
im(Y11) divided by ω



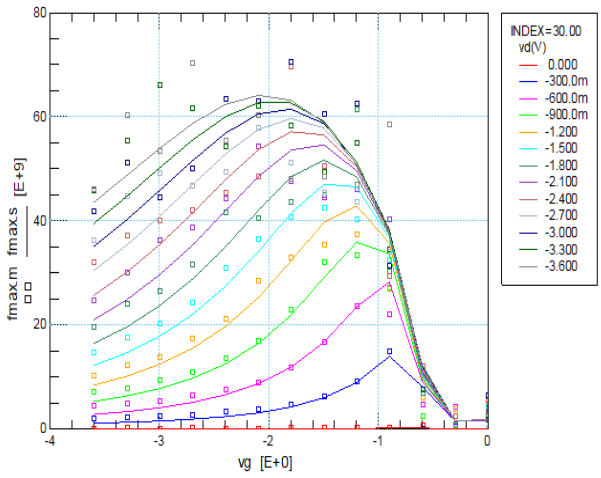
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



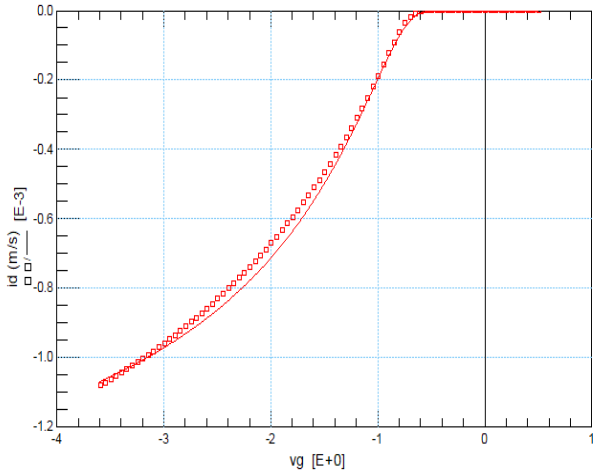
fT_vg = f(Vg,Vd)



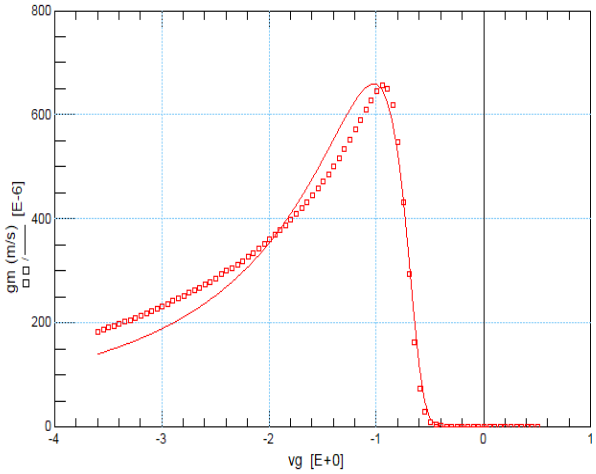
fmax_vg = f(Vg,Vd)

5.8 NF05_WF10u0_L0u4_S453A_D4

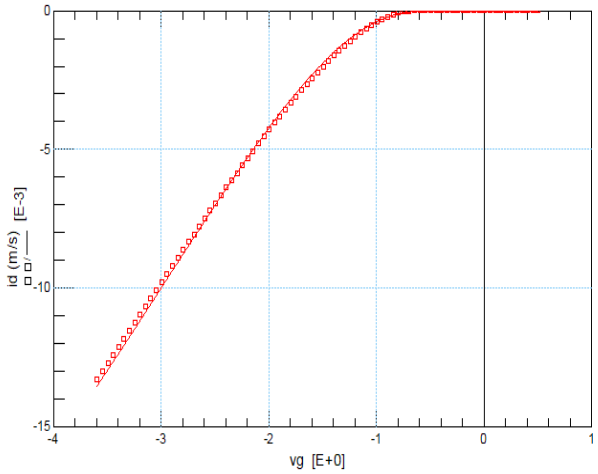
IdVg, T = 27°C



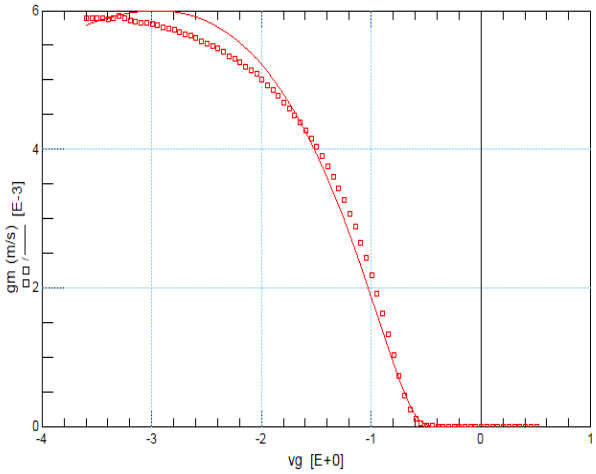
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

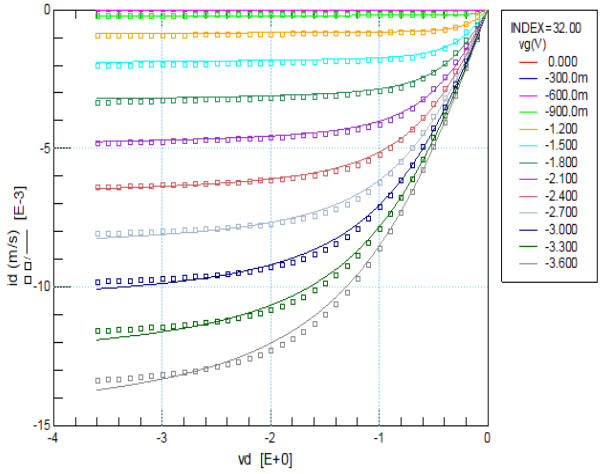


Id = f(Vg) @ Vd = -3.3V

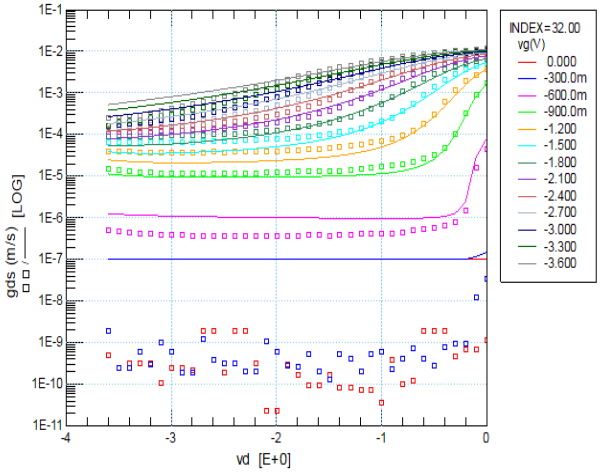


gm = f(Vg) @ Vd = -3.3V

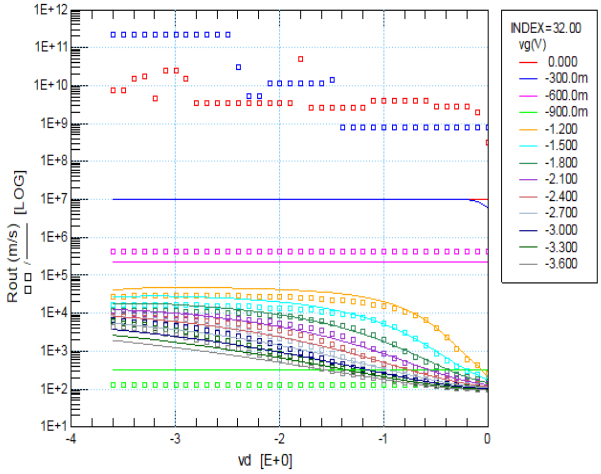
IdVd, T = 27°C



Id = f(Vd)

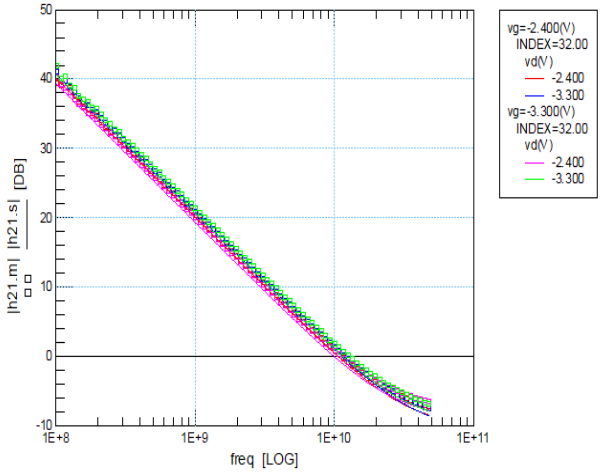


gds = f(Vd)

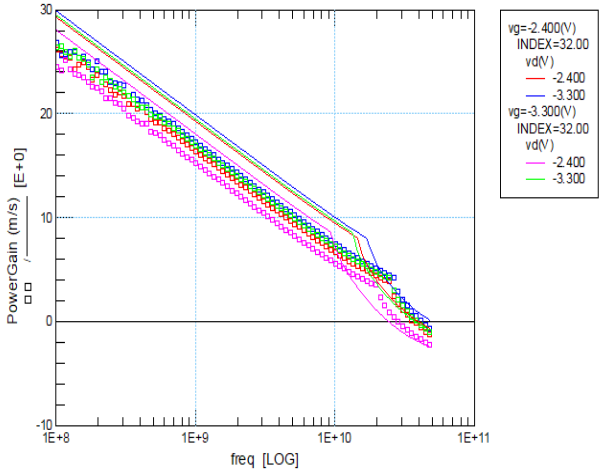


Rout = f(Vd)

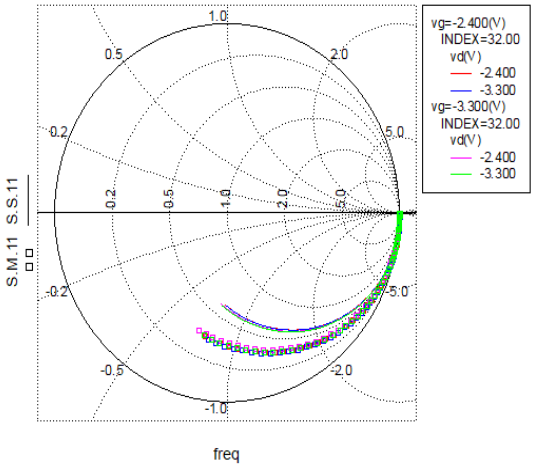
S-Parameter, T = 27°C



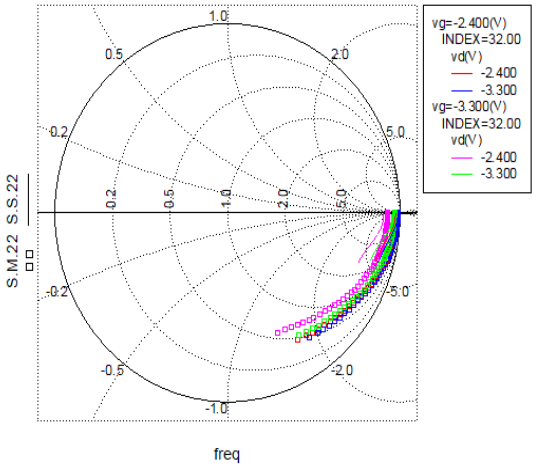
|h21| = f(Vg,Vd)



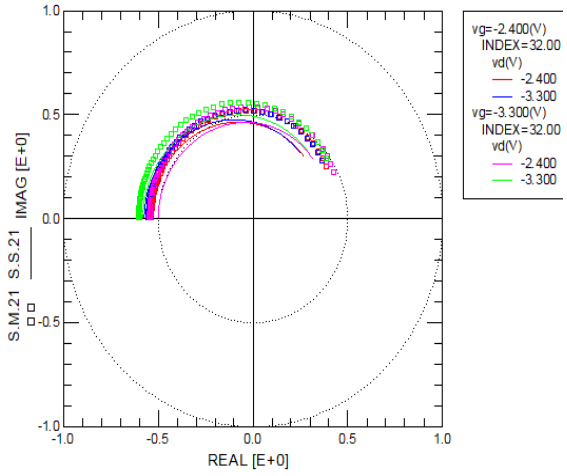
MSG/MAG = f(Vg,Vd)



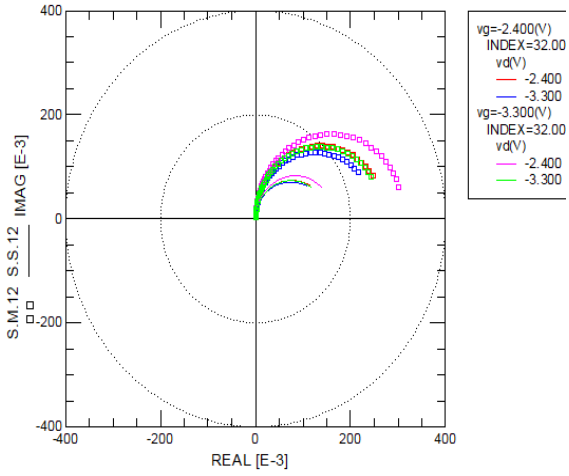
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

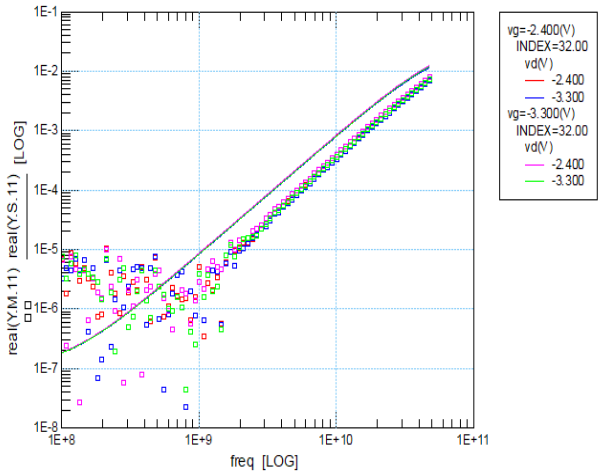


S21 = f(Vg,Vd)

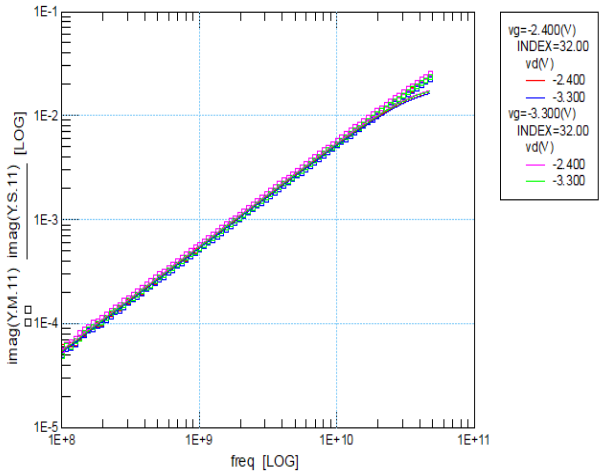


S12 = f(Vg,Vd)

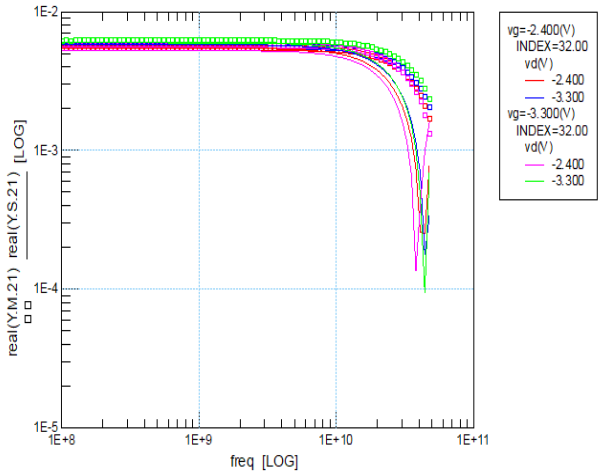
Y-Parameter, T = 27°C



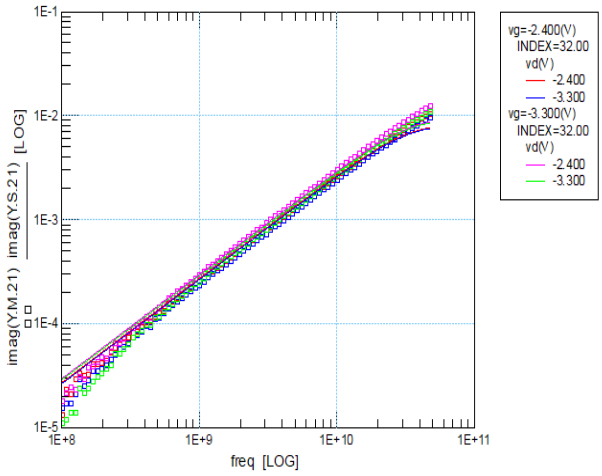
re(Y11) = f(Vg,Vd)



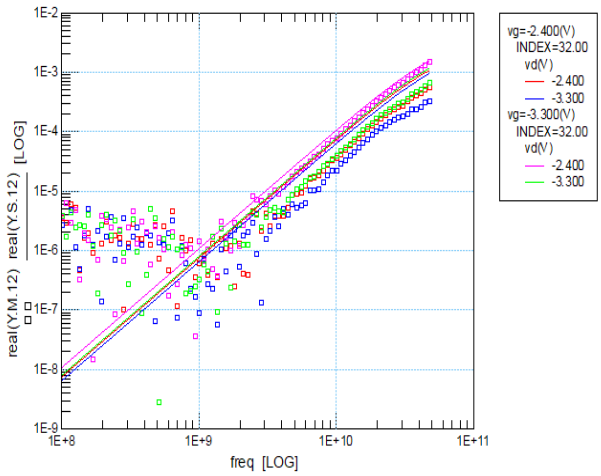
im(Y11) = f(Vg,Vd)



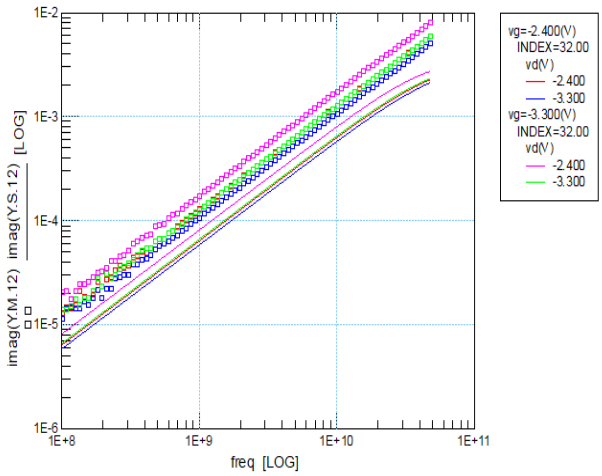
re(Y21) = f(Vg,Vd)



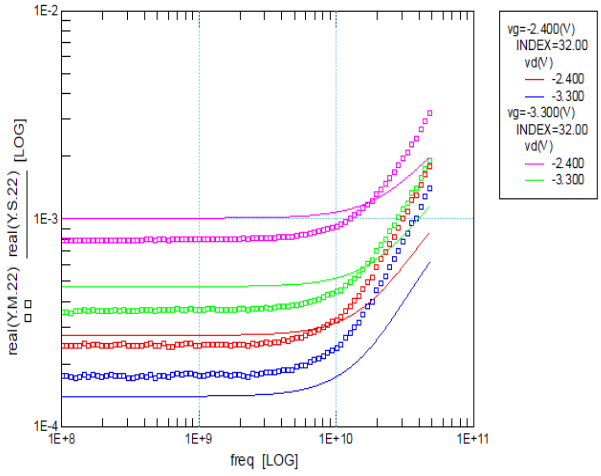
im(Y21) = f(Vg,Vd)



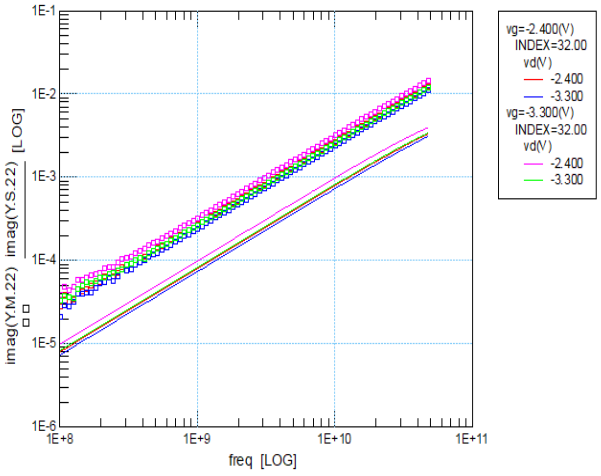
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

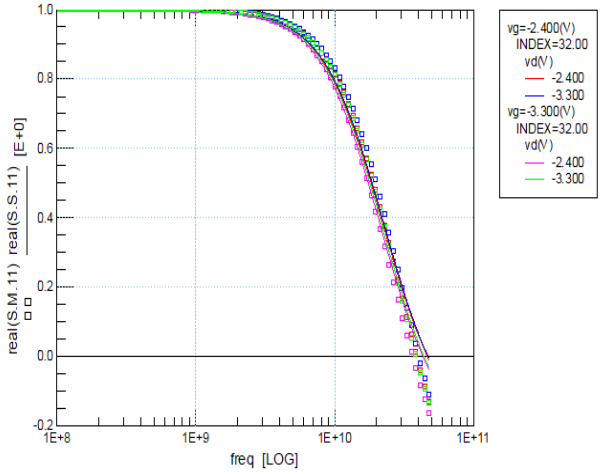


re(Y22) = f(Vg,Vd)

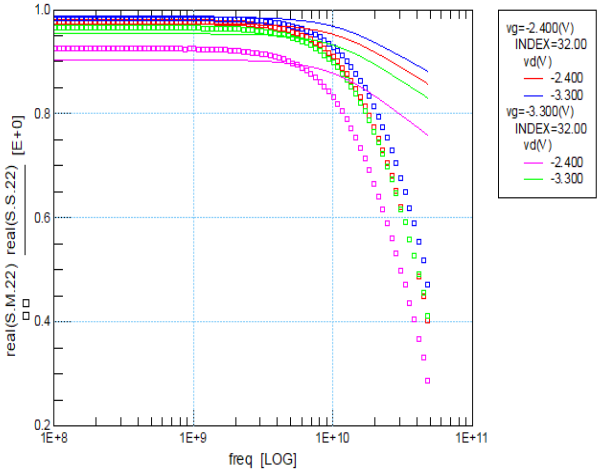


im(Y22) = f(Vg,Vd)

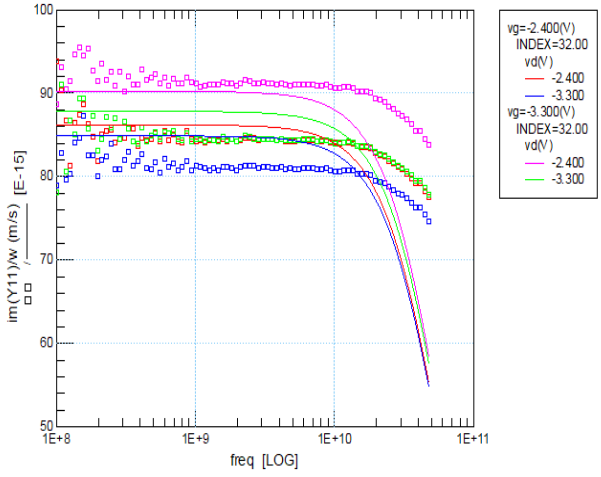
Verify Plots, T = 27°C



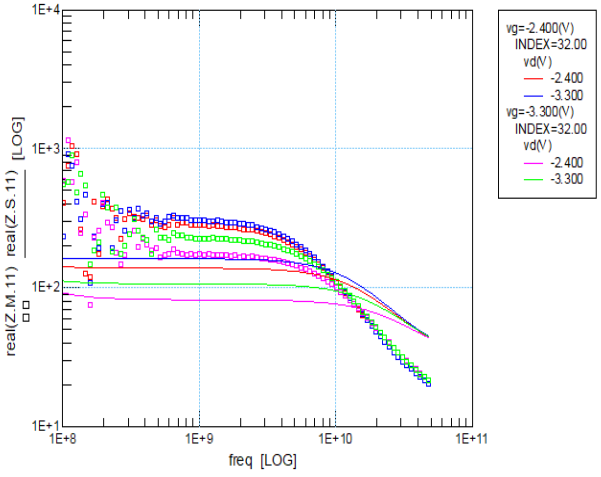
re(S11) = f(Vg,Vd)



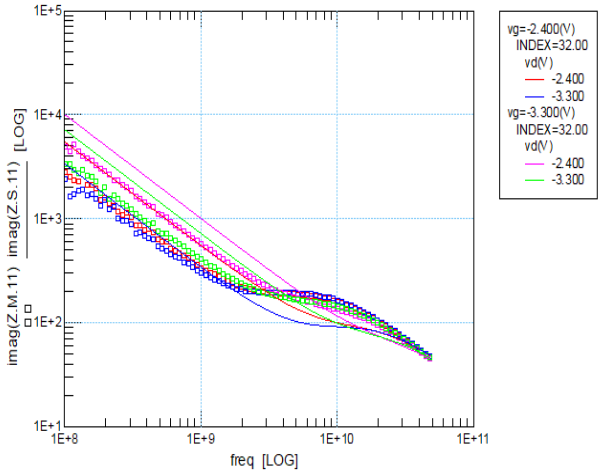
re(S22) = f(Vg,Vd)



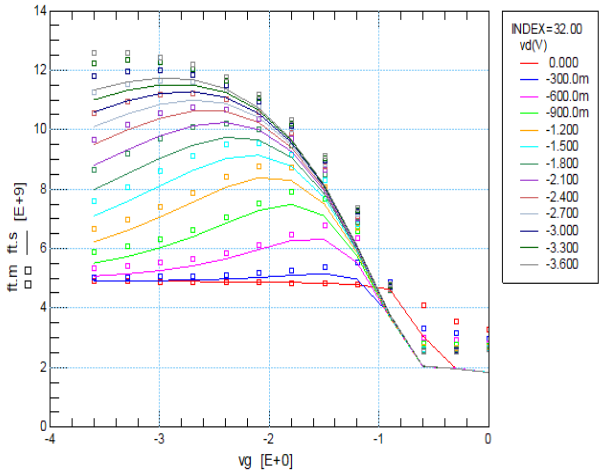
im(Y11) divided by ω



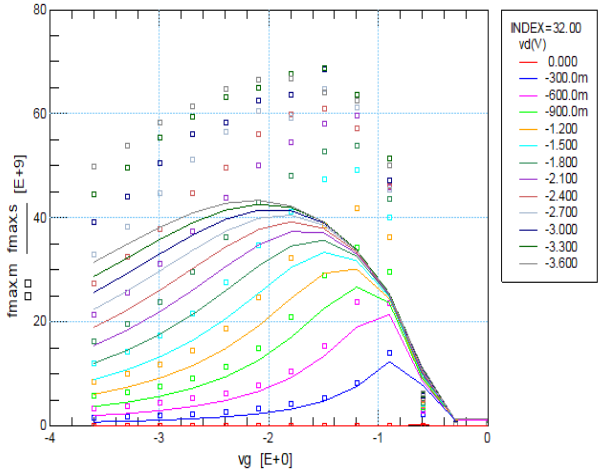
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



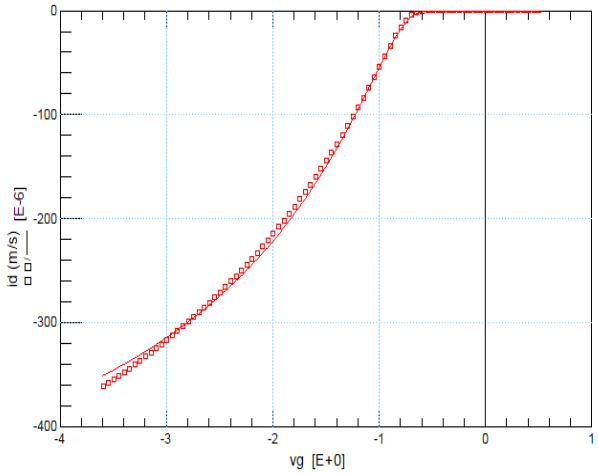
fT_vg = f(Vg,Vd)



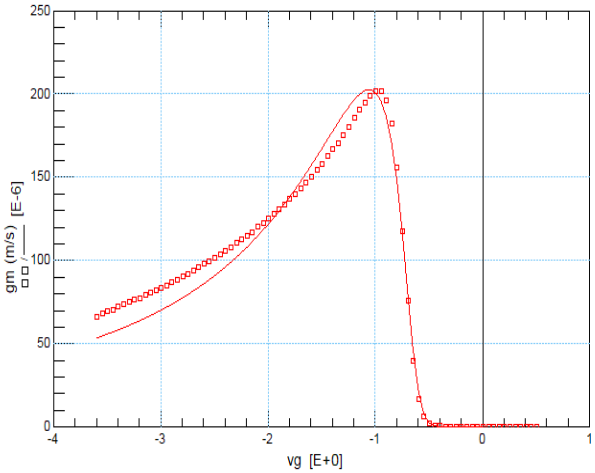
fmax_vg = f(Vg,Vd)

5.9 NF04_WF05u0_L0u5_S453A_D0

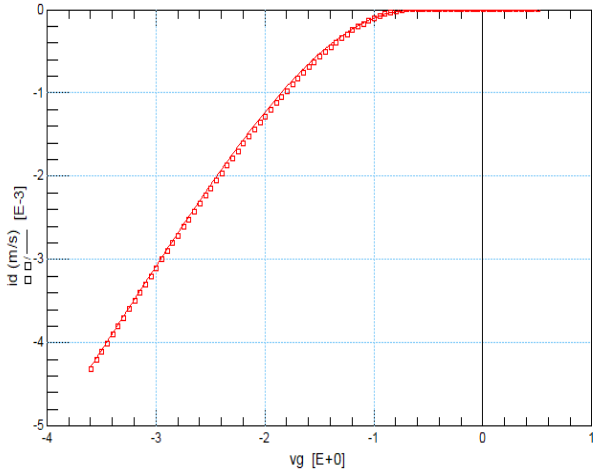
IdVg, T = 27°C



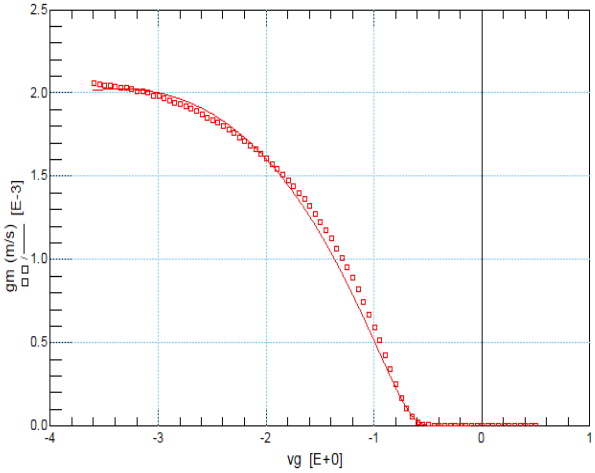
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

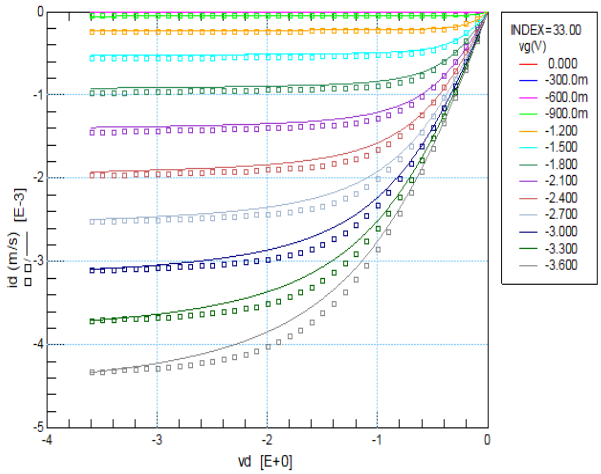


Id = f(Vg) @ Vd = -3.3V

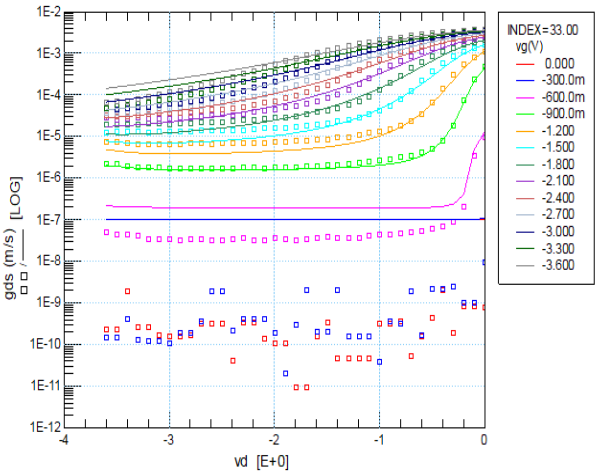


gm = f(Vg) @ Vd = -3.3V

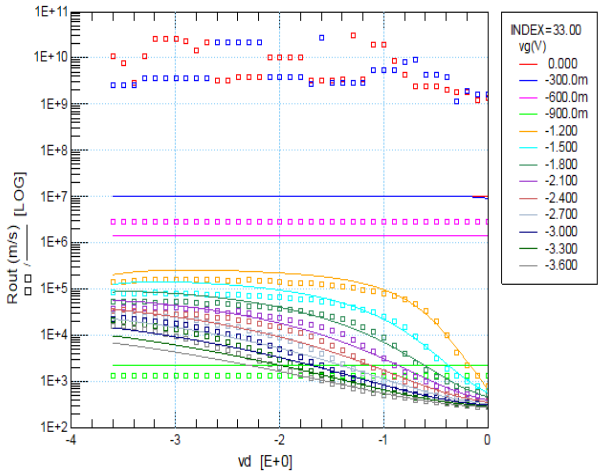
IdVd, T = 27°C



Id = f(Vd)

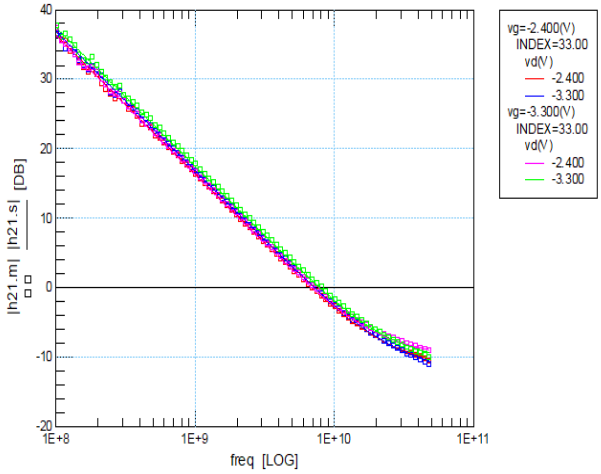


gds = f(Vd)

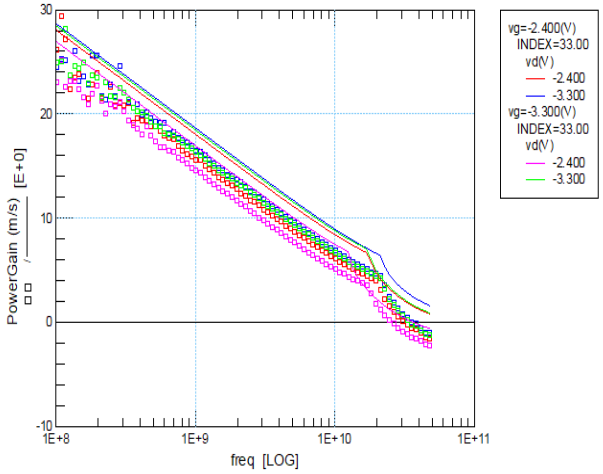


Rout = f(Vd)

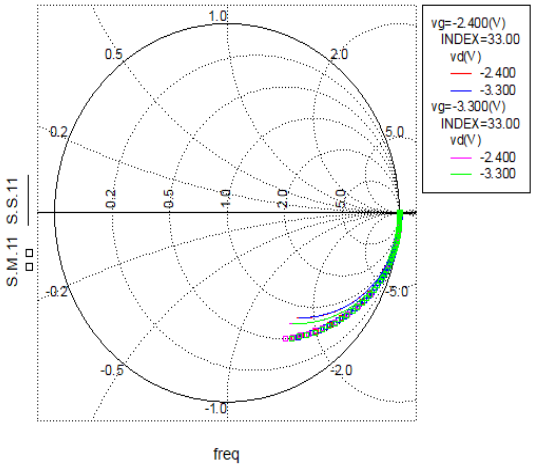
S-Parameter, T = 27°C



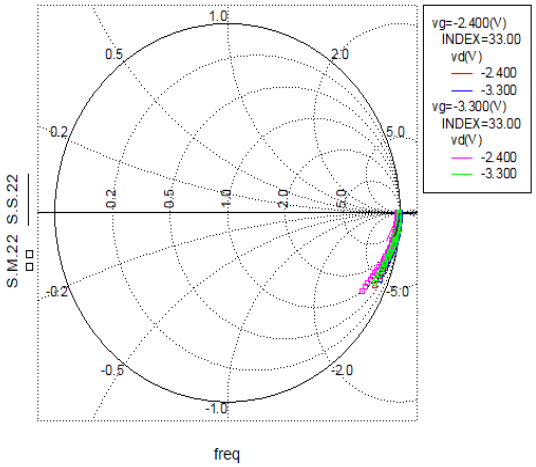
|h21| = f(Vg,Vd)



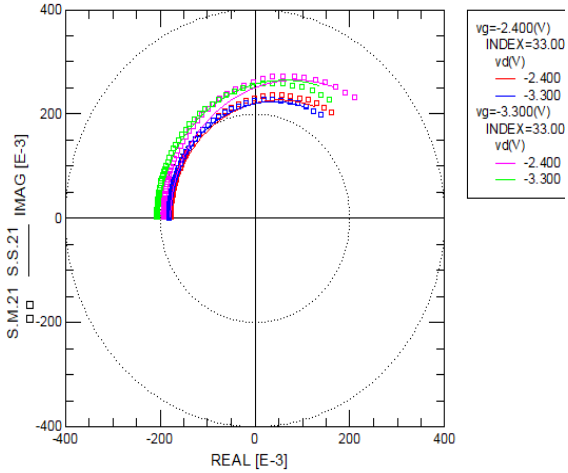
MSG/MAG = f(Vg,Vd)



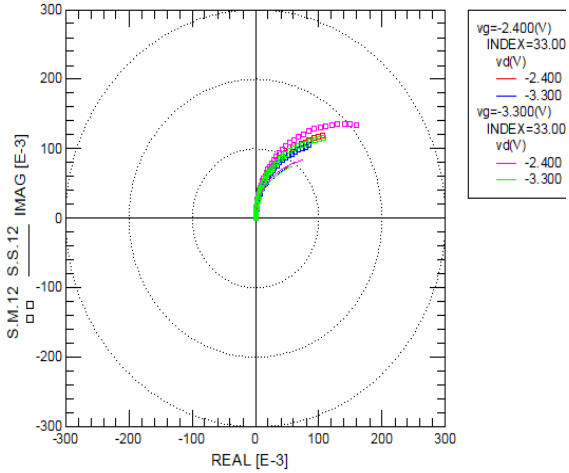
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

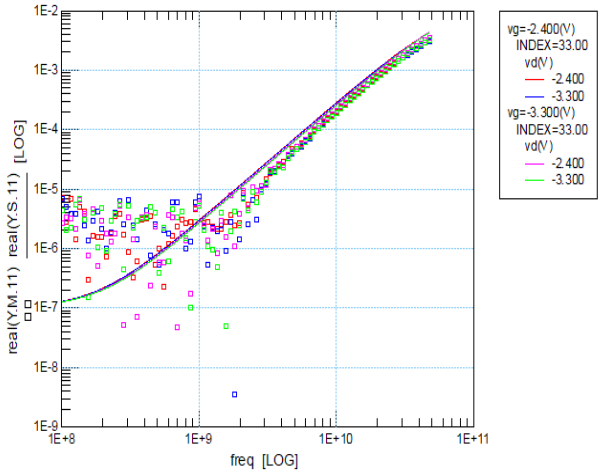


S21 = f(Vg,Vd)

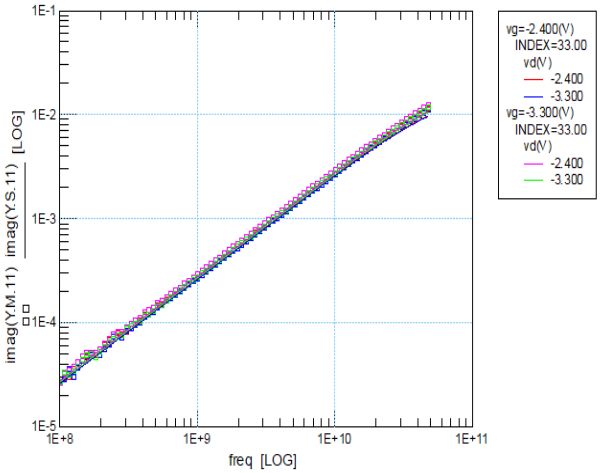


S12 = f(Vg,Vd)

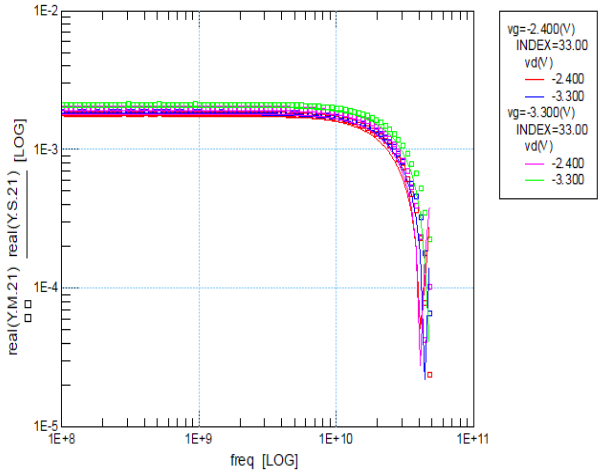
Y-Parameter, T = 27°C



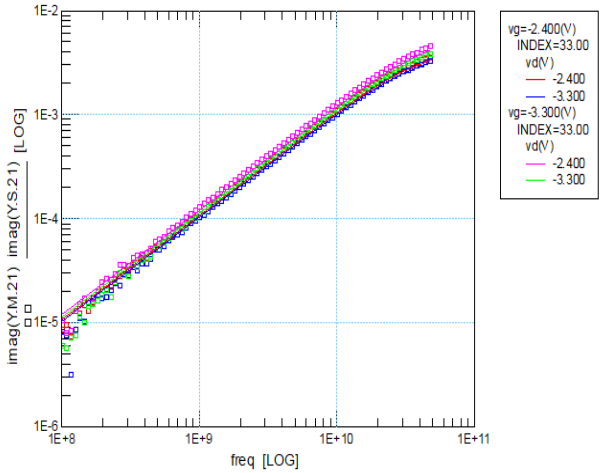
re(Y11) = f(Vg,Vd)



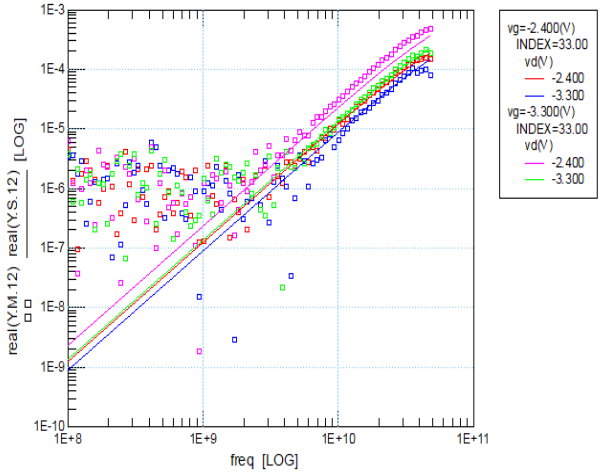
im(Y11) = f(Vg,Vd)



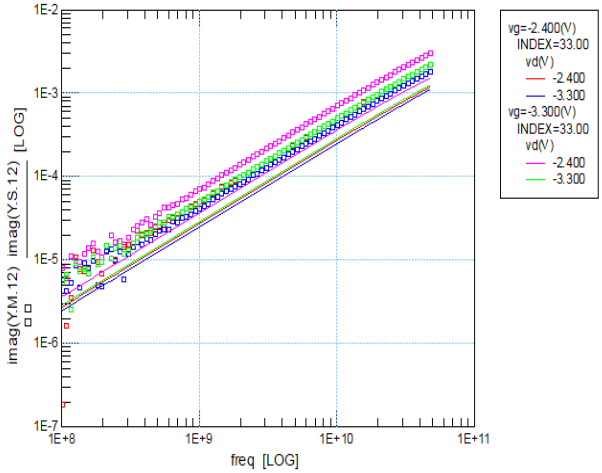
re(Y21) = f(Vg,Vd)



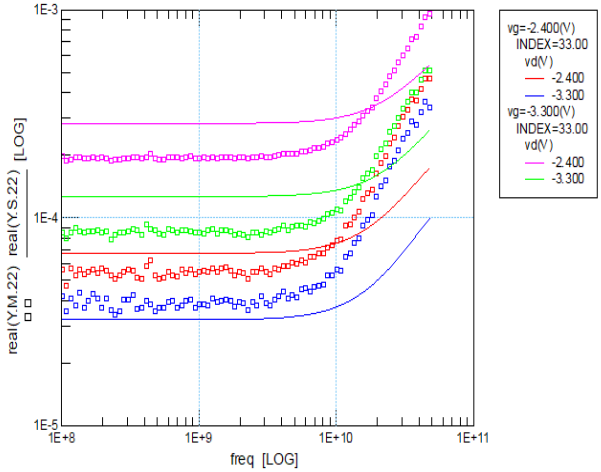
im(Y21) = f(Vg,Vd)



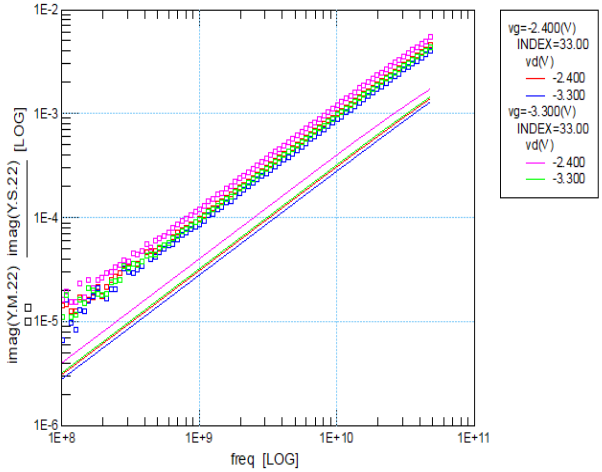
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

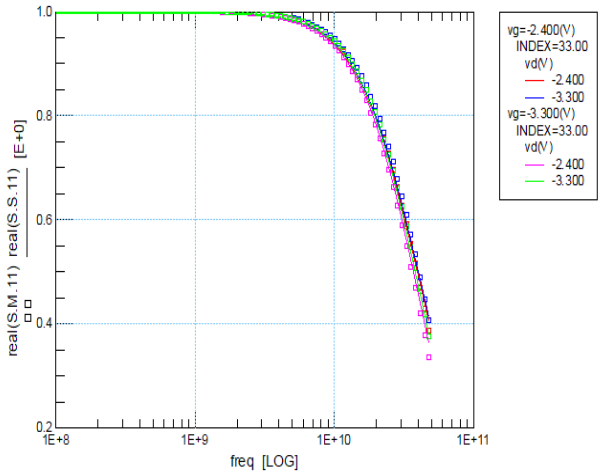


re(Y22) = f(Vg,Vd)

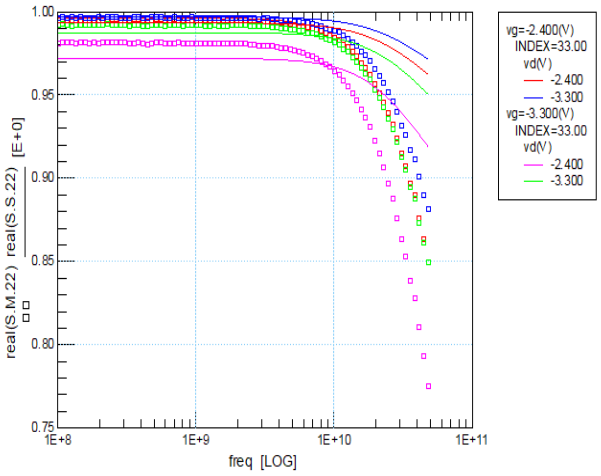


im(Y22) = f(Vg,Vd)

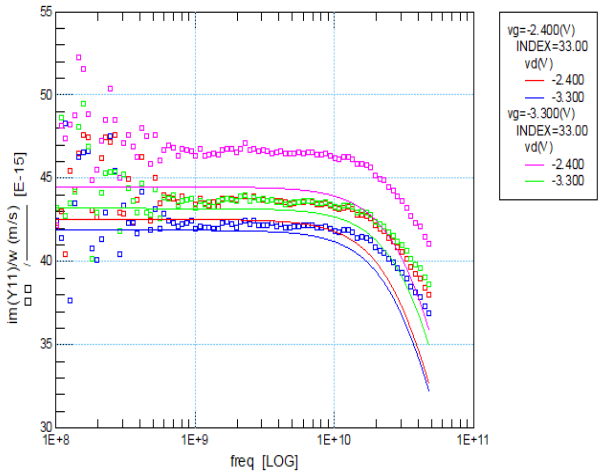
Verify Plots, T = 27°C



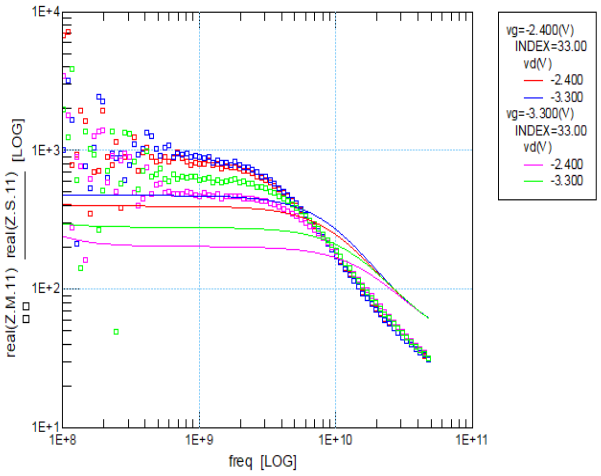
re(S11) = f(Vg,Vd)



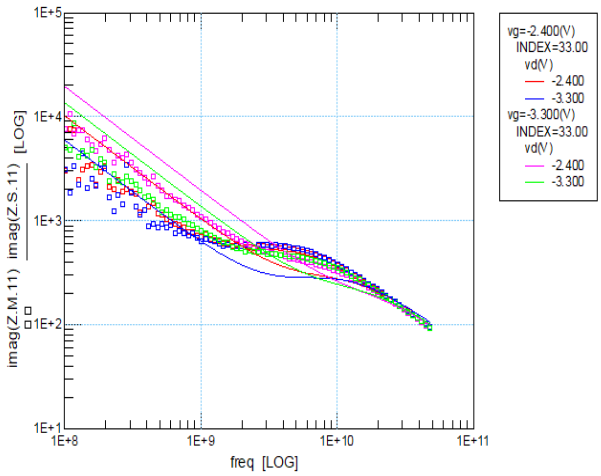
re(S22) = f(Vg,Vd)



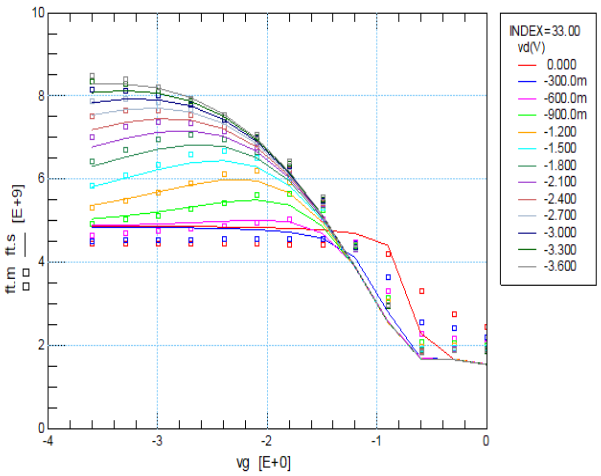
im(Y11) divided by ω



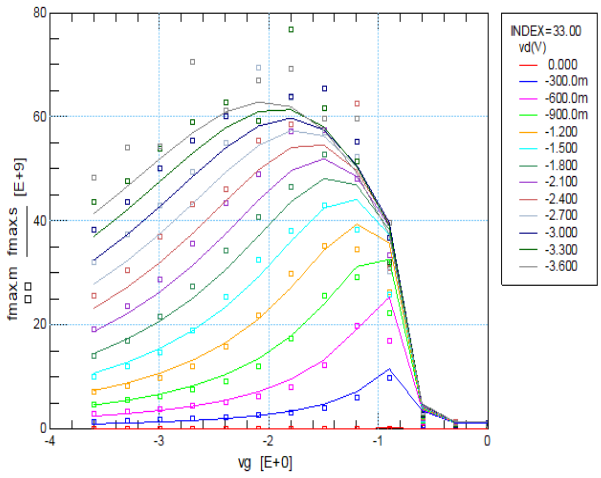
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



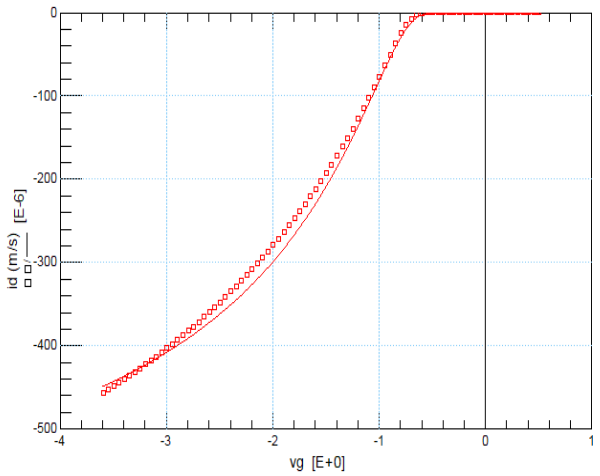
fT_vg = f(Vg,Vd)



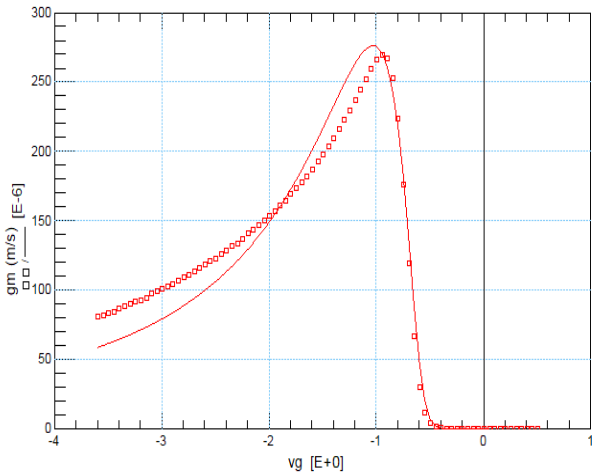
fmax_vg = f(Vg,Vd)

5.10 NF04_WF05u0_L0u4_post_C5

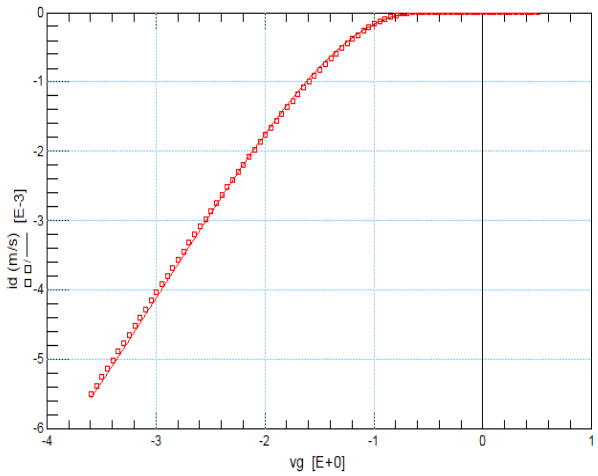
IdVg, T = 27°C



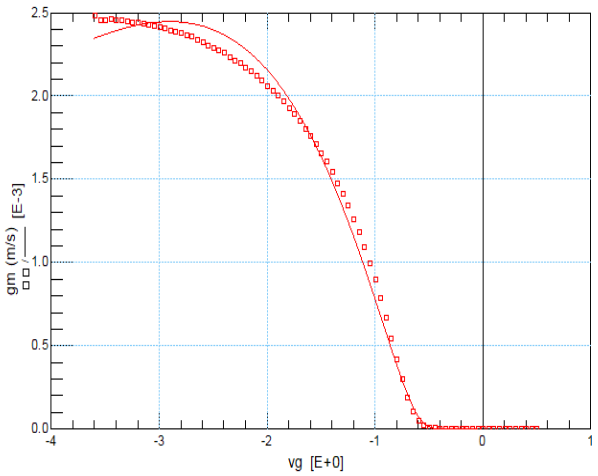
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

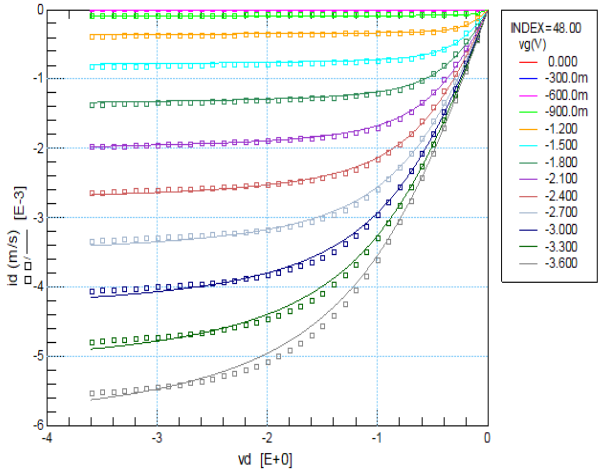


Id = f(Vg) @ Vd = -3.3V

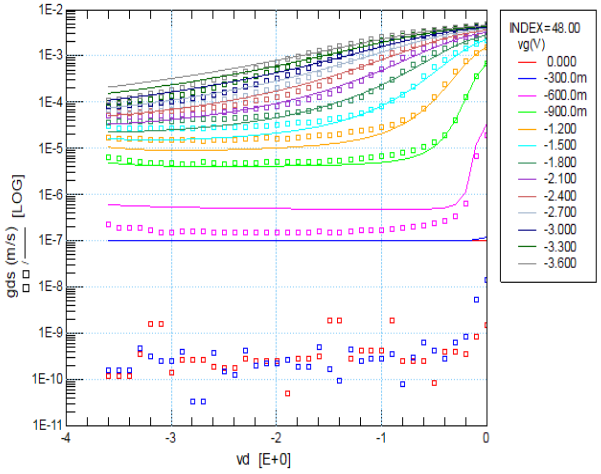


gm = f(Vg) @ Vd = -3.3V

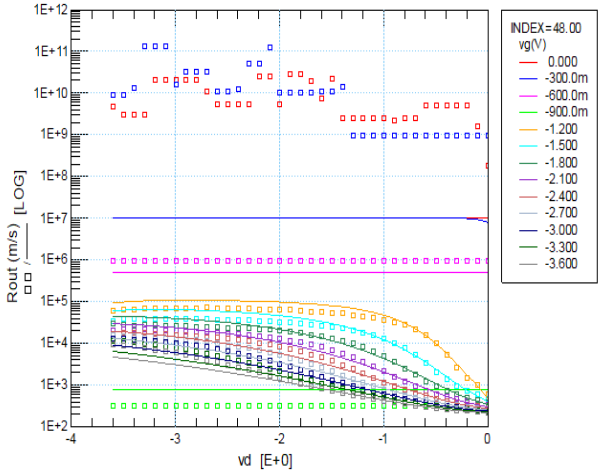
IdVd, T = 27°C



Id = f(Vd)

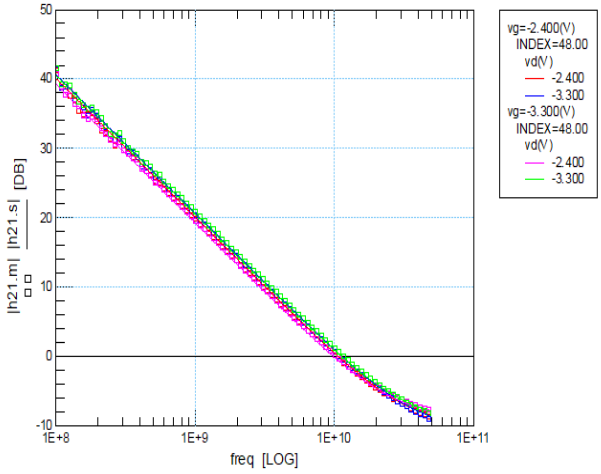


gds = f(Vd)

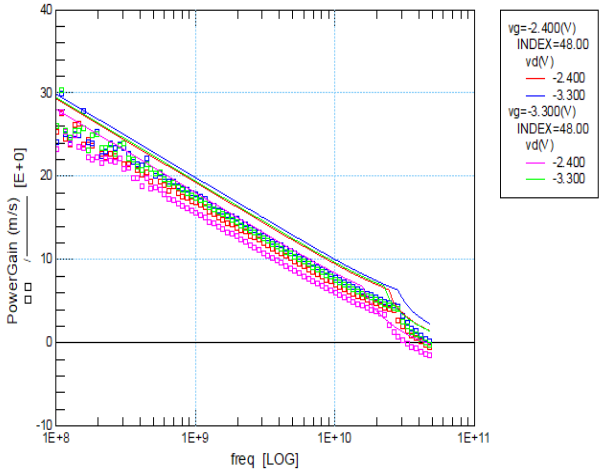


Rout = f(Vd)

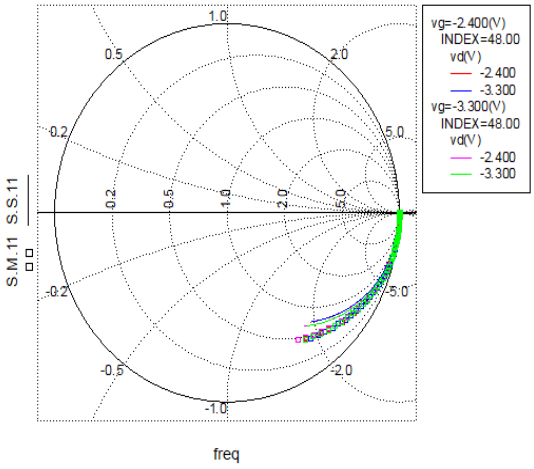
S-Parameter, T = 27°C



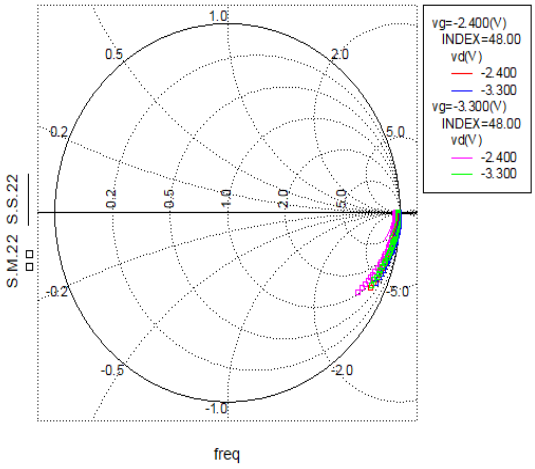
|h21| = f(Vg,Vd)



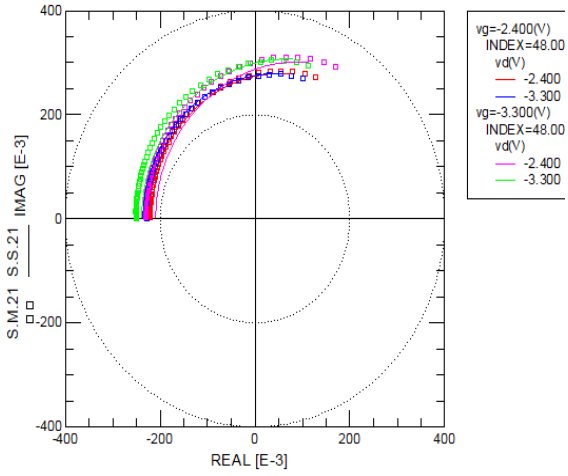
MSG/MAG = f(Vg,Vd)



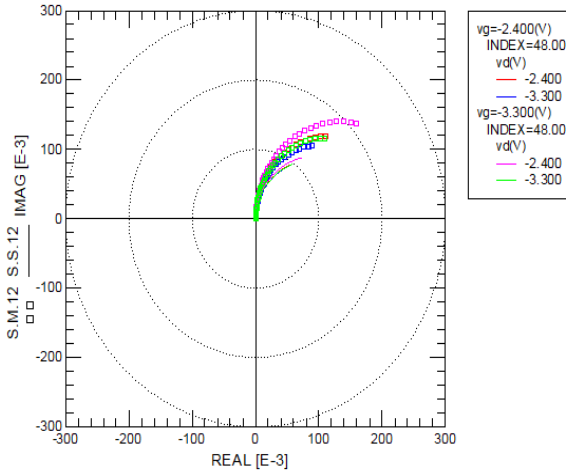
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

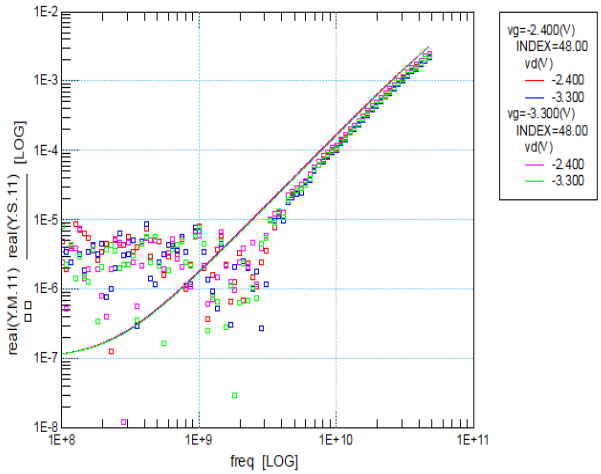


S21 = f(Vg,Vd)

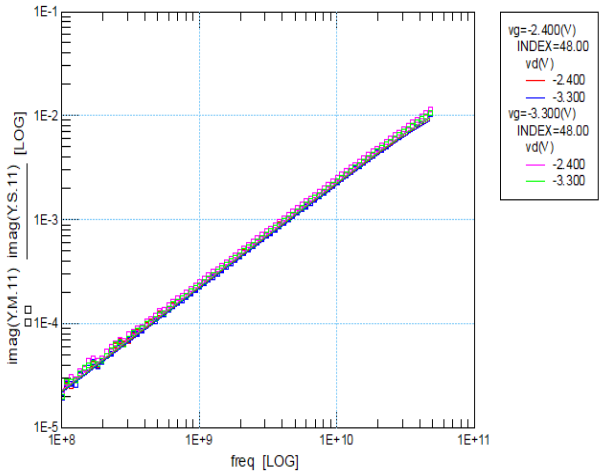


S12 = f(Vg,Vd)

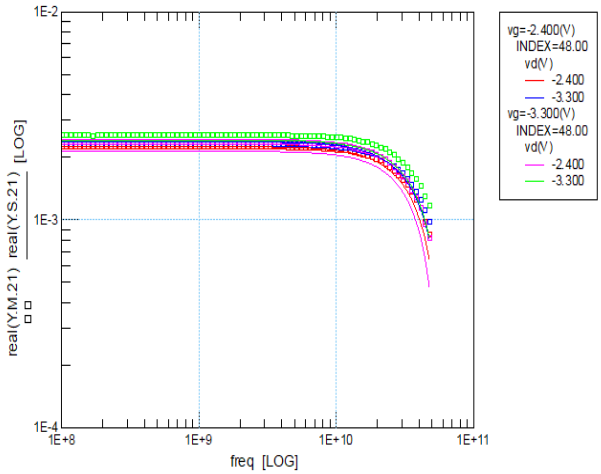
Y-Parameter, T = 27°C



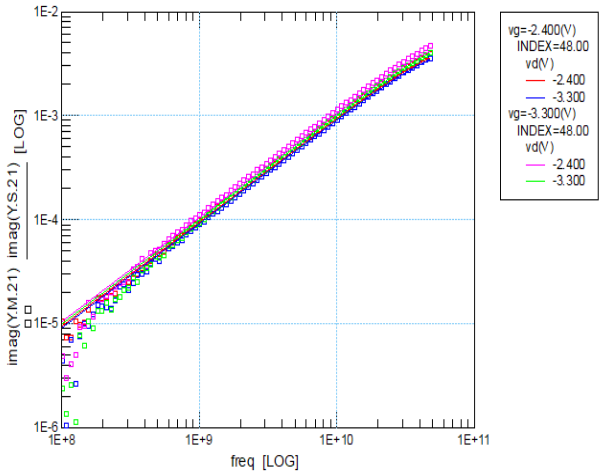
re(Y11) = f(Vg,Vd)



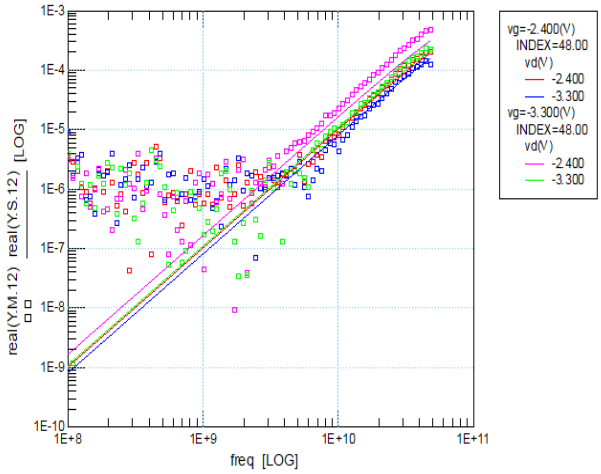
im(Y11) = f(Vg,Vd)



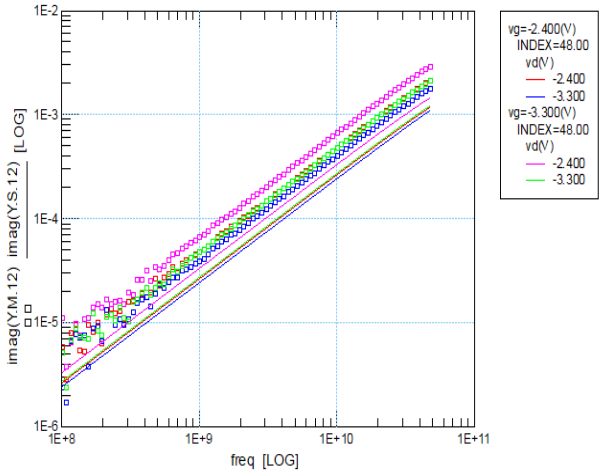
re(Y21) = f(Vg,Vd)



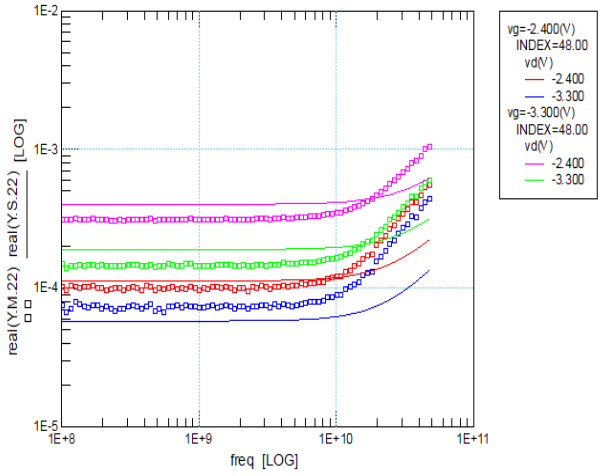
im(Y21) = f(Vg,Vd)



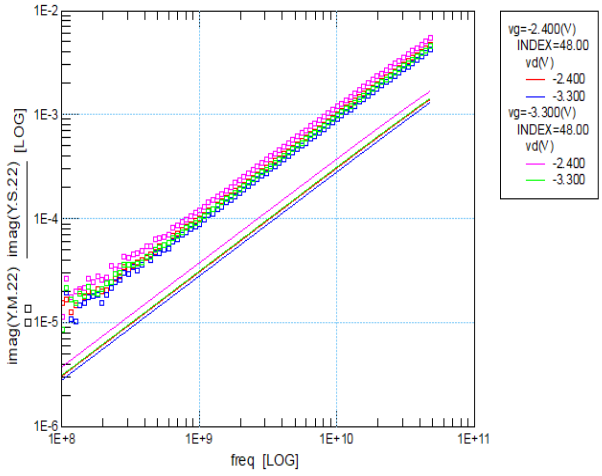
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

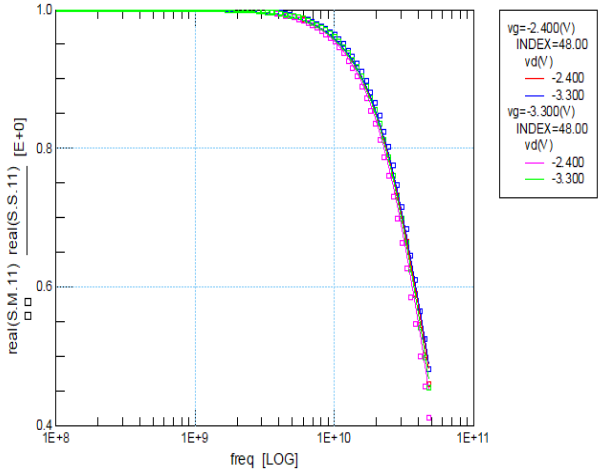


re(Y22) = f(Vg,Vd)

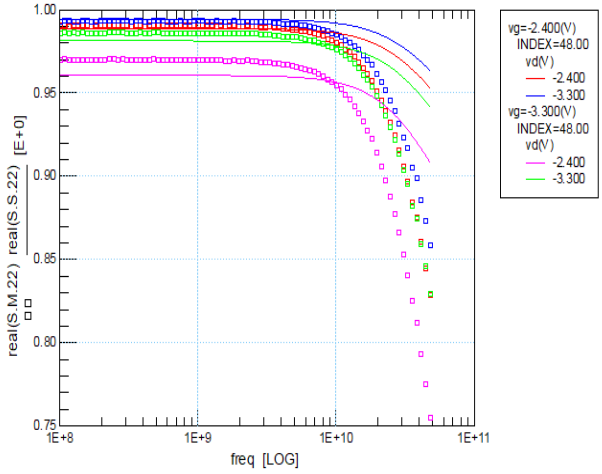


im(Y22) = f(Vg,Vd)

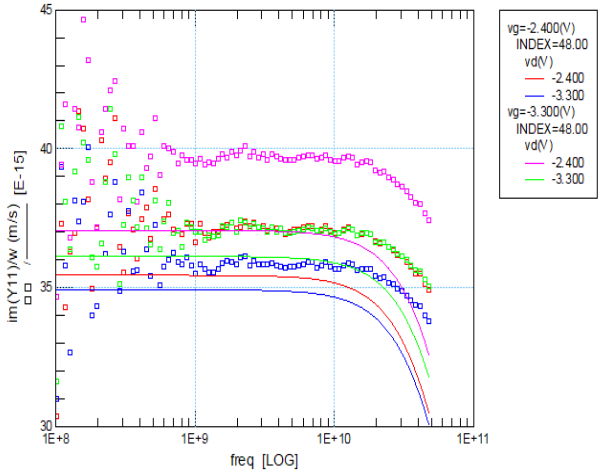
Verify Plots, T = 27°C



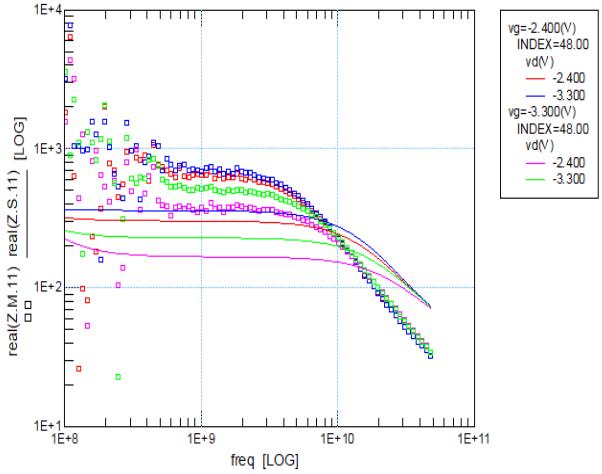
re(S11) = f(Vg,Vd)



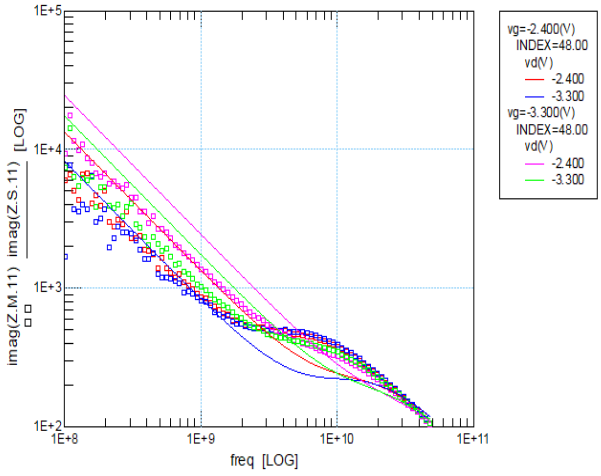
re(S22) = f(Vg,Vd)



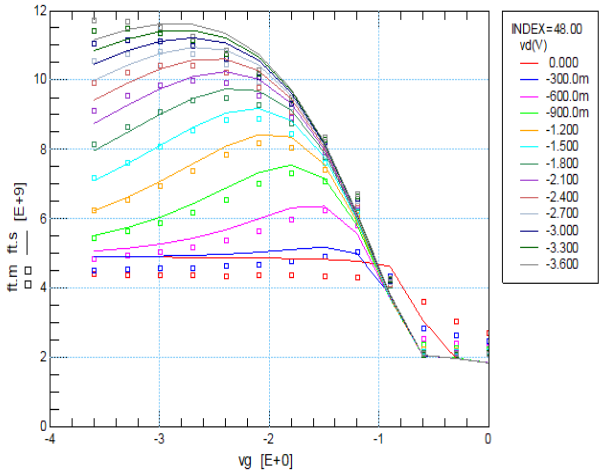
im(Y11) divided by ω



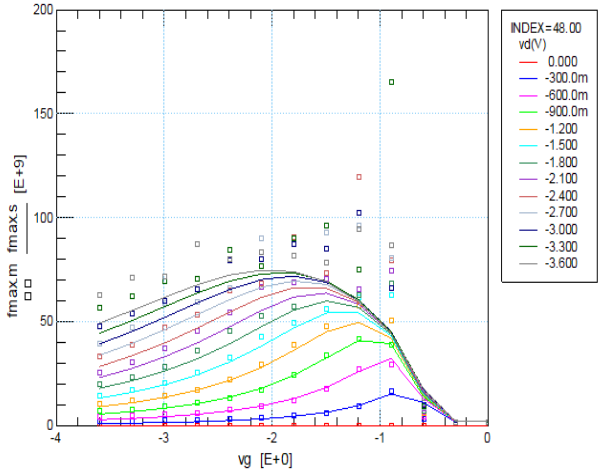
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



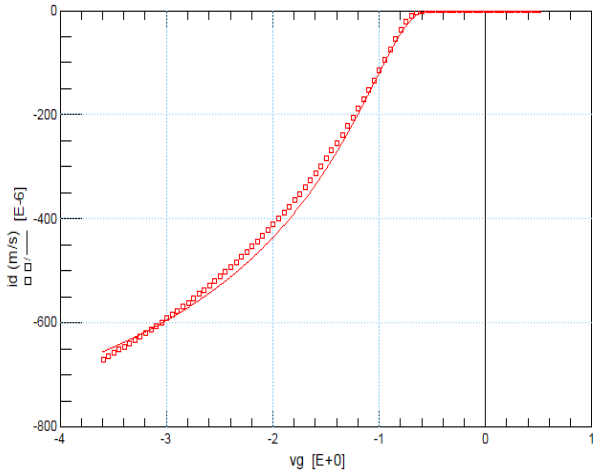
fT_vg = f(Vg,Vd)



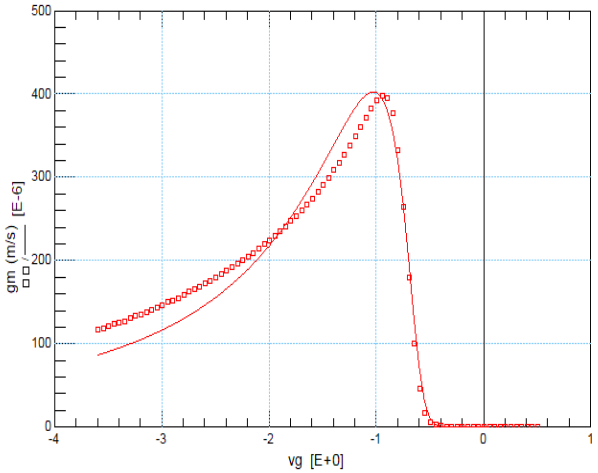
fmax_vg = f(Vg,Vd)

5.11 NF04_WF07u5_L0u4_S453A_D2

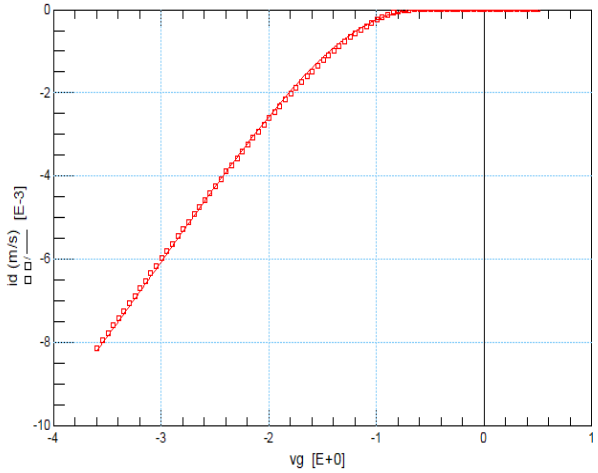
IdVg, T = 27°C



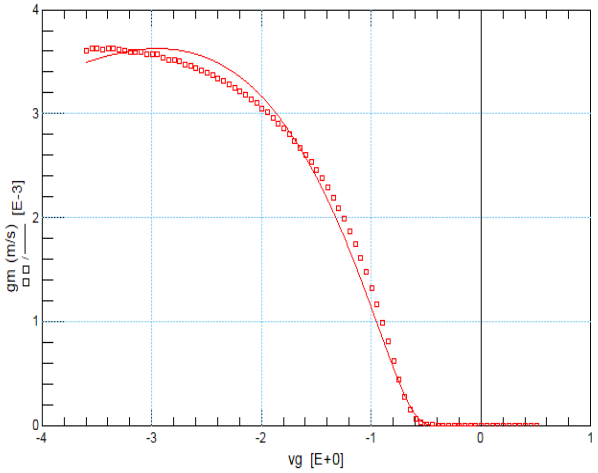
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

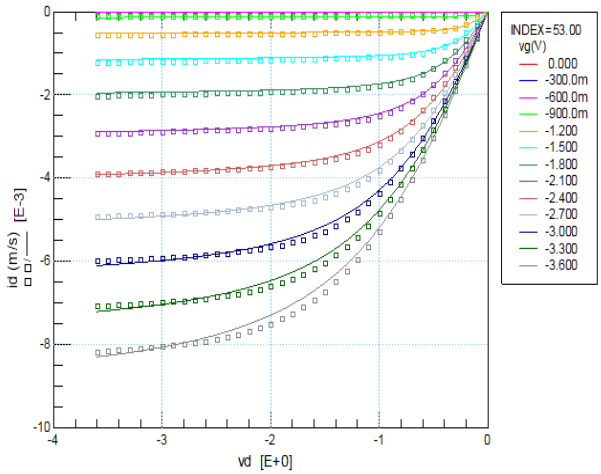


Id = f(Vg) @ Vd = -3.3V

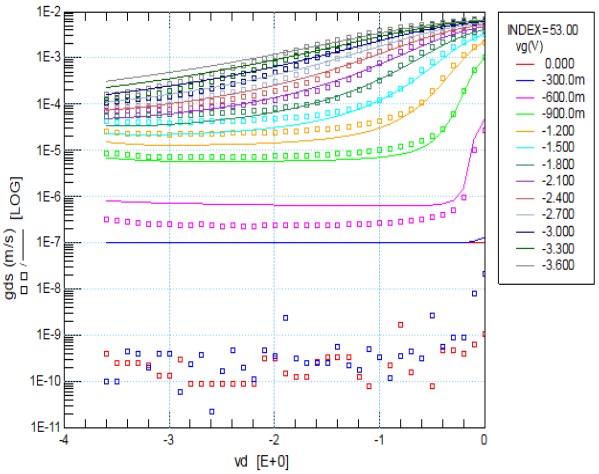


gm = f(Vg) @ Vd = -3.3V

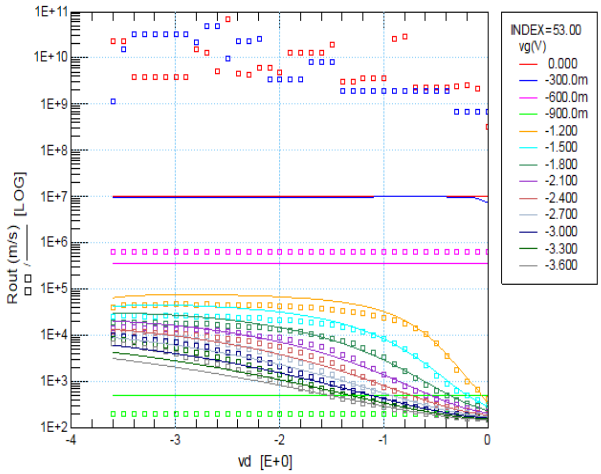
IdVd, T = 27°C



Id = f(Vd)

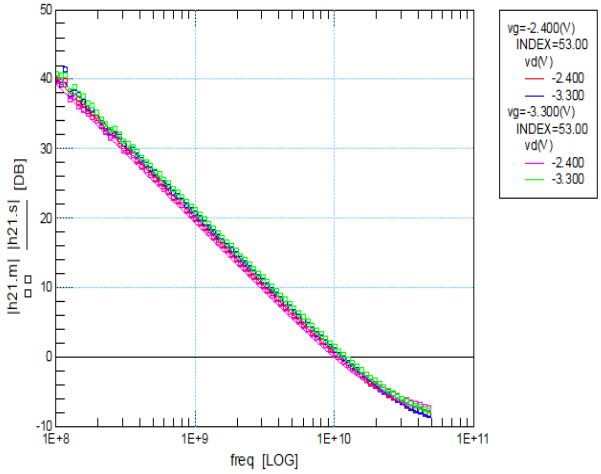


gds = f(Vd)

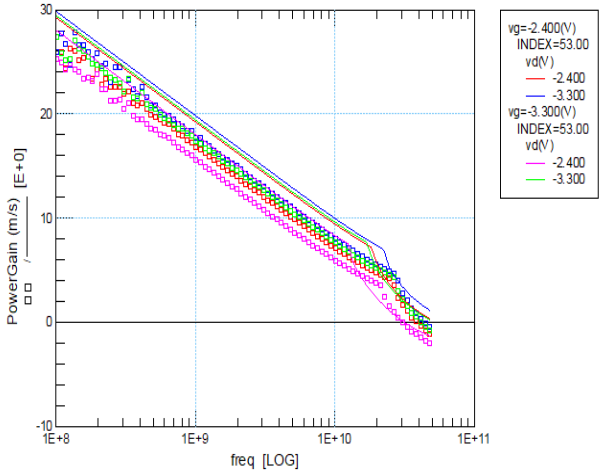


Rout = f(Vd)

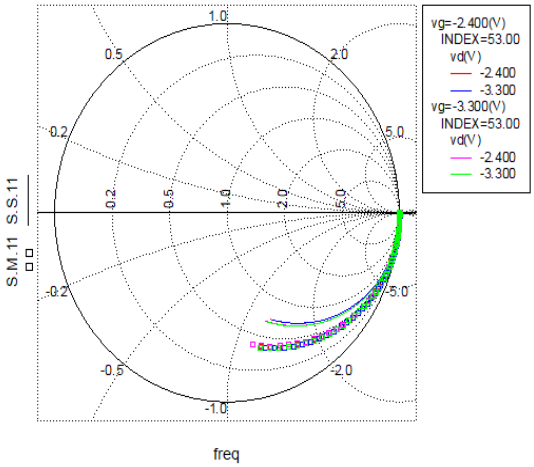
S-Parameter, T = 27°C



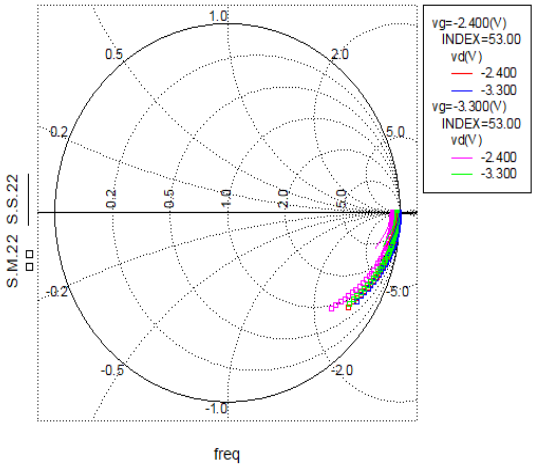
|h21| = f(Vg,Vd)



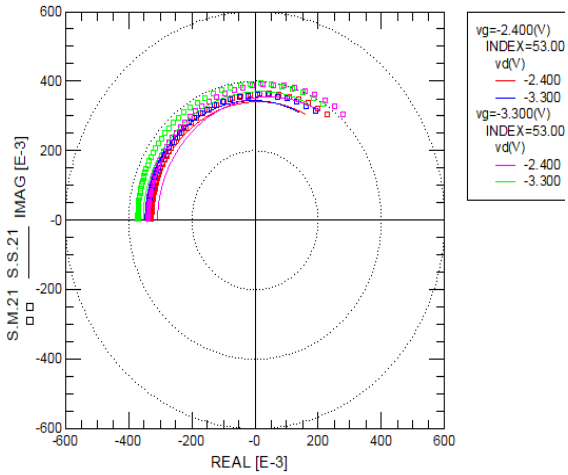
MSG/MAG = f(Vg,Vd)



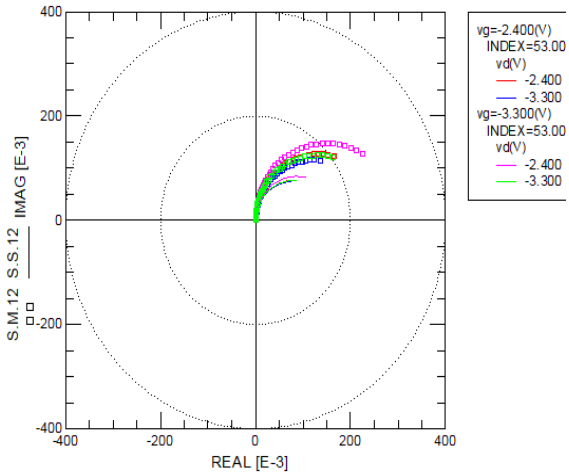
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

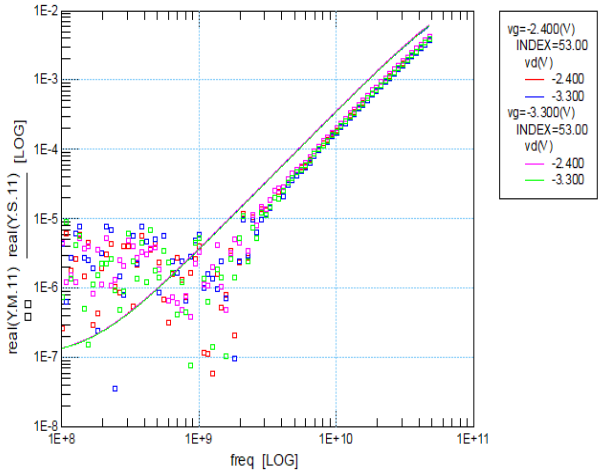


S21 = f(Vg,Vd)

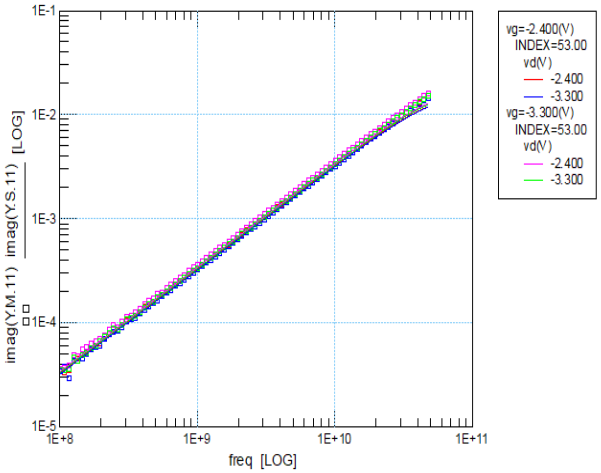


S12 = f(Vg,Vd)

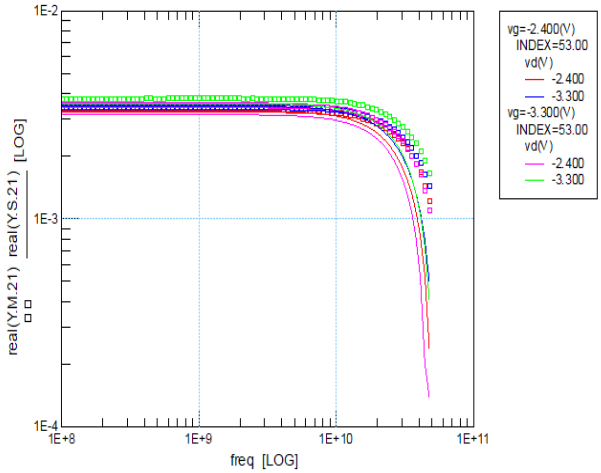
Y-Parameter, T = 27°C



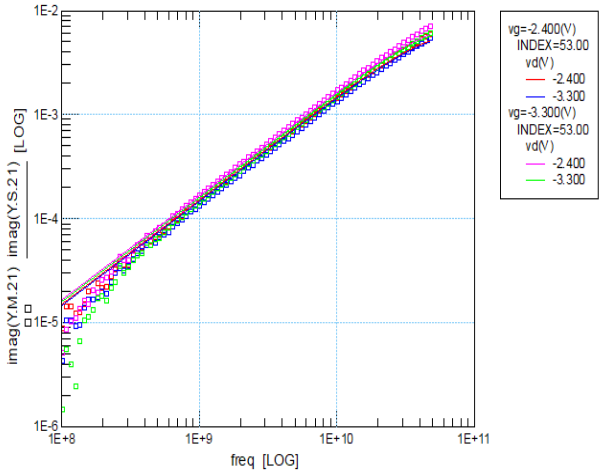
re(Y11) = f(Vg,Vd)



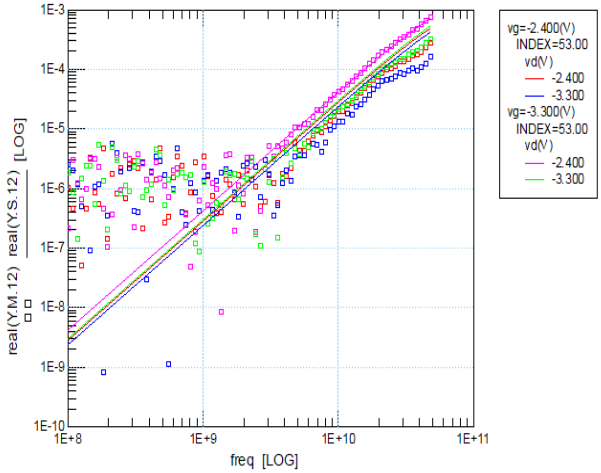
im(Y11) = f(Vg,Vd)



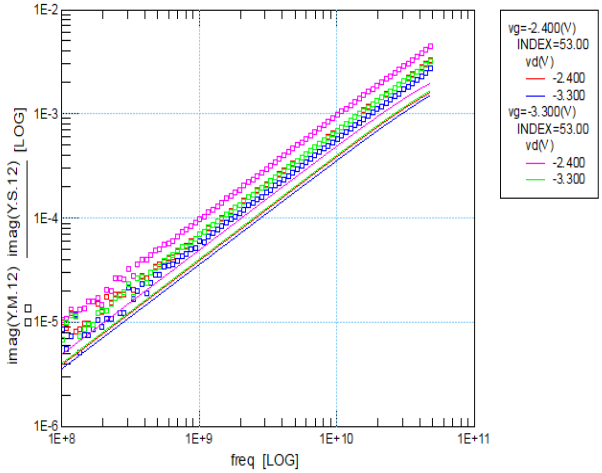
re(Y21) = f(Vg,Vd)



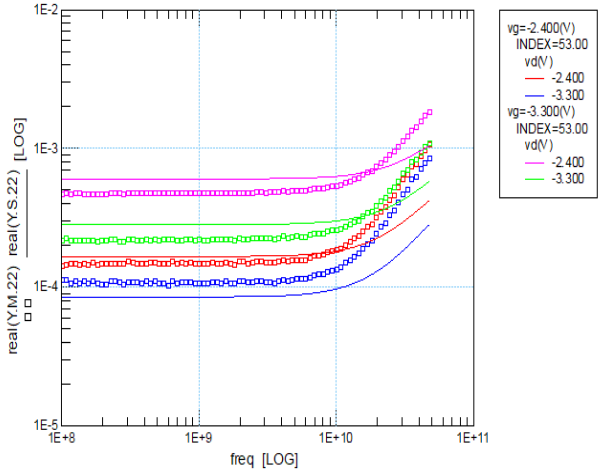
im(Y21) = f(Vg,Vd)



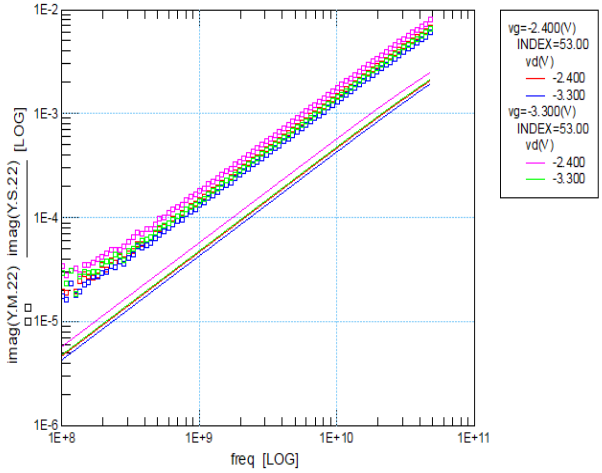
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

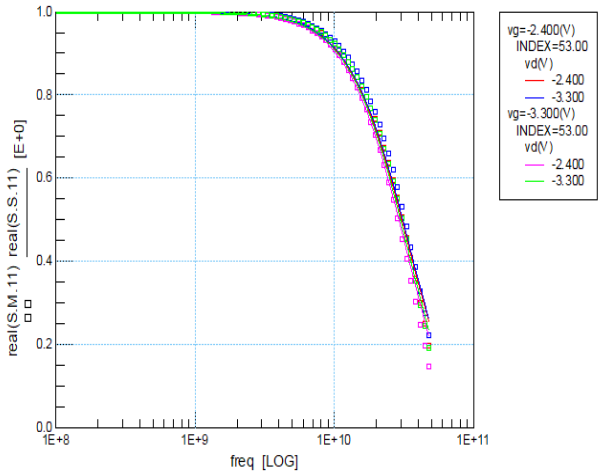


re(Y22) = f(Vg,Vd)

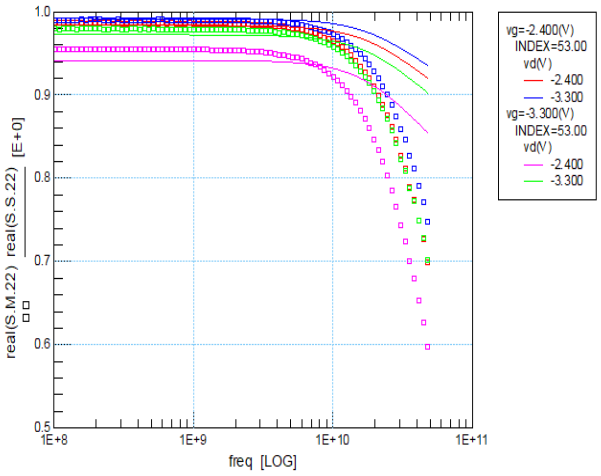


im(Y22) = f(Vg,Vd)

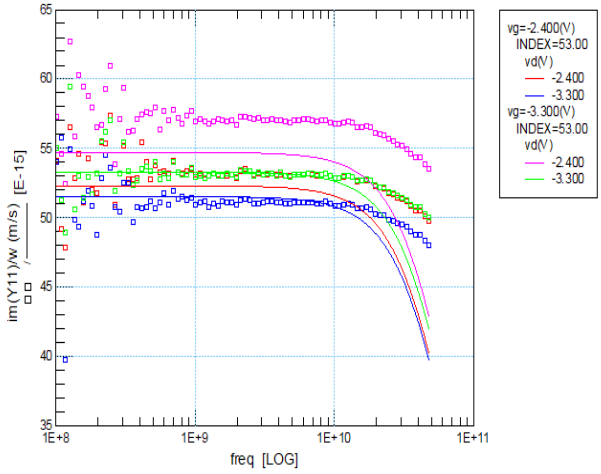
Verify Plots, T = 27°C



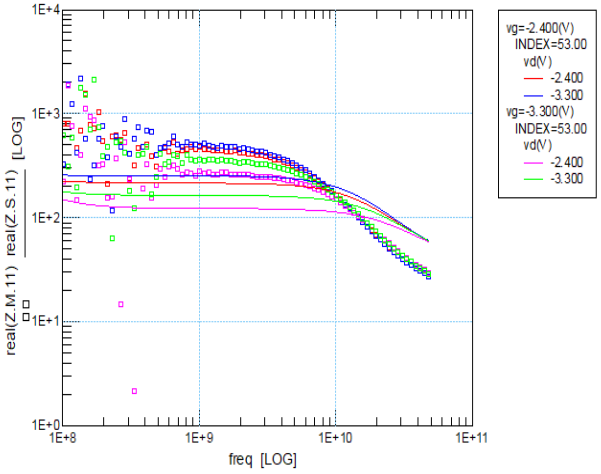
re(S11) = f(Vg,Vd)



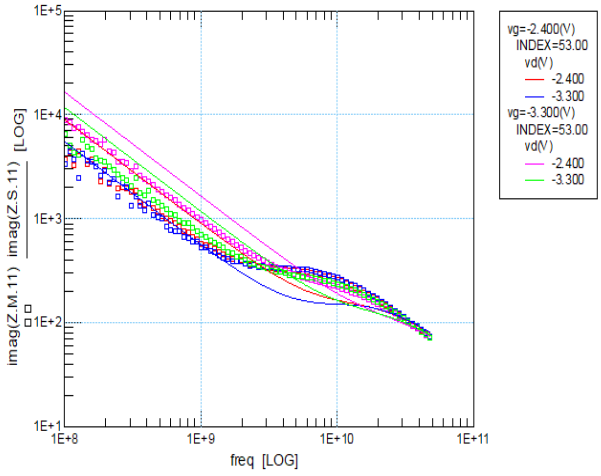
re(S22) = f(Vg,Vd)



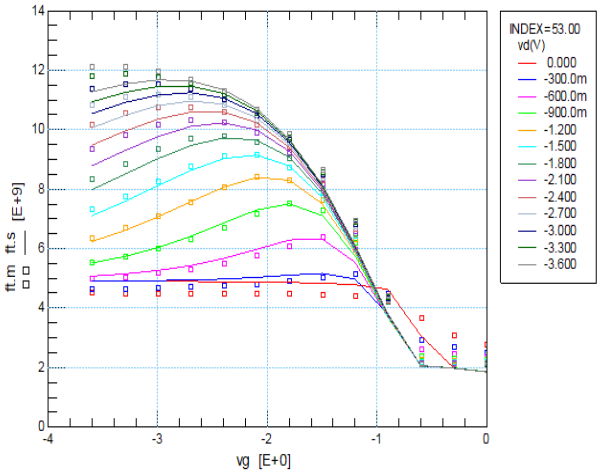
im(Y11) divided by ω



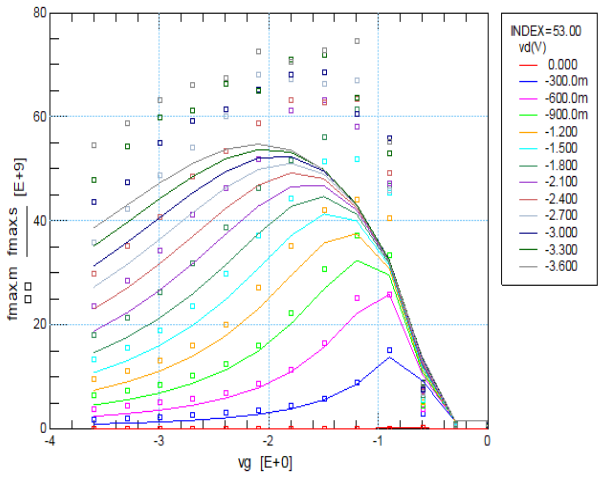
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



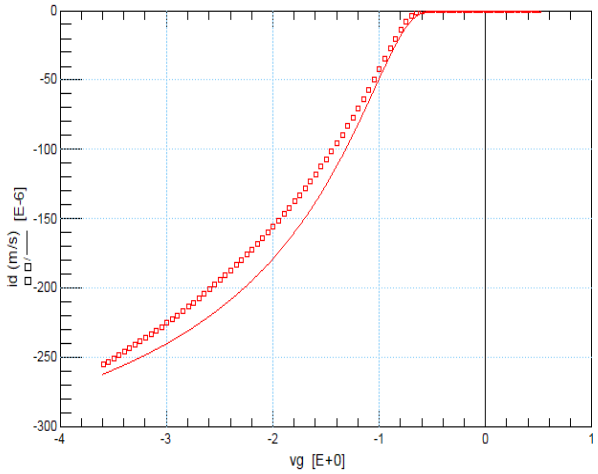
fT_vg = f(Vg,Vd)



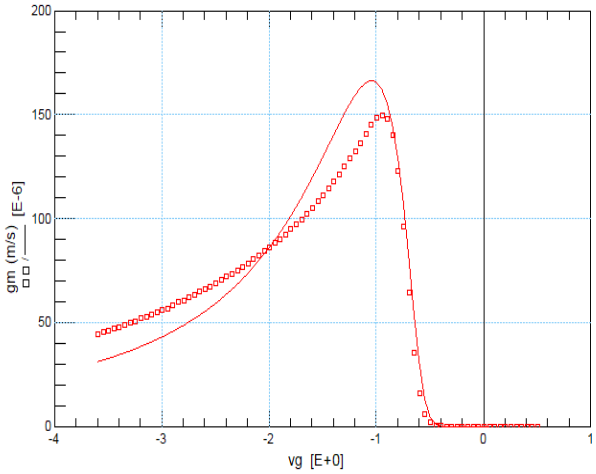
fmax_vg = f(Vg,Vd)

5.12 NF10_WF01u0_L0u4_S453A_E0

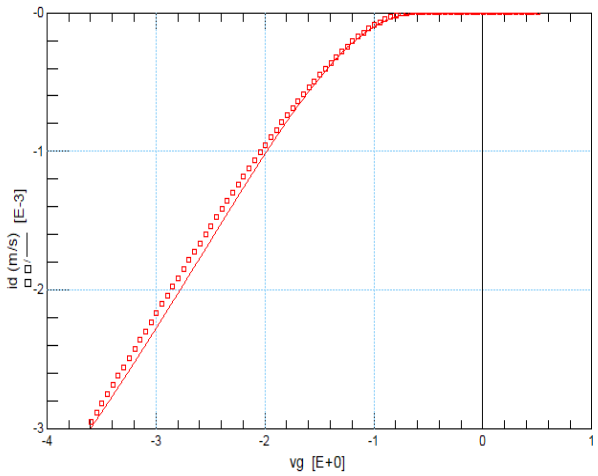
IdVg, T = 27°C



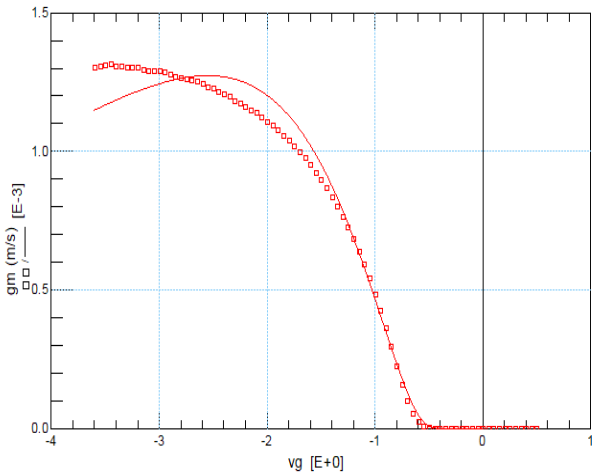
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

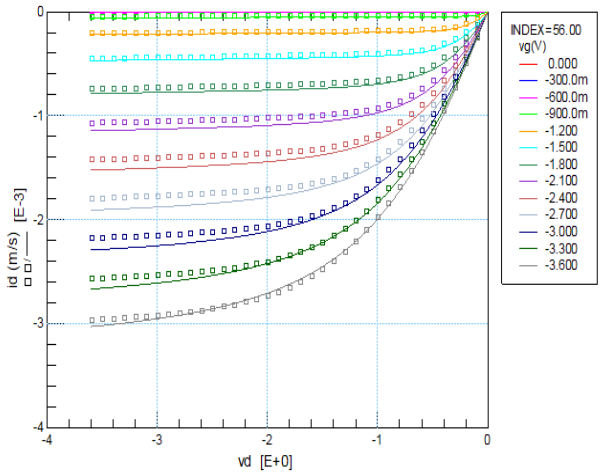


Id = f(Vg) @ Vd = -3.3V

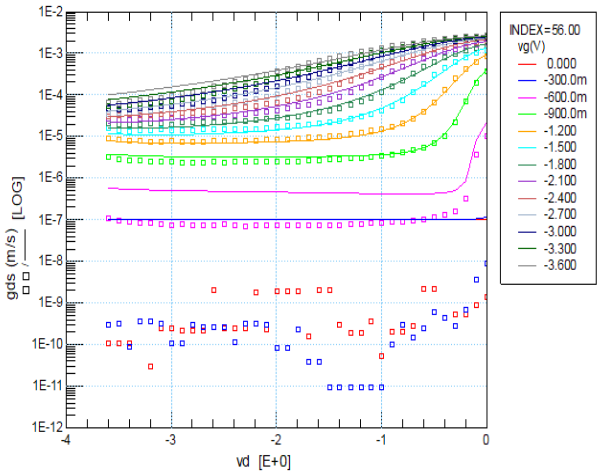


gm = f(Vg) @ Vd = -3.3V

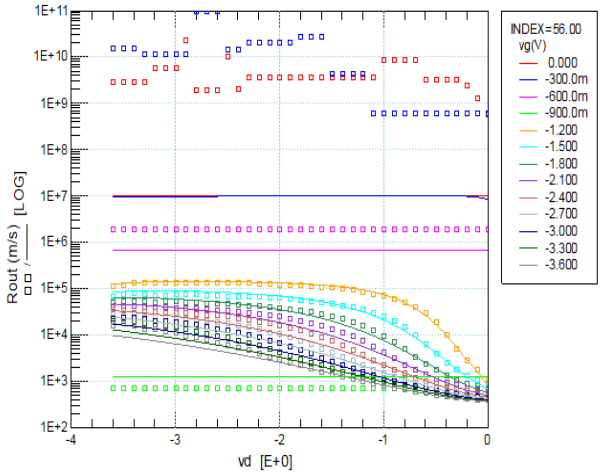
IdVd, T = 27°C



Id = f(Vd)

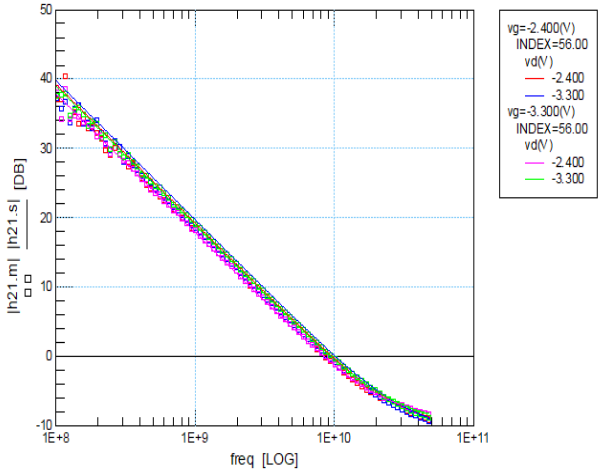


gds = f(Vd)

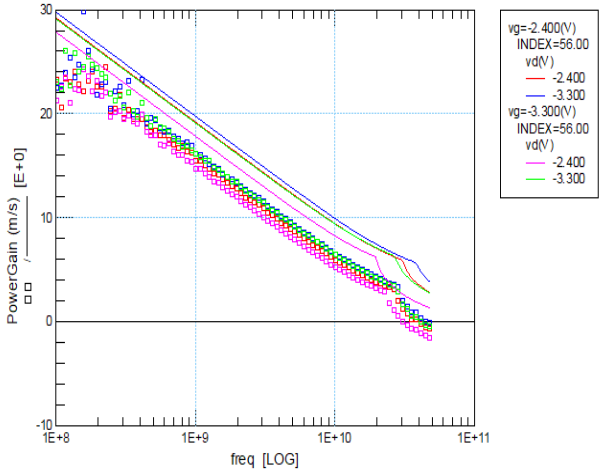


Rout = f(Vd)

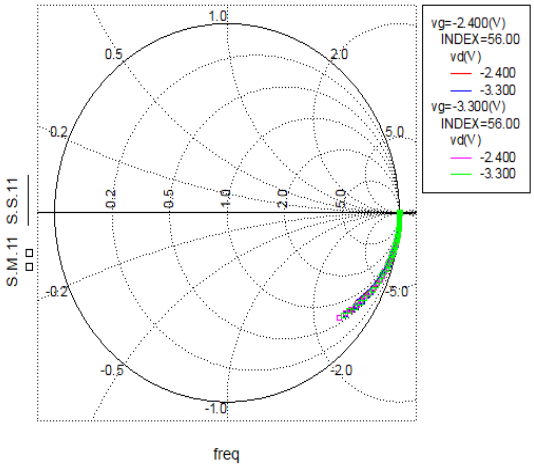
S-Parameter, T = 27°C



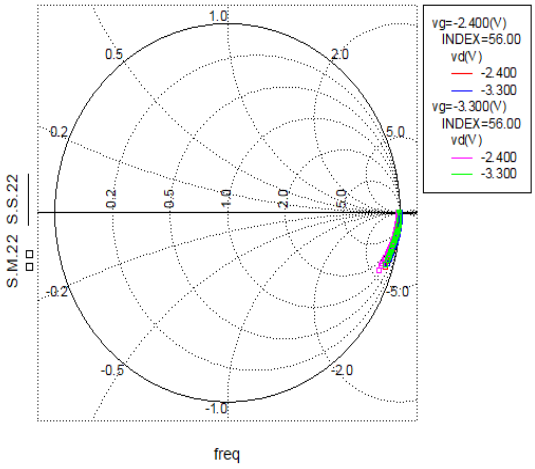
|h21| = f(Vg,Vd)



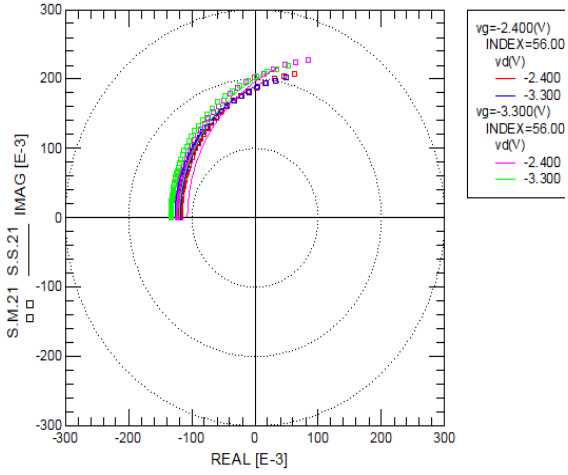
MSG/MAG = f(Vg,Vd)



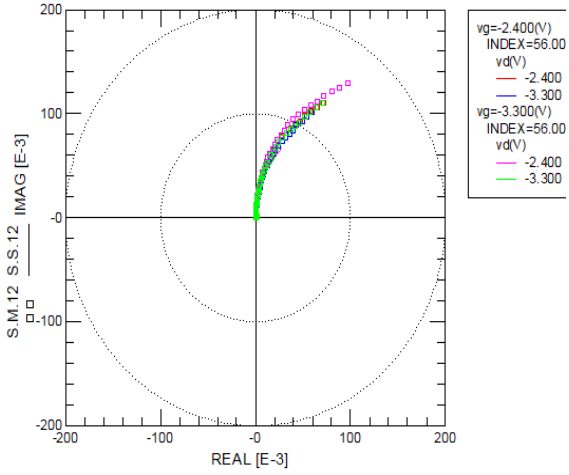
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

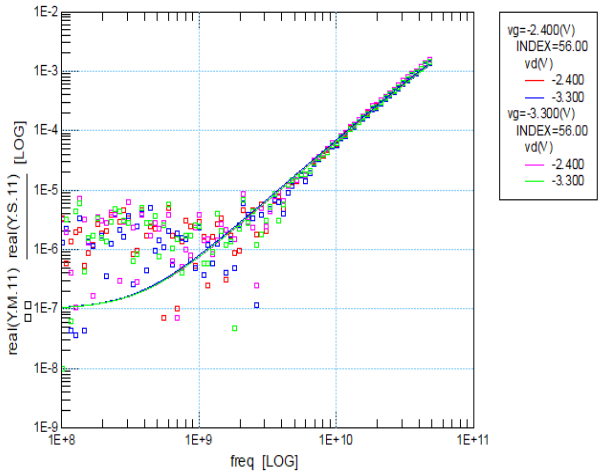


S21 = f(Vg,Vd)

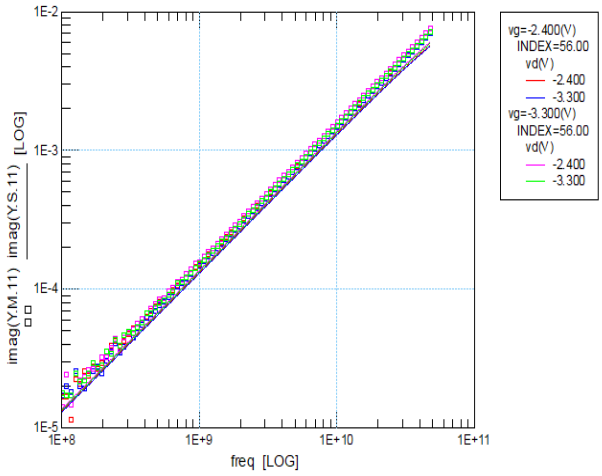


S12 = f(Vg,Vd)

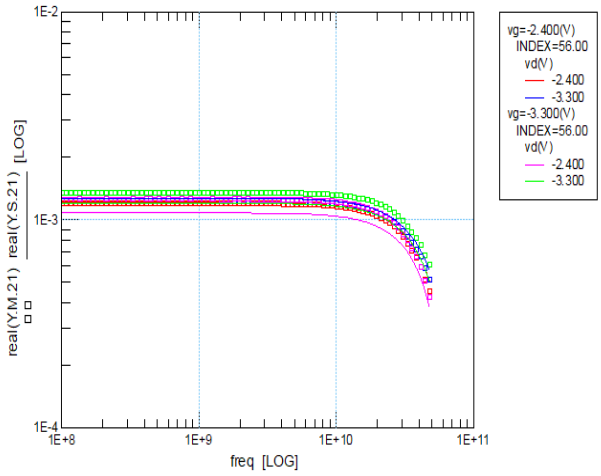
Y-Parameter, T = 27°C



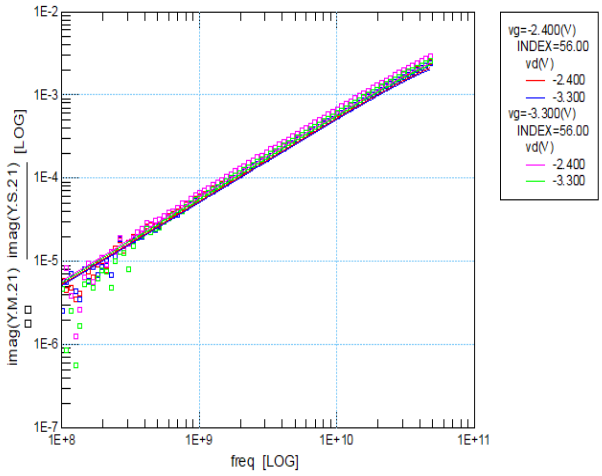
re(Y11) = f(Vg,Vd)



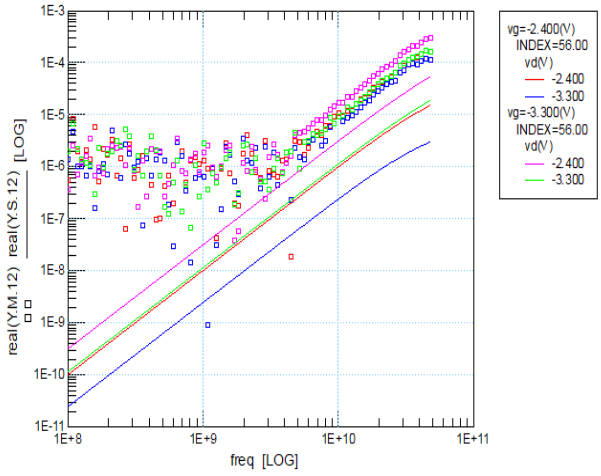
im(Y11) = f(Vg,Vd)



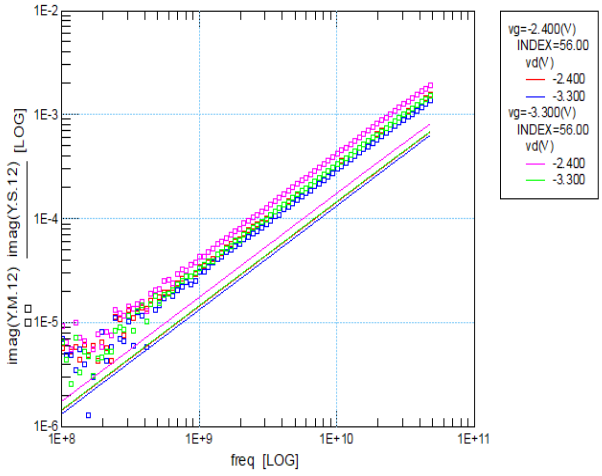
re(Y21) = f(Vg,Vd)



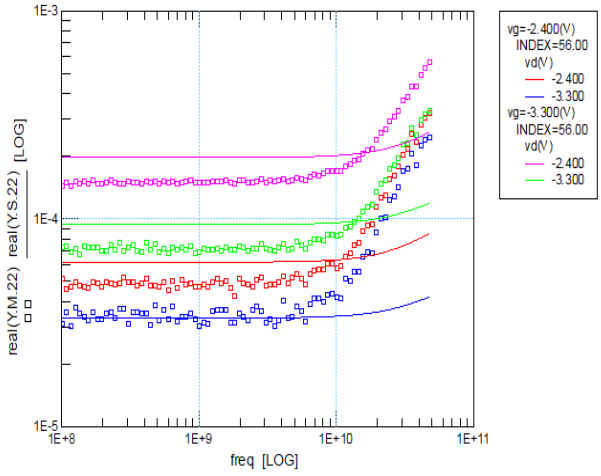
im(Y21) = f(Vg,Vd)



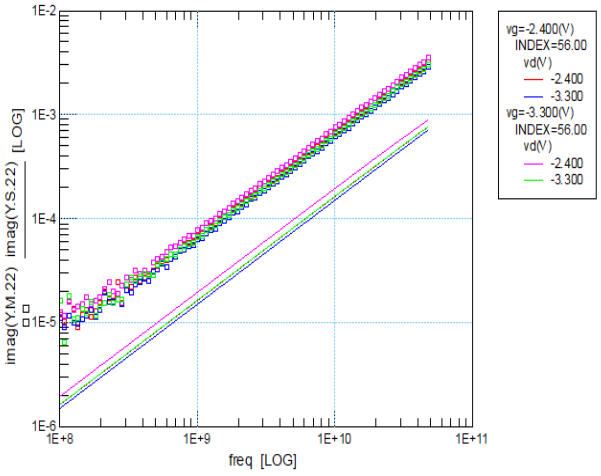
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

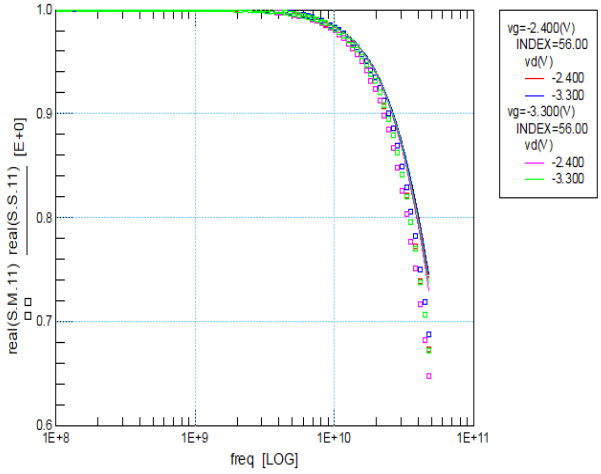


re(Y22) = f(Vg,Vd)

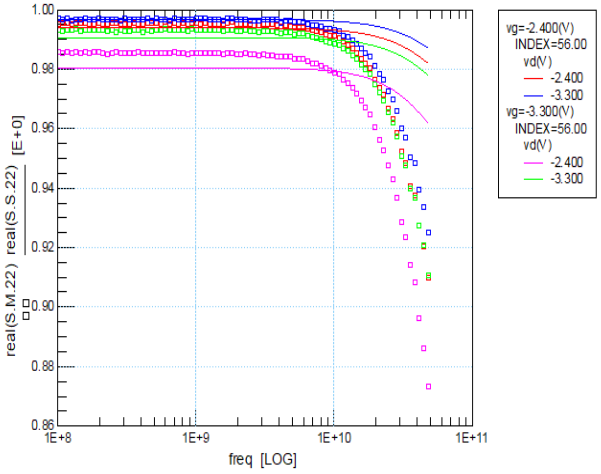


im(Y22) = f(Vg,Vd)

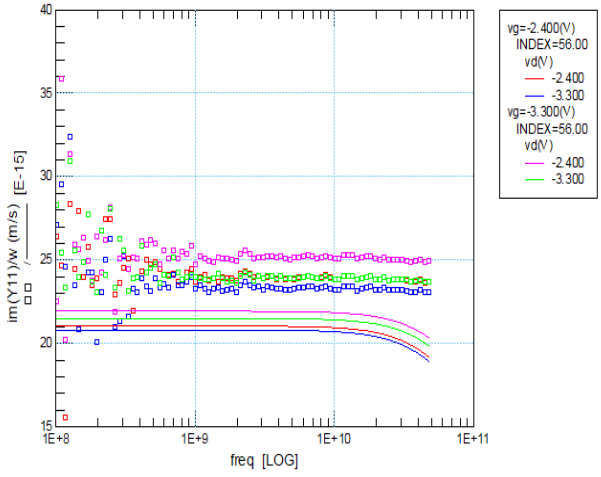
Verify Plots, T = 27°C



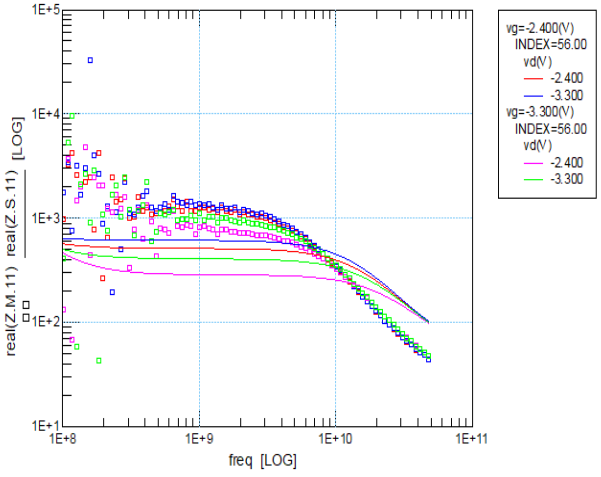
re(S11) = f(Vg,Vd)



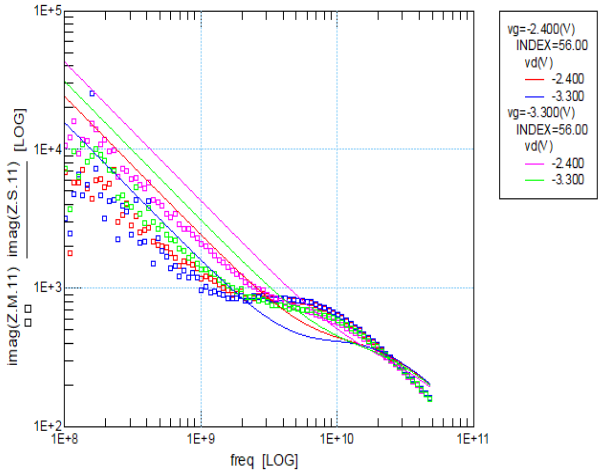
re(S22) = f(Vg,Vd)



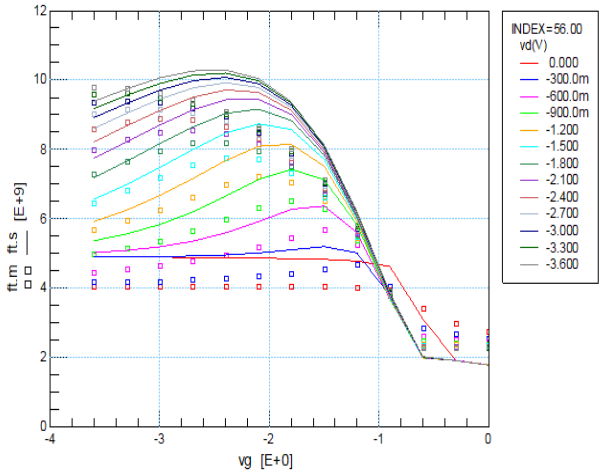
im(Y11) divided by ω



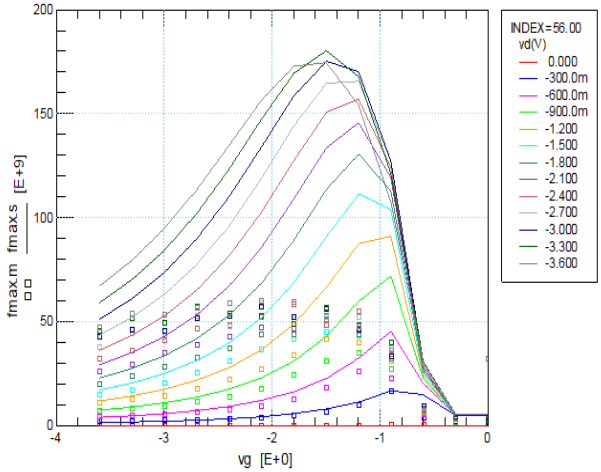
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



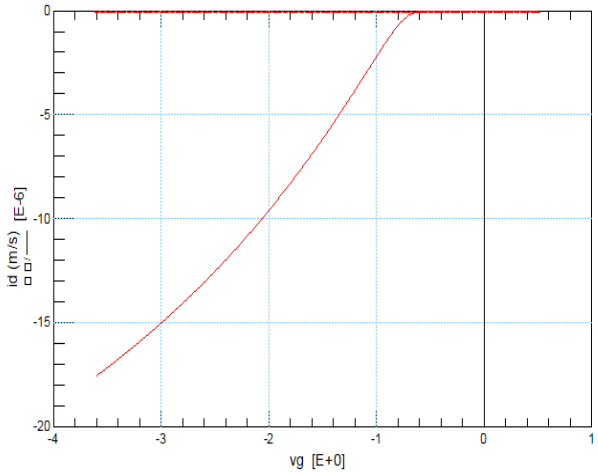
fT_vg = f(Vg,Vd)



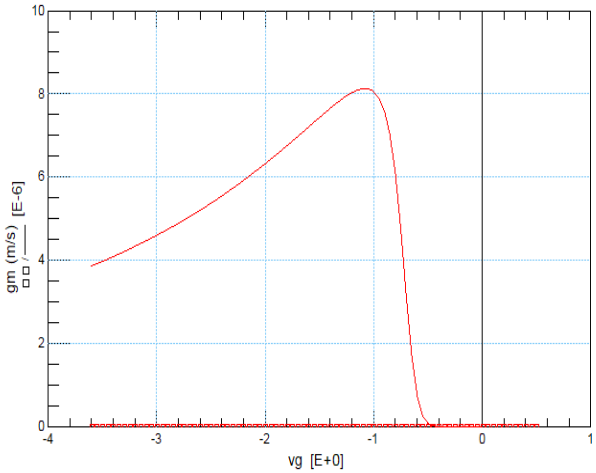
fmax_vg = f(Vg,Vd)

5.13 NF04_WF05u0_L10u0_S453A_21

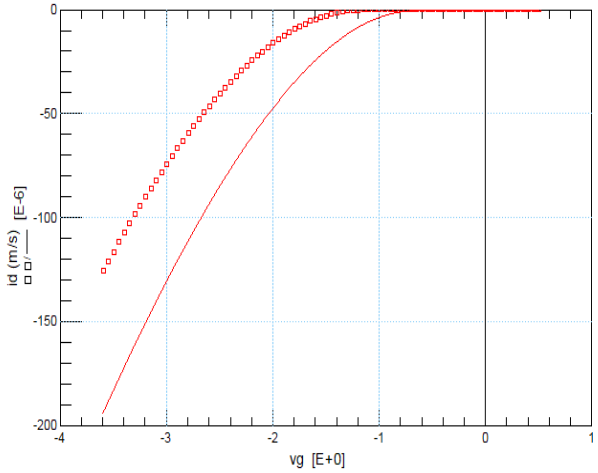
IdVg, T = 27°C



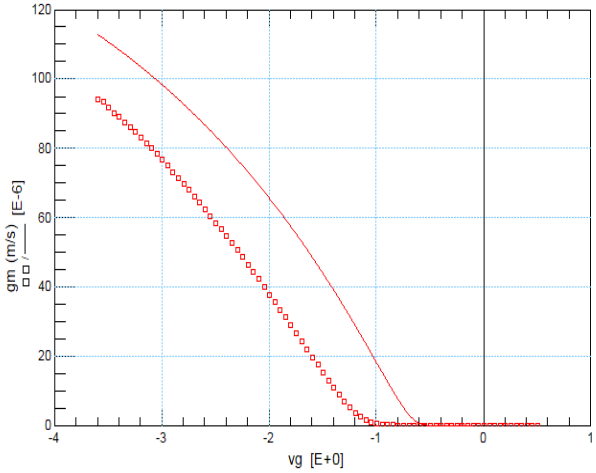
Id = f(Vg) @ Vd = -0.1V



gm = f(Vg) @ Vd = -0.1V

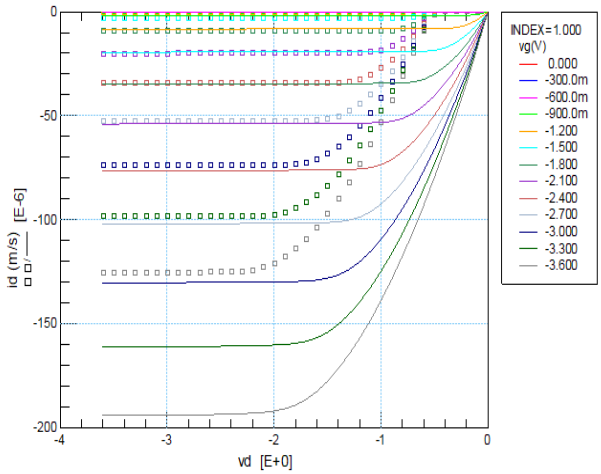


Id = f(Vg) @ Vd = -3.3V

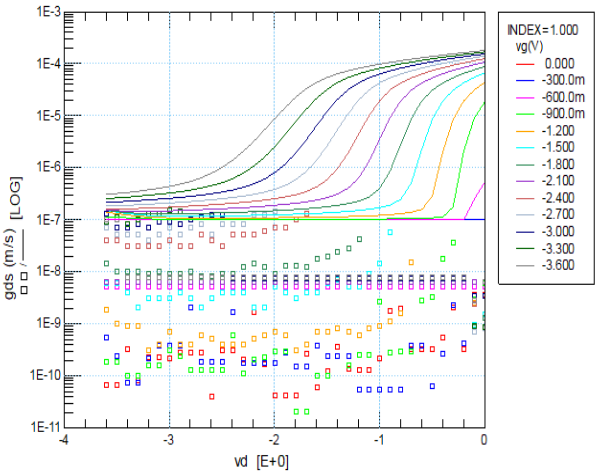


gm = f(Vg) @ Vd = -3.3V

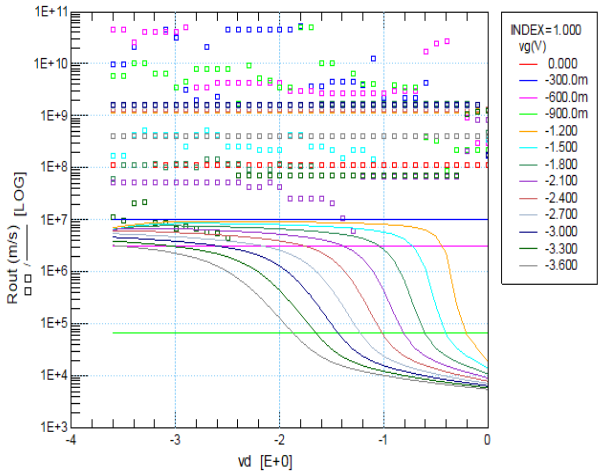
IdVd, T = 27°C



Id = f(Vd)

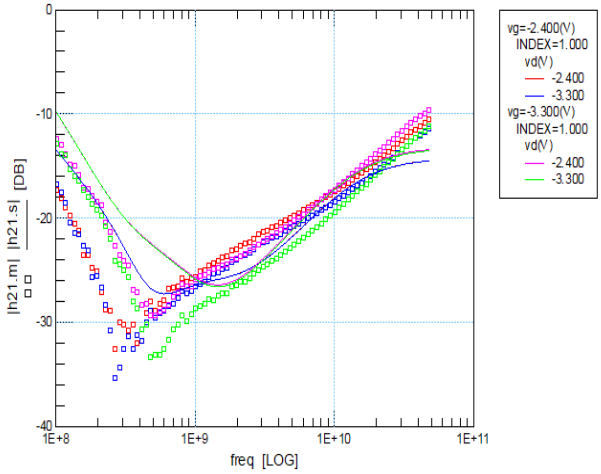


gds = f(Vd)

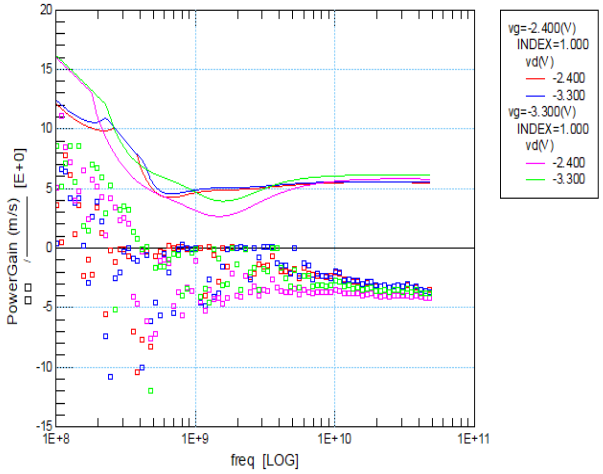


Rout = f(Vd)

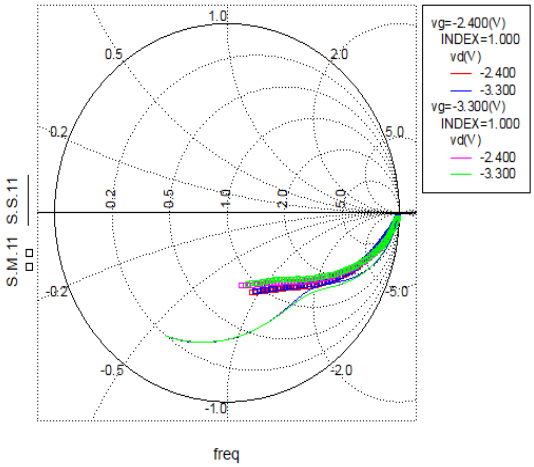
S-Parameter, T = 27°C



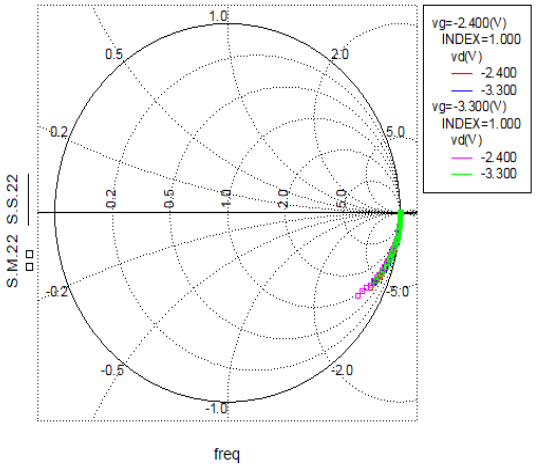
|h21| = f(Vg,Vd)



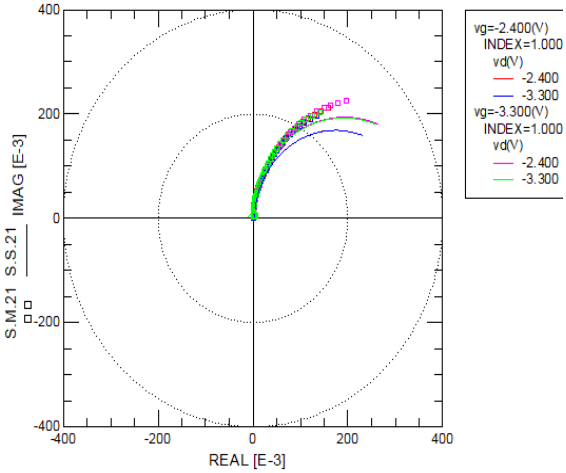
MSG/MAG = f(Vg,Vd)



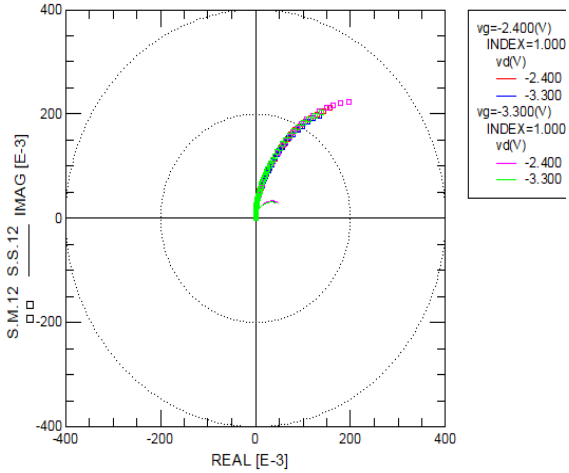
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

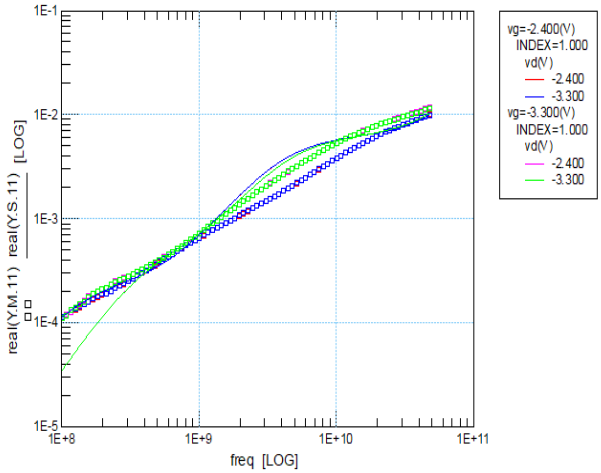


S21 = f(Vg,Vd)

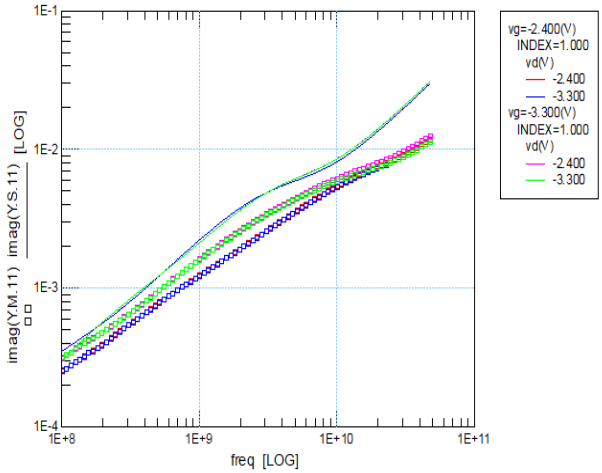


S12 = f(Vg,Vd)

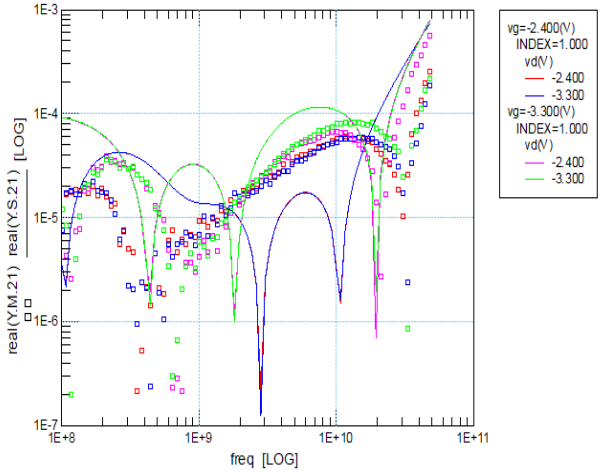
Y-Parameter, T = 27°C



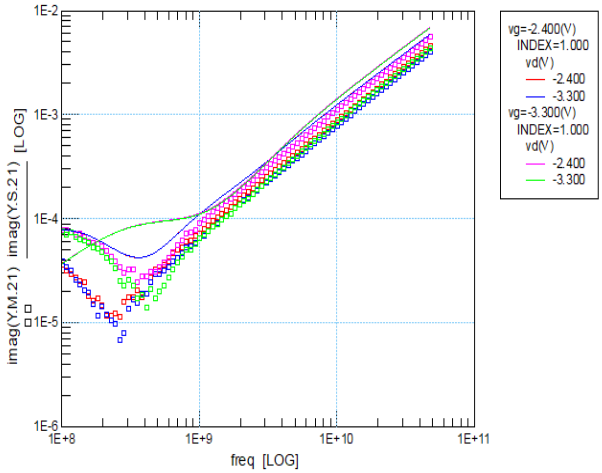
re(Y11) = f(Vg,Vd)



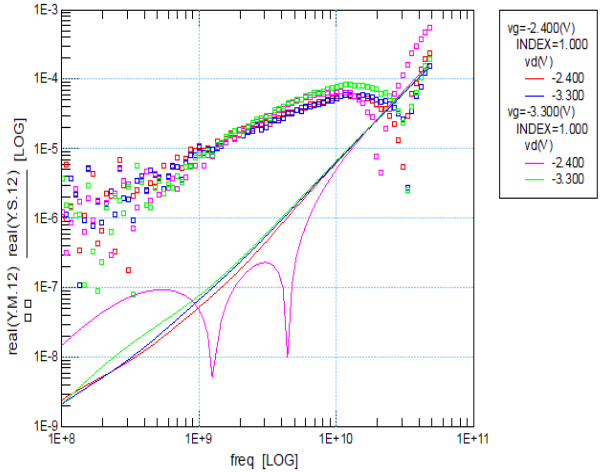
im(Y11) = f(Vg,Vd)



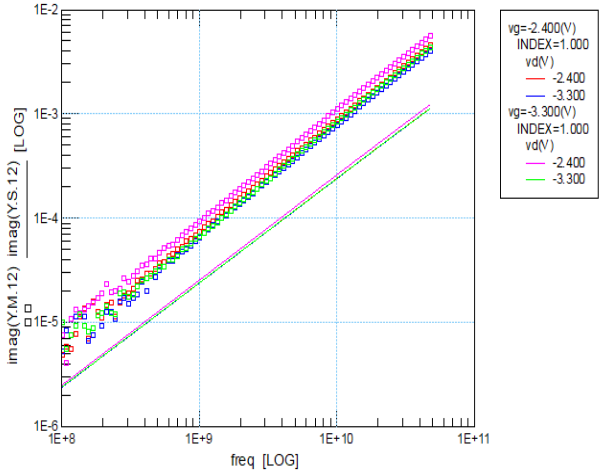
re(Y21) = f(Vg,Vd)



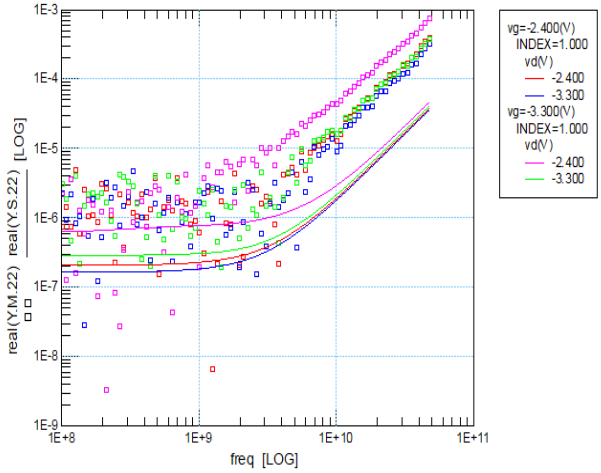
im(Y21) = f(Vg,Vd)



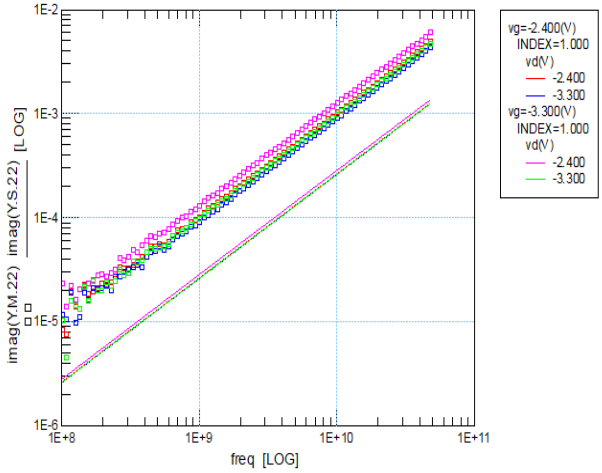
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

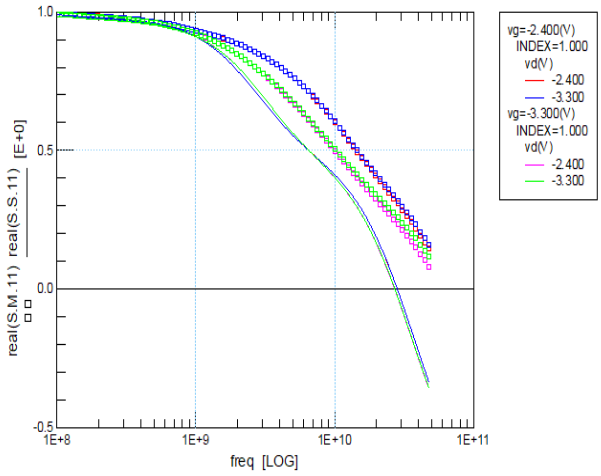


re(Y22) = f(Vg,Vd)

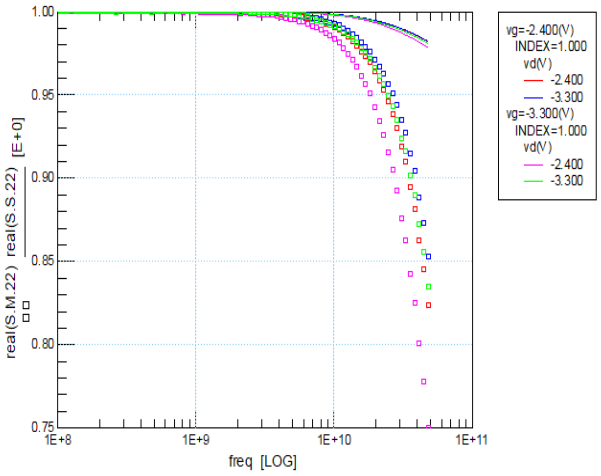


im(Y22) = f(Vg,Vd)

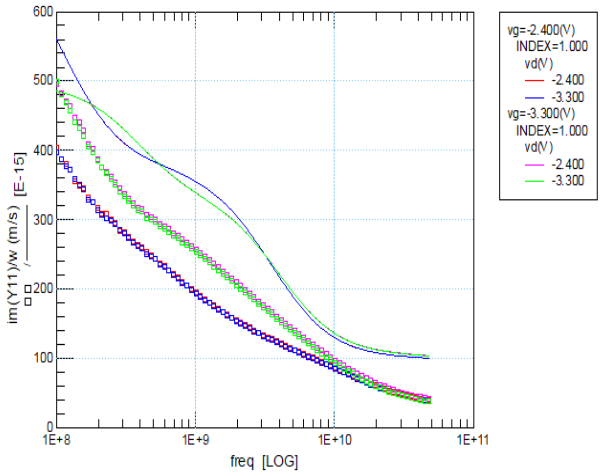
Verify Plots, T = 27°C



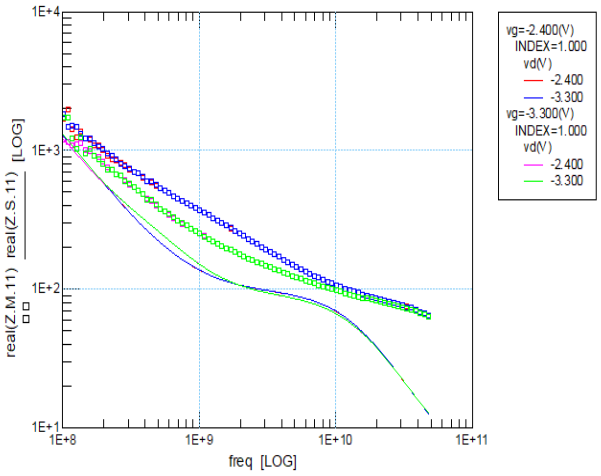
re(S11) = f(Vg,Vd)



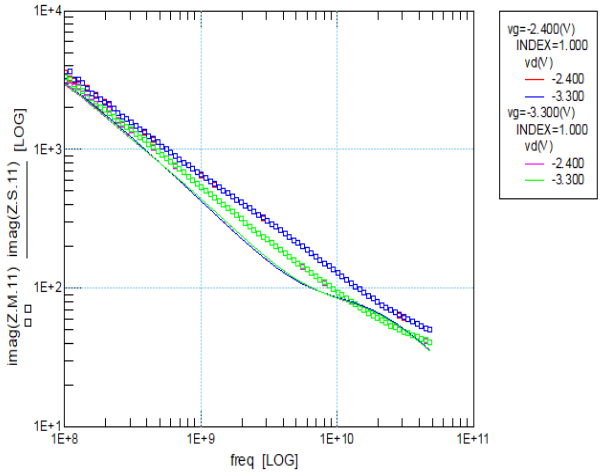
re(S22) = f(Vg,Vd)



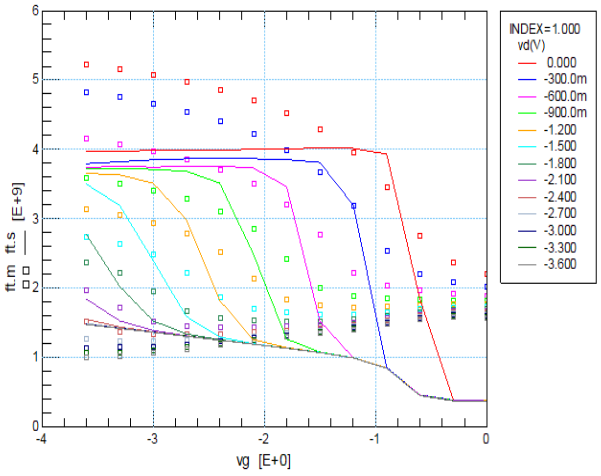
im(Y11) divided by ω



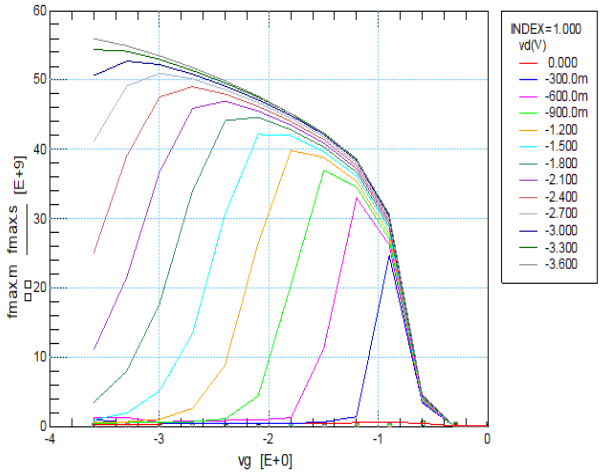
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



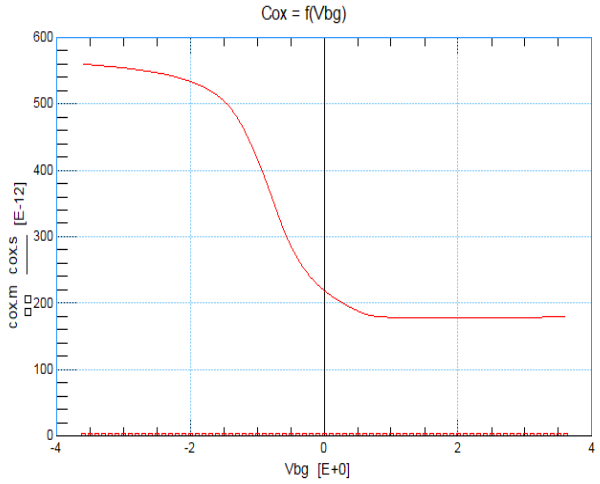
fT_vg = f(Vg,Vd)



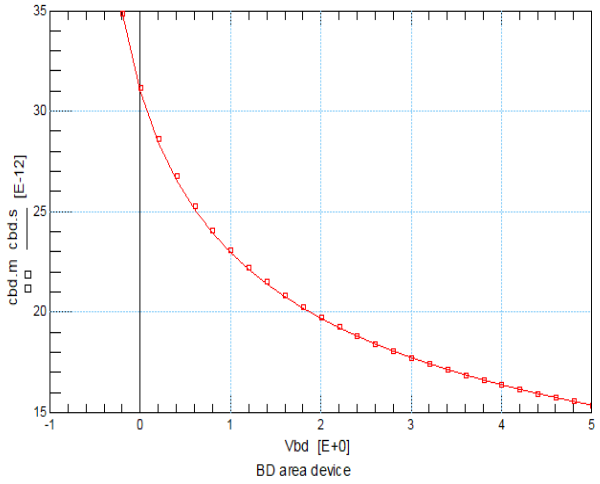
fmax_vg = f(Vg,Vd)

6 Capacitances

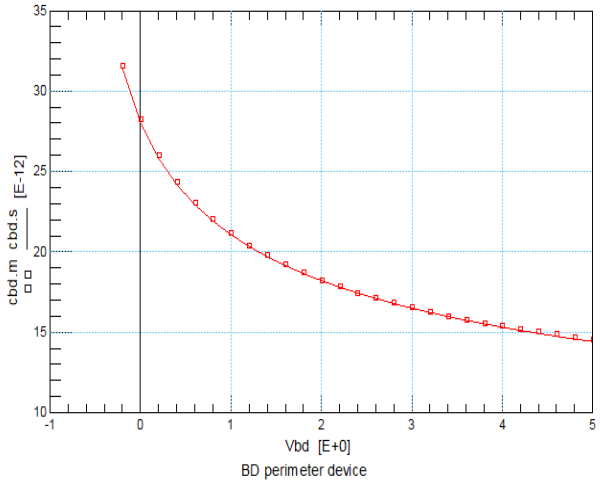
6.1 Gate Oxide Capacitance



6.2 Junction Capacitance Bulk - Drain

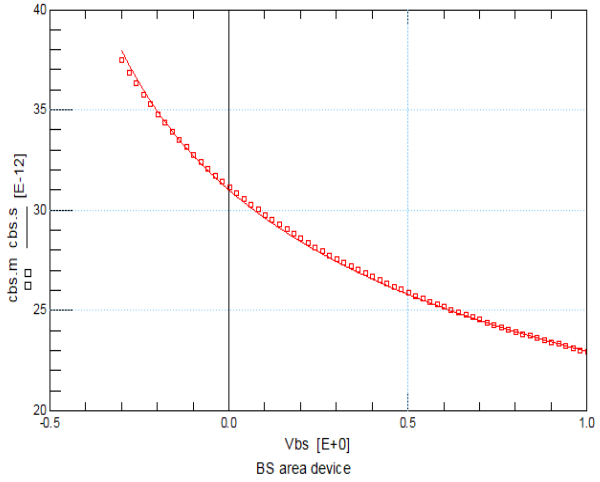


Area

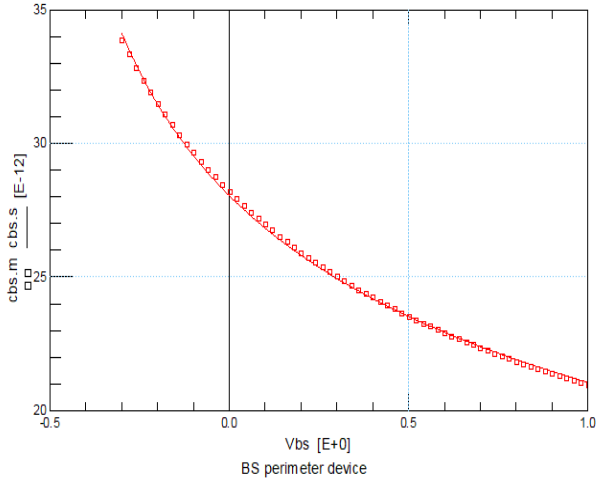


Perimeter

6.3 Junction Capacitance Bulk - Source

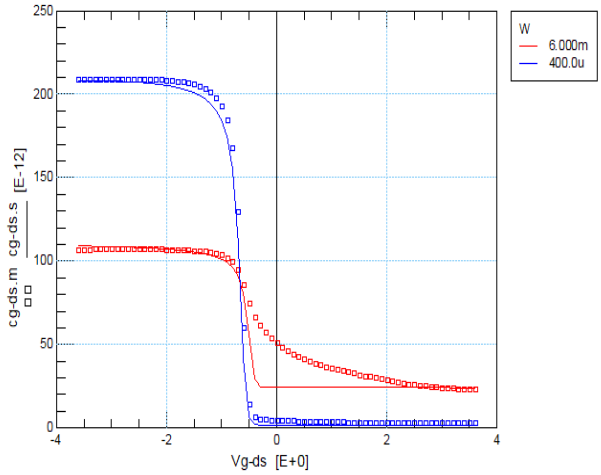


Area

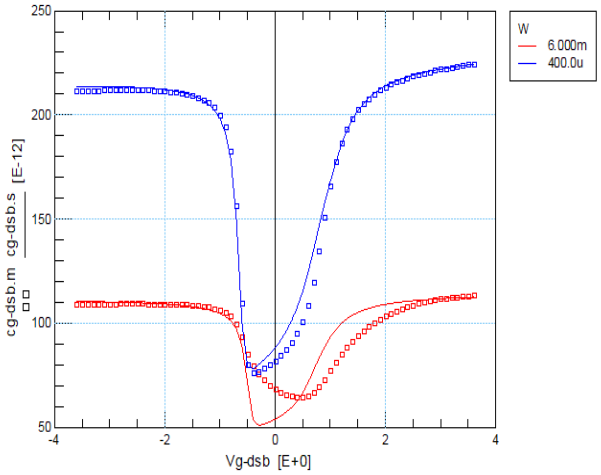


Perimeter

6.4 Overlap Capacitance



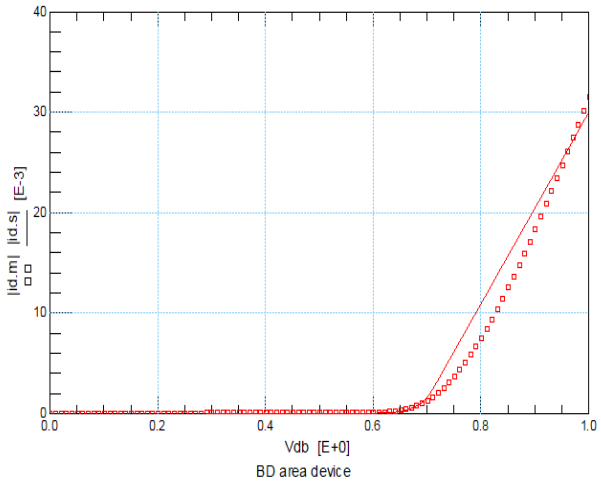
Gate - Drain,Source



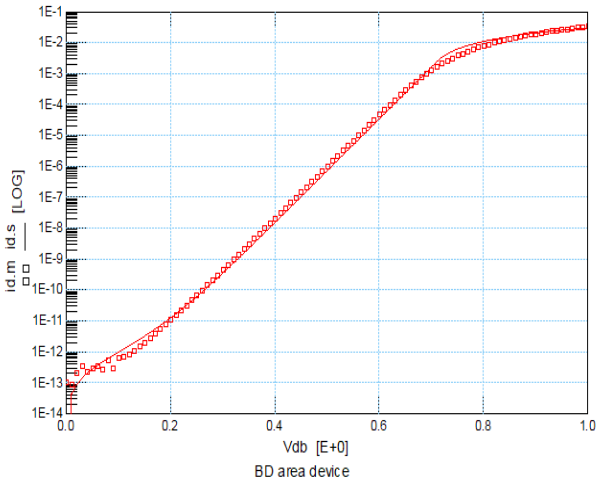
Gate - Drain,Source,Bulk

7 Diodes

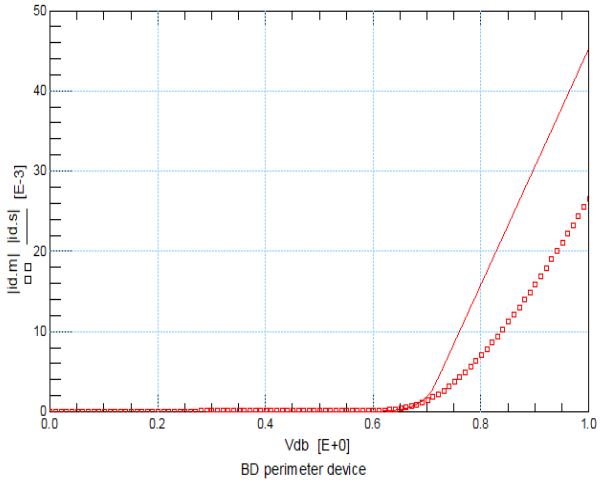
7.1 Diode Bulk - Drain



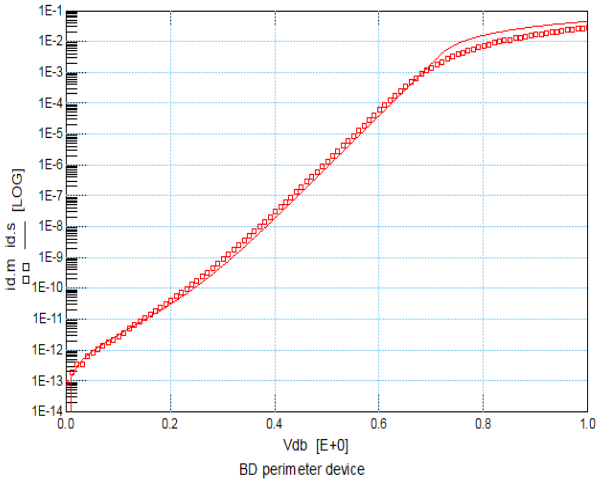
Area



Area

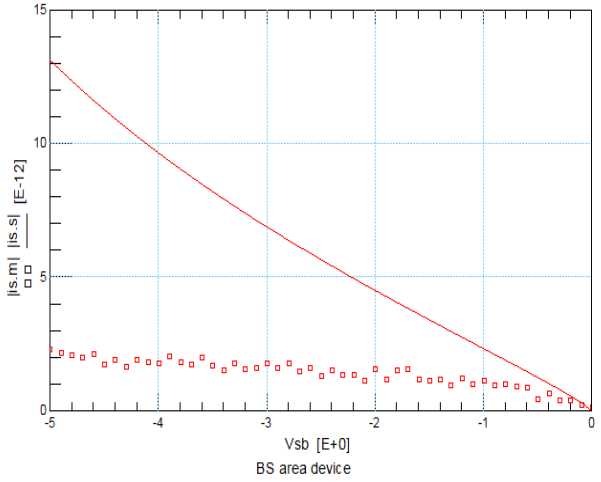


Perimeter

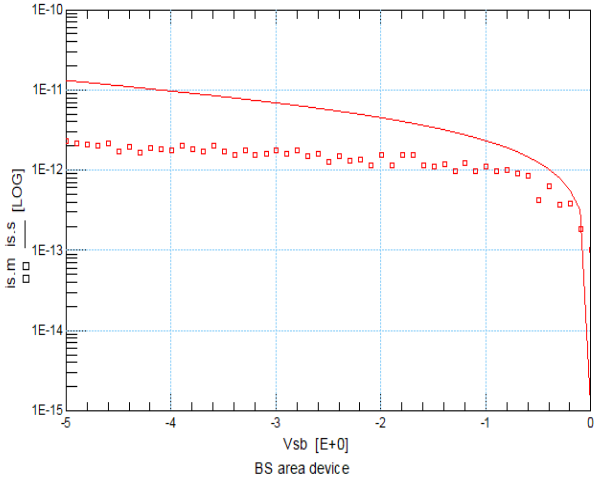


Perimeter

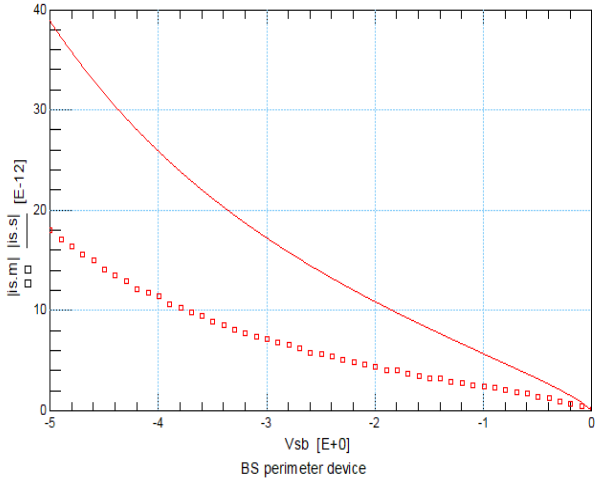
7.2 Diode Bulk - Source



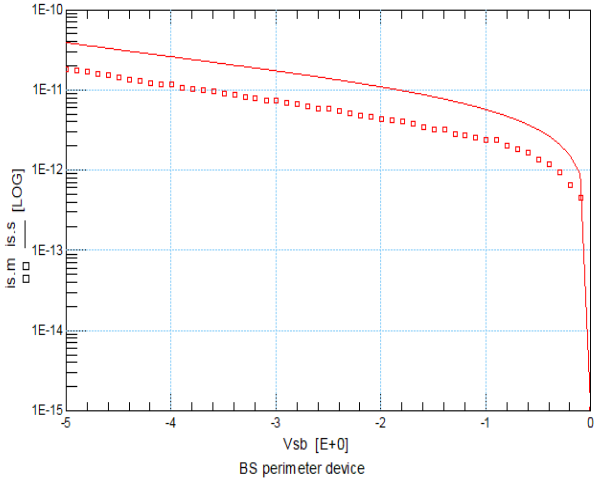
Area



Area



Perimeter

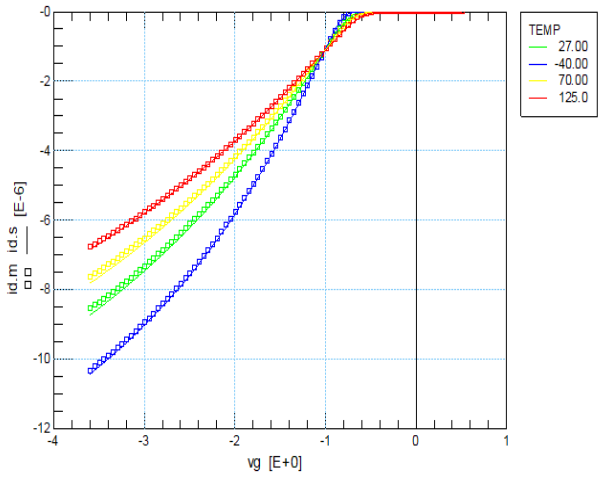


Perimeter

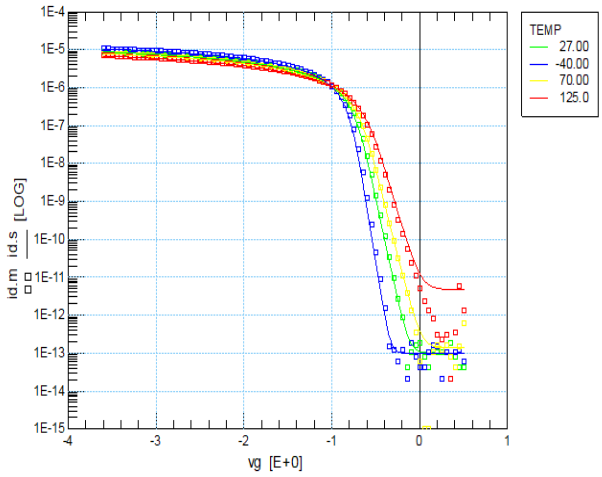
8 Temperature Measurement

8.1 Transistors

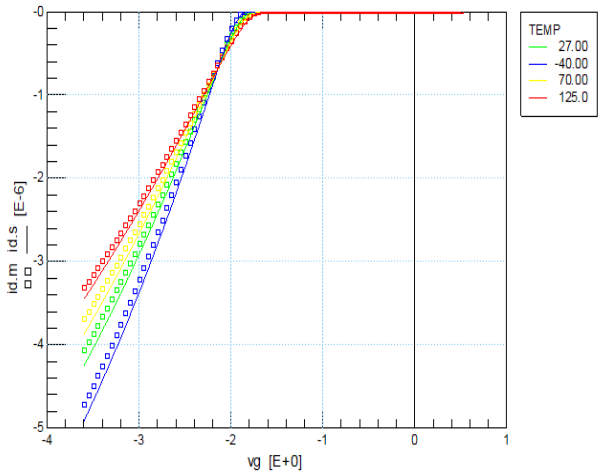
8.1.1 W10u0_L10u0_S562_3



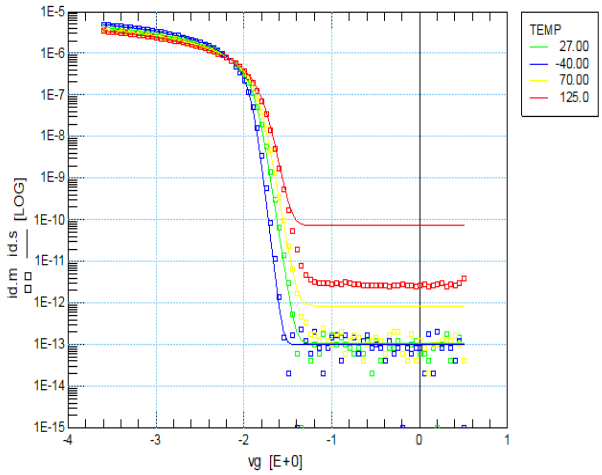
Id = f (vg, T) @ low Vds, Vbs = 0



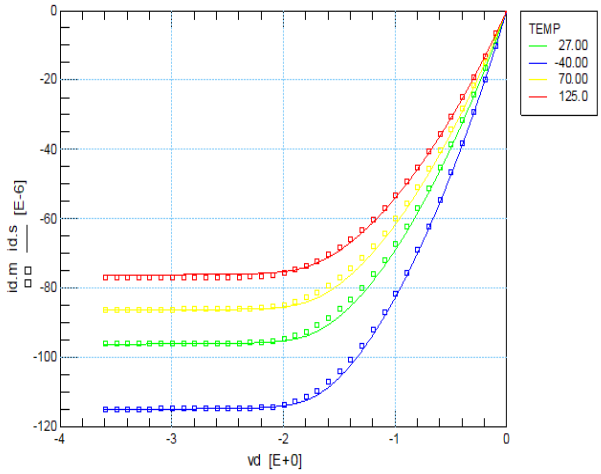
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

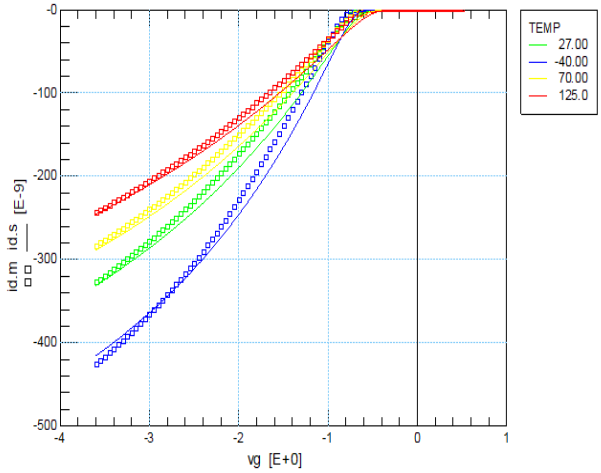


log(Id) = f (vg, T) @ low Vds, Vbs = max

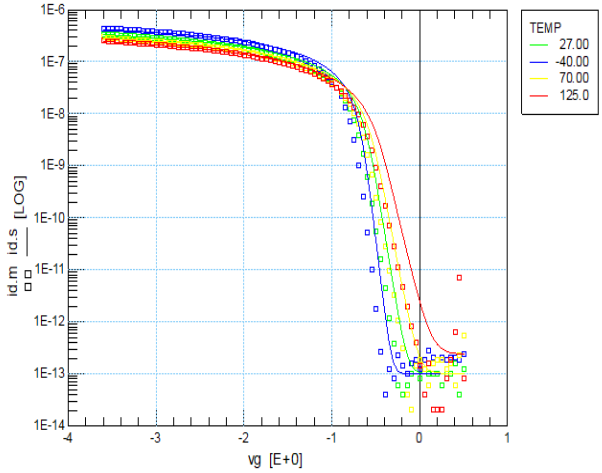


Id = f (Vd, Vg, T) @ Vbs = 0

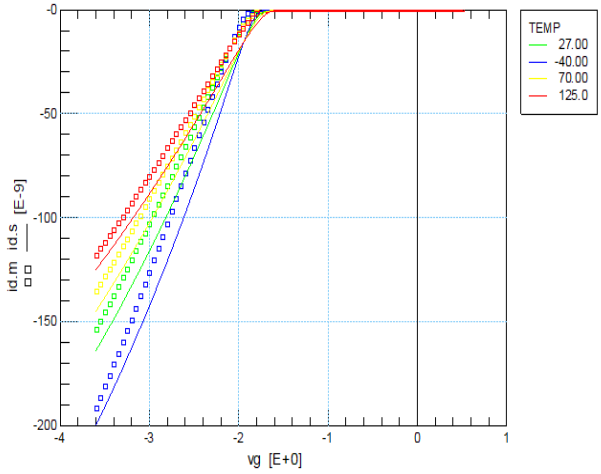
8.1.2 W0u3_L10u0_S555_4



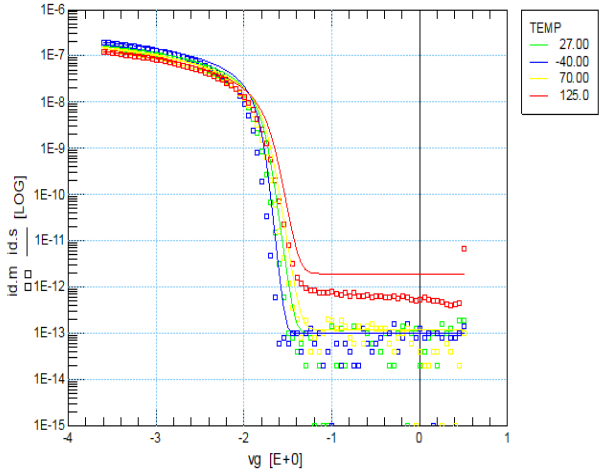
Id = f (vg, T) @ low Vds, Vbs = 0



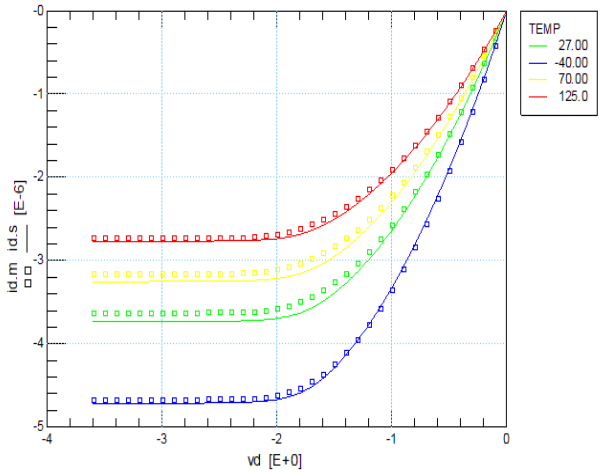
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

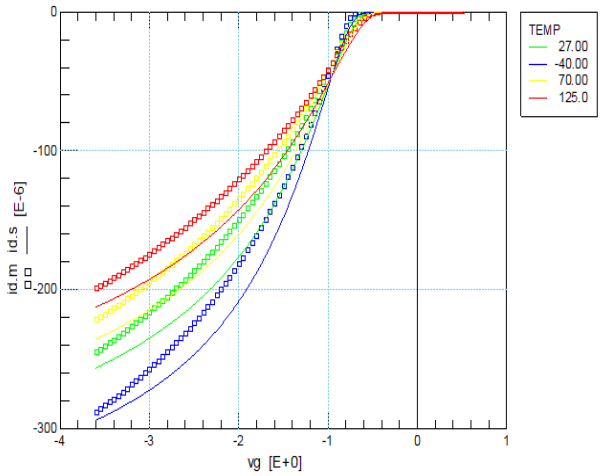


log(Id) = f (vg, T) @ low Vds, Vbs = max

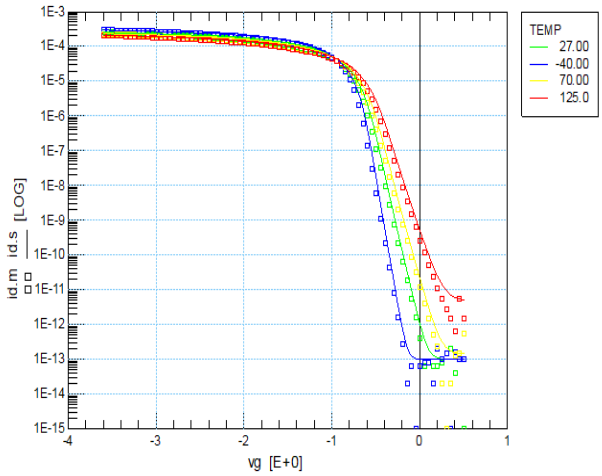


Id = f (Vd, Vg, T) @ Vbs = 0

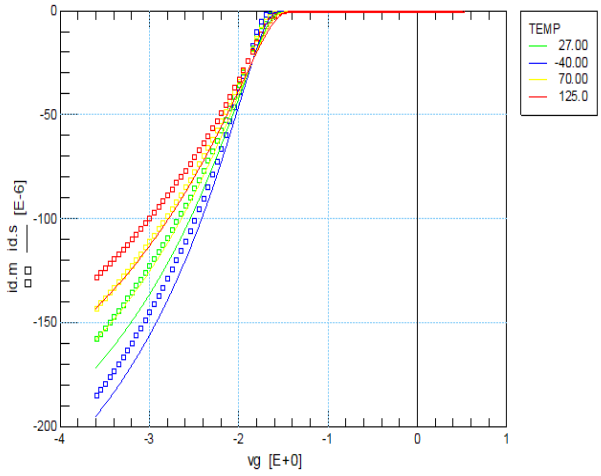
8.1.3 W10u0_L0u35_S561_3



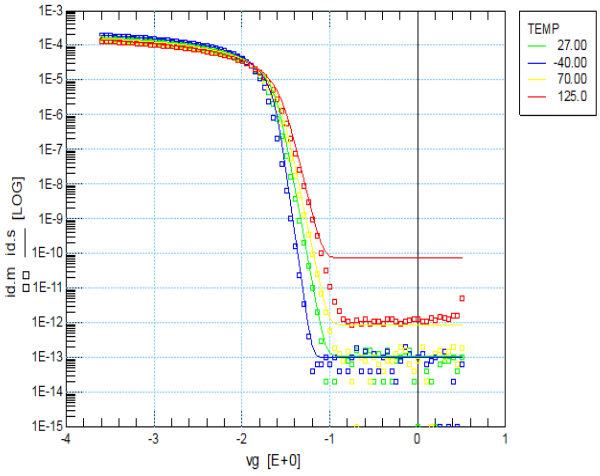
Id = f (vg, T) @ low Vds, Vbs = 0



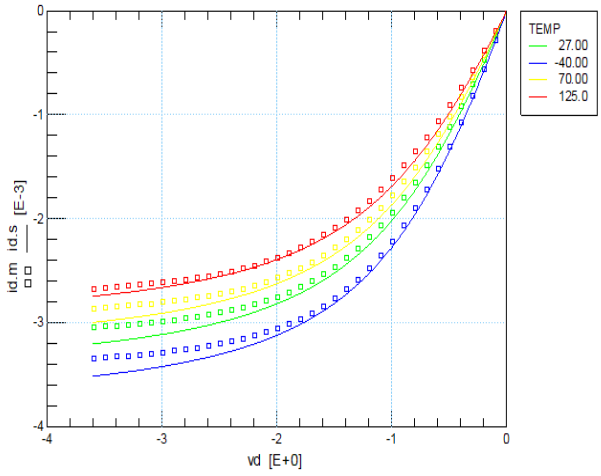
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

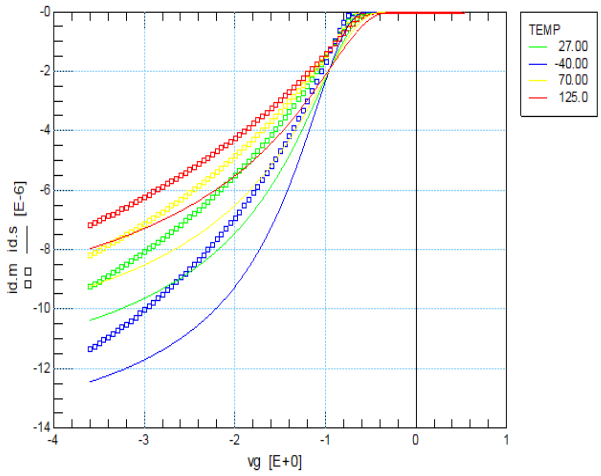


log(Id) = f (vg, T) @ low Vds, Vbs = max

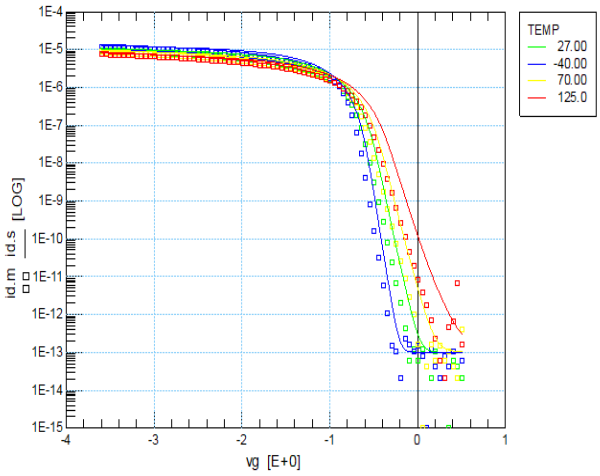


Id = f (Vd, Vg, T) @ Vbs = 0

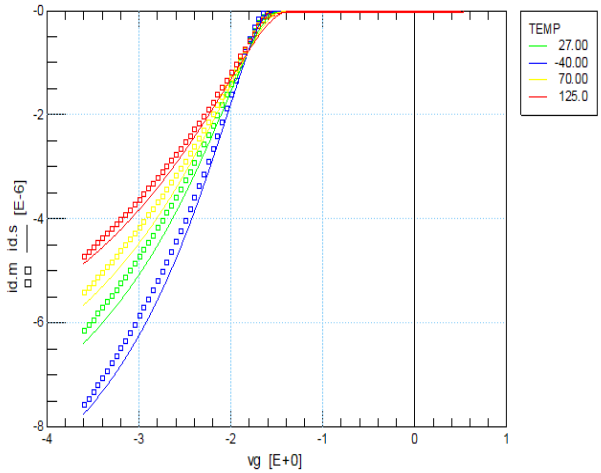
8.1.4 W0u3_L0u35_S565_3



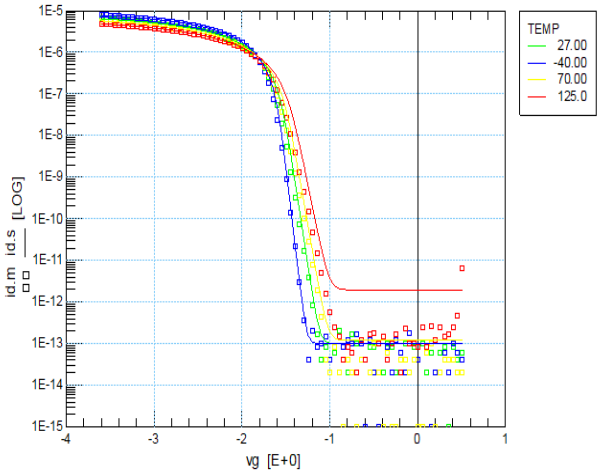
Id = f (vg, T) @ low Vds, Vbs = 0



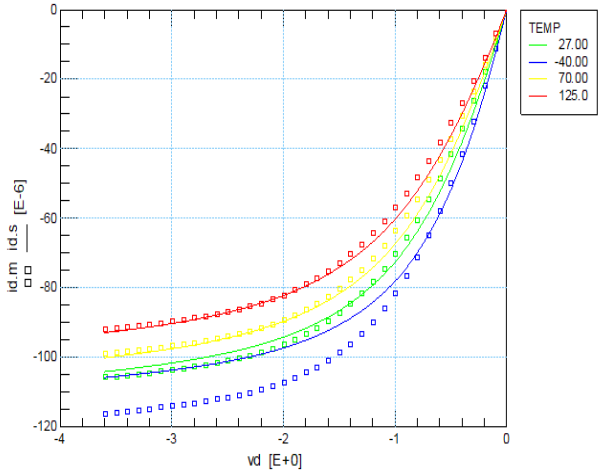
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

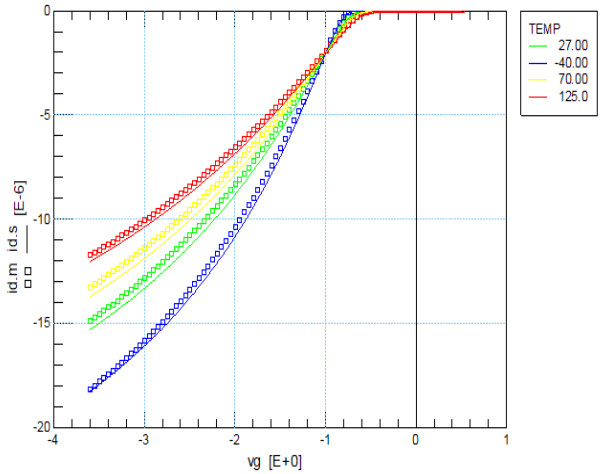


log(Id) = f (vg, T) @ low Vds, Vbs = max

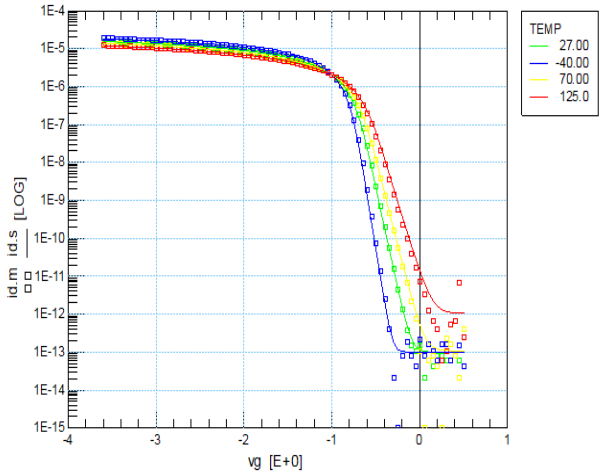


Id = f (Vd, Vg, T) @ Vbs = 0

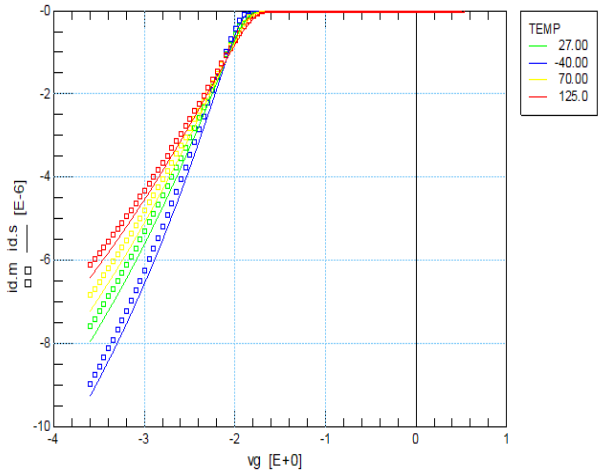
8.1.5 W02u0_L1u2_S563_3



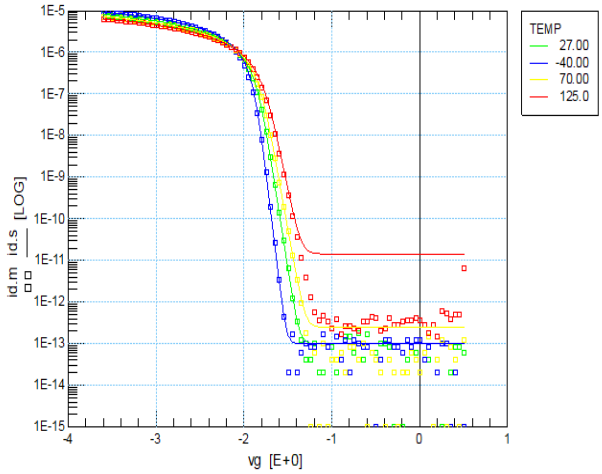
Id = f (vg, T) @ low Vds, Vbs = 0



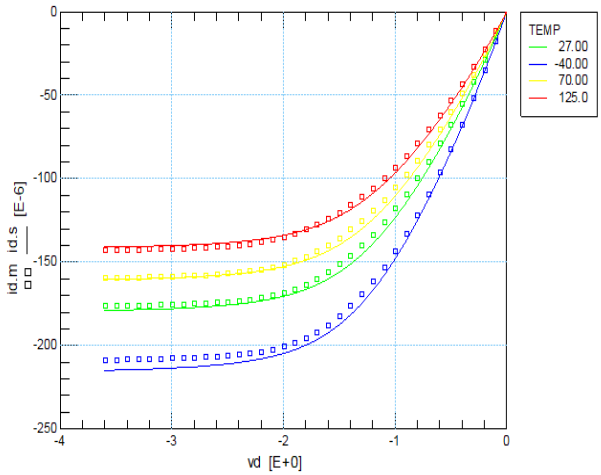
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

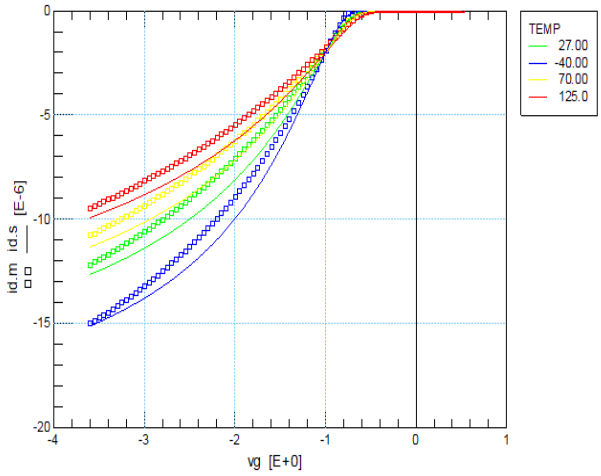


log(Id) = f (vg, T) @ low Vds, Vbs = max

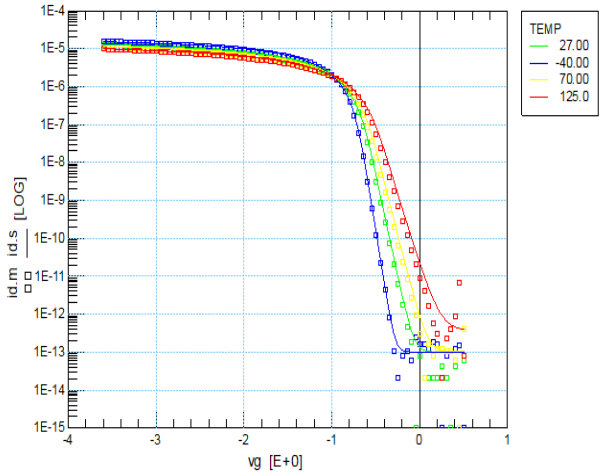


Id = f (Vd, Vg, T) @ Vbs = 0

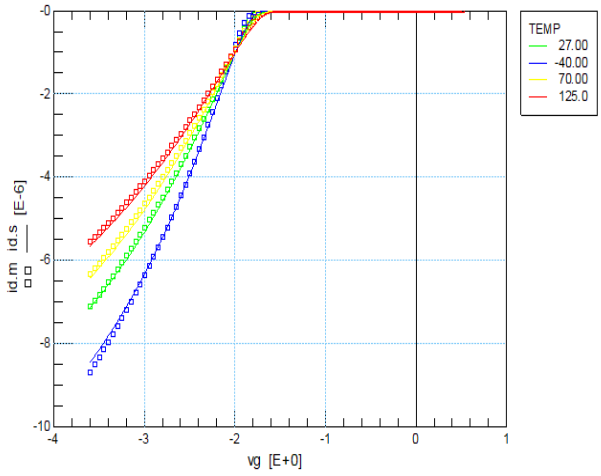
8.1.6 W0u6_L0u5_S564_3



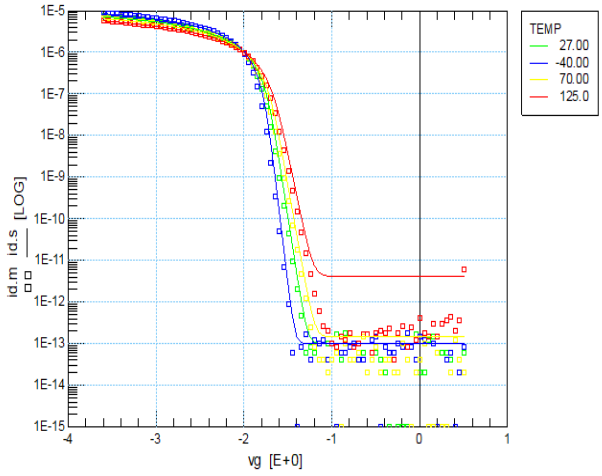
Id = f (vg, T) @ low Vds, Vbs = 0



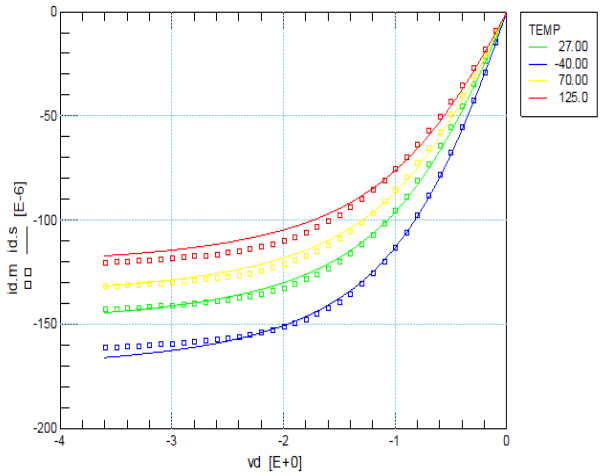
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max



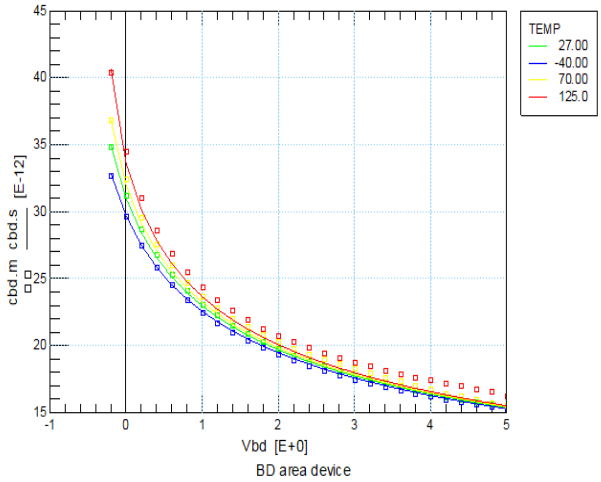
log(Id) = f (vg, T) @ low Vds, Vbs = max



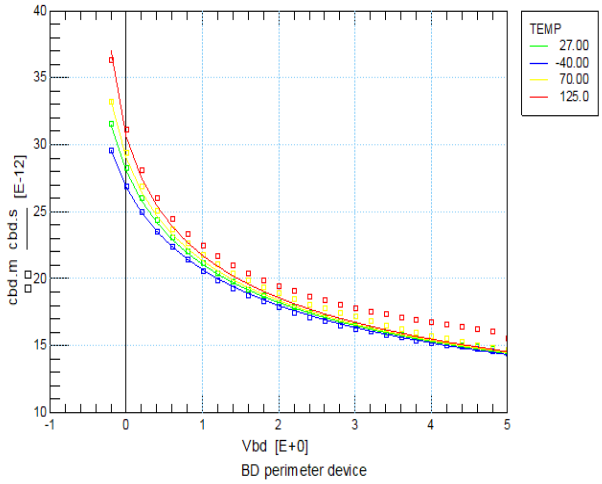
Id = f (Vd, Vg, T) @ Vbs = 0

8.2 Junction Capacitance

8.2.1 Bulk - Drain

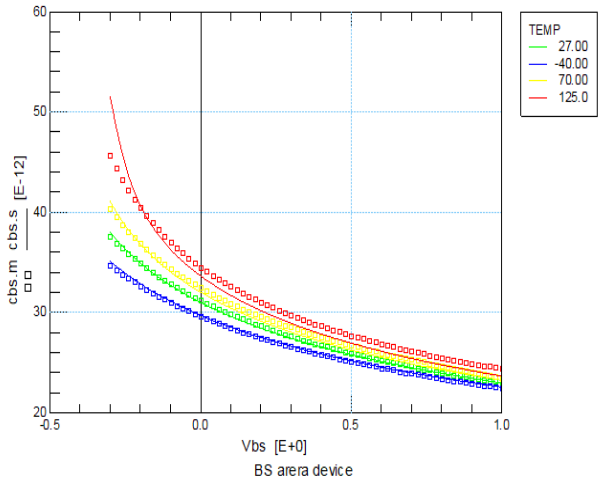


Area

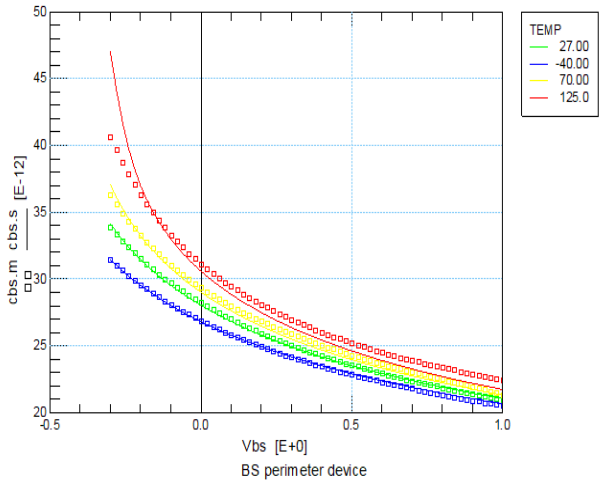


Perimeter

8.2.2 Bulk - Source



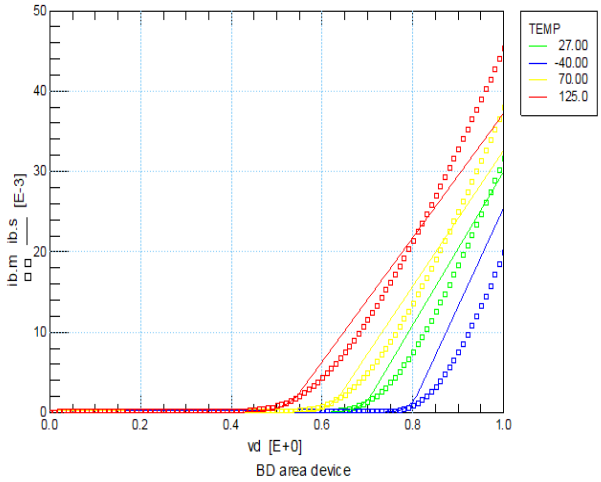
Area



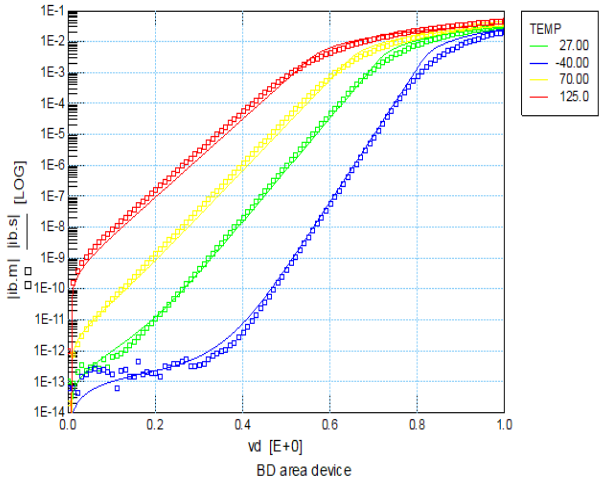
Perimeter

8.3 Parasitic Diodes

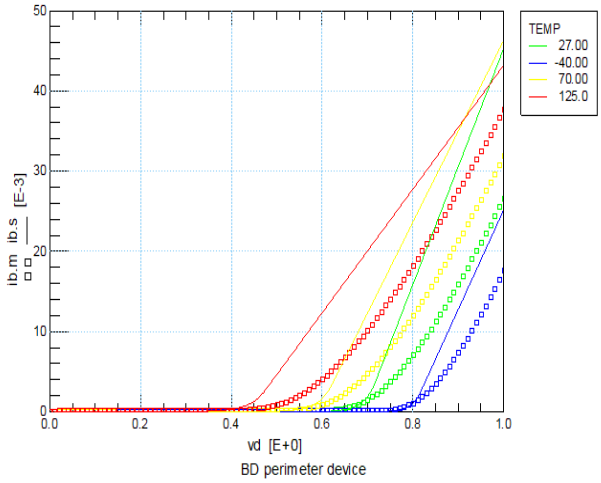
8.3.1 Bulk - Drain



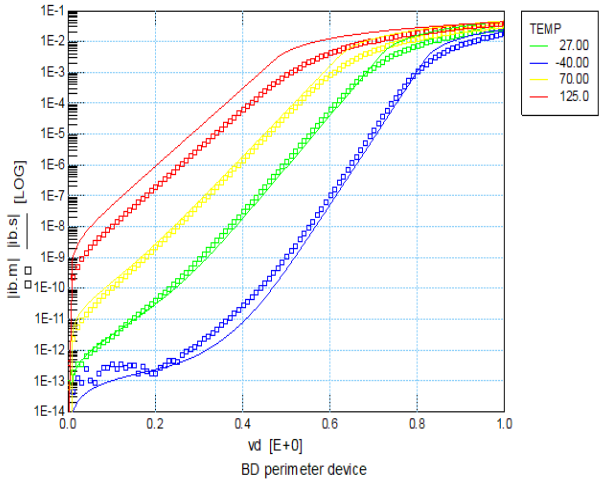
Area



Area

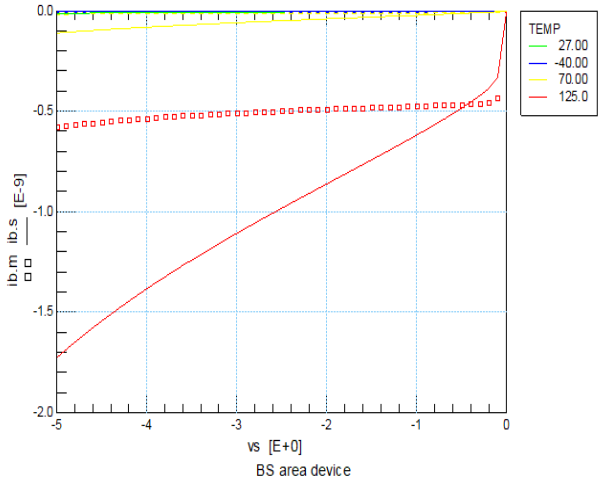


Perimeter

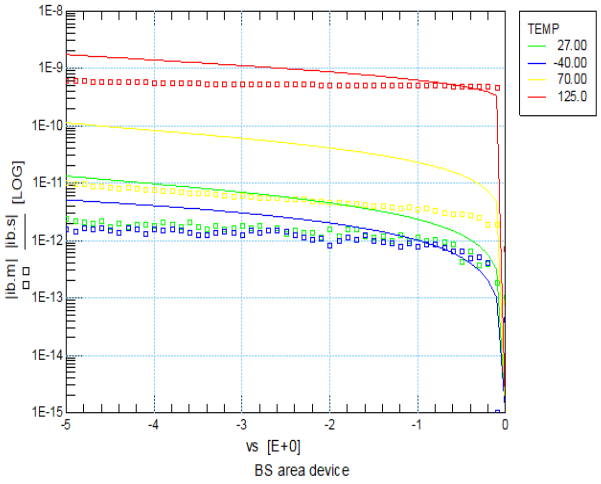


Perimeter

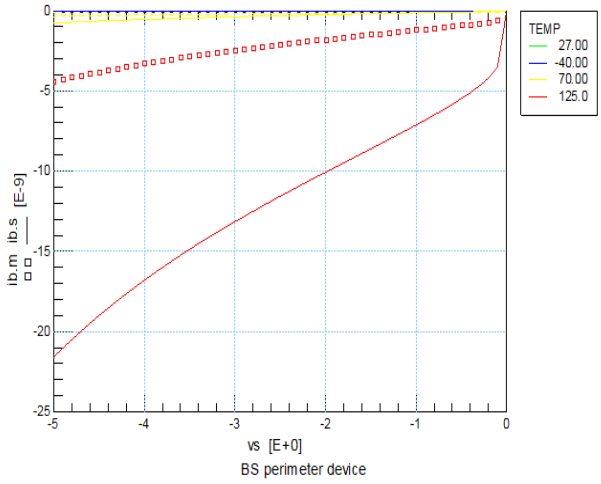
8.3.2 Bulk - Source



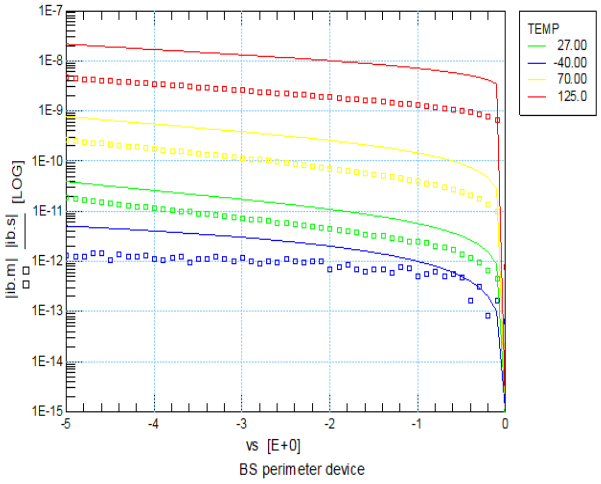
Area



Area



Perimeter



Perimeter

9 Flicker Noise

Flicker noise model parameters are taken from the SG13 simulation models.

9.1 Measurement Conditions

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

9.2 Devices

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

10 Change History

Revision	Device	Changes
190606	All	Initial version
190730	All	Adjust temperature and RF behavior
190924	Summarized Results	Legend added